The Zcash Arboretum

Foundations, deployed protocol, and frontier designs

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Introduction

The Zcash Arboretum is a non-normative guide to the mathematics, cryptography, and engineering of the Zcash protocol. It covers the foundations of Halo 2 and Orchard, deployed consensus and wallet protocols, and designs being explored beyond them. The protocol specification, applicable ZIPs, and consensus rules remain authoritative.

The order is layered. The *Math*, *Crypto*, and *Halo 2* Guides construct the foundations. The *Consensus*, *Ironwood*, and *Wallet* Guides explain the deployed system and its boundaries. The remaining parts examine the unbuilt FlyClient bridge, shielded assets, and threshold authorization.

Three reading paths cover most uses. For prerequisites, begin with the first three parts; readers new to proof systems may start with the worked example that closes the Halo 2 Guide. For the life of a shielded payment, read *Ironwood*, then *Wallet* and *Consensus*. For proposed changes, read the relevant frontier part only after its lower-layer dependencies. Each part restarts its own section numbering so that citations agree with the separately published volume.

Part I
Math Guide: Mathematical foundations for Halo 2 and Zcash Orchard

Abstract

This volume of *The Zcash Arboretum* develops the mathematics underlying the cryptography of Zcash from first principles, assuming only elementary algebra and calculus. It builds a single tower: the notation of sets, functions, and relations in which every later claim is written and spoken; the integers and their remainders modulo a prime, where an extended Euclidean algorithm turns division into multiplication by an inverse; the abstract structures—groups, rings, and fields—that name the laws this arithmetic obeys; polynomials over a field, whose scarcity of roots lets two enormous polynomials be compared at a single random point; the finite fields, their possible sizes, and their cyclic multiplicative structure; the number theory of squares, square roots, and the discrete logarithm; linear algebra and the inner products behind short algebraic receipts for long lists of secrets; the roots of unity that furnish evaluation domains and the fast Fourier transform; elliptic curves, whose point groups have no known subexponential discrete-log attack when chosen appropriately; and finally the probability, asymptotics, and model of computation required to state—and mean—a claim of 128-bit security.

The series consumes three artefacts of this tower: the Pasta curve cycle, constructed in §10; the polynomial machinery on which Halo 2 rests, developed in §5 and §9; and the security-statement language, fixed in §11. A first reading takes the sections in order; a reader after one artefact may enter at its section and follow its backward references down the tower.

1 Notation: sets, functions, and relations

Open the specification of a modern cryptographic protocol — a signature scheme, say, or a scheme for proving a statement without revealing why it is true — and try to read a single line of it aloud. Before any mathematics has happened, the page presents three alphabets at once: double-struck capitals such as $\mathbb{Z}$ and $\mathbb{F}_q$ naming entire systems of numbers, Greek letters such as λ and ρ naming parameters and challenges, and a thicket of operator symbols — $\oplus$, $\|$, $\xleftarrow{\$}$, $[k]P$ — each with a fixed meaning and a spoken name. A reader who cannot pronounce a line cannot discuss it with a colleague, and a reader who mistakes $\subseteq$ for $\in$, or swaps the order of two quantifiers, has misread the claim itself.

This section fixes the small stock of set-theoretic and logical language in which the whole series is written, and it closes by paying the opening debt in full: three reference tables listing every recurring symbol with its spoken form (§1.5). A reader comfortable with elementary mathematics may skim the prose and keep the tables to hand.

1.1 Sets

A *set* is a collection of distinct objects, its *elements*; one writes $x \in A$ for membership and $x \notin A$ for its negation. A set may be given by listing, $\{a, b, c\}$, or by *set-builder* notation $\{x \in A : P(x)\}$ ("the x in A such that $P(x)$"). One writes $A \subseteq B$ when every element of A lies in B, and $A = B$ exactly when $A \subseteq B$ and $B \subseteq A$ — so proving two sets equal means proving two inclusions. The usual operations are union $A \cup B$, intersection $A \cap B$, difference $A \setminus B$, and the empty set $\emptyset$. The *Cartesian product* is $A \times B = \{(a, b) : a \in A,\ b \in B\}$, and A^n is the set of n-tuples over A. For a finite set A, $|A|$ denotes its number of elements. The standard number systems are the naturals $\mathbb{N}$, the integers $\mathbb{Z}$, the rationals $\mathbb{Q}$, the reals $\mathbb{R}$, and the complex numbers $\mathbb{C}$, with $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$; we construct the underlying arithmetic of the prime field $\mathbb{F}_p$ in §2, certify it as a field—and first call it $\mathbb{F}_p$—in §4, and treat general finite fields $\mathbb{F}_q$ in §6.

1.2 Functions

A *function* $f\colon A \to B$ assigns to each $a \in A$ a unique value $f(a) \in B$; A is the *domain* and B the *codomain*. It is *injective* (one-to-one) if $f(a) = f(a')$ implies $a = a'$; *surjective* (onto) if every $b \in B$ equals $f(a)$ for some $a \in A$; and *bijective* if both. The *composition* of $f\colon A \to B$ and $g\colon B \to C$ is $(g \circ f)(a) = g(f(a))$; composition is associative, $h \circ (g \circ f) = (h \circ g) \circ f$, both sides sending a to $h(g(f(a)))$. The *identity* on A is $\mathrm{id}_A\colon A \to A$, $\mathrm{id}_A(a) = a$.

> **Theorem 1.1** (Inverse functions). *A function $f\colon A \to B$ has a two-sided* inverse *$f^{-1}\colon B \to A$, meaning $f^{-1} \circ f = \mathrm{id}_A$ and $f \circ f^{-1} = \mathrm{id}_B$, if and only if f is bijective. When it exists, the inverse is unique.*

Proof. Suppose a two-sided inverse g exists. If $f(a) = f(a')$, applying g gives $a = g(f(a)) = g(f(a')) = a'$, so f is injective; and every $b \in B$ satisfies $b = f(g(b))$, so f is surjective. Conversely, suppose f is bijective. Each $b \in B$ has at least one preimage by surjectivity and at most one by injectivity, so we may define $g(b)$ to be the unique $a \in A$ with $f(a) = b$; then $g(f(a)) = a$ and $f(g(b)) = b$ hold by construction. Finally, if g and g' are both two-sided inverses, associativity of composition gives $g = g \circ \mathrm{id}_B = g \circ (f \circ g') = (g \circ f) \circ g' = \mathrm{id}_A \circ g' = g'$; the inverse is unique, and we write it f^{-1}. □

For $S \subseteq A$ the *image* is $f(S) = \{f(a) : a \in S\}$, and for $T \subseteq B$ the *preimage* is $f^{-1}(T) = \{a \in A : f(a) \in T\}$, defined whether or not f is invertible. The overloading of f^{-1} is harmless: when the inverse function exists, the preimage of T under f is exactly the image of T under f^{-1}.

1.3 Relations and equivalences

A *binary relation* on A is a subset $R \subseteq A \times A$; one writes $a \sim b$ for $(a, b) \in R$. It is an *equivalence relation* if it is *reflexive* ($a \sim a$), *symmetric* ($a \sim b \Rightarrow b \sim a$), and *transitive* ($a \sim b$ and $b \sim c$ imply $a \sim c$). The *equivalence class* of a is $[a] = \{x \in A : x \sim a\}$, and the set of distinct classes is the *quotient* $A/{\sim}$.

> **Proposition 1.2** (Classes partition the set). *Let $\sim$ be an equivalence relation on a set A. The distinct equivalence classes* partition *A: they are nonempty, pairwise disjoint, and their union is A.*

Proof. Reflexivity gives $a \in [a]$ for every $a \in A$, so each class is nonempty and every element lies in some class; the union of the classes is therefore A. For disjointness, suppose the classes $[a]$ and $[b]$ share an element x, and let $y \in [a]$ be arbitrary. From $y \sim a$ and $a \sim x$ (the latter by symmetry from $x \sim a$), transitivity gives $y \sim x$; combining with $x \sim b$ gives $y \sim b$, so $y \in [b]$. Thus $[a] \subseteq [b]$, and exchanging the roles of a and b gives the reverse inclusion; two classes that meet coincide. □

We use this construction twice in the present volume: to form the integers modulo n (§2) and the cosets of a subgroup (§3).

A second kind of relation orders rather than identifies. A *partial order* on A is a relation $\preceq$ that is reflexive, transitive, and *antisymmetric*: $a \preceq b$ and $b \preceq a$ imply $a = b$. Four partial orders recur in the series: the usual $\leq$ on $\mathbb{Z}$; inclusion $\subseteq$ on the subsets of a fixed set, whose antisymmetry is the two-inclusions rule of §1.1; divisibility $\mid$ on $\mathbb{N}$ (§2.2); and the *prefix* relation on finite sequences, under which a sequence precedes every sequence that extends it. Two elements are *comparable* if $a \preceq b$ or $b \preceq a$, and a partial order under which every two elements are comparable is a *total* (or *linear*) order. The order $\leq$ on $\mathbb{Z}$ is total; the other three are not: $\{1\}$ and $\{2\}$ are incomparable under inclusion, as are 2 and 3 under divisibility and the one-term sequences (1) and (2) under prefix.

Every partial order has a *strict* companion, $a \prec b$ meaning $a \preceq b$ and $a \neq b$; the two determine each other.

> **Proposition 1.3** (Strict and non-strict orders)**.** *If $\preceq$ is a partial order on A, its strict companion $\prec$ is* irreflexive *($a \prec a$ holds for no a) and transitive. Conversely, if $\prec$ is irreflexive and transitive, then "$a \prec b$ or $a = b$" defines a partial order on A whose strict companion is $\prec$.*

Proof. Irreflexivity of the companion is the clause $a \neq a$. For transitivity let $a \prec b \prec c$; then $a \preceq c$ because $\preceq$ is transitive, and $a = c$ would give $a \preceq b \preceq a$, hence $a = b$ by antisymmetry, contradicting $a \prec b$. Conversely, the relation "$a \prec b$ or $a = b$" is reflexive by construction and transitive by cases: if either link is an equality the other link carries the conclusion, and otherwise transitivity of $\prec$ does. It is antisymmetric, since $a \neq b$ with both $a \preceq b$ and $b \preceq a$ would mean $a \prec b$ and $b \prec a$, whence $a \prec a$, against irreflexivity. Its strict companion is $\prec$ again: "($a \prec b$ or $a = b$) and $a \neq b$" reduces to $a \prec b$, which already excludes $a = b$ by irreflexivity. □

A strict order $\prec$ is a *strict total order* when its non-strict companion is total, equivalently when for all a, b exactly one of $a \prec b$, $a = b$, $b \prec a$ holds (*trichotomy*): at least one holds by totality, and any two together would give $a \prec a$, by substitution or by transitivity. The usual $<$ on $\mathbb{Z}$ is the model. A function $f\colon A \to B$ between ordered sets is *monotone* (order-preserving) if $a \preceq a'$ implies $f(a) \preceq f(a')$, and *strictly increasing* if $a \prec a'$ implies $f(a) \prec f(a')$. The one further property of an order that the volume assumes, the well-ordering of $\leq$ on $\mathbb{N}$, opens §2.1.

1.4 Logic and proof conventions

We employ the symbols $\forall$ ("for all"), $\exists$ ("there exists"), $\exists!$ ("there exists a unique"), and "$:$" or "s.t." for "such that". Negation swaps the quantifiers: $\neg(\forall x\, P(x))$ is $\exists x\, \neg P(x)$, and $\neg(\exists x\, P(x))$ is $\forall x\, \neg P(x)$. The order of unlike quantifiers matters: $\forall x\, \exists y\, P(x, y)$ (the y may depend on x) is weaker than $\exists y\, \forall x\, P(x, y)$ (one y works for all x).

The following conventions are fixed for the entire series. We prove an implication $P \Rightarrow Q$ directly (assume P, derive Q), by contraposition (assume $\neg Q$, derive $\neg P$), or by contradiction; we prove a statement $\forall x\, P(x)$ by taking an arbitrary x, and $\exists x\, P(x)$ by exhibiting a witness. We prove statements about all natural numbers by induction.

1.5 Symbols and how to say them

The debt incurred at the opening of this section falls due here. The series draws on three alphabets: blackboard-bold letters for the standard number systems and structures, Greek letters, and a handful of relational and operator symbols. Tables 1–3 collect these symbols and the complete Greek alphabet, with common English pronunciations, spelled approximately for reading mathematics aloud. These vary between speakers and regions; the alternatives shown are not exhaustive. A capitalised syllable carries the stress, and a slash separates alternatives. These are English letter names, not a guide to pronouncing Greek words. Each symbol is defined where it is first used; the notes here are reminders, not definitions.

2 The integers and modular arithmetic

How does one divide in a finite world? The proof systems documented in this series do their arithmetic not with the unbounded integers of school algebra but with the finitely many remainders left upon division by a fixed, enormous prime. Every step a prover or verifier takes is an addition, a multiplication, or—the delicate case—a division of such remainders. Adding and multiplying remainders is easy to imagine: compute in $\mathbb{Z}$, then discard multiples of the modulus. But what

Symbol	Name	Said	Typical role in the series
$\mathbb{N}$	blackboard N	"natural numbers"	the naturals
$\mathbb{Z}$	blackboard Z	"the integers"	the integers (German *Zahlen*)
$\mathbb{Q}$	blackboard Q	"the rationals"	the rationals (*quotients*)
$\mathbb{R}$	blackboard R	"the reals"	the real numbers
$\mathbb{C}$	blackboard C	"the complexes"	the complex numbers
$\mathbb{F}_q$	blackboard F	"eff-cue"	the finite field of q elements
$\mathbb{G}$	blackboard G	"group G"	an abstract group
$\mathbb{E}$	blackboard E	"expectation"	expectation of a random variable (§11)
$\mathbb{P}$	blackboard P	"the Pallas curve"	the Pallas group (§10; protocol-spec notation)

Table 1: Blackboard-bold letters. "Blackboard bold" names the double-struck style itself; in speech one usually says only the letter or the system it denotes. Elliptic curves are written italic E or calligraphic $\mathcal{E}$, never blackboard bold. The squared $\mathbb{P}^2$ of §10 is unrelated: it denotes the projective plane.

can it mean to divide, when "one half" names no remainder? The answer developed in this section is that division is multiplication by a *modular inverse*; that a remainder possesses an inverse exactly when a certain greatest common divisor equals one; and that an algorithm already in Euclid's *Elements*, suitably extended, computes the inverse efficiently.

The integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ are the arithmetical bedrock on which every algebraic and cryptographic structure in this monograph ultimately rests. The finite number systems underlying elliptic curves, the scalar arithmetic of Halo 2, and the polynomial commitments of Orchard (short values that bind a prover to entire polynomials, which a verifier then tests by evaluation) all reduce, in the end, to computations with congruence classes of integers. This section develops the theory of the integers from a small set of assumptions and arrives at the integers modulo n—the set $\mathbb{Z}/n\mathbb{Z}$ of congruence classes—together with its units and the algorithms that make arithmetic with them effective. We keep the treatment constructive throughout: every existence statement comes, wherever possible, with an algorithm that produces the object in question, since cryptography requires not merely that inverses and representatives exist but that one can compute them efficiently.

2.1 Well-ordering and the integers

We take the basic algebraic properties of $\mathbb{Z}$ for granted: addition and multiplication are associative and commutative, and multiplication distributes over addition; the integers 0 and 1 are identities for addition and multiplication respectively; every integer a has a negative $-a$; there are no zero divisors (if $ab = 0$ then $a = 0$ or $b = 0$); and $\mathbb{Z}$ carries a total order $\leq$ compatible with the arithmetic. The single deep property we assume axiomatically is the following.

> **Theorem 2.1** (Well-ordering principle). *Every nonempty subset $S \subseteq \mathbb{N} = \{0, 1, 2, \dots\}$ of the natural numbers has a least element: there exists $m \in S$ such that $m \leq s$ for all $s \in S$.*

The well-ordering principle is logically equivalent to the principle of mathematical induction; we use both freely. It is also the engine of this section: nearly every existence result below is, at bottom, an application of well-ordering, in which one exhibits a nonempty set of natural numbers and extracts its minimum. Four instances map the road ahead. The remainder in the division algorithm is the least element of the set $\{a - bk : k \in \mathbb{Z},\ a - bk \geq 0\}$; the Euclidean algorithm terminates because a strictly decreasing sequence of nonnegative integers cannot be infinite; the Bézout identity extracts the least positive element of a set of integer combinations; and existence in the fundamental theorem of arithmetic proceeds by minimal counterexample.

Lower	Capital	Name	Said	Lower	Capital	Name	Said
α	A	alpha	AL-fa	ν	N	nu	nyoo / noo
β	B	beta	BEE-ta / BAY-ta	ξ	Ξ	xi	ksigh / zigh
γ	Γ	gamma	GAM-a	o	O	omicron	OM-i-kron / oh-MY-kron
δ	Δ	delta	DEL-ta	π	Π	pi	pie
ϵ, ε	E	epsilon	EP-si-lon / ep-SIGH-lun	ρ	P	rho	roe
ζ	Z	zeta	ZEE-ta / ZAY-ta	σ	Σ	sigma	SIG-ma
η	H	eta	EE-ta / AY-ta	τ	T	tau	tow / taw
θ	Θ	theta	THEE-ta / THAY-ta	υ	Y	upsilon	UP-si-lon / yoop-SIGH-lun
ι	I	iota	eye-OH-ta	ϕ, φ	Φ	phi	fie / fee
κ	K	kappa	KAP-a	χ	X	chi	kigh
λ	Λ	lambda	LAM-da	ψ	Ψ	psi	sigh / psigh / see / psee
μ	M	mu	mew / moo	ω	Ω	omega	OH-mig-a / oh-MAY-ga

Table 2: The 24 letters of the Greek alphabet in lower and upper case, with common English readings. For β, ζ, η, and θ, the EE forms are usual in British English; the AY forms are common in American English. The spellings "ksigh", "zigh", "kigh", and "psigh" rhyme with "eye"; the *k* in "kigh" is hard, and the *th* in theta is as in "thin". For τ, "tow" rhymes with "cow" and "taw" with "paw". Greek-language readings differ: the Greek name for ξ, for example, is approximately "ksee". Capitals reuse the same names: Δ ("delta"), Σ ("sigma"; the summation sign, and the name of the Σ-protocol family of later volumes), Π ("pi"; the product sign, and a conventional name in those volumes for a protocol or problem), Φ, Ψ, Ω, Θ, Λ.

2.2 Divisibility

Definition 2.2. Let $a, b \in \mathbb{Z}$. One says that *a divides b*, written $a \mid b$, if there exists an integer $c \in \mathbb{Z}$ with $b = ac$. In this case a is called a *divisor* or *factor* of b, and b is a *multiple* of a. If no such c exists one writes $a \nmid b$.

Several immediate consequences of the definition deserve to be recorded; we shall use them constantly and usually without explicit citation.

Proposition 2.3. *Let $a, b, c, d \in \mathbb{Z}$. Then:*

1. *$a \mid a$ (reflexivity), $1 \mid a$, and $a \mid 0$.*
2. *If $a \mid b$ and $b \mid c$ then $a \mid c$ (transitivity).*
3. *If $a \mid b$ and $a \mid c$ then $a \mid (bx + cy)$ for all $x, y \in \mathbb{Z}$; in particular $a \mid (b \pm c)$.*
4. *If $a \mid b$ and $b \neq 0$ then $|a| \leq |b|$.*
5. *If $a \mid b$ and $b \mid a$ then $a = \pm b$.*
6. *$0 \mid b$ if and only if $b = 0$.*

Proof. (1) is immediate: $a = a \cdot 1, a = 1 \cdot a, 0 = a \cdot 0$. For (2), write $b = ae$ and $c = bf$; then $c = a(ef)$, so $a \mid c$. For (3), write $b = ae$ and $c = af$; then $bx + cy = a(ex + fy)$. For (4), write $b = ac$ with $b \neq 0$, so $c \neq 0$ and hence $|c| \geq 1$; then $|b| = |a|\,|c| \geq |a|$. For (5), if either is 0 then by (6) so is the other and the claim holds; otherwise (4) gives $|a| \leq |b|$ and $|b| \leq |a|$, so $|a| = |b|$ and $a = \pm b$. For (6), $0 \mid b$ means $b = 0 \cdot c = 0$; conversely $0 \mid 0$. □

The relation $\mid$ is thus reflexive and transitive; it is a partial order in the sense of §1.3 when restricted to the natural numbers (where (5) becomes $a = b$), but on all of $\mathbb{Z}$ it fails to be antisym-

Symbol	Said	Meaning
$\in$, $\notin$	"in", "not in"	set membership
$\subseteq$, $\subset$	"subset of"	(proper) inclusion
$\cup$, $\cap$	"union", "intersect"	set union and intersection
$A \setminus B$	"A minus B"	set difference
$\times$	"times" / "cross"	Cartesian product
$\mid$, $\nmid$	"divides", "does not divide"	divisibility
$\preceq$, $\prec$	"precedes"	partial order, strict companion
$\equiv$	"congruent to"	congruence, qualified $\pmod{n}$
$a \bmod n$	"a mod n"	remainder of division by n
$:=$, $=:$	"is defined as"	definitional equality
$\mapsto$	"maps to"	the action of a function
$g \circ f$	"g after f"	function composition
$\lfloor x \rfloor$, $\lceil x \rceil$	"floor", "ceiling"	round down, round up
$\log x$, $\log_2 x$, $\log_g h$	"log", "log base two", "log base g"	natural, base-2, and discrete (base g) logarithm
$\langle a, b \rangle$	"inner product"	inner product; $\langle g \rangle$ the subgroup generated by g
$\oplus$	"x-or"	bitwise exclusive or
$x \parallel y$	"x concat y"	concatenation of bit or byte strings
$x \leftarrow D$, $x \xleftarrow{\$} S$	"drawn from"	sampling ($ marks uniform); also assignment
$[k]\,P$	"k times P"	scalar multiplication of a point (§10)
$P^\star$	"P star"	canonical bit-encoding of a point or element
$\cdot$	"dot"	product; argument placeholder, as in $F(\cdot)$
$\lvert A \rvert$	"size of A" / "order of A"	cardinality

Table 3: Relational and operator symbols. The symbol $\oplus$ is read "x-or"; every use in the series is the bitwise exclusive or. The encoding $P^\star$ is protocol-spec notation, defined in later volumes.

metric precisely because of signs. The *units* of $\mathbb{Z}$—the integers dividing 1—are exactly ± 1, and two integers that divide each other are called *associates*; in $\mathbb{Z}$ the associates of a are $\pm a$.

2.3 The division algorithm

Divisibility is the special case of exact division. The fundamental fact that makes the arithmetic of $\mathbb{Z}$ tractable is that one can always divide *with remainder*, and that the quotient and remainder are uniquely determined once a convention for the remainder's range is fixed.

Theorem 2.4 (Division algorithm)**.** *Let $a \in \mathbb{Z}$ and $b \in \mathbb{Z}$ with $b > 0$. Then there exist unique integers q (the* quotient*) and r (the* remainder*) such that*

$$a = bq + r, \qquad 0 \le r < b.$$

Proof. Existence. Consider the set

$$S = \{\, a - bk : k \in \mathbb{Z},\ a - bk \ge 0 \,\} \subseteq \mathbb{N}.$$

First, $S \neq \emptyset$. If $a \ge 0$ then $a = a - b \cdot 0 \in S$. If $a < 0$ then, since $b \ge 1$, taking $k = a$ gives $a - ba = a(1 - b) \ge 0$ (a product of a nonpositive and a nonpositive factor), so $a - ba \in S$. Thus S is a nonempty subset of $\mathbb{N}$, and by the well-ordering principle (Theorem 2.1) it has a least element $r = a - bq$ for some $q \in \mathbb{Z}$. By construction $r \ge 0$. If $r \ge b$, then $r - b = a - b(q + 1) \ge 0$ would lie in S and be strictly smaller than r, contradicting minimality. Hence $0 \le r < b$.

Uniqueness. Suppose $a = bq + r = bq' + r'$ with $0 \le r, r' < b$. Then $b(q - q') = r' - r$, so $b \mid (r' - r)$. But $-b < r' - r < b$, and the only multiple of b strictly between $-b$ and b is 0. Hence $r' = r$, and then $b(q - q') = 0$ with $b \neq 0$ forces $q = q'$. $\square$

Remark 2.5. The hypothesis $b > 0$ is only for definiteness. For general $b \neq 0$ one obtains unique q, r with $a = bq + r$ and $0 \leq r < |b|$, by applying the theorem to $|b|$ and adjusting the sign of q. Note that the remainder convention $0 \leq r < |b|$ differs from the truncation behaviour of many programming languages, where the remainder of a negative dividend may be negative; for the number-theoretic results below the convention above is the appropriate one.

We denote the remainder r produced by the division algorithm by $a \bmod b$, and the quotient q by $\lfloor a/b \rfloor$ when $b > 0$ (the *floor* of the rational number a/b). With this notation, $b \mid a$ holds precisely when $a \bmod b = 0$.

2.4 The greatest common divisor and the Euclidean algorithm

Definition 2.6. A *common divisor* of integers a and b is an integer d with $d \mid a$ and $d \mid b$. The *greatest common divisor* of a and b, not both zero, is the largest such d; it is denoted $\gcd(a, b)$. By convention $\gcd(0, 0) = 0$. Integers a and b are *coprime* (or *relatively prime*) if $\gcd(a, b) = 1$.

The greatest common divisor is well defined when $(a, b) \neq (0, 0)$: the set of common divisors is nonempty (it contains 1) and bounded above (every common divisor of, say, the nonzero a has absolute value at most $|a|$ by Proposition 2.3(4)), so it has a largest element, which is positive since 1 is a common divisor. We establish a much stronger characterisation shortly: the gcd is not merely the numerically largest common divisor but is divisible by every common divisor. The key computational engine is the following observation.

Lemma 2.7. *Let $a, b \in \mathbb{Z}$. For any $q \in \mathbb{Z}$,*

$$\gcd(a, b) = \gcd(b,\ a - qb).$$

In particular, if $a = bq + r$ then $\gcd(a, b) = \gcd(b, r)$, and $\gcd(a, 0) = |a|$.

Proof. It suffices to show the two pairs (a, b) and $(b, a-qb)$ have exactly the same common divisors; equal sets of common divisors have equal greatest elements (and the same coprimality status). If $d \mid a$ and $d \mid b$, then $d \mid (a - qb)$ by Proposition 2.3(3), so d is a common divisor of b and $a - qb$. Conversely, if $d \mid b$ and $d \mid (a - qb)$, then $d \mid ((a - qb) + qb) = a$, so d is a common divisor of a and b. The two sets of common divisors coincide. Finally $\gcd(a, 0) = |a|$ because the common divisors of a and 0 are exactly the divisors of a, the largest of which in absolute value is $|a|$ (and $|a| > 0$ when $a \neq 0$; the case $a = 0$ is the convention $\gcd(0, 0) = 0$). □

Iterating the lemma with the division algorithm yields the oldest nontrivial algorithm in mathematics.

Theorem 2.8 (Euclidean algorithm)**.** *Let a, b be integers with $b \neq 0$. Define a sequence of remainders by $r_0 = |a|$, $r_1 = |b|$, and, as long as $r_i \neq 0$, let $r_{i+1} = r_{i-1} \bmod r_i$, so that*

$$r_{i-1} = r_i q_i + r_{i+1}, \qquad 0 \leq r_{i+1} < r_i.$$

The sequence $r_1 > r_2 > \cdots$ of positive remainders is strictly decreasing, so some $r_{n+1} = 0$; the last nonzero remainder satisfies $r_n = \gcd(a, b)$.

Proof. The remainders $r_1, r_2, \ldots$ are nonnegative integers and, while nonzero, strictly decreasing because $r_{i+1} = r_{i-1} \bmod r_i < r_i$. A strictly decreasing sequence of nonnegative integers cannot be infinite (again by well-ordering: otherwise the set of values would have no least element), so

the algorithm terminates with some $r_{n+1} = 0$ and $r_n > 0$. By Lemma 2.7, each step preserves the gcd:

$$\gcd(a, b) = \gcd(r_0, r_1) = \gcd(r_1, r_2) = \cdots = \gcd(r_n, r_{n+1}) = \gcd(r_n, 0) = r_n,$$

using $\gcd(|a|, |b|) = \gcd(a, b)$ (signs do not affect common divisors). □

Example 2.9. Compute $\gcd(1071, 462)$:

$$\begin{aligned} 1071 &= 2 \cdot 462 + 147, \\ 462 &= 3 \cdot 147 + 21, \\ 147 &= 7 \cdot 21 + 0. \end{aligned}$$

The last nonzero remainder is 21, so $\gcd(1071, 462) = 21$. Indeed $1071 = 21 \cdot 51$ and $462 = 21 \cdot 22$, with $\gcd(51, 22) = 1$.

Remark 2.10. The Euclidean algorithm is efficient. The number of division steps for inputs $a > b > 0$ is $O(\log b)$; the worst case occurs at consecutive Fibonacci numbers (Lamé's theorem), where every quotient q_i equals 1 except the final one (which equals 2) and the remainders shrink as slowly as possible. With schoolbook arithmetic on k-bit inputs the whole computation costs $O(k^2)$ bit operations. This efficiency is what renders gcd-based cryptographic operations—above all modular inversion, below—practical at the sizes used in Halo 2 and Orchard.

2.5 The Bézout identity and the extended Euclidean algorithm

The Euclidean algorithm computes the gcd, but the back-substitution of its equations yields something more powerful: the gcd is an *integer linear combination* of the inputs. This is among the most useful facts in elementary number theory.

Theorem 2.11 (Bézout's identity)**.** *Let $a, b \in \mathbb{Z}$, not both zero, and let $d = \gcd(a, b)$. Then there exist integers $u, v \in \mathbb{Z}$ with*

$$au + bv = d.$$

Moreover d is the smallest *positive integer expressible in the form $ax + by$ with $x, y \in \mathbb{Z}$, and the set of all such combinations is exactly the set of multiples of d:*

$$\{ ax + by : x, y \in \mathbb{Z} \} = d\mathbb{Z} = \{ kd : k \in \mathbb{Z} \}.$$

Proof. Let $I = \{ ax + by : x, y \in \mathbb{Z} \}$. This set contains a and b and is closed under addition and under multiplication by any integer; concretely, if $s = ax_1 + by_1$ and $t = ax_2 + by_2$ lie in I, then for any $m, n \in \mathbb{Z}$ one has

$$ms + nt = a(mx_1 + nx_2) + b(my_1 + ny_2) \in I.$$

Since a, b are not both zero, I contains a positive element (one of $\pm a, \pm b$), so the set $I \cap \mathbb{N}_{>0}$ of positive elements of I is nonempty. By well-ordering it has a least element $g = au + bv > 0$.

We claim that g divides every element of I. Indeed, let $c \in I$ and write $c = gq + r$ with $0 \le r < g$ by the division algorithm. Then $r = c - gq \in I$ because I is closed under the operations just described. By minimality of g among positive elements of I, and $0 \le r < g$, necessarily $r = 0$. Hence $g \mid c$ for every $c \in I$; in particular $g \mid a$ and $g \mid b$, so g is a common divisor of a and b. Conversely, any common divisor e of a and b divides $au + bv = g$ by Proposition 2.3(3). Therefore g is a common divisor that is divisible by every common divisor; being positive, it is in particular the *greatest* common divisor, so $g = d$. The displayed equation $au + bv = d$ follows, d is the least positive element of I, and $I = g\mathbb{Z} = d\mathbb{Z}$ because every element of I is a multiple of g, as shown, and conversely every multiple $kg = a(ku) + b(kv)$ lies in I. □

Remark 2.12. The proof yields a characterisation of the gcd that is independent of the order $\leq$: $d = \gcd(a, b)$ is, up to sign, the unique common divisor of a and b that is divisible by every common divisor. This is the formulation that survives in settings where "largest" may make no sense but "divisible by every common divisor" still does; later sections adopt it when they leave the integers. We call a nonempty subset $I \subseteq \mathbb{Z}$ closed under addition and under multiplication by arbitrary integers an *ideal*; the content of Bézout's identity is that every ideal of $\mathbb{Z}$ has the form $d\mathbb{Z}$ for a single generator d. Ideals recur in section 4, where they drive a general construction whose prototype is the passage from integers to congruence classes carried out later in this section.

The Bézout coefficients u, v are not unique. If $au + bv = d$, then for every $k \in \mathbb{Z}$,

$$a\left(u + k\tfrac{b}{d}\right) + b\left(v - k\tfrac{a}{d}\right) = d,$$

since the added terms $k\frac{ab}{d}$ and $-k\frac{ab}{d}$ cancel; and these are all the solutions. To see this, note first that a/d and b/d are coprime: a common divisor e of both makes de a common divisor of a and b, so $de \mid au + bv = d$ by Proposition 2.3(3), forcing $e = \pm 1$. Now suppose $au' + bv' = d$. Subtracting $au + bv = d$ and dividing by $d > 0$ gives $\frac{a}{d}(u' - u) = -\frac{b}{d}(v' - v)$, so b/d divides $\frac{a}{d}(u' - u)$; by Proposition 2.15(1) below (whose proof rests only on Bézout's identity, already established), b/d divides $u' - u$, say $u' = u + k\frac{b}{d}$ with $k \in \mathbb{Z}$. Substituting back gives $b(v' - v + k\frac{a}{d}) = 0$, so $v' = v - k\frac{a}{d}$ when $b \neq 0$; and when $b = 0$ we have $u' = u$, $a/d = \pm 1$, and $k = (v - v')\,\frac{a}{d}$ places (u', v') in the family. To compute a particular pair (u, v) efficiently, we augment the Euclidean algorithm so that it carries each remainder as an explicit combination of a and b.

Theorem 2.13 (Extended Euclidean algorithm)**.** *There is an algorithm that, on input $a, b \in \mathbb{Z}$ not both zero, outputs $d = \gcd(a, b)$ together with integers u, v satisfying $au + bv = d$, using the same number of division steps as the Euclidean algorithm.*

Proof. Alongside the remainder sequence of Theorem 2.8, maintain two auxiliary sequences (s_i) and (t_i) with the invariant

$$r_i = a\,s_i + b\,t_i \qquad \text{for all } i. \tag{$*$}$$

Initialise (taking $a, b \geq 0$ for clarity; we restore signs at the end)

$$r_0 = a,\ s_0 = 1,\ t_0 = 0; \qquad r_1 = b,\ s_1 = 0,\ t_1 = 1,$$

so $(*)$ holds for $i = 0, 1$. At step $i \geq 1$, having $r_{i-1} = r_i q_i + r_{i+1}$ from the division algorithm, set

$$s_{i+1} = s_{i-1} - q_i s_i, \qquad t_{i+1} = t_{i-1} - q_i t_i.$$

Then

$$a\,s_{i+1} + b\,t_{i+1} = (as_{i-1} + bt_{i-1}) - q_i\,(as_i + bt_i) = r_{i-1} - q_i r_i = r_{i+1},$$

so induction preserves $(*)$. When the algorithm terminates with $r_{n+1} = 0$ and $r_n = d = \gcd(a, b)$, the invariant for $i = n$ reads $d = as_n + bt_n$; output $u = s_n$, $v = t_n$. If $b = 0$, then $a \neq 0$ and termination is immediate with $n = 0$: no division step runs, the last nonzero remainder is $r_0 = a = \gcd(a, 0)$ (Lemma 2.7), and the invariant at $i = 0$ already gives the output $(d, u, v) = (a, 1, 0)$. If $b \neq 0$, termination and the value of d are exactly as in Theorem 2.8. In either case each step does only a constant amount of extra work. For general inputs, run the algorithm on $(|a|, |b|)$ to obtain $|a|\,u' + |b|\,v' = d$ (recall $\gcd(|a|, |b|) = \gcd(a, b)$) and output $u = \operatorname{sgn}(a)\,u'$, $v = \operatorname{sgn}(b)\,v'$, so that $au + bv = d$. □

In practice it is convenient to lay the computation out as a table with columns (r_i, q_i, s_i, t_i), computing each new row from the previous two.

Example 2.14. Run the extended algorithm on $a = 1071$, $b = 462$ (continuing Example 2.9). The quotients are $q_1 = 2, q_2 = 3, q_3 = 7$.

i	r_i	s_i	t_i
0	1071	1	0
1	462	0	1
2	147	1	-2
3	21	-3	7
4	0	22	-51

For instance, row 2 is row 0 minus $q_1 = 2$ times row 1: $s_2 = 1 - 2 \cdot 0 = 1$, $t_2 = 0 - 2 \cdot 1 = -2$, and indeed $1071 \cdot 1 + 462 \cdot (-2) = 1071 - 924 = 147$. The last nonzero remainder is $r_3 = 21 = \gcd(1071, 462)$, with

$$1071 \cdot (-3) + 462 \cdot 7 = -3213 + 3234 = 21.$$

Thus $(u, v) = (-3, 7)$. The final row $(s_4, t_4) = (22, -51) = (b/d, -a/d)$ up to sign records the homogeneous solution: $1071 \cdot 22 + 462 \cdot (-51) = 0$.

A first, decisive application of Bézout's identity is Euclid's lemma, the property of primes that ultimately forces unique factorisation.

Proposition 2.15 (Euclid's lemma)**.** *Let $a, b, c \in \mathbb{Z}$.*

1. *If $c \mid ab$ and $\gcd(c, a) = 1$, then $c \mid b$.*
2. *If $\gcd(a, b) = 1$, $a \mid n$, and $b \mid n$, then $ab \mid n$.*
3. *If $\gcd(a, b) = 1$ and $\gcd(a, c) = 1$, then $\gcd(a, bc) = 1$.*

Proof. (1) By Bézout there are u, v with $cu + av = 1$. Multiply by b: $b = cub + avb = c(ub) + (ab)v$. Since $c \mid ab$, c divides both terms on the right, hence $c \mid b$.

(2) Write $n = ak$. Since $b \mid n = ak$ and $\gcd(b, a) = 1$, part (1) gives $b \mid k$, say $k = bm$. Then $n = a(bm) = (ab)m$, so $ab \mid n$.

(3) By Bézout, $ax_1 + by_1 = 1$ and $ax_2 + cy_2 = 1$ for suitable integers. Multiplying,

$$1 = (ax_1 + by_1)(ax_2 + cy_2) = a\underbrace{(ax_1x_2 + x_1cy_2 + by_1x_2)}_{\in\mathbb{Z}} + bc\,(y_1y_2),$$

which exhibits 1 as an integer combination of a and bc. Hence $\gcd(a, bc)$ divides 1, so it equals 1. □

2.6 Primes and the fundamental theorem of arithmetic

Definition 2.16. An integer $p > 1$ is *prime* if its only positive divisors are 1 and p. An integer $n > 1$ that is not prime is *composite*; thus n is composite iff $n = ab$ for some integers $1 < a, b < n$. The integer 1 is a *unit* and is by convention neither prime nor composite.

The exclusion of 1 from the primes is not arbitrary: it is exactly what makes factorisation into primes unique, since otherwise one could insert arbitrarily many factors of 1. The decisive property of primes is the prime case of Euclid's lemma.

Lemma 2.17 (Euclid's lemma for primes)**.** *Let p be prime and $a, b \in \mathbb{Z}$. If $p \mid ab$ then $p \mid a$ or $p \mid b$. More generally, if $p \mid a_1a_2 \cdots a_k$ then $p \mid a_i$ for some i.*

Proof. The only positive divisors of p are 1 and p, so $\gcd(p,a) \in \{1,p\}$. If $\gcd(p,a) = p$ then $p \mid a$ and the claim holds; otherwise $\gcd(p,a) = 1$, and Proposition 2.15(1) gives $p \mid b$. The general statement follows by induction on k: from $p \mid a_1(a_2 \cdots a_k)$ it follows that $p \mid a_1$ or $p \mid a_2 \cdots a_k$, and the inductive hypothesis handles the latter. □

Theorem 2.18 (Fundamental theorem of arithmetic)**.** *Every integer $n > 1$ can be written as a product of primes,*

$$n = p_1 p_2 \cdots p_k,$$

and this factorisation is unique up to the order of the factors. Equivalently, there is a unique expression

$$n = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r}, \qquad p_1 < p_2 < \cdots < p_r \text{ prime}, \ e_i \geq 1.$$

Proof. Existence. Suppose, for contradiction, that some integer > 1 is not a product of primes, and let n be the least such (well-ordering). Then n is not prime (a prime is trivially a product of one prime), so $n = ab$ with $1 < a, b < n$. By minimality, each of a, b is a product of primes; concatenating these factorisations expresses n as a product of primes, a contradiction. Hence every $n > 1$ factors into primes.

Uniqueness. Suppose

$$p_1 p_2 \cdots p_k = q_1 q_2 \cdots q_\ell$$

are two factorisations of the same integer into primes. The argument proceeds by induction on k. If $k = 0$ the product is the empty product 1, forcing $\ell = 0$ as well (a nonempty product of primes is > 1). For the inductive step, p_1 divides the right-hand product, so by Lemma 2.17 $p_1 \mid q_j$ for some j. Since q_j is prime, its only divisor > 1 is itself, so $p_1 = q_j$. Cancelling this common factor (legitimate as $\mathbb{Z}$ has no zero divisors) yields

$$p_2 \cdots p_k = q_1 \cdots \widehat{q_j} \cdots q_\ell,$$

where the hat denotes omission. By the inductive hypothesis these two factorisations agree up to order, with $k - 1 = \ell - 1$; restoring $p_1 = q_j$ shows the original factorisations agree up to order. Collecting equal primes into prime powers gives the canonical form, whose uniqueness is immediate from the uniqueness of the multiset of primes—the factors counted with multiplicity, without regard to order. □

Remark 2.19. Both halves of the theorem are needed and have different characters. Existence is a statement about the order structure (well-ordering) and would hold in any reasonable factorisation; uniqueness rests squarely on Euclid's lemma, hence on Bézout, hence on the division algorithm. There are number systems with a division-like structure failing Euclid's lemma in which factorisation into irreducibles is *not* unique; the integers avoid this pathology precisely because they admit division with remainder, from which the whole chain above flows.

2.7 Congruences and residue classes

The development now passes from divisibility to the arithmetic of remainders. Fix an integer $n \geq 1$, the *modulus*.

Definition 2.20. Let $n \geq 1$ and $a, b \in \mathbb{Z}$. The integer a is *congruent to b modulo n*, written

$$a \equiv b \pmod{n},$$

if $n \mid (a-b)$, equivalently if a and b leave the same remainder upon division by n.

The two formulations agree: if $a = nq_1 + r_1$ and $b = nq_2 + r_2$ with $0 \le r_1, r_2 < n$, then $a - b = n(q_1 - q_2) + (r_1 - r_2)$, and $n \mid (a-b)$ iff $n \mid (r_1 - r_2)$ iff $r_1 = r_2$ (since $|r_1 - r_2| < n$).

Proposition 2.21. *Fix $n \ge 1$. Congruence modulo n is an equivalence relation on $\mathbb{Z}$ that is moreover compatible with addition and multiplication: if $a \equiv a'$ and $b \equiv b' \pmod{n}$, then*

$$a + b \equiv a' + b' \pmod{n}, \qquad ab \equiv a'b' \pmod{n}.$$

Consequently $a^k \equiv (a')^k \pmod{n}$ for all $k \ge 0$, and one may substitute congruent values freely in any polynomial expression with integer coefficients.

Proof. Reflexivity ($n \mid 0$), symmetry ($n \mid (a-b) \Rightarrow n \mid (b-a)$), and transitivity ($n \mid (a-b)$ and $n \mid (b-c)$ give $n \mid (a-c)$ by Proposition 2.3(3)) are immediate. For compatibility, write $a - a' = ns$ and $b - b' = nt$. Then

$$(a+b) - (a'+b') = n(s+t),$$

and

$$ab - a'b' = ab - a'b + a'b - a'b' = (a-a')b + a'(b-b') = n(sb + a't),$$

so both $n \mid ((a+b) - (a'+b'))$ and $n \mid (ab - a'b')$. The statement about powers and polynomial expressions follows by induction from the multiplicative case. □

Because congruence is an equivalence relation, it partitions $\mathbb{Z}$ into disjoint *congruence classes* (also called *residue classes*).

Definition 2.22. The *congruence class* (or *residue class*) of a modulo n is

$$[a]_n = \{\, b \in \mathbb{Z} : b \equiv a \pmod{n} \,\} = \{\, a + kn : k \in \mathbb{Z} \,\} = a + n\mathbb{Z}.$$

We write the set of all congruence classes as

$$\mathbb{Z}/n\mathbb{Z} = \{\, [a]_n : a \in \mathbb{Z} \,\}.$$

By the division algorithm every integer is congruent modulo n to exactly one of $0, 1, \dots, n-1$, namely its remainder; hence the n classes $[0]_n, [1]_n, \dots, [n-1]_n$ are distinct and exhaust $\mathbb{Z}/n\mathbb{Z}$. The set $\{0, 1, \dots, n-1\}$ is the standard *system of representatives*, called the *least nonnegative residues*.

2.8 Units modulo n and modular inverses

Division is the inverse of multiplication, so the opening problem of the section comes down to a multiplicative question about congruences: for which a does the congruence $ax \equiv 1 \pmod{n}$ admit a solution? The Euclidean machinery characterises the solvable cases completely and computes the solutions.

Definition 2.23. Let $n \ge 1$. A class $[a]_n \in \mathbb{Z}/n\mathbb{Z}$ is a *unit* (or is *invertible*) if there exists an integer x with $ax \equiv 1 \pmod{n}$. Any class $[x]_n$ with this property is the (*modular*) *inverse* of $[a]_n$, written $[a]_n^{-1}$; one also calls x a modular inverse of a modulo n. We write the set of units as

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{\, [a]_n : [a]_n \text{ is a unit} \,\}.$$

The definition is well posed on classes: by Proposition 2.21, if $a' \equiv a$ and $x' \equiv x \pmod{n}$ then $a'x' \equiv ax \pmod{n}$, so whether $ax \equiv 1 \pmod{n}$ holds depends only on the classes $[a]_n$ and $[x]_n$, never on the representatives chosen. The definite article in "the inverse" is justified by the uniqueness part of the next theorem—the payoff of the section.

Theorem 2.24. *Let $n \geq 1$ and $a \in \mathbb{Z}$. The class $[a]_n$ is a unit in $\mathbb{Z}/n\mathbb{Z}$ if and only if $\gcd(a, n) = 1$. When $\gcd(a, n) = 1$, the inverse is unique, and the extended Euclidean algorithm computes a representative: if $au + nv = 1$, then $[a]_n^{-1} = [u]_n$.*

Proof. Suppose $\gcd(a, n) = 1$. By Bézout (Theorem 2.11) there are u, v with $au+nv = 1$. Reducing modulo n, $au \equiv 1 \pmod{n}$, so $[u]_n$ is an inverse of $[a]_n$. The extended Euclidean algorithm (Theorem 2.13) produces such a u explicitly.

Conversely, if $[a]_n$ is a unit, say $ax \equiv 1 \pmod{n}$, then $ax - 1 = nk$ for some k, i.e. $ax - nk = 1$. Any common divisor of a and n divides the left side, hence divides 1; thus $\gcd(a, n) = 1$.

Uniqueness: if $ax \equiv 1$ and $ax' \equiv 1 \pmod{n}$, then $x \equiv x(ax') \equiv (xa)x' \equiv x' \pmod{n}$—a sandwich using only the associativity and commutativity of integer multiplication together with Proposition 2.21—so $[x]_n = [x']_n$. □

Example 2.25. Consider the computation of $[17]_{43}^{-1}$. The extended Euclidean algorithm on $(43, 17)$:

$$\begin{aligned} 43 &= 2 \cdot 17 + 9, \\ 17 &= 1 \cdot 9 + 8, \\ 9 &= 1 \cdot 8 + 1, \\ 8 &= 8 \cdot 1 + 0, \end{aligned}$$

so $\gcd(43, 17) = 1$. Back-substituting:

$$1 = 9 - 8 = 9 - (17 - 9) = 2 \cdot 9 - 17 = 2(43 - 2 \cdot 17) - 17 = 2 \cdot 43 - 5 \cdot 17.$$

Hence $-5 \cdot 17 \equiv 1 \pmod{43}$, and $[17]_{43}^{-1} = [-5]_{43} = [38]_{43}$. Check: $17 \cdot 38 = 646 = 15 \cdot 43 + 1$, so $17 \cdot 38 \equiv 1 \pmod{43}$.

Counting the units will matter as much as finding them: the count defined next carries structural weight in later sections, and we fix it here so that those sections can cite a definition rather than re-derive one.

Definition 2.26 (Euler totient)**.** *Euler's totient function* $\varphi(n)$ is the number of integers a with $1 \leq a \leq n$ and $\gcd(a, n) = 1$. Equivalently, $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$.

The two counts agree: the classes of $1, 2, \ldots, n$ run exactly once through all of $\mathbb{Z}/n\mathbb{Z}$ (the class of n being $[0]_n$), and by Theorem 2.24 the class $[a]_n$ is a unit precisely when $\gcd(a, n) = 1$.

Example 2.27. We have $\varphi(1) = 1$, and $\varphi(p) = p - 1$ for prime p: for $1 \leq a < p$ the positive divisor $\gcd(a, p)$ of p is 1 or p, and $p \nmid a$ rules out p, so every one of the $p - 1$ nonzero classes is a unit. By direct enumeration, $\varphi(12) = |\{1, 5, 7, 11\}| = 4$.

The cheque presented at the head of the section is now cashed. To divide by a in the finite world of remainders modulo n is to multiply by the inverse class $[a]_n^{-1}$; Theorem 2.24 says exactly when that inverse exists—precisely when $\gcd(a, n) = 1$, hence for every nonzero class when the modulus is prime—and the extended Euclidean algorithm computes it in $O(\log n)$ division steps (Remark 2.10). The laws verified along the way—associativity, commutativity, distributivity, identities, negatives, and now inverses—have been stated concretely, for the integers and their congruence classes alone. They are instances of patterns that pervade mathematics; Sections 3 and 4 isolate those patterns, give them their standard names, and re-read $\mathbb{Z}/n\mathbb{Z}$ in their light. Nothing proved there revises anything proved here.

3 Groups

Two parties who have never met wish to agree on a secret value while every message they exchange is read by an eavesdropper. One solution begins with a public set carrying a single public operation and a public starting element g. Each party privately chooses a count of repetitions—say a and b—combines g with itself that many times, and publishes the result. Each then takes the *other's* published element and applies their own secret count of repetitions to it. If repetition is well behaved, both arrive at the element produced by ab repetitions of g, and so share a secret, while the eavesdropper has seen only g and the two published values. Three demands on the operation are implicit. It must *compose*: "a repetitions, then b more" must be unambiguous however the intermediate steps are bracketed. It must *invert*: repetition counts should obey an arithmetic, with an identity and the possibility of undoing a step, not a mere accumulation. And it must *hide* repetition: recovering the secret count from the published value must be hard. The third demand is a computational matter taken up in later volumes; the first two are algebra, and this section isolates exactly the structure that supplies them.

That structure is the *group*, the foundational abstraction of modern algebra. A group axiomatises the bare minimum needed to discuss symmetry and invertible composition: one associative operation with an identity and inverses. Almost every object met later in this series is a group—the integers modulo a prime under addition, the nonzero residues under multiplication, the points of an elliptic curve, the symmetries of a finite field. This section develops group theory from first principles and proves its central structural results in full: Lagrange's theorem, the classification of cyclic groups, the quotient of an abelian group by a subgroup, and the basic theory of homomorphisms and isomorphisms. The payoff for the opening problem is the cyclic-group theory together with Lagrange's theorem: in any finite group, the arithmetic of repetition counts is precisely the modular arithmetic of section 2, the modulus being the order of the repeated element—a divisor of the group's order.

3.1 The group axioms

The first task is to make "combining two elements" precise.

Definition 3.1 (Binary operation)**.** Let S be a set. A *binary operation* on S is a function

$$*\colon S \times S \to S, \qquad (a, b) \mapsto a * b.$$

Closure—that $a * b$ again lies in S—is built into the very statement that $*$ maps *into* S. We name it explicitly nonetheless, because for a subset $H \subseteq S$ the question of whether $*$ restricts to a binary operation on H is exactly the question of whether H is *closed* under $*$; this question drives the theory of subgroups below (§3.4).

Definition 3.2 (Group)**.** A *group* is a pair $(G, *)$ consisting of a set G and a binary operation $*$ on G satisfying three axioms.

(G1) *Associativity.* For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.

(G2) *Identity.* There exists $e \in G$ such that $e * a = a * e = a$ for all $a \in G$.

(G3) *Inverses.* For each $a \in G$ there exists $b \in G$ such that $a * b = b * a = e$, where e is an identity as in (G2).

A set with an operation satisfying only (G1) is a *semigroup*; if it also satisfies (G2) it is a *monoid*. A group is thus a monoid in which every element is invertible.

The number of elements of G, written $|G|$, is the *order* of the group; if $|G|$ is finite, G is a *finite group*, and otherwise an *infinite group*. We frequently say "the group G", leaving the operation understood.

The axioms assert the *existence* of an identity and of inverses, but not, on their face, uniqueness. Uniqueness is a theorem—and a useful one, since it is what licenses the notations e and a^{-1}.

Proposition 3.3. *Let $(G, *)$ be a group.*

1. *The identity element is unique.*
2. *Each $a \in G$ has a unique inverse, denoted a^{-1}.*
3. (Cancellation.) *For all $a, x, y \in G$: if $a * x = a * y$ then $x = y$, and if $x * a = y * a$ then $x = y$.*
4. *For all $a, b \in G$, $(a^{-1})^{-1} = a$ and $(a * b)^{-1} = b^{-1} * a^{-1}$.*

Proof. (1) Suppose e and e' both satisfy (G2). Using that e' is an identity, $e = e * e'$; using that e is an identity, $e * e' = e'$. Hence $e = e'$.

(2) Suppose b and b' are both inverses of a. Then

$$b = b * e = b * (a * b') = (b * a) * b' = e * b' = b',$$

using (G2), the inverse property of b', associativity, and the inverse property of b.

(3) Suppose $a * x = a * y$. Multiplying on the left by a^{-1} and reassociating,

$$x = (a^{-1} * a) * x = a^{-1} * (a * x) = a^{-1} * (a * y) = (a^{-1} * a) * y = y.$$

The right-cancellation law is proved symmetrically, multiplying on the right by a^{-1}.

(4) Since $a * a^{-1} = a^{-1} * a = e$, the element a is an inverse of a^{-1}; by (2) it is *the* inverse, so $(a^{-1})^{-1} = a$. For the product formula, compute

$$(a * b) * (b^{-1} * a^{-1}) = a * (b * b^{-1}) * a^{-1} = a * a^{-1} = e,$$

and symmetrically $(b^{-1} * a^{-1}) * (a * b) = e$; by (2), $(a * b)^{-1} = b^{-1} * a^{-1}$. The order reversal is the "socks–shoes" rule: the inverse of putting on socks, then shoes, is removing shoes, then socks. □

Two notational conventions dominate. *Multiplicative* notation writes the operation as juxtaposition ab (or $a \cdot b$), the identity as 1, the inverse as a^{-1}, and powers as $a^n = \underbrace{a \cdots a}_{n \text{ factors}}$ for $n \geq 1$, with $a^0 = 1$ and $a^{-n} = (a^{-1})^n$. *Additive* notation, reserved almost exclusively for commutative groups, writes $a + b$, the identity as 0, the inverse as $-a$, and na for the n-fold sum. The exponent laws

$$a^m a^n = a^{m+n}, \qquad (a^m)^n = a^{mn}$$

hold for all integers m, n, proved by induction together with Proposition 3.3. Henceforth ab denotes $a * b$, and we drop the symbol $*$ unless clarity demands it.

The power notation presumes something the axioms do not literally grant: that an n-fold product needs no brackets at all. Axiom (G1) speaks only of three factors. The gap is closed once and for all by the next proposition, and thereafter products are written without brackets and powers without comment.

Proposition 3.4 (Generalised associativity)**.** *Let $a_1, \ldots, a_n$, $n \geq 1$, be elements of a group—or of any set with an associative operation. Every way of bracketing the product $a_1 a_2 \cdots a_n$ yields the same*

> *element, namely the* left-normed *product* $L_n = (\cdots((a_1a_2)a_3)\cdots)a_n$. *In particular* a^n *is a single well-defined element, however the n factors are combined.*

Proof. By strong induction on n. For $n \leq 2$ there is only one bracketing, and for $n = 3$ the two bracketings agree by (G1). Let $n \geq 3$ and assume the claim for all shorter products. Any bracketed product P of $a_1, \dots, a_n$ has an outermost multiplication, splitting it as $P = QR$ with Q a bracketed product of $a_1, \dots, a_k$ and R one of $a_{k+1}, \dots, a_n$, for some $1 \leq k < n$. The induction hypothesis evaluates both halves: $Q = L_k$, and R is the left-normed product of $a_{k+1}, \dots, a_n$. If $k = n - 1$, then $R = a_n$ and $P = L_{n-1}a_n = L_n$ directly. If $k < n - 1$, write $R = R'a_n$ with R' the left-normed product of $a_{k+1}, \dots, a_{n-1}$; then (G1) and the induction hypothesis, applied to the bracketed product L_kR' of the $n - 1$ elements $a_1, \dots, a_{n-1}$, give

$$P = L_k(R'a_n) = (L_kR')\,a_n = L_{n-1}\,a_n = L_n.$$

Every bracketing therefore equals L_n. □

> **Definition 3.5** (Abelian group)**.** A group $(G, *)$ is *abelian* (or *commutative*) if $a * b = b * a$ for all $a, b \in G$; otherwise it is *non-abelian*.

The term honours Niels Henrik Abel. In an abelian group products may be freely reordered, and the socks–shoes formula $(ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1}$ loses its order-sensitivity. Convention reserves additive notation for abelian groups precisely because + connotes commutativity.

3.2 First examples

Example 3.6. The pair $(\mathbb{Z}, +)$ is an infinite abelian group: addition is associative and commutative, the identity is 0, and the inverse of n is $-n$. Likewise $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, and $(\mathbb{C}, +)$ are abelian groups. By contrast $(\mathbb{Z}, \cdot)$ is *not* a group: the integer 2 has no multiplicative inverse in $\mathbb{Z}$.

Example 3.7. The nonzero rationals $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$ form an abelian group under multiplication, with identity 1 and inverse $1/a$; likewise $\mathbb{R}^\times$ and $\mathbb{C}^\times$. Zero must be excluded because it has no multiplicative inverse. The pattern is general: for any field F—the reader has met $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$; the abstract notion is defined in section 4—the nonzero elements $F^\times = F \setminus \{0\}$ form an abelian group under multiplication. Indeed, this is part of the very definition of a field.

Congruence classes furnish the central finite examples. Recall from section 2 the set $\mathbb{Z}/n\mathbb{Z}$ of residue classes $[a]_n$ and define operations on classes through representatives,

$$[a]_n + [b]_n = [a + b]_n, \qquad [a]_n\,[b]_n = [ab]_n.$$

By Proposition 2.21 the right-hand sides do not depend on the representatives chosen, so both operations are well defined, and each law of integer arithmetic—associativity, commutativity, the identities $[0]_n$ and $[1]_n$, negatives—passes to classes by computing on representatives. (The full structural statement, that these operations make $\mathbb{Z}/n\mathbb{Z}$ a commutative ring, is Theorem 4.6 in section 4; the present section uses only the fragments just listed.) Addition makes all of $\mathbb{Z}/n\mathbb{Z}$ a group, as Example 3.9 below records. For $n > 1$, multiplication does not—$[0]_n$ has no inverse—but section 2 identified exactly which classes are invertible: the *units*, the classes $[a]_n$ with $\gcd(a, n) = 1$ (Theorem 2.24). The units form a group.

> **Proposition 3.8.** *Under multiplication of classes, the set* $(\mathbb{Z}/n\mathbb{Z})^\times$ *of units of* $\mathbb{Z}/n\mathbb{Z}$ *is an abelian group. Its order is* $|(\mathbb{Z}/n\mathbb{Z})^\times| = \varphi(n)$*, Euler's totient of Definition 2.26: the number of integers a with* $1 \leq a \leq n$ *and* $\gcd(a, n) = 1$.

Proof. Multiplication of classes is associative and commutative with identity $[1]_n$, as derived above from Proposition 2.21 by computing on representatives, and $[1]_n$ is a unit (it is its own inverse). The operation is closed on units: if $[a]_n$ and $[b]_n$ are units with inverses $[a]_n^{-1}$ and $[b]_n^{-1}$, then $[ab]_n$ has inverse $[b]_n^{-1}[a]_n^{-1}$, since $(ab)(b^{-1}a^{-1}) \equiv 1 \pmod{n}$ for any representatives of the inverse classes. Every unit has an inverse by definition, and that inverse is itself a unit (it is invertible, with inverse the original class). Hence $(\mathbb{Z}/n\mathbb{Z})^\times$ is an abelian group. By Theorem 2.24 its elements are exactly the classes $[a]_n$ with $\gcd(a,n) = 1$ and $1 \le a \le n$, of which there are $\varphi(n)$ by Definition 2.26. □

Example 3.9. The pair $(\mathbb{Z}/n\mathbb{Z}, +)$ is an abelian group of order n: the discussion above derived associativity, commutativity, the identity $[0]_n$, and negatives from Proposition 2.21, and the classes number exactly n (the least nonnegative residues, Definition 2.22). The invertible classes form the abelian group $(\mathbb{Z}/n\mathbb{Z})^\times$ of order $\varphi(n)$ under multiplication (Theorem 2.24 and Proposition 3.8). When $n = p$ is prime, every nonzero class is a unit (Example 2.27), so $(\mathbb{Z}/p\mathbb{Z})^\times$ has order $p - 1$—this is Example 3.7 specialised to $\mathbb{Z}/p\mathbb{Z}$, which section 4 shows to be a field and denotes $\mathbb{F}_p$.

Example 3.10. The one-element set $\{e\}$, with its only possible operation $e * e = e$, is the *trivial group*. The group $\mathbb{Z}/2\mathbb{Z} = \{0, 1\}$ is the smallest nontrivial group. A subtler small example is the *Klein four-group* $V = \{e, a, b, c\}$, in which every element composed with itself gives e and the product of any two distinct nonidentity elements is the third; the verification that this table satisfies the axioms is routine. The group V is abelian of order 4.

Although V and $\mathbb{Z}/4\mathbb{Z}$ both have order 4, they are structurally different: in V every element combined with itself returns the identity, while in $\mathbb{Z}/4\mathbb{Z}$ the class 1 must be added to itself four times to reach 0. The language that makes "structurally different" precise is that of *isomorphism* (§3.8); in its terms, V is not isomorphic to $\mathbb{Z}/4\mathbb{Z}$, because V has no element of order 4 in the sense defined next.

3.3 The order of an element

Definition 3.11 (Order of an element)**.** Let G be a group and $a \in G$. The *order* of a, written $\operatorname{ord}(a)$, is the least positive integer n with $a^n = e$, if such an n exists; a then has *finite order*. If no positive power of a equals e, then a has *infinite order* and one writes $\operatorname{ord}(a) = \infty$.

The word "order" now has two meanings—the order $|G|$ of a group and the order $\operatorname{ord}(a)$ of an element. The clash is deliberate and is reconciled by the cyclic-group theory below: $\operatorname{ord}(a)$ equals the order of the smallest subgroup of G containing a (Proposition 3.21).

Proposition 3.12. *Let a have finite order $d = \operatorname{ord}(a)$, and let $m, k \in \mathbb{Z}$.*

1. *$a^m = e$ if and only if $d \mid m$.*
2. *$a^m = a^k$ if and only if $m \equiv k \pmod{d}$.*
3. *The distinct powers of a are exactly $e = a^0, a^1, \dots, a^{d-1}$.*

Proof. (1) If $d \mid m$, write $m = dq$; then $a^m = (a^d)^q = e^q = e$. Conversely, suppose $a^m = e$ and divide with remainder (Theorem 2.4): $m = dq + r$ with $0 \le r < d$. Then

$$e = a^m = (a^d)^q\, a^r = a^r,$$

so r is a nonnegative integer smaller than d with $a^r = e$. Minimality of d among *positive* exponents forces $r = 0$, whence $d \mid m$.

(2) We have $a^m = a^k$ if and only if $a^{m-k} = e$ (multiply by a^{-k} and use the exponent laws), which by (1) holds if and only if $d \mid (m-k)$, i.e. $m \equiv k \pmod{d}$ (Definition 2.20).

(3) By (2) the powers $a^0, a^1, \dots, a^{d-1}$ are pairwise distinct, their exponents being pairwise incongruent modulo d; and every power a^m equals $a^{m \bmod d}$, again by (2). □

The final step of part (1) deserves to be isolated, because it is the recurring device of this section: to show that every witness of some property is a multiple of the *least positive* witness m, take an arbitrary witness t, divide $t = mq + r$ with $0 \le r < m$, show that the remainder r is itself a witness, and conclude $r = 0$ by minimality. It is the same shape as the proof of Bézout's identity (Theorem 2.11), transplanted from the integers into a group, and it reappears twice below: for the least positive power of g lying in a subgroup (Theorem 3.24) and for the least positive element of a subgroup of $\mathbb{Z}$ (Proposition 3.26).

Example 3.13. In $(\mathbb{Z}/n\mathbb{Z}, +)$, read the powers of Definition 3.11 additively: the k-fold sum of the class 1 is the class k, which is 0 exactly when $n \mid k$, so $\operatorname{ord}(1) = n$. More generally $\operatorname{ord}(a) = n/\gcd(a, n)$ for any class a, by Corollary 3.23 below read additively. In $\mathbb{C}^\times$ the element i has order 4: its successive powers are

$$i^1 = i, \qquad i^2 = -1, \qquad i^3 = -i, \qquad i^4 = 1.$$

In $(\mathbb{Z}, +)$ every nonzero element has infinite order: no nonzero multiple of a nonzero integer is 0.

3.4 Subgroups

Definition 3.14 (Subgroup)**.** A subset $H \subseteq G$ of a group $(G, *)$ is a *subgroup*, written $H \le G$, if H is itself a group under the restriction of $*$ to $H \times H$: explicitly, H is nonempty, closed under $*$, contains the identity e of G, and contains a^{-1} whenever it contains a. Every group has the *trivial* subgroup $\{e\}$ and G itself; a subgroup other than G is *proper*.

Two small points keep this definition honest. First, the identity of a subgroup necessarily coincides with the identity of G: if $e_H \in H$ satisfies $e_H * e_H = e_H$, then cancelling e_H in G against $e_H * e = e_H$ (Proposition 3.3(3)) gives $e_H = e$. Second, inverses computed in H agree with those computed in G, by uniqueness of inverses in G. Nothing about H, in other words, depends on whether we regard it as a group in its own right or as a subset of G.

Checking all the group axioms for a subset would be wasteful, since associativity is inherited for free. A single condition suffices.

Proposition 3.15 (One-step subgroup test)**.** *Let G be a group and $H \subseteq G$. Then $H \le G$ if and only if*

1. *H is nonempty, and*
2. *for all $a, b \in H$, the element ab^{-1} lies in H.*

Proof. If $H \le G$, then H is nonempty and, for $a, b \in H$, it contains b^{-1} and hence ab^{-1}; both conditions hold.

Conversely, suppose (1) and (2) hold. Pick $a \in H$ by (1); taking $b = a$ in (2) gives $e = aa^{-1} \in H$. For any $b \in H$, taking the pair (e, b) in (2) gives $b^{-1} = eb^{-1} \in H$, so H is closed under inverses. Finally, for $a, b \in H$ we now know $b^{-1} \in H$, and applying (2) to the pair (a, b^{-1}) gives $a(b^{-1})^{-1} = ab \in H$, so H is closed under the operation. Associativity is inherited from G, so H is a group. □

Remark 3.16. For a *finite* nonempty subset H, closure under the operation alone forces $H \leq G$. Indeed, given $a \in H$, closure puts all the powers $a, a^2, a^3, \ldots$ in H; since H is finite they cannot all be distinct, so $a^i = a^j$ for some $i < j$, giving $a^{j-i} = e \in H$. Then $a^{-1} = a^{j-i-1} \in H$: this is a positive power of a when $j - i \geq 2$, and when $j - i = 1$ we have $a = e$, whose inverse e already lies in H. Thus for finite subsets, closure under multiplication is the only condition to verify.

Example 3.17. The even integers $2\mathbb{Z} = \{2k : k \in \mathbb{Z}\}$ form a subgroup of $(\mathbb{Z}, +)$: the set is nonempty, and $2k - 2l = 2(k - l)$ is again even (the subgroup test, written additively). More generally $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\} \leq \mathbb{Z}$ for every $n \geq 0$, by the same computation—and these are *all* the subgroups of $\mathbb{Z}$, as Proposition 3.26 will show.

Proposition 3.18. *The intersection $\bigcap_{i \in I} H_i$ of any family of subgroups $H_i \leq G$ is a subgroup of G.*

Proof. The identity e lies in every H_i, so the intersection is nonempty. If a and b lie in the intersection, then for each i both lie in H_i, so $ab^{-1} \in H_i$ by Proposition 3.15; hence ab^{-1} lies in the intersection, and the one-step test applies. □

Proposition 3.18 is what makes "the smallest subgroup containing a given set" well defined: the intersection of all subgroups containing the set is itself a subgroup containing the set, and it is contained in every other one.

Definition 3.19 (Generated subgroup)**.** For a subset $S \subseteq G$, the *subgroup generated by* S, written $\langle S \rangle$, is the intersection of all subgroups of G containing S. By Proposition 3.18 it is a subgroup, and by construction it is the smallest subgroup of G containing S.

Concretely, $\langle S \rangle$ consists of all finite products $s_1^{\varepsilon_1} s_2^{\varepsilon_2} \cdots s_k^{\varepsilon_k}$ with $s_i \in S$ and $\varepsilon_i \in \{+1, -1\}$, the empty product being e. Indeed, the set of such products contains S, contains e, and is closed under multiplication (concatenate) and under inversion (reverse the factors and negate the exponents, by the socks–shoes rule), so it is a subgroup containing S; conversely, every subgroup containing S is closed under these operations and so contains every such product. Hence the two descriptions coincide.

3.5 Cyclic groups

Definition 3.20 (Cyclic group)**.** A group G is *cyclic* if there exists $g \in G$ with

$$G = \langle g \rangle = \{ g^k : k \in \mathbb{Z} \};$$

such an element g is a *generator* of G.

The displayed description matches Definition 3.19: the set $\{g^k : k \in \mathbb{Z}\}$ contains g, and it is closed under products and inverses by the exponent laws, so it is a subgroup containing g; conversely any subgroup containing g must contain all its powers. Hence $\langle g \rangle = \{g^k : k \in \mathbb{Z}\}$: for a single element g, the general generated-subgroup construction and the naive one agree. Every cyclic group is abelian, since $g^m g^n = g^{m+n} = g^n g^m$.

Proposition 3.21. *Let $G = \langle g \rangle$ be cyclic.*

1. *If g has infinite order, the powers g^k ($k \in \mathbb{Z}$) are pairwise distinct, so G is infinite.*
2. *If g has finite order d, then $G = \{e, g, g^2, \ldots, g^{d-1}\}$ has exactly d elements.*

In both cases $|\langle g\rangle| = \operatorname{ord}(g)$.

Proof. (1) Suppose $g^i = g^j$ with $i \neq j$; without loss of generality $i > j$. Then $g^{i-j} = e$ with $i - j > 0$, contradicting infinite order. So the powers are pairwise distinct and G is infinite.

(2) This is Proposition 3.12(3): the distinct powers of g are exactly $g^0, g^1, \dots, g^{d-1}$, and these exhaust $G = \{g^k : k \in \mathbb{Z}\}$. □

Example 3.22. The group $(\mathbb{Z}, +)$ is infinite cyclic, generated by 1 and also by -1. The group $(\mathbb{Z}/n\mathbb{Z}, +)$ is cyclic of order n, generated by 1 (Example 3.13). The unit group $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$ is *not* cyclic: each of its four elements squares to 1 ($3^2 = 9 \equiv 1, 5^2 = 25 \equiv 1, 7^2 = 49 \equiv 1 \pmod 8$), so no element has order 4. By contrast $(\mathbb{Z}/5\mathbb{Z})^\times = \{1, 2, 3, 4\}$ is cyclic, generated by 2: its powers run

$$2^1 \equiv 2, \qquad 2^2 \equiv 4, \qquad 2^3 \equiv 3, \qquad 2^4 \equiv 1 \pmod 5,$$

exhausting the group.

The order of a power of g is determined by a gcd, and the formula tells us exactly which powers are again generators.

Corollary 3.23. *Let g have finite order n and let $k \in \mathbb{Z}$. Then*

$$\operatorname{ord}(g^k) = \frac{n}{\gcd(n, k)}.$$

In particular g^k generates $\langle g\rangle$ if and only if $\gcd(n, k) = 1$, so a cyclic group of order n has exactly $\varphi(n)$ generators.

Proof. Set $d = \gcd(n, k)$ and write $k = dk'$, $n = dn'$ with $\gcd(n', k') = 1$. For $m \geq 1$,

$$(g^k)^m = g^{km} = e \iff n \mid km \iff dn' \mid dk'm \iff n' \mid k'm \iff n' \mid m,$$

using Proposition 3.12(1) for the first equivalence and Euclid's lemma (Proposition 2.15(1), with $\gcd(n', k') = 1$) for the last. The least positive such m is $n' = n/d$, which is therefore $\operatorname{ord}(g^k)$.

For the second claim, $\langle g^k\rangle \subseteq \langle g\rangle$ always, and by Proposition 3.21 the subgroup $\langle g^k\rangle$ has $\operatorname{ord}(g^k) = n/\gcd(n, k)$ elements; it equals the n-element set $\langle g\rangle$ exactly when $\gcd(n, k) = 1$. Since the powers $g^1, g^2, \dots, g^n$ run through each element of $\langle g\rangle$ exactly once (Proposition 3.12(3)), the generators correspond to the $k \in \{1, \dots, n\}$ with $\gcd(n, k) = 1$, of which there are $\varphi(n)$ by Definition 2.26. □

The subgroups of a cyclic group can be classified completely. The theorem below is used constantly in the sequel; its part (3) is the origin of the "one subgroup per divisor" picture of $\mathbb{Z}/n\mathbb{Z}$.

Theorem 3.24 (Subgroups of cyclic groups)**.** *Let $G = \langle g\rangle$ be a cyclic group.*

1. *Every subgroup of G is cyclic.*
2. *If G is infinite, its subgroups are exactly $\langle g^m\rangle$ for $m \geq 0$; distinct $m \geq 0$ give distinct subgroups, and for $m \geq 1$ the subgroup $\langle g^m\rangle$ is isomorphic to $(\mathbb{Z}, +)$.*
3. *If $|G| = n$, then for each positive divisor $d \mid n$ there is exactly one subgroup of G of order d, namely $\langle g^{n/d}\rangle$, and these are all the subgroups of G. Subgroups of G thus correspond bijectively to positive divisors of n.*

Proof. (1) Let $H \leq G$. If $H = \{e\}$, then $H = \langle e \rangle$ is cyclic. Otherwise H contains some $g^t \neq e$, so $t \neq 0$; since H also contains $(g^t)^{-1} = g^{-t}$, it contains a power of g with positive exponent. Let m be the least positive integer with $g^m \in H$ (well-ordering, Theorem 2.1). We claim $H = \langle g^m \rangle$. The inclusion $\supseteq$ holds by closure. For $\subseteq$, take any element of H; it is some g^t, since $H \subseteq G = \langle g \rangle$. Divide $t = mq + r$ with $0 \leq r < m$; then

$$g^r = g^{t-mq} = g^t\,(g^m)^{-q} \in H,$$

and minimality of m forces $r = 0$—the division-minimality device again. Hence $g^t = (g^m)^q \in \langle g^m \rangle$.

(2) Let G be infinite. Then g has infinite order, for otherwise G would be finite by Proposition 3.21(2). By part (1) every subgroup is $\{e\} = \langle g^0 \rangle$ or $\langle g^m \rangle$ with $m \geq 1$ least positive, and conversely each $\langle g^m \rangle$ is a subgroup. For distinctness, note that for $m \geq 1$ the positive exponents t with $g^t \in \langle g^m \rangle$ are exactly the positive multiples of m: the elements of $\langle g^m \rangle$ are the powers g^{mk}, and $g^t = g^{mk}$ forces $t = mk$ because distinct exponents give distinct powers (Proposition 3.21(1)). The least such exponent recovers m, so distinct $m \geq 1$ give distinct subgroups, and each differs from $\{e\}$ because $g^m \neq e$. Finally, for $m \geq 1$ the element g^m has infinite order (if $(g^m)^t = g^{mt} = e$ then $mt = 0$, so $t = 0$), so $\langle g^m \rangle$ is an infinite cyclic group and is therefore isomorphic to $(\mathbb{Z}, +)$ by Theorem 3.43(1) below. No circularity arises: the proof of that theorem uses only Propositions 3.12 and 3.21.

(3) Let $|G| = n$, so $\operatorname{ord}(g) = n$ by Proposition 3.21.

Existence. Let $d \mid n$ be a positive divisor. Since n/d divides n, we have $\gcd(n, n/d) = n/d$, so Corollary 3.23 gives

$$\operatorname{ord}(g^{n/d}) = \frac{n}{\gcd(n,\, n/d)} = \frac{n}{n/d} = d,$$

and $\langle g^{n/d} \rangle$ is a subgroup of order d by Proposition 3.21.

Uniqueness. Let $H \leq G$ with $|H| = d$. If $d = 1$ then $H = \{e\} = \langle g^n \rangle = \langle g^{n/1} \rangle$, since $g^n = e$. Otherwise, as in part (1), $H = \langle g^m \rangle$ for the least positive m with $g^m \in H$, and Propositions 3.12 and 3.21 together with Corollary 3.23 give

$$d = |H| = \operatorname{ord}(g^m) = \frac{n}{\gcd(n, m)}, \qquad \text{so} \qquad \gcd(n, m) = \frac{n}{d}.$$

In particular n/d divides m, say $m = (n/d)\,m'$; then $g^m = (g^{n/d})^{m'}$, so $H = \langle g^m \rangle \subseteq \langle g^{n/d} \rangle$. Both sides have exactly d elements, so $H = \langle g^{n/d} \rangle$.

All subgroups arise. Any $H \leq G$ equals some $\langle g^m \rangle$ (or $\{e\}$), whose order $n/\gcd(n, m)$ is a positive divisor d of n; by uniqueness $H = \langle g^{n/d} \rangle$. The assignment $d \mapsto \langle g^{n/d} \rangle$ is therefore surjective onto the set of subgroups, and it is injective because subgroups of different orders are distinct. This is the asserted bijection. □

Remark 3.25. Specialised to $G = \mathbb{Z}/n\mathbb{Z}$ with generator 1 (written additively), Theorem 3.24(3) reads: for each divisor $d \mid n$ there is exactly one subgroup of order d, namely

$$\langle n/d \rangle = \{\, 0,\ n/d,\ 2n/d,\ \ldots,\ (d-1)\,n/d \,\},$$

itself cyclic of order d, hence isomorphic to $\mathbb{Z}/d\mathbb{Z}$ by the classification below. For $n = 12$ the six subgroups correspond to the divisors $1, 2, 3, 4, 6, 12$, and inclusion between subgroups mirrors divisibility between the corresponding divisors: the subgroup lattice of $\mathbb{Z}/12\mathbb{Z}$ is the divisor lattice of 12, with chains such as $1 \mid 2 \mid 4$, $1 \mid 2 \mid 6$, and $3 \mid 6$.

The classification promised in Example 3.17 now follows; it is Theorem 3.24(2) read additively for $G = (\mathbb{Z}, +) = \langle 1 \rangle$, but the direct argument is short enough to give in full.

Proposition 3.26. *Every subgroup of $(\mathbb{Z}, +)$ has the form $n\mathbb{Z}$ for a unique integer $n \geq 0$.*

Proof. Let $H \leq \mathbb{Z}$. If $H = \{0\}$, then $H = 0\mathbb{Z}$. Otherwise H contains a nonzero integer and, being closed under negation, contains a positive one; let n be its least positive element (Theorem 2.1). Closure under addition and negation gives $n\mathbb{Z} \subseteq H$. Conversely, for any $h \in H$, divide $h = nq + r$ with $0 \leq r < n$; then $r = h - nq \in H$, and minimality of n forces $r = 0$, so $h \in n\mathbb{Z}$. Hence $H = n\mathbb{Z}$. For uniqueness, note n is recovered from H as its least positive element (and $n = 0$ exactly for $H = \{0\}$), so distinct $n \geq 0$ give distinct subgroups. □

3.6 Cosets and Lagrange's theorem

A subgroup $H \leq G$ slices G into translated copies of itself, and counting the slices is the mechanism underlying Lagrange's theorem: partition the group into blocks, exhibit a bijection between each block and H, and multiply the block count by the block size. The blocks are the cosets.

Definition 3.27 (Cosets and index)**.** Let $H \leq G$ and $a \in G$. The *left coset* of H by a is

$$aH = \{\, ah : h \in H \,\},$$

and the *right coset* is $Ha = \{ha : h \in H\}$. In additive notation one writes $a + H = \{a + h : h \in H\}$. The set of left cosets is denoted G/H, and the *index* of H in G is $[G : H] = |G/H|$, the number of left cosets.

Lemma 3.28 (Cosets partition the group)**.** *Let $H \leq G$.*

1. *Each $a \in G$ lies in aH; the left cosets cover G.*
2. *Two left cosets are either equal or disjoint: $aH = bH$ if and only if $a^{-1}b \in H$, and otherwise $aH \cap bH = \emptyset$.*
3. *Every left coset has the same cardinality as H: the map $h \mapsto ah$ is a bijection from H onto aH.*

The same statements hold for right cosets, with $ba^{-1} \in H$ as the criterion in (2).

Proof. (1) Since $e \in H$, we have $a = ae \in aH$.

(2) First suppose the cosets meet: $ah_1 = bh_2$ for some $h_1, h_2 \in H$. Then $a^{-1}b = h_1h_2^{-1} \in H$. Conversely, suppose $a^{-1}b = h \in H$, so $b = ah$. Then every element $bh' \in bH$ equals $a(hh') \in aH$, giving $bH \subseteq aH$; and since $b^{-1}a = h^{-1} \in H$, the same argument with the roles of a and b exchanged gives $aH \subseteq bH$. Hence $aH = bH$. Combining the two directions: if two cosets share even one element they coincide, so distinct cosets are disjoint.

(3) The map $\lambda_a \colon H \to aH$, $h \mapsto ah$, is surjective by the definition of aH and injective by left cancellation (Proposition 3.3(3)): $ah = ah'$ implies $h = h'$. Hence $|aH| = |H|$.

For right cosets, the same three arguments apply with sides exchanged; in (2) the criterion becomes $ba^{-1} \in H$. □

Theorem 3.29 (Lagrange's theorem)**.** *Let G be a finite group and $H \leq G$. Then $|H|$ divides $|G|$, and*

$$|G| = [G : H]\,|H|.$$

Proof. By Lemma 3.28, the distinct left cosets of H partition G: they cover G by part (1) and are pairwise disjoint by part (2). Since G is finite there are finitely many of them, say $[G : H] = k$,

and each has exactly $|H|$ elements by part (3). Summing the sizes of the blocks of the partition,

$$|G| = \sum_{\text{distinct cosets } aH} |aH| = k\,|H| = [G : H]\,|H|. \qquad \square$$

Three corollaries follow in quick succession; the third is the theorem this section owes to the rest of the volume.

Corollary 3.30. *In a finite group G, the order of every element divides $|G|$.*

Proof. For $a \in G$, the cyclic subgroup $\langle a \rangle$ has order $\operatorname{ord}(a)$ by Proposition 3.21 (the order is finite, since $\langle a \rangle \subseteq G$ is finite), and Theorem 3.29 makes $|\langle a \rangle|$ a divisor of $|G|$. $\square$

Corollary 3.31. *If G is a finite group of order N, then $a^N = e$ for every $a \in G$.*

Proof. Let $d = \operatorname{ord}(a)$. By Corollary 3.30, $d \mid N$, say $N = dm$; then $a^N = (a^d)^m = e^m = e$. $\square$

Corollary 3.32 (Euler's theorem; Fermat's little theorem)**.** *Let $n \geq 1$ and $a \in \mathbb{Z}$ with* $\gcd(a, n) = 1$. *Then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

In particular, for a prime p and a not divisible by p,

$$a^{p-1} \equiv 1 \pmod{p},$$

and $a^p \equiv a \pmod{p}$ holds for all *integers a.*

Proof. Apply Corollary 3.31 to the group $(\mathbb{Z}/n\mathbb{Z})^\times$, which has order $\varphi(n)$ (Proposition 3.8). For $\gcd(a, n) = 1$ the class $[a]_n$ is a unit (Theorem 2.24), so $[a]_n^{\varphi(n)} = [1]_n$, which is the first congruence. For $n = p$ prime, $\varphi(p) = p - 1$ (Example 2.27), and $p \nmid a$ gives $\gcd(a, p) = 1$, yielding the second congruence. Multiplying it by a gives $a^p \equiv a \pmod{p}$; and when $p \mid a$ both sides are congruent to 0, so the last congruence holds for all a. $\square$

Remark 3.33. The congruences of Corollary 3.32 are proved here, once; section 7 restates them under the heading *Fermat's little theorem and Euler's theorem* and develops their cryptographic role there, citing this proof rather than repeating it.

Corollary 3.34. *Every group G of prime order p is cyclic, hence isomorphic to $\mathbb{Z}/p\mathbb{Z}$, and has no proper nontrivial subgroups.*

Proof. Pick $a \in G$ with $a \neq e$ (possible since $p \geq 2$). The subgroup $\langle a \rangle$ has at least two elements, and by Theorem 3.29 its order divides p; a divisor of p exceeding 1 equals p, so $\langle a \rangle = G$ and G is cyclic. The isomorphism with $\mathbb{Z}/p\mathbb{Z}$ is Theorem 3.43 below. Finally, any subgroup has order 1 or p by Lagrange, i.e. is $\{e\}$ or G. $\square$

3.7 Quotients of abelian groups

Cosets do more than count: for an abelian group they themselves form a group. This single construction underlies the quotient rings of section 4 and, through them, $\mathbb{Z}/n\mathbb{Z}$ and the polynomial quotients used throughout the series; it is all of quotient theory the volume requires.

Proposition 3.35 (Quotient of an abelian group)**.** *Let A be an abelian group, written additively, and let $H \leq A$. The operation*

$$(a + H) + (b + H) = (a + b) + H$$

on the coset set A/H is well defined and makes A/H an abelian group—the quotient group *of A by H—with identity $H = 0 + H$ and inverse $-(a + H) = (-a) + H$. If A is finite, then*

$$|A/H| = [A : H] = \frac{|A|}{|H|}.$$

Proof. The only point of substance is that the operation is well defined: its output is stated in terms of chosen representatives a and b, and we must check that replacing them by other representatives of the same cosets leaves the result unchanged. Suppose then that $a+H = a'+H$ and $b+H = b'+H$. By Lemma 3.28(2), written additively, this means $a' - a \in H$ and $b' - b \in H$. Then

$$(a' + b') - (a + b) = (a' - a) + (b' - b) \in H,$$

since H is closed under addition; applying Lemma 3.28(2) once more, $(a' + b') + H = (a + b) + H$. The operation is well defined. This representative-independence check is the template for every coset-wise construction in the volume; it recurs, in the same words, for quotient-ring multiplication in section 4.

The axioms now pass coset-wise from A: for associativity,

$$((a+H)+(b+H))+(c+H) = ((a+b)+c)+H = (a+(b+c))+H = (a+H)+((b+H)+(c+H));$$

commutativity follows the same way from commutativity in A; the coset $0+H = H$ is an identity; and $(-a)+H$ is an inverse of $a+H$. Hence A/H is an abelian group. When A is finite, the number of cosets is $[A : H] = |A|/|H|$ by Theorem 3.29—the same partition-into-equal-blocks count as before. □

The construction is not new to this volume in disguise: for $A = (\mathbb{Z}, +)$ and $H = n\mathbb{Z}$, Definition 2.22 exhibits each residue class as $[a]_n = a + n\mathbb{Z}$, a coset of $n\mathbb{Z} \leq \mathbb{Z}$, and the coset addition above is exactly addition of classes. The notation $\mathbb{Z}/n\mathbb{Z}$, introduced in section 2 as a bare name for the set of classes, is thus an instance of the general quotient notation A/H, and $(\mathbb{Z}/n\mathbb{Z}, +)$ is the quotient group $\mathbb{Z}/n\mathbb{Z}$ in the sense just constructed.

Remark 3.36. For a non-abelian group G the same construction succeeds exactly for those subgroups $N \leq G$ whose left and right cosets coincide, $gN = Ng$ for all $g \in G$; these are the *normal* subgroups. Commutativity makes every subgroup of an abelian group normal, which is why no extra hypothesis was needed in Proposition 3.35. The present volume deliberately needs only the abelian case, and we do not develop the general theory.

3.8 Homomorphisms and isomorphisms

Groups are compared through the maps that respect their operations.

Definition 3.37 (Group homomorphism)**.** Let $(G, *)$ and $(G', \circ)$ be groups. A function $\phi\colon G \to G'$ is a *group homomorphism* if

$$\phi(a * b) = \phi(a) \circ \phi(b) \qquad \text{for all } a, b \in G.$$

The set of all homomorphisms from G to G' is denoted $\mathrm{Hom}(G, G')$.

Proposition 3.38. *Let $\phi\colon G \to G'$ be a homomorphism, and write e, e' for the identities of G and G'.*

1. $\phi(e) = e'$.
2. $\phi(a^{-1}) = \phi(a)^{-1}$ *for all* $a \in G$.
3. $\phi(a^n) = \phi(a)^n$ *for all* $a \in G$ *and* $n \in \mathbb{Z}$.
4. *The image* $\phi(G)$ *is a subgroup of* G'.

Proof. (1) From $e = e * e$ we get $\phi(e) = \phi(e) \circ \phi(e)$; cancelling $\phi(e)$ in G' (Proposition 3.3(3)) gives $e' = \phi(e)$.

(2) Applying ϕ to $a * a^{-1} = e = a^{-1} * a$ and using (1),

$$\phi(a) \circ \phi(a^{-1}) = e' = \phi(a^{-1}) \circ \phi(a),$$

so $\phi(a^{-1})$ is an inverse of $\phi(a)$; by uniqueness of inverses, $\phi(a^{-1}) = \phi(a)^{-1}$.

(3) For $n \geq 0$, induct: the case $n = 0$ is (1), and $\phi(a^{n+1}) = \phi(a^n * a) = \phi(a)^n \circ \phi(a) = \phi(a)^{n+1}$. For $n < 0$, write $n = -m$ with $m > 0$ and combine with (2): $\phi(a^{-m}) = \phi((a^{-1})^m) = \phi(a^{-1})^m = (\phi(a)^{-1})^m = \phi(a)^{-m}$.

(4) The image is nonempty, containing $e' = \phi(e)$. For $x = \phi(a)$ and $y = \phi(b)$ in $\phi(G)$,

$$x \circ y^{-1} = \phi(a) \circ \phi(b)^{-1} = \phi(a * b^{-1}) \in \phi(G),$$

using (2); the one-step subgroup test (Proposition 3.15) concludes. □

Definition 3.39 (Kernel and image)**.** The *kernel* and *image* of a homomorphism $\phi\colon G \to G'$ are

$$\ker(\phi) = \{\, a \in G : \phi(a) = e' \,\} = \phi^{-1}(\{e'\}), \qquad \operatorname{im}(\phi) = \phi(G) = \{\, \phi(a) : a \in G \,\}.$$

Proposition 3.40. *For any homomorphism $\phi\colon G \to G'$, the kernel is a subgroup,* $\ker(\phi) \leq G$*, and ϕ is injective if and only if* $\ker(\phi) = \{e\}$.

Proof. The kernel contains e by Proposition 3.38(1), so it is nonempty; and for $a, b \in \ker(\phi)$,

$$\phi(ab^{-1}) = \phi(a) \circ \phi(b)^{-1} = e' \circ (e')^{-1} = e',$$

so $ab^{-1} \in \ker(\phi)$; the one-step test gives $\ker(\phi) \leq G$.

If ϕ is injective and $\phi(a) = e' = \phi(e)$, then $a = e$; the kernel is trivial. Conversely, suppose $\ker(\phi) = \{e\}$ and $\phi(a) = \phi(b)$. Then $\phi(ab^{-1}) = \phi(a) \circ \phi(b)^{-1} = e'$, so $ab^{-1} \in \ker(\phi) = \{e\}$, forcing $ab^{-1} = e$, i.e. $a = b$. □

Kernels are exactly the subgroups by which one may quotient: for abelian G, Proposition 3.35 forms $G/\ker(\phi)$ directly, and in general the kernel satisfies the normality condition of Remark 3.36—for $n \in \ker(\phi)$ and $g \in G$, one computes $\phi(gng^{-1}) = \phi(g) \circ e' \circ \phi(g)^{-1} = e'$, so conjugating the kernel by g lands back in the kernel, whence $g\ker(\phi) = \ker(\phi)\,g$.

Definition 3.41 (Isomorphism)**.** An *isomorphism* is a bijective homomorphism $\phi\colon G \to G'$; when one exists, the groups are *isomorphic*, written $G \cong G'$. An isomorphism from G to itself is an *automorphism*. An injective homomorphism is a *monomorphism*; a surjective one is an *epimorphism*.

Proposition 3.42. *The set-theoretic inverse of an isomorphism is an isomorphism, and $\cong$ is an equivalence relation on any collection of groups.*

Proof. Let $\phi\colon G \to G'$ be an isomorphism, with inverse function ϕ^{-1} (which exists since ϕ is a bijection, Theorem 1.1). For $x, y \in G'$, write $x = \phi(a)$ and $y = \phi(b)$; then

$$\phi^{-1}(x \circ y) = \phi^{-1}(\phi(a * b)) = a * b = \phi^{-1}(x) * \phi^{-1}(y),$$

so ϕ^{-1} is a homomorphism, and it is a bijection; hence an isomorphism. For the equivalence relation: reflexivity is witnessed by the identity automorphism id_G; symmetry by the inverse just constructed; and transitivity because a composite of homomorphisms is a homomorphism (apply the defining property twice) and a composite of bijections is a bijection. □

Isomorphic groups are "the same group" for every group-theoretic purpose: any property expressible in terms of the operation alone— orders of elements, the lattice of subgroups, commutativity—transfers across an isomorphism. One therefore classifies groups *up to isomorphism*, and the classification of cyclic groups is the model result of this kind: up to isomorphism there is exactly one cyclic group of each order.

Theorem 3.43 (Classification of cyclic groups)**.** *Every cyclic group is isomorphic to exactly one of the following:*

1. *the infinite cyclic group $(\mathbb{Z}, +)$, if it is infinite;*
2. *the finite cyclic group $(\mathbb{Z}/n\mathbb{Z}, +)$, if it has order n.*

In particular, two cyclic groups are isomorphic if and only if they have the same order.

Proof. Let $G = \langle g \rangle$ and define

$$\phi\colon \mathbb{Z} \to G, \qquad \phi(k) = g^k.$$

The exponent law $g^{k+l} = g^k g^l$ says exactly that ϕ is a homomorphism from $(\mathbb{Z}, +)$ to G, and ϕ is surjective because every element of G is a power of g.

Case 1: g has infinite order. Distinct exponents give distinct powers (Proposition 3.21(1)), so ϕ is injective, hence a bijective homomorphism, and $\mathbb{Z} \cong G$.

Case 2: $\mathrm{ord}(g) = n$. By Proposition 3.12(1),

$$\ker(\phi) = \{\, k \in \mathbb{Z} : g^k = e \,\} = n\mathbb{Z}.$$

Define the induced map

$$\bar{\phi}\colon \mathbb{Z}/n\mathbb{Z} \to G, \qquad \bar{\phi}([k]_n) = g^k.$$

It is well defined: if $[k]_n = [l]_n$ then $n \mid (k - l)$, so $g^k = g^l$ by Proposition 3.12(2). It is a homomorphism: $\bar{\phi}([k]_n + [l]_n) = \bar{\phi}([k + l]_n) = g^{k+l} = \bar{\phi}([k]_n)\,\bar{\phi}([l]_n)$. It is surjective because ϕ is. And it is injective by Proposition 3.40: if $\bar{\phi}([k]_n) = g^k = e$, then $n \mid k$ (Proposition 3.12(1)), so $[k]_n = [0]_n$; the kernel of $\bar{\phi}$ is trivial. Hence $\mathbb{Z}/n\mathbb{Z} \cong G$.

Distinctness. Isomorphic groups have equal order, an isomorphism being in particular a bijection. The group $\mathbb{Z}$ is infinite and the groups $\mathbb{Z}/n\mathbb{Z}$ for $n = 1, 2, 3, \ldots$ have the pairwise distinct finite orders n, so no two of the listed groups are isomorphic, and a cyclic group is isomorphic to exactly one of them—the one matching its order. The final claim follows: cyclic groups of equal order are isomorphic to the same listed group, hence to each other (Proposition 3.42), and isomorphic groups have equal order. □

The cheque presented at the head of the section can now be cashed. Fix any finite group G and a public element $g \in G$, and let $n = \operatorname{ord}(g)$. Associativity makes "$a$ repetitions of g" unambiguous—the element g^a, however bracketed, by generalised associativity (Proposition 3.4)—and the exponent laws give

$$(g^a)^b = g^{ab} = (g^b)^a,$$

so the two parties of the opening protocol, each applying a secret repetition count to the other's published value, really do arrive at a common secret element. Repetition counts obey an arithmetic, not a mere accumulation: by the classification theorem the subgroup $\langle g \rangle$ is a copy of $\mathbb{Z}/n\mathbb{Z}$, so exponents add, negate, and matter only modulo n (Proposition 3.12)—the modular arithmetic of section 2, transplanted into the exponent. Lagrange's theorem locates the modulus: n divides $|G|$, and $a^{|G|} = e$ for every $a \in G$ (Corollary 3.31); in the unit groups $(\mathbb{Z}/n\mathbb{Z})^\times$ this specialises to Euler's congruence $a^{\varphi(n)} \equiv 1 \pmod{n}$ (Corollary 3.32). Composition and inversion, the first two demands of the opening problem, are thereby supplied in full generality. The third demand—that publishing g^a should *hide* a—is not a theorem of algebra at all: it is a statement about computation, and it fails in some groups (in $(\mathbb{Z}/n\mathbb{Z}, +)$, recovering a from ag is a single modular division) while appearing to hold in others. Making the demand precise, and building the groups for which it is credible, occupies the sections that follow and the volumes above them.

4 Rings and fields

An arithmetic circuit—the object a proof system ultimately asks a verifier to check—is a network of gates, each of which either *adds* or *multiplies* two previously computed values. A verifier must therefore calculate in a number system where both operations make sense at once and interact through the familiar laws of school algebra; and at certain moments—inverting a random challenge, cancelling a common factor from both sides of an identity—the verifier must *divide* as well. In what kind of structure can one do both, and when can one also divide? The structures of genuine cryptographic interest all support two interacting operations obeying the arithmetic laws of $\mathbb{Z}$: the integers themselves, the rationals, the residue classes modulo n, polynomials, matrices. The abstract capture of this pattern is the *ring*; the key special case in which division by every nonzero element is possible is the *field*. Fields are the natural arena for linear algebra and for the theory of polynomials, and the finite fields $\mathbb{F}_p$ and $\mathbb{F}_q$ are the computational substrate of essentially all the elliptic-curve and proof-system machinery later in the monograph.

The section also discharges a debt. Section 2 verified, concretely and one law at a time, that congruence classes admit an arithmetic, and promised that the verified laws would receive their structural names in due course. They receive them here: $\mathbb{Z}/n\mathbb{Z}$ is re-read as a ring, indeed as the quotient of the ring $\mathbb{Z}$ by the ideal $n\mathbb{Z}$, and the theorem that $\mathbb{Z}/p\mathbb{Z}$ is a field is stated in the language that the rest of the series uses. As a standing convention we assume from §3 only the theory of *abelian* groups; additive structure is written additively with identity 0, multiplicative structure multiplicatively with identity 1.

4.1 Rings

Definition 4.1 (Ring)**.** A *ring* is a set R equipped with two binary operations $+$ and $\cdot$ such that:

1. $(R, +)$ is an abelian group: addition is associative and commutative, there is an element 0 with $a + 0 = a$ for all a, and every a has an *additive inverse* $-a$ with $a + (-a) = 0$;
2. multiplication is associative: $a \cdot (b \cdot c) = (a \cdot b) \cdot c$;

3. there is a *multiplicative identity* 1 with $1 \cdot a = a \cdot 1 = a$ for all a;
4. multiplication distributes over addition on both sides: $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.

The ring is *commutative* if moreover $a \cdot b = b \cdot a$ for all $a, b \in R$. We write ab for $a \cdot b$, let multiplication bind tighter than addition, and call 0 the *zero* and 1 the *unity* of the ring.

Remark 4.2. Conventions in the literature vary. Some authors omit requirement (3), saying "ring with unity" for the notion above and "rng" for the weaker one. In this monograph *every ring has a* 1, and the structure-preserving maps between rings—the ring homomorphisms of theorem 4.21 below—are required to preserve it. Moreover "ring" frequently means "commutative ring" in the examples that matter to us; commutativity is always stated explicitly when a result depends on it.

The distributive law is the only axiom coupling the two operations, and a surprising amount of everyday algebra follows from it alone. The following identities are used constantly and, after this proposition, without comment.

Proposition 4.3 (Elementary ring arithmetic)**.** *Let R be a ring and $a, b \in R$. Then:*

1. $0 \cdot a = a \cdot 0 = 0$;
2. $(-a)b = a(-b) = -(ab)$;
3. $(-a)(-b) = ab$;
4. *for all integers $m, n \in \mathbb{Z}$, $(ma)(nb) = (mn)(ab)$, where na denotes the n-fold additive multiple ($0a = 0$, $(n + 1)a = na + a$, and $(-n)a = -(na)$, as for any abelian group, §3).*

Proof. (1) From $0 = 0 + 0$ and distributivity, $0 \cdot a = (0 + 0) \cdot a = 0 \cdot a + 0 \cdot a$; adding $-(0 \cdot a)$ to both sides leaves $0 = 0 \cdot a$. The computation for $a \cdot 0$ is symmetric.

(2) By distributivity and (1), $ab + (-a)b = (a + (-a))b = 0 \cdot b = 0$, so $(-a)b$ is the additive inverse of ab, that is, $(-a)b = -(ab)$; likewise $ab + a(-b) = a(b + (-b)) = 0$ gives $a(-b) = -(ab)$.

(3) Applying (2) twice, $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$, the last step because negation is an involution in the abelian group $(R, +)$.

(4) First fix a, b and show $a(nb) = n(ab)$ for $n \geq 0$ by induction: the case $n = 0$ is (1), and $a((n + 1)b) = a(nb + b) = a(nb) + ab = n(ab) + ab = (n + 1)(ab)$ by distributivity and the inductive hypothesis. For $n < 0$ write $n = -m$ with $m > 0$; then $a(nb) = a(-(mb)) = -(a(mb)) = -(m(ab)) = n(ab)$ by (2). The same argument in the first factor gives $(ma)c = m(ac)$ for all c, and combining,

$$(ma)(nb) = m(a(nb)) = m(n(ab)) = (mn)(ab),$$

the final equality being the usual bookkeeping for iterated multiples in an abelian group, which we omit. □

Remark 4.4 (The zero ring). Nothing in theorem 4.1 forces $1 \neq 0$. If $1 = 0$ in a ring R, then every $a \in R$ satisfies $a = 1 \cdot a = 0 \cdot a = 0$ by theorem 4.3(1), so $R = \{0\}$: the *zero ring* (or *trivial ring*), a legitimate ring in which both operations are the only possible ones. Whenever the zero ring must be excluded we impose the assumption $1 \neq 0$ explicitly; the definition of a field below builds $1 \neq 0$ in as a standing assumption.

Example 4.5 (Basic rings)**.**

1. The integers $\mathbb{Z}$ form the prototype commutative ring; much of ring theory consists of isolating which features of $\mathbb{Z}$ survive in generality.
2. The rationals $\mathbb{Q}$, the reals $\mathbb{R}$, and the complexes $\mathbb{C}$ are commutative rings—indeed fields, in the sense of §4.4.

3. The residue classes $\mathbb{Z}/n\mathbb{Z} = \{\overline{0}, \overline{1}, \dots, \overline{n-1}\}$ for $n \geq 1$—in this section's examples we abbreviate the class $[a]_n$ of Definition 2.22 to $\overline{a}$ when the modulus is clear—under the induced operations form a commutative ring whose additive group is the cyclic group of order n (§3); this is theorem 4.6 below. We write $\mathbb{Z}/n\mathbb{Z}$, or $\mathbb{Z}_n$ when no confusion threatens. The case $n = 1$ gives the zero ring of theorem 4.4.

4. The *polynomial ring* $R[x]$ over a commutative ring R consists of the polynomials $a_0 + a_1 x + \cdots + a_d x^d$ with coefficients $a_i \in R$, added coefficient-wise and multiplied by expanding and collecting powers of x; it is a commutative ring with the same 0 and 1 as R (the constant polynomials). The theory of polynomials over a field is developed in its own section later in the volume.

5. The *matrix ring* $\mathrm{M}_n(R)$ of $n \times n$ matrices over a commutative ring R, $n \geq 1$, is a ring under matrix addition and multiplication. It is *not* commutative for $n \geq 2$ and $R \neq \{0\}$, even when R is commutative: in $\mathrm{M}_2(\mathbb{Z})$,

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \text{but} \qquad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

   This is the first important noncommutative example.

6. The *product ring* $R \times S$ of two rings carries the componentwise operations, with zero $(0, 0)$ and identity $(1, 1)$; likewise any finite product $R_1 \times \cdots \times R_k$.

7. The zero ring $\{0\}$.

The third example is the one this monograph computes in, and it is the one whose ring structure was promised rather than named in §2. We now discharge that promise.

Theorem 4.6. *Let $n \geq 1$. The operations*

$$[a]_n + [b]_n := [a + b]_n, \qquad [a]_n \cdot [b]_n := [ab]_n$$

are well defined and make $\mathbb{Z}/n\mathbb{Z}$ a commutative ring with additive identity $[0]_n$ and multiplicative identity $[1]_n$. It has exactly n elements.

Proof. The right-hand sides are defined in terms of chosen representatives a, b, so we must check that different choices give the same class. This is exactly the compatibility half of Proposition 2.21: if $a' \equiv a$ and $b' \equiv b \pmod{n}$, then $a' + b' \equiv a + b$ and $a'b' \equiv ab \pmod{n}$, so $[a' + b']_n = [a + b]_n$ and $[a'b']_n = [ab]_n$. The operations are therefore well defined on classes.

Every ring axiom is now inherited from the corresponding law of $\mathbb{Z}$ by passing to classes. For instance, distributivity:

$$\begin{aligned} [a]_n([b]_n + [c]_n) &= [a]_n[b + c]_n = [a(b + c)]_n = [ab + ac]_n \\ &= [ab]_n + [ac]_n = [a]_n[b]_n + [a]_n[c]_n, \end{aligned}$$

using only the definitions and the distributive law of the integers. Associativity and commutativity of both operations follow by the identical pattern. The class $[0]_n$ is an additive identity and $[1]_n$ a multiplicative one, since $[a + 0]_n = [a]_n$ and $[1 \cdot a]_n = [a]_n$; and $-[a]_n = [-a]_n$ because $[a]_n + [-a]_n = [a + (-a)]_n = [0]_n$.

Finally, by the division algorithm every integer is congruent modulo n to exactly one of $0, 1, \dots, n-1$ (the least nonnegative residues, Definition 2.22 and the discussion following it), so the classes $[0]_n, \dots, [n-1]_n$ are distinct and exhaust $\mathbb{Z}/n\mathbb{Z}$: the ring has exactly n elements. □

4.2 Units, zero divisors, and integral domains

Two features of the integers now deserve names of their own. First, $\mathbb{Z}$ enjoys *cancellation*: a product of nonzero integers is nonzero, so a nonzero common factor may be struck from both sides of an equation. Second, division *within* $\mathbb{Z}$ is nonetheless rare: only ± 1 divide 1. The next two definitions abstract these phenomena—units capture the elements one can divide by, zero divisors the obstruction to cancelling.

Definition 4.7 (Unit)**.** An element u of a ring R is a *unit* (or is *invertible*) if there exists $v \in R$ with $uv = vu = 1$. Such a v is unique: if $uv = vu = 1$ and $uv' = v'u = 1$, then $v = v \cdot 1 = v(uv') = (vu)v' = 1 \cdot v' = v'$. It is called the *inverse* of u and written u^{-1}. The set of units of R is denoted $R^\times$.

Proposition 4.8. *For any ring R, the set $R^\times$ of units is a group under the multiplication of R, called the* group of units (*or* multiplicative group) *of R.*

Proof. The identity 1 is a unit (it is its own inverse), so $1 \in R^\times$. For closure, let $u, w \in R^\times$; then

$$(uw)(w^{-1}u^{-1}) = u(ww^{-1})u^{-1} = u \cdot 1 \cdot u^{-1} = 1,$$

and symmetrically $(w^{-1}u^{-1})(uw) = 1$, so uw is a unit with $(uw)^{-1} = w^{-1}u^{-1}$. Multiplication in $R^\times$ is associative because it is associative in R. Finally each u^{-1} is itself a unit, its inverse being u; so every element of $R^\times$ has an inverse in $R^\times$, and the group axioms are verified. □

Example 4.9 (Units)**.**

1. The units of the integers are $\mathbb{Z}^\times = \{1, -1\}$: if $uv = 1$ in $\mathbb{Z}$ then $|u|\,|v| = 1$ forces $|u| = 1$.
2. In $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ every nonzero element is a unit: $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$, $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$, $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$. Commutative rings with $1 \neq 0$ in which *every* nonzero element is a unit are precisely the fields of §4.4; this is their defining feature. Both qualifiers are needed: the zero ring of theorem 4.4 satisfies the unit condition vacuously, and noncommutative rings satisfying it—the *division rings*—exist but are not fields.
3. In $\mathbb{Z}/n\mathbb{Z}$, the class $\bar{a}$ is a unit if and only if $\gcd(a, n) = 1$, by Theorem 2.24; thus $(\mathbb{Z}/n\mathbb{Z})^\times$ consists of the residue classes coprime to the modulus, and its order is Euler's totient $\varphi(n)$ (Definition 2.26).
4. In the matrix ring, $\mathrm{M}_n(R)^\times$ is the group of invertible $n \times n$ matrices. Over a field it is the *general linear group* GL_n, the matrices of nonzero determinant; the determinant belongs to linear algebra, treated later in the volume.

Definition 4.10 (Zero divisor)**.** A *nonzero* element a of a ring R is a *left zero divisor* if $ab = 0$ for some nonzero $b \in R$, and a *right zero divisor* if $ba = 0$ for some nonzero b. In a commutative ring the two notions coincide and we say simply *zero divisor*. A nonzero element that is neither is called *regular* (or *cancellable*).

The terminology "cancellable" is justified by the following equivalence: zero divisors are exactly the obstruction to dividing out a common factor.

Proposition 4.11. *A nonzero element a of a ring R is not a left zero divisor if and only if it is* left-cancellable*: $ab = ac$ implies $b = c$ for all $b, c \in R$.*

Proof. Suppose a is not a left zero divisor and $ab = ac$. Then $a(b - c) = ab - ac = 0$, using theorem 4.3(2) to expand $a(b + (-c))$; since a is nonzero and not a left zero divisor, the factor $b - c$ must be 0, so $b = c$. Conversely, suppose a is a left zero divisor, say $ab = 0$ with $b \neq 0$. Then $ab = 0 = a \cdot 0$ while $b \neq 0$: cancellation fails. □

Remark 4.12. A unit is never a zero divisor: if $u \in R^\times$ and $ub = 0$, then $b = 1 \cdot b = (u^{-1}u)b = u^{-1}(ub) = u^{-1} \cdot 0 = 0$. The converse fails in general—in $\mathbb{Z}$ every nonzero element is a non-zero-divisor, yet only ± 1 are units—but every nonzero non-zero-divisor is a unit in a finite commutative ring, by the injectivity-implies-surjectivity argument that proves theorem 4.29 below.

Example 4.13 (Zero divisors)**.**

1. In $\mathbb{Z}/6\mathbb{Z}$, $\overline{2} \cdot \overline{3} = \overline{6} = \overline{0}$, so $\overline{2}$ and $\overline{3}$ are zero divisors. In general a nonzero class $\overline{a} \in \mathbb{Z}/n\mathbb{Z}$ is a zero divisor precisely when $d := \gcd(a, n) > 1$: in that case $a \cdot (n/d) = (a/d) \cdot n \equiv 0 \pmod{n}$ while $\overline{n/d} \neq \overline{0}$ (as $0 < n/d < n$); and when $d = 1$ the class is a unit (Example 4.9(3)), hence no zero divisor by theorem 4.12. Thus in $\mathbb{Z}/n\mathbb{Z}$ every nonzero element is either a unit or a zero divisor—a dichotomy revisited from a structural angle by theorem 4.29.
2. In a product ring $R \times S$ with both factors nonzero, $(1, 0)(0, 1) = (0, 0)$: product rings always have zero divisors.
3. In $\mathrm{M}_2(\mathbb{R})$, the diagonal matrices $\mathrm{diag}(1, 0)$ and $\mathrm{diag}(0, 1)$ multiply to the zero matrix; more generally every nonzero *singular* matrix—the standard name for a square matrix that is not invertible—is a zero divisor there.

Definition 4.14 (Integral domain)**.** An *integral domain* (or simply *domain*) is a commutative ring R with $1 \neq 0$ and no zero divisors: $ab = 0$ implies $a = 0$ or $b = 0$. Equivalently, by theorem 4.11, a domain is a nonzero commutative ring in which every nonzero element is cancellable.

The residue rings sort themselves neatly against this definition, and the sorting is governed by primality.

Proposition 4.15. *Let $n \geq 2$. The ring $\mathbb{Z}/n\mathbb{Z}$ has no zero divisors if and only if n is prime; that is, $\mathbb{Z}/n\mathbb{Z}$ is an integral domain if and only if n is prime.*

Proof. If $n = ab$ is composite with $1 < a, b < n$, then $[a]_n[b]_n = [n]_n = [0]_n$ with both factors nonzero (their representatives lie strictly between 0 and n), so $\mathbb{Z}/n\mathbb{Z}$ has zero divisors. Conversely let $n = p$ be prime and suppose $[a]_p[b]_p = [0]_p$, that is, $p \mid ab$. Euclid's lemma for primes (Lemma 2.17) gives $p \mid a$ or $p \mid b$, that is, $[a]_p = [0]_p$ or $[b]_p = [0]_p$. Since $p \geq 2$ we also have $[1]_p \neq [0]_p$, so $\mathbb{Z}/p\mathbb{Z}$ is an integral domain. □

For prime p the ring $\mathbb{Z}/p\mathbb{Z}$ is in fact far better than a domain: every nonzero class is invertible, so it is a *field*—a term defined in §4.4, where the statement is proved as theorem 4.26.

Example 4.16 (Integral domains)**.** The integers $\mathbb{Z}$ form an integral domain—the example the name commemorates. Every field is a domain (theorem 4.28 below). The ring $\mathbb{Z}/n\mathbb{Z}$ is a domain exactly when n is prime (theorem 4.15; the composite case is witnessed concretely by Example 4.13(1)). If R is a domain, so is the polynomial ring $R[x]$: let f and g be nonzero of degrees m and n—the largest indices carrying nonzero coefficients f_m and g_n, the *leading coefficients*. In fg

the coefficient of x^{m+n} is $\sum_{i+j=m+n} f_i g_j$; every term with $i > m$ or $j > n$ has a vanishing factor, and $i \le m, j \le n, i + j = m + n$ force $i = m, j = n$, so the coefficient equals $f_m g_n$, nonzero because R is a domain. Hence $fg \neq 0$. Non-domains include $\mathbb{Z}/6\mathbb{Z}$, product rings with two nonzero factors, and the matrix rings $\mathrm{M}_n(R)$ for $n \ge 2$ (Example 4.13).

4.3 Ideals and quotient rings

We next carry quotient formation from groups to rings. For an abelian group and any subgroup, the cosets themselves form a group (Proposition 3.35); the ring-theoretic analogue asks for the subsets of a ring by which one can quotient while keeping *both* operations, and the answer is the notion of an *ideal*. The treatment here is deliberately brief: we need ideals chiefly to construct $\mathbb{Z}/n\mathbb{Z}$ conceptually and to indicate where the finite fields $\mathbb{F}_q$ come from; a fuller development belongs to commutative algebra.

Definition 4.17 (Ideal)**.** Let R be a commutative ring. An *ideal* of R is a subset $I \subseteq R$ such that

1. I is a subgroup of the additive group $(R, +)$: $0 \in I$, and $a - b \in I$ whenever $a, b \in I$; and
2. I *absorbs* multiplication by arbitrary ring elements: $ra \in I$ whenever $r \in R$ and $a \in I$.

In noncommutative rings one distinguishes left, right, and two-sided ideals according to the side on which absorption holds; commutative rings suffice for this volume, and there the three notions coincide.

Despite containing 0 and being closed under the ring's operations in the sense above, an ideal is almost never a subring: if an ideal I contains 1—or, by the same absorption argument, any unit u, since then $1 = u^{-1}u \in I$—then $r = r \cdot 1 \in I$ for every $r \in R$, so $I = R$. The only ideal containing a unit is the whole ring. This little observation will carry real weight: it is the reason a field has no ideals other than the two trivial ones, and hence no interesting quotients.

Example 4.18 (Ideals)**.**

1. In any commutative ring R, the subsets $\{0\}$ and R are ideals—the *trivial* and *improper* ideals respectively.
2. For $a \in R$, the *principal ideal generated by* a is
$$(a) := aR = \{\, ar : r \in R \,\},$$
the set of multiples of a; absorption holds because $r'(ar) = a(r'r)$. In $\mathbb{Z}$, the principal ideal $(n) = n\mathbb{Z}$ is the set of multiples of n. In fact *every* ideal of $\mathbb{Z}$ is principal: an ideal $I \neq \{0\}$ contains a nonzero element and hence (closing under negation) a positive one, so it contains a least positive element n by well-ordering; for any $a \in I$ the division algorithm gives $a = qn + r$ with $0 \le r < n$, whence $r = a - qn \in I$ by absorption and subgroup closure, forcing $r = 0$ by minimality; so $I = n\mathbb{Z}$. This sharpens the observation of Remark 2.12, where ideals of $\mathbb{Z}$ first appeared as the sets of Bézout combinations.
3. For $a_1, \dots, a_k \in R$, the *ideal generated by* $a_1, \dots, a_k$ is $(a_1, \dots, a_k) = \{\, r_1 a_1 + \cdots + r_k a_k : r_i \in R \,\}$, the smallest ideal containing them all.

An integral domain in which every ideal is principal, as in (2), is called a *principal ideal domain* (PID).

Definition 4.19 (Quotient ring operations)**.** Let I be an ideal of a commutative ring R. Since I is a subgroup of the abelian group $(R, +)$, the additive cosets $a + I$ form the quotient group R/I under coset addition $(a + I) + (b + I) = (a + b) + I$ (Proposition 3.35). Define a multiplication

of cosets by

$$(a + I)(b + I) := ab + I.$$

Proposition 4.20. *With the operations of theorem 4.19, the set R/I is a commutative ring—the* quotient ring *of R by I—with zero $0 + I = I$ and identity $1 + I$. In particular the coset multiplication is well defined.*

Proof. Well-definedness is the point at which absorption is needed. Let $a' = a + s$ and $b' = b + t$ be other representatives of the same cosets, with $s, t \in I$. Then

$$a'b' = (a + s)(b + t) = ab + at + sb + st,$$

and each of at, sb, st lies in I by the absorption property; hence $a'b' - ab \in I$ and $a'b' + I = ab + I$. The product coset is therefore independent of the representatives chosen.

The ring axioms now pass to cosets exactly as in the proof of theorem 4.6. For instance, distributivity reads

$$\begin{aligned}(a + I)((b + I) + (c + I)) &= (a + I)((b + c) + I) = a(b + c) + I \\ &= (ab + ac) + I = (a + I)(b + I) + (a + I)(c + I),\end{aligned}$$

and associativity and commutativity of both operations follow by the same pattern from the corresponding laws in R. The coset $I = 0 + I$ is the additive identity, $1 + I$ the multiplicative identity, and $-(a + I) = (-a) + I$. □

One more definition names the maps under which ring structure is preserved; we need it to say precisely in what sense the quotient construction generalises the passage from $\mathbb{Z}$ to $\mathbb{Z}/n\mathbb{Z}$.

Definition 4.21 (Ring homomorphism)**.** A *ring homomorphism* between rings R and S is a map $\phi\colon R \to S$ with

$$\phi(a + b) = \phi(a) + \phi(b), \qquad \phi(ab) = \phi(a)\phi(b), \qquad \phi(1) = 1$$

for all $a, b \in R$. Its *kernel* is $\ker \phi = \{\, a \in R : \phi(a) = 0 \,\}$.

The kernel of a ring homomorphism out of a commutative ring is always an ideal: it is a subgroup of $(R, +)$ because ϕ is in particular a homomorphism of additive groups, and it absorbs because $a \in \ker \phi$ gives $\phi(ra) = \phi(r)\phi(a) = \phi(r) \cdot 0 = 0$.

Example 4.22. Take $R = \mathbb{Z}$ and $I = n\mathbb{Z}$. The cosets $a + n\mathbb{Z}$ are literally the residue classes $[a]_n$ of Definition 2.22, and the coset operations of theorem 4.19 are exactly residue arithmetic: the quotient construction recovers the ring $\mathbb{Z}/n\mathbb{Z}$ of Example 4.5(3), and is the conceptual origin of modular arithmetic. The same construction with a different ring produces the general finite fields: quotients $\mathbb{F}_p[x]/(f)$ of a polynomial ring by the principal ideal of an *irreducible* polynomial f of degree k—irreducibility, the polynomial analogue of primality, is defined in §5—furnish the finite fields $\mathbb{F}_q$ with $q = p^k$ elements. We record the recipe here and take it up in §6; the curves of Halo 2 and Orchard work over *prime* fields, so the prime case $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ carries the main line of development.

Remark 4.23. Structurally, then, $\mathbb{Z}/n\mathbb{Z}$ is the quotient of the ring $\mathbb{Z}$ by the ideal $n\mathbb{Z} = \{\, kn : k \in \mathbb{Z} \,\}$, and the map

$$\pi\colon \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}, \qquad \pi(a) = [a]_n,$$

is a surjective ring homomorphism with kernel $n\mathbb{Z}$. This is the prototype of every quotient construction used in the volume's algebra: one starts from a ring, singles out an ideal of elements

to be "declared zero", and computes with cosets. On notation: we abbreviate $[a]_n$ to $\bar{a}$ (as in the examples above) or drop the decoration altogether, writing plain a, whenever the modulus is clear from context; $\mathbb{Z}/n\mathbb{Z}$ and $\mathbb{Z}_n$ are used interchangeably; and $\mathbb{Z}_n$ always means this ring, never the n-adic integers (which do not appear in this series).

4.4 Fields

Definition 4.24 (Field)**.** A *field* is a commutative ring F with $1 \neq 0$ in which every nonzero element is a unit. Equivalently, a field is a set F with two operations $+$ and $\cdot$ such that $(F, +)$ is an abelian group with identity 0, the nonzero elements $(F \setminus \{0\}, \cdot)$ form an abelian group with identity 1, and multiplication distributes over addition. Unwinding the definition: in a field one may add, subtract, multiply, and divide by any nonzero element, with $a/b := ab^{-1}$. The group $F^\times = (F \setminus \{0\}, \cdot)$ is the *multiplicative group* of the field. (Two points of the equivalence are not direct transcriptions. Passing from the first formulation to the second, one must show $F \setminus \{0\}$ is closed under multiplication; that is theorem 4.28 below. Passing back, the ring axioms must be checked for products involving 0: distributivity forces $0 \cdot a = a \cdot 0 = 0$ by the computation of theorem 4.3(1), whereupon every product with a zero factor vanishes and the associativity, commutativity, and identity laws extend from $F \setminus \{0\}$ to all of F; and $1 \neq 0$ holds because the group $F \setminus \{0\}$ contains its identity 1.)

Definition 4.25 (Subfield and extension field)**.** A subset $F \subseteq K$ of a field K is a *subfield* when it contains 0 and 1 and is itself a field under the operations of K restricted to F. When F is a subfield of K, the larger field K is called an *extension field* (or simply an *extension*) of F; the two phrases name one relation viewed from its two ends. More generally, an injective ring homomorphism $F \hookrightarrow K$ between fields exhibits K as an extension of the copy of F inside it, and we permit ourselves the usual abuse of identifying F with that copy.

The finite world of §2 already contains the series' most important fields, and the unit criterion proved there identifies them at once.

Corollary 4.26. *If p is prime, then every nonzero class in $\mathbb{Z}/p\mathbb{Z}$ is a unit, so $\mathbb{Z}/p\mathbb{Z}$ is a field; it is denoted $\mathbb{F}_p$. More generally, $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime.*

Proof. Let p be prime and $[a]_p \neq [0]_p$, so that $p \nmid a$. The only positive divisors of p are 1 and p, so $\gcd(a, p) = 1$, and Theorem 2.24 makes $[a]_p$ a unit. Since $[1]_p \neq [0]_p$ for $p \geq 2$, the ring $\mathbb{Z}/p\mathbb{Z}$ is a field. Conversely, if $n = ab$ is composite with $1 < a, b < n$, then $[a]_n \neq [0]_n$ but $\gcd(a, n) = a > 1$, so $[a]_n$ is not a unit (Theorem 2.24 again) and the ring is not a field; and $n = 1$ gives the zero ring, which is not a field because a field has $1 \neq 0$ by definition. □

Example 4.27 (Fields)**.**

1. The rationals $\mathbb{Q}$, the reals $\mathbb{R}$, and the complexes $\mathbb{C}$ are fields. The integers are not: 2 has no inverse in $\mathbb{Z}$.

2. For p prime, $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$ is a field (theorem 4.26; theorem 4.29 below recovers the same fact abstractly). It is called the *prime field* of characteristic p—theorem 4.30 names the invariant—and is fundamental to every later cryptographic construction in the series. Its finite extensions $\mathbb{F}_q$ with $q = p^k$ elements exist for every prime power, arising from the quotient recipe of Example 4.22, and are developed in §6.

3. The subset $\mathbb{Q}(i) = \{a + bi : a, b \in \mathbb{Q}\} \subseteq \mathbb{C}$ is a field: the inverse of a nonzero $a + bi$ is $(a - bi)/(a^2 + b^2)$, which again has rational coordinates. More generally the *number fields*,

obtained by adjoining to $\mathbb{Q}$ roots of polynomial equations with rational coefficients, are fields.

4. For any field F, the *rational functions* $F(x) = \{f/g : f, g \in F[x],\ g \neq 0\}$ form a field under the usual arithmetic of fractions—the field of fractions of the polynomial ring $F[x]$.

Proposition 4.28. *Every field is an integral domain.*

Proof. Let F be a field. Commutativity and $1 \neq 0$ hold by definition. Suppose $ab = 0$ with $a \neq 0$. Then a is a unit, and multiplying by its inverse,

$$b = 1 \cdot b = (a^{-1}a)b = a^{-1}(ab) = a^{-1} \cdot 0 = 0.$$

Hence F has no zero divisors. □

The converse fails: $\mathbb{Z}$ is a domain but not a field (Example 4.16). Finiteness, however, closes the gap entirely, and does so by a counting device worth naming, for it recurs. Call it the *injectivity-implies-surjectivity argument*: to invert an element a of a finite structure, show that multiplication by a is injective, conclude that it is surjective because an injective map from a finite set to itself must be onto, and read off a preimage of 1. The same device proves the claim of theorem 4.12 that in a finite commutative ring every nonzero non-zero-divisor is a unit.

Theorem 4.29. *Every finite integral domain is a field.*

Proof. Let R be a finite integral domain and $a \in R$ nonzero. Consider the multiplication map

$$\mu_a \colon R \to R, \qquad \mu_a(x) = ax.$$

It is injective: if $ax = ay$, then $a(x - y) = 0$, and since R is a domain and $a \neq 0$ this forces $x - y = 0$, that is, $x = y$ (equivalently, a is cancellable by theorem 4.11). An injective map from a finite set to itself is surjective: the image $\mu_a(R)$ has $|R|$ distinct elements and sits inside R, so it is all of R. In particular 1 lies in the image, so $ab = 1$ for some $b \in R$; commutativity gives $ba = 1$ as well, so $b = a^{-1}$ and a is a unit. Since R is a domain, $1 \neq 0$, and every nonzero element has just been shown invertible: R is a field. □

Applied to $\mathbb{Z}/p\mathbb{Z}$—a finite domain by theorem 4.15—the theorem recovers theorem 4.26. The two proofs differ instructively. The pigeonhole argument merely asserts that each inverse *exists*; the extended Euclidean algorithm of §2 *computes* it, in $O(\log p)$ division steps. Cryptography needs both: the abstract statement to reason with, and the algorithm to run.

4.5 The characteristic

Every ring receives a canonical map from the integers, and the behaviour of that map is a fundamental invariant: it measures how much of $\mathbb{Z}$ survives inside the ring.

Definition 4.30 (Characteristic)**.** Let R be a ring, and for $n \in \mathbb{Z}$ let $n \cdot 1$ denote the n-fold additive multiple of 1 (so $n \cdot 1 = 1 + \cdots + 1$ with n summands for $n > 0$, $0 \cdot 1 = 0$, and $(-n) \cdot 1 = -(n \cdot 1)$). The *characteristic* $\operatorname{char}(R)$ is the least positive integer n with $n \cdot 1 = 0$, if such an n exists, and 0 otherwise.

An equivalent formulation is often more useful. The map

$$\iota\colon \mathbb{Z} \to R, \qquad \iota(n) = n \cdot 1,$$

is a ring homomorphism: additivity is the bookkeeping of iterated multiples in the abelian group $(R, +)$, multiplicativity is theorem 4.3(4) with $a = b = 1$—namely $\iota(m)\iota(n) = (m \cdot 1)(n \cdot 1) = (mn)(1{\cdot}1) = \iota(mn)$—and $\iota(1) = 1$. Its kernel is an ideal of $\mathbb{Z}$, hence of the form $n\mathbb{Z}$ for a unique $n \geq 0$ by Example 4.18(2), and $\mathrm{char}(R)$ is exactly this nonnegative generator: $\mathrm{char}(R) = 0$ precisely when ι is injective, so that a faithful copy of $\mathbb{Z}$ sits inside R, and $\mathrm{char}(R) = n > 0$ when the kernel is $n\mathbb{Z}$.

Example 4.31 (Characteristics)**.** The rings $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ all have characteristic 0: no positive multiple of 1 vanishes. For the residue rings, $\mathrm{char}(\mathbb{Z}/n\mathbb{Z}) = n$: the multiple $n{\cdot}\overline{1} = \overline{n} = \overline{0}$ vanishes, and no smaller positive multiple does, since $k{\cdot}\overline{1} = \overline{k} \neq \overline{0}$ for $0 < k < n$. In particular $\mathrm{char}(\mathbb{F}_p) = p$.

> **Theorem 4.32.** *The characteristic of an integral domain—in particular, of a field—is either* 0 *or a prime number.*

Proof. Let R be an integral domain with $\mathrm{char}(R) = n > 0$; we show n is prime. First $n \neq 1$: otherwise $1 \cdot 1 = 1 = 0$, making R the zero ring (theorem 4.4) and contradicting $1 \neq 0$ in a domain. Suppose $n = ab$ with $1 < a, b < n$. By theorem 4.3(4),

$$0 = n \cdot 1 = (ab)(1 \cdot 1) = (a \cdot 1)(b \cdot 1),$$

so $a \cdot 1 = 0$ or $b \cdot 1 = 0$ since R is a domain. Either case exhibits a positive integer smaller than n whose multiple of 1 vanishes, contradicting the minimality of $n = \mathrm{char}(R)$. Hence n admits no such factorisation, and being greater than 1, it is prime. □

Remark 4.33 (The prime subfield). Let F be a field. If $\mathrm{char}(F) = 0$, the homomorphism ι embeds a copy of $\mathbb{Z}$ in F, and dividing—possible in a field—a copy of $\mathbb{Q}$ as well. If $\mathrm{char}(F) = p$, the image of ι is a copy of $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. The smallest subfield so obtained is called the *prime subfield* of F; every field is thus an extension of $\mathbb{Q}$ or of some $\mathbb{F}_p$, and the dichotomy between characteristic zero and characteristic p pervades the subject. The fields $\mathbb{F}_q$ underlying the cryptography treated in this series all have prime characteristic p. There the "freshman's dream" identity $(a + b)^p = a^p + b^p$ actually holds, and gives rise to the Frobenius endomorphism; both are developed in the finite-fields section (theorem 6.13).

The question that opened the section is now answered in full. A verifier that must add and multiply computes in a ring; the laws verified one at a time for $\mathbb{Z}/n\mathbb{Z}$ in §2 are precisely the ring axioms, and the construction that produced them is the quotient of $\mathbb{Z}$ by the ideal $n\mathbb{Z}$. A verifier that must also divide computes in a field, and theorem 4.26 says exactly which moduli provide one: the primes. The payoff theorem, theorem 4.29, closes the circle with a purely structural guarantee: in a finite world there is no daylight between the absence of zero divisors and the presence of division—any nonzero finite commutative ring in which nonzero elements never multiply to zero is automatically a field. The prime fields $\mathbb{F}_p$ certified by these results are the ground on which the volume now builds: polynomials, linear algebra, roots of unity, and elliptic curves are all developed over them in the sections that follow.

5 Polynomials over a field

Throughout this section F denotes a field. Consider a problem that sounds impossible. Two parties each hold a polynomial of enormous degree—millions of coefficients, say, the encoding

of an entire computation—and wish to convince themselves that the two polynomials are equal. Comparing coefficient lists costs as much as reading the computation itself. Could a *single* spot check suffice: pick one value α at random, evaluate both polynomials there, and declare them equal if the two field elements agree? The astonishing answer is yes—with an error probability so small that no adversary can exploit it—and this section develops exactly the algebra that makes the answer honest.

Polynomials are the engine behind finite fields, error-correcting codes, and—most importantly here—the polynomial commitment schemes and *arithmetisation* (the encoding of a computation as identities between polynomials) underlying Halo 2 and Zcash Orchard. The single most exploited fact in that edifice is elementary: a nonzero polynomial of degree d over a field has at most d roots. From this one fact the section extracts Lagrange interpolation and the Schwartz–Zippel principle, the probabilistic heart of every interactive proof we eventually construct. The only prerequisite is the field theory of the preceding section.

5.1 The polynomial ring

One point of principle governs everything that follows, so we state it before the first definition: *a polynomial is not a function*. It is a formal expression—at bottom, a finite list of coefficients—which one may *evaluate* to obtain a function, but which in general carries more information than that function. Over the finite fields of cryptographic interest the distinction is not pedantic but essential, and the closing subsections of this section (§5.9) return to it at length.

Definition 5.1 (Polynomial)**.** A *polynomial* over F in the *indeterminate* X is a formal expression

$$f \;=\; a_0 + a_1 X + a_2 X^2 + \cdots + a_n X^n \;=\; \sum_{i=0}^{n} a_i X^i,$$

where $n \in \mathbb{N} = \{0, 1, 2, \dots\}$ and the *coefficients* $a_0, \dots, a_n$ lie in F. Formally, f is identified with its coefficient sequence $(a_0, a_1, a_2, \dots)$, in which only finitely many entries are nonzero. Two polynomials are equal precisely when their coefficient sequences agree term by term:

$$\sum_i a_i X^i = \sum_i b_i X^i \quad \Longleftrightarrow \quad a_i = b_i \text{ for every } i \in \mathbb{N}.$$

The symbol X is an *indeterminate* (a formal variable): it is not an element of F and stands for nothing in particular; it is merely a placeholder organising the coefficients. The set of all polynomials over F in X is denoted $F[X]$.

Remark 5.2 (The sequence model)*.* The fully rigorous rendering of theorem 5.1 takes $F[X]$ to be the set of all functions $a\colon \mathbb{N} \to F$ that are *eventually zero*: there is some N with $a(i) = 0$ for all $i > N$. The expression $\sum_i a_i X^i$ is then merely suggestive notation for the sequence $(a_0, a_1, \dots)$. We use the two viewpoints interchangeably; the notation is justified because the arithmetic of these sequences, defined next, is exactly the arithmetic the notation suggests.

Definition 5.3 (Ring operations on $F[X]$)**.** Let $f = \sum_i a_i X^i$ and $g = \sum_i b_i X^i$ lie in $F[X]$, with the shorter coefficient sequence zero-padded so that both run over the same index set. Their *sum* and *product* are

$$f + g \;=\; \sum_{i \ge 0} (a_i + b_i)\, X^i, \qquad fg \;=\; \sum_{k \ge 0} c_k X^k \quad \text{where} \quad c_k \;=\; \sum_{i+j=k} a_i b_j \;=\; \sum_{i=0}^{k} a_i b_{k-i}.$$

The coefficient c_k is the *convolution* of the two coefficient sequences. Both defining sums are finite—only finitely many a_i and b_j are nonzero, hence only finitely many c_k are nonzero—so the sum and the product are again polynomials.

The convolution formula is exactly what multiplying out $(\sum_i a_i X^i)(\sum_j b_j X^j)$ and collecting like powers of X gives, using $X^i \cdot X^j = X^{i+j}$. Stated as a convolution, however, it makes no appeal to substituting a value for X: it is a purely formal manipulation of coefficient sequences. The formal/functional distinction announced above is thereby kept intact at the level of the operations themselves.

The verification that these operations obey the ring axioms (theorem 4.1) is a single device applied over and over: to prove an identity between polynomials, compute the coefficient of X^m on each side as a finite sum over index decompositions, then reindex or regroup that sum. No evaluation is ever involved. The associativity step below is the archetype.

Theorem 5.4 ($F[X]$ is a commutative ring)**.** *With the operations of theorem 5.3, the set $F[X]$ is a commutative ring whose additive identity is the zero polynomial $0 = (0, 0, 0, \dots)$ and whose multiplicative identity is the constant polynomial $1 = (1, 0, 0, \dots)$. Moreover the map $c \mapsto (c, 0, 0, \dots)$ embeds F into $F[X]$ as a subring, the* constant polynomials*; in this sense $F \subseteq F[X]$.*

Proof. Additive structure. Addition is coordinatewise, so its commutativity and associativity are inherited directly from the corresponding laws of the additive group of F, the sequence $(0, 0, \dots)$ is an additive identity, and $(-a_0, -a_1, \dots)$ is an additive inverse of $(a_0, a_1, \dots)$. Hence $(F[X], +)$ is an abelian group.

Commutativity of multiplication. The coefficient of X^k in fg is $\sum_{i+j=k} a_i b_j$; reindexing the sum by $(i, j) \mapsto (j, i)$ and using commutativity of multiplication in F,

$$\sum_{i+j=k} a_i b_j = \sum_{i+j=k} b_i a_j,$$

which is the coefficient of X^k in gf.

Associativity of multiplication. Write $h = \sum_l c_l X^l$. The coefficient of X^m in $(fg)h$ is

$$\sum_{k+l=m} \Big(\sum_{i+j=k} a_i b_j \Big) c_l = \sum_{i+j+l=m} a_i b_j c_l,$$

and expanding $f(gh)$ instead produces $\sum_{i+j+l=m} a_i (b_j c_l)$, the same symmetric triple sum by associativity in F. Hence $(fg)h = f(gh)$ coefficient by coefficient.

Distributivity. The coefficient of X^k in $f(g + h)$ is $\sum_{i+j=k} a_i (b_j + c_j)$, which distributivity in F splits as $\sum_{i+j=k} a_i b_j + \sum_{i+j=k} a_i c_j$: the coefficient of X^k in $fg + fh$. The other distributive law follows from this one and commutativity.

Identity. With $1 = (1, 0, 0, \dots)$, the k-th coefficient of $1 \cdot f$ is $\sum_{i+j=k} \mathbf{1}[i = 0]\, a_j = a_k$, where $\mathbf{1}[i = 0]$ denotes the coefficient sequence of 1 (equal to 1 at index 0 and 0 elsewhere); so $1 \cdot f = f$.

Embedding. Let $\iota(c) = (c, 0, 0, \dots)$. The map ι preserves sums coordinatewise, and $\iota(c)\iota(d)$ has k-th coefficient cd for $k = 0$ and 0 otherwise, so $\iota(c)\iota(d) = \iota(cd)$; also $\iota(1) = 1$. Distinct constants give distinct sequences, so ι is an injective ring homomorphism (theorem 4.21) and its image is a subring of $F[X]$ isomorphic to F. □

Remark 5.5 (Scalar multiplication). Multiplying $f = \sum_i a_i X^i$ by a constant $c \in F$, viewed as the constant polynomial $\iota(c)$, gives the *scalar multiple* $cf = \sum_i (ca_i) X^i$. Together with addition, this makes $F[X]$ a *vector space* over F—a structure studied systematically in the linear-algebra section later in the volume; informally, a system closed under addition and under scaling by field elements, in which the monomials $1, X, X^2, X^3, \dots$ play the role of an infinite coordinate system (a *basis*). This linear structure is relied on constantly, for instance whenever we count polynomials of bounded degree.

5.2 Degree

Definition 5.6 (Degree, leading coefficient, monic)**.** Let $f = \sum_i a_i X^i \in F[X]$ be nonzero. The *degree* $\deg f$ is the largest index n with $a_n \neq 0$; the coefficient a_n is the *leading coefficient* $\mathrm{lc}(f)$, the term $a_n X^n$ is the *leading term*, and a_0 is the *constant term*. A nonzero polynomial is *monic* if its leading coefficient is 1. The constant polynomials are the elements of F embedded by theorem 5.4; the nonzero ones have degree 0. The zero polynomial is assigned $\deg 0 = -\infty$, with the conventions $-\infty < n$, $-\infty + n = -\infty$, and $-\infty + (-\infty) = -\infty$ for every $n \in \mathbb{Z}$. Polynomials of degree 1, 2, and 3 are called *linear*, *quadratic*, and *cubic* respectively.

The convention $\deg 0 = -\infty$ is not pedantry: it makes the degree formulas below hold without exceptions, which streamlines every later induction on degree. Induction on degree is thereby set up as a reusable device for the rest of the volume.

Proposition 5.7 (Degree of sums and products)**.** *For all $f, g \in F[X]$,*

$$\deg(f+g) \;\leq\; \max\{\deg f, \deg g\}, \tag{1}$$

with equality whenever $\deg f \neq \deg g$, and

$$\deg(fg) \;=\; \deg f + \deg g. \tag{2}$$

Proof. If either polynomial is zero, both statements reduce to the $-\infty$ conventions of theorem 5.6. So let f and g be nonzero, of degrees m and n with leading coefficients a_m and b_n.

Sum. The coefficient of X^k in $f+g$ is $a_k + b_k$, which vanishes for $k > \max\{m, n\}$; this is (1). If $m > n$, the coefficient of X^m is $a_m + 0 = a_m \neq 0$, giving equality (symmetrically for $n > m$). When $m = n$, cancellation $a_m + b_m = 0$ can lower the degree, which is why the sum bound is only an inequality.

Product. The coefficient $c_{m+n} = \sum_{i+j=m+n} a_i b_j$ has only the surviving term $i = m, j = n$: terms with $i > m$ have $a_i = 0$, and terms with $i < m$ force $j > n$, so $b_j = 0$. Hence $c_{m+n} = a_m b_n$, and the same argument shows $c_k = 0$ for $k > m+n$. Since F is a field it has no zero divisors (theorem 4.28), so $a_m b_n \neq 0$; therefore $\deg(fg) = m + n$ and $\mathrm{lc}(fg) = \mathrm{lc}(f)\,\mathrm{lc}(g)$. □

The product formula (2) is the first place the field hypothesis does real work: it required $a_m b_n \neq 0$, that is, the absence of zero divisors in F. This single fact has sweeping consequences for everything that follows, beginning with the next three corollaries.

Corollary 5.8 ($F[X]$ is an integral domain)**.** *The ring $F[X]$ has no zero divisors: if $fg = 0$, then $f = 0$ or $g = 0$.*

Proof. If f and g are both nonzero, then $\deg(fg) = \deg f + \deg g \geq 0$ by (2), so $fg \neq 0$. Contrapositively, $fg = 0$ forces $f = 0$ or $g = 0$. □

Corollary 5.9 (Cancellation)**.** *Let $f, g, h \in F[X]$ with $h \neq 0$. If $fh = gh$, then $f = g$.*

Proof. From $fh = gh$ we get $(f - g)h = 0$; since $h \neq 0$, theorem 5.8 forces $f - g = 0$. □

Proposition 5.10 (Units of $F[X]$)**.** *A polynomial $f \in F[X]$ is a unit—has a multiplicative inverse in $F[X]$—if and only if it is a nonzero constant. Thus $F[X]^\times = F^\times$.*

Proof. A nonzero constant c has the inverse $c^{-1} \in F \subseteq F[X]$. Conversely, $fg = 1$ gives $\deg f + \deg g = \deg 1 = 0$ by (2); since degrees of nonzero polynomials are nonnegative, $\deg f = \deg g = 0$, and f is a nonzero constant. □

Definition 5.11 (Associates)**.** Two polynomials $f, g \in F[X]$ are *associates* if $f = ug$ for some unit $u \in F^\times$, that is, if they differ only by a nonzero scalar factor.

Every nonzero polynomial f has exactly one monic associate, obtained by dividing through by its leading coefficient: $\mathrm{lc}(f)^{-1} f$. Choosing the monic representative is the standard way to pin down *the* greatest common divisor or *the* irreducible factor uniquely, and we adopt this normalisation below.

5.3 Division with remainder

The arithmetic of $F[X]$ parallels that of the integers $\mathbb{Z}$ with remarkable fidelity: greatest common divisors, a Euclidean algorithm, a Bézout identity, primes (here called irreducibles), and unique factorisation all reappear. The source of the parallel is a single statement, division with remainder, in which *degree* plays the role that absolute value plays in $\mathbb{Z}$.

Theorem 5.12 (Division algorithm)**.** *Let $f, g \in F[X]$ with $g \neq 0$. There exist* unique *polynomials $q, r \in F[X]$ such that*

$$f = qg + r \qquad \textit{and} \qquad \deg r < \deg g.$$

We call q the quotient *and r the* remainder*, and write $q = f \operatorname{div} g$ and $r = f \bmod g$.*

Proof. Existence. Fix g, of degree n with leading coefficient b_n, and induct on $\deg f$. If $\deg f < n$—including $f = 0$ with $\deg f = -\infty$—take $q = 0$ and $r = f$. Otherwise let $m = \deg f \geq n$, with leading coefficient a_m, and form

$$f_1 \;=\; f - \frac{a_m}{b_n} X^{m-n} g.$$

The subtracted term has degree m and leading coefficient $(a_m/b_n)\, b_n = a_m$, so it cancels the leading term of f: either $f_1 = 0$ or $\deg f_1 < m$. By the induction hypothesis $f_1 = q_1 g + r$ with $\deg r < n$, and then $f = (q_1 + (a_m/b_n) X^{m-n}) g + r$. The field hypothesis enters visibly here: division by the leading coefficient b_n requires b_n^{-1} to exist.

Uniqueness. Suppose $f = qg + r = q'g + r'$ with both remainders of degree $< n$. Then $(q - q')g = r' - r$. If $q \neq q'$, then $\deg((q - q')g) = \deg(q - q') + n \geq n$ by (2), whereas $\deg(r' - r) \leq \max\{\deg r', \deg r\} < n$ by (1)—a contradiction. Hence $q = q'$, and then $r = f - qg = r'$. □

Remark 5.13 (Polynomial long division). The existence proof is constructive, and it is precisely *polynomial long division*: repeatedly eliminate the current leading term by subtracting an appropriate monomial multiple of g. Each step strictly lowers the degree of the running remainder, so the process terminates after at most $\deg f - \deg g + 1$ steps when $\deg f \geq \deg g$, and takes no steps when $f = 0$ or $\deg f < \deg g$. The device is reusable in both of its guises: as an algorithm, and as the leading-term-elimination pattern for inductions on degree.

For a concrete instance, divide $f = X^3 + 2X + 1$ by $g = X^2 + 1$ over $\mathbb{Q}$. Subtracting $X \cdot g = X^3 + X$ leaves $X + 1$, whose degree 1 is already below $\deg g = 2$; hence $q = X$ and $r = X + 1$, and indeed

$$X^3 + 2X + 1 \;=\; X\,(X^2 + 1) + (X + 1).$$

One elimination step suffices here.

Definition 5.14 (Divisibility). For $f, g \in F[X]$, we say g *divides* f, written $g \mid f$, if $f = qg$ for some $q \in F[X]$; equivalently, for $g \neq 0$, if the remainder $f \bmod g$ is 0. In that case g is a *divisor* (or *factor*) of f.

5.4 Roots and the factor theorem

The destination of this subsection is the central counting bound of the whole section, stated at once so that the reader knows what the machinery is for: *a nonzero polynomial of degree d over a field has at most d distinct roots* (theorem 5.18 below). Everything the volume later builds on polynomials—the cyclicity of the multiplicative group of a finite field, Lagrange interpolation, the Schwartz–Zippel principle—is load-bearing on this one fact. To reach it we must first say what a root is, and that requires making substitution precise: this is the first place where a polynomial gives rise to a function.

Definition 5.15 (Evaluation and roots). Let $f = \sum_{i=0}^{n} a_i X^i \in F[X]$ and $\alpha \in F$. The *evaluation* of f at α is the field element

$$f(\alpha) \;=\; \sum_{i=0}^{n} a_i \alpha^i \;\in\; F,$$

obtained by substituting α for the indeterminate X. An element $\alpha \in F$ is a *root* (or *zero*) of f if $f(\alpha) = 0$.

Proposition 5.16 (Evaluation is a ring homomorphism). *Fix $\alpha \in F$. The evaluation map $\mathrm{ev}_\alpha \colon F[X] \to F, f \mapsto f(\alpha)$, satisfies*

$$(f+g)(\alpha) = f(\alpha) + g(\alpha), \qquad (fg)(\alpha) = f(\alpha)\, g(\alpha), \qquad 1(\alpha) = 1$$

for all $f, g \in F[X]$; that is, ev_α is a ring homomorphism.

Proof. Additivity is immediate from the coordinatewise definition of addition. For multiplicativity, with $c_k = \sum_{i+j=k} a_i b_j$ as in theorem 5.3,

$$(fg)(\alpha) = \sum_k c_k \alpha^k = \sum_k \sum_{i+j=k} a_i b_j\, \alpha^{i+j} = \Big(\sum_i a_i \alpha^i\Big)\Big(\sum_j b_j \alpha^j\Big) = f(\alpha)\, g(\alpha),$$

the middle equality because expanding the product of the two finite sums and grouping terms by $k = i + j$ produces exactly the double sum. The formal reason evaluation respects the algebra is that the convolution defining polynomial multiplication collapses to ordinary multiplication in F via $\alpha^i \alpha^j = \alpha^{i+j}$. Finally $1(\alpha) = 1$ by direct substitution. □

Theorem 5.17 (Remainder theorem; factor theorem). *Let $f \in F[X]$ and $\alpha \in F$.*

1. (Remainder theorem.) *The remainder of f upon division by the linear polynomial $X - \alpha$ is the constant $f(\alpha)$: there is $q \in F[X]$ with $f = q\,(X - \alpha) + f(\alpha)$.*
2. (Factor theorem.) *The element α is a root of f if and only if $(X - \alpha) \mid f$.*

Proof. (1) Divide f by $X - \alpha$ (theorem 5.12): since $\deg(X - \alpha) = 1$, the remainder has degree < 1, hence is a constant $r \in F$. Evaluating $f = q\,(X - \alpha) + r$ at α via theorem 5.16 gives $f(\alpha) = q(\alpha)(\alpha - \alpha) + r = r$.

(2) By (1) and the uniqueness of the remainder, $(X-\alpha) \mid f$ holds exactly when the remainder $f(\alpha)$ is 0, that is, when α is a root of f. □

The factor theorem is an immediate but powerful consequence of the division algorithm: it converts a statement about *values* (a root) into a statement about *divisibility* (a linear factor). Iterating that conversion proves the promised bound.

Theorem 5.18 (At most d roots)**.** *A nonzero polynomial $f \in F[X]$ of degree d has at most d distinct roots in F.*

Proof. We induct on $d = \deg f$. If $d = 0$, then f is a nonzero constant, which evaluates to itself everywhere and so has $0 \le d$ roots. Let $d \ge 1$. If f has no root in F the claim holds; otherwise let α be a root. By the factor theorem, $f = (X-\alpha)\, q$, and $\deg q = d-1$ by (2). Now let $\beta \neq \alpha$ be any other root of f. Evaluating with theorem 5.16,

$$0 = f(\beta) = (\beta - \alpha)\, q(\beta);$$

since $\beta - \alpha \neq 0$ and F has no zero divisors (theorem 4.28), we get $q(\beta) = 0$. Every root of f other than α is therefore a root of q, and by the induction hypothesis q has at most $d-1$ distinct roots. Hence f has at most $(d-1)+1 = d$. □

Remark 5.19 (Why a field is needed). Both ingredients of the proof—the absence of zero divisors and the degree bookkeeping—are essential. Over a ring with zero divisors the bound fails outright: in $\mathbb{Z}/8\mathbb{Z}$ the quadratic X^2-1 has the four roots $\overline{1}, \overline{3}, \overline{5}, \overline{7}$, since $1^2 = 1$, $3^2 = 9 \equiv 1$, $5^2 = 25 \equiv 1$, and $7^2 = 49 \equiv 1 \pmod 8$—a degree-2 polynomial with four roots. This is why the polynomial methods underlying Halo 2 require a field, and indeed a finite field $\mathbb{F}_p$ or $\mathbb{F}_q$, beneath them.

Corollary 5.20 (Few points determine a low-degree polynomial)**.** *Let $f, g \in F[X]$ both have degree at most d. If $f(\alpha) = g(\alpha)$ for $d+1$ distinct values $\alpha \in F$, then $f = g$ as polynomials.*

Proof. The difference $h = f - g$ has $\deg h \le \max\{\deg f, \deg g\} \le d$ by (1) and vanishes at $d+1$ distinct points. Were h nonzero, it would be a polynomial of degree at most d with $d+1$ distinct roots, contradicting theorem 5.18. Hence $h = 0$, that is, $f = g$. □

The corollary is the deterministic skeleton of two constructions developed later in this section: Lagrange interpolation (§5.7), its constructive converse, and the Schwartz–Zippel principle (§5.10), its probabilistic refinement. The root bound is thus positioned as the load-bearing fact behind the polynomial identity testing and interpolation machinery used by the proving system.

5.5 Greatest common divisors

Because $F[X]$ supports division with remainder, the entire theory of greatest common divisors from elementary number theory (§2) transfers verbatim from $\mathbb{Z}$ to $F[X]$, with degree replacing absolute value as the measure that decreases. We run through the transfer explicitly, both because the statements are used constantly and because the polynomial Euclidean algorithm is itself a tool of the trade.

Definition 5.21 (Greatest common divisor)**.** Let $f, g \in F[X]$, not both zero. A *greatest common divisor* of f and g is a polynomial d such that

1. $d \mid f$ and $d \mid g$; and

2. every common divisor c of f and g satisfies $c \mid d$.

A greatest common divisor is determined only up to multiplication by a unit; the unique *monic* one (theorem 5.23) is singled out and denoted $\gcd(f,g)$. When $\gcd(f,g) = 1$, the polynomials f and g are called *coprime* (or *relatively prime*). Since every polynomial divides 0, we have $\gcd(f,0) = \mathrm{lc}(f)^{-1}f$ for $f \neq 0$.

Lemma 5.22 (Euclidean step). *Let $f,g \in F[X]$ with $g \neq 0$, and let $f = qg + r$ be the division of theorem 5.12. Then the pairs (f,g) and (g,r) have exactly the same common divisors; in particular, their greatest common divisors coincide.*

Proof. If $c \mid f$ and $c \mid g$, then $c \mid (f - qg) = r$; so c is a common divisor of (g,r). Conversely, if $c \mid g$ and $c \mid r$, then $c \mid (qg + r) = f$. The two sets of common divisors are therefore equal, hence so are their maximal elements under divisibility. □

Theorem 5.23 (Euclidean algorithm; Bézout's identity). *Let $f,g \in F[X]$, not both zero. Then $\gcd(f,g)$ exists and is unique as a monic polynomial, and there exist $s,t \in F[X]$ with*

$$\gcd(f,g) \;=\; sf + tg. \tag{3}$$

Concretely: set $r_0 = f$ and $r_1 = g$, and while $r_i \neq 0$ divide

$$r_{i-1} = q_i r_i + r_{i+1}, \qquad \deg r_{i+1} < \deg r_i.$$

The degrees strictly decrease, so some $r_{N+1} = 0$; the last nonzero remainder r_N, made monic, is $\gcd(f,g)$.

Proof. Termination. If $r_1 = 0$, the algorithm stops at once with $N = 0$ and $r_N = f \neq 0$. Otherwise the degrees $\deg r_1 > \deg r_2 > \dots$ form a strictly decreasing sequence of nonnegative integers, which cannot continue forever; so some $r_{N+1} = 0$ with $r_N \neq 0$.

The gcd property. Theorem 5.22, applied at each division, preserves the *set* of common divisors:

$$\begin{aligned}\{\text{common divisors of } (f,g)\} &= \{\text{common divisors of } (r_1,r_2)\}\\ &= \cdots = \{\text{common divisors of } (r_N,0)\},\end{aligned}$$

and the last set is exactly the set of divisors of r_N, since every polynomial divides 0. Thus r_N divides both f and g, and every common divisor of f and g divides r_N: the monic associate $\mathrm{lc}(r_N)^{-1}r_N$ is a monic greatest common divisor.

Uniqueness. Two monic greatest common divisors divide each other, so they differ by a unit $u \in F^\times$ (theorem 5.10); comparing leading coefficients of monic polynomials forces $u = 1$.

Bézout. We show by induction on i that each remainder r_i is an $F[X]$-linear combination of f and g. The bases $r_0 = 1 \cdot f + 0 \cdot g$ and $r_1 = 0 \cdot f + 1 \cdot g$ are trivial, and the recurrence $r_{i+1} = r_{i-1} - q_i r_i$ expresses r_{i+1} as a combination once r_{i-1} and r_i are. Hence $r_N = s_0 f + t_0 g$ for some $s_0, t_0 \in F[X]$; dividing through by $\mathrm{lc}(r_N)$ yields (3). □

Remark 5.24 (Extended Euclidean algorithm). Carrying the linear-combination coefficients along with the remainders, exactly as in the integer case (Theorem 2.13), produces s and t explicitly alongside the gcd. The standard application is inversion in a quotient $F[X]/(m)$ for irreducible m: if $\gcd(f,m) = 1$, then $sf + tm = 1$, so s is the inverse of f modulo m. This is how one divides in the extension fields $\mathbb{F}_q$ constructed in section 6. The polynomial version matters both because it parallels the integer one exactly and because it computes gcds of the polynomials a proof system

manipulates. A caveat on deployed practice: the curves of Halo 2 and Orchard are defined over *prime* fields $\mathbb{F}_p$, not extension fields, and field inversion there uses the integer extended Euclidean algorithm (Theorem 2.13) or Fermat's little theorem—not the polynomial algorithm in $F[X]/(m)$.

Corollary 5.25 (Euclid's lemma in $F[X]$)**.** *Let $p, f, g \in F[X]$.*

1. *If $\gcd(p, f) = 1$ and $p \mid fg$, then $p \mid g$.*
2. *If p is irreducible and $p \mid fg$, then $p \mid f$ or $p \mid g$.*

Here irreducible *is the polynomial analogue of prime, made precise in theorem 5.26 below.*

Proof. (1) Multiplying the Bézout identity $sp + tf = 1$ by g gives $spg + tfg = g$. The polynomial p divides both terms on the left—the second because $p \mid fg$—hence divides g.

(2) Suppose $p \nmid f$. The gcd $\gcd(p, f)$ is a monic divisor of p that is not the monic associate of p (otherwise $p \mid f$). By irreducibility the only monic divisors of p are 1 and its monic associate, so $\gcd(p, f) = 1$ and part (1) applies. □

5.6 Unique factorisation

Definition 5.26 (Irreducible polynomial)**.** A polynomial $p \in F[X]$ is *irreducible over F* if $\deg p \geq 1$ and p admits no factorisation $p = gh$ with both $\deg g \geq 1$ and $\deg h \geq 1$; equivalently, p is nonconstant and its only factorisations are the trivial ones, (unit) × (associate of p). A nonconstant polynomial that is not irreducible is *reducible*.

Remark 5.27 (Irreducibility is relative to the base field). The polynomial $X^2 + 1$ is irreducible over $\mathbb{R}$ (it has no real root, and a real quadratic factors precisely when it has a root, by the test below), yet splits as $(X - i)(X + i)$ over $\mathbb{C}$, and factors as $(X + 1)^2$ over $\mathbb{F}_2$, by direct expansion in characteristic 2. Degree-1 polynomials are irreducible over every field. A useful *rootlessness test*: a polynomial of degree 2 or 3 is irreducible over F if and only if it has no root in F, because any nontrivial factorisation of a quadratic or cubic must include a linear factor, which by the factor theorem (theorem 5.17) supplies a root. The test fails from degree 4 onward: a product of two irreducible quadratics has no roots yet is reducible.

Theorem 5.28 ($F[X]$ is a unique factorisation domain)**.** *Every nonconstant $f \in F[X]$ factors as*

$$f = c\, p_1 p_2 \cdots p_k$$

with $c = \mathrm{lc}(f) \in F^\times$ a constant and each p_i monic irreducible. The factorisation is unique up to order: if also $f = c'\, q_1 \cdots q_l$ with $c' \in F^\times$ and each q_j monic irreducible, then $c = c'$, $k = l$, and the q_j are a reordering of the p_i.

Proof. The proof is templated on the fundamental theorem of arithmetic (Theorem 2.18), with degree in place of absolute value.

Existence, by induction on $\deg f \geq 1$. If f is irreducible, then $f = \mathrm{lc}(f) \cdot p_1$ with p_1 its monic associate. If $f = gh$ is reducible with $1 \leq \deg g, \deg h < \deg f$, the induction hypothesis factors g and h, and multiplying the two factorisations—collecting the two constants into one—factors f.

Uniqueness. Comparing leading coefficients in the two factorisations gives $c = c' = \mathrm{lc}(f)$, since monic factors contribute leading coefficient 1 by (2). Cancelling the constants (theorem 5.9) leaves $p_1 \cdots p_k = q_1 \cdots q_l$; we induct on k. For $k = 1$ the left side is irreducible, so $l = 1$ and $p_1 = q_1$. For $k \geq 2$: p_1 divides $q_1 \cdots q_l$, so iterating Euclid's lemma (theorem 5.25(2)) gives $p_1 \mid q_j$ for

some j. A monic irreducible dividing a monic irreducible equals it: $q_j = p_1 h$ forces $\deg h = 0$ by irreducibility of q_j, and comparing leading coefficients gives $h = 1$. Cancel $p_1 = q_j$ from both sides (theorem 5.9). If $l = 1$, the cancellation leaves $p_2 \cdots p_k = 1$, which is impossible: the left side is a product of $k - 1 \geq 1$ nonconstant polynomials and so has degree at least 1 by (2), while $\deg 1 = 0$. Hence $l \geq 2$, and the induction hypothesis, applied to $p_2 \cdots p_k = \prod_{j' \neq j} q_{j'}$, concludes $k = l$ and matches the remaining factors. □

Remark 5.29 (The structural reason: $F[X]$ is a PID). The deeper explanation of unique factorisation is structural. The degree function and theorem 5.12 make $F[X]$ a *Euclidean domain*—an integral domain with division with remainder relative to a size measure—and every Euclidean domain is a *principal ideal domain* (PID): each ideal (theorem 4.17) of $F[X]$ is $(g) = \{ qg : q \in F[X] \}$ for a single generator g, namely any nonzero element of least degree in a nonzero ideal (the zero ideal is generated by 0), by the same division-and-minimality argument that proved every ideal of $\mathbb{Z}$ principal (Example 4.18(2)). The generator of the ideal $\{ sf + tg : s, t \in F[X] \}$ is a gcd of f and g, and Bézout's identity (3) is the concrete shadow of this principality. The standard chain

$$\text{Euclidean} \implies \text{PID} \implies \text{UFD}$$

has thus been walked explicitly for $F[X]$, ending at theorem 5.28.

5.7 Lagrange interpolation

The corollary "few points determine a low-degree polynomial" (theorem 5.20) says that a polynomial of degree at most d is pinned down by its values at $d + 1$ points. The present subsection proves the constructive converse: *any* $d + 1$ prescribed values at distinct points are attained by exactly one polynomial of degree at most d, and there is an explicit formula for it. This is the algebraic backbone of secret sharing, of Reed–Solomon error-correcting codes, and of the polynomial encodings in Halo 2.

Theorem 5.30 (Lagrange interpolation). *Let $x_0, \dots, x_d \in F$ be distinct (the* nodes*) and let $y_0, \dots, y_d \in F$ be arbitrary targets. There exists a unique $f \in F[X]$ with $\deg f \leq d$ and $f(x_i) = y_i$ for all i, given explicitly by*

$$f(X) = \sum_{i=0}^{d} y_i \, \ell_i(X), \qquad \textit{where} \qquad \ell_i(X) = \prod_{j \neq i} \frac{X - x_j}{x_i - x_j}$$

are the Lagrange basis polynomials *for the nodes.*

Proof. The ℓ_i are genuine polynomials. Each denominator $\prod_{j \neq i}(x_i - x_j)$ is a product of nonzero field elements, the nodes being distinct, hence is an invertible constant; so ℓ_i is a polynomial, of degree exactly d (a product of d linear factors divided by a nonzero constant).

The delta property. Writing δ_{im} for the Kronecker symbol (equal to 1 if $i = m$ and 0 otherwise),

$$\ell_i(x_m) = \delta_{im} \qquad (0 \leq i, m \leq d) : \tag{4}$$

at $m = i$ every factor $(x_i - x_j)/(x_i - x_j)$ is 1, while at $m \neq i$ the factor with $j = m$ has numerator $x_m - x_m = 0$. Consequently $f(x_m) = \sum_i y_i \, \ell_i(x_m) = y_m$ for each m, and $\deg f \leq d$ by (1), each summand having degree at most d.

Uniqueness. Two interpolants of degree at most d agree at the $d + 1$ distinct nodes, hence are equal by theorem 5.20. □

Example 5.31 (A worked interpolation over $\mathbb{Q}$)**.** Find the polynomial of degree at most 2 with $f(0) = 1, f(1) = 3$, and $f(2) = 7$. The basis polynomials for the nodes $0, 1, 2$ are

$$\begin{aligned}
\ell_0 &= \frac{(X-1)(X-2)}{(0-1)(0-2)} = \frac{(X-1)(X-2)}{2}, \\
\ell_1 &= \frac{(X-0)(X-2)}{(1-0)(1-2)} = -X(X-2), \\
\ell_2 &= \frac{(X-0)(X-1)}{(2-0)(2-1)} = \frac{X(X-1)}{2}.
\end{aligned}$$

Hence

$$f \;=\; 1 \cdot \ell_0 + 3 \cdot \ell_1 + 7 \cdot \ell_2 \;=\; \tfrac{1}{2}(X^2 - 3X + 2) + 3(2X - X^2) + \tfrac{7}{2}(X^2 - X) \;=\; X^2 + X + 1,$$

and indeed $f(0) = 1, f(1) = 3, f(2) = 7$.

Remark 5.32 (The evaluation isomorphism)*.* Fix $d + 1$ distinct nodes and let $V = \{f \in F[X] : \deg f \leq d\}$. In the language of the linear-algebra section later in the volume, V is a $(d+1)$-dimensional vector space over F with basis $1, X, \dots, X^d$, and the evaluation map

$$\mathrm{ev}\colon V \to F^{d+1}, \qquad f \longmapsto (f(x_0), \dots, f(x_d)),$$

is F-linear. Its surjectivity is exactly the existence half of theorem 5.30, and its injectivity exactly the uniqueness half; so ev is a linear isomorphism $V \cong F^{d+1}$. The Lagrange basis $\{\ell_i\}$ is precisely the basis of V matched to the standard coordinates of F^{d+1}, by the delta property (4): the polynomial with prescribed values $(y_0, \dots, y_d)$ has coordinates $(y_0, \dots, y_d)$ in the basis $\{\ell_i\}$. This switching between a polynomial's *coefficient representation* and its *evaluation representation* is the conceptual core of fast polynomial arithmetic via the fast Fourier transform, treated with the roots of unity later in the volume, and of the evaluation-domain encodings used throughout Halo 2.

Remark 5.33 (The vanishing polynomial)*.* The companion object to the Lagrange basis is the *vanishing polynomial* of the node set $\{x_0, \dots, x_d\}$,

$$Z(X) \;=\; \prod_{i=0}^{d} (X - x_i),$$

the unique monic polynomial of degree $d + 1$ vanishing at every node. A polynomial g vanishes on the whole node set if and only if $Z \mid g$. One direction is immediate from theorem 5.16. For the other, apply the factor theorem repeatedly: $g(x_0) = 0$ gives $g = (X - x_0)\, g_1$; evaluating at $x_1 \neq x_0$ gives $0 = (x_1 - x_0)\, g_1(x_1)$, so $g_1(x_1) = 0$ since F has no zero divisors, and $g_1 = (X - x_1)\, g_2$; continuing through the remaining nodes strips off one linear factor per node. The repetition is legitimate precisely because the nodes are distinct—the linear factors $X - x_i$ are pairwise coprime. The pair (Lagrange basis, vanishing polynomial) is exactly the data a proof system manipulates when arguing that a committed polynomial agrees with prescribed values on an evaluation domain.

5.8 Multiplicity and the formal derivative

A root can be a root "several times over": the quadratic $X^2 - 2X + 1 = (X - 1)^2$ vanishes at 1 doubly. Counting roots correctly requires making that idea precise, and detecting it requires a derivative—which we define with no limits at all.

Definition 5.34 (Multiplicity of a root)**.** Let $f \in F[X]$ be nonzero and let $\alpha \in F$ be a root of f. The *multiplicity* $\mathrm{mult}_\alpha(f)$ is the largest integer $m \geq 1$ with $(X - \alpha)^m \mid f$. A root of multiplicity

1 is *simple*; a root of multiplicity at least 2 is *repeated* (or *multiple*).

The largest such m exists: the powers $(X-\alpha)^m$ have strictly increasing degrees, so only finitely many can divide the fixed f, and at least one does by the factor theorem. An equivalent characterisation is often handier: $m = \mathrm{mult}_\alpha(f)$ exactly when $f = (X-\alpha)^m g$ with $g(\alpha) \neq 0$. Indeed, writing $f = (X-\alpha)^m g$ with m maximal forces $g(\alpha) \neq 0$, else the factor theorem would split a further factor $X-\alpha$ off g; conversely, if $f = (X-\alpha)^m g$ with $g(\alpha) \neq 0$, then $(X-\alpha)^{m+1} \mid f$ would give $(X-\alpha)^m g = (X-\alpha)^{m+1} h$, and cancelling (theorem 5.9) $g = (X-\alpha)h$, making α a root of g.

The count of roots *with* multiplicity still respects the degree. The proof rests on a small device worth isolating, since it recurs.

Lemma 5.35 (Coprime factors multiply). *Let $f_1, \dots, f_r \in F[X]$ be pairwise coprime ($\gcd(f_i, f_j) = 1$ for $i \neq j$) and suppose each f_i divides f. Then $f_1 f_2 \cdots f_r \mid f$.*

Proof. First let $r = 2$, and write $f = f_1 u = f_2 v$. Bézout (3) gives $sf_1 + tf_2 = 1$, so

$$f \;=\; sf_1 f + tf_2 f \;=\; sf_1(f_2 v) + tf_2(f_1 u) \;=\; f_1 f_2\,(sv + tu),$$

and $f_1 f_2 \mid f$. For $r > 2$, induct: by hypothesis $f_1 \cdots f_{r-1} \mid f$, and the product $f_1 \cdots f_{r-1}$ is coprime to f_r—a nonconstant common divisor would have a monic irreducible factor (theorem 5.28), which by Euclid's lemma (theorem 5.25(2)) would divide some f_i with $i < r$ as well as f_r, contradicting $\gcd(f_i, f_r) = 1$. Now apply the case $r = 2$. □

Proposition 5.36 (Counting roots with multiplicity). *Let $f \in F[X]$ be nonzero of degree d, with all its distinct roots $\alpha_1, \dots, \alpha_r \in F$ of multiplicities $m_1, \dots, m_r$. Then*

$$f \;=\; (X-\alpha_1)^{m_1} \cdots (X-\alpha_r)^{m_r}\, g$$

for some $g \in F[X]$ with no roots in F, and $\sum_i m_i \leq d$. The number of roots counted with multiplicity is thus at most d, with equality if and only if f splits into linear factors over F.

Proof. We first check that the powers $(X-\alpha_i)^{m_i}$ are pairwise coprime. Each $X-\alpha_i$ is monic irreducible (degree 1), and by unique factorisation the monic divisors of $(X-\alpha_i)^{m_i}$ are exactly the powers $(X-\alpha_i)^e$ with $0 \leq e \leq m_i$. For $i \neq j$ the only monic polynomial that is simultaneously a power of $X-\alpha_i$ and of $X-\alpha_j$ is 1, since $X-\alpha_i = X-\alpha_j$ would force $\alpha_i = \alpha_j$; so $\gcd((X-\alpha_i)^{m_i}, (X-\alpha_j)^{m_j}) = 1$.

Each $(X-\alpha_i)^{m_i}$ divides f by the definition of multiplicity, so theorem 5.35 gives $f = (X-\alpha_1)^{m_1} \cdots (X-\alpha_r)^{m_r} g$. Maximality of each m_i gives $g(\alpha_i) \neq 0$: a further factor $X-\alpha_i$ inside g would raise the multiplicity. And g has no other roots either: a root β of g is a root of f, hence one of the α_i. Taking degrees with (2),

$$m_1 + \cdots + m_r + \deg g \;=\; d,$$

so $\sum_i m_i \leq d$, with equality precisely when $\deg g = 0$, that is, when f is a constant times a product of linear factors. □

Definition 5.37 (Formal derivative). The *formal derivative* is the map $D\colon F[X] \to F[X]$ defined on $f = \sum_{i=0}^{n} a_i X^i$ by

$$f' \;=\; Df \;=\; \sum_{i=1}^{n} i\, a_i\, X^{i-1} \;=\; a_1 + 2a_2 X + \cdots + n a_n X^{n-1},$$

where the coefficient $i\,a_i$ means a_i added to itself i times in F—that is, a_i multiplied by the image of the integer i under the canonical map $\mathbb{Z} \to F$ (theorem 4.30). No limits are involved: the definition is purely formal and algebraic, valid over any field, and it is introduced for one purpose—to detect repeated roots without any appeal to analysis.

Proposition 5.38 (Rules of formal differentiation)**.** *For all $f, g \in F[X]$, $c \in F$, $\alpha \in F$, and $m \geq 1$:*

1. (Linearity.) $(f+g)' = f' + g'$ *and* $(cf)' = cf'$.
2. (Product rule.) $(fg)' = f'g + fg'$.
3. (Power rule.) $\big((X-\alpha)^m\big)' = m\,(X-\alpha)^{m-1}$.

Proof. (1) is immediate from the coordinatewise, visibly linear definition of D.

(2) Both sides are *bilinear* in the pair (f, g): by part (1), each is linear in f with g fixed and linear in g with f fixed. It therefore suffices to verify the identity on monomials $f = X^a$, $g = X^b$, since every polynomial is a linear combination of monomials and the general case follows by expanding $f = \sum_a u_a X^a$, $g = \sum_b v_b X^b$ and applying the monomial case termwise. On monomials,

$$(X^a X^b)' = (X^{a+b})' = (a+b)\,X^{a+b-1} = aX^{a-1}X^b + X^a\,bX^{b-1} = (X^a)'X^b + X^a(X^b)',$$

with the convention that a term carrying the integer coefficient 0 (the case $a = 0$ or $b = 0$) is the zero polynomial.

(3) Induct on m. The base is $(X-\alpha)' = 1$. For the step, the product rule gives

$$\big((X-\alpha)^m\big)' = \big((X-\alpha)(X-\alpha)^{m-1}\big)' = (X-\alpha)^{m-1} + (X-\alpha)\,(m-1)(X-\alpha)^{m-2} = m\,(X-\alpha)^{m-1}. \qquad \square$$

Part (3) of the next theorem must exhibit a repeated root inside an extension field that is not given in advance; the following lemma manufactures the required extension. It is the quotient recipe of Example 4.22 carried out for polynomial rings, and the proof is the argument that made $\mathbb{Z}/p\mathbb{Z}$ a field (theorem 4.26), transposed from $\mathbb{Z}$ to $F[X]$.

Lemma 5.39 (Adjoining a root)**.** *Let $h \in F[X]$ be irreducible. Then the quotient ring $K = F[X]/(h)$ is a field; the map $a \mapsto a + (h)$ embeds F into K as a subfield, so that K is an extension of F; and the class $\alpha = X + (h)$ is a root of h in K.*

Proof. The principal ideal $(h) = hF[X]$ is an ideal of the commutative ring $F[X]$ (Example 4.18(2)), so K is a commutative ring under the coset operations (theorem 4.20). Its identity is not zero: $1 + (h) = 0 + (h)$ would say $h \mid 1$, impossible since a factorisation $1 = hk$ has degree $\deg h + \deg k = 0$ by (2), while $\deg h \geq 1$.

Every nonzero class is a unit. Let $f + (h) \neq 0 + (h)$, i.e. $h \nmid f$. The gcd $\gcd(h, f)$ is then a monic divisor of the irreducible h other than its monic associate, hence equals 1, exactly as in the proof of theorem 5.25(2); Bézout's identity (3) supplies $s, t \in F[X]$ with $sf + th = 1$, and reducing modulo (h) gives $(s + (h))(f + (h)) = 1 + (h)$. So K is a field.

The map $\iota(a) = a + (h)$ on constants preserves sums, products, and 1 directly from the coset operations of theorem 4.19, and it is injective: $\iota(a) = 0$ with $a \neq 0$ would make h divide a nonzero constant, again impossible by degrees. We identify F with its image, making K an extension field of F.

Finally, write $h = \sum_i c_i X^i$ and compute in K:

$$h(\alpha) = \sum_i \iota(c_i)\,(X + (h))^i = \Big(\sum_i c_i X^i\Big) + (h) = h + (h) = 0 + (h),$$

so α is a root of h in K. $\square$

Theorem 5.40 (Derivative test for repeated roots; separability). *Let $f \in F[X]$ be nonzero and $\alpha \in F$.*

1. *The element α is a repeated root of f—a root of multiplicity at least 2—if and only if $f(\alpha) = 0$ and $f'(\alpha) = 0$.*
2. *If f has a repeated root in F, then $\gcd(f, f') \neq 1$.*
3. *Call f* separable *if it has no repeated root in any field containing F. Then f is separable if and only if $\gcd(f, f') = 1$ in $F[X]$.*

Proof. (1) Let α be a root of multiplicity $m \geq 1$ and write $f = (X - \alpha)^m g$ with $g(\alpha) \neq 0$. The product and power rules give

$$f' = m(X - \alpha)^{m-1} g + (X - \alpha)^m g' = (X - \alpha)^{m-1} [\, mg + (X - \alpha) g' \,].$$

If $m = 1$, the first factor is 1 and evaluating at α gives $f'(\alpha) = 1 \cdot g(\alpha) = g(\alpha) \neq 0$: a simple root of f is not a root of f'. If $m \geq 2$, then $(X - \alpha)^{m-1}$ divides f' and vanishes at α, so $f'(\alpha) = 0$. Together with the factor theorem for $f(\alpha) = 0$, this proves the equivalence.

(2) A repeated root $\alpha \in F$ is a common root of f and f' by (1), so the factor theorem puts the common factor $X - \alpha$ into both, and $(X - \alpha) \mid \gcd(f, f')$.

(3) The key observation is that *gcds are stable under field extension*: for an extension field K of F (theorem 4.25), the Euclidean algorithm of theorem 5.23 run on $f, f' \in F[X]$ uses only field operations on their coefficients, which lie in F; run inside $K[X]$ on the same inputs it produces the identical quotients and remainders. Hence the gcd computed in $K[X]$ equals the gcd computed in $F[X]$.

Suppose $\gcd(f, f') = 1$ in $F[X]$, and let $K \supseteq F$ be any extension. By stability, $\gcd(f, f') = 1$ in $K[X]$ as well, so by (2) applied over K, the polynomial f has no repeated root in K: f is separable.

Conversely, suppose $d = \gcd(f, f')$ is nonconstant; we exhibit a repeated root in some extension. Let h be a monic irreducible factor of d (theorem 5.28), and let $K = F[X]/(h)$ be the extension field supplied by theorem 5.39, in which h has a root α. The polynomial h divides d, which divides both f and f'; evaluating the factorisations at α (theorem 5.16) gives $f(\alpha) = f'(\alpha) = 0$. By (1), applied over K, the element α is a repeated root of f in K: f is not separable. □

The stability observation in the proof deserves emphasis as a technique: it licenses testing separability—the absence of repeated roots in *every* extension of the base field at once—by a gcd computation carried out entirely in the base field.

Remark 5.41 (The pitfall of positive characteristic). The integer coefficient i in $f' = \sum_i i\, a_i X^{i-1}$ is interpreted *in* F, and may vanish there. Over a field of characteristic p—where $p = 0$ in F, p prime, the setting of $\mathbb{F}_p$ and $\mathbb{F}_q$ (theorem 4.30)—the derivative of X^p is $p\, X^{p-1} = 0$. Thus $f = X^p - 1$ over $\mathbb{F}_p$ has $f' = 0$ and $\gcd(f, f') = \gcd(f, 0) = f \neq 1$; and indeed

$$X^p - 1 = (X - 1)^p \quad \text{in } \mathbb{F}_p[X],$$

a factorisation the finite-fields section establishes through the freshman's-dream identity $(a + b)^p = a^p + b^p$ of characteristic p (theorem 6.13, applied to $(X + (-1))^p$ together with $(-1)^p = -1$ in every $\mathbb{F}_p$): a single root 1 of multiplicity p. This characteristic-p behaviour is a structural feature to respect when working over the finite fields of Orchard, not a pathology to avoid.

5.9 Polynomials versus polynomial functions

The opening of this section insisted that a polynomial is not a function. The debt of that insistence now falls due: we make the relationship precise and show exactly when it is safe—and when it is not—to conflate the two.

Definition 5.42 (Polynomial function)**.** Every $f \in F[X]$ induces a *polynomial function*

$$\tilde{f} \colon F \to F, \qquad \alpha \longmapsto f(\alpha).$$

The assignment $f \mapsto \tilde{f}$ is a ring homomorphism from $F[X]$ to the ring F^F of *all* functions $F \to F$ under pointwise operations ($(u+v)(\alpha) = u(\alpha) + v(\alpha)$ and $(uv)(\alpha) = u(\alpha)v(\alpha)$), by theorem 5.16 applied at every point.

Proposition 5.43 (When formal equals functional)**.** *Let $f, g \in F[X]$.*

1. *If F is infinite, then $\tilde{f} = \tilde{g}$ as functions implies $f = g$ as polynomials. Equivalently, $f \mapsto \tilde{f}$ is injective, and polynomials over an infinite field may be identified with polynomial functions.*
2. *If F is finite, the map is* not *injective. Over $\mathbb{F}_p$, the nonzero polynomial $X^p - X$ induces the zero function, since Fermat's little theorem (Corollary 3.32) gives $\alpha^p = \alpha$ for every $\alpha \in \mathbb{F}_p$.*

Proof. (1) Let $h = f - g$, so that $\tilde{h}$ is the zero function: h vanishes at every point of the infinite set F. Were h nonzero of degree d, it would have at most d roots (theorem 5.18), yet it has infinitely many—a contradiction. Hence $h = 0$.

(2) Let F be finite, with $q = |F|$ elements, and consider the *vanishing-polynomial witness*

$$w(X) \;=\; \prod_{\alpha \in F} (X - \alpha),$$

a monic polynomial of degree q—in particular nonzero as a formal object—which vanishes at every point of F by construction. Thus $\tilde{w} = \tilde{0}$ while $w \neq 0$: the map $f \mapsto \tilde{f}$ identifies distinct polynomials. Over $F = \mathbb{F}_p$ the witness specialises to $X^p - X$: by Fermat's little theorem every element of $\mathbb{F}_p$ is a root of the monic degree-p polynomial $X^p - X$, so theorem 5.36 factors it as $\prod_{\alpha \in \mathbb{F}_p} (X - \alpha)$—the p distinct linear factors exhaust the degree—and $X^p - X$ induces the zero function despite having degree $p \geq 1$. □

Remark 5.44 (Why the formal viewpoint is the correct one). Part (2) of theorem 5.43 is precisely why polynomials are defined as coefficient sequences and not as functions. Cryptography works over finite fields, where the functional view loses information: the polynomials X and X^p induce the same function on $\mathbb{F}_p$, yet have degrees 1 and p. And the *degree* of a polynomial—a property of the formal object, invisible to the induced function over a finite field—is what the security of polynomial commitment schemes rests on. A prover commits to a low-degree *formal* polynomial; the verifier tests it by *evaluation*; and the gap between agreement-as-function at the tested points and identity to the claimed formal polynomial is exactly what the Schwartz–Zippel lemma, next, quantifies.

5.10 The Schwartz–Zippel lemma

Only one preliminary is needed to state the payoff theorem. To draw α *uniformly at random* from a finite nonempty set S means to select each element with equal likelihood $1/|S|$; the *probability* of an event is then the fraction of elements of S producing it, so that $\Pr[f(\alpha) = 0] = |\{\alpha \in S : f(\alpha) = 0\}|/|S|$. The systematic language of probability is developed later in the volume; this counting case is all the theorem needs.

Theorem 5.45 (Univariate Schwartz–Zippel)**.** *Let $f \in F[X]$ be nonzero of degree at most d, let*

> *$S \subseteq F$ be finite and nonempty, and let α be drawn uniformly at random from S. Then*
>
> $$\Pr[f(\alpha) = 0] \;\leq\; \frac{d}{|S|}. \tag{5}$$
>
> *More generally, if $f, g \in F[X]$ are distinct, each of degree at most d, then $\Pr[f(\alpha) = g(\alpha)] \leq d/|S|$.*

Proof. By theorem 5.18, the nonzero f has at most d roots in F, hence at most d in the subset S. Under the uniform draw,

$$\Pr[f(\alpha) = 0] = \frac{|\{\alpha \in S : f(\alpha) = 0\}|}{|S|} \;\leq\; \frac{d}{|S|}.$$

For the general form, apply this to $h = f - g$, which is nonzero and has degree at most d by (1), noting that $f(\alpha) = g(\alpha)$ exactly when $h(\alpha) = 0$. □

Remark 5.46 (The lemma stated plainly, and used). Inequality (5) formalises the slogan: *two low-degree polynomials that agree at many points must be equal*. The deterministic version is theorem 5.20—agreement at $d + 1$ points forces equality—and Schwartz–Zippel is its probabilistic refinement: a *single* random evaluation point detects unequal polynomials of degree at most d, except with probability at most $d/|S|$. With $|S|$ exponentially large—taking $S = \mathbb{F}_q$ with $q \approx 2^{254}$, the 255-bit fields underlying Orchard—and d polynomially bounded, the failure probability is astronomically small; the term *negligible* receives its precise asymptotic meaning later in the volume. This is the foundational soundness argument for polynomial identity testing, and hence for the polynomial commitment schemes and arithmetisation checks of Halo 2: to verify a purported polynomial identity, evaluate both sides at a random challenge and compare.

Remark 5.47 (Multivariate generalisation). The full Schwartz–Zippel lemma extends to polynomials in several indeterminates $f \in F[X_1, \ldots, X_n]$: a nonzero polynomial of total degree d—the largest sum of exponents $i_1 + \cdots + i_n$ over the monomials with nonzero coefficient—vanishes at a uniformly random point of S^n with probability at most $d/|S|$. The univariate case proved here is both the base case of the multivariate inductive proof and the version most directly used: the later volumes invoke only the univariate lemma directly. Where an identity in several random challenges arises—for instance a permutation argument drawing a pair of challenges (β, γ)—the multivariate bound applies and follows from the univariate lemma applied once per variable.

The cheque drawn in the section's opening can now be cashed in full. Two parties hold polynomials f and g of degree at most d over a 255-bit field $\mathbb{F}_q$—even with d in the millions, they agree on a uniformly random challenge $\alpha \in \mathbb{F}_q$ with probability at most $d/q \approx d \cdot 2^{-254}$ unless they are literally the same formal polynomial. One evaluation apiece, one comparison of field elements, and the enormous coefficient lists never change hands: that is the trade the proving system makes on every identity it checks, and the entire cost of the bargain is the elementary fact with which the section began—a nonzero polynomial of degree d over a field has at most d roots.

6 Finite fields

A processor stores a number in a word of fixed width. A 64-bit word can hold exactly 2^{64} distinct values, and the hardware's arithmetic wraps around on overflow: what the machine natively computes is addition and multiplication modulo 2^{64}. For a verifier that must add, multiply, *and divide*, this built-in arithmetic falls at the last hurdle: in $\mathbb{Z}/2^{64}\mathbb{Z}$ no even class has an inverse, since a class is a unit only when its representative is coprime to the modulus (Theorem 2.24). The question this section answers is the natural one to ask next. Among all finite arithmetics—finite sets closed under an addition and a multiplication obeying the ring laws—which ones allow division by *every* nonzero element? In the language of §4: what are the finite fields, what sizes do they come in, and what structure do they carry?

Part of the answer is already in hand. The residue ring $\mathbb{Z}/n\mathbb{Z}$ is a field exactly when n is prime (theorem 4.26), giving one finite field $\mathbb{F}_p$ for every prime p; and the quotient recipe of Example 4.22 promised further examples of prime-power size p^k. This section completes the picture with two theorems. The first, theorem 6.9, is a prohibition: *no other sizes occur*—every finite field has exactly p^k elements for some prime p, so there is no field with 6, 10, or 12 elements. This constrains cardinalities, not binary representation widths: fields of size 2^k exist for every positive k. The second, theorem 6.17, is the section's payoff and one of the most consequential facts in the volume: the multiplicative group $\mathbb{F}_q^\times$ is *cyclic*—the powers of a single well-chosen element sweep out every nonzero value. Cyclicity is what makes discrete logarithms meaningful and what guarantees the roots of unity on which the polynomial machinery of later sections runs.

The route is as follows. We first make the prime field $\mathbb{F}_p$ computational, with two inversion algorithms and the square-and-multiply technique that powers one of them. We then determine the possible sizes of a finite field through the characteristic and the prime subfield, meet the Frobenius map $x \mapsto x^p$—the symmetry peculiar to characteristic p—and finally prove cyclicity by a counting argument of Gauss.

6.1 The prime field and two inversion routes

The field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ certified by theorem 4.26 is the workhorse of the entire series, and the first order of business is to compute in it. Addition, subtraction, and multiplication are residue arithmetic as in §2; the operation that needs an algorithm is inversion. As §4 stressed, an existence proof and an algorithm are different assets, and cryptography needs both. For inversion in $\mathbb{F}_p$ there are two standard routes, with different characters.

Construction 6.1 (Inversion by the extended Euclidean algorithm)**.** Given a with $0 < a < p$, run the extended Euclidean algorithm (Theorem 2.13) on the pair (a, p). Since p is prime and $p \nmid a$, the gcd is 1, and the algorithm outputs integers u, v with $au + pv = 1$. Reducing modulo p gives $au \equiv 1 \pmod{p}$, that is,

$$a^{-1} = [u]_p \quad \text{in } \mathbb{F}_p.$$

The cost is $O(\log p)$ division steps (Theorem 2.13). This is the standard general-purpose method; a worked instance is Example 2.25, which computed $[17]_{43}^{-1} = [38]_{43}$ by exactly this route.

The second route rests on Fermat's little theorem and on a technique for fast exponentiation that is used throughout cryptography, so we record the technique first, in the generality it deserves.

Construction 6.2 (Square-and-multiply)**.** Let a be an element of any structure with an associative multiplication, and let $e \geq 1$ have binary expansion $e = \sum_i e_i 2^i$ with digits $e_i \in \{0, 1\}$. Squaring repeatedly produces the values

$$a^{2^0} = a, \qquad a^{2^{i+1}} = (a^{2^i})^2,$$

and multiplying together the factors indexed by the nonzero digits gives

$$a^e = \prod_{i\,:\,e_i=1} a^{2^i}.$$

The computation uses $\lfloor \log_2 e \rfloor$ squarings and at most $\lfloor \log_2 e \rfloor$ further multiplications—$O(\log e)$ multiplications in all, against the $e - 1$ multiplications of the naive product $a \cdot a \cdots a$. For the

255-bit exponents of later sections the difference is that between a few hundred multiplications and a count that no conceivable computer could finish.

Remark 6.3. Written additively, the same algorithm computes the multiple $e \cdot P$ of a group element P by repeated doubling and selective adding. Under the name *double-and-add* it is the algorithm for scalar multiplication on elliptic curves (theorem 10.20), where the group operation is written $+$; the two names denote one algorithm in two notations.

Construction 6.4 (Inversion by Fermat's little theorem)**.** Let $a \in \mathbb{F}_p$ be nonzero. Fermat's little theorem (Corollary 3.32) gives $a^{p-1} = 1$, hence $a \cdot a^{p-2} = 1$, and so

$$a^{-1} = a^{p-2},$$

computed by square-and-multiply (theorem 6.2) with exponent $p - 2$, at a cost of $O(\log p)$ modular multiplications when $p > 2$. For $p = 2$, the only nonzero element is 1, whose inverse is 1; the exponent is zero and no multiplication is needed.

Remark 6.5 (Constant-time inversion). The two routes are asymptotically comparable, but they differ in a respect invisible to asymptotics and central to practice. In theorem 6.4 the sequence of operations performed—which steps square, which multiply—depends only on the binary digits of the *public* exponent $p - 2$, never on the secret input a; the method is *constant-time-friendly*. The extended Euclidean route, by contrast, performs a sequence of divisions whose lengths and quotients depend on a itself, so its running time and memory-access pattern can leak information about a to an observer timing the computation. Side-channel-resistant implementations therefore often prefer the Fermat route despite its slightly larger constant factor.

6.2 Characteristic, prime subfield, and prime-power order

We turn to the sizes question: which cardinalities can a finite field have? The tool is the characteristic of section 4.5. Recall the canonical homomorphism

$$\iota \colon \mathbb{Z} \to F, \qquad \iota(n) = n \cdot 1_F,$$

whose kernel is an ideal $n\mathbb{Z}$ of $\mathbb{Z}$; the characteristic $\mathrm{char}(F)$ is its nonnegative generator (Definition 4.30).

Theorem 6.6. *Every finite field F has prime characteristic:* $\mathrm{char}(F) = p$ *for some prime p. In particular* $\mathrm{char}(\mathbb{F}_p) = p$.

Proof. The map ι cannot be injective, since its domain $\mathbb{Z}$ is infinite and its codomain F is finite; two distinct integers $m < n$ must satisfy $\iota(m) = \iota(n)$, and then $n - m$ is a nonzero element of $\ker \iota$. Hence $\ker \iota = n\mathbb{Z}$ with $n > 0$, that is, $\mathrm{char}(F) > 0$; and the characteristic of a field, being nonzero, is prime by Theorem 4.32. For $F = \mathbb{F}_p$ itself, Example 4.31 computed $\mathrm{char}(\mathbb{Z}/p\mathbb{Z}) = p$ directly. □

The characteristic locates a canonical copy of $\mathbb{F}_p$ inside every field of characteristic p. Remark 4.33 described this copy informally; we now make it precise, beginning with the notion of a subfield.

Definition 6.7 (Subfield; prime subfield)**.** A *subfield* of a field F is a subset $K \subseteq F$ containing 0 and 1 and closed under addition, negation, multiplication, and inversion of nonzero elements; with the restricted operations, K is itself a field. The *prime subfield* of F is the intersection of

all subfields of F—equivalently, the smallest subfield of F, and equivalently again the subfield generated by 1_F.

Two words on the equivalences. The intersection of any family of subfields is again a subfield, since each defining closure property survives intersection; and the intersection of *all* subfields is contained in every subfield, hence is the smallest one. That this smallest subfield is exactly what 1_F generates—the closure of $\{1_F\}$ under the field operations—follows from the classification below, whose proof exhibits the prime subfield inside the closure of $\{1_F\}$.

Proposition 6.8 (Classification of prime subfields)**.** *Let F be a field. If* $\mathrm{char}(F) = p > 0$*, the prime subfield of F is isomorphic to $\mathbb{F}_p$. If* $\mathrm{char}(F) = 0$*, it is isomorphic to $\mathbb{Q}$.*

Proof. Suppose first that $\mathrm{char}(F) = p > 0$, so that $\ker \iota = p\mathbb{Z}$. Define

$$\bar{\iota}\colon \mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} \to F, \qquad \bar{\iota}([a]_p) = a \cdot 1_F.$$

The map is well defined and injective at a stroke: $a \cdot 1_F = b \cdot 1_F$ holds if and only if $(a-b) \cdot 1_F = 0$, i.e. $a - b \in \ker \iota = p\mathbb{Z}$, i.e. $[a]_p = [b]_p$. It is a ring homomorphism because ι is one (section 4.5) and $\bar{\iota}$ merely re-reads ι on residue classes. Its image $K = \{\, a \cdot 1_F : a \in \mathbb{Z} \,\}$ is therefore a subring of F isomorphic to the field $\mathbb{F}_p$, and K is in fact a subfield: for $[a]_p \neq [0]_p$ the computation $\bar{\iota}([a]_p)\,\bar{\iota}([a]_p^{-1}) = \bar{\iota}([1]_p) = 1_F$ exhibits the inverse of each nonzero element of K inside K. Finally, K lies in every subfield: a subfield contains 1_F, hence all its additive multiples $a \cdot 1_F$ by closure under addition and negation. So K is contained in the intersection of all subfields and is itself a subfield, forcing K to *be* that intersection: the prime subfield is $K \cong \mathbb{F}_p$.

Now suppose $\mathrm{char}(F) = 0$, so that ι is injective and $b \cdot 1_F \neq 0$ for every nonzero integer b. Define

$$j\colon \mathbb{Q} \to F, \qquad j\left(\frac{a}{b}\right) = (a \cdot 1_F)(b \cdot 1_F)^{-1}.$$

The map is well defined: if $a/b = c/d$ with $b, d \neq 0$, then $ad = bc$ in $\mathbb{Z}$, so $(a \cdot 1_F)(d \cdot 1_F) = (b \cdot 1_F)(c \cdot 1_F)$ by applying ι, and multiplying both sides by $(b \cdot 1_F)^{-1}(d \cdot 1_F)^{-1}$ gives $(a \cdot 1_F)(b \cdot 1_F)^{-1} = (c \cdot 1_F)(d \cdot 1_F)^{-1}$. That j preserves sums, products, and 1 is the usual arithmetic of fractions, transported by the homomorphism ι and commutativity. Injectivity is immediate: $j(a/b) = 0$ forces $a \cdot 1_F = 0$, hence $a = 0$. The image of j is thus a subfield of F isomorphic to $\mathbb{Q}$; and it lies in every subfield K, since K contains 1_F, hence every $a \cdot 1_F$, hence—being closed under inversion and multiplication—every $(a \cdot 1_F)(b \cdot 1_F)^{-1}$. As before the image must equal the intersection of all subfields, and the prime subfield is a copy of $\mathbb{Q}$. □

The size constraint now falls out of a counting device worth naming, for it recurs whenever a field sits over a subfield: *vector-space counting over the prime subfield*. Every field F of characteristic p is a vector space over its prime subfield $\mathbb{F}_p$: vectors are the elements of F, vector addition is the field addition, and scalar multiplication $c \cdot x$ for $c \in \mathbb{F}_p \subseteq F$ is the field multiplication, so the vector-space axioms are instances of the ring axioms of F. The proof below uses exactly two facts about vector spaces: a space with a finite spanning set has a basis, and coordinates with respect to a basis are unique. Both are stated and proved in the linear-algebra section (section 8), whose development is independent of the present theorem, so no circularity arises.

Theorem 6.9 (Finite fields have prime-power order)**.** *Every finite field F has cardinality $|F| = p^k$ for some prime p and integer $k \geq 1$, where $p =$* $\mathrm{char}(F)$*.*

Proof. By theorem 6.6 the characteristic of F is a prime p, and by theorem 6.8 the prime subfield of F is a copy of $\mathbb{F}_p$, with which we identify it. As explained above, F is then a vector space over

$\mathbb{F}_p$. It is spanned by the finite set F itself, so it has a finite basis $e_1, \dots, e_k$; and $k \geq 1$ because $1 \neq 0$ makes F nonzero. Every element of F is thus

$$c_1 e_1 + c_2 e_2 + \cdots + c_k e_k$$

for a *unique* tuple $(c_1, \dots, c_k)$ with each $c_i \in \mathbb{F}_p$: existence because the e_i span, uniqueness because coordinates with respect to a basis are unique (section 8). The assignment $(c_1, \dots, c_k) \mapsto \sum_i c_i e_i$ is therefore a bijection from $\mathbb{F}_p^k$ to F, and counting tuples gives $|F| = p \cdot p \cdots p = p^k$. □

Fields of every prime-power size do exist, and are unique; we state the fact in full, but prove only the part the series stands on.

Theorem 6.10 (Existence and uniqueness of $\mathbb{F}_q$)**.** *For every prime power $q = p^k$ there exists a field with exactly q elements, and any two finite fields of the same cardinality are isomorphic. One constructs such a field as the quotient $\mathbb{F}_p[x]/(f)$ of the polynomial ring by the ideal generated by an irreducible polynomial f of degree k—the recipe recorded in Example 4.22.*

Remark 6.11. The theorem is deliberately left unproved, and the omission costs the series nothing. The fields on the critical path of Halo 2 and Orchard are *prime* fields, the case $q = p$ with $k = 1$, and there the theorem is already proved: existence is theorem 4.26, and uniqueness is immediate from the classification, since a field with p elements has order $p^k = p$ with $k = 1$ by theorem 6.9, so it equals its own prime subfield, a copy of $\mathbb{F}_p$ (theorem 6.8). The general case $k \geq 2$ would require a development of polynomial factorisation over field extensions on which nothing later in the series relies, so we state it for orientation only. *Convention, fixed henceforth*: the letter $q = p^k$ denotes a prime power, and $\mathbb{F}_q$ a field with q elements, of characteristic p; results are stated for general $\mathbb{F}_q$ whenever the proof is identical to the prime case, and the reader focused on Halo 2 may read $q = p$ throughout.

6.3 The Frobenius endomorphism

Characteristic p is not merely a bookkeeping invariant; it changes what algebra is true. The most famous instance is the schoolroom error $(a + b)^p = a^p + b^p$, false over $\mathbb{Q}$ and $\mathbb{R}$ for every $p \geq 2$, which in characteristic p becomes a theorem.

The proof expands $(a + b)^p$ by the binomial theorem, so we first fix its coefficients and their two readings.

Lemma 6.12 (Binomial theorem; the coefficients count subsets)**.** *Let R be a commutative ring. For all $a, b \in R$ and every integer $n \geq 0$,*

$$(a + b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k},$$

where the integer coefficient $\binom{n}{k}$ acts as the additive multiple $\binom{n}{k} \cdot x = x + \cdots + x$, equals the number of k-element subsets of an n-element set, and satisfies

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}.$$

Proof. Expand the product $(a + b)^n = (a + b) \cdots (a + b)$ by distributivity without collecting terms: the result is a sum of 2^n products, one for each way of choosing, from every factor, either its a or its b. By commutativity such a product equals $a^k b^{n-k}$ exactly when a was chosen from k of the n factors, and it is determined by the set of those k factors; so the coefficient of $a^k b^{n-k}$ is the number of k-element subsets of the n factors, and collecting the terms gives the displayed sum.

For the formula, count the sequences of k distinct elements of an n-element set in two ways. Choosing the entries in turn, there are n choices for the first, $n-1$ for the second, and $n-k+1$ for the k-th: $n(n-1)\cdots(n-k+1) = n!/(n-k)!$ sequences. Alternatively, each such sequence is a k-element subset together with an ordering of it, and a k-element set has $k!$ orderings — the same count with k in place of n — so there are $k!$ sequences per subset, and the number of subsets is $n!/(k!\,(n-k)!)$. □

Lemma 6.13 (Freshman's dream)**.** *Let R be a commutative ring of prime characteristic p. Then for all $a, b \in R$,*

$$(a+b)^p = a^p + b^p,$$

and more generally $(a+b)^{p^n} = a^{p^n} + b^{p^n}$ for every $n \geq 1$.

Proof. By the binomial theorem (theorem 6.12),

$$(a+b)^p = \sum_{i=0}^{p} \binom{p}{i} a^i b^{p-i},$$

where the integer coefficient $\binom{p}{i}$ acts as the additive multiple $\binom{p}{i} \cdot x = x + \cdots + x$. We claim that $p \mid \binom{p}{i}$ for $0 < i < p$. From the factorial identity of the same lemma

$$\binom{p}{i} i!\,(p-i)! = p!$$

the prime p divides the right-hand side. It does not divide $i!\,(p-i)!$: that product is a product of integers all smaller than p, and if p divided the product it would divide one of the factors by Euclid's lemma for primes (Lemma 2.17), which is impossible. Applying Euclid's lemma to the left-hand side instead, p must divide $\binom{p}{i}$, say $\binom{p}{i} = pm$.

Now let $0 < i < p$ and consider the corresponding term. Writing $\iota(n) = n \cdot 1_R$ for the canonical homomorphism of section 4.5, the coefficient acts as multiplication by $\iota(\binom{p}{i}) = \iota(pm) = \iota(p)\,\iota(m) = 0 \cdot \iota(m) = 0$, since $\mathrm{char}(R) = p$ means exactly $\iota(p) = p \cdot 1_R = 0$. Every middle term of the binomial expansion therefore vanishes, and the terms $i = p$ and $i = 0$ leave $(a+b)^p = a^p + b^p$.

The general case follows by induction on n, the base case $n = 1$ being the identity just proved. For $n \geq 2$, apply the base case to the elements $a^{p^{n-1}}$ and $b^{p^{n-1}}$:

$$(a+b)^{p^n} = ((a+b)^{p^{n-1}})^p = (a^{p^{n-1}} + b^{p^{n-1}})^p = a^{p^n} + b^{p^n},$$

using the inductive hypothesis in the middle step. □

The identity says that in characteristic p, raising to the p-th power respects addition—and it obviously respects multiplication. A map respecting both operations is a homomorphism, and a homomorphism from a structure to itself is called an *endomorphism*; this one has a name of its own.

Definition 6.14 (Frobenius endomorphism)**.** Let F be a field of characteristic $p > 0$. The *Frobenius map* of F is

$$\mathrm{Frob}\colon F \to F, \qquad \mathrm{Frob}(x) = x^p.$$

Proposition 6.15. *Let F be a field of characteristic p. The Frobenius map is a ring homomorphism $F \to F$ fixing the prime subfield pointwise. If F is finite,* Frob *is an automorphism of F—a bijective*

ring homomorphism of F onto itself.

Proof. Multiplicativity is commutativity and associativity alone: $(xy)^p = x^p y^p$. Additivity is the freshman's dream (Lemma 6.13), and $1^p = 1$; so Frob is a ring homomorphism.

Injectivity holds for a reason worth isolating, since it recurs: *every ring homomorphism between fields is injective*. Indeed, if $\phi\colon F \to F'$ had $\phi(x) = 0$ for some $x \neq 0$, then

$$1 = \phi(1) = \phi(x^{-1}x) = \phi(x^{-1})\,\phi(x) = \phi(x^{-1}) \cdot 0 = 0$$

in F', contradicting $1 \neq 0$ in a field. In particular Frob is injective.

The prime subfield is fixed pointwise. By theorem 6.8 its elements are the values $\bar{\iota}([a]_p) = a \cdot 1_F$, and since $\bar{\iota}$ is a ring homomorphism,

$$\mathrm{Frob}(\bar{\iota}([a]_p)) = \bar{\iota}([a]_p)^p = \bar{\iota}([a]_p^p) = \bar{\iota}([a^p]_p) = \bar{\iota}([a]_p),$$

the last step because $a^p \equiv a \pmod{p}$ for *all* integers a (Corollary 3.32).

Finally, let F be finite. An injective map from a finite set to itself is surjective—the injectivity-implies-surjectivity device of section 4.4—so Frob is a bijective ring homomorphism of F onto itself, an automorphism. □

The iterates $\mathrm{Frob}^n(x) = x^{p^n}$ are again homomorphisms, as the general case of the freshman's dream states directly. On the prime field $\mathbb{F}_p$ itself the Frobenius map is the identity; on larger fields of characteristic p it is a genuine, structure-preserving symmetry, and theorem 6.24 below records where it re-enters the cryptographic story.

6.4 Cyclicity of the multiplicative group

The multiplicative group $\mathbb{F}_q^\times$ of a finite field is a finite abelian group of order $q-1$: every one of the $q-1$ nonzero elements is invertible, precisely because $\mathbb{F}_q$ is a field. Finite abelian groups can in general be quite far from cyclic—the unit group $(\mathbb{Z}/8\mathbb{Z})^\times$ of Example 3.22 has order 4 with no element of order 4—so it is a genuinely strong fact that $\mathbb{F}_q^\times$ is always cyclic. The proof is a counting argument of Gauss, and its arithmetic engine is an identity about the totient function φ. Everything we use about φ is its *counting definition* (Definition 2.26): $\varphi(n)$ is the number of integers a with $1 \le a \le n$ and $\gcd(a,n) = 1$. No further totient theory is needed or assumed.

Lemma 6.16 (Gauss's divisor-sum identity)**.** *For every integer $n \ge 1$,*

$$\sum_{d \mid n} \varphi(d) = n,$$

the sum running over the positive divisors d of n.

Proof. Partition the set $\{1, 2, \dots, n\}$ according to the value of $\gcd(a,n)$. For each a the gcd is a positive divisor d of n, so the classes

$$C_d = \{\, a : 1 \le a \le n,\ \gcd(a,n) = d \,\}, \qquad d \mid n,$$

are disjoint and cover $\{1, \dots, n\}$; hence $n = \sum_{d \mid n} |C_d|$.

We claim $|C_d| = \varphi(n/d)$. A member of C_d is divisible by d, so it has the form $a = db$ with $1 \le b \le n/d$; we show that, for such a,

$$\gcd(db, n) = d \quad \Longleftrightarrow \quad \gcd(b, n/d) = 1.$$

For the forward direction, suppose $e = \gcd(b, n/d) > 1$; then de divides both $db = a$ and $d \cdot (n/d) = n$, so de is a common divisor exceeding d and $\gcd(a,n) \ge de > d$. For the converse, suppose

$\gcd(b, n/d) = 1$. The integer d is a common divisor of a and n, so d divides $g = \gcd(a, n)$ by the strongest-common-divisor property (Remark 2.12); write $g = dc$ with $c \geq 1$. From $dc \mid db$ we get $c \mid b$, and from $dc \mid n = d(n/d)$ we get $c \mid n/d$ (cancel d in each divisibility); so c divides $\gcd(b, n/d) = 1$, forcing $c = 1$ and $g = d$.

The members of C_d are therefore exactly the integers db with $1 \leq b \leq n/d$ and $\gcd(b, n/d) = 1$, and by the counting definition of the totient (Definition 2.26) there are precisely $\varphi(n/d)$ of them. Summing,

$$n = \sum_{d \mid n} \varphi\left(\frac{n}{d}\right) = \sum_{d \mid n} \varphi(d),$$

the second equality because $d \mapsto n/d$ is a bijection of the set of positive divisors of n onto itself. □

The theorem now follows from a beautiful squeeze. We count the elements of each possible order twice over—once inside the group, once through the totient—and Gauss's identity leaves the two counts no room to differ.

Theorem 6.17 (Finite subgroups of $F^\times$ are cyclic)**.** *Let F be any field and let $G \leq F^\times$ be a finite subgroup of the multiplicative group, of order N. Then G is cyclic. In particular, for a finite field $\mathbb{F}_q$ the whole multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $q - 1$.*

Proof. For each divisor $d \mid N$, let

$$\psi(d) = |\{\, a \in G : \operatorname{ord}(a) = d \,\}|$$

count the elements of G of order exactly d. Every element of the finite group G has finite order dividing $N = |G|$ (Corollary 3.30), so the sets counted by the $\psi(d)$ partition G and

$$\sum_{d \mid N} \psi(d) = N. \tag{6}$$

The key claim is that *for each $d \mid N$*, either $\psi(d) = 0$ or $\psi(d) = \varphi(d)$. Suppose $\psi(d) > 0$ and choose $a \in G$ of order d. The powers $a^0, a^1, \ldots, a^{d-1}$ are d distinct elements of G (Proposition 3.12(3)), and each satisfies

$$(a^j)^d = (a^d)^j = 1^j = 1,$$

so each is a root in F of the polynomial $X^d - 1 \in F[X]$. That polynomial has degree d, so it has *at most d* roots in F (Theorem 5.18); the d distinct powers of a therefore account for *all* of its roots. Now take any $b \in G$ of order d. Then $b^d = 1$, so b is a root of $X^d - 1$ and hence $b = a^j$ for some $0 \leq j < d$. By Corollary 3.23,

$$\operatorname{ord}(a^j) = \frac{d}{\gcd(d, j)},$$

which equals d exactly when $\gcd(j, d) = 1$. The elements of order d in G are thus precisely the powers a^j with $0 \leq j < d$ and $\gcd(j, d) = 1$, and by the counting definition of the totient there are $\varphi(d)$ of them: $\psi(d) = \varphi(d)$. This proves the claim, and with it the inequality $\psi(d) \leq \varphi(d)$ for every $d \mid N$.

Summing the inequality over the divisors of N and comparing (6) with Gauss's identity (Lemma 6.16),

$$N = \sum_{d \mid N} \psi(d) \;\leq\; \sum_{d \mid N} \varphi(d) = N.$$

The two ends are equal, so the inequality is an equality term by term: $\psi(d) = \varphi(d)$ for *every* divisor $d \mid N$. In particular

$$\psi(N) = \varphi(N) \geq 1,$$

since $a = 1$ always witnesses $\gcd(1, N) = 1$ in the totient count. Hence G contains an element g of order N; its powers form a subgroup of G with N distinct elements (Proposition 3.12(3)), which must be all of G. So $G = \langle g \rangle$ is cyclic.

For the final statement, let $F = \mathbb{F}_q$ be finite. The multiplicative group $\mathbb{F}_q^\times$ has $q - 1$ elements—every nonzero element of a field is a unit—and is a finite subgroup of itself, so it is cyclic of order $q - 1$. □

The equality $\psi(d) = \varphi(d)$ established along the way is worth extracting: it was proved for subgroups of a field's multiplicative group, but it holds in any cyclic group, by the same gcd computation and with no field in sight.

Corollary 6.18 (Element-order census in cyclic groups)**.** *A cyclic group G of order N contains, for each divisor $d \mid N$, exactly $\varphi(d)$ elements of order d, and no elements of any other order. In particular G has exactly $\varphi(N)$ generators.*

Proof. Write $G = \langle g \rangle$ with $\operatorname{ord}(g) = N$; the elements of G are the powers g^j, $0 \le j < N$, each appearing once (Proposition 3.12(3)). By Corollary 3.23, $\operatorname{ord}(g^j) = N/\gcd(N, j)$, which is always a divisor of N; and $\operatorname{ord}(g^j) = d$ if and only if $\gcd(j, N) = N/d$. By the gcd computation in the proof of Lemma 6.16 (applied with $n = N$ and divisor N/d), the exponents $j \in \{1, \dots, N\}$ with $\gcd(j, N) = N/d$ are exactly $j = (N/d)\,t$ with $1 \le t \le d$ and $\gcd(t, d) = 1$, of which there are $\varphi(d)$ by Definition 2.26. The generators are the elements of order N, numbering $\varphi(N)$. □

Definition 6.19 (Primitive root)**.** A generator of the cyclic group $\mathbb{F}_q^\times$ is called a *primitive root* (or *primitive element*) of $\mathbb{F}_q$. Equivalently, an element $g \in \mathbb{F}_q^\times$ is primitive if and only if $\operatorname{ord}(g) = q - 1$.

Existence is theorem 6.17, and the census refines it: $\mathbb{F}_q$ has exactly $\varphi(q - 1)$ primitive roots (theorem 6.18), always at least one.

Remark 6.20. The name has a second reading, developed later in the volume: a primitive root of $\mathbb{F}_q$ is precisely a primitive $(q - 1)$-th root of unity in the sense of theorem 9.3, where the identification is restated once roots of unity have been defined in general. The two vocabularies—group-theoretic "generator of $\mathbb{F}_q^\times$" and polynomial-flavoured "primitive $(q-1)$-th root of unity"—name the same elements.

Example 6.21 (Primitive roots of $\mathbb{F}_7$)**.** The group $\mathbb{F}_7^\times$ has order 6. Testing $g = 3$, the successive powers modulo 7 are

$$3^1 = 3, \quad 3^2 = 2, \quad 3^3 = 6, \quad 3^4 = 4, \quad 3^5 = 5, \quad 3^6 = 1,$$

which run through all of $\{1, \dots, 6\}$: the element 3 is a primitive root of $\mathbb{F}_7$. By contrast 2 is not: its powers are $2, 4, 1$, so $\operatorname{ord}(2) = 3 < 6$. The full roster of primitive roots consists of the powers 3^j with $\gcd(j, 6) = 1$ (Corollary 3.23), namely $j = 1$ and $j = 5$: the primitive roots of $\mathbb{F}_7$ are $3^1 = 3$ and $3^5 = 5$, and there are $\varphi(6) = 2$ of them, as the census predicts.

Cyclicity settles, in one stroke, how many solutions the equation $x^n = 1$ has in a finite field—the question on which the evaluation domains of later sections turn.

Corollary 6.22 (Roots of unity in $\mathbb{F}_q$)**.** *Let $\mathbb{F}_q$ be a finite field and $n \ge 1$. The solutions of $x^n = 1$ in $\mathbb{F}_q^\times$ number exactly $\gcd(n, q - 1)$, and they form the cyclic subgroup of $\mathbb{F}_q^\times$ of that order. In particular, when $n \mid q - 1$ the equation has exactly n solutions.*

Proof. Fix a primitive root g (theorem 6.17) and write each element of $\mathbb{F}_q^\times$ uniquely as $x = g^j$ with $0 \le j \le q-2$ (Proposition 3.12(3)). Then $x^n = g^{jn}$, and $g^{jn} = 1$ holds if and only if $(q-1) \mid jn$ (Proposition 3.12(1)). Set $d = \gcd(n, q-1)$ and write $q-1 = dm$, $n = dn'$ with $\gcd(m, n') = 1$. Then

$$(q-1) \mid jn \iff dm \mid j\,dn' \iff m \mid jn' \iff m \mid j,$$

the last step by Euclid's lemma (Proposition 2.15(1), with $\gcd(m, n') = 1$). The qualifying exponents in $\{0, \dots, q-2\}$ are therefore the multiples of $m = (q-1)/d$, namely $j = tm$ for $t = 0, 1, \dots, d-1$: exactly d of them. The corresponding solutions $x = (g^m)^t$ are exactly the powers of g^m, which by Corollary 3.23 has order $(q-1)/\gcd(q-1, m) = (q-1)/m = d$; so the solution set is the cyclic subgroup $\langle g^m \rangle$ of order $d = \gcd(n, q-1)$. When $n \mid q-1$ we have $d = n$, giving exactly n solutions. □

Remark 6.23 (Two-adicity and proof-system fields). The corollary is precisely what renders a curve such as Pallas or Vesta suitable for Halo 2: one chooses the field size q so that $q-1$ is divisible by a large power of 2, and the corollary then *guarantees* a multiplicative subgroup of 2-power order—the evaluation domain on which the number-theoretic transform, and hence efficient polynomial arithmetic, operates. The construction of these domains and the transform itself are the business of section 9; the curves are introduced in section 10.

Remark 6.24 (Cyclicity and Frobenius in cryptography). The section's two structural discoveries each carry cryptographic weight. First, the cyclicity of $\mathbb{F}_q^\times$ (theorem 6.17) underlies the discrete-logarithm problem—writing every nonzero element as g^j invites the question of recovering j, and the presumed hardness of doing so is a foundational assumption of later volumes—and, through theorem 6.22, guarantees the existence of the roots of unity used in polynomial commitment schemes. Second, the Frobenius endomorphism (theorem 6.14) is not merely abstract: on an elliptic curve over $\mathbb{F}_p$ it induces an endomorphism of the curve whose characteristic polynomial $T^2 - tT + p$ encodes the number of curve points as $N = p + 1 - t$, where t is the Hasse trace—tying the size of the cryptographic group directly to the field-theoretic Frobenius. Both connections are taken up in section 10.

The question that opened the section is now answered in full. A finite arithmetic in which every nonzero element inverts exists only at prime-power sizes—theorem 6.9 forbids every other cardinality—and at each admissible size $q = p^k$ there is exactly one field, up to isomorphism (theorem 6.10). The machine's own wrap-around arithmetic is instructive here: 2^{64} is a prime-power size, so a field with 2^{64} elements exists, but it is *not* the ring $\mathbb{Z}/2^{64}\mathbb{Z}$ of word arithmetic—whose even classes have no inverses—but the polynomial-quotient construction of theorem 6.10, with a different multiplication altogether. The proof systems of this series instead take the prime route: a prime p of roughly 255 bits, the field $\mathbb{F}_p$, inversion by theorem 6.1 or theorem 6.4. The payoff theorem then delivers the structure that everything later rides on: the multiplicative group of the chosen field is a single cycle, generated by one element, with exactly $\varphi(d)$ elements of each order $d \mid q-1$ and a guaranteed subgroup of every size dividing $q-1$. Polynomials over these fields, the subject that the volume develops next into evaluation domains, fast transforms, and commitment schemes, will draw on that guarantee constantly.

7 Number theory for cryptography

An implementation that transmits a point of an elliptic curve—a pair (x, y) of field elements satisfying an equation $y^2 = x^3 + ax + b$, as constructed in section 10—need not send both coordinates. For a given x the equation determines y^2 exactly, hence determines y up to sign, so wire formats send x together with a single bit selecting one of the two candidates, and the transmission halves in size. The receiver is then left with two number-theoretic tasks: decide whether $x^3 + ax + b$ is a square modulo p at all—if not, no point has this x-coordinate—and, if it is, *compute* a square root

of it. Both tasks must run in the time of a few modular exponentiations, on numbers hundreds of bits long. The same constructions make a complementary demand in the opposite direction: they publish powers g^x of a known group element while the exponent x is a secret key, so exponentiation must be easy to perform and yet infeasible to *invert*. This section assembles the classical number theory behind both demands: the totient and the Chinese Remainder Theorem, Fermat's little theorem and Euler's theorem in their role as workhorses of modular exponentiation, multiplicative orders and primitive roots, quadratic residues with the Euler criterion, square-root extraction modulo a prime, and finally the discrete logarithm problem together with the reason *prime* group order is so important for cryptographic hardness.

Throughout, n and m denote positive integers and p denotes a prime. The section builds directly on the arithmetic of section 2, the group theory of section 3, and the cyclicity theorem of section 6; several statements proved there reappear here, restated in the arithmetic language in which cryptography uses them, with proofs reduced to citations.

7.1 Euler's totient function and the Chinese Remainder Theorem

Recall from section 2 that a class $[a]_n \in \mathbb{Z}/n\mathbb{Z}$ is a unit if and only if $\gcd(a, n) = 1$ (Theorem 2.24), that the units form a finite abelian group $(\mathbb{Z}/n\mathbb{Z})^\times$ (Proposition 3.8), and that Euler's totient $\varphi(n)$ counts them: $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$ (Definition 2.26). In particular $\varphi(p) = p - 1$ for a prime p (Example 2.27), so $\mathbb{F}_p^\times$ has order $p - 1$. These counts govern everything that follows: the totient is the exponent in Euler's theorem, the order of the group whose cyclic structure the next subsections exploit, and—through the orders of elliptic-curve groups—a quantity that the parameter choices of Halo 2 and Orchard are built around. What section 2 did not provide is a way to *compute* $\varphi(n)$ beyond direct enumeration. Two facts close the gap: the totient's value on prime powers, and its behaviour on coprime factors.

Proposition 7.1 (Totient of a prime power)**.** *For a prime p and an integer $k \geq 1$,*

$$\varphi(p^k) = p^k - p^{k-1} = p^k\Big(1 - \frac{1}{p}\Big).$$

Proof. Among $1, \dots, p^k$, we claim the integers *not* coprime to p^k are exactly the multiples of p. If $p \mid a$ then $p \mid \gcd(a, p^k)$, so the gcd exceeds 1. Conversely, if $g = \gcd(a, p^k) > 1$, then g has a prime factor ℓ; from $\ell \mid g \mid p^k$ and Euclid's lemma for primes (Lemma 2.17) we get $\ell = p$, and $\ell \mid g \mid a$ gives $p \mid a$. The multiples of p in the range are $p, 2p, \dots, p^{k-1} \cdot p$, of which there are p^{k-1}; subtracting them from the p^k integers in the range leaves $\varphi(p^k) = p^k - p^{k-1}$. □

The behaviour on coprime factors is a consequence of a structural theorem that will be used again, in a quite different role, at the end of the section: a residue modulo a product of two coprime moduli carries exactly the same information as a *pair* of residues, one modulo each factor. The natural home for the statement is the language of rings (section 4): recall the product ring $R \times S$ with componentwise operations (Example 4.5), and call a bijective ring homomorphism (Definition 4.21) a *ring isomorphism*, mirroring Definition 3.41 for groups.

Theorem 7.2 (Chinese Remainder Theorem)**.** *If $\gcd(m, n) = 1$, then the map*

$$\mathbb{Z}/mn\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}, \qquad [a]_{mn} \longmapsto ([a]_m, [a]_n)$$

is a ring isomorphism. It restricts to a group isomorphism

$$(\mathbb{Z}/mn\mathbb{Z})^\times \;\cong\; (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times.$$

Proof. The map is well defined: if $[a]_{mn} = [a']_{mn}$ then $mn \mid a - a'$, hence $m \mid a - a'$ and $n \mid a - a'$, so the two components agree. It is a ring homomorphism because each component is: reduction modulo m (and modulo n) sends sums to sums, products to products, and 1 to 1 (Theorem 4.6), and the product ring's operations are componentwise.

For injectivity it suffices, since the map is in particular a homomorphism of additive groups, to check that the kernel is trivial (Proposition 3.40). Suppose $[a]_m = [0]_m$ and $[a]_n = [0]_n$, that is, $m \mid a$ and $n \mid a$. Since $\gcd(m, n) = 1$, Euclid's lemma (Proposition 2.15(2)) gives $mn \mid a$, i.e. $[a]_{mn} = [0]_{mn}$.

Both sides have exactly mn elements (Theorem 4.6 for each factor), and an injective map between finite sets of equal size is bijective: the mn distinct inputs have mn distinct images, which must exhaust the codomain. Hence the map is a ring isomorphism.

For the multiplicative statement, a ring isomorphism ϕ matches units with units: if $uv = 1$ then $\phi(u)\phi(v) = \phi(1) = 1$, and the same applies to ϕ^{-1}. The units of the product ring are exactly the pairs of units, $(u, v)^{-1} = (u^{-1}, v^{-1})$ existing precisely when both components are invertible. Restricting ϕ to unit groups therefore gives a bijection $(\mathbb{Z}/mn\mathbb{Z})^\times \to (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times$ that respects multiplication: a group isomorphism. □

Corollary 7.3 (Multiplicativity and the product formula)**.** *If* $\gcd(m, n) = 1$ *then* $\varphi(mn) = \varphi(m)\,\varphi(n)$*. Hence if* $n = \prod_i p_i^{k_i}$ *is the prime factorisation of* n*, then*

$$\varphi(n) = \prod_i (p_i^{k_i} - p_i^{k_i - 1}) = n \prod_{p \mid n} \Big(1 - \frac{1}{p}\Big),$$

the last product running over the distinct primes dividing n*.*

Proof. The unit-group isomorphism of Theorem 7.2 is in particular a bijection, and the cardinality of a product of two sets is the product of the cardinalities, so $\varphi(mn) = \varphi(m)\,\varphi(n)$.

For the product formula, the prime powers $p_i^{k_i}$ are pairwise coprime: a common divisor exceeding 1 of two of them would have a prime factor dividing both, which by Euclid's lemma for primes (Lemma 2.17) would equal both p_i and p_j. Moreover each $p_i^{k_i}$ is coprime to the product of the others, by repeated application of Proposition 2.15(3). Induction on the number of factors therefore extends multiplicativity to $\varphi(n) = \prod_i \varphi(p_i^{k_i})$, and Proposition 7.1 evaluates each factor:

$$\varphi(n) = \prod_i (p_i^{k_i} - p_i^{k_i - 1}) = \prod_i p_i^{k_i}\Big(1 - \frac{1}{p_i}\Big) = n \prod_{p \mid n} \Big(1 - \frac{1}{p}\Big). \qquad \square$$

As a sanity check, the product formula gives $\varphi(12) = 12(1 - \frac{1}{2})(1 - \frac{1}{3}) = 4$, agreeing with the direct enumeration $\{1, 5, 7, 11\}$ of Example 2.27. Together with Gauss's divisor-sum identity $\sum_{d \mid n} \varphi(d) = n$, proved as Lemma 6.16, this completes the totient's basic theory.

7.2 Fermat's little theorem and Euler's theorem

The congruences of this subsection were proved once, as Corollary 3.32, where they fell out of Lagrange's theorem in three lines. We restate them here in the arithmetic form in which cryptography uses them; the proofs are citations, not repetitions.

Theorem 7.4 (Euler's theorem)**.** *If* $\gcd(a, n) = 1$*, then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Proof. The hypothesis says that $[a]_n$ is a unit (Theorem 2.24), i.e. an element of the finite group $(\mathbb{Z}/n\mathbb{Z})^\times$, whose order is $\varphi(n)$ (Definition 2.26). By Corollary 3.31—every element g of a finite group of order N satisfies $g^N = e$, a consequence of Lagrange's theorem—we get $[a]_n^{\varphi(n)} = [1]_n$, which is the stated congruence. This is Corollary 3.32 verbatim. □

Corollary 7.5 (Fermat's little theorem)**.** *Let p be prime. If $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$. Moreover $a^p \equiv a \pmod{p}$ for* every *integer a.*

Proof. For $p \nmid a$ we have $\gcd(a, p) = 1$, so Theorem 7.4 with $n = p$ and $\varphi(p) = p-1$ (Example 2.27) gives the first congruence; multiplying it by a gives $a^p \equiv a \pmod{p}$. When $p \mid a$ both a^p and a are congruent to 0, so the second congruence holds for all a. □

Remark 7.6 (Cryptographic use of Euler and Fermat). Euler's theorem underlies RSA: if $ed \equiv 1 \pmod{\varphi(n)}$, say $ed = 1 + k\varphi(n)$, then

$$(a^e)^d = a^{1+k\varphi(n)} = a \cdot \left(a^{\varphi(n)}\right)^k \equiv a \pmod{n}$$

whenever $\gcd(a, n) = 1$: raising to the e-th power is undone by raising to the d-th, which is what makes one exponent a public encryption key and the other its private inverse. The elliptic-curve cryptography of Orchard instead exploits Fermat's little theorem in the field $\mathbb{F}_p$: every nonzero x satisfies $x^{p-1} = 1$, so inversion is computable as $x^{-1} = x^{p-2}$ (theorem 6.4), and Fermat is the engine of the Euler criterion for square roots developed below.

7.3 Multiplicative order and primitive roots

Definition 7.7 (Multiplicative order)**.** Let $\gcd(a, n) = 1$. The *multiplicative order* of a modulo n, written $\mathrm{ord}_n(a)$, is the least positive integer k with $a^k \equiv 1 \pmod{n}$. Such a k exists because $a^{\varphi(n)} \equiv 1$ by Theorem 7.4.

The quantity $\mathrm{ord}_n(a)$ is exactly the order of $[a]_n$ as an element of the group $(\mathbb{Z}/n\mathbb{Z})^\times$, so the order theory of section 3 applies verbatim: $a^m \equiv 1 \pmod{n}$ if and only if $\mathrm{ord}_n(a) \mid m$ (Proposition 3.12(1))—in particular $\mathrm{ord}_n(a) \mid \varphi(n)$, by Euler's theorem—and

$$\mathrm{ord}_n(a^j) = \frac{\mathrm{ord}_n(a)}{\gcd(j, \mathrm{ord}_n(a))}$$

(Corollary 3.23).

Definition 7.8 (Primitive root)**.** A *primitive root* modulo n is an element g with $\mathrm{ord}_n(g) = \varphi(n)$; equivalently, g generates $(\mathbb{Z}/n\mathbb{Z})^\times$ as a cyclic group, so that $\{g^0, g^1, \ldots, g^{\varphi(n)-1}\}$ lists all the units.

Primitive roots need not exist for every modulus. The unit group $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$ has order $\varphi(8) = 4$, yet every one of its elements squares to 1 (Example 3.22): the element orders are 1 for the identity and 2 for each of 3, 5, 7, so no element has order 4 and no primitive root modulo 8 exists. Indeed the unit group is a copy of the Klein four-group of Example 3.10: every element is its own inverse, and the product of two distinct non-identity elements a and b is neither e (which would force $b = a^{-1} = a$) nor a nor b (which would force $b = e$ or $a = e$), so it is the third. The decisive case for cryptography is $n = p$ prime, where the multiplicative group is always cyclic.

Corollary 7.9 ($\mathbb{F}_p^\times$ is cyclic)**.** *For every prime p, the group $\mathbb{F}_p^\times = (\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p-1$. In particular primitive roots modulo p exist, and there are exactly $\varphi(p-1)$ of them; Example 6.21 exhibits the two primitive roots 3 and 5 of $\mathbb{F}_7$.*

Proof. The group $\mathbb{F}_p^\times$ is a finite subgroup, of order $p-1$, of the multiplicative group of the field $\mathbb{F}_p$, hence cyclic by Theorem 6.17. A generator g has order $p-1$, and the powers g^k that again generate are those with $\gcd(k, p-1) = 1$ (Corollary 3.23), numbering $\varphi(p-1)$. □

Remark 7.10 (Cyclicity and prime-order groups). Cyclicity is why the prime-order groups of elliptic-curve cryptography behave exactly like $\mathbb{Z}/q\mathbb{Z}$ under addition once a generator is fixed: a cyclic group of order q is isomorphic to $(\mathbb{Z}/q\mathbb{Z}, +)$ via $g^k \leftrightarrow k$ (Theorem 3.43). For completeness we record, without proof and without needing it later, the full classification: primitive roots modulo n exist exactly when $n \in \{1, 2, 4, p^k, 2p^k\}$ for an odd prime p.

7.4 Quadratic residues and the Euler criterion

We turn to squares modulo a prime. They control whether a curve point with a given x-coordinate exists—the first of the two tasks posed at the head of the section—and they are the objects from which square roots in $\mathbb{F}_p$ will be extracted.

Definition 7.11 (Quadratic residue)**.** Let p be an odd prime and a an integer with $p \nmid a$. We say a is a *quadratic residue* modulo p (a QR) if the congruence $x^2 \equiv a \pmod{p}$ has a solution, and a *quadratic non-residue* (a QNR) otherwise.

Proposition 7.12 (Exactly half are residues)**.** *For an odd prime p, exactly $\frac{p-1}{2}$ of the nonzero residues modulo p are quadratic residues and $\frac{p-1}{2}$ are non-residues. The squaring map $x \mapsto x^2$ is exactly two-to-one on $\mathbb{F}_p^\times$.*

Proof. Consider $\sigma\colon \mathbb{F}_p^\times \to \mathbb{F}_p^\times$, $\sigma(x) = x^2$; its image is exactly the set of quadratic residues. For $x, y \in \mathbb{F}_p^\times$,

$$\sigma(x) = \sigma(y) \iff x^2 - y^2 = 0 \iff (x-y)(x+y) = 0 \iff y = \pm x,$$

the middle equivalence because a field has no zero divisors (Proposition 4.28). Since p is odd, $x \neq -x$ for $x \neq 0$ (otherwise $2x = 0$ with 2 and x nonzero), so every fibre of σ over its image has exactly two elements. Counting, $|\operatorname{im}\sigma| = \frac{p-1}{2}$, and the complement—the non-residues—has the same size. □

Theorem 7.13 (Euler's criterion)**.** *Let p be an odd prime and $p \nmid a$. Then*

$$a^{\frac{p-1}{2}} \equiv \begin{cases} 1 \pmod{p} & \text{if } a \text{ is a quadratic residue,} \\ -1 \pmod{p} & \text{if } a \text{ is a quadratic non-residue.} \end{cases}$$

Proof. First, the value is always ± 1. Fermat's little theorem (Corollary 7.5) gives $(a^{(p-1)/2})^2 = a^{p-1} \equiv 1$, and the only square roots of 1 modulo a prime are ± 1: from $x^2 \equiv 1$ we get $p \mid (x-1)(x+1)$, so $p \mid x-1$ or $p \mid x+1$ by Euclid's lemma for primes (Lemma 2.17), i.e. $x \equiv \pm 1$. Hence $a^{(p-1)/2} \equiv \pm 1 \pmod{p}$.

If a is a residue, say $a \equiv x^2$ with $p \nmid x$, then $a^{(p-1)/2} \equiv x^{p-1} \equiv 1$, again by Fermat. Thus every one of the $\frac{p-1}{2}$ quadratic residues (Proposition 7.12) is a root of the polynomial $t^{(p-1)/2} - 1$ over $\mathbb{F}_p$. That polynomial has degree $\frac{p-1}{2}$, so it has *at most* $\frac{p-1}{2}$ roots (Theorem 5.18); the residues therefore account for all of them. A non-residue a is consequently not a root, yet still satisfies $a^{(p-1)/2} \equiv \pm 1$; the remaining possibility is $a^{(p-1)/2} \equiv -1$. □

The standard notation for quadratic-residue status is the Legendre symbol.

Definition 7.14 (Legendre symbol)**.** For an odd prime p and an integer a, the *Legendre symbol* is

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \mid a, \\ 1 & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } a \text{ is a quadratic non-residue modulo } p. \end{cases}$$

Euler's criterion (Theorem 7.13) then reads, uniformly,

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p},$$

so the symbol is computable by a single modular exponentiation—an $O(\log p)$ affair by square-and-multiply (theorem 6.2). The criterion also shows that the symbol is *completely multiplicative* in a:

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right), \qquad \text{since} \qquad (ab)^{\frac{p-1}{2}} = a^{\frac{p-1}{2}} b^{\frac{p-1}{2}}.$$

Both observations carry weight in the worked example that follows, which verifies a constant deployed in Orchard.

Example 7.15 (13 is a non-square in both Pasta fields)**.** The Pasta hash-to-curve construction used by Orchard—a procedure, taken up in later volumes, that maps arbitrary byte strings to curve points and requires a fixed non-square constant in the base field—fixes $Z = -13$ as a verifiably non-square constant in both base fields. Let p and q be the Pasta primes of section 10; both satisfy $p \equiv q \equiv 1 \pmod 4$. We verify the non-squareness here, working modulo p; the computation modulo q is identical.

First, -1 is a square modulo any prime $\equiv 1 \pmod 4$: the exponent $\frac{p-1}{2}$ is even, so Euler's criterion gives

$$\left(\frac{-1}{p}\right) \equiv (-1)^{\frac{p-1}{2}} = 1.$$

By complete multiplicativity,

$$\left(\frac{-13}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{13}{p}\right) = \left(\frac{13}{p}\right),$$

so -13 is a non-square precisely when 13 is. Whether 13 is a square is settled by one Euler-criterion exponentiation in each field, a computation any computer-algebra system reproduces in milliseconds:

$$13^{\frac{p-1}{2}} \equiv p - 1 \equiv -1 \pmod{p}, \qquad 13^{\frac{q-1}{2}} \equiv q - 1 \equiv -1 \pmod{q},$$

whence $\left(\frac{13}{p}\right) = \left(\frac{13}{q}\right) = -1$. Thus 13, and with it $Z = -13$, is a quadratic non-residue modulo both Pasta primes, exactly as the hash-to-curve design requires.

Example 7.16 (5 is a non-square in the Pallas base field)**.** The same one-line test settles a second constant on which later volumes lean, this time in the curve equation itself. Pallas is the curve $y^2 = x^3 + 5$ over $\mathbb{F}_p$ (section 10), and the arithmetic circuits of Halo 2 encode its point at infinity as the coordinate pair $(0, 0)$ (theorem 10.13). The encoding is unambiguous only if no genuine point has x-coordinate 0, i.e. only if $y^2 = 5$ has no solution in $\mathbb{F}_p$: the question is whether 5 is a square in the *base* field $\mathbb{F}_p$ of Pallas, not in its scalar field $\mathbb{F}_q$. Euler's criterion decides it in one exponentiation:

$$5^{\frac{p-1}{2}} \equiv p - 1 \equiv -1 \pmod{p},$$

so $\left(\frac{5}{p}\right) = -1$. The constant 5 is a quadratic non-residue modulo the Pallas base-field prime, and no point of Pallas has $x = 0$. (The same computation modulo q gives $5^{(q-1)/2} \equiv -1 \pmod{q}$, so Vesta has no such point either.) That no point has $y = 0$ is a different fact—-5 is not a cube in $\mathbb{F}_p$—which theorem 10.26 reads off from the oddness of the group order; the two together are the hypotheses of theorem 10.14.

7.5 Square roots modulo a prime

Deciding that a is a residue is one matter; *producing* an x with $x^2 \equiv a$ is another, and the point decompression of the section's opening needs the production, not just the decision: the receiver must reconstruct the coordinate y from y^2. How hard this is depends on p mod 4.

Proposition 7.17 (The $p \equiv 3 \pmod 4$ shortcut)**.** *Let $p \equiv 3 \pmod 4$ be prime and let a be a quadratic residue modulo p with $p \nmid a$. Then*

$$x \equiv a^{\frac{p+1}{4}} \pmod{p}$$

is a square root of a, and the two square roots of a are $\pm x$.

Proof. Since $p \equiv 3 \pmod 4$, the quantity $\frac{p+1}{4}$ is a positive integer. Compute

$$x^2 = a^{\frac{p+1}{2}} = a^{\frac{p-1}{2}} \cdot a \equiv \left(\frac{a}{p}\right) \cdot a = a \pmod{p},$$

using Euler's criterion (Theorem 7.13) and the hypothesis that a is a residue, so $\left(\frac{a}{p}\right) = 1$. The roots come in the pair $\pm x$ by the two-to-one property of squaring (Proposition 7.12). □

Remark 7.18 (Cost and the Pasta primes)*.* The single-exponentiation shortcut requires $p \equiv 3 \pmod 4$, which is one reason cryptographers often favour such primes. For $p \equiv 5 \pmod 8$ a closed-form variant (Atkin's formula) still exists. In general the cost of square-root extraction grows with the 2-adic valuation s of $p - 1$: writing $p - 1 = 2^s d$ with d odd, the Tonelli–Shanks algorithm below performs up to s corrective iterations. The Pasta primes have $s = 32$, a value chosen deliberately for FFT-friendliness (theorem 9.10), so their implementations use Tonelli–Shanks.

When $p \equiv 1 \pmod 4$ one writes $p - 1 = 2^s d$ with d odd and $s \geq 2$. The algorithm operates inside the subgroup of $\mathbb{F}_p^\times$ of order 2^s—the unique such subgroup, since $\mathbb{F}_p^\times$ is cyclic (Corollary 7.9 and Theorem 3.24(3))—where the obstruction to the naive exponentiation trick lives, and clears that obstruction one power of two at a time.

Theorem 7.19 (Tonelli–Shanks)**.** *Let p be an odd prime, $p - 1 = 2^s d$ with d odd, and let a be a*

> *quadratic residue with $p \nmid a$. Given any quadratic non-residue z modulo p, one can compute a square root of a modulo p using $O(s^2 + \log p)$ modular multiplications.*

Proof. Write $H \leq \mathbb{F}_p^\times$ for the unique cyclic subgroup of order 2^s. Set $c = z^d$. We claim c generates H. By Euler's criterion (Theorem 7.13) $z^{(p-1)/2} \equiv -1$, so

$$c^{2^{s-1}} = z^{2^{s-1}d} = z^{\frac{p-1}{2}} \equiv -1 \neq 1, \qquad c^{2^s} = z^{p-1} \equiv 1$$

by Fermat (Corollary 7.5); hence $\mathrm{ord}_p(c)$ divides 2^s but not 2^{s-1}, forcing $\mathrm{ord}_p(c) = 2^s$ and $\langle c \rangle = H$.

Initialise

$$x = a^{\frac{d+1}{2}}, \qquad t = a^d, \qquad m = s$$

(the exponent $\frac{d+1}{2}$ is an integer because d is odd). We maintain three invariants:

1. $x^2 = a\,t$;
2. the current c has order exactly 2^m;
3. the order of t divides 2^{m-1}.

Initially (1) holds since $x^2 = a^{d+1} = a \cdot a^d$; (2) was just proved; and (3) holds because $t^{2^{s-1}} = a^{(p-1)/2} \equiv 1$ by Euler's criterion (Theorem 7.13), a being a residue.

Now iterate. If $t = 1$, then $x^2 = a$ by (1) and we are done. Otherwise let $2^i = \mathrm{ord}_p(t)$, found as the least $i \geq 1$ with $t^{2^i} = 1$ by repeatedly squaring t; invariant (3) gives $1 \leq i \leq m-1$. Set

$$b = c^{2^{m-i-1}},$$

and update

$$x \leftarrow xb, \qquad t \leftarrow tb^2, \qquad c \leftarrow b^2, \qquad m \leftarrow i.$$

The invariants survive. For (1), $(xb)^2 = x^2b^2 = a\,t\,b^2$, which is a times the new t. For (2), the old c has order 2^m, so $\mathrm{ord}_p(b) = 2^m / \gcd(2^m, 2^{m-i-1}) = 2^{i+1}$ (Corollary 3.23), and the new $c = b^2$ has order 2^i—exactly 2 to the new m. For (3), both the old t and b^2 have order exactly 2^i, so $t^{2^{i-1}}$ and $(b^2)^{2^{i-1}}$ each have order 2; the only element of order 2 in $\mathbb{F}_p^\times$ is -1, because the square roots of 1 are ± 1 (as in the proof of Theorem 7.13); hence

$$(t\,b^2)^{2^{i-1}} = t^{2^{i-1}} (b^2)^{2^{i-1}} = (-1)(-1) = 1,$$

so the new t has order dividing 2^{i-1}, which is 2^{m-1} for the new m.

Each iteration strictly decreases m, from s through a descending chain of nonnegative integers, so after at most s iterations the order of t reaches $2^0 = 1$, i.e. $t = 1$, and x is the desired root. For the cost: locating i and computing b each take at most $m \leq s$ squarings, so the loop costs $O(s)$ multiplications per iteration and $O(s^2)$ in total; the three initial exponentiations, with exponents smaller than p, cost $O(\log p)$ multiplications each by square-and-multiply (theorem 6.2). □

The one ingredient the theorem presupposes is a quadratic non-residue z, and here—for the only time in this section—randomness enters. A uniformly random element of $\mathbb{F}_p^\times$ is a non-residue with probability exactly $\frac{1}{2}$, since precisely half the elements qualify (Proposition 7.12), and each candidate is checked by one Euler-criterion exponentiation (Definition 7.14), i.e. $O(\log p)$ modular multiplications. A handful of random trials therefore finds z almost immediately, and implementations simply hard-code one non-residue per field, verified once—as the constant $Z = -13$ of Example 7.15 is. No deterministic polynomial-time method for finding a non-residue is invoked anywhere in this volume.

Remark 7.20 (Which root, and detecting failure). In every case the square roots of a residue come in a pair $\pm x$ (Proposition 7.12), so a *canonical* root must be pinned down by a convention—for instance "choose the root whose least nonnegative representative is even", as elliptic-curve point encodings do; the compressed point's extra bit selects between the two. If a is *not* a residue, both algorithms above detect the failure, in different ways. The shortcut of Proposition 7.17 still outputs a candidate $a^{(p+1)/4}$, and that candidate fails the final check $x^2 \equiv a$—one squaring suffices to detect failure. The variable-time Tonelli–Shanks algorithm just given cannot complete its update on a non-residue: Euler's criterion gives $a^{(p-1)/2} \equiv -1$, so $t = a^d$ satisfies $t^{2^{s-1}} \equiv -1$ and has order exactly 2^s, and already the first search for the least $i \geq 1$ with $t^{2^i} = 1$ returns $i = m$, which invariant (3) of the proof of Theorem 7.19 forbids and for which the update exponent 2^{m-i-1} is undefined. This variant can report "not a residue" the moment $i = m$ occurs; equivalently it may verify $a^{(p-1)/2} \equiv 1$ up front, at the same $O(\log p)$ cost as the initial exponentiations.

7.6 The discrete logarithm problem

The second demand of the section's opening—exponentiation easy forward, infeasible backward—now takes centre stage. We phrase it for a generic cyclic group, written multiplicatively, because in Orchard the group in question is the group of points of an elliptic curve, written additively, of large prime order; every statement below transfers by renaming the operation.

Definition 7.21 (Discrete logarithm problem). Let $G = \langle g \rangle$ be a finite cyclic group of order N with generator g. Given $h \in G$, the *discrete logarithm* of h to base g, written $\log_g h$, is the unique residue $x \in \mathbb{Z}/N\mathbb{Z}$ with $g^x = h$. The *discrete logarithm problem* (DLP) is to compute x from (G, g, h).

Existence and uniqueness of $\log_g h$ are the content of the classification of cyclic groups: the map $x \mapsto g^x$ is a group isomorphism $(\mathbb{Z}/N\mathbb{Z}, +) \xrightarrow{\sim} G$ (Theorem 3.43, whose map $\bar{\phi}$ this is), so it has a well-defined inverse, and $\log_g$ is that inverse. The two directions could not differ more in cost. Computing g^x from x takes $O(\log N)$ group operations by square-and-multiply (theorem 6.2); computing x from g^x is the DLP, for which no efficient algorithm is known in well-chosen groups. This asymmetry—a bijection cheap in one direction and expensive in the other—is the one-wayness on which elliptic-curve cryptography rests.

Remark 7.22 (Generic complexity and group choice). Two *generic* algorithms solve the DLP in any group of order N, using nothing but the group operation. *Baby-step giant-step* is deterministic: set $m = \lceil \sqrt{N} \rceil$ and write the unknown as $x = im + j$ with $0 \leq i, j < m$; tabulate the "baby steps" g^j for all j, then walk the "giant steps" $h(g^{-m})^i$ for $i = 0, 1, 2, \ldots$ until one equals a tabulated g^j, at which point $hg^{-im} = g^j$ gives $x = im + j$. Both the table and the walk are $O(\sqrt{N})$, so the cost is $O(\sqrt{N})$ time *and* space. *Pollard's rho* (not developed here) is a randomised alternative with the same $O(\sqrt{N})$ expected running time—on a heuristic analysis, made explicit in section 10—but only $O(1)$ space, and is what attackers would run in practice. In the other direction, Shoup's theorem shows that $\Omega(\sqrt{N})$ group operations are *necessary* for any generic algorithm when N is prime, so for generic attacks the square-root cost is exact. Reaching a λ-bit security level—no attack cheaper than 2^λ operations—therefore requires $N \approx 2^{2\lambda}$. This is why the Pallas and Vesta scalar fields have prime order roughly 255 bits long, targeting roughly 128-bit security (section 10). The qualifier *generic* matters: in $(\mathbb{Z}/p\mathbb{Z})^\times$ sub-exponential index-calculus attacks exist, which exploit the ring structure of the integers behind the group, and this is why elliptic-curve groups—where no such attack is known—permit much smaller parameters than multiplicative groups of comparable security.

The next theorem makes the phrase "well-chosen" precise: if the group order N is composite with small prime factors, the DLP falls apart into small pieces, and the Chinese Remainder Theorem—proved in this section for a purely arithmetic purpose—reassembles the pieces into the secret.

Theorem 7.23 (Pohlig–Hellman reduction)**.** *Let $G = \langle g \rangle$ be cyclic of order $N = \prod_{i=1}^{r} q_i^{e_i}$, the q_i distinct primes. The discrete logarithm in G is computable by solving discrete logarithms in subgroups of prime order q_i—e_i of them for each i—and combining the answers with the Chinese Remainder Theorem, at a total cost of*

$$O\Big(\sum_{i=1}^{r} e_i(\log N + \sqrt{q_i}\,)\Big)$$

group operations. When N has a large prime factor $q_{\max}$ the $\sqrt{q_{\max}}$ term dominates; when N is smooth*—a product of primes bounded polynomially in $\log N$—every term is polynomial in $\log N$ and the DLP is easy. Corollary 7.24 draws the design consequence.*

Proof. Let $h = g^x$ with x the unknown. The reduction has two stages.

Stage (a): splitting across coprime factors. For each i set $N_i = N/q_i^{e_i}$ and project:

$$g_i = g^{N_i}, \qquad h_i = h^{N_i}.$$

By Corollary 3.23, $\operatorname{ord}(g_i) = N/\gcd(N, N_i) = N/N_i = q_i^{e_i}$, so g_i generates the unique subgroup of G of order $q_i^{e_i}$ (Theorem 3.24(3)). Moreover $h_i = g^{xN_i} = g_i^x$, and since g_i has order $q_i^{e_i}$, the exponent matters only modulo $q_i^{e_i}$ (Proposition 3.12(2)): writing $x_i = x \bmod q_i^{e_i}$,

$$\log_{g_i} h_i = x_i.$$

Solving the r smaller DLPs yields $x \bmod q_i^{e_i}$ for every i; since the prime powers $q_i^{e_i}$ are pairwise coprime, the Chinese Remainder Theorem (Theorem 7.2, iterated across the factors as in the proof of Corollary 7.3) determines x modulo N uniquely. Each projection costs $O(\log N)$ operations by square-and-multiply.

Stage (b): lifting a prime power digit by digit. Fix one prime power q^e (dropping the subscript i), with $\gamma = g_i$ of order q^e and $\eta = h_i = \gamma^{x'}$, $x' = x_i$. Write x' in base q:

$$x' \equiv x_0 + x_1 q + \cdots + x_{e-1} q^{e-1} \pmod{q^e}, \qquad 0 \le x_j < q.$$

The element $\delta = \gamma^{q^{e-1}}$ has order q (Corollary 3.23 again). Raise η to the power q^{e-1}: every digit beyond the first is multiplied by a multiple of q^e in the exponent and vanishes, leaving

$$\eta^{q^{e-1}} = \gamma^{x' q^{e-1}} = \delta^{x_0},$$

a single DLP in the group $\langle \delta \rangle$ of prime order q, which baby-step giant-step (Remark 7.22) solves in $O(\sqrt{q})$ operations. With $x_0, \dots, x_{j-1}$ in hand, strip them off and repeat: the element $\eta_j = \eta\,\gamma^{-(x_0 + x_1 q + \cdots + x_{j-1} q^{j-1})}$ equals γ raised to $x_j q^j$ + (multiples of q^{j+1}), so

$$\eta_j^{q^{e-1-j}} = \delta^{x_j},$$

and digit x_j is again one order-q DLP. Each of the e digits costs one such DLP plus $O(\log N)$ operations for the exponentiations.

Summing over the digits of all the prime-power factors gives $\sum_i e_i$ order-q_i DLPs at $O(\sqrt{q_i})$ apiece plus $O(\log N)$ bookkeeping for each, which is the stated bound. □

Corollary 7.24 (Why prime order matters). *If N has a small prime factor q, an attacker can recover the discrete logarithm modulo q in $O(\log N + \sqrt{q})$ group operations—$O(\log N)$ to project into the order-q subgroup and $O(\sqrt{q})$ for the small DLP there—leaking part of the secret x. Security therefore requires N to have a large prime factor; the safest and simplest choice is to make N itself a large prime, so that G has no proper nontrivial subgroups and the Pohlig–Hellman reduction offers no purchase at all. This is precisely why the elliptic-curve groups used in Halo 2 and Orchard have large prime order.*

Proof. For the first claim, run stage (a) of Theorem 7.23 for the single factor q: the projections $g^{N/q}$, $h^{N/q}$ cost $O(\log N)$ operations, and the resulting DLP in a group of order q costs $O(\sqrt{q})$ by baby-step giant-step—yielding x mod q. For the second, if N is prime then G has no subgroups besides $\{e\}$ and G (Corollary 3.34), so the reduction's only output is the original problem, and the generic $\Omega(\sqrt{N})$ floor of Remark 7.22 stands at full height. □

Remark 7.25 (Cofactors and small-subgroup attacks). In practice an elliptic curve over $\mathbb{F}_p$ has group order $h \cdot q$, where q is a large prime and h is a small *cofactor*, and cryptographic operations stay confined to the subgroup of prime order q. If a protocol neglects to check that incoming points lie in that subgroup (or fails to clear the cofactor), an adversary can supply a point of low order and run Pohlig–Hellman in the small factor to extract information about the secret modulo h—a real-world *small-subgroup attack*. The Pasta curves of Halo 2 have cofactor $h = 1$: their group orders are themselves prime (the primes recorded in section 10), which eliminates this class of attacks at the level of the group itself rather than by protocol-level checks.

The two demands with which the section opened are now both met. A receiver handed a compressed point (x, bit) decides in one modular exponentiation whether $x^3 + ax + b$ is a square—Euler's criterion—and, when it is, extracts a root by Proposition 7.17 or by Tonelli–Shanks (Theorem 7.19), the transmitted bit selecting between the pair $\pm y$ (Remark 7.20); for the Pasta fields, with their deliberately large 2-adic valuation $s = 32$, Tonelli–Shanks is the deployed route. On the security side, exponentiation runs in $O(\log N)$ operations while its inverse, the discrete logarithm, costs any generic attacker $\Omega(\sqrt{N})$—a floor that composite group orders would undermine through Pohlig–Hellman, and that the prime, roughly 255-bit orders of the Pasta groups preserve in full, with no small subgroup left to attack. The structure of $\mathbb{Z}/q\mathbb{Z}$ for a large prime q thus supplies both the functionality—a field of scalars acting on the group—and the security of the constructions built on it. The curves themselves, and the primes p and q that this section has treated as black boxes, are the business of section 10.

8 Linear algebra over a field

Begin with the engineering problem that shapes the whole section. A prover holds a list of n secret numbers—the intermediate values of some large computation, the *wire values* of an *arithmetic circuit* in the vocabulary of later volumes, with n in the millions, say—and must hand a verifier a *receipt* for the list: a single short object that pins the list down, to be opened or argued about later. Two demands make the problem interesting. The receipt must be *one* element of some fixed algebraic structure, however large n grows; and receipts must *add*—whoever holds the receipts for two lists, together with two scaling factors, must be able to compute the receipt for the scaled sum of the lists without ever seeing the lists themselves. The construction that meets both demands is disarmingly brief: fix n public elements $G_1, \dots, G_n$ of a suitable group, and issue for the list $(a_1, \dots, a_n)$ the single group element $[a_1]G_1 + \cdots + [a_n]G_n$. Whether this can possibly work—what "receipts add" should mean exactly, how one group element can absorb n secrets, and what price in ambiguity the compression exacts—are questions of *linear algebra*, and answering them is the business of this section.

Linear algebra is the study of *vector spaces*—collections of objects that can be added together and scaled by elements of a field—together with the structure-preserving maps between them. It

is the computational backbone of the cryptography this monograph serves: commitments to vectors, the linear constraints that encode arithmetic circuits, the polynomial-evaluation arguments at the heart of the inner-product argument, and the Pedersen-style multiscalar multiplications of Halo 2 and Orchard are all, at bottom, statements about linear maps and (bi)linear forms over a finite field. The section develops the theory from the ground up, assuming only the notion of a field (section 4), and then specialises to the two settings that matter here: the coordinate space F^n, in practice taken over a finite field $\mathbb{F}_q$, and the "mixed" pairing between vectors of scalars and vectors of group elements.

The section also settles two debts contracted below it. Remark 5.5 promised that the linear structure of $F[X]$—closure under addition and scaling, with the monomials as an infinite coordinate system—would be studied systematically in due course; the study happens here. And the proof of Theorem 6.9 borrowed, against a promissory note, exactly two facts about vector spaces: that a space with a finite spanning set has a basis, and that coordinates with respect to a basis are unique. The first is proved in Theorem 8.17; the second follows from linear independence, as recorded in Definition 8.14. These arguments use nothing from the theory of finite fields, so the note is honoured without circularity. Throughout, F denotes an arbitrary field with identities 0 and 1; the field axioms of Definition 4.24 are used freely and without citation.

8.1 Vector spaces and subspaces

Definition 8.1 (Vector space)**.** Let F be a field. A *vector space over F* (an *F-vector space*) is a set V equipped with two operations, *vector addition* $+\colon V \times V \to V$ and *scalar multiplication* $\cdot\colon F \times V \to V$, such that for all $u, v, w \in V$ and all $a, b \in F$:

1. $(V, +)$ is an abelian group: addition is associative and commutative, there is a *zero vector* $\mathbf{0} \in V$ with $v + \mathbf{0} = v$, and every v has an *additive inverse* $-v$ with $v + (-v) = \mathbf{0}$;
2. scalar multiplication is compatible with the multiplication of F: $a \cdot (b \cdot v) = (ab) \cdot v$;
3. the identity of F acts as the identity: $1 \cdot v = v$;
4. scalar multiplication distributes over vector addition: $a \cdot (u + v) = a \cdot u + a \cdot v$;
5. scalar multiplication distributes over scalar addition: $(a + b) \cdot v = a \cdot v + b \cdot v$.

The elements of V are called *vectors*, the elements of F *scalars*. We write av for $a \cdot v$, and we allow the scalar $0 \in F$ and the vector $\mathbf{0} \in V$ to share a symbol whenever context distinguishes them.

A few consequences of the axioms recur so often that we record them once and, thereafter, use them without comment.

Proposition 8.2. *Let V be an F-vector space, $v \in V$, and $a \in F$. Then:*

1. $0 \cdot v = \mathbf{0}$;
2. $a \cdot \mathbf{0} = \mathbf{0}$;
3. $(-1) \cdot v = -v$;
4. *if $av = \mathbf{0}$, then $a = 0$ or $v = \mathbf{0}$.*

Proof. (1) By distributivity over scalar addition, $0 \cdot v = (0 + 0) \cdot v = 0 \cdot v + 0 \cdot v$; adding $-(0 \cdot v)$ to both sides leaves $\mathbf{0} = 0 \cdot v$.

(2) Likewise $a \cdot \mathbf{0} = a \cdot (\mathbf{0} + \mathbf{0}) = a \cdot \mathbf{0} + a \cdot \mathbf{0}$, and cancelling in the abelian group $(V, +)$ gives $a \cdot \mathbf{0} = \mathbf{0}$.

(3) Using the identity axiom, distributivity, and (1),

$$v + (-1) \cdot v = 1 \cdot v + (-1) \cdot v = (1 + (-1)) \cdot v = 0 \cdot v = \mathbf{0},$$

so $(-1) \cdot v$ is the additive inverse of v; by uniqueness of inverses in a group, $(-1) \cdot v = -v$.

(4) Suppose $av = \mathbf{0}$ and $a \neq 0$. Since F is a field, a^{-1} exists, and by the identity and compatibility axioms

$$v = 1 \cdot v = (a^{-1}a) \cdot v = a^{-1} \cdot (av) = a^{-1} \cdot \mathbf{0} = \mathbf{0},$$

the last step by (2). □

Example 8.3 (The coordinate space F^n)**.** The single most important example is the *coordinate space*

$$F^n = \{ (x_1, \dots, x_n) : x_i \in F \},$$

the set of n-tuples of scalars, under componentwise operations:

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n), \qquad a \cdot (x_1, \dots, x_n) = (ax_1, \dots, ax_n).$$

The zero vector is $(0, \dots, 0)$, and every axiom of theorem 8.1 follows from the corresponding field axiom applied in each coordinate. The degenerate case $n = 0$ is the *trivial* (or *zero*) vector space $\{\mathbf{0}\}$, consisting of the empty tuple alone. As a standing convention we write vectors of F^n in boldface and think of them as columns, $\mathbf{x} = (x_1, \dots, x_n)^\mathsf{T}$.

Example 8.4 (Further examples)**.**

1. *Polynomials.* The polynomials $F[X]$ form an F-vector space under addition and the scalar multiplication of Remark 5.5. So does the subset

$$F[X]_{<n} = \{ f \in F[X] : f = 0 \text{ or } \deg f < n \}$$

of polynomials of degree less than n: sums and scalar multiples of such polynomials again have degree less than n (or are zero), by Proposition 5.7, so the operations restrict and the axioms are inherited. This space is exactly the setting in which polynomial commitment schemes operate.

2. *Functions.* For any set S, the set F^S of all functions $S \to F$ is an F-vector space under pointwise operations: $(f + g)(s) = f(s) + g(s)$ and $(af)(s) = af(s)$. Example 8.3 is the special case $S = \{1, \dots, n\}$, a tuple being a function of its index.

3. *Matrices.* The set $F^{m \times n}$ of $m \times n$ arrays of scalars is an F-vector space under entrywise addition and scaling—the case of (2) in which S is the grid of index pairs (i, j).

4. *Field extensions.* If F is a subfield of a larger field K (§4), then K is an F-vector space: vector addition is the addition of K, and scalar multiplication is the multiplication of K restricted to pairs in $F \times K$, so the axioms are instances of the field axioms of K. The dimension of this vector space—in the sense of theorem 8.17 below—is called the *degree* of the extension, written $[K : F]$. For instance $\mathbb{C}$ is an $\mathbb{R}$-vector space of dimension 2, with basis $1, i$, so $[\mathbb{C} : \mathbb{R}] = 2$; and the finite field $\mathbb{F}_q$ with $q = p^k$ is a k-dimensional $\mathbb{F}_p$-vector space, $[\mathbb{F}_q : \mathbb{F}_p] = k$: this is precisely the structure on which the proof of Theorem 6.9 rested.

A vector space often contains smaller vector spaces sitting inside it, sharing its operations; the solution sets of homogeneous linear equations, the spans of the next subsection, and the kernels and images of linear maps are all of this kind.

Definition 8.5 (Subspace)**.** A subset $W \subseteq V$ of an F-vector space V is a (*linear*) *subspace* if W is itself a vector space under the operations inherited from V; equivalently—as the next proposition makes precise—if and only if W is nonempty and closed under the vector-space operations.

Proposition 8.6 (Subspace criterion)**.** *A subset $W \subseteq V$ is a subspace if and only if*

1. $\mathbf{0} \in W$;
2. *W is closed under addition: $u, v \in W$ implies $u + v \in W$; and*
3. *W is closed under scalar multiplication: $a \in F$ and $v \in W$ imply $av \in W$.*

Equivalently, W is a subspace if and only if $W \neq \emptyset$ and $au + bv \in W$ for all $a, b \in F$ and $u, v \in W$.

Proof. If W is a subspace, then (1)–(3) hold: closure (2) and (3) is what it means for the inherited operations to be operations *on* W, and the zero vector of W is the zero vector of V, since a zero vector $\mathbf{0}_W$ of W satisfies $\mathbf{0}_W + \mathbf{0}_W = \mathbf{0}_W$, and cancelling in V gives $\mathbf{0}_W = \mathbf{0}$.

Conversely, suppose (1)–(3) hold. Closure makes the two operations well defined on W. The additive inverse of any $v \in W$ is $(-1)v \in W$, by (3) and Proposition 8.2(3). Every remaining axiom of theorem 8.1 is a universally quantified identity—associativity, commutativity, compatibility, the two distributive laws, the action of 1—and each holds for all elements of V, in particular for all elements of W. Hence W is a vector space.

For the two-scalar form: if W is a subspace, then $au + bv \in W$ by (2) and (3) combined, and $W \ni \mathbf{0}$ is nonempty. Conversely, choosing $(a, b) = (1, 1)$ recovers (2), choosing $(a, b) = (a, 0)$ recovers (3) (using Proposition 8.2(1) to identify $0v$), and $\mathbf{0} = 0v \in W$ for any v in the nonempty W, which is (1). □

Example 8.7. In F^3 the set $W = \{ (x_1, x_2, x_3) : x_1 + x_2 + x_3 = 0 \}$ is a subspace: it contains $\mathbf{0}$, and if the coordinates of $\mathbf{x}$ and $\mathbf{y}$ each sum to 0, then those of $a\mathbf{x} + b\mathbf{y}$ sum to $a \cdot 0 + b \cdot 0 = 0$. By contrast $\{ (x_1, x_2, x_3) : x_1 + x_2 + x_3 = 1 \}$ is *not* a subspace: it does not contain $\mathbf{0}$. The same computation shows that the solution set of any single homogeneous linear equation $c_1x_1 + \cdots + c_nx_n = 0$ (the c_i fixed scalars) is a subspace of F^n, and the solution set of a finite *system* of such equations, being an intersection of subspaces, is a subspace by the next proposition. These solution sets return shortly in structural guise, as the kernels of linear maps (theorem 8.18).

Proposition 8.8 (Intersections and sums)**.** *Let V be an F-vector space.*

1. *The intersection $\bigcap_{i \in I} W_i$ of an arbitrary family of subspaces of V is a subspace.*
2. *The* sum $W_1 + \cdots + W_k = \{ w_1 + \cdots + w_k : w_j \in W_j \}$ *of finitely many subspaces is a subspace, and it is the smallest subspace of V containing every W_j.*

The union *of two subspaces is in general* not *a subspace.*

Proof. (1) Each W_i contains $\mathbf{0}$, so the intersection does. If u, v lie in every W_i and $a, b \in F$, then $au + bv \in W_i$ for every i by Proposition 8.6, hence $au + bv$ lies in the intersection, which is therefore a subspace by the same criterion.

(2) The sum contains $\mathbf{0} = \mathbf{0} + \cdots + \mathbf{0}$, and it contains each W_j (take every other summand $\mathbf{0}$). For closure, take two elements $\sum_j w_j$ and $\sum_j w'_j$ with $w_j, w'_j \in W_j$, and scalars a, b; regrouping,

$$a \sum_j w_j + b \sum_j w'_j = \sum_j (aw_j + bw'_j),$$

and $aw_j + bw'_j \in W_j$ since W_j is a subspace, so the combination lies in the sum. Finally, any subspace U containing every W_j contains each summand w_j of any element of the sum, hence contains the element itself by closure under addition; so the sum is contained in U, and is the smallest such subspace.

For the union, take $V = F^2$ (any field has $|F| > 1$, since $1 \neq 0$) and the two axes $\{(x, 0) : x \in F\}$ and $\{(0, y) : y \in F\}$—subspaces by the criterion. Their union contains $(1, 0)$ and $(0, 1)$ but not the sum $(1, 1) = (1, 0) + (0, 1)$, since $(1, 1)$ lies on neither axis; closure under addition fails. □

8.2 Linear combinations, span, and linear independence

The subspaces met so far were carved out by equations. The complementary way to produce a subspace is to build one up from chosen vectors, taking everything their repeated addition and scaling can reach.

Definition 8.9 (Linear combination and span)**.** Let $S \subseteq V$. A *linear combination* of vectors in S is any finite sum

$$a_1 v_1 + a_2 v_2 + \cdots + a_k v_k, \qquad a_i \in F,\ v_i \in S.$$

The *span* of S, written $\operatorname{span}(S)$ or $\langle S \rangle$, is the set of all linear combinations of finitely many vectors of S; by convention $\operatorname{span}(\emptyset) = \{\mathbf{0}\}$. The set S *spans* (or *generates*) V if $\operatorname{span}(S) = V$, and V is *finitely generated* (synonymously, *finite-dimensional*) if some finite set spans it.

Proposition 8.10. *For any $S \subseteq V$, the span $\operatorname{span}(S)$ is the smallest subspace of V containing S: it is a subspace, it contains S, and it is contained in every subspace of V that contains S.*

Proof. The zero vector is the empty linear combination (or, for nonempty S, the combination $0v$), and a sum of two linear combinations of vectors in S, or a scalar multiple of one, is again such a linear combination—concatenate the lists and rescale the coefficients. By Proposition 8.6, $\operatorname{span}(S)$ is a subspace, and it contains each $v \in S$ as the combination $1 \cdot v$. If W is any subspace containing S, then closure under addition and scaling forces every linear combination of elements of S into W, by induction on the number of summands; hence $\operatorname{span}(S) \subseteq W$. □

Definition 8.11 (Linear independence)**.** A finite list of vectors $v_1, \ldots, v_k \in V$ is *linearly independent* if the only way to express the zero vector as a linear combination of the list is with all coefficients zero:

$$a_1 v_1 + \cdots + a_k v_k = \mathbf{0} \quad \Longrightarrow \quad a_1 = \cdots = a_k = 0.$$

Otherwise the list is *linearly dependent*, and a relation $\sum_i a_i v_i = \mathbf{0}$ with some $a_i \neq 0$ is called a *dependence relation*. An arbitrary—possibly infinite—set $S \subseteq V$ is linearly independent if every finite list of distinct vectors from S is linearly independent.

Remark 8.12. Linear independence captures *non-redundancy*: no vector of an independent list contributes anything the others already provide. In the smallest cases the definition unwinds to familiar statements. A single vector v is independent if and only if $v \neq \mathbf{0}$ (Proposition 8.2(4)). Two vectors are dependent if and only if one is a scalar multiple of the other. In general, a list is dependent exactly when some vector in it lies in the span of the others—the content of the next lemma.

Lemma 8.13 (Dependence and redundancy)**.** *A list $v_1, \dots, v_k$ with $k \geq 1$ is linearly dependent if and only if some v_j is a linear combination of the others. Moreover, if $v_1, \dots, v_k$ is independent but $v_1, \dots, v_k, w$ is dependent, then $w \in \operatorname{span}(v_1, \dots, v_k)$, and the coefficients expressing w are unique.*

Proof. Suppose first that $v_j = \sum_{i \neq j} a_i v_i$. Then

$$\sum_{i \neq j} a_i v_i + (-1)\, v_j = \mathbf{0}$$

is a dependence relation, its coefficient -1 on v_j being nonzero (as $1 \neq 0$ in a field). Conversely, given a dependence relation $\sum_i a_i v_i = \mathbf{0}$ with $a_j \neq 0$, the field supplies a_j^{-1}, and solving for v_j gives $v_j = -a_j^{-1} \sum_{i \neq j} a_i v_i$, a linear combination of the others.

For the second statement, take a dependence relation $\sum_i a_i v_i + bw = \mathbf{0}$ for the extended list. The coefficient b cannot be 0: otherwise the relation would be a nontrivial relation among $v_1, \dots, v_k$ alone, contradicting their independence. Hence $w = -b^{-1} \sum_i a_i v_i \in \operatorname{span}(v_1, \dots, v_k)$. For uniqueness, suppose $w = \sum_i c_i v_i = \sum_i c_i' v_i$; subtracting, $\sum_i (c_i - c_i') v_i = \mathbf{0}$, and independence forces $c_i = c_i'$ for every i. □

8.3 Bases, dimension, and linear maps

Spanning says a list reaches everything; independence says it carries no dead weight. A list with both properties is a coordinate system.

Definition 8.14 (Basis)**.** A *basis* of V is a linearly independent set B that spans V. Every $v \in V$ is then a *unique finite* linear combination of elements of B: existence because B spans, uniqueness because two expressions subtract to a relation among finitely many distinct elements of the independent set (as in Lemma 8.13). For a finite basis $B = (b_1, \dots, b_n)$, write

$$v = c_1 b_1 + \cdots + c_n b_n, \qquad [v]_B = (c_1, \dots, c_n),$$

and call the scalars c_i the *coordinates* of v in B.

Example 8.15 (The two bases in constant use)**.** The *standard basis* of F^n is $e_1, \dots, e_n$, where e_i has a 1 in position i and 0 elsewhere. Both properties reduce to reading off coordinates: the combination $\sum_i c_i e_i$ is the tuple $(c_1, \dots, c_n)$, so every vector is such a combination (spanning) and only the zero combination gives $\mathbf{0}$ (independence). The *monomial basis* of $F[X]_{<n}$ is $1, X, \dots, X^{n-1}$: a polynomial of degree less than n is, by construction, a linear combination of these monomials, and independence is the fact that polynomials are equal exactly when their coefficient sequences agree (Definition 5.1).

That all bases of a given space have the same length is not obvious; it rests on the following exchange principle, which lets an independent list consume a spanning list one slot at a time.

Lemma 8.16 (Steinitz exchange)**.** *If $v_1, \dots, v_m$ are linearly independent in V and $w_1, \dots, w_n$ span V, then $m \leq n$.*

Proof. We prove by induction on $k = 0, 1, \dots, m$ that, after a suitable renumbering of the w_j, the hybrid list

$$v_1, \dots, v_k, w_{k+1}, \dots, w_n$$

spans V. The case $k = 0$ is the hypothesis on the w_j.

For the induction step, suppose the hybrid list at stage $k < m$ spans V. In particular it expresses v_{k+1}:

$$v_{k+1} = \sum_{i \le k} a_i v_i + \sum_{j > k} c_j w_j.$$

Some coefficient c_j with $j > k$ must be nonzero: were they all zero, v_{k+1} would be a linear combination of $v_1, \dots, v_k$, and Lemma 8.13 would make the list $v_1, \dots, v_{k+1}$ dependent, contradicting the independence of the v_i. (This step silently requires a slot $j > k$ to exist, i.e. $k < n$; if instead $k = n$, the hybrid list is $v_1, \dots, v_n$ and the displayed equation already yields the same contradiction.) Renumber so that $c_{k+1} \neq 0$, and solve for the displaced vector:

$$w_{k+1} = c_{k+1}^{-1}\Big(v_{k+1} - \sum_{i \le k} a_i v_i - \sum_{j > k+1} c_j w_j\Big).$$

Every vector of the stage-k list therefore lies in the span of the stage-$(k{+}1)$ list $v_1, \dots, v_{k+1}$, $w_{k+2}, \dots, w_n$ —the v_i and the remaining w_j trivially, w_{k+1} by the display—so by Proposition 8.10 the stage-$(k{+}1)$ list spans V, completing the induction.

Each step trades one w for one v, so the process can run to $k = m$ only if there are at least m slots to trade: $m \le n$. □

Theorem 8.17 (Dimension)**.** *Let V be a finitely generated F-vector space. Then V has a basis, and any two bases of V have the same number of elements. This common number is the* dimension $\dim_F V$ *(written* $\dim V$ *when the field is clear). In particular* $\dim_F F^n = n$ *and* $\dim_F F[X]_{<n} = n$.

Proof. Existence. Let $w_1, \dots, w_n$ be a finite spanning list. While the list is dependent, Lemma 8.13 exhibits some w_j in the span of the others; discarding it leaves the span unchanged, since any combination using w_j can be rewritten without it (substitute and regroup, Proposition 8.10). Each discard shortens the list, so after at most n discards the process halts at an independent spanning list—a basis. (If V is the zero space, the process discards everything and halts at the empty list, a basis of $\{\mathbf{0}\}$ with $\operatorname{span}(\emptyset) = \{\mathbf{0}\}$; the dimension is 0.)

Invariance. Every basis is finite: if a basis contained more than the n elements of a finite spanning list, any $n + 1$ of its elements would contradict Lemma 8.16. Let $b_1, \dots, b_m$ and $b'_1, \dots, b'_n$ be bases. Applying Lemma 8.16 with the first list as the independent one and the second as the spanning one gives $m \le n$; exchanging the roles gives $n \le m$. Hence $m = n$.

The two named spaces have the explicit bases of Example 8.15, each of length n. □

With coordinates in hand, the maps that respect the linear structure come next; they are the section's real subject, for the Pedersen-style commitments developed in the series are such maps.

Definition 8.18 (Linear map, kernel, image)**.** A function $T\colon V \to W$ between F-vector spaces is *linear* (an *F-linear map*) if

$$T(au + bv) = a\,T(u) + b\,T(v) \qquad \text{for all } a, b \in F \text{ and } u, v \in V.$$

Its *kernel* and *image* are

$$\ker T = \{\, v \in V : T(v) = \mathbf{0} \,\} \subseteq V, \qquad \operatorname{im} T = \{\, T(v) : v \in V \,\} \subseteq W.$$

A bijective linear map is an *isomorphism*.

Three basic facts follow at once from the subspace criterion and linearity, and we verify them here once. First, taking $a = b = 0$ gives $T(\mathbf{0}) = \mathbf{0}$, so $\mathbf{0} \in \ker T$ and $\mathbf{0} \in \operatorname{im} T$. Second, both kernel and image are subspaces: if $u, v \in \ker T$ then $T(au + bv) = a\,T(u) + b\,T(v) = \mathbf{0}$, and if

$T(u), T(v) \in \operatorname{im} T$ then $a\,T(u) + b\,T(v) = T(au + bv) \in \operatorname{im} T$; Proposition 8.6 applies to each. Third, T is injective if and only if $\ker T = \{\mathbf{0}\}$: if $T(u) = T(v)$, then $T(u - v) = \mathbf{0}$ by linearity, so a trivial kernel forces $u = v$; conversely an injective T sends only $\mathbf{0}$ to $T(\mathbf{0}) = \mathbf{0}$. We also note that for fixed V and W the set $\operatorname{Hom}(V, W)$ of all linear maps $V \to W$ is itself an F-vector space under the pointwise operations of Example 8.4(2)—a sum or scalar multiple of linear maps is linear, so $\operatorname{Hom}(V, W)$ is a subspace of the space of all functions $V \to W$.

Theorem 8.19 (Rank–nullity)**.** *Let $T\colon V \to W$ be a linear map with V finite-dimensional. Then*

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

Proof. The kernel is a subspace of the finite-dimensional V, and is itself finite-dimensional: an independent list in $\ker T$ has length at most $\dim V$ by Lemma 8.16 (tested against a basis of V, which spans), so a longest independent list $u_1, \dots, u_k$ in $\ker T$ exists, and it spans $\ker T$ —any $u \in \ker T$ outside its span would extend it to a longer independent list by Lemma 8.13. Thus $u_1, \dots, u_k$ is a basis of $\ker T$.

Extend it to a basis of V: while the current independent list fails to span V, adjoin any vector outside its span—independence is preserved by Lemma 8.13—and the process terminates, again because independent lists cannot exceed $\dim V$ in length (Lemma 8.16 against any basis of V). Write the resulting basis as $u_1, \dots, u_k, v_1, \dots, v_r$, so that $\dim V = k + r$.

We claim $T(v_1), \dots, T(v_r)$ is a basis of $\operatorname{im} T$; the theorem follows, since then $\dim \operatorname{im} T = r$ and $\dim V = k + r = \dim \ker T + \dim \operatorname{im} T$.

Spanning. Any element of $\operatorname{im} T$ is $T(v)$ for some $v = \sum_i a_i u_i + \sum_j c_j v_j$, and linearity with $T(u_i) = \mathbf{0}$ gives $T(v) = \sum_j c_j\, T(v_j)$.

Independence. Suppose $\sum_j c_j\, T(v_j) = \mathbf{0}$. By linearity $T(\sum_j c_j v_j) = \mathbf{0}$, so $\sum_j c_j v_j \in \ker T$, and expressing this kernel element in the kernel basis gives $\sum_j c_j v_j = \sum_i a_i u_i$ for some scalars a_i. Rearranged, $\sum_i a_i u_i - \sum_j c_j v_j = \mathbf{0}$ is a linear relation among the combined basis of V, whose independence forces every coefficient—in particular every c_j—to vanish. □

Rank–nullity is the dimension-counting engine of the section. It shows at once that a linear map from a higher-dimensional space into a lower-dimensional one must have a nontrivial kernel—the image cannot have dimension exceeding the target's, so the kernel absorbs the difference—and distinct inputs must therefore collide. That forced collision is exactly what makes vector commitments compressing and only *computationally* binding (Remark 8.29).

8.4 Bilinear forms and the inner product on F^n

The maps considered so far are linear in a single argument. Commitment and proof systems are pervaded by expressions linear in *each* of two arguments—above all the inner product between two vectors. Over $\mathbb{R}$ such products come wrapped in geometry: lengths, angles, positivity. Over the finite fields of cryptographic interest there is no ordering, hence no "positive" and no length; what survives, and what the applications actually use, is the bare algebra. The appropriate level of generality is therefore the *bilinear form*.

Definition 8.20 (Bilinear form)**.** A *bilinear form* on an F-vector space V is a function $\beta\colon V \times V \to F$ that is linear in each argument separately:

$$\beta(au + a'u',\, v) = a\,\beta(u, v) + a'\,\beta(u', v), \qquad \beta(u,\, bv + b'v') = b\,\beta(u, v) + b'\,\beta(u, v'),$$

for all $u, u', v, v' \in V$ and $a, a', b, b' \in F$. The form is *symmetric* if $\beta(u, v) = \beta(v, u)$ for all u, v, and *nondegenerate* if $\beta(u, v) = 0$ for all v implies $u = \mathbf{0}$.

Relative to a fixed basis $b_1, \dots, b_n$ of V, a bilinear form is captured by finitely many scalars: expanding both arguments in the basis and applying bilinearity termwise, $\beta(u,v) = \sum_{i,j} \beta(b_i, b_j)\,[u]_i[v]_j$, so β is determined by—and represented by—the unique $n \times n$ array M with entries $M_{ij} = \beta(b_i, b_j)$, via $\beta(u,v) = [u]_B^{\mathsf{T}} M\,[v]_B$ in matrix shorthand. We record the representation for orientation but make no further use of it; only the special case with M the identity array matters below.

Definition 8.21 (Standard inner product on F^n)**.** The *standard inner product* (or *dot product*) on F^n is

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=1}^{n} a_i b_i = \mathbf{a}^{\mathsf{T}}\mathbf{b}, \qquad \mathbf{a} = (a_1, \dots, a_n),\ \mathbf{b} = (b_1, \dots, b_n).$$

Bilinearity holds because each term $a_i b_i$ is linear in each factor, and symmetry because multiplication in F commutes; the representing array in the standard basis is the identity ($\langle e_i, e_j \rangle$ is 1 if $i = j$ and 0 otherwise). The form is nondegenerate: if $\langle \mathbf{a}, \mathbf{b} \rangle = 0$ for every $\mathbf{b}$, then in particular $a_i = \langle \mathbf{a}, e_i \rangle = 0$ for each i, so $\mathbf{a} = \mathbf{0}$.

Remark 8.22 (No positivity over finite fields). Over $\mathbb{R}$ the standard inner product is *positive definite*: $\langle \mathbf{a}, \mathbf{a} \rangle = \sum_i a_i^2 \geq 0$, with equality only at $\mathbf{0}$. Positivity is what yields the Euclidean norm $\|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle}$ and the Cauchy–Schwarz inequality $|\langle \mathbf{a}, \mathbf{b} \rangle| \leq \|\mathbf{a}\|\,\|\mathbf{b}\|$—the geometric superstructure of real inner products. Over a finite field $\mathbb{F}_q$ there is no ordering compatible with the arithmetic, so "≥ 0" is meaningless, and the superstructure collapses: there exist nonzero *isotropic* vectors $\mathbf{a} \neq \mathbf{0}$ with $\langle \mathbf{a}, \mathbf{a} \rangle = 0$. Over $\mathbb{F}_5$, for instance, the vector $\mathbf{a} = (1, 2)$ has

$$\langle \mathbf{a}, \mathbf{a} \rangle = 1^2 + 2^2 = 1 + 4 = 5 = 0$$

in $\mathbb{F}_5$. What survives the collapse are the algebraically robust properties: bilinearity, symmetry, and nondegeneracy. These are all that the arguments in proof systems rely on, which is why the loss of positivity costs the applications nothing.

Nondegeneracy is not a consolation prize; it powers a dictionary between vectors and the scalar-valued linear maps on them.

Proposition 8.23 (Functionals via the inner product)**.** *For each $\mathbf{a} \in F^n$, the map $\mathbf{b} \mapsto \langle \mathbf{a}, \mathbf{b} \rangle$ is a linear map $F^n \to F$ (a* linear functional*). The assignment*

$$F^n \longrightarrow \mathrm{Hom}(F^n, F), \qquad \mathbf{a} \longmapsto \langle \mathbf{a}, \cdot \rangle,$$

is an isomorphism of F-vector spaces onto the dual space $\mathrm{Hom}(F^n, F)$ *of all linear functionals on F^n. In particular both spaces have dimension n.*

Proof. Each $\langle \mathbf{a}, \cdot \rangle$ is linear in $\mathbf{b}$ by bilinearity, and the assignment itself is linear in $\mathbf{a}$ for the same reason, mapping into the vector space $\mathrm{Hom}(F^n, F)$ of theorem 8.18. It is injective: if $\langle \mathbf{a}, \cdot \rangle$ is the zero functional, then $\langle \mathbf{a}, \mathbf{b} \rangle = 0$ for every $\mathbf{b}$, and nondegeneracy (theorem 8.21) gives $\mathbf{a} = \mathbf{0}$, so the kernel is trivial. It is surjective: given a linear functional φ, set $a_i = \varphi(e_i)$ and $\mathbf{a} = (a_1, \dots, a_n)$; then for every $\mathbf{b} = \sum_i b_i e_i$, linearity of φ gives

$$\varphi(\mathbf{b}) = \sum_i b_i\,\varphi(e_i) = \sum_i a_i b_i = \langle \mathbf{a}, \mathbf{b} \rangle,$$

so $\varphi = \langle \mathbf{a}, \cdot \rangle$. A bijective linear map is an isomorphism, and $\dim \mathrm{Hom}(F^n, F) = \dim F^n = n$ follows since an isomorphism carries a basis to a basis (its inverse is linear, and both directions preserve spanning and independence). $\square$

Remark 8.24 (The dictionary in use). Proposition 8.23 says that "a linear functional on F^n" and "a vector in F^n" are interchangeable: every linear way of producing one scalar from a vector *is* an inner product against a fixed vector, and conversely. The instance to keep in mind is polynomial evaluation. Fix a point $z \in F$; the map sending a polynomial to its value at z is linear in the coefficients, so it must be an inner product against some fixed vector—and indeed, for $p \in F[X]_{<n}$ with coefficient vector $\mathbf{c} = (c_0, c_1, \dots, c_{n-1})$,

$$p(z) = c_0 + c_1 z + \cdots + c_{n-1} z^{n-1} = \langle \mathbf{c}, (1, z, z^2, \dots, z^{n-1}) \rangle.$$

Evaluating a *committed* polynomial at a challenge point is therefore an inner-product claim about the committed coefficient vector; this is precisely the shape of statement an inner-product argument proves (theorem 8.32).

8.5 The mixed inner product with group elements

The commitments used in Halo 2 and Orchard pair a vector of *scalars* with a vector of *group elements*. We isolate the algebra here, working over an abstract group; the group of actual cryptographic interest—the points of an elliptic curve—is constructed in the elliptic-curve section later in the volume, and only its abelian-group structure is needed for everything proved below, so the results transfer verbatim.

Construction 8.25 (Scalar multiplication by $\mathbb{F}_r$ on a prime-order group)**.** Let $(\mathbb{G}, +)$ be a finite abelian group of prime order r, written additively with identity O. For an integer $a \in \mathbb{Z}$ and $G \in \mathbb{G}$, write $[a]G$ for the a-fold multiple of G—the additive reading of the powers of §3: $[0]G = O$, $[a]G = G + \cdots + G$ with a summands for $a > 0$, and $[-a]G = -([a]G)$. The brackets keep the scalar visually separate from the group element. The multiple $[a]G$ depends only on the residue of a modulo r: the order of G divides $|\mathbb{G}| = r$ (Corollary 3.30), and $[a]G = [b]G$ whenever $a \equiv b \pmod{\operatorname{ord}(G)}$ (Proposition 3.12(2), read additively), so congruence modulo r suffices. The operation therefore *descends* to scalars in the field $\mathbb{F}_r = \mathbb{Z}/r\mathbb{Z}$ (theorem 4.26): for $a \in \mathbb{F}_r$ and $G \in \mathbb{G}$, define $[a]G$ using any integer representative of a. It is this descended operation—group elements scaled by elements of a *field*—that the rest of the section uses.

Proposition 8.26 ($\mathbb{G}$ as an $\mathbb{F}_r$-vector space)**.** *Let $\mathbb{G}$ be an abelian group of prime order r, with the scalar multiplication of theorem 8.25. Then $\mathbb{G}$ is an $\mathbb{F}_r$-vector space of dimension* 1*; any element $G \neq O$ is a basis, and every element of $\mathbb{G}$ is $[a]G$ for a unique $a \in \mathbb{F}_r$.*

Proof. The vector-space axioms are the standard laws of multiples. For integers a, b, the identities $[a + b]G = [a]G + [b]G$ and $[a]([b]G) = [ab]G$ are the exponent laws of §3 in additive notation, $[1]G = G$ is the definition, and $[a](G + H) = [a]G + [a]H$ holds in any *abelian* group: for $a > 0$ it is the rearrangement of a summands $G + H$ into a summands G followed by a summands H, legitimate by commutativity and associativity (formally, induction on a), and the cases $a \leq 0$ follow by negating. Each law respects reduction modulo r by theorem 8.25, so the four scalar axioms of theorem 8.1 hold with scalars in $\mathbb{F}_r$, and $(\mathbb{G}, +)$ is an abelian group by hypothesis: $\mathbb{G}$ is an $\mathbb{F}_r$-vector space.

Now let $G \neq O$. By Lagrange's theorem (Theorem 3.29, via Corollary 3.30) the order of G divides the prime r, and it exceeds 1 since G is not the identity; hence $\operatorname{ord}(G) = r$. The r multiples $[0]G, [1]G, \dots, [r-1]G$ are then pairwise distinct (Proposition 3.12(2), additively: $[a]G = [b]G$ forces $a \equiv b \pmod{r}$), and $\mathbb{G}$ has exactly r elements, so the multiples exhaust $\mathbb{G}$. Thus $\{G\}$ spans; and it is independent, since $[a]G = O$ forces $a = 0$ in $\mathbb{F}_r$, again because $\operatorname{ord}(G) = r$. A one-element basis gives $\dim_{\mathbb{F}_r} \mathbb{G} = 1$, and the uniqueness of a in $[a]G$ is the uniqueness of coordinates in a basis (theorem 8.14). □

Definition 8.27 (Mixed inner product, multiscalar multiplication)**.** Let $\mathbf{G} = (G_1, \dots, G_n) \in \mathbb{G}^n$ be a fixed tuple of group elements and $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{F}_r^n$ a vector of scalars. Their *mixed inner product*—in computational contexts a *multiscalar multiplication* (MSM), in cryptographic ones a *Pedersen-style inner product*—is the group element

$$\langle \mathbf{a}, \mathbf{G} \rangle = \sum_{i=1}^{n} [a_i]\, G_i \in \mathbb{G}.$$

We reuse the angle-bracket notation of theorem 8.21 deliberately; the type of the second argument—scalar vector or vector of group elements—determines which product is meant.

Proposition 8.28 (Bilinearity of the mixed product)**.** *Fix $\mathbf{G} \in \mathbb{G}^n$. The map $\mathbf{a} \mapsto \langle \mathbf{a}, \mathbf{G} \rangle$ is $\mathbb{F}_r$-linear from $\mathbb{F}_r^n$ to $\mathbb{G}$:*

$$\langle c\,\mathbf{a} + c'\,\mathbf{a}', \mathbf{G} \rangle = [c]\langle \mathbf{a}, \mathbf{G} \rangle + [c']\langle \mathbf{a}', \mathbf{G} \rangle \qquad \text{for all } c, c' \in \mathbb{F}_r \text{ and } \mathbf{a}, \mathbf{a}' \in \mathbb{F}_r^n.$$

Symmetrically, for fixed $\mathbf{a} \in \mathbb{F}_r^n$ the map $\mathbf{G} \mapsto \langle \mathbf{a}, \mathbf{G} \rangle$ is a group homomorphism $\mathbb{G}^n \to \mathbb{G}$, and is $\mathbb{F}_r$-linear when $\mathbb{G}^n$ is viewed as an $\mathbb{F}_r$-vector space (componentwise, as in Example 8.3 with $\mathbb{G}$ in place of F). Thus $\langle \cdot, \cdot \rangle \colon \mathbb{F}_r^n \times \mathbb{G}^n \to \mathbb{G}$ is bilinear over $\mathbb{F}_r$.

Proof. Both statements are the vector-space laws of $\mathbb{G}$ (Proposition 8.26) applied componentwise. For the first,

$$\langle c\,\mathbf{a} + c'\,\mathbf{a}', \mathbf{G} \rangle = \sum_i [\, ca_i + c'a_i' \,]\, G_i = \sum_i ([c][a_i]G_i + [c'][a_i']G_i) = [c] \sum_i [a_i]G_i + [c'] \sum_i [a_i']G_i,$$

using $[a+b]G = [a]G + [b]G$ and $[ab]G = [a]([b]G)$ in each component and then regrouping the sum in the abelian group $\mathbb{G}$. For the second, linearity in $\mathbf{G}$ follows by the symmetric computation on the other slot of the defining sum, using $[a](G+H) = [a]G + [a]H$ componentwise. □

Remark 8.29 (Why the mixed product is the right abstraction)*.* Read $\langle \mathbf{a}, \mathbf{G} \rangle$ as a *commitment* to the scalar vector $\mathbf{a}$ under the public "basis" $\mathbf{G}$—the receipt of the section's opening problem. Proposition 8.28 is then exactly the *homomorphic* property that the problem demanded: the commitment to $c\,\mathbf{a} + c'\,\mathbf{a}'$ is the $[c]$-, $[c']$-scaled combination of the individual commitments, computable by anyone holding those commitments and the scalars c, c', without knowledge of $\mathbf{a}$ or $\mathbf{a}'$. Provided the G_i are chosen so that no party can exhibit a nontrivial relation $\sum_i [a_i]G_i = O$—a computational hypothesis, the *discrete-logarithm hardness assumption*, made precise in later volumes of this series—the commitment is *binding*: the map $\mathbf{a} \mapsto \langle \mathbf{a}, \mathbf{G} \rangle$ is computationally injective, in the sense that no feasible computation produces two vectors with the same image. Genuinely injective it cannot be once $n \geq 2$: the map is linear from the n-dimensional $\mathbb{F}_r^n$ into the one-dimensional $\mathbb{G}$ (Proposition 8.26), so by rank–nullity (Theorem 8.19) its kernel has dimension at least $n-1$. Collisions exist in abundance; binding is the claim that they cannot be *found*, a computational rather than information-theoretic statement. The classical Pedersen commitment to a single value a with randomness ρ is the special case $n = 2$, with $H \neq O$,

$$[a]G + [\rho]H = \langle (a, \rho), (G, H) \rangle,$$

which is moreover *perfectly hiding*: for fixed a, as ρ ranges uniformly over $\mathbb{F}_r$ the term $[\rho]H$ ranges uniformly over $\mathbb{G}$ (it is a bijection of $\mathbb{G}$, by unique representation in the basis $\{H\}$), so the commitment's distribution is uniform and carries no information about a.

8.6 How this machinery appears in commitment and proof systems

We close by drawing the threads together. Each point below is developed rigorously in later volumes of this series; the remarks here are descriptive, but the linear-algebraic content behind them is already proved in full.

Remark 8.30 (Witnesses, constraints, and assignments). An arithmetic circuit over $\mathbb{F}_q$ becomes, after *arithmetisation*—the encoding of a computation as a system of polynomial constraints on the wire values—a collection of equations that the vector $\mathbf{z} \in \mathbb{F}_q^N$ of all wire values must satisfy. The *linear* part of such a system—the wiring (or "copy") constraints of the PLONK family, which assert that certain wires carry equal values—consists of homogeneous linear equations in the entries of $\mathbf{z}$, organised in practice as arrays of coefficients (one row per equation) acting on the assignment vector. Satisfying the linear constraints confines $\mathbf{z}$ to a subspace of $\mathbb{F}_q^N$ (Example 8.7), or, when public inputs fix some entries, to a translate $\mathbf{z}_0 + W$ of one—an *affine* subspace. The multiplicative constraints are handled separately, by the polynomial machinery of the surrounding sections. Rank–nullity (Theorem 8.19) quantifies the degrees of freedom that remain after the linear constraints are imposed: the dimension of the witness space.

Remark 8.31 (Commitments as linear maps). A Pedersen vector commitment $\mathbf{a} \mapsto \langle \mathbf{a}, \mathbf{G} \rangle$ is a linear map $\mathbb{F}_r^n \to \mathbb{G}$ (Proposition 8.28), and its linearity is the source of every algebraic manipulation a verifier performs: taking known linear combinations of commitments, folding two commitment keys $\mathbf{G}, \mathbf{G}'$ into $\mathbf{G} + [x]\mathbf{G}'$ under a challenge x, and checking claimed openings by re-evaluating the same linear map. A polynomial commitment is the identical map applied to a coefficient vector, and an *evaluation* of the committed polynomial at a point z is the scalar inner product $\langle \mathbf{c}, (1, z, \dots, z^{n-1}) \rangle$ (Remark 8.24).

Remark 8.32 (The inner-product relation). The core statement an inner-product argument proves has the shape: "I know vectors $\mathbf{a}, \mathbf{b} \in \mathbb{F}_r^n$ such that

$$P = \langle \mathbf{a}, \mathbf{G} \rangle + \langle \mathbf{b}, \mathbf{H} \rangle + [\, \langle \mathbf{a}, \mathbf{b} \rangle \,] U$$

for public $\mathbf{G}, \mathbf{H} \in \mathbb{G}^n$ and $U \in \mathbb{G}$." Three of this section's constructions meet in the one equation: two mixed inner products $\langle \mathbf{a}, \mathbf{G} \rangle$ and $\langle \mathbf{b}, \mathbf{H} \rangle$, committing to the witness vectors; one scalar inner product $\langle \mathbf{a}, \mathbf{b} \rangle$, the value being argued; and the bilinearity of all three, which is what permits prover and verifier to *fold* a length-n instance into a length-$n/2$ one and recurse. The argument's soundness ultimately reduces to the nondegeneracy of these forms together with the hardness of finding kernel elements of the commitment map—exactly the linear-algebraic and computational facts assembled above.

The cheque drawn at the head of the section is now cashed. The receipt for a list $(a_1, \dots, a_n)$ of secrets is the mixed inner product $\langle \mathbf{a}, \mathbf{G} \rangle = \sum_i [a_i] G_i$: one group element, whatever the size of n. "Receipts add" means precisely that the receipt map is linear, and that is the payoff theorem, Proposition 8.28. One element can absorb n secrets only because the map compresses an n-dimensional space into a one-dimensional one, and rank–nullity prices the compression exactly: a kernel of dimension at least $n - 1$, an unavoidable reservoir of collisions that Remark 8.29 tames computationally rather than absolutely.

In summary: vector spaces, bases and dimension, linear maps with their kernels and images governed by rank–nullity, bilinear forms, and the two flavours of inner product—scalar with scalar, scalar with group—constitute the linear-algebraic vocabulary for the cryptography ahead. Pedersen-style commitments are linear maps, and their polynomial evaluation openings assert inner products; nondegeneracy and dimension counting supply the relevant algebra. Other commitment constructions, such as hash-based commitments, need not be linear. The next section applies the machinery to the polynomial spaces $F[X]_{<n}$ over structured evaluation domains, and the section on elliptic curves constructs the group $\mathbb{G}$ that the mixed inner product has so far taken as abstract.

9 Roots of unity and evaluation domains

Run a computation for n steps and record, at each step, the value it holds in each of its registers. The record is a table: one column per register, one row per step, every entry an element of a finite field. A proof system of the kind underlying Halo 2 and Zcash Orchard convinces a verifier that such a table obeys its rules—that every row follows from the one before it—without the verifier replaying the computation. Its basic move is to treat each column as the value table of a *polynomial*: a function given by its values at n fixed points is, by Lagrange interpolation (theorem 5.30), the same data as a polynomial of degree less than n, and statements about the table become statements about polynomials—checkable all at once, at a single random point, by the Schwartz–Zippel principle (theorem 5.45). For this plan to be workable two demands must be met. The n evaluation points must be *structured*, so that "every row" conditions, "next row" shifts, and the bookkeeping polynomials attached to the point set stay cheap; and converting between a polynomial's values and its coefficients—and multiplying two polynomials—must cost far less than the n^2 field operations of the schoolbook methods, or provers could never handle tables of practical size. Both demands are met by one choice of point set: the powers of a *root of unity*. Nearly every operation such a proof system performs—committing to a polynomial, checking that two polynomials agree, enforcing that a constraint vanishes everywhere it should, multiplying or interpolating efficiently—is ultimately a computation indexed by the finite multiplicative subgroup those powers form. This section develops the theory of these subgroups and delivers, in the fast Fourier transform, the $O(n \log n)$ arithmetic promised above.

Throughout, $\mathbb{F}$ denotes a field. When $\mathbb{F}$ is to be finite, the notation $\mathbb{F}_q$ denotes the field with q elements (so q is a prime power, theorem 6.9); when we intend a prime field specifically we write $\mathbb{F}_p$. The multiplicative group of nonzero elements is $\mathbb{F}^\times = \mathbb{F} \setminus \{0\}$ (Example 3.7). We reserve the letter n for the order of the domain we shall construct; throughout, $n \geq 1$.

The route is as follows. We first define roots of unity and the primitive ones among them, and settle exactly when a finite field contains them: the condition is $n \mid q - 1$, a direct consequence of the cyclicity of $\mathbb{F}_q^\times$ proved in §6. We then fix the *evaluation domain* H, the subgroup those roots form, and work out its toolkit: the vanishing polynomial $X^n - 1$ and its cosets, the Lagrange basis in closed form, and the evaluation/interpolation isomorphism in matrix form. The isomorphism, read as a matrix, is the number-theoretic transform; the convolution theorem turns polynomial multiplication into entrywise multiplication of value vectors; and the fast Fourier transform computes the transform in $O(n \log n)$ operations. A closing synthesis assembles the pieces into the reason these domains form the substrate of polynomial proof systems.

9.1 Roots of unity

The starting point is the equation $X^n = 1$. Its solutions in a field are the roots of unity of order dividing n.

Definition 9.1 (Root of unity)**.** Let $\mathbb{F}$ be a field and $n \in \mathbb{N}_{>0}$. An element $\omega \in \mathbb{F}$ is an *n-th root of unity* if $\omega^n = 1$. We write

$$\mu_n(\mathbb{F}) = \{\, \omega \in \mathbb{F} : \omega^n = 1 \,\}$$

for the set of all n-th roots of unity in $\mathbb{F}$.

Note that $0 \notin \mu_n(\mathbb{F})$, since $0^n = 0 \neq 1$; every n-th root of unity is therefore a unit, i.e. lies in $\mathbb{F}^\times$. The set $\mu_n(\mathbb{F})$ is not merely a set; it is a group.

Proposition 9.2. *The set $\mu_n(\mathbb{F})$ is a finite subgroup of $\mathbb{F}^\times$ of order at most n.*

Proof. The identity 1 satisfies $1^n = 1$, so $1 \in \mu_n(\mathbb{F})$. If $\omega, \zeta \in \mu_n(\mathbb{F})$, then $(\omega\zeta)^n = \omega^n\zeta^n = 1{\cdot}1 = 1$ because $\mathbb{F}^\times$ is abelian, so $\omega\zeta \in \mu_n(\mathbb{F})$; and $(\omega^{-1})^n = (\omega^n)^{-1} = 1^{-1} = 1$, so $\omega^{-1} \in \mu_n(\mathbb{F})$. Hence $\mu_n(\mathbb{F})$ is a subgroup of $\mathbb{F}^\times$ (theorem 3.14). For the order bound, every element of $\mu_n(\mathbb{F})$ is a root of the polynomial $X^n - 1 \in \mathbb{F}[X]$, which is nonzero of degree n and so has at most n roots in $\mathbb{F}$ (theorem 5.18). Thus $|\mu_n(\mathbb{F})| \le n$. □

The bound $|\mu_n(\mathbb{F})| \le n$ need not be attained. Whether it is attained is precisely the question of the existence of a *primitive* n-th root of unity.

Definition 9.3 (Primitive root of unity)**.** An element $\omega \in \mathbb{F}$ is a *primitive n-th root of unity* if $\omega^n = 1$ and $\omega^k \ne 1$ for every integer k with $1 \le k < n$. Equivalently, ω is a primitive n-th root of unity if its multiplicative order $\operatorname{ord}(\omega)$ (theorem 3.11) equals n.

Here $\operatorname{ord}(\omega)$ is the least positive integer d with $\omega^d = 1$; it exists for any $\omega \in \mu_n(\mathbb{F})$ because n itself is such an exponent and the positive integers are well-ordered. The two formulations in theorem 9.3 agree: $\omega^k \ne 1$ for $1 \le k < n$ together with $\omega^n = 1$ states exactly that the least positive exponent killing ω is n.

Remark 9.4 (Primitive roots revisited)*.* The identification promised in theorem 6.20 can now be stated. A primitive root of a finite field $\mathbb{F}_q$—a generator of the cyclic group $\mathbb{F}_q^\times$, in the sense of theorem 6.19—is an element of multiplicative order $q - 1$, which by theorem 9.3 is precisely a primitive $(q-1)$-th root of unity. The group-theoretic vocabulary "generator of $\mathbb{F}_q^\times$" and the polynomial-flavoured vocabulary "primitive $(q-1)$-th root of unity" name the same elements.

Lemma 9.5 (Order divides exponent)**.** *Let $\omega \in \mathbb{F}^\times$ have finite order $d = \operatorname{ord}(\omega)$. Then for any integer m we have $\omega^m = 1$ if and only if $d \mid m$. In particular, ω is an n-th root of unity if and only if $d \mid n$, and ω is a* primitive *n-th root of unity if and only if $d = n$.*

Proof. The first claim is theorem 3.12(1), read in the abelian group $\mathbb{F}^\times$: if $d \mid m$, write $m = dk$, so $\omega^m = (\omega^d)^k = 1$; conversely, division with remainder $m = qd + r$, $0 \le r < d$, gives $1 = \omega^m = \omega^r$, and minimality of d forces $r = 0$. The remaining clauses instantiate it: $\omega^n = 1 \iff d \mid n$ is the case $m = n$, and primitivity is the case $d = n$ by theorem 9.3. □

The lemma's constant companion is the order-of-a-power formula in a cyclic group, $\operatorname{ord}(g^k) = \operatorname{ord}(g)/\gcd(k, \operatorname{ord}(g))$, proved as Corollary 3.23; we use the pair repeatedly below as the standard device for computing orders.

Proposition 9.6. *If $\omega \in \mathbb{F}$ is a primitive n-th root of unity, then*

$$\mu_n(\mathbb{F}) = \langle \omega \rangle = \{\, 1, \omega, \omega^2, \dots, \omega^{n-1} \,\}$$

is a cyclic group of order exactly n, and these n powers are pairwise distinct.

Proof. The powers $1, \omega, \dots, \omega^{n-1}$ are pairwise distinct: if $\omega^i = \omega^j$ with $0 \le i \le j \le n-1$, then $\omega^{j-i} = 1$ with $0 \le j - i < n$, so primitivity (theorem 9.5, with $d = n$) forces $j - i = 0$. Thus $\langle \omega \rangle$ has exactly n elements, all of which lie in $\mu_n(\mathbb{F})$ since $(\omega^k)^n = (\omega^n)^k = 1$. But $|\mu_n(\mathbb{F})| \le n$ by theorem 9.2, so the inclusion $\langle \omega \rangle \subseteq \mu_n(\mathbb{F})$ is an equality of n-element sets. □

Over the complex numbers a primitive n-th root of unity always exists, namely $\omega = e^{2\pi i/n}$: this is the classical picture of n equally spaced points on the unit circle, with the primitive roots sitting at the angles $2\pi k/n$ for $\gcd(k, n) = 1$. Over a finite field existence is more delicate, and it is exactly the finite case that matters for cryptography.

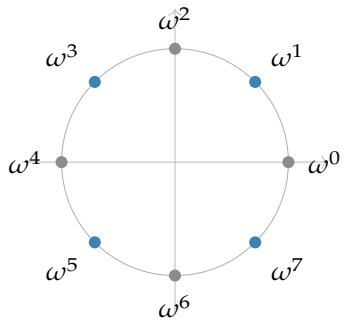


Figure 1: The eight 8-th roots of unity in $\mathbb{C}$, equally spaced on the unit circle at angles $2\pi k/8$, with $\omega = e^{2\pi i/8}$. The four primitive ones ($k = 1, 3, 5, 7$, exactly the k coprime to 8; marked in colour) each generate the whole cycle; the grey points $\omega^0 = 1$, $\omega^2 = i$, $\omega^4 = -1$, $\omega^6 = -i$ have orders $1, 4, 2, 4$ and generate proper subgroups. Over a finite field the same group structure survives with the geometry stripped away—the picture to keep is the cycle, not the circle.

9.2 Existence over finite fields: the condition $n \mid q - 1$

The decisive structural fact about finite fields is that their multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $q - 1$ (theorem 6.17); the existence theorem for roots of unity rests on it.

> **Theorem 9.7** (Existence and count of primitive n-th roots of unity)**.** *Let $\mathbb{F}_q$ be a finite field and $n \in \mathbb{N}_{>0}$ with $\gcd(n, q) = 1$ (automatic when $n < p$, where $p = \operatorname{char} \mathbb{F}_q$; equivalently, in characteristic p, exactly the condition $p \nmid n$). Then:*
>
> 1. *$\mathbb{F}_q$ contains a primitive n-th root of unity if and only if $n \mid q - 1$.*
> 2. *When $n \mid q - 1$, the group $\mu_n(\mathbb{F}_q)$ is cyclic of order exactly n, and the number of primitive n-th roots of unity in $\mathbb{F}_q$ is $\varphi(n)$, where φ is Euler's totient function (theorem 2.26).*

Proof. Let g be a generator of $\mathbb{F}_q^\times$, which is cyclic of order $q - 1$ by theorem 6.17.

($\Rightarrow$) Suppose ω is a primitive n-th root of unity in $\mathbb{F}_q$. Then $\omega \in \mathbb{F}_q^\times$ has order n by theorem 9.3, and by Lagrange's theorem the order of an element divides the group order $q-1$ (Corollary 3.30). Hence $n \mid q - 1$.

($\Leftarrow$) Suppose $n \mid q - 1$, say $q - 1 = nm$. Set $\omega = g^m$. Then $\omega^n = g^{mn} = g^{q-1} = 1$, so $\omega \in \mu_n(\mathbb{F}_q)$. Its order is

$$\operatorname{ord}(\omega) = \operatorname{ord}(g^m) = \frac{\operatorname{ord}(g)}{\gcd(m, \operatorname{ord}(g))} = \frac{q-1}{\gcd(m, q-1)} = \frac{q-1}{m} = n,$$

using the order-of-a-power formula (Corollary 3.23) together with $m \mid q-1$ (so $\gcd(m, q-1) = m$). Thus ω is a primitive n-th root of unity, and by theorem 9.6, $\mu_n(\mathbb{F}_q) = \langle \omega \rangle$ is cyclic of order exactly n.

Count. In the cyclic group $\mu_n(\mathbb{F}_q) = \langle \omega \rangle$ of order n, the element ω^k has order $n/\gcd(k, n)$, again by Corollary 3.23. This equals n precisely when $\gcd(k, n) = 1$, and the number of $k \in \{0, 1, \dots, n-1\}$ with $\gcd(k, n) = 1$ is by definition $\varphi(n)$ (theorem 2.26). Hence there are exactly $\varphi(n)$ primitive n-th roots of unity.

(A side note on the coprimality hypothesis. The condition $\gcd(n, q) = 1$ is what guarantees that $X^n - 1$ is separable—$\gcd(X^n - 1, nX^{n-1}) = 1$ when $n \neq 0$ in $\mathbb{F}_q$—hence has n distinct roots in a splitting field over $\mathbb{F}_q$ (theorem 5.40); that all n of them lie in $\mathbb{F}_q$ itself is exactly the condition $n \mid q - 1$. If $p = \operatorname{char} \mathbb{F}_q$ divided n, then $X^n - 1$ would have repeated roots and $\mu_n(\mathbb{F}_q)$ would be strictly smaller. But $n \mid q-1$ already forces $\gcd(n, q) = 1$, since $\gcd(q-1, q) = 1$, so the hypothesis is in fact subsumed.) $\square$

Remark 9.8. The proof shows that for a finite field the clean statement is simply: $\mathbb{F}_q$ *has a primitive n-th root of unity* $\iff$ $n \mid q-1$, with no separate coprimality side-condition, because $n \mid q-1$ automatically makes n coprime to q. We stated the condition $\gcd(n,q)=1$ explicitly only to flag the role of the characteristic.

Example 9.9. Take $\mathbb{F}_q = \mathbb{F}_p$ with $p=7$. Here $q-1=6$, and 3 is a generator of $\mathbb{F}_7^\times$: its successive powers are $3,2,6,4,5,1$, all six nonzero residues (Example 6.21). For $n=3 \mid 6$, set $m=6/3=2$ and $\omega = 3^2 = 2$. Indeed $2^1=2$, $2^2=4$, $2^3=8\equiv 1$, so $\omega=2$ has order 3 and $\mu_3(\mathbb{F}_7)=\{1,2,4\}$. There are $\varphi(3)=2$ primitive cube roots of unity, namely 2 and 4. By contrast $4 \nmid 6$, so $\mathbb{F}_7$ has *no* primitive 4-th root of unity: $X^4-1=(X^2-1)(X^2+1)$, and X^2+1 is irreducible over $\mathbb{F}_7$ because it has no root there, -1 not being a square modulo 7.

Remark 9.10 (Two-adicity and FFT-friendly primes). For the fast algorithms below it is desirable that n be a power of two, or at least *smooth*—a product of small primes, ideally with many factors of two. Theorem 9.7 shows this is possible exactly when $2^k \mid q-1$. The largest k with $2^k \mid q-1$ is called the *two-adicity* of the field. Designers deliberately choose primes p with high two-adicity—so that $p-1=2^k\cdot c$ for large k—as scalar fields in proof systems precisely so that large smooth evaluation domains exist. Both Pasta primes—those underlying the Pallas and Vesta curves of section 10—satisfy $2^{32} \mid p-1$, with two-adicity exactly 32 (the primes are specified in §5.4.9.6, "Pallas and Vesta", of the Zcash protocol specification), so each Pasta field contains primitive 2^k-th roots of unity for every $k \le 32$, furnishing power-of-two evaluation domains of every size up to 2^{32}; this is what makes the entire machinery of this section available to Halo 2 and Orchard.

9.3 The evaluation domain H and its vanishing polynomial

We now fix the central object of this section.

Definition 9.11 (Evaluation domain)**.** Let $\mathbb{F}$ be a field containing a primitive n-th root of unity ω. The *evaluation domain* of order n is the multiplicative subgroup

$$H = \langle \omega \rangle = \{1, \omega, \omega^2, \dots, \omega^{n-1}\} = \mu_n(\mathbb{F}) \subseteq \mathbb{F}^\times,$$

a cyclic group of order n by theorem 9.6. We index its elements as $\omega^0, \omega^1, \dots, \omega^{n-1}$, and abbreviate ω^i by h_i when convenient. Both $\mathbb{F}$ and ω remain fixed for the rest of the section.

Because H is a multiplicative *group*, it is closed under products and inverses, and $1 \in H$. These elementary facts are exactly what make H substantially more useful than an arbitrary set of n interpolation points: cosets, products, and shifts of H interact algebraically. The single most important polynomial attached to H is the one that vanishes precisely on it.

Definition 9.12 (Vanishing polynomial)**.** The *vanishing polynomial* (or *zerofier*) of H is

$$Z_H(X) = \prod_{h \in H} (X-h) \in \mathbb{F}[X].$$

On a generic n-point set the vanishing polynomial (theorem 5.33) is an unstructured product of n linear factors. On the group H it collapses.

Theorem 9.13 (Closed form and factorisation of the vanishing polynomial)**.** *With* $H = \mu_n(\mathbb{F})$

of order n generated by a primitive n-th root of unity ω,

$$Z_H(X) \;=\; X^n - 1 \;=\; \prod_{i=0}^{n-1}(X - \omega^i).$$

Moreover Z_H is separable: it has n distinct roots and $\gcd(Z_H, Z_H') = 1$.

Proof. Every $h \in H$ satisfies $h^n = 1$, hence is a root of $X^n - 1$. The n elements of H are distinct (theorem 9.6), so $X^n - 1$ has n distinct roots, namely all of H, and factors as $X^n - 1 = c\prod_{h\in H}(X - h)$ for some constant $c \in \mathbb{F}^\times$ by iterating the factor theorem (theorem 5.36, with the degree leaving no room for further factors). Comparing leading coefficients—both $X^n - 1$ and the product are monic—gives $c = 1$, whence $X^n - 1 = \prod_{h\in H}(X - h) = Z_H(X)$.

For separability, the formal derivative is $Z_H'(X) = nX^{n-1}$ (theorem 5.37). The existence of the n distinct roots already forces $\operatorname{char}\mathbb{F} \nmid n$: if $\operatorname{char}\mathbb{F} = \ell$ divided n, say $n = \ell m$, the freshman's dream (theorem 6.13) would give $X^n - 1 = (X^m - 1)^\ell$, a polynomial with at most $m < n$ distinct roots—a contradiction. Hence $n \neq 0$ in $\mathbb{F}$, so nX^{n-1} is nonzero at every element of H, since each is nonzero. Thus it has no root in common with $X^n - 1$, so $\gcd(Z_H, Z_H') = 1$ and Z_H is separable (theorem 5.40). □

The contraposition through the freshman's dream is worth isolating as a device: it converts a *root count* (n distinct n-th roots of unity exist) into a *characteristic condition* ($\operatorname{char}\mathbb{F} \nmid n$), and thereby licenses dividing by n in $\mathbb{F}$—which the closed forms below do constantly.

The closed form $Z_H(X) = X^n - 1$ is the workhorse of polynomial proof systems: testing whether a polynomial f vanishes on all of H reduces to checking divisibility by $X^n - 1$, which one can verify by exhibiting a quotient $t(X)$ with $f(X) = t(X)(X^n - 1)$. Moreover, evaluating Z_H at any point costs a single exponentiation X^n, not n multiplications. We record two structural consequences.

Proposition 9.14 (Cosets and shifts)**.** *Let $\gamma \in \mathbb{F}^\times$ and let $\gamma H = \{\gamma h : h \in H\}$ be the corresponding coset (theorem 3.27). Then the polynomial vanishing on γH is*

$$Z_{\gamma H}(X) = \prod_{h\in H}(X - \gamma h) = X^n - \gamma^n.$$

In particular, distinct cosets of H in $\mathbb{F}^\times$ have vanishing polynomials $X^n - c$ for distinct constants $c = \gamma^n$, and the cosets of H partition $\mathbb{F}^\times$ (theorem 3.28) into translates of H, on each of which X^n is constant.

Proof. Substitute $Y = X/\gamma$:

$$\prod_{h\in H}(X - \gamma h) = \gamma^n \prod_{h\in H}(Y - h) = \gamma^n(Y^n - 1) = \gamma^n((X/\gamma)^n - 1) = X^n - \gamma^n,$$

using theorem 9.13 for the middle equality. The constant is $c = \gamma^n$, and

$$\gamma_1 H = \gamma_2 H \iff \gamma_1/\gamma_2 \in H \iff (\gamma_1/\gamma_2)^n = 1 \iff \gamma_1^n = \gamma_2^n,$$

so distinct cosets give distinct constants c. □

Lemma 9.15 (Finite geometric series)**.** *Let $\mathbb{F}$ be a field, $\zeta \in \mathbb{F}$ with $\zeta \neq 1$, and $n \geq 1$. Then*

$$\sum_{i=0}^{n-1} \zeta^i = \frac{\zeta^n - 1}{\zeta - 1}.$$

Proof. Write $S = 1 + \zeta + \zeta^2 + \cdots + \zeta^{n-1}$. Multiplying by ζ advances every term one slot:

$$\zeta S = \zeta + \zeta^2 + \cdots + \zeta^{n-1} + \zeta^n.$$

In the difference $\zeta S - S$, the terms $\zeta, \zeta^2, \ldots, \zeta^{n-1}$ appear in both lines and cancel; only the two extremes survive:

$$(\zeta - 1)\, S \;=\; \zeta S - S \;=\; \zeta^n - 1.$$

Since $\zeta \neq 1$, the factor $\zeta - 1$ is invertible, and dividing by it yields the claim. □

Proposition 9.16 (Sum of roots of unity)**.** *If $n > 1$ then $\sum_{i=0}^{n-1} \omega^i = 0$, and more generally for any integer k,*

$$\sum_{i=0}^{n-1} \omega^{ki} = \begin{cases} n & \text{if } n \mid k, \\ 0 & \text{if } n \nmid k. \end{cases}$$

Proof. If $n \mid k$, then $\omega^{ki} = (\omega^n)^{ki/n} = 1$ for each i, so the sum is n. If $n \nmid k$, set $\zeta = \omega^k \neq 1$ (since ω has order n, we have $\omega^k = 1 \iff n \mid k$ by theorem 9.5), and theorem 9.15 gives

$$\sum_{i=0}^{n-1} \zeta^i = \frac{\zeta^n - 1}{\zeta - 1} = \frac{(\omega^n)^k - 1}{\zeta - 1} = \frac{1 - 1}{\zeta - 1} = 0,$$

the denominator being nonzero because $\zeta \neq 1$. The first claim is the case $k = 1$, valid for $n > 1$ since then $n \nmid 1$. □

Remark 9.17 (Rotation makes the vanishing inevitable). For $n \nmid k$ there is also a proof with no formula at all. The n terms of $\sum_i \omega^{ki}$ run through the subgroup $\langle \zeta \rangle$ generated by $\zeta = \omega^k$, each element appearing equally often; multiplying the sum by ζ merely rotates that cycle one notch—each term advances to the next, the last wrapping around to the first—so the sum satisfies $\zeta S = S$. A quantity invariant under scaling by something other than 1 can only be zero: $(\zeta - 1)S = 0$ with $\zeta \neq 1$ forces $S = 0$. The balanced cycle of fig. 1 is this argument made visible.

Remark 9.18 (Why "orthogonality"). Call two vectors in $\mathbb{F}^n$ *orthogonal* when their standard inner product (theorem 8.21) vanishes. For each frequency k let $v_k = (1, \omega^k, \omega^{2k}, \ldots, \omega^{(n-1)k})$ be its vector of powers, and let $\bar{v}_k = (1, \omega^{-k}, \omega^{-2k}, \ldots, \omega^{-(n-1)k})$ be the same construction run with ω^{-1} in place of ω: the *mirror* of v_k, walking the cycle in the opposite direction, in the role the complex conjugate plays classically (on the unit circle $\bar{z} = z^{-1}$). Then $\langle \bar{v}_k, v_{k'} \rangle = \sum_i \omega^{i(k'-k)}$, which theorem 9.16 evaluates to 0 for $k \neq k'$ and to n for $k = k'$: distinct frequency vectors are orthogonal, and each pairs with its own mirror to n. (Which factor carries the mirror is a convention: with the roles exchanged the exponent is $k - k'$, and the criterion $n \mid k' - k$ is symmetric, so the verdict is the same; the order here matches the product $\overline{V}\, V$ computed in theorem 9.29.) This is the finite-field counterpart of the orthogonality of the exponentials $\theta \mapsto e^{2\pi i k\theta}$ in classical Fourier analysis, the balanced trip around the cycle replacing the integral over the circle. It is the engine of the inverse-transform proof below (theorem 9.29): there the v_k appear as the columns of the transform matrix, so in the inverse sum every wrong frequency's contribution rides on one of these vanishing pairings and cancels, while the right one survives n-fold and the $1/n$ rescales it.

9.4 The Lagrange basis on H

Attention now passes from the group H to the polynomials we study on it. Since H has exactly n points, values on H pin down polynomials of degree less than n (theorem 5.20); the basis adapted to this situation is the Lagrange basis of §5.7, which on the group H acquires a closed form it has on no generic point set.

Definition 9.19. Let $\mathbb{F}[X]_{<n}$ denote the $\mathbb{F}$-vector space of polynomials of degree strictly less than n, together with the zero polynomial. It has the monomial basis $\{1, X, \dots, X^{n-1}\}$ (theorem 8.14), so $\dim_{\mathbb{F}} \mathbb{F}[X]_{<n} = n$ (theorem 8.17).

Definition 9.20. For each $i \in \{0, 1, \dots, n-1\}$, the i-th *Lagrange basis polynomial* for $H = \{\omega^0, \dots, \omega^{n-1}\}$ is

$$L_i(X) = \prod_{\substack{j=0 \\ j \neq i}}^{n-1} \frac{X - \omega^j}{\omega^i - \omega^j} \in \mathbb{F}[X]_{<n},$$

the basis polynomial ℓ_i of theorem 5.30 for the nodes $\omega^0, \dots, \omega^{n-1}$.

Proposition 9.21 (Interpolation property)**.** *The polynomials $L_0, \dots, L_{n-1}$ satisfy the Kronecker-delta conditions*

$$L_i(\omega^j) = \delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j, \end{cases}$$

they form a basis of $\mathbb{F}[X]_{<n}$, and they give the unique interpolant: for any data $(v_0, \dots, v_{n-1}) \in \mathbb{F}^n$, the polynomial

$$f(X) = \sum_{i=0}^{n-1} v_i L_i(X)$$

is the unique element of $\mathbb{F}[X]_{<n}$ with $f(\omega^i) = v_i$ for all i.

Proof. Each L_i is well defined and of degree $n-1$ because the nodes ω^j are distinct (theorem 9.6), so the denominators $\omega^i - \omega^j$ with $j \neq i$ are nonzero. The delta conditions, the interpolation $f(\omega^j) = v_j$, and the uniqueness are theorem 5.30 (with (4)) applied to these nodes with $d = n-1$; uniqueness rests, as there, on the root count: two interpolants in $\mathbb{F}[X]_{<n}$ differ by a polynomial of degree $< n$ with n distinct roots, which is the zero polynomial (theorem 5.18). It remains to see that the L_i form a basis of $\mathbb{F}[X]_{<n}$. They are linearly independent: if $\sum_i c_i L_i = 0$, evaluation at ω^j yields $c_j = 0$ for every j (theorem 8.11). Being n independent vectors in the n-dimensional space $\mathbb{F}[X]_{<n}$ (theorem 9.19), they form a basis (theorem 8.17). □

The root-count step deserves a name, for it recurs: to prove two polynomials of degree $< n$ equal, show they agree on all n points of H. We use this *root-count device* for the uniqueness above, for theorem 9.24, and for Example 9.25 below.

On a generic point set the Lagrange polynomials admit no closed form simpler than the defining product. On the *group* H they collapse to a compact expression involving the vanishing polynomial.

Theorem 9.22 (Closed form of the Lagrange basis on H)**.** *For $H = \mu_n(\mathbb{F})$ with primitive root ω,*

and for each i,

$$L_i(X) \;=\; \frac{\omega^i}{n} \cdot \frac{X^n - 1}{X - \omega^i} \;=\; \frac{\omega^i\,(X^n - 1)}{n\,(X - \omega^i)}.$$

In particular, dividing out one linear factor and scaling yields all n Lagrange polynomials from the single *polynomial $Z_H(X) = X^n - 1$.*

Proof. Fix i. From theorem 9.13, $Z_H(X) = \prod_{j=0}^{n-1}(X - \omega^j)$, so

$$\frac{Z_H(X)}{X - \omega^i} = \prod_{\substack{j=0\\ j\neq i}}^{n-1}(X - \omega^j),$$

which is exactly the numerator of L_i in theorem 9.20. It remains to evaluate the denominator $\prod_{j\neq i}(\omega^i - \omega^j)$. Writing $Z_H = (X - \omega^i)\,g$ with $g = \prod_{j\neq i}(X - \omega^j)$, the product rule (theorem 5.38) gives $Z_H' = g + (X - \omega^i)\,g'$, so

$$Z_H'(\omega^i) = g(\omega^i) = \prod_{\substack{j=0\\ j\neq i}}^{n-1}(\omega^i - \omega^j)$$

—the derivative at a simple root equals the product of the differences to the other roots, the computation already made for simple roots in the proof of theorem 5.40. Concretely $Z_H'(X) = nX^{n-1}$, so

$$Z_H'(\omega^i) = n\,(\omega^i)^{n-1} = n\,\omega^{in}\,\omega^{-i} = n \cdot 1 \cdot \omega^{-i} = \frac{n}{\omega^i},$$

using $\omega^{in} = (\omega^n)^i = 1$. Therefore

$$L_i(X) = \frac{Z_H(X)/(X - \omega^i)}{Z_H'(\omega^i)} = \frac{(X^n - 1)/(X - \omega^i)}{n/\omega^i} = \frac{\omega^i\,(X^n - 1)}{n\,(X - \omega^i)},$$

as claimed. □

Remark 9.23 (Barycentric form). The closed form renders the *barycentric* interpolation formula on H especially clean. Writing $f(X) = \sum_i v_i L_i(X)$ and substituting theorem 9.22,

$$f(X) = \frac{X^n - 1}{n} \sum_{i=0}^{n-1} \frac{\omega^i\,v_i}{X - \omega^i}.$$

Evaluating f at a point $z \notin H$ thus costs one evaluation of $z^n - 1$ and n terms of the sum, with no per-point polynomial reconstruction. A verifier can use this formula when the values v_i are known, as for public-input polynomials; a commitment to hidden values does not by itself permit this evaluation.

Proposition 9.24 (Lagrange polynomials sum to one)**.** *The Lagrange basis polynomials for H satisfy $\sum_{i=0}^{n-1} L_i(X) = 1$ identically.*

Proof. The polynomial $g(X) = \sum_i L_i(X) - 1$ lies in $\mathbb{F}[X]_{<n}$ and vanishes at every ω^j, because $\sum_i L_i(\omega^j) = \sum_i \delta_{ij} = 1$ (theorem 9.21). By the root-count device—a polynomial of degree $< n$ with n distinct roots is zero (theorem 5.18)—$g \equiv 0$. □

Example 9.25 (The all-ones value vector)**.** The constant polynomial 1 interpolates the data $v_i = 1$ for all i; theorem 9.24 is the statement $\sum_i 1 \cdot L_i = 1$. Likewise the values of X on H are $v_i = \omega^i$, and, for $n \geq 2$, theorem 9.21 gives $\sum_i \omega^i L_i(X) = X$: both sides have degree $< n$ and agree on all of H, so the root-count device applies. We invoke these identities constantly when rewriting monomials in the Lagrange basis.

9.5 Evaluation and interpolation as mutually inverse linear maps

An element of $\mathbb{F}[X]_{<n}$ admits two natural descriptions: by its n coefficients, or by its n values on H. Both are coordinate systems on the same n-dimensional vector space, and passage between them is a linear isomorphism—the specialisation to H of the evaluation isomorphism met for general nodes in theorem 5.32.

Definition 9.26 (Evaluation map)**.** The *evaluation map* on H is the $\mathbb{F}$-linear map (theorem 8.18)

$$\mathrm{ev}_H \colon \mathbb{F}[X]_{<n} \longrightarrow \mathbb{F}^n, \qquad \mathrm{ev}_H(f) = (f(\omega^0), f(\omega^1), \dots, f(\omega^{n-1})).$$

Linearity is immediate from $(af + bg)(\omega^i) = af(\omega^i) + bg(\omega^i)$, an instance of theorem 5.16.

Theorem 9.27 (Evaluation and interpolation are mutually inverse)**.** *The evaluation map* ev_H *is a vector-space isomorphism. Its inverse is the* interpolation map

$$\mathrm{int}_H \colon \mathbb{F}^n \longrightarrow \mathbb{F}[X]_{<n}, \qquad \mathrm{int}_H(v_0, \dots, v_{n-1}) = \sum_{i=0}^{n-1} v_i L_i(X).$$

That is, $\mathrm{int}_H \circ \mathrm{ev}_H = \mathrm{id}_{\mathbb{F}[X]_{<n}}$ *and* $\mathrm{ev}_H \circ \mathrm{int}_H = \mathrm{id}_{\mathbb{F}^n}$.

Proof. Both maps are linear and act between spaces of the same finite dimension n, so verifying one composition would suffice (a one-sided inverse makes ev_H injective, hence surjective by rank–nullity, theorem 8.19); we check both for clarity. For $v \in \mathbb{F}^n$, the polynomial $\mathrm{int}_H(v) = \sum_i v_i L_i$ has value $\sum_i v_i L_i(\omega^j) = v_j$ at ω^j by theorem 9.21, so $\mathrm{ev}_H(\mathrm{int}_H(v)) = v$. For $f \in \mathbb{F}[X]_{<n}$, the polynomial $\mathrm{int}_H(\mathrm{ev}_H(f)) = \sum_i f(\omega^i) L_i$ is, by the uniqueness clause of theorem 9.21, the unique degree-$< n$ polynomial taking the values $f(\omega^i)$ on H—which is f itself. Hence the two maps are inverse; in particular ev_H is bijective and linear, i.e. an isomorphism (theorem 8.18). □

In coordinates, ev_H is given by a matrix, which we name here because it is the bridge to the Fourier transform. The notation is worth fixing once, since §8 did not need it: an $n \times n$ *matrix* over $\mathbb{F}$ is a doubly indexed family $A = (A_{ik})_{0 \le i,k \le n-1}$ of elements of $\mathbb{F}$, acting on a vector $c \in \mathbb{F}^n$ by $(Ac)_i = \sum_k A_{ik} c_k$—the general form, written out through the standard basis (Example 8.15), of a linear map $\mathbb{F}^n \to \mathbb{F}^n$. The product of two matrices is defined so that $(AB)\,c = A(Bc)$, which forces $(AB)_{ij} = \sum_k A_{ik} B_{kj}$; the *identity matrix* I, with entries δ_{ik}, acts as the identity map; and A is *invertible* if some matrix A^{-1} satisfies $A^{-1}A = AA^{-1} = I$.

Definition 9.28 (Vandermonde / DFT matrix)**.** Let $f(X) = \sum_{k=0}^{n-1} c_k X^k$ have coefficient vector $c = (c_0, \dots, c_{n-1})$. Then $\mathrm{ev}_H(f)$ has i-th entry $f(\omega^i) = \sum_k c_k\, \omega^{ik}$. Thus, relative to the monomial basis on $\mathbb{F}[X]_{<n}$ and the standard basis on $\mathbb{F}^n$, the *Vandermonde matrix*

$$V = (\omega^{ik})_{0 \le i,k \le n-1} = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(n-1)} \\ \vdots & & & & \vdots \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & \cdots & \omega^{(n-1)^2} \end{pmatrix}$$

—rows indexed by the evaluation point ω^i, columns by the power k—represents ev_H, so that $\mathrm{ev}_H(f) = Vc$. We call V the (discrete Fourier) transform matrix on H.

Theorem 9.29 (Invertibility and the inverse transform)**.** *The matrix V of theorem 9.28 is invertible, with*

$$(V^{-1})_{k,i} = \frac{1}{n}\,\omega^{-ik}.$$

Equivalently, $V^{-1} = \frac{1}{n}\overline{V}$, where $\overline{V} = (\omega^{-ik})_{i,k}$ is the transform matrix built from ω^{-1} (itself a primitive n-th root of unity). Consequently the coefficient and value representations are related by the mutually inverse formulas

$$c_k = \frac{1}{n}\sum_{i=0}^{n-1} f(\omega^i)\,\omega^{-ik}, \qquad f(\omega^i) = \sum_{k=0}^{n-1} c_k\,\omega^{ik}.$$

Proof. First note that $\omega^{-1} = \omega^{n-1}$ has order n by Corollary 3.23 (with $\gcd(n-1, n) = 1$), so $\overline{V}$ is again a matrix of the shape in theorem 9.28. We show $\overline{V}V = nI$. The (k, k') entry of $\overline{V}V$ is

$$\sum_{i=0}^{n-1} \omega^{-ik}\,\omega^{ik'} = \sum_{i=0}^{n-1} \omega^{i(k'-k)}.$$

By theorem 9.16 this sum is n if $n \mid (k' - k)$ and 0 otherwise; for $k, k' \in \{0, \dots, n-1\}$ we have $n \mid (k' - k) \iff k = k'$. Hence $\overline{V}V = nI$, and the same computation with the roles of the two factors exchanged gives $V\overline{V} = nI$. Since $n \neq 0$ in $\mathbb{F}$ (theorem 9.13), we may divide by n: $V^{-1} = \frac{1}{n}\overline{V}$. The component formulas write out the matrix identities $\mathrm{ev}_H(f) = Vc$ and $c = V^{-1}\,\mathrm{ev}_H(f)$. □

Remark 9.30. Theorem 9.27 and theorem 9.29 are two faces of one statement. The abstract face: evaluation on H is an isomorphism whose inverse is Lagrange interpolation. The concrete face: that isomorphism is the matrix V, and its inverse is, up to the scalar $1/n$, the same matrix with ω replaced by ω^{-1}. The near-symmetry between a transform and its inverse—differing only by the substitution $\omega \mapsto \omega^{-1}$ and a scale $1/n$—is the algebraic reason the fast inverse transform of theorem 9.37 costs exactly as much as the forward one.

9.6 The number-theoretic transform and convolution

The map ev_H, read as the matrix V, is the discrete Fourier transform, but over a finite field rather than over $\mathbb{C}$. Because it involves no analytic structure—no complex exponentials, no convergence—it is called the *number-theoretic transform*.

Definition 9.31 (Number-theoretic transform)**.** Let ω be a primitive n-th root of unity in $\mathbb{F}$. The *number-theoretic transform* (NTT) sends the coefficient vector of a polynomial to its value vector on H: it is the evaluation map ev_H written in coordinates. Explicitly, the NTT of $a = (a_0, \dots, a_{n-1}) \in \mathbb{F}^n$ is $\hat{a} = (\hat{a}_0, \dots, \hat{a}_{n-1})$ with

$$\hat{a}_j = \sum_{k=0}^{n-1} a_k\,\omega^{jk}, \qquad j = 0, \dots, n-1,$$

each entry being the value at ω^j of the polynomial with coefficients a. The *inverse number-theoretic transform* is interpolation—the recovery of the coefficients from the values—and by theorem 9.29 it too is a single explicit sum:

$$a_k = \frac{1}{n}\sum_{j=0}^{n-1} \hat{a}_j\,\omega^{-jk}.$$

The NTT enjoys the same convolution property as the classical DFT, and this is the mechanism underlying fast polynomial multiplication. Recall that the coefficients of a product are a *convolution*: with $f = \sum_i a_i X^i$ and $g = \sum_j b_j X^j$, the coefficient of X^k in fg is $\sum_{i+j=k} a_i b_j$ (theorem 5.3). The *cyclic* variant folds indices modulo n: for $a, b \in \mathbb{F}^n$, the cyclic convolution $a * b \in \mathbb{F}^n$ has entries

$$(a * b)_k = \sum_{\substack{0 \le i,j \le n-1 \\ i+j \equiv k \pmod{n}}} a_i b_j, \qquad k = 0, \dots, n-1.$$

Theorem 9.32 (Convolution theorem). *For $f, g \in \mathbb{F}[X]_{<n}$ with $\deg f + \deg g < n$, let $c = fg$ be their product (of degree $< n$). Then for every $h \in H$,*

$$c(h) = f(h)\, g(h),$$

i.e. pointwise multiplication of value vectors corresponds to polynomial multiplication. Equivalently, writing $a, b \in \mathbb{F}^n$ for the coefficient vectors of f, g (padded with zeros up to length n),

$$\widehat{a * b}_j = \hat{a}_j \hat{b}_j \qquad (j = 0, \dots, n-1),$$

so the NTT turns convolution into entrywise multiplication.

Proof. The first identity is the statement that evaluation at h is a ring homomorphism $\mathbb{F}[X] \to \mathbb{F}$ (theorem 5.16): $(fg)(h) = f(h)\, g(h)$. Since $\deg(fg) < n$, interpolation recovers the product $c = fg$ from its values on the n-point set H (theorem 9.21), so the entrywise product of the value vectors uniquely determines c.

For the convolution statement the plan is: (i) multiplication in $\mathbb{F}[X]/(X^n - 1)$ *is* cyclic convolution of coefficient vectors; (ii) evaluation on H is a ring isomorphism from that quotient onto $\mathbb{F}^n$ with entrywise operations; (iii) the degree bound rules out wrap-around, so the cyclic product is the genuine one. For (i), work in the quotient ring $\mathbb{F}[X]/(X^n - 1)$ (theorem 4.19 and theorem 4.20, applied to the ideal of multiples of $X^n - 1$, theorem 4.17). Every class has a unique representative of degree $< n$, by division with remainder (theorem 5.12); and multiplying two such representatives and reducing replaces each X^k with $k \ge n$ by X^{k-n}, since $X^n \equiv 1$. The coefficient of X^k in the reduced product of the representatives with coefficient vectors a and b is therefore $\sum_{i+j \equiv k \pmod{n}} a_i b_j = (a * b)_k$: multiplication in $\mathbb{F}[X]/(X^n - 1)$ *is* cyclic convolution of coefficient vectors.

Now consider the map $[p] \mapsto \mathrm{ev}_H(p)$ from $\mathbb{F}[X]/(X^n - 1)$ to $\mathbb{F}^n$, where $\mathbb{F}^n$ is a commutative ring under entrywise addition and multiplication (the ring laws hold in each coordinate because they hold in $\mathbb{F}$), with multiplicative identity $(1, \dots, 1)$. The map is well defined: two representatives of a class differ by a multiple of $X^n - 1$, which vanishes on H (theorem 9.13). It is a ring homomorphism (theorem 4.21), because evaluation at each point of H is one (theorem 5.16) and the operations on $\mathbb{F}^n$ are entrywise. And it is bijective: classes correspond to their degree-$< n$ representatives, on which ev_H is a bijection by theorem 9.27. Hence it is a ring isomorphism $\mathbb{F}[X]/(X^n - 1) \xrightarrow{\sim} \mathbb{F}^n$, and it sends the product of the classes of f and g—coefficient vector $a * b$—to the entrywise product of $\hat{a}$ and $\hat{b}$. This is the displayed identity. Finally, when $\deg f + \deg g < n$, the genuine product fg already has degree $< n$, so no folding occurs, the cyclic and ordinary products coincide, and the statement computes fg itself. □

Remark 9.33 (Computational significance). Theorem 9.32 reduces multiplying two degree-$< n/2$ polynomials—a quadratic-cost operation by the schoolbook method—to: transform both (NTT), multiply the value vectors entrywise (n multiplications), and transform back (inverse NTT). If the NTT runs in $O(n \log n)$ operations, the whole multiplication costs $O(n \log n)$. The fast Fourier transform supplies that conditional.

Example 9.34 (The convolution theorem in $\mathbb{F}_{17}$)**.** In $\mathbb{F}_{17}$, the element $\omega = 4$ has $\omega^2 = 16 = -1$, so it is a primitive 4-th root of unity with domain $H = \{1, 4, 16, 13\}$. To multiply $f = 1 + 2X$ and $g = 3 + X$, pad both coefficient vectors to length 4 and transform: $\hat{a} = (3, 9, 16, 10)$ and $\hat{b} = (4, 7, 2, 16)$, the values of f and of g on H. Multiplying entrywise gives $(12, 12, 15, 7)$, the values of fg on H, and the inverse transform returns $(3, 7, 2, 0)$: the coefficients of $fg = 3 + 7X + 2X^2$, agreeing with the schoolbook product. At $n = 4$ nothing is saved; the point is the shape of the pipeline, which becomes decisive once the transforms themselves cost $O(n \log n)$ (§9.7).

9.7 The fast Fourier transform

Computing $\hat{a} = Va$ directly from theorem 9.28 costs $\Theta(n^2)$ field multiplications: each of the n outputs is a sum of n products. The fast Fourier transform (FFT)—here a number-theoretic FFT—computes the same vector in $O(n \log n)$ operations by a divide-and-conquer recursion, provided n is a power of two (or, more generally, smooth). This is where the requirement that H have smooth order, theorem 9.10, becomes indispensable.

Theorem 9.35 (Cooley–Tukey radix-2 step)**.** *Suppose $n = 2m$ and ω is a primitive n-th root of unity in $\mathbb{F}$. Given $f(X) = \sum_{k=0}^{n-1} a_k X^k$, split its coefficients by parity into*

$$f_{\text{even}}(Y) = \sum_{k=0}^{m-1} a_{2k}\, Y^k, \qquad f_{\text{odd}}(Y) = \sum_{k=0}^{m-1} a_{2k+1}\, Y^k,$$

so that $f(X) = f_{\text{even}}(X^2) + X f_{\text{odd}}(X^2)$. Then $\eta := \omega^2$ is a primitive m-th root of unity, and for $j = 0, \dots, m-1$,

$$f(\omega^j) = f_{\text{even}}(\eta^j) + \omega^j f_{\text{odd}}(\eta^j), \tag{7}$$

$$f(\omega^{j+m}) = f_{\text{even}}(\eta^j) - \omega^j f_{\text{odd}}(\eta^j). \tag{8}$$

Thus the length-n NTT of a reduces to two length-m NTTs (of the even and odd coefficient subsequences, over the domain $\langle \eta \rangle$) plus $O(n)$ extra operations.

Proof. The identity $f(X) = f_{\text{even}}(X^2) + X f_{\text{odd}}(X^2)$ regroups the sum $\sum_k a_k X^k$ into even-index and odd-index terms; in the odd terms one factor of X is pulled out, leaving powers X^{2k}. That $\eta = \omega^2$ is a primitive m-th root of unity follows from theorem 9.5 and the order-of-a-power formula (Corollary 3.23): $\operatorname{ord}(\omega^2) = n/\gcd(2, n) = n/2 = m$, using $2 \mid n$.

It remains to evaluate. For (7), put $X = \omega^j$; then $X^2 = \omega^{2j} = \eta^j$, so $f(\omega^j) = f_{\text{even}}(\eta^j) + \omega^j f_{\text{odd}}(\eta^j)$. For (8), put $X = \omega^{j+m}$; then $X^2 = \omega^{2j+2m} = \omega^{2j}\,\omega^n = \eta^j \cdot 1 = \eta^j$ (since $2m = n$ and $\omega^n = 1$), while the linear factor is $\omega^{j+m} = \omega^j\,\omega^m = -\omega^j$. Here $\omega^m = -1$, because ω^m has order $n/\gcd(m, n) = n/m = 2$ (Corollary 3.23), and -1 is the unique element of order 2 in a field: $x^2 = 1$ with $x \neq 1$ forces $(x-1)(x+1) = 0$ and so $x = -1$; and $-1 \neq 1$, since $\operatorname{char} \mathbb{F} \nmid n$ (theorem 9.13) with n even forces $\operatorname{char} \mathbb{F} \neq 2$. Substituting, $f(\omega^{j+m}) = f_{\text{even}}(\eta^j) - \omega^j f_{\text{odd}}(\eta^j)$. □

The half-domain trick just used—$\omega^m = -1$ whenever $n = 2m$, so the second half of the outputs differs from the first only by a sign on the odd part—recurs in every radix-2 argument over an evaluation domain; it is worth remembering alongside the theorem.

Remark 9.36 (The butterfly). Equations (7)–(8) constitute the *butterfly*: from the two subtransform outputs $E_j = f_{\text{even}}(\eta^j)$ and $O_j = f_{\text{odd}}(\eta^j)$ one forms the *twiddle* $t_j = \omega^j O_j$ and reads off the two outputs $E_j + t_j$ and $E_j - t_j$. Pictorially, E_j and O_j sit as two nodes on the left; each is carried to both outputs on the right, the two E_j-edges with weight 1 and the two O_j-edges with weights $+\omega^j$ and $-\omega^j$; the pair of crossing edges forms the × that names the operation. Each butterfly costs one multiplication (the twiddle) and two additions, and there are $n/2$ of them per level.

Theorem 9.37 ($O(n \log n)$ cost of the FFT)**.** *Let $n = 2^t$ and let $T(n)$ be the number of field additions and multiplications the radix-2 recursion of theorem 9.35 uses to compute a length-n NTT. Then*

$$T(n) = 2\,T(n/2) + cn \quad (n > 1), \qquad T(1) = O(1),$$

for a constant c, and therefore $T(n) = O(n \log n)$. The inverse NTT has the same cost, since it is the forward NTT with ω replaced by ω^{-1}, followed by scaling each entry by $1/n$ (theorem 9.29).

Proof. The recursion is exactly theorem 9.35: a length-n transform calls two length-$n/2$ transforms (on the even and odd coefficient subsequences) and combines them with $n/2$ butterflies of $O(1)$ operations each, for the cn overhead. Unrolling, with $n = 2^t$,

$$T(2^t) = 2\,T(2^{t-1}) + c\,2^t.$$

Divide by 2^t and set $S(t) = T(2^t)/2^t$: then $S(t) = S(t-1) + c$, so $S(t) = S(0) + ct = O(1) + ct$, and $T(2^t) = 2^t S(t) = O(2^t) + ct\,2^t = O(t\,2^t)$. With $t = \log_2 n$ this is $T(n) = O(n \log n)$. (Equivalently: the recursion tree has $\log_2 n$ levels, each performing $\Theta(n)$ total work across all subproblems at that level.) The inverse-transform claim is theorem 9.29: the inverse is the same Cooley–Tukey recursion with ω^{-1} in place of ω (also a primitive n-th root of unity), plus a final pass of n divisions by n, all within $O(n \log n)$. □

Remark 9.38 (Mixed radix and smoothness). The hypothesis that enables fast transforms is that n be *smooth*—a product of small primes—and, crucially, that the field actually *contain* a primitive n-th root of unity, i.e. $n \mid q - 1$ (theorem 9.7). A power-of-two n dividing $q - 1$ is the optimal case: pure radix-2 with the cleanest butterfly.

Example 9.39 (A length-4 transform)**.** Take $n = 4$ with primitive 4-th root ω (so $\omega^2 = -1$ and $\omega^4 = 1$, and $\eta = \omega^2 = -1$ is the primitive 2nd root). For $f = a_0 + a_1 X + a_2 X^2 + a_3 X^3$, the two length-2 subtransforms over the domain $\{1, -1\}$—each itself a single butterfly, with twiddle $\eta^0 = 1$—are

$$E = (a_0 + a_2,\ a_0 - a_2), \qquad O = (a_1 + a_3,\ a_1 - a_3),$$

and the butterflies give

$$\hat{a}_0 = E_0 + \omega^0 O_0, \quad \hat{a}_2 = E_0 - \omega^0 O_0, \quad \hat{a}_1 = E_1 + \omega^1 O_1, \quad \hat{a}_3 = E_1 - \omega^1 O_1.$$

(Note the output indexing: outputs j and $j+m$ pair up, per (7)–(8).) This uses 2 subtransforms of 2 additions each plus 2 butterflies of 2 additions each ($n/2 = 2$ per level, theorem 9.36)—together 2 twiddle multiplications and 8 additions—versus 16 coefficient-products and 12 additions for the direct 4×4 matrix multiply; and the gap widens as n grows.

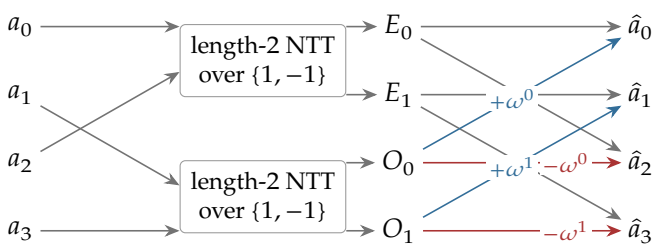


Figure 2: The length-4 transform of theorem 9.39 as a circuit. The inputs split by parity into two length-2 subtransforms, whose outputs E_0, E_1 and O_0, O_1 feed the two butterflies of theorem 9.36: each E_j is carried to outputs $\hat{a}_j$ and $\hat{a}_{j+2}$ with weight 1, each O_j with weights $+\omega^j$ and $-\omega^j$ (eq. (7)–eq. (8)); the crossing pair of weighted edges is the × that names the butterfly.

9.8 The role of smooth multiplicative domains in polynomial proof systems

This section closes by assembling the pieces into the reason these domains form the substrate of Halo 2 and Orchard. A proof system of the polynomial type encodes a computation as a system of polynomial identities and convinces a verifier that those identities hold. The evaluation domain $H = \mu_n$ is where the encoding lives—the n-step table of the section's opening is laid out on the n points of H—and every property proved above is load-bearing.

Remark 9.40 (The roles played by H, Z_H, the Lagrange basis, and the FFT).

- **Witnesses as evaluations on H.** A prover lays out the trace of a computation as the values of one or more polynomials on the n points of H. By theorem 9.27 this is the same data as a polynomial of degree $< n$; the prover may move freely between the value (Lagrange) representation, natural for stating constraints row by row, and the coefficient representation, natural for committing and for low-degree testing. The change of representation is the (inverse) NTT.

- **Constraints as divisibility by Z_H.** "Constraint polynomial C holds on every row" means C vanishes on all of H, which by theorem 9.13 means exactly that $Z_H = X^n - 1$ divides C. The prover demonstrates this by exhibiting a quotient t with $C(X) = t(X)\,(X^n - 1)$. The inexpensive closed form $Z_H(X) = X^n - 1$—one exponentiation to evaluate—is what makes this check cheap at a random point. Theorem 9.14 extends the same technique to cosets γH, used to separate quotient and remainder domains.

- **Lagrange selectors.** The Lagrange polynomials L_i of theorem 9.20 are *selectors*: L_i is 1 on row i and 0 elsewhere (theorem 9.21), so boundary and instance constraints (e.g. "the first row equals the public input") become $L_0 \cdot (\dots)$. The closed form and the barycentric formula (theorem 9.22, theorem 9.23) let the verifier evaluate these at a random challenge in $O(n)$ or even $O(\log n)$ field operations rather than reconstructing polynomials.

- **Group structure for permutations and shifts.** Because H is a cyclic *group* (theorem 9.11), multiplication by ω is the cyclic "next-row" shift $\omega^i \mapsto \omega^{i+1}$. Constraints relating a row to its neighbour become $f(\omega X)$ versus $f(X)$; permutation arguments—the heart of Halo 2's copy constraints—rest on the action of the group on itself and on its cosets. An arbitrary set of n points offers none of this.

- **Smoothness for speed.** Provers manipulate polynomials of size n (often after blowing the domain up by a small constant factor for low-degree testing). Every transform between representations, every multiplication, every coset evaluation is an NTT. By theorem 9.37 these cost $O(n \log n)$ instead of $O(n^2)$—the difference between feasible and infeasible at the sizes used in practice—*provided* n is smooth and $\mathbb{F}_q$ contains a primitive n-th root of unity. By theorem 9.7 that requires $n \mid q - 1$; by theorem 9.10 this is why high-two-adicity primes are chosen for the scalar fields of the Pallas and Vesta curves (section 10).

In summary: the existence condition $n \mid q - 1$ (theorem 9.7) supplies the group H; the group supplies the clean vanishing polynomial (theorem 9.13), the closed-form Lagrange basis (theorem 9.22), and the evaluation/interpolation isomorphism (theorem 9.27); and smoothness of n supplies the $O(n \log n)$ FFT (theorem 9.37). Together these discharge both demands of the section's opening—a structured point set on which an execution trace can live, and $O(n \log n)$ multiplication and interpolation on it—and they are exactly the ingredients a polynomial proof system requires, which is why such systems rest on smooth multiplicative evaluation domains and on the fields that contain them.

10 Elliptic curves

The multiplicative groups of the preceding sections harbour a structural weakness. As theorem 7.22 recorded, the discrete logarithm in $\mathbb{F}_p^\times$ does not resist attack at the generic square-root cost: the *index-calculus* family of algorithms exploits the fact that residues come from ordinary integers, and integers factor into primes, to solve the problem in subexponential time. The practical consequence is severe. To push the cheapest known attack on $\mathbb{F}_p^\times$ beyond 2^{128} operations, the prime p must run to *thousands* of bits, and every exponentiation drags that bulk along. What cryptography wants instead is a group of roughly 2^{255} elements for which no attack faster than the square-root bound is known—a group without known exploitable arithmetic behind its elements, where the generic $O(\sqrt{N})$ attacks of theorem 7.22 remain the best known. This section constructs the standard supplier of such groups.

An *elliptic curve* is, concretely, the solution set of a certain cubic equation in two variables, together with one extra "point at infinity". Its notable feature is that this solution set carries the structure of an abelian group, defined by a strikingly geometric rule; and that inverting scalar multiplication in this group is hard—the discrete logarithm problem in a well-chosen elliptic-curve group appears to require exponential time. The combination of clean algebraic structure with a hard computational problem is exactly what cryptography requires.

Throughout the section we work over a field K whose characteristic (theorem 4.30) is neither 2 nor 3. In the applications K is a finite prime field $\mathbb{F}_p$ with p a large prime, but the basic theory is cleaner over a general field, and we specialise when needed. The reader should keep in mind the two *Pasta curves*

$$E_p : y^2 = x^3 + 5 \quad \text{over } \mathbb{F}_p, \qquad E_q : y^2 = x^3 + 5 \quad \text{over } \mathbb{F}_q,$$

where p and q are the two 255-bit primes

$$p = 2^{254} + 0x224698fc094cf91b992d30ed00000001,$$
$$q = 2^{254} + 0x224698fc0994a8dd8c46eb2100000001.$$

These are the primes that theorem 6.23 and theorem 7.18 handled as black boxes; here they finally acquire their curves. The constants are the deployed ones: the primes and the curve equations $y^2 = x^3 + 5$ are specified in §5.4.9.6 ("Pallas and Vesta") of the Zcash protocol specification. The curve E_p, defined over $\mathbb{F}_p$, is called *Pallas*; the curve E_q, defined over $\mathbb{F}_q$, is called *Vesta*. Together they form a *cycle*: the number of $\mathbb{F}_p$-points of Pallas equals q, and the number of $\mathbb{F}_q$-points of Vesta equals p—a coincidence of design, not chance, whose meaning unfolds across the section. The equation $y^2 = x^3 + 5$ serves as the running example throughout, with a toy curve over $\mathbb{F}_{11}$ developed in parallel for hand computation.

10.1 Weierstrass equations

Definition 10.1 (Short Weierstrass equation)**.** Let K be a field of characteristic $\neq 2, 3$, and let $a, b \in K$. The *short Weierstrass equation* with coefficients a, b is the equation

$$y^2 = x^3 + ax + b. \tag{9}$$

The elements a and b are the *Weierstrass coefficients*.

Over a field of characteristic $\neq 2, 3$, an invertible affine change of variables always reduces the most general cubic of this shape, $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$, to the form (9): completing the square in y (possible since $2 \neq 0$) removes the xy and y terms but changes the quadratic coefficient to $a_2 + a_1^2/4$; translating with the substitution $x = X - (a_2 + a_1^2/4)/3$ (possible since $3 \neq 0$) removes the X^2 term. This is precisely why characteristics 2 and 3 are excluded: those

cases require the longer Weierstrass form, and they are not relevant to the Pasta curves, which live over large prime fields.

Not every choice of a, b gives a usable curve. We must exclude the cubics whose graph has a self-intersection or a cusp, because the group law defined below breaks down at such a singular point.

Definition 10.2 (Discriminant and nonsingularity)**.** The *discriminant* of the short Weierstrass equation (9) is

$$\Delta = -16\,(4a^3 + 27b^2) \in K. \tag{10}$$

The equation, and the cubic $f(x) = x^3 + ax + b$, is *nonsingular* (or *smooth*) if $\Delta \neq 0$, equivalently $4a^3 + 27b^2 \neq 0$. (The factor -16 is a convention inherited from the general Weierstrass form; over our fields it is nonzero and affects nothing.)

The quantity $4a^3 + 27b^2$ is exactly the obstruction to f having distinct roots. The following lemma makes this explicit by a computation reusable for any depressed cubic—a cubic with no square term—and is the engine of the smoothness criterion.

Lemma 10.3 (Discriminant of a depressed cubic)**.** *Let $f(x) = x^3+ax+b = (x-r_1)(x-r_2)(x-r_3)$ with r_1, r_2, r_3 in some field containing K. Then*

$$\prod_{i<j}(r_i - r_j)^2 = -(4a^3 + 27b^2).$$

Proof. Expanding $(x-r_1)(x-r_2)(x-r_3)$ and comparing coefficients with x^3+ax+b gives Vieta's relations for the depressed cubic:

$$r_1 + r_2 + r_3 = 0, \qquad r_1r_2 + r_1r_3 + r_2r_3 = a, \qquad r_1r_2r_3 = -b.$$

Differentiating $f(x) = \prod_i (x - r_i)$ by the product rule and evaluating at a root kills every term but one: $f'(r_i) = \prod_{j\neq i}(r_i - r_j)$. Multiplying the three evaluations pairs each factor $(r_i - r_j)$ with $(r_j - r_i)$:

$$\prod_i f'(r_i) = \prod_{i<j}(r_i - r_j)(r_j - r_i) = (-1)^3 \prod_{i<j}(r_i - r_j)^2.$$

It remains to compute $\prod_i f'(r_i) = \prod_i (3r_i^2 + a)$. Expanding the product and collecting the elementary symmetric functions of the squares r_i^2,

$$\prod_i (3r_i^2 + a) = 27\prod_i r_i^2 + 9a\sum_{i<j} r_i^2 r_j^2 + 3a^2\sum_i r_i^2 + a^3.$$

Vieta's relations evaluate each symmetric function: with $e_1 = 0$, $e_2 = a$, $e_3 = -b$,

$$\sum_i r_i^2 = e_1^2 - 2e_2 = -2a, \qquad \sum_{i<j} r_i^2 r_j^2 = e_2^2 - 2e_1e_3 = a^2, \qquad \prod_i r_i^2 = e_3^2 = b^2.$$

Substituting, $\prod_i f'(r_i) = 27b^2 + 9a^3 - 6a^3 + a^3 = 4a^3 + 27b^2$, and the two displays together give $\prod_{i<j}(r_i - r_j)^2 = -(4a^3 + 27b^2)$. □

Proposition 10.4 (Smoothness, four ways)**.** *Let $f(x) = x^3 + ax + b \in K[x]$. The following are equivalent.*

1. *The quantity $4a^3 + 27b^2$ of theorem 10.2 is nonzero.*

2. *The cubic f has no repeated root in the algebraic closure $\overline{K}$—the smallest extension field of K in which every nonconstant polynomial over K factors into linear factors.*

3. *The polynomials f and f' are coprime in $\overline{K}[x]$.*

4. *There is no point $(x_0, y_0) \in \overline{K}^2$ satisfying the curve equation $y^2 = f(x)$ together with the two partial-derivative equations $2y = 0$ and $f'(x) = 0$.*

Proof. Equivalence (1) $\Leftrightarrow$ (2): over $\overline{K}$ the monic cubic factors as $f = (x - r_1)(x - r_2)(x - r_3)$, and theorem 10.3 gives $\prod_{i<j}(r_i - r_j)^2 = -(4a^3 + 27b^2)$. The left side vanishes exactly when two roots coincide, so f has a repeated root if and only if $4a^3 + 27b^2 = 0$.

Equivalence (2) $\Leftrightarrow$ (3): this is the derivative test of theorem 5.40, applied over $\overline{K}$ where f splits: a repeated root of f is a common root of f and f', hence a common factor $(x - \alpha)$, and conversely a nontrivial $\gcd(f, f')$ has a root in $\overline{K}$ which is then a repeated root of f.

Equivalence (4) $\Leftrightarrow$ (2): a singular point of the affine curve $y^2 = f(x)$ is a point where both partial derivatives of $g(x, y) = y^2 - f(x)$ vanish, namely $\partial g/\partial y = 2y = 0$ and $\partial g/\partial x = -f'(x) = 0$. Since $\mathrm{char}(K) \neq 2$, the first equation forces $y_0 = 0$; the curve equation then gives $f(x_0) = 0$, which combined with $f'(x_0) = 0$ says x_0 is a repeated root of f (theorem 5.40(1)). Conversely a repeated root x_0 of f yields the singular point $(x_0, 0)$. □

Definition 10.5 (Elliptic curve and rational points)**.** An *elliptic curve* over K is a short Weierstrass equation (9) with nonzero discriminant, $4a^3 + 27b^2 \neq 0$. For any field extension $L \supseteq K$, the set of *L-rational affine points* is

$$E^{\mathrm{aff}}(L) = \{\, (x, y) \in L \times L : y^2 = x^3 + ax + b \,\},$$

and the set of *L-rational points* is

$$E(L) = E^{\mathrm{aff}}(L) \cup \{O\},$$

where O is a single formal symbol called the *point at infinity*. When $L = K$ one writes simply E for the curve and $E(K)$ for its rational points.

Remark 10.6 (The Pasta curves are smooth). For the Pasta curves $a = 0$ and $b = 5$, so $4a^3 + 27b^2 = 27 \cdot 25 = 675 = 3^3 \cdot 5^2$. The curve is therefore nonsingular whenever the characteristic does not divide 675, i.e. whenever the prime is neither 3 nor 5; the Pasta primes are enormous, so this is automatic. More generally, any curve $y^2 = x^3 + b$ with $b \neq 0$ is nonsingular away from characteristics 2 and 3.

10.2 The point at infinity, geometrically

The extra point O is not decoration; it is the identity element of the group to come, and it has an honest geometric home. To see it, view the affine plane as sitting inside the *projective plane*. A point of the projective plane $\mathbb{P}^2(K)$ is an equivalence class $[X : Y : Z]$ of nonzero triples $(X, Y, Z) \in K^3 \setminus \{0\}$, where $(X, Y, Z) \sim (\lambda X, \lambda Y, \lambda Z)$ for every $\lambda \in K^\times$. The affine points (x, y) embed as $[x : y : 1]$; the points with $Z = 0$ form the *line at infinity*, the extra "directions". Homogenising the Weierstrass equation (9) by setting $x = X/Z$, $y = Y/Z$ and clearing denominators gives the *projective Weierstrass equation*

$$Y^2 Z = X^3 + aXZ^2 + bZ^3. \tag{11}$$

Setting $Z = 0$ in (11) forces $X^3 = 0$, hence $X = 0$, leaving the *single* projective point $[0 : 1 : 0]$. This is the point at infinity O. It lies infinitely far up in the vertical direction, and every vertical

line $x = \text{const}$ passes through it. The reader may safely picture $\mathcal{O}$ as one point pinned at the top (and bottom) of the y-axis, where all vertical lines meet; the projective equation above is the licence for that picture.

10.3 The chord-and-tangent group law: geometry

We now describe the procedure for "adding" two points of $E(K)$. The construction is purely geometric, and it rests on a single fact: a line meets a cubic curve in exactly three points, counted with multiplicity. The point at infinity is engineered precisely so that the counting is always exact.

Definition 10.7 (The group law, geometric form)**.** Let $P, Q \in E(K)$. Define $P + Q$ as follows.

- If $P = \mathcal{O}$, set $P + Q = Q$; if $Q = \mathcal{O}$, set $P + Q = P$.
- Otherwise $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ are affine. Draw the line ℓ through P and Q; if $P = Q$, let ℓ be the tangent line to the curve at P. The line ℓ meets the curve in a third point R', counted with multiplicity, where a vertical ℓ has third point $\mathcal{O}$. Define $P + Q$ to be the *reflection of R' across the x-axis*: if $R' = (x_3, -y_3)$ then $P + Q = (x_3, y_3)$, and if $R' = \mathcal{O}$ then $P + Q = \mathcal{O}$.

The reflection step deserves a comment, because it is what converts "the three intersection points of a line sum to a fixed thing" into a genuine group law with $\mathcal{O}$ as identity. The underlying rule is this: *the three intersection points of any line with the curve add up to $\mathcal{O}$.* (Stated here as a guiding picture; every use of it below is discharged by an explicit computation with the formulas of §10.4.) Reflection across the x-axis sends an affine point (x, y) to $(x, -y)$, which—as the next subsection confirms—is its additive inverse; and it sends $\mathcal{O}$ to $\mathcal{O}$. To convert the geometry into formulas one must (i) find the third intersection point of a line with the cubic and (ii) handle the degenerate cases. The key algebraic device for (i) is Vieta's relation between the roots of a cubic and its coefficients, already deployed in theorem 10.3.

10.4 Explicit affine formulas

Fix an elliptic curve $E : y^2 = x^3 + ax + b$ over K, and let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ denote affine points.

Definition 10.8 (Negation)**.** Define

$$-\mathcal{O} = \mathcal{O}, \qquad -(x_1, y_1) = (x_1, -y_1).$$

The reflected point $(x_1, -y_1)$ is again on the curve, since the curve equation depends on y only through y^2. Geometrically, the line through P and $-P$ is the vertical line $x = x_1$, which meets the curve at P, $-P$, and $\mathcal{O}$.

Construction 10.9 (Chord addition)**.** Suppose $P, Q \neq \mathcal{O}$ and $x_1 \neq x_2$. The line through P and Q is $y = \lambda x + \mu$ with

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1}, \qquad \mu = y_1 - \lambda x_1.$$

Substituting the line into the curve equation, $(\lambda x + \mu)^2 = x^3 + ax + b$, and collecting terms gives

$$g(x) := x^3 - \lambda^2 x^2 + (a - 2\lambda\mu)\, x + (b - \mu^2) = 0.$$

Both x_1 and x_2 are roots of the monic cubic g: each point lies on the line and on the curve, so $g(x_i) = f(x_i) - (\lambda x_i + \mu)^2 = f(x_i) - y_i^2 = 0$. Since $x_1 \neq x_2$, factoring out $(x - x_1)(x - x_2)$ (theorem 5.17, applied twice) leaves a monic linear factor $x - x_3$ with $x_3 \in K$: the x-coordinate of the third intersection point R'. By Vieta, the sum of the roots equals the negative of the x^2-coefficient—so $x_1 + x_2 + x_3 = \lambda^2$, and the unknown root is read off with no factoring at all. Hence

$$x_3 = \lambda^2 - x_1 - x_2, \qquad y_3 = \lambda\,(x_1 - x_3) - y_1, \tag{12}$$

and $P + Q = (x_3, y_3)$. The y-formula already incorporates the reflection across the x-axis: the third intersection point is $(x_3, \lambda x_3 + \mu)$, and reflecting gives $y_3 = -(\lambda x_3 + \mu) = -(\lambda x_3 + y_1 - \lambda x_1) = \lambda(x_1 - x_3) - y_1$.

Construction 10.10 (Tangent doubling)**.** Suppose $P = Q = (x_1, y_1)$ with $y_1 \neq 0$. Implicit differentiation of $y^2 = x^3 + ax + b$ yields the tangent slope: from $2y\,dy = (3x^2 + a)\,dx$,

$$\lambda = \frac{3x_1^2 + a}{2y_1}.$$

With $\mu = y_1 - \lambda x_1$, substituting $y = \lambda x + \mu$ into the curve equation produces the same monic cubic $g(x) = x^3 + ax + b - (\lambda x + \mu)^2$ as in theorem 10.9, and here x_1 is a *double* root. The slogan "tangency counts the intersection twice" is a one-line check: $g(x_1) = f(x_1) - y_1^2 = 0$, and

$$g'(x_1) = 3x_1^2 + a - 2\lambda\,(\lambda x_1 + \mu) = 3x_1^2 + a - 2\lambda y_1 = 0$$

by the choice of λ, so x_1 is a repeated root of g (theorem 5.40(1)). Factoring $(x - x_1)^2$ out of g leaves a monic linear factor $x - x_3$ with $x_3 \in K$, and Vieta now reads $x_1 + x_1 + x_3 = \lambda^2$, hence

$$x_3 = \lambda^2 - 2x_1, \qquad y_3 = \lambda\,(x_1 - x_3) - y_1, \qquad [2]P = (x_3, y_3). \tag{13}$$

The remaining cases are exactly those in which the line is vertical.

- If $P = \mathcal{O}$, then $P + Q = Q$; symmetrically if $Q = \mathcal{O}$.
- If $x_1 = x_2$ but $y_1 = -y_2$ (so $Q = -P$, including the subcase $y_1 = y_2 = 0$), the line is vertical, the third point is $\mathcal{O}$, and $P + Q = \mathcal{O}$.
- If $P = Q$ and $y_1 = 0$, the tangent is vertical (the slope formula in theorem 10.10 has zero denominator), so $[2]P = \mathcal{O}$. These are exactly the points of order two, studied in section 10.9.

Every case is covered: either an input is $\mathcal{O}$; or both inputs are affine with $x_1 \neq x_2$ (use (12)); or both are affine with $x_1 = x_2$, in which case either $y_1 = -y_2$ (result $\mathcal{O}$) or $y_1 = y_2 \neq 0$ (use (13)).

Proposition 10.11 (Closure)**.** *For all $P, Q \in E(K)$, the point $P + Q$ computed by the case analysis above lies in $E(K)$. In particular $E(K)$ is closed under $+$.*

Proof. If either input is $\mathcal{O}$, the result is the other input, which lies in $E(K)$ by hypothesis; if the result is $\mathcal{O}$, it lies in $E(K)$ by definition. Otherwise (x_3, y_3) is produced by (12) or (13), and in both constructions x_3 was exhibited as a root of the cubic $g(x) = x^3 + ax + b - (\lambda x + \mu)^2$—as the third root beside x_1, x_2 in theorem 10.9, and beside the double root x_1 in theorem 10.10. The equation $g(x_3) = 0$ states precisely that the third intersection point $(x_3, \lambda x_3 + \mu) = (x_3, -y_3)$ satisfies the curve equation; and since the curve equation is invariant under $y \mapsto -y$, the reflected point (x_3, y_3) is on the curve too, so $P + Q \in E^{\mathrm{aff}}(K) \subseteq E(K)$. For rationality, observe that all coordinates are rational expressions in x_1, x_2, y_1, y_2, a whose denominators are $x_2 - x_1$ or $2y_1$, nonzero in the cases where each formula is invoked; so $x_3, y_3 \in K$. □

Remark 10.12 (Complete versus incomplete addition). The chord formula (12) is *incomplete*: it fails whenever $x_1 = x_2$—the doubling and inverse cases—and an implementation must branch into the case analysis above. *Complete* addition formulas, valid for *all* input pairs including $P = Q$ and $P = -Q$, exist for the curves used here, at the cost of more field operations per addition; theorem 10.13 exhibits the deployed set and theorem 10.14 proves it complete. The distinction is load-bearing in later volumes. Halo 2's elliptic-curve gadgets provide complete addition and use incomplete gates only where their callers justify the exclusions; the Sinsemilla hash of Orchard deliberately uses the *incomplete* formulas nondeterministically—an exceptional case does not violate the constraints but leaves the affected output unconstrained (the prover may choose the intermediate slope freely), which the specification models by letting the hash return $\perp$ and weakening the proved statement accordingly; its Merkle path validity check is even permitted to pass on the resulting sentinel values. Safety rests not on the proof system but on Theorem 5.4.4 of the Zcash protocol specification: any input reaching an exceptional case of the incomplete addition immediately yields a nontrivial discrete-logarithm relation among the Sinsemilla generators, so under discrete-log hardness on Pallas (theorem 10.31) such inputs are infeasible to find, for honest and adversarial provers alike.

An implementation branches; a circuit cannot. In a proof system the group law is checked by a fixed list of polynomial identities in the input, output, and auxiliary cells of a row, evaluated on every row alike, so the case analysis of this subsection has to be folded into algebra: each branch must appear as a factor that vanishes exactly when the branch is not taken. The deployed gate achieves this with five auxiliary cells, four of them "inverse-or-zero" hints that let a polynomial test a field element for zero.

Construction 10.13 (Complete addition as deployed). Let $E : y^2 = x^3 + b$ over $\mathbb{F}_p$ with $p > 3$, and encode the point at infinity as the pair $(0, 0)$, so that every element of $E(\mathbb{F}_p)$ is a pair of field elements. For $z \in \mathbb{F}_p$ let $\mathrm{inv}_0(z)$ be z^{-1} if $z \neq 0$ and 0 if $z = 0$. For encoded inputs $P = (x_p, y_p)$ and $Q = (x_q, y_q)$ the prover fills five auxiliary cells,

$$\alpha = \mathrm{inv}_0(x_q - x_p), \qquad \beta = \mathrm{inv}_0(x_p), \qquad \gamma = \mathrm{inv}_0(x_q), \qquad \delta = \begin{cases} \mathrm{inv}_0(y_q + y_p) & \text{if } x_q = x_p, \\ 0 & \text{otherwise,} \end{cases}$$

$$\lambda = \begin{cases} (y_q - y_p)\,\alpha & \text{if } x_q \neq x_p, \\ 3x_p^2\,(2y_p)^{-1} & \text{if } x_q = x_p \text{ and } y_p \neq 0, \\ 0 & \text{otherwise,} \end{cases}$$

and an output pair (x_r, y_r): it is Q if $x_p = 0$; else P if $x_q = 0$; else $(0, 0)$ if $x_q = x_p$ and $y_q = -y_p$; else $x_r = \lambda^2 - x_p - x_q$ and $y_r = \lambda(x_p - x_r) - y_p$. The verifier accepts the row iff the twelve

identities

$$(x_q - x_p)((x_q - x_p)\lambda - (y_q - y_p)) = 0, \tag{C1}$$
$$(1 - (x_q - x_p)\alpha)(2y_p\lambda - 3x_p^2) = 0, \tag{C2}$$
$$x_p x_q (x_q - x_p)\,(\lambda^2 - x_p - x_q - x_r) = 0, \tag{C3a}$$
$$x_p x_q (x_q - x_p)\,(\lambda(x_p - x_r) - y_p - y_r) = 0, \tag{C3b}$$
$$x_p x_q (y_q + y_p)\,(\lambda^2 - x_p - x_q - x_r) = 0, \tag{C3c}$$
$$x_p x_q (y_q + y_p)\,(\lambda(x_p - x_r) - y_p - y_r) = 0, \tag{C3d}$$
$$(1 - x_p\beta)\,(x_r - x_q) = 0, \tag{C4a}$$
$$(1 - x_p\beta)\,(y_r - y_q) = 0, \tag{C4b}$$
$$(1 - x_q\gamma)\,(x_r - x_p) = 0, \tag{C5a}$$
$$(1 - x_q\gamma)\,(y_r - y_p) = 0, \tag{C5b}$$
$$(1 - (x_q - x_p)\alpha - (y_q + y_p)\delta)\,x_r = 0, \tag{C6a}$$
$$(1 - (x_q - x_p)\alpha - (y_q + y_p)\delta)\,y_r = 0 \tag{C6b}$$

hold in $\mathbb{F}_p$. These are, in this order, the twelve constraints of the gate `complete addition` in the `halo2_gadgets` crate (`src/ecc/chip/add.rs`, `create_gate`), each carrying there one further factor, a per-row cell equal to 1 on the rows where the gate applies and 0 elsewhere, which switches the list on or off; the cell assignments are those of `assign_region` in the same file, and the circuit field is the Pallas base field $\mathbb{F}_p$.

Proposition 10.14 (The gate is complete and sound)**.** *Suppose no affine point of $E(\mathbb{F}_p)$ has $x = 0$ and none has $y = 0$—for $y^2 = x^3 + b$, that b is a non-square and $-b$ a non-cube in $\mathbb{F}_p$. Then for all $P, Q \in E(\mathbb{F}_p)$, encoded as in theorem 10.13:*

1. (completeness) *the prover's cells satisfy (C1)–(C6b), and the output pair encodes $P + Q$;*
2. (soundness) *every $(\lambda, \alpha, \beta, \gamma, \delta, x_r, y_r) \in \mathbb{F}_p^7$ satisfying (C1)–(C6b) has (x_r, y_r) equal to the encoding of $P + Q$.*

This is what complete *means for a gate: the twelve identities are satisfiable for every input pair—doublings, inverse pairs, and the identity included—and the output they admit is unique, so they are the graph of the total function $+\colon E(\mathbb{F}_p) \times E(\mathbb{F}_p) \to E(\mathbb{F}_p)$, with no case left to a branch outside the constraint system.*

Proof. Under the hypothesis an encoded pair has $x = 0$ iff it is $\mathcal{O}$, and two affine points with the same x-coordinate have $y_q = \pm y_p$ with $y_p \neq 0$, so they are equal or inverse and not both. One observation drives every case: for $z \in \mathbb{F}_p$ and any cell value h, the factor $1 - zh$ equals 1 when $z = 0$, whatever h is, while for $z \neq 0$ the honest choice $h = z^{-1}$ makes it 0. A constraint of the shape $(1 - zh)\,t = 0$ therefore forces $t = 0$ exactly when $z = 0$ and is discharged by the honest hint otherwise. Five cases exhaust the input pairs.

Case $P = \mathcal{O}$. Then $x_p = 0$, so (C4a)–(C4b) force $(x_r, y_r) = (x_q, y_q)$, the encoding of $\mathcal{O} + Q = Q$; that is soundness. The prover outputs Q by the first clause, and its cells satisfy the rest: (C1) reads $x_q(x_q\lambda - y_q) = 0$ and holds for $\lambda = y_q\,\mathrm{inv}_0(x_q)$; (C2) and (C3a)–(C3d) carry a factor x_p or $2y_p\lambda - 3x_p^2$, both zero; (C5a)–(C5b) are discharged by $\gamma = x_q^{-1}$ if $Q \neq \mathcal{O}$, and read $(x_r, y_r) = (0, 0) = (x_p, y_p)$ if $Q = \mathcal{O}$; and (C6a)–(C6b) have first factor $1 - x_q\alpha - 0$, which is 0 for $\alpha = x_q^{-1}$ when $Q \neq \mathcal{O}$, while for $Q = \mathcal{O}$ the other factor $x_r = y_r = 0$ vanishes.

Case $Q = \mathcal{O}$, $P \neq \mathcal{O}$. Now $x_q = 0 \neq x_p$, so (C5a)–(C5b) force $(x_r, y_r) = (x_p, y_p)$, the encoding of P: soundness. The prover outputs P by the second clause, with $\alpha = (-x_p)^{-1}$ and $\lambda = (-y_p)\alpha = y_p/x_p$; then (C1) reads $(-x_p)(-x_p\lambda + y_p) = 0$ and holds; (C2) and (C6a)–(C6b) carry the factor $1 - (x_q - x_p)\alpha = 0$, the latter with $\delta = 0$; (C3a)–(C3d) carry $x_q = 0$; and (C4a)–(C4b) are discharged by $\beta = x_p^{-1}$.

Case P, Q affine with $x_q \neq x_p$. Constraint (C1) forces λ to be the chord slope $(y_q - y_p)/(x_q - x_p)$, and since $x_p x_q (x_q - x_p) \neq 0$, (C3a)–(C3b) force (x_r, y_r) to be the chord formula (12)—the encoding of $P+Q$, which is affine because $x_p \neq x_q$. That is soundness. The prover's λ is that slope and its output that formula, by the fourth clause; (C3c)–(C3d) contain the same two vanishing factors; (C2), (C4a)–(C4b), (C5a)–(C5b) are discharged by the honest α, β, γ; and (C6a)–(C6b) have first factor $1 - 1 - 0 = 0$, since $\delta = 0$.

Case $Q = -P$, P affine. Here $x_q = x_p$ and $y_q + y_p = 0$, so both hint-bearing products in (C6a)–(C6b) are 0 whatever α and δ are, and the constraints force $x_r = y_r = 0$: the encoding of $P + (-P) = \mathcal{O}$, which is soundness. The prover outputs $(0, 0)$ by the third clause; (C1) and (C3a)–(C3b) carry $x_q - x_p = 0$, and (C3c)–(C3d) carry $y_q + y_p = 0$; (C2), whose first factor is now 1, needs $2y_p\lambda = 3x_p^2$, which the honest $\lambda = 3x_p^2(2y_p)^{-1}$ supplies ($y_p \neq 0$); and (C4a)–(C5b) are discharged by $\beta = \gamma = x_p^{-1}$.

Case $Q = P$ affine. Again $x_q = x_p$ fixes $(x_q - x_p)\alpha = 0$, so (C2) forces λ to be the tangent slope $3x_p^2/(2y_p)$, and since $x_p x_q (y_q + y_p) = 2x_p^2 y_p \neq 0$, (C3c)–(C3d) force (x_r, y_r) to be the doubling formula (13)—the encoding of $[2]P$, affine because $y_p \neq 0$. That is soundness. The prover's λ is that slope and its output that formula, by the fourth clause; (C1) and (C3a)–(C3b) carry $x_q - x_p = 0$; (C4a)–(C5b) are discharged by $\beta = \gamma = x_p^{-1}$; and (C6a)–(C6b) have first factor $1 - 0 - 2y_p\delta = 0$ for the honest $\delta = (2y_p)^{-1}$. □

Example 10.15 (The gate on the toy curve)**.** Pallas satisfies both hypotheses of theorem 10.14: 5 is a non-square in $\mathbb{F}_p$ (theorem 7.16) and -5 is not a cube (theorem 10.26). The toy curve $y^2 = x^3 + 5$ over $\mathbb{F}_{11}$ of theorem 10.38 satisfies neither—$5 = 4^2$ gives the points $(0, \pm 4)$, and $(8, 0)$ has order 2—which makes it the right place to watch the gate work and to see the hypotheses earn their keep. For each input pair considered below, a script determines every output pair (x_r, y_r) that *some* assignment of the five auxiliary cells in $\mathbb{F}_{11}$ makes the twelve identities accept—the search over $\alpha, \beta, \gamma, \delta$ factored by which identities mention each: β only (C4), γ only (C5), α and δ only (C2) and (C6)—so soundness is checked against all provers, not only the honest one. On the nine points $\mathcal{O}$, $(4, \pm 5)$, $(5, \pm 3)$, $(6, \pm 1)$, $(10, \pm 2)$ with $x, y \neq 0$, all 81 ordered pairs—eight doublings, eight inverse pairs, seventeen involving $\mathcal{O}$—admit exactly one output, the encoding of the true sum. Three of them, worked: for $[2](5, 3)$ the honest cells are $\lambda = 3 \cdot 5^2 \cdot 6^{-1} = 9 \cdot 2 = 7$ and $\delta = 6^{-1} = 2$, giving $(x_r, y_r) = (49 - 10,\ 7(5 - 6) - 3) = (6, 1)$, the value of (13); for the inverse pair $(5, 3) + (5, 8)$ the prover supplies the same $\lambda = 7$ but $\delta = \mathrm{inv}_0(11) = 0$, and (C6a)–(C6b) force $(0, 0)$; for $\mathcal{O} + (5, 3)$ the cells are $\alpha = 5^{-1} = 9$, $\lambda = 3 \cdot 9 = 5$, $\beta = 0$, and (C4a)–(C4b) force $(5, 3)$. Off the admissible set the gate fails as the proof predicts. With $P = (0, 4)$ and $Q = (5, 3)$ the identities admit only the output $(5, 3)$—the coordinate $x_p = 0$ is read as $\mathcal{O}$—whereas $P + Q = (10, 9)$ (theorem 10.38); the pair $(0, 4) + (0, 7)$ admits no output at all, (C4b) demanding $y_r = 7$ and (C5b) $y_r = 4$; and $(8, 0) + (8, 0)$ admits none either, since (C2) reduces to $3 \cdot 8^2 = 0$, false in $\mathbb{F}_{11}$. On Pallas none of the three inputs exists, and the gate is complete.

10.5 Identity, inverses, commutativity, and associativity

The destination of the whole construction is the following theorem.

> **Theorem 10.16** (Group law)**.** *Let E be an elliptic curve over a field K. The set $E(K)$ with the chord-and-tangent operation $+$ of theorem 10.7 is an abelian group with identity $\mathcal{O}$.*

Closure is theorem 10.11. The next proposition supplies every remaining axiom except one.

Proposition 10.17 (Identity, inverses, commutativity)**.** *The operation $+$ on $E(K)$ has the following properties.*

1. *The point $\mathcal{O}$ is a two-sided identity: $P + \mathcal{O} = \mathcal{O} + P = P$ for all P.*
2. *Every P has the two-sided inverse $-P$ of theorem 10.8: $P + (-P) = \mathcal{O}$.*
3. *The operation is commutative: $P + Q = Q + P$.*

Proof. (1) is the first clause of theorem 10.7. (2): for $P = (x, y)$ with $y \neq 0$, the line through P and $-P = (x, -y)$ is vertical, so its third point is $\mathcal{O}$ and $P + (-P) = \mathcal{O}$ by the vertical-line case; for $y = 0$ one has $P = -P$ and $[2]P = \mathcal{O}$ directly by the vertical-tangent case; and $\mathcal{O} + (-\mathcal{O}) = \mathcal{O} + \mathcal{O} = \mathcal{O}$. (3): the chord through P and Q does not depend on the order in which the two points are named, and neither does the tangent in the doubling case; algebraically, the formulas (12) are symmetric under the swap $(x_1, y_1) \leftrightarrow (x_2, y_2)$—the slope λ is unchanged, and $x_3 = \lambda^2 - x_1 - x_2$ is symmetric in x_1, x_2. □

The one axiom not yet established is *associativity*, $(P + Q) + R = P + (Q + R)$, and it is the deep and surprising part of the theory: the definition gives no reason why reflecting third intersection points should be an associative operation, and a brute-force verification through the explicit formulas splinters into a large number of coordinate cases, each demanding lengthy rational-function identities. We sketch the conceptual proof, which exhibits the reason associativity holds.

Proof of associativity (sketch). The key tool is the classical *Cayley–Bacharach theorem* for plane cubics: if two cubics with no common component meet in nine points counted with intersection multiplicity, then any third cubic passing through eight of them with the required multiplicities also passes through the ninth. To prove $(P + Q) + R = P + (Q + R)$, one writes both sides via the chord-and-tangent construction and arranges the relevant intersections among the curve E and two degenerate cubics, each a product of three suitably chosen lines. Cayley–Bacharach forces the two candidate sums to coincide; its intersection-multiplicity formulation includes tangencies and coincident points. Translating "reflect across the x-axis" into "the three points of any line sum to $\mathcal{O}$" renders the bookkeeping uniform.

A complete, case-free treatment identifies $E(K)$ with the degree-zero part $\mathrm{Pic}^0(E)$ of the divisor class group of the curve via the map $P \mapsto [P] - [\mathcal{O}]$, a bijection that intertwines the two operations; the group law on $\mathrm{Pic}^0(E)$ is manifestly associative, because it is inherited from addition of divisors modulo principal divisors. Either route yields a genuine proof; both lie beyond the elementary scope of this volume, so we take associativity as established. □

Remark 10.18 (The notation $[n]P$). For $n \in \mathbb{Z}$ and $P \in E(K)$, we write $[n]P$ for the n-fold sum $P + \cdots + P$, with $[0]P = \mathcal{O}$ and $[-n]P = -([n]P)$. This makes the abelian group $E(K)$ a $\mathbb{Z}$-module: integers act on points. The map $n \mapsto [n]P$ is the *scalar multiplication* of theorem 10.20—not any coordinate-wise product.

10.6 Projective and Jacobian coordinates

The affine formulas (12) and (13) each require a field *inversion*, by $x_2 - x_1$ or $2y_1$. In a large prime field an inversion costs roughly as much as dozens of multiplications, whether computed by the extended Euclidean algorithm (theorem 6.1) or as z^{p-2} by Fermat (theorem 6.4). When a long chain of group operations arises, as in the scalar multiplication of the next subsection, it pays to defer all inversions to the very end by carrying points in a redundant coordinate system whose group law uses only additions, subtractions, and multiplications.

Remark 10.19 (Inversion-free coordinate systems). Two redundant systems serve this purpose. In (*homogeneous*) *projective coordinates*, a triple $[X : Y : Z]$ with $Z \neq 0$ represents the affine point $(X/Z, Y/Z)$ on the projective curve (11), with $\mathcal{O} = [0 : 1 : 0]$. In *Jacobian coordinates*, a triple $(X : Y : Z)$ with $Z \neq 0$ represents $(X/Z^2, Y/Z^3)$, under the equivalence $(X, Y, Z) \sim (\lambda^2 X, \lambda^3 Y, \lambda Z)$; the curve equation becomes $Y^2 = X^3 + aXZ^4 + bZ^6$. In either system the addition and doubling formulas can be rewritten with no divisions at all, at the cost of more multiplications per operation; a long chain of group operations then defers every inversion to a single final recovery of affine coordinates—compute Z^{-1} once, then the needed powers of it by multiplication. Jacobian coordinates usually yield the faster doubling and are the standard choice in cryptographic libraries, including the implementation Halo 2 uses: the `pasta_curves` crate carries its curve points in Jacobian coordinates. For curves $y^2 = x^3 + b$ with $a = 0$, the term aZ^4 in the Jacobian doubling formulas vanishes, making doubling especially cheap—one reason the Pasta curves take $a = 0$.

10.7 Scalar multiplication and double-and-add

The fundamental operation of elliptic-curve cryptography is *scalar multiplication*: given a point P and an integer $n \geq 0$, compute

$$[n]P = \underbrace{P + P + \cdots + P}_{n \text{ times}}.$$

Computing this by $n - 1$ successive additions is infeasible when n has hundreds of bits. Instead one uses the binary expansion of n, exactly as in the square-and-multiply technique of theorem 6.2; as theorem 6.3 promised, the additive rendition of that algorithm is the following.

Definition 10.20 (Double-and-add). If $n = 0$, return $\mathcal{O}$. Otherwise write $n = \sum_{i=0}^{k-1} n_i 2^i$ with digits $n_i \in \{0, 1\}$ and top digit $n_{k-1} = 1$. The *left-to-right double-and-add* algorithm computes $[n]P$ as follows: set $R \leftarrow P$; for $i = k - 2$ down to 0, set $R \leftarrow [2]R$, and if $n_i = 1$ set $R \leftarrow R + P$; return R.

Proposition 10.21. *For $n \geq 1$, double-and-add returns $[n]P$ using exactly $k - 1$ doublings and at most $k - 1$ additions, where $k = \lfloor \log_2 n \rfloor + 1$ is the bit length of n. For $n = 0$, it returns $\mathcal{O}$ without a group operation.*

Proof. The case $n = 0$ is immediate. Let $n \geq 1$, and let R_j denote the value of R after bits $n_{k-1}, \ldots, n_j$ have been incorporated. We claim $R_j = [\lfloor n/2^j \rfloor]P$, by downward induction on j. At initialization, $R_{k-1} = P = [\lfloor n/2^{k-1} \rfloor]P$ because the top bit is 1. Assuming $R_{j+1} = [\lfloor n/2^{j+1} \rfloor]P$, the next iteration computes

$$[2]R_{j+1} + [n_j]P = [\, 2\lfloor n/2^{j+1} \rfloor + n_j \,]P = [\lfloor n/2^j \rfloor]P,$$

where the last step is the digit identity $2\lfloor n/2^{j+1} \rfloor + n_j = \lfloor n/2^j \rfloor$: stripping the last bit of $\lfloor n/2^j \rfloor$ and re-appending it recovers the number. At $j = 0$ the invariant reads $R_0 = [n]P$, which is what the algorithm returns. For the counts, the loop has exactly $k - 1$ iterations, each performing one doubling and at most one addition. □

Scalar multiplication by an n-bit scalar therefore costs $O(n)$ group operations—$O(n)$ field operations up to constant factors, or $O(n \log^2 p)$ bit operations with schoolbook field arithmetic. This efficiency of the *forward* map, contrasted with the apparent hardness of the inverse problem (§10.11), is the foundation of the cryptography.

10.8 The group structure of $E(\mathbb{F}_p)$

Now specialise $K = \mathbb{F}_p$, the prime field with p elements (p prime, $p > 3$). Since $\mathbb{F}_p$ is finite, so is $E(\mathbb{F}_p)$: each of the p values of x admits at most two values of y (theorem 5.18, applied to $y^2 - f(x)$ as a polynomial in y), so with $\mathcal{O}$ there are at most $2p + 1$ points. By theorem 10.16, the set $E(\mathbb{F}_p)$ is thus a *finite abelian group*. One of the central theorems of the subject governs its order.

Theorem 10.22 (Hasse). *Let E be an elliptic curve over $\mathbb{F}_p$. Write*

$$\#E(\mathbb{F}_p) = p + 1 - t,$$

which defines *the integer t, called the* trace of Frobenius. *Then*

$$|t| \le 2\sqrt{p};$$

equivalently, $\#E(\mathbb{F}_p)$ lies in the Hasse interval $[\,p + 1 - 2\sqrt{p},\ p + 1 + 2\sqrt{p}\,]$.

Proof (sketch). The count has an exact character-sum expression. For fixed x, the number of $y \in \mathbb{F}_p$ with $y^2 = f(x)$ is $1 + \left(\frac{f(x)}{p}\right)$ when $f(x) \neq 0$—two solutions for a nonzero square, none for a non-square (theorem 7.12)—and exactly $1 = 1 + \left(\frac{0}{p}\right)$ when $f(x) = 0$, using the Legendre symbol of theorem 7.14 with the convention $\left(\frac{0}{p}\right) = 0$. Summing over x,

$$\#E^{\mathrm{aff}}(\mathbb{F}_p) = \sum_{x \in \mathbb{F}_p} \left(1 + \left(\frac{x^3 + ax + b}{p}\right)\right) = p + \sum_{x \in \mathbb{F}_p} \left(\frac{x^3 + ax + b}{p}\right),$$

so $t = -\sum_x \left(\frac{x^3+ax+b}{p}\right)$. Heuristically, as x ranges over $\mathbb{F}_p$ the value $f(x)$ is a nonzero square about half the time (two points), a non-square about half the time (none), and zero occasionally (one point); the p summands ± 1 behave like a random walk, and Hasse's bound $|t| \le 2\sqrt{p}$ is exactly square-root cancellation in this character sum.

The conceptual proof realises t as the trace of the p-power *Frobenius endomorphism* $\pi\colon (x, y) \mapsto (x^p, y^p)$ of the curve—the curve-level shadow of the field automorphism of theorem 6.14. One shows that π satisfies the characteristic equation $\pi^2 - [t]\pi + [p] = 0$ in the endomorphism ring of E, and that the associated quadratic form (the degree) is positive definite; the discriminant condition $t^2 - 4p \le 0$ for the complex roots of the characteristic polynomial then yields $|t| \le 2\sqrt{p}$. The full argument requires the theory of isogenies and the Weil pairing, beyond the present scope. □

Hasse's theorem says the order of $E(\mathbb{F}_p)$ is *very close* to p: the deviation from $p + 1$ is at most $2\sqrt{p}$, a relative error of order $p^{-1/2}$. For cryptography this is crucial twice over. It guarantees with no brute-force count that the group is large—about p elements, so about 255 bits for the Pasta curves—and it anchors the point-counting algorithms (Schoof–Elkies–Atkin) that compute the *exact* order in polynomial time, which is how curve designers certify their group orders.

For the cryptographic application one wants $E(\mathbb{F}_p)$ to contain a large subgroup of *prime* order. The general structure theory of $E(\mathbb{F}_p)$ as an abelian group is not needed in this series, because the Pasta curves are designed so that $\#E(\mathbb{F}_p)$ is itself prime: a group of prime order is cyclic with no proper nontrivial subgroups (theorem 3.34), so $E(\mathbb{F}_p)$ is cyclic of prime order—the optimal case.

10.9 Torsion, the cofactor, and prime-order subgroups

Definition 10.23 (Order of a point and torsion)**.** The *order* of a point $P \in E(\mathbb{F}_p)$ is its order as a group element (theorem 3.11): the least integer $m \geq 1$ with $[m]P = \mathcal{O}$. It exists and divides $\#E(\mathbb{F}_p)$, by Lagrange's theorem (theorem 3.29 via theorem 3.30). For an integer $m \geq 1$, the *m-torsion subgroup* is

$$E[m] = \{ P \in E(\overline{\mathbb{F}_p}) : [m]P = \mathcal{O} \},$$

the points over the algebraic closure killed by m; its rational part is $E(\mathbb{F}_p)[m] = E[m] \cap E(\mathbb{F}_p)$.

Definition 10.24 (Cofactor)**.** Suppose $\#E(\mathbb{F}_p) = h \cdot r$, where r is the (large) prime to be used and h is the remaining factor. The integer h is the *cofactor*, and r is the order of the *prime-order subgroup*: a point P of order r generates a cyclic subgroup $\langle P \rangle \cong \mathbb{Z}/r\mathbb{Z}$, and this subgroup is where the cryptography lives.

A nontrivial cofactor $h > 1$ is undesirable: it admits *small-subgroup* points, of order dividing h, which can leak information or break protocol assumptions—the small-subgroup attacks of theorem 7.25. Implementations therefore either *clear the cofactor*, multiplying incoming points by h, or use prime-order curves with $h = 1$. The Pasta curves have cofactor 1: the orders $\#E_p(\mathbb{F}_p) = q$ and $\#E_q(\mathbb{F}_q) = p$ are themselves prime, so $E_p(\mathbb{F}_p) \cong \mathbb{Z}/q\mathbb{Z}$ and $E_q(\mathbb{F}_q) \cong \mathbb{Z}/p\mathbb{Z}$ are cyclic of prime order and no cofactor clearing is ever needed. This is one of the principal design advantages of the Pasta cycle for proof systems.

Proposition 10.25 (Two-torsion)**.** *The affine points of order* 2 *in* $E(\mathbb{F}_p)$ *are exactly the points* $(x_0, 0)$ *with* $f(x_0) = x_0^3 + ax_0 + b = 0$*. Hence* $E(\mathbb{F}_p)$ *has* 0, 1, *or* 3 *points of order* 2 *according as* f *has* 0, 1, *or* 3 *roots in* $\mathbb{F}_p$*; and* $\#E(\mathbb{F}_p)$ *is even if and only if* f *has a root in* $\mathbb{F}_p$*.*

Proof. A point $P = (x, y)$ has order 2 iff $P = -P$ and $P \neq \mathcal{O}$, i.e. $y = -y$, i.e. $y = 0$ (since $\mathrm{char}(\mathbb{F}_p) \neq 2$); the curve equation then forces $f(x) = 0$. Conversely each root x_0 of f in $\mathbb{F}_p$ gives the order-2 point $(x_0, 0)$.

Next, the number N of roots of f in $\mathbb{F}_p$ can only be 0, 1, or 3: two roots force a third. Suppose f has distinct roots $r_1 \neq r_2$ in $\mathbb{F}_p$. The factor theorem (theorem 5.17) gives $f = (x - r_1)h$; evaluating at r_2 yields $0 = (r_2 - r_1)\, h(r_2)$, so $h(r_2) = 0$, and a second application factors the monic cubic completely, $f = (x - r_1)(x - r_2)(x - r_3)$ with $r_3 \in \mathbb{F}_p$—explicitly $r_3 = -r_1 - r_2$ by Vieta, the depressed cubic having no x^2 term. Nonsingularity says f has no repeated root even over $\overline{\mathbb{F}_p}$ (theorem 10.4(2)), so $r_3 \notin \{r_1, r_2\}$, and f has exactly three roots (theorem 5.18 caps the count). Hence $N \in \{0, 1, 3\}$, matching the trichotomy of the statement. Together with $\mathcal{O}$ the order-2 points form the rational 2-torsion subgroup $E(\mathbb{F}_p)[2]$, which is accordingly trivial, $\mathbb{Z}/2\mathbb{Z}$, or $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

For the parity claim, consider negation $P \mapsto -P$, an *involution*—a self-inverse map—of the finite set $E(\mathbb{F}_p)$. Its fixed points are exactly $\mathcal{O}$ and the points $(x_0, 0)$; every other point pairs off with its distinct negative into a 2-element orbit. Counting the set by orbits,

$$\#E(\mathbb{F}_p) \equiv 1 + N \pmod 2,$$

where $N \in \{0, 1, 3\}$ is the number of roots of f in $\mathbb{F}_p$. Hence $\#E(\mathbb{F}_p)$ is even iff N is odd, i.e. iff $N \geq 1$, i.e. iff f has a root in $\mathbb{F}_p$. (The device is worth remembering: an involution reads off a group's cardinality modulo 2 from its fixed points alone.) □

Remark 10.26 (No two-torsion on the Pasta curves). For $y^2 = x^3 + 5$ over $\mathbb{F}_p$, the curve has a point of order 2 iff $x^3 + 5 = 0$ has a solution, i.e. iff -5 is a cube in $\mathbb{F}_p$. For the Pasta primes the order $\#E_p(\mathbb{F}_p) = q$ is odd (it is prime), so there are *no* rational 2-torsion points; consistently, -5 is not a cube in $\mathbb{F}_p$.

10.10 Base fields, scalar fields, and the Pasta cycle

Step back from the geometry and ask what a machine actually does when it works on $E(\mathbb{F}_p)$. Every operation of the group law—the slope $\lambda = (y_2 - y_1)(x_2 - x_1)^{-1}$, the squarings, and the inversions needed to add and to double—is arithmetic in $\mathbb{F}_p$, performed with exactly the prime-field algorithms of section 6.1 (extended-Euclid or Fermat inversion). The field $\mathbb{F}_p$ is called the *base field* (or coordinate field) of the curve: it is where the coordinates live.

Scalars live elsewhere. By theorem 10.22 the group $E(\mathbb{F}_p)$ is a finite abelian group of order $N = \#E(\mathbb{F}_p)$ close to p; for cryptography one selects a curve whose order has a large prime factor r, and keys and signatures reside in the subgroup of order r (theorem 10.24). A scalar multiplication $P \mapsto [s]P$ on a point of order r depends only on $s \bmod r$, since $[r]P = \mathcal{O}$ (theorem 3.12); scalars therefore live modulo r—and since r is prime, they form a *field* (theorem 4.26).

Definition 10.27 (Base field and scalar field)**.** For an elliptic curve used in cryptography, with a prime-order subgroup of order r:

- the *base field* $\mathbb{F}_p$ is the field over which the curve is defined and in which point coordinates lie;
- the *scalar field* $\mathbb{F}_r = \mathbb{Z}/r\mathbb{Z}$ is the field of scalars by which one multiplies points, i.e. the integers modulo the subgroup order r.

Both are prime fields, but in general $p \neq r$.

Remark 10.28 (Two fields at once, and the Pasta cycle). A typical elliptic-curve operation manipulates two distinct prime fields simultaneously: the exponent arithmetic runs in $\mathbb{F}_r$ while the coordinate arithmetic runs in $\mathbb{F}_p$. Halo 2's Pallas and Vesta curves (the "Pasta" pair) exploit this deliberately. They form a *cycle of curves*: the base field of one is the scalar field of the other and vice versa,

$$\#E_p(\mathbb{F}_p) = q, \qquad \#E_q(\mathbb{F}_q) = p,$$

so the scalar field of Pallas is $\mathbb{F}_q$, the base field of Vesta, and symmetrically (§5.4.9.6 of the Zcash protocol specification). This lets a proof system express the verifier of one curve's arithmetic *natively* in the field of the other, enabling the recursive proof composition at the heart of Halo 2. Moreover both primes are chosen $\equiv 1 \pmod{2^{32}}$, so by theorem 6.22 each field contains a multiplicative subgroup of order 2^{32}, furnishing the high-order roots of unity that the number-theoretic transform of section 9 needs for fast polynomial multiplication.

10.11 The elliptic-curve discrete logarithm problem

The section's opening problem now returns in its proper habitat. The discrete logarithm problem of theorem 7.21 was stated for an abstract cyclic group; the elliptic-curve instance reads as follows, in additive notation.

Definition 10.29 (ECDLP)**.** Let $P \in E(\mathbb{F}_p)$ be a point of large prime order r, generating $G = \langle P \rangle$. The *elliptic-curve discrete logarithm problem* (ECDLP) is: given P and a point $Q \in G$, find the unique integer $n \in \{0, 1, \dots, r-1\}$ with $Q = [n]P$. One writes $n = \log_P Q$.

The map $n \mapsto [n]P$ is a group isomorphism $\mathbb{Z}/r\mathbb{Z} \to G$ (theorem 3.43), so a discrete logarithm always exists and is unique; the entire content of the problem is *computational*. The forward map is fast by double-and-add (theorem 10.21); no efficient algorithm is known to invert it on a well-chosen curve. What is known is the generic pair of attacks, which transfer verbatim from theorem 7.22 to the additive setting.

Proposition 10.30 (Square-root attacks)**.** *Baby-step giant-step solves the ECDLP in a group of prime order r in $O(\sqrt{r})$ group operations and $O(\sqrt{r})$ space. Pollard's rho method matches the $O(\sqrt{r})$ bound in expectation with only $O(1)$ space, under a modelling assumption on its pseudo-random walk that is stated explicitly in the proof and is unproved for every concrete walk.*

Proof. For baby-step giant-step, set $m = \lceil \sqrt{r} \rceil$ and write the unknown logarithm as $n = im + j$ with $0 \le i, j < m$; since $m^2 \ge r$, every $n \in \{0, \dots, r-1\}$ has such a decomposition. Tabulate the *baby steps* $[j]P$ for $j = 0, \dots, m-1$ (a table of m points, built with $m-1$ additions), compute $S = [m]P$ once, and walk the *giant steps* $Q - [i]S$ for $i = 0, 1, 2, \dots$, testing each against the table. At $i = \lfloor n/m \rfloor$ the walk hits $Q - [im]P = [n - im]P = [j]P$, a table entry; the collision reveals i and j, hence $n = im + j$. Both the table and the walk cost $O(m) = O(\sqrt{r})$ group operations, and the table holds $O(\sqrt{r})$ points.

Pollard's rho eliminates the table, at the price of an explicitly heuristic analysis. The algorithm takes a pseudo-random walk on G whose every position is maintained in the form $[a]P + [b]Q$ with known coefficients a, b, started from such a combination with random coefficients; the walk's next step depends only on the current point, so once the walk revisits a point it cycles forever. A repeat is therefore detected with $O(1)$ memory: run a second copy of the walk at double speed, and the two copies meet at a common point inside the cycle within a constant factor of the first repeat time. *Modelling assumption*: the successive positions of the walk behave like independent uniform samples from G. No proof of this is known for any concrete walk—the assumption is what "well-designed" means, and experiment supports it—and the rest of the argument is conditional on it. Under the assumption, the first k positions are pairwise distinct with probability at most $e^{-k(k-1)/(2r)}$ by the birthday bound—the forward reference theorem 11.18 is quantitative machinery from the closing section—so the number T of steps to the first repeat satisfies $\Pr[T > k] \le e^{-k(k-1)/(2r)}$. The tail-sum formula (theorem 11.33) converts the tail bound into an expectation bound: with $m = \lceil \sqrt{r} \rceil$ and $r \ge 4$,

$$\mathbb{E}[T] = \sum_{k \ge 1} \Pr[T \ge k] \le m \sum_{j \ge 0} \Pr[T > jm] \le m\Big(1 + \sum_{j \ge 1} e^{-j/4}\Big) = O(\sqrt{r}),$$

where the middle step groups the indices k into blocks of length m, bounding each block by its first term, and the last uses $jm(jm-1) \ge j^2 m(m-1) \ge j^2(r - \sqrt{r}) \ge j^2 r/2$ for $j \ge 1$, so that $\Pr[T > jm] \le e^{-j^2/4} \le e^{-j/4}$, a convergent geometric tail. A repeat presents one group element with two representations, $[a]P + [b]Q = [a']P + [b']Q$. If $b' \not\equiv b \pmod{r}$, this rearranges to $[a - a']P = [b' - b]Q = [(b' - b)\, n]P$, whence

$$n \equiv (a - a')(b' - b)^{-1} \pmod{r},$$

the inverse existing because r is prime. If instead $b' \equiv b$, then $[a]P = [a']P$ forces $a \equiv a'$: the two representations coincide, the collision reveals nothing, and the walk is restarted with fresh random starting coefficients; that such degenerate collisions are rare is a further part of the heuristic, borne out in practice. Both algorithms run in time exponential in the bit length $\log_2 r$. □

Remark 10.31 (Best known attacks on well-chosen curves). For well-chosen curves, no unrestricted classical attack faster than $O(\sqrt{r})$ is known; this is not an unconditional lower bound for all classical algorithms. In the classical generic-group model, however, Shoup's matching lower bound is proved (theorem 7.22). For $r \approx 2^{254}$—the 255-bit Pasta orders—this is about 2^{127} operations, infeasible. Curves with special structure can be weaker, and the known weak classes are avoided by design. *Anomalous* curves, those with $\#E(\mathbb{F}_p) = p$, admit a polynomial-time attack through the formal group. The MOV/Frey–Rück attack transfers the ECDLP into a finite-field discrete logarithm when the *embedding degree*—the least $k \ge 1$ with $r \mid p^k - 1$, i.e. the degree (theorem 8.4; the dimension of the extension over its base) of the smallest extension of $\mathbb{F}_p$ whose multiplicative group contains a subgroup of order r—is small, where index calculus

applies. Cryptographically secure curves, the Pasta curves included, are chosen with a large prime subgroup order, a large embedding degree, and $\#E(\mathbb{F}_p) \neq p$, defeating every known subexponential attack.

Remark 10.32 (The index-calculus contrast). The contrast with the multiplicative groups $\mathbb{F}_p^\times$ is the economic heart of the subject. In $\mathbb{F}_p^\times$ the index-calculus method solves the discrete logarithm in subexponential time $L_p[1/3]$—shorthand for $2^{\Theta(\lambda^{1/3}(\log\lambda)^{2/3})}$ operations in the bit length $\lambda = \log_2 p$, a notation defined precisely in theorem 11.60 of the closing section—exploiting the integer arithmetic behind the residues (theorem 7.22); those groups must therefore be made very large. On a well-chosen elliptic curve no index calculus is known—the points offer no analogue of "factoring into small primes"—so the generic $O(\sqrt{r})$ bound stands, and much smaller groups suffice: about 256 bits for 128-bit security, versus thousands of bits for $\mathbb{F}_p^\times$. Smaller groups mean faster arithmetic and shorter keys, which is why elliptic curves dominate modern protocols, including Halo 2.

10.12 The j-invariant and the GLV endomorphism

Two final pieces of structure round out the theory: a single quantity that classifies curves up to isomorphism, and an "extra symmetry" available exactly when that quantity vanishes—which the Pasta curves arrange on purpose.

Definition 10.33 (j-invariant)**.** The *j-invariant* of the curve $y^2 = x^3+ax+b$ (with $4a^3+27b^2 \neq 0$) is

$$j = 1728\,\frac{4a^3}{4a^3+27b^2} \in K,$$

where $1728 = 12^3$ is a normalising constant of historical origin.

Proposition 10.34. *Two elliptic curves over $\overline{K}$ are isomorphic over $\overline{K}$ if and only if they have the same j-invariant. The admissible coordinate changes preserving short Weierstrass form are $(x,y) \mapsto (u^2x, u^3y)$ for $u \in \overline{K}^\times$, under which $(a,b) \mapsto (u^4a, u^6b)$; the combination j is invariant.*

Proof (*sketch*). A direct computation shows that the only isomorphisms between short Weierstrass curves fixing $\mathcal{O}$ are the scalings $(x,y) \mapsto (u^2x, u^3y)$. Substituting $x = u^{-2}x'$, $y = u^{-3}y'$ into $y^2 = x^3 + ax + b$ and clearing u^6 gives $y'^2 = x'^3 + u^4a\,x' + u^6b$, which is the stated action on coefficients. Under it, numerator and denominator of $4a^3/(4a^3+27b^2)$ both scale by u^{12}, so j is unchanged. Conversely, given two curves with equal j-invariants one solves for a suitable u transforming one coefficient pair into the other, with separate easy cases $j = 0$ (both a vanish) and $j = 1728$ (both b vanish). □

Remark 10.35 (The Pasta curves have $j = 0$). For $y^2 = x^3 + b$ one has $a = 0$, so $j = 0$; both Pasta curves have $j = 0$. Curves with $j = 0$ are special: they admit an extra automorphism of order 3, which underlies the efficient endomorphism described next. (At the other extreme, $b = 0$ gives $j = 1728$ and an order-4 automorphism.) The vanishing $j = 0$ is deliberate in the Pasta design: it is what makes the Gallant–Lambert–Vanstone (GLV) speedup available, roughly halving the doubling count in scalar multiplication.

Proposition 10.36 (The GLV endomorphism for $y^2 = x^3 + b$)**.** *Suppose $\mathbb{F}_p$ contains a primitive cube root of unity ω, i.e. $p \equiv 1 \pmod 3$ (theorem 6.22), so $\omega \in \mathbb{F}_p$ satisfies $\omega^2 + \omega + 1 = 0$. On*

> *$E : y^2 = x^3 + b$, the map*
> $$\phi(x, y) = (\omega x,\ y), \qquad \phi(\mathcal{O}) = \mathcal{O},$$
> *is a group endomorphism of $E(\mathbb{F}_p)$. Let $G = \langle P \rangle$ be a subgroup of prime order r with $r \equiv 1 \pmod 3$ and $\phi(G) \subseteq G$. Then ϕ acts on G as multiplication by a scalar: there is $\lambda \in \mathbb{Z}/r\mathbb{Z}$ with $\lambda^2 + \lambda + 1 \equiv 0 \pmod r$ and*
> $$\phi(Q) = [\lambda]\, Q \quad \textit{for all } Q \in G.$$
> *For the Pasta curves the hypothesis $\phi(G) \subseteq G$ is automatic, since $E(\mathbb{F}_p)$ itself is the prime-order group G; and $r \equiv 1 \pmod 3$ holds for both orders.*

Proof. The map lands on the curve. If $y^2 = x^3 + b$ then $y^2 = (\omega x)^3 + b$, because $\omega^3 = 1$; so $(\omega x, y) \in E(\mathbb{F}_p)$.

The map is a homomorphism. We check $\phi(P_1 + P_2) = \phi(P_1) + \phi(P_2)$ against the case analysis of §10.4. The $\mathcal{O}$ cases are trivial, and $\phi(-P) = -\phi(P)$ since negation touches only y; in particular the inverse case $\phi(P) + \phi(-P) = \mathcal{O} = \phi(\mathcal{O})$ is respected, and ϕ preserves the case split, since it preserves equality of x-coordinates. For a chord $(x_1 \neq x_2)$, the slope transforms as

$$\lambda' = \frac{y_2 - y_1}{\omega x_2 - \omega x_1} = \omega^{-1}\lambda = \omega^2\lambda,$$

using $\omega^{-1} = \omega^2$. Then, by (12) and $\omega^4 = \omega$,

$$x_3' = \lambda'^2 - \omega x_1 - \omega x_2 = \omega^4\lambda^2 - \omega x_1 - \omega x_2 = \omega\,(\lambda^2 - x_1 - x_2) = \omega x_3,$$

$$y_3' = \lambda'(\omega x_1 - x_3') - y_1 = \omega^2\lambda \cdot \omega(x_1 - x_3) - y_1 = \lambda(x_1 - x_3) - y_1 = y_3,$$

so $\phi(P_1) + \phi(P_2) = (\omega x_3, y_3) = \phi(P_1 + P_2)$. The tangent case runs identically with $\lambda' = 3(\omega x_1)^2/(2y_1) = \omega^2\lambda$ in (13).

The relation $\phi^2 + \phi + \mathrm{id} = 0$. Let $P = (x, y)$ be affine; we show $P + \phi(P) + \phi^2(P) = \mathcal{O}$ directly from the formulas of §10.4.

First let $x \neq 0$. The points $P = (x, y)$, $\phi(P) = (\omega x, y)$, $\phi^2(P) = (\omega^2 x, y)$ share their y-coordinate and have pairwise distinct x-coordinates, since $1, \omega, \omega^2$ are distinct and $x \neq 0$. The chord formula (12) applies to $P + \phi(P)$, with slope $\lambda = (y - y)/(\omega x - x) = 0$:

$$x_3 = 0 - x - \omega x = -(1 + \omega)\, x = \omega^2 x, \qquad y_3 = 0 \cdot (x - x_3) - y = -y,$$

using $1 + \omega + \omega^2 = 0$. Hence $P + \phi(P) = (\omega^2 x, -y) = -\phi^2(P)$ (theorem 10.8), and $P + \phi(P) + \phi^2(P) = \mathcal{O}$.

If instead $x = 0$, the three points coincide: $\phi(P) = \phi^2(P) = P$. Here $y \neq 0$, since $x = y = 0$ on the curve would force $b = 0$, which nonsingularity $27b^2 \neq 0$ forbids; so the doubling formula (13) applies, with $a = 0$:

$$\lambda = \frac{3 \cdot 0^2 + 0}{2y} = 0, \qquad x_3 = 0 - 2 \cdot 0 = 0, \qquad y_3 = 0 \cdot (0 - 0) - y = -y,$$

so $[2]P = (0, -y) = -P$, hence $P + \phi(P) + \phi^2(P) = [3]P = \mathcal{O}$ in this case too. (The three points P, $\phi(P)$, $\phi^2(P)$ are exactly the intersections of the horizontal line through P with the curve, the roots of $X^3 = y^2 - b$ being x, ωx, $\omega^2 x$; the computation just performed is the "three points of a line sum to $\mathcal{O}$" picture made rigorous, including the triple intersection at $x = 0$.)

With $\phi(\mathcal{O}) = \mathcal{O}$ the identity holds on all of $E(\mathbb{F}_p)$; as an identity of endomorphisms, $\phi^2 + \phi + \mathrm{id} = 0$. Incidentally $\phi^3 = \mathrm{id}$, since $\omega^3 = 1$; and whenever $\#E(\mathbb{F}_p) > 3$—as always at cryptographic sizes, the Pasta curves included—ϕ has order exactly 3, because at most two affine points, $(0, \pm y_0)$ with $y_0^2 = b$, share the coordinate $x = 0$, so some point of $E(\mathbb{F}_p)$ has $x \neq 0$ and ϕ moves it. The proviso is not vacuous: $y^2 = x^3 + 4$ over $\mathbb{F}_7$ has only the three points $\mathcal{O}$, $(0, 2)$, $(0, 5)$, and there $\phi = \mathrm{id}$.

Action on G as a scalar. The subgroup $G \cong \mathbb{Z}/r\mathbb{Z}$ is cyclic with generator P, and $\phi(G) \subseteq G$ by hypothesis, so $\phi(P) = [\lambda]P$ for some $\lambda \in \mathbb{Z}/r\mathbb{Z}$; then for any $Q = [m]P \in G$, the homomorphism property gives $\phi(Q) = [m]\phi(P) = [m\lambda]P = [\lambda]Q$. (This is the statement $\mathrm{Hom}(\mathbb{Z}/r\mathbb{Z}, \mathbb{Z}/r\mathbb{Z}) \cong \mathbb{Z}/r\mathbb{Z}$: an endomorphism of a cyclic group is determined by, and equal to multiplication by, the image of a generator.) Applying the endomorphism relation to P,

$$[\lambda^2 + \lambda + 1]P = \mathcal{O} \quad \Longrightarrow \quad \lambda^2 + \lambda + 1 \equiv 0 \pmod{r},$$

since P has order r. Consistently, this congruence is solvable exactly when $\mathbb{F}_r$ contains a primitive cube root of unity, i.e. when $r \equiv 1 \pmod 3$ (theorem 6.22 again)—the hypothesis of the statement. □

Remark 10.37 (How GLV halves the doubling count). Computing ϕ costs a single field multiplication $x \mapsto \omega x$, negligible against a group operation. One therefore decomposes a scalar $n \in \mathbb{Z}/r\mathbb{Z}$ as

$$[n]P = [n_1]P + [n_2]\,\phi(P), \qquad n \equiv n_1 + n_2\lambda \pmod{r},$$

where n_1, n_2 can each be taken only about $\frac{1}{2}\log_2 r$ bits long—such a pair is found by lattice reduction in the lattice of pairs (c_1, c_2) with $c_1 + c_2\lambda \equiv 0 \pmod{r}$. The two half-length scalar multiplications are then run *simultaneously*, sharing their doublings (Straus–Shamir's trick): one double-and-add loop of half the length, adding P, $\phi(P)$, or both according to the bits of n_1 and n_2. The result costs about half the doublings of a single full-length double-and-add—a substantial saving, and its availability is one reason the Pasta curves take the form $y^2 = x^3 + b$ over primes $p \equiv 1 \pmod 3$.

10.13 A worked toy example

The entire machinery now runs on a curve small enough for hand computation.

Example 10.38 (The curve $y^2 = x^3+5$ over $\mathbb{F}_{11}$)**.** Take $p = 11$: then $p > 3$, and $4\cdot 0+27\cdot 5^2 = 675 \equiv 4 \not\equiv 0 \pmod{11}$, so the curve $E : y^2 = x^3 + 5$ is nonsingular over $\mathbb{F}_{11}$ (theorem 10.5). Squaring the residues $0, 1, \ldots, 10$ yields the squares $\{0, 1, 3, 4, 5, 9\}$, so the nonzero quadratic residues are $\{1, 3, 4, 5, 9\}$. Tabulating $f(x) = x^3 + 5$ mod 11 and reading off the y-values:

x	x^3 mod 11	$f(x) = x^3 + 5$	y with $y^2 = f(x)$
0	0	5	± 4 ($4^2 = 16 \equiv 5$)
1	1	6	none ($6 \notin$ QR)
2	8	2	none
3	5	10	none
4	9	3	± 5 ($5^2 = 25 \equiv 3$)
5	4	9	± 3
6	7	1	± 1
7	2	7	none
8	6	0	0
9	3	8	none
10	10	4	± 2

Counting: the five values $x \in \{0, 4, 5, 6, 10\}$ contribute two points each (10 points), $x = 8$ contributes the single point $(8, 0)$, and $\mathcal{O}$ completes the census:

$$\#E(\mathbb{F}_{11}) = 10 + 1 + 1 = 12.$$

Hasse's bound checks out: $p + 1 = 12$, so the trace is $t = 0$, and indeed $|t| = 0 \leq 2\sqrt{11} \approx 6.63$. The point $(8, 0)$ has order 2, consistent with theorem 10.25 and the even order 12; here $-5 \equiv 6$ *is* a cube, namely $8^3 = 512 \equiv 6$.

An addition. Let $P = (0,4)$ and $Q = (5,3)$. Since $x_1 = 0 \neq 5 = x_2$, the chord formula (12) applies:

$$\lambda = \frac{3-4}{5-0} = (-1)\cdot 5^{-1} = (-1)\cdot 9 = -9 \equiv 2 \pmod{11},$$

using $5^{-1} = 9$ (check: $5 \cdot 9 = 45 \equiv 1$). Then

$$x_3 = \lambda^2 - x_1 - x_2 = 4 - 0 - 5 = -1 \equiv 10, \qquad y_3 = \lambda(x_1 - x_3) - y_1 = 2\,(0-10) - 4 = -24 \equiv 9,$$

so $P + Q = (10,9)$. The result is on the curve: $9^2 = 81 \equiv 4$ and $10^3 + 5 = 1005 \equiv 4 \pmod{11}$, as required.

A doubling. Take $P = (0,4)$. With $a = 0$, formula (13) gives

$$\lambda = \frac{3x_1^2 + a}{2y_1} = \frac{0}{8} = 0, \qquad x_3 = \lambda^2 - 2x_1 = 0, \qquad y_3 = \lambda(x_1 - x_3) - y_1 = -4 \equiv 7,$$

so $[2](0,4) = (0,7) = -(0,4)$. Hence $[3](0,4) = \mathcal{O}$: the point $(0,4)$ has order 3.

The GLV endomorphism. Since $11 \equiv 2 \pmod 3$, the field $\mathbb{F}_{11}$ contains *no* primitive cube root of unity (theorem 6.22 requires $3 \mid p-1$), so the GLV map is not available over $\mathbb{F}_{11}$. The Pasta primes do satisfy $p \equiv q \equiv 1 \pmod 3$, so GLV applies there. For a small GLV-friendly illustration take instead $p = 13 \equiv 1 \pmod 3$, where $\omega = 3$ satisfies $3^2 + 3 + 1 = 13 \equiv 0 \pmod{13}$; then $\phi(x,y) = (3x,y)$ is a nontrivial endomorphism of $y^2 = x^3 + b$ over $\mathbb{F}_{13}$.

10.14 Pallas and Vesta assembled

Every property developed in this section was chosen with the closing example in mind. Here the pieces assemble.

Example 10.39 (The Pasta curves)**.** For the running example $y^2 = x^3 + 5$, with p and q the 255-bit primes of the section opening:

- *Pallas*, over $\mathbb{F}_p$: the order is $\#E_p(\mathbb{F}_p) = q$, a prime, so the group is cyclic, $E_p(\mathbb{F}_p) \cong \mathbb{Z}/q\mathbb{Z}$ (theorem 3.34), with cofactor $h = 1$ (theorem 10.24). The trace is $t = p + 1 - q$, an 87-bit (negative) integer, far inside the Hasse bound $2\sqrt{p} \approx 2^{128}$ of theorem 10.22.
- *Vesta*, over $\mathbb{F}_q$: the order is $\#E_q(\mathbb{F}_q) = p$, a prime, so $E_q(\mathbb{F}_q) \cong \mathbb{Z}/p\mathbb{Z}$, again with cofactor 1.
- Both curves have $j = 0$ (theorem 10.35), and both primes satisfy $p \equiv q \equiv 1 \pmod 3$, so each curve admits the order-3 endomorphism $\phi(x,y) = (\omega x, y)$ of theorem 10.36 and supports the GLV speedup of theorem 10.37.
- Both fields have two-adicity 32 in the sense of theorem 9.10: 2^{32} divides $p-1$ and $q-1$, so each field carries the multiplicative subgroups of order 2^{32} promised by theorem 10.28—the evaluation domains of the fast polynomial arithmetic of section 9.
- The cycle property $\#E_p(\mathbb{F}_p) = q$ and $\#E_q(\mathbb{F}_q) = p$ means the scalar field of each curve is the base field of the other (theorem 10.27): the base field of Pallas is $\mathbb{F}_p$ and its scalar field is $\mathbb{F}_q$, matching the base field of Vesta, and symmetrically. This is exactly what lets recursive proof systems chain proofs across the two curves without expensive non-native field arithmetic (theorem 10.28). The cycle is a deployed fact: the `pasta_curves` crate wires the scalar field of Pallas to be $\mathbb{F}_q$ and that of Vesta to be $\mathbb{F}_p$, as specified in §5.4.9.6 of the Zcash protocol specification.

The discrete logarithm problem in $E_p(\mathbb{F}_p)$ and $E_q(\mathbb{F}_q)$ is believed to require about 2^{127} operations—$\sqrt{r}$ for the 255-bit orders $r \approx 2^{254}$ (theorem 10.30)—with no faster attack known (theorem 10.31), just below the 128-bit security level targeted by Halo 2 and Zcash Orchard.

The cheque from the section's opening is hereby cashed. The demand was a roughly 255-bit group for which no attack faster than the square-root bound is known, and each Pasta curve supplies one: a cyclic group of prime 255-bit order (theorem 10.22 and theorem 3.34), carrying a fast forward map (theorem 10.21, accelerated by theorem 10.37) whose inversion—the ECDLP—resists everything known except the generic $O(\sqrt{r})$ attacks of theorem 10.30. Where $\mathbb{F}_p^\times$ needs thousands of bits to outrun index calculus, the curve group needs 255; and the two groups even interlock as a cycle, each one's scalars living in the other's coordinates. Later volumes build on precisely this pair.

11 Probability, asymptotics, and computation

A deployed signature scheme is typically advertised as offering "128-bit security". The question this section exists to answer is: what, precisely, does that phrase assert, and about whom? It cannot assert that forgery is impossible. A forger may simply guess the secret key: keys are finite strings, so the guess succeeds with some positive probability, and a forger with astronomical luck wins outright. Any honest guarantee is therefore a statement of *probability* — a bound on the chance that the forger wins. Nor can it speak of one forger only: a guarantee against yesterday's attack says nothing about tomorrow's, so the bound must quantify over *every* attacker whose resources fall within stated limits. Making the sentence "every efficient forger succeeds with at most a stated probability" precise takes three intertwined languages: the language of *probability*, which permits reasoning about random keys, random challenges, and the probability that an attack succeeds; the language of *asymptotics*, which describes how resources and success probabilities scale; and the language of *computation*, which fixes what an efficient algorithm is and what it means for a problem to be hard. This section develops all three from elementary foundations and closes by cashing the opening cheque: the final subsection states exactly what "128-bit security" asserts, and about whom. The only prerequisites are familiarity with finite sets, sums, and elementary calculus — limits, the exponential function, and, for the closing subsection's density-defined laws, the improper Riemann integral; everything else is constructed here.

Until the closing subsection, all probability spaces are finite. This is no real restriction for the cryptographic material: keys, messages, group elements, bounded-length messages, and the random coins of an algorithm with a fixed running-time bound live in finite sets, and finite spaces need no measure theory. Unbounded retry loops need the extension below even when they terminate with probability one. The closing subsection (§11.7) then extends the theory by exactly the rungs later volumes need — countable additivity, continuity along monotone events, a waiting-time toolkit, and laws defined by densities — declaring its one measure-theoretic debt in place (Remark 11.31).

11.1 Finite probability spaces and events

The basic object is a finite set of outcomes together with an assignment of weights summing to one.

Definition 11.1 (Finite probability space)**.** A *finite probability space* is a pair $(\Omega, \Pr)$ where Ω is a nonempty finite set, called the *sample space*, and $\Pr\colon \Omega \to [0,1]$ is a function, called the *probability mass function*, satisfying the normalisation condition

$$\sum_{\omega \in \Omega} \Pr[\omega] = 1.$$

An element $\omega \in \Omega$ is an *outcome* (or *elementary event*).

Definition 11.2 (Event)**.** An *event* is a subset $A \subseteq \Omega$. Its *probability* is

$$\Pr[A] = \sum_{\omega \in A} \Pr[\omega],$$

with the convention $\Pr[\emptyset] = 0$ (an empty sum). The event A *occurs* on outcome ω if $\omega \in A$.

Because Ω is finite, every subset is an event. This convenience is worth flagging: on an uncountable sample space the general theory specifies a family of admissible events together with its probability assignment, rather than automatically admitting all subsets; finiteness spares us that machinery, and Remark 11.31 marks the one place this series touches it. The following elementary facts hold, and we use them freely.

Proposition 11.3 (Basic axioms)**.** *For any finite probability space* $(\Omega, \Pr)$ *and events* $A, B \subseteq \Omega$*:*

1. $0 \le \Pr[A] \le 1$, $\Pr[\emptyset] = 0$, *and* $\Pr[\Omega] = 1$.
2. (Complement.) $\Pr[\Omega \setminus A] = 1 - \Pr[A]$.
3. (Monotonicity.) *If* $A \subseteq B$ *then* $\Pr[A] \le \Pr[B]$.
4. (Inclusion–exclusion for two events.) $\Pr[A \cup B] = \Pr[A] + \Pr[B] - \Pr[A \cap B]$.
5. (Finite additivity.) *If* $A_1, \dots, A_n$ *are pairwise disjoint then* $\Pr\left[\bigcup_{i=1}^n A_i\right] = \sum_{i=1}^n \Pr[A_i]$.

Proof. Every claim follows by manipulating the defining sums. For (1), each summand $\Pr[\omega]$ lies in $[0,1]$ and the total is 1 by normalisation; an event's probability is a partial sum of nonnegative terms, hence lies in $[0,1]$. For (2), the sets A and $\Omega \setminus A$ partition Ω, so $\Pr[A] + \Pr[\Omega \setminus A] = \sum_{\omega \in \Omega} \Pr[\omega] = 1$. For (3), splitting the sum over B gives $\Pr[B] = \sum_{\omega \in A} \Pr[\omega] + \sum_{\omega \in B \setminus A} \Pr[\omega] \ge \Pr[A]$, the second term being a sum of nonnegative numbers. For (5), disjointness means each ω in the union lies in exactly one A_i, so the double sum $\sum_i \sum_{\omega \in A_i} \Pr[\omega]$ counts each $\omega \in \bigcup_i A_i$ exactly once. For (4), write $A \cup B = A \sqcup (B \setminus A)$ (disjoint), so by (5) $\Pr[A \cup B] = \Pr[A] + \Pr[B \setminus A]$; also $B = (A \cap B) \sqcup (B \setminus A)$, giving $\Pr[B \setminus A] = \Pr[B] - \Pr[A \cap B]$. Substituting yields the claim. □

The single most important example, ubiquitous in cryptography, is the uniform distribution.

Definition 11.4 (Uniform distribution)**.** Let S be a nonempty finite set. The *uniform distribution* on S is the probability space $(S, \Pr)$ with $\Pr[\omega] = 1/|S|$ for every $\omega \in S$. The notation $x \leftarrow S$ (or $x \xleftarrow{\$} S$) means that x is sampled according to the uniform distribution on S. Under the uniform distribution, for any event $A \subseteq S$,

$$\Pr[A] = \frac{|A|}{|S|}.$$

Thus, for uniform sampling, probability is exactly the ratio of favourable outcomes to total outcomes, recovering the classical counting notion of probability.

11.2 Conditional probability and independence

Frequently we learn that one event has occurred and must update the probability of another accordingly. Conditioning captures this.

Definition 11.5 (Conditional probability)**.** Let A, B be events with $\Pr[B] > 0$. The *conditional probability of A given B* is

$$\Pr[A \mid B] = \frac{\Pr[A \cap B]}{\Pr[B]}.$$

When $\Pr[B] = 0$ the quantity is left undefined.

Conditioning on B restricts the sample space to B and renormalises so that B has probability 1: the map $\omega \mapsto \Pr[\{\omega\} \mid B]$ is itself a probability mass function on B, and $\Pr[A \mid B]$ is the probability it assigns to $A \cap B$. Two immediate and indispensable consequences follow.

Proposition 11.6 (Chain rule and total probability)**.** *Let $A_1, \dots, A_n$ be events with $\Pr[A_1 \cap \cdots \cap A_{n-1}] > 0$. Then*

$$\Pr[A_1 \cap \cdots \cap A_n] = \Pr[A_1]\ \Pr[A_2 \mid A_1]\ \Pr[A_3 \mid A_1 \cap A_2] \cdots \Pr[A_n \mid A_1 \cap \cdots \cap A_{n-1}].$$

Moreover, if $B_1, \dots, B_k$ partition Ω (pairwise disjoint with union Ω) and each $\Pr[B_i] > 0$, then for any event A,

$$\Pr[A] = \sum_{i=1}^{k} \Pr[A \mid B_i]\ \Pr[B_i].$$

Proof. The chain rule follows by telescoping: each factor $\Pr[A_j \mid A_1 \cap \cdots \cap A_{j-1}]$ equals $\Pr[A_1 \cap \cdots \cap A_j] / \Pr[A_1 \cap \cdots \cap A_{j-1}]$ by theorem 11.5, and the product collapses to $\Pr[A_1 \cap \cdots \cap A_n]/1$. (Monotonicity ensures that all the intermediate intersections have positive probability, so every factor is defined.) For total probability, the events $A \cap B_i$ are pairwise disjoint with union A (since the B_i partition Ω), so finite additivity and $\Pr[A \cap B_i] = \Pr[A \mid B_i]\ \Pr[B_i]$ give $\Pr[A] = \sum_i \Pr[A \cap B_i] = \sum_i \Pr[A \mid B_i]\ \Pr[B_i]$. □

Independence is the formal statement that one event carries no information about another.

Definition 11.7 (Independence of events)**.** Two events A, B are *independent* if

$$\Pr[A \cap B] = \Pr[A]\ \Pr[B].$$

A family $\{A_i\}_{i \in I}$ of events is *mutually independent* if for every finite subset $J \subseteq I$,

$$\Pr\Big[\bigcap_{i \in J} A_i\Big] = \prod_{i \in J} \Pr[A_i].$$

Remark 11.8. When $\Pr[B] > 0$, independence of A and B is equivalent to $\Pr[A \mid B] = \Pr[A]$: conditioning on B does not change the probability of A. Mutual independence is strictly stronger than pairwise independence. The standard counterexample: toss two fair coins, and let A be "the first is heads", B "the second is heads", and C "the two coins differ". Each pair is independent, yet $\Pr[A \cap B \cap C] = 0 \neq \frac{1}{8} = \Pr[A]\ \Pr[B]\ \Pr[C]$, so the three are not mutually independent.

11.3 Random variables and expectation

A random variable assigns a value to each outcome; it summarises a random experiment by a number (or a vector, a string, a group element).

Definition 11.9 (Random variable)**.** Let $(\Omega, \Pr)$ be a finite probability space and T any set. A (T-*valued*) *random variable* is a function $X\colon \Omega \to T$. When $T \subseteq \mathbb{R}$, we call X *real-valued*. For $t \in T$,

$$\Pr[X = t] = \Pr[\{\omega \in \Omega : X(\omega) = t\}],$$

and more generally $\Pr[X \in B]$ denotes the probability that X takes a value in $B \subseteq T$. The function $t \mapsto \Pr[X = t]$ on the finite image of X is the *distribution* (or *law*) of X.

Definition 11.10 (Independent random variables)**.** Random variables $X_1, \dots, X_n$ (with values in sets $T_1, \dots, T_n$) are *mutually independent* if for all $t_1 \in T_1, \dots, t_n \in T_n$,

$$\Pr[X_1 = t_1, \dots, X_n = t_n] = \prod_{i=1}^{n} \Pr[X_i = t_i].$$

Definition 11.11 (Expectation)**.** Let $X\colon \Omega \to \mathbb{R}$ be a real-valued random variable on a finite space. Its *expectation* (or *mean*) is

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega)\ \Pr[\omega] = \sum_{t \in \mathrm{im}(X)} t \cdot \Pr[X = t],$$

where $\mathrm{im}(X)$ denotes the (finite) image of X. The two expressions agree by grouping outcomes according to their X-value.

A frequently used bridge between the two notions is the following: the expectation of the indicator of an event is exactly its probability.

Definition 11.12 (Indicator)**.** For an event A, its *indicator random variable* $\mathbf{1}_A : \Omega \to \{0, 1\}$ takes the value $\mathbf{1}_A(\omega) = 1$ if $\omega \in A$ and 0 otherwise.

Proposition 11.13 (Indicator expectation)**.** *For any event A, $\mathbb{E}[\mathbf{1}_A] = \Pr[A]$.*

Proof. By theorem 11.11, $\mathbb{E}[\mathbf{1}_A] = \sum_\omega \mathbf{1}_A(\omega)\ \Pr[\omega] = \sum_{\omega \in A} \Pr[\omega] = \Pr[A]$, since the indicator vanishes off A. □

The principal property of expectation is its linearity, which holds with no independence assumption whatsoever; this absence of any independence hypothesis is what makes expectation so much more tractable than probability.

Theorem 11.14 (Linearity of expectation)**.** *Let $X, Y\colon \Omega \to \mathbb{R}$ be random variables on a finite space and $a, b \in \mathbb{R}$. Then*

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y].$$

More generally, for random variables $X_1, \dots, X_n$ and scalars $a_1, \dots, a_n$,

$$\mathbb{E}\Big[\sum_{i=1}^{n} a_i X_i\Big] = \sum_{i=1}^{n} a_i\,\mathbb{E}[X_i].$$

This holds whether or not the X_i are independent.

Proof. Directly from the definition,

$$\begin{aligned}\mathbb{E}[aX + bY] &= \sum_{\omega\in\Omega} (aX(\omega) + bY(\omega)) \Pr[\omega] \\ &= a \sum_{\omega} X(\omega) \Pr[\omega] + b \sum_{\omega} Y(\omega) \Pr[\omega] \\ &= a\,\mathbb{E}[X] + b\,\mathbb{E}[Y],\end{aligned}$$

splitting the finite sum and extracting the constants. The n-term statement follows by induction on n. □

A one-line consequence converts an expectation bound into a probability bound; it underlies every averaging ("heavy-row") argument in the series.

Proposition 11.15 (Markov's inequality)**.** *Let $X \geq 0$ be a real-valued random variable and $a > 0$. Then $\Pr[X \geq a] \leq \mathbb{E}[X]/a$.*

Proof. Every outcome contributes nonnegatively to $\mathbb{E}[X]$, and each outcome with $X(\omega) \geq a$ contributes at least $a \Pr[\omega]$; hence $\mathbb{E}[X] \geq a \Pr[X \geq a]$. □

11.4 The union bound and a birthday calculation

The payoff of this subsection, stated up front, is the square-root law that sets the output lengths of cryptographic hash functions: among roughly $\sqrt{N}$ uniform samples from a set of size N, two are likely to coincide, so an n-bit hash resists collision-finding only up to about $2^{n/2}$ work, and output lengths are chosen twice the desired security level. The engine of the calculation is the most frequently used inequality in cryptographic proofs: the union bound. It is crude, it requires no independence, and it almost always suffices.

Theorem 11.16 (Union bound / Boole's inequality)**.** *For any events $A_1, \dots, A_n$ in a finite probability space,*

$$\Pr\Big[\bigcup_{i=1}^{n} A_i\Big] \leq \sum_{i=1}^{n} \Pr[A_i].$$

Proof. Disjointify the union into "first occurrence" events: $B_1 = A_1$ and $B_i = A_i \setminus (A_1 \cup \cdots \cup A_{i-1})$ for $i \geq 2$. The B_i are pairwise disjoint, $B_i \subseteq A_i$, and $\bigcup_i B_i = \bigcup_i A_i$. By finite additivity and monotonicity (theorem 11.3),

$$\Pr\Big[\bigcup_i A_i\Big] = \Pr\Big[\bigcup_i B_i\Big] = \sum_i \Pr[B_i] \leq \sum_i \Pr[A_i].$$ □

The disjointification device — replacing events by pairwise disjoint pieces with the same union — recurs throughout the section: the "first occurrence" form above proves the union bound, and a telescoping form for increasing chains proves the continuity theorem of §11.7. In each case it turns a subadditivity or limit claim into an exact additivity computation. The complementary inequality for conjunctions is equally useful and follows by taking complements.

Corollary 11.17 (Union bound for the good event)**.** *If each event A_i holds except with probability at most ε_i (that is, $\Pr[\Omega \setminus A_i] \leq \varepsilon_i$), then all of them hold simultaneously except with probability at*

most $\sum_i \varepsilon_i$*:*

$$\Pr\Big[\bigcap_{i=1}^{n} A_i\Big] \geq 1 - \sum_{i=1}^{n} \varepsilon_i.$$

Proof. By De Morgan, $\Omega \setminus \bigcap_i A_i = \bigcup_i (\Omega \setminus A_i)$. Apply the union bound to the complements and take the complement of both sides. □

We now derive the *birthday bound*, which governs how soon collisions appear when sampling uniformly.

Proposition 11.18 (Birthday bound)**.** *Sample $x_1, \dots, x_k$ independently and uniformly from a set of size N, and let C be the event that some two of them collide, i.e. $x_i = x_j$ for some $i < j$. Then*

$$\Pr[C] \leq \binom{k}{2}\frac{1}{N} = \frac{k(k-1)}{2N},$$

and, on the other side,

$$\Pr[C] = 1 - \prod_{i=0}^{k-1}\Big(1 - \frac{i}{N}\Big) \geq 1 - e^{-k(k-1)/(2N)}.$$

In particular a collision becomes likely once $k = \Theta(\sqrt{N})$.

Proof. Upper bound. For $i < j$ let A_{ij} be the event $x_i = x_j$. Since x_i and x_j are independent and uniform, $\Pr[A_{ij}] = \sum_v \Pr[x_i = v]\Pr[x_j = v] = N \cdot (1/N)^2 = 1/N$. Now $C = \bigcup_{i<j} A_{ij}$, a union of $\binom{k}{2}$ events, one per two-element subset of the indices (theorem 6.12), so the union bound (theorem 11.16) gives $\Pr[C] \leq \binom{k}{2}/N$.

Lower bound. The complement $\overline{C}$ is the event that all k samples are distinct. If $k > N$, the pigeonhole principle forces a collision, so $\Pr[\overline{C}] = 0$; the product also vanishes, through its factor $1 - N/N$, so the stated equality holds and the exponential bound is trivial. Let then $k \leq N$, and for $0 \leq i \leq k$ let D_i be the event that the first i samples are pairwise distinct ($D_0 = D_1 = \Omega$, and $D_k = \overline{C}$). We show $\Pr[D_i] = \prod_{j=0}^{i-1}(1 - j/N)$ by induction on $i \leq k$; every factor is positive, since $j \leq k - 1 \leq N - 1$, so in particular each D_i has positive probability and conditioning on it is legitimate (theorem 11.5). For the step, the events pinning the first i samples to specific distinct values partition D_i, each with probability $N^{-i} > 0$ by independence and uniformity; given any one of them, the next sample avoids the i values taken with probability $1 - i/N$, again by independence and uniformity. Averaging over the cells (total probability, theorem 11.6) gives $\Pr[D_{i+1} \mid D_i] = 1 - i/N$, and $\Pr[D_{i+1}] = \Pr[D_{i+1} \mid D_i]\Pr[D_i]$ completes the induction. Taking $i = k$ yields the stated equality, in which every factor is nonnegative, so the inequality $1 - x \leq e^{-x}$ applies factor by factor. (For $0 \leq x < 1$ this inequality follows from Bernoulli's inequality, $(1 - x/n)^n \geq 1 - x$ for $n > x$, by letting $n \to \infty$ in the limit definition of e^{-x}; for $x \geq 1$ the left side is nonpositive.) Thus

$$\Pr[\overline{C}] = \prod_{i=0}^{k-1}\Big(1 - \frac{i}{N}\Big) \leq \prod_{i=0}^{k-1} e^{-i/N} = \exp\Big(-\tfrac{1}{N}\sum_{i=0}^{k-1} i\Big) = e^{-k(k-1)/(2N)},$$

summing the exponents via $\sum_{i=0}^{k-1} i = k(k-1)/2$. Taking complements gives $\Pr[C] = 1 - \Pr[\overline{C}] \geq 1 - e^{-k(k-1)/(2N)}$. □

Example 11.19. With $N = 365$ and $k = 23$ — the classical birthday party — the exact product gives $\Pr[C] = 0.5073$, already a majority chance, and the proposition brackets it as $0.5000 \leq$

$\Pr[C] \leq 0.6932$. The threshold is visible at small scale too: with $N = 16$, the fifth sample (just past $\sqrt{16} = 4$) tips the odds, $\Pr[C] = 0.5001$ at $k = 5$.

Remark 11.20. The threshold $k \approx \sqrt{N}$ is the reason an n-bit hash function (so $N = 2^n$) offers only about $2^{n/2}$ collision resistance: an adversary expects a collision after roughly $2^{n/2}$ random evaluations — for $N = 2^{256}$, that is $\sqrt{2^{256}} = 2^{128}$. For this reason one chooses output lengths twice the desired security level. The same square-root phenomenon recurs in random-walk discrete-log attacks (Pollard's rho, theorem 10.30), so designers size the curves used in Halo 2 and Orchard so that $\sqrt{N}$ is itself astronomically large.

11.5 Statistical distance

Comparing distributions — for example a real key-generation procedure against an idealised uniform one — requires a notion of how far apart two distributions are. The appropriate notion for "no statistical test can distinguish them well" is statistical distance.

Definition 11.21 (Statistical distance)**.** Let P and Q be probability distributions on the same finite set S (i.e. probability mass functions $P, Q\colon S \to [0,1]$). Their *statistical distance* (or *total variation distance*) is

$$\Delta(P,Q) = \frac{1}{2}\sum_{s\in S}|P(s) - Q(s)|.$$

For random variables X, Y with values in S, $\Delta(X,Y)$ denotes the statistical distance of their distributions.

The factor $\frac{1}{2}$ makes Δ range over $[0,1]$ and gives it a clean interpretation as the maximum advantage of a distinguisher, which part (3) of the next theorem establishes.

Theorem 11.22 (Properties of statistical distance)**.** *Let P, Q, R be distributions on a finite set S. Then:*

1. $0 \leq \Delta(P,Q) \leq 1$; *and* $\Delta(P,Q) = 0$ *if and only if* $P = Q$.
2. (Metric.) Δ *is symmetric and satisfies the triangle inequality* $\Delta(P,R) \leq \Delta(P,Q) + \Delta(Q,R)$.
3. (Variational characterisation.) $\Delta(P,Q) = \max_{A\subseteq S}(P(A) - Q(A)) = \max_{A\subseteq S}|P(A) - Q(A)|$, *where* $P(A) = \sum_{s\in A} P(s)$.
4. (Data processing / post-processing.) *For any (possibly randomised) function f applied to a sample,* $\Delta(f(P), f(Q)) \leq \Delta(P,Q)$. *In particular no algorithm can increase statistical distance.*

Proof. (1) Nonnegativity is clear. For the upper bound, split S into $S^+ = \{s : P(s) \geq Q(s)\}$ and $S^- = S \setminus S^+$. Since $\sum_s P(s) = \sum_s Q(s) = 1$, we have $\sum_s (P(s) - Q(s)) = 0$, so $\sum_{s\in S^+}(P(s) - Q(s)) = \sum_{s\in S^-}(Q(s) - P(s)) =: D \geq 0$. Hence $\Delta(P,Q) = \frac{1}{2}(\sum_{S^+}(P - Q) + \sum_{S^-}(Q - P)) = D$. As $D = \sum_{S^+}(P(s) - Q(s)) \leq \sum_{S^+} P(s) \leq 1$, we get $\Delta \leq 1$. If $\Delta = 0$ then every $|P(s) - Q(s)| = 0$, so $P = Q$; the converse is immediate.

(2) Symmetry is evident. For the triangle inequality, $|P(s) - R(s)| \leq |P(s) - Q(s)| + |Q(s) - R(s)|$ termwise by the triangle inequality on $\mathbb{R}$; summing over s and halving gives the claim.

(3) With D as above, take $A = S^+$: then $P(A) - Q(A) = \sum_{S^+}(P - Q) = D = \Delta(P,Q)$, so the maximum is at least Δ. Conversely, for *any* $A \subseteq S$, $P(A) - Q(A) = \sum_{s\in A}(P(s) - Q(s)) \leq \sum_{s\in A\cap S^+}(P(s) - Q(s)) \leq D = \Delta$, since dropping the negative terms only increases the sum and extending from $A \cap S^+$ to all of S^+ increases it further. Thus the maximum over A of $P(A) - Q(A)$ is

exactly Δ; replacing A by its complement swaps the sign, giving the same value for $\max_A |P(A) - Q(A)|$.

(4) First take f deterministic, $f\colon S \to U$. For $u \in U$, $f(P)(u) = P(f^{-1}(u))$. Then

$$\Delta(f(P), f(Q)) = \tfrac{1}{2}\sum_u |P(f^{-1}(u)) - Q(f^{-1}(u))| \le \tfrac{1}{2}\sum_u \sum_{s \in f^{-1}(u)} |P(s) - Q(s)| = \Delta(P, Q),$$

using $|\sum_s a_s| \le \sum_s |a_s|$ on each fibre, and that the fibres $f^{-1}(u)$ partition S. A randomised f is a deterministic function of the sample s together with independent fresh coins r; apply the deterministic case to the joint variable (s, r), noting that the two joint distributions $P \times U_r$ and $Q \times U_r$ have the same statistical distance $\Delta(P, Q)$, the shared coin distribution contributing nothing. □

Remark 11.23 (Distinguishing interpretation). Part (3) states that $\Delta(P, Q)$ is exactly the best advantage of any test — even a computationally unbounded one — that, given a single sample, must guess whether it came from P or Q: the optimal test outputs "P" on the set $A = S^+$, and its advantage $\Pr_P[A] - \Pr_Q[A]$ equals $\Delta(P, Q)$. Part (4) is what makes statistical distance compose in proofs: applying the same procedure to two close inputs keeps the outputs close. If $\Delta(P, Q) \le \varepsilon$, then P and Q are interchangeable in any analysis at the cost of an additive ε in every probability. When ε is negligible (theorem 11.61 below), we call P and Q *statistically indistinguishable*.

11.6 Uniform sampling and the bias of modular reduction

Cryptography requires uniform elements of finite sets throughout: a uniform scalar in $\mathbb{Z}/q\mathbb{Z}$ (section 2), a uniform field element of $\mathbb{F}_p$ (section 6), a uniform point in a group. The most tractable source of randomness, however, is a stream of independent uniform bits. We must therefore convert uniform bitstrings into uniform elements of a target set S, and we must understand the bias the conversion introduces.

If $|S| = 2^\ell$ is a power of two, the conversion is exact: read ℓ uniform bits as a number in $\{0, \dots, 2^\ell - 1\}$ and index into S. The nontrivial case arises when $|S|$ is not a power of two — most importantly when $S = \mathbb{Z}/q\mathbb{Z}$ for a prime q, the scalar field of an elliptic curve (section 10). A common and elementary method is *reduce-modulo*: sample a long uniform integer and reduce it mod q. The result is not perfectly uniform — some residues receive one more preimage than others — but the bias is negligible provided the integer is long enough. The next proposition makes this precise.

Proposition 11.24 (Bias of modular reduction). *Let $q \ge 2$ be an integer and let $K = 2^L$ for a nonnegative integer L. Sample U uniformly from $\{0, 1, \dots, K-1\}$ (equivalently, read L uniform bits as an integer), and set $R = U \bmod q \in \{0, 1, \dots, q-1\}$. Let $\mathcal{U}_q$ denote the uniform distribution on $\{0, \dots, q-1\}$. Then the statistical distance of R from uniform satisfies*

$$\Delta(R, \mathcal{U}_q) \le \frac{q}{K} = \frac{q}{2^L}.$$

Consequently, if $q < 2^n$ and one takes $L = n + s$ bits, then $\Delta(R, \mathcal{U}_q) < 2^{-s}$.

Proof. Write $K = mq + t$ with $m = \lfloor K/q \rfloor$ and remainder $t = K \bmod q$, so $0 \le t < q$. For a residue $r \in \{0, \dots, q-1\}$, the number of integers in $\{0, \dots, K-1\}$ congruent to r modulo q is $m + 1$ if $r < t$ and m if $r \ge t$. Hence

$$\Pr[R = r] = \begin{cases} (m+1)/K, & r < t, \\ m/K, & r \ge t. \end{cases}$$

Comparing with the target probability $1/q$ and using $m/K \le 1/q \le (m+1)/K$ (since $mq \le K \le (m+1)q$),

$$\left|\Pr[R = r] - \tfrac{1}{q}\right| \le \frac{m+1}{K} - \frac{m}{K} = \frac{1}{K} \qquad \text{for every } r.$$

Therefore

$$\Delta(R, \mathcal{U}_q) = \frac{1}{2}\sum_{r=0}^{q-1}\left|\Pr[R = r] - \tfrac{1}{q}\right| \le \frac{1}{2}\cdot q \cdot \frac{1}{K} = \frac{q}{2K} \le \frac{q}{K}.$$

(The computation actually yields the sharper bound $q/(2K)$; q/K is the usual stated form.) For the final claim, $q < 2^n$ and $K = 2^{n+s}$ give $q/K < 2^n/2^{n+s} = 2^{-s}$. □

Example 11.25. Reduce $L = 6$ uniform bits modulo $q = 5$. Here $K = 64 = 12 \cdot 5 + 4$, so the four residues $r < 4$ receive 13 preimages each and the residue $r = 4$ receives 12; the statistical distance from uniform works out to exactly $1/80$, within the guarantee 2^{-3} of the proposition (as $q < 2^3$ and $s = 6 - 3 = 3$).

Remark 11.26 (Sampling scalars in practice). The consequence to remember: to sample a scalar mod a ~256-bit prime q with statistical distance below 2^{-128} from uniform, draw $256 + 128 = 384$ uniform bits and reduce. The bias 2^{-s}, with the slack s set to the security parameter, is negligible (theorem 11.61), so the reduce-modulo sample is statistically indistinguishable from a perfectly uniform scalar, and by the data-processing inequality (theorem 11.22(4)) substituting one for the other anywhere costs at most 2^{-128} per use. An alternative, *rejection sampling* — draw $\lceil \log_2 q \rceil$ bits, accept if the result is $< q$, else retry — gives *exactly* uniform output at the cost of a variable but (with overwhelming probability) small number of retries; for this series' purposes the negligible bias of reduce-modulo is acceptable and simpler to analyse. The Orchard specification employs exactly this reduce-modulo sampling when deriving scalars from hash outputs: its ToScalar reduces the full 512-bit PRFexpand output modulo the 255-bit group order — a slack of 257 bits, comfortably beyond the 128 of the worked example (Zcash protocol specification §4.2.3 and §5.4.2).

11.7 Beyond finite spaces: countable additivity, limits, and densities

Everything so far lives on a finite sample space, and for the cryptographic material of the upper volumes that is enough: keys, challenges, and transcripts are finite strings. Two uses need more. A process watched for as long as it takes — a random walk pursued until it first hits a barrier — has no finite horizon, and a waiting time measured in continuous units has no finite sample space at all. This subsection extends the finite theory by exactly the rungs those uses require: countable additivity, continuity along monotone events, a toolkit for waiting times under stopping rules, laws defined by densities, and a toolkit of clocks and tail bounds built on them.

Definition 11.27 (Countable probability space)**.** A *countable probability space* is a pair $(\Omega, \Pr)$ in which Ω is a countable set and $\Pr\colon \Omega \to [0,1]$ satisfies $\sum_{\omega\in\Omega} \Pr[\omega] = 1$; the sum is independent of the enumeration order because its terms are nonnegative. Events are arbitrary subsets $A \subseteq \Omega$, with $\Pr[A] = \sum_{\omega\in A} \Pr[\omega]$, and Definitions 11.2–11.11 carry over with "finite" read as "countable" (a random variable's image may now be countably infinite), with one exception: theorem 11.4 does not — equal weights cannot sum to 1 on a countably infinite set, so the uniform distribution remains a strictly finite notion. Expectation additionally requires its defining sum to converge absolutely.

Proposition 11.28 (Countable additivity)**.** *Let* $(\Omega, \Pr)$ *be a countable probability space and*

> *$A_1, A_2, \ldots$ pairwise disjoint events. Then*
>
> $$\Pr\Big[\bigcup_{n\geq 1} A_n\Big] = \sum_{n\geq 1} \Pr[A_n],$$
>
> *and in particular the basic axioms of theorem 11.3 hold unchanged.*

Proof. Every outcome of the union lies in exactly one A_n, so both sides sum the same nonnegative terms $\Pr[\omega]$, grouped differently; a series of nonnegative terms may be summed in any order, or in groups, without changing its value. □

The free regrouping of nonnegative series invoked here — reorder, regroup, exchange double sums, recount by multiplicity — is the workhorse of every infinite-space argument in this subsection: it proves countable additivity above, the tail-sum formula below (where each term is counted by its multiplicity), and the exchange step in Wald's identity.

> **Theorem 11.29** (Continuity along monotone events)**.** *Let* Pr *be a probability assignment, on any sample space, defined for a family of events closed under complement and countable union, and countably additive on it. If $A_1 \subseteq A_2 \subseteq \cdots$ is an increasing chain of events, then*
>
> $$\Pr\Big[\bigcup_{n\geq 1} A_n\Big] = \lim_{n\to\infty} \Pr[A_n],$$
>
> *and dually $\Pr[\bigcap_n B_n] = \lim_n \Pr[B_n]$ for a decreasing chain $B_1 \supseteq B_2 \supseteq \cdots$. On a countable space the hypothesis holds automatically, by theorem 11.28.*

Proof. Disjointify by telescoping: set $D_1 = A_1$ and $D_n = A_n \setminus A_{n-1}$ for $n \geq 2$. The D_n are pairwise disjoint, $\bigcup_{k\leq n} D_k = A_n$, and $\bigcup_n D_n = \bigcup_n A_n$. Countable additivity gives

$$\Pr\Big[\bigcup_n A_n\Big] = \sum_{n\geq 1} \Pr[D_n] = \lim_{n\to\infty} \sum_{k=1}^{n} \Pr[D_k] = \lim_{n\to\infty} \Pr[A_n],$$

the final step by finite additivity. The decreasing case follows by complementation. □

> **Definition 11.30** (Infinite sequence of independent trials)**.** Fix a finite outcome set T and a distribution p on T. An *infinite sequence of independent trials* with law p is a sequence of random variables $X_1, X_2, \ldots$ such that every prefix $(X_1, \ldots, X_n)$ consists of n mutually independent p-distributed trials in the sense of theorem 11.10 — equivalently, the prefix's law is that of the finite probability space T^n carrying the product weights $\Pr[(t_1, \ldots, t_n)] = p(t_1) \cdots p(t_n)$. An event is *finitely determined* if membership in it is decided by some fixed finite prefix; its probability is the finite-space value. An event of the form "some prefix satisfies P" is the union of an increasing chain of finitely determined events, and theorem 11.29 assigns it the limit of the prefix probabilities.

Remark 11.31 (What is taken as given). Theorem 11.30 presumes that a single, countably additive probability assignment on infinite sequences exists that restricts correctly to every finite prefix at once. That existence statement is the Kolmogorov extension theorem, and its proof belongs to measure theory, which lies off this series' critical path: the space of infinite sequences over at least two possible outcomes is uncountable, so it escapes theorem 11.27, and the extension specifies the admissible events. We therefore take it — with its countable additivity, hence theorem 11.29 — as a foundation stone rather than a theorem. On infinite-sequence spaces, only finitely determined events and their monotone limits appear in this series, and for those the values are forced by the

finite theory together with continuity; the density-defined laws of theorem 11.32 carry the analogous debt, their interval and joint-event probabilities being taken to cohere as the stated integrals. The independent families of theorem 11.40 extend that debt in two ways: joint events of several density-defined quantities are taken to be iterated integrals, in whichever order of integration is convenient — the exchange of order for nonnegative integrands is the Fubini–Tonelli theorem, likewise measure-theoretic — and the gap sequences of theorem 11.44 are infinite families, of which only finite prefixes enter any event this series uses, except through countable unions over the values of an arrival count — $N(t)$ of theorem 11.44, or the count X of theorem 11.51, each event "count $= x$" being a joint event of finitely many gaps. For any set B of values, the event "count $\in B$" — among them the event that the count is finite, and the tail events of theorem 11.49 — is taken to have probability $\sum_{x\in B} \Pr[\text{count} = x]$, by the countable additivity assumed above.

Definition 11.32 (Law defined by a density)**.** A real-valued random quantity X has *density* $f\colon \mathbb{R} \to \mathbb{R}_{\geq 0}$ if $\int_{-\infty}^{\infty} f(t)\,dt = 1$ and

$$\Pr[a \leq X \leq b] = \int_a^b f(t)\,dt \qquad \text{for all } a \leq b;$$

improper Riemann integrals suffice throughout this series. The *distribution function* is $F(t) = \Pr[X \leq t] = \int_{-\infty}^{t} f$, so $\Pr[X > t] = 1 - F(t)$, and single points carry probability zero. The expectation is $\mathbb{E}[X] = \int_{-\infty}^{\infty} t f(t)\,dt$ when this integral converges absolutely. Two density-defined quantities X, Y are *independent* if $\Pr[X \in I,\ Y \in J] = \Pr[X \in I]\ \Pr[Y \in J]$ for all intervals I, J; probabilities of joint events are then taken to factor through iterated integrals of the product $f_X(s) f_Y(t)$.

11.7.1 The waiting-time toolkit: tail sums, geometric trials, and Wald's identity

Waiting times are the extension's chief dividend. The three short results of this toolkit — used by later volumes' extractor and race analyses — follow from countable expectation and finitely determined events alone. One convention first: for a variable with values in $\mathbb{N} \cup \{\infty\}$, the defining sum of the expectation has nonnegative terms, and we permit the value $+\infty$ when it diverges.

Lemma 11.33 (Tail-sum formula)**.** *Let T be a random variable with values in $\mathbb{N} \cup \{\infty\}$ and $\Pr[T = \infty] = 0$. Then $\mathbb{E}[T] = \sum_{k\geq 1} \Pr[T \geq k]$, either side being infinite exactly when the other is.*

Proof. Expanding $\Pr[T \geq k] = \sum_{m\geq k} \Pr[T = m]$ and regrouping the doubly indexed nonnegative series, each term $\Pr[T = m]$ is counted once for every $k \leq m$, that is, m times; nonnegative series may be regrouped freely. □

Proposition 11.34 (Geometric waiting time)**.** *In an infinite sequence of independent trials, each succeeding with probability $\varepsilon > 0$, let T be the index of the first success. Then $\Pr[T \geq k] = (1-\varepsilon)^{k-1}$ and $\mathbb{E}[T] = 1/\varepsilon$.*

Proof. The event $T \geq k$ says the first $k-1$ trials all fail — a finitely determined event of probability $(1-\varepsilon)^{k-1}$; in particular $\Pr[T = \infty] = \lim_k (1-\varepsilon)^{k-1} = 0$ by continuity (theorem 11.29). The tail-sum formula then gives $\mathbb{E}[T] = \sum_{k\geq 1} (1-\varepsilon)^{k-1} = 1/\varepsilon$, a geometric series. □

Proposition 11.35 (Wald's identity)**.** *Let $X_1, X_2, \dots$ be an infinite sequence of independent trials with common law, each $X_i \geq 0$ with finite mean, and let T be a* stopping rule*: a random index such that, for every k, whether $T = k$ is decided by the first k trials alone. If $\mathbb{E}[T] < \infty$, then*

$$\mathbb{E}\Big[\sum_{i=1}^{T} X_i\Big] = \mathbb{E}[T] \cdot \mathbb{E}[X_1].$$

Proof. Write $S = \sum_{i=1}^{T} X_i$ and $S_k = \sum_{i=1}^{k} X_i$, so that $S = S_k$ on the event $T = k$. The hypothesis $\mathbb{E}[T] < \infty$ forces $\Pr[T = \infty] = 0$, by the convention preceding theorem 11.33. The variable S takes countably many values — on $\{T = k\}$ it agrees with S_k, whose values run over a finite set, the trials having finitely many outcomes — and each event $\{S = s\}$ is the disjoint countable union $\bigcup_k \{T = k,\ S_k = s\}$ of finitely determined events, so its probability is assigned by countable additivity (theorem 11.31). By $\mathbb{E}[S]$ we mean $\sum_s s \Pr[S = s]$, the expectation of this countably supported distribution, a sum of nonnegative terms.

Expanding each $\Pr[S = s]$ and regrouping the doubly indexed nonnegative series,

$$\mathbb{E}[S] = \sum_s s \sum_{k \geq 1} \Pr[T = k,\ S_k = s] = \sum_{k \geq 1} \mathbb{E}[S_k \mathbf{1}_{T=k}],$$

each inner expectation living on the finite prefix space of the first k trials (theorem 11.30), where linearity (theorem 11.14) expands it as $\sum_{i=1}^{k} \mathbb{E}[X_i \mathbf{1}_{T=k}]$. Regrouping the nonnegative double series once more,

$$\mathbb{E}[S] = \sum_{i \geq 1} \sum_{k \geq i} \mathbb{E}[X_i \mathbf{1}_{T=k}].$$

Fix i and let $n \geq i$. The events $T = k$ (for $k \leq n$), $T > n$, and $T \geq i$ are all decided by the first n trials, and on that prefix space the pointwise identity $\mathbf{1}_{T \geq i} = \sum_{k=i}^{n} \mathbf{1}_{T=k} + \mathbf{1}_{T>n}$ holds, so linearity gives

$$\sum_{k=i}^{n} \mathbb{E}[X_i \mathbf{1}_{T=k}] = \mathbb{E}[X_i \mathbf{1}_{T \geq i}] - \mathbb{E}[X_i \mathbf{1}_{T>n}].$$

The trials take finitely many values, so $X_i \leq M$ for a constant M, and $0 \leq \mathbb{E}[X_i \mathbf{1}_{T>n}] \leq M \Pr[T > n] \to 0$ as $n \to \infty$: the terms $\Pr[T \geq k]$ of the convergent series $\mathbb{E}[T]$ tend to zero. Hence $\sum_{k \geq i} \mathbb{E}[X_i \mathbf{1}_{T=k}] = \mathbb{E}[X_i \mathbf{1}_{T \geq i}]$.

Now the crux factorisation. The event $T \geq i$ — the negation of $T \leq i - 1$ — is decided by the first $i - 1$ trials, so its indicator is a function $g(t_1, \dots, t_{i-1})$ of the first $i - 1$ trial values, and on the prefix space of the first i trials the expectation splits along the product weights of theorem 11.30: with p the common law,

$$\mathbb{E}[X_i \mathbf{1}_{T \geq i}] = \sum_{t_1, \dots, t_i} g(t_1, \dots, t_{i-1})\ t_i\, p(t_1) \cdots p(t_i) = \Pr[T \geq i] \cdot \mathbb{E}[X_1],$$

the double sum factoring into the sum over $(t_1, \dots, t_{i-1})$, which totals $\Pr[T \geq i]$, times $\sum_{t_i} t_i\, p(t_i) = \mathbb{E}[X_1]$. Assembling the pieces and applying the tail-sum formula (theorem 11.33),

$$\mathbb{E}[S] = \sum_{i \geq 1} \mathbb{E}[X_1] \Pr[T \geq i] = \mathbb{E}[X_1] \cdot \mathbb{E}[T]. \qquad \square$$

11.7.2 The clock toolkit: exponential clocks, Poisson counts, and the Chernoff method

The second dividend is continuous time. A block-discovery process is modelled by *clocks*: each party's next block arrives after a random delay, a delay does not remember how long it has already run, and two parties race. A light client that samples a chain needs a tail bound — how likely is

an adversary to have found cL blocks while the honest chain found L? — and a wallet that spaces its network queries needs a delay law with a prescribed mean and no memory. This unit builds the clocks, counts their arrivals, and proves the tail bounds those uses invoke, in the forms they invoke them. Everything reduces to density-defined laws (theorem 11.32), elementary calculus, and Markov's inequality with a free parameter.

Definition 11.36 (Exponential law)**.** A random quantity X is *exponential with rate* $\lambda > 0$, written $X \sim \mathrm{Exp}(\lambda)$, if it has the density $f(t) = \lambda e^{-\lambda t}$ for $t \geq 0$ and $f(t) = 0$ for $t < 0$.

Lemma 11.37 (Tail, mean, and memorylessness)**.** *Let* $X \sim \mathrm{Exp}(\lambda)$*. Then* $\Pr[X > t] = e^{-\lambda t}$ *for all* $t \geq 0$, $\mathbb{E}[X] = 1/\lambda$*, and*

$$\Pr[X > s + t \mid X > s] = \Pr[X > t] \qquad \textit{for all } s, t \geq 0,$$

conditional probability being the ratio of theorem 11.5: having waited without an arrival teaches nothing about the remaining wait. Conversely, a nonnegative quantity with $\Pr[X > t] = e^{-\lambda t}$ *for all* $t \geq 0$ *is exponential with rate* λ*.*

Proof. The density integrates to 1, and $\Pr[X > t] = 1 - \int_0^t \lambda e^{-\lambda s}\, ds = e^{-\lambda t}$. Integrating by parts,

$$\mathbb{E}[X] = \int_0^\infty t\lambda e^{-\lambda t}\, dt = [-te^{-\lambda t}]_0^\infty + \int_0^\infty e^{-\lambda t}\, dt = \frac{1}{\lambda}.$$

Since $X > s+t$ implies $X > s$, the conditional probability is $e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t}$. For the converse, the events $X > a$ and $X > b$ ($a \leq b$) differ by $a < X \leq b$, so $\Pr[a < X \leq b] = e^{-\lambda a} - e^{-\lambda b} = \int_a^b \lambda e^{-\lambda t}\, dt$; and the point a carries probability at most $e^{-\lambda(a-\delta)} - e^{-\lambda a}$ for every $\delta > 0$, hence zero, so $\Pr[a \leq X \leq b]$ is the same integral. □

Remark 11.38 (Where exponential clocks come from)*.* The exponential law is the continuous limit of the geometric waiting time (theorem 11.34). If trials succeed with a small probability ε and each occupies time ε/λ, the first success arrives at time $\varepsilon T/\lambda$, and for integer-valued T

$$\Pr[\varepsilon T/\lambda > t] = \Pr[T \geq \lfloor \lambda t/\varepsilon \rfloor + 1] = (1-\varepsilon)^{\lfloor \lambda t/\varepsilon \rfloor} \longrightarrow e^{-\lambda t} \qquad (\varepsilon \to 0),$$

because $(1-\varepsilon)^{1/\varepsilon} \to e^{-1}$ and the floor alters the exponent by less than one. Whether a proof of work fits this picture — each candidate independent of the last — is a question about the trials, which the volumes on consensus take up.

Example 11.39 (A jittered schedule)**.** A wallet that must poll each of n addresses about once a day waits $\Delta \sim \mathrm{Exp}(1/T)$ between polls, with $T = 86400/n$ seconds: $\mathbb{E}[\Delta] = T$ by theorem 11.37, one expected poll per address per day. For $n = 3$, $T = 28\,800$ seconds, eight hours; the gap exceeds a full day with probability $e^{-3} = 0.0498$, and exceeds twice its mean with probability $e^{-2} = 0.1353$ whatever T is. By memorylessness, an observer who has seen no poll for a while has learnt nothing about when the next one comes.

Two clocks race, and clocks are restarted; both need joint events of several density-defined quantities, which theorem 11.32 provides only for pairs.

Definition 11.40 (Independent family of density-defined quantities)**.** Density-defined quantities $X_1, \dots, X_n$ with densities $f_1, \dots, f_n$ form an *independent family* if $\Pr[X_1 \in I_1, \dots, X_n \in I_n] = \prod_i \Pr[X_i \in I_i]$ for all intervals $I_1, \dots, I_n$. As in theorem 11.32, the probability of a joint event $(X_1, \dots, X_n) \in R$, for a region $R \subseteq \mathbb{R}^n$ cut out by finitely many inequalities, is then taken to be

the iterated integral of $f_1(s_1)\cdots f_n(s_n)$ over R, in whichever order of integration is convenient (theorem 11.31). In particular a quantity that is a function of one block of the family is independent of a quantity that is a function of a disjoint block: integrate the two blocks separately. A sequence $X_1, X_2, \dots$ is an independent family if every finite prefix is one, in the manner of theorem 11.30.

Lemma 11.41 (Integrating out one quantity). *Let U be a density-defined quantity, with density f_U, that is a function of the first k members of an independent family $X_1, \dots, X_n$, and write $W = (X_{k+1}, \dots, X_n)$. Let R be a region in the (U, W)-coordinates whose section $\{s : (s, w) \in R\}$ is an interval for every fixed value w of W. Then*

$$\Pr[(U, W) \in R] = \int_{-\infty}^{\infty} f_U(s)\ \Pr[W \in R_s]\, ds, \qquad R_s = \{w : (s, w) \in R\},$$

the inner probability being a joint event of $X_{k+1}, \dots, X_n$ alone.

Proof. Integrate the product density over the first k coordinates first, holding w fixed: the result is the probability that U lies in the interval $\{s : (s, w) \in R\}$, which by theorem 11.32 equals $\int f_U(s)\, \mathbf{1}[(s, w) \in R]\, ds$. What remains is the integral of $f_{k+1}\cdots f_n$ against this function of w; exchanging the order and integrating over w first leaves, for each s, the inner probability $\Pr[W \in R_s]$. □

Lemma 11.42 (Density of a sum). *Let U, V be independent density-defined quantities, nonnegative (their densities vanish on the negative axis). Then $U + V$ is density-defined, with density the* convolution *$(f_U * f_V)(x) = \int_0^x f_U(u) f_V(x - u)\, du$ for $x \geq 0$.*

Proof. For $0 \leq a \leq b$, the event $a \leq U + V \leq b$ is the region $\{(u, v) : u, v \geq 0,\ a \leq u + v \leq b\}$, so

$$\Pr[a \leq U + V \leq b] = \int_0^{\infty} f_U(u) \int_{\max(a-u,0)}^{\max(b-u,0)} f_V(v)\, dv\, du = \int_0^{\infty} f_U(u) \int_{\max(a,u)}^{\max(b,u)} f_V(x - u)\, dx\, du,$$

substituting $x = u + v$ in the inner integral. Exchanging the order, the region becomes $a \leq x \leq b$, $0 \leq u \leq x$, and the right-hand side is $\int_a^b (f_U * f_V)(x)\, dx$. With $a = 0$ and $b = \infty$ the left-hand side is 1, so $f_U * f_V$ is a density. □

Lemma 11.43 (Two-clock race). *Let $X \sim \mathrm{Exp}(\lambda)$ and $Y \sim \mathrm{Exp}(\mu)$ be independent. Then, for every $u \geq 0$,*

$$\Pr[X < Y,\ X \leq u] = \frac{\lambda}{\lambda + \mu}(1 - e^{-(\lambda+\mu)u}),$$

symmetrically with the roles of (X, λ) and (Y, μ) exchanged, and $\Pr[X = Y] = 0$. Consequently $\Pr[X < Y] = \lambda/(\lambda + \mu)$; the winning time $\min(X, Y)$ is exponential with rate $\lambda + \mu$; and the winner is independent of the winning time,

$$\Pr[X < Y,\ \min(X, Y) \leq u] = \Pr[X < Y] \cdot \Pr[\min(X, Y) \leq u] \qquad \textit{for all } u \geq 0.$$

Proof. The event is the region $0 \leq s \leq u$, $t > s$ in the (X, Y)-plane. Integrating over t first (theorem 11.40),

$$\Pr[X < Y,\ X \leq u] = \int_0^u \lambda e^{-\lambda s}\ \Pr[Y > s]\, ds = \int_0^u \lambda e^{-(\lambda+\mu)s}\, ds = \frac{\lambda}{\lambda + \mu}(1 - e^{-(\lambda+\mu)u}).$$

The diagonal $s = t$ has inner integral zero, so $\Pr[X = Y] = 0$, and $u = \infty$ gives $\Pr[X < Y]$. On the event $X < Y$ the winning time is X, so the displayed formula is $\Pr[X < Y,\ \min(X, Y) \le u]$; adding its mirror for $Y < X$ gives $\Pr[\min(X, Y) \le u] = 1 - e^{-(\lambda+\mu)u}$, whence $\min(X, Y)$ is exponential with rate $\lambda + \mu$ by the characterisation in theorem 11.37, and the formula is exactly the product $\Pr[X < Y] \cdot \Pr[\min(X, Y) \le u]$. Differencing at two values of u extends the factorisation from half-lines to intervals. □

Definition 11.44 (Exponential clock process)**.** An *exponential clock of rate* λ — a *Poisson process*, once theorem 11.47 is in hand — is an independent family $G_1, G_2, \ldots$ of $\mathrm{Exp}(\lambda)$ quantities, the *gaps*. Its *arrival times* are the partial sums $S_n = G_1 + \cdots + G_n$, with $S_0 = 0$, and its *arrival count* up to time $t \ge 0$ is $N(t) = \sup\{n : S_n \le t\}$, the number of arrivals in $[0, t]$, a value in $\mathbb{N} \cup \{\infty\}$ that theorem 11.47 shows finite with probability one. The event $N(t) = n$ is $S_n \le t < S_{n+1}$, a joint event of the first $n + 1$ gaps.

Lemma 11.45 (Arrival-time density)**.** *The n-th arrival time S_n of an exponential clock of rate λ has density $f_n(s) = \lambda^n s^{n-1} e^{-\lambda s}/(n-1)!$ for $s \ge 0$.*

Proof. Induction on n: f_1 is the exponential density, and $S_{n+1} = S_n + G_{n+1}$ with S_n a function of the first n gaps, hence independent of G_{n+1} (theorem 11.40), so theorem 11.42 gives

$$f_{n+1}(s) = \int_0^s \frac{\lambda^n u^{n-1} e^{-\lambda u}}{(n-1)!}\, \lambda e^{-\lambda(s-u)}\, du = \frac{\lambda^{n+1} e^{-\lambda s}}{(n-1)!} \int_0^s u^{n-1}\, du = \frac{\lambda^{n+1} s^n e^{-\lambda s}}{n!}.$$ □

Definition 11.46 (Poisson law)**.** A random variable K with values in $\mathbb{N}$ is *Poisson with mean* $\nu > 0$ if $\Pr[K = k] = e^{-\nu}\nu^k/k!$ for every $k \ge 0$. The weights sum to 1 by the exponential series $e^\nu = \sum_{k\ge0} \nu^k/k!$, and $\mathbb{E}[K] = \sum_{k\ge1} k\, e^{-\nu}\nu^k/k! = \nu e^{-\nu} \sum_{k\ge1} \nu^{k-1}/(k-1)! = \nu$, so the parameter is indeed the mean.

An arrival count is a function of the gaps of a clock family, so it lives on no finite or countable space, and the expectation of theorem 11.11 does not literally apply to it. One convention serves every such count in this unit — $N(t)$ above and the count X of theorem 11.51 below. Let X be a variable with values in $\mathbb{N} \cup \{\infty\}$ that is a function of the gaps, each event $X = x$ with x finite being a joint event of finitely many gaps, with the probability of theorem 11.40. Events $X \in B$, for sets B of finite values, have the probabilities $\sum_{x\in B} \Pr[X = x]$ (theorem 11.31), and when X is finite with probability 1 its *mean* is defined by the law,

$$\mathbb{E}[X] = \sum_{x\ge0} x \Pr[X = x],$$

a sum of nonnegative terms, permitted to be $+\infty$ as in the convention preceding theorem 11.33. On a finite or countable space this sum is the second expression of theorem 11.11, the expectation regrouped by value (theorem 11.27), so the two definitions agree whenever both apply.

Theorem 11.47 (Poisson arrival counts)**.** *For an exponential clock of rate λ and every $t \ge 0$, the arrival count $N(t)$ is Poisson with mean λt:*

$$\Pr[N(t) = n] = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, \qquad n = 0, 1, 2, \ldots$$

In particular $N(t)$ is finite with probability 1 and $\mathbb{E}[N(t)] = \lambda t$: a rate-$\lambda$ clock delivers λ arrivals per

unit time on average.

Proof. For $n = 0$ the event is $G_1 > t$, of probability $e^{-\lambda t}$. For $n \geq 1$ the event $S_n \leq t < S_n + G_{n+1}$ is, in the (S_n, G_{n+1})-coordinates, the region $s \leq t$, $g > t - s$, whose section at fixed g is the interval $(t - g, t]$; the quantity S_n is a function of the first n gaps with the density of theorem 11.45, so theorem 11.41 gives

$$\Pr[N(t) = n] = \int_0^t \frac{\lambda^n s^{n-1} e^{-\lambda s}}{(n-1)!} \Pr[G_{n+1} > t - s]\, ds = \frac{\lambda^n e^{-\lambda t}}{(n-1)!} \int_0^t s^{n-1}\, ds = e^{-\lambda t} \frac{(\lambda t)^n}{n!}.$$

The event $N(t) < \infty$ is the disjoint union of the events $N(t) = n$, so its probability is the sum of the weights (theorem 11.31), which is 1 (theorem 11.46); and the mean — the sum over the law, by the convention above — is the one computed in theorem 11.46. □

The tail bounds now. Markov's inequality applied to X itself decays only like $1/a$; applied to e^{tX} it decays exponentially, and the free parameter t is then tuned to the threshold. That is the whole of the Chernoff method.

Definition 11.48 (Moment generating function)**.** For a real-valued random variable X with finite or countable image — one on a finite or countable space, or an arrival count of a clock family in the sense of the convention preceding theorem 11.47, restricted to its finite values (the value ∞ has probability zero there), whose law is countably additive over its values: $\Pr[X \in B] = \sum_{x \in B} \Pr[X = x]$ for every set B of values (theorem 11.31) — and a real t, the *moment generating function* is

$$M_X(t) = \sum_x e^{tx} \Pr[X = x] \in (0, \infty],$$

the sum running over the finite values of X; its terms are positive, so it either converges or is $+\infty$. On a countable space it is $\mathbb{E}[e^{tX}]$, the expectation regrouped by value, and $M_X(0) = \sum_x \Pr[X = x] = 1$.

Theorem 11.49 (The Chernoff method)**.** *Let X be as above, a real, and $t > 0$ with $M_X(t) < \infty$. Then*

$$\Pr[X \geq a] \leq e^{-ta} M_X(t),$$

and hence $\Pr[X \geq a] \leq \inf_t e^{-ta} M_X(t)$, the infimum over the $t > 0$ at which M_X is finite.

Proof. The events $X = x$ are disjoint, so $\Pr[X \geq a] = \sum_{x \geq a} \Pr[X = x]$ by countable additivity (theorem 11.28, or the clause of theorem 11.48 for a clock family). Since $t > 0$, each term with $x \geq a$ satisfies $\Pr[X = x] \leq e^{t(x-a)} \Pr[X = x]$, and summing gives $\Pr[X \geq a] \leq e^{-ta} \sum_x e^{tx} \Pr[X = x]$, the terms with $x < a$ being nonnegative. This is Markov's inequality (theorem 11.15) applied to e^{tX}, written on the law of X. □

Corollary 11.50 (Chernoff bound for a Poisson variable)**.** *Let K be Poisson with mean ν and let $a > \nu$. Then $M_K(t) = \exp(\nu(e^t - 1))$ for every real t, and*

$$\Pr[K \geq a] \leq e^{a-\nu} \Big(\frac{\nu}{a}\Big)^a.$$

In particular, for $\nu = \mu L$ and $a = cL$ with $c > \mu$, the bound reads $e^{L(c-\mu)}\, (c/\mu)^{-cL}$.

Proof. Expanding, $M_K(t) = \sum_k e^{tk} e^{-\nu} \nu^k / k! = e^{-\nu} \sum_k (\nu e^t)^k / k! = e^{-\nu} e^{\nu e^t}$. The exponent $\phi(t) = -ta + \nu(e^t - 1)$ of the Chernoff bound has $\phi'(t) = -a + \nu e^t$ and $\phi''(t) = \nu e^t > 0$, so it is minimised at $e^t = a/\nu$, a positive t exactly when $a > \nu$; there $\phi = -a \log(a/\nu) + a - \nu$, with log the natural logarithm — base-free log is natural throughout, base two being written $\log_2$ — and exponentiating gives the bound. Substituting $\nu = \mu L$ and $a = cL$ gives $e^{cL - \mu L}(\mu/c)^{cL}$. □

Theorem 11.51 (Negative-binomial arrival count)**.** *Let two exponential clocks be independent — their gaps together form one independent family — with rates λ, the* honest *clock, and μ, the* adversary *clock, and let X be the number of adversary arrivals strictly before the L-th honest arrival, for a fixed $L \geq 1$. Then X is finite with probability 1, with the* negative binomial *law*

$$\Pr[X = m] = \binom{m + L - 1}{m} p^L q^m, \qquad p = \frac{\lambda}{\lambda + \mu}, \quad q = \frac{\mu}{\lambda + \mu}, \qquad m = 0, 1, 2, \dots,$$

of mean $Lq/p = L\mu/\lambda$, and, for every real t with $\mu e^t < \lambda + \mu$,

$$M_X(t) = \Big(\frac{\lambda}{\lambda + \mu - \mu e^t}\Big)^L;$$

with the honest rate normalised to 1 this is $M_X(t) = (1/(1 + \mu - \mu e^t))^L$. Only the ratio μ/λ enters.

Proof. Write S_L for the L-th honest arrival time, a function of the honest gaps with the density f_L of theorem 11.45, and N_A for the adversary's arrival count. The event $X = m$ differs from $N_A(S_L) = m$ only on the events $S_L = A_1 + \cdots + A_m$ and $S_L = A_1 + \cdots + A_{m+1}$, with A_i the adversary gaps; each of these, in the coordinates $(S_L, A_1, \dots, A_{m+1})$, has a single point as its section at fixed adversary gaps, hence probability zero (theorem 11.32), so $\Pr[X = m] = \Pr[N_A(S_L) = m]$. The latter event is the region $A_1 + \cdots + A_m \leq S_L < A_1 + \cdots + A_{m+1}$, whose section at fixed adversary gaps is an interval. By theorem 11.41 and theorem 11.47,

$$\begin{aligned}\Pr[X = m] &= \int_0^\infty f_L(s)\, \Pr[N_A(s) = m]\, ds = \frac{\lambda^L \mu^m}{(L-1)!\, m!} \int_0^\infty s^{m+L-1} e^{-(\lambda+\mu)s}\, ds \\ &= \frac{\lambda^L \mu^m}{(L-1)!\, m!} \cdot \frac{(m + L - 1)!}{(\lambda + \mu)^{m+L}},\end{aligned}$$

using $\int_0^\infty s^k e^{-\alpha s}\, ds = k!/\alpha^{k+1}$ for $\alpha > 0$, which k integrations by parts establish. Rearranging the factorials gives the binomial coefficient and the powers of p and q.

For the generating function, set $x = qe^t$, so that $0 \leq x < 1$ exactly when $\mu e^t < \lambda + \mu$. Then $M_X(t) = p^L \sum_{m \geq 0} \binom{m+L-1}{m} x^m$, and we claim the series is $(1 - x)^{-L}$, by induction on L. For $L = 1$ it is the geometric series. For the step, write $T_L(n) = \sum_{m=0}^n \binom{m+L-1}{m} x^m$ for the partial sums. Pascal's rule $\binom{m+L}{m} = \binom{m+L-1}{m-1} + \binom{m+L-1}{m}$ — sort the m-subsets of $m + L$ items, which the coefficients count (theorem 6.12), by whether they contain a distinguished item — gives, on subtracting $x\, T_{L+1}(n)$ from $T_{L+1}(n)$ term by term,

$$(1 - x)\, T_{L+1}(n) = T_L(n) - \binom{n + L}{n} x^{n+1}.$$

Hence $T_{L+1}(n) \leq T_L(n)/(1 - x) \leq (1 - x)^{-L-1}$ by the induction hypothesis: the partial sums $T_{L+1}(n)$ increase and are bounded, so the series converges, and its terms $\binom{n+L}{n} x^n$ tend to zero. Letting $n \to \infty$ in the display gives $(1 - x) \sum_{m \geq 0} \binom{m+L}{m} x^m = (1 - x)^{-L}$, the claim for $L + 1$. Hence $M_X(t) = (p/(1 - qe^t))^L$, which is the displayed form after multiplying numerator and denominator by $\lambda + \mu$. At $t = 0$ this gives $\sum_m \Pr[X = m] = 1$; the event $X < \infty$ is the disjoint

union of the events $X = m$, so its probability is this sum (theorem 11.31), and X is finite with probability 1. For the mean — the sum over the law, by the convention preceding theorem 11.47 — $m\binom{m+L-1}{m} = L\binom{m+L-1}{m-1}$, so $\mathbb{E}[X] = Lq\,p^L \sum_{k\geq 0}\binom{k+L}{k}q^k = Lq\,p^L(1-q)^{-L-1} = Lq/p$ by the same series with $L+1$ in place of L. □

Remark 11.52 (The coin-sequence form). The same law arises without clocks. In an infinite sequence of independent trials (theorem 11.30) succeeding with probability p, the number of failures strictly before the L-th success equals m exactly when the $(m+L)$-th trial succeeds and, among the first $m+L-1$, precisely m fail: $\binom{m+L-1}{m}$ orders, one per choice of the m failing positions (theorem 6.12), each of probability $p^{L-1}q^m \cdot p$. The two-clock race and memorylessness (theorem 11.43 and theorem 11.37) are what identify the clock model's attribution sequence — which clock fires next — with such a coin sequence of success probability $\lambda/(\lambda+\mu)$; later volumes carry out that identification.

Theorem 11.53 (Chernoff bound for the arrival count). *In the setting of theorem 11.51 with honest rate* 1 *and adversary rate μ, for every $c > \mu$,*

$$\Pr[X \geq cL] \leq \Big[\Big(\frac{1+c}{1+\mu}\Big)^{1+c}\Big(\frac{\mu}{c}\Big)^{c}\Big]^L,$$

the Chernoff bound optimised at $e^t = c(1+\mu)/(\mu(1+c))$. The base in brackets is strictly less than 1*, so the bound decays exponentially in L.*

Proof. The count X is finite with probability one with values in $\mathbb{N}$ (theorem 11.51) and has a law countably additive over its values (theorem 11.31), so theorem 11.49 applies: with theorem 11.51, $\Pr[X \geq cL] \leq e^{-tcL}M_X(t) = e^{L\phi(t)}$ with $\phi(t) = -tc - \log(1+\mu-\mu e^t)$, for $0 < t < \log((1+\mu)/\mu)$. On that interval

$$\phi'(t) = -c + \frac{\mu e^t}{1+\mu-\mu e^t}, \qquad \phi''(t) = \frac{\mu e^t(1+\mu)}{(1+\mu-\mu e^t)^2} > 0,$$

so ϕ' is strictly increasing (positive derivative) and vanishes at most once, namely where $\mu e^t(1+c) = c(1+\mu)$, that is, at $e^{t^*} = c(1+\mu)/(\mu(1+c))$; this lies in the interval because $c > \mu$ gives $e^{t^*} > 1$ and $c/(1+c) < 1$ gives $e^{t^*} < (1+\mu)/\mu$. Thus ϕ decreases before t^* and increases after it, and $\phi(t^*)$ is its minimum on the interval. At t^*, $1+\mu-\mu e^{t^*} = (1+\mu)/(1+c)$, so $e^{\phi(t^*)} = (e^{t^*})^{-c}\,(1+c)/(1+\mu)$, which is the bracketed base. Finally $\phi(0) = 0$ and ϕ decreases on $[0, t^*]$, since $\phi'(0) = \mu - c < 0$, so $\phi(t^*) < 0$: the base is below 1. □

Example 11.54 (The Poisson expression is not a bound for the arrival count). With the honest rate 1, the arrival count X has mean μL, the same as the Poisson variable of theorem 11.50 with $\nu = \mu L$, which counts adversary arrivals in a window of fixed length L; the two laws differ because the window here ends at the random time S_L. Substituting one for the other is tempting and wrong. At $\mu = 1/2$ and $c = 3/4$, the exact tail $\Pr[X \geq cL]$ (summing theorem 11.51), the bound of theorem 11.53, and the Poisson expression $e^{L(c-\mu)}(c/\mu)^{-cL}$ are

L	exact tail	clock bound	Poisson expression
40	$5.05 \cdot 10^{-2}$	$2.53 \cdot 10^{-1}$	$1.15 \cdot 10^{-1}$
200	$1.09 \cdot 10^{-4}$	$1.04 \cdot 10^{-3}$	$2.00 \cdot 10^{-5}$
400	$8.23 \cdot 10^{-8}$	$1.08 \cdot 10^{-6}$	$4.00 \cdot 10^{-10}$

At $L = 200$ and $L = 400$ the Poisson expression lies *below* the exact tail, so it is not an upper bound for the arrival count; its base per unit of L is 0.9473, against 0.9662 for the valid bound. A bound stated for the wrong law is not a bound.

Later volumes build block-discovery and extraction analyses directly on this base: block clocks are the exponential clocks above, their races and arrival counts the lemmas of this unit, block-attribution sequences are infinite sequences of independent trials, extractor running times are the waiting times of the previous unit (the geometric waiting time and Wald's identity), and the limit arguments instantiate theorem 11.29.

11.8 Asymptotic notation

The discussion now turns from probability to growth rates. To address efficiency and hardness, we must compare how functions of an integer parameter grow as that parameter tends to infinity. The Bachmann–Landau notation makes such comparisons precise. Throughout, functions map $\mathbb{N}$ (or a cofinite subset of it) to the nonnegative reals $\mathbb{R}_{\geq 0}$, and "for sufficiently large n" means "for all n beyond some threshold n_0".

Definition 11.55 (Big-O, Omega, Theta, little-o, little-omega)**.** Let $f, g\colon \mathbb{N} \to \mathbb{R}_{\geq 0}$ with g eventually positive.

- $f = O(g)$ (read "f is big-O of g", an asymptotic upper bound) if there exist constants $c > 0$ and n_0 with $f(n) \leq c\,g(n)$ for all $n \geq n_0$.
- $f = \Omega(g)$ (asymptotic lower bound) if there exist $c > 0$ and n_0 with $f(n) \geq c\,g(n)$ for all $n \geq n_0$; equivalently, $f = \Omega(g)$ if and only if $g = O(f)$.
- $f = \Theta(g)$ (tight bound) if $f = O(g)$ and $f = \Omega(g)$; i.e. there are $c_1, c_2 > 0$ and n_0 with $c_1 g(n) \leq f(n) \leq c_2 g(n)$ for $n \geq n_0$.
- $f = o(g)$ (read "little-o", strictly smaller order) if for every $c > 0$ there is n_0 with $f(n) \leq c\,g(n)$ for all $n \geq n_0$; equivalently $\lim_{n\to\infty} f(n)/g(n) = 0$.
- $f = \omega(g)$ (strictly larger order) if $g = o(f)$; equivalently $\lim_{n\to\infty} f(n)/g(n) = \infty$.

Remark 11.56. The "=" in "$f = O(g)$" is traditional but denotes set membership: $O(g)$ is the class of all functions bounded above by a constant multiple of g, and $f = O(g)$ means $f \in O(g)$. One never reads such equations right to left. We follow the customary abuse of notation.

Example 11.57. The polynomial $3n^2 + 10n + 7$ is $\Theta(n^2)$: for $n \geq 1$ one has $3n^2 \leq 3n^2 + 10n + 7 \leq 20n^2$, with equality on the right at $n = 1$, so the constant 20 is exact there. Also $n^2 = o(n^3)$, since $n^2/n^3 = 1/n \to 0$; $\log_2 n = o(n)$; $n^{100} = o(2^n)$ (any polynomial is little-o of any exponential with base > 1); and $n! = \omega(2^n)$. A cautionary pair: $2^n = o(2^{n+1})$ is *false*, but $2^n = \Theta(2^{n+1})$ is true, since $2^{n+1} = 2 \cdot 2^n$ is within a constant factor.

Proposition 11.58 (Useful closure facts)**.** *Let $f_1 = O(g_1)$ and $f_2 = O(g_2)$. Then $f_1 + f_2 = O(\max(g_1, g_2)) = O(g_1 + g_2)$ and $f_1 f_2 = O(g_1 g_2)$. The same closure statements hold with O replaced uniformly by Ω or Θ. Moreover $O(\cdot)$ is transitive: $f = O(g)$ and $g = O(h)$ imply $f = O(h)$.*

Proof. Choose witnesses (c_1, n_1) and (c_2, n_2) for the two hypotheses and let $n_0 = \max(n_1, n_2)$. For $n \geq n_0$, $f_1(n) + f_2(n) \leq c_1 g_1(n) + c_2 g_2(n) \leq (c_1 + c_2)\max(g_1(n), g_2(n))$, and $\max(g_1, g_2) \leq g_1 + g_2 \leq 2\max(g_1, g_2)$, so the two right-hand forms agree up to a constant; this proves the sum rule. For products, $f_1(n) f_2(n) \leq c_1 c_2\, g_1(n) g_2(n)$. Transitivity: $f \leq c\,g$ and $g \leq c'h$ eventually give $f \leq cc'\,h$ eventually. The Ω statements follow by the equivalence $f = \Omega(g) \Leftrightarrow g = O(f)$, and Θ by combining the two. □

11.9 Polynomial, exponential, and negligible functions

The payoff of this subsection is a closure theorem (theorem 11.63): the class of *negligible* functions — those shrinking faster than the reciprocal of every polynomial — is stable under exactly the operations that security proofs perform, addition and multiplication by polynomials, which is why "negligible" is the right formalisation of "ignorable". We first classify the growth rates relevant to cryptography. Fix a distinguished variable $\lambda \in \mathbb{N}$, the *security parameter* (discussed in detail in §11.11); informally, larger λ means more security and larger keys. Resources (running time, key length) and success probabilities are functions of λ.

Definition 11.59 (Polynomial and super-polynomial)**.** A function $f : \mathbb{N} \to \mathbb{R}_{\geq 0}$ is *polynomially bounded*, written $f = \mathrm{poly}(\lambda)$, if there is a constant c (and a threshold) with $f(\lambda) \leq \lambda^c + c$ for all large λ; equivalently, $f(\lambda) = O(\lambda^c)$ for some constant c. It is *super-polynomial* if it grows faster than every polynomial, i.e. $f(\lambda) = \omega(\lambda^c)$ for every constant c (equivalently, $\lambda^c = o(f(\lambda))$ for all c).

Definition 11.60 (Exponential and sub-exponential)**.** A function f is *exponential* if $f(\lambda) = 2^{\Omega(\lambda)}$; the prototypical case is $f(\lambda) = 2^{c\lambda}$ for a constant $c > 0$. A super-polynomial function with $f(\lambda) = 2^{o(\lambda)}$ is *sub-exponential*; examples are $\lambda^{\log_2 \lambda} = 2^{(\log_2 \lambda)^2}$ and the index-calculus running time $L_p[1/3] = 2^{\Theta(\lambda^{1/3}(\log \lambda)^{2/3})}$ in the bit length $\lambda = \log_2 p$ (theorem 7.22). Every exponential function is super-polynomial; the converse fails by these examples.

The central cryptographic notion is that of a negligible function. A negligible probability may be ignored because no efficient (polynomially many) repetition of an experiment can amplify it to a noticeable chance — the closure theorem below makes this precise.

Definition 11.61 (Negligible function)**.** A function $\nu : \mathbb{N} \to \mathbb{R}_{\geq 0}$ is *negligible*, written $\nu = \mathrm{negl}(\lambda)$, if for every positive constant c there exists λ_0 such that

$$\nu(\lambda) < \frac{1}{\lambda^c} \qquad \text{for all } \lambda \geq \lambda_0.$$

Equivalently, $\nu(\lambda) = o(\lambda^{-c})$ for every constant $c > 0$; equivalently, $\lambda^c \nu(\lambda) \to 0$ as $\lambda \to \infty$ for every constant c. A function is *non-negligible* if it is not negligible, i.e. $f(\lambda) \geq \lambda^{-c}$ for infinitely many λ, for some constant c; it is *noticeable* if there exist c and λ_0 with $f(\lambda) \geq \lambda^{-c}$ for all $\lambda \geq \lambda_0$ (noticeable implies non-negligible, but not conversely); it is *overwhelming* if $1 - f$ is negligible.

Example 11.62. The functions $2^{-\lambda}$, $2^{-\sqrt{\lambda}}$, and $\lambda^{-\log \lambda}$ are negligible. For $2^{-\lambda}$: given any c, the product $2^{-\lambda}\lambda^c = \lambda^c/2^\lambda$ tends to 0 (a polynomial over an exponential), meeting the definition. By contrast, $1/\lambda^{100}$ is *not* negligible — taking $c = 101$ shows $1/\lambda^{100} > 1/\lambda^{101}$ for all $\lambda \geq 2$ — it is noticeable, even though it tends to 0. This is the key subtlety: tending to zero does not suffice; a negligible function must beat every inverse polynomial.

Proposition 11.63 (Closure properties of negligible functions)**.** *Let ν, μ be negligible and let p be a polynomially bounded function. Then:*

1. *$\nu + \mu$ is negligible (the class is closed under addition, hence under any fixed finite sum).*
2. *$p \cdot \nu$ is negligible (a polynomial times a negligible function is negligible).*

3. *If $f(\lambda) \le \nu(\lambda)$ for all large λ, then f is negligible (domination by a negligible function).*
4. *Summing $p(\lambda)$ functions, each bounded by a single negligible function ν, yields a negligible function; in particular a union bound over polynomially many negligible events is negligible.*

Proof. The mechanism throughout is an exponent shift: to beat λ^{-c}, invoke negligibility at a strictly larger exponent, so that the surplus absorbs the combining factor. The same shift is the standard mechanism behind hybrid and union-bound steps in security proofs.

(1) Fix c. Applying the definition at the exponent $c+1$, there is a threshold beyond which $\nu(\lambda), \mu(\lambda) < \lambda^{-(c+1)} \le \frac{1}{2}\lambda^{-c}$ (the latter once $\lambda \ge 2$); adding the two bounds gives $\nu(\lambda) + \mu(\lambda) < \lambda^{-c}$ for all large λ.

(2) Suppose $p(\lambda) \le \lambda^d$ for large λ (and some constant d). Fix c. By negligibility applied at the exponent $c+d$, $\nu(\lambda) < \lambda^{-(c+d)}$ for large λ, whence $p(\lambda)\nu(\lambda) < \lambda^d\lambda^{-(c+d)} = \lambda^{-c}$ for large λ. Hence $p\nu$ is negligible.

(3) Immediate: if $f \le \nu$ eventually and $\nu < \lambda^{-c}$ eventually, then $f < \lambda^{-c}$ eventually, for every c.

(4) Write the sum $\Sigma(\lambda) = \sum_{i=1}^{p(\lambda)} \nu_i(\lambda)$, where each $\nu_i \le \nu$ by hypothesis (in security proofs all the ν_i arise from one bound). Then $\Sigma(\lambda) \le p(\lambda)\,\nu(\lambda)$, which is negligible by (2); conclude by (3). □

Remark 11.64 (Why negligible is the threshold of security). Suppose an adversary wins a game with negligible probability $\nu(\lambda)$. If it repeats the attack $p(\lambda)$ times (polynomially many, the most an efficient adversary can afford), the probability that some repetition succeeds is, by the union bound, at most $p(\lambda)\nu(\lambda)$ — still negligible by theorem 11.63(2). Thus negligible success probability remains negligible under any efficient amplification: it is genuinely ignorable. Conversely, a noticeable success probability $\ge\lambda^{-c}$ can be boosted to a constant by λ^c repetitions. The negligible/non-negligible split is the formal backbone of the phrase "the scheme is secure except with negligible probability": security asserts a negligible advantage, and an attack means a non-negligible one. (A merely non-negligible — not noticeable — success probability is boosted to a constant only on the infinitely many λ where the inverse-polynomial bound holds, which already suffices to contradict a security definition.)

11.10 Algorithms, running time, and PPT

It remains to make "efficient algorithm" and "infeasible" precise. We adopt the standard model: algorithms are Turing machines (equivalently, up to polynomial factors, programs in any reasonable model of computation), inputs and outputs are finite binary strings, and the measured resource is the number of elementary steps as a function of input length. Write $\{0,1\}^*$ for the set of all finite binary strings and $|x|$ for the length of $x \in \{0,1\}^*$.

Definition 11.65 (Deterministic polynomial-time algorithm)**.** An algorithm M runs in (*deterministic*) *polynomial time* if there is a polynomial p such that, on every input $x \in \{0,1\}^*$, M halts within at most $p(|x|)$ steps and produces an output. The class of decision problems solvable by such machines is P.

Determinism is insufficient for cryptography: key generation and sampling require randomness. We model randomness by granting the machine an auxiliary tape of independent uniform random bits (the "coins").

Definition 11.66 (Probabilistic algorithm and PPT)**.** A *probabilistic* (or *randomised*) algorithm M is a Turing machine with an additional read-only *random tape* initialised with independent

uniformly random bits $r \in \{0,1\}^{\mathbb{N}}$. For a fixed input x, the output $M(x)$ is a random variable over the choice of r; write $M(x;r)$ to display the coins explicitly, and then $M(x;\cdot)$ is a deterministic function. The algorithm runs in *probabilistic polynomial time* (it is *PPT*) if there is a polynomial p such that, for every input x *and every choice of coins* r, M halts within $p(|x|)$ steps. (Bounding the time for every r, not merely in expectation, is the standard "strict" notion and ensures M ever reads only $p(|x|)$ coins.)

Remark 11.67 (Non-uniformity and circuits). There are two standard ways to formalise an efficient adversary. The *uniform* model is a single PPT machine that works for all input lengths, as above. The *non-uniform* model allows a separate algorithm (equivalently, a Boolean circuit) of size $\mathrm{poly}(\lambda)$ for each λ, which may be hard-wired with an arbitrary polynomial-length "advice" string depending on λ. Non-uniform adversaries are at least as powerful as uniform ones and are often more convenient in reduction proofs, because the advice can encode a "best" attack. The asymptotic theory of this section is identical for both models; security definitions in the Halo 2/Orchard setting typically state security against non-uniform PPT adversaries, the more conservative choice.

With efficiency pinned down, we can state the strong hardness notion used in cryptography. The guiding principle is the Cobham–Edmonds thesis: "efficiently computable" means "computable in polynomial time". Feasibility and the infeasibility notion below are useful opposite regimes, but they are not logical negations: intermediate success probabilities can fall into neither class.

Definition 11.68 (Computational infeasibility)**.** A computational task parameterised by the security parameter λ is (*computationally*) *feasible* if there is a PPT algorithm solving it with overwhelming success probability. It is *infeasible* if for *every* PPT algorithm $\mathcal{A}$ the probability that $\mathcal{A}$ solves it is negligible in λ:

$$\Pr[\mathcal{A} \text{ solves the task on parameter } \lambda] = \mathrm{negl}(\lambda),$$

where the probability is over the task's randomness and $\mathcal{A}$'s coins. Equivalently: no efficient algorithm succeeds with non-negligible probability.

Remark 11.69 (Reading a hardness assumption). Cryptographic assumptions are infeasibility statements. For instance, the discrete-logarithm assumption in a family of groups $\{\mathbb{G}_\lambda\}$ of order $q(\lambda)$ asserts: for every PPT $\mathcal{A}$,

$$\Pr[\mathcal{A}(g, g^x) = x] = \mathrm{negl}(\lambda),$$

the probability taken over a uniformly chosen generator g of $\mathbb{G}_\lambda$, a uniform $x \leftarrow \mathbb{Z}/q\mathbb{Z}$, and $\mathcal{A}$'s coins. The entire edifice of Halo 2 and Orchard rests on such asymptotic infeasibility assumptions; choosing a parameter size does not prove them. Separately, concrete analysis asks how much work the best known attacks need at one fixed size. Generic discrete-log attacks take $\Theta(\sqrt{q})$ group operations by the birthday/rho phenomenon of theorem 11.20; absent a faster structure-specific attack, an order near $2^{2\lambda}$ therefore targets about λ bits of generic-group security. The Pasta orders are just above 2^{254}, giving roughly 2^{127} generic work. The asymptotic language permits the terms "negligible" and "polynomial"; concrete estimates remain conditional on the attacks presently known.

11.11 The security parameter

The section closes by making explicit the parameter that threads through every preceding definition, and by paying its opening debt.

Definition 11.70 (Security parameter)**.** The *security parameter* is a positive integer $\lambda \in \mathbb{N}$ supplied to all algorithms of a cryptographic scheme, conventionally in unary as the string 1^λ (so that "polynomial in the input length" coincides with "polynomial in λ"). It governs simultaneously:

- the *sizes* of objects — key lengths and encodings, each $\mathrm{poly}(\lambda)$; in a generic discrete-log instantiation one commonly chooses a group order near $2^{2\lambda}$;
- the *running time* of honest algorithms, which must be $\mathrm{poly}(\lambda)$; and
- the *security level*: every PPT adversary's advantage must be $\mathrm{negl}(\lambda)$.

Remark 11.71 (Why unary, and what λ buys). Passing 1^λ (a string of λ ones, of length λ) rather than the integer λ in binary (length $\log_2 \lambda$) is a deliberate convention: it forces "polynomial time in the input length" to mean polynomial in λ itself, so an honest key generator may take time $\mathrm{poly}(\lambda)$ to produce λ-bit keys. In a scheme satisfying the security definition, increasing λ raises honest cost only polynomially while every PPT adversary's success remains negligible; in common concrete families, the best known attack cost grows exponentially. Asymptotic security is the statement that this gap holds for all sufficiently large λ; the practitioner then fixes a single value (commonly $\lambda = 128$, targeting 2^{128}-operation attacks) and checks the concrete estimates against that target — the just-over-2^{254} Pallas/Vesta group orders give roughly 2^{127} generic group operations (section 10; Zcash protocol specification §5.4.9.6), hash outputs of $\geq$256 bits per the birthday bound of theorem 11.18, and sampling slack of at least 128 extra bits per theorem 11.24 (the deployed ToScalar of theorem 11.26 reduces 512 bits, a slack of 257). Every "negligible" in the remainder of the series is implicitly a function of this λ.

The opening cheque can now be cashed, with its two layers kept separate. Asymptotic infeasibility (theorem 11.68) quantifies over every probabilistic polynomial-time forger — uniform or hard-wired with advice (theorem 11.67) — and requires its success to be negligible in λ, a bound that no polynomial number of repetitions amplifies to a noticeable chance (theorem 11.64). A concrete "128-bit" estimate instead says that the best known applicable attack at a fixed parameter size costs on the order of 2^{128} operations; it is an attack estimate, not a theorem about every possible algorithm. Pasta's just-over-2^{254} group orders give roughly 2^{127} generic group operations and are conventionally placed near that target. The forger of the opening paragraph may still guess a key and win with astronomical luck; the formal promise is negligible success against every PPT forger, conditional on the stated cryptographic assumptions.

Part II
Crypto Guide: Cryptographic primitives from scratch

Abstract

This volume of *The Zcash Arboretum* assembles the cryptographic toolbox: the primitives from which a shielded protocol is built, each defined, constructed, and proved against a named enemy. It assumes the *Math Guide*, whose groups, finite fields, elliptic curves, probability, and model of computation are used throughout without re-derivation.

Every guarantee here is a claim about an adversary, and every security definition is a game: a challenger, an efficient adversary, and an advantage that must be negligible. A proof is a reduction that turns any winning adversary into a solver for a long-attacked computational problem, with the reduction's loss accounted for down to a concrete bit-security budget. Under that discipline the volume denies, in turn: the cryptanalyst who would recover the exponent behind a public group element, and with it every key; the forger by coincidence, who would collide two documents into one digest or replay a digest outside its role; the peeker and the equivocator on either side of a sealed envelope, who would read a commitment early or open it to a different value; the oracle distinguisher, who would tell a short secret key from blind chance; the owner of the channel, who would read, splice, or counterfeit what passes over it; the woman in the middle, who would terminate both ends of a "secure" conversation at her own desk; the forger of signatures, who would exhibit an authorisation the signer never made; the cheating prover and the curious verifier, who would certify a falsehood or squeeze a transcript for the witness it hides; and the teller of lies about sets, who would prove membership in a collection that never recorded it. What survives their attentions— hash functions, commitments, pseudorandom functions, authenticated encryption, key agreement, signatures, zero-knowledge arguments, and Merkle trees—is the toolbox the deployed protocol spends.

1 The provable-security model

Every claim this volume makes ends, once unwound, in a sentence about an enemy. No forger produces a valid signature on a message the signer never signed; no observer tells an encryption of m_0 from an encryption of m_1; no prover opens one commitment to two different values. Before any such sentence can be a theorem, the enemy must be a mathematical object: something with a precisely delimited repertoire, about which a universally quantified statement can be proved. The preceding volume supplied the algebraic raw material—groups in which discrete logarithms appear hard, finite fields, elliptic curves—but an object is not yet a guarantee. This section builds the adversary, and around it the framework in which "the scheme is secure" becomes a claim with mathematical content.

The central methodological idea is the following. One almost never knows how to prove unconditionally that a cryptographic scheme is secure; such a proof would typically imply $\mathsf{P} \neq \mathsf{NP}$—that problems whose solutions can be checked in polynomial time are not all solvable in polynomial time—and a great deal more. Cryptography therefore adopts a *relativised* standard of proof. One isolates a small number of computational problems—factoring, discrete logarithm, decisional Diffie–Hellman, and the like—that have resisted decades of attack, and *reduces* the security of the schemes to the conjectured hardness of those problems. A proof of security is then a proof of an implication: *if* the underlying problem is hard, *then* the scheme is secure. Equivalently, in its contrapositive and more operationally useful form: *any* adversary that breaks the scheme can be mechanically transformed into an algorithm that solves the underlying hard problem. Since no such algorithm is believed to exist, one concludes that no such adversary exists.

To render any of this rigorous, we must pin down four things: what counts as a "scheme", parametrised by a notion of input size; what counts as an "adversary"; what it means for an

adversary to "break" a scheme, measured by a quantity called its *advantage*; and what it means for one problem to reduce to another. We develop these in turn, followed by the two idealisations under which proofs proceed (the standard model and the random oracle model), and finally the translation between the clean asymptotic language and the concrete "n-bit security" numbers that practitioners actually quote.

1.1 Adversaries and the security parameter

The vocabulary of efficiency and smallness is the *Math Guide*'s, fixed in its closing section; we recall it here in the form this volume uses and extend it with the refinements that are specifically cryptographic.

The *security parameter* $\lambda \in \mathbb{N}$ is a positive integer that controls the sizes of all objects in a cryptographic scheme—key lengths, group orders, output lengths—and is conventionally supplied to every algorithm in unary, written 1^λ, the string of λ ones, so that an algorithm running in time polynomial in the length of its input automatically receives time polynomial in λ (*Math Guide*, §"The security parameter"). Every scheme in this monograph is accordingly not a single object but an infinite *family* of objects indexed by λ.

Two questions immediately arise: why a family, and why unary.

The reason for a *family* is that security is inherently asymptotic. There is no such thing as an absolutely unbreakable scheme with fixed finite key length: an adversary with enough time can always enumerate a finite key space. What we can hope for is that as λ grows, the resources required to break the scheme grow far faster than the resources required to use it. The honest parties' costs (key generation, encryption, signing) should grow *slowly* in λ—polynomially—while the best attack's cost should grow *quickly*—super-polynomially, ideally exponentially. The security parameter is the dial that separates these two growth rates, and security is a statement about the limit $\lambda \to \infty$.

The reason for *unary* encoding, 1^λ rather than the $\lceil \log_2 \lambda \rceil$-bit binary representation of λ, is a bookkeeping convenience that aligns two notions of "polynomial time". Complexity theory measures running time as a function of input *length*. If λ came in binary it would have length $\Theta(\log \lambda)$, and an algorithm "polynomial in its input length" would get only $\mathrm{poly}(\log \lambda)$ steps—not even enough to write down a λ-bit key. Padding the parameter to length λ ensures that "polynomial in the input length" coincides with the cryptographically meaningful "polynomial in λ". This is purely a normalisation; nothing of substance depends on it.

Remark 1.1. Because everything is indexed by λ, every quantity we discuss is really a *function* of λ: the adversary's success probability, its running time, the key length. A single number—"the advantage is at most 2^{-128}"—either fixes a concrete λ or describes the function's behaviour for the λ of interest. Keeping the dependence on λ explicit, at least mentally, prevents nearly every confusion in this subject.

Having fixed how scheme sizes scale, we fix what an attacker may do. An algorithm $\mathcal{A}$ is *probabilistic polynomial-time* (PPT) if it is a probabilistic Turing machine—one equipped with a read-only tape of independent uniformly random bits, its *random tape*—and there is a polynomial p such that on every input x, *and for every setting of the random tape*, $\mathcal{A}$ halts within $p(|x|)$ steps: the strict, worst-case notion, exactly as fixed in the *Math Guide* (§"Algorithms, running time, and PPT"). Write $y \leftarrow \mathcal{A}(x)$ for the random variable obtained by running $\mathcal{A}$ on x with freshly sampled randomness, and $y := \mathcal{A}(x; r)$ for the deterministic output when the random tape is fixed to the string r. The honest algorithms and the adversary alike are modelled as PPT machines.

Two of the modelling choices folded into that sentence warrant comment.

Why polynomial time? Polynomial time is the standard mathematical abstraction of "feasible computation". It is robust—closed under composition, and invariant across all reasonable deterministic computational models (the strong Church–Turing thesis)—which makes it a clean class to quantify over. Crucially, we require the *honest* parties to run in polynomial time too; a

scheme is useful only if it can be used efficiently. The security claim is then an asymmetry between a polynomial-time honest world and a quantified statement that *no* polynomial-time attack succeeds.

Why probabilistic? Randomness is not a luxury but a necessity in cryptography. Deterministic encryption leaks whether two ciphertexts encrypt the same message; key generation must sample secrets unpredictably. We therefore grant the adversary randomness too, both for fairness and because a deterministic adversary is the special case where the random tape is ignored.

The adversary class itself admits a strengthening that the *Math Guide* mentions (§"Algorithms, running time, and PPT") and this volume adopts as standard.

Definition 1.2 (Non-uniform PPT)**.** A *non-uniform* PPT adversary is a family $\{\mathcal{A}_\lambda\}_{\lambda\in\mathbb{N}}$ of probabilistic circuits (or, equivalently, Turing machines each supplied with an *advice string* z_λ of length $\mathrm{poly}(\lambda)$) such that a single polynomial in λ bounds the size of $\mathcal{A}_\lambda$. The advice may depend arbitrarily on λ but not on the particular inputs drawn at run time.

Remark 1.3. The distinction between uniform adversaries (a single machine that reads 1^λ and works for all λ) and non-uniform ones (a separate circuit per λ, or a uniform machine with per-λ advice) is a genuine one. Non-uniformity grants the adversary free precomputed advice tailored to each parameter size— for example, a short hint about the group of the day. Security against non-uniform adversaries is the stronger and now-standard requirement, partly because non-uniform models compose more cleanly under reduction (one may "hard-wire" a fixed good choice of randomness or auxiliary input). All definitions below can be read in either flavour; we state hardness assumptions against non-uniform adversaries when it matters and otherwise leave the choice implicit.

Remark 1.4 (Expected versus strict polynomial time, and quantum adversaries). Two refinements recur. First, some definitions allow *expected* polynomial-time adversaries, bounding the average rather than worst-case running time; this matters for certain extraction arguments in zero-knowledge, and we flag it where it arises. Second, for post-quantum security one replaces PPT by *quantum* polynomial-time (QPT)—polynomial-size circuits of quantum gates followed by a measurement, a model this volume does not formalise; the consequences for the assumptions this volume relies on are taken up under "The quantum caveat: Shor's algorithm" in the next section. The structural definitions of this section (advantage, indistinguishability, reduction) are agnostic to which class of adversaries one quantifies over; only the underlying hardness assumptions change.

1.2 Distribution ensembles and indistinguishability

The most basic thing an adversary can attempt is to *tell two situations apart*—a real protocol transcript from a simulated one, an encryption of m_0 from an encryption of m_1, a pseudorandom string from a truly random one. The most basic security goal is that it cannot. To formalise "look the same" we first formalise the two situations as *ensembles* of probability distributions, then define three grades of sameness.

Definition 1.5 (Probability ensemble)**.** A *probability ensemble* is a sequence $X = \{X_\lambda\}_{\lambda\in\mathbb{N}}$ of probability distributions (equivalently, random variables), one for each value of the security parameter, where X_λ is supported on binary strings whose length is bounded by a polynomial in λ. There is often an additional index—an input a—written $\{X_\lambda(a)\}$; for clarity we suppress it unless needed.

The natural metric on a single pair of distributions is the statistical distance of the *Math Guide* (§"Statistical distance"), which we recall in the slightly wider generality this volume needs: supports there are finite, while ensembles as just defined live on countable sets of strings.

Definition 1.6 (Statistical distance)**.** Let X, Y be discrete random variables over a common finite or countable set S. Their *statistical distance* (or total variation distance) is

$$\Delta(X, Y) \;:=\; \frac{1}{2}\sum_{s \in S} |\Pr[X = s] - \Pr[Y = s]|.$$

Proposition 1.7 (Variational characterisation)**.** *For X, Y over S,*

$$\Delta(X, Y) \;=\; \max_{T \subseteq S}(\Pr[X \in T] - \Pr[Y \in T]) \;=\; \max_{D}(\Pr[D(X) = 1] - \Pr[D(Y) = 1]),$$

where the last maximum ranges over all (even computationally unbounded, deterministic) decision functions $D : S \to \{0, 1\}$. In particular, no procedure whatsoever— efficient or not—can distinguish X from Y with advantage exceeding $\Delta(X, Y)$.

Proof. The set form is the variational characterisation in the statistical-distance theorem of the *Math Guide* (§"Statistical distance"), whose argument is unchanged over a countable S. Only the reformulation over decision functions is new: a deterministic D is the indicator of the set $T = D^{-1}(1)$, so the maximum over D equals the maximum over T; a randomised D is a convex combination of deterministic ones and cannot do better. □

Proposition 1.8 (Properties of statistical distance)**.** *The statistical distance is a metric on probability distributions: $0 \le \Delta(X, Y) \le 1$, it is symmetric, $\Delta(X, Y) = 0$ iff X and Y are identically distributed, and it satisfies the triangle inequality $\Delta(X, Z) \le \Delta(X, Y) + \Delta(Y, Z)$. Moreover it never increases under (possibly randomised) post-processing: for any function—indeed any randomised algorithm—f,*

$$\Delta(f(X), f(Y)) \;\le\; \Delta(X, Y).$$

Proof. These are the metric and data-processing parts of the statistical-distance theorem of the *Math Guide* (§"Statistical distance"), again with the argument unchanged over a countable S. □

The weakest of the three grades below needs the asymptotic vocabulary of the *Math Guide* (§"Polynomial, exponential, and negligible functions"), which we recall in one breath. A function $f : \mathbb{N} \to \mathbb{R}_{\ge 0}$ is *negligible*, written $f = \mathrm{negl}(\lambda)$, if for every $c \in \mathbb{N}$ there exists λ_0 such that $f(\lambda) < \lambda^{-c}$ for all $\lambda \ge \lambda_0$—it decays faster than the reciprocal of every polynomial. A function is *non-negligible* if it is not negligible; it is *noticeable* if there exists c with $f(\lambda) \ge \lambda^{-c}$ for all sufficiently large λ (noticeable functions are non-negligible, but not conversely); it is *overwhelming* if $1 - f$ is negligible. The class of negligible functions is closed under addition and under multiplication by any polynomial—the closure property, proved there, on which the hybrid argument of the next subsection depends.

Definition 1.9 (Perfect, statistical, and computational indistinguishability)**.** Let $X = \{X_\lambda\}$ and $Y = \{Y_\lambda\}$ be probability ensembles.

- The ensembles X and Y are *perfectly indistinguishable*, written $X \equiv Y$, if X_λ and Y_λ are *identically distributed* for every λ; equivalently $\Delta(X_\lambda, Y_\lambda) = 0$ for all λ.
- The ensembles X and Y are *statistically indistinguishable*, written $X \approx_s Y$, if their statistical distance is negligible:
$$\Delta(X_\lambda, Y_\lambda) \;=\; \mathrm{negl}(\lambda).$$

- The ensembles X and Y are *computationally indistinguishable*, written $X \approx_c Y$, if for every PPT algorithm D (the *distinguisher*), the function

$$\mathrm{Adv}^{\mathrm{dist}}_{X,Y}(D,\lambda) \;:=\; \left|\Pr[D(1^\lambda, X_\lambda) = 1] - \Pr[D(1^\lambda, Y_\lambda) = 1]\right|$$

  is negligible in λ. The probabilities are over the draw from the ensemble and over D's internal randomness.

These three notions sit in a strict hierarchy. The implications are immediate from the definitions and the variational characterisation; standard examples witness the strictness. The second example needs two primitives, which we define here in the form the volume uses throughout; write U_n for the uniform distribution on $\{0,1\}^n$. A *one-way function* is a polynomial-time computable $h\colon \{0,1\}^* \to \{0,1\}^*$ that no PPT algorithm inverts on a random input: for every PPT $\mathcal{A}$,

$$\Pr[h(\mathcal{A}(1^\lambda, h(x))) = h(x)] \;=\; \mathrm{negl}(\lambda) \qquad (x \leftarrow U_\lambda).$$

A *pseudorandom generator* (PRG) is a polynomial-time computable $G\colon \{0,1\}^\lambda \to \{0,1\}^{2\lambda}$ whose output on a uniform seed is computationally indistinguishable from uniform, $\{G(U_\lambda)\} \approx_c \{U_{2\lambda}\}$; the doubling stretch is a normalisation. Whether one-way functions exist is open, which is why the second strictness below is stated conditionally.

Proposition 1.10 (Hierarchy)**.** *For all ensembles X, Y:*

$$X \equiv Y \implies X \approx_s Y \implies X \approx_c Y.$$

The first implication is strict: there exist ensembles that are statistically but not perfectly indistinguishable. The second is strict provided one-way functions exist: under that assumption there exist ensembles that are computationally but not statistically indistinguishable.

Proof. If $X \equiv Y$ then $\Delta(X_\lambda, Y_\lambda) = 0$, which is negligible, so $X \approx_s Y$. If $X \approx_s Y$, then for any—in particular any PPT—distinguisher D, Proposition 1.7 gives $\mathrm{Adv}^{\mathrm{dist}}_{X,Y}(D,\lambda) \le \Delta(X_\lambda, Y_\lambda) = \mathrm{negl}(\lambda)$, so $X \approx_c Y$.

For strictness of the first implication, let X_λ be uniform on $\{0,1\}^\lambda$ and let Y_λ be uniform on $\{0,1\}^\lambda \setminus \{0^\lambda\}$. They are not identically distributed (the point 0^λ has probabilities $2^{-\lambda}$ and 0), so not perfectly indistinguishable, yet $\Delta(X_\lambda, Y_\lambda) = 2^{-\lambda} = \mathrm{negl}(\lambda)$, so they are statistically indistinguishable.

For strictness of the second, assume one-way functions exist; then a PRG exists—the Håstad–Impagliazzo–Levin–Luby theorem, together with the routine lemma that extends any stretch to the doubling one, both used without proof—and furnishes an ensemble: let G be a PRG, let $X_\lambda := G(U_\lambda)$, and let $Y_\lambda := U_{2\lambda}$. By definition of a PRG, $X \approx_c Y$. But X_λ is supported on at most 2^λ strings out of $2^{2\lambda}$, so a set T equal to the image of G has $\Pr[X_\lambda \in T] = 1$ while $\Pr[Y_\lambda \in T] \le 2^\lambda/2^{2\lambda} = 2^{-\lambda}$; hence $\Delta(X_\lambda, Y_\lambda) \ge 1 - 2^{-\lambda}$, which is not negligible. Thus $X \approx_c Y$ but $X \not\approx_s Y$. □

Remark 1.11 (The price and the prize of computational indistinguishability). The strictness example is instructive. Statistically the PRG output and a random string are almost as far apart as two distributions can be—distance nearly 1—because the PRG image is a vanishing fraction of the codomain. An unbounded distinguisher simply checks membership in the image and wins. The entire content of computational indistinguishability is that *finding* that image membership test is infeasible. This is the bargain at the heart of cryptography: we trade the impossible (information-theoretic security with short keys) for the merely conjectured-hard (computational security), and in exchange we obtain the ability to encrypt long messages under short keys, build

public-key primitives, and so on. Statistical and perfect security, when attainable, are unconditional and need no hardness assumption; computational security buys vastly more functionality at the cost of resting on an assumption.

1.3 The hybrid argument

Indistinguishability would be of little use if every application had to be argued from scratch. Two lemmas make it composable: efficient post-processing cannot create a distinguishing edge, and a polynomially long chain of pairwise-indistinguishable steps cannot accumulate one. The first is a reduction in miniature—the template for every reduction in this volume; the second, the hybrid argument, is the workhorse of the entire proof literature.

Lemma 1.12 (Closure under efficient post-processing)**.** *If $X \approx_c Y$ and M is any PPT algorithm, then the ensembles $\{M(1^\lambda, X_\lambda)\}$ and $\{M(1^\lambda, Y_\lambda)\}$ satisfy $\{M(X_\lambda)\} \approx_c \{M(Y_\lambda)\}$.*

Proof. Suppose some PPT distinguisher D separates $\{M(X_\lambda)\}$ from $\{M(Y_\lambda)\}$ with advantage $\delta(\lambda)$. Define $D' := D \circ M$, i.e. $D'(1^\lambda, z) := D(1^\lambda, M(1^\lambda, z))$. Since M and D are both PPT, so is D'. Then

$$\mathrm{Adv}^{\mathrm{dist}}_{X,Y}(D', \lambda) = |\Pr[D(M(X_\lambda)) = 1] - \Pr[D(M(Y_\lambda)) = 1]| = \delta(\lambda).$$

By $X \approx_c Y$ the left side is negligible, so δ is negligible. □

Many proofs must compare two ensembles that are far apart but connected by a chain of pairwise-close intermediate ensembles. The hybrid argument turns such a chain into a single conclusion. It is essentially the triangle inequality, but with a refinement—the uniformity hypothesis—that explains why the number of hybrids must be polynomial and that careless statements omit.

Lemma 1.13 (Hybrid lemma)**.** *Let $t = t(\lambda)$ be polynomially bounded, and let $H^0_\lambda, H^1_\lambda, \dots, H^t_\lambda$ be a sequence of distributions (the* hybrids*). Suppose that for every PPT distinguisher D the neighbouring advantages*

$$\alpha_i(D, \lambda) := |\Pr[D(H^i_\lambda) = 1] - \Pr[D(H^{i+1}_\lambda) = 1]|$$

are uniformly *negligible: a single negligible function ν (depending on D) satisfies $\alpha_i(D, \lambda) \le \nu(\lambda)$ for all $i \in \{0, \dots, t(\lambda) - 1\}$ simultaneously. Equivalently, $\max_{0 \le i < t(\lambda)} \alpha_i(D, \lambda) = \mathrm{negl}(\lambda)$, i.e. $\alpha_{i(\lambda)}(D, \lambda)$ is negligible along* every *index function $i(\lambda)$—arbitrary, not merely efficiently computable, since the maximising index need not be efficient. Then $\{H^0_\lambda\}$ and $\{H^t_\lambda\}$ are computationally indistinguishable, i.e. for every PPT D, $|\Pr[D(H^0_\lambda) = 1] - \Pr[D(H^t_\lambda) = 1]| = \mathrm{negl}(\lambda)$.*

Proof. Fix a PPT distinguisher D and let ν be the uniform bound the hypothesis supplies for D. By the triangle inequality applied to the telescoping sum,

$$|\Pr[D(H^0_\lambda) = 1] - \Pr[D(H^t_\lambda) = 1]| = \Big|\sum_{i=0}^{t-1} (\Pr[D(H^i_\lambda) = 1] - \Pr[D(H^{i+1}_\lambda) = 1])\Big| \le \sum_{i=0}^{t-1} \alpha_i(D, \lambda),$$

and the hypothesis bounds the sum by $t(\lambda) \cdot \nu(\lambda)$, which is negligible since t is a polynomial (closure of the negligible class under polynomial multiplication). Hence $\{H^0\} \approx_c \{H^t\}$. □

Uniformity in i is essential, and no argument derives it from the weaker hypothesis that each α_i is negligible for every *fixed* i: there the threshold beyond which the i-th term is small can depend on i, and the dependence cannot be removed. A countermodel makes this concrete: take $t(\lambda) = \lambda + 1$ and let H^i_λ be the point mass at 0 if $i \le \lambda$ and the point mass at 1 otherwise. For

every fixed i the advantage $\alpha_i(D,\lambda)$ is nonzero only at $\lambda = i$, hence negligible; yet H_λ^0 and $H_\lambda^{t(\lambda)}$ are the point masses at 0 and 1, perfectly distinguishable.

Applications usually obtain the conclusion by a stronger, structural route. Suppose the chain arises from a *single* underlying assumption $X \approx_c Y$: neighbouring hybrids H_λ^i and H_λ^{i+1} differ only in that one designated component is drawn from X_λ in the former and from Y_λ in the latter, and each hybrid is efficiently samplable given $(1^\lambda, i)$ and a sample of that component. Then the *random-index* distinguisher D' for the pair (X, Y), on challenge z, samples i uniformly from $\{0, \dots, t-1\}$, builds the hybrid with z in the designated position, and runs D; writing $p_i := \Pr[D(H_\lambda^i) = 1]$, its distinguishing advantage equals $|\frac{1}{t}\sum_{i=0}^{t-1}(p_i - p_{i+1})| = |p_0 - p_t|/t$. Since D' is a single PPT machine, the assumption makes its advantage negligible, and $|p_0 - p_t| \leq t(\lambda) \cdot \mathrm{Adv}(D') = \mathrm{negl}(\lambda)$. Note that this argument bounds D's total advantage directly, bypassing the lemma's uniform per-step hypothesis rather than discharging it; the lemma itself covers the general case in which no such common structure is available.

Remark 1.14. The hybrid argument is precisely where the "polynomially many" constraint earns its keep. A negligible per-step advantage multiplied by a polynomial number of steps remains negligible; a negligible per-step advantage multiplied by, say, 2^λ steps need not. This is why constructions that would require exponentially many hybrids do not yield security proofs, and why the closure of negligibility under polynomial factors is not a technicality but the load-bearing wall of the whole edifice.

1.4 Security as a game

We can now state what it means to "break" a scheme. The modern idiom expresses every security definition as a *game* (or experiment) played between a *challenger* and an adversary $\mathcal{A}$. The challenger, following a fixed protocol, sets up the scheme, answers whatever queries the adversary may pose, and at the end declares the adversary a winner or loser by outputting a single bit. Security says that no PPT adversary wins by more than it could by trivial guessing.

Definition 1.15 (Security game and advantage)**.** A *security game* (or *experiment*) G is a PPT interactive algorithm, the challenger, that takes 1^λ, interacts with an adversary $\mathcal{A}$, and finally outputs a bit $\mathsf{G}^{\mathcal{A}}(\lambda) \in \{0,1\}$, where 1 denotes "adversary wins". The *advantage* of $\mathcal{A}$ is a measure of how much better than trivial its winning probability is; its precise form depends on the type of game:

- *Distinguishing* (*indistinguishability*) *games*, where the challenger secretly chooses a uniform bit $b \in \{0,1\}$ and $\mathcal{A}$ outputs a guess b'; the adversary wins iff $b' = b$. Here

$$\mathrm{Adv}_{\mathsf{G}}(\mathcal{A},\lambda) := \left|\Pr[\mathsf{G}^{\mathcal{A}}(\lambda) = 1] - \tfrac{1}{2}\right| = \tfrac{1}{2}|\Pr[b' = 1 \mid b = 1] - \Pr[b' = 1 \mid b = 0]|.$$

  Trivial guessing achieves winning probability 1/2, hence advantage 0.

- *Search* (*unforgeability/inversion*) *games*, where the adversary must produce a specific kind of object—a forgery, a preimage, a discrete logarithm—and wins iff it succeeds. Here the baseline winning probability is (essentially) 0, and one simply sets

$$\mathrm{Adv}_{\mathsf{G}}(\mathcal{A},\lambda) := \Pr[\mathsf{G}^{\mathcal{A}}(\lambda) = 1].$$

Definition 1.16 (Security relative to a game)**.** A scheme is *secure* with respect to the game G if for every PPT adversary $\mathcal{A}$,

$$\mathrm{Adv}_{\mathsf{G}}(\mathcal{A},\lambda) = \mathrm{negl}(\lambda).$$

Two remarks on the shape of this definition are in order. First, the quantifier order is essential: *for every* PPT adversary, the advantage is negligible. For each fixed adversary there may be a different negligible bound; the claim is universal over the entire class. Second, the use of negl on the right is what makes the definition asymptotic and clean: it is robust to the polynomial slop that reductions introduce, and it captures "the adversary does no better than the trivial strategy, up to an amount that vanishes faster than any inverse polynomial".

Example 1.17 (Pseudorandomness, as a game)**.** To anchor the abstraction with machinery already on the table, consider the distinguishing game for a pseudorandom generator $G : \{0,1\}^\lambda \to \{0,1\}^{2\lambda}$—the primitive that witnessed strictness in Proposition 1.10:

1. The challenger samples $b \leftarrow \{0,1\}$ uniformly. If $b = 0$ it samples a seed $s \leftarrow \{0,1\}^\lambda$ and sends $z := G(s)$; if $b = 1$ it sends a uniform $z \leftarrow \{0,1\}^{2\lambda}$.
2. The adversary $\mathcal{A}$ outputs a guess b'; the game outputs 1 iff $b' = b$.

The generator is secure iff $\mathrm{Adv}(\mathcal{A},\lambda) = |\Pr[b' = b] - \frac{1}{2}|$ is negligible for every PPT $\mathcal{A}$. Here the game packaging and the ensemble packaging coincide: the two situations are the fixed ensembles $\{G(U_\lambda)\}$ and $\{U_{2\lambda}\}$, and the displayed identity in Definition 1.15 shows the game advantage of $\mathcal{A}$ is exactly half its distinguishing advantage in the sense of Definition 1.9—so the game is secure iff the ensembles are computationally indistinguishable. The game form earns its keep beyond this case: when the adversary chooses part of the challenge or queries oracles adaptively, the interaction no longer collapses to a comparison of two fixed ensembles, and the game is the more general packaging—the shape the security definitions of the later sections take.

Remark 1.18 (Game-hopping). Once we phrase security as a game, proofs become *sequences of games*. One starts from the real security game and rewrites it through a series of games $\mathsf{G}_0, \mathsf{G}_1, \ldots, \mathsf{G}_k$, each differing from the last by a small, justifiable change, until reaching a final game in which the adversary provably has zero (or trivially negligible) advantage. Each hop is justified by exactly one of three moves: the two games are identical unless some negligible-probability "bad" event occurs, so their winning probabilities differ by at most $\Pr[\mathrm{bad}]$ (on the complement of that event the two runs coincide); or distinguishing the two games would break a hardness assumption (a reduction, below); or the change is purely syntactic. Bounding the total advantage is then a telescoping sum across hops—a structured cousin of the hybrid argument (Lemma 1.13). This discipline keeps otherwise sprawling proofs honest and auditable.

1.5 Reductions

The engine of provable security is the reduction. A *reduction* is an efficient transformation that converts a successful adversary against a scheme into a successful algorithm against a problem believed to be hard. Its existence proves the implication "Π is hard $\Rightarrow$ X is secure" by establishing the contrapositive "X is broken $\Rightarrow$ Π is solved".

Definition 1.19 (Computational problem and hardness)**.** A *computational problem* Π consists of a PPT instance generator $\mathrm{Inst}(1^\lambda)$ producing a problem instance together with any side information, and a relation deciding when a candidate output *solves* the instance. The *success probability* of an algorithm $\mathcal{B}$ is

$$\mathrm{Succ}_\Pi(\mathcal{B},\lambda) := \Pr[\mathcal{B}(1^\lambda, \mathrm{Inst}(1^\lambda)) \text{ solves the instance}].$$

The problem Π is *hard* if $\mathrm{Succ}_\Pi(\mathcal{B},\lambda) = \mathrm{negl}(\lambda)$ for every PPT $\mathcal{B}$ (for a search problem), or if every PPT $\mathcal{B}$ decides the instance with at most negligible advantage, $\mathrm{Adv}(\mathcal{B},\lambda) = \mathrm{negl}(\lambda)$ in the sense of Definition 1.15 (for a decision problem).

Definition 1.20 (Black-box reduction)**.** Let X be a scheme with security game G_X and Π a computational problem. A (*black-box*) *reduction* from breaking X to solving Π is a PPT oracle algorithm $\mathcal{R}$ with the following per-adversary guarantee: for *every* adversary $\mathcal{A}$, writing $\varepsilon_X(\lambda) := \mathrm{Adv}_{\mathsf{G}_X}(\mathcal{A},\lambda)$ for its advantage, the algorithm $\mathcal{R}^{\mathcal{A}}$—which runs $\mathcal{A}$ as a subroutine, simulating $\mathcal{A}$'s game internally—solves Π with success probability

$$\mathrm{Succ}_{\Pi}(\mathcal{R}^{\mathcal{A}},\lambda) \;\geq\; f(\varepsilon_X(\lambda),\lambda),$$

where f is a function fixed by the reduction (not by $\mathcal{A}$) that maps every non-negligible ε_X to a non-negligible bound. The running time of $\mathcal{R}$ is measured with each call to $\mathcal{A}$ counted as one step, so "PPT" bounds $\mathcal{R}$'s own work, and hence its number of calls to $\mathcal{A}$, by a polynomial in λ; the running time of $\mathcal{R}^{\mathcal{A}}$ is then polynomial whenever $\mathcal{A}$'s is, however many times $\mathcal{R}$ runs $\mathcal{A}$. How far f degrades ε_X is the reduction's *loss*, discussed below. The reduction is *black-box* because $\mathcal{R}$ uses $\mathcal{A}$ only through its input/output interface, making no use of $\mathcal{A}$'s code.

The structure of a reduction is always the same and is worth stating as a recipe. To prove "if Π is hard then X is secure":

1. Assume toward the contrapositive a PPT adversary $\mathcal{A}$ that breaks X with non-negligible advantage ε_X.
2. Construct a PPT algorithm $\mathcal{R}$ that, given a random instance of Π, *embeds* that instance into a simulated security game for $\mathcal{A}$. The simulation must be faithful: the view $\mathcal{R}$ presents to $\mathcal{A}$ must follow (perfectly, statistically, or computationally) the distribution of the real game, so that $\mathcal{A}$'s advantage carries over.
3. Run $\mathcal{A}$. From whatever $\mathcal{A}$ does to win its game, $\mathcal{R}$ *extracts* a solution to the Π-instance.
4. Conclude that $\mathcal{R}$ solves Π with non-negligible probability, contradicting the hardness of Π. Therefore no such $\mathcal{A}$ exists, and X is secure.

A complete, genuine reduction illustrates the point.

Theorem 1.21 (Hard-core security of the Diffie–Hellman-style key, schematically)**.** *Let* $\Pi =$ CDH *be the computational Diffie–Hellman problem in a cyclic group* $\mathbb{G} = \langle g \rangle$ *of prime order* q*, the setting of the* Math Guide (*§"The discrete logarithm problem"*)*: given* (g, g^a, g^b) *for uniform* $a, b \in \mathbb{Z}/q\mathbb{Z}$*, compute* g^{ab}*. Consider the "key-recovery" game* G_X *for a key-exchange scheme in which the adversary, given the public transcript* (g^a, g^b)*, wins iff it outputs the shared key* g^{ab}*. If* CDH *is hard, then no PPT adversary wins* G_X *with noticeable probability.*

Proof. Suppose $\mathcal{A}$ is a PPT adversary that, on input a transcript (g^a, g^b), outputs g^{ab} with probability $\varepsilon_X(\lambda) = \mathrm{Adv}_{\mathsf{G}_X}(\mathcal{A},\lambda)$. Construct $\mathcal{R}$ solving CDH. On input a CDH instance $(g, A, B) = (g, g^a, g^b)$:

1. The reduction $\mathcal{R}$ forms the simulated transcript (A, B) and feeds it to $\mathcal{A}$ as the public transcript of G_X.
2. Because (a, b) are uniform in the CDH instance *and* in the real game, the view of $\mathcal{A}$ follows *identically* the distribution of its view in G_X (the simulation is perfect; no loss in faithfulness).
3. When $\mathcal{A}$ outputs a candidate key K, $\mathcal{R}$ outputs K as its CDH answer.

Since the simulated and real views are identically distributed, $\mathcal{A}$ outputs $K = g^{ab}$ with exactly probability $\varepsilon_X(\lambda)$; whenever it does, $\mathcal{R}$ returns the correct CDH solution. Hence

$$\mathrm{Succ}_{\mathrm{CDH}}(\mathcal{R}, \lambda) \;=\; \varepsilon_X(\lambda).$$

The running time of $\mathcal{R}$ is that of $\mathcal{A}$ plus a constant amount of group bookkeeping, hence polynomial. If ε_X were noticeable, $\mathcal{R}$ would solve CDH with noticeable probability, contradicting its hardness. Therefore $\varepsilon_X(\lambda)$ is not noticeable, as claimed. In fact, since $\mathcal{R}$ is PPT and CDH is hard (Definition 1.19, search branch), $\varepsilon_X(\lambda) = \mathrm{Succ}_{\mathrm{CDH}}(\mathcal{R}, \lambda) = \mathrm{negl}(\lambda)$. □

This example is deliberately *tight*: the reduction loses nothing, calls the adversary once, and converts advantage ε_X into success probability exactly ε_X. Most reductions are not so fortunate, which leads to the quantitative anatomy of a reduction.

Definition 1.22 (Reduction loss and tightness)**.** Consider a reduction $\mathcal{R}$ that transforms an adversary running in time t with advantage ε into a Π-solver running in time $t' = t + t_{\mathcal{R}}$ with success probability ε'. The *security loss* is the factor

$$L(\lambda) \;:=\; \frac{\varepsilon/t}{\varepsilon'/t'} \;=\; \frac{\varepsilon}{\varepsilon'} \cdot \frac{t'}{t},$$

the ratio of the solver's work-per-success to the adversary's work-per-success. A reduction is *tight* if $L(\lambda) = O(1)$ (a small constant, ideally 1), and *loose* if $L(\lambda)$ grows with λ (e.g. $L = q_H$, the number of random-oracle queries (Definition 1.26 below), or $L = q_s$, the number of signature queries, or $L = \lambda$). Two common sources of loss are: an *advantage gap* $\varepsilon' \ll \varepsilon$ (the solver succeeds far less often than the adversary, e.g. because the reduction must *guess* where to embed its instance among q queries, succeeding only with probability $1/q$); and a *time blow-up* $t' \gg t$ (the reduction runs the adversary many times, as in rewinding/forking arguments).

Remark 1.23 (Why loss matters concretely). Asymptotically, loss is invisible: a polynomial loss factor turns a negligible advantage into a negligible advantage and a hard problem into security, without qualification. Concretely, however, loss carries a cost. Suppose a scheme reduces to a problem believed to offer 128 bits of security, but the reduction loses a factor $L = 2^{40}$ (not unusual: $q_H \approx 2^{60}$ random-oracle queries times a forking-lemma square-root loss can be much worse). Then the *proof* guarantees only $128 - 40 = 88$ bits of security for the scheme. To restore a 128-bit guarantee one must enlarge the group so that the *underlying* problem offers 168 bits—larger keys, slower operations. A tight reduction lets the scheme inherit the full strength of the assumption at the smallest parameters; a loose one imposes a permanent cost on every user. This is why tightness is a first-class design goal in modern cryptography, not a theoretical nicety.

Remark 1.24 (Rewinding and the forking lemma, in brief). A recurrent source of large loss is *rewinding*: running the adversary, recording its behaviour, then restarting it from a saved state with fresh randomness on some branch to obtain a second, correlated transcript. From two transcripts that share a prefix but diverge afterward one can often extract a secret (e.g. two valid signatures on the same commitment yield the signing key, or two accepting proof transcripts yield a witness). The *forking lemma*, stated as the general forking lemma under "Security in the random oracle model and the forking lemma", quantifies the success probability of obtaining such a pair: if the adversary succeeds with probability ε after at most q random-oracle queries whose answers are drawn from a set of size h, the forked run succeeds with probability at least $\varepsilon(\varepsilon/q - 1/h)$, i.e. on the order of ε^2/q. The square in ε is a quadratic security loss: to keep ε^2/q noticeable one needs ε noticeably large, roughly halving the bit-security extracted. Such arguments—central to the analyses of Schnorr-type and Fiat–Shamir signatures that this volume builds toward—are the archetype of loose but still valid reductions.

1.6 The standard model and the random oracle model

Reductions reduce security to assumptions. *Which* assumptions one is willing to make defines the *model* in which a proof resides. Two models predominate.

Definition 1.25 (Standard model)**.** A proof resides in the *standard model* if it assumes only the computational hardness of explicitly stated problems (factoring, discrete logarithm, the decisional Diffie–Hellman problem of the next section, and so on) and grants the adversary no idealised access to any component of the scheme. Every primitive, including every hash function, is a concrete algorithm whose code the adversary may inspect and exploit.

Definition 1.26 (Random oracle model)**.** A proof resides in the *random oracle model* (ROM) if it idealises one or more hash functions as a *random oracle*: a publicly accessible function $\mathcal{O}$ that, on each distinct input, returns an *independent, uniformly random* output of the appropriate length, consistently answering repeated queries with the same value. The oracle is available to the honest parties, to the adversary, and—crucially—to the reduction. No party holds the "code" of $\mathcal{O}$; it can only be queried.

The random oracle model is an idealisation: real hash functions are fixed, public algorithms, not truly random functions. The motivation for adopting it is that the reduction gains two capabilities that the standard model withholds and that render many otherwise-intractable proofs tractable.

Remark 1.27 (The powers of the random oracle: observability and programmability)*.* In the ROM the reduction *simulates* the oracle for the adversary, and this yields two additional capabilities. *Observability*: the reduction sees every query the adversary makes to $\mathcal{O}$. Since the adversary, to use a hash value meaningfully, must query it, the reduction learns the adversary's intermediate computations—for instance the preimage the adversary is about to hash. *Programmability*: the reduction may choose each oracle answer on the fly, subject only to the answers appearing uniform and consistent. It can therefore *plant* its hard-problem instance inside an oracle response—e.g. set $\mathcal{O}(x) := B$ where B is part of a CDH challenge—so that the adversary's success necessarily reveals the solution. Neither capability exists for a concrete standard-model hash function, which fixes its outputs in advance and hides its internal queries.

Example 1.28 (The benefit of the ROM: full-domain hash, sketched)**.** In the "full domain hash" signature, the signer signs a message m as $\sigma = H(m)^d \bmod N$, where (N, d) is an RSA private key—a modulus $N = pq$ with secret prime factors and a secret exponent d paired with a public e so that $x^{ed} \equiv x \pmod{N}$ for x coprime to N, by Euler's theorem (*Math Guide*, §"Fermat's little theorem and Euler's theorem"); thus $x \mapsto x^e$ is a permutation of the unit group $(\mathbb{Z}/N\mathbb{Z})^\times$ that only the holder of d can invert—and H is a hash function mapping messages into $(\mathbb{Z}/N\mathbb{Z})^\times$. (Restricting to units costs nothing: a nonzero non-unit of $\mathbb{Z}/N\mathbb{Z}$ shares a factor with N, so stumbling on one factors N. Hash functions appear here informally, as this sketch needs them; their formal syntax and security games are the business of the next-but-one section.) To prove unforgeability one reduces to RSA inversion: given a challenge y (find y^d), the reduction programs the oracle so that for a random subset of messages $H(m) := y \cdot r_m^e$ (embedding the challenge) and for the rest $H(m) := r_m^e$ (where it knows the "signature" r_m); with each r_m drawn uniformly from $(\mathbb{Z}/N\mathbb{Z})^\times$, the programmed values are distributed exactly as honest oracle answers on H's range. When the forger outputs a forgery on a message where the reduction embedded the challenge, the reduction extracts y^d. This proof has *no known standard-model analogue* for the plain scheme: it relies essentially on programming H. The cost is a security loss (the reduction must guess which message the forger attacks, incurring a factor q_H unless cleverly optimised)— connecting back to Definition 1.22.

Remark 1.29 (The status of the ROM: heuristic, not theorem). A proof in the ROM is a proof about an *idealised* world: deploying the scheme instantiates $\mathcal{O}$ with a concrete hash function, and the proof does *not* formally transfer—a ROM proof rules out all attacks that treat the hash as a black box, while offering no guarantee against attacks on the specific structure of the chosen hash. The case for trusting it runs in three steps: prove the scheme secure against a perfect random oracle; instantiate $\mathcal{O}$ with a hash believed to have no exploitable structure; and observe that any remaining attack must exploit structure the random oracle lacks, so would be a direct break of the presumed structure-free hash. The gap is nonetheless a theorem: Canetti, Goldreich, and Halevi exhibit contrived schemes provably secure in the ROM yet insecure under *every* efficiently computable instantiation, so no concrete hash soundly instantiates the oracle for all schemes. No natural deployed scheme is known to suffer this fate, and the working stance of this volume is the practical one: a ROM proof is strong positive evidence, not a guarantee, and a standard-model proof is preferred when one is available at comparable cost. One caveat looks forward. A quantum adversary (Remark 1.4) queries the oracle *in superposition*, a single query addressing a weighted combination of all inputs at once; in the resulting quantum random oracle model (QROM) the two powers of Remark 1.27 become subtler and classical ROM theorems do not automatically carry over, so post-quantum claims for arguments made non-interactive by hashing the transcript (the Fiat–Shamir transform, developed under "The Fiat–Shamir transform: from interactive to non-interactive") must be made in that model.

Remark 1.30 (Relatives: CRS, generic-group, and algebraic-group models). Between and beyond these poles lie further idealisations relevant to the proof systems this volume targets. The *common reference string* (CRS) model posits a trusted public string sampled from a prescribed distribution, used by many succinct proof systems. The *generic group model* (GGM) idealises the group dual to the manner in which the ROM idealises a hash: the adversary may only perform group operations through an oracle on opaque "handles", never exploiting the bit representation of group elements; it is the natural setting for lower bounds on discrete-log-type problems. The *algebraic group model* (AGM) is an intermediate, milder idealisation requiring the adversary to "explain" every group element it outputs as a known combination of its inputs. These models trade realism for provability along the same axis as the ROM, and the constructions ahead invoke them by name where they require them.

1.7 Concrete security and the bit-security budget

The asymptotic language—"negligible", "polynomial", "for all sufficiently large λ"—is mathematically clean but operationally silent. It establishes that a scheme is eventually secure; it does not establish whether *this* deployment with *these* parameters resists an adversary with *that* budget. Concrete (or exact) security supplies the missing numbers.

Definition 1.31 (Concrete security: (t, ε)-security)**.** Fix the security parameter to a concrete value (so all sizes are fixed). A scheme is (t, ε)-*secure* for a game G if every adversary running in time at most t has advantage $\mathrm{Adv}_{\mathsf{G}}(\mathcal{A}) \leq \varepsilon$. More refined statements bound the advantage as an explicit function of the adversary's resources, e.g. time t, number of queries q, and memory s: $\mathrm{Adv}_{\mathsf{G}}(\mathcal{A}) \leq f(t, q, s)$.

Definition 1.32 (n-bit security)**.** A scheme (or problem) is commonly said to offer *n bits of security* when the cheapest known attack achieving constant success probability (say at least $\frac{1}{2}$) costs on the order of 2^n elementary operations. A proof may justify the stronger, explicit claim that every adversary running in time t has advantage at most about $t/2^n$, but that linear advantage bound is not implied merely by the absence of a faster known attack. Thus "128-bit

> security" is a concrete estimate, relative to a specified attack model and state of knowledge, that breaking the scheme costs roughly 2^{128} work.

The relationship to the asymptotic picture is the following: one chooses the security parameter λ *precisely so that* the resulting concrete scheme delivers the desired number of bits of security. In the cleanest case the two coincide, $n = \lambda$, but in general one must account for the actual cost of the best attack on the underlying problem.

Example 1.33 (Why $\lambda \neq n$ for discrete logarithm)**.** For a generic-group or hash-based primitive, doubling the output length doubles the bit security: a hash with $2n$-bit output offers n-bit collision resistance—by the birthday bound (*Math Guide*, §"The union bound and a birthday calculation")—and $2n$-bit preimage resistance, so the parameter and the security level track simply. For discrete-logarithm-based primitives the accounting differs and instructs. In a group of prime order $q \approx 2^\ell$, generic attacks (baby-step/giant-step, Pollard's rho) solve discrete logarithm in time $\approx \sqrt{q} = 2^{\ell/2}$ (*Math Guide*, §"The discrete logarithm problem"). Hence a prime-order elliptic-curve group whose order is 256 bits offers only about 128 bits of security—one halves the bit length of the group order to obtain the generic security level. For discrete logarithm in a finite-field multiplicative group, sub-exponential index-calculus attacks additionally force the ambient field modulus into the thousands of bits for 128-bit security. This is precisely why elliptic-curve groups, for which no comparable sub-exponential classical attack is known, are preferred: they deliver n-bit security at $\ell \approx 2n$ rather than thousands of bits. The single number n conceals a problem-specific translation from all parameter sizes to the security level. The working groups of this volume are priced by exactly this arithmetic: the Pasta curves assembled in the *Math Guide* (§"Pallas and Vesta assembled") have prime order close to 2^{254}, so the square-root rule prices generic discrete logarithm there at about 2^{127} operations; the next section audits them criterion by criterion under "Instantiation on elliptic curves; the Pasta curves".

Remark 1.34 (Reading a bit-security budget end to end). Combining the preceding components yields the practitioner's parameter-selection recipe. Decide the target security level n (commonly 128). Identify the underlying hard problem and the cost of its best attack as a function of the parameter (e.g. $2^{\ell/2}$ for elliptic-curve discrete log), and set the parameter so the problem itself offers n bits (here $\ell = 2n = 256$). Then subtract the reduction's security loss (Definition 1.22): if the proof loses a factor $L = 2^k$ (that is, k bits), the problem must offer $n + k$ bits for the *scheme* to offer n, so one enlarges the parameter accordingly—or, with a tight reduction, one need not. Finally, sanity-check against the ROM caveat (Remark 1.29): a ROM proof bounds only black-box attacks on the hash, so the concrete hash must additionally exhibit no exploitable structure. The clean asymptotic theorem "the scheme is secure if the problem is hard" thus unfolds, through the quantitative machinery of this section, into a concrete choice of bytes.

Remark 1.35 (A note on what security does and does not promise). A caution that frames everything above is in order. A proof of security is a proof *relative to a model*: a definition of the goal (the game), a class of adversaries (PPT, possibly non-uniform or quantum), and a set of assumptions (the hard problems, and any idealisation such as the ROM). It guarantees nothing outside that model. Side-channel leakage (timing, power), faulty randomness, implementation bugs, broken assumptions, and goals the game did not capture all lie beyond its reach. The value of the framework is not that it renders schemes unconditionally safe—nothing can—but that it renders the *conditions* of safety explicit, falsifiable, and modular, so that progress on assumptions and attacks translates directly into recalibrated, auditable guarantees. Everything this monograph builds rests on this discipline.

2 Hardness assumptions

Every secret in this volume is an exponent. A spending key, the trapdoor of a commitment, the nonce of a signature—each lives in $\mathbb{Z}/q\mathbb{Z}$ and faces the world only through a group element g^x.

The adversary of this section is the patient cryptanalyst who collects those elements from the public record and asks for nothing more than the exponent behind one of them: an adversary who computes discrete logarithms recovers signing keys from verification keys, opens and re-opens commitments at will, and strips the blinding from every rerandomised key the later sections construct. What she can *forge* is everything; what she must be *denied* is a single inversion. This section assembles the family of assumptions that deny it: the discrete logarithm problem itself, the ladder of Diffie–Hellman variants above it, the generic-attack bounds that calibrate how hard the problem can possibly be, the composite-order failure mode that squanders that hardness, the elliptic curves on which the remainder of the volume instantiates the family, and the quantum caveat under which the whole edifice stands. Throughout, the lodestar is a single number: the security parameter λ, read as the bit-length of security demanded. The recurring conclusion is that to obtain λ bits of security against the best generic attacks the group must have *prime* order q with $\log_2 q \geq 2\lambda$. Notation is fixed once: $G = \langle g \rangle$ denotes a cyclic group of prime order q, written multiplicatively until §2.6 switches to the additive notation of elliptic curves.

2.1 The discrete logarithm problem

The stage is a finite cyclic group. Recall from the *Math Guide* (§"The discrete logarithm problem") that a group G is *cyclic* if $G = \langle g \rangle = \{ g^k : k \in \mathbb{Z} \}$ for some *generator* g; written additively, as we shall write elliptic curves, this reads $G = \{ kg : k \in \mathbb{Z} \}$. If $|G| = q$ then $\mathbb{Z}/q\mathbb{Z} \cong G$ via $k \mapsto g^k$, an isomorphism trivial to evaluate forwards—repeated squaring costs $O(\log k)$ group operations—but conjecturally infeasible to invert. That asymmetry *is* the discrete logarithm problem.

Definition 2.1 (Discrete logarithm)**.** Let $G = \langle g \rangle$ be a cyclic group of order q, written multiplicatively. For $h \in G$, the *discrete logarithm* of h to the base g, denoted $\log_g h$, is the unique residue $x \in \mathbb{Z}/q\mathbb{Z}$ with $g^x = h$. Existence follows from $G = \langle g \rangle$ and $\mathrm{ord}(g) = q$; uniqueness because $g^x = g^{x'}$ forces $g^{x-x'} = 1$, whence $q \mid x - x'$.

Definition 2.2 (Discrete logarithm problem, DLP)**.** Fix a family $\{G_\lambda\}$ of cyclic groups, where $G_\lambda = \langle g_\lambda \rangle$ has *prime* order q_λ with $\log_2 q_\lambda = \Theta(\lambda)$, together with a sampler that on input 1^λ outputs a description of $(G_\lambda, g_\lambda, q_\lambda)$. The game $\mathsf{DLOG}^{\mathcal{A}}(\lambda)$ runs as follows.

1. The challenger runs the sampler, draws $x \xleftarrow{\$} \mathbb{Z}/q_\lambda\mathbb{Z}$ uniformly, and sends $(G_\lambda, g_\lambda, q_\lambda, h = g_\lambda^x)$ to the adversary $\mathcal{A}$.
2. The adversary outputs $x' \in \mathbb{Z}/q_\lambda\mathbb{Z}$.
3. The game outputs 1 iff $g_\lambda^{x'} = h$.

The *DLP assumption* for the family states that for every PPT $\mathcal{A}$,

$$\Pr[\mathsf{DLOG}^{\mathcal{A}}(\lambda) = 1] \in \mathrm{negl}(\lambda).$$

The order is restricted to be prime for reasons that §2.5 makes emphatic: in prime order every non-identity element is a generator, there are no proper nontrivial subgroups through which information can leak, and the problem cannot be split across factors of the order. For now we record the convenience that prime order affords immediately.

Proposition 2.3 (Random self-reducibility of DLP)**.** *In a cyclic group $G = \langle g \rangle$ of prime order q, the*

> *discrete logarithm problem is* random self-reducible*: an algorithm solving a fraction ε of instances, over uniformly random h, converts into one solving* any *fixed instance with probability ε per trial. Consequently worst-case and average-case hardness coincide.*

Proof. Let $\mathcal{A}$ succeed on a uniform target with probability ε. Given a fixed challenge $h = g^x$, draw $r \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$ and form $h' = h \cdot g^r = g^{x+r}$. Addition modulo q is a bijection and r is uniform, so h' is uniform in G and independent of h; hence $\mathcal{A}(h')$ returns $x + r \bmod q$ with probability ε, and subtracting r recovers x. Each candidate is checked by testing $g^x \stackrel{?}{=} h$, so failures are recognised. Fresh randomness r per trial makes trials independent: t of them succeed with probability $1 - (1-\varepsilon)^t$, overwhelming once $t = \omega(\varepsilon^{-1} \log \lambda)$. □

Random self-reducibility is the formal licence to speak of "the" hardness of DLP in a group: an average-case solver also solves every fixed instance after rerandomisation. This does not literally make every instance difficult—the logarithm of the identity, for one, is immediate.

2.2 Diffie–Hellman: computational and decisional

The discrete logarithm problem asks to invert exponentiation. Many protocols—the Diffie–Hellman key exchange and ElGamal encryption are the historical motivations—need only a weaker-looking guarantee about the *product* of two exponents. Suppose Alice publishes g^a and Bob publishes g^b; their shared secret is g^{ab}, which each computes from the other's public value and their own exponent. The eavesdropper sees g, g^a, g^b and wants g^{ab}.

Definition 2.4 (Computational Diffie–Hellman, CDH)**.** With notation as in Definition 2.2, the game $\mathsf{CDH}^{\mathcal{A}}(\lambda)$ runs as follows: the challenger draws $a, b \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$ independently and sends (g, g^a, g^b); the adversary outputs $z \in G$; the game outputs 1 iff $z = g^{ab}$. The *CDH assumption* states that for every PPT $\mathcal{A}$,

$$\Pr_{a,b}[\mathsf{CDH}^{\mathcal{A}}(\lambda) = 1] \in \mathrm{negl}(\lambda).$$

The computational guarantee delivers the shared key but not its secrecy: an adversary may be unable to *produce* g^{ab} yet perfectly able to *distinguish* it from a random group element, and that alone already breaks, say, the indistinguishability of an ElGamal ciphertext. The decisional variant excludes it.

Definition 2.5 (Decisional Diffie–Hellman, DDH)**.** The *decisional Diffie–Hellman problem* is to distinguish two distributions over G^3,

$$\mathcal{D}_{\mathrm{dh}} = (g^a, g^b, g^{ab}), \qquad \mathcal{D}_{\mathrm{rand}} = (g^a, g^b, g^c),$$

with a, b, c independent and uniform in $\mathbb{Z}/q\mathbb{Z}$; this is the distinguishing game of Definition 1.15, the challenger handing over a sample of one or the other according to its hidden bit. The *DDH assumption* states that every PPT distinguisher D has negligible advantage

$$\mathrm{Adv}_{\mathrm{ddh}}(D) = \left|\Pr_{a,b}[D(g^a, g^b, g^{ab}) = 1] - \Pr_{a,b,c}[D(g^a, g^b, g^c) = 1]\right| \in \mathrm{negl}(\lambda).$$

A triple (g^a, g^b, g^c) with $c \equiv ab \pmod q$ is a *Diffie–Hellman triple*; the assumption says no efficient distinguisher recognises such triples.

Remark 2.6 (DDH admits a random self-reduction). Rerandomisation works for DDH as it did for DLP, though the correct map takes some care. Given a target triple $(u, v, w) = (g^a, g^b, g^c)$, choose $\alpha, \gamma \xleftarrow{\$} (\mathbb{Z}/q\mathbb{Z})^\times$ and $\beta, \delta \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$, and form

$$(u', v', w') = (u^\alpha g^\beta,\ v^\gamma g^\delta,\ w^{\alpha\gamma}\, u^{\alpha\delta}\, v^{\beta\gamma}\, g^{\beta\delta}).$$

Writing $A = \alpha a + \beta$, $B = \gamma b + \delta$, and $E = c - ab$, the new exponents are A, B, and $AB + \alpha\gamma E$, with A and B uniform and independent. A DH triple ($E = 0$) therefore becomes a fresh uniform DH triple; a non-DH triple becomes uniform conditional on being non-DH, because $\alpha\gamma E$ is uniform in $(\mathbb{Z}/q\mathbb{Z})^\times$. The latter distribution lies at statistical distance (Definition 1.6) $1/q$ from $\mathcal{D}_{\text{rand}}$, whose third component completes a DH triple by accident with probability $1/q$. A noticeable distinguishing advantage on random instances thus transfers to every fixed triple, up to this negligible term.

2.3 The assumption hierarchy

The three problems form a chain. Write $A \leq B$ for "A reduces to B": an oracle solving B yields an efficient solver for A; equivalently, B is at least as hard as A, so that if B is easy then A is easy; equivalently again, the assumption "A is hard" is the *stronger* commitment, since it implies "B is hard". The reductions of this subsection give

$$\text{DDH} \ \leq\ \text{CDH} \ \leq\ \text{DLP},$$

ordering the problems by increasing hardness and the assumptions by decreasing strength: assuming DDH hard is the strongest commitment, assuming DLP hard the weakest.

Theorem 2.7 (DLP is at least as hard as CDH)**.** *If DLP is solvable in PPT, then CDH is solvable in PPT. Equivalently, the CDH assumption implies the DLP assumption.*

Proof. Suppose $\mathcal{A}$ solves DLP. Given a CDH instance (g, g^a, g^b), run $\mathcal{A}$ on (g, g^a) to recover a, then compute $(g^b)^a = g^{ab}$ by fast exponentiation. The cost is one DLP call plus $O(\log q)$ group operations, and only one of the two exponents ever needed inverting. □

Theorem 2.8 (CDH is at least as hard as DDH)**.** *If CDH is solvable in PPT with non-negligible success probability, then DDH is solvable in PPT with non-negligible advantage; if the success probability is noticeable, rerandomisation drives the advantage overwhelmingly close to* 1*. Equivalently, the DDH assumption implies the CDH assumption.*

Proof. Suppose $\mathcal{A}$ solves CDH with probability ε. Build a distinguisher D: on input (g^a, g^b, z), call $\mathcal{A}(g, g^a, g^b)$ to obtain a candidate w, and output 1 iff $w = z$. On a Diffie–Hellman triple, $z = g^{ab}$, so D outputs 1 whenever $\mathcal{A}$ succeeds—probability ε. On a random triple, $z = g^c$ with c uniform and independent of a, b, while w depends only on (g^a, g^b); the event $w = z$ therefore has probability exactly $1/q$. Hence

$$\text{Adv}_{\text{ddh}}(D) \ \geq\ \varepsilon - \frac{1}{q},$$

non-negligible whenever ε is, since $1/q$ is negligible.

For the amplified form let ε be noticeable. The distinguisher D' rerandomises its input t times independently by the map of Remark 2.6, runs D with fresh coins on each copy, and outputs 1 iff *some* copy answered 1. A DH input becomes t independent uniform DH triples, each accepted with probability ε, so D' accepts with probability at least $1 - (1-\varepsilon)^t$. A sample of $\mathcal{D}_{\text{rand}}$ fails to be a DH triple except with probability $1/q$, and then each copy's third component is uniform over

the $q-1$ non-completing values given the first two, so each copy is accepted with probability at most $1/(q-1)$ and D' accepts with probability at most $1/q + t/(q-1)$. With $t = \omega(\varepsilon^{-1}\log\lambda)$, polynomial because ε is noticeable, the advantage of D' is $1 - \mathrm{negl}(\lambda)$. □

Example 2.9 (DDH can be easy while CDH and DLP are hard)**.** Some elliptic-curve groups carry an efficiently computable, non-degenerate *bilinear pairing* $e\colon G \times G \to G_T$ with $e(g^a, g^b) = e(g,g)^{ab}$. On such a group DDH is *easy*: to test whether (g^a, g^b, z) is a DH triple, check

$$e(g^a, g^b) \stackrel{?}{=} e(g, z),$$

which holds iff $\log_g z \equiv ab \pmod{q}$. Yet CDH—and a fortiori DLP—may remain hard, because the pairing reveals g^{ab} only inside the *target* group G_T, never as an element of G. Such groups, called *gap groups*, separate DDH from CDH and found pairing-based cryptography. The moral is that DDH is a genuinely stronger assumption: choosing a "DDH group" means choosing a group that rules out pairings of this convenient kind on G itself.

Remark 2.10 (The converses are open)*.* The reverse reductions are not known in general. Whether CDH implies DLP—whether the ability to compute g^{ab} can be leveraged to extract a—is the long-standing Diffie–Hellman versus discrete log question; partial results of den Boer and Maurer establish equivalence when a suitable auxiliary group of *smooth* order—an order that is a product of small primes—is available, but no unconditional equivalence is known. In the other direction DDH may be strictly easier than CDH, as Example 2.9 shows. Practice therefore assumes exactly the strength a protocol requires: signatures and key exchange often need only CDH, while ElGamal-style encryption needs DDH for its ciphertexts to hide the plaintext.

2.4 Generic algorithms and the square-root barrier

How hard can the discrete logarithm problem possibly be? In a group with no exploitable structure beyond the group law, the best known algorithms are *generic*: they manipulate group elements only through multiplication, inversion, and equality testing, never inspecting the bit-representation of an element. Shoup's idealisation makes the class precise: a random injective *labelling* $\sigma\colon \mathbb{Z}/q\mathbb{Z} \to S$ encodes the element g^k as the opaque string $\sigma(k)$, and the algorithm may only ask an oracle for $\sigma(i+j)$ given $\sigma(i)$ and $\sigma(j)$, and compare labels for equality. Two generic algorithms achieve $O(\sqrt{q})$ group operations; a matching lower bound then shows that, generically, nothing does better.

Baby-step giant-step

> **Proposition 2.11** (Baby-step giant-step, BSGS)**.** *In a cyclic group of order q, any discrete logarithm can be computed in $O(\sqrt{q})$ group operations and $O(\sqrt{q})$ stored group elements, deterministically and unconditionally.*

Proof. Let $m = \lceil\sqrt{q}\rceil$. Since $m^2 \geq q$, every $x \in \{0, \dots, q-1\}$ decomposes as

$$x = i \cdot m + j, \qquad 0 \leq i < m, \quad 0 \leq j < m,$$

and the target relation $g^x = h$ becomes $g^j = h \cdot (g^{-m})^i$. *Baby steps:* compute and store the table $\{(j, g^j) : 0 \leq j < m\}$, indexed by group element in a hash map—$m-1$ multiplications. *Giant steps:* set $u = g^{-m}$ (one inversion plus one exponentiation, $O(\log q)$ operations) and for $i = 0, 1, 2, \dots$ compute $h \cdot u^i$ incrementally, looking each value up in the table. The decomposition guarantees some pair (i, j) matches, yielding $x = im + j$ after at most m giant steps. Total: $O(\sqrt{q})$ operations and $O(\sqrt{q})$ memory. □

Remark 2.12 (Memory is the bottleneck). The $O(\sqrt{q})$ *memory* of BSGS, not its time, is what bites in practice: at the 128-bit security level no attacker can store $\sqrt{q} \approx 2^{128}$ table entries. Pollard's rho method matches the time bound with negligible memory, which is why rho, rather than BSGS, is the realistic generic attack.

Pollard's rho

> **Proposition 2.13** (Pollard's rho, sketch). *In a cyclic group of prime order q there is a randomised algorithm computing discrete logarithms in expected $O(\sqrt{q})$ group operations and $O(1)$ storage.*

Proof. Sketch. Walk pseudo-randomly on G, keeping with each visited element its known representation $g^\alpha h^\beta$. Partition G into three roughly equal sets S_1, S_2, S_3 by an easily computed function of the element's label, and from $P = g^\alpha h^\beta$ step to

$$f(P) = \begin{cases} h \cdot P = g^\alpha h^{\beta+1}, & P \in S_1, \\ P^2 = g^{2\alpha} h^{2\beta}, & P \in S_2, \\ g \cdot P = g^{\alpha+1} h^\beta, & P \in S_3, \end{cases}$$

each step updating $(\alpha, \beta) \in (\mathbb{Z}/q\mathbb{Z})^2$ explicitly at $O(1)$ cost. The sequence eventually cycles—a tail leading into a loop, the shape of the letter ρ. Modelling f as a random function on q elements, the birthday bound (*Math Guide*, §"The union bound and a birthday calculation") produces a collision $P_i = P_j$, $i \neq j$, within $O(\sqrt{q})$ steps in expectation. A collision gives $g^{\alpha_i} h^{\beta_i} = g^{\alpha_j} h^{\beta_j}$, hence $g^{\alpha_i - \alpha_j} = h^{\beta_j - \beta_i}$; writing $h = g^x$,

$$\alpha_i - \alpha_j \equiv x\,(\beta_j - \beta_i) \pmod{q},$$

and if $\beta_j \not\equiv \beta_i$ the difference is invertible modulo the *prime* q, so $x \equiv (\alpha_i - \alpha_j)(\beta_j - \beta_i)^{-1} \pmod{q}$. Floyd's tortoise-and-hare cycle detection, or Brent's improvement, finds the collision with $O(1)$ memory, never storing the trajectory. The exceptional case $\beta_j \equiv \beta_i$ occurs with probability $O(1/q)$; re-randomising the start point handles it. □

Remark 2.14 (Cost, not time, and constant factors). Pollard rho parallelises almost perfectly via the van Oorschot–Wiener method of *distinguished points*: each of P processors walks independently and reports to a shared table only the points whose label satisfies a cheap predicate, say a run of leading zero bits; two walks that ever meet coincide from then on and are detected at the next distinguished point, so the processors achieve $\sqrt{q}/P$ wall-clock time with modest communication. The security-relevant measure is therefore the *cost* of the attack—the total operation count $\sqrt{q}$, read as the attacker's budget—not the time on any single machine. A further constant-factor saving applies in groups admitting the $\pm$ automorphism $P \mapsto -P$, as elliptic curves do: identifying P with $-P$ halves the effective search space and lowers the expected count from $\sqrt{\pi q/2}$ to $\sqrt{\pi q/4}$, a factor $\sqrt{2}$ that leaves the $\Theta(\sqrt{q})$ scaling untouched. The constant is the birthday integral: a random walk on N points is still collision-free after k steps with probability $\prod_{i<k}(1 - i/N) \approx e^{-k^2/2N}$ (the product in the birthday calculation cited above), and summing over k gives an expected first-collision index of $\int_0^\infty e^{-k^2/2N}\,dk = \sqrt{\pi N/2}$.

The generic lower bound

The two attacks are not merely the best known; up to constants they are the best *possible* for any generic algorithm. This is Shoup's theorem, the formal statement of the square-root barrier.

Theorem 2.15 (Shoup's generic lower bound)**.** *Let q be prime. Any generic algorithm that makes at most n queries to the group oracle outputs the discrete logarithm of a uniformly random challenge with probability at most*

$$\frac{\binom{n+2}{2} + 1}{q} = O\left(\frac{n^2}{q}\right).$$

In particular, constant success probability requires $n = \Omega(\sqrt{q})$ queries.

Proof. Sketch—the linear-polynomial technique. In the idealisation, the challenge logarithm is an indeterminate X drawn uniformly from $\mathbb{Z}/q\mathbb{Z}$. By induction on the queries, every element the algorithm forms is σ of a *known linear polynomial* $a_i + b_i X \bmod q$ whose coefficients the algorithm chose. Treat X formally: as long as the polynomials produced are pairwise distinct as functions of X, the labels seen are consistent with X taking any value, so the algorithm has learnt nothing and can only guess, succeeding with probability $1/q$. Information leaks only through an accidental collision: two distinct polynomials agreeing at the secret x, that is, $(a_i - a_j) + (b_i - b_j)\, x \equiv 0 \pmod{q}$. For a fixed pair this nonzero linear equation has at most one root modulo the prime q—the degree-one case of the Schwartz–Zippel lemma (*Math Guide*, §"The Schwartz–Zippel lemma")—so it holds with probability at most $1/q$ over uniform x. With n queries plus the two inputs g and h there are at most $\binom{n+2}{2}$ pairs, and the union bound (*Math Guide*, §"The union bound and a birthday calculation") caps the collision probability at $\binom{n+2}{2}/q$. Adding the $1/q$ guess yields the bound; solving $n^2/q = \Omega(1)$ gives $n = \Omega(\sqrt{q})$. □

Corollary 2.16 (Why $\log_2 q \geq 2\lambda$)**.** *To force every generic attacker to spend at least 2^λ group operations—λ bits of security against generic attacks—the prime group order must satisfy*

$$\sqrt{q} \gtrsim 2^\lambda \qquad \Longleftrightarrow \qquad \log_2 q \geq 2\lambda.$$

Proof. Propositions 2.11 and 2.13 show that $\Theta(\sqrt{q})$ operations suffice for a generic attacker, and Theorem 2.15 shows no generic attacker does asymptotically better. Equating the attacker's cost $\sqrt{q}$ with the target work factor 2^λ gives $q \approx 2^{2\lambda}$; demanding at least this much is $\log_2 q \geq 2\lambda$. □

Example 2.17 (The 128-bit target)**.** For $\lambda = 128$ the corollary demands a prime group order with $\log_2 q \geq 256$. This explains the standardised 256-bit elliptic curves: NIST P-256 (secp256r1) has a 256-bit field and a 256-bit group order, so $\sqrt{q} \approx 2^{128.0}$. The Pasta curves of §2.6, with 255-bit orders just above 2^{254} and about 126 bits of security, sit just below this class. Contrast the multiplicative group $\mathbb{F}_p^\times$: there the sub-exponential index calculus of Remark 2.18 forces $\log_2 p$ into the thousands of bits for the same λ. Elliptic curves earn their keep precisely because no sub-exponential, generic-beating algorithm is known for them.

Remark 2.18 (Why the barrier is special to elliptic curves)*.* The square-root barrier governs *generic* groups only; a concrete representation can leak structure that beats it. The multiplicative group $\mathbb{F}_p^\times$ is the cautionary case: *index calculus* exploits the factorisation of integers into small primes—smooth numbers—to solve DLP in sub-exponential time

$$\exp(O((\log p)^{1/3} (\log\log p)^{2/3})),$$

far below $\sqrt{p}$. No analogue is known for the points of a well-chosen elliptic curve over $\mathbb{F}_p$: there is no useful notion of a "small" point over which to factor. This non-generic gap is why elliptic-curve groups can live at 256 bits where $\mathbb{F}_p^\times$ must reach about 3072 bits for the same security. It also means the rule $\log_2 q \geq 2\lambda$ protects against generic attacks *only*; a curve must additionally be screened against the special-purpose attacks of Definition 2.24.

2.5 Pohlig–Hellman and the necessity of large prime order

The square-root barrier priced the attack at $\sqrt{q}$ for *prime* q. If the group order factors, the discrete logarithm splits into independent pieces the size of the factors, and the difficulty collapses to that of the *largest prime* dividing the order. This is the Pohlig–Hellman reduction—the reason Definition 2.2 insisted on prime order.

Theorem 2.19 (Pohlig–Hellman)**.** *Let $G = \langle g \rangle$ have order $N = \prod_{i=1}^{r} p_i^{e_i}$ with the p_i distinct primes. Then a discrete logarithm in G can be computed in*

$$O\left(\sum_{i=1}^{r} e_i(\log N + \sqrt{p_i}\,)\right)$$

group operations. The term $\sqrt{p_{\max}}$ dominates, where $p_{\max}$ is the largest prime dividing N.

Proof. We compute $x = \log_g h$ modulo each prime power $p_i^{e_i}$ and recombine by the Chinese Remainder Theorem, the moduli being pairwise coprime with product N.

Reduction to prime-power order. Fix $p = p_i$, $e = e_i$, and put $g_i = g^{N/p^e}$, $h_i = h^{N/p^e}$. Then g_i has order exactly p^e, and $\log_{g_i} h_i \equiv x \pmod{p^e}$: raising to N/p^e kills the part of x coprime to p's power and leaves $x \bmod p^e$. It therefore suffices to solve DLP in $\langle g_i \rangle$ of order p^e.

Reduction to prime order, digit by digit. Write the unknown in base p:

$$x \equiv x_0 + x_1 p + \cdots + x_{e-1} p^{e-1} \pmod{p^e}, \qquad 0 \le x_k < p.$$

Let $\gamma = g_i^{p^{e-1}}$, an element of order exactly p. Raising h_i to the power p^{e-1} gives

$$h_i^{p^{e-1}} = g_i^{x p^{e-1}} = \gamma^{x_0},$$

because every higher digit contributes a multiple of p^e to the exponent and annihilates γ. Thus x_0 is a single discrete logarithm in a group of prime order p, solvable in $O(\sqrt{p})$ operations by BSGS or rho (Propositions 2.11 and 2.13). Having x_0, set $h_i^{(1)} = h_i \cdot g_i^{-x_0}$; its logarithm is divisible by p, and the same step at exponent p^{e-2} yields x_1. Iterating recovers all e digits with e prime-order DLPs of cost $O(\sqrt{p})$ each, plus $O(e \log N)$ operations of exponentiation. Summing over i and applying the CRT gives the stated bound. □

Example 2.20 (A composite-order trap)**.** Suppose, contrary to good practice, a group has order $N = 2^{256} - 1$. The order factors completely into

$$\begin{aligned} 2^{256} - 1 = 3 \cdot 5 \cdot 17 \cdot 257 \cdot 641 \cdot 65537 \cdot 274177 \cdot 6700417 \cdot 67280421310721 \\ \cdot 59649589127497217 \cdot 5704689200685129054721, \end{aligned}$$

a product of small primes—it is *smooth*. Despite N having 256 bits, the largest prime factor is only about $2^{72.3}$, far below 2^{128}, so Pohlig–Hellman with rho in the largest subgroup solves DLP in roughly $\sqrt{p_{\max}} \approx 2^{36.1}$ operations—about 2^{36}, nowhere near the apparent 128-bit level. A 256-bit order is necessary but *not* sufficient: it must be (almost) prime.

Corollary 2.21 (Design rule for the group order)**.** *For DLP-based security at level λ, the group order must be divisible by a prime q with $\log_2 q \ge 2\lambda$, and the cryptographic group must be, or be restricted to, the subgroup of order q. When the ambient group has order $N = h \cdot q$ with small* cofactor *h and large prime q, one works in the order-q subgroup; clearing the cofactor—multiplying incoming ele-*

ments by h, or checking subgroup membership—prevents small-subgroup attacks that would otherwise leak x mod *h via Pohlig–Hellman on the order-h part.*

Remark 2.22 (Prime order, revisited). Here is the deeper justification for "prime order" in Definition 2.2. Prime order q makes Pohlig–Hellman vacuous (the only factor is q itself), makes every non-identity element a generator (no element lands in a weak proper subgroup), and removes cofactor bookkeeping entirely. The Pasta curves below are engineered to have prime order: each group is its own prime-order subgroup, with cofactor 1.

2.6 Instantiation on elliptic curves; the Pasta curves

We now fix the concrete groups the remainder of the volume uses. Recall from the *Math Guide* (§"Elliptic curves", with the group law in §"The group structure of $E(\mathbb{F}_p)$") that an elliptic curve over $\mathbb{F}_p$, for $p > 3$ prime, in short Weierstrass form is

$$E\colon y^2 = x^3 + ax + b, \qquad a, b \in \mathbb{F}_p, \quad 4a^3 + 27b^2 \neq 0,$$

and that its set of $\mathbb{F}_p$-points,

$$E(\mathbb{F}_p) = \{(x, y) \in \mathbb{F}_p^2 : y^2 = x^3 + ax + b\} \cup \{O\},$$

equipped with chord-and-tangent addition and the point at infinity O as identity, forms a finite abelian group. The discrete logarithm problem in this group—given P and $Q = [x]P$, find x—is the *elliptic-curve discrete logarithm problem* (ECDLP), stated in its own habitat in the *Math Guide* (§"The elliptic-curve discrete logarithm problem"); CDH and DDH transcribe verbatim with multiplicative notation replaced by the additive scalar multiplication $x \mapsto [x]P$, giving ECCDH and ECDDH. The group size is pinned down by the Hasse bound, likewise recalled.

Theorem 2.23 (Hasse bound, recalled)**.** *For E over $\mathbb{F}_p$* (Math Guide, *§"The group structure of $E(\mathbb{F}_p)$"*),

$$|\#E(\mathbb{F}_p) - (p+1)| \;\leq\; 2\sqrt{p}, \qquad \textit{i.e.} \qquad \#E(\mathbb{F}_p) = p + 1 - t, \quad |t| \leq 2\sqrt{p},$$

where t is the trace of Frobenius. *The group order is thus $p + 1$ up to a deviation of order $\sqrt{p}$.*

The parameter recipe for λ-bit security follows: choose p of about 2λ bits, so that $\#E(\mathbb{F}_p) \approx p \approx 2^{2\lambda}$, and insist by Corollary 2.21 that $\#E(\mathbb{F}_p)$ be prime—cofactor 1. One then assumes ECDLP, ECCDH, and ECDDH at the level Corollary 2.16 predicts, *provided* the curve avoids the known structural attacks that beat the generic bound. The screening criteria are as follows.

Definition 2.24 (Secure-curve criteria)**.** A curve $E/\mathbb{F}_p$ of prime order $q = \#E(\mathbb{F}_p)$ is admissible for λ-bit DLP security if:

1. (*Size / generic.*) The order satisfies $\log_2 q \geq 2\lambda$, so Pollard rho costs at least 2^λ operations (Corollary 2.16).
2. (*Prime order.*) The order q is prime, so Pohlig–Hellman is vacuous and the cofactor is 1 (Corollary 2.21).
3. (*Anti-MOV / large embedding degree.*) The multiplicative order k of p modulo q—the *embedding degree*—is large. The MOV/Frey–Rück attack uses a pairing to embed $E(\mathbb{F}_p)$ into $\mathbb{F}_{p^k}^\times$ and runs the sub-exponential index calculus of Remark 2.18 there; the threat is real only for small k, as on *supersingular* curves—over a prime field $p > 3$, those with

trace $t = 0$, so that $\#E(\mathbb{F}_p) = p + 1$ divides $p^2 - 1$ and $k \leq 2$. A curve with $t \neq 0$ is *ordinary*.

4. (*Anti-Smart / not anomalous.*) The order avoids $q = p$: on an *anomalous* curve, $\#E(\mathbb{F}_p) = p$ with trace $t = 1$, the Smart–Satoh–Araki–Semaev attack solves ECDLP in polynomial time.

5. (*Twist security.*) The *quadratic twist* E'—the curve $dy^2 = x^3 + ax + b$ for a fixed nonsquare $d \in \mathbb{F}_p$; every $x \in \mathbb{F}_p$ is the x-coordinate of a point of E or of E'—should also have large prime order, so that computation cannot be pushed onto a weak twist by a fault or an invalid-point injection.

Definition 2.25 (The Pasta curves)**.** The *Pasta* curves are a pair, *Pallas* and *Vesta*, in short Weierstrass form

$$y^2 = x^3 + 5$$

(so $a = 0$, $b = 5$, a j-invariant-0 shape), defined over two 255-bit primes p and q arranged in a *cycle*:

$$\#\text{Pallas} = q \ \ (\text{Pallas defined over } \mathbb{F}_p), \qquad \#\text{Vesta} = p \ \ (\text{Vesta defined over } \mathbb{F}_q).$$

The order of each curve equals the field of definition of the other. Both primes are congruent to 1 mod 2^{32}, giving their multiplicative groups large 2-adic valuation for fast FFTs, and each curve has cofactor 1: the whole group is of prime order. The primes themselves are assembled digit by digit in the *Math Guide* (§"Pallas and Vesta assembled"), and the field/scalar bookkeeping of the cycle in §"Base fields, scalar fields, and the Pasta cycle" there.

Remark 2.26 (Why a cycle, and what it buys). The defining property—#Pallas $= q$, the base field of Vesta, and #Vesta $= p$, the base field of Pallas—makes (Pallas, Vesta) an *amicable*, or cyclic, pair: one curve's scalar field is the other's coordinate field. A proof system can therefore use Pallas-arithmetic to verify statements about Vesta-arithmetic and vice versa without non-native field emulation; this is the mechanism behind recursive proof composition in Halo 2, whose recursion alternates between the two curves. The cycle is an *efficiency* property, not a security one: it does not enter the hardness assumptions, and each curve is screened independently against Definition 2.24.

Proposition 2.27 (The Pasta curves against the criteria)**.** *The curves Pallas and Vesta satisfy criteria (1)–(4) of Definition 2.24 at $\lambda \approx 126$. Criterion (5) fails for both curves—neither twist has prime order, and the Pallas twist falls below the SafeCurves 2^{100} twist-attack cost threshold—but the failure is rendered moot by mandatory point validation.*

Proof. Sketch, criterion by criterion.

(1) Each order is a 255-bit prime just above 2^{254}, so $\log_2 q \approx 254.0$ and the bare generic cost is $\sqrt{q} \approx 2^{127.0}$. Plain Pollard rho expects about $\sqrt{\pi q/2} \approx 2^{127.3}$ operations; the $\pm$ automorphism speedup of Remark 2.14 lowers this to $\sqrt{\pi q/4} \approx 2^{126.8}$; and the order-6 automorphism group of these j-invariant-0 curves—the factor $\sqrt{m}$ for an order-m automorphism group of Remark 2.28, here an extra $\sqrt{3}$—lowers it to $\sqrt{\pi q/12} \approx 2^{126.0}$. Hence "about 126 bits" of security, the same range as the 128-bit curves of Example 2.17.

(2) Both orders are prime by construction: cofactor 1, Pohlig–Hellman vacuous.

(3) Both curves are ordinary: the trace $t = p + 1 - q$ is odd, so $t \neq 0$. The embedding degree k of each is enormous. For Pallas, $q - 1 = 2^{32} \cdot 3^2 \cdot 1709 \cdot 24859 \cdot C$ with C a 194-bit composite,

and $p^{(q-1)/32} \not\equiv 1$ and $p^{(q-1)/\ell} \not\equiv 1 \pmod{q}$ for $\ell \in \{3, 1709, 24859\}$, so the order k of p modulo q is a multiple of $2^{28} \cdot 3^2 \cdot 1709 \cdot 24859 > 2^{56}$; for Vesta, $p - 1 = 2^{32} \cdot 3 \cdot 463 \cdot C'$, and $q^{(p-1)/4} \not\equiv 1$, $q^{(p-1)/\ell} \not\equiv 1 \pmod{p}$ for $\ell \in \{3, 463\}$ make k a multiple of $2^{31} \cdot 3 \cdot 463 > 2^{41}$. Either extension field is astronomically beyond any feasible index-calculus target; MOV/Frey–Rück does not apply.

(4) Neither curve is anomalous: $\#\text{Pallas} = q \neq p$ and $\#\text{Vesta} = p \neq q$, so the trace $t = p + 1 - q \neq 1$ and the Smart attack does not apply.

(5) Twist security is the criterion the pair does not meet. The Pasta search deliberately ignored it—the curves were generated with the search flag `--ignoretwist` of the `amicable.sage` search recorded in the README of the `zcash/pasta` repository—and neither twist has prime order. Each $x \in \mathbb{F}_p$ supplies two affine points in total, on E or on E' (one on each when $x^3 + b = 0$), so $\#E + \#E' = 2(p + 1)$ and $\#E' = 2(p + 1) - \#E = p + 1 + t$; hence $\#\text{Pallas}' = 2(p + 1) - q$ and $\#\text{Vesta}' = 2(q + 1) - p$, which factor as

$$\#\text{Pallas}' = 3^2 \cdot 7 \cdot 8191 \cdot 85021 \cdot 11540602760389 \cdot P_{176},$$
$$\#\text{Vesta}' = 3^3 \cdot 7 \cdot 13 \cdot 2851 \cdot 30097 \cdot P_{217},$$

with P_{176} a 176-bit prime and P_{217} a 217-bit prime. The best attack on the Pallas twist therefore costs about $\sqrt{P_{176}} \approx 2^{87.6}$, below the SafeCurves 2^{100} twist-security threshold; the Vesta twist clears the threshold at about $2^{108.2}$. The failure is moot in our setting: twist attacks require an implementation that accepts an attacker-supplied x-coordinate without validation (an x-only ladder) or that skips its on-curve checks, and the deployed arithmetic does neither. The `pasta_curves` crate carries points in full coordinates and has no x-only code path; its deserialisation (`from_bytes`) reconstructs y by taking the square root of $x^3 + 5$, failing exactly when x lies on the twist, where $x^3 + 5$ is a non-square, and even `from_bytes_unchecked` delegates to the checked path because a compressed encoding cannot skip the curve check, while points built from raw coordinates are gated on `is_on_curve`. This is the behaviour protocol specification § 5.4.9.6 mandates: its decoding map returns $\perp$ (failure) when $x^3 + b$ has no square root, and an implementation must not assume that the root exists or that the encoding lies on the curve. Orchard deserialises its points through this routine, and cofactor 1 leaves no small subgroup on the curve itself. Invalid-curve and twist-point injection are therefore excluded by *validation*, not by twist structure.

The standard generic analysis governs, and the security level is about 126 bits. □

Remark 2.28 (j-invariant 0 and endomorphisms). The shape $y^2 = x^3 + b$ gives j-invariant 0, and with it an extra automorphism of order 3 (*Math Guide*, §"The j-invariant and the GLV endomorphism"): for ω a primitive cube root of unity in $\mathbb{F}_p$, the cheap endomorphism $\phi(x, y) = (\omega x, y)$ acts on the group as multiplication by a fixed scalar ζ with $\zeta^2 + \zeta + 1 \equiv 0 \pmod{q}$ (the λ of the *Math Guide*'s proposition), enabling the GLV speedup for honest scalar multiplication. The same endomorphism hands the attacker a slightly larger automorphism group, and with it the constant-factor Pollard-rho speedup used in Proposition 2.27—a factor $\sqrt{m}$ for an order-m automorphism group—but that is a small constant and does not change the $\Theta(\sqrt{q})$ scaling. Beyond that constant, the extra automorphism is not known to weaken ECDLP; the danger lies only in small embedding degree and in anomalousness, both excluded above.

2.7 The quantum caveat: Shor's algorithm

Every assumption above holds the adversary to classical PPT computation. Against a large fault-tolerant quantum computer the picture changes qualitatively, and honesty requires stating exactly what breaks and what survives.

Theorem 2.29 (Shor, informal; stated without proof)**.** *There is a quantum algorithm that, given*

> *a cyclic group $G = \langle g \rangle$ of order q with efficient group operations and a target $h = g^x$, outputs x in time polynomial in* $\log q$*, succeeding with overwhelming probability. Consequently DLP—and with it CDH and DDH—falls in quantum polynomial time in* any *group, elliptic-curve groups included.*

We do not prove the theorem; the shape of the algorithm is nevertheless worth recording. Discrete logarithm is an instance of the *hidden subgroup problem* over $(\mathbb{Z}/q\mathbb{Z})^2$. The function $f(\alpha, \beta) = g^\alpha h^{-\beta} = g^{\alpha - x\beta}$ is constant exactly on the cosets of

$$H = \{(\alpha, \beta) : \alpha \equiv x\beta \pmod{q}\} = \langle (x, 1) \rangle \subseteq (\mathbb{Z}/q\mathbb{Z})^2,$$

a line of slope x. The algorithm prepares a uniform superposition over $(\mathbb{Z}/q\mathbb{Z})^2$, evaluates f into a second register, and applies the quantum Fourier transform—the reversible change of basis into the characters of $(\mathbb{Z}/q\mathbb{Z})^2$, the homomorphisms $\chi_{u,v}(\alpha, \beta) = e^{2\pi i(u\alpha + v\beta)/q}$ into the unit circle; measurement then samples characters trivial on H, i.e. pairs (u, v) with $ux + v \equiv 0 \pmod{q}$, and a handful of samples reveal the slope x by classical linear algebra modulo q. The whole procedure uses $O(\mathrm{poly}(\log q))$ qubits and gates, and it touches the group only through the black-box map $(\alpha, \beta) \mapsto g^\alpha h^{-\beta}$, so it is agnostic to representation: elliptic-curve DLP falls exactly as $\mathbb{F}_p^\times$ DLP does, and integer factoring falls to the order-finding variant.

Remark 2.30 (No contradiction with the square-root barrier). Shoup's lower bound (Theorem 2.15) assumes the generic model for *classical* algorithms making sequential, individual oracle queries. Shor's algorithm is non-generic in exactly the relevant sense: it queries the group function in superposition and exploits the periodicity the Fourier transform exposes, abilities the classical generic model does not grant. The two results coexist: the barrier bounds classical generic attackers; Shor is a quantum, non-generic attacker.

Remark 2.31 (What Shor breaks and what survives). Shor's algorithm breaks the entire discrete-log family and integer factoring in polynomial time: RSA, finite-field Diffie–Hellman, and elliptic-curve cryptography are all quantum-broken. What it does *not* break:

- *Symmetric primitives and hash functions.* Grover's algorithm searches a space of size 2^n in about $2^{n/2}$ quantum queries—a quadratic, not exponential, speedup—so a cipher with a 2λ-bit key, or an $n = 2\lambda$-bit hash against preimages, retains λ bits of *quantum* security. Generic quantum collision finding instead costs $\Theta(2^{n/3})$ queries, so in the query metric an output of about 3λ bits is needed for λ-bit quantum collision security—though the algorithm needs comparably much quantum memory, and whether it beats classical parallel collision search in realistic cost models is debated. Doubling therefore suffices for key search and preimages, but for collisions only under that contested cost reading.
- *Post-quantum assumptions.* Hardness assumptions based on neither discrete logarithms nor factoring are not known to fall to Shor and underlie post-quantum cryptography.

The discrete-log assumptions of this volume are therefore adopted under the standing premise that no cryptographically relevant quantum computer exists. Within that premise, and against all classical attacks, the Pasta curves deliver the approximately 126-bit security of Proposition 2.27—the ground on which the Halo 2 constructions to come are built.

2.8 Standing assumptions

Remark 2.32 (Standing assumptions for the sequel). We collect the premises the rest of the volume relies on. For each Pasta curve $C \in \{\text{Pallas}, \text{Vesta}\}$ with prime order q_C and generator g_C:

1. ECDLP is hard in $\langle g_C \rangle$: no classical PPT adversary finds x from $[x]g_C$ with non-negligible probability—concretely, every known attack costs at least 2^{126} operations (Proposition 2.27).

2. ECCDH and, where needed, ECDDH hold in $\langle g_C \rangle$ (Definitions 2.4 and 2.5, transcribed additively).
3. The adversary is classical and probabilistic polynomial-time (Remark 2.31).

By the hierarchy of §2.3, assuming ECDDH is the strongest commitment and implies the rest; each later construction invokes, by name, the weakest assumption it requires.

3 Hash functions and the random oracle model

The adversary of this section forges by *coincidence*. She prepares two contracts—one benign, one ruinous—whose digests agree, has the benign one signed, and produces the ruinous one in court; or she takes a digest minted for one purpose, say a key-derivation output, and replays it in another role, say a commitment, exploiting the fact that both are interchangeable bit strings; or she simply walks a pseudo-random path through the digest space until, square-root-of-the-space steps later, two inputs collide. Denying her these coincidences is the business of the hash function: a function compressing arbitrary data into a short fixed-length tag. It underlies digital signatures, commitment schemes, message authentication, key derivation, proof-of-work, Merkle trees, and the Fiat–Shamir transform behind Halo 2's non-interactive arguments—so many purposes for one primitive because an *ideal* hash acts like a deterministic source of fresh randomness keyed by its input. This section builds the primitive from the ground up: its syntax; the three classical security notions and the generic birthday attack that bounds them; the random oracle model; domain separation; hashing into a finite field and onto an elliptic curve; the arithmetisation-friendly hash Poseidon that Orchard evaluates inside its circuit; and finally a register of the transparent layer's SHA-256-derived hashes.

3.1 Syntax

Notation for strings is fixed once. The set of all finite binary strings is $\{0,1\}^*$; the strings of length exactly n form $\{0,1\}^n$; the length of a string x is $|x|$; and concatenation is written $x \parallel y$.

Definition 3.1 (Hash function)**.** A *hash function* with output length $n \in \mathbb{N}$ is a function

$$H\colon \{0,1\}^* \longrightarrow \{0,1\}^n.$$

Elements of the codomain are called *digests*, *hashes*, or *tags*. The function is *compressing* on a domain $D \subseteq \{0,1\}^*$ if D contains strings longer than n; when the domain is all of $\{0,1\}^*$, the function is overwhelmingly many-to-one.

A single fixed function such as SHA-256 is what practice deploys, but a foundational wrinkle appears the moment one tries to say what its security *means*. For any fixed compressing H there *exist* inputs $x \neq x'$ with $H(x) = H(x')$—the pigeonhole principle guarantees it—and therefore a tiny program that simply outputs one such hardcoded pair also exists. The asymptotic statement "no efficient adversary finds a collision" is consequently *false* for every fixed compressing function: the hardcoding program is efficient and always succeeds. (Writing that program down is another matter—nobody knows a collision for SHA-256—but existence suffices to falsify the universally quantified claim.) The standard remedy indexes the function by a key.

Definition 3.2 (Keyed hash family)**.** A *keyed hash family* (or *hash function ensemble*) is a function

$$H\colon \mathcal{K} \times \{0,1\}^* \longrightarrow \{0,1\}^n,$$

where $\mathcal{K}$ is a finite key space. A key $s \in \mathcal{K}$ is sampled *publicly* at random, fixed, and we write $H_s(\cdot) := H(s, \cdot)$ for the family member in force.

The key is *not* secret. Its role is to index which member of the family the adversary must attack: a hardcoded collision for one member is useless against a freshly sampled other, so "no efficient adversary finds a collision for a random member" is a meaningful asymptotic statement.

Remark 3.3 (Convention: keyed definitions, unkeyed deployments). The convention of this volume follows the standard resolution of the mismatch between theory and practice. Security notions are stated for the keyed family, where the definitions are clean; every concrete construction we deploy is unkeyed; and the key is mentally identified with "the choice of standard"—the sampling of s happened, once and in public, when the function was designed. No formal theorem bridges the two readings; the gap is the same one the random oracle model of §3.4 inhabits, and we flag it rather than hide it.

Definition 3.4 (Extendable-output function)**.** An *extendable-output function* (XOF) is a primitive producing a digest of caller-chosen length: syntactically a function

$$H\colon \{0,1\}^* \times \mathbb{N} \longrightarrow \{0,1\}^*, \qquad |H(x,\ell)| = \ell,$$

whose outputs are *prefix-consistent*: for $\ell \le \ell'$, the string $H(x,\ell)$ is a prefix of $H(x,\ell')$, so a longer request extends a shorter one as a stream.

The canonical XOFs are SHAKE128 and SHAKE256 from the SHA-3 family. An XOF is the natural tool wherever a *controlled number* of uniform bits is needed rather than a fixed digest—above all when hashing into a finite field or onto a curve, where the output must be reduced modulo a prime; §3.6 returns to this.

3.2 Security notions: preimage, second-preimage, and collision resistance

Each of the three classical security notions is a game between a challenger and a PPT adversary $\mathcal{A}$, in the template of Definition 1.15; the keyed family H is secure in a given sense if every PPT adversary wins the corresponding game with probability negligible in the security parameter λ. All sizes are functions of λ: the output length $n = n(\lambda)$, the key length, and a polynomially bounded challenge input length $m = m(\lambda)$ from which the games sample their random challenge $x \xleftarrow{\$} \{0,1\}^m$. Recall that a function $\varepsilon(\lambda)$ is negligible, $\varepsilon \in \mathrm{negl}(\lambda)$, if it decays faster than the reciprocal of every polynomial (*Math Guide*, §"Polynomial, exponential, and negligible functions"; recalled in §1.2). The three notions form a strict hierarchy, and no single definition of a "good" hash exists; which notion an application needs is a per-application question that Remark 3.10 takes up.

Definition 3.5 (Preimage resistance)**.** The game $\mathsf{PRE}_H^{\mathcal{A}}(\lambda)$ runs as follows.

1. The challenger samples $s \xleftarrow{\$} \mathcal{K}$ and $x \xleftarrow{\$} \{0,1\}^m$, computes $y := H_s(x)$, and sends (s, y) to the adversary.
2. The adversary outputs $x' \in \{0,1\}^*$.
3. The game outputs 1 iff $H_s(x') = y$.

The family H is *preimage resistant* (or *one-way*) if for every PPT $\mathcal{A}$,

$$\mathrm{Adv}_H^{\mathrm{pre}}(\mathcal{A},\lambda) := \Pr[\mathsf{PRE}_H^{\mathcal{A}}(\lambda) = 1] \in \mathrm{negl}(\lambda).$$

Note the winning condition: the adversary, given the key and the digest of a random message, must output *any* string hashing to y. Recovering the original x is not required; any preimage wins.

Definition 3.6 (Second-preimage resistance)**.** The game $\mathsf{SEC}_H^{\mathcal{A}}(\lambda)$ runs as follows.

1. The challenger samples $s \xleftarrow{\$} \mathcal{K}$ and $x \xleftarrow{\$} \{0,1\}^m$, and sends (s,x) to the adversary.
2. The adversary outputs $x' \in \{0,1\}^*$.
3. The game outputs 1 iff $x' \neq x$ and $H_s(x') = H_s(x)$.

The family H is *second-preimage resistant* if for every PPT $\mathcal{A}$,

$$\mathrm{Adv}_H^{\mathrm{sec}}(\mathcal{A},\lambda) := \Pr[\mathsf{SEC}_H^{\mathcal{A}}(\lambda) = 1] \in \mathrm{negl}(\lambda).$$

Here the challenger *gives* the adversary a specific random message; the adversary must produce a different message colliding with it.

Definition 3.7 (Collision resistance)**.** The game $\mathsf{COL}_H^{\mathcal{A}}(\lambda)$ runs as follows.

1. The challenger samples $s \xleftarrow{\$} \mathcal{K}$ and sends s to the adversary.
2. The adversary outputs a pair (x,x').
3. The game outputs 1 iff $x \neq x'$ and $H_s(x) = H_s(x')$.

The family H is *collision resistant* if for every PPT $\mathcal{A}$,

$$\mathrm{Adv}_H^{\mathrm{col}}(\mathcal{A},\lambda) := \Pr[\mathsf{COL}_H^{\mathcal{A}}(\lambda) = 1] \in \mathrm{negl}(\lambda).$$

The three games differ in *who chooses what*. In the preimage game the challenger chooses the message and reveals only its digest; in the second-preimage game the challenger chooses the message and reveals it; in the collision game the adversary chooses *both* messages and need only make them collide. More adversarial choice means an easier game and hence a stronger security demand on the function; collision resistance, granting the most choice, is the strongest of the three.

Proposition 3.8 (Hierarchy of notions)**.** *Let H be a keyed hash family that is compressing, with challenge input length $m > n$. Then:*

1. *collision resistance implies second-preimage resistance, with no loss; and*
2. *second-preimage resistance implies preimage resistance up to an additive loss of 2^{n-m}, which is negligible—so the implication holds outright—whenever H is sufficiently compressing, $m - n = \omega(\log \lambda)$.*

Neither implication reverses in general.

Proof. (1) From an adversary $\mathcal{A}$ against second-preimage resistance, construct $\mathcal{B}$ against collision resistance: on input the key s, the reduction $\mathcal{B}$ samples $x \xleftarrow{\$} \{0,1\}^m$ *itself*, runs $x' \leftarrow \mathcal{A}(s,x)$—

the simulated view is exactly that of SEC, since $\mathcal{B}$ samples x from the same distribution as the challenger—and outputs the pair (x, x'). Whenever $\mathcal{A}$ succeeds, $x' \neq x$ and $H_s(x') = H_s(x)$, which is precisely a collision. Hence $\mathrm{Adv}_H^{\mathrm{col}}(\mathcal{B}, \lambda) \geq \mathrm{Adv}_H^{\mathrm{sec}}(\mathcal{A}, \lambda)$: the reduction is tight, and if the left side is negligible so is the right.

(2) Toward the contrapositive, let $\mathcal{A}$ be an inverter; we build a second-preimage finder $\mathcal{B}$. On input (s, x), the reduction computes $y := H_s(x)$, runs $x' \leftarrow \mathcal{A}(s, y)$—again a perfect simulation of PRE, since x is uniform and y is its digest—and outputs x'. Whenever $\mathcal{A}$ succeeds, $H_s(x') = y = H_s(x)$; what remains is to argue $x' \neq x$ with good probability. Fix $\mathcal{A}$'s coins, so that its output on each digest is a single string $x' = \mathcal{A}(s, y)$. An input x can satisfy $x = \mathcal{A}(s, H_s(x))$ only if x is *the* output assigned to its own digest, so each of the 2^n digests contributes at most one such input, and at most 2^n inputs in all are "fixed points" of the composed map. Over uniform $x \xleftarrow{\$} \{0,1\}^m$, therefore, $\Pr[x' = x] \leq 2^n/2^m = 2^{n-m}$; averaging over $\mathcal{A}$'s coins preserves the bound. Hence

$$\mathrm{Adv}_H^{\mathrm{sec}}(\mathcal{B}, \lambda) \;\geq\; \mathrm{Adv}_H^{\mathrm{pre}}(\mathcal{A}, \lambda) - 2^{n-m}.$$

The loss is *additive*, not multiplicative: nothing bounds $\Pr[x' \neq x]$ *conditioned* on success, because $\mathcal{A}$'s successes might concentrate on small fibres where returning x itself is likely. Sufficient compression, $m - n = \omega(\log \lambda)$, makes 2^{n-m} negligible and the implication unconditional.

Non-reversal is witnessed by Example 3.9 below. □

Example 3.9 (Preimage-resistant but not collision-resistant)**.** Take any one-way family H and define $H'_s(x) := H_s(\bar{x})$, where $\bar{x}$ deletes the last bit of x. The family H' inherits one-wayness: a preimage challenge (s, y) for H is distributed exactly as one for H' at one bit greater challenge length, and an H'-inverter's output x'' satisfies $H_s(\overline{x''}) = y$, so deleting its last bit hands the reduction a preimage under H. Yet the pair $x \| 0, x \| 1$ is an immediate collision for every x, found without any computation. Hence preimage resistance does not imply collision resistance. The other separations are constructed similarly; the implications of Proposition 3.8 are genuinely one-directional.

Remark 3.10 (Which application needs which notion). Password storage—keeping a digest of each password so that a stolen table does not reveal them—is the natural home of *preimage* resistance: the thief holds digests, and any preimage lets her log in. (The notion alone is not the whole story there: passwords are guessable, so deployments hash each together with a unique random string, its *salt*, through a deliberately slow function, against offline guessing.) A signature scheme that signs $H(m)$, where an adversary may later try to substitute a forged document for the honestly chosen m, needs *second-preimage* resistance: the message is fixed and honest, and the attacker must collide with it. A scheme in which the *signer herself* may cheat—preparing two contracts with the same digest, signing the benign one, presenting the malicious one—needs full *collision* resistance, because there the adversary controls both messages. Commitment schemes, Merkle trees, and the Fiat–Shamir transform all fall into this last category, which is why collision resistance is the headline property of a cryptographic hash.

3.3 The birthday bound on generic collision finding

How hard can collision finding be? The generic attack—evaluate the hash on many inputs and wait for a coincidence—succeeds far sooner than intuition suggests, and its cost is governed by the birthday calculation of the *Math Guide*.

> **Lemma 3.11** (Birthday bound for hashing)**.** *Let $H\colon \{0,1\}^* \to \{0,1\}^n$ and $N = 2^n$. Evaluate H on q distinct inputs, in the idealised model where the q digests behave as independent uniform samples*

from $\{0,1\}^n$, and let C be the event that two digests coincide. Then

$$\Pr[C] \;=\; 1 - \prod_{i=0}^{q-1}\Big(1 - \frac{i}{N}\Big),$$

with the two-sided estimate

$$1 - e^{-q(q-1)/(2N)} \;\le\; \Pr[C] \;\le\; \frac{q(q-1)}{2N} \;\le\; \frac{q^2}{2N},$$

and the lower bound gives $\Pr[C] \ge 1/2$ *once* $q \ge 1.18\sqrt{N}$. *A collision therefore appears with constant probability after about* $\sqrt{N} = 2^{n/2}$ *evaluations.*

Proof. The exact product and both estimates are the birthday-bound proposition of the *Math Guide* (§"The union bound and a birthday calculation"), instantiated with set size $N = 2^n$ and sample count $k = q$; we do not re-derive them. Only the half-probability threshold is new. Setting the exponent of the lower bound equal to $-\ln 2$ gives $q(q-1) = 2N \ln 2$, i.e.

$$q \;\approx\; \sqrt{2\ln 2}\,\sqrt{N} \;\approx\; 1.18\sqrt{N},$$

the constant being $\sqrt{2\ln 2} = 1.1774\ldots$; for $q \ge 1.18\sqrt{N}$ the exponential falls below $1/2$ and $\Pr[C] \ge 1/2$. □

Remark 3.12 (Reading the birthday bound). Three consequences deserve emphasis.

The square-root law. An n-bit digest yields only $n/2$ bits of collision security in the sense of Definition 1.32. For the conventional 128-bit collision security one must take $n = 256$—which is why SHA-256 (and not "SHA-128", which does not exist) is the workhorse digest size, and why the transcript hashes of Fiat–Shamir-style protocols must carry at least 256-bit output.

The attack is practical. The naive attack stores all $q \approx 2^{n/2}$ digests and sorts—$2^{n/2}$ memory, the same obstruction that made baby-step giant-step unrealistic in Remark 2.12. But collision finding, like the discrete-logarithm walk of Proposition 2.13, admits cycle finding: iterate $x \mapsto H(x)$ and detect the cycle with Floyd's or Brent's algorithm, in the manner of Pollard's rho, or run the van Oorschot–Wiener parallel collision search across many machines. These achieve the same $2^{n/2}$ time in essentially constant memory, so the birthday attack is a genuine budget item, not a thought experiment.

Preimages cost the square. The best generic attack against preimage and second-preimage resistance is brute force: about 2^n evaluations to hit a prescribed digest. Those notions therefore enjoy full n-bit generic security—twice the collision level—because the attacker cannot exploit birthday coincidences among self-chosen pairs; the collision game is the only one that pays the square-root discount.

3.4 The random oracle model

The model section introduced the random oracle model as one of the two idealisations under which proofs proceed (Definition 1.26), stated its two powers informally (Remark 1.27), and fixed its status as a heuristic with known limits (Remark 1.29). With hash syntax now in hand we can define the idealised object precisely and *prove* the powers.

Definition 3.13 (Random oracle)**.** A *random oracle* with output length n is a function $\mathcal{O}\colon \{0,1\}^* \to \{0,1\}^n$ drawn uniformly at random from the set of all such functions. Equivalently—and this is the form every proof uses—the oracle is realised *lazily*: a table, initially empty, is maintained; on query x, if x is in the table the stored value is returned, and

otherwise a fresh $\mathcal{O}(x) \xleftarrow{\$} \{0,1\}^n$ is sampled, recorded, and returned.

Three features characterise the object: outputs on distinct inputs are independent and uniform; outputs on repeated inputs are consistent; and access is *only* by querying—the oracle has no description shorter than its (exponentially long) truth table. A scheme lives in the random oracle model (ROM) when every party, honest or adversarial, interacts with a single shared $\mathcal{O}$ in place of a concrete hash, and security is proved relative to the random choice of $\mathcal{O}$; on deployment a concrete hash—SHA-256, say, or BLAKE2b—*instantiates* the oracle.

Proposition 3.14 (The two powers of the ROM)**.** *In a ROM security proof, the reduction, which simulates $\mathcal{O}$ for the adversary, gains two capabilities that no concrete hash affords.*

1. Observability (extractability). *The reduction sees every input the adversary queries. Since querying is the only way to learn anything about $\mathcal{O}(x)$, if the adversary's output depends on $\mathcal{O}(x)$ then the reduction holds x.*

2. Programmability. *On a not-yet-queried input x, the reduction may choose the response $\mathcal{O}(x)$ to be any value of its liking, provided the value is distributed uniformly; the simulation remains perfect, and the reduction can thereby* plant *a challenge—a value it must invert, say—inside an oracle answer.*

Proof. Both capabilities are immediate from the lazy realisation of Definition 3.13, because in the simulation the reduction *is* the table-keeper. Every query arrives at the reduction before an answer exists, which is observability. And on a fresh input the honest table-keeper would sample the answer uniformly and independently of the adversary's view so far; a reduction that instead inserts any value *with that same distribution* produces a view identical to the honest one, while the table keeps repeated queries consistent. That is programmability. □

Remark 3.15 (Why Fiat–Shamir needs the ROM). The paradigmatic consumer of both powers at once is the Fiat–Shamir transform—the compiler that renders Halo 2's interactive protocol non-interactive, developed in this volume under "From identification to signature via the Fiat–Shamir transform" and in full generality under "The Fiat–Shamir transform: from interactive to non-interactive". The transform replaces the verifier's random challenge e by $e = \mathcal{O}(\text{transcript so far})$. Soundness of the resulting non-interactive argument is proved by *rewinding*: the reduction observes (capability 1) the transcript the prover commits to, then *reprograms* (capability 2) the oracle to return a fresh independent challenge on that same transcript, extracting two accepting transcripts that share a prefix and, from them, a witness—the forking technique previewed in Remark 1.24. Neither step is available against a concrete hash, whose outputs the reduction can neither observe selectively nor rewrite. This is the proof that lives *only* in the ROM, and the reason the model is indispensable to the cryptography this monograph targets.

3.5 Domain separation and personalisation

One physical hash serves many purposes in a deployed protocol: it derives keys, commits to notes, tags spent notes, seeds Fiat–Shamir challenges. Digests, however, are interchangeable bit strings, and an output produced for one purpose can be *replayed* as another unless the purposes are cryptographically insulated. Domain separation is the discipline that insulates them.

Definition 3.16 (Domain separation)**.** Let H be a hash modelled as a random oracle, and let $\{\tau_i\}_{i \in I}$ be a set of distinct, *prefix-free* domain tags (also called labels or personalisation strings), one per intended use: no τ_i is a prefix of another—as when all tags have equal length, or when

each is length-prefixed. The *use-i hash* is

$$H^{(i)}(x) \;:=\; H(\tau_i \,\|\, x).$$

Proposition 3.17 (Domain separation yields independent oracles)**.** *If H is a random oracle and the tags $\{\tau_i\}$ are prefix-free, then the family $\{H^{(i)}\}_{i \in I}$ is distributed exactly as a collection of* independent *random oracles. Consequently a digest obtained from $H^{(i)}$ is—except with the negligible probability of a generic collision—useless against any use $j \neq i$.*

Proof. Prefix-freeness makes the domains $\tau_i \,\|\, \{0,1\}^*$ pairwise disjoint: a string with prefix τ_i cannot also have prefix τ_j unless one tag is a prefix of the other. Inputs queried through different $H^{(i)}$ are therefore never equal. A random oracle assigns independent uniform values to distinct inputs, so restricting H to the disjoint slices yields, per slice, a uniformly random function, independent across slices—precisely a collection of independent random oracles. The replay conclusion follows: the value $H^{(i)}(x)$ reveals nothing about any $H^{(j)}(\cdot)$ with $j \neq i$. □

Three idioms implement the discipline in deployed practice; Zcash uses all three.

1. *Context tagging.* Bind an ASCII string naming the use into every call—prepended, as in Definition 3.16, or in any framing that keeps the tag unambiguously delimited from the data. The Orchard tag inventory is of this kind: the strings `z.cash:Orchard-cv`, `z.cash:Orchard-NoteCommit`, and `z.cash:Orchard-CommitIvk` separate the value commitment, note commitment, and key commitment uses of one group-hash (protocol specification §§ 5.4.8.3–5.4.8.4), and `z.cash:Orchard-gd` separates diversified key generation (protocol specification § 5.4.1.6). The deployed framing mirrors the definition's rather than copying it: the curve hash of §3.7 *appends* its domain tag after the message, closed by a byte carrying the tag's length (protocol specification § 5.4.9.8); the length byte makes the tag self-delimiting, so distinct tags still induce disjoint effective inputs—all the proof of Proposition 3.17 uses. The tags reach that hash by two routes, and the commitment tags by both: `-cv` and `-gd` serve directly as its domain tag, while each commitment tag, suffixed `-M`, is hashed as the *message* under the fixed domain tag `z.cash:SinsemillaQ` to derive one of the commitment's two base points and, suffixed `-r`, serves directly as the domain tag deriving the other (protocol specification §§ 5.4.1.9 and 5.4.8.4).

2. *Native personalisation.* BLAKE2b's parameter block carries a dedicated 16-byte personalisation field that is mixed into the initialisation vector, so distinct personalisations separate the primitive's effective inputs before a single message byte is processed—the field for which Zcash favours BLAKE2 for its oracles. With BLAKE2b modelled as a random oracle of the pair (personalisation, message), the fixed-length field is a prefix-free tag and Proposition 3.17 makes the resulting domains independent; a concrete BLAKE2b instantiation does not make that independence a fact "by construction". The deployed personalisations include `Zcash_ExpandSeed` for key expansion (protocol specification § 5.4.2), `Zcash_OrchardKDF` for the note-encryption key-derivation function (KDF, the object of §6.6; protocol specification § 5.4.5.6), and `Zcash_Orchardock` for the outgoing cipher key (protocol specification § 5.4.2)—all exactly 16 bytes.

3. *Suffix and framing bits in the SHA-3 family.* FIPS 202 reserves domain-separation suffix bits appended to the message before padding: 01 for SHA-3 hashing and 1111 for the SHAKE XOFs. SHA3-256 and SHAKE256 share the same underlying permutation, yet the framed inputs differ, so their effective inputs are disjoint—though their finite digests can still coincide by chance.

The unifying requirement behind all three idioms is that distinct uses induce *disjoint effective inputs* to the underlying primitive. An ad hoc tag that happens to be a prefix of another silently merges two domains and reopens the replay attack—which is why Definition 3.16 demands prefix-freeness rather than mere distinctness. The Merkle trees of the final section apply the same discipline *within* a single structure, tagging tree layers to foreclose a forgery that confuses leaves with internal nodes; see "Layer and domain separation" there.

3.6 Hashing to a field element

The primitives ahead—commitments over the Pasta curves, Fiat–Shamir challenges, key derivations—repeatedly need to hash a message *to a field element*: a near-uniform element of $\mathbb{F}_p$ rather than a bit string. The obstacle is arithmetic: a hash emits uniform *bits*, and reducing n uniform bits modulo a prime p is not uniform on $\mathbb{F}_p$, because 2^n is not a multiple of p—some residues receive one more preimage than others. The remedy is to hash *wide* and reduce.

Construction 3.18 (Expand-then-reduce)**.** Let p be a prime of $L = \lceil \log_2 p \rceil$ bits, and fix a XOF, or a domain-separated hash, expand, producing arbitrary-length near-uniform output. To map a message m to $\mathbb{F}_p$, with a domain tag τ and a security slack of k bits (commonly $k = 128$):

1. compute $b \leftarrow \mathsf{expand}(\tau \,\|\, m,\ L + k)$, a string of $L + k$ near-uniform bits;
2. interpret b as an integer $B \in \{0, \dots, 2^{L+k} - 1\}$ and output $B \bmod p \in \mathbb{F}_p$.

Proposition 3.19 (Near-uniformity of expand-then-reduce)**.** *If* expand *is a random oracle, the output of Construction 3.18 lies within statistical distance (Definition 1.6) at most*

$$\frac{p}{2^{L+k+1}} \;\le\; 2^{-k}$$

of the uniform distribution on $\mathbb{F}_p$. With $k = 128$, no algorithm whatsoever—efficient or not—distinguishes the output from a uniform field element with advantage better than 2^{-128}.

Proof. The integer B is uniform on $\{0, \dots, K - 1\}$ with $K = 2^{L+k}$, and the bias of reducing a uniform B modulo p is exactly the situation of the modular-reduction proposition of the *Math Guide* (§"Uniform sampling and the bias of modular reduction"): each residue's probability differs from $1/p$ by at most $1/K$, because the preimage counts $\lfloor K/p \rfloor$ and $\lceil K/p \rceil$ differ by one, and summing over the p residues gives statistical distance at most $p/(2K) = p/2^{L+k+1}$—the sharper form proved there. Since $p < 2^L$, this is less than $2^{-(k+1)} \le 2^{-k}$. The final claim is the variational characterisation of statistical distance (Proposition 1.7): no decision procedure exceeds the distance. □

The slack is not optional. With $k = 0$—a single L-bit block reduced modulo p—the bias can be macroscopic: for p just above a power of two, $K = 2^L$ is barely twice p, the residues beyond the wrap-around point receive only one preimage while the rest receive two, and each deprived residue carries as little as half its uniform mass. The k extra bits are exactly what drives this imbalance to negligibility.

Deployed practice packages the construction. The IETF hash-to-curve specification (RFC 9380) defines hash_to_field$(m, count)$, producing *count* field elements by calling an expansion routine—expand_message_xmd over a fixed-output hash, or expand_message_xof over a XOF—with a *mandatory* domain-separation tag, and reducing each chunk of

$$L_{\mathrm{RFC}} \;=\; \left\lceil (\lceil \log_2 p \rceil + k)/8 \right\rceil \text{ bytes}$$

modulo p, where k is the suite's target security level in bits. The RFC's own suites for 255-bit fields—the curve25519 family—set $k = 128$, hence 48-byte chunks. The Zcash suite for Pallas and Vesta is more generous: it takes $k = 256$, reducing each element from a full 64-byte BLAKE2b-512 digest, so the slack of Proposition 3.19 is realised with room to spare (the `hash_to_field` routine expands with BLAKE2b in the XMD pattern under the suite tag `{tag}-{curve}_XMD:BLAKE2b_SSWU_RO_` and reduces two 64-byte chunks each via `from_uniform_bytes`; protocol specification § 5.4.9.8). This is the concrete recipe for mapping identities to field elements, and for producing the field elements that the next subsection feeds to a curve. Halo 2's deployed Fiat–Shamir transcript uses the same wide-output-then-reduce principle: each challenge is the 64-byte digest of a BLAKE2b state personalised with `Halo2-Transcript`, reduced by `from_uniform_bytes`; the order in which that state absorbs messages and squeezes challenges is the *Halo 2 Guide*'s, §"The complete Halo 2 protocol".

3.7 Hashing to a curve point

The volume ahead also needs to hash *onto an elliptic curve*: given a message, produce a point of $E(\mathbb{F}_p)$ that behaves like a uniform group element—the raw material for commitment generators whose discrete logarithms nobody knows. Two requirements pull against each other. The output should be near-uniform in the relevant subgroup, so that the map can stand in for a random oracle into the group in security arguments; and the map must run in *constant time*, because its input is often secret-adjacent.

The naive approach fails the second requirement instructively. Hash the message to a candidate $x \in \mathbb{F}_p$ (Construction 3.18); test whether $g(x) = x^3 + ax + b$ is a quadratic residue, so that a y-coordinate exists; if not, increment x and retry. This *try-and-increment* loop is correct and roughly uniform, but the number of trials depends on the message: about half of all x fail the residue test, since the squares are half of $\mathbb{F}_p^\times$ (*Math Guide*, §"Quadratic residues and the Euler criterion", whose Legendre symbol $\left(\frac{g(x)}{p}\right)$ is exactly the test), so the running time leaks information about m—an avenue for side-channel attacks. Constant-time hashing to a curve requires a map that succeeds on the *first* try, for every input.

Construction 3.20 (Simplified Shallue–van de Woestijne–Ulas map)**.** Let $E\colon y^2 = g(x) = x^3 + ax + b$ over $\mathbb{F}_p$ with $ab \neq 0$, and choose, as RFC 9380 requires, a constant $Z \in \mathbb{F}_p$ such that Z is a nonsquare, $Z \neq -1$, the polynomial $g(X) - Z$ is irreducible over $\mathbb{F}_p$, and $g(b/(Za))$ is a square. (The RFC selects Z mechanically, as the candidate of smallest absolute value meeting these conditions—a nothing-up-my-sleeve constant with no hidden design freedom.) The map $\mathrm{SWU}\colon \mathbb{F}_p \to E(\mathbb{F}_p)$ sends u to a point as follows. Compute the two candidate x-coordinates

$$x_1 = \frac{-b}{a}\Big(1 + \frac{1}{Z^2u^4 + Zu^2}\Big), \qquad x_2 = Zu^2x_1,$$

with the fallback $x_1 = b/(Za)$ when the denominator $Z^2u^4 + Zu^2$ vanishes (as at $u = 0$). Set $x = x_1$ if $g(x_1)$ is a square and $x = x_2$ otherwise; let y be a square root of $g(x)$ (*Math Guide*, §"Square roots modulo a prime"), its sign fixed by a convention—the deployed map matches the parity of y to that of u—and output (x, y).

The construction contains no trial loop, hence runs in constant time; what needs proof is that one of the two candidates always works.

Proposition 3.21 (Correctness of the branch selection)**.** *Under the parameter conditions of Construction 3.20, the map is well-defined for every $u \in \mathbb{F}_p$: the selected value $g(x)$ is a square, so* $\mathrm{SWU}(u)$

> *is a genuine point of* $E(\mathbb{F}_p)$.

Proof sketch. Suppose first that $D = Z^2u^4 + Zu^2 \neq 0$. The key algebraic identity is

$$g(x_2) \;=\; Z^3u^6\, g(x_1),$$

a polynomial identity in u: substitute $x_2 = Zu^2x_1$ into g and simplify using the formula for x_1 (we take the computation as given). If $g(x_1) = 0$, the first candidate already works, with $y = 0$. Otherwise apply multiplicativity of the Legendre symbol (*Math Guide*, §"Quadratic residues and the Euler criterion"):

$$\left(\frac{g(x_2)}{p}\right) = \left(\frac{Z^3u^6}{p}\right)\left(\frac{g(x_1)}{p}\right) = \left(\frac{Z}{p}\right)\left(\frac{(Zu^3)^2}{p}\right)\left(\frac{g(x_1)}{p}\right) = \left(\frac{Z}{p}\right)\left(\frac{g(x_1)}{p}\right),$$

since $Z^2u^6 = (Zu^3)^2$ is a nonzero square, contributing $+1$ (note that $D \neq 0$ forces $u \neq 0$). Because Z is a nonsquare, $\left(\frac{Z}{p}\right) = -1$, so the two symbols are opposite and the branch selection picks the square. If $D = 0$, the prescribed fallback has $g(x_1) = g(b/(Za))$, a square by the last condition on Z. Thus every input produces a valid affine point. □

A single SWU evaluation is not the end of the pipeline: the map is not surjective onto $E(\mathbb{F}_p)$, and its image is not uniformly covered—it reaches only part of the curve, with a lumpy distribution. The standardised pipeline repairs both defects.

Construction 3.22 (The hash-to-curve pipeline)**.** The RFC 9380 map $\mathsf{hash_to_curve}\colon \{0,1\}^* \to E(\mathbb{F}_p)$ composes four steps:

1. hash to the field: $\mathsf{hash_to_field}(m, 2)$ produces two independent near-uniform field elements u_0, u_1 (§3.6);
2. map each through SWU: $Q_0 = \mathrm{SWU}(u_0)$, $Q_1 = \mathrm{SWU}(u_1)$;
3. sum on the curve: $R = Q_0 + Q_1$. Summing two independent SWU images provably repairs the lumpiness: the sum lies within negligible statistical distance of uniform on $E(\mathbb{F}_p)$, the theorem on which the pipeline's random-oracle claim rests;
4. clear the cofactor: multiply R by $h = \#E(\mathbb{F}_p)/r$ to land in the prime-order subgroup of order r where the cryptography occurs.

The result is a constant-time hash from messages to group elements which, under the hash model and suite conditions of RFC 9380, can instantiate a random oracle into the group. For the Pasta curves the last step is vacuous—their cofactor is 1 (Definition 2.25).

Remark 3.23 (The isogeny detour forced by j-invariant 0). For the Pasta curves one step needs a detour. The SWU formula for x_1 divides by a, and its numerator carries the factor $-b$: with $a = 0$ the formula is undefined, while $b = 0$ makes x_1 identically zero with $g(x_1) = 0$, collapsing the map to the constant point $(0, 0)$. The construction therefore requires $ab \neq 0$—and Pallas and Vesta have $a = 0$ (the j-invariant-0 shape $y^2 = x^3 + 5$ of Definition 2.25; curves with $b = 0$, $j = 1728$, are excluded symmetrically). The obstruction is intrinsic, not notational: the j-invariant is an isomorphism invariant, and a Weierstrass change of variables rescales (a, b) to (u^4a, u^6b) (*Math Guide*, §"The j-invariant and the GLV endomorphism"), so *no* model of these curves attains $ab \neq 0$. The standardised remedy evaluates simplified SWU on an *isogenous* auxiliary curve $E'\colon y^2 = x^3 + a'x + b'$ with $a'b' \neq 0$, then transports the result through an *isogeny* $E' \to E$: a nonconstant map between curves given by polynomial coordinate formulas and respecting the group law. The deployed implementation applies a degree-3 isogeny, and—because an isogeny is a group homomorphism—sums $Q_0 + Q_1$ on E' first and transports the single sum (protocol specification § 5.4.9.8).

The culminating application ties together collision resistance, the random-oracle idealisation, and hashing to a curve.

Construction 3.24 (Nothing-up-my-sleeve generators)**.** To generate public generators $G_1, \ldots, G_k$ of a vector commitment (the Pedersen vector commitment of Construction 4.22 in the next section) so that no one knows any discrete-logarithm relation among them, derive each as

$$G_i \;=\; \mathsf{hash_to_curve}(\tau \,\|\, i)$$

from a fixed public domain tag τ—often a human-readable protocol name—and the index i. Because the seed is a transparent, auditable string with no hidden parameters, no party could have engineered a backdoor relation in advance. This is exactly how Halo 2 produces the generators of its Pedersen vector and inner-product commitments (the latter Construction 4.38): every generator G_i is `hash_to_curve("Halo2-Parameters")` applied to a five-byte message—a zero byte, then the four-byte little-endian encoding of i—the leading zero keeping the G_i disjoint from the two auxiliary generators the same tag derives from the one-byte messages 1 and 2, and Orchard's value commitment derives its two bases from the tag `z.cash:Orchard-cv` with the one-byte messages `"v"` and `"r"`.

Remark 3.25 (Why NUMS generators are binding-safe). Knowledge of a discrete-logarithm relation $\sum_i a_i G_i = 0$ with not all a_i zero would let a committer open one commitment two ways, destroying the binding property that the next section defines and relies on. The NUMS derivation forecloses this: the generators are outputs of a function modelled as a random oracle into the group, so they are distributed as independent uniform group elements, and finding a discrete-logarithm relation among independent uniform elements *reduces* to computing a discrete logarithm on E, hard under the standing assumptions of Remark 2.32; the reduction, which embeds its challenge in every generator, is Theorem 4.23 of the next section. Hence no efficient party, the scheme designer included, can know such a relation.

3.8 Arithmetisation-friendly hashing: Poseidon

The hashes of this section so far are *bit-oriented*: SHA-2 and BLAKE2 shuffle and XOR 32- and 64-bit words. Inside an arithmetic circuit over a large prime field $\mathbb{F}_p$—the setting of proof systems such as Halo 2, whose cost model, developed in the *Halo 2 Guide*, §"From computation to a grid of constraints", prices each field multiplication as a constraint—such a hash is prohibitively expensive: every Boolean operation must be re-expressed through field arithmetic and range checks, and the tens of thousands of resulting constraints dominate the circuit. Arithmetisation-friendly hashes invert the design: the round function is a handful of *low-degree field operations*, native to the circuit, incurring few constraints. Poseidon (Grassi, Khovratovich, Rechberger, Roy, and Schofnegger, 2019) is the instance Orchard employs, and it arrives via a mode of operation worth defining in general.

Construction 3.26 (The sponge construction)**.** Fix a permutation $f\colon \mathbb{F}_p^t \to \mathbb{F}_p^t$ on a state of t field elements. Split the state into a *rate* of r elements—the only part input is added into and output is read from—and a *capacity* of $c = t - r$ elements, never accessed directly: the security reserve. To hash, *absorb* the message into the rate, r elements at a time, applying f between blocks; then *squeeze* output from the rate, applying f between reads. Collision and preimage resistance hold up to approximately $\sqrt{p^c}$ work, as for a random sponge: the birthday bound of Lemma 3.11 applied to the p^c capacity states the adversary cannot see.

Construction 3.27 (The Poseidon permutation)**.** The permutation f is a sequence of rounds, each comprising three steps:

1. *add round constants*: add a fixed vector in $\mathbb{F}_p^t$ to the state;
2. *S-box*: apply $x \mapsto x^\alpha$ to state elements, where $\alpha \geq 3$ is the smallest integer with $\gcd(\alpha, p-1) = 1$, so that the power map is a bijection on $\mathbb{F}_p$; over the Pasta fields $\alpha = 5$, because $3 \mid p-1$ there rules out $\alpha = 3$;
3. *mix*: multiply the state by a fixed maximum-distance-separable (MDS) matrix $M \in \mathbb{F}_p^{t\times t}$—one whose every square submatrix is invertible, giving the fastest possible diffusion of any element's change across the whole state.

The cost-saving principle (the HADES strategy) distinguishes two round types: a few *full* rounds at the start and end apply the S-box to all t elements, while in the many *partial* rounds between them the S-box touches a single element—the MDS mix still diffusing it across the state. Round counts are chosen to stand against the known algebraic (Gröbner-basis) attacks.

Remark 3.28 (Circuit cost). Inside a circuit, the only non-linear operation in all of Poseidon is $x \mapsto x^5$, verified by a couple of multiplication gates per S-box; the round-constant additions and the MDS multiplication are linear and almost free. A 2-to-1 Poseidon hash costs a few hundred constraints, against tens of thousands for a bit-oriented hash—the difference between a practical circuit and an impractical one.

Example 3.29 (The Orchard Poseidon instance)**.** Orchard fixes $t = 3$ (rate 2, capacity 1), $\alpha = 5$, and 8 full plus 56 partial rounds, over the Pallas base field, at the 128-bit security level—the parametrisation named `P128Pow5T3` in the deployed source (protocol specification § 5.4.1.10). It is used in *constant-input-length* mode as a two-input hash:

$$\mathsf{PoseidonHash}(x, y) \;=\; f([\,x,\ y,\ 2^{65}\,])_0,$$

the first component of the state after a single permutation of the input padded by the fixed capacity value $2^{65} = 2 \cdot 2^{64}$, which encodes the input length 2 into the capacity so that no message padding is required (the `ConstantLength` domain sets the initial capacity element to $L \cdot 2^{64}$). Keyed by its first argument, this hash serves as the pseudorandom function

$$F_{\mathsf{nk}}(\rho) \;=\; \mathsf{PoseidonHash}(\mathsf{nk}, \rho)$$

—the PRF vocabulary is owned by "Pseudorandom functions and permutations" later in this volume—deployed as $\mathsf{PRF}^{\mathsf{nfOrchard}}$ (protocol specification § 5.4.2), from which Orchard constructs the nullifier—a construction developed in the *Ironwood Guide*, §"The nullifier"; this volume supplies only the PRF. One contrast deserves note: unlike the Pedersen and Sinsemilla hashes constructed in the next section, whose collision resistance reduces to a clean assumption (discrete log), Poseidon's security rests on cryptanalytic confidence in its round function—the same epistemic footing as SHA-256, not the same as a reduction.

3.9 The transparent layer's hashes

The Orchard shielded protocol hashes with BLAKE2b, Sinsemilla, and Poseidon. The transparent layer, inherited from Bitcoin, hashes with SHA-256 and two functions derived from it, none needing machinery beyond §3.2 and the birthday bound of §3.3; this register fixes their names, domains, and deployed uses for the volumes above.

- *RIPEMD-160.* A 160-bit iterated (Merkle–Damgård, the construction of §5.3) hash, $\{0,1\}^* \to \{0,1\}^{160}$, older than Bitcoin and inherited with its address format, whose collision

resistance sits at the 80-bit birthday bound of Lemma 3.11. Zcash uses it only inside hash160 below and the script opcodes OP_RIPEMD160 and OP_HASH160.

- *The address hash* hash160 := RIPEMD160 ∘ SHA256. A pay-to-public-key-hash (P2PKH) address carries the 20-byte hash160 of the 33-byte compressed encoding of an ECDSA public key (the scheme is registered in §8.10), and a pay-to-script-hash (P2SH) address the hash160 of a *redeem script*, the spending condition the spender later reveals and satisfies (protocol specification § 5.6.1.1). The P2PKH output script—the condition a spend of the output must satisfy—is OP_DUP OP_HASH160 ⟨h⟩ OP_EQUALVERIFY OP_CHECKSIG, which recomputes the hash from the public key the spender reveals; wallets compute it through the zcash_transparent crate, and a partially created transaction (PCZT), the multi-party construction format of the *Wallet Guide*, §"Partially created transactions (PCZT)", carries on each transparent input preimage maps keyed by RIPEMD-160, SHA-256, hash160, and hash256 digests.
- *Double SHA-256* (*SHA-256d, or* hash256). The composition $x \mapsto \mathrm{SHA256}(\mathrm{SHA256}(x))$ (protocol specification § 5.4.1.1), whose outer application also closes the length-extension property of plain SHA-256 (§5.3). It hashes block headers; its leading four bytes are the checksum that Base58Check, the string encoding of transparent addresses, appends to the raw address bytes before converting to base 58 (protocol specification § 5.6.1.1) and the payload checksum in every peer-to-peer message header; and it is the legacy transaction identifier: the txid of a version-4-or-earlier transaction is SHA-256d of its serialisation (protocol specification § 7.1.1), whereas a version-5 txid is the BLAKE2b digest tree of ZIP 244.

The toolbox now holds its first primitive in full: a hash with three calibrated security notions, an idealisation that licenses proofs, a discipline that insulates its many uses, and maps carrying its output into fields, onto curves, and inside circuits. The next section spends this capital immediately: commitments—the sealed envelopes of the protocol—are built from exactly the hashes, generators, and hardness assumptions now on the table.

4 Commitment schemes

Two adversaries stalk this section, one on each side of a sealed envelope. Before the envelope is opened, the adversary is the *peeker*: the receiver, who holds the sealed commitment and tries to distinguish which of two candidate messages it contains—to read the bid before the auction closes, the vote before the tally. After the envelope has been delivered, the adversary is the *equivocator*: the sender, who tries to open the same commitment to a message other than the one sealed inside—to change the bid once the rival's is public. A commitment scheme is the cryptographic sealed, tamper-evident envelope that defeats both: once delivered, the sender can no longer alter the message (*binding*), yet the recipient cannot read it until it is unsealed (*hiding*).

The primitive is the natural next step after hashing: the two constructions at the heart of this section, Pedersen and Sinsemilla, need nothing beyond the discrete-logarithm hardness of §2 and the nothing-up-my-sleeve generators that §3 showed how to derive, and nearly everything built later consumes them. Classical applications set the scene: coin-flipping over a telephone (each party commits to a bit before either reveals), zero-knowledge proofs (the prover commits to its secrets and proves statements about the sealed values), and verifiable secret sharing. Centrally for this volume, the *polynomial* commitments of §4.9 are the cryptographic engine of modern succinct argument systems, including the one deployed in Orchard: the value commitments checked by binding signatures, the note commitments accumulated in Merkle trees, and the witness polynomials of the Halo 2 proof system are all commitments in the precise sense defined here.

The plan of the section follows the two adversaries. We first fix the syntax of a commitment scheme and the three models of who may be trusted to set it up. We then define hiding and binding as games in the tradition of §1.4, each in three quantitative flavours—perfect, statistical,

computational—and prove that the two perfections are jointly unattainable: every scheme must choose which adversary to defeat unconditionally. The constructions that follow are ordered by increasing algebraic richness: hash-based commitments (opaque but transparent and plausibly post-quantum), the Pedersen commitment and its vector variant (perfectly hiding, computationally binding under discrete log, and—decisively—linear), Sinsemilla (a commitment whose internal operations are curve operations, built to be cheap inside a proof), then polynomial commitment schemes, contrasting the pairing-based KZG family with the inner-product family that Zcash deploys, and the bridge from commitments to arguments. A closing subsection returns to the classical application named above—secret sharing—and builds its verifiable form from coefficient commitments, for the threshold signatures of the *FROST Guide*.

4.1 The sealed envelope

The envelope metaphor repays being taken slowly, because each of its physical properties has an exact formal counterpart. The sender writes a message on a card, seals it in the envelope, and hands the envelope over. From that moment the receiver *possesses* the commitment: whatever the message was, it is now fixed—the sender cannot swap the card, because the envelope is in the receiver's hands and any tampering would show. Yet the receiver *learns* nothing: the envelope is opaque. Later, at a time of the protocol's choosing, the sender opens the envelope—in the formalism, transmits the message together with the randomness used to seal it—and the receiver checks that the revealed message is consistent with the envelope received earlier.

The two guarantees are directed at different parties and different times. Hiding protects the *sender* against a curious *receiver* during the window between commit and reveal; binding protects the *receiver* against a dishonest *sender* at reveal time. A scheme that is hiding but not binding is an envelope with a false bottom: private, but worthless as evidence. A scheme that is binding but not hiding is a glass envelope: tamper-evident, but no secret survives it. The games of §4.3 formalise exactly these two failure modes, and the theorem of §4.4 shows that against computationally unbounded adversaries one of the two protections must give way.

4.2 Syntax

Definition 4.1 (Commitment scheme)**.** A *commitment scheme* Π for a message space $\mathcal{M}$ and randomness space $\mathcal{R}$ is a triple of algorithms $(\mathsf{Setup}, \mathsf{Commit}, \mathsf{Open})$:

- The algorithm $\mathsf{Setup}(1^\lambda)$ is PPT (§1.1) and outputs *public parameters* pp, which implicitly determine the message space $\mathcal{M}_{pp}$ and randomness space $\mathcal{R}_{pp}$ and are an implicit input to the remaining algorithms.
- The algorithm $\mathsf{Commit}(pp, m; r)$, for $m \in \mathcal{M}_{pp}$ and $r \in \mathcal{R}_{pp}$, is deterministic given r and outputs a *commitment* c. We write $\mathsf{Commit}(pp, m)$, without an explicit r, for the probabilistic algorithm that samples $r \xleftarrow{\$} \mathcal{R}_{pp}$ uniformly and returns r alongside c.
- The algorithm $\mathsf{Open}(pp, c, m, r)$ is deterministic and outputs a bit: 1 (accept) or 0 (reject).

Correctness requires that honestly produced commitments open: for all pp in the support of $\mathsf{Setup}(1^\lambda)$, all $m \in \mathcal{M}_{pp}$, and all $r \in \mathcal{R}_{pp}$,

$$\mathsf{Open}(pp,\ \mathsf{Commit}(pp, m; r),\ m,\ r) = 1.$$

The pair (m, r) is called the *opening* (or *decommitment*) of c. Every scheme in this volume is *canonically opening*: $\mathsf{Open}(pp, c, m, r)$ merely recomputes $\mathsf{Commit}(pp, m; r)$ and checks equality with c. This is the adopted convention unless stated otherwise. For a canonically opening

scheme correctness holds automatically, and a valid opening of c is precisely a pair (m, r) with $\mathrm{Commit}(pp, m; r) = c$—a reformulation the proofs below use constantly.

Remark 4.2 (The two phases, and a third concern). A commitment scheme is used in two temporally separated phases. In the *commit phase* the sender transmits c and nothing else; in the *reveal phase* the sender transmits the opening (m, r) and the receiver runs Open. The two security properties attach to the two phases: hiding constrains what the receiver can learn *after the commit phase*, while binding constrains what the sender can do *in the reveal phase*. A third concern precedes both: the *setup phase* that produces pp. Who runs Setup, and whether that party can be trusted, proves decisive for some of the constructions ahead (Remark 4.3).

Remark 4.3 (Three setup models). Commitment schemes divide by how much trust their parameters demand.

1. *Common reference string* (*CRS*). A trusted party runs Setup and is assumed to *erase* the randomness behind it. The parameters have internal structure, and knowledge of the discarded randomness—the *trapdoor*—would break the scheme. Of the schemes treated in this section, only KZG (§4.9) genuinely resides here.

2. *Uniform random string* (*URS*), *or transparent setup.* The parameters consist only of public coins: uniformly random group elements with no known relations among them. There is no trapdoor and no trusted party—anyone can re-derive the parameters from a public seed. This is the regime of the Pedersen and inner-product commitments below. A Pedersen generator derived by hashing to the curve (Construction 3.24) is transparent even though the textbook presentation phrases Setup as sampling and discarding a secret.

3. *Plain.* No parameters at all, or only a standardised hash function—the home of the hash-based commitment below.

Transparent setup is highly desirable: a flawed or subverted trusted setup can silently destroy binding, as the KZG trapdoor demonstrates concretely in §4.9.

4.3 Hiding and binding

Both properties are games in the sense of Definition 1.15. The hiding game arms the peeker: she chooses the two candidate messages herself, receives a commitment to one of them, and must say which.

Definition 4.4 (Hiding game and hiding)**.** Let Π be a commitment scheme and $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ an adversary. For $b \in \{0, 1\}$, the game $\mathsf{Hide}^{\mathcal{A}}_{\Pi}(\lambda, b)$ runs as follows.

1. The challenger runs $pp \leftarrow \mathrm{Setup}(1^\lambda)$.
2. The adversary $\mathcal{A}_1(pp)$ outputs two messages $m_0, m_1 \in \mathcal{M}_{pp}$ and state st.
3. The challenger samples $r \xleftarrow{\$} \mathcal{R}_{pp}$ and computes $c = \mathrm{Commit}(pp, m_b; r)$.
4. The adversary $\mathcal{A}_2(st, c)$ outputs a bit b', which is the output of the experiment.

The *hiding advantage* of $\mathcal{A}$ is

$$\mathrm{Adv}^{\mathrm{hide}}_{\Pi}(\mathcal{A}, \lambda) := \left|\Pr[\mathsf{Hide}^{\mathcal{A}}_{\Pi}(\lambda, 0) = 1] - \Pr[\mathsf{Hide}^{\mathcal{A}}_{\Pi}(\lambda, 1) = 1]\right|.$$

The scheme is *computationally hiding* if the advantage is negligible (§1.2) for every PPT $\mathcal{A}$; *statistically hiding* if it is negligible even for computationally unbounded $\mathcal{A}$; and *perfectly hiding* if it is exactly 0 for every adversary and every λ.

Proposition 4.5 (Distributional characterisation of hiding)**.** *A commitment scheme* Π *is perfectly hiding if and only if for every pp in the support of* Setup *and every pair* $m_0, m_1 \in \mathcal{M}_{pp}$*, the random variables*

$$\mathrm{Commit}(pp, m_0; r) \quad \textit{and} \quad \mathrm{Commit}(pp, m_1; r), \qquad r \xleftarrow{\$} \mathcal{R}_{pp},$$

are identically distributed. It is statistically hiding if the statistical distance (Definition 1.6) between them is negligible. (Only the perfect clause is an equivalence: parameters of negligible Setup *probability could exhibit large distance without granting any adversary noticeable advantage.)*

Proof. Fix pp and m_0, m_1, and write D_m for the distribution of $\mathrm{Commit}(pp, m; r)$ over uniform r. By the variational characterisation of statistical distance (Proposition 1.7), $\Delta(D_{m_0}, D_{m_1})$ equals *exactly* the maximum advantage, over all decision procedures—efficient or not—of telling D_{m_0} from D_{m_1}. An adversary who outputs the pair (m_0, m_1) on parameters pp and applies the optimal test therefore achieves hiding advantage $\Delta(D_{m_0}, D_{m_1})$ whenever Setup produces pp; conversely no adversary can exceed the expectation of this quantity over pp. Advantage exactly 0 for all adversaries is thus equivalent to $\Delta = 0$—identical distributions—for every attainable pp and every pair, and negligible statistical distance yields negligible advantage for unbounded adversaries. This coincidence of the game formulation with the distance formulation is precisely why statistical hiding was *defined* through the unbounded-adversary game. □

The binding game arms the equivocator. He is not handed an honest commitment; he manufactures his own, together with two openings.

Definition 4.6 (Binding game and binding)**.** For a commitment scheme Π and adversary $\mathcal{A}$, the game $\mathsf{Bind}^{\mathcal{A}}_{\Pi}(\lambda)$ runs as follows.

1. The challenger runs $pp \leftarrow \mathrm{Setup}(1^\lambda)$.
2. The adversary $\mathcal{A}(pp)$ outputs a commitment c and two openings (m_0, r_0) and (m_1, r_1).
3. The game outputs 1 iff $m_0 \neq m_1$ and $\mathrm{Open}(pp, c, m_0, r_0) = \mathrm{Open}(pp, c, m_1, r_1) = 1$.

The *binding advantage* of $\mathcal{A}$ is

$$\mathrm{Adv}^{\mathrm{bind}}_{\Pi}(\mathcal{A}, \lambda) := \Pr[\mathsf{Bind}^{\mathcal{A}}_{\Pi}(\lambda) = 1],$$

a search game in the sense of Definition 1.15. The scheme is *computationally*, *statistically*, or *perfectly binding* according as the advantage is negligible for every PPT adversary, negligible for every unbounded adversary, or exactly 0 for every adversary and every λ.

For a canonically opening scheme the binding game has a useful collision reformulation: for fixed pp, two openings collide precisely when $\mathrm{Commit}(pp, m_0; r_0) = \mathrm{Commit}(pp, m_1; r_1)$ with $m_0 \neq m_1$. Perfect binding then says that distinct messages never share a commitment value at all—the message m is a deterministic function of c. Perfect binding is thus equivalent to the map $c \mapsto m$ being well defined on all attainable commitments, a formulation the counting argument of Remark 4.9 exploits.

Remark 4.7 (The asymmetry of the quantifiers). The two games quantify over different objects. Hiding quantifies over what can be *extracted from* a single honestly produced commitment: the

challenger seals the envelope, and the adversary only looks. Binding quantifies over what can be *manufactured*: the adversary builds an ambiguous commitment of its own choosing, under no obligation to follow Commit honestly. A scheme can be strong in one respect and weak in the other, and the two properties actively pull in opposite directions—the more commitment values a message can take (good for hiding), the more opportunities for two messages to share one (bad for binding). The next subsection makes this tension a theorem.

4.4 Incompatibility of perfect hiding and perfect binding

> **Theorem 4.8** (No scheme is perfectly hiding and perfectly binding)**.** *Let Π be a canonically opening commitment scheme with $|\mathcal{M}_{pp}| \geq 2$. If Π is perfectly hiding, then it is not perfectly binding: indeed, a computationally unbounded adversary wins the binding game with probability 1. Hence no nontrivial commitment scheme is simultaneously perfectly hiding and perfectly binding.*

Proof. Fix any pp in the support of $\mathsf{Setup}(1^\lambda)$ and any two distinct messages $m_0, m_1 \in \mathcal{M}_{pp}$. For a message m, let

$$C_m := \{\, \mathsf{Commit}(pp, m; r) : r \in \mathcal{R}_{pp} \,\}$$

be the set of attainable commitments to m, and let D_m be the distribution of $\mathsf{Commit}(pp, m; r)$ for uniform r. Perfect hiding makes D_{m_0} and D_{m_1} identical (Proposition 4.5); identical distributions have identical supports, so $C_{m_0} = C_{m_1}$. Pick any c in this common support. By definition of the supports there exist $r_0, r_1 \in \mathcal{R}_{pp}$ with

$$\mathsf{Commit}(pp, m_0; r_0) = c = \mathsf{Commit}(pp, m_1; r_1),$$

and then $(c, (m_0, r_0), (m_1, r_1))$ wins the binding game: both openings are valid by canonical opening, and $m_0 \neq m_1$. The double opening *exists* for every attainable pp; an unbounded adversary *finds* it by enumerating $\mathcal{R}_{pp}$, so it wins Bind with probability 1, giving $\mathrm{Adv}^{\mathsf{bind}}_{\Pi} = 1 \neq 0$. □

Remark 4.9 (The impossibility as a counting fact). Information-theoretically the theorem is a statement about how many bits the commitment carries. Perfect binding forces the map $c \mapsto m$ to be well defined, so c determines m completely: the commitment carries at least $\log_2|\mathcal{M}|$ bits of information about the message. Perfect hiding forces the distribution of c to be independent of m: the commitment carries *zero* bits. The two demands are compatible only when $\log_2|\mathcal{M}| = 0$, that is, $|\mathcal{M}| = 1$—when there is nothing to commit to. Every real scheme therefore chooses which property to make perfect (or statistical) and settles for the other being computational. The Pedersen commitment of §4.6 is perfectly hiding and computationally binding; the hash-based commitment of §4.5, read in the random oracle model, is computationally binding (from collision resistance) and computationally hiding.

Remark 4.10 (Two regimes, two failure modes). The choice is not merely aesthetic; it decides *how* the scheme fails if its assumption falls. If binding is only computational, an adversary with enough power—one able to compute discrete logarithms, say—could *retroactively change* a committed value, and the soundness of any protocol built on top degrades to a computational assumption. If hiding is only computational, such an adversary could *retroactively learn* a committed value: privacy degrades the same way, and everlasting secrecy is lost the day the assumption breaks, even for commitments published long before. Which failure is worse is a judgement about the application: whether integrity or long-term confidentiality is the greater concern. A ledger that publishes its commitments forever has reason to care about the second failure mode; a protocol whose soundness guards live funds has reason to care about the first.

4.5 Hash-based commitments

The simplest construction seals the envelope with a hash. Recall from §3 the two properties it needs: a hash function $H\colon \{0,1\}^* \to \{0,1\}^n$ is *collision resistant* (Definition 3.7) if no PPT adversary finds $x \neq x'$ with $H(x) = H(x')$ except with negligible probability, and *preimage resistant* (one-way) if, given $H(x)$ for random x of appropriate length, no PPT adversary finds any x' with $H(x') = H(x)$ except with negligible probability.

Construction 4.11 (Hash commitment)**.** Let $H\colon \{0,1\}^* \to \{0,1\}^n$ be a hash function. The *hash commitment* has message space $\mathcal{M} = \{0,1\}^*$ and randomness space $\mathcal{R} = \{0,1\}^\ell$ for a randomness length $\ell = \ell(\lambda)$ (for example $\ell = 2n$). Let $\mathsf{enc}(m,r)$ be an injective, self-delimiting encoding of the pair—for example a canonical length encoding of m, followed by m, followed by the fixed-length string r. Define

$$\mathrm{Commit}(m;r) := H(\mathsf{enc}(m,r)),$$

opened canonically. There are no nontrivial public parameters beyond the choice of H—the scheme lives in the plain model of Remark 4.3.

Proposition 4.12 (Binding of the hash commitment)**.** *If H is collision resistant, the hash commitment is computationally binding.*

Proof. Suppose $\mathcal{A}$ wins the binding game: it outputs c with valid openings (m_0, r_0), (m_1, r_1) and $m_0 \neq m_1$, so $H(\mathsf{enc}(m_0, r_0)) = c = H(\mathsf{enc}(m_1, r_1))$. Injectivity of enc makes the two hash inputs distinct strings—this is exactly what the self-delimiting encoding buys; without it, a boundary shift could present distinct pairs as the same string. The adversary has therefore produced an H-collision, and the reduction that runs $\mathcal{A}$ and outputs the pair $(\mathsf{enc}(m_0, r_0),\ \mathsf{enc}(m_1, r_1))$ wins the collision game with exactly the binding advantage of $\mathcal{A}$, in essentially the same time. □

Proposition 4.13 (Hiding of the hash commitment, in the ROM)**.** *Model H as a random oracle (Definition 3.13). For messages of length* $\mathrm{poly}(\lambda)$ *and randomness length $\ell = \omega(\log \lambda)$, the hash commitment is computationally hiding: an adversary making at most q oracle queries has hiding advantage at most $q \cdot 2^{-\ell}$.*

Proof. The challenge commitment is $c = H(\mathsf{enc}(m_b, r))$ for uniform $r \in \{0,1\}^\ell$. Because the oracle's outputs are independent and uniform, c is a fresh uniform string—carrying no information about b—*unless* the adversary queries the oracle at the exact point $\mathsf{enc}(m_b, r)$. Before any such query the randomness r is information-theoretically hidden: the adversary's view is independent of it. Each of the q queries therefore hits that point with probability at most $2^{-\ell}$, and the union bound (*Math Guide*, §"The union bound and a birthday calculation") caps the probability of any hit by $q \cdot 2^{-\ell}$. Conditioned on no hit, the adversary's view is identically distributed in the two worlds $b = 0$ and $b = 1$, so the advantage is at most $q \cdot 2^{-\ell}$, negligible for $\ell = \omega(\log \lambda)$ and polynomial q. □

Remark 4.14 (Why the randomness is essential). Dropping r is fatal. The deterministic commitment $c = H(m)$ fails hiding for every low-entropy message space: to distinguish a commitment to m_0 from one to m_1, the peeker simply hashes her two candidates and compares. Nothing about collision resistance or one-wayness prevents this—the attack never inverts anything. The salt r contributes ℓ uniformly random bits, so the hash input is unpredictable even when the message

is one of two known values. The same phenomenon renders deterministic encryption insecure for low-entropy plaintexts, and the same cure—randomisation—applies.

Remark 4.15 (Strength profile). The hash commitment has two virtues. It is *transparent*: no trusted setup, indeed no parameters, and no algebraic structure for a subverted setup to poison. And it is *plausibly post-quantum*: generic quantum collision finding takes $\Theta(2^{n/3})$ queries against an n-bit hash (Remark 2.31), so increasing the output length compensates for the loss. Its weakness is the flip side of its opacity: the commitment is an unstructured bit string. One cannot add two hash commitments and obtain a commitment to the sum of the messages; no algebraic relation on commitments reflects any relation on the committed values. The Pedersen construction repairs exactly this defect, at the price of resting on discrete log.

4.6 The Pedersen commitment

We change algebraic scenery. Let $\mathbb{G}$ be a cyclic group of prime order q, written *additively*: for a point $P \in \mathbb{G}$ and scalar $a \in \mathbb{F}_q$, the scalar multiple is $[a]P$, with $[0]P = \mathcal{O}$ the identity, and $a \mapsto [a]P$ is a group homomorphism $\mathbb{F}_q \to \mathbb{G}$. Concretely, $\mathbb{G}$ is the group of $\mathbb{F}_p$-points of a prime-order elliptic curve (*Math Guide*, §"The group structure of $E(\mathbb{F}_p)$"), and in this volume the Pallas curve of §2.6; but the only structural facts the constructions use are that $\mathbb{G}$ is isomorphic to $\mathbb{F}_q$ as a group and that discrete logarithms in $\mathbb{G}$ are hard. Formally, a *group generator* $\mathcal{G}$ on input 1^λ outputs $(\mathbb{G}, q, G)$ with q a $\Theta(\lambda)$-bit prime and G a generator, and the DLog assumption for $\mathcal{G}$ states that no PPT adversary, given $(\mathbb{G}, q, G, [a]G)$ for uniform $a \xleftarrow{\$} \mathbb{F}_q$, outputs a except with negligible probability—Definition 2.2, transcribed additively as in §2.6.

Construction 4.16 (Pedersen commitment). The algorithm $\mathrm{Setup}(1^\lambda)$ runs $(\mathbb{G}, q, G) \leftarrow \mathcal{G}(1^\lambda)$, samples $s \xleftarrow{\$} \mathbb{F}_q^\times$, sets $H := [s]G$, outputs $pp = (\mathbb{G}, q, G, H)$, and *discards* s. The message and randomness spaces are $\mathcal{M} = \mathcal{R} = \mathbb{F}_q$, and

$$\mathrm{Commit}(pp, m; r) := [m]G + [r]H,$$

opened canonically.

The parameters are two generators G and H whose relative discrete logarithm $s = \log_G H$ is unknown to *all* parties, the committer included. Everything hinges on this one unknown scalar: knowledge of s breaks binding (Remark 4.19), while perfect hiding, as the proof below shows, does not even require ignorance of s. In deployment the sampled-and-discarded s of the textbook presentation is replaced by deriving H through hashing to the curve (Construction 3.24): nobody ever *holds* s, and the setup is transparent (Remark 4.3).

Theorem 4.17 (Perfect hiding). *The Pedersen commitment is perfectly hiding.*

Proof. Fix any attainable $pp = (\mathbb{G}, q, G, H)$; since $s \in \mathbb{F}_q^\times$, the point H is not $\mathcal{O}$. Because $\mathbb{G}$ has prime order, the nonzero H generates $\mathbb{G}$, so $r \mapsto [r]H$ is a bijection $\mathbb{F}_q \to \mathbb{G}$. For uniform r, therefore, $[r]H$ is uniform on $\mathbb{G}$—and so is its translate $[m]G + [r]H$, for *every* message m: adding the fixed element $[m]G$ permutes the group and leaves the uniform distribution invariant. The commitment distribution is uniform on $\mathbb{G}$ regardless of m, so for any m_0, m_1 the two distributions coincide exactly, and every adversary's advantage is 0 by Proposition 4.5. □

Theorem 4.18 (Computational binding under DLog). *If the DLog assumption holds for $\mathcal{G}$, the Pedersen commitment is computationally binding. Concretely, from any adversary $\mathcal{A}$ that wins the*

> *binding game with probability ε one constructs $\mathcal{B}$ computing discrete logarithms with probability $(1 - 1/q)\,\varepsilon \geq \varepsilon - 1/q$, in essentially the same running time.*

Proof. First the algebra. A double opening gives $[m_0]G + [r_0]H = [m_1]G + [r_1]H$ with $m_0 \neq m_1$, so

$$[m_0 - m_1]G = [r_1 - r_0]H.$$

The case $r_1 = r_0$ is impossible: it would force $[m_0 - m_1]G = \mathcal{O}$, and since G has order q this means $m_0 \equiv m_1 \pmod{q}$, contradicting $m_0 \neq m_1$ in $\mathbb{F}_q$. With $r_1 - r_0$ invertible modulo the prime q, the relation solves for the hidden discrete logarithm:

$$s = \log_G H = (m_0 - m_1)(r_1 - r_0)^{-1} \bmod q.$$

A double opening *is* the discrete log of H.

Now the reduction. On a DLog challenge $(\mathbb{G}, q, G, Q)$, the algorithm $\mathcal{B}$ sets $H := Q$, hands $pp = (\mathbb{G}, q, G, Q)$ to $\mathcal{A}$, and, when $\mathcal{A}$ returns a double opening, outputs $(m_0 - m_1)(r_1 - r_0)^{-1} \bmod q = \log_G Q$. One distribution gap needs accounting: in the real scheme $H = [s]G$ with $s \xleftarrow{\$} \mathbb{F}_q^\times$, so H is uniform on $\mathbb{G} \setminus \{\mathcal{O}\}$, whereas the challenge $Q = [a]G$ has a uniform on all of $\mathbb{F}_q$, so Q is uniform on all of $\mathbb{G}$. The two distributions differ only on the event $Q = \mathcal{O}$, of probability $1/q$—their statistical distance is exactly $1/q$—and on that event $\mathcal{A}$ cannot win anyway (with $H = \mathcal{O}$ the relation above forces $m_0 = m_1$). Conditioned on $Q \neq \mathcal{O}$, the simulation is perfect, so

$$\mathrm{Adv}^{\mathrm{dlog}}(\mathcal{B}) = \left(1 - \tfrac{1}{q}\right) \mathrm{Adv}^{\mathrm{bind}}_{\Pi}(\mathcal{A}) \;\geq\; \varepsilon - \tfrac{1}{q},$$

and $\mathcal{B}$'s overhead over $\mathcal{A}$ is a constant number of field operations. □

The same embed-and-extract pattern, scaled up to many generators, proves the vector generalisation (Theorem 4.23).

Remark 4.19 (Equivocation with the trapdoor). Any party who knows $s = \log_G H$ can open a Pedersen commitment to *any* message. Given $c = [m]G + [r]H$ and a target m', set

$$r' := r + (m - m')\, s^{-1} \bmod q;$$

then $[m']G + [r']H = [m']G + [r]H + [(m - m')]G = c$. This trapdoor property is not a defect of the construction—it is precisely why discarding s in Setup is mandatory, and it is a feature the theory exploits deliberately: zero-knowledge simulators use exactly this equivocation to open commitments consistently with a transcript produced out of order. The same phenomenon priced the setup models of Remark 4.3: a commitment scheme with a trapdoor is only as binding as the claim that nobody holds it.

> **Theorem 4.20** (Additive homomorphism). *For all $m_0, m_1, r_0, r_1 \in \mathbb{F}_q$,*
>
> $$\mathrm{Commit}(m_0; r_0) + \mathrm{Commit}(m_1; r_1) = \mathrm{Commit}(m_0 + m_1;\, r_0 + r_1),$$
>
> *and more generally, for all scalars $\alpha, \beta \in \mathbb{F}_q$,*
>
> $$[\alpha]\,\mathrm{Commit}(m_0; r_0) + [\beta]\,\mathrm{Commit}(m_1; r_1) = \mathrm{Commit}(\alpha m_0 + \beta m_1;\, \alpha r_0 + \beta r_1).$$

Proof. Expand and regroup, using commutativity of $+$ in $\mathbb{G}$, the identity $[a]P + [b]P = [a + b]P$, and the linearity $[\alpha]([m]G + [r]H) = [\alpha m]G + [\alpha r]H$ of scalar multiplication:

$$\begin{aligned}[\alpha]([m_0]G + [r_0]H) + [\beta]([m_1]G + [r_1]H) &= [\alpha m_0]G + [\beta m_1]G + [\alpha r_0]H + [\beta r_1]H \\ &= [\alpha m_0 + \beta m_1]G + [\alpha r_0 + \beta r_1]H.\end{aligned}$$ □

Remark 4.21 (Value balancing: the homomorphism deployed). The homomorphism is what makes Pedersen commitments the right tool for *hidden arithmetic*, and Zcash's value balancing is the canonical deployment. Each transaction value v_i is committed as $C_i = [v_i]G + [r_i]H$; a transaction balances iff the sum of its input values equals the sum of its output values. By Theorem 4.20, the verifier can check this *without learning any* v_i: the value components cancel in $\sum C_{\text{in}} - \sum C_{\text{out}}$ precisely when the value sums agree *modulo the group order* q, in which case the difference equals $[r]H$ for the net randomness r—a commitment to 0. Cancellation alone certifies balance only modulo q—value sums differing by a multiple of q cancel too—and the deployed protocol closes the gap with range checks carried inside the proofs: the values of the notes (the shielded value records) spent and created by each Action (Orchard's combined spend-and-output unit) are proved to lie in $\{0, \dots, 2^{64} - 1\}$ (the Action statement, protocol specification § 4.18.4; enforced by the note-commitment gadget's `ValueCanonicity` gate), the committed net value is the difference of two such values, and the number of summands is bounded, so the sums cannot wrap (§ 4.14). The prover does not reveal r; it proves *knowledge* of r by using $[r]H$ as the verification key of a signature over the transaction—the *binding signature*, treated with the completed balance argument in §8.9. In deployed Orchard the instantiation is $\mathsf{ValueCommit}(v; \mathsf{rcv}) = [v]V + [\mathsf{rcv}]R$ with hashed-to-curve generators V, R (protocol specification § 5.4.8.3); each Action publishes a commitment to its *net* value change, and the verifier folds the published commitments with a commitment to the transparent value balance to obtain the binding verification key. The homomorphism turns an arithmetic relation on hidden values into a group equation on public commitments—a template the rest of the volume reuses repeatedly.

4.7 Pedersen vector commitments

A single group element can seal far more than a single scalar.

Construction 4.22 (Pedersen vector commitment)**.** On input 1^λ and a length $n = \text{poly}(\lambda)$, Setup outputs $(\mathbb{G}, q)$ together with generators $G_1, \dots, G_n, H \xleftarrow{\$} \mathbb{G} \setminus \{O\}$, sampled so that no nontrivial discrete-log relation among them is known to any party—in practice each generator is derived from a public seed by hashing into the curve, rejecting the identity if it occurs, a nothing-up-my-sleeve procedure (Construction 3.24) that makes the setup transparent. The message space is $\mathbb{F}_q^n$, the randomness space $\mathbb{F}_q$, and for $\mathbf{m} = (m_1, \dots, m_n)$,

$$\text{Commit}(pp, \mathbf{m}; r) := \sum_{i=1}^{n} [m_i]G_i + [r]H \;=\; \langle \mathbf{m}, \mathbf{G} \rangle + [r]H,$$

where $\langle \mathbf{m}, \mathbf{G} \rangle := \sum_i [m_i]G_i$ is the multiscalar (mixed) inner product of the message vector with the generator vector $\mathbf{G} = (G_1, \dots, G_n)$. Opening is canonical.

Theorem 4.23 (Properties of the vector commitment)**.** *The Pedersen vector commitment is perfectly hiding, and computationally binding under the DLog assumption in* $\mathbb{G}$.

Proof. Hiding. Exactly as in Theorem 4.17: for uniform r the blinding term $[r]H$ is uniform on $\mathbb{G}$, so the commitment is uniform on $\mathbb{G}$ regardless of the message vector.

Binding. A double opening with $\mathbf{m} \neq \mathbf{m}'$ gives $\langle \mathbf{m}, \mathbf{G} \rangle + [r]H = \langle \mathbf{m}', \mathbf{G} \rangle + [r']H$, that is, a *nontrivial discrete-log relation*

$$\sum_{i=1}^{n} [m_i - m_i']\, G_i + [r - r']\, H = O$$

with not all coefficients zero. The reduction $\mathcal{B}$ must extract a single logarithm from a relation among $n+1$ generators, and it does so by embedding its challenge in *every* generator. On challenge $Q = [s]G$, it samples $u_i, w_i \xleftarrow{\$} \mathbb{F}_q$ for each i and $u_H, w_H \xleftarrow{\$} \mathbb{F}_q$, sets

$$G_i := [u_i]G + [w_i]Q, \qquad H := [u_H]G + [w_H]Q,$$

and resamples any pair whose generator lands on $\mathcal{O}$. Each result is uniform on $\mathbb{G} \setminus \{\mathcal{O}\}$—the $[u_i]G$ term alone makes the pre-rejection point uniform on $\mathbb{G}$—so the simulated parameters follow the real distribution. Crucially, the parameters reveal nothing about the w-coordinates: for every candidate value of w_i there is exactly one u_i consistent with the observed G_i, so conditioned on the adversary's entire view the w_i and w_H remain independent and uniform. Write the recovered relation as $\sum_i [a_i]G_i + [a_H]H = \mathcal{O}$ with coefficient vector $(a_1, \dots, a_n, a_H) \neq \mathbf{0}$. Substituting the embeddings and collecting the G- and Q-components gives

$$c_0 + c_1 s \equiv 0 \pmod{q}, \qquad c_0 = \sum_i a_i u_i + a_H u_H, \quad c_1 = \sum_i a_i w_i + a_H w_H.$$

Since the coefficient vector is nonzero and the w-coordinates are uniform and independent of it, c_1 is uniform on $\mathbb{F}_q$, hence nonzero with probability $1 - 1/q$; in that case $s = -c_0\, c_1^{-1} \bmod q$, and $\mathcal{B}$ outputs the challenge logarithm. Altogether $\mathrm{Adv}^{\mathrm{dlog}}(\mathcal{B}) \geq (1 - 1/q)\, \mathrm{Adv}^{\mathrm{bind}}(\mathcal{A})$. □

Remark 4.24 (Compression and the cardinality trade-off). The vector commitment is *compressing*: n field elements (plus the randomness) map to a single group element, exponentially shorter than the message for large n. Compression is incompatible with statistical binding for a counting reason: there are q^n message vectors and only q commitment values, so colliding openings exist in overwhelming abundance. This is exactly why binding is only computational—an unbounded adversary finds a colliding opening by solving the underlying discrete-log relation—and it is the trade-off of Theorem 4.8 made visible at the level of cardinalities: a compressing commitment *cannot* determine its message, so it might as well hide it perfectly.

Remark 4.25 (Linearity survives). The vector commitment remains additively homomorphic in both message and randomness, componentwise: $\mathrm{Commit}(\mathbf{m}; r) + \mathrm{Commit}(\mathbf{m}'; r') = \mathrm{Commit}(\mathbf{m} + \mathbf{m}'; r + r')$, by the same regrouping as Theorem 4.20. This linearity, applied to a *committed* vector against *public* coefficient vectors, is the foundation of the inner-product arguments of §4.9, which prove statements about $\langle \mathbf{m}, \mathbf{b} \rangle$ for public $\mathbf{b}$ while revealing only a single committed group element.

4.8 Sinsemilla: an algebraic hash-based commitment

Between the opaque hash commitment of §4.5 and the fully linear Pedersen commitment sits an intermediate species: *algebraic hash functions*, collision-resistant hashes whose internal operations are elliptic-curve group operations. The point of the species is proof cost. A zero-knowledge circuit that must verify a hash evaluation pays for every internal operation; when those operations are curve additions in the very field the proof system works over, the verification is cheap, while collision resistance still reduces to the discrete logarithm. Sinsemilla, deployed in Zcash Orchard, is the archetype: a Pedersen-style hash over a sequence of message chunks. We build up to it through its ancestor.

Construction 4.26 (Pedersen hash)**.** Split the message M into k chunks $m_1, \dots, m_k$, each interpreted as a scalar in a bounded range, and fix k independent generators $P_1, \dots, P_k \in \mathbb{G}$ with

unknown mutual discrete logarithms. Define

$$\mathsf{PedersenHash}(M) := \sum_{j=1}^{k} [m_j]P_j \in \mathbb{G}.$$

This is the unblinded Pedersen vector commitment map used as a hash. The output is a group element; composing with a fixed encoding of group elements as bit strings yields an ordinary hash.

Proposition 4.27 (Collision resistance of the Pedersen hash)**.** *Suppose no efficient algorithm finds a nontrivial relation $\sum_j [a_j]P_j = \mathcal{O}$ with not all $a_j = 0$ and each $|a_j|$ within the allowed chunk range—hardness that follows from DLog when the P_j are independent uniform generators. Then* PedersenHash *is collision resistant.*

Proof. A collision $M \neq M'$, with chunk decompositions (m_j) and (m'_j), gives $\sum_j [m_j - m'_j]\, P_j = \mathcal{O}$ with some coefficient $m_j - m'_j \neq 0$—a nontrivial relation with coefficients in the allowed range. The reduction from relation-finding to DLog is the every-generator embedding of Theorem 4.23: set each $P_j = [u_j]G + [w_j]Q$ for the challenge Q, and read the unknown logarithm off the recovered relation. □

The Pedersen hash needs one independent generator per chunk, which is costly for long inputs: the generator table grows linearly with the maximum message length. Sinsemilla reorganises the computation into an iterated two-operand form that reuses a small fixed generator table—shared by all chunk positions—while preserving the discrete-log collision reduction.

Construction 4.28 (Sinsemilla hash)**.** Split the message M into pieces $m_1, \ldots, m_k$, each of fixed width $w \leq 32$ bits, and identify a piece $v \in \{0,1\}^w$ with its integer value, bits read little-endian (the specification's LEBS2IP). The public parameters are a base point $Q \in \mathbb{G}$ and a table $S\colon \{0,1\}^w \to \mathbb{G}$ assigning to each possible piece value its own generator, both obtained by hashing to the curve. Write $\mathsf{GroupHash}(D, M)$ for the hash-to-curve pipeline of Construction 3.22 applied to the byte string M under the domain-separation tag built from the byte string D—for Pallas the RFC 9380 suite tag $D \,\|\, \texttt{"-pallas_XMD:BLAKE2b_SSWU_RO_"}$ of §3.6 (protocol specification § 5.4.9.8)—and write $\mathrm{LE}_{32}(j)$ for the four-byte little-endian encoding of an integer $0 \leq j < 2^{32}$. The table is

$$S(v) := \mathsf{GroupHash}(\texttt{"z.cash:SinsemillaS"}, \mathrm{LE}_{32}(v)), \qquad v \in \{0,1\}^w,$$

with 2^w entries shared by every instance, and the base point is personalised (§3.5) to the enclosing use—a byte string D naming it, such as a note-commitment or Merkle-hash domain—by taking that string as the message:

$$Q := \mathsf{GroupHash}(\texttt{"z.cash:SinsemillaQ"}, D).$$

Both derivations are nothing-up-my-sleeve (Construction 3.24), so no party knows any discrete-log relation among $\{S(v)\}_v \cup \{Q\}$ (Remark 3.25). The deployed instantiation takes $w = 10$ (the specification's k) over Pallas: the `sinsemilla` crate fixes the two personalisation strings and derives Q in `HashDomain::new`; the table ships precomputed as `SINSEMILLA_S`, and the crate's test `sinsemilla_s` re-derives every entry from `u32::to_le_bytes` of the index—exactly LE_{32}; the circuit loads that table as its lookup; protocol specification § 5.4.1.9.

The hash iterates over the pieces with a running accumulator $\mathrm{Acc} \in \mathbb{G}$:

$$\mathrm{Acc}_0 := Q, \qquad \mathrm{Acc}_j := \mathrm{Acc}_{j-1} + (\mathrm{Acc}_{j-1} + S(m_j)) = [2]\mathrm{Acc}_{j-1} + S(m_j),$$

and outputs the point $\mathsf{SinsemillaHashToPoint}(M) := \mathrm{Acc}_k$. The step costs two *incomplete* curve additions—first $\mathrm{Acc}_{j-1} + S(m_j)$, then the result plus Acc_{j-1}; no doubling is ever performed, and the addition formula is undefined when its operands are equal or are mutual negatives (*Math Guide*, §"Elliptic curves")—and this pair of additions is the operation the circuit verifies in very few constraints. Unrolling the recurrence gives the closed form

$$\mathsf{SinsemillaHashToPoint}(M) = \sum_{j=1}^{k} [2^{k-j}]S(m_j) + [2^k]Q :$$

the integer coefficients are powers of two determined by position alone, and the message enters only through *which* generator $S(m_j)$ each piece selects.

Proposition 4.29 (Collision resistance of Sinsemilla)**.** *Model the table entries $S(v)$ and the base point Q as independent uniform generators (the nothing-up-my-sleeve hash-to-curve derivation covers Q as well as the table). Then any collision in* SinsemillaHashToPoint*, on inputs for which its incomplete additions are defined, yields a nontrivial discrete-log relation among $\{S(v)\}_v \cup \{Q\}$, and any input on which an incomplete addition is* undefined *yields one directly; either way a discrete logarithm follows via the every-generator embedding of Theorem 4.23 extended to Q. The point hash is thus collision resistant under the DLog assumption in $\mathbb{G}$, the exceptional cases of incomplete addition covered by the same reduction rather than excluded.*

Proof sketch. Let $M \neq M'$ collide. Suppose first that they have the same piece count k. Subtracting the unrolled forms cancels the $[2^k]Q$ terms and leaves

$$\sum_{j=1}^{k} [2^{k-j}](S(m_j) - S(m'_j)) = \mathcal{O}.$$

Regroup over the distinct table generators: for each value $v \in \{0,1\}^w$, the net coefficient on $S(v)$ is

$$c_v = \sum_{j\,:\,m_j=v} 2^{k-j} - \sum_{j\,:\,m'_j=v} 2^{k-j},$$

and the relation reads $\sum_v [c_v]\, S(v) = \mathcal{O}$. Each c_v is a difference of two subset-sums of the distinct powers $2^{k-1}, \ldots, 2^0$, hence an integer with $|c_v| < 2^k$. For *admissible* message lengths—those with 2^k below the group order q, a finite design-time check—such a coefficient vanishes modulo q only if it vanishes as an integer; and $c_v = 0$ forces, by uniqueness of binary expansions, the two subset-sums to be equal as sets of powers, that is, the positions carrying value v in M and in M' to coincide. Coincidence for every v would force $M = M'$; since $M \neq M'$, some $c_v \neq 0$ and the relation is nontrivial.

If instead the piece counts differ, say $k > k'$, the Q-constants do not cancel: the relation acquires the coefficient $2^k - 2^{k'}$ on Q, nonzero and below the group order for admissible lengths, so the relation is again nontrivial. In either case, embedding a DLog challenge in every table generator *and* in Q recovers the challenge logarithm exactly as in Proposition 4.27.

Finally, the exceptional cases. An incomplete addition is undefined only when its operands are equal or are mutual negatives, so an input reaching one presents a coincidence $[\alpha]\mathrm{Acc}_{j-1} + S(m_j) = \mathcal{O}$ with $\alpha \in \{-1, 1, 2\}$. Substituting the unrolled form of Acc_{j-1} makes this an explicit relation among Q and the table generators, with coefficient $\alpha\, 2^{j-1} \neq 0$ on Q and all magnitudes

at most 2^k—below the group order for admissible lengths. An input that merely triggers an exceptional case therefore hands over a nontrivial relation itself, and the same embedding extracts the challenge logarithm from it; the protocol specification proves exactly this for the deployed instantiation (§ 5.4.1.9). □

The deployed hash differs from the point hash in two ways. First, the deployed SinsemillaHash applies the Pallas coordinate extractor to $\mathsf{SinsemillaHashToPoint}(M)$, returning its x-coordinate rather than the point itself. Collision resistance survives the extraction: equal extracted x-coordinates mean the two points are equal or are negatives of one another, and either signed equality yields the same kind of nontrivial generator relation as in the proof above. Second, the final piece is zero-padded to the piece width—the `Pad` iterator of the `sinsemilla` crate pads the message bits to a multiple of the piece width before chunking—so two bit strings of different lengths within the same final piece present identical piece sequences, and the deployed hash claims collision resistance only *between inputs of a fixed length for a given personalisation*. This is exactly the property the protocol specification requires and no more (protocol specification § 5.4.1.9: collision resistance is demanded "between inputs of fixed length, for a given personalization input"); each personalisation domain hashes inputs of a single length, as the Merkle-tree hash MerkleCRH of §10 illustrates. The variable-length argument in the proof above applies to the piece-aligned point presentation, where the piece sequence is the message.

The step from hash to commitment is the Pedersen blinding term, verbatim.

Construction 4.30 (SinsemillaCommit)**.** Fix one further independent generator H', hashed to the curve with its own domain string, so that no discrete-log relation between H' and $\{S(v)\}_v \cup \{Q\}$ is known. Define, for $r \xleftarrow{\$} \mathbb{F}_q$,

$$\mathsf{SinsemillaCommit}(m; r) := \mathsf{SinsemillaHashToPoint}(m) + [r]H'.$$

Hiding is inherited from Pedersen verbatim: by the argument of Theorem 4.17, $[r]H'$ is uniform on $\mathbb{G}$ for uniform r, so the commitment distribution is independent of m, and the scheme is perfectly hiding. Binding rests on discrete log: two valid openings to distinct messages yield either a SinsemillaHashToPoint collision—excluded by Proposition 4.29—or a nontrivial discrete-log relation between H' and the hash generators.

Orchard's note commitment NoteCommit instantiates exactly this map: the `orchard` crate derives it through the `sinsemilla` crate's `CommitDomain::commit` (protocol specification § 5.4.8.4). Its key commitment CommitIvk composes the same map with x-coordinate extraction—the specification's SinsemillaShortCommit—returning a base-field element rather than a point.

Remark 4.31 (Why Sinsemilla exists: cost inside a proof). Sinsemilla's advantage is not speed outside a circuit—ordinary hashes are faster there—but verification cost *inside* one. In a SNARK (§9) the prover re-expresses every computational step as polynomial constraints; a bit-oriented hash such as SHA-256, all Boolean choppings and modular additions foreign to the proof field, costs tens of thousands of constraints per evaluation, whereas Sinsemilla's per-piece cost is a handful of curve operations chosen to lie in the very field the proof system already works over. The result is a collision-resistant hash and commitment—used for note commitments and Merkle trees in Orchard—simultaneously cheap to prove and grounded in the same discrete-log hardness as the surrounding protocol. The presentation here is the conceptual core; the *Ironwood Guide* takes up the concrete Orchard instantiation—its per-use personalisation strings and the in-circuit handling of incomplete addition.

4.9 Polynomial commitment schemes

The final refinement commits not to a value or a vector but to a *function*—a polynomial—and supports proving statements about its values. This is the primitive on which the succinct argument systems of §9 stand.

Definition 4.32 (Polynomial commitment scheme)**.** Fix a field $\mathbb{F}$ and a degree bound d. A *polynomial commitment scheme* (PCS) is a tuple $(\mathsf{Setup}, \mathsf{Commit}, \mathsf{Open}, \mathsf{Eval}, \mathsf{Verify})$:

- The algorithm $\mathsf{Setup}(1^\lambda, d)$ outputs public parameters pp supporting degree at most d;
- the algorithm $\mathsf{Commit}(pp, f; r)$ outputs a commitment C to a polynomial $f \in \mathbb{F}[X]_{\leq d}$, optionally with hiding randomness r;
- the algorithm $\mathsf{Open}(pp, C, f, r)$ outputs a bit, verifying that C commits to the *whole* polynomial f;
- the algorithm $\mathsf{Eval}(pp, f, r, z)$, for an evaluation point $z \in \mathbb{F}$, outputs a claimed value $y = f(z)$ together with an *evaluation proof* π;
- the algorithm $\mathsf{Verify}(pp, C, z, y, \pi)$ outputs a bit, checking the evaluation proof against the commitment.

Beyond the hiding and binding of the underlying commitment (Definitions 4.4 and 4.6), a PCS must satisfy *evaluation binding*: no PPT prover can produce a commitment C, a point z, and two accepting proofs for distinct claimed values $y \neq y'$ at z, except with negligible probability. Stronger schemes additionally satisfy *extractability* (proof of knowledge): from any successful prover one can extract an actual polynomial f of degree at most d consistent with all the evaluations it gets accepted.

Remark 4.33 (Rigidity, and why evaluation binding is separate). Why should one short commitment support probing at arbitrary points? Because polynomials are *rigid*: a nonzero polynomial of degree at most d has at most d roots (*Math Guide*, §"The Schwartz–Zippel lemma"), so two distinct polynomials of degree at most d agree on at most d points. Answering a different value at even a single point is the behaviour of a prover holding a *different polynomial*—there is no wiggle room between functions. But a warning is in order: an accepting evaluation proof is *not* an opening of C, so evaluation binding does *not* follow from the binding of Commit. It is a separate cryptographic requirement, and each construction below proves it from its own assumption—a strong Diffie–Hellman assumption for KZG, discrete log via a folding extractor for the inner-product scheme. The central design problem of a PCS is making the evaluation proof π short and fast to verify; committing to the coefficient list with any old commitment is easy, and useless.

KZG: evaluation in the exponent

The pairing-based scheme of Kate, Zaverucha, and Goldberg achieves constant-size proofs by hiding a secret evaluation point inside the parameters.

Construction 4.34 (KZG commitment)**.** Let $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ be groups of prime order $|\mathbb{F}|$ equipped with a nondegenerate bilinear pairing $e\colon \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ (Example 2.9), with generators $G \in \mathbb{G}_1$ and $\hat{G} \in \mathbb{G}_2$. The algorithm Setup samples a secret $\tau \xleftarrow{\$} \mathbb{F}$, publishes the *structured reference string*

$$pp = ([\tau^0]G,\ [\tau^1]G,\ \ldots,\ [\tau^d]G,\ \hat{G},\ [\tau]\hat{G}),$$

and *discards* τ. The commitment to $f = \sum_{i=0}^{d} a_i X^i$ is evaluation in the exponent:

$$\mathrm{Commit}(pp, f) := \sum_{i=0}^{d} [a_i]([\tau^i]G) = [f(\tau)]G,$$

computable from pp by anyone, without knowledge of τ.

Construction 4.35 (KZG evaluation proof)**.** To prove $f(z) = y$, the prover uses the factor theorem: z is a root of $f(X) - y$, so the quotient

$$q(X) := \frac{f(X) - y}{X - z} \in \mathbb{F}[X]_{\leq d-1}$$

is a genuine polynomial *precisely because* $f(z) - y = 0$. The proof is the commitment to the quotient, $\pi := [q(\tau)]G$, computed from the reference string. The verifier checks the single pairing equation

$$e(C - [y]G,\ \hat{G}) \stackrel{?}{=} e(\pi,\ [\tau]\hat{G} - [z]\hat{G}),$$

which, by bilinearity, is exactly the in-the-exponent form of the polynomial identity $f(\tau) - y = q(\tau)\,(\tau - z)$.

Proposition 4.36 (KZG evaluation binding, sketched)**.** *Evaluation binding of KZG rests on the d-strong Diffie–Hellman assumption in the bilinear group: given the powers* $[\tau^i]G$*, it is infeasible to produce* $[(\tau - z)^{-1}]G$ *for any* z *of one's choosing.*

Proof idea. Two accepting proofs π, π' for distinct values $y \neq y'$ at the same point z satisfy the two pairing equations; subtracting them (in the exponent) gives $\pi - \pi' = [(y' - y)(\tau - z)^{-1}]G$, so the forger's output yields

$$[(\tau - z)^{-1}]G = [(y' - y)^{-1}](\pi - \pi'),$$

exactly the element the assumption declares infeasible. □

Remark 4.37 (The KZG trade). What KZG buys: commitments and evaluation proofs of *constant* size (one group element each) and *constant* verification time (two pairings), all independent of the degree d. What it costs is twofold. First, the setup is trusted in the strongest sense: the secret τ is a global trapdoor. Anyone who learns τ can forge accepting evaluation proofs for *false* statements—for any commitment C, any point z, and any desired value y, the element $\pi := [(\tau - z)^{-1}](C - [y]G)$ passes the pairing check, no knowledge of any polynomial required. This is the promised demonstration that a subverted setup silently destroys binding (Remark 4.3); evaluation binding survives only if τ is provably destroyed, in practice by a multi-party *powers-of-tau* ceremony whose trapdoor stays secret as long as at least one participant is honest. Second, the construction requires pairing-friendly curves, a more delicate and less efficient setting than the ordinary curves of §2.6. Binding is computational either way, resting on the bilinear assumption—consistent with Theorem 4.8.

The inner-product alternative

The transparent alternative commits to the coefficient vector with the Pedersen vector commitment and replaces the pairing check by a recursive argument.

Construction 4.38 (IPA-style polynomial commitment)**.** Let $\mathbf{G} = (G_0, \dots, G_d)$ and H be transparent generators with unknown mutual discrete logs, derived from a public seed (Construction 4.22). Commit to $f = \sum_{i=0}^{d} a_i X^i$ by the Pedersen vector commitment of its coefficient vector $\mathbf{a} = (a_0, \dots, a_d)$:

$$\mathrm{Commit}(pp, f; r) := \sum_{i=0}^{d} [a_i] G_i + [r]H = \langle \mathbf{a}, \mathbf{G} \rangle + [r]H.$$

No pairings, no trusted setup.

Evaluation is an inner product: writing $\mathbf{z} = (1, z, z^2, \dots, z^d)$ for the public vector of powers of the query point,

$$f(z) = \sum_{i=0}^{d} a_i z^i = \langle \mathbf{a}, \mathbf{z} \rangle,$$

so proving an evaluation reduces to proving an inner-product relation about a committed vector against a public vector. The *inner-product argument* (IPA) proves it recursively: each round folds the two halves of the coefficient vector—and of the generator vector—into vectors of half the dimension by taking a random linear combination dictated by a verifier challenge, sending two group elements per round (the cross-term commitments L_j and R_j) to account for the mixed terms; after $\log_2(d+1)$ rounds a single scalar remains, and the verifier checks one final equation tying it to the folded commitment and the folded generators.

Proposition 4.39 (Properties of the IPA-based PCS)**.** *The inner-product polynomial commitment has: transparent setup (public-coin parameters, no trapdoor); evaluation proofs of $O(\log d)$ group elements; and $O(d)$-group-operation verification, batchable across proofs. Its binding and evaluation binding reduce to the discrete logarithm assumption in $\mathbb{G}$.*

Proof idea. Binding of the commitment itself is Theorem 4.23. For evaluation binding and extractability, one shows the folding protocol is *tree special-sound*: from a suitable tree of accepting transcripts, branching over the verifier's challenge at each round, an extractor works back up the recursion and computes a coefficient vector $\mathbf{a}$ consistent with the commitment and the claimed inner product—a notion this sketch uses only informally and §9 defines precisely. Two distinct extracted openings of the same commitment would yield a nontrivial discrete-log relation among $\mathbf{G}, H$, contradicting DLog by the every-generator embedding. The $O(\log d)$ proof size is the halving recursion; the $O(d)$ verification is the final multiscalar multiplication over the unfolded generators. □

Remark 4.40 (KZG versus IPA, and the Halo lineage). The two families mark out the design space. KZG gives constant proof size and constant verification, at the price of a structured trusted setup and pairing-friendly curves; the IPA gives transparent setup over a plain curve and rests on nothing beyond discrete log, at the price of logarithmic proofs and linear-time verification. The Halo 2 proof system deployed in Zcash Orchard builds on the inner-product family precisely to avoid trusted setup, and recovers practical verification cost through *batching*: the deployed verifier folds the final multiscalar multiplications of many proofs into one shared multiexponentiation—each proof's deferred check is scaled by a fresh random factor and accumulated, and a single combined check is evaluated once. Batching amortises the per-proof group operations across a batch, though per-proof verifier work remains linear in d. The Halo lineage's accumulation technique extends the same deferral across *recursively composed* proofs, one proof attesting to another's deferred check—the capability the Pasta cycle of Remark 2.26 was designed to serve—but deployed Zcash does not use recursion: each Orchard proof is verified (in batch) directly.

Remark 4.41 (Hiding variants). As stated, the KZG commitment is binding but not hiding: it is a deterministic function of f, so committing twice to the same polynomial is visible, and a low-entropy polynomial can be found by search—the failure of Remark 4.14 in new clothing. That is acceptable when the polynomial encodes only public data. Whenever it encodes secrets—as the witness polynomials of a zero-knowledge proof do—randomness is required: KZG is augmented with a blinding summand $[r][\gamma]G$ using an extra setup element $[\gamma]G$, while the IPA commitment of Construction 4.38 already carries its $[r]H$ and is perfectly hiding by Theorem 4.23. The resulting regime is the perfect-hiding/computational-binding profile of Pedersen (Theorems 4.17 and 4.18), lifted from field elements to polynomials.

From commitments to arguments

The polynomial commitment is where the section's main line hands over to the proof systems of §9, and the hand-over can be stated in one paragraph. A polynomial commitment scheme turns an information-theoretic proof object—a polynomial identity holding over a large field—into a succinct cryptographic argument. The prover commits to its witness polynomials; the verifier challenges at random points; evaluation proofs then convince the verifier that the committed polynomials satisfy the required identities, without the polynomials ever being revealed. Soundness is supplied by Schwartz–Zippel (*Math Guide*, §"The Schwartz–Zippel lemma"): a nonzero polynomial of low degree is almost surely nonzero at a random point, so an identity that holds at the challenge almost surely holds identically. Binding and succinctness are supplied by the commitment: the prover is locked to its polynomials *before* seeing the challenge, and the verifier handles only short commitments and proofs in place of the polynomials themselves. How this compilation is carried out in full—and what "argument of knowledge" precisely means—is the business of §9, whose SNARK compilation subsection develops the evaluation-argument role sketched here.

4.10 Secret sharing

The section opened by naming verifiable secret sharing among the classical applications of commitments; it closes by constructing it, because the threshold signatures of the *FROST Guide* stand on it and nothing below that volume otherwise provides it. The setting is a prime field $\mathbb{F}_q$—in the sequel the scalar field of the group $\mathbb{G}$ of §4.6—and the tool is Lagrange interpolation (*Math Guide*, §"Lagrange interpolation"): a polynomial of degree at most $t-1$ over $\mathbb{F}_q$ is determined by its values at any t distinct points, and any t prescribed values at distinct points are attained by exactly one such polynomial.

Construction 4.42 (Shamir t-of-n secret sharing). Fix a threshold t and a party count n with $1 \le t \le n < q$. To share a secret $s \in \mathbb{F}_q$, the dealer samples coefficients $a_1, \dots, a_{t-1} \xleftarrow{\$} \mathbb{F}_q$ independently and uniformly and forms

$$f(X) := s + a_1 X + a_2 X^2 + \cdots + a_{t-1} X^{t-1} \in \mathbb{F}_q[X], \qquad f(0) = s,$$

a polynomial of degree at most $t-1$ with constant term the secret. Party $i \in \{1, \dots, n\}$ receives the *share* $s_i := f(i)$, sent privately; the identifiers $1, \dots, n$ are distinct nonzero elements of $\mathbb{F}_q$ because $n < q$. *Reconstruction* from the shares of any set I of t parties is interpolation at 0:

$$s = f(0) = \sum_{i \in I} \lambda_{I,i} s_i, \qquad \lambda_{I,i} := \prod_{j \in I,\, j \neq i} \frac{j}{j-i},$$

where $\lambda_{I,i} = \ell_i(0)$ is the Lagrange basis polynomial for the nodes I evaluated at 0, computable from the identifiers alone.

Correctness of reconstruction is the interpolation theorem: the t points $(i, s_i)_{i \in I}$ determine a unique polynomial of degree at most $t-1$, which is f, and the Lagrange formula $f(X) = \sum_{i \in I} s_i \, \ell_i(X)$ evaluated at $X = 0$ gives the displayed sum. The coalition may be chosen after the shares are dealt, since the weights depend only on I.

Theorem 4.43 (Perfect secrecy below the threshold)**.** *Let I be any set of at most $t-1$ identifiers. For every secret $s \in \mathbb{F}_q$, the shares $(s_i)_{i \in I}$ of Construction 4.42 are jointly uniform on $\mathbb{F}_q^{|I|}$. In particular their distribution does not depend on s: fewer than t shares are independent of the secret, and an adversary of unbounded power holding them has advantage exactly 0 at telling any two candidate secrets apart.*

Proof. It suffices to treat $|I| = t-1$, since fewer shares are a projection of these. Fix s and I, and consider the map

$$\Sigma_s \colon \mathbb{F}_q^{t-1} \to \mathbb{F}_q^{t-1}, \qquad (a_1, \dots, a_{t-1}) \mapsto (f(i))_{i \in I},$$

from the dealer's coefficient vector to the coalition's shares. It is injective: two coefficient vectors with the same shares give two polynomials of degree at most $t-1$ that agree at the t distinct points 0 (where both equal s) and I, hence are equal by uniqueness of interpolation, so the coefficient vectors coincide. It is surjective: for any target $(y_i)_{i \in I}$, Lagrange interpolation through the t points $(0, s)$ and $(i, y_i)_{i \in I}$ produces a polynomial of degree at most $t-1$ with constant term s, whose remaining coefficients are a preimage. So Σ_s is a bijection—for *every* s—and it carries the uniform distribution on coefficient vectors to the uniform distribution on $\mathbb{F}_q^{t-1}$. The share vector is therefore uniform whatever the secret, and for any $s_0 \neq s_1$ the two share distributions coincide exactly; by the variational characterisation of statistical distance (Proposition 1.7) no procedure, efficient or not, distinguishes them. □

The theorem is the perfect-hiding half of a commitment (Proposition 4.5 in different clothing): the coalition's view is a uniformly random point of $\mathbb{F}_q^{t-1}$, consistent with every candidate secret through exactly one polynomial each. What Shamir sharing lacks is the binding half. Nothing stops a dishonest dealer from handing out values that lie on no common polynomial of degree at most $t-1$, so that different coalitions of t parties reconstruct different "secrets"—and a party receiving a share cannot tell. Verifiable secret sharing repairs this by having the dealer commit to the polynomial.

Construction 4.44 (Feldman verifiable secret sharing)**.** Let $(\mathbb{G}, q, G)$ be as in §4.6, so that $\mathbb{F}_q$ is the scalar field of $\mathbb{G}$. The dealer shares s as in Construction 4.42, with $f(X) = \sum_{k=0}^{t-1} a_k X^k$ and $a_0 = s$, and additionally *broadcasts* the coefficient commitments

$$\Phi_k := [a_k]\, G \in \mathbb{G}, \qquad k = 0, \dots, t-1,$$

the coefficients in the exponent. Party i, on receiving its private share s_i, accepts it iff

$$[s_i]\, G \stackrel{?}{=} \sum_{k=0}^{t-1} [i^k] \Phi_k.$$

An honest share passes: since $a \mapsto [a]G$ is a homomorphism $\mathbb{F}_q \to \mathbb{G}$, $\sum_k [i^k]\Phi_k = \sum_k [a_k i^k] G = [f(i)]G$.

Proposition 4.45 (What the Feldman check certifies). *Because G generates the prime-order group $\mathbb{G}$, the map $a \mapsto [a]G$ is a bijection $\mathbb{F}_q \to \mathbb{G}$, so the broadcast vector $(\Phi_0, \dots, \Phi_{t-1})$—however the dealer chose it—determines a unique polynomial $f^*(X) := \sum_k a_k^* X^k$ with $a_k^* = \log_G \Phi_k$, of degree at most $t-1$. Party i's check passes iff $s_i = f^*(i)$. Consequently every coalition of t parties holding accepted shares reconstructs the same polynomial f^* and the same secret $s^* = f^*(0) = \log_G \Phi_0$, and a party handed an inconsistent share detects it alone and with certainty. The check certifies nothing about the value of s^*, which the dealer chooses freely.*

Proof. Both claims are injectivity of $a \mapsto [a]G$. The right-hand side of the check equals $[f^*(i)]G$ by the homomorphism, so the equation $[s_i]G = [f^*(i)]G$ holds iff $s_i = f^*(i)$. Accepted shares are thus evaluations of the one polynomial f^*, and t of them interpolate to it by uniqueness. □

In the language of this section, the vector $(\Phi_k)_k$ is a *perfectly binding* commitment to the polynomial f—the coefficient list committed entry by entry under the bijection $a \mapsto [a]G$—and the share check is an evaluation check against it, verifiable by anyone because it is linear in the coefficients. Perfect binding costs hiding, as Theorem 4.8 demands.

Remark 4.46 (The leak, and where it is harmless). Feldman sharing publishes $\Phi_0 = [s]G$. Secrecy below the threshold is no longer perfect: from the broadcast alone an unbounded adversary computes $s = \log_G \Phi_0$. For an efficient coalition of $t-1$ parties the leak is exactly $[s]G$ and nothing more: its entire view—shares and broadcast—is computable from $[s]G$ and $t-1$ uniformly random field elements, because the shares are uniform by Theorem 4.43 and the Φ_k are then determined by interpolating in the exponent through $(0, [s]G)$ and the points $(i, [s_i]G)$. Recovering s from that view is the discrete-logarithm problem (Definition 2.2). The leak is acceptable exactly when s is itself a discrete-logarithm secret key whose public key $[s]G$ is published anyway; then Φ_0 *is* the public key and nothing new escapes. When the secret must stay perfectly hidden, Pedersen verifiable secret sharing replaces each Φ_k by the Pedersen commitment $[a_k]G + [b_k]H$ of Construction 4.16, restoring perfect hiding at the price of computational binding—off this volume's path and not constructed here. Threshold signing is the consumer: in the *FROST Guide* each signer holds a Shamir share of a signing key, the coefficient commitments broadcast during key generation are Feldman's, and the Lagrange weights $\lambda_{I,i}$ recombine per-signer contributions into a single signature.

5 Pseudorandom functions, block ciphers, and format-preserving encryption

The adversary of this section sits before a black box. She may submit any input she likes and read off the answer; she may let each query depend on everything the box has said so far; and when she has asked all she cares to ask, she must announce a single bit: does a short secret key drive the box, or blind chance? If no efficient strategy tells the difference, then λ bits of key have bought a perfect counterfeit of an object too large to exist in the world—a uniformly random function—and everything downstream that needs random-looking values on demand can draw them from that key. If she *can* tell, forgery follows at once. Key a Merkle–Damgård hash by naive concatenation and she extends an authentication tag she was shown onto a message nobody authenticated, without ever learning the key. Derive the diversifiers behind Orchard's shielded addresses by any map she can predict or invert, and she links every address of one wallet to its siblings. One adversary—the *oracle distinguisher*—thus carries the whole section, and every definition in it is the same distinguishing game at a different strength: pseudorandom functions against random functions, pseudorandom permutations against random permutations, strong pseudorandom permutations against a two-sided oracle. The switching lemma prices the substitution of one ideal object for the other; AES supplies the deployed permutation; keyed

BLAKE2 supplies the PRFs this section inventories; and format-preserving encryption closes the section by manufacturing, from a PRF that is not invertible, the keyed bijection on an 11-byte format that Orchard's address derivation demands.

Notation is that of the provable-security model. Strings of length ℓ form $\{0,1\}^\ell$, all finite strings form $\{0,1\}^*$, and $|x|$ is the length of x; for a finite set S, the notation $x \xleftarrow{\$} S$ draws x uniformly from S, independently of everything else. Adversaries are PPT machines in the security parameter λ (§1.1), a negligible function is $\mathrm{negl}(\lambda)$ (recalled in §1.2), and every security notion below is a distinguishing game in the sense of Definition 1.15, scored by the difference

$$\mathrm{Adv}(\mathcal{A}) \;=\; |\Pr[\mathcal{A} \text{ outputs 1 in } \mathsf{Exp}_1] - \Pr[\mathcal{A} \text{ outputs 1 in } \mathsf{Exp}_0]|$$

between the two experiments the game distinguishes. One device remains to be formalised: the black box itself.

Definition 5.1 (Oracle access)**.** For a function f, the notation $\mathcal{A}^f$ denotes an adversary granted *oracle access* to f: a black box to which $\mathcal{A}$ may submit inputs x of its choice, receiving $f(x)$ in a single computation step, and from which it learns nothing about f beyond the answers to the queries it makes. The adversary's running time always bounds the number of its oracle queries, which is hence polynomial in λ.

5.1 Pseudorandom functions and permutations

Pseudorandomness is measured against an ideal object, which must come first: the "truly random function" has to be made precise before anything can be said to imitate it.

Definition 5.2 (Random function)**.** Let $\mathsf{Func}(D,R)$ denote the set of all functions from a finite set D to a finite set R. A *random function* from D to R is an element $F \xleftarrow{\$} \mathsf{Func}(D,R)$, chosen uniformly at random. Equivalently, the values $\{F(x)\}_{x\in D}$ are independent and uniformly distributed over R.

There are $|R|^{|D|}$ functions in $\mathsf{Func}(D,R)$, an astronomically large set for the domains of interest: for $D = R = \{0,1\}^{128}$ the count is $2^{128\cdot 2^{128}}$. No such function can be stored or transmitted—its truth table is the only description it has, and the table does not fit in the universe. The essential property of a pseudorandom function is that it replaces this object by a short key while preserving its observable behaviour. The clean mental model of the ideal object is *lazy sampling*, exactly as for the random oracle of Definition 3.13: the oracle keeps a table, initially empty; on a fresh query x it samples $F(x) \xleftarrow{\$} R$, records the pair, and returns it; on a repeated query it returns the recorded value. This produces exactly the distribution of Definition 5.2 while only ever materialising the points actually queried.

Definition 5.3 (Keyed function)**.** A *keyed function* is a deterministic, efficiently computable map

$$F\colon \{0,1\}^\lambda \times \{0,1\}^{\ell(\lambda)} \longrightarrow \{0,1\}^{m(\lambda)}, \qquad (k,x) \longmapsto F_k(x),$$

where the input length ℓ and output length m are polynomially bounded functions of the key length λ. Fixing the key k yields the function $F_k\colon \{0,1\}^{\ell(\lambda)} \to \{0,1\}^{m(\lambda)}$.

Definition 5.4 (Pseudorandom function)**.** Let F be a keyed function. The distinguishing game $\mathsf{PRF}_F^{\mathcal{A}}(\lambda)$ runs as follows.

1. The challenger samples a bit $b \xleftarrow{\$} \{0,1\}$. If $b = 1$ it samples $k \xleftarrow{\$} \{0,1\}^\lambda$ and sets $\mathcal{O} := F_k$; if $b = 0$ it sets $\mathcal{O} := R$ for $R \xleftarrow{\$} \mathsf{Func}(\ell(\lambda), m(\lambda))$, realised by lazy sampling, where $\mathsf{Func}(\ell, m) := \mathsf{Func}(\{0,1\}^\ell, \{0,1\}^m)$.
2. The adversary $\mathcal{A}(1^\lambda)$, receiving the security parameter in unary to convey the parameter size, queries $\mathcal{O}$ adaptively.
3. The adversary outputs a bit b'; the game outputs 1 iff $b' = b$.

The family F is a *pseudorandom function* (PRF) if for every PPT adversary $\mathcal{A}$,

$$\mathrm{Adv}_F^{\mathrm{prf}}(\mathcal{A}, \lambda) := \Big| \Pr_{k \xleftarrow{\$} \{0,1\}^\lambda} [\mathcal{A}^{F_k}(1^\lambda) = 1] - \Pr_{R \xleftarrow{\$} \mathsf{Func}(\ell(\lambda), m(\lambda))} [\mathcal{A}^{R}(1^\lambda) = 1] \Big| \in \mathrm{negl}(\lambda).$$

The two probabilities are the adversary's output distributions in the worlds $b = 1$ and $b = 0$, so this difference equals twice the bit-guessing advantage $|\Pr[\mathsf{PRF}_F^{\mathcal{A}}(\lambda) = 1] - \frac{1}{2}|$ of Definition 1.15; the factor of two is immaterial to negligibility, and the difference form is the one every bound below is stated in.

Stated informally: an adversary handed a black box that is either F_k for a secret random key or a freshly sampled truly random function, and allowed to query it adaptively, cannot tell which it holds. Two features of the definition carry the weight. First, the algorithm defining F is public; only the sampled key is hidden. The adversary obtains oracle access to the resulting keyed instance F_k through Definition 5.1—never the key, only its input–output behaviour. Second, the comparison object is the gigantic, unstorable random function; the PRF compresses it into λ bits while remaining indistinguishable to every efficient observer.

That last qualifier is not removable: pseudorandomness is an inherently computational notion. Information theory cannot afford the trade. As k ranges over its 2^λ values, the keyed function F_k takes at most 2^λ distinct values in $\mathsf{Func}(D, R)$—a vanishing fraction of all $|R|^{|D|}$ functions—so the two distributions of Definition 5.4 are statistically almost as far apart as distributions can be, and an *unbounded* adversary could in principle distinguish them. Security holds only against computationally bounded distinguishers, in precisely the manner of the PRG separation in the proof of Proposition 1.10.

Remark 5.5 (Necessity of adaptivity). The adversary chooses its queries; it need not commit to them in advance, and later queries may depend on earlier answers. This *adaptive* access is the strongest reasonable model and the one downstream constructions rely on. A weaker non-adaptive definition, in which the adversary fixes all queries before seeing any answer, is genuinely weaker—families exist that pass every non-adaptive test yet fall to an adaptive distinguisher—and is insufficient for many applications.

A block cipher is not merely a keyed function but a keyed *permutation*: each key must yield an invertible map, or decryption would be impossible. The relevant ideal object is therefore a random permutation.

Definition 5.6 (Keyed permutation)**.** A keyed function $F\colon \{0,1\}^\lambda \times \{0,1\}^\ell \to \{0,1\}^\ell$ with equal input and output length is a *keyed permutation* if for every key k the map F_k is a bijection of $\{0,1\}^\ell$, and both F_k and F_k^{-1} are efficiently computable given k. The set of all permutations of $\{0,1\}^\ell$ is $\mathsf{Perm}(\ell)$, of cardinality $(2^\ell)!$.

Definition 5.7 (Pseudorandom permutation)**.** Let F be a keyed permutation. The game $\mathsf{PRP}_F^{\mathcal{A}}(\lambda)$ is the game of Definition 5.4 with the ideal world replaced: for $b = 0$ the challenger sets $\mathcal{O} := P$ for $P \xleftarrow{\$} \mathsf{Perm}(\ell)$, a uniformly random permutation. The family F is a *pseudorandom permutation* (PRP) if for every PPT adversary $\mathcal{A}$,

$$\mathrm{Adv}_F^{\mathrm{prp}}(\mathcal{A},\lambda) := \Big|\Pr_k[\mathcal{A}^{F_k}(1^\lambda) = 1] - \Pr_{P \xleftarrow{\$} \mathsf{Perm}(\ell)}[\mathcal{A}^P(1^\lambda) = 1]\Big| \in \mathrm{negl}(\lambda).$$

Definition 5.8 (Strong pseudorandom permutation)**.** A keyed permutation F is a *strong pseudorandom permutation* (SPRP) if no PPT adversary distinguishes even when granted oracle access to *both* the permutation and its inverse: for every PPT $\mathcal{A}$,

$$\mathrm{Adv}_F^{\mathrm{sprp}}(\mathcal{A},\lambda) := \Big|\Pr_k[\mathcal{A}^{F_k,F_k^{-1}}(1^\lambda) = 1] - \Pr_{P \xleftarrow{\$} \mathsf{Perm}(\ell)}[\mathcal{A}^{P,P^{-1}}(1^\lambda) = 1]\Big| \in \mathrm{negl}(\lambda).$$

The distinction between a PRP and an SPRP is exactly the availability of a decryption oracle. A block cipher deployed where the attacker can obtain decryptions of chosen ciphertexts must be an SPRP; this is the natural security target for a block cipher, and the one the next subsection adopts.

From the outside, the two ideal objects are nearly the same thing. Query a length-preserving oracle on q distinct inputs. If the oracle is a random function $R \in \mathsf{Func}(\ell,\ell)$, the answers are q independent uniform strings, and output collisions occur. If the oracle is a random permutation $P \in \mathsf{Perm}(\ell)$, the answers are q *distinct* uniform strings, and no collision can occur. This is the *only* statistical difference visible to any observer, and it is the entire content of the switching lemma.

Lemma 5.9 (PRP/PRF switching lemma)**.** *Let $\ell \geq 1$ and let $\mathcal{A}$ be any adversary—even a computationally unbounded one—making at most q queries to its oracle. Then*

$$\Big|\Pr_{P \xleftarrow{\$} \mathsf{Perm}(\ell)}[\mathcal{A}^P = 1] - \Pr_{R \xleftarrow{\$} \mathsf{Func}(\ell,\ell)}[\mathcal{A}^R = 1]\Big| \leq \frac{q(q-1)}{2^{\ell+1}}.$$

We use the lemma as a statement only. Its proof is a standard game-hopping argument (Remark 1.18) bounding the probability of the single distinguishing event just identified—an output collision among the q answers—and the bound is the birthday-type count $\binom{q}{2}/2^\ell$ familiar from Lemma 3.11, whose underlying calculation is the *Math Guide*'s (§"The union bound and a birthday calculation").

The consequence is a licence: a secure PRP on a large domain is also a secure PRF whenever the number of queries stays well below the birthday bound $2^{\ell/2}$, because the adversary's total advantage against the function world exceeds its advantage against the permutation world by at most $q(q-1)/2^{\ell+1}$. For AES, with $\ell = 128$, the switching term $q^2/2^{129}$ is negligible for any realistic query budget. This is the licence by which the FF1 construction of §5.4 treats AES—a permutation—as the PRF its round function requires.

5.2 Block ciphers and AES

Definition 5.10 (Block cipher)**.** A *block cipher* with block length b and key length κ is a keyed permutation

$$E\colon \{0,1\}^\kappa \times \{0,1\}^b \longrightarrow \{0,1\}^b,$$

together with its inverse $D = E^{-1}$ satisfying $D_k(E_k(x)) = x$ for all k, x. Here E_k is *encryption* and D_k *decryption* under key k. The security goal is that E be a strong pseudorandom permutation (Definition 5.8).

A block cipher is a concrete, fast, fixed-parameter instantiation of the SPRP abstraction, and the fixing of parameters changes the language of its security claims. The asymptotic PRP definition speaks of a family indexed by λ; a real block cipher fixes b and κ once and for all—$b = 128$ and $\kappa \in \{128, 192, 256\}$ for AES—and offers *concrete* security in the resource-explicit style of Definition 1.31: an explicit function of the adversary's resources bounds the advantage of the best known attacks.

Definition 5.11 (Concrete (t, q, ε)-security)**.** A block cipher E is a (t, q, ε)*-secure* PRP if every adversary running in time at most t and making at most q oracle queries has $\mathrm{Adv}_E^{\mathrm{prp}}(\mathcal{A}) \leq \varepsilon$; analogously for SPRP and PRF security.

Construction 5.12 (AES)**.** The Rijndael cipher of Daemen and Rijmen, standardised as AES in FIPS 197, has block length $b = 128$ bits and key lengths $\kappa \in \{128, 192, 256\}$, with $N_r \in \{10, 12, 14\}$ rounds respectively. Structurally it is a *substitution–permutation network* (SPN): the 128-bit state, viewed as a 4×4 array of bytes—elements of $\mathbb{F}_{2^8}$—is first whitened by an initial round-key XOR, then passes through N_r rounds built from four layers:

1. **SubBytes**, a nonlinear byte substitution—the *S-box*—built on field inversion $a \mapsto a^{-1}$ in $\mathbb{F}_{2^8}$ (composed with a fixed affine map), chosen so that every input difference maps to any given output difference with only small probability, and every linear approximation of it holds with probability close to one half;
2. **ShiftRows**, a transposition of bytes within the array;
3. **MixColumns**, an invertible column-wise linear mix; and
4. **AddRoundKey**, the XOR of a 128-bit round key that a key schedule expands from k.

The first $N_r - 1$ rounds compose all four layers in this order; the final round omits MixColumns. The two permutation layers together guarantee that after two rounds every output byte depends on every input byte. Each layer—the initial whitening included—is individually invertible, so E_k is a permutation as Definition 5.10 requires, and decryption applies the inverses in reverse order.

Remark 5.13 (Confusion, diffusion, and the status of AES security)**.** The SPN design is the modern embodiment of Shannon's two principles. The nonlinear S-box supplies *confusion*: making the relationship between key and ciphertext as complex as possible. The ShiftRows and MixColumns layers supply *diffusion*: spreading the influence of each input bit over the whole output. Iterating the two for ten or more rounds is what gives AES its empirical resistance to all known attacks. No proof that AES is a PRP exists; like every practical block cipher, its security is a *conjecture*, corroborated by decades of failed cryptanalysis but resting on no reduction to a cleaner problem. Every theorem downstream therefore takes "E is a PRP/SPRP" as an assumption and reasons from it—the same epistemic footing on which Remark 1.29 placed the concrete hash functions.

5.3 PRFs from hash functions: length extension and keyed BLAKE2

Block ciphers are one source of PRFs; cryptographic hash functions are another, often more convenient one, being unkeyed and ubiquitous. Here the oracle distinguisher does not merely

distinguish—she forges. The naive keying of a hash H by prepending the key,

$$F_k(m) \;=\; H(k \,\|\, m),$$

is unsafe for the iterated Merkle–Damgård hashes (MD5, SHA-1, SHA-256)—which pad the message into blocks, fold the blocks one at a time into a fixed-width chaining state through a compression function, and emit the final state—because of the *length-extension attack*: from $H(k \,\|\, m)$ and $|m|$ alone, the adversary computes $H(k \,\|\, m \,\|\, \mathrm{pad} \,\|\, m')$ for a suffix m' of her choosing, *without knowing* k. The reason is structural: a Merkle–Damgård digest is exactly the internal chaining state after absorbing $k \,\|\, m$, so the published output is a resumable snapshot of the iteration, and the attacker simply continues absorbing blocks from where the honest computation stopped. Used as a message authentication code (MAC)—a keyed function whose output tag certifies a message, defined under "Message authentication codes" in the next section—the naive construction thus surrenders a valid tag on $m \,\|\, \mathrm{pad} \,\|\, m'$, a message the key holder never authenticated.

The classical repair is HMAC, the nested double call

$$\mathrm{HMAC}_k(m) \;=\; H((k \oplus \mathrm{opad}) \,\|\, H((k \oplus \mathrm{ipad}) \,\|\, m))$$

for two fixed pad constants: the inner hash absorbs the message under one key-derived value, and the outer hash re-keys the fixed-length result, so the emitted digest is no longer a resumable chaining state. Bellare's analysis shows that HMAC is a PRF whenever the underlying compression function is, up to a birthday term $q^2/2^\ell$ in the chaining-state length ℓ; as a PRF it is in particular a secure message authentication code, and it is the building block of the standardised key-derivation function HKDF, noted in passing under "Key-derivation functions" in the next section. This summary is all we need, because the shielded protocol uses HMAC nowhere: its hash of choice admits direct keying.

Modern hash functions of the sponge family (Construction 3.26) and of the HAIFA family—the Merkle–Damgård variant whose compression function also takes the count of bits hashed so far, the lineage of BLAKE2—do not suffer length extension and so admit a far simpler keying. The BLAKE2 design is built to be keyed *natively*: the construction loads the key length into its parameter block and prepends the key, padded to a full block, as the first message block, and its compression function mixes in a block counter and finalisation flags. (Its successor BLAKE3 is natively keyed too, by a different mechanism: the key words replace the initial state, and a dedicated flag marks the keyed mode.) One simply writes

$$F_k(m) \;=\; \mathrm{BLAKE2}(m;\ \mathrm{key} = k)$$

and obtains a PRF directly—no nested double call. No length-extension attack remains to repair: the finalisation flag, XORed into the state for the last block, means the emitted digest is *not* the raw chaining value and cannot be used to resume the computation.

Remark 5.14 (Zcash's use of keyed and personalised BLAKE2). The Zcash protocol uses BLAKE2b and BLAKE2s pervasively, exploiting two parameters of the BLAKE2 design: the *key*, for PRF and MAC use as above, and the *personalisation* string, a domain-separation tag baked into the initial state—16 bytes for BLAKE2b, 8 bytes for BLAKE2s. Personalisation makes the same input hashed for two purposes—nullifier derivation versus note-commitment derivation, say—yield independent-looking outputs, so each use site is effectively an independent PRF: the practical realisation of the independent-oracle-per-context idiom that Proposition 3.17 proved for tagged random oracles, obtained here before a single message byte is processed and cheaper and cleaner than HMAC, because BLAKE2 provides keying and domain separation as first-class features. The deployed inventory spans both widths. On the BLAKE2b side, the key-expansion PRF $\mathsf{PRF}^{\mathsf{expand}}$ is BLAKE2b-512 personalised with `Zcash_ExpandSeed` (protocol specification §"Pseudo Random Functions"), evaluated on the spending key to derive Orchard's key tree; the note-encryption KDF is personalised with `Zcash_OrchardKDF` (§"Orchard Key Derivation")

and the outgoing cipher key with `Zcash_Orchardock` (§"Pseudo Random Functions")—all 16-byte tags. On the BLAKE2s side, the nullifier PRF and fixed-base generators of Sapling—the shielded protocol preceding Orchard—use 8-byte personalisations such as `Zcash_nf` and `Zcash_PH` (§"Pseudo Random Functions" and §"Pedersen Hash Function").

5.4 Format-preserving encryption and FF1

All constructions so far operate on bit strings of a length the cipher's block size dictates. The deployed need that closes this section does not. Orchard derives the *diversifiers* behind its shielded addresses—one 11-byte value per address, many addresses per spending key—by encrypting a sequential address index with a keyed bijection on the set of 11-byte strings, so that the published diversifiers of one wallet look like unrelated random values rather than the consecutive integers they conceal. An adversary who could invert the map, or merely distinguish it from a random permutation of the format, would link a wallet's addresses to one another. What is needed is precisely a block cipher whose "block" is an exotic finite set; the classical commercial need—encrypting a 16-digit card number into a 16-digit card number so that legacy databases, field-length constraints, and checksum formats continue to accept it—is the same problem. This is the problem *format-preserving encryption* (FPE) solves.

Definition 5.15 (Format-preserving encryption)**.** Let $\mathcal{X}$ be a finite set, the *format*—for example $\mathcal{X} = \{0, 1, \dots, 9\}^{16}$, the 16-digit strings. A *format-preserving encryption scheme* on $\mathcal{X}$ is a keyed permutation $\mathcal{E}\colon \{0,1\}^\kappa \times \mathcal{X} \to \mathcal{X}$ such that for every key k the map $\mathcal{E}_k\colon \mathcal{X} \to \mathcal{X}$ is a bijection with efficiently computable inverse $\mathcal{D}_k$. Security requires $\mathcal{E}$ to be a (strong) pseudorandom permutation on $\mathcal{X}$—Definition 5.7 with $\mathsf{Perm}(\mathcal{X})$ in place of $\mathsf{Perm}(\{0,1\}^\ell)$: no efficient adversary distinguishes $\mathcal{E}_k$ from a uniformly random permutation of $\mathcal{X}$, even with access to the inverse.

The defining feature is that the domain $\mathcal{X}$ is *arbitrary*—in particular $|\mathcal{X}|$ need not be a power of two, so a plain block cipher does not apply. What is required is a keyed pseudorandom *bijection* on a set whose size is, say, 10^{16}. Determinism is forced, and it separates FPE from the randomised encryption modes of the next section: the format $\mathcal{X}$ has no room to store a per-message random value, so FPE is the deterministic, format-preserving analogue of a block cipher on the alphabet of $\mathcal{X}$. It necessarily leaks equality of plaintexts, as any deterministic cipher must; this is the unavoidable cost of preserving format with no expansion, and it is acceptable for the tokenisation use cases FPE targets—and immaterial for diversifier derivation, where each index is encrypted once.

Formats reduce to integers. Any format of the shape "strings of length ν over an alphabet of radix ρ" is in bijection with $\{0, 1, \dots, \rho^\nu - 1\}$ by reading the string as a base-ρ numeral, so it suffices to build a keyed pseudorandom permutation of $\mathbb{Z}/M\mathbb{Z}$ for an arbitrary modulus $M = \rho^\nu$. The FF1 construction below splits $M = a \cdot b$ into two factors of nearly equal size and runs a *Feistel network* whose two halves live in $\mathbb{Z}/a\mathbb{Z}$ and $\mathbb{Z}/b\mathbb{Z}$. The Feistel network is the structural device that turns a round function—not necessarily invertible—into a permutation; we present the classical bitwise case first, then the modular generalisation FF1 uses.

Definition 5.16 (Balanced Feistel network)**.** Let $F_1, \dots, F_r\colon \{0,1\}^w \to \{0,1\}^w$ be arbitrary *round functions*. The r-round Feistel network $\Psi[F_1, \dots, F_r]$ is the map on $\{0,1\}^{2w}$ defined as follows: write the input as (L_0, R_0) with $L_0, R_0 \in \{0,1\}^w$, iterate

$$L_i = R_{i-1}, \qquad R_i = L_{i-1} \oplus F_i(R_{i-1}), \qquad i = 1, \dots, r,$$

and output (L_r, R_r).

Proposition 5.17 (Feistel is a bijection regardless of the round functions)**.** *For any round functions $F_1, \dots, F_r$—injective or not—the map $\Psi[F_1, \dots, F_r]$ is a bijection of $\{0,1\}^{2w}$, inverted by running the rounds backwards:*

$$R_{i-1} = L_i, \qquad L_{i-1} = R_i \oplus F_i(L_i), \qquad i = r, \dots, 1.$$

Proof. It suffices to show each round is invertible, the composition of bijections being a bijection. The i-th round sends $(L_{i-1}, R_{i-1}) \mapsto (R_{i-1}, L_{i-1} \oplus F_i(R_{i-1}))$. Given the output (L_i, R_i), one recovers $R_{i-1} = L_i$ directly, and then $L_{i-1} = R_i \oplus F_i(R_{i-1}) = R_i \oplus F_i(L_i)$, using that XOR is its own inverse. The recovery uses only *forward* evaluations of F_i—at no point is an inverse of the round function required, which is the crucial point permitting a permutation to be built out of a one-way-ish PRF. The displayed backward recursion is exactly this inversion applied round by round. □

This proposition is the structural heart of format-preserving encryption: it manufactures a *keyed bijection* from a keyed function that is not itself a bijection. In FF1 the round functions derive from a PRF—AES iterated in cipher-block-chaining fashion, as the construction below specifies—so the network is invertible without ever inverting the PRF: AES is only ever evaluated in the forward direction, in encryption and decryption alike.

Bijectivity is free; pseudorandomness is not. The Luby–Rackoff theorem is the classical statement that enough Feistel rounds over a PRF yield a pseudorandom permutation.

Theorem 5.18 (Luby–Rackoff)**.** *Let the round functions $F_1, \dots, F_r$ be independent PRFs on $\{0,1\}^w$. Against an adversary making q queries, the Feistel network Ψ is:*

1. *with $r = 3$ rounds, a pseudorandom permutation (forward queries only); and*
2. *with $r = 4$ rounds, a* strong *pseudorandom permutation (forward and inverse queries);*

in each case with distinguishing advantage $O(q^2/2^w)$ above the PRF advantage of the round functions.

We use the theorem as a statement, without proof, but the shape of the argument is worth recording, because it explains both the birthday term and the round counts. One first replaces each PRF round function by a truly random function, at the cost of r PRF advantage terms. For three rounds one then shows that, unless two of the at most q queries collide in an intermediate Feistel value—a birthday event of probability $O(q^2/2^w)$—the middle round's random function is evaluated at distinct inputs, so it produces independent uniform values that perfectly mask the relationship between input and output halves, and the network's answers are distributed exactly as a random *function*'s—independent and uniform on $\{0,1\}^{2w}$. A final application of the switching lemma (Lemma 5.9), at a further cost of $q(q-1)/2^{2w+1}$, carries the comparison from the random function to the random permutation; the term is absorbed by the same $O(q^2/2^w)$ bound. The fourth round adds the symmetry needed to resist inverse queries as well, upgrading PRP to SPRP. The full argument is a transcript analysis bounding the probability of any internal collision; the leftover term is the stated birthday bound.

Remark 5.19 (Why FF1 uses ten rounds rather than four). Theorem 5.18 gives security only up to $q \approx 2^{w/2}$ queries, and in the FPE setting w is tiny—one half of a 16-digit number—so the birthday bound $q^2/2^w$ is weak. The standardised FF1 therefore prescribes *ten* rounds rather than the minimal four, as a security margin against the more powerful attacks known for small-domain Feistel networks, which can sometimes beat the generic birthday bound. The classification of this design claim is worth making explicit: the round count itself is *specified*—NIST SP 800-38G fixes ten rounds—while a security proof matching the margin is an *open problem*; the analysis that exists (Luby–Rackoff and its small-domain refinements) does not certify the full strength the margin

is hoped to buy. What the round count does *not* affect is invertibility, which Proposition 5.17 guarantees for any round count and any round function.

Construction 5.20 (The FF1 mode of NIST SP 800-38G)**.** The FF1 mode instantiates an *unbalanced, modular, tweaked* Feistel network of ten rounds to build a PRP on $\mathbb{Z}/M\mathbb{Z}$ with $M = \rho^\nu$, the radix-ρ numeral reading of a length-ν string. It supports a *tweak* T: a non-secret string that selects, in effect, one permutation from a family indexed by T, so that the same key encrypts the same plaintext differently under different tweaks—useful for binding a token to its context. Its operation:

1. **Split.** Set $u = \lfloor \nu/2 \rfloor$ and $v = \nu - u$. Split the plaintext numeral string X into a left part A of u symbols and a right part B of v symbols, with integer values in $\{0, \dots, \rho^u - 1\}$ and $\{0, \dots, \rho^v - 1\}$ respectively. The two halves live in the moduli $a = \rho^u$ and $b = \rho^v$; the network is unbalanced when ν is odd.

2. **Round function from a PRF.** For each round $i = 0, 1, \dots, 9$, assemble a byte string Q encoding the tweak T, the round index i, and the current right half B; prepend a fixed parameter block P encoding the radix, the lengths, and the tweak length; and apply $\mathrm{PRF}(P \parallel Q)$, where PRF is AES in CBC-MAC mode: set $Y_0 = 0^{128}$, iterate $Y_j = \mathrm{AES}_k(Y_{j-1} \oplus X_j)$ over the 16-byte blocks X_j of $P \parallel Q$, and output the final block Y_m (SP 800-38G, Algorithm 6). The iteration chains AES through an evolving state exactly as the Merkle–Damgård iteration of §5.3 chains a compression function; emitting the final state is safe *here* because the PRF's inputs form a *prefix-free* set: the leading block P encodes the lengths of everything that follows, so two distinct inputs either differ already in their first block or agree on all lengths, and in neither case is one a proper prefix of the other. Prefix-free inputs are precisely the setting in which plain CBC-MAC is proved to be a PRF (a result of Petrank and Rackoff), and no length-extension query can arise. Further AES calls then expand the result to a long enough digest, which is read as an integer y.

3. **Modular Feistel step.** Update the halves by the modular analogue of Definition 5.16:

$$C = (A + y) \bmod m, \qquad A \leftarrow B, \qquad B \leftarrow C,$$

   where the modulus m alternates between $a = \rho^u$ and $b = \rho^v$ with the parity of the round, so that the sum lands in the correct half's domain; modular addition replaces XOR.

4. **Output.** After ten rounds, concatenate the final A and B and render the result back as a length-ν radix-ρ string.

Decryption runs the rounds in reverse, replacing modular addition by modular subtraction, $A \leftarrow (C - y) \bmod m$, exactly mirroring the backward recursion of Proposition 5.17. One quantitative detail of step 2 deserves note: the standard sizes the expanded digest at least four bytes beyond what the larger half requires, so the integer y ranges over at least 2^{32} multiples of the modulus, and the bias of the reduction $y \bmod m$ away from uniform is bounded by the modular-reduction bias of the *Math Guide* (§"Uniform sampling and the bias of modular reduction") at below 2^{-33} per round.

Proposition 5.21 (FF1 is a keyed bijection on $\mathcal{X}$)**.** *For every key k and tweak T, the FF1 map $\mathcal{E}_{k,T} \colon \mathcal{X} \to \mathcal{X}$ is a bijection of the format set $\mathcal{X}$, and $\mathcal{D}_{k,T}$ is its two-sided inverse.*

Proof. Each round is the map $(A, B) \mapsto (B, (A + y) \bmod m)$, where $y = y(i, T, B)$ depends only on the round index, the tweak, and the right half B—*not* on A. Fix B, i, and T; then y is a constant

and $A \mapsto (A + y) \bmod m$ is a translation of $\mathbb{Z}/m\mathbb{Z}$, hence a bijection, inverted by subtracting the same constant. The full round is therefore invertible: from the output (B, C) recover B directly, recompute $y = y(i, T, B)$ by a *forward* PRF evaluation, and set $A = (C - y) \bmod m$. Each even-indexed round is a bijection of $\mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}$ onto $\mathbb{Z}/b\mathbb{Z} \times \mathbb{Z}/a\mathbb{Z}$ and each odd-indexed round a bijection back, the two products coinciding when ν is even; the composition of the ten rounds, an even number, is a bijection of $\mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}$. The numeral map $(A, B) \mapsto A \cdot b + B$ identifies $\mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}$ with $\mathbb{Z}/M\mathbb{Z}$ as a bijection *of sets only*, not of groups: the factors satisfy $\gcd(a, b) = \rho^{\min(u,v)} > 1$, so the Chinese remainder theorem does not apply, and only bijectivity is used. Transporting through this map and the numeral bijection $\mathcal{X} \cong \mathbb{Z}/M\mathbb{Z}$, the composition $\mathcal{E}_{k,T}$ is a bijection of $\mathcal{X}$. Running the rounds backward with subtraction inverts it on both sides, so $\mathcal{D}_{k,T} = \mathcal{E}_{k,T}^{-1}$. □

Remark 5.22 (Bijectivity from algebra, security from the PRF). The construction separates its two obligations cleanly, and the separation is why Feistel is the standard route to FPE. Invertibility comes for free from the structure: Propositions 5.17 and 5.21 hold for *any* round function whatsoever, including a broken or constant one. The PRF—AES—is responsible not for invertibility but for *pseudorandomness*: making $\mathcal{E}_{k,T}$ look like a uniformly random permutation of $\mathcal{X}$ in the sense of Definition 5.15, per the Luby–Rackoff *heuristic*—Theorem 5.18 adapted, beyond what it literally proves, to the modular, multi-round, tweaked setting.

Remark 5.23 (The domain-size caveat, and the single Orchard use). Two facts complete the picture, one cautionary and one deployed.

The domain-size caveat is real. The published SP 800-38G requires $\rho^\nu \geq 100$ and recommends $\rho^\nu \geq 10^6$—its draft Revision 1 would make 10^6 mandatory—because for very small domains an attacker can recover or distinguish the keyed permutation by exhaustively cataloguing input–output pairs, and known message-recovery attacks on small-domain FPE (Bellare–Hoang–Tessaro and successors) erode the security margin. For the moderately large domains FPE is meant for, and with its ten rounds, FF1 is the NIST-standardised, AES-backed realisation of a keyed pseudorandom bijection on an arbitrary finite format.

The single use of FF1 in Orchard is the diversifier derivation with which this subsection opened: FF1-AES256 maps an address index to the 11-byte diversifier embedded in a shielded address, so that one spending key yields many mutually unlinkable addresses. The deployed instance (protocol specification §"Pseudo Random Permutations") fixes radix $\rho = 2$, length $\nu = 88$, and the empty tweak, and reads the 11-byte index as a little-endian bit string, so the domain is the 2^{88} strings of 88 bits—comfortably beyond the caveat above. The address built on the diversifier, and the case for a keyed permutation rather than a hash in this role, are the business of the *Ironwood Guide* (§"Diversified addresses").

The toolbox has gained its keyed compartment: one distinguishing game at three strengths, a standard permutation conjectured to win it, a natively keyed hash that turns domain-separated digests into independent PRFs, and a Feistel construction that bends a PRF into a bijection on any finite format. The next section spends these tools on the channel itself: symmetric encryption, authentication, and the authenticated-encryption composition that carries every shielded note, each built by running a PRF in a mode that converts its indistinguishability into secrecy or integrity.

6 Symmetric encryption, AEAD, and key derivation

The adversary of this section owns the channel. Every ciphertext passes through her hands, and she attacks on two fronts at once. As an eavesdropper she tries to *distinguish*: to tell an encryption of one note plaintext from an encryption of another, or to recognise that two ciphertexts hide the same message. As a forger she tries to *modify*: a stream cipher's ciphertext is XOR-malleable, so flipping a ciphertext bit flips the corresponding plaintext bit without any key material at all; she

can also shift bytes between a packet's public header and its encrypted body, replay a ciphertext under a different context, or manufacture a wholly new ciphertext that the receiver accepts. She is patient enough to exploit a single reused pad—two ciphertexts under one pad hand her the XOR of the plaintexts—and mathematically literate enough to exploit a key that is merely high-entropy rather than uniform. This section arms the honest parties against all of it, one guarantee at a time: confidentiality from the one-time pad through the nonce-based stream cipher ChaCha20 and the chosen-plaintext game; integrity from message authentication codes through universal hashing and Carter–Wegman; their composition as authenticated encryption with associated data, culminating in the ChaCha20-Poly1305 scheme that Zcash's note-encryption layer deploys; and finally key derivation, the manufacture of the uniform key that every theorem in between takes as its hypothesis.

Throughout, an *adversary* is a probabilistic algorithm $\mathcal{A}$; the notation $y \leftarrow \mathcal{A}(x)$ denotes running $\mathcal{A}$ on input x and assigning its output to y, and $x \xleftarrow{\$} S$ denotes sampling x uniformly from the finite set S. Every adversary in the computational definitions is probabilistic polynomial time (PPT) in the security parameter λ, and $\mathrm{negl}(\lambda)$ denotes a negligible function (*Math Guide*, §"Polynomial, exponential, and negligible functions"; recalled in §1.2).

The one-time pad and EAV security. The information-theoretic baseline is the *one-time pad*: to encrypt $m \in \{0,1\}^L$ under a uniformly random key $k \in \{0,1\}^L$ used exactly once, output $c = m \oplus k$. Since $m \oplus k$ is uniform whatever m is, the ciphertext distribution is independent of the message, and even an unbounded eavesdropper learns nothing—Shannon's *perfect secrecy*. The price is severe on both counts the name advertises: the key must be as long as the message, and it must never be reused, for two ciphertexts under one pad reveal $c \oplus c' = m \oplus m'$—the *two-time-pad* failure, which the adversary of the opening paragraph collects for free. The computational relaxation is *EAV security* (security against an eavesdropper): an adversary who chooses two equal-length messages m_0, m_1 and sees a single ciphertext $\mathrm{Enc}_k(m_b)$ for a hidden uniform bit b guesses b with advantage at most $\mathrm{negl}(\lambda)$—the one-ciphertext, no-oracle restriction of the chosen-plaintext game formalised in §6.2. Replacing the truly random pad by the output of a pseudorandom generator stretched from a short seed—the *pseudo-one-time pad*—achieves EAV security with a fixed-length key, at the cost of resting on a pseudorandomness assumption.

6.1 Stream ciphers and ChaCha20

A *stream cipher* is the practical instantiation of the pseudo-one-time pad: a keyed, seekable generator producing a *keystream* that XORs into the plaintext. To encrypt many messages under one key without violating the one-time-pad no-reuse rule, modern stream ciphers take an additional public input called a *nonce* (number used once) and behave like a pseudorandom function of the nonce, in the sense of Section 5.

Definition 6.1 (Nonce-based stream cipher)**.** A *nonce-based stream cipher* is a deterministic function

$$\mathrm{KS} : \mathcal{K} \times \mathcal{N} \times \mathbb{N} \to \{0,1\}^*$$

producing, for key k, nonce ν, and length L, a keystream $\mathrm{KS}(k,\nu,L) \in \{0,1\}^L$. Encryption of $m \in \{0,1\}^L$ is $\mathrm{Enc}_k(\nu, m) = m \oplus \mathrm{KS}(k,\nu,|m|)$ with ciphertext (ν, c); decryption recomputes the keystream and XORs it off. The security goal is that for distinct nonces the keystreams are jointly indistinguishable from independent uniform strings, provided no nonce repeats under a fixed key.

The stream cipher Zcash deploys—inside the AEAD of §6.5, in the note-encryption layer—is ChaCha20, designed by Bernstein and standardised in RFC 8439. Its core is a reversible mixing of four 32-bit words built only from modular addition, XOR, and rotation (an "ARX" design),

which avoids data-dependent table lookups and hence the timing side channels that table-based ciphers must engineer around.

Construction 6.2 (ChaCha20)**.** The ChaCha20 state is a 4×4 matrix of 32-bit words, viewed as a vector $(x_0, \ldots, x_{15}) \in (\mathbb{Z}/2^{32}\mathbb{Z})^{16}$; the symbol $+$ denotes addition modulo 2^{32}, $\oplus$ XOR, and $x \lll r$ left rotation by r bits. The *quarter-round* $\mathrm{QR}(a, b, c, d)$ updates four words by

$$\begin{aligned} a &\mathrel{+}= b; \quad d \mathrel{\oplus}= a; \quad d \lll= 16; \\ c &\mathrel{+}= d; \quad b \mathrel{\oplus}= c; \quad b \lll= 12; \\ a &\mathrel{+}= b; \quad d \mathrel{\oplus}= a; \quad d \lll= 8; \\ c &\mathrel{+}= d; \quad b \mathrel{\oplus}= c; \quad b \lll= 7. \end{aligned}$$

The state is initialised from the 128-bit constant "`expand 32-byte k`" (words x_0–x_3), a 256-bit key (x_4–x_{11}), a 32-bit block counter (x_{12}), and a 96-bit nonce (x_{13}–x_{15}). One *double round* applies QR to the four columns and then to the four diagonals; the 20 rounds of ChaCha20 are ten double rounds. Writing X for the initial state and $R(X)$ for the state after the rounds, the 64-byte keystream block is the *feed-forward* sum $R(X) + X$ (word-wise, modulo 2^{32}), serialised in little-endian order.

Remark 6.3 (Feed-forward and counter mode)*.* The rounds R form a bijection—each quarter-round is invertible, being a composition of invertible word operations—so without the final addition the block function would be an invertible permutation, and an adversary holding a keystream block could run the rounds backwards. The feed-forward sum $R(X) + X$ destroys invertibility; it is the standard Davies–Meyer-style move for turning a permutation into a one-way mixing function. Incrementing the block counter x_{12} produces successive 64-byte keystream blocks, so $\mathrm{KS}(k, \nu, L)$ is the concatenation of the blocks for counters $0, 1, 2, \ldots$; this *counter mode* is what makes the cipher seekable and parallelisable. The security status is a conjecture, in the same heuristic position as every concrete cipher of Section 5: ChaCha20 is *conjectured* to be a secure PRF of the pair (nonce, counter), and no attack better than brute force over the 256-bit key is known. Reusing a (k, ν) pair reproduces the same keystream and re-creates the two-time-pad failure; nonce uniqueness is a hard requirement, not a hygiene recommendation.

6.2 Chosen-plaintext security

EAV security models an adversary who sees a single ciphertext. The adversary of this section is stronger: she can influence which messages get encrypted—by triggering payments, injecting plaintext into a session, or simply guessing—and she observes many ciphertexts. The corresponding notion is security against *chosen-plaintext attack* (CPA), modelled by granting her oracle access to the encryption algorithm, in the game template of Definition 1.15.

Definition 6.4 (CPA security)**.** For a symmetric encryption scheme Π with key generator Gen, adversary $\mathcal{A}$, and bit $b \in \{0, 1\}$, the game $\mathsf{CPA}^{\mathcal{A}}_{\Pi}(\lambda, b)$ runs as follows.

1. The challenger samples $k \leftarrow \mathrm{Gen}(1^\lambda)$ and gives $\mathcal{A}$ oracle access to $\mathrm{Enc}_k(\cdot)$.
2. At some point $\mathcal{A}$ submits a challenge pair (m_0, m_1) of equal-length messages and receives $c^\star \leftarrow \mathrm{Enc}_k(m_b)$.
3. The adversary may continue querying the oracle and finally outputs a bit b', which is the output of the game.

The scheme Π is *CPA-secure* if for every PPT $\mathcal{A}$,

$$\mathrm{Adv}^{\mathrm{cpa}}_{\Pi}(\mathcal{A},\lambda) := |\Pr[\mathsf{CPA}^{\mathcal{A}}_{\Pi}(\lambda,0)=1] - \Pr[\mathsf{CPA}^{\mathcal{A}}_{\Pi}(\lambda,1)=1]| \in \mathrm{negl}(\lambda).$$

The first consequence of giving the adversary an encryption oracle is a structural one: encryption cannot be a function of the message alone.

Proposition 6.5 (Determinism precludes CPA security)**.** *No scheme whose* Enc *is deterministic and stateless is CPA-secure.*

Proof. The adversary queries the oracle on two distinct messages $m_0 \neq m_1$, recording $c_0 = \mathrm{Enc}_k(m_0)$ and $c_1 = \mathrm{Enc}_k(m_1)$. Decryption correctness forces $c_0 \neq c_1$: a single ciphertext cannot decrypt to both messages. She submits the challenge (m_0, m_1), receives $c^\star = \mathrm{Enc}_k(m_b)$, and outputs 0 if $c^\star = c_0$ and 1 otherwise. Since Enc is deterministic, $c^\star = c_b$, and since $c_0 \neq c_1$ the comparison identifies b: the guess is always correct and the advantage is 1. □

Consequently a CPA-secure scheme must be randomised or nonce-based. A nonce-based stream cipher used with a fresh nonce per encryption—random, or drawn from a counter—attains CPA security under the PRF conjecture for its keystream generator, because each encryption is then effectively a fresh one-time pad. We record the notion here and defer the routine PRF-based proof to the AEAD analysis (Theorem 6.20), where confidentiality and integrity are treated together.

Remark 6.6 (Semantic security). Goldwasser and Micali's original notion, *semantic security*, demands that whatever an efficient adversary can compute about the plaintext from the ciphertext, she could have computed without the ciphertext, from the message length alone. Semantic security is provably equivalent to the indistinguishability notions above; we use indistinguishability throughout because it is far easier to manipulate in reductions. The content is the same: the ciphertext is computationally useless for learning anything about the message beyond its length.

6.3 Message authentication codes

CPA security protects confidentiality and says nothing about integrity, and for the XOR-based stream ciphers above the gap is not hypothetical: flipping a bit of the ciphertext flips the corresponding plaintext bit, so the channel-owning adversary can convert a payment of one amount into a payment of another without ever touching the key. Detecting tampering requires a second primitive, keyed like the cipher but aimed at the forger rather than the eavesdropper: the *message authentication code*. Here we build this symmetric, shared-key form of authentication, the form that the AEAD needs.

Definition 6.7 (Message authentication code)**.** A *message authentication code* (MAC) is a pair $(\mathrm{Mac}, \mathrm{Vrfy})$ with key space $\mathcal{K}$: algorithm $\mathrm{Mac}_k(m)$ outputs a tag t, and $\mathrm{Vrfy}_k(m,t) \in \{0,1\}$ accepts (1) or rejects (0). Correctness requires $\mathrm{Vrfy}_k(m, \mathrm{Mac}_k(m)) = 1$ for all k, m. When Mac is deterministic, canonical verification recomputes the tag and checks equality.

Definition 6.8 (Existential and strong unforgeability)**.** For a MAC Π and adversary $\mathcal{A}$, the game $\mathsf{Forge}^{\mathcal{A}}_{\Pi}(\lambda)$ runs as follows.

1. The challenger samples $k \xleftarrow{\$} \mathcal{K}$ and gives $\mathcal{A}$ oracle access to $\mathrm{Mac}_k(\cdot)$; let Q be the set of messages queried.

2. The adversary outputs a pair $(m^\star, t^\star)$; the game outputs 1 iff $\mathrm{Vrfy}_k(m^\star, t^\star) = 1$ and $m^\star \notin Q$.

The MAC is *existentially unforgeable under chosen-message attack* (EUF-CMA) if for every PPT $\mathcal{A}$,

$$\mathrm{Adv}^{\mathrm{forge}}_{\Pi}(\mathcal{A}, \lambda) := \Pr[\mathsf{Forge}^{\mathcal{A}}_{\Pi}(\lambda) = 1] \in \mathrm{negl}(\lambda).$$

A *strongly* unforgeable (sUF-CMA) MAC additionally forbids producing a new valid tag on an already-queried message: the adversary wins on $(m^\star, t^\star)$ as long as the oracle never returned that exact *pair*. We write $\mathrm{Adv}^{\text{s-forge}}_{\Pi}(\mathcal{A}, \lambda)$ for the advantage in this variant.

The MAC Zcash uses, Poly1305, is not a pseudorandom-function MAC but a *one-time, universal-hash* MAC: it provides information-theoretic security per nonce and is extremely fast, at the cost of a key that authenticates only a single message. The enabling abstraction is a hash family whose *differences* are unpredictable.

Definition 6.9 (ε-almost-Δ-universal hash family)**.** Let $(\mathcal{G}, +)$ be a finite abelian group. A family of functions $\{h_r : \mathcal{X} \to \mathcal{G}\}_{r \in \mathcal{R}}$ is *ε-almost-Δ-universal* (ε-AΔU) if for all distinct $x, x' \in \mathcal{X}$ and every $\delta \in \mathcal{G}$,

$$\Pr_{r \xleftarrow{\$} \mathcal{R}} [\, h_r(x) - h_r(x') = \delta \,] \le \varepsilon.$$

Taking $\delta = 0$ recovers ordinary ε-universality—low collision probability—while the Δ condition controls *additive differences*, which is what makes the one-time MAC below secure. With $\mathcal{G} = \{0,1\}^n$ under XOR this is the classical almost-XOR-universality; Poly1305 takes $\mathcal{G} = \mathbb{Z}/2^{128}\mathbb{Z}$ under addition modulo 2^{128}.

The Carter–Wegman paradigm turns any AΔU family into a one-time MAC by masking the hash value with a fresh uniform pad.

Theorem 6.10 (Carter–Wegman one-time MAC)**.** *Let $\{h_r\}$ be ε-AΔU into $(\mathcal{G}, +)$. Define a MAC with key (r, s), where $r \xleftarrow{\$} \mathcal{R}$ and the pad $s \xleftarrow{\$} \mathcal{G}$, by $\mathrm{Mac}_{r,s}(m) = h_r(m) + s$. If (r, s) authenticates at most one message, then any adversary—even a computationally unbounded one—who sees one message–tag pair (m, t) forges a valid pair $(m^\star, t^\star)$ with $m^\star \neq m$ with probability at most ε.*

Proof. The adversary observes $t = h_r(m) + s$ and must output $(m^\star, t^\star)$ with $m^\star \neq m$ and $t^\star = h_r(m^\star) + s$. Subtracting the two tag equations eliminates the pad: the forgery succeeds iff

$$h_r(m^\star) - h_r(m) = t^\star - t.$$

Condition on the adversary's view. The pad s is uniform and independent of r, so given the single observed tag t, every value of $h_r(m)$ is equally consistent—each candidate value determines a unique s—and the observation therefore leaks nothing about r: the posterior on r remains uniform on $\mathcal{R}$. The adversary's difference $\delta := t^\star - t$ and distinct pair $(m, m^\star)$ are thus fixed independently of r, and the AΔU property gives $\Pr[\, h_r(m^\star) - h_r(m) = \delta \,] \le \varepsilon$. □

The MAC Poly1305 instantiates the paradigm with a polynomial-evaluation hash over the prime field $\mathbb{F}_p$ for $p = 2^{130} - 5$ (prime-field arithmetic as in the *Math Guide*, §"The prime field and two inversion routes").

Construction 6.11 (Poly1305)**.** Fix the prime $p = 2^{130} - 5$ and work in $\mathbb{F}_p$. The key is a pair (r, s): r is a 128-bit value subjected to a fixed bit-*clamping* (the top four bits of bytes $3, 7, 11, 15$ and

the bottom two bits of bytes $4, 8, 12$ are forced to zero), and s is a 128-bit pad. To authenticate a message, split it into ℓ blocks $c_1, \dots, c_\ell$ of at most 16 bytes; encode each block as an integer in $[0, 2^{128})$ and add 2^{8b}, where b is the block's byte length—this sets a 1 bit just above the block's bytes, making every coefficient nonzero and the block encoding injective in both content and byte length. Interpreting r and each c_i as elements of $\mathbb{F}_p$, the hash is the polynomial evaluation

$$h_r(m) \;=\; \sum_{i=1}^{\ell} c_i\, r^{\ell-i+1} \;\in\; \mathbb{F}_p,$$

computed by Horner's rule as $(\cdots((c_1 r + c_2)r + c_3)r \cdots)r$. The tag is

$$\mathrm{Mac}_{r,s}(m) \;=\; ((h_r(m) \bmod p) + s) \bmod 2^{128},$$

the field hash reduced and added to the pad modulo 2^{128}.

Proposition 6.12 (Poly1305 is almost-Δ-universal)**.** *Restricted to messages of at most L bytes—hence at most $\ell_{\max} = \lceil L/16 \rceil$ blocks—the reduced Poly1305 hash family $\{m \mapsto h_r(m) \bmod p\}$, read into $\mathbb{Z}/2^{128}\mathbb{Z}$, is ε-almost-Δ-universal with respect to addition modulo 2^{128}, with*

$$\varepsilon \;\le\; \frac{8\,\ell_{\max}}{2^{106}}.$$

The loss over the ideal $\ell_{\max} \cdot p^{-1}$ accounts for the clamping of r—which restricts r to a subset of size 2^{106}—and for the final reduction modulo 2^{128}.

Proof. Sketch. Fix two distinct messages m, m' with block encodings (c_i) and (c'_j), padded to a common length $\ell \le \ell_{\max}$ by leading zero coefficients without changing their values. The block encoding guarantees that $m \neq m'$ yields distinct coefficient vectors: genuine coefficients are nonzero, so differing block counts set a nonzero coefficient against a zero padding one, while equal block counts leave two blocks differing in content or in final-block byte length, hence in encoding. The difference of hashes is

$$h_r(m) - h_r(m') \;=\; \sum_{i=1}^{\ell} (c_i - c'_i)\, r^{\ell-i+1} \;=:\; P(r) \;\in\; \mathbb{F}_p,$$

a nonzero polynomial in r of degree at most ℓ. Now count the ways a target difference $\delta \in \mathbb{Z}/2^{128}\mathbb{Z}$ can be hit. The reduced hash values lie in $[0, p)$, so their integer difference lies in the interval $(-p, p)$, which has length $2p < 8 \cdot 2^{128}$; each residue class modulo 2^{128} therefore contains at most $\lceil 2p/2^{128} \rceil = 8$ candidate integer differences u. For each such u, the equation $P(r) = u$ in $\mathbb{F}_p$ has at most ℓ roots—a polynomial of degree $\le \ell$ over a field has at most ℓ roots (*Math Guide*, §"Roots and the factor theorem")—so at most 8ℓ values of r produce the difference δ. The clamped r is uniform over a set of 2^{106} values (clamping zeroes 22 of the 128 bits), so it lands on one of those roots with probability at most $8\ell/2^{106} \le 8\ell_{\max}/2^{106}$. □

The bound is comfortably small in practice: for messages of 16 384 bytes ($\ell_{\max} = 1024$ blocks) it gives $\varepsilon \le 8 \cdot 1024/2^{106} = 2^{-93}$, against the ideal $\ell_{\max}/p \approx 2^{-120}$—the clamping and reduction losses cost 27 bits of the margin and leave it enormous. Combining Proposition 6.12 with the Carter–Wegman Theorem 6.10: Poly1305 with a fresh, uniformly random pad s per message—uniformity, not mere unpredictability, is what the theorem's pad-elimination step uses—is a one-time MAC with forgery probability at most $8\ell_{\max}/2^{106}$ per verification attempt. In the AEAD below, ChaCha20 itself generates the one-time key (r, s) pseudorandomly from the nonce—uniform once the cipher is idealised as a random function—so the per-nonce one-time guarantee is exactly what is needed.

6.4 Authenticated encryption with associated data

We now combine the two guarantees. The composite goal is *authenticated encryption*: the adversary should be unable to learn anything about plaintexts (CPA security) *and* unable to produce any new ciphertext that decrypts to anything other than the distinguished rejection symbol $\perp$, a value distinct from every message (ciphertext integrity). Real protocols additionally require ciphertexts to bind to public context—packet headers, version bytes, an ephemeral public key—that travels in the clear but must be authenticated; this is the *associated data*.

Definition 6.13 (AEAD scheme)**.** An *authenticated encryption with associated data* (AEAD) scheme is a pair $(\mathrm{Enc}, \mathrm{Dec})$ with key space $\mathcal{K}$, nonce space $\mathcal{N}$, and associated-data space $\mathcal{D}$: algorithm $\mathrm{Enc}_k(\nu, a, m)$ returns a ciphertext c, and $\mathrm{Dec}_k(\nu, a, c)$ returns a message or $\perp$. Correctness: $\mathrm{Dec}_k(\nu, a, \mathrm{Enc}_k(\nu, a, m)) = m$ for all k, ν, a, m. The scheme authenticates but does not encrypt the associated data a.

Definition 6.14 (Ciphertext integrity, INT-CTXT)**.** For an AEAD scheme Π and adversary $\mathcal{A}$, the game $\mathsf{Ctxt}_{\Pi}^{\mathcal{A}}(\lambda)$ runs as follows.

1. The challenger samples $k \xleftarrow{\$} \mathcal{K}$ and gives $\mathcal{A}$ two oracles: an encryption oracle $\mathrm{Enc}_k(\cdot, \cdot, \cdot)$, recording the set Q of returned triples (ν, a, c), and a verification oracle that on (ν, a, c) returns 1 if $\mathrm{Dec}_k(\nu, a, c) \neq \perp$ and 0 otherwise. The adversary must never repeat a nonce to the *encryption* oracle (the *nonce-respecting* model); verification queries may reuse nonces freely.
2. The game outputs 1 iff some verification query $(\nu^\star, a^\star, c^\star) \notin Q$ is answered 1.

The scheme Π has *ciphertext integrity* if for every PPT $\mathcal{A}$, $\Pr[\mathsf{Ctxt}_{\Pi}^{\mathcal{A}}(\lambda) = 1] \in \mathrm{negl}(\lambda)$.

Definition 6.15 (Authenticated encryption)**.** An AEAD scheme is *secure* (is *an AEAD*) if it is both CPA-secure and has ciphertext integrity (Definition 6.14), where the CPA game of Definition 6.4 is run with $k \xleftarrow{\$} \mathcal{K}$ and extended so that the adversary supplies the nonce and associated data of every oracle query and of the challenge, subject to the *nonce-respecting* restriction—a restriction on the adversary, not a property of the scheme—that no nonce is used twice across oracle queries and challenge. The restriction is necessary: an AEAD may encrypt deterministically given (ν, a, m)—ChaCha20-Poly1305 does—and then querying the oracle at (ν, a, m_0) and submitting the challenge (ν, a, m_0, m_1) would identify the bit by comparing ciphertexts.

The standard route from a CPA-secure cipher and a strongly unforgeable MAC to an AEAD is *encrypt-then-MAC*: encrypt the message, compute a MAC over the ciphertext together with the associated data and nonce, and append the tag; decryption verifies the tag *before* decrypting, and any tag failure returns $\perp$. The contrasting orders—MAC-then-encrypt and encrypt-and-MAC—do *not* generically achieve authenticated encryption; encrypt-then-MAC does.

Theorem 6.16 (Encrypt-then-MAC yields authenticated encryption)**.** *Let* $\Pi_E = (\mathrm{Enc}, \mathrm{Dec})$ *be a CPA-secure encryption scheme and* $\Pi_M = (\mathrm{Mac}, \mathrm{Vrfy})$ *a strongly unforgeable* (*sUF-CMA*) *MAC,*

with independent *keys* k_E, k_M. *Define the composite scheme* Π' *by*

$$\mathrm{Enc}'_{k_E,k_M}(\nu, a, m) = (c, t), \qquad c \leftarrow \mathrm{Enc}_{k_E}(\nu, m), \quad t \leftarrow \mathrm{Mac}_{k_M}(\nu \,\|\, a \,\|\, c),$$

with Dec' *returning* $\perp$ *unless* $\mathrm{Vrfy}_{k_M}(\nu\|a\|c,\, t) = 1$, *in which case it returns* $\mathrm{Dec}_{k_E}(\nu, c)$. *Then* Π' *is a secure AEAD: it is CPA-secure and has ciphertext integrity.*

Proof. Ciphertext integrity. Suppose a PPT $\mathcal{A}$ making at most q_v verification queries wins the game: some fresh query $(\nu^\star, a^\star, (c^\star, t^\star)) \notin Q$ decrypts successfully, i.e. $\mathrm{Vrfy}_{k_M}(\nu^\star\|a^\star\|c^\star, t^\star) = 1$. Each encryption-oracle call on (ν, a, m) returned (c, t) and corresponds to one MAC query on the string $w = \nu\|a\|c$ returning tag t. Because the framing $\nu\|a\|c$ parses uniquely (the component lengths are fixed or prepended), the freshness of $(\nu^\star, a^\star, c^\star)$ means the MAC oracle never produced the string–tag pair $(w^\star, t^\star)$ with $w^\star := \nu^\star\|a^\star\|c^\star$: either $w^\star$ was never queried, or it was queried and returned a different tag. Either way $(w^\star, t^\star)$ is a fresh valid pair—a *strong* forgery against Π_M. The reduction $\mathcal{B}$ runs $\mathcal{A}$, answering encryption queries with its own MAC oracle and a self-chosen k_E. Verification queries $\mathcal{B}$ cannot answer, lacking k_M, so it guesses: it samples a uniform index $i \le q_v$ in advance, answers each fresh verification query before the i-th with 0 (queries on triples in Q are answered 1, correctly), halts at the i-th fresh query, and outputs that query's $(w^\star, t^\star)$ as its forgery. On the event that $\mathcal{A}$ wins and i hits her *first* successful fresh query—the guess is right with probability $1/q_v$—every earlier 0 answer was correct, the simulation is perfect, and $\mathcal{B}$ wins, so

$$\Pr[\mathrm{Ctxt}^{\mathcal{A}}_{\Pi'}(\lambda) = 1] \;\le\; q_v \cdot \mathrm{Adv}^{\text{s-forge}}_{\Pi_M}(\mathcal{B}, \lambda) \;\in\; \mathrm{negl}(\lambda),$$

the factor q_v being polynomial.

CPA security. The tag is computed from c (together with k_M, ν, a, and the MAC's own coins) and so carries no information about the challenge bit beyond what c already carries. Formally, the reduction $\mathcal{B}'$ samples k_M itself, forwards $\mathcal{A}$'s oracle and challenge queries to its own Π_E oracle to obtain the c-components, and appends tags it computes with k_M. This simulates Π' perfectly, so $\mathrm{Adv}^{\text{cpa}}_{\Pi'}(\mathcal{A}, \lambda) = \mathrm{Adv}^{\text{cpa}}_{\Pi_E}(\mathcal{B}', \lambda) \in \mathrm{negl}(\lambda)$. □

Remark 6.17 (Why the ordering matters). Encrypt-then-MAC lets the receiver reject forged ciphertexts *without decrypting*, and this is what blocks chosen-ciphertext and padding-oracle attacks: a rejected ciphertext never reaches the decryption logic, so the decryption logic cannot leak. By contrast, MAC-then-encrypt verifies only after decrypting, exposing the decryption routine to attacker-chosen ciphertexts—the historical source of a long line of padding-oracle breaks. Ciphertext integrity is also exactly the property that upgrades CPA security to chosen-ciphertext (CCA) security, the CPA game of Definition 6.4 with a decryption oracle added that refuses only the challenge ciphertext: an authenticated-encryption scheme is automatically CCA-secure, because that oracle is useless to the adversary—every ciphertext she did not legitimately obtain decrypts to $\perp$ except with negligible probability.

6.5 The ChaCha20-Poly1305 AEAD

We now assemble the concrete scheme of RFC 8439, the AEAD by which Zcash encrypts every shielded note ciphertext and outgoing ciphertext (protocol specification, §"Symmetric Encryption"). It is an encrypt-then-MAC composition of the ChaCha20 stream cipher with the Poly1305 one-time MAC, with one economical twist: the cipher derives *both* the keystream and the one-time MAC key from the single (k, ν) input.

Construction 6.18 (ChaCha20-Poly1305)**.** Fix a 256-bit key k and a 96-bit nonce ν. Encryption of a message m with associated data a proceeds as follows.

1. *One-time MAC key.* Run ChaCha20 with key k, nonce ν, and block counter 0; take the first 32 bytes of the keystream block as the Poly1305 one-time key (r, s)—clamp the first 16 bytes to form r; the next 16 are s. Discard the rest of this block.

2. *Encrypt.* Run ChaCha20 with key k and nonce ν starting at block counter 1, producing the keystream KS_1 (the keystream of Remark 6.3 taken from counter 1 onward), and set $c = m \oplus \mathrm{KS}_1(k, \nu, |m|)$.

3. *Authenticate.* Form the Poly1305 input by concatenating: a padded with zeros to a 16-byte boundary, c padded with zeros to a 16-byte boundary, the 8-byte little-endian length of a, and the 8-byte little-endian length of c. Compute the tag $t = \mathrm{Poly1305}_{r,s}$(that input).

The ciphertext is (c, t). Decryption recomputes (r, s) and the tag from (k, ν, a, c), compares it to t in *constant time*, returns $\perp$ on mismatch, and otherwise outputs $m = c \oplus \mathrm{KS}_1(k, \nu, |c|)$.

Remark 6.19 (Length encoding prevents splicing). The trailing 8-byte lengths of a and c are essential. Without them, the zero-padding that aligns a and c to 16-byte boundaries would let an adversary shift bytes between the associated data and the ciphertext, or pad-extend one of them, while leaving the Poly1305 input string unchanged—breaking the binding between a and c even though every authenticated byte is intact. Appending the explicit lengths makes the MAC input an injective encoding of the pair (a, c), the same injectivity that the per-block length encoding supplies within one message in the AΔU analysis (Proposition 6.12).

Theorem 6.20 (Security of ChaCha20-Poly1305)**.** *Model ChaCha20 as a pseudorandom function $F_k(\nu, j)$ from (nonce, block counter) pairs to 64-byte blocks. In the nonce-respecting model, ChaCha20-Poly1305 is a secure AEAD. Concretely, consider a PPT adversary making q encryption queries and q_v verification queries, each of whose Poly1305 inputs—padded associated data, padded ciphertext, and the length block—comprises at most $\ell_{\max}$ sixteen-byte blocks, $\ell_{\max} = \lceil |a|/16 \rceil + \lceil |c|/16 \rceil + 1$. Her CPA advantage is at most twice the best PRF-distinguishing advantage against F at comparable cost, and she forges—wins the INT-CTXT game—with probability at most*

$$\mathrm{Adv}_F^{\mathrm{prf}}(\mathcal{B}, \lambda) \;+\; \frac{8\, q_v\, \ell_{\max}}{2^{106}},$$

where $\mathcal{B}$ is a PRF distinguisher of comparable cost.

Proof. Sketch. Replace F_k by a truly random function Φ from (nonce, counter) pairs to 64-byte blocks. Any noticeable change in the adversary's success probability yields a PRF distinguisher of comparable cost—it simulates the whole game using its function oracle—so each replacement costs one $\mathrm{Adv}_F^{\mathrm{prf}}$ term; the CPA game is played twice (once per challenge bit), whence the factor two there. We analyse the idealised scheme.

Confidentiality. Since the adversary is nonce-respecting, every encryption query uses a fresh ν, and the keystream blocks $\Phi(\nu, 1), \Phi(\nu, 2), \ldots$ are uniform and independent of everything else in her view; the ciphertext $c = m \oplus \mathrm{KS}$ is a genuine one-time pad and reveals nothing about m beyond its length. The tag is a deterministic function of c, a, ν and the (independent) key (r, s), and adds no information. The idealised CPA advantage is therefore zero.

Integrity. For each nonce ν, the block $\Phi(\nu, 0)[0{:}32]$ is uniform and independent across distinct nonces; clamping its first half samples r from Poly1305's prescribed restricted set, while the second half s stays uniform, so the resulting one-time keys are independent across nonces. Because counter 0 is reserved for this key while encryption consumes counters ≥ 1, each one-time key is also independent of the keystream that produced its ciphertext. Per nonce, the setting is exactly the Carter–Wegman one-time MAC of Theorem 6.10 instantiated with the ε-AΔU Poly1305 hash of Proposition 6.12, $\varepsilon \leq 8\ell_{\max}/2^{106}$. Consider any single fresh verification query $(\nu^\star, a^\star, c^\star, t^\star)$. If

$\nu^\star$ was never used for encryption, the pad s for $\nu^\star$ is uniform and entirely unknown, so the tag is correct with probability 2^{-128}. If $\nu^\star$ equals the nonce of some encryption query—nonce reuse is allowed in *verification* queries, only encryption is nonce-respecting— then the adversary has seen exactly one tag under that (r, s), and freshness of $(a^\star, c^\star)$ makes this a forgery on a new message under a one-time key, succeeding with probability at most ε by Theorem 6.10. The union bound over the q_v verification queries (*Math Guide*, §"The union bound and a birthday calculation") gives forgery probability at most $8q_v\ell_{\max}/2^{106}$ in the idealised scheme; adding back the PRF-switching cost yields the stated bound. □

Remark 6.21 (Single key, two roles). The scheme reuses one 256-bit key for both encryption and authentication, seemingly violating the independent-keys hypothesis of Theorem 6.16. Reserving block counter 0 for the Poly1305 key and counters ≥ 1 for the keystream is exactly what restores independence: once the cipher is idealised as a truly random function—the switch the proof of Theorem 6.20 pays for with a PRF-distinguishing term—outputs at distinct counters are independent, so (r, s) is independent of the keystream and the two-key analysis applies. For the PRF itself the independence is computational, not literal. This is a recurring design pattern—domain separation by a counter or label, the same discipline as §3.5, lets one master key safely play several roles.

Remark 6.22 (Wrong-key rejection and key commitment). The upper volumes consume this AEAD in a *trial-decryption* pattern: a receiver detects the notes addressed to it by attempting decryption of every ciphertext under each of its keys, relying on decryption under a wrong key to return $\perp$ (protocol specification, §"Symmetric Encryption"). For *honestly produced* ciphertexts the scheme delivers this. Fix a ciphertext $(c, t) = \mathrm{Enc}_k(\nu, a, m)$ and a key k' independent of k, and idealise the cipher under k and under k' as independent random functions, as in the proof of Theorem 6.20. The one-time key (r', s') that decryption under k' derives from block $(\nu, 0)$ is then uniform and independent of (c, t): the pad s' alone makes the recomputed tag match t with probability exactly 2^{-128}, and even conditioned on one exposed tag under (r', s'), Theorem 6.10 bounds acceptance by the AΔU slack $8\ell_{\max}/2^{106}$ of Proposition 6.12, with $\ell_{\max}$ as in Theorem 6.20. Decryption under an independent key therefore rejects except with probability at most $2^{-128} + 8\ell_{\max}/2^{106}$ per attempt, plus two PRF-distinguishing terms for the two idealisations.

The guarantee stops at honest senders, and the boundary deserves care. CPA security and INT-CTXT are *single-key* notions: nothing in Definition 6.15 constrains a ciphertext crafted by a *malicious sender* who knows several keys. The scheme is in fact not *key-committing*—the separate notion demanding that no ciphertext verify under two distinct keys—and a sender who chooses keys k_1, k_2 herself can craft a single (c, t) that decrypts validly, to different plaintexts, under both: with (r_i, s_i) the one-time key each k_i derives, the hash h_{r_i} of Construction 6.11 is linear in the blocks of c, so fixing t, leaving two blocks of c free, and solving over $\mathbb{F}_p$ the two linear equations that pin the hash under r_i to $t - s_i$ (retrying until both solutions encode valid blocks) makes both keys accept (c, t)—the multi-key collision of Len, Grubbs, and Ristenpart's partitioning-oracle attack. The upper volumes' trial-decryption arguments therefore never rest on the tag alone against adversarial senders: their note-plaintext consistency checks—recomputing the note commitment from the decrypted plaintext and comparing it against the commitment the transaction carries—are what close this gap.

6.6 Key-derivation functions

Theorem 6.20 assumes a uniformly random 256-bit key. In practice keys may come from sources that are *high-entropy but not uniform*: a Diffie–Hellman shared secret is a structured group encoding, not a uniform bit string, and a hardware noise source may spread its unpredictability unevenly across its bits. (A passphrase is a different, usually *low*-entropy source, and needs a deliberately costly password KDF rather than the machinery of this subsection.) The adversary is owed nothing here—if the honest parties feed a non-uniform key into a scheme proved secure

for uniform keys, the proof simply does not apply, and structure in the key is structure she may exploit. A *key-derivation function* (KDF) converts an appropriate source secret into one or more pseudorandom keys. The right entropy measure is worst-case, not average-case.

Definition 6.23 (Min-entropy)**.** The *min-entropy* of a random variable X is

$$H_\infty(X) \;=\; -\log_2 \max_x \Pr[X = x];$$

equivalently, the best chance of guessing X in one try is $2^{-H_\infty(X)}$. Given side information Z, the *conditional min-entropy* is

$$\tilde{H}_\infty(X \mid Z) \;=\; -\log_2 \mathbb{E}_{z \leftarrow Z} \max_x \Pr[X = x \mid Z = z],$$

so that $2^{-\tilde{H}_\infty(X|Z)}$ is the best chance of guessing X from Z. Recall the *statistical distance* $\Delta(X, Y) = \frac{1}{2}\sum_u |\Pr[X = u] - \Pr[Y = u]|$ of Definition 1.6 (*Math Guide*, §"Statistical distance"); we say X is *ϵ-close to uniform* on a set S if $\Delta(X, U_S) \le \epsilon$.

Definition 6.24 (Key-derivation function and its security)**.** A *key-derivation function* is a function

$$\mathrm{KDF} : \mathcal{X} \times \mathcal{I} \times \{0,1\}^* \times \mathbb{N} \to \{0,1\}^*$$

taking a source secret (a sample of a random variable Σ over $\mathcal{X}$), a non-secret *salt* $\mathrm{slt} \in \mathcal{I}$, an *info* or context string ctx, and a desired output length L. The security goal must specify the source model. For an arbitrary source of conditional min-entropy at least γ given the adversary's side information, an extractor-style guarantee uses a salt sampled uniformly and independently of the source and requires the pair $(\mathrm{slt}, \mathrm{KDF}(\Sigma, \mathrm{slt}, \mathrm{ctx}, L))$ to be computationally indistinguishable from (slt, U_L), even given ctx and that side information. No fixed deterministic unsalted function can provide this guarantee for *every* high-min-entropy source; an unsalted KDF therefore rests on a narrower premise, such as input keying material that is already computationally pseudorandom, or that is hard to query in a random-oracle model.

The notion a KDF targets is that of a seeded randomness extractor whose output need only *look* uniform.

Definition 6.25 (Strong computational extractor)**.** A keyed function $\mathrm{Ext} : \mathcal{I} \times \mathcal{X} \to \{0,1\}^n$ is a *strong (γ, ϵ)-computational extractor* if for every source Σ over $\mathcal{X}$ of conditional min-entropy at least γ given auxiliary information Z, and a uniformly random salt $\mathrm{slt} \xleftarrow{\$} \mathcal{I}$, the pair $(\mathrm{slt}, \mathrm{Ext}_{\mathrm{slt}}(\Sigma))$ is computationally indistinguishable from (slt, U_n), even given Z, with distinguishing advantage at most ϵ. "Strong" means the salt is revealed to the distinguisher.

Information theory can realise the notion unconditionally; we state the classical result *without proof*, since the deployed mechanism below rests on random-oracle modelling instead.

Theorem 6.26 (Leftover hash lemma; stated without proof)**.** *Let $\{g_r : \mathcal{X} \to \{0,1\}^n\}$ be a 2^{-n}-universal hash family with a uniform public seed R. For any X with $H_\infty(X) \ge \gamma$,*

$$\Delta((R, g_R(X)),\ (R, U_n)) \;\le\; \tfrac{1}{2}\sqrt{2^{n-\gamma}}.$$

Thus investing $\gamma \ge n + 2\log_2(1/\epsilon)$ bits of min-entropy buys n output bits that are ϵ-close to uniform—

> *a strong (γ, ϵ)-extractor in the statistical, not merely computational, sense.*

HKDF, in passing. The standardised general-purpose KDF is Krawczyk's HKDF (RFC 5869), built from HMAC (§5.3) in an *extract-then-expand* architecture: a salted HMAC call first distils suitable non-uniform input keying material into a short pseudorandom key (the extractor stage, justified along leftover-hash lines), and a counter-chained sequence of HMAC calls then expands that key, under a context string, up to RFC 5869's limit of 255 hash-output-length blocks (the expander stage, justified by HMAC's PRF security). The resulting KDF is a standard choice across many protocol source models and output lengths, though its precise guarantee still depends on assumptions about the source and about HMAC. Zcash does not use it—indeed the shielded protocol uses HMAC nowhere: neither the `orchard` nor the `zcash_note_encryption` crate carries an HMAC dependency (HMAC enters the wallet stack through the transparent BIP-32 derivation and BIP-39 mnemonic layers and, transitively, through the optional Tor client, never through a shielded crate)—because its requirements are narrower and a single hash call meets them in the random-oracle analysis.

The Zcash KDF. Orchard derives the note-encryption key by one personalised BLAKE2b call:

$$K_{\text{enc}} = \text{KDF}(\mathsf{sharedSecret}, \mathsf{epk}) = \text{BLAKE2b-256}(\texttt{"Zcash_OrchardKDF"};\ \mathsf{sharedSecret} \,\|\, \mathsf{epk}),$$

where sharedSecret is the Diffie–Hellman shared point of the note-encryption key agreement, epk is the sender's ephemeral public key, and the quoted string occupies BLAKE2b's 16-byte personalisation field (the function `kdf_orchard` hashes the serialised shared point followed by the ephemeral-key bytes with a 32-byte output; protocol specification, §"Orchard Key Derivation"). The analysis models this personalised hash as a random oracle: by the domain-separation principle of §3.5 (Proposition 3.17, via the native-personalisation idiom catalogued there), the personalisation makes it an oracle independent of every other hash use in the protocol, and the derived key is uniform in the adversary's view *unless* she queries that oracle at the hidden sharedSecret—a query whose argument she must first compute from the two public keys, an instance of the computational Diffie–Hellman problem of §2.2. Folding epk into the input binds the derived key to this particular ephemeral, the role the context string plays in HKDF. No salt is used because the claim is this narrower Diffie–Hellman-plus-random-oracle one—the unsalted premise anticipated in Definition 6.24—not extraction from every high-min-entropy source.

Remark 6.27 (The role of the KDF in the larger protocol). In the Zcash note-encryption layer, a Diffie–Hellman exchange on the Pallas curve produces a shared group element, and the single BLAKE2b call above condenses that element into the symmetric key fed to ChaCha20-Poly1305. The two halves of this section meet exactly here: key derivation manufactures the uniform key that the AEAD security theorem (Theorem 6.20) takes as its hypothesis, and the chain from a non-uniform shared secret to an authenticated ciphertext is complete. Section 7 assembles that chain end to end as hybrid public-key encryption.

7 Key agreement

Two parties who have never met share no secret, and the adversary of this section intends to keep it that way. She owns the channel between them. As an eavesdropper she records the two group elements they exchange and tries to *distinguish* the key they derive from a random string—a key she can distinguish is a key whose every later use leaks. As an active attacker she does better than listen: she intercepts each party's contribution, substitutes her own, and runs one honest-looking session with each victim, so that both "secure channels" terminate at her desk and she reads and rewrites every message in transit—forging, as we shall prove, without violating any hardness assumption at all. And on a public ledger her seat is permanent: every note ciphertext

is published for ever, and she may scan them all at leisure, holding each against every address she can enumerate. This section builds the machinery that denies her all three postures: the Diffie–Hellman protocol atop the exponentiation asymmetry of Section 2; the static/ephemeral taxonomy that turns an interactive exchange into a one-shot flow; the key-agreement scheme abstraction that the deployed code programs against; passive security, which we prove equal to the DDH advantage *exactly*; the man-in-the-middle attack and the three external mechanisms of authentication; the elliptic-curve instantiation with its validation duties; the hybrid composition with a key-derivation function and an AEAD; key privacy, which keeps a ciphertext silent about its recipient; and the deployed endpoint, shielded-note delivery as hybrid public-key encryption carried in band on the ledger.

7.1 The Diffie–Hellman protocol

The raw material is the asymmetry established in Section 2. In a cyclic group $G = \langle g \rangle$ of order q, the map

$$\exp_g \colon \mathbb{Z}/q\mathbb{Z} \to G, \qquad x \mapsto g^x,$$

is a group isomorphism—it carries addition of exponents to multiplication of elements—that is cheap to evaluate, at $O(\log q)$ group operations by repeated squaring, but believed infeasible to invert in the groups this volume uses: inverting it is precisely the discrete logarithm problem (Definition 2.2). Everything in this section depends only on the group structure and the hardness of certain problems in it, never on the presentation of the group; the reference instances are a large prime-order subgroup of $\mathbb{F}_p^\times$ and the elliptic-curve point groups of §7.6. Diffie and Hellman's observation is that the asymmetry alone already lets two strangers manufacture a shared secret in public.

Construction 7.1 (Diffie–Hellman key exchange)**.** Fix public parameters (G, g, q) with $G = \langle g \rangle$ of order q, known to all parties (and to the adversary).

1. Alice samples $a \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$ uniformly, computes $A = g^a$, and sends A to Bob.
2. Bob samples $b \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$ uniformly, computes $B = g^b$, and sends B to Alice.
3. Alice computes $k_A = B^a$; Bob computes $k_B = A^b$.

The value $k_A = k_B$ is the *shared secret*; the transmitted elements A and B are the *public shares* (public keys), and the exponents a and b are the *private keys*, never sent.

Proposition 7.2 (Correctness of Diffie–Hellman)**.** *In Construction 7.1 both parties compute the same group element, $k_A = k_B = g^{ab}$.*

Proof. Unwinding the definitions,

$$k_A = B^a = (g^b)^a = g^{ba} = g^{ab} = (g^a)^b = A^b = k_B,$$

where $ba = ab$ because integer multiplication is commutative, and exponents may be read modulo q because g has order q. □

The shared secret g^{ab} is thus a well-defined function of the transcript (A, B): it can be computed by anyone who knows *one* of the two private keys, and—conjecturally—by no one who knows only the public shares. Making that conjecture precise, and seeing exactly how much it buys, is the business of §7.4. First, a structural precaution.

Remark 7.3 (Small subgroups and the prime-order rule). If G has small subgroups, an adversary can exploit them: by substituting for a public share an element of small order she confines the victim's computed secret to a small subgroup and learns the victim's exponent modulo a small factor of q—the Pohlig–Hellman leakage of Corollary 2.21 in protocol form. The remedy is the standing rule of Section 2: prefer G of *prime* order q. Then the only subgroups are $\{1\}$ and G itself, every non-identity element is a generator, and no nontrivial confinement is possible. When the natural ambient group does not have prime order—$\mathbb{F}_p^\times$ has composite order $p-1$—one works inside a prime-order subgroup and has each recipient *validate* incoming elements as members: check $B^q = 1$ and $B \neq 1$ before exponentiating. The elliptic-curve analogue of this validation appears in Remark 7.19.

7.2 Static and ephemeral keys

Nothing in Construction 7.1 says how long a key pair lives, and the protocol changes character with the answer.

Definition 7.4 (Static and ephemeral keys)**.** A Diffie–Hellman key pair (x, g^x) is *static* (long-term) if it is generated once and reused across many sessions, typically published and bound to an identity; it is *ephemeral* if it is generated freshly for a single session and discarded immediately afterwards.

Three protocol shapes result, and all three matter to what follows.

- *Ephemeral–ephemeral* (*DHE*). Both parties use fresh keys. Once a and b are erased, no one—the parties included—can recover g^{ab}: this is *forward secrecy*, the guarantee that a later compromise of all stored long-term state cannot decrypt recorded past sessions, because the only secrets that ever existed were the deleted ephemerals.
- *Static–ephemeral.* The recipient Bob holds a static pair $(b, B = g^b)$, with B known in advance; Alice contributes a fresh ephemeral (a, A). The shared secret g^{ab} then requires *no interaction from Bob at agreement time*: Alice computes $g^{ab} = B^a$ from the published B, attaches A, and sends everything in a single flow; Bob recovers $g^{ab} = A^b$ whenever he next comes online. This is exactly the shape of hybrid encryption (§7.7) and of shielded-note delivery (§7.9): sender online once, recipient possibly offline, no round trip.
- *Static–static.* Both keys are long-term, so g^{ab} is a fixed function of the two identities—the same in every session. There is no forward secrecy, and the never-varying secret must be combined with fresh per-session randomness (a nonce) before it may key any symmetric scheme; using it directly would reuse one key across all time.

The static–ephemeral case is the conceptual bridge from key exchange, an interactive protocol, to public-key encryption, a one-shot algorithm. A static public key $B = g^b$ functions exactly as an encryption public key: anyone can derive a secret shared with its owner by sending a single ephemeral share. The rest of the section develops that bridge until, by §7.9, it carries deployed traffic.

7.3 The key-agreement scheme abstraction

Deployed code does not manipulate exponents; it programs against an interface. We therefore package the non-interactive (static–ephemeral) pattern as a scheme with named algorithms, so that later constructions—and the deployed note-encryption machinery—can invoke key agreement without caring which group sits underneath.

Definition 7.5 (Key-agreement scheme)**.** A (non-interactive) *key-agreement scheme* over a public parameter set pp consists of a private-key space $\mathcal{S}$, a public-key space $\mathcal{P}$, a shared-secret space $\mathcal{K}$, and three algorithms:

- $\mathrm{KeyGen}(\mathsf{pp}) \to s$, randomised, outputting a private key $s \in \mathcal{S}$;
- $\mathrm{DerivePublic}(\mathsf{pp}, s) \to p$, deterministic, mapping a private key to its public key $p \in \mathcal{P}$;
- $\mathrm{Agree}(\mathsf{pp}, s, p') \to k$, deterministic, mapping one's own private key and a counterparty's public key to a shared secret $k \in \mathcal{K}$.

Correctness demands that for all pp and all s, t in the support of $\mathrm{KeyGen}(\mathsf{pp})$,

$$\mathrm{Agree}(\mathsf{pp}, s, \mathrm{DerivePublic}(\mathsf{pp}, t)) = \mathrm{Agree}(\mathsf{pp}, t, \mathrm{DerivePublic}(\mathsf{pp}, s)). \tag{1}$$

Construction 7.6 (Diffie–Hellman as a key-agreement scheme)**.** Fix $\mathsf{pp} = (G, g, q)$ with $G = \langle g \rangle$ of prime order q. The scheme $\mathrm{DH}_{G,g}$ takes $\mathcal{S} = \mathbb{Z}/q\mathbb{Z}$ (or $(\mathbb{Z}/q\mathbb{Z}) \setminus \{0\}$, to avoid the degenerate case whose public key is the identity), $\mathcal{P} = \mathcal{K} = G$, and

$$\mathrm{KeyGen}\colon a \xleftarrow{\$} \mathcal{S}, \qquad \mathrm{DerivePublic}(\mathsf{pp}, a) = g^a, \qquad \mathrm{Agree}(\mathsf{pp}, a, B) = B^a.$$

Correctness is exactly Proposition 7.2: $\mathrm{Agree}(\mathsf{pp}, s, g^t) = (g^t)^s = g^{ts} = g^{st} = (g^s)^t = \mathrm{Agree}(\mathsf{pp}, t, g^s)$.

Remark 7.7 (The deployed interface generalises the base point). The Zcash note-encryption machinery programs against precisely this abstraction, with one generalisation: the deployed DerivePublic takes the base point as an *explicit argument* rather than fixing a protocol-wide generator, so that Sapling and Orchard derive keys over per-address *diversified bases* (protocol specification §"Key Agreement"; the `zcash_note_encryption` crate's `Domain` trait's `ka_derive_public`, `ka_agree_enc`, and `ka_agree_dec` operate on abstract associated types, and Orchard's implementation of `derive_public` in the `orchard` crate receives the base g_d as a parameter). Those bases are put to work in §7.9, and the address machinery that produces them is detailed in the *Ironwood Guide* (§"Diversified addresses"). What matters here is that the interface still invokes an abstract key agreement and never inspects the underlying group—which is why the same construction has transported across three groups, from Sprout's Curve25519 to Sapling's Jubjub to Orchard's Pallas.

Remark 7.8 (The abstraction boundary, and what falls outside it). Any concrete scheme satisfying equation (1) can be swapped in without changing the surrounding logic: a prime-field group, an elliptic curve, or a post-quantum non-interactive key agreement such as the CSIDH group action. A *key-encapsulation mechanism* (KEM), by contrast, cannot satisfy the correctness equation at all: encapsulation requires the recipient's public key before it can produce its ciphertext, so no counterparty-independent DerivePublic exists. The static–ephemeral one-flow *pattern* does generalise to any KEM—the encapsulation ciphertext plays the role of the ephemeral public key—but at the cost of changing the interface the surrounding logic programs against.

7.4 Passive security and the Diffie–Hellman assumptions

The weakest adversary is the eavesdropper: she reads the channel but does not alter it, and her entire view of an honest session is the transcript (g^a, g^b). Both formalisations of her task were laid down in §2.2. The *computational* problem is to produce the shared secret: writing $\mathrm{Adv}_{\mathrm{cdh}}(\mathcal{A})$

for the success probability of $\mathcal{A}$ in the CDH game of Definition 2.4,

$$\mathrm{Adv}_{\mathrm{cdh}}(\mathcal{A}) = \Pr_{a,b}[\mathcal{A}(g^a, g^b) = g^{ab}].$$

The *decisional* problem is to distinguish (g^a, g^b, g^{ab}) from (g^a, g^b, g^c) with advantage $\mathrm{Adv}_{\mathrm{ddh}}(\mathcal{A})$ (Definition 2.5). The hierarchy DDH $\leq$ CDH $\leq$ DLP was established in §2.3 (Theorems 2.7 and 2.8), together with the pairing-based separation of DDH from CDH (Example 2.9). The distinction is not pedantry. Hardness of CDH asserts only that the *whole* secret cannot be computed; keying a symmetric cipher requires that the secret be indistinguishable from a uniform group element, so that no individual bit or predicate of it leaks. The following classical break shows the gap is real in the most familiar group.

Remark 7.9 (A concrete DDH break: the Legendre symbol). Take $G = \mathbb{F}_p^\times$, the *full* multiplicative group, with generator g. The Legendre symbol $\left(\frac{h}{p}\right)$ is efficiently computable by Euler's criterion $h^{(p-1)/2} \bmod p$ and is multiplicative (*Math Guide*, §"Quadratic residues and the Euler criterion"). A generator of $\mathbb{F}_p^\times$ is a nonsquare, so $\left(\frac{g^a}{p}\right) = (-1)^a$: the symbol of a public share reveals the *parity* of its exponent. The parities of a and b determine the parity of ab, hence $\left(\frac{g^{ab}}{p}\right) = (-1)^{ab}$ is determined by the two observed symbols. The distinguisher computes the predicted symbol of the shared secret from (g^a, g^b), compares it with the symbol of the third component, and outputs "Diffie–Hellman" on a match. On a genuine triple the match is certain; on a random triple the third component's symbol is an independent uniform sign, matching with probability 1/2. The distinguishing advantage is the constant 1/2—even though CDH in $\mathbb{F}_p^\times$ is believed hard. The CDH/DDH gap therefore needs no pairings; the oldest group in the subject exhibits it. The cure is the prime-order rule of Remark 7.3: in a subgroup of odd prime order q, every element h is a square ($h = (h^{(q+1)/2})^2$), the symbol is identically $+1$, and the leak vanishes. This is one of the two reasons—the other being Pohlig–Hellman—that DDH is only ever assumed in prime-order groups.

With the assumption in place, passive security is not merely implied by DDH; it *is* DDH, advantage for advantage. We state the game in the volume's tradition and prove the identification.

Definition 7.10 (Passive security of a key-agreement scheme)**.** For a key-agreement scheme Σ over pp and an adversary $\mathcal{A}$, the game $\mathsf{KA\text{-}IND}_\Sigma^{\mathcal{A}}(\lambda)$ runs as follows.

1. The challenger samples $s, t \leftarrow \mathsf{KeyGen}(\mathsf{pp})$ independently and computes the transcript $(\mathsf{DerivePublic}(\mathsf{pp}, s), \mathsf{DerivePublic}(\mathsf{pp}, t))$.
2. It sets $k_0 = \mathsf{Agree}(\mathsf{pp}, s, \mathsf{DerivePublic}(\mathsf{pp}, t))$, samples $k_1 \xleftarrow{\$} \mathcal{K}$ uniformly, draws a hidden bit $\beta \xleftarrow{\$} \{0, 1\}$, and sends the transcript together with k_β.
3. The adversary outputs a bit β'.

The advantage is $\mathrm{Adv}_\Sigma^{\text{ka-ind}}(\mathcal{A}) = |\Pr[\beta' = 1 \mid \beta = 0] - \Pr[\beta' = 1 \mid \beta = 1]|$; the scheme is *passively secure* if every PPT adversary's advantage is negligible in λ.

Theorem 7.11 (Passive security of Diffie–Hellman)**.** *For the scheme* $\mathrm{DH}_{G,g}$ *of Construction 7.6 with* KeyGen *uniform on* $\mathbb{Z}/q\mathbb{Z}$*, every adversary* $\mathcal{A}$ *satisfies*

$$\mathrm{Adv}_{\mathrm{DH}_{G,g}}^{\text{ka-ind}}(\mathcal{A}) = \mathrm{Adv}_{\mathrm{ddh}}(\mathcal{A}),$$

identically and with running time preserved exactly. Under the DDH assumption the shared secret of an honest ephemeral–ephemeral session is therefore pseudorandom given the transcript: a passive adversary learns nothing about it that she could not have guessed about a random group element.

Proof. The two games are the same game. In KA-IND for $\mathrm{DH}_{G,g}$, the transcript is (g^a, g^b) with a, b uniform and independent; the real key is $k_0 = g^{ab}$; and the uniform alternative $k_1 \xleftarrow{\$} G$ is exactly g^c for uniform independent c, because $\exp_g$ is a bijection. The challenge (g^a, g^b, k_β) is thus verbatim a sample of $\mathcal{D}_{\mathrm{dh}}$ or $\mathcal{D}_{\mathrm{rand}}$ according to β—the DDH distinguishing game of Definition 2.5. The identity map converts each adversary into the other, so the advantages coincide and no running time is added. □

Remark 7.12 (Why pseudorandomness, not mere unpredictability). Every use of a Diffie–Hellman secret in this volume feeds it into a key-derivation function to produce a symmetric key. What should the group deliver to that function? Hardness of CDH guarantees the secret is hard to produce *in full*, but not that it lacks efficiently predictable structure: the Legendre leak of Remark 7.9 is exactly such structure. Hardness of DDH guarantees the secret is as good as a random group element (Theorem 7.11)—no predicate of it is efficiently computable from the transcript. In practice one often assumes only CDH and models the hash-based KDF as a random oracle: the derived key is then uniform unless the adversary queries the oracle at the hidden shared secret, a route made precise in Theorem 7.24. Either way the design goal is the same: a derived symmetric key indistinguishable from uniform, because that is the hypothesis every theorem of Section 6 consumes.

7.5 Active attacks and the role of authentication

Passive security is the most the bare protocol can offer, and the gap between it and what a channel needs is best exhibited as a construction. The adversary now controls the wire.

Definition 7.13 (Man-in-the-middle attack on unauthenticated Diffie–Hellman)**.** An active adversary $\mathcal{M}$ controlling the channel between Alice and Bob runs two independent Diffie–Hellman sessions, one with each victim. She samples $m \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$ and computes g^m. When Alice sends $A = g^a$ intended for Bob, $\mathcal{M}$ intercepts it and forwards g^m to Bob in its place; when Bob replies $B = g^b$, she intercepts it and forwards g^m to Alice. Alice computes $k_A = (g^m)^a = g^{am}$, which $\mathcal{M}$ also computes as A^m; symmetrically Bob computes $k_B = (g^m)^b = g^{bm}$, which $\mathcal{M}$ computes as B^m.

Proposition 7.14 (The attack defeats unauthenticated Diffie–Hellman)**.** *After the attack of Definition 7.13, Alice and Bob each hold a secret they believe is shared with the other but is in fact shared with $\mathcal{M}$; and $\mathcal{M}$, knowing both g^{am} and g^{bm}, can decrypt, read, modify, and re-encrypt every subsequent message in each direction, transparently to both parties. No assumption on G—not even ideal hardness of the discrete logarithm problem—prevents this.*

Proof. Correctness of Diffie–Hellman (Proposition 7.2), applied once per session, gives agreement between Alice and $\mathcal{M}$ on g^{am} and between Bob and $\mathcal{M}$ on g^{bm}. The adversary knows her own exponent m and received both A and B, so she computes both secrets directly—no problem instance is ever inverted, which is why no hardness assumption is relevant. Neither victim verifies the *origin* of the group element they received; each transcript is a syntactically valid run of Construction 7.1, so nothing in either party's view distinguishes the attacked execution from an honest one. With k_A and k_B in hand, $\mathcal{M}$ decrypts traffic from Alice under k_A, re-encrypts it under k_B for Bob, and symmetrically in reverse, editing at will in between. □

The lesson deserves stating structurally: *passive security concerns the secrecy of the shared secret; active security additionally requires the authenticity of the public shares*. Confidentiality of a channel is worthless if the channel is established with the wrong party. The bare protocol (Construction 7.1) provides no authentication—nothing in g^a or g^b testifies to who sent it—so the remedy must bind each public share to its sender's identity, letting the recipient reject a substituted share. Every such binding is *external* to the key exchange itself.

Remark 7.15 (Three mechanisms supply authentication). All deployed remedies fall into three families, each external to the bare exchange.

1. *Signed Diffie–Hellman.* Each party signs its ephemeral share with a long-term signing key whose verification key is certified—by a certificate authority (a third party whose signature on the key the parties already trust) or by trust-on-first-use (accepting the first key seen and rejecting later changes). The adversary of Proposition 7.14 cannot forge Alice's signature on g^m, so her substitution is rejected. Digital signatures are the subject of Section 8; this composition is the structure of authenticated TLS.

2. *Static keys that are themselves authenticated.* If Bob's static public key $B = g^b$ reaches Alice authentically—a trusted directory, a pinned key—then a static–ephemeral exchange toward B already authenticates Bob to Alice: only the holder of b can complete it. Note the direction: this authenticates the *recipient*, not the sender. It is the model relevant to public-key encryption, and to shielded-note delivery, where the recipient's address *is* an authenticated public key (§7.9).

3. *Out-of-band verification.* The parties compare a short fingerprint of the exchanged public keys over a channel that is authentic even if low-bandwidth—digits read aloud on a call, a QR code scanned in person.

In every case the key-agreement security goal remains the passive guarantee of Theorem 7.11; authentication is a separable layer composed around the exchange, not a property of the group operation.

7.6 Elliptic-curve Diffie–Hellman

The development so far is deliberately group-agnostic; we now instantiate it on the groups the deployed protocol actually uses. Recall from the *Math Guide* (§"Elliptic curves") that an elliptic curve E over a field F of characteristic neither 2 nor 3 is given in short Weierstrass form $y^2 = x^3 + ax + b$ with nonzero discriminant, and that its F-rational points $E(F)$ form a finite abelian group under chord-and-tangent addition with the point at infinity O as identity (*Math Guide*, §"The group structure of $E(\mathbb{F}_p)$"). The group is written additively, so exponentiation $g \mapsto g^x$ becomes *scalar multiplication* $P \mapsto [x]P$, computable in $O(\log x)$ point additions by double-and-add. Elliptic-curve Diffie–Hellman is the transliteration.

Construction 7.16 (ECDH)**.** Fix an elliptic curve E/F, a base point $P \in E(F)$ of large prime order q generating $G = \langle P \rangle$, and cofactor $h = |E(F)|/q$. In the abstraction of Definition 7.5:

$$\text{KeyGen}\colon a \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}, \qquad \text{DerivePublic}(a) = [a]P, \qquad \text{Agree}(a, Q) = [a]Q.$$

The shared secret of parties with private keys a and b is $[a]([b]P) = [ab]P = [b]([a]P)$. The associated hard problems are the discrete-logarithm, CDH, and DDH definitions of Section 2 (Definitions 2.2, 2.4, and 2.5) read with $[x]P$ in place of g^x—the problems ECDLP (*Math Guide*, §"The elliptic-curve discrete logarithm problem"), ECCDH, and ECDDH of §2.6.

Proposition 7.17 (Correctness of ECDH)**.** *The scheme of Construction 7.16 satisfies the key-agreement correctness condition* (1)*.*

Proof. Scalar multiplication is cyclic-group exponentiation in additive dress: the group is abelian, and $\mathbb{Z}$ acts on $\langle P \rangle$ through $\mathbb{Z}/q\mathbb{Z}$ because $[q]P = \mathcal{O}$, so $[x]([y]P) = [xy]P$ for all integers x, y. Hence $\mathrm{Agree}(a, [b]P) = [ab]P = [ba]P = \mathrm{Agree}(b, [a]P)$, which is condition (1). □

Remark 7.18 (Why elliptic curves)*.* Well-chosen elliptic-curve groups admit only the generic $O(\sqrt{q})$ attacks of §2.4, whereas index calculus solves the discrete logarithm in $\mathbb{F}_p^\times$ in subexponential time (Remark 2.18); a 256-bit curve therefore delivers the security level that a multi-thousand-bit prime field requires (Example 2.17). That efficiency, together with the availability of curves engineered for cheap in-circuit arithmetic, is why the Pallas curve of §2.6 underlies Orchard.

Remark 7.19 (Subgroup and curve validation)*.* A recipient computing $[a]Q$ on an attacker-supplied point Q must guard against two pitfalls, the elliptic-curve forms of the validation duty in Remark 7.3.

1. *Invalid-curve attack.* The received coordinates may satisfy not E's equation but that of a different, weaker curve which shares the same addition formulas (the chord and tangent slopes of the *Math Guide*, §"Explicit affine formulas", involve a but never b); scalar multiplication then runs happily in the wrong group, whose order may be smooth. The implementation must check the curve equation on every incoming point.

2. *Small-subgroup attack.* When the cofactor $h > 1$, an adversary submits a point Q of small order dividing h, confining $[a]Q$ to a tiny subgroup and learning a modulo a small factor. The defences are *cofactor clearing*—multiply incoming points by h—or choosing a curve with $h = 1$.

Prime-order curves such as Pallas sidestep the cofactor issue entirely: with $h = 1$ there is no small subgroup to hit, and only the on-curve check remains. The deployed arithmetic enforces exactly that check at deserialisation, as documented in Proposition 2.27, whose compressed encoding cannot even represent a point off the curve.

7.7 Hybrid public-key encryption

Three obstacles separate the Diffie–Hellman shared group element from a usable encryption scheme. First, the shared secret is a *structured group element*—a curve point with coordinates satisfying an equation—not a uniform string of key bits. Second, the exchange by itself transports no message: it manufactures a secret, and stops. Third, we want the static–ephemeral non-interactive flow of §7.2, so that the recipient need not be online. All three are resolved by one composition: key agreement for the secret, a key-derivation function for uniformity, an authenticated cipher for the message—*hybrid public-key encryption* (HPKE). The two symmetric ingredients were built in Section 6; we recall them in the specialised shape the composition needs.

Definition 7.20 (KDF, recalled and specialised)**.** A *key-derivation function* here is a deterministic function

$$\mathrm{KDF}\colon \mathcal{K} \times \{0,1\}^* \to \{0,1\}^\ell,$$

mapping input keying material $k \in \mathcal{K}$ together with a context string to an ℓ-bit key. This specialises Definition 6.24 (§6.6) by omitting the salt and fixing the output length ℓ. For Theorem 7.24 we model the function as a random oracle (§3.4): distinct inputs then have independent uniform outputs, and security follows provided the adversary never queries the oracle at the hidden Diffie–Hellman secret and context. This is a source-specific random-oracle

guarantee, not a generic claim that a deterministic function extracts uniform bits from every high-min-entropy source.

Definition 7.21 (AEAD, recalled)**.** An *authenticated encryption scheme with associated data* with key space $\{0,1\}^\ell$—matching the KDF output—is a pair $(\mathrm{Enc}, \mathrm{Dec})$ with $\mathrm{Enc}_K(N, ad, m) = c$ and $\mathrm{Dec}_K(N, ad, c) \in \{m\} \cup \{\perp\}$, where N is a nonce, ad is associated data authenticated but not encrypted, and $\perp$ denotes rejection (Definition 6.13). Correctness: $\mathrm{Dec}_K(N, ad, \mathrm{Enc}_K(N, ad, m)) = m$. Security is authenticated-encryption security (Definition 6.15): encryptions of any two equal-length messages are CPA-indistinguishable (Definition 6.4), *and* no adversary can produce a fresh ciphertext decrypting to anything but $\perp$ (ciphertext integrity, Definition 6.14)— confidentiality and integrity simultaneously.

Construction 7.22 (DH-based hybrid public-key encryption)**.** Fix ECDH over $G = \langle P \rangle$ of prime order q (Construction 7.16, in the interface of Definition 7.5), a secure KDF, and a secure AEAD as above. The recipient holds a static key pair $(b, B = [b]P)$ with B published. To *encrypt* a message m to B:

1. sample an ephemeral $e \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$ and set $E_{\mathrm{pk}} = [e]P$;
2. compute the shared secret $S = \mathrm{Agree}(e, B) = [eb]P$;
3. derive the symmetric key $K = \mathrm{KDF}(S, \mathit{context})$, where *context* includes E_{pk} and the protocol's labels;
4. output (E_{pk}, c) with $c = \mathrm{Enc}_K(N, ad, m)$ for a fixed or derived nonce N—fixed is safe because each K is used once.

To *decrypt* (E_{pk}, c) with b: compute $S' = [b]E_{\mathrm{pk}}$, derive $K' = \mathrm{KDF}(S', \mathit{context})$ with E_{pk} read from the ciphertext, and output $\mathrm{Dec}_{K'}(N, ad, c)$, which is m on an honest ciphertext; on a tampered one, ciphertext integrity makes the output $\perp$ except with negligible probability. Anyone holding B can encrypt; only the holder of b can decrypt.

Theorem 7.23 (Correctness of hybrid encryption)**.** *For every message m and every honestly generated ciphertext (E_{pk}, c) of Construction 7.22, decryption returns m.*

Proof. Correctness of ECDH (Proposition 7.17) gives $S' = [b]([e]P) = [be]P = [eb]P = [e]B = S$. The KDF is deterministic and both sides supply the same context—the recipient recovers E_{pk} from the ciphertext itself—so $K' = K$. Correctness of the AEAD (Definition 7.21) finishes: $\mathrm{Dec}_K(N, ad, \mathrm{Enc}_K(N, ad, m)) = m$. □

Theorem 7.24 (Passive security of hybrid encryption)**.** *Model the KDF as a random oracle (Definition 3.13) and let the AEAD be secure in the sense of Definition 6.15. Consider a passive adversary $\mathcal{A}$ who sees the public key B and a single challenge ciphertext: she names two equal-length messages m_0, m_1, receives the encryption of m_β for a hidden uniform bit β, and guesses β, with CPA advantage $\mathrm{Adv}_{\mathrm{cpa}}(\mathcal{A})$ defined as in Definition 6.4. Under CDH in G no PPT adversary has non-negligible*

> *advantage; concretely, if $\mathcal{A}$ makes at most q_H random-oracle queries, then*
>
> $$\mathrm{Adv}_{\mathrm{cpa}}(\mathcal{A}) \;\le\; 2\,q_H \cdot \mathrm{Adv}_{\mathrm{cdh}}(\mathcal{B}) \;+\; \mathrm{Adv}_{\mathrm{ae}}(\mathcal{C}),$$
>
> *for explicit reductions $\mathcal{B}$ against CDH and $\mathcal{C}$ against the AEAD's authenticated-encryption security (of which the proof consumes only the confidentiality half, so $\mathrm{Adv}_{\mathrm{ae}}(\mathcal{C})$ is the CPA advantage of Definition 6.4 read against the AEAD), each running in essentially the same time as $\mathcal{A}$.*

Proof. The proof is a game hop.

Game 0 is the real CPA game: the challenge ciphertext is (E_{pk}, c) with $E_{\mathrm{pk}} = [e]P$, $S = [eb]P$, $K = \mathrm{KDF}(S, \mathit{context})$, and $c = \mathrm{Enc}_K(N, \mathit{ad}, m_\beta)$. Say $\mathcal{A}$ *wins* a game when her guess β' equals β.

Game 1 replaces K by an independent uniform ℓ-bit string, leaving everything else unchanged.

The hop. In the random oracle model the two games proceed identically unless the adversary queries the oracle at exactly the point $(S, \mathit{context}) = ([eb]P, \mathit{context})$: until that query, the oracle's value there is a uniform string she has never seen, which is precisely what Game 1 hands out. But producing $S = [eb]P$ from the view $(B, E_{\mathrm{pk}}) = ([b]P, [e]P)$ is exactly solving CDH. The reduction $\mathcal{B}$ embeds its CDH challenge as (B, E_{pk}) and simulates Game 1 around it: it draws β and a uniform ℓ-bit key K itself, encrypts m_β under K to form the challenge ciphertext, and implements the oracle by lazy sampling—a fresh uniform string for each new query, the stored answer on a repeated one, so that the oracle remains a consistent function. The simulation is a perfect run of Game 1, and the critical query occurs in it with the same probability as in Game 0, the two games being identical until it happens. One obstruction remains: in a pairing-free group $\mathcal{B}$ cannot *recognise* which of the queries carries $[eb]P$ (Example 2.9 shows recognising it is the DDH-test a gap group would supply). So $\mathcal{B}$ picks one of the at most q_H queries uniformly at random and outputs its first component; whenever the critical query occurs, $\mathcal{B}$'s choice is correct with probability at least $1/q_H$. Hence

$$|\Pr[\mathcal{A} \text{ wins Game } 0] - \Pr[\mathcal{A} \text{ wins Game } 1]| \;\le\; q_H \cdot \mathrm{Adv}_{\mathrm{cdh}}(\mathcal{B}).$$

The factor q_H is a genuine guessing-type security loss, in the tightness taxonomy of Definition 1.22. Assuming *gap*-CDH—CDH hardness even given a DDH-decision oracle, which would let $\mathcal{B}$ test each query—restores tightness; we state the loose bound because it is what plain CDH buys, and honesty about the loss is part of the bound.

Game 1 analysed. The AEAD key is now uniform and independent of everything else in the adversary's view, which is exactly the hypothesis of the AEAD's confidentiality game. The reduction $\mathcal{C}$ forwards (m_0, m_1) to its AEAD challenger, embeds the returned ciphertext as c alongside a self-generated E_{pk}, and relays $\mathcal{A}$'s guess; the AEAD challenger's hidden bit plays β, so $|\Pr[\mathcal{A} \text{ wins Game } 1] - \frac{1}{2}| = \frac{1}{2}\,\mathrm{Adv}_{\mathrm{ae}}(\mathcal{C})$.

Accounting. The difference-form advantage of Definition 6.4 is twice the bit-guessing advantage $|\Pr[\mathcal{A} \text{ wins}] - \frac{1}{2}|$ of Definition 1.15, so the triangle inequality across the hop gives

$$\mathrm{Adv}_{\mathrm{cpa}}(\mathcal{A}) \;\le\; 2\,(q_H \cdot \mathrm{Adv}_{\mathrm{cdh}}(\mathcal{B}) + \tfrac{1}{2}\,\mathrm{Adv}_{\mathrm{ae}}(\mathcal{C})) \;=\; 2\,q_H \cdot \mathrm{Adv}_{\mathrm{cdh}}(\mathcal{B}) + \mathrm{Adv}_{\mathrm{ae}}(\mathcal{C}),$$

the stated bound.

Alternative hypothesis. The random oracle can be traded away, but not for Definition 6.24 alone: that guarantee is stated for a uniform independent salt, which the specialised KDF of Definition 7.20 omits, and the definition denies any generic unsalted guarantee. The trade needs an explicit hypothesis on the deployed function—the narrower premise Definition 6.24 allows for unsalted use: for a *uniform* group element S, the output $\mathrm{KDF}(S, \mathit{context})$ is computationally indistinguishable from a uniform ℓ-bit string given the context. Under that hypothesis and DDH in place of CDH, Theorem 7.11 first replaces the shared secret by a uniform group element at cost $\mathrm{Adv}_{\mathrm{ddh}}$, and the stated indistinguishability then replaces K by a uniform string—no oracle and no q_H factor, at the price of the stronger decisional assumption on the group and an explicit pseudorandomness assumption on the KDF. □

Remark 7.25 (Every ingredient is necessary). Dropping any piece of Construction 7.22 breaks the composition.

- *Without the KDF*, one would key the cipher with the raw group element S. Its bit-representation is non-uniform—curve points do not fill the encoding space—and in non-DDH groups it is partially predictable, the Legendre-symbol leakage of Remark 7.9 being the classical instance. The KDF standardises the format, extracts uniform bits, and, by folding E_{pk} into its context, ties the derived key to this particular ephemeral share.
- *Without AEAD integrity*, a passively confidential but malleable cipher would let an active adversary tamper with c undetectably—the XOR-malleability of Section 6's stream ciphers is total. Ciphertext integrity (Definition 6.14) makes any modification decrypt to $\perp$ except with negligible probability, giving the symmetric payload its chosen-ciphertext robustness.
- *Without the ephemeral key*—a static–static variant—every message to B would begin from the same shared secret. The fresh per-message ephemeral, with E_{pk} bound into the KDF context, gives each ciphertext separate key material, so nothing relies on nonce uniqueness under a reused key. What it does *not* supply is forward secrecy against later compromise of the recipient's static key b: from a recorded E_{pk}, an attacker who learns b recomputes $[b]E_{\mathrm{pk}}$ and with it the ciphertext's key. Erasing e buys only the sender-side guarantee that the key derives from no stored secret of the sender.

7.8 Key privacy

Theorem 7.24 answers the eavesdropper and leaves the scanner untouched. Her posture is different: she holds a published ciphertext against every public key she can enumerate and asks not what it says but *to whom*. Confidentiality is silent on the question—a scheme may hide the message perfectly and still stamp the recipient's key on every ciphertext. The property that denies her is *key privacy*, due to Bellare, Boldyreva, Desai, and Pointcheval. We state it for a generic public-key scheme, then prove it for the bare static–ephemeral flow with the message folded in—ElGamal encryption, the shape whose key privacy the upper volumes consume.

Definition 7.26 (Key privacy under chosen-plaintext attack, IK-CPA). A *public-key encryption scheme* Π over pp consists of a randomised $\mathrm{KeyGen}(\mathsf{pp}) \to (\mathsf{sk}, \mathsf{pk})$, a randomised $\mathrm{Enc}_{\mathsf{pk}}(m) \to c$, and a deterministic $\mathrm{Dec}_{\mathsf{sk}}(c)$ with $\mathrm{Dec}_{\mathsf{sk}}(\mathrm{Enc}_{\mathsf{pk}}(m)) = m$ for every key pair in the support of KeyGen and every message m. For an adversary $\mathcal{A}$ the game $\mathsf{IK\text{-}CPA}^{\mathcal{A}}_{\Pi}(\lambda)$ runs as follows.

1. The challenger samples $(\mathsf{sk}_0, \mathsf{pk}_0), (\mathsf{sk}_1, \mathsf{pk}_1) \leftarrow \mathrm{KeyGen}(\mathsf{pp})$ independently and sends $(\mathsf{pk}_0, \mathsf{pk}_1)$.
2. The adversary outputs a message m.
3. The challenger draws a hidden bit $\beta \xleftarrow{\$} \{0,1\}$ and sends $c \leftarrow \mathrm{Enc}_{\mathsf{pk}_\beta}(m)$.
4. The adversary outputs a bit β'.

The advantage is $\mathrm{Adv}^{\text{ik-cpa}}_{\Pi}(\mathcal{A}) = |\Pr[\beta' = 1 \mid \beta = 0] - \Pr[\beta' = 1 \mid \beta = 1]|$; the scheme is *key-private* (IK-CPA) if every PPT adversary's advantage is negligible in λ. The message is the adversary's own, so nothing about it is hidden; what the game hides is the key.

Construction 7.27 (ElGamal encryption). Fix $\mathsf{pp} = (G, P, q)$ as in Construction 7.16, written

additively, and take group elements as messages; write EG for the scheme. A key pair carries its own base H: in the *fixed-base* variant $H = P$; in the *per-key-base* variant KeyGen samples $H \xleftarrow{\$} G \setminus \{O\}$, the generalised interface of Remark 7.7.

- KeyGen: choose H as above and $x \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$; output $\mathsf{sk} = x$ and $\mathsf{pk} = (H, Y)$ with $Y = [x]H$.
- $\mathrm{Enc}_{(H,Y)}(M)$: sample $r \xleftarrow{\$} \mathbb{Z}/q\mathbb{Z}$; output $(R, C) = ([r]H,\ [r]Y + M)$.
- $\mathrm{Dec}_x(R, C)$: output $C - [x]R$.

Correctness is the scalar identity in the proof of Proposition 7.17, valid for any base in G: $[x]R = [x]([r]H) = [r]([x]H) = [r]Y$, so $C - [x]R = M$. The scheme is the static–ephemeral flow of §7.2 with the shared secret $[rx]H$ spent as a one-time pad on a group element; the fixed-base variant is ElGamal's original scheme transposed from $\mathbb{F}_p^\times$ to a prime-order group.

Theorem 7.28 (ElGamal is key-private under DDH)**.** *For ElGamal over G (Construction 7.27, either base variant), every adversary $\mathcal{A}$ yields two distinguishers $\mathcal{B}_0, \mathcal{B}_1$ against DDH in G (Definition 2.5, read with $[x]P$ in place of g^x as in Construction 7.16), each running in the time of $\mathcal{A}$ plus one key generation and four scalar multiplications, with*

$$\mathrm{Adv}^{\text{ik-cpa}}_{\mathrm{EG}}(\mathcal{A}) \;\le\; \mathrm{Adv}_{\mathrm{ddh}}(\mathcal{B}_0) + \mathrm{Adv}_{\mathrm{ddh}}(\mathcal{B}_1).$$

Under the DDH assumption ElGamal is therefore IK-CPA: a ciphertext is computationally independent of the public key it was made for.

Proof. The proof is a hybrid through a game that uses no key at all. Write H_0 and H_1 for the IK-CPA game with β fixed to 0 and to 1, and p_0, p_1 for the probability that $\mathcal{A}$ outputs 1 in each, so that $\mathrm{Adv}^{\text{ik-cpa}}_{\mathrm{EG}}(\mathcal{A}) = |p_0 - p_1|$. Let H_* be the game whose challenger, after the same key generation and the same message M from $\mathcal{A}$, answers with $(R, C) = (U_1, U_2 + M)$ for $U_1, U_2 \xleftarrow{\$} G$ independent and uniform, and let p_* be the corresponding probability. The challenge of H_* is computed without reference to either key, and the triangle inequality $|p_0 - p_1| \le |p_0 - p_*| + |p_* - p_1|$ reduces the claim to bounding each hop by a DDH advantage.

The hop $\mathsf{H}_0 \to \mathsf{H}_*$. The distinguisher $\mathcal{B}_0$ receives a triple $(X, Y, Z) = ([a]P, [b]P, [c]P)$ with $c = ab$ or c uniform. It samples $s \xleftarrow{\$} (\mathbb{Z}/q\mathbb{Z})^\times$ in the per-key-base variant and sets $s = 1$ in the fixed-base variant, plants the challenge in key 0 as

$$H_0 = [s]P, \qquad Y_0 = [s]X = [a]H_0,$$

generates $(x_1, (H_1, Y_1))$ honestly, and hands $\mathcal{A}$ the pair $((H_0, Y_0), (H_1, Y_1))$. For invertible s the base $[s]P$ is uniform on $G \setminus \{O\}$, and $Y_0 = [a]H_0$ with a uniform is an honest public key whose private key a the distinguisher never learns—nor needs, the game offering no decryption. On receiving M it replies with

$$(R, C) = ([s]Y,\ [s]Z + M)$$

and outputs $\mathcal{A}$'s bit. If $c = ab$, then $R = [sb]P = [b]H_0$ and $[s]Z = [sab]P = [b]([a]H_0) = [b]Y_0$, so (R, C) is an honest encryption under key 0 with ephemeral $r = b$ uniform: the view of $\mathcal{A}$ is exactly H_0. If c is uniform, then $[s]Z = [sc]P$ is uniform in G and independent of everything else because s is invertible, and $R = [b]H_0$ is uniform in G and independent of everything else because H_0 generates G and b is independent of a, c, s, of key 1, and of M, which $\mathcal{A}$ chose before seeing R: the view is exactly H_*. Hence $|p_0 - p_*| = \mathrm{Adv}_{\mathrm{ddh}}(\mathcal{B}_0)$.

The hop $\mathsf{H}_* \to \mathsf{H}_1$ is the same with the challenge planted in key 1 and key 0 generated honestly, giving $\mathcal{B}_1$ with $|p_* - p_1| = \mathrm{Adv}_{\mathrm{ddh}}(\mathcal{B}_1)$. Each distinguisher performs one honest key generation and the four scalar multiplications $[s]P, [s]X, [s]Y, [s]Z$ beyond running $\mathcal{A}$. Summing the two hops gives the bound. □

Remark 7.29 (What key privacy buys upstairs). The hybrid transfers verbatim to Construction 7.22: planting the challenge in the recipient's key and the ephemeral share exactly as above makes the shared secret S uniform, and once it is, the pair (E_{pk}, c) with $c = \mathrm{Enc}_{\mathrm{KDF}(S,\mathit{context})}(N, \mathit{ad}, m)$ is computed without reference to the recipient's key—*provided* the KDF context names no recipient key, which it must not. The deployed KDF takes $(\mathsf{sharedSecret}, \mathsf{epk})$ (§6.6): the shared secret is the keying material, the context is epk under a personalisation label, and no key of the recipient enters. The protocol specification requires the composed note-encryption scheme to be key-private (§"Key Derivation"): a chain scanner holding a note ciphertext against every address she can enumerate learns nothing about which one it was sent to. The second use is the group-element share of diversified-address unlinkability. A diversified address $(G_d, \mathsf{pk_d}) = (G_d, [\mathsf{ivk}]\, G_d)$ (Remark 7.30) is a per-key-base ElGamal public key, and a fresh address of the same ivk over a new base $G_{d'}$ has, when that base is modelled as a uniform group element, the distribution of an encryption of $\mathcal{O}$ under that key; telling which of two authorities a third address belongs to is then the IK-CPA game, and Theorem 7.28 answers it. The protocol specification sets up that correspondence, modelling the group hash behind the base as a random oracle (§"$\mathsf{DiversifyHash}^{\mathsf{Sapling}}$ and $\mathsf{DiversifyHash}^{\mathsf{Orchard}}$ Hash Functions"); the *Wallet Guide* states the resulting unlinkability requirement (§"Diversifier indices").

7.9 Shielded-note delivery as deployed hybrid encryption

The chain assembled in this section is not an analogy for what the deployed protocol does; it is the deployed protocol, run in band on the public ledger.

Remark 7.30 (Note delivery is HPKE on the ledger). A shielded address embeds a static public key $\mathsf{pk_d} = [\mathsf{ivk}]\, G_d$ on the protocol's curve (Pallas, in Orchard), where the private scalar ivk is the recipient's incoming viewing key. The base G_d is not a protocol-wide generator but a per-address *diversified base*, the hash-to-curve image (§3.7) of the address's diversifier—the generalised interface of Remark 7.7 exists precisely to accommodate it (`DiversifiedTransmissionKey` is derived as $\mathsf{pk_d} = [\mathsf{ivk}]\, g_d$); the *Ironwood Guide* develops the key hierarchy (§"Viewing keys") and address encoding (§"Diversified addresses") behind these bases. The sender runs Construction 7.22 over that base, with one deployed refinement: the ephemeral scalar esk is not sampled directly but derived by $\mathsf{PRF}^{\mathsf{expand}}$ (Remark 5.14) from the note's random seed (ZIP 212; `RandomSeed::esk_inner` applies `PrfExpand::ORCHARD_ESK`), so that the uniform-ephemeral hypothesis of Theorem 7.24 is met through the PRF's pseudorandomness and the recipient can later re-derive esk from the delivered seed. The sender publishes $\mathsf{epk} = [\mathsf{esk}]\, G_d$ in the transaction, derives

$$K = \mathrm{KDF}([\mathsf{esk} \cdot \mathsf{ivk}]\, G_d,\ \mathsf{epk})$$

by the personalised BLAKE2b call of §6.6, and AEAD-encrypts the note plaintext—a version byte, the recipient's diversifier, the value, the note's random seed, and the memo—under K with $\mathsf{ChaCha20\text{-}Poly1305}$ (Construction 6.18), attaching the ciphertext to the transaction. The nonce is fixed to zero and the associated data is empty, safe because each K keys exactly one encryption (protocol specification §"Encryption (Sapling and Orchard)"; the `zcash_note_encryption` crate's `encrypt_note_plaintext`, which chains `ka_agree_enc`, `kdf`, and `ChaCha20Poly1305` with the zero nonce; the Orchard instantiation of the `Domain` trait is in the `orchard` crate).

Chain scanners see (epk, c) in every shielded output, but lacking ivk they learn nothing (Theorem 7.24); the recipient *trial-decrypts*, deriving $K' = \mathrm{KDF}([\mathsf{ivk}]\, \mathsf{epk}, \mathsf{epk})$ for each incoming transaction and keeping those outputs whose AEAD tag verifies *and* whose plaintext passes two consistency checks: the note commitment recomputed from it must match the commitment on chain,

and the esk re-derived from its random seed must reproduce the published epk—the notes addressed to them (protocol specification §"Decryption using an Incoming Viewing Key (Sapling and Orchard)"). Non-interactivity is essential, and it is exactly the static–ephemeral flow of §7.2: the recipient need never have been online when the note was sent. Authenticity of the static key—the concern of §7.5—is supplied by mechanism (2) of Remark 7.15: the address is part of the recipient's verified payment credential, so the man-in-the-middle substitution has nowhere to stand; and, as noted there, this authenticates the recipient rather than the sender. The complementary guarantees on the note's *contents*—that the value is well-formed and conserved, that the commitment on chain matches the plaintext delivered—come not from the encryption but from the accompanying zero-knowledge proof.

The primitive toolbox now covers the delivery of a secret to a party who was never online to negotiate it. What travels alongside that ciphertext—the authorising signatures over the transaction that carries it—is the business of the next section.

8 Digital signatures

The adversary of this section is the *forger*. It watches the network accept spend after spend, each authorised by a signature; it may coax the legitimate signer into signing any messages of its choosing; and it wins if it can then exhibit one signature the signer never made—a spend of coins it does not control, an authorisation conjured from public data alone. Every verifier in a consensus network checks every signature, so the forger's target is public and its victory is total: a single forged spend authorisation steals, and a single forged balance certificate mints value from nothing. The primitive that defeats it, the digital signature, is the last cryptographic object of this volume built directly on the discrete logarithm; it will also turn out, in its Schnorr form, to be the first *proof of knowledge* we construct—the bridge to the proof systems of Section 9.

8.1 Signatures versus MACs

The message authentication codes of §6.3 already defeat a forger, but only between two parties who share a secret key: verification runs Vrfy_k, so whoever can *check* a tag can also *mint* one. Authentication by MAC is therefore private and repudiable—a verifier cannot exhibit the tag to a third party as evidence, since the verifier could have produced it. A blockchain inverts every one of these constraints. A transaction's authorisation must be checked by every full node, none of whom the signer has met and none of whom may be capable of forging it; and the check must be transferable, so that a block, once validated, convinces everyone else.

A *digital signature* meets these demands by splitting the key. Signing uses a secret key *sk*; verification uses a matching *public* key *pk* that may be published to the world. Anybody can verify; only the holder of *sk* can sign; and because verification needs no secret, a valid signature is *non-repudiable* evidence that the holder of *sk* signed. The price is computational: where the Carter–Wegman MAC of §6.3 was information-theoretically secure, every signature scheme in this section rests on the hardness of the discrete logarithm.

Zcash's Orchard protocol uses two closely related signature schemes, both Schnorr-type schemes over an elliptic curve, and both derived from the one identification protocol this section develops (protocol specification § 5.4.7):

- the *spend authorisation signature*, whose key pair can be *re-randomised* so that successive uses are unlinkable (protocol specification § 4.15), and
- the *binding signature*, which simultaneously certifies the consistency of a transaction's value commitments and proves knowledge of a discrete logarithm (protocol specification § 4.14).

Conventions for this section. Throughout, $\mathbb{G}$ is a cyclic group of prime order q, written additively, with a fixed generator G; scalar multiplication is $[x]P$ for $x \in \mathbb{F}_q$ and $P \in \mathbb{G}$. Concretely $\mathbb{G}$ is the Pallas group of §2.6, whose order is the 255-bit prime q, so scalars live in the field $\mathbb{F}_q$. Because $\mathbb{G}$ has prime order and G generates it, the map $x \mapsto [x]G$ is a bijection $\mathbb{F}_q \to \mathbb{G}$: the discrete logarithm of any $P \in \mathbb{G}$ to base G always *exists* and is *unique*. Hardness of computing it is the DLog assumption of §2.1, transcribed additively.

8.2 Syntax and security goal

Definition 8.1 (Digital signature scheme)**.** A *digital signature scheme* is a triple of PPT algorithms $\Pi = (\mathsf{KeyGen}, \mathsf{Sign}, \mathsf{Verify})$ with, for each security parameter λ, a message space $\mathcal{M}_\lambda$:

- $\mathsf{KeyGen}(1^\lambda)$ outputs a key pair (pk, sk);
- $\mathsf{Sign}(sk, m)$, for $m \in \mathcal{M}_\lambda$, outputs a signature σ;
- $\mathsf{Verify}(pk, m, \sigma)$ is deterministic and outputs a bit.

Correctness requires that for every λ, every (pk, sk) in the support of $\mathsf{KeyGen}(1^\lambda)$, and every $m \in \mathcal{M}_\lambda$,

$$\Pr[\mathsf{Verify}(pk, m, \mathsf{Sign}(sk, m)) = 1] = 1,$$

the probability over the coins of Sign. Probability-1 correctness is called *perfect*; a scheme achieving only overwhelming-probability correctness tolerates a $\mathsf{negl}(\lambda)$ failure rate. Every scheme in this section is perfectly correct.

The security goal transcribes the MAC forgery game of Definition 6.8 into the public-key setting: the adversary now *receives* the verification key, and its oracle access models a signer who can be induced to sign arbitrary messages.

Definition 8.2 (Existential unforgeability under chosen-message attack)**.** For a signature scheme Π and adversary $\mathcal{A}$, the game $\mathsf{Forge}^{\mathcal{A}}_{\Pi}(\lambda)$ runs as follows.

1. The challenger runs $(pk, sk) \leftarrow \mathsf{KeyGen}(1^\lambda)$.
2. The adversary $\mathcal{A}$ runs on input pk with oracle access to $\mathsf{Sign}(sk, \cdot)$; let Q be the set of messages it submits to the oracle, and let $(m^\star, \sigma^\star)$ be its output.
3. The game outputs 1 iff $\mathsf{Verify}(pk, m^\star, \sigma^\star) = 1$ and $m^\star \notin Q$.

Scheme Π is *existentially unforgeable under chosen-message attack* (EUF-CMA) if for every PPT $\mathcal{A}$,

$$\mathsf{Adv}^{\mathsf{forge}}_{\Pi}(\mathcal{A}, \lambda) := \Pr[\mathsf{Forge}^{\mathcal{A}}_{\Pi}(\lambda) = 1] \in \mathsf{negl}(\lambda).$$

Two features of the definition deserve emphasis. The oracle access is *adaptive* and bounded only by the adversary's polynomial running time: it may choose each query after seeing all previous signatures. And the forgery is *existential*: the adversary wins with a valid signature on *any* message never submitted to the oracle, however meaningless—the definition does not ask the forged message to be useful, so security under it protects even messages the signer would consider absurd.

Definition 8.3 (Strong unforgeability)**.** The *strong* variant, sUF-CMA, replaces the winning condition $m^\star \notin Q$ by: the pair $(m^\star, \sigma^\star)$ was not among the message–signature pairs returned

by the oracle.

Strong unforgeability forbids producing even a *new* signature on an *already-signed* message. Plain EUF-CMA permits a scheme in which, given one valid signature, anyone can cheaply compute a second, different valid signature on the same message; such *malleability* is harmless when a signature only certifies origin, but fatal whenever a signature doubles as a unique token—certain anti-replay mechanisms, for instance, identify a message by its signature and break if signatures are malleable.

8.3 The hash-and-sign paradigm

The textbook mechanisms—RSA, Schnorr, ECDSA—sign only fixed-size inputs: an element of $\mathbb{Z}/N\mathbb{Z}$ for RSA, a scalar in $\mathbb{F}_q$ for Schnorr and ECDSA. Arbitrary messages are accommodated by hashing first, and the composition is provably safe.

Theorem 8.4 (Hash-and-sign). *Let Π' be a signature scheme with message space $\mathcal{Y}_\lambda$, and let $H\colon \{0,1\}^* \to \mathcal{Y}_\lambda$ be drawn from a collision-resistant family (Definition 3.7). Define Π with message space $\{0,1\}^*$ by*

$$\mathrm{Sign}(sk, m) := \mathrm{Sign}'(sk, H(m)), \qquad \mathrm{Verify}(pk, m, \sigma) := \mathrm{Verify}'(pk, H(m), \sigma).$$

If Π' is EUF-CMA and H is collision resistant, then Π is EUF-CMA; concretely, for every adversary $\mathcal{A}$ against Π there are comparably efficient $\mathcal{B}_1$ against H and $\mathcal{B}_2$ against Π' with

$$\Pr[\mathcal{A}\ \textit{wins}] \ \le\ \mathrm{Adv}_H^{\mathrm{col}}(\mathcal{B}_1) + \mathrm{Adv}_{\Pi'}^{\mathrm{forge}}(\mathcal{B}_2).$$

Proof. Let $(m^\star, \sigma^\star)$ be a valid forgery of $\mathcal{A}$ on a fresh message $m^\star \notin Q$, and split on the digest. *Case 1: the digest is reused*, i.e. $H(m^\star) = H(m)$ for some queried $m \in Q$. Then $m^\star \neq m$ but their digests agree, so the pair $(m^\star, m)$ is a collision in H; the reduction $\mathcal{B}_1$, which generates a key pair with KeyGen$'$ itself and answers $\mathcal{A}$'s signing queries by signing digests under that secret key—the collision game grants no signing oracle—outputs it. *Case 2: the digest is fresh*, i.e. $H(m^\star)$ differs from the digest of every queried message. Then $(H(m^\star), \sigma^\star)$ is a valid forgery against Π' on a message never submitted to *its* signing oracle, and $\mathcal{B}_2$ outputs it. The two cases are exhaustive, so the union bound on the winning event gives the stated inequality. □

Remark 8.5 (What "hash-and-sign" hides). In all the Schnorr-type schemes below, the hash absorbs more than the message: the public key and the per-signature commitment value enter the hash alongside m. Hashing the public key prevents key-substitution (duplicate-signature) attacks, in which a signature valid under one key is exhibited as valid under another; hashing the commitment is exactly what the Fiat–Shamir transform of §8.5 requires. "Hash-and-sign" in practice therefore means hashing more than the message; Theorem 8.4 captures only the message-compression part of the discipline.

8.4 The Schnorr identification protocol

Every scheme in the rest of this section is a costume worn by a single interactive protocol, in which a prover convinces a verifier that it *knows* the discrete logarithm of a public point— without revealing anything about it. We develop the protocol self-contained, with concrete statements and proofs; the general vocabulary it exemplifies (*Sigma-protocols*, of which it is the canonical instance: three-move, public-coin, honest-verifier zero-knowledge (HVZK) proofs of knowledge) is deliberately deferred to Section 9, where it is developed for arbitrary relations.

The object of the protocol is the discrete-log relation

$$R_{\mathrm{dl}} = \{\, (X, x) \in \mathbb{G} \times \mathbb{F}_q \ :\ X = [x]G \,\} :$$

the *statement* is a public point X, the *witness* its discrete logarithm x.

Construction 8.6 (Schnorr identification)**.** Fix $\mathbb{G} = \langle G \rangle$ of prime order q. The prover holds a secret key $x \in \mathbb{F}_q$; both parties hold the public key $X = [x]G$. The protocol has three moves.

1. *Commitment.* The prover samples $r \xleftarrow{\$} \mathbb{F}_q$ uniformly and sends $R = [r]G$. We call R the *commitment* (or *nonce point*) and r the *nonce*.
2. *Challenge.* The verifier samples $c \xleftarrow{\$} \mathbb{F}_q$ uniformly and sends c.
3. *Response.* The prover sends $s = r + cx$ in $\mathbb{F}_q$.

The verifier accepts iff

$$[s]G \;=\; R + [c]X. \tag{2}$$

A triple (R, c, s) satisfying (2) is an *accepting transcript* for X. The verifier's only message is a uniformly random value chosen independently of R, so the protocol is *public-coin*: the verifier keeps no secrets and exercises no discretion.

Three properties make this protocol what it is: honest runs always accept; answering two distinct challenges on one commitment is possible only for a party who knows x; and a transcript reveals nothing about x. We prove each in turn.

Proposition 8.7 (Perfect completeness)**.** *If prover and verifier are honest, the verifier accepts with probability* 1.

Proof. With $R = [r]G$, $X = [x]G$ and $s = r + cx$, linearity of scalar multiplication gives

$$[s]G = [r + cx]G = [r]G + [cx]G = R + [c]([x]G) = R + [c]X,$$

which is exactly (2). The computation holds identically for every choice of r and c. □

Theorem 8.8 (2-special soundness)**.** *There is an efficient* extractor *that, given two accepting transcripts (R, c, s) and (R, c', s') for the same statement X with the same commitment R but distinct challenges $c \neq c'$, outputs the discrete logarithm x of X.*

Proof. Both transcripts satisfy (2); subtracting the two equations in $\mathbb{G}$ cancels R:

$$[s - s']G \;=\; [c - c']X \;=\; [(c - c')x]G.$$

Since G generates a group of prime order q, equality of the two multiples forces the equality of scalars $s - s' = (c - c')x$ in $\mathbb{F}_q$. The hypothesis $c \neq c'$ makes $c - c'$ a nonzero element of the field $\mathbb{F}_q$, hence invertible, so the extractor outputs

$$x \;=\; (s - s')\,(c - c')^{-1} \in \mathbb{F}_q.$$

Verification that $X = [x]G$ is direct, and the whole computation costs a single field inversion. □

Theorem 8.8 is the central mechanism of this section. It makes the protocol a proof *of knowledge*: below, it converts a convincing prover into an extractable witness; after the Fiat–Shamir transform it is the source of unforgeability; and in §8.6 it reappears with its sign reversed, as the reason a signer must never reuse a nonce.

Remark 8.9 (From special soundness to knowledge, by rewinding). Suppose a (possibly dishonest) prover makes the verifier accept with probability ε noticeably larger than $1/q$—the probability of simply guessing the single challenge for which a prepared response works. Run the prover once: record its commitment R, feed it a uniform challenge c, and note whether its response accepts. Now *rewind* it to the moment just after it sent R—restore its internal state, which is possible because we run the prover as a subroutine—and feed a fresh independent challenge c'. A standard averaging argument shows that with probability roughly ε^2, both runs accept and $c \neq c'$; the argument is the *heavy-row lemma*, an instance of Markov's inequality applied to the matrix whose rows are the prover's random coins and whose columns are challenges (*Math Guide*, §"Random variables and expectation"). The two accepting transcripts share R, so the extractor of Theorem 8.8 recovers x. Making the verifier accept with non-trivial probability is therefore, up to rewinding, the same as *knowing* x. The same rewinding argument, sharpened into the forking lemma, carries the signature analysis of §8.6.

> **Theorem 8.10** (Special honest-verifier zero knowledge). *There is an efficient simulator* Sim *that, on input a statement $X \in \mathbb{G}$ and a challenge $c \in \mathbb{F}_q$, outputs an accepting transcript (R, c, s) distributed* identically *to an honest execution conditioned on the challenge being c. Consequently, for a uniform challenge, the distribution of simulated transcripts coincides exactly with the distribution of real ones.*

Proof. The simulator works *backwards*: it samples $s \xleftarrow{\$} \mathbb{F}_q$ uniformly, sets

$$R := [s]G - [c]X,$$

and outputs (R, c, s)—accepting by construction, since (2) rearranges to exactly this definition of R.

Compare distributions with the challenge fixed to c. In a real honest transcript, r is uniform on $\mathbb{F}_q$, $R = [r]G$, and $s = r + cx$; the map $r \mapsto r + cx$ is a translation of $\mathbb{F}_q$, hence a bijection, so s is uniform on $\mathbb{F}_q$, and $R = [s - cx]G = [s]G - [c]X$ is the same deterministic function of s that the simulator uses. The real transcript *is*: sample s uniform, set $R = [s]G - [c]X$—precisely the simulated distribution. The two distributions are identical, not merely statistically close, and averaging over a uniform c preserves the identity. □

Remark 8.11 (Why honest-verifier zero knowledge suffices). The simulator of Theorem 8.10 matches the real distribution only when the challenge is sampled independently of R. A cheating verifier could instead choose c as a function of R, and against such a verifier the simulator—which commits to R only *after* seeing c—no longer obviously works. For our purposes honest-verifier zero knowledge is exactly enough: after the Fiat–Shamir transform the challenge is a *hash* of R, and in the random oracle model that hash behaves like an honest, R-independent coin. The deeper point of the theorem is what the simulator's existence *proves*: since anyone, without x, can manufacture transcripts with the true distribution, a transcript carries no information about x beyond what the statement X already reveals.

8.5 From identification to signature via the Fiat–Shamir transform

An identification protocol convinces one verifier, once, interactively. A signature must convince everyone, forever, on its own. The *Fiat–Shamir transform* removes the interaction by making the prover compute the challenge itself—as a hash of everything the verifier would have seen before issuing it, with the message folded in so that the resulting object certifies m. The transform is generic: it converts *any* Sigma-protocol into a non-interactive proof, and, with a message in the hash, into a signature. This subsection owns only the Schnorr instance; the general transform, its subtleties and its failure modes are the business of Section 9 (§"The Fiat–Shamir transform: from interactive to non-interactive").

Construction 8.12 (Schnorr signature)**.** Fix $\mathbb{G} = \langle G \rangle$ of prime order q and a hash function $H\colon \{0,1\}^* \to \mathbb{F}_q$. Messages are arbitrary bit strings.

- $\mathrm{KeyGen}(1^\lambda)$: sample $x \xleftarrow{\$} \mathbb{F}_q$, set $X = [x]G$, output $(pk, sk) = (X, x)$.
- $\mathrm{Sign}(x, m)$: sample $r \xleftarrow{\$} \mathbb{F}_q$, set $R = [r]G$, compute the challenge
$$c = H(R \,\|\, X \,\|\, m),$$
set $s = r + cx$ in $\mathbb{F}_q$, and output $\sigma = (R, s)$.
- $\mathrm{Verify}(X, m, (R, s))$: recompute $c = H(R \,\|\, X \,\|\, m)$ and accept iff $[s]G = R + [c]X$.

Proposition 8.13 (Perfect correctness)**.** *The Schnorr signature scheme is perfectly correct.*

Proof. For an honest signature (R, s) with $R = [r]G$, $c = H(R \,\|\, X \,\|\, m)$ and $s = r + cx$, the verifier recomputes the same c—the challenge is a deterministic function of (R, X, m), all of which the verifier holds—and the check $[s]G = R + [c]X$ holds by the completeness computation of Proposition 8.7. □

Remark 8.14 (Two conventions)*.* Two presentational choices in Construction 8.12 deserve note.

1. The signature transmits (R, s) and omits c, which the verifier recomputes. The space-saving variant transmits (c, s) instead: the verifier recovers $R = [s]G - [c]X$ (the simulator's equation, read as reconstruction) and checks $H(R \,\|\, X \,\|\, m) = c$. The two are interchangeable; the deployed schemes below use the (R, s) form.
2. The hash takes the public key X alongside R and m—*key-prefixing*. The single-key unforgeability proof of §8.6 does not require it, but it forecloses attacks that relate signatures under different keys (Remark 8.5), it is standard in modern instantiations such as EdDSA, and it earns its keep in the re-randomised setting of §8.8.

8.6 Security in the random oracle model and the forking lemma

What must a forger of Schnorr signatures accomplish? A valid pair $(R^\star, s^\star)$ on a fresh message is an accepting transcript of the identification protocol whose challenge is $H(R^\star \| X \| m^\star)$—a value the forger cannot choose freely if the hash is honest. The security argument makes "honest" precise by idealising H, then turns the rewinding of Remark 8.9 into a quantitative tool.

Definition 8.15 (Random oracle, general range)**.** The random oracle of Definition 3.13 generalises verbatim to an arbitrary finite range: a random oracle with range Y is a function $O\colon \{0,1\}^* \to Y$ drawn uniformly from the set of all such functions, realised lazily by sampling each new answer $O(z) \xleftarrow{\$} Y$ on first query. For Schnorr signatures the range is the challenge space $Y = \mathbb{F}_q$, and the ROM for signatures models $H(\cdot) = O(\cdot)$, with every party—signer, verifier, forger, and reduction—sharing oracle access (Definition 1.26).

The model remains a heuristic, with the uninstantiability caveat and the working stance of Remark 1.29: a ROM proof rules out all attacks that treat the hash as a black box—the overwhelming majority—and it is the standard yardstick for Fiat–Shamir signatures. The reductions below exploit both powers of Proposition 3.14: *programmability* (the reduction chooses oracle answers

on the fly, provided they appear uniform) and *observability* (the reduction sees every query the adversary makes).

The first tool disposes of the signing oracle: in the ROM, the zero-knowledge simulator of Theorem 8.10 answers signing queries without the secret key.

Lemma 8.16 (Signature simulation)**.** *In the ROM there is an efficient simulator that, given only the public key X and the ability to program H, answers signing queries with signatures distributed identically to genuine ones, except that it aborts with probability at most $q_s(q_s + q_h)/q$ over an attack making q_s signing and q_h hash queries.*

Proof. To sign m without x: sample $c \xleftarrow{\$} \mathbb{F}_q$ and $s \xleftarrow{\$} \mathbb{F}_q$ uniformly, set $R := [s]G - [c]X$—the HVZK simulator of Theorem 8.10—then *program*

$$H(R \parallel X \parallel m) := c,$$

and output (R, s). The signature verifies by construction, and by Theorem 8.10 its distribution matches a real signature's exactly: in both cases s is uniform, and c is uniform because a fresh oracle answer is.

Programming fails only if the entry $H(R \parallel X \parallel m)$ is already defined—by an earlier adversary hash query or an earlier simulated signature. The point $R = [s]G - [c]X$ is a fixed translate of $[s]G$ for uniform s, hence uniform over the q elements of $\mathbb{G}$; at most $q_h + q_s$ oracle entries exist at any point of the attack, so a given signing query collides with probability at most $(q_h + q_s)/q$, and the union bound over q_s signing queries gives the stated abort probability. Conditioned on no abort, the adversary's view is identical to the real game. □

Lemma 8.16 reduces forgery to a pure discrete-log problem: the reduction can play the whole EUF-CMA game knowing only X. What remains is to convert one forgery into *two* accepting transcripts on the same commitment—the input that Theorem 8.8 demands. The quantitative engine is the forking lemma, in the game-based form due to Bellare and Neven.

Theorem 8.17 (General forking lemma)**.** *Fix an integer $Q \geq 1$, a finite set C with $|C| = h \geq 2$, and a randomised input generator* Gen*. Let B be a randomised algorithm that on input y and challenges $c_1, \dots, c_Q \in C$ returns a pair (I, out) with $I \in \{0, 1, \dots, Q\}$, where $I = 0$ means failure, and let*

$$acc := \Pr[I \geq 1 : y \leftarrow \mathrm{Gen};\ c_1, \dots, c_Q \xleftarrow{\$} C;\ (I, out) \leftarrow B(y, c_1, \dots, c_Q)].$$

Define the forking algorithm F_B on input y: pick coins ρ for B and $c_1, \dots, c_Q \xleftarrow{\$} C$; run $(I, out) \leftarrow B(y, c_1, \dots, c_Q; \rho)$; if $I = 0$, return $\perp$; otherwise resample $c'_I, \dots, c'_Q \xleftarrow{\$} C$ afresh and run $(I', out') \leftarrow B(y, c_1, \dots, c_{I-1}, c'_I, \dots, c'_Q; \rho)$ with the same *coins ρ; if $I' = I$ and $c_I \neq c'_I$, return (I, out, out'), else $\perp$. Then*

$$frk := \Pr[F_B(y) \neq \perp : y \leftarrow \mathrm{Gen}] \ \geq\ acc\left(\frac{acc}{Q} - \frac{1}{h}\right),$$

the probability also over the coins of F_B.

Proof idea. Condition on the coins ρ and the prefix $c_1, \dots, c_{I-1}$ up to the forking point. For each fixed prefix there is (averaging over the suffix) a set S of *good* values of c_I on which B succeeds with index I; write $p = |S|/h$. The fork succeeds at this prefix iff the first draw c_I and the independent re-draw c'_I both land in S and differ: two independent uniform draws land in S together with probability p^2, and subtracting the diagonal of equal draws costs at most $p \cdot 1/h$, so the per-prefix fork probability is at least $p(p - 1/h)$. Taking expectations over prefixes, $\mathbb{E}[p^2] \geq (\mathbb{E}[p])^2$, since

$\mathbb{E}[p^2] - (\mathbb{E}[p])^2 = \mathbb{E}[(p - \mathbb{E}[p])^2] \geq 0$ by linearity of expectation (*Math Guide*, §"Random variables and expectation"); this squares the average success probability. The index I ranging over Q possible positions introduces the factor $1/Q$, and collecting terms yields $frk \geq acc^2/Q - acc/h$. The squaring step is where the *squared* probability—and hence the tightness loss examined below—enters. □

Theorem 8.18 (EUF-CMA security of Schnorr signatures in the ROM)**.** *Model H as a random oracle with range $\mathbb{F}_q$. If an adversary $\mathcal{A}$ runs in time t, makes at most q_h hash queries and q_s signing queries, and forges with probability ε, then there is an algorithm $\mathcal{D}$ running in time approximately $2t$ that computes discrete logarithms in $\mathbb{G}$ with probability*

$$\varepsilon' \geq \frac{\varepsilon^2}{q_h + q_s + 1} - \frac{q_h + q_s + 1}{q} - \frac{q_s(q_h + q_s)}{q}.$$

In particular, if DLog is hard in $\mathbb{G}$, the Schnorr signature scheme is EUF-CMA in the ROM.

Proof structure. Wrap $\mathcal{A}$ as the algorithm B of Theorem 8.17, with input $y = X$—the DLog challenge, so Gen outputs $X = [x]G$ for uniform x—challenge set $C = \mathbb{F}_q$ (so $h = q$), and $Q = q_h + q_s + 1$ prepared challenges. Algorithm B runs $\mathcal{A}$ on $pk = X$, answering the j-th *distinct* hash query with c_j from its challenge list and answering signing queries by the simulator of Lemma 8.16, whose programming aborts cost the $q_s(q_h + q_s)/q$ term.

Suppose $\mathcal{A}$ outputs a valid fresh forgery $(m^\star, (R^\star, s^\star))$, and let $c^\star = H(R^\star \| X \| m^\star)$. The critical hash input $R^\star \| X \| m^\star$ must actually have been queried during the run: otherwise $c^\star$ is a uniform value $\mathcal{A}$ has never seen, and the verification equation $[s^\star]G = R^\star + [c^\star]X$ then holds only with probability $1/q$ over the oracle's choice—the $(q_h + q_s + 1)/q$ term collects these guessing events (the $+1$ covering the query B itself makes to verify the forgery). So B outputs the index I of the critical query together with $out = (R^\star, c^\star, s^\star, m^\star)$, and $acc \geq \varepsilon$ minus the loss terms above.

Now fork. Both runs of F_B share the coins ρ and the challenge prefix $c_1, \ldots, c_{I-1}$, which together determine $\mathcal{A}$'s behaviour up to the moment it produces the critical query; both runs therefore query the *same* critical input $R^\star \| X \| m^\star$ at index I, but receive different answers $c^\star \neq c'^\star$. Success of the fork yields two accepting transcripts $(R^\star, c^\star, s^\star)$ and $(R^\star, c'^\star, s'^\star)$ with the same commitment and distinct challenges, whereupon the extractor of Theorem 8.8 computes

$$x = (s^\star - s'^\star)(c^\star - c'^\star)^{-1},$$

the discrete logarithm of X. Substituting acc, $h = q$ and $Q = q_h + q_s + 1$ into $frk \geq acc^2/Q - acc/h$ gives the stated bound; the running time is two runs of $\mathcal{A}$ plus bookkeeping. □

Example 8.19 (The tightness loss, in numbers)**.** The dominant term of Theorem 8.18 is $\varepsilon' \gtrsim \varepsilon^2/q_h$: the reduction runs the forger twice and *squares* its success probability, divided by the number of hash queries. The reduction is therefore far from tight (Definition 1.22). Concretely, a forger succeeding with probability $\varepsilon = 2^{-40}$ after $q_h = 2^{60}$ hash queries yields a guaranteed DLog-solver advantage of only about

$$\varepsilon^2/q_h = 2^{-80}/2^{60} = 2^{-140},$$

a factor 2^{100} weaker than ε itself. The guarantee runs in the useful direction—any forger yields *some* DLog solver, so DLog hardness still implies security—but the quantitative content is feeble. Two lessons follow, and they must be kept apart. The conservative lesson (Remark 1.23): a designer who wants *this theorem* to underwrite a target signature-security level must provision $\mathbb{G}$ with roughly twice as many bits of DLog security as the target, since the reduction hands back only the square of the forger's advantage; and the squaring is intrinsic to the rewinding route—it enters at the squaring step of Theorem 8.17, so any analysis that runs the forger twice

and extracts by special soundness pays it. What the theorem does *not* establish is a matching attack: the loss belongs to the reduction, not provably to the scheme, so doubled parameters are the price of this proof, not a demonstrated necessity, and parameter selection in practice weighs concrete attack models and query bounds as well. Tighter reductions exist once the rewinding route is abandoned—under the *one-more* discrete logarithm assumption (that no efficient adversary, given $\ell + 1$ random elements of $\mathbb{G}$ and ℓ queries to a discrete-logarithm oracle, outputs all $\ell + 1$ logarithms), under DDH (Definition 2.5) together with key-prefixing, or in the idealised algebraic models of Remark 1.30.

Proposition 8.20 (Nonce reuse is fatal)**.** *Suppose a signer produces two Schnorr signatures (R, s) and (R, s') on distinct messages $m \neq m'$ with the* same *commitment R—a nonce r reused outright, or drawn twice from a faulty source. Then, except with probability $1/q$ over the challenge hash, anyone can compute the secret key from the two signatures.*

Proof. The challenges $c = H(R \| X \| m)$ and $c' = H(R \| X \| m')$ differ except with probability $1/q$, the chance that the random oracle answers the two distinct inputs identically. Given $c \neq c'$, the pairs (R, c, s) and (R, c', s') are two accepting transcripts sharing a commitment with distinct challenges, and the extractor of Theorem 8.8—run now by the *adversary*—outputs $x = (s - s')(c - c')^{-1}$. □

Special soundness, the source of unforgeability, is thus also its dark mirror: the security of the scheme rests entirely on the nonce being fresh and unpredictable for every signature. This is not a theoretical worry—it is the classic operational failure of deployed Schnorr-family schemes—and it motivates deriving nonces *deterministically* from the secret key and message, the defining idea of EdDSA, to which we now turn.

8.7 RedDSA: re-randomisable Schnorr for Orchard

Construction 8.21 (EdDSA, schematically)**.** EdDSA retains the Schnorr signature algebra of Construction 8.12 unchanged and hardens two implementation surfaces.

1. *Deterministic nonces.* The nonce is derived as a hash of a secret prefix (part of the expanded secret key) and the message, rather than sampled fresh: no faulty random-number generator can repeat a nonce across distinct messages and trigger Proposition 8.20.
2. *Key-prefixing.* The challenge hash absorbs the public key alongside R and m, as in Construction 8.12 and for the reasons of Remark 8.14.

In the ROM the deterministic nonce is, to anyone ignorant of the secret prefix, a fresh uniform value per message, so the unforgeability proof of Theorem 8.18 carries over essentially unchanged.

Zcash's signature scheme generalises EdDSA in two directions, and the generalisations are the whole point: the base point becomes a parameter, and the key becomes shiftable.

Construction 8.22 (RedDSA)**.** RedDSA is the Schnorr/EdDSA scheme with the base point $P \in \mathbb{G}$ a *parameter* of the scheme: different uses sign with respect to different generators.

- KeyGen: sample $x \xleftarrow{\$} \mathbb{F}_q$, output secret key x and public key $X = [x]P$.

- $\mathrm{Sign}(x, m)$: sample a fresh random byte string T and derive the nonce $r = H(T \parallel X \parallel m)$. This is *randomised*, hash-based nonce generation: unlike EdDSA's derivation, it is not deterministic from the signing key and message, so its safety rests on T being fresh and unpredictable, the hashing ensuring only that a repeated T cannot repeat a nonce across distinct messages. Set $R = [r]P$, $c = H(R \parallel X \parallel m) \in \mathbb{F}_q$, $s = r + cx$, and output $\sigma = (R, s)$.
- $\mathrm{Verify}(X, m, (R, s))$: accept iff $[s]P = R + [c]X$ with $c = H(R \parallel X \parallel m)$.

For a fixed base point, RedDSA *is* Schnorr/EdDSA, and its EUF-CMA security in the ROM is Theorem 8.18 with G replaced by P.

Remark 8.23 (RedDSA as deployed). The deployed scheme follows Construction 8.22 line for line (protocol specification § 5.4.7). The randomiser T is 80 uniform bytes, and both hashes are the function $\mathsf{H}^{\circledast}$: BLAKE2b-512 personalised with `Zcash_RedPallasH`, its 64-byte output reduced to $\mathbb{F}_q$ by wide reduction— the expand-then-reduce discipline of Construction 3.18, with the 512-bit width making the modular bias negligible. The nonce hash absorbs $T \parallel X \parallel m$ and the challenge hash $R \parallel X \parallel m$, key-prefixed in both cases.

Orchard instantiates RedDSA *twice*, both times as *RedPallas*—RedDSA over the Pallas curve—and the two instantiations differ *only* in the base point:

- the *spend authorisation* base point, derived by hashing to the curve with domain `z.cash:Orchard` and input `G`, for the re-randomisable signature of §8.8; and
- the *value-commitment randomness* base point, derived with domain `z.cash:Orchard-cv` and input `r`, for the binding signature of §8.9. This is byte-for-byte the randomness base of Orchard's value commitment— an identity that §8.9 shows is the entire design.

The curve varies only across protocol generations: Sapling's *RedJubjub* is the same scheme over the Jubjub curve; the curve, its two base points and the $\mathsf{H}^{\circledast}$ personalisation (`Zcash_RedJubjubH` against `Zcash_RedPallasH`) are all that distinguish the two.

8.8 Key re-randomisation and unlinkability

A shielded protocol faces a linkage problem that no ordinary signature scheme addresses. Spending a note requires proving spend authority, and the obvious design—publish the spender's public key and a signature under it—would stamp every spend by the same wallet with the same key, linking them all. Orchard's answer exploits the algebraic structure of the RedDSA public key: because the key is a homomorphic image of the secret, it can be *shifted* into a fresh disguise for every spend.

Definition 8.24 (Key re-randomisation). Fix a base point P. Given a secret key $x \in \mathbb{F}_q$ with public key $X = [x]P$, and a *randomiser* $\alpha \in \mathbb{F}_q$, the re-randomised signing and verification keys are

$$rsk := x + \alpha \in \mathbb{F}_q, \qquad rk := X + [\alpha]P \in \mathbb{G}.$$

Orchard publishes a fresh rk per spend, signs with the shifted secret key rsk, and proves in zero knowledge that rk is a valid re-randomisation of an authorised public key.

Proposition 8.25 (Re-randomisation consistency). *The pair (rsk, rk) is a valid RedDSA key pair: $rk = [rsk]P$, so a RedDSA signature under rsk verifies under rk.*

Proof. Linearity of scalar multiplication gives

$$[rsk]P = [x + \alpha]P = [x]P + [\alpha]P = X + [\alpha]P = rk.$$

The pair is thus exactly KeyGen-shaped, and the correctness of signing and verification (Proposition 8.13) applies verbatim. □

The enabling structure deserves a sentence of its own. The public key is a homomorphic image $[\cdot]P$ of the scalar, so shifting the secret by α shifts the public key by $[\alpha]P$—a quantity computable from α and P alone, *without* the secret key. A verifier, or a circuit, can therefore check the relationship between X and rk given α, while only the spender who knows x can sign under the shifted key.

> **Theorem 8.26** (Perfect unlinkability of re-randomised keys)**.** *Fix a base point P. For any public key $X \in \mathbb{G}$, if $\alpha \xleftarrow{\$} \mathbb{F}_q$ is uniform then $rk = X + [\alpha]P$ is uniformly distributed over $\mathbb{G}$. Consequently, in the game where an adversary—of unbounded computational power, and permitted to choose two public keys X_0, X_1 itself—receives $rk = X_b + [\alpha]P$ for a challenger's uniform bit b and fresh uniform α, and must guess b: the distributions of rk for $b = 0$ and $b = 1$ are* identical*, and no adversary guesses b with probability better than* $1/2$.

Proof. The point P generates $\mathbb{G}$, which has prime order q, so $\alpha \mapsto [\alpha]P$ is a bijection $\mathbb{F}_q \to \mathbb{G}$; for uniform α, the point $[\alpha]P$ is uniform over $\mathbb{G}$. Translation by the fixed element X is a bijection of $\mathbb{G}$ and carries the uniform distribution to itself, so $rk = X + [\alpha]P$ is uniform over $\mathbb{G}$—for *every* X, hence independently of X. The two conditional distributions of rk coincide exactly, so the adversary's view is statistically independent of b; no test distinguishes identical distributions, and the best any distinguisher can do is guess, succeeding with probability exactly $1/2$. □

The guarantee is information-theoretic—perfect, resting on no computational assumption—but its *precondition* matters: α must be sampled uniformly and freshly per use, and must be kept secret. In Orchard the transaction builder samples $\alpha \xleftarrow{\$} \mathbb{F}_q$ afresh for each spend and reveals it only to the proving circuit and the signer.

Remark 8.27 (Scope of unlinkability). Theorem 8.26 covers only the re-randomised *key*. Full unlinkability of a spend further requires that the accompanying signature and zero-knowledge proof leak nothing beyond their statements, that the randomiser α stays secret, and that no other transaction field correlates two spends. In Orchard the protocol publishes rk with each Action (the unit that spends one note and creates one) while proving, inside the Halo 2 circuit, knowledge of a witness for $rk = X + [\alpha]P$ with X a public key whose holder is authorised to spend the note (Orchard's *spend validating key*)—the circuit constrains rk as a public input—and the network verifies the spend authorisation signature against the public rk. Signing under rk requires $rsk = x + \alpha$, that is, knowledge of *both* x and α; only the legitimate key-holder in possession of this spend's α can sign.

Remark 8.28 (Unforgeability under re-randomised keys, and the key-prefixing subtlety). Does unforgeability survive re-randomisation? A forgery (R, s) under rk satisfies $[s]P = R + [c]\,rk$ with $c = H(R \,\|\, rk \,\|\, m)$. One is tempted to subtract $[\alpha]P$ from the verification equation: $[s - c\alpha]P = R + [c]X$, apparently a forgery under the original key X. But the identity holds with $c = H(R \,\|\, rk \,\|\, m)$, while the base verifier recomputes $H(R \,\|\, X \,\|\, m)$—the plain subtraction converts forgeries only for the *non-key-prefixed* variant of Schnorr. For key-prefixed RedDSA one argues in the ROM by either of two routes. (i) The pair $(rsk, rk) = (x + \alpha,\ X + [\alpha]P)$ is itself a base-scheme key pair (Proposition 8.25) whose public key is uniform (Theorem 8.26), so a forger against rk is *literally* a forger against the base scheme at the key rk; the reduction, which knows α, recovers $x = rsk - \alpha$ from the extracted rsk. (ii) When the adversary sees signatures under several randomisers α_i, one re-runs the forking argument of Theorem 8.18 on the critical query $H(R^\star \,\|\, rk \,\|\, m^\star)$, simulating signing under the remaining keys by oracle programming (Lemma 8.16). A pitfall marks the boundary: a naive translation that programs $H(R \,\|\, rk \,\|\, m) := H_{\text{base}}(R \,\|\, X \,\|\, m)$ must be a carefully injective do-

main swap, and it fails outright with multiple randomisers—distinct rk_i-prefixed inputs would collide on a single X-prefixed point, a detectable deviation from a random oracle.

8.9 Binding signatures as a proof of knowledge of a discrete logarithm

The second RedPallas instantiation guards a different treasure: not who may spend, but whether value is conserved. A shielded transaction hides its amounts inside commitments; the forger of this subsection is a counterfeiter who would arrange commitments whose hidden values do not balance—minting money—while still producing whatever certificate the protocol demands. The defence consumes the homomorphism established in §4.6 and turns this section's central lesson—a Schnorr signature is a proof of knowledge of a discrete logarithm—from analysis into design.

The value-commitment setup. Shielded protocols commit to value with the homomorphic Pedersen commitment

$$\mathrm{cm}(v, \rho) \;=\; [v]V + [\rho]P,$$

where V and P are *independent* generators of $\mathbb{G}$—independent meaning that no party knows the discrete logarithm of one to the base of the other, arranged by deriving both by hashing to the curve (Construction 3.24)—and ρ is a random blinding scalar. (In the notation of the `orchard` crate and of Remark 4.21, the randomness base here called P is the generator R; this section reserves R for the signature nonce point. The base P is exactly the binding base point of Remark 8.23.) By Theorem 4.20, the group sum of commitments commits to the sum of values with the sum of blindings.

Sapling commits *per note*: a transaction carries input commitments $\mathrm{cm}(v_i^{\mathrm{in}}, \rho_i^{\mathrm{in}})$ and output commitments $\mathrm{cm}(v_j^{\mathrm{out}}, \rho_j^{\mathrm{out}})$, and the net commitment is

$$B \;=\; \sum_i \mathrm{cm}(v_i^{\mathrm{in}}, \rho_i^{\mathrm{in}}) \;-\; \sum_j \mathrm{cm}(v_j^{\mathrm{out}}, \rho_j^{\mathrm{out}}).$$

Orchard has *no* per-note value commitments: each Action—the unit that simultaneously spends a note of value v^{old} and creates a note of value v^{new}—commits once to its *net* value change, publishing

$$\mathrm{cv}^{\mathrm{net}} = \mathrm{cm}(v^{\mathrm{old}} - v^{\mathrm{new}}, \rho) = [v^{\mathrm{old}} - v^{\mathrm{new}}]V + [\rho]P,$$

and $B = \sum_i \mathrm{cv}^{\mathrm{net},i}$ over the transaction's Actions (protocol specification § 5.4.8.3). The Action structure itself—how spends and outputs are fused into these units and assembled into a transaction—is the business of the *Ironwood Guide*. Either way, the homomorphism collapses the pile:

$$B \;=\; [\Delta_v]V + [\bar{\rho}]P,$$

where Δ_v is the net value— $\sum_i v_i^{\mathrm{in}} - \sum_j v_j^{\mathrm{out}}$ in Sapling, $\sum_i (v_i^{\mathrm{old}} - v_i^{\mathrm{new}})$ in Orchard— and $\bar{\rho}$ is the net blinding. The transaction balances precisely when Δ_v equals the public balancing value v^{bal}; for a transaction with internal balance $\Delta_v = 0$, the V-component vanishes and $B = [\bar{\rho}]P$. Three observations set up the construction. Every element of $\mathbb{G}$ is some multiple of P, since P generates the prime-order group, so what distinguishes a balanced transaction is not the *existence* of a representation $B = [\beta]P$ but the ability to compute one. When the values balance, the signer knows the multiplier: it is the net blinding $\bar{\rho}$, assembled from the individual blindings. When they do not, computing any P-multiple representation of B from the published commitments would require knowing $\log_P V$, assumed hard.

Definition 8.29 (Binding signature)**.** With B and $\bar{\rho}$ as above and the transaction balanced, so that $B = [\bar{\rho}]P$, set the *binding verification key* $bvk := B$ and the *binding signing key* $bsk := \bar{\rho}$.

The binding signature is a RedDSA signature over the base point P, with signing key $\bar{\rho}$ and verification key B, on a message that is the hash of the transaction (the *sighash*). The verifier *recomputes* B from the publicly listed value commitments and the public balancing value, then checks $[s]P = R + [c]B$ with $c = H(R \,\|\, B \,\|\, \text{sighash})$.

Theorem 8.30 (The binding signature is a proof of knowledge of $\log_P B$)**.** *In the ROM, a valid binding signature on a transaction is a proof of knowledge of the discrete logarithm of the net commitment B to the base P; combined with the soundness of the accompanying zero-knowledge proofs (per Action in Orchard, per note in Sapling), which fix* and range-check *the committed values, it forces the transaction to balance.*

Proof. Part 1: proof of knowledge. The binding signature is exactly a Fiat–Shamir Schnorr proof for the relation R_{dl} with base P and statement B. By special soundness (Theorem 8.8) together with the forking argument of Theorem 8.18— specialised to a single statement and *no* signing oracle—any algorithm producing a valid signature under B with non-negligible probability can be rewound to two accepting transcripts (R, c, s), (R, c', s') with $c \neq c'$, from which the extractor recovers $\bar{\rho} = (s - s')(c - c')^{-1}$ with $B = [\bar{\rho}]P$. Producing a valid binding signature is therefore equivalent, up to the standard rewinding loss, to *knowing* $\log_P B$.

Part 2: knowledge forces balance. Write $B = [\Delta_v]V + [\bar{\rho}']P$ for the *true* net value and net blinding that the commitments imply; these are well defined once the accompanying proofs fix the committed openings—in Orchard the per-Action circuit fixes v^{old}, v^{new} and the opening of $\mathrm{cv}^{\mathrm{net}}$, in Sapling the per-note proofs fix each v_i, ρ_i. If the signer knows $\bar{\rho}$ with $B = [\bar{\rho}]P$, subtracting the two expressions for B gives

$$[\Delta_v]V \;=\; [\bar{\rho} - \bar{\rho}']P,$$

where Δ_v enters as a scalar, that is, as its residue in $\mathbb{F}_q$. Whenever $\Delta_v \not\equiv 0 \pmod{q}$, that residue is a nonzero element of $\mathbb{F}_q$, hence invertible, so

$$V \;=\; [(\bar{\rho} - \bar{\rho}')\,\Delta_v^{-1}]P$$

—exhibiting $\log_P V$ and contradicting the independence of V and P. Therefore $\Delta_v \equiv 0 \pmod{q}$.

Part 3: from congruence to equality. Balance in $\mathbb{F}_q$ alone is not yet economic balance: an integer net value equal to a nonzero multiple of q would pass the test above while minting q units at a stroke. The gap is closed by the *range checks* in the same accompanying proofs: in Orchard the per-Action circuit constrains v^{old} and v^{new} to $\{0, \ldots, 2^{64} - 1\}$ (the Action statement, protocol specification § 4.18.4), and in Sapling the per-note proofs constrain each v_i likewise. A transaction with n Actions (count notes instead for Sapling) therefore has $|\Delta_v| \leq n\,(2^{64} - 1)$ as an integer, vastly below the 255-bit q for any transaction that can be serialised, and the only multiple of q in that range is 0. Hence $\Delta_v = 0$ over $\mathbb{Z}$, not merely in $\mathbb{F}_q$. □

In deployment the transaction is not internally balanced: it declares a public balancing value v^{bal}, and the verifier signs off against $B - [v^{\mathrm{bal}}]V$, folding the public value into the verification key—the `orchard` crate's `Bundle::binding_validating_key` subtracts a commitment to v^{bal} with zero blinding from the sum of the Actions' $\mathrm{cv}^{\mathrm{net}}$ (protocol specification § 4.14). The argument of Theorem 8.30 applies verbatim to the folded key and forces $\Delta_v \equiv v^{\mathrm{bal}} \pmod{q}$; the balancing value is encoded as a signed 64-bit integer, so the range checks promote the congruence to the integer equality $\Delta_v = v^{\mathrm{bal}}$ exactly as in Part 3.

Remark 8.31 (Why a signature, and not just a proof). The single pair (R, s) accomplishes three things at once. First, it proves knowledge of $\bar{\rho}$, certifying balance, by Theorem 8.30. Second, because the Fiat–Shamir challenge hashes the transaction sighash, the same signature *authenticates* the transaction: altering any signed field changes c and invalidates (R, s), so the binding signature doubles as an integrity check binding the shielded components together—the origin of the

name. Third, the construction needs no trusted setup and no separate circuit for the balance check: it is pure elliptic-curve arithmetic that anyone can verify. The homomorphism of the commitment turns the arithmetic statement "the values balance" into the algebraic statement "B is a known multiple of P", and the Schnorr signature is the off-the-shelf proof of knowledge for precisely that statement—the purest illustration in the protocol that a digital signature *is* a proof of knowledge of a discrete logarithm.

Remark 8.32 (One scheme, two meanings). Set the two RedPallas uses side by side. The spend authorisation signature verifies under a re-randomised key $rk = X + [\alpha]P$, chosen to *hide* which long-term key authorised the spend: the signing key $rsk = x + \alpha$ encodes authority, and α buys unlinkability (Theorem 8.26). The binding signature verifies under a key $B = [\bar{\rho}]P$ that the spender does *not* choose: the transaction's own commitments force it, the "secret key" $\bar{\rho}$ is the net blinding factor, and signing proves balance (Theorem 8.30). Both are the same three lines of Schnorr arithmetic; what differs is the *meaning* assigned to the discrete logarithm—authority in one case, value-conservation in the other. The unifying lesson of this section: once a public key is the image of a secret under $x \mapsto [x]P$, a Schnorr signature is a compact, non-interactive proof of knowing that secret, and the designer chooses what knowing it *means*.

8.10 The transparent layer's schemes: ECDSA and BIP-340 over secp256k1

Two signature schemes in the series live outside the Pallas group: the transparent layer authorises spends with ECDSA, and the issuance bundle of Zcash Shielded Assets is authorised with a BIP-340 Schnorr signature. Both run over secp256k1, and both are registered here at the level a consumer needs—syntax, deployed conventions, and where their security stands—rather than constructed from scratch.

The curve secp256k1 is the short Weierstrass curve $y^2 = x^3 + 7$ over the prime field $\mathbb{F}_p$ with $p = 2^{256} - 2^{32} - 977$, with a fixed base point G of prime order n (a 256-bit prime) and cofactor 1; every constant is a named parameter of SEC 2. In SEC 2's terminology it is a *Koblitz curve*, one with $a = 0$, a choice that buys implementation speed and plays no role in what follows. The group law, point encodings, and the discrete-logarithm problem on such a curve are the general machinery of the *Math Guide*, §"Elliptic curves" and §"The elliptic-curve discrete logarithm problem"; the hardness assumption is Definition 2.1 in the group $\langle G \rangle$.

Construction 8.33 (ECDSA over secp256k1, schematically)**.** Let $e \in \mathbb{Z}/n\mathbb{Z}$ be the integer read from the 32-byte digest of the message, and write $x(R)$ for the x-coordinate of a point, read as an integer; the secret scalar is written d, since x is taken.

- KeyGen: sample $d \xleftarrow{\$} [1, n-1]$, set $X = [d]G$, output (X, d).
- Sign(d, e): sample $k \xleftarrow{\$} [1, n-1]$, set $R = [k]G$ and $r = x(R) \bmod n$, then $s = k^{-1}(e + rd) \bmod n$; restart if $r = 0$ or $s = 0$; output (r, s).
- Verify$(X, e, (r, s))$: require $r, s \in [1, n-1]$, compute $R' = [es^{-1}]G + [rs^{-1}]X$, and accept iff R' is not the identity and $x(R') \equiv r \pmod{n}$.

Correctness: $R' = [s^{-1}(e + rd)]G = [k]G = R$. The nonce k must be fresh and secret, for the reason of Proposition 8.20 in different algebra: two signatures (r, s) and (r, s') under one k on distinct digests $e \neq e'$ give $s - s' = k^{-1}(e - e')$, hence $k = (e - e')(s - s')^{-1}$ and then $d = (sk - e)\, r^{-1}$.

Where Schnorr's challenge is a hash of R, ECDSA's is the coordinate r itself, so the scheme is not a Fiat–Shamir transform and the forking-lemma route to Theorem 8.18 does not apply. Unforgeability analyses exist—in the generic group model, and in the ROM under additional as-

sumptions on the map $R \mapsto x(R) \bmod n$—but they lie outside this volume's path: the transparent layer inherits ECDSA's security from Bitcoin's track record rather than from a reduction.

Remark 8.34 (Deployment conventions: the sighash, DER, and low s). Three conventions fix how the transparent layer uses Construction 8.33, which protocol specification § 4.1.7 names as the script signature scheme.

1. *The message is the sighash.* The digest e is the 32-byte SIGHASH transaction hash of protocol specification § 4.10 (ZIP 244 for version-5 transactions), so the sighash algorithm plays the role of the hash in Theorem 8.4. Signer and verifier both feed it in as a digest.

2. *DER encoding and the sighash-type byte.* On the wire a signature is the DER encoding of (r, s)—an ASN.1 `SEQUENCE` of two minimally encoded `INTEGER`s—followed by one trailing byte, the *sighash type*, selecting which parts of the transaction the sighash commits to (SIGHASH_ALL for every signature the transaction builder emits; a signer working from a partially created transaction honours whichever valid type the input records). The signer appends the byte; the verifier splits it off and enforces strict DER unconditionally, the BIP-66 rule, so a malformed encoding fails script evaluation. The public key travels as its 33-byte compressed encoding, the bytes that `hash160` commits to (§3.9).

3. *Low-s normalisation.* If (r, s) verifies then so does $(r, n - s)$, so a third party can re-encode a valid signature and change a pre-ZIP-244 transaction id; the normalisation takes the representative $s \leq (n-1)/2$ (BIP-62), which the deployed signer always emits (`secp256k1` crate, `sign_ecdsa`, whose libsecp256k1 backend documents the created signature as always in lower-s form) and which script evaluation checks only under the `LowS` flag, one the consensus check does not set (`zcash_script` crate, `src/external/pubkey.rs`, `check_low_s`; `zebra-script` crate, `src/lib.rs`).

Construction 8.35 (BIP-340 Schnorr over secp256k1, schematically)**.** Write $H_\tau(m) = \mathrm{SHA256}(\mathrm{SHA256}(\tau) \,\|\, \mathrm{SHA256}(\tau) \,\|\, m)$ for the *tagged hash* with ASCII tag τ—domain separation in the sense of Definition 3.16, the doubled tag digest making the prefix a whole 64-byte SHA-256 block—and read digests as integers modulo n.

- *Keys.* Sample $d \xleftarrow{\$} [1, n-1]$ and set $X = [d]G$. The public key is the 32-byte coordinate $x(X)$ alone (*x-only*): the verifier reconstructs X as the point with that abscissa and *even* y, and the signer replaces d by $n - d$ when $y(X)$ is odd, keeping the pair consistent.

- $\mathrm{Sign}(d, m)$: derive $k = H_{\texttt{BIP0340/nonce}}(t \,\|\, x(X) \,\|\, m) \bmod n$ with $t = d \oplus H_{\texttt{BIP0340/aux}}(a)$ for auxiliary randomness a; set $R = [k]G$, negating k when $y(R)$ is odd; compute $e = H_{\texttt{BIP0340/challenge}}(x(R) \,\|\, x(X) \,\|\, m) \bmod n$ and $s = k + ed \bmod n$; output the 64 bytes $(x(R), s)$.

- $\mathrm{Verify}(x(X), m, (r, s))$: lift $x(X)$ to the even-y point X, recompute e, set $R' = [s]G - [e]X$, and accept iff R' is not the identity, $y(R')$ is even, and $x(R') = r$.

Remark 8.36 (BIP-340 against Construction 8.12; the ZSA consumer). The algebra is the Schnorr signature of Construction 8.12 in its (R, s) form with key-prefixing (Remark 8.14), under $\mathbb{G} = \langle G \rangle$ of order n and $H = H_{\texttt{BIP0340/challenge}}$; the unforgeability proof of §8.6 carries over with that tagged hash modelled as the random oracle. What differs is convention: points travel by abscissa alone with parity fixed by the even-y rule, saving a byte per key and the parity of R per signature and letting the same 32 bytes serve as key encoding and as hash input; the challenge hash is domain-separated by tag; and the nonce is deterministic with optional auxiliary randomness, as in Construction 8.21. The in-series consumer is the issuance authorisation signature of

Zcash Shielded Assets: ZIP 227 (*specified*, in a draft ZIP) instantiates IssueAuthSig as BIP-340 over secp256k1, prefixes both the validating-key encoding and the signature with a scheme byte `0x00`, and signs the transaction's sighash. The reference implementation, not yet deployed, is the QED-it fork of the `orchard` crate (commit `cf801a5`), `src/issuance/auth.rs`, whose `ZSASchnorr` signs through the `secp256k1` crate's BIP-340 routine with zeroed auxiliary randomness. The issuance protocol itself is the business of the *ZSA Guide*.

9 Interactive proofs, zero knowledge, and SNARKs

The adversary of this section wears two faces, one on each side of a conversation. Facing the verifier stands the *cheating prover*: a party with no witness—no spendable note, no authorisation, no satisfying assignment—who would nevertheless talk the verifier into accepting, and thereby mint value, spend another's coins, or certify a false computation to a network in which every validator takes the proof's word for the hidden data. Facing the prover stands the *curious verifier*: a party who follows the protocol, or deviates from it arbitrarily, in order to squeeze from the interaction more than the single bit it is owed—to distinguish which witness the prover holds, and so unmask the spender, link the transactions, or read the amounts a shielded protocol exists to hide. The two threats pull in opposite directions: defeating the first means the transcript must pin the prover down; defeating the second means the transcript must reveal nothing. That both can be defeated at once—and, more remarkably still, by a proof *shorter than the statement it certifies*—is the subject of this section, and the reason a private transaction can be checked by millions of verifiers who learn nothing from it but its validity.

9.1 Three questions

Three questions organise everything that follows.

Can a prover convince a sceptical verifier that a statement is true? Not by assertion—the verifier trusts nothing—but by a protocol at whose end the verifier is justified in accepting, because no cheating strategy could have survived it. The answer is the theory of *interactive proofs* and *arguments*, refined into *proofs of knowledge*: protocols after which the verifier is entitled to conclude not merely that a witness exists but that the prover *possesses* one.

Can the prover convince without revealing why the statement is true? The witness—the factorisation, the colouring, the discrete logarithm, the transcript of the computation—must stay hidden; the verifier must finish the protocol knowing the statement holds and nothing else. The answer is *zero knowledge*, made precise by the simulation paradigm.

Can the proof be small—dwarfed by the statement it certifies—and cheap to check? A blockchain verifier cannot afford to re-execute every computation it accepts, nor even to read a proof as long as the computation's description. The answer is the *SNARK*: the succinct non-interactive argument of knowledge.

The development answers the questions in order and culminates in the architecture underlying the Halo 2 proving system deployed in Orchard: a *polynomial interactive oracle proof*, compiled into an argument by a *polynomial commitment scheme* (§4.9) and made non-interactive by the *Fiat–Shamir transform*. Every earlier section of this volume feeds in: the games and reductions of §1, the discrete logarithm of §2, the random oracle of §3, the commitments of §4, and the Schnorr protocol of §8.4, which will turn out to have been this section's canonical example all along.

Conventions are those of §1 throughout: λ is the security parameter, adversaries are PPT (non-uniform where it matters), and negl denotes a negligible function. Proof systems introduce one re-indexing: the objects of study are indexed by an *instance* x rather than by 1^λ, and asymptotic statements are measured in the instance length $|x|$—for the instances this volume cares about, $|x|$ and λ are polynomially related, so the two scales agree up to the polynomial slop the definitions already absorb.

9.2 Languages, relations, and witnesses

What is a "statement", and what does it mean to hold the "why" of one? The vocabulary is that of languages and relations over bit strings.

Definition 9.1 (Language, relation, witness)**.** Fix the alphabet $\{0,1\}$. A *language* is a set $L \subseteq \{0,1\}^*$. A *binary relation* is a set $R \subseteq \{0,1\}^* \times \{0,1\}^*$ of pairs (x,w), where x is called the *instance* (or *statement*) and w a *witness* for x. The relation *induces* the language

$$L_R := \{\, x \,:\, \exists\, w \text{ with } (x,w) \in R \,\},$$

and for each instance x the (possibly empty) *witness set* is $R(x) := \{\, w \,:\, (x,w) \in R \,\}$. The relation R is an *NP-relation* if it is

1. *polynomially balanced*: there is a polynomial p with $|w| \le p(|x|)$ for every $(x,w) \in R$; and
2. *polynomial-time decidable*: a deterministic algorithm decides, in time polynomial in $|x|$, whether a given pair (x,w) belongs to R.

The class NP is exactly the class of languages induced this way: a language L lies in NP if and only if $L = L_R$ for some NP-relation R. This certificate characterisation of NP we take from complexity theory without proof; it is the bridge over which every "statement with a short checkable reason" enters the theory. Membership in L_R is existentially quantified—*some* witness exists—while holding an element of $R(x)$ is possession of the reason itself. The gap between the two quantifiers is where this section lives.

Example 9.2 (Three running relations)**.** Three relations recur through the section, in increasing order of expressive power.

1. *Discrete logarithm.* In the setting of §8.4—a cyclic group $\mathbb{G} = \langle G \rangle$ of prime order q, written additively—the relation is

   $$R_{\mathrm{dl}} = \{\, (X,x) \in \mathbb{G} \times \mathbb{F}_q \,:\, X = [x]G \,\}.$$

   Because $x \mapsto [x]G$ is a bijection $\mathbb{F}_q \to \mathbb{G}$, every instance $X \in \mathbb{G}$ has exactly one witness: the induced language is *trivial*—all of $\mathbb{G}$—and membership is vacuous. What is hard, under the DLP assumption of §2.1, is to *produce* the witness. This is the prototype of why proving membership differs from proving knowledge.

2. *Graph three-colouring.* With instances encoding finite graphs $G = (V,E)$,

   $$R_{3\mathrm{col}} = \{\, (G,c) \,:\, c\colon V \to \{1,2,3\} \text{ and } c(u) \neq c(v) \text{ for all } \{u,v\} \in E \,\},$$

   the relation of *proper* three-colourings. The induced language $L_{3\mathrm{col}}$ is NP-complete—every NP language reduces to it in polynomial time, a notion and fact we take from complexity theory without proof—so a zero-knowledge protocol for it yields one for *every* NP language by reduction (§9.8).

3. *Circuit satisfiability.* With instances encoding circuits,

   $$R_{\mathrm{sat}} = \{\, (C,w) \,:\, C \text{ a Boolean (or arithmetic) circuit and } C(w) = 1 \,\}.$$

   This is the universal target of practical proof systems: a computation is first *arithmetised* into the satisfiability of an arithmetic circuit over a finite field $\mathbb{F}$, and the prover proves knowledge of a satisfying assignment (§9.8).

The discrete-logarithm example already shows that "$x \in L$" is too weak a target for cryptography: every group element is some multiple of G, so a protocol establishing mere membership establishes nothing. The content of the Schnorr protocol is that the prover *knows* the scalar—and formalising "knows" for an algorithm, an object with no mental states, is the central conceptual move of this section (§9.4).

9.3 Interactive protocols, proofs, and arguments

Definition 9.3 (Interactive protocol)**.** An *interactive protocol* is a pair (P, V) of algorithms, the *prover* and the *verifier*. Both receive a common input x; the prover additionally receives a private input w (a witness, or an arbitrary auxiliary string); each party holds its own private random tape. The parties alternate sending messages $a_1, a_2, \ldots$ over a number of rounds polynomial in $|x|$; after the last message the verifier outputs a single bit, written

$$\langle P(w), V\rangle(x) \;\in\; \{0,1\},$$

with 1 for *accept*—a random variable over both parties' coins. The *transcript* of an execution is the full sequence of exchanged messages (together with x, and with the verifier's coins when relevant). The protocol is *public-coin* if every message the verifier sends is a fresh uniformly random string that it also reveals—the verifier's messages *are* its coins—and its final decision is a deterministic function of the transcript. The verifier is required to run in probabilistic polynomial time, always.

The last clause is the asymmetry on which everything turns. The verifier is the party being *protected*, so its power is fixed at PPT once and for all: a guarantee that only holds against verifiers with super-polynomial resources would protect nobody. The prover is the party whose power we *vary*, and that variation is exactly the distinction between proofs and arguments.

Definition 9.4 (Interactive proof and interactive argument)**.** Let $L \subseteq \{0,1\}^*$ be a language, and let $c, s\colon \mathbb{N} \to [0,1]$. The protocol (P, V) is an *interactive proof system* for L with *completeness error* c and *soundness error* s if:

1. *Completeness.* For every $x \in L$ there is a prover input w with

$$\Pr[\langle P(w), V\rangle(x) = 1] \;\geq\; 1 - c(|x|).$$

2. *Soundness.* For every $x \notin L$ and *every* prover strategy P^*—of *unbounded* computational power, with arbitrary auxiliary input z—

$$\Pr[\langle P^*(z), V\rangle(x) = 1] \;\leq\; s(|x|).$$

The soundness clause is a game in the sense of §1.4: the honest verifier is the challenger, the prover strategy is the adversary, and the win condition is an accept on a false instance. The two thresholds must be bounded apart,

$$c(|x|) + s(|x|) \;\leq\; 1 - \frac{1}{\mathrm{poly}(|x|)},$$

and in the protocols of this volume both errors are negligible (often $c = 0$). The class of languages admitting such proof systems— with the honest prover unbounded—is IP. Every protocol this volume constructs is for a language $L = L_R$ induced by an NP-relation R, with the

honest prover required to run in polynomial time given a witness $w \in R(x)$ as its private input; the unbounded honest prover matters only for the complexity-theoretic statements below.

An *interactive argument system* (or *computationally sound* proof) relaxes only the soundness quantifier: the guarantee holds against every *PPT* cheating prover P^*, that is, for $x \notin L$ and PPT P^* with arbitrary auxiliary input z, $\Pr[\langle P^*(z), V\rangle(x) = 1] \leq s(|x|)$, where s is required to be negligible. Soundness of an argument typically rests on a computational hardness assumption.

Remark 9.5 (Proofs versus arguments: neither dominates). A proof resists an unbounded cheating prover: its soundness is information-theoretic and unconditional. An argument resists only efficient adversaries, and an unbounded prover can typically break one outright—by breaking the binding of a commitment the protocol relies on (Theorem 4.8 showed perfectly hiding commitments have this exposure), or simply by brute-forcing a discrete logarithm. In exchange, arguments achieve properties provably impossible for proofs, chief among them *succinctness*: a sound interactive proof for an NP-complete language whose communication is much smaller than the witness would let an unbounded prover compress NP certificates below what complexity theory is believed to allow, collapsing classes believed distinct, whereas succinct *arguments* exist under standard assumptions (§9.8). For cryptographic applications—all of the SNARK literature included—the prover is a real machine with real resource bounds, and the argument notion is exactly right: bounded-prover soundness is the price of succinctness, and its gateway.

Why interaction, and why randomness? Strip both away and see what is left. With a deterministic verifier and a single prover message, the system proves exactly the class NP: the one message is a certificate, the verifier's check is the relation's decision procedure—and there is no hope of zero knowledge, since the certificate is, in spirit, the witness itself, and the verifier could replay the prover's analysis at leisure. Verifier randomness changes the game: a cheating prover must commit to its messages *before* knowing the challenge, so a random spot-check catches what a predictable one never would. Interaction lets the verifier probe adaptively, round after round. Together they buy real power: the equality IP = PSPACE—interactive proofs capture exactly the languages decidable with polynomial memory and unrestricted running time, a class believed far larger than NP—is a landmark of complexity theory (Lund, Fortnow, Karloff, and Nisan; Shamir) that we cite without proof. For this section the sharper point is qualitative: interaction and randomness are what make zero knowledge and knowledge extraction possible at all. A single static certificate can be neither hidden nor rewound.

Proposition 9.6 (Soundness amplification by repetition). *Let (P, V) be an interactive proof (or argument) for L with completeness error c and soundness error s, and let $k = k(|x|)$ be polynomially bounded. The protocol that runs k independent instances and accepts only if all k accept has completeness error at most $k \cdot c$ and soundness error at most s^k. Sequential repetition achieves this always; parallel repetition achieves it for proofs and, with care, for many arguments. In particular, if $s \leq 1 - \delta$ for a noticeable function δ, then choosing k with $\delta(|x|) \cdot k(|x|) = \omega(\log|x|)$—always possible with k polynomial—drives the soundness error below every inverse polynomial, so a base protocol needs only a noticeable soundness* gap.

Proof. For completeness, the honest prover fails at least one of the k runs with probability at most $k \cdot c$ by the union bound (*Math Guide*, §"The union bound and a birthday calculation"). For soundness under sequential repetition, fix $x \notin L$ and any cheating strategy; conditioned on the transcripts of the first $i - 1$ runs, the i-th run is an execution against a fresh independently-random verifier, so it accepts with probability at most s whatever state the prover carries forward; multiplying the conditional bounds gives s^k. (For parallel repetition of proofs the same bound holds by a more careful argument, which we omit; for arguments parallel repetition can fail to amplify in general, and we use only the sequential form.) Finally $s^k \leq (1 - \delta)^k \leq e^{-\delta k}$, which is negligible once $\delta k = \omega(\log|x|)$. □

9.4 Knowledge soundness and extractors

Soundness, as defined, is silent exactly where cryptography speaks. For the relation R_{dl} every instance $X \in \mathbb{G}$ lies in the induced language, so the soundness clause of Definition 9.4—quantified over $x \notin L$—holds *vacuously*: a verifier that always accepts is "sound" for the trivial language. The assurance actually wanted is that a convincing prover *possesses* the discrete logarithm. But possession is not a property an algorithm wears on its sleeve; the definition reads it operationally. A prover knows a witness if an efficient procedure, granted full control over the prover—the ability to run it, feed it messages, and *rewind* it to an earlier state to try different continuations—can compute the witness from it. The procedure is called the *extractor*, and what it extracts is the meaning of "knows".

Definition 9.7 (Knowledge soundness; proof of knowledge)**.** Let (P, V) be an interactive protocol and R an NP-relation. The protocol is a *proof of knowledge* for R with *knowledge error* $\kappa\colon \mathbb{N} \to [0,1]$ if there exist a probabilistic oracle algorithm $\mathcal{E}$ (the *extractor*) and a polynomial q such that for every prover strategy P^*—possibly cheating, possibly computationally unbounded—and every instance x, writing

$$\varepsilon(x) \;:=\; \Pr[\langle P^*, V\rangle(x) = 1]$$

for the probability that P^* convinces the honest verifier, the following holds whenever $\varepsilon(x) > \kappa(x)$: the extractor $\mathcal{E}$, given x and oracle access to P^* *with the ability to rewind*—to run P^* from any point with the same coins, feeding verifier messages of $\mathcal{E}$'s choosing—outputs a witness $w \in R(x)$ within expected running time

$$\frac{q(|x|)}{\varepsilon(x) - \kappa(x)}.$$

If the guarantee is required only for *PPT* prover strategies P^*, the protocol is an *argument of knowledge*.

Three consequences can be read straight off the definition. The extraction cost may blow up as ε approaches κ: the bound $q(|x|)/(\varepsilon - \kappa)$ is useful precisely when the prover's success is noticeably above the knowledge error. A prover succeeding with probability at most κ need yield nothing at all—κ is the "free" success rate, typically the probability of blind guessing, below which conviction carries no evidence of knowledge. And any prover noticeably above κ surrenders the witness: whatever code and coins produce conviction can be mined, efficiently, for the very object the conviction asserts. Knowledge is thereby defined as *efficient computability relative to the prover's own code and coins*—an algorithm knows what can be efficiently computed from it. Note the expected-time clause: extractors are the standing reason this volume's adversary class admits expected polynomial time (Remark 1.4).

Remark 9.8 (The rewinding justification). No live counterparty could rewind the prover, and none ever does: the extractor is a *thought experiment*, never executed during an honest run of the protocol. It exists to give the security claim its meaning—and, in security proofs, its teeth. In a reduction the extractor *is* the reduction: the cheating prover is a subroutine the reduction runs through its input/output interface alone (Definition 1.20), and black-box access already includes choosing the subroutine's random tape and restarting it from the top with the same coins—so replaying it under different challenges is not science fiction but the ordinary right of a machine over its own subroutines, no reading of its code required. The same device runs in both directions across this section: the *simulator* of §9.5 rewinds a verifier to fake transcripts without a witness, and the extractor rewinds a prover to mine real witnesses from conviction. Rewinding is the single most important technical device in the theory.

Proposition 9.9 (Knowledge soundness implies soundness)**.** *A proof of knowledge for R with knowledge error κ is an interactive proof for L_R with soundness error $s(x) \leq \kappa(x)$. The converse fails: a protocol can be sound without being a proof of knowledge.*

Proof. Fix $x \notin L_R$, so that $R(x) = \emptyset$. If some prover strategy P^* achieved $\varepsilon(x) > \kappa(x)$, the extractor run on P^* would halt in finite expected time and output some $w \in R(x)$— impossible, as the witness set is empty. Hence $\varepsilon(x) \leq \kappa(x)$ for every P^*, which is the soundness bound. For the failure of the converse: soundness constrains only false instances, so a protocol for a trivial language—such as R_{dl}'s, where no false instances exist—is automatically sound however it behaves, while nothing forces a witness to be extractable from it. One may be convinced of a true statement by a prover who does not know why it is true; knowledge soundness is strictly stronger, and it—not mere soundness—is the notion cryptographic applications actually rely on (§9.8). □

9.5 The simulation paradigm and zero knowledge

What should "the verifier learned nothing" mean? The definitional move—often called the most influential idea in modern cryptography—is the *simulation paradigm*: the verifier learned nothing beyond the truth of the statement if it could have produced, entirely on its own and without ever talking to the prover, a transcript indistinguishable from the real one. Whatever a party can manufacture for itself it did not learn from the conversation; if the whole conversation is reproducible without the witness, the conversation taught nothing about the witness. The three grades of indistinguishability—perfect, statistical, computational—carry over verbatim from §1.2 (Definitions 1.6 and 1.9, with the unbounded-distinguisher bound of Proposition 1.7 and the hierarchy of Proposition 1.10) under one re-indexing: ensembles are indexed by instances x (and auxiliary inputs) rather than by the security parameter alone, and negligibility is measured in $|x|$.

Definition 9.10 (Zero knowledge)**.** Let (P, V) be an interactive proof or argument for $L = L_R$. For a verifier strategy V^*, an instance x, a witness $w \in R(x)$, and an auxiliary input z, the *view*

$$\mathrm{View}_{V^*}(P(w), V^*(z))(x)$$

is the random variable consisting of x, the random tape of V^*, and every message V^* receives—everything the verifier sees. The protocol is *zero knowledge* if there exists a PPT simulator Sim such that for every $x \in L$, every witness $w \in R(x)$, and every auxiliary input z, the ensembles

$$\{\mathrm{Sim}(x, z)\} \quad \text{and} \quad \{\mathrm{View}_{V^*}(P(w), V^*(z))(x)\}$$

are indistinguishable; the grade of indistinguishability—perfect, statistical, computational—gives *perfect* (PZK), *statistical* (SZK), or *computational* (CZK) zero knowledge. Two regimes for the quantifier over V^*:

1. *Honest-verifier zero knowledge* (HVZK): the guarantee is required only for $V^* = V$, the verifier that follows the protocol, with z empty; the simulator must reproduce the distribution of honest-execution transcripts.

2. *Malicious-verifier (auxiliary-input) zero knowledge*: the guarantee holds for *every* PPT verifier strategy V^*, deviating arbitrarily, with arbitrary auxiliary input z. Here Sim may use V^* as a subroutine *and rewind it*, and is permitted expected polynomial running time (Remark 1.4).

Why does the simulator's existence suffice? The simulator receives x—and z—but *not* the

witness w. If its output is nevertheless indistinguishable from the real view, then the real view carries no efficiently extractable information about w beyond the fact $x \in L$: any property of w an efficient verifier could compute from the interaction, it could compute equally well from the simulator's output, which was manufactured without w—otherwise the computation would itself be a distinguisher. The simulator wields a power the prover lacks: it may rewind V^*, retrying until the verifier's challenges happen to align with a transcript preparable without the witness. That power is legitimate for exactly the reason rewinding always is (Remark 9.8): the simulator is the analyst's tool, holding the verifier as a subroutine, not a participant in any real execution.

The auxiliary input z models the prior knowledge a malicious verifier may bring to the conversation—the history of earlier protocol runs, partial information about the witness, anything. Quantifying over all z is what makes zero knowledge *compose*: a protocol that is zero knowledge for every auxiliary input remains zero knowledge when run as a subroutine of a larger system, whose surrounding state simply becomes the z of the analysis. Practice therefore adopts the auxiliary-input formulation, and so does this volume.

9.6 Sigma-protocols

The general definitions are now in place, and one protocol shape realises them with remarkable economy: three moves, public coins, and algebra doing all the work. The shape is named for the capital Σ its message flow traces on the page.

Definition 9.11 (Sigma-protocol)**.** Let R be an NP-relation. A *Sigma-protocol* for R with *challenge space* C is a three-move public-coin protocol on common input x, the prover holding w with $(x, w) \in R$:

1. *Commitment.* The prover sends a first message a (the *commitment*, or announcement), computed from (x, w) and fresh randomness.
2. *Challenge.* The verifier sends $e \xleftarrow{\$} C$, uniformly random.
3. *Response.* The prover sends z, computed from (x, w, a, e) and its randomness.

The verifier decides by a deterministic predicate $\mathrm{Verify}(x, a, e, z) \in \{0, 1\}$. Three properties are required.

- *Completeness.* An honest prover holding $(x, w) \in R$ convinces the honest verifier with probability 1.
- *Special soundness.* There is a PPT algorithm that, given x and any two accepting transcripts (a, e, z) and (a, e', z') with the *same* commitment a but *distinct* challenges $e \neq e'$, outputs a witness $w \in R(x)$.
- *Special honest-verifier zero knowledge* (sHVZK). There is a PPT simulator that, on input x and *any* challenge $e \in C$, outputs a pair (a, z) such that $\mathrm{Verify}(x, a, e, z) = 1$ and the triple (a, e, z) is distributed exactly—or statistically, or computationally, close to—a real honest transcript conditioned on the challenge being e. The simulator chooses *a after* fixing e; reversing the honest order is what makes simulation possible without w.

Two accepting transcripts sharing a commitment but differing in challenge are called a *collision*. Special soundness says a collision *is* a witness: it is the local, two-transcript reformulation of knowledge extraction, and the engine of the theorem that follows.

Theorem 9.12 (Special soundness yields knowledge soundness)**.** *Let (P, V) be a Sigma-protocol for R with challenge space C of size $N := |C|$. Then (P, V) is a proof of knowledge for R with knowledge error $\kappa = 1/N$.*

Proof. We construct an extractor establishing the slightly weaker knowledge error $2/N$, with expected running time $O(1/(\varepsilon - 2/N))$ prover invocations whenever the prover's success probability satisfies $\varepsilon > 2/N$; the closing paragraph addresses the gap to the optimal $1/N$.

The prover as a matrix. Fix x and a prover strategy P^*; fold any auxiliary input into its code. The behaviour of P^* is a deterministic function of its random tape ρ: fixing ρ fixes the commitment $a(\rho)$, and then each challenge e determines the response $z(\rho, e)$. Form the Boolean matrix

$$H[\rho, e] \;:=\; \begin{cases} 1 & \text{if } \mathrm{Verify}(x,\, a(\rho),\, e,\, z(\rho, e)) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

with one row per random tape and one column per challenge; then $\varepsilon = \Pr_{\rho,e}[H[\rho, e] = 1]$, and a collision is two 1-entries in a single row.

The extractor. Step 1. Run P^* with fresh ρ and a uniform challenge e, repeatedly, until an accepting transcript (a, e, z) appears. The number of trials is geometric with success probability ε, so this costs $1/\varepsilon$ invocations in expectation—the geometric waiting time (*Math Guide*, §"Beyond finite spaces: countable additivity, limits, and densities"). *Step 2.* Alternate two kinds of trial: (i) rewind P^* to the moment just after it emitted a—the tape ρ stays fixed—and feed one fresh uniform challenge $e' \neq e$; (ii) make one entirely fresh probe, as in Step 1. If a kind-(i) trial accepts, a collision $(a, e, z), (a, e', z')$ is in hand: go to Step 3. If a kind-(ii) trial accepts, *abandon* the current row and continue Step 2 from the new accepting transcript (a fresh *phase*). *Step 3.* Apply the special-soundness algorithm of Definition 9.11 to the collision and output the witness.

Analysis. Call a row ρ *heavy* if its acceptance density $\delta_\rho := \Pr_e[H[\rho, e] = 1]$ is at least $\varepsilon/2$. The heavy-row lemma—a Markov-style averaging—states that an accepting entry lies in a heavy row with probability at least $1/2$: the total accepting mass contributed by light rows is less than $\varepsilon/2$ (each light row contributes less than $\varepsilon/2$ of its own row measure), so conditioned on acceptance the light rows carry less than half the mass. Every phase opens with a fresh accepting transcript distributed exactly as Step 1's (the opening transcript is either Step 1's or a kind-(ii) success, both conditioned draws from the accepting entries), so the lemma applies at every phase, not only the first.

Each alternation of Step 2 contains a kind-(ii) trial, which accepts with probability ε; the number of alternations in a phase is therefore dominated by a geometric variable with mean $1/\varepsilon$, and a phase costs $O(1/\varepsilon)$ invocations in expectation. This interleaving is not an optimisation but a necessity: a row whose only accepting challenge is the already-used e—any row with $\delta_\rho \leq 1/N$—contains no collision at all, and an extractor that only rewound would sit in such a row for unbounded expected time. The fresh probes cap every phase.

Now condition on the phase's row being heavy. A kind-(i) trial draws e' uniformly from $C \setminus \{e\}$ and accepts with probability at least

$$\frac{\delta_\rho N - 1}{N - 1} \;\geq\; \delta_\rho - \frac{1}{N} \;\geq\; \frac{\varepsilon}{2} - \frac{1}{N},$$

which is positive precisely when $\varepsilon > 2/N$. The phase ends at its first success of either kind, and the chance that this first success is a collision rather than a row-switch is at least

$$\frac{\varepsilon/2 - 1/N}{(\varepsilon/2 - 1/N) + \varepsilon} \;=\; \Omega\!\left(\frac{\varepsilon - 2/N}{\varepsilon}\right).$$

Un-conditioning costs the heavy-row factor $1/2$, so each phase yields a collision with probability $\Omega((\varepsilon - 2/N)/\varepsilon)$, and the number of phases is dominated by a geometric variable with mean

$O(\varepsilon/(\varepsilon - 2/N))$. By Wald's identity—the expected total is the expected number of phases times the expected cost per phase (*Math Guide*, same section)—the extractor's expected number of prover invocations is

$$O\left(\frac{\varepsilon}{\varepsilon - 2/N}\right) \cdot O\left(\frac{1}{\varepsilon}\right) = O\left(\frac{1}{\varepsilon - 2/N}\right),$$

of the form $q(|x|)/(\varepsilon - 2/N)$ demanded by Definition 9.7 with $\kappa = 2/N$.

Closing the factor of two. The threshold $\delta_\rho \geq \varepsilon/2$ defining heaviness is an artefact of this exposition, and it is exactly what costs the extra $1/N$ in the knowledge error. Sharper accounting removes it: Damgård's refined heavy-row analysis in his Sigma-protocol lecture notes, and, in full generality, the k-special-soundness extractor of Attema, Cramer, and Kohl, which achieves knowledge error exactly $(k-1)/N$—here $k = 2$, giving the optimal $\kappa = 1/N$ that matches the soundness error. We cite these rather than reproduce them; every use in this volume tolerates the factor of two. □

Remark 9.13 (Knowledge error equals soundness error, and how to size the challenge space). The knowledge error $1/N$ of Theorem 9.12 is no accident of the analysis: an explicit attack meets it. A prover with no witness guesses the challenge in advance: it picks $e^* \in C$, runs the sHVZK simulator on (x, e^*) to obtain an accepting pair (a, z)—for a protocol such as Schnorr's the simulator's algebra works for *every* instance—and sends a. If the verifier's challenge happens to equal e^*, which occurs with probability exactly $1/N$, the prepared z is accepted. One challenge is thus always answerable without a witness, so $1/N$ is also the protocol's soundness error, and Theorem 9.12 says the two coincide. One caveat keeps the lower bound honest: under Definition 9.7 an extractor facing this guessing prover ($\varepsilon = 1/N$) is granted expected time $q(|x|)/(1/N - \kappa)$, so for exponential N a claimed κ below $1/N$ is not formally refuted—the definition would hand the extractor a super-polynomial budget, enough to brute-force a witness unaided. The attack does rule out any noticeably smaller κ once the extractor is required to run in strict polynomial time and witnesses are genuinely hard to compute: such an extractor, run on this witness-free prover, would otherwise mine a witness from nothing—for R_{dl}, solving the discrete-logarithm problem itself. To make the error negligible there are two routes: take C exponentially large—for instance $C = \mathbb{F}_q$ for a λ-bit prime q, so $N = q$—or repeat a small-challenge protocol $\omega(\log \lambda)$ times (Proposition 9.6). Large challenge spaces are why a *single* round of Schnorr already has negligible knowledge error, while a protocol with a polynomially small challenge space needs many repetitions.

Multi-round protocols generalise the collision to a tree. The succinct arguments of §9.8 run many challenge rounds, and their extractors need more than two transcripts.

Definition 9.14 (Tree special soundness)**.** Let (P, V) be a $(2\mu + 1)$-move public-coin protocol for R in which the prover speaks first and the verifier issues μ uniform challenges from a space C, and let $k_1, \ldots, k_\mu$ be positive integers. A $(k_1, \ldots, k_\mu)$*-tree of accepting transcripts* for x is a tree of depth μ in which every node at level i has k_i children labelled by *pairwise distinct* challenges for round i, every root-to-leaf path carries the prover messages of one accepting transcript, and transcripts passing through a common node share their prefix up to that node. The protocol is $(k_1, \ldots, k_\mu)$*-special-sound* if a PPT algorithm computes a witness $w \in R(x)$ from any such tree. A Sigma-protocol is the case $\mu = 1$, $k_1 = 2$.

Extraction generalises phase by phase: under the hypotheses of a tree-extraction theorem—the Attema–Cramer–Kohl analysis cited in the proof of Theorem 9.12 is the general form—rewinding the prover round by round assembles the required tree of $\prod_i k_i$ accepting transcripts: the protocol is a proof of knowledge with knowledge error growing from $1/|C|$ to the order of $(\sum_i (k_i - 1))/|C|$, and with extraction cost polynomial provided the tree size $\prod_i k_i$ is. Both degradations are benign in the regime the compiled protocols of this section occupy: C exponentially large, μ logarithmically bounded, and the k_i constant, so the error stays negligible

and $\prod_i k_i$ polynomial. This is the notion under which the inner-product argument's extractor operates (Proposition 4.39): each of the $\log_2 d$ folding rounds contributes one level of branching to the tree, three accepting children at every node. Each round's accepting relation is linear in three unknown parent terms with coefficients that are powers of the challenge, so three suitably distinct challenges make that system invertible. The *Halo 2 Guide* carries out that extraction for the deployed protocol, whose IPA normalisation involves each round's challenge through the coefficients u^{-1}, 1, u: the extractor accordingly needs pairwise distinct *nonzero* challenges at every node—a negligible-fraction exclusion absorbed into the knowledge error—and not the pairwise distinct squares ($u \neq \pm u'$) that the alternative $u^{-2}, 1, u^2$ normalisation would demand.

> **Proposition 9.15** (sHVZK implies HVZK)**.** *A Sigma-protocol satisfying special honest-verifier zero knowledge is honest-verifier zero knowledge in the sense of Definition 9.10, with the same grade (perfect, statistical, or computational).*

Proof. The honest view is a triple (a, e, z) in which e is uniform on C. The HVZK simulator draws $e \xleftarrow{\$} C$ itself, invokes the sHVZK simulator on (x, e) to obtain (a, z), and outputs (a, e, z). For each fixed e, the simulated pair (a, z) is distributed as a real run conditioned on that challenge—exactly, or up to the stated grade of closeness—and the simulator's e has the same uniform marginal as the honest verifier's; averaging the conditional distributions over uniform e therefore reproduces the honest view, with any statistical or computational slack preserved under the averaging. □

Example 9.16 (Schnorr: the canonical Sigma-protocol)**.** The Schnorr identification protocol of §8.4 is a Sigma-protocol for the discrete-logarithm relation R_{dl} of Example 9.2, and the properties proved there, self-contained, are exactly the clauses of Definition 9.11. The commitment is $R = [r]G$; the challenge is a uniform $c \in \mathbb{F}_q$; the response is $s = r + cx$; the predicate is $[s]G = R + [c]X$. Perfect completeness is Proposition 8.7; 2-special soundness is Theorem 8.8, whose collision algebra yields the witness $x = (s - s')(c - c')^{-1}$; special HVZK is Theorem 8.10, whose simulator samples s and solves $R = [s]G - [c]X$—choosing the commitment after the challenge, as the definition prescribes. The challenge space is $\mathbb{F}_q$, so $N = q$ and the knowledge error $1/q$ is negligible in a single round (Remark 9.13). By Theorem 9.12 and Proposition 9.15, Schnorr's protocol is a perfect-HVZK proof of knowledge of a discrete logarithm.

Remark 9.17 (From honest to malicious verifiers: three remedies). Sigma-protocols are honest-verifier zero knowledge only. A malicious V^* may choose e as a function of the commitment a, and the sHVZK simulator—which fixes e before producing a—does not obviously cope: nothing lets it predict which challenge the verifier's function will return for a commitment it has yet to manufacture. Three standard remedies close the gap. (1) *Guess and rewind*, for small challenge spaces: the simulator guesses $e^* \xleftarrow{\$} C$, runs the sHVZK simulator on (x, e^*), and feeds the resulting a to V^*; if the verifier's challenge equals e^*, the round is simulated, and otherwise the simulator rewinds V^* and guesses afresh. The honest prover fixes a before any challenge exists, so the simulated a carries (almost) no information about the guess and each attempt succeeds with probability close to $1/N$: for polynomially large N the expected $O(N)$ attempts stay polynomial, and sequential repetition of the small-challenge protocol—simulated round by round, the auxiliary input carrying the history—is full malicious-verifier zero knowledge, at the price of many rounds. For exponentially large N the guess never lands, and repetition does not generically upgrade honest-verifier to malicious-verifier zero knowledge. (2) *Challenge commitment*: the verifier commits to e before seeing a (a coin-tossing step using the commitments of §4), so e cannot depend on a; under the additional conditions such a compiler imposes for simulation, the simulator rewinds past the opening. (3) *The Fiat–Shamir transform*: remove the verifier's choice altogether by deriving e from a hash of the transcript—under the hypotheses of Theorem 9.20, in the programmable random oracle model, the resulting *non-interactive* protocol is zero knowledge against *all* verifiers, since there is no live challenge left to skew. For the SNARK pipeline route (3)

is the one that matters, and honest-verifier zero knowledge is exactly the property the transform requires. It is next.

9.7 The Fiat–Shamir transform: from interactive to non-interactive

In a public-coin protocol the verifier contributes nothing but fresh randomness: its messages are coin tosses, its decision a deterministic function of the transcript (Definition 9.3). Every such message can therefore be replaced by a value that *anyone* can compute—a hash of everything sent so far—and the interaction collapses into a single proof string, verifiable by recomputing the hashes. The prover no longer talks to a live, possibly malicious verifier but to a fixed public function, which is why honest-verifier zero knowledge upgrades to full non-interactive zero knowledge. The cost is that the hash must be idealised: the analysis lives in the random oracle model, with $H\colon \{0,1\}^* \to C$ a random oracle whose range is the challenge space (Definition 8.15), and it exploits both powers established in Proposition 3.14: *observability*—the reduction sees every query the adversary makes—and *programmability*—the reduction may fix the oracle's output at a freshly queried point to any value, provided answers stay consistent and uniformly distributed.

Definition 9.18 (The Fiat–Shamir transform)**.** Let (P, V) be a public-coin protocol for R with challenge space C, and let H be a random oracle with range C. The *Fiat–Shamir transform* $\mathrm{FS}[(P, V), H]$ is the non-interactive system defined, in the Sigma case, by:

- *Prover.* Compute the commitment a as in (P, V); set $e := H(x, a)$; compute the response z; output the proof $\pi := (a, z)$ (equivalently (a, e, z)).
- *Verifier.* Given x and $\pi = (a, z)$, recompute $e := H(x, a)$ and accept iff $\mathrm{Verify}(x, a, e, z) = 1$.

A multi-round public-coin protocol applies the rule per round: the i-th challenge is

$$e_i \;:=\; H(x,\, a_1,\, e_1,\, \ldots,\, a_i),$$

each challenge hashing the *entire* transcript prefix. To bind a message m to the proof—turning it into a signature—one includes m in the hash: $e := H(x, a, m)$.

Remark 9.19 (The pitfall: weak Fiat–Shamir). The definition's insistence that each challenge hash *all* prior messages, the instance x included, is load-bearing, and omitting any part of it has broken deployed systems. Hashing $e := H(a)$ without x—"weak Fiat–Shamir"—lets an attacker fix (a, e, z) first and choose the instance *afterwards*, searching for an x that the frozen challenge happens to validate: the interactive protocol's soundness quantified over provers who commit before the challenge, and an unhashed x re-opens exactly that order of quantifiers, yielding proofs of false statements. In multi-round protocols, hashing only the latest message rather than the running prefix breaks soundness the same way, one round at a time. The maxim: *the challenge must be a hash of everything the verifier would have seen up to the moment it speaks.*

Theorem 9.20 (Fiat–Shamir security in the ROM)**.** *Let (P, V) be a Sigma-protocol for R with challenge space C of size N, special soundness, and special HVZK; assume further that, conditioned on (x, e), the first message output by the sHVZK simulator has min-entropy $\omega(\log \lambda)$ (min-entropy H_∞: Definition 6.23). In the random oracle model, $\Pi = \mathrm{FS}[(P, V), H]$ is a non-interactive argument for R that is:*

1. perfectly complete*, inheriting the protocol's completeness;*
2. knowledge sound*: from any PPT prover P^* that makes at most Q random-oracle queries and*

> *outputs an accepting proof for x with probability ε, an extractor obtains a witness $w \in R(x)$ with probability at least $\varepsilon(\varepsilon/Q - 1/N)$ up to constant factors—non-negligible whenever ε is non-negligible and N is super-polynomial; and*
>
> 3. non-interactive zero knowledge*: a PPT simulator that controls the random oracle produces, without the witness, proofs indistinguishable from real ones—for all verifiers, including malicious ones.*

Proof. Completeness. The honest prover computes $e = H(x, a)$ and an honest response; the verifier recomputes the very same e from the same inputs and runs the same predicate, so the protocol's probability-1 completeness transfers verbatim.

Zero knowledge, by programming. The simulator, on input x, draws $e \xleftarrow{\$} C$, runs the sHVZK simulator on (x, e) to obtain an accepting pair (a, z), and then *programs* the oracle:

$$H(x, a) \;:=\; e.$$

Programming fails only if (x, a) was queried before this moment; by hypothesis the simulated first message has, conditioned on (x, e), min-entropy $\omega(\log \lambda)$, so a given prior query collides with a with probability $2^{-H_\infty(a)}$, and a union bound over the adversary's prior queries makes the failure probability negligible. Conditioned on no failure, the programmed answer is a fresh uniform value—exactly what an unprogrammed oracle would have returned—so the oracle's joint distribution is untouched, and by sHVZK the output (a, z) is distributed as a real proof. The programming is invisible to every verifier, and there is no live challenge for a malicious verifier to skew: its malice is moot, and honest-verifier simulation delivers full non-interactive zero knowledge.

Knowledge soundness, by forking. Let P^* output an accepting proof (a, z) for x with probability ε. With overwhelming probability a successful P^* actually queried H at (x, a): otherwise the challenge $e = H(x, a)$ used by the verifier is a uniform value the prover never saw, and the proof verifies with probability at most $1/N$. Say the critical query is the j-th of the at most Q queries. The extractor runs P^* once, recording the transcript of oracle answers and the index j of the critical query; it then *rewinds* P^* to just before the j-th query and re-runs it with the same coins, answering the first $j - 1$ queries identically—so the commitment a is unchanged—and the j-th and subsequent queries with fresh uniform values, the j-th being $e' \xleftarrow{\$} C$. This is precisely the experiment of the general forking lemma (Theorem 8.17, with $h = N$): if P^* succeeds with probability ε over Q query slots, then with probability at least $\varepsilon(\varepsilon/Q - 1/N)$ both runs succeed at the same index *and* $e \neq e'$. Conditioned on that event, the two runs yield accepting transcripts (a, e, z) and (a, e', z') with a common commitment and distinct challenges—a collision—and the special-soundness algorithm of Definition 9.11 outputs $w \in R(x)$. □

Remark 9.21 (Caveats, and the refinements SNARKs need). The random oracle is an idealisation: a real hash function is fixed, public, and not random, and there exist contrived protocols secure in the ROM yet insecure under *every* concrete instantiation. The transform is nonetheless trusted in practice, with the working stance of Remark 1.29, and it underlies essentially every deployed non-interactive proof and signature—Fiat–Shamir-compiled SNARKs included. Two refinements matter for the compiled arguments of §9.8. First, Theorem 9.20 covers the one-challenge Sigma case only, and does not by itself prove security for a multi-round protocol such as Halo 2's: there the extractor must fork the transcript at *each* round, its success probability degrading with the round count and with the protocol's soundness structure, and special soundness must generalise to tree special soundness (Definition 9.14)—a separate multi-round theorem, whose hypotheses must match the protocol at hand. Second, when the underlying interactive protocol is itself only computationally sound—an argument—the ROM extraction *layers on top of* the computational soundness reduction rather than replacing it. There is also a gap between proofs that merely observe the oracle and proofs that program it (the non-programmable ROM

is strictly weaker as a proof technique); deployed systems adopt the programmable model—the zero-knowledge clause above, which programs the oracle outright, already shows why—and an analysis must state which programmable-oracle (and, where used, algebraic) model it works in.

Example 9.22 (Schnorr signatures, recovered)**.** Applying Definition 9.18 to Schnorr's Sigma-protocol (Example 9.16) with a message m folded into the challenge hash yields, symbol for symbol, the Schnorr signature scheme of Construction 8.12; and its EUF-CMA security in the ROM (Theorem 8.18) is established by exactly the forking argument in the proof of Theorem 9.20, augmented with a signing-oracle simulation. The signatures section proved the instance; this section supplies the general statement it instantiates. The slogan: *proof of knowledge + Fiat–Shamir = signature*.

9.8 SNARKs: succinct non-interactive arguments of knowledge

The Sigma-protocol paradigm reaches all of NP: a commit–challenge–reveal proof of knowledge for the NP-complete relation $R_{3\mathrm{col}}$ of Example 9.2—zero knowledge under one-way functions alone, a result of Goldreich, Micali, and Wigderson taken without proof—extends to every NP language by reduction. The generic reduction is hopelessly inefficient, however, its proofs polynomially *longer* than the computation they certify, so practical systems abandon it and arithmetise the computation directly.

A blockchain verifier operates under constraints the classical theory never contemplated. It cannot interact—a transaction is broadcast once and verified by parties who will never speak to its author, years later, from a proof string sitting in a block. It cannot afford to re-execute—every full node checks every proof, so verification must cost less than the computation being certified, by orders of magnitude. And it must resist a prover with every incentive to cheat: an accepting proof *is* money. The object meeting all three constraints at once is the SNARK.

Definition 9.23 (Non-interactive argument)**.** A *non-interactive argument* for an NP-relation R is a triple of PPT algorithms:

- $\mathrm{Setup}(1^\lambda, R) \to srs$, producing a *structured reference string*—the public parameters. In the random oracle model all three algorithms additionally access one globally sampled oracle; the oracle is not part of the *srs*—a PPT setup cannot output an infinite object—though public parameters may be derived through it. A setup holding no secret trapdoor is *transparent*: no secret is ever held, so none must be trusted to have been discarded (Remark 4.3).
- $\mathrm{Prove}(srs, x, w) \to \pi$, for $(x, w) \in R$.
- $\mathrm{Verify}(srs, x, \pi) \to \{0, 1\}$.

The system is *complete* if honestly generated proofs always verify. A *preprocessing* argument lets Setup depend on the circuit or relation and output, alongside the proving parameters, a short *verification key*, amortising the verifier's per-statement work across all statements of that shape.

Definition 9.24 (SNARK)**.** A *succinct non-interactive argument of knowledge* (SNARK) for an NP-relation R is a non-interactive argument that is additionally:

1. *Knowledge sound.* For every PPT prover P^* there is a PPT extractor $\mathcal{E}$—in the ROM, ob-

serving P^*'s oracle queries and rewinding or forking it—such that, on the same setup,

$$\Pr[\mathrm{Verify}(srs, x, \pi) = 1 \ \wedge \ (x, \mathcal{E}^{P^*}) \notin R] \ \leq \ \mathrm{negl}(\lambda),$$

the probability over the setup and both algorithms' coins: an accepting proof implies an extractable witness, except with negligible probability.

2. *Succinct.* The proof size $|\pi|$ and the verifier's running time are *sublinear* in the size of the computation proved—typically $|\pi| = O(\mathrm{poly}(\lambda))$ or $O(\mathrm{poly}(\lambda)\log|C|)$, and verification $O(\mathrm{poly}(\lambda)\log|C|)$ for a circuit C, independent of, or polylogarithmic in, the witness and circuit sizes. (Some of the literature reserves "succinct" for polylogarithmic proofs; the constant-size proofs of pairing-based schemes are the strongest form.)

With zero knowledge—the non-interactive, ROM form of Definition 9.10, as delivered in the Sigma case by Theorem 9.20, clause 3—it is a *zk-SNARK*.

Remark 9.25 (Why these systems are arguments). The "AR" in SNARK is earned by necessity, not by preference. Remark 9.5 showed that a succinct *proof* for an NP-complete language is essentially ruled out: a transcript shorter than the witness cannot information-theoretically pin the witness down, and an unbounded prover could otherwise compress certificates below what complexity theory is believed to allow. Succinct *arguments* escape because a computationally bounded prover cannot *find* a convincing-but-false short proof even though one may exist—succinctness and computational soundness arise together, by necessity. The constructions below realise the necessity at a specific spot: their polynomial commitments are binding only against *efficient* adversaries, under cryptographic assumptions, and an unbounded prover could equivocate on a commitment and defeat extraction. It is at that commitment layer that the information-theoretic core turns into an argument.

What does *knowledge* soundness buy an application? Consider a private transaction whose statement reads: *there exists a note I am authorised to spend, and input and output values balance*. Plain soundness guarantees only that *some* satisfying witness exists—and for such existential statements about a large anonymity set, witnesses exist in abundance: other people's notes satisfy the existential quantifier perfectly well. Soundness alone would not prevent a prover holding no spendable note from proving the existential statement, should she somehow come by a proof. Knowledge soundness closes the gap: whoever produced the accepting proof *actually holds* a concrete witness—a real spendable note, a real authorisation—that the extractor could recover from her. The application may then act—release funds, grant access—exactly as though the witness had been presented in the clear, while it stays hidden. Soundness protects against false statements; knowledge soundness protects against provers who do not know what they claim. Almost every cryptographic use, and certainly every monetary one, requires the latter (Proposition 9.9).

The pairing-based route

The first SNARKs to reach deployment, and the ones the deployed Sapling protocol still uses, verify with a bilinear pairing.

Definition 9.26 (Bilinear pairing). Let $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ be cyclic groups of prime order r, the first two written additively with generators G and $\hat{G}$. A *bilinear pairing* is a map $e\colon \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ that is

1. *bilinear*: $e([a]P, [b]Q) = e(P, Q)^{ab}$ for all $P \in \mathbb{G}_1$, $Q \in \mathbb{G}_2$, and $a, b \in \mathbb{F}_r$;

2. *non-degenerate*: $e(G, \hat{G}) \neq 1$, so that $e(G, \hat{G})$ generates $\mathbb{G}_T$; and
3. *efficiently computable*.

Bilinearity lets a verifier test one multiplication of hidden scalars—whether $c = ab$, given $[a]G$, $[b]\hat{G}$, and $[c]G$—without learning any of them (Example 2.9); the KZG check of Construction 4.35 is such a test. The groups come from *pairing-friendly* elliptic curves, chosen with a small embedding degree (the quantity of Definition 2.24, there required large) so that $\mathbb{G}_T$ lives in a manageable extension field; the curve of the deployed Sapling protocol is BLS12-381, a Barreto–Lynn–Scott curve of embedding degree twelve (protocol specification § 5.4.9.2). The deployed system treats the pairing as a black box, and so does this volume.

The lineage is short. The BCTV14 family (Ben-Sasson, Chiesa, Tromer, and Virza) is a preprocessing argument in the sense of Definition 9.23: the setup encodes one fixed arithmetic circuit into a structured reference string of group elements, the prover's work is a handful of multiscalar multiplications over that string, and the proof is a constant number of group elements—eight, in the deployed encoding (protocol specification § 5.4.10.1)—checked by a constant number of pairing equations, whatever the circuit's size. Groth's refinement (Groth16) cuts the proof to *three* group elements, $\pi_A, \pi_C \in \mathbb{G}_1$ and $\pi_B \in \mathbb{G}_2$, and verification to a single pairing-product equation: three pairings—π_A against π_B, π_C against a fixed verification-key element, and a public-input-weighted combination of verification-key elements against another—compared with one pairing precomputed once per circuit. Deployed Zcash uses Groth16 over BLS12-381 for every Sapling spend and output proof (protocol specification § 5.4.10.2; proofs produced and verified by the `bellman` crate's `groth16` module, consumed through the `zcash_proofs` crate). PLONK, the third generation, is the recipe of Theorem 9.28 below with the KZG scheme of Construction 4.34 as its commitment: the gate-and-permutation arithmetisation Halo 2 also uses, each polynomial opening checked by two pairings, so proof size and verification cost stay constant. Constant-size proofs and constant-time verification, independent of the circuit, are the family's signature—the strongest form of the succinctness clause of Definition 9.24.

The price is the setup. The reference string is produced from secret scalars—the evaluation point τ of Construction 4.34 and, in Groth16, further secrets that tie the proof elements to one circuit—which must be *destroyed* once the parameters are published: they are the *toxic waste*. Whoever retains them forges accepting proofs of false statements at will, exactly as the trapdoor τ forges KZG openings (Remark 4.37), and no verifier can tell: knowledge soundness collapses silently. For Groth16 the waste is *per circuit*—every new statement shape needs a fresh setup—whereas PLONK's KZG string is universal, one powers-of-τ string serving every circuit up to a size bound, but a trusted one still. The mitigation is a *multiparty ceremony*: participants contribute secret randomness in turn, each re-randomising the running parameters and destroying its own share, so that the waste stays unknown as long as *one* participant was honest. Zcash ran such ceremonies for the parameters of its earlier proof systems—the BCTV14 Sprout circuit and the Groth16 Sapling circuits (protocol specification §§ 5.7–5.8)—and each remains sound only on that one-honest-participant assumption.

The other route through the design space keeps the discrete logarithm and nothing else. The inner-product commitment of Construction 4.38 needs no pairing and no trusted setup—its generators are hashed into existence, with no secret behind them—at the price of logarithmic proofs and a linear-time opening check (Proposition 4.39); it is the road the *Halo 2 Guide* takes, and the one the rest of this subsection paves.

Polynomial interactive oracle proofs

Modern SNARKs are built in two layers—one information-theoretic, one cryptographic—and the information-theoretic layer speaks the language of polynomials.

Definition 9.27 (Polynomial interactive oracle proof)**.** Fix a finite field $\mathbb{F}$. A *polynomial interactive oracle proof* (Poly-IOP) for a relation R over $\mathbb{F}$ is a public-coin interactive protocol in which, in place of ordinary messages, the prover in each round sends an *oracle* for a polynomial $p_i \in \mathbb{F}[X]$ of some bounded degree; the verifier sends uniformly random field-element challenges, and may *query* each polynomial oracle at points of its choice, receiving the evaluation $p_i(\zeta)$. After the interaction the verifier accepts or rejects as a function of its challenges and the few evaluations it queried. Completeness and (knowledge) soundness are as for interactive proofs, with the soundness analysis purely algebraic—resting, invariably, on the Schwartz–Zippel lemma: a nonzero polynomial of degree at most d over $\mathbb{F}$ vanishes at a uniformly random point with probability at most $d/|\mathbb{F}|$ (*Math Guide*, §"The Schwartz–Zippel lemma"). A good Poly-IOP verifier makes only $O(1)$ or $O(\log)$ queries, each a single evaluation.

The oracle is an idealisation—no real prover can hand over an opaque box answering evaluation queries honestly—but a productive one: it isolates the algebra from the cryptography. Soundness of the Poly-IOP layer is *unconditional*, resting on no assumption beyond the rigidity of low-degree polynomials; the cryptography enters only when the oracles are instantiated.

Arithmetisation is how a computation becomes a Poly-IOP statement. The claim "I know w with $C(w) = 1$" (the relation R_{sat} of Example 9.2) is encoded as a small set of polynomial identities over $\mathbb{F}$ that hold if and only if the computation is correct. The computation is first expressed as a *constraint system*—the customisable *PLONKish* gate-and-permutation form used by Halo 2—whose satisfaction is equivalent to a handful of polynomials satisfying identities on a chosen evaluation domain, a fixed finite set of points with one point per constraint row. In the PLONKish form, polynomials encoding the wire values a, b, c and fixed selector polynomials q_L, q_R, q_M, q_O, q_C chosen at circuit-definition time must satisfy the gate identity

$$q_L \cdot a \;+\; q_R \cdot b \;+\; q_M \cdot ab \;+\; q_O \cdot c \;+\; q_C \;=\; 0$$

at every point of the domain—selectors switching each row among addition, multiplication, and custom behaviours—while a *permutation argument* enforces the copy constraints wiring one gate's output to another's input. One more step turns the on-domain requirement into a spot-checkable one: the identity is demanded only on the domain H, and a polynomial vanishes on all of H exactly when it is divisible by the vanishing polynomial $Z_H := \prod_{h \in H}(X - h)$ (*Math Guide*, §"Roots and the factor theorem"), so the prover also sends an oracle for the quotient t with

$$q_L \cdot a + q_R \cdot b + q_M \cdot ab + q_O \cdot c + q_C \;=\; t \cdot Z_H$$

as an identity of polynomials—one that must now hold at *every* point of $\mathbb{F}$. Checking it at a single random point is the Poly-IOP: the verifier queries the oracles at a random ζ, checks the displayed equation there, and by Schwartz–Zippel a false statement survives with probability at most $d/|\mathbb{F}|$, with d bounding the degree of either side.

The polynomial commitment scheme is the cryptographic half. The PCS of Definition 4.32 supplies exactly what the idealised oracle promised: a short, binding commitment to a polynomial, with short proofs—verifiable against the commitment alone—that $p(\zeta) = v$; evaluation binding forbids two accepted values at one point, and the *extractability* the definition names is what SNARK compilation consumes: a prover whose evaluation proofs verify must "know" an underlying polynomial. Three instantiations mark out the design space: KZG (Construction 4.34)—pairing-based, constant-size, trusted setup; the inner-product-argument commitment over a single elliptic curve (Construction 4.38)—transparent, no trusted setup, logarithmic-size openings, the choice made by Halo 2; and the hash-based transparent schemes, outside this volume's deployed path and mentioned only to bound the territory.

Theorem 9.28 (The compilation recipe, informal)**.** *Let* PIOP *be a public-coin polynomial IOP for R, complete and knowledge-sound, and let* PC *be a polynomial commitment scheme that is correct, evaluation-binding, and extractable. Compile as follows: wherever the prover would send a polynomial oracle p_i, it sends* $\mathrm{Commit}(p_i)$ *instead; wherever the verifier would query $p_i(\zeta)$, the prover sends the claimed value together with a* PC *evaluation proof, which the verifier checks against the commitment. Under compatible composition hypotheses the result is a public-coin interactive argument of knowledge for R, and a suitable multi-round Fiat–Shamir theorem in the ROM* (*Definition 9.18 gives the transform: every challenge a hash of the entire transcript so far*) *makes it non-interactive. Its proof is succinct; under the strict Definition 9.24 it is a SNARK only if verification is also sublinear in the proved computation. If, additionally, the Poly-IOP is properly blinded—random masking terms on the prover's polynomials, hiding commitments* (*Remark 4.41*)*—and the corresponding simulation theorem applies, the result is zero knowledge.*

Proof sketch: layered extraction and the succinctness account. Completeness transfers verbatim: honest commitments open to honest evaluations, and the honest Poly-IOP verifier accepts.

Knowledge soundness is *layered*, one extractor feeding another. Given a successful prover against the compiled argument, the SNARK extractor first invokes PC *extractability* on each commitment, recovering actual polynomials $\hat{p}_i$ of the right degrees consistent with every evaluation the prover got accepted; this step is where the cryptographic assumption lives, and why the compiled object is an *argument*—a prover that could equivocate on its commitments would defeat it, and only computational binding rules that out. The extractor then hands the $\hat{p}_i$ to the Poly-IOP's *unconditional* knowledge extractor, which is sound against *any* polynomials—in particular the extracted ones—and outputs $w \in R(x)$; evaluation binding guarantees the prover could not have answered queries inconsistently with the committed $\hat{p}_i$, so the algebraic soundness analysis applies to the extracted polynomials as run. Fiat–Shamir then converts the public-coin interactive argument to a non-interactive one in the ROM by the multi-round generalisation of Theorem 9.20: challenges become transcript hashes (Remark 9.19 governing what they must absorb), observability and forking supply the rewinding that the layered extractor needs—tree-style, per Definition 9.14—and the honest-verifier zero knowledge of the blinded protocol becomes full non-interactive zero knowledge by oracle programming.

Succinctness is an accounting exercise, and the account is paid entirely by PC. The proof consists of a constant or logarithmic number of commitments and evaluation proofs—short by construction, sublinear in the circuit, independent of the witness—so the compiled *proof* is succinct whenever the commitment scheme's objects are. The *verifier* is sublinear only when the evaluation checks are themselves succinct, as for KZG's two pairings; with the inner-product commitment each evaluation check costs $O(d)$ group operations (Proposition 4.39), so the compiled verifier runs in time linear in the circuit while the proof stays succinct. The compiled object therefore satisfies every clause of Definition 9.24 when the PC openings verify succinctly, and otherwise yields succinct proofs with linear-time verification—the trade Halo 2 accepts and then recovers through batching. □

Remark 9.29 (Halo 2 is the recipe, instantiated). The proving system deployed in Orchard (protocol specification § 5.4.10.3) instantiates the recipe of Theorem 9.28, and each ingredient can be pointed to in the deployed source. The Poly-IOP is a PLONKish arithmetisation—custom gates declared as polynomial expressions, lookup arguments asserting that witness values lie in a declared table, and a permutation argument enforcing copy constraints. The polynomial commitment is the inner-product-argument scheme over a single elliptic curve, *transparent*—the generators are derived by hashing to the curve from the fixed domain string `Halo2-Parameters`, so the setup holds no trapdoor—with logarithmic-size openings. Fiat–Shamir over a concrete hash makes the argument non-interactive: the transcript is a running BLAKE2b state into which every prover message is absorbed, domain-separated by type, and out of which every challenge is squeezed—the entire-prefix discipline of Definition 9.18 enforced by construction.

The linear cost of each IPA opening check (Proposition 4.39) makes the deployed verifier's work linear in the commitment dimension: Halo 2 has succinct *proofs* but does not meet the strict sublinear-verifier clause of Definition 9.24, and it is called a zk-SNARK under the broader convention that asks succinctness of the proof alone. The system's signature innovation—the technique that named *Halo*—is *accumulation*: the one expensive step, the $O(d)$ final check of each IPA opening, is deferred and passed along rather than paid inside a circuit, one proof attesting to the verification of another—which is what makes *recursive* composition possible without a trusted setup. Deployed Zcash does not use recursion: each Orchard proof is verified directly, and what the deployed verifier takes from the Halo idea is the deferral alone, exploited for *batching*—many proofs' deferred final checks collapsing into one multiscalar multiplication (consumed for Orchard bundles as described in Remark 4.40). These claims rest on model and composition hypotheses beyond the generic recipe (Remark 9.21).

Three load-bearing ideas from this section carry forward into everything the upper volumes build. *Rewinding* (Remark 9.8) is the universal device: the same cloning of an algorithm's state powers the extractor that defines knowledge and the simulator that defines zero knowledge. *The layered split* of Theorem 9.28—an unconditionally sound information-theoretic core, wrapped in a cryptographically enforced commitment layer—is the architecture of every modern proof system, and the right frame for auditing one: algebraic bugs and cryptographic bugs live in different layers. And *knowledge soundness*, not mere soundness, is what an application stakes its security on: a shielded protocol does not care that a spendable note exists; it cares that the prover has one.

10 Merkle trees and commitments to sets

The adversary of this final section tells lies about a set. Her first lie is *inclusion*: she claims that a value sits in a committed collection it never entered—a payment out of thin air, spent against a ledger that never recorded it. Her second lie is *equivocation*: she publishes one short commitment and later produces two different collections that both explain it, so that the commitment pinned down nothing. Her third role is not a lie but a gaze: as an *observer* she watches an honest member prove that it belongs, and learns *which* member spoke—enough, in a payment system, to trace every coin. This section defeats all three with a single construction, built from nothing beyond the collision-resistant hash of §3, and closes the volume by spending everything the earlier sections banked: the commitments of §4 become the leaves, the algebraic hashes of §3 and §4 become the internal nodes, and the zero-knowledge arguments of §9 hide the path from leaf to root.

The plan runs from public to private. We first fix the two statements a verifier might want proved about a large collection, and recall the compression function and its collision game. The binary Merkle tree follows, with its root as a binding commitment to the whole list; a worked depth-two tree keeps every index honest. Authentication paths then give logarithmic-size membership proofs, complete and sound; the soundness proof introduces the one technique this section reuses until the end—extracting a hash collision from the first level at which two disagreeing computations agree. Layer and domain separation keep that extraction honest against structural confusion between leaves and nodes, with a deployed failure as the cautionary tale. Append-only trees and the frontier make the structure grow in logarithmic time; the vector-commitment abstraction names exactly what the root achieves; and zero-knowledge paths convert the public proof into a private one. The capstone assembles an Orchard shielded spend end to end.

10.1 Two statements about a set

The setting is asymmetric. A prover holds a large collection of values—transactions in a block, public keys in a directory, records in a log; for this volume, the ever-growing set of note commitments in a shielded payment system. A verifier wants conviction about that collection without storing it, indeed without ever seeing most of it. Two statements cover the uses that matter here.

- *Membership.* "Value v occurs at position i of the list"—or, forgetting the position, "v belongs to the set."
- *Integrity.* "The collection is exactly the one committed to earlier, unaltered."

What is wanted is a single short *commitment* that determines the whole collection, together with short *membership proofs*: "short" meaning of size independent of the number of elements, or at worst logarithmic in it. A commitment of that shape makes integrity a one-comparison check, and membership a small certificate that travels with the value it certifies. The construction achieving this—the *Merkle tree*, due to Ralph Merkle—needs nothing more than a collision-resistant hash function, and it is the membership backbone of Orchard: every shielded spend proves, against a published root, that some note commitment lies in the tree.

Notation is as fixed in §3.1 and carried throughout the volume: $\{0,1\}^*$ is the set of finite binary strings, $\{0,1\}^n$ the strings of length exactly n, $x \| y$ concatenation, and λ the security parameter. All algorithms are PPT in λ, and $\mathrm{negl}(\lambda)$ denotes a negligible function (*Math Guide*, §"Polynomial, exponential, and negligible functions").

10.2 The compression function and collision resistance

The tree consumes one ingredient, which we recall in the keyed form that makes its security a meaningful asymptotic statement.

Definition 10.1 (Compression function)**.** Let $H\colon \mathcal{K} \times \{0,1\}^* \to \{0,1\}^\ell$ be a keyed hash family exactly as in Definition 3.2: a key $s \in \mathcal{K}$ is sampled publicly at random and fixed, the digest length is $\ell = \ell(\lambda)$, and $H_s := H(s,\cdot)$ is the family member in force. When the input length is restricted to a fixed multiple of the output length, the family is a *compression function*; a *two-to-one* compression function has signature

$$h_s\colon \{0,1\}^{2\ell} \longrightarrow \{0,1\}^\ell,$$

taking two digests to one.

The key s models the choice of a concrete function from a family, which is what rescues collision resistance from the hardcoded-collision triviality discussed in §3.1: for a single fixed function a colliding pair exists and some small program outputs it, so only the keyed formulation supports a clean quantifier. In practice one deploys a fixed standardised function—or, inside arithmetic circuits, an algebraic hash in the sense of §3.8 and §4.8—and we suppress s from the notation, writing h and H plain.

The security property the tree needs is collision resistance, exactly as in Definition 3.7 (§3.2): the game $\mathsf{COL}_H^{\mathcal{A}}$ hands the adversary the key s, the adversary outputs a pair (x, x'), and it wins if $x \neq x'$ while $H_s(x) = H_s(x')$; the family is collision resistant if every PPT adversary wins with probability $\mathrm{negl}(\lambda)$. This is the strongest of the three classical hash notions (Proposition 3.8), and the generic birthday attack caps an ℓ-bit digest at $\ell/2$ bits of collision security (Lemma 3.11)—whence the 256-bit outputs used at the 128-bit level here. Merkle trees require *full* collision resistance, not merely second-preimage resistance, for a reason the soundness theorem (Theorem 10.12) makes exact: the party who assembles the tree may itself be the adversary, choosing every leaf, which is precisely the both-inputs-adversarial regime that Remark 3.10 assigns to collision resistance.

10.3 Binary Merkle trees

The construction hashes the collection pairwise, level by level, until one digest remains.

Definition 10.2 (Binary Merkle tree)**.** Fix a two-to-one compression function $h\colon \{0,1\}^{2\ell} \to \{0,1\}^{\ell}$, a depth $d \in \mathbb{N}$, and an ordered list $L = (L_0, \dots, L_{2^d-1})$ of 2^d leaf values in $\{0,1\}^{\ell}$. The *Merkle tree of depth d over L* is the complete binary tree with 2^d leaves whose nodes carry labels in $\{0,1\}^{\ell}$, defined level by level from the leaves (level 0) to the root (level d) by

$$N_{0,j} = L_j \quad (0 \le j < 2^d), \qquad N_{t+1,j} = h(N_{t,2j} \,\|\, N_{t,2j+1}) \quad (0 \le t < d,\ 0 \le j < 2^{d-t-1}). \tag{3}$$

Level t has 2^{d-t} nodes, and $N_{t,j}$ is the j-th node from the left at level t. The unique level-d node is the *root*: $\mathsf{root}(L) := N_{d,0}$.

Example 10.3 (A depth-two tree)**.** With $d = 2$ and leaves L_0, L_1, L_2, L_3, the tree has seven nodes: the four leaves, the two level-one nodes

$$N_{1,0} = h(L_0 \,\|\, L_1), \qquad N_{1,1} = h(L_2 \,\|\, L_3),$$

and the root

$$\mathsf{root}(L) = N_{2,0} = h(\, h(L_0 \,\|\, L_1) \,\|\, h(L_2 \,\|\, L_3)\,).$$

The single ℓ-bit root summarises the whole list of four ℓ-bit strings; the example carries through the next two subsections.

A list whose length m is not a power of two is handled by fixing the depth d with $2^d \ge m$ and padding the remaining $2^d - m$ leaf slots with a canonical value—the all-zero string, say, or the hash of the empty string. Padding must not create ambiguity: two different lists must never yield the same padded leaf sequence, else the root fails to determine the list before any cryptography is consulted. Binding the true length m into the root's computation—one further compression of the top node with an encoding of m—removes any such ambiguity. An alternative is the *unbalanced* tree, in which a node with a single child passes its child's label upward unchanged; it is convenient but subtle, and we return to it alongside second-preimage attacks in Remark 10.16.

The root is short; the claim that it *commits* to the list is a security property, stated as a game in the tradition of Definition 1.15.

Definition 10.4 (Merkle commitment; binding)**.** The *Merkle commitment* to a list L of 2^d leaves is $\mathsf{root}(L) \in \{0,1\}^{\ell}$. The game $\mathsf{BIND}^{\mathcal{A}}_{H,d}(\lambda)$ runs as follows.

1. The challenger samples $s \xleftarrow{\$} \mathcal{K}$, fixing the compression function $h := H_s$, and sends s to the adversary.
2. The adversary outputs two lists L, L', each of 2^d leaves in $\{0,1\}^{\ell}$.
3. The game outputs 1 iff $L \neq L'$ and $\mathsf{root}(L) = \mathsf{root}(L')$.

The commitment is *binding* if every PPT adversary wins with probability $\mathrm{negl}(\lambda)$.

Proposition 10.5 (Binding of the root)**.** *If h is collision resistant, the Merkle commitment is binding. Concretely, there is a deterministic procedure that, from any two distinct equal-length lists $L \neq L'$ with $\mathsf{root}(L) = \mathsf{root}(L')$, extracts a collision for h in time linear in the tree size—one root-to-leaf walk of $O(d)$ hash comparisons. The reduction is tight: an adversary winning* BIND *with probability ε yields a collision finder succeeding with the same probability ε.*

Proof. Build both trees. The lists differ at some leaf index j_0, so $N_{0,j_0} \neq N'_{0,j_0}$, writing N and N' for the labels of the two trees. Follow the leaf-to-root path

$$N_{0,j_0},\ N_{1,\lfloor j_0/2 \rfloor},\ N_{2,\lfloor j_0/4 \rfloor},\ \dots,\ N_{d,0}$$

in both trees. The labels differ at level 0 and agree at level d—the common root—so there is a *smallest* level t at which they first agree: with $j = \lfloor j_0/2^t \rfloor$,

$$N_{t,j} = N'_{t,j}, \qquad \text{yet} \qquad N_{t-1,2j} \neq N'_{t-1,2j} \text{ or } N_{t-1,2j+1} \neq N'_{t-1,2j+1},$$

since one of the two children at level $t-1$ is the path node, at which the labels still disagree by minimality of t. By the recurrence (3),

$$h(N_{t-1,2j} \,\|\, N_{t-1,2j+1}) = N_{t,j} = N'_{t,j} = h(N'_{t-1,2j} \,\|\, N'_{t-1,2j+1}),$$

while the two 2ℓ-bit inputs differ in at least one half: a collision for h, located after at most d comparisons along one path. The reduction wrapping this extractor runs the BIND adversary once and extracts whenever it wins, so collision resistance forces the winning probability to be negligible. □

The proof's engine deserves its name now, because the section reuses it almost verbatim twice more: *a discrepancy at the leaves and agreement at the root must, by this pigeonhole-like descent, manifest a collision somewhere on the path between them.* There is a smallest level at which two disagreeing computations agree; the compression call producing that first agreement has two differing preimages. All three security theorems of this section are this one sentence wearing different inputs.

10.4 Authentication paths and membership proofs

The root alone certifies integrity; membership needs a certificate relating one leaf to the root. The certificate is the list of siblings along the leaf's path upward—everything the verifier needs to recompute the root from the leaf, and nothing more.

Definition 10.6 (Authentication path)**.** Let a depth-d tree over L be given, and let i be a leaf index, $0 \leq i < 2^d$, with binary expansion $i = b_{d-1}b_{d-2}\cdots b_0$, so that bit b_t records whether the level-t node on the leaf-to-root path is a left child ($b_t = 0$) or a right child ($b_t = 1$). The *authentication path* (also *Merkle proof* or *membership witness*) for index i is

$$\pi_i = ((s_0, b_0), (s_1, b_1), \dots, (s_{d-1}, b_{d-1})),$$

where s_t is the label of the *sibling* of the level-t node on the path. Explicitly, with $j_t = \lfloor i/2^t \rfloor$ the path node's index at level t,

$$s_t = \begin{cases} N_{t,j_t+1} & \text{if } b_t = 0 \quad \text{(sibling on the right)}, \\ N_{t,j_t-1} & \text{if } b_t = 1 \quad \text{(sibling on the left)}. \end{cases}$$

The path has exactly d entries.

Definition 10.7 (Path verification)**.** The verifier $\mathsf{Verify}(rt, i, v, \pi_i)$, on a root rt, an index i, a claimed leaf value $v \in \{0,1\}^\ell$, and a path $\pi_i = ((s_0, b_0), \dots, (s_{d-1}, b_{d-1}))$, proceeds in two stages. First the *index-consistency check*: reject unless, for every t, the bit b_t carried in π_i equals bit t of the binary expansion of i. Then set $a_0 := v$ and fold the path upward: for $t = 0, \dots, d-1$,

$$a_{t+1} := \begin{cases} h(a_t \,\|\, s_t) & \text{if } b_t = 0, \\ h(s_t \,\|\, a_t) & \text{if } b_t = 1. \end{cases} \tag{4}$$

Accept iff $a_d = rt$. We call a_d *the root recomputed from* (i, v, π_i). The bit b_t places the running

value on the side matching the argument order of the tree-building recurrence: a left child is hashed on the left.

Proposition 10.8 (Completeness)**.** *For every list L, every index i, and π_i the genuine authentication path for i in the tree over L,*

$$\mathsf{Verify}(\mathsf{root}(L),\, i,\, L_i,\, \pi_i) = \mathsf{accept}.$$

Proof. The index-consistency check passes by construction, the genuine path carrying exactly the bits of i. We show by induction on t that $a_t = N_{t,\lfloor i/2^t \rfloor}$, the label of the level-t node on the leaf-to-root path. The base is $a_0 = L_i = N_{0,i}$. For the step, the level-t path node and its sibling s_t are precisely the two children of the level-$(t+1)$ path node, and the fold (4) places a_t on the side dictated by b_t with s_t on the other side—reproducing exactly the two arguments of h in the recurrence (3), so $a_{t+1} = N_{t+1,\lfloor i/2^{t+1} \rfloor}$. At $t = d$ this reads $a_d = N_{d,0} = \mathsf{root}(L)$, and the verifier accepts. □

Proposition 10.9 (Logarithmic proof size)**.** *An authentication path in a depth-d tree consists of d sibling labels and d direction bits, $d\ell + d$ bits in all, and verification performs exactly d evaluations of h. Choosing the minimal sufficient depth $d = \lceil \log_2 m \rceil$ for a list of $m \geq 2$ leaves (any $2^d \geq m$ suffices) yields proofs of $\lceil \log_2 m \rceil (\ell + 1)$ bits—logarithmic in the length of the list.*

Proof. One sibling label, one direction bit, and one compression per level, over the d levels above the leaves. □

The concrete scale is worth pausing on. For a set of a billion elements—$m = 2^{30} = 1\,073\,741\,824$—a membership proof consists of 30 hash values and 30 direction bits. With a 256-bit hash the sibling labels occupy $30 \cdot 256 = 7680$ bits, which is 960 bytes; direction bits included, 964 bytes—under a kilobyte, however the billion elements are arranged. (The byte counts are the output of the volume's companion script.) This logarithmic succinctness is the central objective: the verifier's storage is one digest, the certificate fits in a packet, and neither grows meaningfully as the set does.

Example 10.10 (A depth-two membership proof)**.** Continue Example 10.3, and prove membership of L_2 at index $i = 2 = (10)_2$, so $b_0 = 0$ and $b_1 = 1$. At level 0 the path node $N_{0,2} = L_2$ is a left child, with sibling $s_0 = N_{0,3} = L_3$. At level 1 the path node $N_{1,1} = h(L_2 \,\|\, L_3)$ is a right child, with sibling $s_1 = N_{1,0} = h(L_0 \,\|\, L_1)$. The path is

$$\pi_2 = ((L_3, 0),\ (h(L_0 \,\|\, L_1), 1)).$$

Verification folds upward: $a_1 = h(L_2 \,\|\, L_3)$, then $a_2 = h(h(L_0 \,\|\, L_1) \,\|\, a_1) = \mathsf{root}(L)$, and the verifier accepts. Note what the verifier never saw: neither L_0 nor L_1 themselves, only their digest $h(L_0 \,\|\, L_1)$—yet it is convinced that L_2 sits at index 2 of the committed list.

10.5 Soundness: forging a path implies a collision

Completeness says honest paths verify; the security question is whether dishonest ones can. The threat is the inclusion lie of the section's opening: an adversary who convinces the verifier, against the genuine root, that position i holds a value it does not.

Definition 10.11 (Path forgery)**.** Fix a list L of 2^d leaves with root $rt = \mathsf{root}(L)$. The game $\mathsf{PATHFORGE}^{\mathcal{A}}_{H,L}(\lambda)$ runs as follows.

1. The challenger samples $s \xleftarrow{\$} \mathcal{K}$ to fix $h := H_s$, builds the tree over L, and sends s together with L—equivalently, the whole tree—to the adversary.
2. The adversary outputs an index i and a pair (v^*, π^*).
3. The game outputs 1 iff $\mathsf{Verify}(rt, i, v^*, \pi^*) = \mathsf{accept}$ and $v^* \neq L_i$.

A winning output is a *path forgery at index i*: the verifier accepts, against the honestly computed root, a leaf value different from the genuine one at position i.

Theorem 10.12 (Soundness of Merkle membership proofs)**.** *Let h be collision resistant. Then no PPT adversary, given the list L and the whole tree, produces a path forgery except with probability* $\mathrm{negl}(\lambda)$*. In constructive form: there is a deterministic extractor that from any list L, index i, and accepting forgery (v^*, π^*) with $v^* \neq L_i$ outputs a collision for h, in time linear in the tree size—recomputing the genuine leaf-to-root labels from L dominates; given the built tree, one verifier run plus a level-by-level comparison, $O(d)$ work in all, completes the extraction.*

Proof. Run the verifier's fold (4) on (i, v^*, π^*), obtaining the chain $a_0^* = v^*, a_1^*, \dots, a_d^* = rt$ (the final equality because the forgery is accepting). Alongside it place the genuine chain $a_t = N_{t,\lfloor i/2^t \rfloor}$, the labels of the leaf-to-root path in the honest tree; by the induction of Proposition 10.8 this is the chain the fold produces on the genuine inputs, and $a_d = rt$.

The two chains disagree at the bottom, $a_0^* = v^* \neq L_i = a_0$, and agree at the top, $a_d^* = a_d = rt$. Hence

$$t^* := \min\{\, t : a_t^* = a_t \,\} \in \{1, \dots, d\}$$

exists, and $a_{t^*-1}^* \neq a_{t^*-1}$ by minimality. Because the forgery is accepting, the index-consistency check of Definition 10.7 forces the direction bits carried in π^* to equal the bits of i—the same bits the genuine path carries—so at level $t^* - 1$ *both* chains place their running value on the same side of the compression call: both compute $h(\cdot \,\|\, \text{sibling})$ if $b_{t^*-1} = 0$, both $h(\text{sibling} \,\|\, \cdot)$ if $b_{t^*-1} = 1$. Form the two 2ℓ-bit preimages accordingly,

$$X^* = \begin{cases} a_{t^*-1}^* \,\|\, s_{t^*-1}^* & \text{if } b_{t^*-1} = 0, \\ s_{t^*-1}^* \,\|\, a_{t^*-1}^* & \text{if } b_{t^*-1} = 1, \end{cases} \qquad X = \begin{cases} a_{t^*-1} \,\|\, s_{t^*-1} & \text{if } b_{t^*-1} = 0, \\ s_{t^*-1} \,\|\, a_{t^*-1} & \text{if } b_{t^*-1} = 1, \end{cases}$$

where $s_{t^*-1}^*$ is the sibling claimed in π^* and s_{t^*-1} the genuine sibling. Then

$$h(X^*) = a_{t^*}^* = a_{t^*} = h(X), \qquad X^* \neq X,$$

the inequality because the two preimages differ in the half occupied by the running value. The pair (X^*, X) is a collision for h, found deterministically in $O(d)$ work once the genuine chain is at hand—a read-off from the built tree, or $2^d - 1$ hashes from the bare list L. □

The reduction wrapping the extractor is immediate. If a PPT adversary $\mathcal{A}$ produces path forgeries with probability $\varepsilon(\lambda)$, the collision finder $\mathcal{B}$ that receives the key, samples or receives L, runs $\mathcal{A}$, and applies the extractor outputs a collision whenever $\mathcal{A}$ succeeds—hence with probability $\varepsilon(\lambda)$, at an overhead of building the tree over L, which $\mathcal{B}$ does in any case to play the challenger, plus one verifier run and $O(d)$ comparisons. Collision resistance of h therefore forces $\varepsilon(\lambda) \in \mathrm{negl}(\lambda)$, and the reduction is tight: no advantage is lost in the translation.

Remark 10.13 (Scope: a fixed honest root). Theorem 10.12 is a statement about a *fixed honest root*: relative to a root that genuinely is $\mathsf{root}(L)$, the only leaf value admitting an accepting path at position i is L_i, up to a collision in h. It does not by itself prevent an adversary who is free to *choose* the root from committing to a maliciously crafted tree; that threat is what binding (Proposition 10.5)

rules out. With an untrusted party supplying the root, both properties are necessary and they divide the labour cleanly: binding ensures the root determines at most one list, and path soundness ensures openings are faithful to it. Together they make the root a sound, binding commitment with succinct openings—the abstraction §10.8 names, where a single argument will deliver both faces at once.

10.6 Layer and domain separation

The soundness theorem has a structural blind spot, inherited from an innocent-looking convenience. Suppose the leaves are arbitrary-length data, hashed into $\{0,1\}^\ell$ by the *same* function used internally. Then an internal label $h(A \parallel B)$ is itself a well-formed leaf value—it has the right length—and one string can play both roles. An adversary exploits the ambiguity directly: she presents the internal node $N_{1,0} = h(L_0 \parallel L_1)$ as if it were a *leaf* near the top of a shallower tree, equipped with the correspondingly shorter authentication path, and a verifier who does not know the depth accepts. The root now has a second list explaining it—a genuine second-preimage attack, with no collision of h anywhere in sight. Absent a way for a label to assert "I am a leaf" or "I am an internal node", the leaf-to-root structure that Theorem 10.12 relies on collapses.

The repair is the domain-separation discipline of §3.5, applied *within* a single structure: distinct structural positions get distinct tags.

Definition 10.14 (Layer and domain separation)**.** A Merkle construction has *domain separation* if the function applied at distinct structural positions is forced to differ, by prepending a position-dependent tag to every hash input. Concretely, from a variable-input-length hash H define

$$\mathsf{LeafHash}(v) := H(\texttt{0x00} \parallel v), \qquad \mathsf{NodeHash}_t(A, B) := H(\texttt{0x01} \parallel \langle t \rangle \parallel A \parallel B),$$

where the byte tags `0x00` and `0x01` distinguish the *layer kind*—leaf versus internal node—and $\langle t \rangle$ is a fixed-length encoding of the level index, distinguishing internal nodes by depth. The tree recurrence becomes

$$N_{0,j} = \mathsf{LeafHash}(L_j), \qquad N_{t+1,j} = \mathsf{NodeHash}_{t+1}(N_{t,2j}, N_{t,2j+1}),$$

and the verifier's fold applies the same tagged functions at the same positions.

Proposition 10.15 (Separation forces within-domain collisions)**.** *With the tagging of Definition 10.14, any collision extracted by the binding proof (Proposition 10.5) or the soundness proof (Theorem 10.12) is a collision for H between two inputs* sharing the same leading tag*: a leaf preimage against a leaf preimage, or a level-t node preimage against a level-t node preimage. In particular, no accepting forgery can substitute an internal node for a leaf, or a node at one level for a node at another, without exhibiting a within-domain collision of H—which collision resistance forbids.*

Proof sketch. The extractors of both proofs locate the smallest level at which two disagreeing computations agree, and collide the two preimages of the compression call at that position. Under the tagged recurrence, the verifier's recomputation applies the position's prescribed function exactly as the honest tree does: at the leaves both preimages pass through $\mathsf{LeafHash}$, at level $t \geq 1$ both pass through $\mathsf{NodeHash}_t$. Both colliding preimages therefore begin with the same tag—`0x00`, or $\texttt{0x01} \parallel \langle t \rangle$—and the collision lies within a single domain. A cross-domain confusion, by contrast, would require two inputs with *different* leading tags and equal H-images, which is again an H-collision; either way the adversary has broken collision resistance, and the shallower-tree

attack above is foreclosed because a leaf preimage and a node preimage can no longer be the same string. □

Remark 10.16 (Unbalanced trees, padding, and a deployed failure). The subtleties are not hypothetical. Bitcoin's transaction Merkle tree handles an odd level by *duplicating* its last node before hashing pairwise, and the duplication rule is not injective: a list of transactions and the same list with its tail duplicated can yield the same root. This is CVE-2012-2459, a consensus-splitting vulnerability—an attacker could present two *different* transaction lists sharing a root, one valid and one invalid, and nodes could be made to cache the invalid one as permanently rejected (documented at length in Zcash's inherited validator: `zebra-chain` crate, `src/block/merkle.rs`, which reproduces the upstream warning that the flawed algorithm must not be reused; the deployed defence—rejecting any block whose transaction hashes are not unique—lives in `zebra-consensus`, `src/block/check.rs`). The lesson generalises to three rules. First, layer-separate leaves from internal nodes, as Definition 10.14 prescribes. Second, make every padding or duplication rule *injective* on admissible lists—for instance by binding the true leaf count into the root's computation. Third, feed NodeHash only fixed-length inputs, so that no length-extension structure of the underlying hash (§3.5) can intervene between layers.

Remark 10.17 (Algebraic hashes in the tree; Orchard's MerkleCRH). When the Merkle path is verified inside a zero-knowledge circuit (§10.9), arithmetic gates over $\mathbb{F}_p$ evaluate the compression function, and the choice of h dominates the cost. A bit-oriented hash such as SHA-256 costs tens of thousands of constraints per call; one therefore uses an *algebraic* hash with compression $h\colon \mathbb{F}_p^2 \to \mathbb{F}_p$. Poseidon (§3.8) is a low-degree permutation-based map computable in few constraints; Sinsemilla (§4.8) is cheap in constraints through lookups and incomplete curve additions, *not* through low degree. For Poseidon, layer and domain separation come not from byte tags but from distinct field constants—initial capacity values or round constants—fed into the permutation; the tree's security argument is identical, with "collision" read as a collision of the field-valued compression function.

Orchard's Merkle hash MerkleCRH retains the tag mechanism of Definition 10.14 in encoded form. Each internal call prepends the 10-bit encoded layer index to the two 255-bit child encodings—a single fixed input length of 520 bits (a total the volume's companion script checks), exactly the fixed-length-per-personalisation regime in which the deployed Sinsemilla hash is collision resistant (§4.8)—with *all* layers sharing one Sinsemilla instance (`MerkleHashOrchard::combine`, prepending `i2lebsp_k(level)` to two `L_ORCHARD_MERKLE` = 255-bit child encodings; protocol specification § 5.4.1.3). Separation between *distinct* Sinsemilla instances comes from the personalisation string—here `z.cash:Orchard-MerkleCRH`—fed to hash-to-curve to derive the instance-specific base point Q, while the generator table S is shared across all instances (`HashDomain::new` derives Q from the personalisation via the `z.cash:SinsemillaQ` hash-to-curve domain; the table derives from the fixed `z.cash:SinsemillaS`). Sinsemilla is collision resistant under a discrete-logarithm assumption (Proposition 4.29), so the tree's hashing and its commitment role fuse into one Pedersen-like map: the same structure that compresses the children also binds them.

10.7 Append-only and incrementally updatable trees

A shielded pool's set of note commitments only ever grows, and it grows one leaf at a time. Rebuilding a depth-32 tree from scratch on every insertion would cost 2^{32} hashes; the structure of the tree permits the update in d hashes and $O(d)$ state, and the state has a name.

Definition 10.18 (Append-only Merkle tree). Fix a maximum depth d, giving capacity 2^d, and the level-indexed NodeHash functions of Definition 10.14. Leaf values here enter as level-0 labels directly, with no LeafHash: the leaves are already digests, and the fixed depth together

with the level tags (Remark 10.17) carries the layer separation—the convention of deployed Orchard, whose leaf is the extracted coordinate itself (Example 10.31). An *append-only* (or *incremental*) *Merkle tree* maintains a position counter m—the number of leaves inserted so far, $0 \le m \le 2^d$—and supports one mutating operation $\mathsf{Append}(v)$, which places v at leaf index m and increments m. Empty leaf slots $m, \dots, 2^d - 1$ carry a fixed *empty-leaf* value $\bot_0$, and the labels of entirely empty subtrees follow by precomputation:

$$\bot_{t+1} := \mathsf{NodeHash}_{t+1}(\bot_t, \bot_t), \qquad 0 \le t < d.$$

The *root after m appends*, written root_m, is the root of the depth-d tree, under this recurrence, whose first m level-0 labels are the inserted values and whose remaining ones are $\bot_0$.

Definition 10.19 (Frontier)**.** The *frontier* of an append-only tree holding m leaves is the minimal state needed to (i) compute the current root and (ii) update incrementally on the next Append. For each level $t \in \{0, \dots, d-1\}$ it holds at most one *carry*: the label of the left child of the next node to be completed at level $t+1$, retained precisely when leaf index m has a 1 bit at position t—a completed left sibling awaiting its right neighbour. Equivalently, writing m in binary as $b_{d-1} \cdots b_0$, the frontier stores, for each t with $b_t = 1$, the already-finalised left-sibling label f_t at level t along the rightmost filled path. Its size is at most d labels—one per set bit of m, hence $O(\log m)$ state.

The right picture is arithmetic. The binary representation of m decomposes the inserted leaves into completed perfect subtrees of distinct heights—one of height t for each set bit b_t—and the frontier holds exactly the roots of those subtrees: the pieces that are "complete and waiting for a right neighbour". An append is then the Merkle analogue of incrementing a binary counter: a carry at level t corresponds to combining two height-t subtrees into one of height $t+1$, and the chain of carries is as long as the run of trailing 1-bits of m.

Proposition 10.20 (Incremental append in logarithmic time)**.** *From the frontier of an append-only tree holding $m < 2^d$ leaves, $\mathsf{Append}(v)$ computes the new frontier and the new root root_{m+1} using at most d evaluations of $\mathsf{NodeHash}$ and $O(d)$ additional work, while storing only $O(d)$ labels throughout.*

Proof. The algorithm is carry propagation followed by an upward fold. Set the running label $c \leftarrow v$ at level $t \leftarrow 0$. While bit t of m is 1, the new path node at level t is a right child whose left sibling is the stored carry f_t: set $c \leftarrow \mathsf{NodeHash}_{t+1}(f_t, c)$, clear the carry f_t, and increment t. When bit t of m is 0, store c as the new carry f_t and stop. Writing z for the number of trailing 1-bits of m, the carry phase costs exactly z hashes and deposits f_z at level z—the root of the just-completed height-z subtree containing the new leaf, matching the binary increment $m \mapsto m+1$, which clears z trailing ones and sets bit z.

To obtain root_{m+1}, fold upward from level z with the just-deposited f_z as the running value—it is the path label at level z, not a sibling. For $t = z, \dots, d-1$, combine the running value with the stored carry f_t as left sibling when bit t of m is 1—those carries lie above the cleared run and survive the append untouched—and with the precomputed empty-subtree label $\bot_t$ as right sibling otherwise; at level z itself bit z of m is 0, so the first combination is with $\bot_z$, never with the deposited carry. One hash per level makes $d - z$ hashes in all (zero if $z = d$, when the tree has just filled). The total is exactly $z + (d - z) = d$ evaluations of $\mathsf{NodeHash}$, and the retained state is the at-most-d carries: precisely the frontier for $m + 1$ leaves. □

Example 10.21 (The frontier as a binary counter)**.** Take depth $d = 3$ (capacity 8) and insert v_0, v_1, v_2 in turn; the frontier evolves exactly as the binary counter of the leaf count m.

- After v_0, $m = 1 = (001)_2$: one completed height-0 subtree; the frontier is $\{f_0 = v_0\}$.

- After v_1, $m = 2 = (010)_2$: leaf v_1 at index 1 is a right child, triggering one carry $c = \mathsf{NodeHash}_1(v_0, v_1)$—a completed height-1 subtree; the frontier is $\{f_1 = c\}$, with f_0 cleared.
- After v_2, $m = 3 = (011)_2$: leaf v_2 at index 2 is a left child and becomes a new height-0 carry; the frontier is $\{f_1 = \mathsf{NodeHash}_1(v_0, v_1),\ f_0 = v_2\}$ —exactly the two set bits of 3.

Each append touched only the carry chain to the new leaf, never the whole tree. (The trace involves no numeric hash values and follows mechanically from the algorithm of Proposition 10.20; the volume's companion script replays it symbolically and checks the occupied frontier levels against the set bits of m at every step.)

Appending changes the root, which raises a question the definition of soundness so far does not answer: which root is a proof valid against?

Definition 10.22 (History binding)**.** An append-only tree is *history-binding* when the surrounding protocol treats every historical root as an acceptable commitment in its own right: each membership proof verifies against the root it was formed for, and appending further leaves never invalidates a previously valid $(root, leaf, path)$ triple for that root. The soundness of each individual proof is exactly Theorem 10.12 applied to the corresponding fixed root.

Deployed Orchard is history-binding by construction. The chain records each block's note-commitment-tree root—its *anchor* —and a spend may prove membership against *any* recorded historical anchor: consensus requires the anchor named by a transaction to be the root of the Orchard note-commitment tree as it stood at the end of some earlier block, checked by membership in the set of recorded anchors (protocol specification § 3.7, 'Action Transfers and their Descriptions'). Per-proof soundness is then the fixed-root soundness theorem, once per anchor. The frontier is likewise the deployed representation of the growing tree (`incrementalmerkletree` crate, types `Frontier` and `NonEmptyFrontier`; the empty-subtree labels are the `EMPTY_ROOTS` table of `orchard`).

Remark 10.23 (Why append-only, and what it costs). Deletion or in-place update could be supported at $O(\log m)$ per operation, given the relevant siblings; the restriction to appends is a design choice, made for a monotonicity it buys. Once a leaf is inserted it is never removed, so a membership proof valid against the root at insertion time remains valid against *that historical root* forever. This is exactly what a shielded pool requires—a note, once added, can always be spent later by proving against some past anchor—and it spares the spender from tracking a moving target as the tree grows under everyone else's insertions. The cost falls on the verifier, who must accept a window of historical roots rather than a single current one; the previous paragraph is that window, deployed.

10.8 Vector commitments

The section has been proving properties of one construction; it pays to name the abstraction the construction inhabits, both to state precisely what the root achieves and to mark the boundary with its relatives. The abstraction is a commitment to an *ordered vector* that opens one position at a time. (Its unordered sibling—the *cryptographic accumulator*, which certifies set membership with no notion of position—exists in the literature, but Orchard never needs it: notes occupy positions, and the position is part of what the spend circuit witnesses. We do not develop it.)

Definition 10.24 (Vector commitment)**.** A *vector commitment* is a tuple of PPT algorithms $(\mathsf{Setup}, \mathsf{Commit}, \mathsf{Open}, \mathsf{Vf})$:

- $\mathsf{Setup}(1^\lambda, n) \to pp$ produces public parameters for vectors of length n;

- $\mathsf{Commit}(pp, (v_0, \ldots, v_{n-1})) \to (C, aux)$ outputs a short commitment and auxiliary state;
- $\mathsf{Open}(pp, i, aux) \to \pi_i$ produces a proof that position i holds v_i;
- $\mathsf{Vf}(pp, C, i, v, \pi) \to \{\mathsf{accept}, \mathsf{reject}\}$ verifies an opening.

The scheme is *correct* if honest openings always verify, and *succinct* if $|C|$ and $|\pi_i|$ are bounded by a fixed polynomial in λ and $\log n$. It is *position-binding* if every PPT adversary wins the following game with probability $\mathrm{negl}(\lambda)$: given pp, the adversary outputs (C, i, v, v', π, π') and wins iff $v \neq v'$ while both $\mathsf{Vf}(pp, C, i, v, \pi)$ and $\mathsf{Vf}(pp, C, i, v', \pi')$ accept. Note that the adversary chooses C itself: binding must hold even for maliciously formed commitments, with no honest committer anywhere in the game.

Theorem 10.25 (Merkle trees are succinct position-binding vector commitments)**.** *Instantiate the syntax by:* Setup *samples h (and fixes d with $n = 2^d$);* Commit *computes the tree and outputs $C = \mathsf{root}$ with aux the full tree;* Open *returns the authentication path; and* $\mathsf{Vf} = \mathsf{Verify}$. *If h is collision resistant, this is a correct, succinct, position-binding vector commitment, with commitment size ℓ and opening size $O(\ell \log n)$.*

Proof. Correctness is completeness (Proposition 10.8), and the size bounds are Proposition 10.9. For position-binding, suppose an adversary outputs two accepting openings (v, π) and (v', π') at the same index i under the same root C, with $v \neq v'$. Run the verifier's fold (4) on each, obtaining two recomputed chains that disagree at the bottom ($v \neq v'$) and agree at the top (both reach C, since both openings accept). The index-consistency check forces both paths to carry the direction bits of i, so at every level both chains place their running value on the same side of the compression call. The first-level-of-agreement extraction of Theorem 10.12 now applies verbatim to this pair of chains and yields two distinct 2ℓ-bit inputs with the same h-image: a collision. The argument needed *no honest reference tree*—it pits the two forged openings against each other—so it binds even a maliciously chosen root, which is the strong form of position-binding the definition demands. □

Remark 10.26 (Position-binding versus path soundness). The two theorems answer different verifiers. Path soundness (Theorem 10.12) fixes an *honest* root and forbids opening a position to anything but its true value; it speaks to a verifier who trusts where the root came from. The vector-commitment theorem (Theorem 10.25) forbids opening a single position two *different* ways even under an *adversarial* root; it speaks to a verifier who received the root from an untrusted source, and it is what "commitment" should mean—C determines at most one value per position, whoever made C. Both reduce to the same first-level-of-agreement collision extraction; the only difference is which pair of chains is collided, forged-against-genuine in the one case and forged-against-forged in the other.

10.9 Privacy-preserving membership via zero-knowledge paths

The Merkle membership proof is succinct but not private—and for a shielded payment system, not private is fatal. Revealing the authentication path of §10.4 discloses the leaf value itself (here, the note commitment being spent) and the index i at which it sits, and even the sibling labels can leak structural information about the neighbourhood of the leaf. An observer correlating spends with the appends that created their leaves links payments end to end: spending a note must not reveal *which* note is spent. The remedy is to prove the *existence* of a valid path in zero knowledge, treating the entire path as a private witness and publishing only the root.

Definition 10.27 (The membership relation)**.** For the domain-separated Merkle construction of depth d, with leaves entering as level-0 labels under the convention of Definition 10.18, define

$$R_{\text{mem}} := \{ (rt\,;\,(v, i, \pi)) : \mathsf{Verify}(rt, i, v, \pi) = \mathsf{accept} \},$$

with the *statement* the public root rt and the *witness* comprising the leaf v, its index i, and the authentication path π. A zero-knowledge argument for R_{mem} lets a prover convince a verifier that *some* leaf lies in the tree with root rt while revealing none of v, i, or π. We assume a zk-SNARK as §9 develops it (Definition 9.24): *completeness* (honest provers convince), *knowledge soundness* (an accepting prover "knows" a witness, formalised by an extractor), and *zero knowledge* (the proof reveals nothing beyond the truth of the statement, formalised by a simulator).

Construction 10.28 (In-circuit path verification)**.** The Merkle-path circuit C_{mem} of depth d takes as public input the root rt, and as private (witness) inputs the leaf v, the direction bits $b_0, \dots, b_{d-1}$, and the siblings $s_0, \dots, s_{d-1}$. It implements the d compression calls of the fold (4)—the level-t call being the position's prescribed function, $\mathsf{NodeHash}_{t+1}$ in the domain-separated construction, with no $\mathsf{LeafHash}$ call since the leaf enters as the level-0 label (Definition 10.18)—each preceded by the conditional swap that selects the argument order (a_t, s_t) or (s_t, a_t) according to b_t, and enforces the single output constraint $a_d = rt$. Its size is d times the cost of one compression call plus $O(d)$ constraints for the swaps. The position enters *only* through the bits $b_0, \dots, b_{d-1}$—the binary decomposition of i—so the index-consistency check of Definition 10.7 is enforced structurally: there is no separate index to disagree with its bits. Deployed Orchard matches this shape exactly: the circuit's Merkle gadget decomposes the witnessed position into little-endian bits and applies a conditional swap per layer (protocol specification § 4.9, 'Merkle Path Validity').

Theorem 10.29 (Privacy-preserving membership)**.** *Let h be collision resistant and let (P, V) be a zk-SNARK for R_{mem}—equivalently, for satisfiability of C_{mem}. Then the protocol "the prover sends a SNARK proof for the statement rt; the verifier checks it" is a membership argument that is:*

1. complete*: a prover holding a genuine leaf and its path produces an accepting proof;*
2. sound*: any prover producing an accepting proof for an honest root $rt = \mathsf{root}(L)$ knows—via the SNARK extractor—a witness (v, i, π) with $\mathsf{Verify}(rt, i, v, \pi) = \mathsf{accept}$, and v equals the genuine leaf L_i unless a collision in h is thereby found: the proven leaf is a real member of the committed list;*
3. private*: the proof reveals nothing about v, i, or π beyond the fact that some valid leaf exists under rt.*

Proof idea. Completeness composes SNARK completeness with tree completeness (Proposition 10.8): the genuine (L_i, i, π_i) satisfies C_{mem}, and the honest prover proves it. Soundness composes the SNARK knowledge extractor with the path-forgery-to-collision extractor of Theorem 10.12: from an accepting prover, the SNARK extractor produces (v, i, π) accepted by Verify; if $v \neq L_i$, that output is precisely a path forgery, hence yields a collision for h, hence occurs with probability $\mathrm{negl}(\lambda)$. Zero knowledge carries over verbatim from the SNARK simulator, which produces from rt alone a proof indistinguishable from the real one—so the real proof cannot be conveying v, i, or π. □

Remark 10.30 (Public root, private path; deployment trade-offs). The asymmetry in R_{mem} is the essential design point. The root must be *public* so that the verifier checks the proof against a state

everyone agrees on—a previously published note-commitment-tree anchor—and soundness is relative to that public root, exactly as Remark 10.13 delimits. The path must be *private* so that observers cannot link the spend to a particular leaf. SNARK succinctness adds the final ingredient: the verifier checks a proof whose size is essentially independent of d—polylogarithmic—even though the witness contains d siblings, so the construction *compresses* the membership proof while simultaneously *hiding* it. This is the precise sense in which a Merkle root over note commitments, opened in zero knowledge, is a privacy-preserving membership primitive.

Deployment chooses where to pay for that verification. Pairing-based schemes such as KZG (§4.9) achieve constant-size proofs with fast verification, at the price of a trusted setup. Halo 2's IPA-based proofs grow logarithmically in size, but their verification is *linear* in the circuit size; the deployment amortises it by batch-verifying many proofs in a single multiexponentiation. The accumulation technique of the Halo lineage—which would make per-proof verifier work sublinear by deferring the linear check across recursively composed proofs—is not deployed: each Orchard proof is verified, in batch, directly.

10.10 A shielded spend, end to end

The volume closes by assembling the pieces. Every primitive below has been constructed in a previous section or in this one; the shielded spend is their composition, and the description follows the deployed Orchard sources.

Example 10.31 (An Orchard shielded spend)**.** Each shielded output of a transaction appends a *note commitment* cm—itself a hiding, binding Sinsemilla commitment to the note's value, recipient, and randomness (§4.8)—as a leaf of an append-only Merkle tree built with the Sinsemilla compression function of Remark 10.17. Strictly, the leaf is the extracted x-coordinate

$$\mathsf{cm}_x = \mathsf{ExtractP}(\mathsf{cm}),$$

a 255-bit Pallas base-field element (*Math Guide*, §"Pallas and Vesta assembled") matching the child encodings of Remark 10.17. The protocol periodically publishes the current root—the *anchor*—on chain, one per block.

To spend a note, its owner:

1. locates the leaf cm_x in the public tree and reads off its authentication path π and position i (`MerklePath::root` recomputes the anchor from exactly these data);
2. chooses a recent anchor $rt = \mathsf{root}_m$ whose tree already contains the leaf—any recorded historical anchor serves, per §10.7;
3. proves in zero knowledge, per Theorem 10.29, the statement rt with witness (cm_x, i, π): membership of the note commitment under the anchor, well-formedness of the note opening, and correct derivation of a *nullifier*—a deterministic tag whose on-chain uniqueness prevents double-spending, computed from the note and the spender's nullifier key on $\mathsf{PRF}^{\mathsf{nfOrchard}}$ of §3.8 by a construction of the *Ironwood Guide*, §"The nullifier" (the recomputed root is constrained to equal the public anchor, and the instance carries the anchor and the nullifier; protocol specification § 4.9, 'Merkle Path Validity', and § 4.18.4, 'Action Statement');
4. publishes the proof and the nullifier—but neither cm, nor i, nor π.

By Theorem 10.29, the spend convinces the network that the spender owns *some* note genuinely added to the tree—soundness resting on the collision resistance of Sinsemilla (Proposition 4.29) and the knowledge soundness of the proof system—while the network learns nothing about *which* note. (One refinement of the deployed circuit is worth recording: the root-equals-anchor constraint is enforced for notes of nonzero value, while zero-valued dummy spends may witness

an arbitrary path—by design, so that every action carries a spend whether or not a real note backs it.) The Merkle root is thus a succinct, binding, privacy-preserving commitment to the set of all notes ever created.

The spend consumes the whole volume. The note commitment is the hiding and binding envelope of §4; the tree hash and its personalised domains are the discipline of §3 instantiated by Sinsemilla; the nullifier is derived under the spender's nullifier key from a PRF of §5, instantiated by Poseidon (§3.8), by the construction of the *Ironwood Guide*, §"The nullifier"; the proof is a zk-SNARK of §9 for the relation R_{mem} extended with the note's opening; and the authorising signatures over the spend—re-randomised RedDSA for spend authority, the binding signature for value balance—are those of §8. How the pieces sit inside a transaction, per Action, with the key hierarchy and note-delivery encryption around them, is the business of the *Ironwood Guide*.

Part III
Halo 2 Guide: The Halo 2 proof system, from arithmetisation to recursion

Abstract

This volume of *The Zcash Arboretum* develops the proof system that Orchard uses: Halo 2, the argument by which the author of a shielded Action convinces every validator that the Action is well formed while disclosing nothing about the note, the key, or the Merkle path behind it. It assumes the *Math Guide*, whose finite fields, polynomials and evaluation domains, elliptic curves, and probability are used throughout without re-derivation, and the *Crypto Guide*, whose definitions it instantiates rather than restates: arguments of knowledge and zero knowledge, the Fiat–Shamir transform, Pedersen vector commitments, and the polynomial-commitment interface with its compilation recipe.

The running example is knowledge of x with $x^3 + x + 5 = 35$ over the field $\mathbb{F}_{97}$, a size at which every number of the example can be checked by hand; every number is computed by script. The volume reduces the example to a grid of gates, the grid to polynomials, the polynomials to a single identity, and the identity to one evaluation at a random point, at which the prover must open a commitment without sending the committed polynomial. The inner-product argument (IPA) that performs the opening has a section of its own: the problem stated without machinery, one fold written out step by step, the recursion from the prover's and the verifier's side, the verifier's structured scalars, the extraction from three transcripts that proves the argument knowledge sound under the discrete-logarithm assumption on Vesta, with the hash that derives the generators modelled as a random oracle, the blinders and what they do not hide, and the running example opened on a curve of prime order 97. The deployed proof system follows: the Action circuit's arithmetisation, the binding of the public statement, the Fiat–Shamir transcript, the reduction of many openings to one, the complete protocol, and its zero knowledge. The volume then analyses the knowledge soundness of the Fiat–Shamir-compiled argument, lists the verifier's checks on the running example in the order he performs them, and closes with recursive composition and accumulation, which Orchard does not deploy.

1 Introduction

1.1 Scope and layering

This is the third volume of *The Zcash Arboretum*, and it develops the cryptography behind Zcash's Orchard protocol: the proof system Halo 2, with which every Orchard Action carries a zero-knowledge proof of its own well-formedness. The protocol specification fixes that use: Halo 2 is used with the Vesta curve (the *Math Guide*, §"Pallas and Vesta assembled") to prove and verify the Orchard Action statement (protocol specification § 4.1.13, "Zero-Knowledge Proving System"), and the proving system is defined by reference to the halo2 book (§ 5.4.10.3, "Halo 2"). The statement itself (§ 4.18.4, "Action Statement (Orchard)") is read in §1.2. Actions of the Orchard pool and of the Ironwood pool, the second shielded pool of Orchard-protocol notes, prove this statement, extended at NU6.3 by one public bit, disableCrossAddress, that when set forces the new note's address to equal the spent note's, with one circuit and one verifying key (protocol specification, § 4.6, "Action Descriptions"; ZIP 229; ZIP 258), so the proof system and proof format described here serve both pools.

The two volumes below this one built its material from the ground up. The *Math Guide* constructed the mathematics: finite fields, elliptic curves, polynomials and evaluation domains, probability. The *Crypto Guide* constructed the cryptographic primitives: hash functions, commitments, Σ-protocols and the Fiat–Shamir transform, zero knowledge, polynomial commitment schemes. This volume assumes all of these and defines none of them afresh; it assembles them into a proof system. In particular the *Crypto Guide* owns every definition this volume instantiates: the non-interactive argument and the SNARK, together with the polynomial interactive

oracle proof and the compilation recipe that turns one into the other (§"SNARKs: succinct non-interactive arguments of knowledge"); the Fiat–Shamir transform (§"The Fiat–Shamir transform: from interactive to non-interactive"); the Pedersen vector commitment (§"Pedersen vector commitments"); and the polynomial-commitment interface, with its sketch of an inner-product opening (§"Polynomial commitment schemes"). Where a lower volume's result is needed it is cited at the point of use, by italic volume name and section title: the volumes are separate documents, so there is no cross-document reference, and this volume forward-references only its own later sections.

1.2 The running example and the security notions

The running example is the statement

$$\text{knowledge of } x \in \mathbb{F}_{97} \text{ with } x^3 + x + 5 = 35,$$

one of whose witnesses (the *Crypto Guide*, §"Languages, relations, and witnesses") is $x = 3$, since $27 + 3 + 5 = 35$; the others in $\mathbb{F}_{97}$ are 18 and 76, and the prover of the example holds $x = 3$. It is arithmetised in §2, committed and opened in §3, bound to its public input 35 and compiled into a proof string in §4, and verified in §6. Its field arithmetic, that of the claim, the grid, the polynomials and every scalar, is in the prime field $\mathbb{F}_{97}$ (the *Math Guide*, §"The prime field and two inversion routes"), a size at which every product can be checked by hand; the commitments of §3 are points of a curve of order 97 whose coordinates lie in a second prime field, $\mathbb{F}_{79}$. Every number printed is computed by script, and the script's output is the source of the text.

A proof system for such a statement is required to have three properties, each a definition of the *Crypto Guide* (§"Interactive proofs, zero knowledge, and SNARKs") used here without restatement. *Overwhelming completeness*: an honest prover holding a witness is accepted with overwhelming probability, in the sense of the *Math Guide* (§"Polynomial, exponential, and negligible functions"). *Knowledge soundness* (the *Crypto Guide*, §"Knowledge soundness and extractors"): an efficient extractor, given rewindable access to any prover the verifier accepts with non-negligible probability, outputs a valid witness. *Zero knowledge* (the *Crypto Guide*, §"The simulation paradigm and zero knowledge"): a simulator given only the public input produces transcripts indistinguishable from those of real executions.

The volume establishes these properties to different extents. Design-stage claims are classified in three grades: *specified*, when the protocol specification or a ZIP states the property; *designed-but-unspecified*, when the design intends it but no specification, ZIP or theorem of this volume states it for the deployed system; and *open problem*, when no design is known to achieve it. The Halo 2 protocol is complete except with negligible probability: for each random challenge, a number of values at most proportional to the size of the circuit, out of the $|\mathbb{F}|$ possible, makes undefined a division that the verifier or the honest prover performs; §4.6 lists these degenerate challenges once the protocol is defined. The inner-product opening argument is knowledge sound under the discrete-logarithm assumption on Vesta, with the hash that derives its generators modelled as a random oracle (Theorem 3.5 and, for the deployed opening, Corollary 3.13). Knowledge soundness of the deployed non-interactive argument rests on an analysis in the algebraic group model and the random-oracle model whose instantiation to the deployed transcript is designed-but-unspecified (§5). Zero knowledge is recorded in §4.6 from the halo2 book's claim for the interactive protocol, under the composition and Fiat–Shamir hypotheses stated there.

The running example has the form of the Orchard Action statement. The prover of an Action shows knowledge of a note, the keys that control it, and a Merkle path such that the note's commitment is well formed, the path, unless the spent value is zero, leads from that commitment to the published tree root (the *Crypto Guide*, §"Merkle trees and commitments to sets"), the net value commitment is a Pedersen commitment to the old value minus the new value (the *Crypto Guide*, §"The Pedersen commitment"), the nullifier, the unique tag whose publication marks the note spent, is computed correctly, and the key that authorises the spend is a re-randomisation of the

owner's key (the *Crypto Guide*, §"Key re-randomisation and unlinkability"). The protocol specification states this as nine conditions on a public input and a private witness (§ 4.18.4, "Action Statement (Orchard)"), the five just paraphrased and four further: that the spent note's address is derived from the same keys, that the new note's commitment is well formed, and that two public bits, when zero, force the spent value and the created value, respectively, to zero. From NU6.3 the circuit checks a tenth condition on a further public bit, which restricts cross-address transfers and is set for every Orchard-pool Action (§ 4.6; ZIP 258). Each condition unfolds into small arithmetic facts over the field $\mathbb{F}_{p_{\mathsf{Pallas}}}$ (the *Math Guide*, §"Base fields, scalar fields, and the Pasta cycle"): the note commitment alone is a Sinsemilla commitment, a Sinsemilla hash to a point plus a blinding multiple of a fixed generator (the *Crypto Guide*, §"Sinsemilla: an algebraic hash-based commitment"), whose hash is evaluated in the circuit with the help of a lookup table. Together the conditions are laid out on the $2^{11} = 2048$ rows of the deployed Action circuit; the lookup table and the row count are given in §4, "From statement to circuit: arithmetisation in practice". The running example itself is a conjunction of *four* arithmetic facts, $x \cdot x = x^2$, $x^2 \cdot x = x^3$, $x^3 + x = s$ and $s + 5 = 35$, one per row of the grid that §2 builds. Both the toy and the Action circuit are instances of the constraint system that §2 defines; the toy's constraint on each row reads that row alone and consults no table.

The chain of reductions. Five reductions connect the claim to the checks the verifier performs.

1. **Computation → grid** (§2, "From computation to a grid of constraints"). The claim is rewritten as a table of rows, each row one elementary gate ($\times$ or $+$), with copy constraints between rows forcing shared values.
2. **Grid → polynomials** (§2, "From the grid to polynomials"). Each column of the table is interpolated into a polynomial (the *Math Guide*, §"Lagrange interpolation"). The condition that every row obeys its gate becomes the condition that one polynomial $G(X)$ is zero at every row.
3. **Vanishing → identity** (§2, "From vanishing to a single identity"). The statement that G is zero at every row becomes the algebraic identity $G(X) = t(X)\, Z_H(X)$ for some quotient t, where Z_H is the vanishing polynomial of the set H of row names (the *Math Guide*, §"The evaluation domain H and its vanishing polynomial"). A trace that satisfies every gate yields such an identity, and a trace that violates a gate cannot. The copy constraints are not part of G; the permutation argument enforces them (item 5).
4. **Identity → one random check** (§2, "Why checking one random point is a proof"). The verifier tests the identity at a single random point z. A false identity holds at a uniformly random point with probability at most its degree divided by the field size (the *Math Guide*, §"The Schwartz–Zippel lemma").
5. **Random check → committed openings.** A binding commitment (the *Crypto Guide*, §"Hiding and binding") prevents the prover from changing her polynomials after z is drawn (§2, "Why the prover cannot wriggle: binding commitments"), and the inner-product argument (IPA) of §3 opens a commitment at z without sending the polynomial; the permutation argument enforces the copy constraints (§2, "Enforcing the wiring: the permutation argument"); a dedicated public column, the instance column, fixes the public input 35 (§4, "Instance columns and binding the public statement"); and blinding hides the witness (§4, "Zero knowledge: hiding the witness in Halo 2").

Section 6 composes the five reductions into one chain of implications, from the verifier's accepted checks back to the claim.

The scale caveat, stated once. The field $\mathbb{F}_{97}$ is a toy. Over Orchard's field $\mathbb{F}_{p_{\mathsf{Pallas}}}$, whose prime has 255 bits, a *fixed* false identity of degree around 2^{14} (the degree of the deployed constraint

polynomial, bounded in §4, "The complete Halo 2 protocol") survives one fresh uniform evaluation challenge with probability around $2^{14}/2^{254} = 2^{-240}$, in place of the visible 3/97 that §2 exhibits for the toy's cheat. Every such number in this volume is one Schwartz–Zippel term: the probability that one random point hides one fixed lie. It is not the security level of the complete proof system, which also rests on the binding of the commitments, the extractor of the opening argument, and the loss of the Fiat–Shamir transform, treated in §3, §4 and §5. The caveat is stated here and not re-derived; every probability printed later is read with it.

1.3 Outline

The sections follow the running example; Figure 1 shows the section in which each stage is built.

Section 2 takes the example from the computation to committed polynomials: the grid, the column polynomials, the single identity, the random point, and then the general form of the constraint system, PLONKish arithmetisation, of which the toy is an instance, followed by the permutation argument for the copy constraints and the lookup argument for facts that no low-degree gate states economically. It ends with the problem a vector commitment alone does not solve: proving the value of a committed polynomial at z without sending the polynomial, in fewer group elements than the polynomial has coefficients.

Section 3 solves that problem with the inner-product argument. It states the inner-product problem without machinery, works one fold with every algebra step written out, gives the recursion with both parties' computations side by side, makes the verifier's work explicit, proves the argument knowledge sound by extraction from three transcripts under the discrete-logarithm assumption with uniform generators, separates what the blinders hide from what they do not, and opens the example's own polynomial through every round on a curve of prime order 97, by script. Its last subsection gives Halo 2's parameters, the deployed form of the argument as a change of variables, the bytes of a proof, the deferred multi-scalar multiplication and the batching that amortises it, and accumulation, which Orchard does not deploy.

Section 4 gives the deployed proof system: the Action circuit's arithmetisation, the instance column that binds the public statement, the Fiat–Shamir transform that replaces the verifier's challenges by hash outputs, the reduction of many openings to one, the complete protocol in seven phases, its zero knowledge, and the compiled-polynomial-IOP pattern the protocol instantiates.

Section 5 treats knowledge soundness of the non-interactive argument: the extractor of §3 when the challenges are hash outputs, the algebraic group model in which the halo2 book proves its tight bound for the analysed protocol, and the gap between concrete and asymptotic security.

Section 6 lists the verifier's checks on the running example in the order he performs them, the chain of implications back to the prover's knowledge of advice satisfying every gate and every copy constraint, and the same chain at Orchard scale.

Section 7 treats recursive proof composition, accumulation schemes, the Halo trick, and the cycle of curves that makes native recursion efficient. Orchard proves with Halo 2 on the Pasta curves but does not recurse, and the section states at the outset that Orchard does not deploy its material.

2 Arithmetisation: from a computation to committed polynomials

This section carries the running example of §1 — the prover knows $x = 3$ with $x^3 + x + 5 = 35$ over $\mathbb{F}_{97}$ — from a computation to a handful of polynomials that a verifier can query at one point, toy first and general second. It ends with the property the argument has not yet provided: that the queried polynomials are fixed before the evaluation point is chosen.

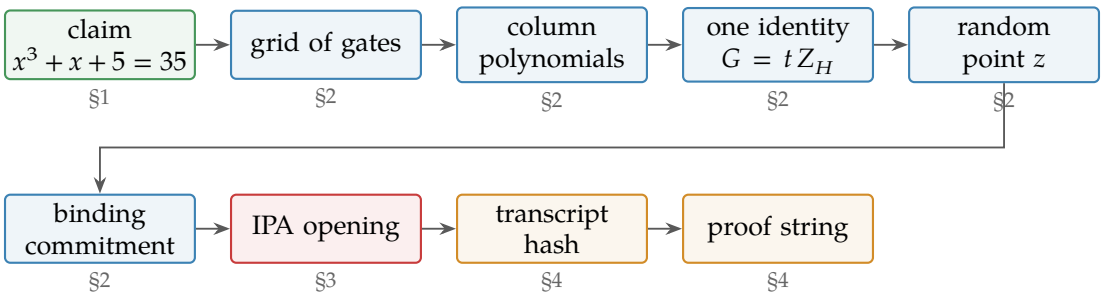


Figure 1: The running example's path through the volume, each stage tagged with the section that owns it. The claim is fixed in this section (green). The grid of gates, the column polynomials, the single identity $G = t Z_H$, the random point z and the binding commitment are built in §2 (blue). The commitment is opened by the inner-product argument in §3 (red). The transcript hash, whose outputs replace the verifier's challenges, and the proof string are produced in §4 (amber). Section 6 reads the chain backwards, from the accepted proof string to the claim.

2.1 From computation to a grid of constraints

The verifier cannot evaluate the computation without the witness. Arithmetisation dismantles the computation into *local* facts, each involving only a handful of values, arranged so that "every local fact holds and the values are consistently shared" is equivalent to "the whole computation ran correctly". The dialect of arithmetisation Halo 2 uses is called *PLONKish*; §2.5 defines it once the toy has shown what it generalises.

Execution trace. Evaluate $x^3 + x + 5$ the way a machine would, naming every intermediate value:

$$x^2 = x \cdot x, \qquad x^3 = x^2 \cdot x, \qquad s = x^3 + x, \qquad s + 5 = 35.$$

Four elementary operations, two multiplications and two additions, and each becomes one row of a table. A row has three *wires* a, b, c (two inputs and an output) and a bank of five *selectors* q_M, q_L, q_R, q_O, q_C that switch on whichever gate the row performs. The universal gate equation, imposed at every row, is

$$q_M\, a\, b + q_L\, a + q_R\, b + q_O\, c + q_C + \mathrm{PI} = 0, \tag{1}$$

where PI carries the negated public input on that row (here -35, for the target 35). A multiplication gate $a \cdot b = c$ is the choice $q_M = 1$, $q_O = -1$ and the rest zero; an addition gate $a + b = c$ is $q_L = q_R = 1$, $q_O = -1$. Filling in the four operations gives the *execution trace* of Table 1. Definition 2.7 generalises this fixed selector bank to arbitrary custom gates and lookup arguments (the halo2 book's *PLONKish Arithmetization* chapter); §4 gives the form of the Action circuit's gates, and one in full, under "From statement to circuit: arithmetisation in practice".

row	a	b	c	q_M	q_L	q_R	q_O	q_C	PI	gate performed
0	3	3	9	1	0	0	-1	0	0	$x \cdot x = x^2$
1	9	3	27	1	0	0	-1	0	0	$x^2 \cdot x = x^3$
2	27	3	30	0	1	1	-1	0	0	$x^3 + x = s$
3	30	0	0	0	1	0	0	5	-35	$s + 5 = 35$

Table 1: The execution trace for $x = 3$, all entries in $\mathbb{F}_{97}$ (so $-1 = 96$ and $-35 = 62$). Every row satisfies the gate equation (1): row 0 reads $1 \cdot 3 \cdot 3 + (-1) \cdot 9 = 0$ and row 3 reads $1 \cdot 30 + 5 + (-35) = 0$. The selector columns $q_\bullet$ are fixed circuit data and PI is the negated public input, both known to the verifier; the wires a, b, c are the prover's *advice*, her secret trace.

Copy constraints. The table as printed does not yet encode the computation. Nothing forces the x in row 0 to equal the x in rows 1 and 2, nor the output $x^2 = 9$ of row 0 to be the input of row 1. A cheating prover could put $x = 3$ in row 0 and $x = 8$ in row 2 and satisfy each gate *in isolation* without the rows forming one computation. The missing assertions are that certain cells are *equal*, the *copy constraints* (also called *equality constraints*):

$$\underbrace{a_0 = b_0 = b_1 = b_2}_{\text{all are } x=3}, \qquad \underbrace{c_0 = a_1}_{x^2=9}, \qquad \underbrace{c_1 = a_2}_{x^3=27}, \qquad \underbrace{c_2 = a_3}_{s=30},$$

where a_j denotes the cell of column a in row j. Figure 2 draws them as wires. They turn a list of unrelated gates into a single dataflow; enforcing them is an argument in its own right, the permutation argument of §2.6.

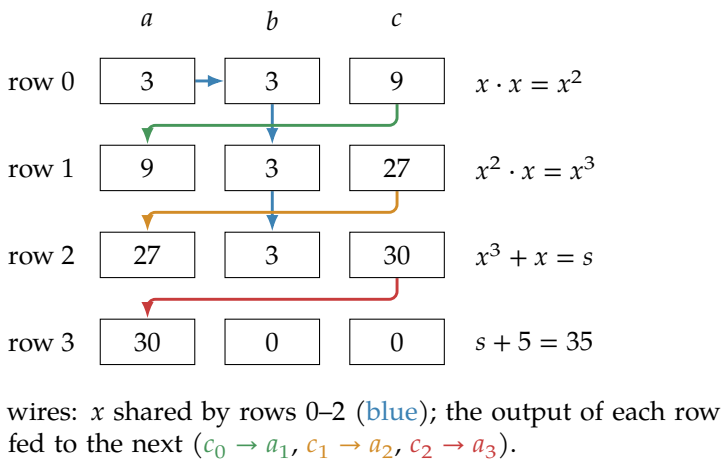


Figure 2: The four-row trace of Table 1 with the copy constraints drawn as wires. Each wire colour is one class of cells that must hold a single value: the blue wire ties the four occurrences of x, and the three coloured wires carry each row's output c_j into the next row's input a_{j+1}. The gates check rows; the wires make the rows one computation.

Satisfiability of the grid. The claim is now equivalent to a purely combinatorial assertion: *there exist values for the advice cells a, b, c such that (i) every row satisfies the gate equation (1) with the fixed selectors, and (ii) every copy constraint holds.* Knowing x is exactly knowing how to fill the advice columns. The verifier knows the fixed columns and the public input; the prover must convince him that satisfying advice exists while revealing none of it. Everything that follows is machinery for doing precisely that.

2.2 From the grid to polynomials

The grid has finitely many rows, and the aim is a statement about all rows at once that the verifier can check at *one* point. A polynomial of degree less than n is determined by its values at n distinct points (*Math Guide*, §"Lagrange interpolation"), and the rest of the section relies on this.

An index set for the rows. The table has $n = 4$ rows, and four distinct field elements must name them. The chosen four are the 4th roots of unity. A *primitive n-th root of unity* is an element ω with $\omega^n = 1$ and no smaller positive power equal to 1 (*Math Guide*, §"Roots of unity"), and the *evaluation domain* it generates is $H = \{\omega^0, \dots, \omega^{n-1}\}$, a multiplicative subgroup of $\mathbb{F}^\times$ of order n (*Math Guide*, §"The evaluation domain H and its vanishing polynomial"). In $\mathbb{F}_{97}$ take $\omega = 22$: then $\omega^2 = 96 = -1$ and $\omega^4 = 1$, so

$$H = \{\omega^0, \omega^1, \omega^2, \omega^3\} = \{1, 22, 96, 75\} = \{1, 22, -1, -22\},$$

and row i lives at the point ω^i. Roots of unity are chosen because they form a cyclic group: the row after ω^i is $\omega \cdot \omega^i$, so reading a column at the next row is evaluating its polynomial at ωX; the polynomial vanishing on H is $X^n - 1$; and interpolation over H is a fast Fourier transform (*Math Guide*, §"The fast Fourier transform"). The gate check of this subsection alone would hold over any four distinct points.

Columns become polynomials. Each column of the table is four numbers, one per row. Interpolate: for the advice column a there is a unique polynomial $a(X)$ of degree < 4 with

$$a(\omega^0) = 3, \quad a(\omega^1) = 9, \quad a(\omega^2) = 27, \quad a(\omega^3) = 30$$

(*Math Guide*, §"Lagrange interpolation"), and likewise $b(X)$, $c(X)$, the publicly known selector polynomials q_M, q_L, q_R, q_O, q_C and the public-input polynomial PI, each of degree < 4. Computed over $\mathbb{F}_{97}$, the nine interpolants are

$$\begin{aligned} a(X) &= 90 + 61X + 22X^2 + 24X^3, & q_M(X) &= 49 + 19X + 30X^3, \\ b(X) &= 75 + 32X + 25X^2 + 65X^3, & q_L(X) &= 49 + 78X + 67X^3, \\ c(X) &= 65 + 16X + 3X^2 + 22X^3, & q_R(X) &= 73 + 24X + 73X^2 + 24X^3, \\ \mathrm{PI}(X) &= 64 + 50X + 33X^2 + 47X^3, & q_O(X) &= 72 + 54X + 24X^2 + 43X^3, \\ & & q_C(X) &= 74 + 76X + 23X^2 + 21X^3, \end{aligned}$$

and each returns its column at the rows: b takes the values $(3, 3, 3, 0)$ on H, q_O the values $(96, 96, 96, 0)$, and PI the values $(0, 0, 0, 62)$. Interpolation is thus a bijection between columns of four values and polynomials of degree < 4: the encoding loses no information.

The gate polynomial. Assemble the left-hand side of the gate equation (1) with the polynomials in place of the per-row numbers.

> **Theorem 2.1** (The gate polynomial)**.** *Let*
>
> $$G(X) := q_M(X)\, a(X)\, b(X) + q_L(X)\, a(X) + q_R(X)\, b(X) + q_O(X)\, c(X) + q_C(X) + \mathrm{PI}(X). \quad (2)$$
>
> *Then* $\deg G \leq 9$*, and* $G(\omega^i) = 0$ *for every row* i *if and only if every gate of the trace is satisfied.*

Proof. The term $q_M ab$ multiplies three polynomials of degree at most 3, and every other term has degree at most 6, so $\deg G \leq 9$. Evaluate G at the label ω^i of row i: each column polynomial returns that row's value there, so $G(\omega^i)$ is exactly the left-hand side of (1) for row i. It is zero precisely when gate i holds, and the conjunction over i gives the claim. □

For the honest trace,

$$G(X) = 36 + 27X + 54X^2 + 50X^3 + 77X^4 + 24X^5 + 43X^6 + 47X^7 + 81X^8 + 46X^9,$$

of degree exactly 9, and it vanishes at all four points of H by construction. "Every gate is satisfied" has become "G vanishes on H"; the copy constraints remain separate. Four separate checks are now one statement about one polynomial.

2.3 From vanishing to a single identity

The condition "G vanishes on H" is still a statement about four points. This subsection replaces it by a polynomial identity that refers to no individual row.

The vanishing polynomial. The *vanishing polynomial* of H is the monic polynomial (leading coefficient 1) whose roots are exactly the points of H,

$$Z_H(X) := \prod_{i=0}^{3}(X - \omega^i) = X^4 - 1,$$

where the closed form $X^n - 1$ holds for every evaluation domain of order n (*Math Guide*, §"The evaluation domain H and its vanishing polynomial"): each ω^i is a root of $X^n - 1$, there are n distinct ones, and comparing leading coefficients finishes. The closed form lets both parties evaluate Z_H at any point with $O(\log n)$ multiplications.

> **Theorem 2.2** (Gate satisfaction is a quotient)**.** *Every gate of the trace holds if and only if there is a polynomial $t(X)$ with $G(X) = t(X)\, Z_H(X)$.*

Proof. By Theorem 2.1, every gate holds if and only if G vanishes on H. The factor theorem (*Math Guide*, §"Roots and the factor theorem") gives $G(\omega^i) = 0$ if and only if $(X - \omega^i) \mid G$; and because the four roots are distinct, G vanishes at all of them if and only if the product of the four linear factors divides G (*Math Guide*, §"Lagrange interpolation", its remark on the vanishing polynomial: a polynomial vanishes on a set of distinct nodes if and only if the product of the corresponding linear factors divides it). Thus

$$G \text{ vanishes on } H \iff (X - \omega^i) \mid G \text{ for each } i \iff Z_H(X) \mid G(X), \tag{3}$$

and divisibility is the existence of the quotient t. □

Satisfaction of the *gate* constraints is precisely the existence of t; correct wiring still requires the copy constraints. For the honest trace the division comes out even: G has degree 9, Z_H has degree 4, and

$$t(X) = \frac{G(X)}{X^4 - 1} = 61 + 70X + 43X^2 + 47X^3 + 81X^4 + 46X^5$$

is a polynomial of degree 5 with remainder zero, as it must be.

Example 2.3 (Why a cheat cannot produce t)**.** Suppose the advice is wrong: some gate fails. Then G does not vanish on all of H, so by (3) Z_H does not divide G, and *no* polynomial t satisfies $G = t\, Z_H$. The dividing line is exact: the quotient exists if and only if every gate is satisfied. Concretely, take a prover who claims $x^2 = 10$ and writes $c_0 = 10$ in row 0 of Table 1 instead of 9. Row 0's gate now reads $3 \cdot 3 - 10 = -1 \neq 0$, so the tampered gate polynomial G^* has $G^*(\omega^0) = -1 = 96$ while still vanishing at the other three rows:

$$G^*(X) = 54 + 10X + 43X^2 + 74X^3 + 83X^4 + 65X^5 + 78X^6 + 47X^7 + 81X^8 + 46X^9.$$

Dividing G^* by Z_H with remainder (*Math Guide*, §"Division with remainder") gives the quotient $t^*(X) = 67 + 14X + 78X^2 + 47X^3 + 81X^4 + 46X^5$ and a nonzero remainder. This Euclidean quotient is an illustrative choice for t^*, not a claim that it maximises the cheat's chances. Whatever t^* she offers as the quotient, the *gap polynomial* $D(X) := G^*(X) - t^*(X)\, Z_H(X)$ is nonzero, since Z_H does not divide G^*. For the Euclidean quotient D is the remainder,

$$D(X) = 24\,(1 + X + X^2 + X^3), \tag{4}$$

of degree 3. The identity $G^* = t^*\, Z_H$ is false, D is the difference of its two sides, and §2.4 shows how the verifier detects $D \neq 0$.

2.4 Why checking one random point is a proof

The prover wishes to convince the verifier that $G(X) = t(X)\,Z_H(X)$ as polynomials, with G built from her secret advice. The verifier does not read the polynomials: they encode the witness, and reading them would cost time at least linear in the number of rows. Instead, the two run a four-step protocol. A *commitment* is, for now, a short string that fixes a value without revealing it (*Crypto Guide*, §"Commitment schemes"); §2.9 says what a commitment to a polynomial must do and which one Halo 2 uses.

Construction 2.4 (The random-point check)**.** 1. The prover commits to her polynomials a, b, c and to the quotient t; the selector and public-input polynomials the verifier already knows.

2. The verifier picks a point $z \in \mathbb{F}$ *uniformly at random* and sends it.
3. The prover *opens* her commitments at z: she returns the field elements $a(z), b(z), c(z), t(z)$ together with short proofs that these are the true evaluations of the committed polynomials.
4. The verifier forms $G(z)$ from the openings and the self-evaluated public polynomials via (2), computes $Z_H(z) = z^4 - 1$, and checks the single field equation

$$G(z) \stackrel{?}{=} t(z)\,Z_H(z). \tag{5}$$

He accepts if and only if (5) holds.

The honest case (completeness). If the prover is honest then $G = t\,Z_H$ holds as polynomials, hence at every point, hence at z. With the honest trace and the draw $z = 20$,

$$Z_H(20) = 20^4 - 1 = 46, \qquad t(20) = 67, \qquad G(20) = 75 = 67 \cdot 46 = t(20)\,Z_H(20),$$

and the verifier accepts. He accepts every time, for every z.

The dishonest case (soundness). Soundness rests on the fact that a nonzero polynomial of low degree has few roots.

Theorem 2.5 (Soundness of the random-point check)**.** *Suppose the trace is invalid, so that the prover's gate polynomial G^* does not vanish on all of H, and let t^* be any polynomial she commits to as the quotient. Set $D(X) := G^*(X) - t^*(X)\,Z_H(X)$. Then D is a nonzero polynomial, and if the prover's polynomials a, b, c (which determine G^*) and t^* are fixed before z is drawn and the verifier receives their true evaluations at z,*

$$\Pr_z[\textit{the verifier accepts}] = \Pr_z[D(z) = 0] \le \frac{\deg D}{|\mathbb{F}|}.$$

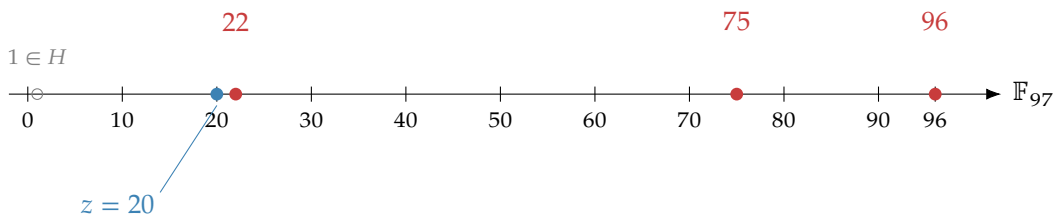
Proof. Were D the zero polynomial, $G^* = t^*\,Z_H$ would be divisible by Z_H and so, by (3), vanish on all of H, which it does not. Hence $D \neq 0$ for every choice of t^*. By hypothesis a, b, c and t^*, and with them G^* and D, are fixed before z is drawn (the verifier never receives G^* itself; from the evaluations at z he reconstructs only the single number $G^*(z)$). The check (5) passes at z exactly when $G^*(z) - t^*(z)\,Z_H(z) = D(z) = 0$, that is, exactly when z is a root of D. A nonzero polynomial of degree at most $\deg D$ has at most $\deg D$ roots, so a uniformly random z is one of them with probability at most $\deg D/|\mathbb{F}|$ (*Math Guide*, §"The Schwartz–Zippel lemma"). □

The bound $\deg D/|\mathbb{F}|$ is negligible when $|\mathbb{F}|$ is exponential in the security parameter and $\deg D$ is polynomial in it. The verifier does not recompute the trace; he tests one polynomial identity at one random point.

Example 2.6 (The random-point check on the toy)**.** For the $x^2 = 10$ cheat of Example 2.3, $D(X) = 24(1 + X + X^2 + X^3)$ has degree 3 and so at most 3 roots; in $\mathbb{F}_{97}$ its roots are exactly $\{22, 96, 75\}$, the three rows where the tampered gate still held (Figure 3). This particular quotient choice slips past only if the random z lands in that three-element set:

$$\Pr[\text{cheat accepted}] = \frac{3}{97}.$$

At the actual draw $z = 20 \notin \{22, 96, 75\}$ the openings give $G^*(20) = 20$ but $t^*(20)\, Z_H(20) = 64$; since $20 \neq 64$, the verifier rejects.



filled: the roots of the gap polynomial D, the set $H \setminus \{1\}$, where the tampered identity still holds; $D(z) = 0$ on 3 of the 97 points only. blue: the draw $z = 20$, where $D(20) \neq 0$ and the check rejects, as it does at every unmarked point.

Figure 3: The gap polynomial $D = 24(1 + X + X^2 + X^3)$ of the $x^2 = 10$ cheat over $\mathbb{F}_{97}$: its three roots $22, 75, 96$ marked in red, the point $1 \in H$ (not a root) in grey, and the verifier's draw $z = 20$ landing elsewhere. The check (5) passes only at a root of D.

Scale. The fraction $3/97$ is large only because the field is a toy, and it concerns one particular quotient; the next paragraph bounds every cheat by $11/97$. What such a term means at the deployed field size is the scale caveat of §1.2.

Bounding the degree. The argument needs $\deg D$ small relative to $|\mathbb{F}|$, otherwise "$\deg D$ roots out of $|\mathbb{F}|$" is no constraint at all. The commitment scheme enforces a degree bound on every committed polynomial (*Crypto Guide*, §"Polynomial commitment schemes"), so the prover cannot commit to a t^* of unbounded degree. The bound follows the degree the commitment admits, not the honest degree. The wires are committed at degree below $n = 4$, so $\deg G^* \leq 9$; committed as the two pieces of degree below n that the deployed rule below prescribes, t^* has degree at most 7, so $\deg(t^* Z_H) \leq 11$ and $\deg D \leq 11$. Every cheat is therefore accepted with probability at most $11/97$, and some cheat attains it, where the illustrative quotient reaches $3/97$: the t^* of degree 7 that agrees with G^*/Z_H at the eight points $2, \ldots, 9$ gives D the eleven roots $2, \ldots, 9, 22, 75, 96$. This is why a circuit's gates are kept to low degree, and why the quotient t, of degree 5 here and up to a few times n in general, is split into degree-bounded pieces. In the deployed system the quotient is cut into pieces of degree below n, each committed separately (the halo2 book's *Vanishing argument* chapter); "The complete Halo 2 protocol" in §4 gives the count.

2.5 PLONKish arithmetisation

The toy has met every ingredient concretely: an evaluation domain H, a generator ω, the vanishing polynomial Z_H, columns interpolated into polynomials of degree $< n$, and a gate expressed as a polynomial identity on H. The general object can now be named.

Definition 2.7 (PLONKish constraint system)**.** Fix a prime field $\mathbb{F}$ and an integer k with $2^k \mid |\mathbb{F}| - 1$, so that $\mathbb{F}^\times$ contains a primitive 2^k-th root of unity ω (*Math Guide*, §"Existence over finite fields: the condition $n \mid q - 1$"), and let $H := \{\omega^0, \dots, \omega^{n-1}\} \subset \mathbb{F}^\times$ be the multiplicative subgroup of size $n := 2^k$ that ω generates. Its vanishing polynomial is $Z_H(X) := X^n - 1$, zero exactly on H, so a congruence $p(X) \equiv 0 \pmod{Z_H(X)}$ means "p is zero at every row ω^i", that is, $Z_H \mid p$. A *PLONKish circuit* $\mathcal{C}$ over H consists of:

1. a finite set Col of columns, each of n cells, one per row, partitioned into *fixed* columns (pre-processed circuit data, known to both parties), *advice* columns (the prover's witness) and *instance* columns (public input); a *cell* is an element of $H \times \mathrm{Col}$, and the circuit assigns every cell of a fixed column its value;
2. a maximum constraint degree $d_{\max}$;
3. a finite set $R \subset \mathbb{Z}$ of *rotations* containing 0; a *column expression* is a polynomial over $\mathbb{F}$ in the values $v(\omega^r X)$ for $v \in \mathrm{Col}$ and $r \in R$, where the value of v at rotation r at row i is the cell of v in row $i + r$ taken modulo n;
4. a finite sequence of *custom gates* $g_j = 0$, each g_j a column expression of degree at most $d_{\max}$;
5. a set $H_{\mathrm{act}} \subseteq H$ of *active* rows: all of H in the toy, and in the deployed system the first $u < n$ rows $\omega^0, \dots, \omega^{u-1}$, the trailing rows being reserved for a boundary row and blinding (Remark 2.17 fixes u);
6. a subset $E \subseteq H_{\mathrm{act}} \times \mathrm{Col}$ of *equality-constrained* cells together with a permutation σ of E (a bijection $E \to E$) whose cycles, the orbits $\{\kappa, \sigma(\kappa), \sigma^2(\kappa), \dots\}$ of the cells $\kappa \in E$, are the classes of cells required to hold a common value;
7. a set of *lookup arguments*, each an input tuple $(A_0, \dots, A_{m-1})$ of column expressions and a table tuple $(S_0, \dots, S_{m-1})$ of columns.

An *assignment* gives every cell a value in $\mathbb{F}$, the fixed cells their circuit values; write v_κ for the value of cell κ. It is *satisfying* if every gate vanishes at every row of H (in the deployed system the selectors are zero outside H_{act}), $v_\kappa = v_{\sigma(\kappa)}$ for every $\kappa \in E$, and for every lookup and every active row ω^i the tuple $(A_0(\omega^i), \dots, A_{m-1}(\omega^i))$ equals $(S_0(\omega^{i'}), \dots, S_{m-1}(\omega^{i'}))$ for some active row $\omega^{i'}$. The *relation* $\mathcal{R}_{\mathcal{C}}$ of the circuit is the set of pairs (instance-column values, advice-column values) of the satisfying assignments.

The letter d is reserved, from §3 on, for the length of a committed vector; the constraint degree is therefore written $d_{\max}$ throughout this volume.

Vanilla PLONK as the special case. The original PLONK system of Gabizon, Williamson and Ciobotaru uses a single fixed five-selector universal gate,

$$q_M(X)\, a(X)\, b(X) + q_L(X)\, a(X) + q_R(X)\, b(X) + q_O(X)\, c(X) + q_C(X) + \mathrm{PI}(X) \equiv 0 \pmod{Z_H(X)},$$

with three wire polynomials a, b, c, selector polynomials q_M, q_L, q_R, q_O, q_C and the public-input polynomial PI, carrying the negated public inputs; the toy's gate (1) is exactly this gate, with PI held in an instance column. PLONKish generalises it by permitting designer-chosen gates of higher arity and degree, each switched by its own selector $q_i(X)$, a fixed column that vanishes on the rows where the gate is inactive. In the toy the five selectors are fixed columns, a, b, c are advice, PI is the instance column, the single gate has degree 3 and reads rotation 0 only, and the copy constraints of §2.1 are the equality set E with σ cycling each class; there are no lookups until §2.7. Figure 4 shows the general shape.

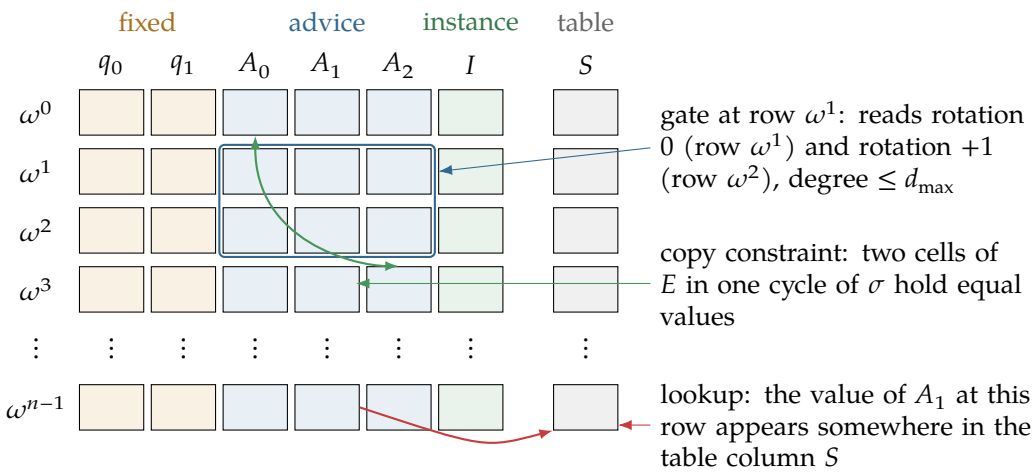


Figure 4: The PLONKish grid of Definition 2.7: columns by kind (fixed q_0, q_1; advice A_0, A_1, A_2; instance I; a fixed table column S), rows indexed by the domain H. The outlined gate at row ω^1 reads its columns at rotations 0 and +1; the green double arrow is one cycle of the equality permutation σ joining two advice cells; the red arrow is a lookup asserting that an advice value appears in S.

Columns as committed polynomials. With the Lagrange basis $\ell_j(X)$ of H, defined by $\ell_j(\omega^j) = 1$ and $\ell_j(\omega^{j'}) = 0$ for $j' \neq j$ (*Math Guide*, §"The Lagrange basis on H"), a column with cells $v_0, \ldots, v_{n-1}$ is the polynomial $v(X) = \sum_j v_j \ell_j(X)$ of degree $< n$, exactly as the toy's columns were interpolated in §2.2. The prover commits to each advice polynomial, and the fixed polynomials are committed once for all proofs, with the inner-product polynomial commitment of §3. Every argument that follows — gates, permutation, lookup, vanishing — is a polynomial identity on these committed polynomials, verified by querying them at random points exactly as §2.4 queried G and t.

2.6 Enforcing the wiring: the permutation argument

The copy constraints of §2.1 remain to be enforced. The gate identity of §§2.2–2.4 checks each row separately; it does not assert that a_0, b_0, b_1, b_2 are the same value x, or that row 1's input is row 0's output, and a prover could satisfy every gate with rows that do not form one computation. The *permutation argument* reduces the equalities to the equality of two products, which is checked at random challenges.

Equalities as a permutation. Group the cells into the classes that must share a value, $\{a_0, b_0, b_1, b_2\}$, $\{c_0, a_1\}$, $\{c_1, a_2\}$, $\{c_2, a_3\}$, and let σ be the permutation of the twelve cells that cycles each class in the order written and fixes b_3 and c_3:

$$a_0 \to b_0 \to b_1 \to b_2 \to a_0, \qquad c_0 \leftrightarrow a_1, \qquad c_1 \leftrightarrow a_2, \qquad c_2 \leftrightarrow a_3.$$

Give every cell a distinct *label*: the cell in row j of the a-, b-, c-column gets ω^j, $k_1\omega^j$, $k_2\omega^j$ respectively, where k_1, k_2 place the three columns in disjoint cosets of H so that no two labels collide. Two cosets of a subgroup are equal or disjoint (*Math Guide*, §"Cosets and Lagrange's theorem"), so it suffices that k_1, k_2 and k_2/k_1 all lie outside H; in $\mathbb{F}_{97}$ take $k_1 = 2$, $k_2 = 3$:

$$H = \{1, 22, 96, 75\}, \qquad 2H = \{2, 44, 95, 53\}, \qquad 3H = \{3, 66, 94, 31\}.$$

Write λ_{cell} for a cell's label and v_{cell} for its value. Asserting "cells in a class are equal" is then equivalent to asserting that the *multiset* (a finite collection in which an element may occur more

than once; two multisets are equal when every element occurs equally often in both) of pairs $\{(v_{\text{cell}}, \lambda_{\text{cell}})\}$ is unchanged when each label is replaced by the label of the cell's image under σ: a pair (v, λ) moves to (v, λ_σ), and the moved pair was already present exactly when the cell at λ_σ carried the same value v. Figure 5 draws the cycles with their labels. Because labels are distinct, comparing two multisets of pairs is a job for a *product*: compress each pair to the linear form $v + \beta\lambda + \gamma$ with verifier challenges β, γ, and compare

$$\prod_{\text{cells}} (v_{\text{cell}} + \beta\lambda_{\text{cell}} + \gamma) \quad \text{against} \quad \prod_{\text{cells}} (v_{\text{cell}} + \beta\lambda_{\sigma(\text{cell})} + \gamma). \tag{6}$$

If the wiring is honest the two products run over the same multiset of factors and agree identically in β, γ. If a copy constraint is false the difference is a nonzero polynomial in β, γ, which can still vanish at one particular challenge pair, but only rarely; the converse is probabilistic, not an "if and only if" for fixed challenges. Theorem 2.11 below makes both halves precise.

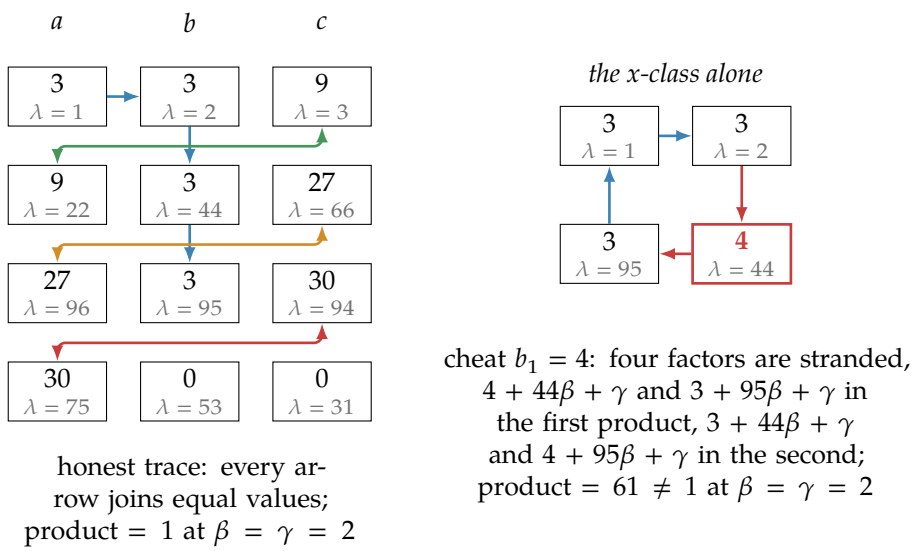


Figure 5: Left: the four copy classes of the toy as cycles of σ on the grid, every cell showing its value and its label λ (column a in H, column b in $2H$, column c in $3H$); the arrow closing the x-class cycle, from b_2 back to a_0, is omitted. Right: the x-class with b_1 tampered to 4; the two arrows leaving and entering the tampered cell now join unequal values, so four factors of (6), two in each product, have no partner in the other product.

The running product. A product over all cells is not a local fact, so the prover commits to a *running product*: a column Z_{perm} with $Z_{\text{perm}}(\omega^0) = 1$ whose row $j+1$ equals row j times the ratio of row j's factors in (6), numerator over denominator. After the last row the product has run over every cell, and because $\omega^n = \omega^0$ the row-to-row rule at the last row forces the full product to equal $Z_{\text{perm}}(\omega^0) = 1$ again: the grand product is 1 exactly when the running product closes its loop. The deployed circuit imposes the rule cross-multiplied, "next row times denominator equals this row times numerator", not as a field division, so a zero factor never makes the constraint undefined; the prover commits to Z_{perm} and the verifier checks the boundary value and the row-to-row update at the random point, exactly as he checked the gate identity (the halo2 book's *Permutation argument* chapter). The identities are written out in Construction 2.10.

A single wired pair. Strip the product down to a single wired pair, $c_0 = a_1$ (both should be 9), with labels $\lambda_{c_0} = 3$ and $\lambda_{a_1} = 22$ that σ swaps. Their factors satisfy

$$(9 + \beta\lambda_{c_0} + \gamma)(9 + \beta\lambda_{a_1} + \gamma) = (9 + \beta\lambda_{a_1} + \gamma)(9 + \beta\lambda_{c_0} + \gamma),$$

because the shared value 9 makes the two left-hand factors a mere reordering of the two right-hand ones: the labels may permute under σ precisely because the values agree. Now let the prover cheat the copy, $a_1 = 8 \neq 9$. The factor $8 + \beta\lambda_{a_1} + \gamma$ on the left is matched on the right only by $8 + \beta\lambda_{c_0} + \gamma$, which differs since the labels differ; the two products differ as polynomials in β, γ. A single unequal value thus leaves a factor without a partner, and the two products differ.

Example 2.8 (The permutation check on the toy)**.** For the honest trace with $\beta = \gamma = 2$, the ratio of the two products in (6) over all twelve cells is exactly 1. Tamper one wire, setting $b_1 = 4$ and so breaking the copy $b_1 = x = 3$, and the same ratio at the same challenges becomes $61 \neq 1$: the inconsistent dataflow is exposed without any value being revealed.

The general argument. Halo 2 states the argument for any number of columns, and the statement needs one lemma on two-variable polynomials. The Schwartz–Zippel lemma of the *Math Guide* (§"The Schwartz–Zippel lemma") is univariate; that section's remark on the multivariate generalisation states the several-variable bound without proof, as a consequence of the univariate lemma applied once per variable, and Lemma 2.9 carries out that derivation for two variables.

Lemma 2.9 (Two variables)**.** *Let $P \in \mathbb{F}[\beta, \gamma]$ be nonzero, of degree at most N in each variable separately. For β, γ drawn independently and uniformly from $\mathbb{F}$, $\Pr[P(\beta, \gamma) = 0] \leq 2N/|\mathbb{F}|$.*

Proof. Write $P = \sum_e c_e(\beta)\,\gamma^e$ with coefficients $c_e \in \mathbb{F}[\beta]$ of degree at most N. Since $P \neq 0$, some $c_{e_0} \neq 0$, and $c_{e_0}(\beta) = 0$ with probability at most $N/|\mathbb{F}|$ (*Math Guide*, §"The Schwartz–Zippel lemma"). When $c_{e_0}(\beta) \neq 0$, the polynomial $P(\beta, \cdot) \in \mathbb{F}[\gamma]$ is nonzero of degree at most N, so it vanishes at the independent uniform γ with probability at most $N/|\mathbb{F}|$. The union bound (*Math Guide*, §"The union bound and a birthday calculation") adds the two. □

Construction 2.10 (Permutation argument over m columns)**.** Let $v_0(X), \ldots, v_{m-1}(X)$ be the interpolants of the m columns that participate in equality constraints, and let $N = mn$ be the number of participating cells. Label the cell (i, j) (column i, row j) by $\delta_i\,\omega^j$, where $\delta_0, \ldots, \delta_{m-1} \in \mathbb{F}^\times$ are chosen so that the cosets $\delta_i H$ are pairwise disjoint and all N labels are distinct; the deployed choice is $\delta_i = \delta^i$ for an element δ whose multiplicative order is the odd part of $|\mathbb{F}^\times|$ (the halo2 book's *Permutation argument* chapter). Preprocessing fixes the *permuted-label polynomials* $s_i(X)$ with $s_i(\omega^j) = \delta_{i'}\,\omega^{j'}$ whenever $\sigma(i, j) = (i', j')$; they are preprocessed like the fixed columns but are not circuit columns, and the verifying key — the verifier's preprocessed description of the circuit, holding commitments to every fixed column and every permuted-label polynomial — contains their commitments. After the columns are committed the verifier sends challenges $\beta, \gamma \in \mathbb{F}$, and the prover commits to a running-product polynomial $Z_{\text{perm}}(X)$ and proves the two identities, both modulo $Z_H(X)$,

$$\begin{aligned} \ell_0(X)\,(1 - Z_{\text{perm}}(X)) &\equiv 0, \\ Z_{\text{perm}}(\omega X) \prod_{i=0}^{m-1} (v_i(X) + \beta\, s_i(X) + \gamma) - Z_{\text{perm}}(X) \prod_{i=0}^{m-1} (v_i(X) + \beta\,\delta_i X + \gamma) &\equiv 0, \end{aligned} \tag{7}$$

which together assert

$$\prod_{i,j} (v_i(\omega^j) + \beta\,\delta_i\,\omega^j + \gamma) = \prod_{i,j} (v_i(\omega^j) + \beta\, s_i(\omega^j) + \gamma). \tag{8}$$

> The verifier opens Z_{perm} at the random point z and at ωz, and checks the boundary and the cross-multiplied update.

The first identity fixes $Z_{\text{perm}}(\omega^0) = 1$; the second, holding at every row j, gives $Z_{\text{perm}}(\omega^{j+1}) = Z_{\text{perm}}(\omega^j) \cdot \prod_i (v_i + \beta\delta_i\omega^j + \gamma) / \prod_i (v_i + \beta s_i(\omega^j) + \gamma)$ whenever the denominator is nonzero, and at $j = n-1$ it returns to row ω^0, whence (8). The toy is the case $m = 3$ with $(\delta_0, \delta_1, \delta_2) = (1, 2, 3)$, which is not of the deployed form δ^i.

> **Theorem 2.11** (Soundness of the permutation argument)**.** *The assignment satisfies every equality constraint (cells related by σ carry equal values) if and only if the two sides of (8) are equal as polynomials in β and γ. Consequently the honest prover always satisfies (8), and if some constraint is violated the two sides agree at uniformly random β, γ with probability at most $2N/|\mathbb{F}|$, where N is the number of participating cells.*

Proof. Index cells by κ, write v_κ and λ_κ for value and label, and set

$$P(\beta, \gamma) := \prod_\kappa (\gamma + v_\kappa + \beta\lambda_\kappa) - \prod_\kappa (\gamma + v_\kappa + \beta\lambda_{\sigma(\kappa)}) \in \mathbb{F}[\beta][\gamma].$$

Each side is a product of monic linear polynomials in γ over the ring $\mathbb{F}[\beta]$, with roots $r_\kappa := -(v_\kappa + \beta\lambda_\kappa)$ and $r'_\kappa := -(v_\kappa + \beta\lambda_{\sigma(\kappa)})$; two such roots are equal in $\mathbb{F}[\beta]$ exactly when their value–label pairs coincide, since a polynomial in β is determined by its coefficients.

If every constraint holds then $v_{\sigma(\kappa)} = v_\kappa$ for all κ, so $r'_\kappa = -(v_{\sigma(\kappa)} + \beta\lambda_{\sigma(\kappa)}) = r_{\sigma(\kappa)}$, and the second product is the first with its factors reindexed by the bijection σ: $P = 0$.

If some constraint fails, pick κ_0 with $v_{\sigma(\kappa_0)} \neq v_{\kappa_0}$. The pair $(v_{\kappa_0}, \lambda_{\sigma(\kappa_0)})$ occurs among the roots r'; among the roots r the label $\lambda_{\sigma(\kappa_0)}$ occurs exactly once, paired with $v_{\sigma(\kappa_0)} \neq v_{\kappa_0}$, so the multisets of roots differ. Cancel every common factor from both products; a factor $(\gamma - r)$ then remains on one side, say the first, and r is not a root of the other side. Substitute $\gamma := r$, a ring homomorphism $\mathbb{F}[\beta][\gamma] \to \mathbb{F}[\beta]$ (*Math Guide*, §"Roots and the factor theorem", evaluation is a ring homomorphism; its proof holds verbatim over the coefficient ring $\mathbb{F}[\beta]$): the first product becomes 0 and the second becomes a product of nonzero elements of $\mathbb{F}[\beta]$, which is nonzero because $\mathbb{F}[\beta]$ is an integral domain (*Math Guide*, §"Degree", Corollary "$F[X]$ is an integral domain"). Hence $P \neq 0$. Its degree in each variable is at most N, and Lemma 2.9 bounds the probability that the random challenges hit a zero of P by $2N/|\mathbb{F}|$. □

The honest prover's ratio is 1 whenever the denominator is nonzero. The deployed boundary rule of Remark 2.17 admits the final value 0, and the error bound changes accordingly.

> **Corollary 2.12** (Permutation argument under the deployed boundary rule)**.** *Let the columns be fixed before β, γ, drawn uniformly, and let the running products be constrained on the active rows $\omega^0, \ldots, \omega^{u-1}$ only, with $Z_{\text{perm}}(\omega^0) = 1$ and the value at the next row ω^u permitted to be 0 or 1 (the deployed rule; Remark 2.17 fixes $u < n$ and gives the identities). If some copy constraint fails, the probability over β, γ that some running products satisfy the update and boundary identities on the active rows is at most $3N/|\mathbb{F}|$, with N the number of participating cells.*

Proof. Suppose some running products satisfy the identities and no numerator factor $v_\kappa + \beta\lambda_\kappa + \gamma$ vanishes. Then induction from $Z_{\text{perm}}(\omega^0) = 1$ keeps every running value nonzero, so the boundary value is 1, the updates telescope, and the two products over the active rows are equal; by Theorem 2.11 that happens with probability at most $2N/|\mathbb{F}|$. For each β at most N values of γ make a numerator factor vanish, which adds $N/|\mathbb{F}|$, and the union bound gives $2N/|\mathbb{F}| + N/|\mathbb{F}| = 3N/|\mathbb{F}|$. □

Cost. Excluding the final commitment check, the verifier's work for the argument is $O(m + \log n)$: evaluating the two identities (7) at the random point z needs the $2m$ linear factors, and $\ell_0(z) = (z^n - 1)/(n\,(z-1))$ costs $O(\log n)$ multiplications by the closed form of the Lagrange basis (*Math Guide*, §"The Lagrange basis on H"). The complete deployed verifier still carries one check linear in n, the multi-scalar multiplication (MSM; *Math Guide*, §"The mixed inner product with group elements") that closes every opening; §3 develops it under "The verifier's work: structured scalars and the final generator".

The deployed chunking. The update identity in (7) has degree $m + 1$ in the committed polynomials, and the active-row factor of Remark 2.17, which masks the rows reserved for blinding, adds one more; so a single product accommodates at most $m \leq d_{\max} - 2$ columns. Columns beyond that are split into chunks of at most $d_{\max} - 2$, each with its own running product, and a boundary identity $\ell_0(X)\,(Z_{\mathrm{perm},a}(X) - Z_{\mathrm{perm},a-1}(\omega^u X)) \equiv 0$ copies the value at the boundary row ω^u, the first row after the active rows $\omega^0, \dots, \omega^{u-1}$, from one product to the start of the next (the halo2 book's *Permutation argument* chapter). The column layout is implementation-specific: the protocol specification fixes the Action statement, not its circuit. The deployed Action circuit, whose implementation is `orchard`, has fifteen equality-enabled columns (ten advice, one instance, four fixed) and maximum constraint degree $d_{\max} = 9$, and therefore uses three running products of 7, 7 and 1 columns, not one.

2.7 Proving a value sits in a table: lookups

Some constraints admit no low-degree gate on the constrained cells alone. "This cell is a ten-bit number" or "this chunk indexes an entry of the Sinsemilla generator table" (Sinsemilla is the hash of the *Crypto Guide*, §"Sinsemilla: an algebraic hash-based commitment") has no low-degree formula in the cell itself: written as one polynomial gate, "$v \in \{0, \dots, 2^{10}-1\}$" is $\prod_{i=0}^{2^{10}-1}(v-i) = 0$, a gate of degree 2^{10}, and a table of arbitrary entries does no better. A decomposition into ten boolean-constrained bits keeps the degree at 2 but spends ten further cells on every such value. The single gate would exceed the degree bound of §2.4: in the toy field its degree 2^{10} exceeds $|\mathbb{F}| = 97$, so the bound $\deg D/|\mathbb{F}|$ is vacuous, and at any field size it multiplies the degree of the quotient the prover must split, commit and open (§2.3). The Action circuit uses range and table constraints throughout.

Membership of every entry of a column A in a table column S is, like a copy constraint, a statement about *multisets*, but it is not a permutation: A may repeat one table entry and omit others. The prover therefore commits to A', a rearrangement of A in which equal values occupy contiguous *runs*, and to S', a rearrangement of S in which the first row of each run of A' holds the matching table value. Two *row-local* conditions then give membership: each row of A' either begins a run, $A'_i = S'_i$, so that its value occurs in S, or continues one, $A'_i = A'_{i-1}$. The rearrangements are certified by the running product of §2.6 without labels: labelled factors fix one wiring σ, whereas here the rearrangements are the prover's choice, so a single running product with unlabelled factors, row by row $(A'_i + \beta)(S'_i + \gamma)$ against $(A_i + \beta)(S_i + \gamma)$, certifies that each primed column is a rearrangement of its original. Induction on the row index then gives membership for every row.

Example 2.13 (A two-bit range check on the toy grid)**.** Constrain cells to two-bit values with the table $S = (0, 1, 2, 3)$ and the column $A = (2, 3, 2, 2)$. Grouped, $A' = (2, 2, 2, 3)$; align $S' = (2, 0, 1, 3)$, a rearrangement of S placing 2 beside the first run and 3 beside the second. Row 0 starts a run ($2 = 2$), rows 1 and 2 continue it (2 above), row 3 starts a run ($3 = 3$): every check is row-local and every polynomial low-degree, however large the table (Figure 6). The unlabelled product certifies the rearrangements: at the challenges $(\beta, \gamma) = (5, 7)$ both $\prod_i (A'_i + \beta)(S'_i + \gamma)$ and $\prod_i (A_i + \beta)(S_i + \gamma)$ equal 82. The forged column $A = (2, 7, 2, 2)$ fails the row checks: its 7 must start a run in any arrangement A', and no rearrangement of a table without a 7 can put a 7

beside it. Over all 4 arrangements of the forged column and all 24 rearrangements of the table, the row checks pass for none.

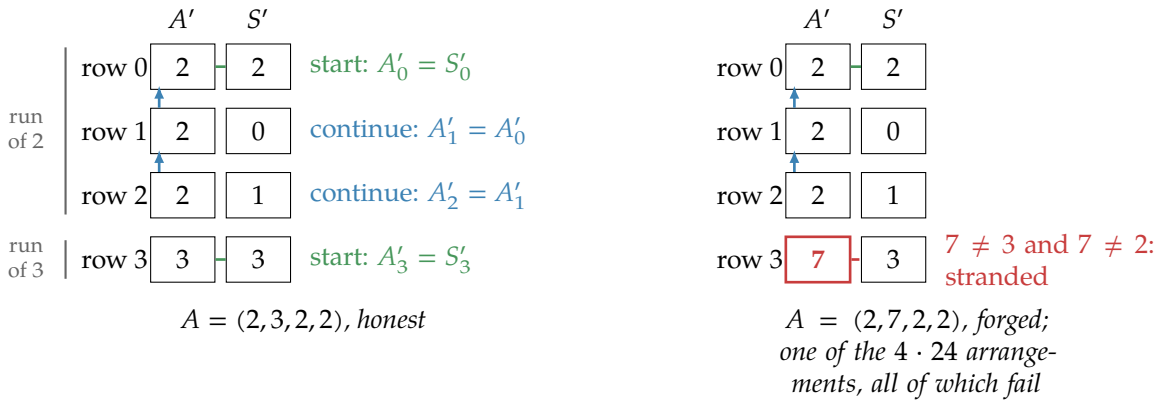


Figure 6: The lookup of Example 2.13. Left: the grouped column A' beside the aligned table S'; a run starts where $A'_i = S'_i$ (green bar) and continues where $A'_i = A'_{i-1}$ (blue arrow to the row above). Right: the forged column with a 7; the 7 starts a run, but no entry of S' equals 7 and the row above holds 2, so both row-local checks fail at row 3.

The Action circuit uses lookups of both kinds: it range-constrains each ten-bit word of a decomposition by a lookup into the table $\{0, \ldots, 2^{10} - 1\}$ (the halo2 book's *Decomposition* chapter), and it evaluates Sinsemilla by a lookup into a fixed table whose 2^{10} rows hold an index i and the two coordinates of the i-th Sinsemilla generator (the halo2 book's *Sinsemilla* chapter; the Orchard book's *Circuit* chapter records the deployed uses of both). A table fact costs a few committed columns and one more running product, and the gates stay low-degree. The form of the Action circuit's gates, one gate in full, and its three lookups are given in §4 under "From statement to circuit: arithmetisation in practice"; the exact identities, the boundary row, and multi-column tables come next.

2.8 The lookup argument

Construction 2.14 (Lookup argument)**.** The objective is that every entry of the column $A(X)$ on the active rows appears as some entry of the column $S(X)$ (the halo2 book's *Lookup argument* chapter). The prover supplies polynomials $A'(X)$ and $S'(X)$ where (i) A' is a permutation of A with like values grouped into contiguous runs, and (ii) S' is a permutation of S in which the first row of each run of like values in A' holds the matching value. The prover commits to A' and S'; after receiving challenges β, γ she commits to a running-product column $Z_{\text{lookup}}(X)$ and proves the four identities, all modulo $Z_H(X)$,

$$\begin{aligned} \ell_0(X)\,(1 - Z_{\text{lookup}}(X)) &\equiv 0, \\ Z_{\text{lookup}}(\omega X)\,(A'(X) + \beta)(S'(X) + \gamma) - Z_{\text{lookup}}(X)\,(A(X) + \beta)(S(X) + \gamma) &\equiv 0, \\ \ell_0(X)\,(A'(X) - S'(X)) &\equiv 0, \\ (A'(X) - S'(X))(A'(X) - A'(\omega^{-1}X)) &\equiv 0, \end{aligned} \tag{9}$$

the second and fourth additionally masked by the active-row factor of Remark 2.17.

Theorem 2.15 (Soundness of the lookup argument)**.** *Let the columns A, S, A', S' be fixed before β, γ are drawn. If the four identities (9) hold on the active rows, with the boundary conditions of Remark 2.17, then except with probability at most $4n/|\mathbb{F}|$ over the challenges β, γ, every value of A on the active rows equals the value of S at some active row.*

Proof. Let the active rows be $0, \dots, u-1$ (Remark 2.17), and write $A_i := A(\omega^i)$, likewise for S, A', S', and $Z_i := Z_{\text{lookup}}(\omega^i)$. Suppose first that no numerator factor $A_i + \beta$ or $S_i + \gamma$ with $i < u$ vanishes. The columns are fixed before the challenges, so each of the u values $-A_i$ equals β with probability $1/|\mathbb{F}|$, and likewise for γ; by the union bound the supposition fails with probability at most $2u/|\mathbb{F}|$. Under it, the second identity at row $i < u$ reads $Z_{i+1}\,(A'_i + \beta)(S'_i + \gamma) = Z_i\,(A_i + \beta)(S_i + \gamma)$, and induction from $Z_0 = 1$ gives $Z_{i+1} \neq 0$ and $(A'_i + \beta)(S'_i + \gamma) \neq 0$ for every $i < u$. Hence $Z_u \neq 0$, the boundary condition forces $Z_u = 1$, and multiplying the identities over $i < u$ and cancelling the nonzero Z_i gives

$$\prod_{i<u}(A'_i + \beta)(S'_i + \gamma) = \prod_{i<u}(A_i + \beta)(S_i + \gamma).$$

Set $f(\beta) := \prod_i (A'_i + \beta)$, $g(\beta) := \prod_i (A_i + \beta)$, $f'(\gamma) := \prod_i (S'_i + \gamma)$, $g'(\gamma) := \prod_i (S_i + \gamma)$ and $P := ff' - gg' \in \mathbb{F}[\beta, \gamma]$. If A' is a permutation of A and S' of S then $f = g$ and $f' = g'$, so $P = 0$. If instead, say, A' is not a permutation of A, then the multisets of roots $\{-A'_i\}$ and $\{-A_i\}$ differ, so $f \neq g$ by the cancel-and-evaluate argument in the proof of Theorem 2.11 (now over the field $\mathbb{F}$ itself); comparing the coefficients of the top power of γ in $ff' = gg'$, both f' and g' being monic of the same degree, would force $f = g$, so $P \neq 0$, of degree at most u in each variable, and Lemma 2.9 bounds the probability that the challenges hide the difference by $2u/|\mathbb{F}|$. The case of S' is symmetric. The two terms together are at most $2u/|\mathbb{F}| + 2u/|\mathbb{F}| \leq 4n/|\mathbb{F}|$.

So, with that probability excepted, A' is a rearrangement of A and S' of S on the active rows. The third identity fixes row 0: $A'_0 = S'_0$, a table value. The fourth, at every later active row i, says $(A'_i - S'_i)(A'_i - A'_{i-1}) = 0$: either $A'_i = S'_i$, a table value, or $A'_i = A'_{i-1}$. (At row 0 the reference $A'(\omega^{-1}X)$ wraps around to the last row and carries no information, which is why the third identity is needed there.) By induction on i, every A'_i equals the entry of S' at some active row, hence that of S at some active row; and since every entry of A on the active rows occurs in A' there, the claim follows. □

Multi-column tables. A lookup of m input expressions $A_0, \dots, A_{m-1}$ against m table columns $S_0, \dots, S_{m-1}$ is reduced to a single column by a random linear combination with a challenge θ drawn after the input and table columns are committed:

$$A_{\mathrm{c}}(X) := \sum_{j=0}^{m-1} \theta^{m-1-j} A_j(X), \qquad S_{\mathrm{c}}(X) := \sum_{j=0}^{m-1} \theta^{m-1-j} S_j(X), \tag{10}$$

and Construction 2.14 runs on A_{c} and S_{c}.

Proposition 2.16 (Compression of a multi-column lookup)**.** *Let θ be drawn uniformly after all input and table columns are fixed. If the input tuple at some active row equals the table tuple at no active row, then the compressed input A_{c} at that row equals S_{c} at some active row with probability at most $n(m-1)/|\mathbb{F}|$.*

Proof. For each active table row the difference of the compressed values is a polynomial in θ of degree at most $m - 1$ whose coefficients are the differences of the tuple entries, not all zero; it is therefore nonzero, and the random θ zeroes it with probability at most $(m-1)/|\mathbb{F}|$ (*Math Guide*, §"The Schwartz–Zippel lemma"). A union bound over the at most n active rows gives $n(m-1)/|\mathbb{F}|$. □

Variable tables, where S is itself an advice column, and range checks, where S enumerates the allowed range, are special cases of the same argument.

Remark 2.17 (Active rows and blinding). The lookup identities, like the permutation identities of §2.6, hold on the *active* rows only. The deployed system withholds the trailing rows of every column from them: a boundary row and, after it, n_{blind} blinding rows, which in every witness column (advice, and the permuted and running-product columns of the arguments) hold uniform random values, for the reason "Zero knowledge: hiding the witness in Halo 2" in §4 gives; fixed and instance columns carry no random values. Let $u := n - n_{\text{blind}} - 1$. Rows $0, \dots, u-1$ are the active rows, row u is the boundary row, and rows $u+1, \dots, n-1$ are the blinding rows. (The letter u indexes a row here; in §3 it denotes a round challenge.) Let $\ell_{\text{blind}}(X)$ be the sum of the Lagrange basis polynomials of the n_{blind} blinding rows, and let $\ell_{\text{last}}(X) := \ell_u(X)$ select the boundary row. Both are computed directly from the domain by both parties; they are not committed circuit columns. Each product-update identity, and the run identity of (9), is multiplied by the factor $(1 - (\ell_{\text{last}}(X) + \ell_{\text{blind}}(X)))$, which is 1 on the active rows and 0 elsewhere, so the random rows do not violate the active-row constraints. The wrap-around that closed the loop in Construction 2.10 is gone, so the running product is pinned at the boundary row instead, by $\ell_{\text{last}}(X)\,(Z(X)^2 - Z(X)) \equiv 0$, which allows $Z(\omega^u) \in \{0, 1\}$ (the halo2 book's *Lookup argument* and *Permutation argument* chapters). The value 0 is permitted so that a vanishing *numerator* factor ($A_i + \beta$ or $S_i + \gamma$; in the permutation argument $v_\kappa + \beta\,\lambda_\kappa + \gamma$) still leaves the honest prover a satisfying assignment: she sets the product to 0 from the next row on. The prover cannot force this event, because β, γ are drawn after her columns are committed. A *denominator* factor that vanishes at an earlier row than every vanishing numerator factor admits no satisfying running product: the update at that row j reads $0 = Z(\omega^j)$ times a nonzero numerator, and $Z(\omega^j) \neq 0$. This happens with probability at most $2u/|\mathbb{F}|$ for a lookup and $N/|\mathbb{F}|$ for the permutation argument, so completeness is overwhelming, not perfect. A boundary value of 0 is not a certificate of multiset equality, so soundness also excludes the challenges at which a numerator factor vanishes: the bound of Theorem 2.15 includes that term, and Corollary 2.12 adds it for the permutation argument.

2.9 Why the prover cannot wriggle: binding commitments

The random-point argument of §2.4 rests on a hypothesis: Theorem 2.5 assumed the prover's polynomials were fixed before she saw z. If she could choose a, b, c, t *after* learning z, she would pick numbers making (5) hold and reveal nothing real. A polynomial commitment enforces that hypothesis.

What a commitment must do. A commitment to a polynomial f is a short string $C = \mathsf{Commit}(f)$ with two properties, both developed in the *Crypto Guide*, §"Hiding and binding":

- **Binding.** Having published C, the prover cannot later open it as two different polynomials; C pins down f. Committing first, then receiving z, then opening, leaves her answering for the *one* polynomial she fixed in advance, the situation Theorem 2.5 assumed.
- **Hiding.** The string C reveals nothing about f. This is what will let the proof be zero-knowledge, under "Zero knowledge: hiding the witness in Halo 2" in §4.

A *polynomial* commitment adds an evaluation proof: given C and a point z, the prover can prove "$f(z) = v$" for the committed f without exposing f, and no prover can produce accepting proofs of two different values at the same point (*evaluation binding*, *Crypto Guide*, §"Polynomial commitment schemes"). Theorem 2.5 concerns the idealised protocol, in which the verifier receives the true evaluations of polynomials fixed in advance. Compiled with a polynomial commitment scheme that is evaluation-binding and extractable, the four-step protocol of §2.4 becomes, under

compatible composition hypotheses, an argument of knowledge (*Crypto Guide*, §"SNARKs: succinct non-interactive arguments of knowledge", Theorem "The compilation recipe, informal"): by its layered extraction, a prover who passes either answers with the true evaluations of polynomials fixed before z, where Theorem 2.5 applies, or breaks the scheme's evaluation binding or extractability. This volume asserts no bound on the compiled error as a sum of $\deg D/|\mathbb{F}|$ and a commitment term (Remark 4.11, §5). For Halo 2 the extractability of the openings is Theorem 3.5 and, for the deployed opening, Corollary 3.13, under the discrete-logarithm assumption on Vesta, with the generator hash modelled as a random oracle.

How Halo 2 commits. Encode f of degree $< n$ by its coefficient vector $\mathbf{f} = (f_0, \dots, f_{n-1})$; the running example's advice polynomial becomes $\mathbf{a} = (90, 61, 22, 24)$. Let $\mathbb{G}$ be a group of prime order $|\mathbb{F}|$ (deployed: Vesta, of order p_{Pallas}), and let $G_0, \dots, G_{n-1}, H \in \mathbb{G}$ be generators derived from a public hash, with $\mathbf{G} := (G_0, \dots, G_{n-1})$; here H is a group element, not the evaluation domain. The commitment is the single group element

$$C = \langle \mathbf{f}, \mathbf{G} \rangle + [r]H = \sum_i [f_i]\, G_i + [r]H,$$

the Pedersen vector commitment of the coefficient vector with blinder $r \in \mathbb{F}$ (*Crypto Guide*, §"Pedersen vector commitments"; the mixed inner product of a scalar vector with a vector of group elements is the *Math Guide*'s, §"The mixed inner product with group elements"). The commitment is perfectly hiding for uniform r and computationally binding under the discrete-logarithm assumption in $\mathbb{G}$, with the hash modelled as a random oracle into $\mathbb{G}$ (*Crypto Guide*, §"Pedersen vector commitments", Theorem "Properties of the vector commitment"): two openings of one C would exhibit a nontrivial discrete-logarithm relation among the G_i and H. Proving $f(z) = v$ is proving a statement about an inner product, since

$$f(z) = \sum_i f_i z^i = \langle \mathbf{f}, (1, z, z^2, \dots, z^{n-1}) \rangle,$$

and the *inner-product argument* does this in a logarithmic number of rounds, each halving the vectors, until the claim is trivial: the evaluation proof is $O(\log n)$ group elements, and it needs *no trusted setup*, because the generators come from a public hash rather than from anyone's secret (*Crypto Guide*, §"Polynomial commitment schemes", the inner-product alternative). Halo 2 uses exactly this one polynomial commitment scheme and no other: the inner-product argument over a group with no pairing (the halo2 book, which the protocol specification names as the definition of the system, § 5.4.10.3, "Halo 2"; its *Inner product argument* chapter), on the Vesta curve, where the protocol specification places the Action statement's proofs (protocol specification, § 5.4.9.6, "Pallas and Vesta"). The parameters, the generators and the deployed byte layout are given at the end of §3.

Opening a vector commitment. For the argument of this section the commitment must bind, so that the prover answers for polynomials fixed before z is drawn. A vector commitment alone does not provide step 3 of the random-point check. The only opening the Pedersen construction offers is the whole vector together with its blinder, $n + 1$ field elements: sending it reveals f, and it is no shorter than f. The verifier needs $\langle \mathbf{f}, (1, z, \dots, z^{n-1}) \rangle = v$ established for a committed $\mathbf{f}$ he never sees, in fewer than n group elements. That is the inner-product argument of §3.

3 The inner-product polynomial commitment

This section constructs the inner-product polynomial commitment that opens the commitments of §2.9 at the verifier's point. It states the problem, performs one halving step, recurses it into the

full argument with both parties' tables, gives the verifier's closing computation, proves knowledge soundness under the discrete-logarithm assumption, proves honest-verifier zero knowledge of the masked opening, runs the running example's polynomial through every round on a curve of prime order 97, and in the last subsection gives the parameters, the normalisation, the opening, the bytes and the deferred check of the deployed Halo 2. Beyond the *Math Guide*, the section relies on these parts of the *Crypto Guide*: the Pedersen vector commitment (§"Pedersen vector commitments"), extractors and the tree extraction of Sigma-protocols (§"Knowledge soundness and extractors", §"The Schnorr identification protocol", §"Sigma-protocols"), the simulation paradigm (§"The simulation paradigm and zero knowledge"), hashing to a curve (§"Hashing to a curve point") and the standing assumptions (§"Standing assumptions"). From this volume, the generic argument takes only the cue of §2.9; the toy reuses the running example's advice polynomial and random point (§2.2, §2.4), restated where used; the last subsection takes the deployed circuit's field from §1.2 and the advice, permutation, lookup and quotient objects from §2, and forward-references §4 for the row count and the remaining objects of the proof; the other references to §2 mark where the argument is used.

3.1 The problem: an inner product in fewer than d group elements

The prover holds a vector $\mathbf{a} = (a_0, \dots, a_{d-1})$ of d field elements. She has published one group element,

$$P = \langle \mathbf{a}, \mathbf{G} \rangle + [r]H = \sum_{i=0}^{d-1} [a_i]\, G_i + [r]H,$$

the Pedersen vector commitment of $\mathbf{a}$ with blinder r under the public generators $\mathbf{G} = (G_0, \dots, G_{d-1})$ and H (*Crypto Guide*, §"Pedersen vector commitments"; the mixed inner product $\langle \mathbf{a}, \mathbf{G} \rangle$ of a scalar vector with a vector of group elements is the *Math Guide*'s, §"The mixed inner product with group elements"). The verifier holds P, a public vector $\mathbf{b} = (b_0, \dots, b_{d-1})$ of his own, and a number v the prover claims. He wants to be sure that

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=0}^{d-1} a_i\, b_i = v.$$

There is a trivial protocol: the prover sends $\mathbf{a}$ and r; the verifier recomputes P from them, which the binding of the commitment makes a meaningful check, and then computes $\langle \mathbf{a}, \mathbf{b} \rangle$ himself. It costs $d+1$ field elements and reveals $\mathbf{a}$. Neither is acceptable here: the vector is the prover's secret advice, and the verifier of §2.4 asks for evaluations rather than polynomials because a polynomial is long, so a proof as long as the vector gains nothing. The problem is therefore:

> *Convince the verifier that $\langle \mathbf{a}, \mathbf{b} \rangle = v$ for the vector $\mathbf{a}$ committed in P, using fewer than d group elements, and without revealing $\mathbf{a}$.*

Why this is polynomial evaluation. Let $p(X) = a_0 + a_1 X + \cdots + a_{d-1} X^{d-1}$ be the polynomial with coefficient vector $\mathbf{a}$, and let x be a point. Then

$$p(x) = \sum_{i=0}^{d-1} a_i\, x^i = \langle \mathbf{a}, (1, x, x^2, \dots, x^{d-1}) \rangle.$$

Evaluating a committed polynomial is therefore *exactly* the problem above, with the public vector $\mathbf{b}(x) := (1, x, x^2, \dots, x^{d-1})$ of powers of the point. The running example makes it concrete. The advice polynomial of §2.2 is $a(X) = 90 + 61X + 22X^2 + 24X^3$ over $\mathbb{F}_{97}$, so $\mathbf{a} = (90, 61, 22, 24)$ and $d = 4$; the random point of §2.4 is $z = 20$, so $\mathbf{b}(20) = (1, 20, 20^2, 20^3) = (1, 20, 12, 46)$ in $\mathbb{F}_{97}$; and the value the verifier needs is $a(20) = \langle \mathbf{a}, \mathbf{b}(20) \rangle = 59$. Everything the random-point check of §2.4 asks of the prover, "open a, b, c and t at z", is four instances of this one problem.

Where the difficulty sits. The commitment is linear in the vector: adding two commitments commits to the sum of their vectors, and scaling one by λ commits to $\lambda\mathbf{a}$ (*Crypto Guide*, §"Pedersen vector commitments", the remark that linearity survives). The inner product is linear in $\mathbf{a}$ too. But the two linear maps land in different places, one in the group and one in the field, and by the hiding of the commitment no public operation takes P to $\langle \mathbf{a}, \mathbf{b}\rangle$. A claim about a vector of length one is checked by sending the scalar and its blinder. The argument that follows reduces the length by *halving*. Each halving step turns a claim about vectors of length 2^j into a claim of the same shape about vectors of length 2^{j-1}, at a price of two group elements, and after $k = \log_2 d$ steps the claim is about a single scalar. The argument uses exactly this linearity.

The relation. Throughout, $\mathbb{G}$ is a cyclic group of prime order p and $\mathbb{F} = \mathbb{F}_p$ its scalar field, so that $\mathbb{G}$ is a one-dimensional $\mathbb{F}$-vector space (*Math Guide*, §"The mixed inner product with group elements").

Definition 3.1 (The opening relation)**.** Fix public generators $\mathbf{G} = (G_0, \dots, G_{d-1})$ and H in $\mathbb{G} \setminus \{O\}$. To open the committed polynomial p at $x \in \mathbb{F}$ with claimed value v is to prove membership in

$$\mathcal{R}_{\mathrm{IPA}} := \Big\{((P, x, v);\ (\mathbf{a}, r)) : P = \langle \mathbf{a}, \mathbf{G}\rangle + [r]H,\ v = \langle \mathbf{a}, \mathbf{b}(x)\rangle,\ \mathbf{a} \in \mathbb{F}^d\Big\},$$
$$\mathbf{b}(x) := (1, x, \dots, x^{d-1}),$$

where (P, x, v) is the public statement and $(\mathbf{a}, r)$ the witness. The condition $\mathbf{a} \in \mathbb{F}^d$ is the degree bound $\deg p \le d - 1$: a commitment under d generators can only hold d coefficients.

Notation for the section. The following conventions hold from here to the end of §3.

- The vector length is $d = 2^k$, and d is also the exclusive degree bound and the number of generators. The round count is $k = \log_2 d$. The letter n, the row count of Definition 2.7, is not used until the deployed parameters at the end of the section, where $d = n$; the constraint degree stays $d_{\max}$ as §2.5 fixed it. The *Crypto Guide*'s sketch of this argument indexes its generators from 0 to d and bounds the degree by d inclusively; nothing changes but the labels.
- Generators are indexed from 0, $\mathbf{G} = (G_0, \dots, G_{d-1})$, so that the coefficient a_i of X^i multiplies G_i; the *Crypto Guide*'s "Pedersen vector commitments" indexes its generators $G_1, \dots, G_n$ instead, a shift of labels only.
- The letter H names the blinding generator $H \in \mathbb{G}$ of the Pedersen commitment; the evaluation domain $H \subset \mathbb{F}$ of §2 does not appear in this section. The letter p names the group order, and with an argument, $p(X)$, the committed polynomial.
- The letter z is the running example's opening point, $z = 20$, and x the generic one. The deployed protocol has a second challenge which its description also calls z; this volume writes it ζ.
- The structured scalars $s_i \in \mathbb{F}$ of §3.4 are scalars; the permuted-label polynomials $s_i(X)$ of Construction 2.10 are polynomials. Type and argument disambiguate. The mask $s(X)$ of §3.6 is one further polynomial, with coefficient vector $\mathbf{m}$.
- In the byte count of §3.8, a is the number of Actions, an integer, not the vector $\mathbf{a}$.
- A vector of length 2^j splits into its *low half*, the first 2^{j-1} entries, and its *high half*, the last 2^{j-1}: $\mathbf{a} = (\mathbf{a}_{\mathrm{lo}}, \mathbf{a}_{\mathrm{hi}})$, and likewise for $\mathbf{G}$ and $\mathbf{b}$.

3.2 One fold, every step written out

One halving step is called a *fold*. This subsection performs one fold on a vector of length d.

Carry the claim inside the group element. The commitment P says nothing about $\mathbf{b}$ or v. Bring them in: after P is fixed, the verifier samples a further generator U uniformly from $\mathbb{G} \setminus \{O\}$ and sends it. Both parties form

$$C := P + [v]\,U. \tag{11}$$

If the prover is honest, $C = \langle \mathbf{a}, \mathbf{G}\rangle + [r]H + [\langle \mathbf{a}, \mathbf{b}\rangle]\,U$: a single group element carrying three kinds of information on three independent sets of generators, the vector on $\mathbf{G}$, the blinder on H, and the inner product on U. The claim "P commits to $\mathbf{a}$ and $\langle \mathbf{a}, \mathbf{b}\rangle = v$" has become the claim that C has this one *shape*, for some $\mathbf{a}$ and r, with the U-coefficient tied to the $\mathbf{G}$-coefficients by the inner product with the public $\mathbf{b}$. The fold preserves exactly this shape while halving the vectors.

Split into halves. Write $\mathbf{a} = (\mathbf{a}_{\mathrm{lo}}, \mathbf{a}_{\mathrm{hi}})$, $\mathbf{G} = (\mathbf{G}_{\mathrm{lo}}, \mathbf{G}_{\mathrm{hi}})$ and $\mathbf{b} = (\mathbf{b}_{\mathrm{lo}}, \mathbf{b}_{\mathrm{hi}})$, each half of length $d/2$. An inner product splits along the halves:

$$\langle \mathbf{a}, \mathbf{G}\rangle = \langle \mathbf{a}_{\mathrm{lo}}, \mathbf{G}_{\mathrm{lo}}\rangle + \langle \mathbf{a}_{\mathrm{hi}}, \mathbf{G}_{\mathrm{hi}}\rangle, \qquad \langle \mathbf{a}, \mathbf{b}\rangle = \langle \mathbf{a}_{\mathrm{lo}}, \mathbf{b}_{\mathrm{lo}}\rangle + \langle \mathbf{a}_{\mathrm{hi}}, \mathbf{b}_{\mathrm{hi}}\rangle.$$

The fold. Let $u \in \mathbb{F}^\times$ be a nonzero challenge, and define the *folded* vectors of length $d/2$,

$$\mathbf{a}' := u\,\mathbf{a}_{\mathrm{lo}} + u^{-1}\,\mathbf{a}_{\mathrm{hi}}, \qquad \mathbf{G}' := u^{-1}\,\mathbf{G}_{\mathrm{lo}} + u\,\mathbf{G}_{\mathrm{hi}}, \qquad \mathbf{b}' := u^{-1}\,\mathbf{b}_{\mathrm{lo}} + u\,\mathbf{b}_{\mathrm{hi}}, \tag{12}$$

entry by entry: $a'_i = u\,a_i + u^{-1}\,a_{i+d/2}$, $G'_i = [u^{-1}]G_i + [u]G_{i+d/2}$ and $b'_i = u^{-1}b_i + u\,b_{i+d/2}$ for $0 \le i < d/2$. The coefficient vector is folded with the weights (u, u^{-1}) and the generator and evaluation vectors with the *inverse* weights (u^{-1}, u); this is the symmetric fold of the Halo paper and of Bulletproofs, and the reason for the inverse weights appears in the next computation.

The inner product of the folded vectors. Expand $\langle \mathbf{a}', \mathbf{G}'\rangle$ using bilinearity of the mixed inner product (*Math Guide*, §"The mixed inner product with group elements"), one term per pair of halves:

$$\begin{aligned}
\langle \mathbf{a}', \mathbf{G}'\rangle &= \langle u\,\mathbf{a}_{\mathrm{lo}} + u^{-1}\mathbf{a}_{\mathrm{hi}},\, u^{-1}\mathbf{G}_{\mathrm{lo}} + u\,\mathbf{G}_{\mathrm{hi}}\rangle \\
&= u\,u^{-1}\langle \mathbf{a}_{\mathrm{lo}}, \mathbf{G}_{\mathrm{lo}}\rangle + u\,u\,\langle \mathbf{a}_{\mathrm{lo}}, \mathbf{G}_{\mathrm{hi}}\rangle + u^{-1}u^{-1}\langle \mathbf{a}_{\mathrm{hi}}, \mathbf{G}_{\mathrm{lo}}\rangle + u^{-1}u\,\langle \mathbf{a}_{\mathrm{hi}}, \mathbf{G}_{\mathrm{hi}}\rangle \\
&= \underbrace{\langle \mathbf{a}_{\mathrm{lo}}, \mathbf{G}_{\mathrm{lo}}\rangle + \langle \mathbf{a}_{\mathrm{hi}}, \mathbf{G}_{\mathrm{hi}}\rangle}_{=\langle \mathbf{a}, \mathbf{G}\rangle} + u^2\,\langle \mathbf{a}_{\mathrm{lo}}, \mathbf{G}_{\mathrm{hi}}\rangle + u^{-2}\,\langle \mathbf{a}_{\mathrm{hi}}, \mathbf{G}_{\mathrm{lo}}\rangle.
\end{aligned}$$

The two *diagonal* terms, low with low and high with high, have weight $u\,u^{-1} = 1$ and reassemble the original $\langle \mathbf{a}, \mathbf{G}\rangle$: that is what the inverse weights were for. The two *cross* terms, low with high and high with low, survive with the weights u^2 and u^{-2}. The same expansion for the scalar inner product, with $\mathbf{b}'$ folded by the same weights as $\mathbf{G}'$, gives

$$\langle \mathbf{a}', \mathbf{b}'\rangle = \langle \mathbf{a}, \mathbf{b}\rangle + u^2\,\langle \mathbf{a}_{\mathrm{lo}}, \mathbf{b}_{\mathrm{hi}}\rangle + u^{-2}\,\langle \mathbf{a}_{\mathrm{hi}}, \mathbf{b}_{\mathrm{lo}}\rangle.$$

The cross terms do not depend on the challenge. The four cross inner products $\langle \mathbf{a}_{\mathrm{lo}}, \mathbf{G}_{\mathrm{hi}}\rangle$, $\langle \mathbf{a}_{\mathrm{hi}}, \mathbf{G}_{\mathrm{lo}}\rangle$, $\langle \mathbf{a}_{\mathrm{lo}}, \mathbf{b}_{\mathrm{hi}}\rangle$ and $\langle \mathbf{a}_{\mathrm{hi}}, \mathbf{b}_{\mathrm{lo}}\rangle$ involve only the halves, not u. The prover can therefore commit to them *before* the challenge is drawn. She samples two fresh blinders $\ell, \rho \in \mathbb{F}$ and sends the two group elements

$$L := \langle \mathbf{a}_{\mathrm{lo}}, \mathbf{G}_{\mathrm{hi}}\rangle + [\ell]H + [\langle \mathbf{a}_{\mathrm{lo}}, \mathbf{b}_{\mathrm{hi}}\rangle]\,U, \qquad R := \langle \mathbf{a}_{\mathrm{hi}}, \mathbf{G}_{\mathrm{lo}}\rangle + [\rho]H + [\langle \mathbf{a}_{\mathrm{hi}}, \mathbf{b}_{\mathrm{lo}}\rangle]\,U, \tag{13}$$

each of the same three-part shape as C: a vector part on the generators, a blinder on H, and the matching scalar inner product on U. Only then does the verifier draw u.

The fold identity. Now add the cross terms to C with the weights the expansion produced and regroup by generator set:

$$\begin{aligned} C + [u^2]L + [u^{-2}]R &= \langle \mathbf{a}, \mathbf{G}\rangle + [r]H + [\langle \mathbf{a}, \mathbf{b}\rangle]\, U \\ &\quad + u^2\big(\langle \mathbf{a}_{\mathrm{lo}}, \mathbf{G}_{\mathrm{hi}}\rangle + [\ell]H + [\langle \mathbf{a}_{\mathrm{lo}}, \mathbf{b}_{\mathrm{hi}}\rangle]\, U\big) \\ &\quad + u^{-2}\big(\langle \mathbf{a}_{\mathrm{hi}}, \mathbf{G}_{\mathrm{lo}}\rangle + [\rho]H + [\langle \mathbf{a}_{\mathrm{hi}}, \mathbf{b}_{\mathrm{lo}}\rangle]\, U\big) \\ &= \big(\langle \mathbf{a}, \mathbf{G}\rangle + u^2\langle \mathbf{a}_{\mathrm{lo}}, \mathbf{G}_{\mathrm{hi}}\rangle + u^{-2}\langle \mathbf{a}_{\mathrm{hi}}, \mathbf{G}_{\mathrm{lo}}\rangle\big) \\ &\quad + [\, r + u^2\ell + u^{-2}\rho\,]H + \big[\langle \mathbf{a}, \mathbf{b}\rangle + u^2\langle \mathbf{a}_{\mathrm{lo}}, \mathbf{b}_{\mathrm{hi}}\rangle + u^{-2}\langle \mathbf{a}_{\mathrm{hi}}, \mathbf{b}_{\mathrm{lo}}\rangle\big]\, U \\ &= \langle \mathbf{a}', \mathbf{G}'\rangle + [r']H + [\langle \mathbf{a}', \mathbf{b}'\rangle]\, U, \end{aligned}$$

where $r' := r + u^2\ell + u^{-2}\rho$ and the last line substitutes the two expansions. Written as one equation, the *fold identity* is

$$C' := C + [u^2]L + [u^{-2}]R = \langle \mathbf{a}', \mathbf{G}'\rangle + [r']H + [\langle \mathbf{a}', \mathbf{b}'\rangle]\, U. \tag{14}$$

What has been achieved. Compare the two sides of (14) with (11). The element C' has the same shape as C, over vectors of half the length: its $\mathbf{G}'$-part is the folded coefficient vector, its H-part a blinder, and its U-coefficient the inner product of the folded vectors. The verifier can compute C', $\mathbf{G}'$ and $\mathbf{b}'$ entirely on his own from C, L, R, u and the public data. The prover computes $\mathbf{a}'$ and r'. The claim about $(\mathbf{a}, \mathbf{b}, \mathbf{G})$ of length d has been replaced by a claim about $(\mathbf{a}', \mathbf{b}', \mathbf{G}')$ of length $d/2$, and the exchange cost exactly two group elements and one field element of challenge. Figure 7 shows the fold and identity (14). Only completeness is claimed here; that a prover who cannot open C cannot open C' either is proved in §3.5.

The evaluation vector folds into a power vector. When $\mathbf{b} = \mathbf{b}(x)$, the high half is the low half scaled: $\mathbf{b}_{\mathrm{hi}} = x^{d/2}\,\mathbf{b}_{\mathrm{lo}}$, since $b_{i+d/2} = x^{i+d/2} = x^{d/2}x^i$. Hence

$$\mathbf{b}' = u^{-1}\mathbf{b}_{\mathrm{lo}} + u\,x^{d/2}\mathbf{b}_{\mathrm{lo}} = (u^{-1} + u\,x^{d/2})\,(1, x, \ldots, x^{d/2-1}) :$$

the folded evaluation vector is again a vector of powers of x, scaled by the scalar $u^{-1} + ux^{d/2}$. The verifier never needs to store $\mathbf{b}'$; he needs one scalar per round. This is the seed of the closed form g of §3.4.

The fold at $d = 4$, symbolically. For the running example's length, the halves are $\mathbf{a}_{\mathrm{lo}} = (a_0, a_1)$, $\mathbf{a}_{\mathrm{hi}} = (a_2, a_3)$, $\mathbf{G}_{\mathrm{lo}} = (G_0, G_1)$, $\mathbf{G}_{\mathrm{hi}} = (G_2, G_3)$, $\mathbf{b}_{\mathrm{lo}} = (1, x)$ and $\mathbf{b}_{\mathrm{hi}} = (x^2, x^3)$, so the cross terms of (13) read

$$\begin{aligned} L &= [a_0]G_2 + [a_1]G_3 + [\ell]H + [a_0x^2 + a_1x^3]\, U, \\ R &= [a_2]G_0 + [a_3]G_1 + [\rho]H + [a_2 + a_3x]\, U, \end{aligned}$$

and the folded vectors are $\mathbf{a}' = (ua_0 + u^{-1}a_2,\ ua_1 + u^{-1}a_3)$, $\mathbf{G}' = ([u^{-1}]G_0 + [u]G_2,\ [u^{-1}]G_1 + [u]G_3)$ and $\mathbf{b}' = (u^{-1} + ux^2)\,(1, x)$. Section 3.7 fills these in with numbers.

3.3 The recursion: k rounds to a single scalar

The folded claim of §3.2 has the shape of the original claim, so it can be folded again. After $k = \log_2 d$ folds the vectors have length one, and a length-one claim is checked by sending the scalar and its blinder. That is the whole protocol; this subsection writes it out as a construction, numbers the rounds, and lays the two parties' work side by side.

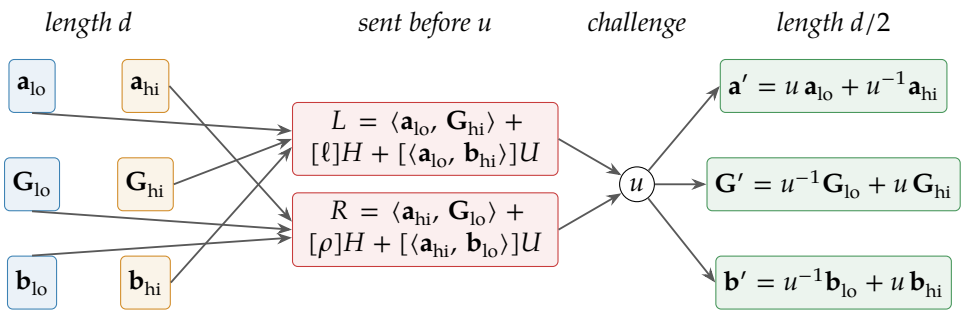


Figure 7: One fold. The three vectors of length d (left; low halves in blue, high halves in amber) determine the two cross terms L and R (red): L from $\mathbf{a}_{\mathrm{lo}}$, $\mathbf{G}_{\mathrm{hi}}$ and $\mathbf{b}_{\mathrm{hi}}$, R from the complementary halves, both sent before the challenge u is drawn. The challenge then folds all three vectors to length $d/2$ (green). The identity beneath is equation (14): the folded commitment has the same three-part shape as the original, over the folded vectors, with the cross terms absorbed at weights u^2 and u^{-2}.

Construction 3.2 (IPA-based polynomial commitment)**.** Let $\mathbb{G}$ be a cyclic group of prime order p in which the discrete logarithm problem is hard (*Crypto Guide*, §"Standing assumptions"), and let $\mathbb{F} = \mathbb{F}_p$. Choose $d = 2^k$ and sample the generators $\mathbf{G} = (G_0, \dots, G_{d-1})$ and H independently and uniformly from $\mathbb{G} \setminus \{O\}$, as in the *Crypto Guide*'s Construction "Pedersen vector commitment" (§"Pedersen vector commitments"). Deployed generators are instead hash-to-curve outputs on public labels, uniform when hash-to-curve is modelled as a random oracle into $\mathbb{G}$ (*Crypto Guide*, §"Hashing to a curve point", Remark "Why NUMS generators are binding-safe").

Commitment. For $p(X) = \sum_{i=0}^{d-1} a_i X^i \in \mathbb{F}[X]$ and a blinder $r \in \mathbb{F}$,

$$\mathsf{Commit}(p; r) := \langle \mathbf{a}, \mathbf{G} \rangle + [r]H \in \mathbb{G}.$$

Opening. To prove $((P, x, v); (\mathbf{a}, r)) \in \mathcal{R}_{\mathrm{IPA}}$, the two parties proceed as follows.

1. **Setup.** The verifier samples U uniformly from $\mathbb{G} \setminus \{O\}$ and sends it. Both parties set $C^{(k)} := P + [v]U$, $\mathbf{G}^{(k)} := \mathbf{G}$ and $\mathbf{b}^{(k)} := \mathbf{b}(x)$; the prover sets $\mathbf{a}^{(k)} := \mathbf{a}$ and $r^{(k)} := r$.

2. **Rounds.** For $j = k, k-1, \dots, 1$, the vectors $\mathbf{a}^{(j)}, \mathbf{G}^{(j)}, \mathbf{b}^{(j)}$ have length 2^j and are split into halves as in §3.2.

   (a) The prover samples independent uniform $\ell_j, \rho_j \in \mathbb{F}$ and sends

   $$L_j := \langle \mathbf{a}^{(j)}_{\mathrm{lo}}, \mathbf{G}^{(j)}_{\mathrm{hi}} \rangle + [\ell_j]H + [\langle \mathbf{a}^{(j)}_{\mathrm{lo}}, \mathbf{b}^{(j)}_{\mathrm{hi}} \rangle]\,U,$$
   $$R_j := \langle \mathbf{a}^{(j)}_{\mathrm{hi}}, \mathbf{G}^{(j)}_{\mathrm{lo}} \rangle + [\rho_j]H + [\langle \mathbf{a}^{(j)}_{\mathrm{hi}}, \mathbf{b}^{(j)}_{\mathrm{lo}} \rangle]\,U.$$

   (b) The verifier sends a uniform $u_j \in \mathbb{F}^\times$.

   (c) Both parties fold by (12):

   $$\mathbf{G}^{(j-1)} := u_j^{-1}\mathbf{G}^{(j)}_{\mathrm{lo}} + u_j\,\mathbf{G}^{(j)}_{\mathrm{hi}}, \qquad \mathbf{b}^{(j-1)} := u_j^{-1}\mathbf{b}^{(j)}_{\mathrm{lo}} + u_j\,\mathbf{b}^{(j)}_{\mathrm{hi}},$$
   $$C^{(j-1)} := C^{(j)} + [u_j^2]L_j + [u_j^{-2}]R_j;$$

the prover also folds $\mathbf{a}^{(j-1)} := u_j\,\mathbf{a}_{\mathrm{lo}}^{(j)} + u_j^{-1}\mathbf{a}_{\mathrm{hi}}^{(j)}$ and $r^{(j-1)} := r^{(j)} + u_j^2\ell_j + u_j^{-2}\rho_j$.

3. **Final message.** After round 1 every vector has length one. Write $a := \mathbf{a}^{(0)}$, $G^{(0)} := \mathbf{G}^{(0)}$ and $b^{(0)} := \mathbf{b}^{(0)}$ for the single entries. The prover sends the scalar a and the aggregated blinder

$$r^* := r^{(0)} = r + \sum_{j=1}^{k}(u_j^2\,\ell_j + u_j^{-2}\rho_j).$$

4. **Check.** The verifier accepts if and only if

$$P + [v]U + \sum_{j=1}^{k}([u_j^2]L_j + [u_j^{-2}]R_j) \;\stackrel{?}{=}\; [a]\,G^{(0)} + [r^*]H + [a\cdot b^{(0)}]\,U, \tag{15}$$

where the left-hand side is $C^{(0)}$ and $G^{(0)}$, $b^{(0)}$ are computed as §3.4 describes.

The round index descends so that round j begins with vectors of length 2^j; round k is the first round and round 1 the last. The scheme instantiates the polynomial-commitment interface of the *Crypto Guide* (§"Polynomial commitment schemes", the inner-product alternative), and the opening protocol is the inner-product argument of Bootle, Cerulli, Chaidos, Groth and Petit as refined in Bulletproofs by Bünz, Bootle, Boneh, Poelstra, Wuille and Maxwell, in the symmetric normalisation those papers use.

Completeness. The honest prover always passes. By induction on the rounds,

$$C^{(j)} = \langle \mathbf{a}^{(j)}, \mathbf{G}^{(j)}\rangle + [r^{(j)}]H + [\langle \mathbf{a}^{(j)}, \mathbf{b}^{(j)}\rangle]\,U \qquad \text{for } j = k, k-1, \dots, 0: \tag{16}$$

at $j = k$ this is (11) with the honest P, and the step from j to $j-1$ is the fold identity (14) applied to the round's vectors, challenge and blinders. At $j = 0$ the vectors have one entry each, the inner products are products of scalars, and (16) reads $C^{(0)} = [a]G^{(0)} + [r^*]H + [a\,b^{(0)}]U$, which is (15). The only way an honest run can fail is a challenge $u_j = 0$, which the verifier excludes.

Proof size. The prover sends two group elements per round and two field elements at the end: $2k$ group elements plus $O(1)$, against the $d + 1 = 2^k + 1$ field elements of the trivial protocol. The verifier keeps only the challenges $u_1, \dots, u_k$ and the received points; he need not fold $\mathbf{G}$ round by round, as the next subsection shows. Figure 8 draws the k rounds, and Tables 2 and 3 list, one row per message, what each party holds, computes and sends, so that a reader can play either role; §3.7 refills the same rows with numbers.

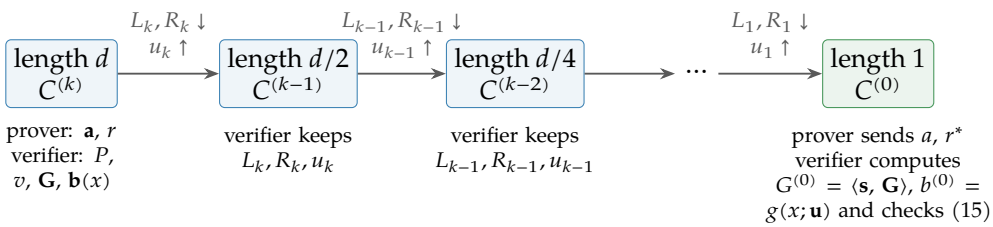


Figure 8: The k rounds of Construction 3.2. Each round halves the vector length: the prover sends two points (↓), the verifier answers with one challenge (↑) and keeps the three values. After k rounds one scalar and one aggregated blinder close the argument; the verifier's closing work is the subject of §3.4.

step	receives / holds	computes and sends
setup	holds $\mathbf{a}$, r, x, v; receives U	sets $\mathbf{a}^{(k)} := \mathbf{a}$, $\mathbf{G}^{(k)} := \mathbf{G}$, $\mathbf{b}^{(k)} := \mathbf{b}(x)$, $r^{(k)} := r$
round j	vectors of length 2^j, split into halves	samples ℓ_j, ρ_j; sends L_j, R_j as in Construction 3.2
	receives u_j	folds $\mathbf{a}^{(j-1)} = u_j \mathbf{a}_{\mathrm{lo}}^{(j)} + u_j^{-1} \mathbf{a}_{\mathrm{hi}}^{(j)}$, $\mathbf{G}^{(j-1)}$, $\mathbf{b}^{(j-1)}$; $r^{(j-1)} = r^{(j)} + u_j^2 \ell_j + u_j^{-2} \rho_j$
final	one entry each: $a = \mathbf{a}^{(0)}$, $r^* = r^{(0)}$	sends a, r^*

Table 2: The prover's side of Construction 3.2, one row per message. The round row repeats for $j = k, \dots, 1$.

step	receives / holds	computes and sends
setup	holds P, x, v, $\mathbf{G}$, H	samples U and sends it; sets $C^{(k)} := P + [v]U$
round j	receives L_j, R_j	samples $u_j \in \mathbb{F}^\times$ and sends it; keeps (L_j, R_j, u_j); optionally $C^{(j-1)} := C^{(j)} + [u_j^2]L_j + [u_j^{-2}]R_j$
final	receives a, r^*	computes $\mathbf{s}$ and $G^{(0)} = \langle \mathbf{s}, \mathbf{G} \rangle$ (one length-d multi-scalar multiplication), $b^{(0)} = g(x; u_1, \dots, u_k)$ ($O(k)$ field operations), and checks (15)

Table 3: The verifier's side of Construction 3.2, one row per message, with the closing computations of §3.4. The round row repeats for $j = k, \dots, 1$; folding C round by round is optional because (15) sums the cross terms directly.

3.4 The verifier's work: structured scalars and the final generator

The check (15) needs two quantities the verifier has not yet been told how to obtain cheaply: the final generator $G^{(0)}$ and the final evaluation scalar $b^{(0)}$. Folding $\mathbf{G}$ round by round costs $d + d/2 + \cdots + 2 = 2d - 2$ scalar multiplications (two per entry of each folded vector), and folding $\mathbf{b}$ likewise many field operations. Both have closed forms in the challenges.

Theorem 3.3 (Structured scalars and the polynomial g)**.** *For $0 \le i < d$ write the binary expansion $i = \sum_{\ell=0}^{k-1} \beta_\ell 2^\ell$ with digits $\beta_\ell \in \{0,1\}$, and define the* structured scalars

$$s_i := \prod_{\ell=1}^{k} u_\ell^{(-1)^{1-\beta_{\ell-1}}} \qquad (\textit{the factor is } u_\ell \textit{ when } \beta_{\ell-1} = 1 \textit{ and } u_\ell^{-1} \textit{ when } \beta_{\ell-1} = 0). \tag{17}$$

Then, after the k folds of Construction 3.2,

$$G^{(0)} = \langle \mathbf{s}, \mathbf{G} \rangle = \sum_{i=0}^{d-1} [s_i]\, G_i, \qquad b^{(0)} = \langle \mathbf{s}, \mathbf{b}(x) \rangle = g(x; u_1, \dots, u_k), \tag{18}$$

where

$$g(X; u_1, \dots, u_k) := \prod_{\ell=1}^{k} (u_\ell^{-1} + u_\ell X^{2^{\ell-1}}), \tag{19}$$

a polynomial of degree $d-1$ in X.

Proof. Follow one original generator G_m through the folds. Claim: after the rounds $k, k-1, \dots, j+1$ have been performed, G_m contributes to exactly one entry of $\mathbf{G}^{(j)}$, the entry at position $m \bmod 2^j$, with coefficient $\prod_{\ell=j+1}^{k} u_\ell^{(-1)^{1-\beta_{\ell-1}}}$, where $\beta_{\ell-1}$ is the digit of m at position $\ell - 1$. At $j = k$ nothing has happened and the claim says G_m sits at position m with coefficient 1. For the step from j to $j-1$: round j folds the length-2^j vector by $G_i^{(j-1)} = [u_j^{-1}]G_i^{(j)} + [u_j]G_{i+2^{j-1}}^{(j)}$ for $0 \le i < 2^{j-1}$. The position $m \bmod 2^j$ lies in the low half $[0, 2^{j-1})$ exactly when the digit β_{j-1} of m is 0, in which case it moves to position $m \bmod 2^{j-1}$ with the extra factor u_j^{-1}; and it lies in the high half exactly when $\beta_{j-1} = 1$, in which case it is the entry $i + 2^{j-1}$ with $i = m \bmod 2^{j-1}$ and moves to position i with the extra factor u_j. Either way the new coefficient is the old one times $u_j^{(-1)^{1-\beta_{j-1}}}$, which is the claim for $j-1$. At $j = 0$ every G_m sits in the single entry $G^{(0)}$ with coefficient s_m, which is the first half of (18). The vector $\mathbf{b}$ is folded with the same weights, so the same argument gives $b^{(0)} = \sum_m s_m b_m = \langle \mathbf{s}, \mathbf{b}(x) \rangle$.

For the closed form, expand the product (19): choosing from the ℓ-th factor either u_ℓ^{-1} (when $\beta_{\ell-1} = 0$) or $u_\ell X^{2^{\ell-1}}$ (when $\beta_{\ell-1} = 1$) produces, for each m, the monomial $s_m X^{\sum_{\ell=1}^{k} \beta_{\ell-1} 2^{\ell-1}} = s_m X^m$, and every $m \in [0, d)$ arises from exactly one choice. Hence $g(X) = \sum_m s_m X^m$, whose value at x is $\langle \mathbf{s}, \mathbf{b}(x) \rangle$, and whose degree is $d-1$ since the top monomial has coefficient $s_{d-1} = \prod_\ell u_\ell \neq 0$. □

At $d = 4$ the scalars are $s_0 = u_1^{-1}u_2^{-1}$, $s_1 = u_1 u_2^{-1}$, $s_2 = u_1^{-1} u_2$ and $s_3 = u_1 u_2$, and $g(x; u_1, u_2) = (u_1^{-1} + u_1 x)(u_2^{-1} + u_2 x^2)$: the digit of weight 1 selects the exponent of u_1, the round that folded pairs of neighbours, and the digit of weight 2 selects the exponent of u_2, the round that folded the two halves of the whole vector.

The final equation, and why it checks. The verifier's closing work is now explicit. He evaluates $b^{(0)} = g(x; \mathbf{u})$ from the challenges; he computes $G^{(0)} = \langle \mathbf{s}, \mathbf{G} \rangle$, one multi-scalar multiplication of length d; he adds up the left-hand side of (15) from P, U, the $2k$ received points and the

challenges, $2k+1$ scalar multiplications; and he compares it with the right-hand side, three more scalar multiplications. For the honest prover the two sides agree by the invariant (16). That a dishonest prover cannot make them agree except by knowing an opening is §3.5.

The cost of the final check. Evaluating g at x costs $3k-2$ field multiplications: k products $u_\ell\, x^{2^{\ell-1}}$, $k-1$ squarings to produce the powers $x, x^2, x^4, \dots$, and $k-1$ multiplications of the factors together; at $k = 2$ that is four multiplications, and at Orchard's $k = 11$ it is 31. But $G^{(0)}$ is a linear combination of *all* d generators, and computing it is a multi-scalar multiplication of length d, at Orchard's $d = 2^{11}$ some 2048 scalar multiplications: the one part of the verifier's work that is linear in the size of the committed vector, and the check "linear in n" that §2.6 deferred to this subsection. Everything else the verifier does is logarithmic. The asymmetry has a name in the closed form: $G^{(0)} = \langle \mathbf{s}, \mathbf{G}\rangle$ is nothing but $\mathsf{Commit}(g; 0)$, the unblinded commitment to the polynomial g whose value the verifier just computed in $O(k)$ operations. Checking a *value* of g is cheap; checking its *commitment* is not. This asymmetry is the seed of the accumulation scheme of §7, and of the deferral described at the end of this section.

3.5 Why it is binding: extraction from three transcripts

This subsection shows that *any* prover that makes the verifier of Construction 3.2 accept knows an opening $(\mathbf{a}, r)$ of P with $\langle \mathbf{a}, \mathbf{b}(x)\rangle = v$, in the operational sense of the *Crypto Guide* (§"Knowledge soundness and extractors"): an extractor that runs the prover as a subroutine, rewinds it to an earlier state and feeds it fresh challenges computes the pair from its answers. The mechanism is the one that makes Schnorr's identification protocol a proof of knowledge (*Crypto Guide*, §"The Schnorr identification protocol"): a single transcript reveals nothing, but two accepting transcripts with the same first message and distinct challenges are two linear equations, and subtracting them cancels the unknown commitment and leaves the witness. Here each round has three unknown group elements instead of one, so three transcripts are needed per round, and the k rounds stack the three-way branching into a tree, the tree special soundness of the *Crypto Guide* (§"Sigma-protocols"). The single-round step is pure linear algebra and is proved in full; the tree is an induction on it; the discrete-logarithm assumption enters once, at the root.

Representations. Fix a level j with generator vector $\mathbf{G}^{(j)}$ of length 2^j. A *representation* of an element $Q \in \mathbb{G}$ over $(\mathbf{G}^{(j)}, H, U)$ is a triple $(\mathbf{x}; y; w) \in \mathbb{F}^{2^j} \times \mathbb{F} \times \mathbb{F}$ with

$$Q = \langle \mathbf{x}, \mathbf{G}^{(j)}\rangle + [y]H + [w]U.$$

Call the representation *well formed* at level j if its U-coordinate is the inner product of its $\mathbf{G}$-coordinates with the round's evaluation vector, $w = \langle \mathbf{x}, \mathbf{b}^{(j)}\rangle$. The invariant (16) says exactly that the honest prover holds a well-formed representation $(\mathbf{a}^{(j)}; r^{(j)}; \langle \mathbf{a}^{(j)}, \mathbf{b}^{(j)}\rangle)$ of $C^{(j)}$ at every level. A representation over folded generators lifts to one over the generators of the level above: if $Q = \langle \mathbf{a}', \mathbf{G}'\rangle + [y]H + [w]U$ with $\mathbf{G}' = u^{-1}\mathbf{G}_{\mathrm{lo}} + u\,\mathbf{G}_{\mathrm{hi}}$, then by bilinearity

$$Q = \langle (u^{-1}\mathbf{a}',\ u\,\mathbf{a}'), \mathbf{G}\rangle + [y]H + [w]U, \tag{20}$$

and the lift preserves well-formedness, since $\langle (u^{-1}\mathbf{a}', u\mathbf{a}'), \mathbf{b}\rangle = \langle \mathbf{a}', u^{-1}\mathbf{b}_{\mathrm{lo}} + u\,\mathbf{b}_{\mathrm{hi}}\rangle = \langle \mathbf{a}', \mathbf{b}'\rangle$. Under the discrete-logarithm assumption a prover can know at most one representation of any Q: two of them differ by a nontrivial relation among $\mathbf{G}$, H and U, which is the binding argument of the *Crypto Guide* (§"Pedersen vector commitments"). Nothing below uses this until the root is reached.

> **Lemma 3.4** (Single-round recovery)**.** *Fix one round of Construction 3.2 at level j: the verifier's element $C := C^{(j)}$, the generators $\mathbf{G} := \mathbf{G}^{(j)}$ and evaluation vector $\mathbf{b} := \mathbf{b}^{(j)}$, and the prover's cross*

terms L, R. Let u, u', u'' be three challenges for this round, and suppose that for each $\nu \in \{u, u', u''\}$ a representation $(\mathbf{x}_\nu; y_\nu; w_\nu)$ of the folded element $C + [\nu^2]L + [\nu^{-2}]R$ over $(\mathbf{G}, H, U)$ is known. If the squares u^2, u'^2, u''^2 are pairwise distinct, then representations $(\mathbf{x}_L; y_L; w_L)$, $(\mathbf{x}_C; y_C; w_C)$ and $(\mathbf{x}_R; y_R; w_R)$ of L, C and R over $(\mathbf{G}, H, U)$ are obtained by solving one 3×3 linear system with rows $(\nu^2, 1, \nu^{-2})$, and they satisfy, for each ν,

$$\mathbf{x}_\nu = \nu^2 \mathbf{x}_L + \mathbf{x}_C + \nu^{-2}\mathbf{x}_R, \qquad y_\nu = \nu^2 y_L + y_C + \nu^{-2} y_R, \qquad w_\nu = \nu^2 w_L + w_C + \nu^{-2} w_R. \quad (21)$$

If the three given representations are well formed at level j, so is the recovered representation of C.

Proof. Write the three known representations as vectors $\mathbf{Y}_\nu := (\mathbf{x}_\nu, y_\nu, w_\nu) \in \mathbb{F}^{2^j+2}$ and the three unknown ones as $\mathbf{X}_L, \mathbf{X}_C, \mathbf{X}_R$. The three group equations

$$[\nu^2]L + C + [\nu^{-2}]R = \langle \mathbf{x}_\nu, \mathbf{G}\rangle + [y_\nu]H + [w_\nu]U, \qquad \nu \in \{u, u', u''\},$$

suggest the linear system

$$M \begin{pmatrix} \mathbf{X}_L \\ \mathbf{X}_C \\ \mathbf{X}_R \end{pmatrix} = \begin{pmatrix} \mathbf{Y}_u \\ \mathbf{Y}_{u'} \\ \mathbf{Y}_{u''} \end{pmatrix}, \qquad M := \begin{pmatrix} u^2 & 1 & u^{-2} \\ u'^2 & 1 & u'^{-2} \\ u''^2 & 1 & u''^{-2} \end{pmatrix},$$

to be solved coordinate by coordinate. Multiplying the row of ν by the nonzero scalar ν^2 turns it into $(\nu^4, \nu^2, 1)$, which does not change the kernel. If $M(c_0, c_1, c_2)^\top = 0$, the polynomial $c_0 W^2 + c_1 W + c_2$ of degree at most 2 vanishes at $W = u^2, u'^2, u''^2$; when these squares are pairwise distinct it is zero (*Math Guide*, §"Roots and the factor theorem", Theorem "At most d roots"), so M is invertible. When two squares coincide, two rows of M coincide and M is singular. Define $(\mathbf{X}_L, \mathbf{X}_C, \mathbf{X}_R)$ as the unique solution. Because it solves the system exactly, $\mathbf{Y}_\nu = \nu^2 \mathbf{X}_L + \mathbf{X}_C + \nu^{-2}\mathbf{X}_R$ for each ν, which is (21) read coordinate by coordinate. That the solutions are representations of L, C and R follows by taking the same linear combinations of the group equations: let $(\alpha, \alpha', \alpha'')$ be the middle row of M^{-1}, so that $\alpha(u^2, 1, u^{-2}) + \alpha'(u'^2, 1, u'^{-2}) + \alpha''(u''^2, 1, u''^{-2}) = (0, 1, 0)$ and $\mathbf{X}_C = \alpha \mathbf{Y}_u + \alpha' \mathbf{Y}_{u'} + \alpha'' \mathbf{Y}_{u''}$. Adding α times the first group equation, α' times the second and α'' times the third gives, on the left, $[0]L + [1]C + [0]R = C$, and on the right, by bilinearity, $\langle \mathbf{x}_C, \mathbf{G}\rangle + [y_C]H + [w_C]U$. The first and third rows of M^{-1} do the same for L and R. For well-formedness, set $\delta_\nu := w_\nu - \langle \mathbf{x}_\nu, \mathbf{b}\rangle$ and $\delta_L := w_L - \langle \mathbf{x}_L, \mathbf{b}\rangle$, $\delta_C := w_C - \langle \mathbf{x}_C, \mathbf{b}\rangle$, $\delta_R := w_R - \langle \mathbf{x}_R, \mathbf{b}\rangle$. Subtracting the inner product of the first identity of (21) with $\mathbf{b}$ from the third gives $\delta_\nu = \nu^2 \delta_L + \delta_C + \nu^{-2}\delta_R$ for each ν, that is, $M(\delta_L, \delta_C, \delta_R)^\top = (\delta_u, \delta_{u'}, \delta_{u''})^\top$. If the given representations are well formed the right-hand side is zero, and M invertible forces $\delta_L = \delta_C = \delta_R = 0$; in particular $w_C = \langle \mathbf{x}_C, \mathbf{b}\rangle$. □

Theorem 3.5 (Knowledge soundness of the opening protocol)**.** *Let $\mathbb{G}$ be a group of prime order in which the discrete logarithm problem is hard, and let $\mathbf{G}$ and H be uniform in $\mathbb{G} \setminus \{O\}$ as in Construction 3.2 (for hashed generators: hash-to-curve modelled as a random oracle into $\mathbb{G}$). The opening protocol of Construction 3.2 is knowledge sound for $\mathcal{R}_{\mathrm{IPA}}$: there is an extractor which, given rewindable access to any prover that makes the verifier accept with non-negligible probability, runs in expected polynomial time and outputs either a witness $(\mathbf{a}, r)$ with $P = \langle \mathbf{a}, \mathbf{G}\rangle + [r]H$ and $\langle \mathbf{a}, \mathbf{b}(x)\rangle = v$, or a nontrivial discrete-logarithm relation among $\mathbf{G}$, H and the verifier's generators U, U'. Its knowledge error is $O(\log d/|\mathbb{F}|)$.*

Proof. The proof is given at the level of this volume: the single-round step is Lemma 3.4, the tree is an induction on it, and the rewinding that supplies the sibling transcripts is the *Crypto Guide*'s (§"The Schnorr identification protocol", the remark on extraction by rewinding, and §"Sigma-protocols", the tree-extraction theorem cited there, which also gives the expected running time). The trees it assembles have children with pairwise distinct challenges (the *Crypto Guide*'s Definition "Tree special soundness"), while Lemma 3.4 needs pairwise distinct squares. A square

has at most two square roots, so any five pairwise distinct challenges contain three with pairwise distinct squares: the rewinding assembles five children per node, and the extractor uses three of them. From a tree the extractor computes a witness or, at the root, a discrete-logarithm relation; that definition asks for a witness alone, and the difference is where the assumption enters.

Leaves. Rewind the prover to the moment after it sent (L_1, R_1) and obtain three accepting final messages (a, r^*) for three challenges u_1 with pairwise distinct squares. An accepting final message is, by (15), a representation of $C^{(0)}$ over $(G^{(0)}, H, U)$, namely $(a; r^*; a\, b^{(0)})$, and it is well formed at level 0: its U-coordinate is the inner product of a with the length-one vector $b^{(0)}$. Lift it to level 1 by (20): $\mathbf{x} = (u_1^{-1}a, u_1 a)$.

One node. The three lifted leaves are well-formed representations of the three folded elements $C^{(1)} + [u_1^2]L_1 + [u_1^{-2}]R_1$ over $(\mathbf{G}^{(1)}, H, U)$. Lemma 3.4 at the round-1 node returns a well-formed representation of $C^{(1)}$ (and representations of L_1 and R_1, which are not needed further).

Induction. Rewind to the moment after (L_2, R_2), obtain three challenges u_2 with pairwise distinct squares, and under each of them repeat the two steps above. That yields, after lifting to level 2, three well-formed representations of the elements $C^{(2)} + [u_2^2]L_2 + [u_2^{-2}]R_2$, and the lemma at the round-2 node returns a well-formed representation of $C^{(2)}$. Continuing upward, after k levels the extractor holds a well-formed representation $(\mathbf{x}; y; w)$ of $C^{(k)} = P + [v]U$ over $(\mathbf{G}, H, U)$, with $w = \langle \mathbf{x}, \mathbf{b}(x)\rangle$. The tree of transcripts consumed has three children at every node and depth k, hence $3^k = d^{\log_2 3}$ leaves (Figure 9).

The root, where the assumption enters. The root representation says $P = \langle \mathbf{x}, \mathbf{G}\rangle + [y]H + [w-v]U$. If $w = v$ then $(\mathbf{x}, y)$ is a witness: $P = \langle \mathbf{x}, \mathbf{G}\rangle + [y]H$ and $\langle \mathbf{x}, \mathbf{b}(x)\rangle = w = v$. If $w \neq v$, the extractor rewinds once more, to the setup step at which the verifier sampled U, and repeats the whole extraction under an independent sample U', obtaining $P = \langle \mathbf{x}', \mathbf{G}\rangle + [y']H + [w' - v]U'$. If $w' = v$ the witness is $(\mathbf{x}', y')$. Otherwise subtracting the two expressions for P gives

$$[w - v]\, U - [w' - v]\, U' + \langle \mathbf{x} - \mathbf{x}', \mathbf{G}\rangle + [y - y']H = O$$

with the nonzero coefficient $w - v$ on U: a nontrivial relation among the independent uniform generators $\mathbf{G}$, H, U, U', which the every-generator embedding of the *Crypto Guide* (§"Pedersen vector commitments", the proof of binding) turns into a discrete logarithm, the reduction embedding its challenge in the public generators and in the verifier's samples alike. This is the only place the assumption is used.

The error. The figure does not come from a union bound over the tree. At each node the extractor needs three siblings whose challenges avoid $O(1)$ values: $\pm\nu$ for each earlier sibling's challenge ν, so that the squares stay pairwise distinct. Each of the $k = \log_2 d$ rounds therefore contributes $O(1/|\mathbb{F}|)$, and the rounds together with the resampling of U give $O(\log d/|\mathbb{F}|)$, the theorem's figure. Collisions among the challenges resampled inside the 3^k-leaf tree cost the extractor a rewind and a fresh sample; they affect its expected running time, not the knowledge error. A naive union bound over the roughly 3^k sibling draws would bound the tree-wide collision probability by $O(d^{\log_2 3}/|\mathbb{F}|)$, still negligible, but that is not the source of the figure. The weaker $O(d/|\mathbb{F}|)$ sometimes quoted for this argument is implied by the proved bound; a larger protocol whose own error is already linear in d loses nothing by quoting it. □

What the theorem does not say. The theorem concerns the interactive protocol; the compiled argument is treated in §5. In the algebraic group model the extractor does not rewind for the representations at all: an algebraic prover hands them over with every group element it sends, and the tree collapses to a single transcript (§5, "The algebraic group model"). After the Fiat–Shamir transform has replaced the verifier's challenges by hashes, the rewinding above must be re-proved against a prover who can query the hash adaptively and resume from any earlier transcript state (for one round, the *Crypto Guide*'s forking lemma, §"Security in the random oracle model and the forking lemma"; the multi-round case in §5), and its cost cannot be obtained by multiplying every node of the tree by an independent $1/\varepsilon$ factor. Nor does the theorem restate

the binding of the commitment itself: that a single P cannot be opened as two vectors is the *Crypto Guide*'s theorem, cited in §2.9 and not reproved here.

The assumption, and what breaks it. The extractor's alternative output is a discrete-logarithm relation among $\mathbf{G}$, H and the verifier's U, U', so the argument is binding as long as discrete logarithms are hard in $\mathbb{G}$ and the generators are uniform; hashed generators are uniform when hash-to-curve is modelled as a random oracle into $\mathbb{G}$ (*Crypto Guide*, §"Hashing to a curve point", Remark "Why NUMS generators are binding-safe"), a hypothesis of Theorem 3.5. For the deployed instance the group is the Vesta curve (§3.8), and the assumption that carries the argument is the discrete logarithm on Vesta, the first of the *Crypto Guide*'s standing assumptions (§"Standing assumptions"), against which the generic attacks cost about 2^{126} operations (§"Instantiation on elliptic curves; the Pasta curves"); a quantum adversary running Shor's algorithm lies outside those assumptions, which fix a classical adversary. Suppose the assumption fails: an algorithm computes discrete logarithms on Vesta. Then it computes the logarithms $\eta_0, \dots, \eta_{d-1}, \eta_H$ of the hashed generators to one base, and every commitment $P = [\pi]$ base opens to every vector: for any $\tilde{\mathbf{a}}$ whatsoever, the blinder $\tilde{r} := (\pi - \langle \tilde{\mathbf{a}}, \boldsymbol{\eta} \rangle)/\eta_H$ satisfies $\langle \tilde{\mathbf{a}}, \mathbf{G} \rangle + [\tilde{r}]H = P$, and the honest protocol run on $(\tilde{\mathbf{a}}, \tilde{r})$ convinces the verifier of $\langle \tilde{\mathbf{a}}, \mathbf{b}(x) \rangle$, a value the cheat chose. One commitment then opens to two vectors and to two values at one point; binding and evaluation binding fall together, and with them the premise of Theorem 2.5 that the polynomials were fixed before z was drawn.

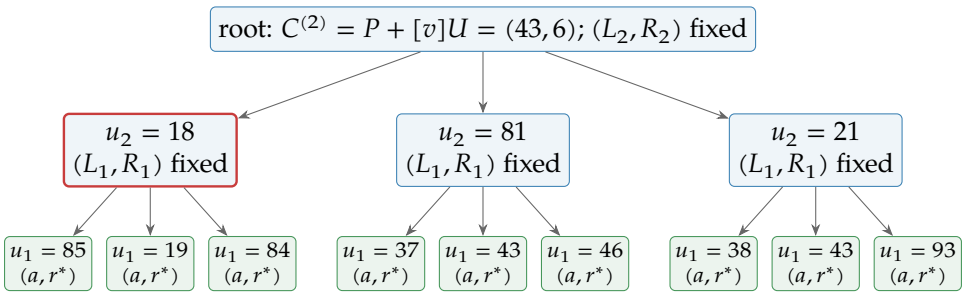
Example 3.6 (The extractor run on the toy)**.** The extractor is run against an honest prover for the running example's vector $\mathbf{a} = (90, 61, 22, 24)$ opened at $z = 20$ with $v = 59$, on the prime-order-97 curve and generators of §3.7, with the blinder $r = 88$, so $P = (51, 31)$ and $C^{(2)} = P + [59]U = (43, 6)$. It fixes (L_2, R_2), branches over three round-2 challenges $u_2 \in \{18, 81, 21\}$, and under each fixes (L_1, R_1) and branches over three round-1 challenges: $\{85, 19, 84\}$, $\{37, 43, 46\}$ and $\{38, 43, 93\}$, nine leaves in all. At the node $u_2 = 18$ the 3×3 system has determinant 67 and returns the representation $\mathbf{a}^{(1)} = (80, 0)$ with blinder 3 and U-coordinate $40 = \langle \mathbf{a}^{(1)}, \mathbf{b}^{(1)} \rangle$, well formed as the lemma promises; the other two nodes return $(50, 41)$ and $(85, 35)$ likewise. At the root the system in u_2 has determinant 75 and returns $\mathbf{a} = (90, 61, 22, 24)$, $r = 88$ and U-coordinate $59 = v$: the extracted witness is the committed vector, its blinder, and the true value. Figure 9 draws the tree with the node $u_2 = 18$ called out.

3.6 Zero knowledge: what the blinders hide and what they do not

The transcript of one opening consists of group elements, challenges and two final scalars. The group elements are hidden by the blinders: the commitment blinder r makes P uniform in $\mathbb{G}$ whatever $\mathbf{a}$ is (perfect hiding, *Crypto Guide*, §"The Pedersen commitment" and §"Pedersen vector commitments"), and the round blinders ℓ_j, ρ_j make every L_j and R_j uniform and independent of everything else for the same reason, since each carries its own fresh $[\ell_j]H$ or $[\rho_j]H$. The aggregated blinder r^* is uniform too, being r plus a combination of the fresh ℓ_j and ρ_j. The final scalar is not hidden by these blinders.

Remark 3.7 (Folding alone is not zero knowledge). The final scalar $a = \mathbf{a}^{(0)}$ is a linear form in the committed coefficients whose coefficients the verifier knows. Follow one coefficient a_i through the k folds exactly as the proof of Theorem 3.3 followed G_m: the coefficient side folds with the weights (u_j, u_j^{-1}), the *inverse* of the generator side's (u_j^{-1}, u_j), so a_i reaches the single final entry multiplied by $\prod_\ell u_\ell^{(-1)^{\beta_{\ell-1}}}$, the factor u_ℓ when the digit $\beta_{\ell-1}$ of i is 0 and u_ℓ^{-1} when it is 1. Writing $\mathbf{t}$ for this vector, $t_i = s_i^{-1}$ entry by entry, and

$$\mathbf{a}^{(0)} = \langle \mathbf{a}, \mathbf{t} \rangle, \qquad t_i := \prod_{\ell=1}^{k} u_\ell^{(-1)^{\beta_{\ell-1}}} = s_i^{-1}; \qquad \text{at } d = 4: \ \mathbf{t} = (u_1 u_2,\ u_1^{-1} u_2,\ u_1 u_2^{-1},\ u_1^{-1} u_2^{-1}). \tag{22}$$



at the node $u_2 = 18$: three leaves lifted by (20) to $\mathbf{x} = (u_1^{-1}a, u_1 a)$; rows $(u_1^2, 1, u_1^{-2})$ for $u_1 \in \{85, 19, 84\}$, determinant 67; solution: $\mathbf{a}^{(1)} = (80, 0)$, blinder 3, U-coordinate $40 = \langle \mathbf{a}^{(1)}, \mathbf{b}^{(1)} \rangle$.
at the root: rows $(u_2^2, 1, u_2^{-2})$ for $u_2 \in \{18, 81, 21\}$, determinant 75; solution: $\mathbf{a} = (90, 61, 22, 24)$, $r = 88$, U-coordinate $59 = v$.

Figure 9: The extractor's tree of accepting transcripts for $d = 4$ (Example 3.6): every node branches over three challenges, every root-to-leaf path is one accepting run of Construction 3.2, and each node's 3×3 system of Lemma 3.4 turns its three children's representations into its own. The red node's system and the root's are written out beneath.

In the toy of §3.7, with $u_2 = 52$ and $u_1 = 20$, the vector is $\mathbf{t} = (70, 22, 75, 79)$, the entrywise inverse of the structured scalars $\mathbf{s} = (79, 75, 22, 70)$, and the final scalar 33 is $\langle (90, 61, 22, 24), (70, 22, 75, 79) \rangle$. One opening thus hands the verifier one linear equation in the four secret coefficients; every further opening of the same commitment under fresh challenges hands him another, and d of them determine $\mathbf{a}$ outright. Four openings of the same vector, with challenge pairs $(u_2, u_1) = (63, 29)$, $(79, 7)$, $(34, 1)$ and $(49, 74)$, end in the scalars 88, 22, 40 and 14, and solving the 4×4 system recovers $(90, 61, 22, 24)$. Blinding the commitment and the round points hides those points, not this scalar. The masked opening below therefore does not fold $\mathbf{a}$ itself: it first masks the polynomial by an independently random polynomial vanishing at the opening point, so that the linear form is taken of a vector the verifier cannot relate to $\mathbf{a}$. The mask's commitment S and the challenge that follows it are essential to the zero-knowledge argument below.

The masked opening. The masked opening runs Construction 3.2 on a masked statement.

1. **Mask.** Before any folding, the prover samples a polynomial $s(X) = \sum_{i=0}^{d-1} m_i X^i$ of degree below d with $s(x) = 0$: she draws the coefficient vector $\mathbf{m}$ uniformly from $\mathbb{F}^d$ and then replaces m_0 by $m_0 - s(x)$, which makes $\mathbf{m}$ uniform on the hyperplane $\{\mathbf{m} : \langle \mathbf{m}, \mathbf{b}(x) \rangle = 0\}$. She samples a blinder r_s and sends
$$S := \mathsf{Commit}(s; r_s) = \langle \mathbf{m}, \mathbf{G} \rangle + [r_s]H.$$
2. **Challenge.** The verifier sends a uniform $\xi \in \mathbb{F}$.
3. **Opening.** By linearity of the commitment (*Crypto Guide*, §"Pedersen vector commitments", the remark that linearity survives),
$$P + [\xi]S = \mathsf{Commit}(p + \xi s; r + \xi r_s), \tag{23}$$
a commitment to the coefficient vector $\mathbf{a} + \xi\,\mathbf{m}$ of the polynomial $p + \xi s$, whose value at x is $p(x) + \xi\, s(x) = v$. The parties run Construction 3.2 on the statement $(P + [\xi]S, x, v)$ with the witness $(\mathbf{a} + \xi\,\mathbf{m}, r + \xi r_s)$, the verifier sampling U in its setup step, after S and ξ.

4. **Final message and check.** The prover sends the final scalar $c := (\mathbf{a} + \xi\,\mathbf{m})^{(0)}$ and the aggregated blinder $f := r + \xi r_s + \sum_{j=1}^{k}(u_j^2\ell_j + u_j^{-2}\rho_j)$. The verifier accepts if and only if

$$P + [\xi]S + [v]U + \sum_{j=1}^{k}([u_j^2]L_j + [u_j^{-2}]R_j) \stackrel{?}{=} [c]\,G^{(0)} + [f]H + [c \cdot b^{(0)}]\,U, \tag{24}$$

which is (15) for the masked statement.

Completeness is that of Construction 3.2, since the witness satisfies $\mathcal{R}_{\mathrm{IPA}}$ for the masked statement; the mask leaves the opened value unchanged because $s(x) = 0$. The transcript of one opening is

$$(S;\ L_k, R_k, \dots, L_1, R_1;\ c, f) \qquad \text{with the challenges} \qquad (\xi, U, u_k, \dots, u_1).$$

The hiding argument. The final scalar is the linear form (22), taken of the masked vector:

$$c = \langle \mathbf{a} + \xi\,\mathbf{m},\ \mathbf{t}\rangle = \langle \mathbf{a},\ \mathbf{t}\rangle + \xi\,\langle \mathbf{m},\ \mathbf{t}\rangle. \tag{25}$$

In the toy, with $\mathbf{m} = (66, 23, 23, 88)$, for which $s(20) = 0$, and $\xi = 29$, the masked vector is $(64, 49, 10, 54)$, its inner product with $\mathbf{b}(20)$ is 59, and under the challenges $u_2 = 52$, $u_1 = 20$ of the unmasked run the final scalar is $c = 1 = \langle (64, 49, 10, 54),\ \mathbf{t}\rangle$ with $\mathbf{t} = (70, 22, 75, 79)$; a second mask $\mathbf{m}' = (32, 68, 64, 88)$ under the same challenges ends in $c' = 75$, and both runs are accepted (§3.7). The same form, on a different masked vector, gives an unrelated scalar.

> **Proposition 3.8** (Honest-verifier zero knowledge of the masked opening)**.** *Let $\mathbb{G}$ have prime order p, and let the verifier draw ξ uniformly from $\mathbb{F}$, U uniformly from $\mathbb{G} \setminus \{O\}$ and each u_j uniformly from $\mathbb{F}^\times$. There is a simulator which, given (P, x, v) but not $(\mathbf{a}, r)$, outputs challenges and a transcript whose distribution lies within statistical distance $1/|\mathbb{F}| + (2/(|\mathbb{F}| - 1))^k$* (Math Guide, *§"Statistical distance") of the honest one. The masked opening is therefore statistical honest-verifier zero knowledge, not perfect, in the sense of the* Crypto Guide (*§"The simulation paradigm and zero knowledge"*).

Proof. Call a challenge tuple *good* if $\xi \neq 0$ and $\mathbf{t}$ is not a multiple of $\mathbf{b}(x)$. The vector $\mathbf{m}$ is uniform on the hyperplane $\langle \mathbf{m},\ \mathbf{b}(x)\rangle = 0$, a subspace of dimension $d-1$. The linear functional $\mathbf{m} \mapsto \langle \mathbf{m},\ \mathbf{t}\rangle$ (*Math Guide*, §"Bilinear forms and the inner product on F^n") restricted to that subspace is nonzero unless $\mathbf{t}$ is a scalar multiple of $\mathbf{b}(x)$. If $\mathbf{t} = \lambda\,\mathbf{b}(x)$, the first entries force $\lambda = t_0 = \prod_\ell u_\ell$, and the entry at $i = 2^{\ell-1}$, whose only nonzero digit is $\beta_{\ell-1}$, reads $t_0\,u_\ell^{-2} = t_0\,x^{2^{\ell-1}}$, that is, $u_\ell^{-2} = x^{2^{\ell-1}}$; each of these k equations has at most two solutions u_ℓ, so at most 2^k challenge tuples among $(|\mathbb{F}| - 1)^k$ are excluded. A nonzero linear functional on a vector space is onto $\mathbb{F}$, and its fibres are the cosets of its kernel, all of equal size, so it carries the uniform distribution to the uniform distribution on $\mathbb{F}$. For good challenges $\xi\langle \mathbf{m},\ \mathbf{t}\rangle$ is therefore uniform, and by (25) so is c, independently of $\mathbf{a}$.

In an honest run with good challenges, S and every L_j, R_j are uniform and independent (fresh blinders), c is uniform and independent of them (the mask), and f is the unique scalar that makes (24) hold once everything else is fixed, H generating $\mathbb{G}$. The simulator produces the same distribution in the reverse order, the way the Schnorr simulator chooses its commitment after the challenge (*Crypto Guide*, §"Sigma-protocols"): it draws the challenges as the verifier does, draws S, all L_j and all R_j but R_1 uniformly from $\mathbb{G}$, draws c and f uniformly from $\mathbb{F}$, and sets R_1 to the one point that satisfies (24), namely $R_1 := [u_1^2](\text{right-hand side} - \text{every other term of the left})$. Both distributions are uniform on the solution set of the one equation: the honest run parametrises that set by the $2k+1$ points and c, with f determined; the simulator parametrises it by $2k$ points and the two scalars, with R_1 determined; and each parametrisation is a bijection onto that set, with p^{2k+2} points on each side since $\mathbb{G}$ and $\mathbb{F}$ both have p elements, because the determined coordinate is unique in each case. For good challenges the simulated transcript is therefore distributed exactly as the real one. The challenges have the same distribution in both, and the tuples that are not good have probability at most $1/|\mathbb{F}| + 2^k/(|\mathbb{F}| - 1)^k$; the statistical distance is at most that probability. □

The gap, stated. This argues one opening in isolation. The proof system opens many committed polynomials at once, under shared challenges, and the mask is one of several blinding devices it uses; whether the joint transcript reveals nothing is a separate question, addressed under "Zero knowledge: hiding the witness in Halo 2" in §4, where the mask, in the deployed form of §3.8, reappears as the last of those devices.

3.7 A complete toy opening, carried by script

The toy opening commits the running example's advice polynomial, $a(X) = 90 + 61X + 22X^2 + 24X^3$ of §2.2, and opens it at the random point $z = 20$ of §2.4 with $v = a(20) = 59$: the opening the proof system needs, of a column polynomial at the verifier's random point, never of the statement polynomial at the running example's secret witness 3. Every value below is computed by script and can be recomputed by hand in $\mathbb{F}_{97}$ and on a curve with 97 points.

A genuine group of order 97. The vector $\mathbf{a}$ lives in $\mathbb{F}_{97}^4$, so the commitment group must have order 97: scalars of a prime-order group live in the field of its order (*Math Guide*, §"Base fields, scalar fields, and the Pasta cycle"), exactly as the deployed circuit's field is the scalar field of the deployed commitment curve. The group $(\mathbb{F}_{97}, +)$ has order 97 but trivially computable discrete logarithms, a division, so the toy uses an elliptic curve. For a curve over a prime field $\mathbb{F}_q$ to have exactly 97 points, the Hasse bound (*Math Guide*, §"The group structure of $E(\mathbb{F}_p)$") confines the prime q to $|q + 1 - 97| \leq 2\sqrt{q}$: $q \in \{79, 83, 89, 97, 101, 103, 107, 109, 113\}$. The case $q = 97$ is excluded on purpose: a curve with as many points as its base field has elements is *anomalous*, and its discrete logarithm is easy (*Math Guide*, §"The elliptic-curve discrete logarithm problem"). The first short Weierstrass curve (*Math Guide*, §"Weierstrass equations") in lexicographic order of (q, A, B) with exactly 97 points is

$$E\colon y^2 = x^3 + 3 \quad \text{over } \mathbb{F}_{79}, \qquad \#E(\mathbb{F}_{79}) = 97,$$

counted point by point; 97 is prime, so the group is cyclic and every point but $\mathcal{O}$ generates it, and its trace $79 + 1 - 97 = -17$ is not 1. It is the same kind of object as the *Math Guide*'s worked curve $y^2 = x^3 + 5$ over $\mathbb{F}_{11}$ (§"A worked toy example"), one size up. With 97 points every discrete logarithm on it is a short search; the toy exhibits the algebra of the argument, never its hardness.

Generators from a hash. The generators are derived from public labels by hashing to the curve, as the deployed ones are: the label and a counter are hashed to a candidate x-coordinate, and the counter is incremented until $x^3 + 3$ is a square in $\mathbb{F}_{79}$. This is a variant of the try-and-increment loop of the *Crypto Guide* (§"Hashing to a curve point"), which increments x itself; its constant-time refinement is irrelevant to a toy. Indexed from 0 so that the coefficient a_i multiplies G_i, the generators are

$$G_0 = (12, 25), \quad G_1 = (17, 27), \quad G_2 = (71, 26), \quad G_3 = (49, 10), \quad H = (78, 9), \quad U = (53, 26).$$

The hashed point U stands in for the verifier's uniform sample of Construction 3.2. No party chose any of them, so no relation among them is known in advance; a single discrete logarithm between two of them would give one, since $H = [h]G_0$ yields $[h]G_0 - H = \mathcal{O}$. On this curve such a logarithm is a short search; on Vesta its infeasibility is the assumption of §3.5.

The instance. The prover holds $\mathbf{a} = (90, 61, 22, 24)$, which takes the values $(3, 9, 27, 30)$ on the domain of §2.2, and the blinder $r = 10$; her commitment is

$$P = [90]G_0 + [61]G_1 + [22]G_2 + [24]G_3 + [10]H = (39, 25).$$

The verifier holds P, the point $z = 20$, hence $\mathbf{b}(20) = (1, 20, 20^2, 20^3) = (1, 20, 12, 46)$ in $\mathbb{F}_{97}$, and the claimed value $v = 59 = \langle \mathbf{a}, \mathbf{b}(20) \rangle$. With $d = 4$ there are $k = 2$ rounds, numbered 2 then 1.

The symmetric opening, both sides. Tables 4 and 5 refill the rows of Tables 2 and 3 with numbers, listing each party's inputs and computations; the verifier's table never uses $\mathbf{a}$.

step	receives / holds	computes and sends
setup	holds $\mathbf{a} = (90, 61, 22, 24)$, $r = 10$, $z = 20$, $v = 59$; receives $U = (53, 26)$	$\mathbf{a}^{(2)} := \mathbf{a}$, $\mathbf{G}^{(2)} := (G_0, G_1, G_2, G_3)$, $\mathbf{b}^{(2)} := (1, 20, 12, 46)$, $r^{(2)} := 10$
round 2	halves $\mathbf{a}_{\text{lo}} = (90, 61)$, $\mathbf{a}_{\text{hi}} = (22, 24)$; $\mathbf{b}_{\text{lo}} = (1, 20)$, $\mathbf{b}_{\text{hi}} = (12, 46)$	samples $\ell_2 = 44$, $\rho_2 = 16$; $\langle \mathbf{a}_{\text{lo}}, \mathbf{b}_{\text{hi}} \rangle = 6$, $\langle \mathbf{a}_{\text{hi}}, \mathbf{b}_{\text{lo}} \rangle = 17$; sends $L_2 = [90]G_2 + [61]G_3 + [44]H + [6]U = (74, 6)$, $R_2 = [22]G_0 + [24]G_1 + [16]H + [17]U = (38, 72)$
	receives $u_2 = 52$ ($u_2^{-1} = 28$)	$\mathbf{a}^{(1)} = 52\,(90, 61) + 28\,(22, 24) = (58, 61)$; $\mathbf{b}^{(1)} = 28\,(1, 20) + 52\,(12, 46) = (70, 42)$; $\mathbf{G}^{(1)} = ([28]G_0 + [52]G_2,\ [28]G_1 + [52]G_3) = ((54, 40), (30, 71))$; $r^{(1)} = 10 + 52^2 \cdot 44 + 28^2 \cdot 16 = 95$
round 1	halves $a_{\text{lo}} = 58$, $a_{\text{hi}} = 61$; $b_{\text{lo}} = 70$, $b_{\text{hi}} = 42$	samples $\ell_1 = 12$, $\rho_1 = 37$; $a_{\text{lo}} b_{\text{hi}} = 11$, $a_{\text{hi}} b_{\text{lo}} = 2$; sends $L_1 = [58]G_1^{(1)} + [12]H + [11]U = (28, 25)$, $R_1 = [61]G_0^{(1)} + [37]H + [2]U = (39, 25)$
	receives $u_1 = 20$ ($u_1^{-1} = 34$)	$a^{(0)} = 20 \cdot 58 + 34 \cdot 61 = 33$; $b^{(0)} = 34 \cdot 70 + 20 \cdot 42 = 19$; $G^{(0)} = [34]G_0^{(1)} + [20]G_1^{(1)} = (12, 25)$; $r^{(0)} = 95 + 20^2 \cdot 12 + 34^2 \cdot 37 = 40$
final	$a = 33$, $r^* = 40$	sends $a = 33$, $r^* = 40$

Table 4: The prover's side of the toy opening (Table 2 with numbers): $\mathbf{a} = (90, 61, 22, 24)$ opened at $z = 20$ on the 97-point curve $y^2 = x^3 + 3$ over $\mathbb{F}_{79}$, all scalars in $\mathbb{F}_{97}$. Points are given as (x, y) with coordinates in $\mathbb{F}_{79}$.

step	receives / holds	computes and sends
setup	holds $P = (39, 25)$, $z = 20$, $v = 59$, $\mathbf{G}$, $H = (78, 9)$	sends $U = (53, 26)$, a hashed point standing in for his sample; $C^{(2)} := P + [59]U = (47, 39)$
round 2	receives $L_2 = (74, 6)$, $R_2 = (38, 72)$	samples $u_2 = 52$ and sends it; keeps $(L_2, R_2, 52)$; $C^{(1)} := C^{(2)} + [52^2]L_2 + [28^2]R_2 = (72, 23)$
round 1	receives $L_1 = (28, 25)$, $R_1 = (39, 25)$	samples $u_1 = 20$ and sends it; keeps $(L_1, R_1, 20)$; $C^{(0)} := C^{(1)} + [20^2]L_1 + [34^2]R_1 = (54, 39)$
final	receives $a = 33$, $r^* = 40$	$\mathbf{s} = (34 \cdot 28,\ 20 \cdot 28,\ 34 \cdot 52,\ 20 \cdot 52) = (79, 75, 22, 70)$; $G^{(0)} = [79]G_0 + [75]G_1 + [22]G_2 + [70]G_3 = (12, 25)$; $b^{(0)} = g(20; 20, 52) = (34 + 20 \cdot 20)(28 + 52 \cdot 12) = 19$; right-hand side $[33]G^{(0)} + [40]H + [33 \cdot 19]U = (54, 39) = C^{(0)}$: **accept**

Table 5: The verifier's side of the same opening (Table 3 with numbers). He never sees $\mathbf{a}$, r, ℓ_j or ρ_j; the structured scalars and $g(20)$ come from the two challenges alone, and $G^{(0)}$ is his one length-4 multi-scalar multiplication.

The verifier's equation (15) closes: the left-hand side $P + [59]U + [52^2]L_2 + [28^2]R_2 + [20^2]L_1 + [34^2]R_1$ is $(54, 39)$, and so is the right-hand side $[33]G^{(0)} + [40]H + [33 \cdot 19]U$. Each fold identity holds along the way: after round 2, $(72, 23) = \langle \mathbf{a}^{(1)}, \mathbf{G}^{(1)} \rangle + [95]H + [\langle \mathbf{a}^{(1)}, \mathbf{b}^{(1)} \rangle]U$, and after round 1, $(54, 39) = [33]\,(12, 25) + [40]H + [33 \cdot 19]U$. The closed forms of Theorem 3.3 agree with the folds: $\langle \mathbf{s}, \mathbf{G} \rangle = (12, 25)$ is the $G^{(0)}$ the prover reached by folding, and $g(20; u_1, u_2) = (u_1^{-1} + u_1 z)(u_2^{-1} + u_2 z^2) = 19$ is her $b^{(0)}$. In a group of 97 elements collisions among the printed points are frequent, and nothing follows from any of them. Here $G^{(0)}$ happens to equal G_0 and R_1 happens to equal P; the masked run below and the runs of §3.8 show more.

A false value is rejected. Let the prover claim $v' = 60$ instead of 59. The prover's messages are those of the honest run; only $C^{(2)} = P + [60]U$ changes, by one more U. The left-hand side becomes $(62, 15)$ while the right-hand side stays $(54, 39)$, and the verifier rejects. By Theorem 3.5, a prover accepted on $v' = 60$ would yield an opening of P with value 60 and hence, beside the honest opening, a discrete-logarithm relation among the generators; on a curve of 97 points such relations are a short search, and the theorem gives no security there.

The masked opening of the same polynomial. The same P is then opened at the same z with the masked opening of §3.6. The prover samples $\mathbf{m} = (66, 23, 23, 88)$, for which $s(20) = 0$, and $r_s = 95$, and sends $S = \langle \mathbf{m}, \mathbf{G} \rangle + [95]H = (70, 71)$; the verifier answers $\xi = 29$. The masked vector $\mathbf{a} + 29\,\mathbf{m} = (64, 49, 10, 54)$ has inner product 59 with $\mathbf{b}(20)$, and the rounds run on it with the point U, the challenges $u_2 = 52$, $u_1 = 20$ and the round blinders of the unmasked run: the prover sends $L_2 = (36, 7)$, $R_2 = (12, 25)$, $L_1 = (51, 31)$ and $R_1 = (72, 23)$, and ends in $c = 1$ and $f = 79$. The final generator $G^{(0)} = (12, 25)$ and the final evaluation scalar $b^{(0)} = 19$ are those of the unmasked run, since they depend on the challenges alone. The verifier's equation (24) closes at $(46, 13)$ on both sides, and the false value $v' = 60$ moves the left-hand side to $(56, 70)$: rejected. A second mask $\mathbf{m}' = (32, 68, 64, 88)$ under the same challenges ends in $c' = 75$ and is accepted as well.

3.8 Halo 2's specifics: parameters, generators from a hash, the deferred multi-scalar multiplication, and the proof bytes

Everything above holds for any prime-order group and any $d = 2^k$. This subsection, and only this one, fixes the group, the size, the generators and the normalisation of the deployed Halo 2, states its opening, and counts the bytes an Orchard proof spends on the argument. Nothing earlier in the section depends on it.

The group and the size. The Orchard Action circuit does its arithmetic in $\mathbb{F}_{p_{\mathsf{Pallas}}}$, the base field of the Pallas curve, because the statement manipulates Pallas points and their coordinates live there (§1.2; *Math Guide*, §"Base fields, scalar fields, and the Pasta cycle"). Its column polynomials therefore have coefficients in $\mathbb{F}_{p_{\mathsf{Pallas}}}$, and a Pedersen commitment to such a vector needs a group whose scalar field is $\mathbb{F}_{p_{\mathsf{Pallas}}}$: that group is the Vesta curve, whose order is p_{Pallas} (protocol specification, § 5.4.9.6, "Pallas and Vesta"). The specification uses Halo 2 with the Vesta curve to prove and verify the Action statement (§ 4.1.13, "Zero-Knowledge Proving System"), so every P, L_j, R_j, S and $G^{(0)}$ of this section is a Vesta point and every scalar a Pallas base-field element. The other half of the Pasta cycle, Pallas's scalar field being Vesta's base field, is needed only by the optional native recursion of §7; Orchard's verifier never uses it. The committed polynomials have degree below the row count $n = 2^{11} = 2048$ of the Action circuit (Definition 2.7; the count is given under "From statement to circuit: arithmetisation in practice" in §4), so here $d = n = 2^{11}$ and the argument runs $k = 11$ rounds.

Generators from a hash. The 2048 vector generators, the blinding generator H and the inner-product generator U are all outputs of hash-to-curve on one fixed domain-separation label, the vector generators indexed by their position and H and U by two further tags (*Crypto Guide*, §"Hashing to a curve point", Construction "Nothing-up-my-sleeve generators"). With hash-to-curve modelled as a random oracle into $\mathbb{G}$ they are independent and uniform, and a relation among them yields a discrete logarithm on Vesta (the same section, Remark "Why NUMS generators are binding-safe"). No secret is generated in producing them, unlike the trapdoor of KZG (*Crypto Guide*, §"Polynomial commitment schemes", under "KZG: evaluation in the exponent"), the point the closing subsubsection returns to. Because U is a fixed parameter rather than a verifier's sample drawn after each P, the deployed opening re-randomises it per opening with a challenge ζ ("The deployed opening in full" below).

Challenges from the transcript. The interactive verifier of Construction 3.2 draws u_j uniformly from $\mathbb{F}^\times$, and the halo2 book's *Protocol Description* excludes zero challenges. The deployed verifier is a hash ("Removing the verifier: the Fiat–Shamir transform" in §4), and the transcript of the `halo2_proofs` implementation squeezes every challenge, ξ, ζ and each u_j, as a full field element with no rejection of zero. A zero u_j would make the fold's inversion fail, so the honest prover succeeds unless one of the k challenges is zero, an event of probability $1/|\mathbb{F}|$ per challenge, about 2^{-254}: completeness of the deployed argument is overwhelming, not literally perfect, in the sense fixed in §1.2.

Remark 3.9 (The deployed IPA normalisation). The deployed opening, stated in full below, is the masked opening of §3.6 with three changes. It folds with the weights $(1, u_j^{-1})$ on the coefficient vector and $(1, u_j)$ on $\mathbf{b}$ and $\mathbf{G}$, with cross terms $L_j = \langle \mathbf{a}_{\mathrm{hi}}, \mathbf{G}_{\mathrm{lo}} \rangle + \cdots$ and $R_j = \langle \mathbf{a}_{\mathrm{lo}}, \mathbf{G}_{\mathrm{hi}} \rangle + \cdots$, the halves swapped relative to (13), and the verifier's accumulation $[u_j^{-1}]L_j + [u_j]R_j$; it scales the cross terms' U-coordinates by a further challenge ζ drawn after ξ; and it moves the claimed value into the commitment as $-[v]G_0$ in place of $+[v]U$. Its final scalars are c and f, and its check is (29). The halo2 book, which the protocol specification (§ 5.4.10.3, "Halo 2") cites for the proving system, states this opening in its chapter *Protocol Description*. Its blinding generator W is this volume's H, its U is U, and its second challenge z is ζ, renamed because $z = 20$ is the running example's opening point; the book's comparison with the scheme of Bünz, Chiesa, Mishra and Spooner (chapter *Comparison to other work*) writes H, U and z for the same three objects. The chapter numbers the round challenges from the first round, $u_0, \dots, u_{k-1}$, where this volume's descending index gives $u_k, \dots, u_1$, and states the verifier's evaluation polynomial (its κ) as

$$\begin{aligned} g_{\mathrm{dep}}(X) &= \prod_{i=0}^{k-1} (1 + u_{k-1-i}\, X^{2^i}) \quad \text{in its numbering,} \\ g_{\mathrm{dep}}(X) &= \prod_{\ell=1}^{k} (1 + u_\ell\, X^{2^{\ell-1}}) \qquad \text{in this volume's,} \end{aligned} \tag{26}$$

with structured scalars $s_i = \prod_{\ell:\beta_{\ell-1}=1} u_\ell$, beside the symmetric $g(X) = \prod_{\ell=1}^{k} (u_\ell^{-1} + u_\ell X^{2^{\ell-1}})$ of (19) with $s_i = \prod_\ell u_\ell^{(-1)^{1-\beta_{\ell-1}}}$.

The change of variables. The two folds are one argument. Run both on the same vector, the same generators and the same evaluation vector, with the deployed challenge of every round the square of the symmetric one, $u_{\mathrm{dep}} := u_{\mathrm{sym}}^2$. In one round, writing u for u_{sym},

$$\begin{aligned} \mathbf{a}'_{\mathrm{sym}} &= u\, \mathbf{a}_{\mathrm{lo}} + u^{-1} \mathbf{a}_{\mathrm{hi}} = u\, (\mathbf{a}_{\mathrm{lo}} + u^{-2} \mathbf{a}_{\mathrm{hi}}) = u\, \mathbf{a}'_{\mathrm{dep}}, \\ \mathbf{G}'_{\mathrm{sym}} &= u^{-1} \mathbf{G}_{\mathrm{lo}} + u\, \mathbf{G}_{\mathrm{hi}} = u^{-1} (\mathbf{G}_{\mathrm{lo}} + u^2 \mathbf{G}_{\mathrm{hi}}) = u^{-1} \mathbf{G}'_{\mathrm{dep}}, \\ \mathbf{b}'_{\mathrm{sym}} &= u^{-1} \mathbf{b}_{\mathrm{lo}} + u\, \mathbf{b}_{\mathrm{hi}} = u^{-1} (\mathbf{b}_{\mathrm{lo}} + u^2 \mathbf{b}_{\mathrm{hi}}) = u^{-1} \mathbf{b}'_{\mathrm{dep}}, \end{aligned}$$

so $\mathbf{a}'_{\mathrm{dep}} = u^{-1} \mathbf{a}'_{\mathrm{sym}}$, $\mathbf{G}'_{\mathrm{dep}} = u\, \mathbf{G}'_{\mathrm{sym}}$ and $\mathbf{b}'_{\mathrm{dep}} = u\, \mathbf{b}'_{\mathrm{sym}}$, and the scalings cancel in every inner product: $\langle \mathbf{a}'_{\mathrm{dep}}, \mathbf{G}'_{\mathrm{dep}} \rangle = \langle \mathbf{a}'_{\mathrm{sym}}, \mathbf{G}'_{\mathrm{sym}} \rangle$ and $\langle \mathbf{a}'_{\mathrm{dep}}, \mathbf{b}'_{\mathrm{dep}} \rangle = \langle \mathbf{a}'_{\mathrm{sym}}, \mathbf{b}'_{\mathrm{sym}} \rangle$. The cross terms agree, up to their H-blinders, once the naming is unwound: the deployed L carries $\langle \mathbf{a}_{\mathrm{hi}}, \mathbf{G}_{\mathrm{lo}} \rangle$, which the symmetric presentation calls R, and its weight $u_{\mathrm{dep}}^{-1} = u^{-2}$ is the symmetric weight on R; likewise for the other cross term. The single-round form holds for one round on identical inputs; over several rounds the change of variables is *cumulative*, because each round's inputs already carry the previous rounds' factors:

$$\mathbf{a}_{\mathrm{dep}}^{(j)} = \Big(\prod_{\ell>j} u_\ell\Big)^{-1} \mathbf{a}_{\mathrm{sym}}^{(j)}, \qquad \mathbf{G}_{\mathrm{dep}}^{(j)} = \Big(\prod_{\ell>j} u_\ell\Big) \mathbf{G}_{\mathrm{sym}}^{(j)}, \qquad \mathbf{b}_{\mathrm{dep}}^{(j)} = \Big(\prod_{\ell>j} u_\ell\Big) \mathbf{b}_{\mathrm{sym}}^{(j)}, \tag{27}$$

with u_ℓ the symmetric challenges of the rounds already run. Each deployed factor of (26) is a

symmetric factor scaled, $1 + u^2 X^m = u\,(u^{-1} + u\,X^m)$, so after k rounds

$$g_{\text{dep}} = \Big(\prod_{\ell=1}^{k} u_\ell\Big)\, g_{\text{sym}}, \qquad \mathbf{s}_{\text{dep}} = \Big(\prod_{\ell=1}^{k} u_\ell\Big)\, \mathbf{s}_{\text{sym}}, \qquad a^{(0)}_{\text{dep}} = \Big(\prod_{\ell=1}^{k} u_\ell\Big)^{-1} a^{(0)}_{\text{sym}},$$

while $G^{(0)}$ and $b^{(0)}$ carry the product itself, so that $[a^{(0)}]G^{(0)}$ and $a^{(0)}\,b^{(0)}$ agree in both normalisations, and the two verifier equations coincide once each run attaches ℓ_j and ρ_j to the same cross product. Table 6 prints both runs on the toy's data and keeps the blinder names instead, so the deployed run attaches each blinder to the other cross product, and its equation differs from the symmetric one only in the H-term ($r^* = 33$ against 40); with the blinders exchanged, the deployed run also closes at $(54, 39)$ with $r^* = 40$. What the substitution does *not* do is identify the challenge distributions: a uniform deployed challenge is not the square of a uniform symmetric one ("Extraction in the deployed rows" below), so the deployed rows are analysed with their own lemma, and the deployed opening is knowledge sound by the argument of §3.5 with Lemma 3.10 at every node and two branchings, over ξ and ζ, in place of the resampled U at the root: Corollary 3.13.

	symmetric fold	deployed fold	relation
round 2: challenge	$u_2 = 52$	$u_2 = 52^2 = 85$	
cross terms sent	$L_2 = (74, 6)$, $R_2 = (38, 72)$	$L_2 = (78, 70)$, $R_2 = (71, 26)$	halves swapped
$\mathbf{a}^{(1)}$	$(58, 61)$	$(72, 59)$	$52^{-1}\cdot$
$\mathbf{G}^{(1)}$	$((54, 40), (30, 71))$	$((53, 53), (41, 6))$	$52\cdot$
$\mathbf{b}^{(1)}$	$(70, 42)$	$(51, 50)$	$52\cdot$
round 1: challenge	$u_1 = 20$	$u_1 = 20^2 = 12$	
cross terms sent	$L_1 = (28, 25)$, $R_1 = (39, 25)$	$L_1 = (43, 6)$, $R_1 = (70, 71)$	halves swapped
$a^{(0)}$	33	85	$(52 \cdot 20)^{-1} = 70^{-1}\cdot$
$G^{(0)}$	$(12, 25)$	$(34, 53)$	$70\cdot$
$b^{(0)}$	19	69	$70\cdot$
structured scalars $\mathbf{s}$	$(79, 75, 22, 70)$	$(1, 12, 85, 50)$	$70\cdot$
$g(20)$	$(u_1^{-1} + u_1 z)(u_2^{-1} + u_2 z^2) = 19$	$(1 + u_1 z)(1 + u_2 z^2) = 69$	$70\cdot$
aggregated blinder r^*	40	33	
verifier's equation	$(54, 39) = (54, 39)$	$(5, 72) = (5, 72)$	same $[a^{(0)}]G^{(0)}$, $a^{(0)}b^{(0)}$

Table 6: The two folds on the same $\mathbf{a} = (90, 61, 22, 24)$, the same generators and the same $\mathbf{b}(20)$, with $u_{\text{dep}} = u_{\text{sym}}^2$ in every round and the same round blinders. The last column is the cumulative factor of (27): after round 2 it is 52, after round 1 it is $52 \cdot 20 = 70$ in $\mathbb{F}_{97}$. The cross terms differ as points because the deployed naming attaches each blinder to the other cross product; every folded quantity differs by the stated factor only.

Extraction in the deployed rows. The deployed fold changes the rows of the extractor's linear system.

> **Lemma 3.10** (Single-round recovery in the deployed rows)**.** *Fix one round at level j as in Lemma 3.4, but let the round fold the coefficient vector with the weights $(1, u^{-1})$ and the generator and evaluation vectors with the weights $(1, u)$,*
>
> $$\mathbf{a}' := \mathbf{a}_{\text{lo}} + u^{-1}\mathbf{a}_{\text{hi}}, \qquad \mathbf{G}' := \mathbf{G}_{\text{lo}} + u\,\mathbf{G}_{\text{hi}}, \qquad \mathbf{b}' := \mathbf{b}_{\text{lo}} + u\,\mathbf{b}_{\text{hi}},$$
>
> *with the cross terms named with the halves swapped relative to* (13)*,*
>
> $$L := \langle \mathbf{a}_{\text{hi}}, \mathbf{G}_{\text{lo}} \rangle + [\ell]H + [\langle \mathbf{a}_{\text{hi}}, \mathbf{b}_{\text{lo}} \rangle]\, U, \qquad R := \langle \mathbf{a}_{\text{lo}}, \mathbf{G}_{\text{hi}} \rangle + [\rho]H + [\langle \mathbf{a}_{\text{lo}}, \mathbf{b}_{\text{hi}} \rangle]\, U,$$

> *and the folded commitment $C' := C + [u^{-1}]L + [u]R$, so that the fold identity reads $C' = \langle \mathbf{a}', \mathbf{G}' \rangle + [r + u^{-1}\ell + u\rho]H + [\langle \mathbf{a}', \mathbf{b}' \rangle]U$ by the same expansion as in §3.2. Then the conclusions of Lemma 3.4 hold with rows $(\nu^{-1}, 1, \nu)$ and the identities $\mathbf{x}_\nu = \nu^{-1}\mathbf{x}_L + \mathbf{x}_C + \nu\,\mathbf{x}_R$ (likewise for y, w), provided only that u, u', u'' are pairwise distinct and nonzero. The deployed cross terms also scale their U-coordinates by a challenge $\zeta \neq 0$ drawn before the rounds ("The deployed opening in full" below), which the lemma covers verbatim with U replaced by $[\zeta]U$.*

Proof. The fold identity is the expansion of §3.2 with the weights changed: the diagonal terms of $\langle \mathbf{a}_{\mathrm{lo}} + u^{-1}\mathbf{a}_{\mathrm{hi}}, \mathbf{G}_{\mathrm{lo}} + u\,\mathbf{G}_{\mathrm{hi}} \rangle$ carry the weights $1 \cdot 1$ and $u^{-1}u = 1$, and the cross terms $\langle \mathbf{a}_{\mathrm{hi}}, \mathbf{G}_{\mathrm{lo}} \rangle$ and $\langle \mathbf{a}_{\mathrm{lo}}, \mathbf{G}_{\mathrm{hi}} \rangle$ carry u^{-1} and u, which is why they are absorbed as $[u^{-1}]L + [u]R$ with L holding the high-with-low product. The lift (20) becomes $\mathbf{x} = (\mathbf{a}', u\,\mathbf{a}')$, and preserves well-formedness since $\langle (\mathbf{a}', u\mathbf{a}'), \mathbf{b} \rangle = \langle \mathbf{a}', \mathbf{b}_{\mathrm{lo}} + u\mathbf{b}_{\mathrm{hi}} \rangle$. The matrix now has rows $(\nu^{-1}, 1, \nu)$; multiplying the row of ν by $\nu \neq 0$ gives $(1, \nu, \nu^2)$, and the same root count, now for a polynomial of degree at most 2 in ν itself, shows that the matrix is invertible exactly when the challenges are pairwise distinct. Everything else is as in the proof of Lemma 3.4. □

The two row shapes impose different hypotheses: two challenges $u' = -u$ have equal squares, so a triple containing them makes the symmetric matrix singular, while the deployed matrix remains invertible whenever the three challenges are distinct and nonzero. Over $\mathbb{F}_{97}$ the triple $(3, 94, 5)$, in which $94 = -3$, has squares $(9, 9, 25)$: the symmetric rows are $(9, 1, 54)$, $(9, 1, 54)$ and $(25, 1, 66)$, two of them equal, with determinant 0; the deployed rows are $(65, 1, 3)$, $(32, 1, 94)$ and $(39, 1, 5)$, with determinant $28 \neq 0$. Remark 3.9 shows that the deployed fold is the symmetric fold under the substitution $u_{\mathrm{dep}} = u_{\mathrm{sym}}^2$, but that relabelling does not carry the hypotheses across: the squaring map $\mathbb{F}^\times \to \mathbb{F}^\times$ is two-to-one onto the squares, so a uniform deployed challenge is not the square of a uniform symmetric one, and the condition "pairwise distinct squares" is a condition on three symmetric challenges that has no counterpart for three uniform deployed ones. Each shape is therefore analysed with its own rows. In the proof of Theorem 3.5 with Lemma 3.10 in place of Lemma 3.4, three pairwise distinct nonzero challenges suffice at each node, and each sibling avoids the earlier siblings' challenges and 0.

Example 3.11 (The extractor in the deployed rows)**.** On the data of Example 3.6, the extractor with rows $(u^{-1}, 1, u)$ and the fold of Lemma 3.10 branches over its own challenges: $u_2 \in \{78, 83, 56\}$ at the root, and under them the round-1 triples $\{31, 71, 92\}$, $\{91, 50, 47\}$ and $\{95, 14, 96\}$. The node systems have determinants 70, 41 and 33 and the root system 54, and the extractor recovers the same $\mathbf{a} = (90, 61, 22, 24)$, $r = 88$ and U-coordinate $59 = v$.

The deployed opening in full. The deployed opening is the masked opening of §3.6 with the deployed fold, a further challenge ζ, and the claimed value moved into the commitment. Let $\mathbf{e}_0 := (1, 0, \dots, 0)$, and write $\mathbf{t}$ for the coefficient-side vector and $\mathbf{s}$ for the structured scalars of the deployed fold throughout the rest of this subsection.

1. **Mask.** The prover samples $s(X)$ with $s(x) = 0$ and r_s, and sends $S := \mathsf{Commit}(s; r_s) = \langle \mathbf{m}, \mathbf{G} \rangle + [r_s]H$, as in §3.6.
2. **Two challenges.** The verifier sends a uniform ξ, and then a uniform ζ.
3. **The masked commitment.** Both parties replace P by

$$P' := P + [\xi]S - [v]\,G_0, \tag{28}$$

which, by linearity of the commitment (*Crypto Guide*, §"Pedersen vector commitments", the remark that linearity survives), commits with blinder $r + \xi r_s$ to the coefficient vector

$$\mathbf{a}' := \mathbf{a} + \xi\,\mathbf{m} - v\,\mathbf{e}_0$$

of the polynomial $p + \xi s - v$, whose value at x is $p(x) + \xi\, s(x) - v = 0$. The claim "$p(x) = v$" has become the claim "$P'$ commits to a vector whose inner product with $\mathbf{b}(x)$ is 0": the value is folded into the commitment, and the opening claims the value 0.

4. **Rounds.** The parties run the k rounds on $(\mathbf{a}', \mathbf{G}, \mathbf{b}(x))$ with the fold of Lemma 3.10, weights $(1, u_j^{-1})$ on the coefficients and $(1, u_j)$ on generators and evaluation vector, and with the U-coordinates of the cross terms scaled by ζ:

$$L_j := \langle \mathbf{a}'^{(j)}_{\mathrm{hi}},\, \mathbf{G}^{(j)}_{\mathrm{lo}} \rangle + [\ell_j]H + [\zeta\, \langle \mathbf{a}'^{(j)}_{\mathrm{hi}},\, \mathbf{b}^{(j)}_{\mathrm{lo}} \rangle]\, U,$$
$$R_j := \langle \mathbf{a}'^{(j)}_{\mathrm{lo}},\, \mathbf{G}^{(j)}_{\mathrm{hi}} \rangle + [\rho_j]H + [\zeta\, \langle \mathbf{a}'^{(j)}_{\mathrm{lo}},\, \mathbf{b}^{(j)}_{\mathrm{hi}} \rangle]\, U;$$

the verifier accumulates $[u_j^{-1}]L_j + [u_j]R_j$, and the prover's blinder accumulates $f := r + \xi r_s + \sum_j (u_j^{-1}\ell_j + u_j\rho_j)$.

5. **Final message and check.** The prover sends the final scalars $c := \mathbf{a}'^{(0)}$ and f. The verifier accepts if and only if

$$P + [\xi]S - [v]G_0 + \sum_{j=1}^{k} ([u_j^{-1}]L_j + [u_j]R_j) \stackrel{?}{=} [c]\, G^{(0)} + [c \cdot b^{(0)} \cdot \zeta]\, U + [f]H, \qquad (29)$$

where $G^{(0)} = \langle \mathbf{s}, \mathbf{G} \rangle$ and $b^{(0)} = \langle \mathbf{s}, \mathbf{b}(x) \rangle = g_{\mathrm{dep}}(x)$ for the structured scalars $s_i = \prod_{\ell:\beta_{\ell-1}=1} u_\ell$ of (26).

Completeness is the invariant (16) again, with $\mathbf{a}'$ in place of $\mathbf{a}$, the value 0 in place of v, and every U-coordinate scaled by ζ; the final U-coordinate is $\zeta\langle \mathbf{a}'^{(0)}, \mathbf{b}^{(0)} \rangle = c\, b^{(0)}\zeta$. Subtracting $[v]G_0$ moves the claimed value into the commitment, so that the argument opens to 0 and the check carries no term in v on U; the mask leaves the opened value unchanged in either form because $s(x) = 0$. The halo2 book gives the reason in its comparison with the scheme of Bünz, Chiesa, Mishra and Spooner, which adds the value on the inner-product generator instead (chapter *Comparison to other work*): G_0 is a fixed base, so $[v]G_0$ is the cheaper multiplication inside a recursive verifier. The factor ζ has a reason of its own. The deployed U is a fixed public generator rather than a verifier's sample drawn after P ("Generators from a hash" above): a prover who knew U in advance could plant a U-component in P or S, and drawing ζ after both re-randomises the generator that carries the inner product, so that a planted component would have to have anticipated ζ; Corollary 3.13 makes this precise.

Hiding in the deployed opening. The final scalar is the linear form of (22) for this fold, taken of the masked vector:

$$c = \langle \mathbf{a} + \xi\, \mathbf{m} - v\, \mathbf{e}_0,\, \mathbf{t} \rangle = \langle \mathbf{a} - v\, \mathbf{e}_0,\, \mathbf{t} \rangle + \xi\, \langle \mathbf{m}, \mathbf{t} \rangle, \qquad t_i = \prod_{\ell:\beta_{\ell-1}=1} u_\ell^{-1}, \qquad (30)$$

the coefficient side folding with the inverse weights of the generator side as always. Unmasked, the deployed fold leaks this form as the symmetric one does (Remark 3.7): four deployed openings of $(90, 61, 22, 24)$ without the mask, with challenge pairs $(u_2, u_1) = (16, 85)$, $(8, 16)$, $(69, 95)$ and $(11, 41)$, end in the scalars 70, 24, 73 and 63, and solving the 4×4 system recovers the vector. Masked, with $\mathbf{m} = (66, 23, 23, 88)$, $\xi = 29$ and the challenges $u_2 = 85$, $u_1 = 12$, the masked vector is $(5, 49, 10, 54)$, its inner product with $\mathbf{b}(20)$ is 0, and the final scalar is $c = 20 = \langle (5, 49, 10, 54), \mathbf{t} \rangle$ with $\mathbf{t} = (1, 89, 8, 33)$; a second mask $\mathbf{m}' = (32, 68, 64, 88)$ under the same challenges ends in $c' = 46$, and both runs are accepted. The transcript of one opening is

$$(S;\ L_k, R_k, \ldots, L_1, R_1;\ c, f) \qquad \text{with the challenges} \qquad (\xi, \zeta, u_k, \ldots, u_1).$$

Corollary 3.12 (Honest-verifier zero knowledge of the deployed opening)**.** *Let $\mathbb{G}$ have prime order p, and let the verifier draw ξ, ζ uniformly from $\mathbb{F}$ and each u_j uniformly from $\mathbb{F}^\times$. There is a simulator which, given (P, x, v) but not $(\mathbf{a}, r)$, outputs challenges and a transcript of the deployed opening whose distribution lies within statistical distance $1/|\mathbb{F}| + (|\mathbb{F}| - 1)^{-k}$ of the honest one.*

Proof. The proof of Proposition 3.8, with the deployed $\mathbf{t}$ and with (29) in place of (24). Call a challenge tuple *good* if $\xi \neq 0$ and $\mathbf{t}$ is not a multiple of $\mathbf{b}(x)$; now $\mathbf{t} = \lambda\, \mathbf{b}(x)$ forces $\lambda = t_0 = 1$ and then $u_\ell^{-1} = x^{2^{\ell-1}}$ for every ℓ, at most one challenge tuple among $(|\mathbb{F}| - 1)^k$. For good challenges c is uniform and independent of $\mathbf{a}$ by (30) and the argument of that proof. The simulator draws the challenges, S, every L_j and every R_j but R_1, and c and f uniformly, and sets R_1 to the one point that satisfies (29), namely $R_1 := [u_1^{-1}](\text{right-hand side} - \text{every other term of the left})$; the counting of that proof shows that the two distributions agree for good challenges, and the tuples that are not good have probability at most $1/|\mathbb{F}| + (|\mathbb{F}| - 1)^{-k}$. □

The deployed opening of the toy polynomial. The toy's P of §3.7 is opened at $z = 20$ with the deployed opening, using the round challenges $u_2 = 85$ and $u_1 = 12$ of Table 6; Table 7 lists both sides with every intermediate value. The verifier's equation (29) closes at $(5, 72)$ on both sides, with $g(20) = (1 + 12 \cdot 20)(1 + 85 \cdot 20^2) = 69$ and $\mathbf{s} = (1, 12, 85, 12 \cdot 85) = (1, 12, 85, 50)$ in the deployed shape, and the false value $v' = 60$ moves the left-hand side to $(67, 31)$ against $(5, 72)$ on the right: rejected once more. The deployed run and the masked symmetric run of §3.7 open the same commitment to the same value with entirely different numbers; Remark 3.9 relates the two folds. The unmasked deployed run of Table 6 and this masked opening both close at $(5, 72)$, and that run's $R_1 = (70, 71)$ equals the mask commitment S: further collisions of the kind noted in §3.7, accidents of a 97-element group, not consequences of the mask.

Corollary 3.13 (Knowledge soundness of the deployed opening)**.** *Let $\mathbb{G}$, $\mathbf{G}$ and H be as in Theorem 3.5, and let U be a fixed generator, uniform in $\mathbb{G} \setminus \{O\}$ like them. The deployed opening, with the check (29), is knowledge sound for $\mathcal{R}_{\mathrm{IPA}}$: there is an extractor which, given rewindable access to any prover that makes the verifier accept with non-negligible probability, runs in expected polynomial time and outputs either a witness $(\mathbf{a}, r)$ for (P, x, v) or a nontrivial discrete-logarithm relation among $\mathbf{G}$, H and U. Its knowledge error is $O(\log d/|\mathbb{F}|)$.*

Proof. The extractor branches over two values $\xi_1 \neq \xi_2$ after S, under each over two nonzero values $\zeta_1 \neq \zeta_2$, and under each of those over the rounds as in the proof of Theorem 3.5, with three pairwise distinct nonzero challenges per node; the tree has $4 \cdot 3^k$ leaves, and the rewinding and running time are as in that proof.

Rounds. Fix ξ and $\zeta \neq 0$, and write $U_\zeta := [\zeta]U$. With U_ζ in place of U the rounds are those of Lemma 3.10, so the induction in the proof of Theorem 3.5 returns a representation $(\mathbf{x}; y; w)$ of $P' = P + [\xi]S - [v]G_0$ over $(\mathbf{G}, H, U_\zeta)$ with $w = \langle \mathbf{x}, \mathbf{b}(x)\rangle$, that is, the representation $(\mathbf{x}; y; \zeta w)$ over $(\mathbf{G}, H, U)$.

The challenge ζ. Under one ξ, the values ζ_1 and ζ_2 give representations $(\mathbf{x}_1; y_1; \zeta_1 w_1)$ and $(\mathbf{x}_2; y_2; \zeta_2 w_2)$ of the same P' over $(\mathbf{G}, H, U)$. If they differ, their difference is a nontrivial relation among $\mathbf{G}$, H and U. Otherwise $\mathbf{x}_1 = \mathbf{x}_2 =: \mathbf{x}$, so $w_1 = w_2 =: w$ and $(\zeta_1 - \zeta_2)\, w = 0$, whence $w = 0$: $P' = \langle \mathbf{x}, \mathbf{G}\rangle + [y]H$ with $\langle \mathbf{x}, \mathbf{b}(x)\rangle = 0$.

The challenge ξ. The values ξ_1 and ξ_2 follow the same S and give $P + [\xi_i]S - [v]G_0 = \langle \mathbf{x}_i, \mathbf{G}\rangle + [y_i]H$ with $\langle \mathbf{x}_i, \mathbf{b}(x)\rangle = 0$ for $i = 1, 2$. Subtracting, $S = \langle \mathbf{m}, \mathbf{G}\rangle + [r_s]H$ with $\mathbf{m} := (\mathbf{x}_1 - \mathbf{x}_2)/(\xi_1 - \xi_2)$, $r_s := (y_1 - y_2)/(\xi_1 - \xi_2)$ and $\langle \mathbf{m}, \mathbf{b}(x)\rangle = 0$. Substituting back, $P = \langle \mathbf{a}, \mathbf{G}\rangle + [r]H$ with $\mathbf{a} := \mathbf{x}_1 - \xi_1 \mathbf{m} + v\, \mathbf{e}_0$ and $r := y_1 - \xi_1 r_s$, and $\langle \mathbf{a}, \mathbf{b}(x)\rangle = 0 - 0 + v \cdot 1 = v$, the first entry of $\mathbf{b}(x)$ being 1. This is the witness.

step	prover	verifier
mask	samples $\mathbf{m} = (66, 23, 23, 88)$, for which $s(20) = 0$, and $r_s = 95$; sends $S = \langle \mathbf{m}, \mathbf{G} \rangle + [95]H = (70, 71)$	receives S; sends $\xi = 29$, then $\zeta = 12$
masked vector	$\mathbf{a}' = \mathbf{a} + 29\,\mathbf{m} - 59\,\mathbf{e}_0 = (5, 49, 10, 54)$; $\langle \mathbf{a}', \mathbf{b}(20) \rangle = 0$; blinder $f := 10 + 29 \cdot 95$	$P' := P + [29]S - [59]G_0$
round 2	halves $\mathbf{a}'_{\text{lo}} = (5, 49)$, $\mathbf{a}'_{\text{hi}} = (10, 54)$; $\ell_2 = 44$, $\rho_2 = 92$; cross terms $\langle \mathbf{a}'_{\text{hi}}, \mathbf{G}_{\text{lo}} \rangle = [10]G_0 + [54]G_1 = (34, 53)$, $\langle \mathbf{a}'_{\text{hi}}, \mathbf{b}_{\text{lo}} \rangle = 23$, $\langle \mathbf{a}'_{\text{lo}}, \mathbf{G}_{\text{hi}} \rangle = [5]G_2 + [49]G_3 = (54, 39)$, $\langle \mathbf{a}'_{\text{lo}}, \mathbf{b}_{\text{hi}} \rangle = 83$; sends $L_2 = (34, 53) + [44]H + [12 \cdot 23]U = (17, 52)$, $R_2 = (54, 39) + [92]H + [12 \cdot 83]U = (47, 40)$ folds $\mathbf{a}'^{(1)} = (5, 49) + 8\,(10, 54) = (85, 93)$, $\mathbf{b}^{(1)} = (1, 20) + 85\,(12, 46) = (51, 50)$, $\mathbf{G}^{(1)} = (G_0 + [85]G_2,\ G_1 + [85]G_3) = ((53, 53), (41, 6))$; $f = 10 + 29 \cdot 95 + 8 \cdot 44 + 85 \cdot 92 = 73$	sends $u_2 = 85$ ($85^{-1} = 8$); keeps $(L_2, R_2, 85)$; $P'^{(1)} := P' + [8]L_2 + [85]R_2 = (64, 5)$
round 1	halves $a'_{\text{lo}} = 85$, $a'_{\text{hi}} = 93$; $\ell_1 = 95$, $\rho_1 = 56$; cross terms $[93]G_0^{(1)} = (58, 8)$, $93 \cdot 51 = 87$, $[85]G_1^{(1)} = (24, 9)$, $85 \cdot 50 = 79$; sends $L_1 = (58, 8) + [95]H + [12 \cdot 87]U = (46, 13)$, $R_1 = (24, 9) + [56]H + [12 \cdot 79]U = (24, 70)$ folds $c = a'^{(0)} = 85 + 89 \cdot 93 = 20$, $b^{(0)} = 51 + 12 \cdot 50 = 69$, $G^{(0)} = G_0^{(1)} + [12]G_1^{(1)} = (34, 53)$; $f = 73 + 89 \cdot 95 + 12 \cdot 56 = 82$	sends $u_1 = 12$ ($12^{-1} = 89$); keeps $(L_1, R_1, 12)$; $P'^{(0)} := P'^{(1)} + [89]L_1 + [12]R_1 = (5, 72)$
final	sends $c = \mathbf{a}'^{(0)} = 20$, $f = 82$	$g(20) = (1 + 12 \cdot 20)(1 + 85 \cdot 20^2) = 69$; $\mathbf{s} = (1, 12, 85, 50)$, $G^{(0)} = \langle \mathbf{s}, \mathbf{G} \rangle = (34, 53)$; left: $P' + [85^{-1}]L_2 + [85]R_2 + [12^{-1}]L_1 + [12]R_1 = (5, 72)$; right: $[20]G^{(0)} + [20 \cdot 69 \cdot 12]U + [82]H = (5, 72)$: **accept**; with $v' = 60$: left $(67, 31) \neq (5, 72)$: **reject**

Table 7: The deployed opening of the same $P = (39, 25)$ at $z = 20$ with the deployed mask: the fold of Lemma 3.10, the cross terms named with the halves swapped, the U-coordinates scaled by ζ, the value 0 opened for $P + [\xi]S - [v]G_0$, and the final scalars c and f. The round challenges are those of Table 6.

The error. Beyond the $O(\log d/|\mathbb{F}|)$ of the rounds, the two branchings at the root and the exclusion of $\zeta = 0$ each cost $O(1/|\mathbb{F}|)$. A relation among $\mathbf{G}$, H and U yields a discrete logarithm by the every-generator embedding, as at the root of Theorem 3.5. □

The bytes. An Orchard proof is a byte string. The proofs of the $a \geq 1$ Action descriptions that one transaction carries in one pool, here called a *bundle*, are aggregated into a single proof (protocol specification, § 4.6, "Action Descriptions"); a transaction with Actions in both the Orchard and the Ironwood pool carries one such proof for each. Its length is $2720 + 2272\,a$ bytes (ZIP 225, the `sizeProofsOrchard` field; ZIP 229, the fields `sizeProofsOrchard` and `sizeProofsIronwood`; the consensus rule is the protocol specification's, § 7.5, "Action Description Encoding and Consensus"), 4992 bytes for a single Action. Every Vesta point and every field element in it occupies 32 bytes. Table 8 lists by kind, from the per-object counts of the Action circuit whose verifying key the consensus rules fix (protocol specification, § 4.6, "Action Descriptions"), every one of those bytes: a per-Action block of 22 points and 49 scalars, repeated for each Action, and a shared block of 33 points and 52 scalars. The inner-product argument's share is the last three rows of the shared block: the blinding commitment S, the $k = 11$ pairs (L_j, R_j), which are 22 points, and the two scalars c and f: 23 points, 736 bytes, or 800 bytes with the scalars, once per bundle, because the complete protocol collapses every opening of every polynomial into one opening ("The multipoint opening argument" in §4). This is the only part of the proof that grows with n, and it grows as $2\log_2 n + 1$ points. Everything else grows with the circuit's structure and not with n: one commitment per advice column, two permuted-column commitments and one running product per lookup, the permutation running products and quotient chunks, the multipoint commitment, and one field element per claimed evaluation.

3.8.1 The "Halo" in Halo 2: deferring the expensive check

The verifier of this section is logarithmic everywhere but in one place. He reads $2k+1$ points and two scalars, evaluates g in $3k-2$ field multiplications, and forms the left-hand side of (29) with $2k+2$ scalar multiplications ($[\xi]S$, $[v]G_0$ and the $2k$ round terms); then he needs $G^{(0)} = \langle \mathbf{s}, \mathbf{G} \rangle$, one combination of *all* 2048 generators, group arithmetic proportional to the size of the whole circuit (§3.4). Two ways of amortising that cost follow, batch verification and accumulation, the second not deployed by Orchard; between them, the paragraph "Transparency: no trusted setup" records why the linear cost is accepted.

Batch verification. The consensus rules require each Action proof to be valid for the verifying key of the Orchard circuit (protocol specification, § 4.6, "Action Descriptions"); validity includes the check (29) with its multi-scalar multiplication of length d, and the specification composes no proofs, so the verifier pays that multiplication itself rather than passing it on. Several proofs can share it (*Crypto Guide*, §"Polynomial commitment schemes", Remark "KZG versus IPA, and the Halo lineage"). Written as one multi-scalar multiplication that must evaluate to $\mathcal{O}$, equation (29) has $d+2k+4$ terms: the generators $G_0, \dots, G_{d-1}$, U, H, P, S and the $2k$ points L_j, R_j. The equations $E_1, \dots, E_b$ of b proofs are checked together by testing $\sum_i [\mu_i]E_i = \mathcal{O}$ for fresh uniform $\mu_i \in \mathbb{F}$ drawn by the verifier.

> **Lemma 3.14** (Randomised batch check)**.** *Let $\mathbb{G}$ have prime order, let $E_1, \dots, E_b \in \mathbb{G}$, and let $\mu_1, \dots, \mu_b \in \mathbb{F}$ be independent and uniform. If some $E_i \neq \mathcal{O}$, then* $\Pr[\sum_j [\mu_j]E_j = \mathcal{O}] = 1/|\mathbb{F}|$.

Proof. Fix i with $E_i \neq \mathcal{O}$ and condition on every μ_j with $j \neq i$. Since $\mathbb{G}$ has prime order, E_i generates it, so $\mu \mapsto [\mu]E_i$ is a bijection $\mathbb{F} \to \mathbb{G}$, and the sum equals $\mathcal{O}$ for exactly one value of μ_i. □

kind	count	bytes	what it is
Per-Action block, once per Action			
point	10	320	advice column commitments, one per advice column
point	3	96	permuted lookup input commitments A', one per lookup (Construction 2.14)
point	3	96	permuted lookup table commitments S', one per lookup (Construction 2.14; unrelated to the mask commitment S)
point	3	96	permutation running-product commitments: the 15 equality-enabled columns in chunks of 7 (§2.6)
point	3	96	lookup running-product commitments, one per lookup
scalar	1	32	instance column evaluation at x
scalar	25	800	$25 = 5 \cdot 2 + 5 \cdot 3$ advice evaluations, one per queried rotation of each advice column (§4.4)
scalar	8	256	permutation product evaluations: 3 products at x and ωx, the first 2 also at $\omega^{-6}x$, the rotation to the boundary row $n - n_{\text{blind}} - 1$ of Remark 2.17, with $n_{\text{blind}} = 5$ blinding rows (§4.6)
scalar	15	480	lookup evaluations: product at x and ωx; A' at x and $\omega^{-1}x$; S' at x
total		2272	22 points and 49 scalars
Shared block, once per bundle			
point	1	32	random polynomial commitment: a random polynomial added to blind the quotient (the complete protocol, §4)
point	8	256	quotient chunk commitments: constraint degree $d_{\max} = 9$, so $d_{\max} - 1 = 8$ chunks of degree below n (Theorem 4.7)
scalar	29	928	fixed column evaluations at x, one per fixed column
scalar	15	480	permuted-label polynomial evaluations at x, one per equality-enabled column (Construction 2.10)
scalar	1	32	random polynomial evaluation at x
point	1	32	multipoint polynomial commitment: the one polynomial into which every query collapses (Construction 4.4)
scalar	5	160	multipoint evaluations: the value of each query group's combined polynomial at the reduction's fresh challenge point, one per distinct set of query points; five sets (§4.4)
point	1	32	**IPA:** the blinding commitment S
point	22	704	**IPA:** the $k = 11$ pairs (L_j, R_j)
scalar	2	64	**IPA:** the final scalars c and f
total		2720	33 points and 52 scalars

Table 8: The bytes of an Orchard proof, $2720 + 2272\,a$ for a Actions (ZIP 225, ZIP 229), laid out as points and scalars of 32 bytes each. The three rows marked **IPA** are this section's: 23 points and 2 scalars, 800 bytes, shared by the whole bundle. The advice, permutation, lookup and quotient rows are the objects of §2; the random polynomial is constructed under "The complete Halo 2 protocol" and the multipoint polynomial under "The multipoint opening argument", both in §4.

Because $\mathbf{G}$, U and H are shared, their terms merge, and b equations combine into one multi-scalar multiplication of $d + 2 + b(2k + 2)$ terms, against $b(d + 2k + 4)$ when checked separately. In the complete protocol P is itself a combination of commitments (§4); those of the verifying key are shared as well, and each further proof adds its own points, the $22a + 33$ of Table 8, and the instance commitments computed from its public inputs, one per Action (§4.2). The $d = 2048$ generator terms are paid once per batch. A batch check is not accumulation: it emits nothing that a later proof could consume, and it verifies the batch and stops. Both take linear combinations of deferred equations; their interfaces and guarantees differ, and the accumulation scheme of Bünz, Chiesa, Mishra and Spooner, sketched below under "Accumulation (not deployed by Orchard)", is not part of the specified protocol.

Transparency: no trusted setup. The generators are outputs of a public hash-to-curve map on fixed inputs (this subsection, "Generators from a hash"); the scheme has no trapdoor and requires no trusted setup, unlike KZG, whose trapdoor permits forgeries (*Crypto Guide*, §"Polynomial commitment schemes", under "KZG: evaluation in the exponent"). The price is the $2k+1$ points of Table 8, against the constant-size opening proof of KZG, and the linear verifier. ZIP 224's motivation names the generation of the structured reference string of Zcash's earlier proving system a point of risk within the protocol and asks for a proving system that does not require one (ZIP 224, Motivation).

Accumulation (not deployed by Orchard). In a recursive verifier, a circuit that verifies a proof, the multi-scalar multiplication $G^{(0)} = \langle \mathbf{s}, \mathbf{G} \rangle = \mathsf{Commit}(g; 0)$ of §3.4 would cost d scalar multiplications inside the circuit. Instead the claim is combined with a running claim of the same shape, $G'^{(0)} = \mathsf{Commit}(g'; 0)$, by a random linear combination with a challenge ρ drawn after both (a verifier challenge, not a round blinder ρ_j of this section; §7 keeps the letter): $G^{(0)} + [\rho]\, G'^{(0)} = \mathsf{Commit}(g + \rho\, g'; 0)$ holds for all ρ when both inputs hold and for at most one ρ otherwise. The combined claim, returned to a single tuple by a prover-assisted opening, is the *accumulator* carried along the chain of proofs, and a single multi-scalar multiplication outside any circuit at the end discharges all combined claims. The soundness of an accumulation scheme for a closely related inner-product commitment is cited in §7 from Bünz, Chiesa, Mishra and Spooner, under the extractability of its openings, which they conjecture from the binding of the Pedersen commitment; the cycle of curves that keeps the recursive arithmetic in the right fields is the other half of Pasta. Both are developed in §7, under "Accumulation and the Halo trick" and "A cycle of curves for efficient native recursion", where recursion and accumulation over the cycle are classified designed-but-unspecified in the sense of §1.2: the protocol specification uses Halo 2 only to prove and verify Action statements and composes no proofs.

4 The Halo 2 proof system

This section defines the Halo 2 proof system as deployed for the Orchard Action statement: the Action circuit, the instance column, the transcript, the multipoint opening, the protocol, and its zero knowledge. It combines the arithmetisation of §2, which reduces a circuit to committed polynomials and identities among them (the gates, the permutation argument for the wiring, the lookup argument for table membership), with the polynomial commitment of §3, which opens a committed polynomial at a point in logarithmically many group elements and is binding under the discrete logarithm on Vesta. The abstractions the *Crypto Guide* owns, the non-interactive argument and the SNARK, the polynomial interactive oracle proof, the Fiat–Shamir transform and the polynomial-commitment interface, are recalled at the point of use. Halo 2 is a *transparent* PLONKish argument: the parameters of its commitment scheme are derived by hashing to the curve and carry no trapdoor (§3.8), and for the Orchard Action circuit its polynomial commitment works over Vesta. The protocol specification defines Action proofs as proofs of the Halo 2

proving system described in the halo2 book (protocol specification, § 5.4.10.3, "Halo 2"); this section cites the book, by chapter, as the specification of the protocol, and names an implementation only for a behaviour that neither document fixes.

The subsections treat, in order, the Action circuit as a PLONKish circuit (its gates, one gate in full, its lookup arguments and its column polynomials); the instance column; the Fiat–Shamir transform and the deployed transcript; the multipoint opening argument; the complete protocol in seven phases; its zero knowledge; and the compiled-polynomial-IOP pattern the protocol instantiates. Recursion is not deployed by Orchard and is treated in §7.

Notation. The constraint degree is $d_{\max}$, as Definition 2.7 fixed it; the letter d is the committed vector length of §3, which equals the row count n throughout this section. The letter H carries three meanings; two come from the sections this one joins: §2 writes H for the evaluation domain and §3 for the blinding generator of the Pedersen commitment. Here the domain appears only through its vanishing polynomial $Z_H(X) = X^n - 1$, its Lagrange basis ℓ_j and its rows ω^j; a bare H inside a commitment formula is the blinding generator, as in §3. In §4.3 and the security statements the letter H also names the transcript hash, modelled as a random oracle, as in the *Crypto Guide*; there it is applied to arguments, $H(\cdot)$, or called the hash or the oracle. The letter x is the evaluation challenge of phase 5; the running example's witness is written by its value 3 here and in §6.

4.1 From statement to circuit: arithmetisation in practice

Definition 2.7 says what a PLONKish circuit is; the deployed instance is the Orchard Action circuit. The protocol specification fixes the statement it proves, the Action statement (protocol specification, § 4.18.4, "Action Statement (Orchard)"), and requires every Action proof to verify under the verifying key identified by the current Orchard circuit version (protocol specification, § 4.6, "Action Descriptions", consensus rules). No specification or ZIP fixes the parameters of the circuit behind that key; they are those of its implementation, `orchard`. As a PLONKish circuit in the sense of Definition 2.7 it has the field $\mathbb{F}_{p_{\mathsf{Pallas}}}$, $n = 2^{11}$ rows, ten advice columns, one instance column and twenty-nine fixed columns, among them its selector columns and the three columns of its lookup table; the constraint-degree bound $d_{\max} = 9$; and three lookup arguments into that table, of 2^{10} rows. Its custom gates encode the conditions of the Action statement, using among others complete and incomplete addition on Pallas, the distinction the *Math Guide* draws in §"Explicit affine formulas" (its remark on complete versus incomplete addition). This subsection gives the form of the gates, one gate in full, the three lookup arguments, and the column polynomials on which every argument of §2 operates.

Key and witness. The verifying key is a deterministic function of the commitment parameters and the circuit description: the values of the fixed columns, the gates, the lookup arguments and the permutation σ of Definition 2.7. For the Action circuit it is a fixed consensus parameter, the key the consensus rule above identifies. The prover's witness is the content of the advice columns. A copy constraint places two cells in one cycle of σ (Construction 2.10), and the public inputs are bound to advice cells in this way (§4.2).

4.1.1 Custom gates

Every custom gate of the deployed circuit, a gate in the sense of Definition 2.7, has the form

$$q \cdot c(v_1(\omega^{r_1}X), \dots, v_s(\omega^{r_s}X)) = 0,$$

where $v_1, \dots, v_s$ are columns read at rotations $r_1, \dots, r_s$, c is a polynomial, and the *selector* q is a polynomial in the fixed columns that is zero on every row where the gate does not apply (the

halo2 book's *PLONKish Arithmetization* chapter). The gate is required to vanish at every row; on the rows where q is zero, the blinding rows among them, it holds whatever the cells contain. The universal gate of vanilla PLONK is the special case of §2.5.

4.1.2 The incomplete-addition gate

The incomplete-addition gate of the Action circuit constrains $R = P + Q$ for affine points $P = (x_p, y_p)$ and $Q = (x_q, y_q)$ of Pallas with $x_p \neq x_q$. It has one selector q_{add} and four equality-enabled advice columns x_p, y_p, x_{qr} and y_{qr}; it reads x_p, y_p, x_q, y_q at the current row and x_r, y_r at the next row of the *same* two columns x_{qr}, y_{qr}, so that Q occupies the selector row and R the row below it. Its two constraints are

$$q_{\text{add}} \cdot ((x_r + x_q + x_p)(x_p - x_q)^2 - (y_p - y_q)^2) = 0, \tag{31}$$

$$q_{\text{add}} \cdot ((y_r + y_q)(x_p - x_q) - (y_p - y_q)(x_q - x_r)) = 0 \tag{32}$$

(the halo2 book's *Incomplete and complete addition* chapter).

Theorem 4.1 (The gate is the chord formula with the slope eliminated)**.** *Let $P = (x_p, y_p)$ and $Q = (x_q, y_q)$ be affine points of a short Weierstrass curve with $x_p \neq x_q$, and set $\lambda := (y_p - y_q)/(x_p - x_q)$. For a pair $(x_r, y_r) \in \mathbb{F}^2$ the following are equivalent:*

1. *the bracketed polynomials of* (31) *and* (32) *both vanish;*
2. *$x_r + x_q + x_p = \lambda^2$ and $y_r + y_q = \lambda\,(x_q - x_r)$;*
3. *$(x_r, y_r) = P + Q$ by the chord formulas of the* Math Guide (*§"Explicit affine formulas"*), *$x_r = \lambda^2 - x_p - x_q$ and $y_r = \lambda\,(x_p - x_r) - y_p$.*

Proof. (i) $\iff$ (ii). Since $x_p - x_q \neq 0$, dividing the first bracket by $(x_p - x_q)^2$ and the second by $(x_p - x_q)$ are invertible operations: the first bracket vanishes exactly when $x_r + x_q + x_p = (y_p - y_q)^2/(x_p - x_q)^2 = \lambda^2$, and the second exactly when $y_r + y_q = \lambda\,(x_q - x_r)$. The gate stores the cleared forms because a constraint must be a polynomial in the cells, and a division is not.

(ii) $\iff$ (iii). The x-equations are the same equation rearranged. For the y-equations, the chord formula uses the slope of the line through P and Q, which is λ whichever endpoint is subtracted from which, and the identity $\lambda\,(x_p - x_q) = y_p - y_q$ gives $\lambda x_p - y_p = \lambda x_q - y_q$; hence $\lambda\,(x_p - x_r) - y_p = \lambda x_q - y_q - \lambda x_r = \lambda\,(x_q - x_r) - y_q$, which is (ii)'s y-equation with y_q moved across. □

On the 97-point toy curve of §3.7, $y^2 = x^3 + 3$ over $\mathbb{F}_{79}$, the chord formulas give $P + Q = (58, 71)$ for $P = (12, 25)$ and $Q = (71, 53)$, and both brackets are 0 at $(x_r, y_r) = (58, 71)$. On Pallas, with $P = (-1, 2)$ and $Q = [2]P$, both brackets are again 0 at the chord sum.

Degrees. The bracket of (31) has degree three in the cells and that of (32) degree two; with a selector column taking values in $\{0, 1\}$ the constraints have degrees four and three. The deployed key combines selectors into shared fixed columns (the halo2 book's *Selector combining* chapter), which can raise both degrees, keeping them within the deployed bound $d_{\max} = 9$.

Layout. The selector q_{add} is 1 on row i, which holds P and Q, each coordinate copy-constrained to the cell it came from; row $i + 1$ holds R (Table 9). The slope λ occupies no cell: the identities (31)–(32) verify the sum without it, which is what Theorem 4.1 says. By that theorem the two constraints force $(x_r, y_r) = P + Q$, a point of the curve, and $R \neq \mathcal{O}$ because $x_p \neq x_q$ excludes $Q = -P$.

row	x_p	y_p	x_{qr}	y_{qr}	q_{add}
i	x_p	y_p	x_q	y_q	1
$i+1$	–	–	x_r	y_r	–

Table 9: The cells of one incomplete addition: two rows of four advice columns and the selector. The four input cells x_p, y_p, x_q, y_q on the selector row i are copy-constrained to the cells they came from; the result cells x_r, y_r occupy the x_{qr}, y_{qr} columns of row $i+1$, where the gate's next-rotation queries find them. Dashes mark cells the gate does not use. The gate does not read q_{add} on row $i+1$. When additions are chained, as in fixed-base scalar multiplication, row $i+1$ is the selector row of the next addition, and R is that addition's Q.

The excluded inputs. If $Q = P$, both brackets vanish identically, since every term carries a factor $x_p - x_q$ or $y_p - y_q$, and (x_r, y_r) is unconstrained. If $Q = -P \neq \mathcal{O}$, the bracket of (31) reduces to $-(2y_p)^2$, which is nonzero because Pallas, of prime order, has no point of order two, and no assignment satisfies the gate. The Action circuit applies this gate only to non-identity inputs with $x_p \neq x_q$: it sums the window multiples of fixed-base scalar multiplication for every window but the last, an offset in the window table excluding the identity and the doubling case there, and adds the last window, whose sum can be the identity or a doubling, with the complete-addition gate (the halo2 book's *Fixed-base scalar multiplication* chapter).

4.1.3 Lookup arguments of the Action circuit

A gate asserts only polynomial identities. "This value has ten bits" and "this point is the j-th Sinsemilla generator" are set memberships, and §2.7 showed why no low-degree gate on the cell alone states them; the circuit states them as lookup arguments. Each is one instance of Construction 2.14, the input expressions playing $A_0, \ldots, A_{m-1}$ and the table columns $S_0, \ldots, S_{m-1}$, compressed by the challenge θ as in (10) whenever $m > 1$.

The table. The Action circuit has three lookup arguments, one range check and two Sinsemilla generator lookups, and all three use one table. It holds, for $0 \leq j < 2^{10}$,

$$\text{row } j: \quad (j,\ X(S[j]),\ Y(S[j])), \qquad S[j] := \texttt{GroupHash}(\texttt{"z.cash:SinsemillaS"},\ \mathrm{LE}_{32}(j)),$$

in three table columns $(t_{\text{idx}}, t_x, t_y)$, where $\mathrm{LE}_\ell(v)$ is the ℓ-bit little-endian encoding of the integer v, the generator table S is defined by the protocol specification (§ 5.4.1.9, "Sinsemilla Hash Function"), and the hash GroupHash that derives it is the *Crypto Guide*'s (§"Sinsemilla: an algebraic hash-based commitment"; the hash-to-curve pipeline is §"Hashing to a curve point"). The first two rows are:

row	coordinates of $S[j]$, hexadecimal
$j = 0$	$X =$ `0db5218b e6881f0f 1431d4ea 7d4afc7b 29a05baf bede62b5 5a91eb91 2044ea5f`
	$Y =$ `2f0f40c2 f152a01c 9caf6629 8493d5d0 944a041c 2e65ba01 17c24f76 bf8e6483`
$j = 1$	$X =$ `211112b4 b3e19518 c8fd33eb 39175404 e67700ca 24649b8f ce4ae33e a108af91`
	$Y =$ `06c59939 93adb03b a38a3e79 cd5a35fe b43c744a 670e19bc 1d83c293 f810c5ee`

Unused table rows. On every active row $j \geq 2^{10}$ (Remark 2.17), each of t_{idx}, t_x and t_y holds its row-0 value, so that every active row of the table is a genuine entry, $(0, X(S[0]), Y(S[0]))$ or $(j, X(S[j]), Y(S[j]))$. Rows left at 0 would add the tuple $(0, 0, 0)$, which is no entry, since row 0 is the only entry with index 0 and $X(S[0]) \neq 0$, and which an input tuple could then match.

The range check. The range-check lookup is

$$q_{\text{lookup}} \cdot \left(q_{\text{running}} \cdot (z_i - 2^{10} z_{i+1}) + (1 - q_{\text{running}}) \cdot z_i\right) \longmapsto t_{\text{idx}}, \tag{33}$$

with z_i and z_{i+1} the running-sum advice column at the current and next rotations. With $q_{\text{running}} = 1$ the input is a running-sum step: the prover witnesses $z_{i+1} = (z_i - a_i) \cdot 2^{-10}$ on successive rows, so the expression recovers the limb $a_i = z_i - 2^{10} z_{i+1}$, and membership in $t_{\text{idx}} = \{0, \dots, 2^{10} - 1\}$ asserts that each limb has ten bits. With $q_{\text{running}} = 0$ the cell itself is looked up. Every range check of the Action circuit is either an instance of this lookup, a limb of fewer than ten bits being looked up together with its shift to ten bits, or, for a range of $R \leq 8$ values, the gate $q \cdot w(w-1) \cdots (w - (R-1)) = 0$; the three-bit windows of fixed-base scalar multiplication and the single bits use the gate (the halo2 book's *Decomposition* chapter).

The Sinsemilla generator lookup. The Sinsemilla lookup is the three-column instance

$$\left(q_{S1} \cdot m_{i+1},\ \ q_{S1} \cdot x_p + (1 - q_{S1}) \cdot X(S[0]),\ \ q_{S1} \cdot y_p + (1 - q_{S1}) \cdot Y(S[0]) \right) \longmapsto (t_{\text{idx}}, t_x, t_y), \tag{34}$$

where $m_{i+1} = z_i - 2^{10} \cdot q_{\text{run}} \cdot z_{i+1}$ recovers the ten-bit message word from the hash's own running-sum decomposition of its message, and $q_{\text{run}} := q_{S2} - q_{S2}(q_{S2} - 1) = q_{S2}(2 - q_{S2})$ is an expression in a fixed column $q_{S2} \in \{0, 1, 2\}$ that evaluates to 1 on the interior rows of a message piece and to 0 on its final row, where the trailing running sum is implicitly zero. The coordinate y_p is not a cell but a polynomial expression in the queried cells equal to the y-coordinate of the generator added (the halo2 book's *Sinsemilla* chapter).

Expression-valued inputs. Lookup inputs may be arbitrary expressions over queried cells, not merely cells. The Sinsemilla lookup asserts that the generator added is $S[m_{i+1}]$ with no advice cell holding that generator's y-coordinate. The $(1 - q_{S1})$ branches set the input tuple of every row with $q_{S1} = 0$ to table row 0, so the assertion of Construction 2.14, that every active row's input tuple appears in the table, holds with nothing to exempt.

4.1.4 Column polynomials and the quotient

Every column, advice, fixed and instance, is a vector of $n = 2^k$ field elements, and its polynomial is its interpolant over the rows, of degree $< n$: the column polynomials of §2.5. The trailing rows of each advice column hold uniformly random field elements, the blinding rows of §4.6, on which every selector is zero and which Remark 2.17 exempts from the permutation and lookup arguments.

The quotient. Each gate polynomial is evaluated with the committed column polynomials substituted for the queried cells, a query at rotation r reading $v(\omega^r X)$, and the resulting polynomials, together with the permutation identities (7) and the lookup identities (9), are folded with powers of a challenge y and divided by $Z_H(X) = X^n - 1$. The result is the quotient $h(X)$ of phase 4 of "The complete Halo 2 protocol" (§4.5), where the folding and the division are written out (the halo2 book's *Vanishing argument* chapter).

The running example. The running example of §2.2 stands at the same point: it interpolated a four-row trace into a, b, c, the five selector polynomials and PI, and assembled the one polynomial $G(X)$ of degree 9 whose divisibility by $Z_H = X^4 - 1$, with the quotient t of degree 5, the rest of the protocol certifies.

The deployed parameters. For the Action circuit $k = 11$, $n = 2^{11} = 2048$ and $d_{\max} = 9$, the bound holding for its gates and for its permutation and lookup identities. The quotient therefore splits into $d_{\max} - 1 = 8$ chunks of degree below n, derived in §4.5. The two arguments this construction leans on are the ones §2 proved: the permutation argument of §2.6 stands behind every copy constraint, and the lookup argument of §2.8 behind the lookups (33) and (34).

4.2 Instance columns and binding the public statement

Definition 2.7 lists instance columns beside fixed and advice columns, and the arguments of §2 treat them like any other column. This subsection states how the instance column binds a proof to its public statement.

The public input of the running example. In the running example the target 35 is public, so it is not advice but *instance* data: it sits in the public-input polynomial $\mathrm{PI}(X)$ of (2), which the verifier builds himself from the column $(0, 0, 0, -35)$; its interpolant $64+50X+33X^2+47X^3$ of §2.2 takes the value $-35 = 62$ on row 3 and 0 elsewhere. A prover who claimed that $X^3 + X + 5 = 36$ has a root would face a *different* G, one with -36 in place of -35, and her honest trace would no longer vanish on the rows: the fourth gate would read $30+5-36 \neq 0$. The public input enters the polynomial G whose divisibility Theorem 2.5 tests, and an accepted proof attests the statement for the verifier's public values.

Definition 4.2 (Instance polynomial)**.** An *instance column* is a public-input column: its n entries $\boldsymbol{\pi} = (\pi_0, \dots, \pi_{n-1}) \in \mathbb{F}^n$ are not part of the witness, and both parties agree on them before verification. Like every other column it is interpolated over the rows in the Lagrange basis,

$$\pi(X) := \sum_{j=0}^{n-1} \pi_j \, \ell_j(X) \in \mathbb{F}[X], \qquad \deg \pi < n,$$

with $\ell_j(\omega^j) = 1$ and $\ell_j(\omega^{j'}) = 0$ for $j' \neq j$ (§2.5; the *Math Guide*, §"The Lagrange basis on H"). The public values occupy designated rows, and every other π_j is 0.

Binding by copy constraints. The deployed circuit binds the public values through the permutation argument of §2.6, not through a gate. The instance column is equality-enabled, so it participates in the permutation σ; it is one of the fifteen columns counted under "The deployed chunking" there, beside the ten advice and four fixed ones. Each designated instance cell is copy-constrained to an advice cell that the circuit's constraints read (the halo2 book's *Permutation argument* chapter for the mechanism), so a satisfying assignment exists only when those advice cells hold the public values the verifier supplied (Figure 10). The alternative, a dedicated gate $q(X)(a(X) - \pi(X)) \equiv 0 \pmod{Z_H(X)}$ with a selector q active exactly on the designated rows, is not used: the copy constraint reuses the permutation argument, at no extra gate degree.

The instance commitment. The verifier does not take $\pi(X)$ from the prover. *Both* parties compute the commitment to each instance polynomial from the instance values (the halo2 book's *Circuit commitments* chapter), with the public blinding factor 1 in place of a random one so that the two computations agree, and absorb it into the transcript before the first challenge is drawn, in phase 1 of §4.5; every Fiat–Shamir challenge thereby depends on the public input. The book fixes the factor 1 for the fixed columns only; for the instance columns it is the choice of the implementation, `halo2_proofs`. Thereafter the instance column is treated like any other: its claimed evaluation $\pi(x)$ appears among the phase-5 evaluations, enters the permutation identities as one

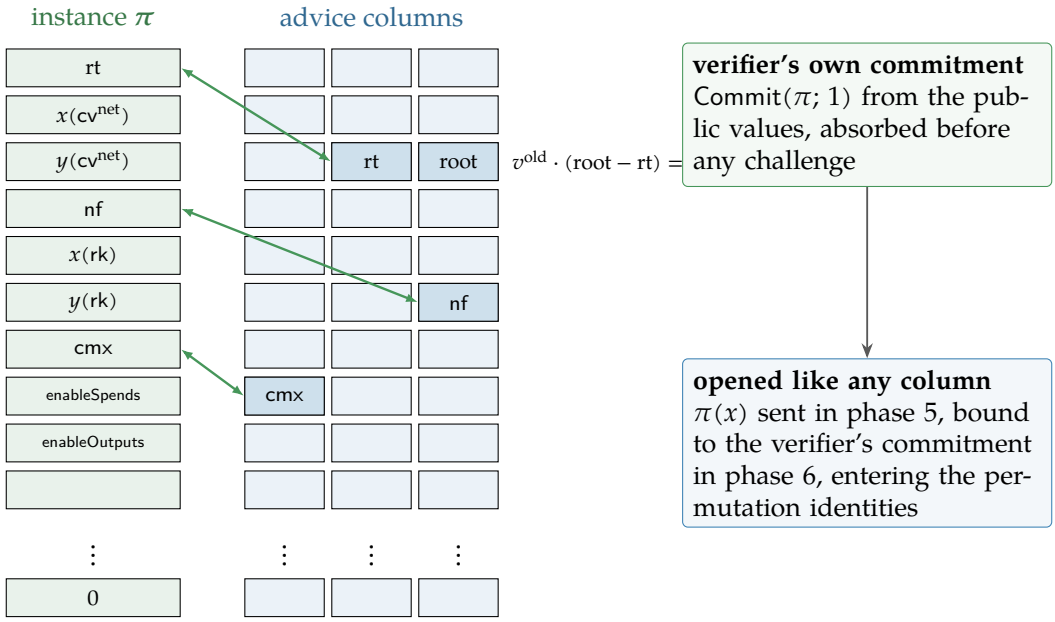


Figure 10: The instance column of the Action circuit. Rows 0 to 8 hold the primary input in the encoding of the protocol specification (§ 4.18.4), the curve points $\mathsf{cv}^{\mathsf{net}}$ and rk as two coordinates each; the further input disableCrossAddress of § 4.6 occupies a row that no normative text fixes, drawn empty; every other row is 0. Each designated cell is copy-constrained to an advice cell the circuit's constraints read; three such constraints are drawn (green double arrows): nf and cmx are copied to the cells where the circuit computes them, rt to a cell that a gate ties to the computed Merkle root. The verifier commits to the column himself, with the fixed blinding factor 1, and absorbs that commitment before the first challenge.

of the v_i of Construction 2.10, and the multipoint opening of phase 6 binds it to the instance commitment. Because the verifier computed that commitment himself, from the instance values he verifies against, the prover has no freedom over π: an evaluation inconsistent with the verifier's own values fails the opening argument, by the evaluation binding recalled in §2.9.

The public input of the Action statement. The Action statement's public input is the tuple $(\mathsf{rt}, \mathsf{cv}^{\mathsf{net}}, \mathsf{nf}, \mathsf{rk}, \mathsf{cmx}, \mathsf{enableSpends}, \mathsf{enableOutputs})$ (protocol specification, § 4.18.4, "Action Statement (Orchard)"): the anchor rt, the root of the note-commitment tree the spent note is proved against (the *Crypto Guide*, §"Append-only and incrementally updatable trees"); the net value commitment $\mathsf{cv}^{\mathsf{net}}$; the nullifier nf; the randomised verification key rk; the extracted commitment $\mathsf{cmx} = \mathsf{Extract}_{\mathbb{P}}(\mathsf{cm})$ of the output note (the *Crypto Guide*, §"A shielded spend, end to end"); and the flags enableSpends and enableOutputs, which permit non-zero-valued spends and outputs. The primary-input note of the same section encodes the tuple as the sequence

$$[\mathsf{rt}, x(\mathsf{cv}^{\mathsf{net}}), y(\mathsf{cv}^{\mathsf{net}}), \mathsf{nf}, x(\mathsf{rk}), y(\mathsf{rk}), \mathsf{cmx}, \mathsf{enableSpends}, \mathsf{enableOutputs}] \in \mathbb{F}^9,$$

each integer read as an element of $\mathbb{F} = \mathbb{F}_{p_{\mathsf{Pallas}}}$. The Action description specifies the proof with an eighth component, $\mathsf{disableCrossAddress} := 1 - \mathsf{enableCrossAddress}$, where enableCrossAddress permits the output note's receiver to differ from the input note's (protocol specification, § 4.6, "Action Descriptions"); from NU6.3 the proof of every Action of either pool is verified with this further public input, which is 1 for every Orchard-pool Action (ZIP 258). Rows 0 to 8 of the instance column hold the nine values above; no normative text fixes the row of the further input, and every other row is 0 (Figure 10). Each is copy-constrained to an advice cell the circuit's constraints read: $\mathsf{cv}^{\mathsf{net}}$, nf, rk and cmx to the cells where the circuit computes them; rt to a cell tied to the computed Merkle root by the gate $v^{\mathrm{old}}(\mathrm{root} - \mathsf{rt}) = 0$, which encodes the statement's condition that $v^{\mathrm{old}} = 0$ or the Merkle path is valid; the two enable flags to cells read by the gates $v^{\mathrm{old}}(1 - \mathsf{enableSpends}) = 0$ and $v^{\mathrm{new}}(1 - \mathsf{enableOutputs}) = 0$, which encode, for flags in $\{0, 1\}$, the conditions $v^{\mathrm{old}} = 0$ or $\mathsf{enableSpends} = 1$, and $v^{\mathrm{new}} = 0$ or $\mathsf{enableOutputs} = 1$; and disableCrossAddress to a cell read by a gate that, when it is nonzero, equates the coordinates of the receivers $(\mathsf{g_d}, \mathsf{pk_d})$ of the two notes. That condition is the meaning § 4.6 gives the flag; the list of conditions in § 4.18.4 does not yet include it.

Binding to the instance. The verifier computes the instance commitment from the public values π itself, and the copy constraints tie the designated cells to the advice cells the constraints read; so a witness extracted from an accepting proof satisfies the relation at π, not at other public values π' the prover may target. The probabilistic conclusion, that a prover who knows no witness for π, whatever witnesses she holds for other instances, convinces the verifier with at most negligible probability, holds under the knowledge soundness whose status §5 classifies: designed-but-unspecified for the deployed transcript, resting on the discrete logarithm on Vesta or on the algebraic group model, with the transcript hash modelled as a random oracle. Under that hypothesis an Action proof that verifies against an anchor, a nullifier and a value commitment certifies knowledge of a witness consistent with those values.

4.3 Removing the verifier: the Fiat–Shamir transform

The protocols of §§2–3 are public-coin: every verifier message is uniformly random and independent of the prover's messages, a field element (the point z of the random-point check, the pair β, γ of the permutation and lookup arguments, the challenges u_j of each opening round) or, in Construction 3.2, the group element U; and the verifier's decision is a deterministic function of the public input and the transcript. Their soundness requires only that each challenge be drawn after the commitments it tests are fixed.

The transform. The Fiat–Shamir transform replaces the i-th challenge c_i by

$$c_i := H(\mathsf{x}, m_1, c_1, \dots, c_{i-1}, m_i),$$

the hash of the public statement x and of the transcript prefix, the prover's messages $m_1, \dots, m_i$ and the earlier challenges, under a fixed hash function H (the *Crypto Guide*, §"The Fiat–Shamir transform: from interactive to non-interactive"). The ordering of §2.9 is enforced because each commitment is an input to the hash that derives the challenge testing it. The toy of §2.4 drew $z = 20$ as an interactive challenge; the deployed prover derives every challenge from a hash. The two kinds of challenge are kept apart in what follows: the interactive one is uniform by assumption, the hashed one only in the model in which the hash is a random oracle.

Grinding. A prover against the compiled argument may vary the inputs of H and inspect the resulting challenges. The bound on that freedom uses round-by-round soundness (Canetti, Chen, Holmgren, Lombardi, Rothblum, Rothblum and Wichs, *Fiat–Shamir: From Practice to Theory*), which Remark 4.11 recalls.

> **Proposition 4.3** (Fiat–Shamir of a round-by-round sound protocol)**.** *Let Π be a public-coin interactive protocol whose partial transcripts for a false statement carry a* doomed *labelling with error ε: the empty transcript is doomed, a doomed complete transcript is rejected, and from a doomed partial transcript, whatever the prover's next message, the next challenge leaves it doomed except with probability at most ε. With H modelled as a random oracle, a prover making at most Q distinct oracle queries, the verifier's recomputations counted among them, makes the verifier of the Fiat–Shamir transform of Π accept with probability at most $Q\varepsilon$.*

Proof. Each query is a transcript prefix followed by a prover message, and its answer is uniform and independent of everything before the query. The verifier accepts only a transcript that is not doomed, and the empty transcript is doomed, so acceptance requires some query on a doomed prefix whose answer leaves the doomed set; each query does so with probability at most ε, and the union bound over the at most Q queries gives $Q\varepsilon$. □

The proposition has the form of the halo2 book's extraction theorem (*Protocol Description* chapter, "Witness-extended Emulation", its term $q\epsilon$) and of the state-restoration analysis of Ghoshal and Tessaro that §5 cites, where its application to the deployed transcript is classified. For the toy's identity check, a single challenge with error $11/97$ (§2.4), ten queries exhaust the bound. This volume establishes no round-by-round error for Halo 2, whose binding is computational. At Orchard scale the identity check's term is about 2^{-240} (one Schwartz–Zippel term, in the sense §1.2 fixed), and budgets of 2^{40}, 2^{64} and 2^{80} queries multiply that term alone to 2^{-200}, 2^{-176} and 2^{-160}; these agree with the $Q\varepsilon_{\mathrm{bad}}$ of the halo2 book's algebraic-group-model bound quoted in §5.2, proved for the abstract protocol, whose transfer to the deployed transcript §5 classifies. The permutation argument's term over (β, γ), $3N/|\mathbb{F}|$ with $N = 15 \cdot 2042$ participating cells (Corollary 2.12), is larger, about $2^{-237.5}$.

The hash must take the whole transcript prefix as input: a statement or commitment left out can be chosen after the challenge is seen, the "weak Fiat–Shamir" failure the *Crypto Guide* records in §"The Fiat–Shamir transform: from interactive to non-interactive". The bound holds with H modelled as a random oracle, a heuristic. The *Crypto Guide* proves the one-round theorem, with its forking-lemma extractor and its programmed-oracle simulator, and states that a multi-round protocol such as this one needs a multi-round theorem whose hypotheses match the protocol at hand; §5 weighs that theorem against the deployed transcript.

Non-interactivity. The compiled argument is non-interactive: the proof is one static object, commitments, evaluations and the opening's scalars, and verification recomputes the challenges

by the same hash and runs the same checks. The protocol specification defines a Halo 2 proof as a byte sequence (protocol specification, § 5.4.10.3, "Halo 2", and its "Encoding of Halo 2 Proofs"), so a full node verifies an Action proof from the transaction alone.

The deployed transcript. Neither the protocol specification nor the halo2 book fixes the transcript that instantiates the transform; it is that of the implementation, `halo2_proofs`, and is as follows. The state is a BLAKE2b-512 hash state with the sixteen-byte personalisation `Halo2-Transcript` (the *Crypto Guide*, §"Domain separation and personalisation"). Absorbing a point $P = (x, y) \neq \mathcal{O}$ of Vesta appends the bytes $\texttt{0x01} \,\|\, \mathrm{LE}_{256}(x) \,\|\, \mathrm{LE}_{256}(y)$; the identity is not absorbable. Absorbing a scalar $s \in \mathbb{F}$ appends $\texttt{0x02} \,\|\, \mathrm{LE}_{256}(s)$. Squeezing a challenge appends `0x00` to the state, which keeps it, and returns the 64-byte digest of the state read as a little-endian integer and reduced modulo p_{Pallas}, with no rejection of any value (§3.8). The proof carries each point as its 32-byte compressed encoding, the x-coordinate with the parity of y in the top bit (protocol specification, § 5.4.9.6, "Pallas and Vesta"), and each scalar as $\mathrm{LE}_{256}(s)$. Before any prover message both parties absorb, as a scalar, a digest of the verifying key, its BLAKE2b hash reduced into $\mathbb{F}$, and then the instance commitments of §4.2, which therefore never appear in the proof. Every prover message is then absorbed in the order the protocol fixes, and each challenge is squeezed at its prescribed point, after everything it must depend on. The proof is the sequence of the prover's messages in the phase order of Table 10, counted by kind in Table 8; the verifier replays the absorptions and recomputes every challenge. Figure 12 in §4.5 draws the timeline.

4.4 The multipoint opening argument

When the verifier has drawn his evaluation point x, the prover has committed to several dozen polynomials and claimed their values at x and at rotations of x: a running product at x and ωx, a permuted lookup input at x and $\omega^{-1}x$, an advice column at every rotation its gates read. Opening each claim with the argument of §3 would cost $2k + 1 = 23$ points per claim, and a one-Action proof makes 95 claims about 69 committed polynomials, the 94 evaluations at x and its rotations in Table 8 and the quotient's ($49a + 46$ claims for a Actions). Halo 2 reduces all of them to a single opening of a single polynomial at a single point. The reduction combines random linear combinations, division by vanishing polynomials and one random-point check. The letters q and f below follow the halo2 book's *Multipoint opening argument* chapter and are unrelated to the selector polynomials $q_\bullet$ of §2, which are among the polynomials being folded; nor is this f the final blinder f of the deployed opening (§3.8), which phase 7 sends. The book's *Protocol Description* chapter writes the values $q_i(x_3)$ as u, the letter this volume reserves for the round challenges of the inner-product argument; they are written $\bar{q}_i$ here. The index i ranges over sets of query points, not over columns.

Construction 4.4 (Multipoint opening)**.** Let $p_1, \dots, p_N$ be polynomials of degree $< n$ with commitments $C_1, \dots, C_N$ under Construction 3.2, each p_ν queried at a finite set $T_\nu \subset \mathbb{F}$ of points, and let $\hat{p}_\nu(t)$ for $t \in T_\nu$ be the values the prover has claimed. Let $T^{(1)}, \dots, T^{(m)}$ be the distinct sets among the T_ν and $I_i := \{\nu : T_\nu = T^{(i)}\}$ the i-th group, enumerated as $\nu_{i,0}, \nu_{i,1}, \dots$; write $Z_i(X) := \prod_{t \in T^{(i)}} (X - t)$.

1. **Fold each group (x_1).** The verifier sends $x_1 \in \mathbb{F}$. For each group i the prover forms $q_i(X) := \sum_e x_1^e \, p_{\nu_{i,e}}(X)$, the verifier forms the same combination of commitments, $Q_i := \sum_e [x_1^e] \, C_{\nu_{i,e}}$, which commits to q_i by linearity (*Crypto Guide*, §"Pedersen vector commitments"), and both form the combined claimed values $\hat{q}_i(t) := \sum_e x_1^e \, \hat{p}_{\nu_{i,e}}(t)$ for $t \in T^{(i)}$. Both interpolate $r_i(X)$, the unique polynomial of degree $< |T^{(i)}|$ with $r_i(t) = \hat{q}_i(t)$ on $T^{(i)}$ (the *Math Guide*,

§"Lagrange interpolation"); the verifier can do this from the claimed values alone.

2. **Divide out the points (x_2).** The verifier sends $x_2 \in \mathbb{F}$. The prover forms the quotients $f_i(X) := (q_i(X) - r_i(X))/Z_i(X)$, which are polynomials exactly when the combined values $\hat{q}_i(t)$ equal $q_i(t)$ for every $t \in T^{(i)}$, in particular when every claimed value is true, and combines them into
$$f(X) := \sum_{i=1}^{m} x_2^{i-1} f_i(X),$$
samples a fresh blinder r_f and sends $F := \mathsf{Commit}(f; r_f)$.
3. **A fresh point (x_3).** The verifier sends $x_3 \in \mathbb{F}$. The prover sends $\bar{q}_i := q_i(x_3)$ for $i = 1, \dots, m$.
4. **The value f must take.** The verifier computes
$$v_f := \sum_{i=1}^{m} x_2^{i-1} \frac{\bar{q}_i - r_i(x_3)}{Z_i(x_3)},$$
the value of $f(x_3)$ if every f_i is a polynomial and every $\bar{q}_i$ is honest; he can, since r_i and Z_i are his to evaluate, and he rejects if some $Z_i(x_3)$ is 0.
5. **One polynomial (x_4).** The verifier sends $x_4 \in \mathbb{F}$. Both parties form
$$f^*(X) := f(X) + \sum_{i=1}^{m} x_4^i q_i(X), \qquad F^* := F + \sum_{i=1}^{m} [x_4^i]\, Q_i, \qquad v^* := v_f + \sum_{i=1}^{m} x_4^i \bar{q}_i,$$
and run one opening of Construction 3.2 on the statement (F^*, x_3, v^*), the prover's witness being the coefficient vector of f^* and the blinder $r_f + \sum_i x_4^i r_{Q_i}$, where r_{Q_i} is the blinder of Q_i, the x_1-combination of the blinders of the polynomials in group i. The verifier accepts the original claims if and only if that opening accepts.

Two remarks on the shape. Nothing is sent between x_1 and x_2, so the deployed transcript squeezes them back to back; they are two challenges because they play two roles, one keeping the polynomials of a group linearly independent and one keeping the quotients of different groups independent. And the reduction is free for the verifier in group operations except for the combinations of commitments, which are a few scalar multiplications per committed polynomial; the one expensive step, the final multi-scalar multiplication, is paid once, inside the single opening. Figure 11 draws the data flow.

Completeness. If every claimed value is true, then $q_i - r_i$ vanishes on $T^{(i)}$ and is divisible by Z_i (the *Math Guide*, §"Roots and the factor theorem"), each f_i is a polynomial of degree $< n$, so is f, and $f(x_3) = v_f$ whenever no $Z_i(x_3)$ is 0; $f^*(x_3) = v^*$ then holds, and the honest opening of §3 accepts. The verifier's rejection when x_3 is one of the query points, an event of probability at most $|\bigcup_i T^{(i)}|/|\mathbb{F}|$, is the reduction's only completeness defect.

Lemma 4.5 (Soundness of the multipoint reduction)**.** *In Construction 4.4, let the committed polynomials p_ν and the claimed values be fixed before x_1 is drawn, let $\hat{f}$ be a polynomial of degree $< n$ committed in F, fixed before x_3 is drawn, let $x_1, \dots, x_4$ be independent and uniform in $\mathbb{F}$, and let $Z(X) := \prod_{t \in \bigcup_i T^{(i)}} (X - t)$. If some claimed value is false, $\hat{p}_{\nu_0}(t_0) \neq p_{\nu_0}(t_0)$ with ν_0 in group i_0, then the probability that the verifier computes v_f and the polynomial $\hat{f} + \sum_i x_4^i q_i$ committed in F^* takes the value v^* at x_3 is at most*
$$\frac{|I_{i_0}| + 2m + n - 1 + \deg Z}{|\mathbb{F}|}.$$

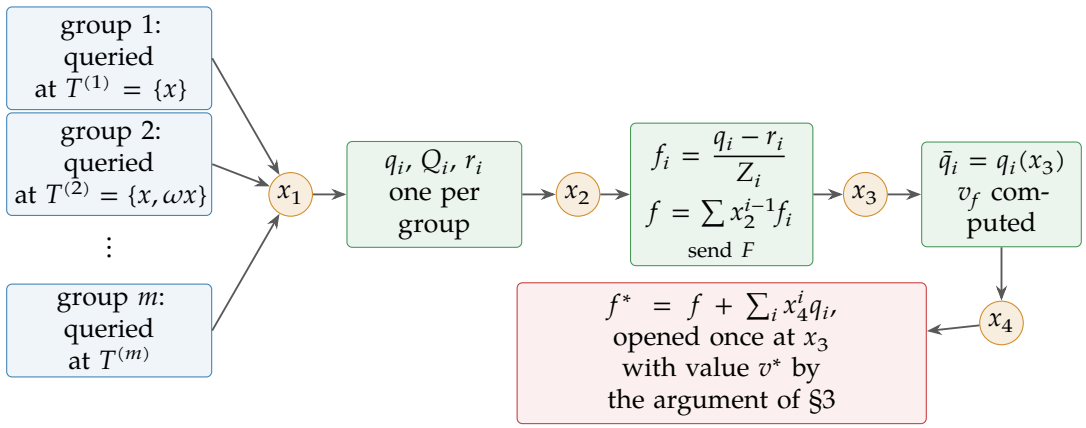


Figure 11: The multipoint reduction of Construction 4.4. Committed polynomials are grouped by the set of points at which they are queried (blue), folded within each group by x_1 into q_i, quotiented by the group's vanishing factor into f_i, combined across groups by x_2 into f, which is committed as F; after the fresh point x_3 the prover sends the $q_i(x_3)$, and x_4 folds f and the q_i into f^* (red), whose single opening at x_3 discharges every original claim. Challenges are amber.

Proof. The argument takes one Schwartz–Zippel term per challenge (the *Math Guide*, §"The Schwartz–Zippel lemma"). *The fold.* The difference $\hat{q}_{i_0}(t_0) - q_{i_0}(t_0) = \sum_e x_1^e (\hat{p}_{\nu_{i_0,e}}(t_0) - p_{\nu_{i_0,e}}(t_0))$ is a nonzero polynomial in x_1 of degree below $|I_{i_0}|$, fixed before x_1 is drawn, so except with probability $|I_{i_0}|/|\mathbb{F}|$ the folded claim is false: $r_{i_0}(t_0) \neq q_{i_0}(t_0)$. *The combination.* Each $T^{(i)}$ is contained in $\bigcup_{i'} T^{(i')}$, so each Z/Z_i is a polynomial. The polynomial

$$M(X) := \hat{f}(X)\, Z(X) - \sum_{i=1}^{m} x_2^{i-1} (q_i(X) - r_i(X)) \frac{Z(X)}{Z_i(X)}$$

is fixed once x_2 is drawn and F is sent, before x_3. At t_0 the term $\hat{f}(t_0)\, Z(t_0)$ and every term with $t_0 \notin T^{(i)}$ vanish, and

$$M(t_0) = - \sum_{i\,:\, t_0 \in T^{(i)}} x_2^{i-1} (q_i(t_0) - r_i(t_0)) \prod_{t \notin T^{(i)}} (t_0 - t),$$

the product over $t \in \bigcup_{i'} T^{(i')} \setminus T^{(i)}$: a polynomial in x_2 of degree below m, fixed before x_2 is drawn, whose coefficient at $x_2^{i_0-1}$ is nonzero. Except with probability $m/|\mathbb{F}|$ over x_2, then, $M(t_0) \neq 0$ and M is a nonzero polynomial of degree at most $n - 1 + \deg Z$. *The last fold.* Acceptance means $(\hat{f}(x_3) - v_f) + \sum_i x_4^i (q_i(x_3) - \bar{q}_i) = 0$, a polynomial equation in x_4 of degree at most m whose coefficients are fixed before x_4 is drawn. Unless every coefficient is zero, that is, unless $\hat{f}(x_3) = v_f$ and $\bar{q}_i = q_i(x_3)$ for every i, a uniform x_4 satisfies it with probability at most $m/|\mathbb{F}|$. *The fresh point.* If $\hat{f}(x_3) = v_f$ and every $\bar{q}_i = q_i(x_3)$, then, since v_f is computed only when $Z(x_3) \neq 0$, multiplying the first equation by $Z(x_3)$ gives $M(x_3) = 0$, which a uniform x_3 achieves for a nonzero M with probability at most $(n - 1 + \deg Z)/|\mathbb{F}|$. A union bound over the four terms gives the stated bound. □

The bound is one more Schwartz–Zippel allowance, of the order of $n/|\mathbb{F}|$. The opening's own knowledge error, $O(\log d/|\mathbb{F}|)$ by Corollary 3.13, lies below it, and that corollary's witness-or-relation form supplies the binding opening the lemma assumes.

The deployed instance. In the Action circuit the point sets are five. Every fixed, instance and permuted-label polynomial, the random polynomial of phase 4 and the collapsed quotient are queried at $\{x\}$; the advice columns, five at $\{x, \omega x\}$ and five at $\{\omega^{-1}x, x, \omega x\}$, according to the rotations their gates read; the permuted lookup input A' at $\{x, \omega^{-1}x\}$ and the permuted table S' at

$\{x\}$; the lookup running products and the last permutation running product at $\{x, \omega x\}$; and the first two permutation running products at $\{x, \omega x, \omega^{-6}x\}$, the last rotation reaching the boundary row of Remark 2.17 that chains one product into the next. The sets $\{x\}$, $\{x, \omega x\}$, $\{x, \omega^{-1}x\}$, $\{\omega^{-1}x, x, \omega x\}$ and $\{x, \omega x, \omega^{-6}x\}$ give the five values $\bar{q}_i$, the multipoint evaluations of Table 8, and the whole reduction adds one point and five scalars to the proof.

4.5 The complete Halo 2 protocol

The pieces are now assembled into the Halo 2 protocol (the halo2 book's *Protocol Description* chapter, instantiated with the arguments of its *Proving system* chapters), in seven phases. Throughout, $\mathbb{F}$ is the constraint field, $\mathbb{F}_{p_{\mathsf{Pallas}}}$ for Orchard; $\mathbb{G}$ is the commitment group, the Vesta curve, whose order is p_{Pallas} (§3.8); the rows are $\omega^0, \dots, \omega^{n-1}$ for a primitive n-th root of unity ω, with $n = 2^k$; and every commitment is $\mathsf{Commit}(p; r) = \langle \mathbf{p}, \mathbf{G} \rangle + [r]H$ of Construction 3.2, H the blinding generator. The protocol has a preprocessed setup and seven phases: (1) advice commitments; (2) lookup compression, challenge θ; (3) running products, challenges β, γ; (4) the quotient, challenge y; (5) the evaluation challenge x; (6) the multipoint opening, challenges x_1, x_2, x_3, x_4; (7) the inner-product opening. Every challenge is squeezed from the transcript of §4.3 at the point named; "the verifier sends" below means that. Table 10 lists the messages with their Orchard counts and Figure 12 draws the timeline.

Setup. The circuit fixes the columns, gates, lookup arguments and the equality permutation σ, as §4.1 described; its selectors and lookup tables are fixed columns. Key generation interpolates every fixed column and every permuted-label polynomial $s_i(X)$ of Construction 2.10 and commits to them with the fixed blinding factor 1 (for the fixed columns, the halo2 book's *Circuit commitments* chapter); these commitments, with the circuit description, form the verifying key. The transparent parameters of the commitment scheme, the n generators $\mathbf{G}$, the blinding generator H and the inner-product generator U hashed to the curve as §3.8 described, together with the domain data, are held separately. All of it is reproducible from public algorithms and the circuit; no secret setup trapdoor exists. Verification is a deterministic function of the parameters, the verifying key, the instance values and the proof.

Phase 1: advice commitments. Both parties first absorb the digest of the verifying key and the instance commitments of §4.2 into the transcript, so that every subsequent challenge depends on the circuit and on the public input. The prover constructs the advice columns $\mathbf{w}_1, \dots, \mathbf{w}_{n_a} \in \mathbb{F}^n$, which with the fixed and instance columns form a satisfying assignment in the sense of Definition 2.7, fills the reserved trailing rows of each with uniformly random field elements, the blinding rows of §4.6, interpolates each to $w_i(X)$ of degree $< n$, and sends $\mathsf{Commit}(w_i; r_i)$ with fresh blinders r_i. The verifier absorbs the commitments.

Phase 2: lookup compression, challenge θ. The verifier sends θ. For each lookup the prover compresses the input expressions and table columns by powers of θ as in (10), forms the permuted columns A' and S' of Construction 2.14, and sends their two commitments; the verifier absorbs them.

Phase 3: running products, challenges β, γ. The verifier sends β and γ. The prover commits to the permutation running products, one per chunk of $d_{\max} - 2$ columns as "The deployed chunking" in §2.6 described, and to one lookup running product Z_{lookup} per lookup, and sends the commitments; the verifier absorbs them. The order is not a convenience. The permuted columns A' and S' must be fixed *before* the challenges that randomise the products are drawn, since Theorem 2.15 bounds the prover's success probability only over challenges drawn after her columns; a prover who saw β, γ first could choose the permutation against them.

Phase 4: the quotient, challenge y. Before y is drawn the prover commits to a uniformly random polynomial of degree $< n$, sends the commitment, and the verifier absorbs it; its role is the zero-knowledge one explained in §4.6. The verifier then sends y. The prover assembles the *gate equation*, one polynomial

$$C(X) := \sum_{t=0}^{T-1} y^t g_t(X),$$

the y-weighted combination of the T constraint polynomials g_t of the circuit: each constraint of each custom gate with its selector factor, the permutation identities (7) and the chunk-boundary identities, the lookup identities (9), and the boundary identities of Remark 2.17, each masked by its active-row factor where that remark requires. If every constraint holds then every g_t vanishes on the rows, so C does, and $Z_H \mid C$ by (3): the *quotient* $h(X) := C(X)/Z_H(X) = C(X)/(X^n - 1)$ is a polynomial. If some constraint fails at some row, Lemma 4.6 bounds the probability that C nonetheless vanishes on the rows, since y is drawn after every polynomial in the g_t is committed and after θ, β and γ.

Lemma 4.6 (Batching by y)**.** *Let $g_0, \dots, g_{T-1} \in \mathbb{F}[X]$ be fixed, and let y be uniform in $\mathbb{F}$. If some g_t does not vanish on the rows, then $C = \sum_{t=0}^{T-1} y^t g_t$ vanishes on the rows with probability at most $(T-1)/|\mathbb{F}|$.*

Proof. Let $g_{t_0}(\omega^j) \neq 0$. Then $C(\omega^j) = \sum_t y^t g_t(\omega^j)$ is a polynomial in y of degree at most $T-1$ with the nonzero coefficient $g_{t_0}(\omega^j)$, so it vanishes for at most $T-1$ values of y (the *Math Guide*, §"The Schwartz–Zippel lemma"). □

Theorem 4.7 (Degree of the quotient)**.** *Let every column polynomial and every permuted-label polynomial s_i have degree $< n$, and every constraint polynomial degree at most $d_{\max}$, counting as degree one each column polynomial at a rotation, each s_i, each of ℓ_0, ℓ_{last} and ℓ_{blind}, and X (as in "The deployed chunking" of §2.6). Then*

$$\deg h \;\le\; d_{\max}(n-1) - n \;=\; (d_{\max} - 1)(n-1) - 1.$$

Proof. A constraint of degree at most $d_{\max}$ is a sum of products of at most $d_{\max}$ such factors, each of degree at most $n-1$ (rotations do not change degrees); so $\deg g_t \le d_{\max}(n-1)$ and $\deg C \le d_{\max}(n-1)$. Division by Z_H, of degree n, subtracts n. The two forms agree since $(d_{\max} - 1)(n-1) - 1 = d_{\max}(n-1) - (n-1) - 1 = d_{\max}(n-1) - n$. □

For the deployed $d_{\max} = 9$ and $n = 2048$ the bound is $\deg h \le 16375$, far above the degree bound $n - 1 = 2047$ that the commitment of Construction 3.2 supports with $d = n$ generators, so h cannot be committed as one polynomial. The prover therefore uses a fixed proof shape of $d_{\max} - 1$ *chunks* $h_0, \dots, h_{d_{\max}-2}$, each of degree $< n$, with

$$h(X) = \sum_{i=0}^{d_{\max}-2} X^{ni} h_i(X),$$

retaining leading zero chunks when the actual degree is smaller, and commits to each chunk separately with its own blinder; there are 8 chunks for Orchard. The count is right because h has at most $(d_{\max}-1)(n-1) = (d_{\max}-1)n - (d_{\max}-1) < (d_{\max}-1)n$ coefficients. Once the evaluation point x is drawn, the verifier recombines the chunk *commitments*, weighted by powers of x^n, into one commitment $\sum_i [x^{ni}]\,\mathsf{Commit}(h_i)$ to the polynomial $\sum_i x^{ni} h_i(X)$, whose value at x is $h(x)$; no chunk is ever opened on its own (the halo2 book's *Protocol Description* chapter, steps 5 to 8, and note 1 of its "Zero-knowledge and Completeness").

Phase 5: the evaluation challenge x. The verifier sends x. The prover evaluates every committed polynomial the verifier's identities query, *except* the quotient chunks, at x and at the rotated points the identities reference, ωx for the running products and for any gate reading the next row, $\omega^{-1}x$ for the permuted lookup input A' and for gates reading the previous row, and $\omega^{-6}x$, the deployed boundary row, for all but the last permutation product, and sends the claimed values as field elements. The deployed proof also carries the evaluations at x of the fixed columns and of the permuted-label polynomials, which the same opening binds to the verifying key's commitments, and of the random polynomial, bound to its phase-4 commitment. No evaluation of h is sent: the verifier computes the expected quotient value $C(x)/(x^n - 1)$ from the sent evaluations, evaluating ℓ_0, ℓ_{last}, ℓ_{blind} and Z_H at x himself, and enforces it against the collapsed chunk commitment inside the multipoint opening of phases 6 and 7, where the claimed evaluations themselves are bound to their commitments as well. The gate equation is thus checked as the identity $C(x) = h(x)\, Z_H(x)$, the deployed form of (5).

Phase 6: the multipoint opening, challenges $x_1, \dots, x_4$. The parties run Construction 4.4 on every committed polynomial with a claimed evaluation: the advice columns, the instance column, the fixed and permuted-label polynomials, the permuted lookup columns, the running products, the random polynomial, and the collapsed quotient with its expected value, the last two joining the $\{x\}$ group. The prover sends the commitment F after x_1, x_2 and the five values $\bar{q}_i$ after x_3; after x_4 both parties hold the statement (F^*, x_3, v^*).

Phase 7: the inner-product opening. The parties execute the deployed opening of §3.8, the masked argument in the normalisation of Remark 3.9, on (F^*, x_3, v^*). The prover sends the mask commitment S; the verifier sends ξ and ζ; then for $k = 11$ rounds the prover sends the pair (L_j, R_j) and the verifier answers with u_j; finally the prover sends the scalars c and f. The verifier accepts if and only if the deployed equation (29) holds with $P := F^*$, $v := v^*$ and $x := x_3$, its one multi-scalar multiplication of length n included. Its symmetric counterpart is the masked check (24), which differs from it by the three changes of that remark: the fold normalisation, a change of variables; the factor ζ on U; and the value carried as $+[v]U$ rather than $-[v]G_0$. Neither it nor (15) is the deployed equation. This is the proof's end; Table 8 counts its bytes by kind, and nothing else is sent.

Accumulation, not deployed. In the recursive variant of Halo 2 the verifier is itself a circuit, and that circuit defers the one expensive step of phase 7, the claim $G^{(0)} = \langle \mathbf{s}, \mathbf{G} \rangle = \mathsf{Commit}(g_{\text{dep}}; 0)$ of (29), the deployed form (Remark 3.9) of the claim $G^{(0)} = \mathsf{Commit}(g; 0)$ of §3.4, into an accumulator folded with the accumulators of earlier proofs; a single decider at the end verifies all accumulated claims. Orchard does not do this. The protocol specification uses Halo 2 only to prove and verify Action statements and composes no proofs (protocol specification, § 4.1.13, "Zero-Knowledge Proving System"), the Orchard design proposal states that it makes no use of the recursion (ZIP 224), and the verifier pays every deferred multiplication itself; several proofs may share it by the randomised batch check of Lemma 3.14. The variant is developed as the coda of §7.

Costs. The prover computes one multi-scalar multiplication of length n per commitment she sends before the opening, $22a + 10$ for a bundle of a Actions: per Action the ten advice columns, the six permuted lookup columns and the three permutation and three lookup products; shared, the random polynomial, the eight quotient chunks and the multipoint polynomial f. She also computes the a instance commitments, as the verifier does (§4.2). The opening adds one of length n, for its mask commitment S, and the $2k$ cross terms of its rounds, two in each, of lengths $n/2, n/4, \dots, 1$ in successive rounds. The verifier performs $O(a + \log n)$ field and group operations for a proof of a Actions: he commits to the a instance columns (at most ten nonzero values each),

recomputes the challenges, evaluates each Action's gate equation at x, combines the $23a + 46$ commitments of phase 6, and runs the k rounds of phase 7; to this he adds the single deferred multi-scalar multiplication of length n, which is the linear term that §3.8.1 discussed and that a batch verifier amortises across proofs with fresh random scalars; that batching is distinct from the accumulation above.

phase	prover sends	verifier squeezes	Orchard count
before 1	— (both absorb the key digest and the instance commitments)	—	0 bytes
1	advice commitments	—	10 points per Action
2	after θ, the A' and S' commitments, per lookup	θ	$3 + 3$ points per Action
3	after β, γ, the permutation products and the lookup products	β, γ	$3 + 3$ points per Action
4	random polynomial commitment; then, after y, the quotient chunks	y	$1 + 8$ points, shared
5	after x, the claimed evaluations at x and its rotations	x	49 scalars per Action; 45 shared
6	after x_1, x_2, the commitment F; then, after x_3, the values $\bar{q}_i$	x_1, x_2; x_3; then x_4	1 point, 5 scalars, shared
7	S; then, after ξ, ζ, the pairs (L_j, R_j) for $j = k, \ldots, 1$, each followed by u_j; then c, f	ξ, ζ; u_j after each pair	$1 + 22$ points, 2 scalars, shared

Table 10: The seven phases as absorb-and-squeeze events, with the Orchard counts of Table 8. Each challenge is listed in the phase that draws it, and the prover column states which messages follow it. The per-Action messages repeat for each of the a Actions of a bundle and the shared ones appear once, giving $2720 + 2272\,a$ bytes. The phase-5 scalars per Action are the instance evaluation, 25 advice evaluations, 8 permutation-product and 15 lookup evaluations; the shared 45 are 29 fixed, 15 permuted-label and 1 random-polynomial evaluation.

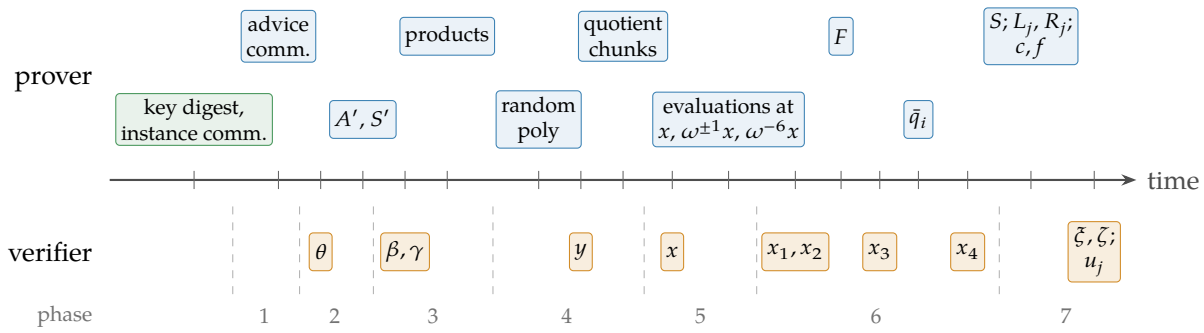


Figure 12: The transcript of §4.3 as a timeline. Prover messages (blue, staggered for legibility) are absorbed above the line; the data both parties hold in advance (green) is absorbed but never sent; challenges (amber) are squeezed below the line, each after everything it must depend on; dashed lines separate the phases of Table 10. The proof is the sequence of blue boxes read left to right.

4.6 Zero knowledge: hiding the witness in Halo 2

Knowledge soundness, classified in §5, concerns what a convincing prover knows; zero knowledge concerns what the proof reveals. Halo 2 is zero-knowledge through two devices, hiding commitments and blinded polynomials, assembled into a simulator. This subsection states what each device achieves on its own, and states the claim for the whole protocol as a cited proposition with its hypotheses.

Evaluations of the running example. The prover of the running example sends commitments and the evaluations $a(20), b(20), c(20), t(20)$ at the random point. An evaluation is a linear equation in the secret coefficients: $a(20) = 90 + 61 \cdot 20 + 22 \cdot 20^2 + 24 \cdot 20^3 = 59$ is one equation in the four unknowns $(90, 61, 22, 24)$, and evaluations at four distinct points determine the column, hence the secret witness 3. Halo 2 therefore blinds: the witness trace reserves independent random rows, and every commitment carries an independent random blinding term. Adding random coefficients to a column polynomial would not do: a column polynomial is determined by its n row values, and a random coefficient on X^i changes every row, breaking the gates; a random multiple of the Lagrange polynomial ℓ_j of a reserved row changes row j alone, which is what a blinding row is. The random values violate no constraint: every selector is zero on the reserved rows, so the gates hold there, and the permutation and lookup identities carry the active-row factor of Remark 2.17. Each column's openings are then uniform (Lemma 4.8); the joint statement is the proposition of "The simulator" below, whose simulator is one in the sense of the *Crypto Guide* (§"The simulation paradigm and zero knowledge").

Hiding commitments. The commitment of Construction 3.2 is *perfectly hiding*. The prover commits a polynomial p as

$$\mathsf{Commit}(p; r) = \langle \mathbf{p}, \mathbf{G} \rangle + [r]\, H, \qquad r \text{ uniform in } \mathbb{F},$$

with $\mathbf{p}$ the coefficient vector, $\mathbf{G}$ the generator vector, and H the blinding generator of §3, of no known discrete-logarithm relation to $\mathbf{G}$; the term $[r]H$ makes the commitment a uniform group element for every p, since H generates $\mathbb{G}$, so a commitment by itself leaks nothing about what was committed. This is the perfect hiding of the Pedersen commitment, proved in the *Crypto Guide* (§"The Pedersen commitment" and §"Hiding and binding") and recalled in §2.9. Every commitment the prover sends in phases 1 to 6, the advice columns, the permuted lookup columns A' and S', the permutation and lookup running products, the random polynomial, the quotient chunks and the multipoint polynomial f, has this form with a fresh r (the halo2 book's *Protocol Description* chapter, which blinds every group element of the protocol, its note 4). The verifying key's commitments and the instance commitments carry the fixed blinding factor 1 instead: they commit public data both parties recompute (§4.5, §4.2), so there is nothing to hide, and a random blinder would prevent the two computations from agreeing.

Blinding rows. The prover opens commitments, and an opening reveals evaluations. To keep those evaluations from revealing the witness, the last n_{blind} rows of the table, the *blinding rows*, carry no circuit data: no gate applies there, and each advice column holds uniformly random field elements on them. By Lemma 4.8 below, a column opened at m distinct points outside the rows needs $n_{\text{blind}} \geq m$. The halo2 book's *Protocol Description* chapter accordingly requires $n_e + 1$ random evaluations of a polynomial opened at n_e rotations of x, the extra one covering its opening at the point x_3 of the multipoint reduction. In the Action circuit a polynomial is opened at most four times: at three rotations of x, for some advice columns and the first two permutation products, and at x_3. The deployed circuit takes $n_{\text{blind}} = 5$, one more than that, a parameter fixed by the implementation, `halo2_proofs`; with the boundary row of Remark 2.17 its usable rows number $n - (n_{\text{blind}} + 1) = 2042$. The permutation and lookup running products are blinded the same way, their trailing rows random, and the remark's active-row factor exempts every one of those rows from every argument, so the random values do not violate a constraint (Figure 13; the halo2 book's *Permutation argument* and *Lookup argument* chapters, their "Zero-knowledge adjustment" sections).

Lemma 4.8 (Openings of a column with blinding rows)**.** *Let $v(X) = \sum_{j=0}^{n-1} v_j\, \ell_j(X)$ be a column polynomial over the rows $\omega^0, \ldots, \omega^{n-1}$, let B be a set of n_{blind} row indices, the blinding rows, and*

> *let $z_1, \dots, z_m \in \mathbb{F}$ be distinct points, none of them a row. With the values on the other rows held fixed, the map $(v_j)_{j \in B} \mapsto (v(z_1), \dots, v(z_m))$ from $\mathbb{F}^{n_{\text{blind}}}$ to $\mathbb{F}^m$ is affine, and its linear part has rank* $\min(m, n_{\text{blind}})$ (Math Guide, §*"Bases, dimension, and linear maps"*). *Consequently, if $m \le n_{\text{blind}}$ and the blinding values are uniform and independent, the m evaluations are jointly uniform on $\mathbb{F}^m$ and independent of the values on the other rows.*

Proof. The map is $v(z_i) = \sum_{j \notin B} v_j \, \ell_j(z_i) + \sum_{j \in B} v_j \, \ell_j(z_i)$, a constant plus the linear map with matrix $L := [\ell_j(z_i)]_{i \le m, j \in B}$. The Lagrange basis on the rows has the closed form $\ell_j(X) = \omega^j (X^n - 1)/(n(X - \omega^j))$ (the *Math Guide*, §"The Lagrange basis on H"), so $L = D_1 K D_2$ with $D_1 := \operatorname{diag}(z_i^n - 1)$, nonzero on the diagonal because no z_i is a row, $D_2 := \operatorname{diag}(\omega^j/n)$, nonzero because n is invertible in $\mathbb{F}$, and $K := [1/(z_i - \omega^j)]$. Hence $\operatorname{rank} L = \operatorname{rank} K$. Every square submatrix of K is nonsingular: take rows I and columns J with $|I| = |J| = s$, and suppose coefficients c_j, $j \in J$, not all zero, satisfy $\sum_{j \in J} c_j/(z_i - \omega^j) = 0$ for all $i \in I$. Over the common denominator, $\sum_{j \in J} c_j/(X - \omega^j) = N(X)/\prod_{j \in J}(X - \omega^j)$ with $N(X) := \sum_{j \in J} c_j \prod_{j' \in J, j' \ne j}(X - \omega^{j'})$ of degree $< s$; the polynomial N vanishes at the s distinct points z_i, so it is the zero polynomial (the *Math Guide*, §"Roots and the factor theorem"), and then $N(\omega^j) = c_j \prod_{j' \ne j}(\omega^j - \omega^{j'}) = 0$ forces $c_j = 0$ for every $j \in J$, since the rows are distinct: a contradiction. Thus K has an invertible square submatrix of size $\min(m, n_{\text{blind}})$, and its rank, at most $\min(m, n_{\text{blind}})$, equals $\min(m, n_{\text{blind}})$.

If $m \le n_{\text{blind}}$ the linear part is onto $\mathbb{F}^m$, so the affine map is onto, and every fibre is a coset of the same kernel, of the same size $|\mathbb{F}|^{n_{\text{blind}} - m}$. A uniform point of $\mathbb{F}^{n_{\text{blind}}}$ therefore lands in each fibre with the same probability: the image is uniform on $\mathbb{F}^m$. The constant term depends on the other rows but not the distribution, so the evaluations are independent of them. □

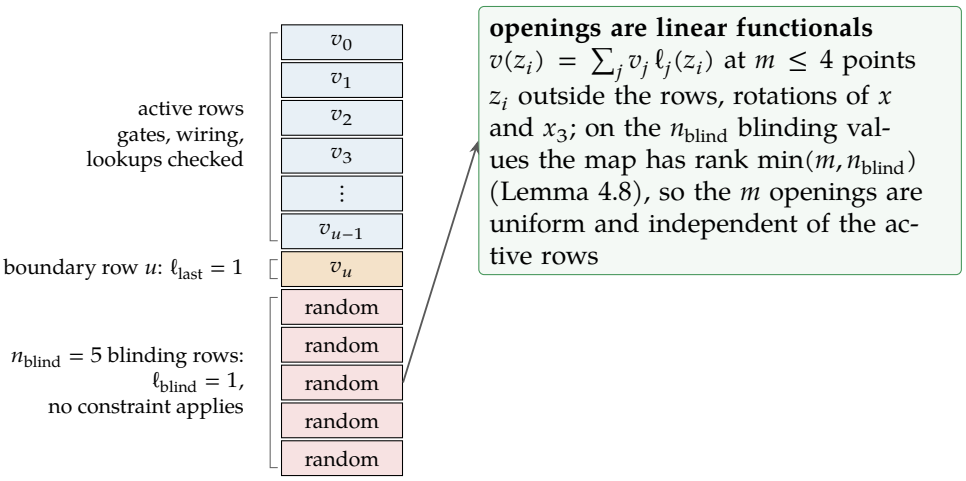


Figure 13: One column of the deployed table: the $u = n - 6 = 2042$ active rows $0, \dots, u - 1$ (blue) on which every argument is checked, the boundary row u selected by ℓ_{last} (amber) that pins the running products, and the $n_{\text{blind}} = 5$ blinding rows (red) selected by ℓ_{blind}, filled with uniform random values and exempted from every constraint by the active-row factor of Remark 2.17. Each opening of the column at a point outside the rows is a linear functional of all n row values, and Lemma 4.8 says the blinding rows alone make at most n_{blind} such openings jointly uniform.

On the toy domain $\{1, 22, 96, 75\}$ of §2.2, the rank of the lemma's map was computed for every choice of $m \le 3$ distinct points among the 93 outside it and for 2000 random choices of four: with $n_{\text{blind}} = 2$ blinding rows the rank is 1 for one point and 2 for two, and the map is onto; for three points the rank stays 2 and it is not; with $n_{\text{blind}} = 3$ the ranks are $1, 2, 3$ for one, two and three points, onto each time, and 3 for four points, no longer onto. The threshold is exact: the map is a bijection at $m = n_{\text{blind}}$ and fails first at $m = n_{\text{blind}} + 1$, so $n_{\text{blind}} \ge m$ is the condition, and the deployed $n_{\text{blind}} = 5$ exceeds the at most $m = 4$ openings of any column by one. One hypothesis of

the lemma is that the points lie outside the rows: an opening *at* a row would return a cell, witness or blinder, and the halo2 book's *Protocol Description* chapter accordingly excludes challenges in the domain (its note 5). The deployed transcript squeezes full field elements with no rejection, and Corollary 4.10 bounds the effect.

Scope of the lemma. It is a *marginal* statement about one column. The proof opens many related polynomials under shared challenges, and the joint distribution of all their evaluations, subject to the public relations among them, the gate equation, the running-product updates, the lookup identities, is not a consequence of the lemma applied column by column. The composition is addressed under "The simulator" below, and its status is stated there.

The quotient and the random polynomial. The quotient chunks admit no blinding-row treatment: C determines them entirely, and the rows of h mean nothing. Their commitment blinders, together with the blinding-row randomness they inherit through C, protect them instead, and their single collapsed opening at x is the publicly computable value $C(x)/(x^n - 1)$, which reveals nothing the verifier did not already hold. There is one more place the quotient could leak: the multipoint reduction evaluates its $\{x\}$-group polynomial q_1 at the fresh point x_3, and q_1 contains the collapsed quotient, whose value at x_3 is not a public function of anything. This is why phase 4 commits a uniformly random polynomial before y and opens it at x alongside the quotient: it joins the same group with the same x_1-weighting, so $q_1(x_3)$ carries a uniform summand, and the value $\bar{q}_1$ the prover sends is uniform whatever the quotient's value at x_3 is. The protocol commits that random polynomial for this reason (the halo2 book's *Protocol Description* chapter, step 3 and note 2).

Composition. In each phase the verifier sees only two kinds of object: hiding commitments, which are uniform group elements, and a bounded number of evaluations of the committed polynomials at the transcript's challenge points. The blinding rows supply randomness which, by Lemma 4.8, makes the openings of each column uniform on its own; a full composition proof must show that this randomness masks the *joint* evaluations subject to the public relations, and must account for the correlations among advice, products, quotient and multipoint polynomials. The final inner-product opening uses the deployed mask of §3.8, the random polynomial vanishing at x_3 whose commitment S opens phase 7, in addition to the commitment and round blinders, because Remark 3.7 showed that a final commitment blinder alone would not hide the folded coefficient; the opening of one polynomial was proved statistically honest-verifier zero knowledge there, and that proof is one ingredient of the composition, not the composition.

The simulator. Zero knowledge requires a simulator for the *complete joint transcript*, given only the public input. The usual proof samples suitably blinded commitments and evaluations satisfying the verifier's equations and then programs the random oracle so that the Fiat–Shamir challenges come out as required, in the manner of the *Crypto Guide*'s one-round simulator (§"The Fiat–Shamir transform: from interactive to non-interactive"). For the interactive protocol the halo2 book exhibits such a simulator and claims the following; this volume records the claim and does not reprove it.

> **Proposition 4.9** (Zero knowledge of the interactive protocol, cited)**.** *The interactive Halo 2 protocol is perfect special honest-verifier zero knowledge, the simulator being given the verifier's challenges in advance (the* Crypto Guide, *§"Sigma-protocols"), for challenge tuples in which every challenge is nonzero and outside the rows and every factor* $1 + u_\ell x_3^{2^{\ell-1}}$ *of* $g_{\mathrm{dep}}(x_3)$ *in* (26) *is nonzero (the halo2 book's* Protocol Description *chapter, "Zero-knowledge and Completeness"). The simulator commits*

> *to uniformly random polynomials in place of every polynomial the prover commits in phases 1 to 3 (the book's a_i: the advice columns, the permuted lookup columns and the running products), the quotient chunks and the multipoint polynomial, and uses its foreknowledge of ξ to choose the mask $s(X)$ so that $p + \xi s - v$ vanishes at x_3.*

> **Corollary 4.10** (Unrestricted challenges)**.** *Let every challenge be uniform in $\mathbb{F}$ with no value excluded, as the deployed transcript draws them. Then the transcripts of the interactive protocol and of the simulator of Proposition 4.9, run on the same challenges, are within statistical distance $N_{\mathrm{ch}}\,(n+2)/|\mathbb{F}|$, where $N_{\mathrm{ch}} = 22$ is the number of challenges, $\theta, \beta, \gamma, y, x, x_1, \ldots, x_4, \xi, \zeta$ and the $k = 11$ values u_ℓ; for $n = 2^{11}$ this is $45100/p_{\mathsf{Pallas}} \approx 2^{-238.5}$.*

Proof. Each challenge has at most $n + 2$ excluded values: 0, the n rows, and, for u_ℓ, the value that makes its factor of $g_{\mathrm{dep}}(x_3)$ zero. The challenges have the same distribution in both experiments; on the event that none takes an excluded value the two transcripts are identically distributed by Proposition 4.9, and by a union bound that event fails with probability at most $N_{\mathrm{ch}}\,(n + 2)/|\mathbb{F}|$, which therefore bounds the statistical distance. □

Perfect hiding of each commitment and Lemma 4.8 are ingredients of that argument, not by themselves a proof that the joint distribution matches a real execution. For the compiled argument, the *Crypto Guide*'s compilation recipe (Theorem "The compilation recipe, informal", §"SNARKs: succinct non-interactive arguments of knowledge") yields zero knowledge when the polynomial IOP is properly blinded and the corresponding simulation theorem applies; for a multi-round protocol that is a multi-round simulation theorem in the programmable random-oracle model, which this volume neither proves nor cites from a source that proves it for this protocol. Non-interactive zero knowledge of the deployed proof is therefore stated, not proved.

The three properties. Subject to the hypotheses just named, Halo 2 targets three guarantees for the relation $\mathcal{R}_C$ of a PLONKish circuit (Definition 2.7), the three that §1.2 demanded of a proof. *Overwhelming completeness*: an honest prover holding a satisfying witness convinces the verifier except on negligible transcript degeneracies, a zero inner-product challenge (§3.8, probability $1/|\mathbb{F}|$ per challenge), a point x_3 among the query points of the multipoint reduction ("Completeness" in §4.4), an evaluation point x on a row, where the verifier's division by $x^n - 1$ in phase 5 is undefined (probability $n/|\mathbb{F}|$), or challenges β, γ at which a running-product denominator factor vanishes at an earlier row than any numerator factor does (Remark 2.17; probability at most $N/|\mathbb{F}|$ for the permutation argument on N participating cells and $2u/|\mathbb{F}|$ per lookup on u active rows). *Knowledge soundness*: an efficient extractor, given rewindable access to any prover the verifier accepts with non-negligible probability, outputs a satisfying witness, so that an accepted proof certifies that one is *known*. Corollary 3.13 supplies the extractor for the opening layer; §5 lifts it to the non-interactive argument, whose challenges are transcript hashes, and classifies what that lifting establishes. *Zero knowledge*: a witness-free simulator reproduces the proof, exactly for restricted challenges (Proposition 4.9) and within the distance of Corollary 4.10 for the deployed ones. Knowledge soundness is the property of Action proofs that the Orchard balance and nullifier arguments consume, together with the binding signature (protocol specification, § 4.14, "Balance and Binding Signature (Orchard)") and the consensus nullifier rules (protocol specification, § 3.9, "Nullifier Sets"); zero knowledge is the property the privacy argument consumes.

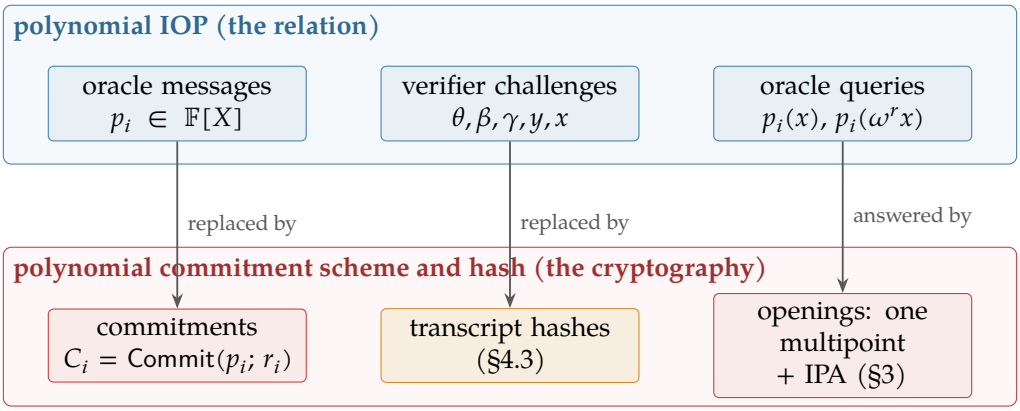
4.7 Halo 2 as a compiled polynomial IOP

The protocol of §4.5 factors into two layers: a *polynomial interactive oracle proof* for the relation, and a polynomial commitment scheme, in the sense of the *Crypto Guide* (§"Polynomial commitment

schemes"), that realises the proof's oracles. Every phase of the protocol belongs to one of the two layers.

The polynomial IOP, recalled. The *Crypto Guide* defines the object (§"SNARKs: succinct non-interactive arguments of knowledge", under "Polynomial interactive oracle proofs"): a public-coin interactive protocol in which, in place of ordinary messages, the prover in each round sends one or more *oracles* for polynomials $p_i \in \mathbb{F}[X]$ of bounded degree, the verifier answers with uniformly random field elements, and at the end the verifier *queries* each oracle at a small number of points, receiving the evaluations, and accepts or rejects as a deterministic function of the challenges, the evaluations and the public input. Completeness, soundness and knowledge soundness are those of interactive arguments, with oracle access in place of reading the messages, and the soundness analysis is purely algebraic, resting on the Schwartz–Zippel lemma. The one respect in which a polynomial IOP differs from a vanilla interactive argument is that the verifier never reads a prover message in full: it queries a few evaluations of each oracle. That is what renders the verifier succinct even when the polynomials have large degree; the message length is never read, only as many field elements as there are queries.

The compiler, recalled. The *Crypto Guide*'s compilation recipe (the same section) turns a polynomial IOP into a non-interactive argument with a polynomial commitment scheme (Setup, Commit, Open, Eval, Verify) and a random oracle H. Setup runs the commitment scheme's setup. The prover simulates the IOP, replacing each oracle message p_i by its commitment C_i, derives every verifier challenge from the transcript by the Fiat–Shamir transform of §4.3, and supplies, for each of the IOP's queries, the claimed evaluation together with an evaluation proof (Eval). The verifier parses the transcript, recomputes the challenges, checks each evaluation proof (Verify), and runs the IOP's accept-or-reject check on the opened evaluations (Figure 14).



phases 1–5 of §4.5 are the upper layer compiled; phases 6–7 are the lower layer's openings

Figure 14: The compiler. The polynomial IOP (blue) is the information-theoretic layer: oracles, public-coin challenges and evaluation queries, sound by the Schwartz–Zippel lemma alone. The cryptographic layer (red and amber) realises it: each oracle becomes a commitment, each challenge a transcript hash, each query an opening, and Halo 2 collapses every opening into one. The two layers are separable: the same IOP compiles with any commitment scheme meeting the hypotheses of Remark 4.11.

PLONK as a polynomial IOP. The IOP that Halo 2 compiles is PLONK's, in the PLONKish generality of §2.5. Its oracles, sent by the prover, are one polynomial per advice column; for each lookup, the two permuted columns A' and S'; the permutation running products, one per chunk,

and one lookup running product per lookup; and the quotient h, split into chunks below the degree bound. The fixed columns, the selectors among them, and the permuted-label polynomials are *preprocessed*, part of the verifying key rather than prover messages, which is why they are absent from the list. The verifier's challenges, in order, are θ for the lookup compression, β and γ for the permutation and lookup products, y for the combination of all constraints into one gate equation, and x for the evaluation point. The final check is the gate equation $C(x) = h(x)\, Z_H(x)$, with C the y-combination of every gate, permutation and lookup constraint polynomial and Z_H the vanishing polynomial of the rows, which the verifier evaluates from the oracles' values at x and at the rotations its constraints read: ωx or $\omega^{-1}x$ for an adjacent row, and $\omega^{-6}x$, the rotation to the boundary row of Remark 2.17, for the chunk-boundary identities of the permutation argument.

The map onto the phases. Phases 1 to 5 of §4.5 are this IOP compiled, with one addition: the commitments to the advice, the permuted lookup columns, the running products and the quotient chunks are the oracle messages replaced by commitments, and the random polynomial of phase 4, committed and evaluated at x, is the compiler's addition for zero knowledge (§4.6), not an IOP message; the challenges are the IOP's challenges squeezed from the transcript, and the claimed evaluations of phase 5 answer the IOP's oracle queries. Phases 6 and 7 bind those answers to the commitments: the multipoint reduction of §4.4, with its challenges $x_1, \dots, x_4$ and combined polynomial f^*, followed by the single inner-product opening of f^*. That is the bridge between the abstract compiler and the concrete protocol, and the phase numbering is the one Table 10 uses. Under the hypotheses of the *Crypto Guide*'s compilation recipe (Theorem "The compilation recipe, informal") and of Remark 4.11, the compiled protocol has the IOP's relation and the commitment scheme's assumptions, transparency, proof size and verifier cost.

Remark 4.11 (The compilation principle, and what it does not say). Soundness of the compiled argument is not the sum of two numbers, and the notions under which it is proved deserve their names. A polynomial IOP is *round-by-round* knowledge sound, informally, when its partial transcripts admit a "doomed" labelling that a prover message cannot escape except with bounded probability over the next verifier challenge; a related *state-restoration* notion lets a prover restore an earlier verifier state and redraw its challenge, which is precisely what a prover attacking a transcript hash can do. Precise equivalence or implication theorems between the two notions require additional conditions on the protocol and on the extractor, and this volume does not identify them. The *Crypto Guide*'s compilation recipe (Theorem "The compilation recipe, informal", §"SNARKs: succinct non-interactive arguments of knowledge") assumes a complete, knowledge-sound public-coin polynomial IOP; a correct, evaluation-binding and extractable commitment scheme; compatible composition hypotheses; and a suitable multi-round Fiat–Shamir theorem in the random-oracle model. For a multi-round protocol such as Halo 2 this volume reads those hypotheses as three: the IOP has a round-by-round or state-restoration knowledge-soundness theorem; the commitment scheme is extractable and evaluation-binding in a form compatible with it; and the transcript satisfies the hypotheses of an appropriate multi-round Fiat–Shamir theorem. The conclusion is that the compiled protocol is a non-interactive argument of knowledge in the random-oracle model, by layered extraction: the commitment scheme's extractor recovers the committed polynomials, converting the argument's prover into an IOP prover; the multi-round Fiat–Shamir theorem transfers the round-by-round or state-restoration guarantee to the non-interactive setting; and evaluation binding ensures that the opened evaluations are those of the extracted polynomials. What it does *not* say is that the knowledge error is determined by two scalars, the IOP's knowledge error and the commitment scheme's extraction error, added together: the exact error and the query loss depend on the definitions and on the extraction model, and §5 names the theorems that fix them for the analysed protocol and classifies their instantiation to the deployed transcript.

Delegated results. The subsection proves nothing afresh. The commitment interface (Setup, Commit, Open, Eval, Verify) with its hiding, binding and extraction properties is the *Crypto Guide*'s (§"Polynomial commitment schemes"); the Fiat–Shamir transform and its random-oracle security are the same volume's (§"The Fiat–Shamir transform: from interactive to non-interactive"); the definitions of completeness, soundness and knowledge soundness for interactive arguments used above are its (§"Interactive proofs, zero knowledge, and SNARKs" and §"Knowledge soundness and extractors"); and the Schwartz–Zippel lemma and the vanishing polynomial of the rows are the *Math Guide*'s (§"The Schwartz–Zippel lemma" and §"The evaluation domain H and its vanishing polynomial"). The volume adds the instantiation: which oracles, which challenges, which openings, and the one commitment scheme that realises them.

5 A rigorous treatment of knowledge soundness

Theorem 3.5 and Corollary 3.13 proved the inner-product opening knowledge sound for a verifier who draws his challenges: an extractor that runs the prover as a subroutine, rewinds it and feeds it fresh challenges recovers the committed vector or a nontrivial discrete-logarithm relation among the generators, which the hardness of the discrete logarithm on Vesta excludes. The deployed verifier draws nothing; every challenge is a hash of the transcript (§4.3), and Remark 4.11 named the hypotheses under which a compiled argument inherits its layers' guarantees without fixing them. This section determines what the extractor establishes when every challenge is a hash of the transcript. It combines the rewinding-extractor framework of the *Crypto Guide*, the definition of knowledge soundness (§"Knowledge soundness and extractors"), the tree extraction of Sigma-protocols (§"Sigma-protocols") and the Fiat–Shamir theorem with its stated caveats (§"The Fiat–Shamir transform: from interactive to non-interactive"), with the algebraic group model, named among the idealisations of §"The standard model and the random oracle model". It lifts the interactive extractor to the non-interactive argument in two layers and names the loss of the naive lifting; it states the model in which the halo2 book proves its extraction theorem and classifies what that theorem does and does not establish for the deployed transcript; and it closes on the gap between concrete and asymptotic security. The running-time vocabulary, polynomial, negligible and PPT, is the *Math Guide*'s (§"Polynomial, exponential, and negligible functions" and §"Algorithms, running time, and PPT"); expected polynomial time is the *Crypto Guide*'s (§"Adversaries and the security parameter", the remark on expected versus strict polynomial time); tightness and loss are the *Crypto Guide*'s (§"Reductions", §"Concrete security and the bit-security budget"); the assumption is its standing one (§"Standing assumptions").

5.1 From the interactive extractor to the non-interactive argument

What the interactive extractor costs. Theorem 3.5 extracts from a tree of accepting transcripts with three children at every node and one level per round: 3^k leaves for $k = \log_2 d$ rounds, that is $d^{\log_2 3}$ with $\log_2 3 = 1.58496\ldots$, polynomial in d though more than a binary tree would cost. At the deployed $d = 2^{11} = 2048$ the tree has $3^{11} = 177147$ leaves against the $2^{11} = 2048$ a binary tree would have; the extractor runs in expected polynomial time, the rewinding that assembles the tree being the *Crypto Guide*'s (§"Sigma-protocols", the tree-extraction theorem cited there). Its knowledge error is the $O(\log d/|\mathbb{F}|)$ of Theorem 3.5, one $O(1/|\mathbb{F}|)$ term per round for the challenges the siblings must avoid. The scale of these terms is worth fixing once: the proof system's own allowances are linear in $n = d$, the $18431/|\mathbb{F}| \approx 2^{-239.8}$ of the identity check (§2.4), whose difference polynomial $C - h\,Z_H$ has degree at most $d_{\max}\,n - 1 = 18431$ at the deployed $d_{\max} = 9$ ($\deg C \le d_{\max}(n-1)$ by the proof of Theorem 4.7, and the $d_{\max} - 1$ committed chunks of the quotient have degree $< n$ each, §4.5); the order-$n/|\mathbb{F}|$ term of the multipoint reduction (Lemma 4.5); and $d/|\mathbb{F}| = 2^{11}/2^{254} = 2^{-243}$ is that linear allowance for the opening. The opening's proved error sits below it, and quoting the linear figure for the opening loses nothing in a system whose

other terms are already linear in d, as the proof of Theorem 3.5 noted. Every one of these is a Schwartz–Zippel term, not a security level, in the sense fixed in §1.2.

Layered extraction. The compiled argument of §4.7 was assembled in two layers, an information-theoretic polynomial IOP and a cryptographic commitment scheme, and its extractor is assembled in the same two layers, run backwards (Figure 15).

The lower layer recovers polynomials. The prover has sent commitments C_ν and, in phases 6 and 7, one opening of the combined $F^* = F + \sum_i [x_4^i]\, Q_i$ with $Q_i = \sum_e [x_1^e]\, C_{\nu_{i,e}}$ (Construction 4.4). Phase 7 runs the deployed opening of §3.8, and the extractor of Corollary 3.13, the tree extraction of Theorem 3.5 with two further levels, ξ and ζ, the latter in the root role of the resampled U, returns the coefficient vector behind F^*, a representation of F^* over $(\mathbf{G}, H)$, or a nontrivial discrete-logarithm relation among $\mathbf{G}$, H and U. To reach the individual commitments the extractor rewinds over the combining challenges as well: the vectors behind F^* for $m+1$ distinct values of x_4 are $m+1$ linear equations in the $m+1$ unknown vectors behind $F, Q_1, \dots, Q_m$, whose coefficient matrix has the rows $(1, x_4, \dots, x_4^m)$ and is invertible because a nonzero polynomial of degree at most m has at most m roots (*Math Guide*, §"Roots and the factor theorem"), the argument of the proof of Lemma 3.4; the vectors behind Q_i for $|I_i|$ distinct values of x_1 likewise yield the vectors behind each C_ν in group i. One C_ν is the collapsed quotient $\sum_i [x^{ni}]\, \mathsf{Commit}(h_i)$ of §4.5; its vectors for $d_{\max} - 1$ values of x with pairwise distinct x^n yield the vectors behind the chunks h_i, by the same argument with x^n in place of x_4. The output is one polynomial of degree $< n$ per commitment the prover sent, and two different vectors extracted for one commitment would be a nontrivial discrete-logarithm relation among $\mathbf{G}$ and H, exactly as at the root of Theorem 3.5. This layer is the only place the discrete logarithm on Vesta enters, and it enters in only one way: the extractor either recovers the polynomials or exhibits a relation.

The upper layer recovers the witness. With the polynomials in hand, the prover's claimed evaluations are the true evaluations of fixed polynomials except with the Schwartz–Zippel allowance of Lemma 4.5, since a claimed value that differs from the extracted polynomial's value survives the reduction only with that probability; and "the true evaluations of fixed polynomials" is precisely the hypothesis under which §2 proved its theorems. Theorem 2.5 turns the accepted identity $C(x) = h(x)\, Z_H(x)$ into divisibility of C, Lemma 4.6 splits it into the vanishing of every constraint polynomial on the rows, Theorem 2.11 with Corollary 2.12 turns the running products into the wiring, and Theorem 2.15 with Proposition 2.16 turns the lookup identities into table membership, each with its stated term over $|\mathbb{F}|$; Theorem 2.2 reads gate satisfaction off divisibility. The witness is then read off directly: the advice columns are the values of the extracted advice polynomials on the rows. Nothing in this layer uses an assumption; the IOP's extractor outputs the oracles, which the layer below supplies. The consequence runs both ways. A break of the discrete logarithm on Vesta breaks the lower layer and nothing else, in the way §3.5 described, one commitment opening to any vector; and no assumption can rescue an identity that the upper layer does not actually enforce, because the upper layer's errors are counting statements about polynomials.

What is proved and what is described. In the interactive setting the composed extractor is a tree extractor in the sense of the *Crypto Guide* with a wider tree: three children at each of the k inner-product levels, two at the ζ level and two at the ξ level, $m+1$ at the x_4 level, $\max_i |I_i|$ at the x_1 level, one x_1 serving every group, and, at the x level, $(d_{\max} - 2)\, n + 1 = 14337$ children: each value of x^n has at most n preimages, so only that many distinct challenges guarantee the $d_{\max} - 1$ pairwise distinct x^n the quotient step needs, as Theorem 3.5 took five children for three distinct squares. The two ζ children must moreover be nonzero, as in Corollary 3.13. Each polynomial the upper layer consumes is fixed by a binding commitment sent before every challenge that tests it: the advice before θ, the permuted lookup columns A' and S' before β and γ, the running products and the random polynomial before y, the quotient chunks before x (§4.5); the polynomial behind F and the mask behind S, committed after x, serve only inside the lower layer; and all of the consumed ones precede x, the first challenge at which the tree branches, so the upper

layer's Schwartz–Zippel arguments apply on every branch without further rewinding. The inner-product part of that tree is Corollary 3.13, proved; the theorems of the upper layer are proved in §2; the accounting that joins them into one tree-extraction statement with one knowledge error is the compilation principle of Remark 4.11, whose hypotheses this volume names and does not discharge. In particular the compiled error is not asserted to be the sum of the layers' errors.

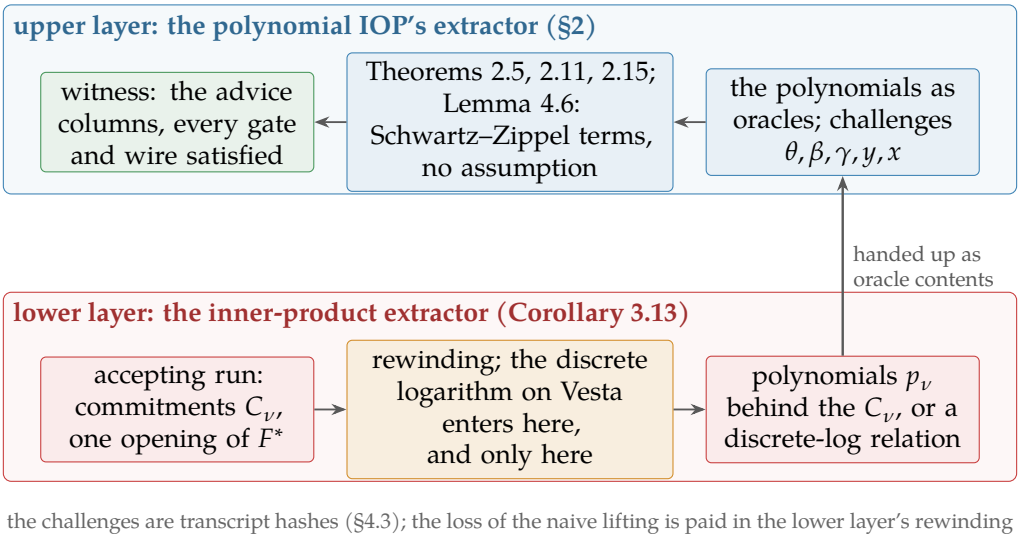


Figure 15: Layered extraction. The lower layer (red) is the cryptographic one: from an accepting run, the extractor of Corollary 3.13, rewinding over the opening's challenges, the combining challenges of the multipoint reduction and the evaluation challenge x, recovers the polynomial behind every commitment or exhibits a discrete-logarithm relation; the assumption (amber) lives here and nowhere else. The upper layer (blue) is the information-theoretic one: with the polynomials as oracle contents, the theorems of §2 read the witness (green) off them with Schwartz–Zippel terms only.

The naive lifting, and its loss. Everything above is interactive: the extractor chose the challenges it fed the prover. In the deployed argument the challenges are hashes, and the extractor of the *Crypto Guide*'s Fiat–Shamir theorem obtains its second transcript by *forking*: it finds, among the prover's at most Q hash queries, the critical one whose answer became the challenge, rewinds to it, and answers it afresh, succeeding with probability about $\varepsilon(\varepsilon/Q - 1/|\mathbb{F}|)$, so that the extractor's success is the prover's squared and divided by Q (§"The Fiat–Shamir transform: from interactive to non-interactive", the knowledge-soundness clause of its theorem). That theorem covers one challenge. The tree of Theorem 3.5 needs three siblings at each of k levels, and a sibling at level j must share the transcript prefix through level $j-1$, which in the hashed setting means agreeing with the prover's prefix and its earlier hash answers: in a transcript hash, changing one message redraws *all* later challenges, so siblings at round j can be found only among the prover's own queries. Let μ be the number of verifier challenges of the protocol being lifted. The simplest lifting guesses, among the at most Q queries, the μ that answered them, and runs the interactive extractor against the interactive prover that the guess defines; the guess is right with probability at least $Q^{-\mu}$, so the knowledge error is multiplied by Q^μ. Attema, Fehr and Klooß (*Fiat–Shamir Transformation of Multi-Round Interactive Proofs*) record this Q^μ as the loss for a general $(2\mu+1)$-move protocol; nesting the one-round forking level by level fares worse, since each level squares the success probability before dividing by Q. For the opening alone μ exceeds $k = \log_2 d$, the choice of U (in the deployed opening, ξ and ζ) being a challenge too, and μ grows once the proof system's own challenges are counted. The loss belongs to this analysis. For $(k_1, \ldots, k_\mu)$-special-sound protocols (*Crypto Guide*, §"Sigma-protocols", the definition of tree special soundness) the

same authors prove, in the random oracle model and with no idealisation of the group, a knowledge error that grows only linearly in Q. The rounds of the opening's extraction tree have that shape, but its root step, which rewinds the choice of U (in the deployed opening, ξ and ζ), and its alternative output, a discrete-logarithm relation, are not checked here against that theorem's hypotheses, and this volume does not assess that route.

With the deployed $k = \log_2 d = 11$, so $\mu > 11$ for the opening alone, and a query budget of $Q = 2^{80}$ hash evaluations, the factor Q^μ exceeds $Q^{11} = 2^{880}$; multiplied by the linear allowance $d/|\mathbb{F}| = 2^{-243}$ it exceeds 2^{637}, a bound above 1, which is no bound at all. Nor does the proved $O(\log d/|\mathbb{F}|)$ help: the challenge space is a 255-bit field with fewer than 2^{255} elements, so 2^{880} exceeds $|\mathbb{F}|$ itself, and a factor larger than $|\mathbb{F}|$ turns every term of the form $c/|\mathbb{F}|$ with $c \geq 1$ into a number above 1. Asymptotically the loss falls elsewhere. With Q and d polynomial in the security parameter λ (*Crypto Guide*, §"Adversaries and the security parameter") and $|\mathbb{F}| = 2^{\Theta(\lambda)}$, the factor $Q^\mu = 2^{O(\log^2 \lambda)}$ is super-polynomial and sub-exponential in the *Math Guide*'s classification (§"Polynomial, exponential, and negligible functions"). The Schwartz–Zippel terms survive it: $Q^\mu\, O(\log d)/|\mathbb{F}| = 2^{O(\log^2 \lambda) - \Theta(\lambda)}$ is negligible. The reduction does not survive it: the guessing extractor succeeds only with probability reduced by Q^μ, or runs Q^μ times longer to compensate, and so is not PPT, and the prover's success is then bounded only by Q^μ times the discrete-logarithm advantage, which need not be negligible when that advantage is merely negligible. The interactive extractor is polynomial, $d^{\log_2 3}$ transcripts in expected polynomial time; the looseness arises in the naive analysis of the passage to the non-interactive argument.

What the tighter notions are for. Remark 4.11 named the two notions built to avoid this: *round-by-round* knowledge soundness, which labels partial transcripts as doomed and bounds the probability that one verifier challenge lifts the doom, and *state-restoration* soundness, which lets the prover restore an earlier verifier state and redraw its challenge, precisely the freedom a prover attacking a transcript hash has. Each exists to obtain a bound on the compiled argument that loses a factor linear in Q rather than Q^μ; no equivalence between them is asserted here, as none was there. Claiming a single factor Q for a compiled protocol nevertheless requires the precise theorem's state-restoration, transcript, challenge and extraction hypotheses to hold of that protocol, and this volume does not assert a factor-Q bound for Halo 2's compiled protocol. The next subsection names the model in which the halo2 book proves such a bound, in the state-restoration form, for its abstract interactive protocol, what that analysis assumes, and why its instantiation to the deployed transcript remains unwritten.

5.2 The algebraic group model

The nested forking of the previous subsection rewinds because the extractor has no other way to learn what is behind a group element: three accepting continuations are the price of one representation. The *algebraic group model* of Fuchsbauer, Kiltz and Loss (*The Algebraic Group Model and its Applications*) removes the need by restricting the adversary's group operations, and it enables tighter analyses for protocols that suit it. It is not by itself a theorem eliminating every Fiat–Shamir loss; what it eliminates is the rewinding, and what remains must still be proved.

Definition 5.1 (Algebraic adversary)**.** An *algebraic adversary* against a group $\mathbb{G}$ is a PPT algorithm that, whenever it outputs a group element $P \in \mathbb{G}$, additionally outputs a vector $\mathbf{a}$ of scalars with

$$P = \langle \mathbf{a}, \mathbf{Y} \rangle,$$

where $\mathbf{Y}$ is the vector of group elements it has previously seen: the generators it was given and the elements it has received. The vector $\mathbf{a}$ is the *representation* of P.

Where the model sits. The algebraic group model sits between the standard model, in which the adversary is any PPT algorithm and every primitive is concrete code it may inspect (*Crypto Guide*, §"The standard model and the random oracle model"), and the generic group model, in which the adversary may perform group operations only through an oracle on opaque handles and never exploits the representation of an element (the same section, the remark on the generic and algebraic group models). An algebraic adversary is less restricted than a generic one, since it may compute on the bit representation of group elements however it likes; it must only be able to explain each element it outputs as a combination of elements it has seen. The model is an idealisation (the same remark): a proof in it says nothing about an adversary that outputs a group element it cannot explain. Figure 16 draws the three models as nested restrictions on the adversary and places this section's analyses on them; the random oracle model, which idealises the hash rather than the group, is an independent axis, and an analysis may combine either group model with it.

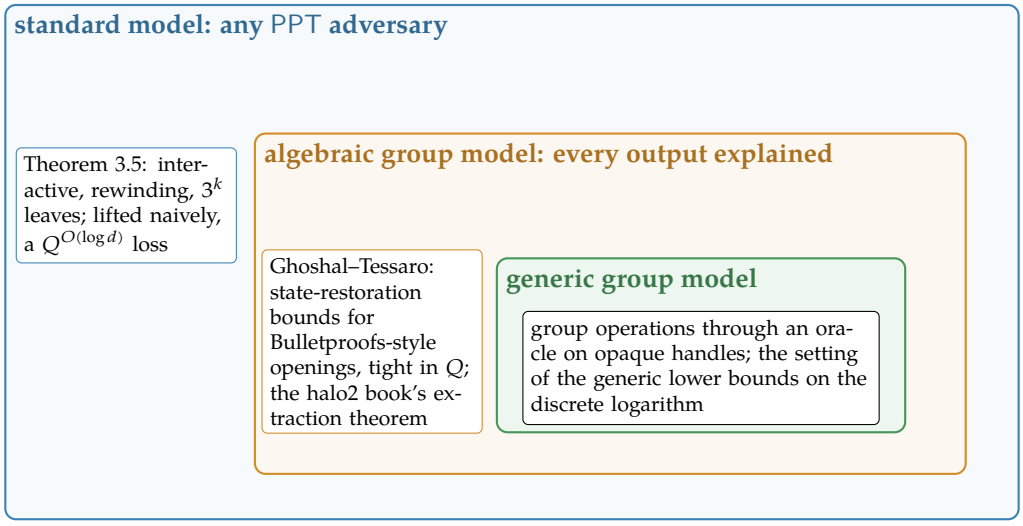


Figure 16: Three models as nested restrictions on the adversary. Every adversary admitted by an inner ring is admitted by the rings around it, so a proof in an outer ring is the stronger statement and a bound in an inner ring is the easier to make tight. The interactive extractor of Theorem 3.5 lives in the standard model for uniform generators; the tight state-restoration bounds of Ghoshal and Tessaro, and the halo2 book's extraction theorem that invokes them, live in the algebraic group model (amber). The generic group model (green) is where the discrete logarithm's own lower bounds are proved.

Online extraction in the algebraic group model. In this model the prover outputs every group element with its representation, and the extractor reads the witness off the representations of a single accepting transcript instead of assembling it from rewound ones; no rewinding occurs, and the price is the assumption that the prover is algebraic. For the deployed opening, equation (29), the halo2 book's *Protocol Description* chapter carries this out ("Witness-extended Emulation", the definition of the bad-challenge sets): comparing the two sides of the accepting equation coordinate by coordinate either exhibits a nontrivial discrete-logarithm relation among the generators or, unless some challenge falls in a set of *bad challenges* fixed by the messages before it, forces the representation of P to open to the claimed value. The analysed protocol draws every challenge from one space $\mathcal{C}$ of field elements, none zero or a row, and each round challenge u_j further so that its factor $1 + u_j x_3^{2^{j-1}}$ of $g_{\mathrm{dep}}(x_3)$ in (26) is nonzero. Each round challenge u_j has at most $4n$ bad values in $\mathcal{C}$, and ζ and ξ have one each; the book bounds the bad sets of the proof system's other challenges, $x_1, \ldots, x_4$, x and those of the committed rounds, by the degrees of the identities they test, the largest being $n_g(n-1)$, the bound on the constraint polynomial's degree in each

indeterminate, provided $n_g(n-1) \geq 4n$, as at the deployed $n_g = 9$, where $18423 > 4n = 8192$; the book admits $n_g \geq 4$, for which this can fail.

Ghoshal and Tessaro, and the deployed stance. Ghoshal and Tessaro (*Tight State-Restoration Soundness in the Algebraic Group Model*) carry this analysis out for Bulletproofs-style inner-product arguments: they prove tight state-restoration soundness in the algebraic group model and derive Fiat–Shamir security bounds from it under their stated hypotheses, the transfer from the state-restoration game to the compiled argument being their theorem. The deployed analysis follows them. The protocol specification defines the proving system by reference to the halo2 book (protocol specification, § 5.4.10.3, "Halo 2"), and the book's *Protocol Description* chapter fixes the framework: it adopts state-restoration soundness because the concrete protocol is realised through the Fiat–Shamir transformation and a cheating prover can fork the transcript at will, and it invokes Ghoshal and Tessaro's theorem for the passage from that game to soundness of the compiled argument. Its abstract interactive protocol is stated for a relation of the form "a fixed multivariate constraint polynomial in the column polynomials and the challenges vanishes on the rows", with the column polynomials as the witness, over public parameters of $n+2$ group elements, $\mathbf{G}$, U and the blinding generator H (written W there).

Definition 5.2 (State-restoration witness-extended emulation)**.** Let Π be a public-coin interactive argument for a relation $\mathcal{R}$ with r challenges from a space $\mathcal{C}$. A *state-restoration prover* $\mathcal{P}^*$, given the public parameters, outputs a statement ϕ and then queries an oracle on pairs (τ, a), where τ is a transcript prefix the oracle has recorded (initially only the empty one) and a is a next prover message; it may thus extend any earlier prefix. In the *real* game the oracle answers a message of rounds 1 to r with a fresh uniform $c \in \mathcal{C}$ and records (τ, a, c), and answers a final message with the verifier's decision on (τ, a), recording (τ, a). In the *ideal* game an algorithm $\mathcal{E}$, given the public parameters and ϕ and keeping state across queries, receives each query (τ, a), for an algebraic prover together with the representations (Definition 5.1) of the group elements in a, returns the answer, which is recorded with the query, and at the end outputs w. For a distinguisher $\mathcal{D}$ given the recorded transcripts,

$$\mathrm{Adv}^{\mathrm{srwee}}_{\Pi,\mathcal{R}}(\mathcal{P}^*, \mathcal{D}, \mathcal{E}) := \Pr[E_{\mathrm{real}}] - \Pr[E_{\mathrm{ideal}}],$$

where E_{real} is the event that $\mathcal{D}$ outputs 1 in the real game, and E_{ideal} the event that $\mathcal{D}$ outputs 1 in the ideal game and, if an accepting transcript was recorded, $(\phi, w) \in \mathcal{R}$.

Theorem 5.3 (Extraction for the abstract protocol; interface)**.** *Let Π be the halo2 book's abstract interactive argument for its relation $\mathcal{R}$, over a group $\mathbb{G}$ of prime order with scalar field $\mathbb{F}$, for n rows, a positive integer n_g with $n_g(n-1) \geq 4n$ and a constraint polynomial of degree at most $n_g(n-1)$ in each indeterminate, with challenges from $\mathcal{C}$. There is an extractor $\mathcal{E}$ such that for every non-uniform algebraic state-restoration prover $\mathcal{P}^*$ making at most Q queries to its oracle there is an adversary $\mathcal{B}$ with*

$$\mathrm{Adv}^{\mathrm{srwee}}_{\Pi,\mathcal{R}}(\mathcal{P}^*, \mathcal{D}, \mathcal{E}) \;\leq\; Q\,\varepsilon_{\mathrm{bad}} + \mathrm{Adv}^{\mathrm{dlrel}}_{\mathbb{G},n+2}(\mathcal{B}), \qquad \varepsilon_{\mathrm{bad}} \leq \frac{n_g(n-1)}{|\mathcal{C}|},$$

for every computationally unbounded distinguisher $\mathcal{D}$. Here $\varepsilon_{\mathrm{bad}}$ bounds the fraction of bad challenges after every partial transcript, and $\mathrm{Adv}^{\mathrm{dlrel}}_{\mathbb{G},n+2}(\mathcal{B})$ is the probability that $\mathcal{B}$, given $n+2$ uniform elements of $\mathbb{G}$, outputs a nontrivial discrete-logarithm relation among them.

The theorem is quoted as an interface and not proved in this volume: it is the halo2 book's (*Protocol Description*, "Witness-extended Emulation"), proved there by instantiating Theorem 1 of Ghoshal and Tessaro with the bad-challenge sets described above. Its extractor works online,

from the representations of a single transcript, and never rewinds. At the deployed $n_g = d_{\max} = 9$ and $n = 2^{11}$, $\varepsilon_{\text{bad}} \leq 18423/|\mathbb{C}| \approx 2^{-239.8}$, and $Q\,\varepsilon_{\text{bad}} \approx 2^{-159.8}$ at $Q = 2^{80}$ queries.

The instantiation gap, classified. Applying those bounds to the deployed Halo 2 transcript requires matching the precise folding, masking, challenge distribution and composition assumptions of the analysed protocol to the deployed one, and this volume does not prove that instantiation theorem. Four mismatches are visible from the earlier sections. The book's abstract relation sees the permutation and lookup arguments of §2 only through the constraint polynomial and its committed columns; the Orchard circuit's actual gates and table columns (§4.1) and its chunked running products ("The deployed chunking" in §2.6) instantiate that abstraction and must be checked against its degree hypotheses. The book's challenges exclude zero, the rows and, for each u_j, the value that zeroes its factor of $g_{\text{dep}}(x_3)$, while the deployed transcript squeezes full field elements with no rejection (§3.8), so the deployed challenge distribution differs from the analysed one by the excluded events. The analysed folding is the deployed normalisation of Remark 3.9, and the masking is the deployed mask of §3.8, so those two match; the hash, however, is BLAKE2b (§4.3), and the transfer theorem's random oracle must be instantiated by it. And the book's theorem is for the interactive argument against a state-restoration prover, the compiled bound following by the transfer theorem under its hypotheses. In the classification of §1.2, the status of the transfer is *designed-but-unspecified*: the bounds exist for the analysed protocol and the design intends them for the deployed one, but the theorem that carries them across is not written, here or in the specification.

5.3 Concrete versus asymptotic security

The *Crypto Guide* distinguishes a reduction's existence from its tightness (§"Reductions", the definition of loss and the remark on why loss matters concretely) and reads security as a budget of bits (§"Concrete security and the bit-security budget"): a scheme inherits the strength of its assumption only up to the loss of the reduction that connects them. Two tightness gaps separate the proved statements from the deployment.

The first gap: the naive lifting. The naive Fiat–Shamir knowledge-soundness analysis of the opening loses the factor $Q^{\mu} > Q^{\log d}$ of §5.1, exponential in the round count $\log d$ with base the query count Q: above 2^{880} at the deployed size for a 2^{80}-query prover, against a challenge space of fewer than 2^{255} elements. The interactive extraction itself is polynomial, $d^{\log_2 3} = 177147$ transcripts at $d = 2048$ in expected polynomial time, with the proved knowledge error $O(\log d/|\mathbb{F}|)$ of Theorem 3.5. The gap lies in the naive analysis of the lifting, and the notions of §5.1 and the model of §5.2 exist to close it.

The second gap: the model. Each analysis cited in this section that avoids the naive loss relies on an idealisation: the halo2 book's extraction theorem (Theorem 5.3) on the algebraic group model, and its passage to the compiled argument on the random oracle model as well; the special-soundness analysis of Attema, Fehr and Klooß on the random oracle model alone. No standard-model tight bound is claimed for the deployment, by this volume or by the analyses it cites; this is a statement about the design's analyses, not a specification claim, and the specification makes none. Read against the bit-security budget, the consequence is concrete. The discrete-logarithm layer offers about 2^{126} operations of classical security (§3.5; *Crypto Guide*, §"Standing assumptions"); a reduction tight in Q lets the argument inherit that strength, less the algebraic term $Q\,\varepsilon_{\text{bad}}$ of Theorem 5.3, about $2^{-159.8}$ at 2^{80} queries, which is not the bottleneck; a reduction losing 2^{880} lets it inherit nothing. For the analysed protocol the number is therefore the assumption's, in the algebraic group and random oracle models; for the deployed argument it is so only through the instantiation that §5.2 classifies designed-but-unspecified.

The stance. Proved without idealisation of the adversary or of the transcript hash are the interactive statement, Theorem 3.5, for uniform generators (the deployed hashed generators add its hypothesis that hash-to-curve is a random oracle), and the information-theoretic layer of §2; described, with the hypotheses named, are the layered extraction that joins them and the naive lifting's loss; cited, with its instantiation to the deployed transcript classified designed-but-unspecified, is the halo2 book's bound in the algebraic group model.

6 The verifier's decision and its implication chain

This section adds no theorem: it lists the verifier's checks in the order he performs them, assembles the implication chain from an accepted proof back to the claim, each link labelled by the statement that proves it and by its error term, repeats the chain for the Orchard Action circuit, and maps the five reductions of §1.2 to the formal statements that own them. Nothing here depends on the recursion of §7, which Orchard does not deploy.

6.1 The verifier's checks, read in order

What he holds. The verifier of the toy holds the fixed data of the circuit, the five selector polynomials and the permuted-label polynomials of Construction 2.10; the public input, the column $(0, 0, 0, -35)$ interpolated to $\mathrm{PI}(X)$ (§4.2); and the proof. The proof is a byte string (§4.3): commitments, claimed evaluations, and one opening, in the order of the phases of §4.5 (Table 10), counted by kind in Table 8. He never sees a column polynomial, a blinder or the advice; he sees curve points and field elements, and he performs the checks below in this order.

1. **The challenges.** He first commits to the instance column himself, from the public input and with the fixed blinder 1 (§4.2), so that the public input never comes from the prover. He then replays the transcript (§4.3): he absorbs the key digest, that instance commitment and each prover message in turn, and squeezes every challenge at its prescribed point. In the toy the point $z = 20$ was drawn by decree; in the deployed protocol the evaluation point x is a hash of everything before it.

2. **The identity at the point.** From the claimed evaluations and the fixed polynomials he evaluates himself, he forms both sides of the gate identity. In the toy this is the one field equation (5): $G(20) = 75$ against $t(20)\, Z_H(20) = 67 \cdot 46 = 75$, accepted; the $c_0 = 10$ cheat of Example 2.3 gives 20 against 64, rejected. In the deployed protocol the identity is $C(x) = h(x)\, Z_H(x)$, with C the y-combination of every constraint polynomial and h the chunked quotient, and it is enforced by opening the collapsed quotient commitment at x to the value $C(x)/Z_H(x)$ that the verifier computes himself from the evaluations of phase 5, inside the multipoint opening (phases 6 and 7 of §4.5).

3. **The wiring.** He evaluates the boundary and update identities of the permutation argument (7) at the point and, in the deployed form, the chunk-boundary identities of §2.6 and the boundary identity $\ell_{\mathrm{last}}\,(Z^2 - Z)$ of Remark 2.17. In the toy the ratio of the two products (6) at $\beta = \gamma = 2$ is 1 for the honest trace and 61 for the tampered $b_1 = 4$ (Example 2.8).

4. **The tables.** He evaluates the four lookup identities (9) and, in the deployed form, the boundary identity $\ell_{\mathrm{last}}\,(Z^2 - Z)$ of Remark 2.17 at the point, five for each lookup argument. The toy's grid has none; the two-bit check of Example 2.13 showed the shape, with both products equal to 82 and the forged 7 stranded in every arrangement. In the deployed protocol items 3 and 4 are not separate checks: their identities are summands of C and are enforced by the one check of item 2; only the toy checks them separately.

5. **The openings.** Every evaluation he received for the three checks above was *claimed*. He now binds each claim to its commitment: the multipoint reduction of Construction 4.4 folds every

claim into one statement, and the inner-product equation (29) discharges it with one multi-scalar multiplication of length n. In the toy, the opening of $a(X)$ at $z = 20$ to the value 59 closes at $(54, 39)$ on both sides (Table 5) and the false value 60 moves the left-hand side to $(62, 15)$; the deployed opening of the same commitment closes at $(5, 72)$ and rejects 60 likewise (Table 7). Among the claims is $\pi(x)$, which the opening binds to his own instance commitment (§4.2).

If every check passes he accepts. The accepted checks imply the following chain, read backwards from the last check to the first; Figure 17 draws it.

The chain, read backwards.

1. *The identity check passed at his random point.* Because the commitment binds, the polynomials behind the openings were fixed before the point was chosen: the Pedersen vector commitment is binding under the discrete logarithm on Vesta (the *Crypto Guide*, §"Pedersen vector commitments", recalled in §2.9), and the opening argument is knowledge sound (Theorem 3.5 and Corollary 3.13), so from any prover who passes the opening an extractor recovers the committed vector and the value it opens to, or else a discrete-logarithm relation among the hashed generators; its knowledge error is $O(\log d/|\mathbb{F}|)$, and the multipoint reduction that fed it adds one Schwartz–Zippel allowance of order $n/|\mathbb{F}|$ (Lemma 4.5). With a hash in place of coins the ordering holds by construction, every commitment being an input of the hash that produced the challenge testing it (§4.3). Because the point was random and a false identity disagrees with the true one at all but $\deg D$ points, passing supports $G(X) = t(X)\, Z_H(X)$ as polynomials except with the error $\deg D/|\mathbb{F}|$ of Theorem 2.5 for this fixed check: $11/97$ for the toy's degree budget, $3/97$ for the particular cheat of Example 2.6, about 2^{-240} at Orchard scale, one term in the sense §1.2 fixed. In the deployed form the identity is $C = h\, Z_H$, and the challenge y that combined every constraint into C costs one more term, $(T-1)/|\mathbb{F}|$ for T constraint polynomials (Lemma 4.6).

2. *The identity holds as polynomials.* Then $Z_H \mid G$, so G vanishes on every row (Theorem 2.2, equivalence (3)), so every gate equation holds at every row (Theorem 2.1). Both steps are exact equivalences with no error term: in the toy, the four rows of Table 1 each satisfy (1). In the deployed form every constraint polynomial combined into C vanishes on the active rows: every custom gate, and the permutation and lookup identities that the next two links consume.

3. *The permutation identities passed.* Except with the error $2N/|\mathbb{F}|$ of Theorem 2.11 over β, γ, with N the number of participating cells, twelve in the toy, and, in the deployed form, $N/|\mathbb{F}|$ more for the challenges at which a product factor vanishes (Corollary 2.12), the copy constraints hold: the per-row gates are wired into one coherent computation, the witness 3 of row 0 is the witness of rows 1 and 2, and each row's output is the next row's input. The lookup identities passed likewise: except with the error $4n/|\mathbb{F}|$ of Theorem 2.15, and $n(m-1)/|\mathbb{F}|$ more for the θ-compression of (10) (Proposition 2.16) when a lookup has $m > 1$ columns, every looked-up value is in its table.

4. *The public input fixed the target at* 35. In the toy, 35 enters the gate polynomial G through $\mathrm{PI}(X)$, which the verifier interpolates himself (§4.2), so it enters the identity of link 1. In the deployed circuit the instance polynomial $\pi(X)$ of Definition 4.2 is the verifier's own: he committed to it from the values he intends to verify against, and its designated cells are copy-constrained to the advice cells that carry the same values, so the permutation of link 3 forces agreement. There is no error term of its own. The relation certified is the one at the public input the verifier supplied (§4.2).

Together with the observation that closed §2.1, that filling the advice columns *is* knowing the witness, because those cells are the wires carrying it and its powers, the four links say: there exists an assignment of the advice columns, a value of the witness, making the entire arithmetisation of $X^3 + X + 5 = 35$ hold, at the public 35.

From existence to knowledge. The chain so far establishes that satisfying advice *exists*. That the prover *knows* it is the extraction question. The extractor of Theorem 3.5 and Corollary 3.13 recovers, from any prover who passes the opening, the committed vector and its blinder: it establishes knowledge of committed polynomial data, and nothing more. It does not by itself prove that those polynomials encode a satisfying circuit witness; that is the upper layer's task in the layered extraction of Remark 4.11, where the commitment scheme's extractor turns commitments into polynomials and the polynomial IOP's extractor turns polynomials into a witness. Under the composed hypotheses named there, a round-by-round or state-restoration knowledge-soundness theorem for the IOP, extractability and evaluation binding for the commitment scheme, and a multi-round Fiat–Shamir theorem for the transcript, an extractor for the *full* protocol obtains such a witness; §5 weighs those hypotheses against the deployed transcript, and in particular states in which model the tight bound is proved.

What he has not learnt. For the interactive protocol and an honest verifier the transcript reveals nothing beyond the truth of the statement (Proposition 4.9, cited, and Corollary 4.10); for the non-interactive deployed proof this is stated, not proved (§4.6). Every commitment the prover sent is a uniform group element (perfect hiding, recalled in §4.6); the verifying key's and the instance commitments carry the fixed blinding factor 1 and commit only public data; the evaluations of each blinded column are uniform by Lemma 4.8; the final scalar of the opening, which folding alone would have leaked as a linear form of the coefficients (Remark 3.7), is masked by the random polynomial $s(X)$ vanishing at the opening point (§3.6, in the deployed form of §3.8); and the joint transcript is simulated by the witness-free simulator of Proposition 4.9, within the distance of Corollary 4.10 for the deployed challenges. The toy has no blinding rows and is not zero knowledge: its claimed value $a(20) = 59$ alone singles out the witness 3 among the three roots $3, 18, 76$ of $X^3 + X + 5 = 35$ in $\mathbb{F}_{97}$, and indeed among all 97 candidates; the blinding rows of Lemma 4.8 are what make the deployed evaluations uniform. The deployed mask makes the opening's own scalar a fresh uniform value: $c = 20$ under one mask (Table 7) and $c' = 46$ under another (§3.8), both accepted.

The error budget. The verifier has checked a small transcript, $2720 + 2272\,a$ bytes for a Actions (Table 8), and a logarithmic opening of $2k + 1 = 23$ points. Each link of the chain carries its own error term, which bounds the failure of that link alone: the identity's $\deg D/|\mathbb{F}|$ (Theorem 2.5); the y-combination's $(T-1)/|\mathbb{F}|$ (Lemma 4.6); the permutation's $2N/|\mathbb{F}|$ (Theorem 2.11) and $N/|\mathbb{F}|$ for the challenges at which a product factor vanishes (Corollary 2.12); the lookup's $4n/|\mathbb{F}|$ (Theorem 2.15) and the θ-compression's $n(m-1)/|\mathbb{F}|$ (Proposition 2.16); the multipoint reduction's allowance of order $n/|\mathbb{F}|$ (Lemma 4.5); and the opening's $O(\log d/|\mathbb{F}|)$ (Theorem 3.5 and Corollary 3.13). Combining them into one knowledge error for the protocol is the compilation principle of Remark 4.11, whose hypotheses §5 names and does not discharge; no sum of the terms is asserted. Beyond these terms lie the loss of replacing coins by a hash, of which the grinding route alone multiplies a term by the query budget Q, so that 2^{-240} becomes 2^{-176} after 2^{64} attempts (§4.3, Proposition 4.3), while the multi-round transform's full loss is weighed in §5, which asserts no factor-Q bound for the compiled protocol; and the assumptions beneath the chain, the discrete logarithm on Vesta that makes the commitments bind and the extractor's alternative output infeasible (Theorem 3.5 and Corollary 3.13), the random-oracle model for the transcript hash, and the extraction model in which the non-interactive bound is proved (§5). The isolated 2^{-240} is not the verdict's error: it is one Schwartz–Zippel term for one fixed lie, read as §1.2 fixed.

6.2 The Action circuit at deployed scale

Nothing in the chain used that the statement was small. Replace the four gates by the constraints of the Orchard Action statement, the nine conditions the protocol specification imposes on the

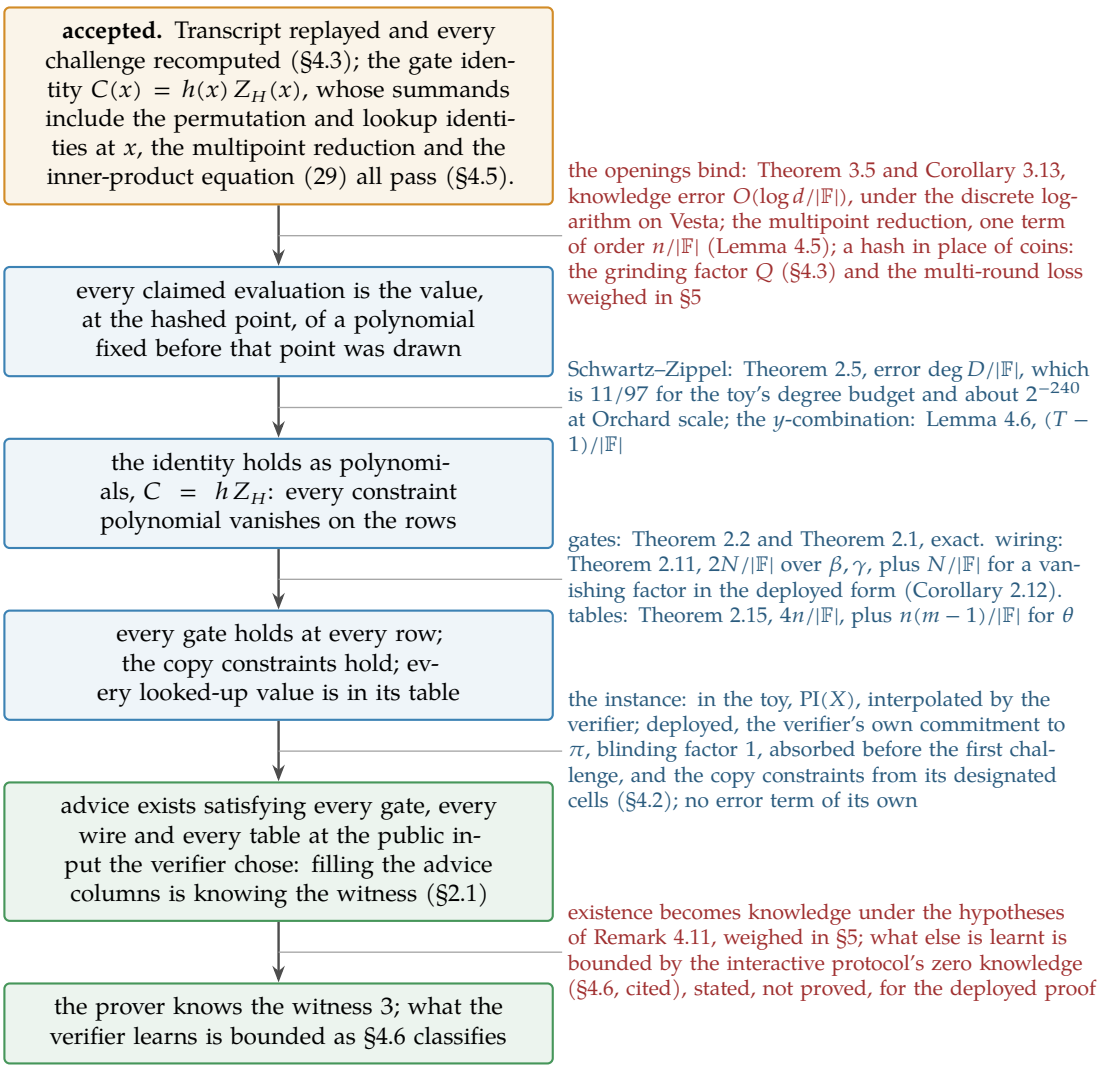


Figure 17: The implication chain from the verifier's accepted checks (amber) back to the claim (green), read downwards; each arrow carries the statement that proves it and its error term. Blue labels are algebraic links whose only losses are Schwartz–Zippel terms or none; red labels are the two links that rest on an assumption or a model hypothesis, the binding of the openings at the top and the extraction of the witness at the bottom.

public input and the witness (protocol specification, § 4.18.4, "Action Statement (Orchard)"): the integrity of the old and new note commitments, the Merkle path from the spent commitment to the anchor unless the spent value is 0 (the *Crypto Guide*, §"Merkle trees and commitments to sets"), the opening of the net value commitment, a Pedersen commitment to the value difference (the *Crypto Guide*, §"The Pedersen commitment"), the nullifier's derivation, the spend authority, that the randomised verification key rk is the re-randomisation of the spend validating key ak by a witnessed randomiser (the *Crypto Guide*, §"Key re-randomisation and unlinkability"), the integrity of the diversified address, that the spent note's address $(\mathsf{g_d}, \mathsf{pk_d})$ satisfies $\mathsf{pk_d} = [\mathsf{ivk}]\,\mathsf{g_d}$ for the incoming viewing key ivk derived in the witness from ak and further key material (the *Crypto Guide*, §"Shielded-note delivery as deployed hybrid encryption") unless that derivation fails, and the two enable flags, which force the spent value to 0 unless $\mathsf{enableSpends} = 1$ and the created value to 0 unless $\mathsf{enableOutputs} = 1$. From NU6.3 the circuit checks a tenth condition on the further public bit $\mathsf{disableCrossAddress}$ (§ 4.6; ZIP 258): when it is set, the receivers $(\mathsf{g_d}, \mathsf{pk_d})$ of the spent and created notes coincide (§4.2). Each condition unfolds into the gates of the deployed circuit (§4.1), and the grid grows accordingly (Table 11): the four rows become $2^{11} = 2048$; the

one universal gate of degree 3 becomes custom gates reading rotations, under the degree bound $d_{\max} = 9$, the incomplete addition of Theorem 4.1 among them; the quotient t of degree 5 becomes h of degree up to 16375, cut into 8 chunks (Theorem 4.7); the four-class permutation on twelve cells becomes fifteen equality-enabled columns in three running products; a Sinsemilla generator table of 2^{10} rows, reached by three lookup arguments, joins the permutation (§4.1.3; the hash is the *Crypto Guide*'s, §"Sinsemilla: an algebraic hash-based commitment"); the four openings at z become 49 claimed evaluations per Action and 45 shared (Table 10), collapsed into one opening of 23 points; and the point $z = 20$ drawn by decree becomes an x squeezed from the BLAKE2b transcript. The spine is identical: arithmetise into a grid, interpolate the columns, compress "all rows correct" into one identity, $G = t\,Z_H$ in the toy and $C = h\,Z_H$ in the deployed protocol, commit, and test at a random point.

	the toy	the Orchard Action
field	$\mathbb{F}_{97}$	$\mathbb{F}_{p_{\mathsf{Pallas}}}$, a prime of 255 bits
commitment group	$y^2 = x^3 + 3$ over $\mathbb{F}_{79}$, 97 points (§3.7)	Vesta, of order p_{Pallas} (§3.8)
rows	$n = 4$, $H = \{1, 22, 96, 75\}$	$n = 2^{11} = 2048$
columns	3 advice, 5 selectors, 1 instance	10 advice, 29 fixed, 1 instance (§4.1)
gates	one universal gate (1), degree 3, rotation 0	custom gates at rotations, $d_{\max} = 9$ (Definition 2.7; the deployed value, §4.1.4)
identity	$G = t\,Z_H$; $\deg G = 9$, $\deg t = 5$	$C = h\,Z_H$ with C the y-combination; $\deg h \leq 16375$, 8 chunks (Theorem 4.7)
wiring	a permutation of twelve cells, four copy classes (b_3, c_3 fixed), one running product (§2.6)	fifteen equality-enabled columns, three running products
table facts	none on the grid; Example 2.13	three lookup arguments into a 2^{10}-row Sinsemilla table (§4.1.3)
the point	$z = 20$, by decree	x squeezed from the BLAKE2b transcript (§4.3)
one-term error	$11/97$; $3/97$ for the $c_0 = 10$ cheat	about 2^{-240}; 2^{-176} after 2^{64} grinding attempts
openings	a, b, c, t at z; four cross-term points and S per opening (§3.7)	49 claimed evaluations per Action and 45 shared, one opening of 23 points (Tables 10 and 8; the collapse, Construction 4.4)
proof	—	$2720 + 2272\,a$ bytes, 4992 for one Action (Table 8)

Table 11: The toy and the deployed Action circuit, object by object. Every row is the same object at two scales; every number is one already printed in the section cited, where it is computed by script or, for the 2^{10} rows of the Sinsemilla table, recorded from the definition of Sinsemilla.

The algebraic core of Action verification. For the verification of an Action proof that the consensus rules require (protocol specification, § 4.6, "Action Descriptions"), the randomised polynomial-identity test of link 1 is the algebraic core: a Schwartz–Zippel argument on one identity at one hashed point. The full verifier of §4.5 evaluates that identity's permutation and lookup summands from the claimed evaluations, binds every claimed evaluation through the multipoint reduction, recomputes every transcript challenge, and closes the single inner-product equation with its one multi-scalar multiplication of length n, before concluding that the circuit's constraints hold at the public input it verified against, the anchor, the net value commitment, the nullifier, the randomised verification key, the extracted note commitment and the flags of §4.2. Under the knowledge soundness whose status §5 classifies (designed-but-unspecified for the deployed transcript), an accepted Action proof certifies, for a known witness, that the constraints of the circuit whose verifying key the consensus rules fix (protocol specification, § 4.6) hold. That these imply the conditions of the Action statement (protocol specification, § 4.18.4) is a property of that circuit, which this volume does not verify and which the specification records as having failed for the Orchard circuit deployed before NU6.2 (the same section); value balance additionally requires the binding signature (protocol specification, § 4.14, "Balance and Binding

Signature (Orchard)"), and privacy rests also on components outside the proof, such as note encryption.

What the formal treatment supplies. Each of the five reductions of §1.2, "The chain of reductions", is owned by a formal statement. Table 12 lists, reduction by reduction, what the toy exhibited and which statement carries it; the last two rows point beyond the spine, to the extractor that turns existence into knowledge and to the recursion, which Orchard does not deploy and on which nothing above depends.

reduction	the toy exhibits	the statement that owns it
computation → grid	Table 1, Figure 2; knowing the witness is filling the advice	Definition 2.7 (§2.1, §2.5)
grid → polynomials	the nine interpolants; G of degree 9 vanishing on H	Theorem 2.1 (§2.2)
vanishing → identity	$t = G/Z_H$ of degree 5; the gap $D = 24(1 + X + X^2 + X^3)$	Theorem 2.2 (§2.3)
identity → one random check	$z = 20$ accepted; 3/97 and 11/97; Figure 3	Theorem 2.5 (§2.4); Lemma 4.6 and Theorem 4.7 (§4.5)
random check → committed openings		
binding	$\mathbf{a} = (90, 61, 22, 24)$ sealed in $P = (39, 25)$	the *Crypto Guide*, §"Pedersen vector commitments", recalled in §2.9
the opening	Tables 4 and 5; $v' = 60$ rejected	Construction 3.2, Theorem 3.3, Lemma 3.4, Theorem 3.5, Corollary 3.13 (§3)
the wiring	product 1 honest, 61 with $b_1 = 4$	Theorem 2.11 (§2.6)
the tables	$82 = 82$; the forged 7 stranded	Theorem 2.15 (§2.8)
the public 35	$\mathrm{PI} = (0, 0, 0, -35)$	Definition 4.2 (§4.2)
the removed verifier	z by decree against x by hash; $Q\varepsilon$	the *Crypto Guide*, §"The Fiat–Shamir transform: from interactive to non-interactive", applied in §4.3
many openings, one	—	Construction 4.4 and Lemma 4.5 (§4.4)
hiding	the leaked linear form; $c = 20$ against $c' = 46$	Remark 3.7 and the mask (§3.6, §3.8); Lemma 4.8 (§4.6)
the pattern	—	Remark 4.11 and "The map onto the phases" (§4.7)
"the prover knows the witness"	—	the extractor in the algebraic group model and the Fiat–Shamir loss (§5, "The algebraic group model")
beyond the spine	—	the accumulation fold, Proposition 7.5, the accumulation scheme cited as Theorem 7.4, and recursion over the cycle of curves, none deployed by Orchard, which uses one direction of the cycle only (§7.3, "Accumulation and the Halo trick"; §7.4, "The half Orchard uses")

Table 12: The map from the worked example to the formal statements that own it, in the order of the reductions of §1.2. The fifth reduction, from the random check to committed openings, is split into its parts.

7 Recursive proof composition and accumulation schemes

This section states recursive proof composition and accumulation for the inner-product argument; Orchard deploys neither. The protocol specification uses Halo 2 for one purpose, to prove and verify Action statements (protocol specification, § 4.1.13, "Zero-Knowledge Proving System"), and requires every Action proof to be valid (§ 4.6, "Action Descriptions"); the design proposal that introduced the pool states that it makes no use of the proof system's support for recur-

sive proofs (ZIP 224, under "Proving system"). A verifier holding several proofs may still check them together. The complete verification equation E_i of each proof is a multi-scalar multiplication over all d generators that should evaluate to $\mathcal{O}$ (§3.8.1, "Batch verification"); for independent uniform $\mu_1, \dots, \mu_b \in \mathbb{F}$ the verifier of b proofs checks $\sum_i [\mu_i] E_i = \mathcal{O}$ instead. If some $E_j \neq \mathcal{O}$, then E_j generates the group of prime order, so for any fixed values of the other factors exactly one μ_j satisfies the combined check, and a false batch passes with probability $1/|\mathbb{F}|$ (Lemma 3.14). Batch verification verifies the batch and emits nothing; it is not the accumulation developed here. In the classification of §1.2, the Pasta curves and the use Orchard makes of them (§7.4) are *specified*; recursion and accumulation over those curves are *designed-but-unspecified*: their constructions are published and the halo2 book describes them (*Recursion* chapter), and ZIP 224 leaves them to future protocol updates, but no specification defines a recursive statement or an accumulation rule for Zcash. Theorem 7.4 and Proposition 7.5 below are theorems, the first cited and the second proved.

7.1 Recursion and its two classical obstacles

The setting. Consider a computation iterated many times,

$$z_m = f^{(m)}(z_0) := f(f(\cdots f(z_0)\cdots)),$$

one step f applied m times to an initial state, and suppose a verifier wants to be sure that z_m is the honest result without recomputing the m steps. Valiant's *incrementally verifiable computation* asks for a proof of z_m that is updated step by step, each update costing about one step's worth of proving and the finished proof no larger than the proof of a single step. The more general setting, in which computations branch and merge and each step consumes the proofs of its predecessors, is the *proof-carrying data* of Chiesa and Tromer: every message in a distributed computation carries a proof that it was produced correctly from messages that themselves carried such proofs.

The construction. The natural construction is *recursive proof composition*: proving, inside one proof, that another proof verifies. Write $\mathcal{V}$ for the verifier of a non-interactive argument (*Crypto Guide*, §"SNARKs: succinct non-interactive arguments of knowledge"), and suppose its algorithm is itself arithmetised, in the sense of §2, into a PLONKish circuit of Definition 2.7, the *verifier circuit*: its instance columns hold the statement and its advice columns hold a proof, and its gates are satisfiable exactly when $\mathcal{V}$ accepts that proof of that statement. At step $i+1$ the prover proves the compound statement

> *"$z_{i+1} = f(z_i)$, and there exists a proof π_i that the verifier circuit accepts for the statement of step i."*

The proof π_i is advice of the step-$(i+1)$ circuit, never sent onward; the proof π_{i+1} that results attests to step $i+1$ and, through the embedded verifier, to π_i, hence to step i, and so on down to z_0. One proof attests to the entire history, and the cost of producing π_{i+1} is the cost of one step of f plus one run of the verifier circuit, independent of how long the history is. The requirement on which everything rests is the one just used: the argument's verifier must be efficiently expressible as a circuit. A verifier whose circuit is as large as the computation it checks would make each step as expensive as the whole history, and the construction would buy nothing.

Two obstacles. The classical instantiations meet that requirement badly, in two different ways.

1. *Pairings in a circuit, and a ceremony.* The pairing-based systems, Groth16, which Sapling deploys, and PLONK compiled with the KZG commitment (*Crypto Guide*, §"SNARKs: succinct non-interactive arguments of knowledge" for Groth16 and the bilinear pairing; §"Polynomial commitment schemes", under "KZG: evaluation in the exponent"), have constant-size proofs

and verifiers that run in time independent of the circuit. But those verifiers evaluate pairings, and a pairing lands in a target group that lives in an extension field $\mathbb{F}_{p^e}$ of the curve's base field, of degree e equal to the curve's embedding degree (*Crypto Guide*, §"Instantiation on elliptic curves; the Pasta curves", the large-embedding-degree criterion). A circuit whose native arithmetic is one prime field must emulate arithmetic in another field with many gates per operation, so a pairing is costly to express inside a circuit. These particular systems also need a trusted setup, the ceremony that §3.8.1 contrasted with hashed generators.

2. *A linear-time verifier.* The transparent systems built on the inner-product argument of §3 need no pairing and no ceremony, but their verifier is linear in the size of the committed vectors: the one multi-scalar multiplication $G^{(0)} = \langle \mathbf{s}, \mathbf{G} \rangle$ of length d that §3.4 singled out. Inside a verifier circuit that multiplication is d scalar multiplications of curve points, each a double-and-add with $\ell - 1$ doublings for an ℓ-bit scalar (*Math Guide*, §"Scalar multiplication and double-and-add"), so the verifier circuit grows linearly in d, the length of the committed vectors of the proof it verifies. A linear-time check dominates the verifier circuit and negates the per-step economy that recursion was meant to deliver.

Deferred verification. Halo, the construction of Bowe, Grigg and Hopwood (*Halo: Recursive Proof Composition without a Trusted Setup*) from which Halo 2 descends, *defers* the linear-time part of verification instead of recomputing it at every step, and uses the inner-product argument, in which no pairing occurs. Bünz, Chiesa, Mishra and Spooner formalised the deferral as an *accumulation scheme* (*Proof-Carrying Data from Accumulation Schemes*; BCMS20), the notion the next subsection defines; their Theorem 7.1 is quoted below, without proof, as Theorem 7.4.

7.2 Accumulation schemes

The idea is to separate what a verifier must check at each step from what can be checked once, at the end of a chain. Fix a *predicate* Φ, a property of an input q decidable in polynomial time, for example "this inner-product opening is correct" of a transcript of Construction 3.2. An accumulation scheme replaces a sequence of checks of Φ by a sequence of accumulation steps and one run of a decider.

Definition 7.1 (Accumulation scheme)**.** Let λ be the security parameter (*Crypto Guide*, §"Adversaries and the security parameter") and Φ a predicate on inputs q. An *accumulation scheme* for Φ is a triple $(\mathsf{P}, \mathsf{V}, \mathsf{D})$ of polynomial-time algorithms with access to a common random oracle, whose *accumulators* are bit strings.

1. The *accumulation prover* P, on input q and an old accumulator $\mathsf{acc}_{\mathrm{old}}$, outputs a new accumulator acc and an accumulation proof π_{V}.
2. The *accumulation verifier* V, on input $(q, \mathsf{acc}_{\mathrm{old}}, \mathsf{acc}, \pi_{\mathsf{V}})$, outputs a bit; the bit 1 asserts that acc correctly accumulates q and $\mathsf{acc}_{\mathrm{old}}$.
3. The *decider* D, on input an accumulator, outputs a bit; an accumulator is *valid* if D outputs 1 on it.

The scheme is *complete* if, whenever $\Phi(q) = 1$, $\mathsf{D}(\mathsf{acc}_{\mathrm{old}}) = 1$ and $(\mathsf{acc}, \pi_{\mathsf{V}})$ is an output of P on $(q, \mathsf{acc}_{\mathrm{old}})$, both $\mathsf{V}(q, \mathsf{acc}_{\mathrm{old}}, \mathsf{acc}, \pi_{\mathsf{V}}) = 1$ and $\mathsf{D}(\mathsf{acc}) = 1$. It is *sound* if for every adversary of size polynomial in λ that outputs $(q, \mathsf{acc}_{\mathrm{old}}, \mathsf{acc}, \pi_{\mathsf{V}})$ the probability, over the random oracle and the adversary's coins, of

$$\mathsf{V}(q, \mathsf{acc}_{\mathrm{old}}, \mathsf{acc}, \pi_{\mathsf{V}}) = 1 \ \wedge \ \mathsf{D}(\mathsf{acc}) = 1 \ \wedge \ (\Phi(q) = 0 \ \vee \ \mathsf{D}(\mathsf{acc}_{\mathrm{old}}) = 0)$$

is negligible in λ.

The definition is that of BCMS20 (Section 4.1) with one input and one old accumulator per step; theirs admits any number of each and adds a generator and an indexer, which fix the public parameters and keys left implicit here. For recursion the size of an accumulator must also be bounded independently of the number of inputs accumulated into it (BCMS20, Definition 5.1).

Soundness carries the checks along a chain.

Lemma 7.2 (Soundness along a chain)**.** *Let* $(\mathsf{P}, \mathsf{V}, \mathsf{D})$ *be sound in the sense of Definition 7.1, let* acc_0 *be valid, and let* m *be polynomial in* λ*. For any efficient algorithm producing* $q_1, \dots, q_m$ *and* $\mathsf{acc}_1, \dots, \mathsf{acc}_m$*, where* acc_i *accumulates* q_i *and* acc_{i-1}*, such that* V *accepts every step and* $\mathsf{D}(\mathsf{acc}_m) = 1$*, the probability that* $\Phi(q_i) = 0$ *for some* i *is at most* m *times the soundness error of the adversary that outputs step* i *for a uniform* $i \in \{1, \dots, m\}$*; it is therefore negligible.*

Proof. If some q_i violates Φ, then either acc_i is valid and step i exhibits the event of the soundness definition, or some later step $j > i$ turns an invalid acc_{j-1} into a valid acc_j and exhibits it. Let $\mathcal{A}$ be the algorithm that produces the chain and outputs step i for a uniform $i \in \{1, \dots, m\}$; since the bad event forces some step to exhibit the soundness event, its probability is at most m times the soundness error of $\mathcal{A}$, which is negligible because m is polynomial (*Math Guide*, §"Polynomial, exponential, and negligible functions", the closure properties). □

The verifier of each step runs the accumulation verifier and never the decider. In a recursive chain the accumulation verifier runs inside the circuit of the step's proof, so the argument also needs the knowledge soundness of each step's proof, to recover what that circuit checked; BCMS20 prove the resulting proof-carrying data secure for computations of constant depth, which for a chain means constant length, because the extractor is applied recursively once per level (Theorem 5.2 and Remark 5.3).

For an inner-product polynomial commitment such a scheme is known under the hypothesis of Theorem 7.4 below, and it resolves the second obstacle of §7.1 where it arose: the verifier circuit runs the $O(\log d)$ part of the opening check, the rounds, the scalars and the small combinations of points, and accumulates the $O(d)$ part, the claim about $G^{(0)}$, into a running accumulator; the decider pays that multi-scalar multiplication once, at the end of the chain. The next subsection defines the accumulator, states the scheme as an interface theorem, and works the fold on the toy curve.

7.3 Accumulation and the Halo trick

The asymmetry, recalled. Everything the verifier of Construction 3.2 does to decide the check (15), or its deployed form (29), is logarithmic in d: $3k - 2$ field multiplications for $g(x; \mathbf{u})$, a combination of the $2k$ received points and a handful of fixed ones with $O(k)$ scalar multiplications, and a comparison of two curve points. The single exception is the evaluation

$$G^{(0)} = \langle \mathbf{s}, \mathbf{G} \rangle = \mathsf{Commit}(g(X; u_1, \dots, u_k); 0),$$

a multi-scalar multiplication of length d over *all* the generators, which §3.4 named the one linear step and recognised as the unblinded commitment to the polynomial g whose *value* the verifier had just computed cheaply. Verifying an Action proof as the consensus rules require includes that step for every proof; batch verification, recalled at the start of this section, evaluates one combined multiplication for the proofs of a batch, and the batch is verified and nothing survives it. Accumulation takes the same kind of linear combination and does something else with it: it carries the combined claim forward as data, unverified, and pays for it once at the end of a chain. Figure 18 sets the two side by side; the rest of this subsection is the right-hand panel, and none of it is deployed by Orchard.

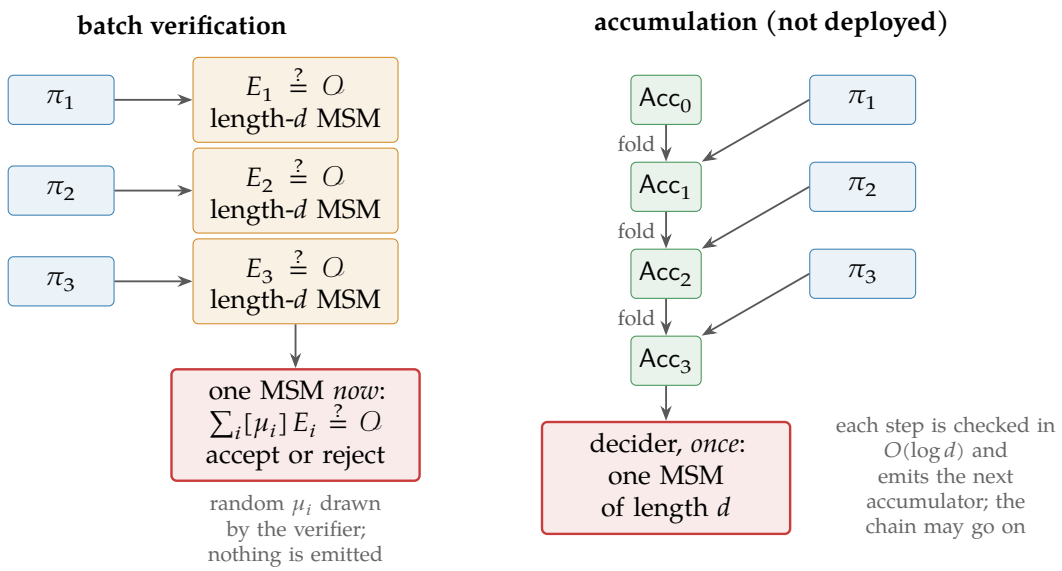


Figure 18: Left: batch verification. Each proof's complete verification equation E_i is a multi-scalar multiplication over the d generators (amber); the verifier combines them with fresh random factors and evaluates one multiplication immediately (red), then accepts or rejects. Right: accumulation. Each proof's deferred claim is accumulated into a running accumulator (green), each step being checked at logarithmic cost; the accumulator is emitted for the next step, and a decider evaluates one length-d multiplication at the end of the chain. Both take random linear combinations; only the right-hand scheme carries a claim forward.

The atomic accumulator. The claim to be deferred has a fixed shape, and the accumulator is that shape written down.

Definition 7.3 (Atomic accumulator for the inner-product argument)**.** An *accumulator instance* is a tuple

$$\mathsf{Acc} := (G^{(0)}, u_1, \dots, u_k) \in \mathbb{G} \times (\mathbb{F}^\times)^k,$$

a curve point and the k round challenges of one opening. It is *valid* if and only if

$$G^{(0)} = \langle \mathbf{s}(\mathbf{u}), \mathbf{G} \rangle = \mathsf{Commit}(g(X; u_1, \dots, u_k); 0), \tag{35}$$

with $\mathbf{s}(\mathbf{u})$ and g the structured scalars and the polynomial of Theorem 3.3, in the symmetric normalisation of Construction 3.2. The deferred claim concerns the *unblinded* commitment: the blinder of the opened commitment was discharged by the $[r^*]H$ term of (15), or the $[f]H$ term of (29), and plays no part in it. From here on the unary $\mathsf{Commit}(\cdot)$ abbreviates $\mathsf{Commit}(\cdot\,; 0)$.

The deployed analogue replaces $\mathbf{s}(\mathbf{u})$ and g by their forms in Remark 3.9; Proposition 7.5 and, with verifier-drawn challenges, the closure argument below hold for it verbatim, with Corollary 3.13 in place of Theorem 3.5, since they use only the linearity of the commitment, the degree of g and the knowledge soundness of the opening.

A verifier who is handed a point $G^{(0)}$ together with the challenges can complete the check (15) in $O(\log d)$ operations *provided* the tuple is valid; validity is precisely what he has not paid for. The scheme quoted next lets him defer that payment.

Theorem 7.4 (Accumulation for the inner-product argument; interface)**.** *Let* $\mathrm{PC}_{\mathrm{DL}}$ *be the inner-product polynomial commitment of BCMS20 (Appendix A.2), for polynomials of degree below d over a group* $\mathbb{G}$ *of prime order, and suppose that* $\mathrm{PC}_{\mathrm{DL}}$ *is a polynomial commitment scheme in the random oracle model, its openings in particular being extractable. Then (BCMS20, Theorem 7.1) there is an accumulation scheme in the sense of Definition 7.1, in the random oracle model, for the predicate "this* $\mathrm{PC}_{\mathrm{DL}}$ *opening is correct", whose accumulators are themselves* $\mathrm{PC}_{\mathrm{DL}}$ *openings, such that:*

1. *the accumulation prover* P *runs in time dominated by* $O(d)$ *scalar multiplications in* $\mathbb{G}$*;*
2. *the accumulation verifier* V *runs in time dominated by* $O(\log d)$ *scalar multiplications in* $\mathbb{G}$*;*
3. *the decider* D *is the opening check of* $\mathrm{PC}_{\mathrm{DL}}$*: a succinct part, which yields a tuple of the form of Definition 7.3 in the normalisation of Remark 3.9, followed by the validity check* (35) *of that tuple, one multi-scalar multiplication of length d.*

The theorem is quoted as an interface and not proved in this volume. Its hypothesis, the extractability of $\mathrm{PC}_{\mathrm{DL}}$ openings, is stated by BCMS20 as a conjecture from the binding of the Pedersen commitment, the extractor they record for the Fiat–Shamir-compiled opening running in superpolynomial time (BCMS20, Appendix A.3.2). The scheme $\mathrm{PC}_{\mathrm{DL}}$ is neither Construction 3.2, whose fold is symmetric, nor the deployed opening of Remark 3.9, which blinds every commitment and subtracts $[v]\,G_0$ where $\mathrm{PC}_{\mathrm{DL}}$ adds a multiple of its inner-product generator (the halo2 book's *Comparison to other work* chapter); no theorem transferring it to either is proved here or cited, and in the classification of §1.2 that transfer, like accumulation over these curves, is designed-but-unspecified. What an accumulation step does for Construction 3.2, and why its random combination is sound, is worked out below at the level the volume proves things.

The accumulation step in the verifier circuit. Consider a Halo 2 verifier circuit, expressed over the scalar field of one Pasta curve and verifying a proof produced over the other curve, an arrangement whose necessity §7.4 explains. At step i of a chain the circuit receives the previous proof π_{i-1}, whose opening is an instance of Construction 3.2, and the accumulator Acc_{i-1} exposed as a public input of π_{i-1}, of which it uses the instance on the curve π_{i-1} is committed on (see below), and does three things.

1. *Run the succinct part.* It recomputes the challenges $\mathbf{u}'$ of the opening from the transcript, evaluates $g(x;\mathbf{u}')$ with $O(k)$ scalar operations, and forms the left-hand side of (15) and every term of the right-hand side except the one on the final generator, here written $G'^{(0)}$, with $O(k)$ point operations. The point operations are native, since the points of π_{i-1} have coordinates in the circuit's field. The challenges, $g(x;\mathbf{u}')$ and the scalar products lie in the scalar field of the curve π_{i-1} is committed on, the other Pasta field; they are computed non-natively, or exposed as public inputs and checked by the next circuit of the chain, which is native to that field ("What the cycle does and does not buy" in §7.4; the halo2 book's *Recursion* chapter).
2. *Witness the expensive part.* It does not compute $G'^{(0)}$. The prover of the step supplies the claimed point as advice, and the circuit uses it to complete the check, which is then passed *conditionally on the validity of the fresh tuple* $(G'^{(0)}, \mathbf{u}')$ of that opening.
3. *Accumulate.* The step now holds two claims of the shape (35): the inherited instance $(G^{(0)}, \mathbf{u})$ of Acc_{i-1} and the fresh $(G'^{(0)}, \mathbf{u}')$. For a challenge ρ derived by the random oracle from both tuples, the two reduce to the single claim

$$G^{(0)} + [\rho]\,G'^{(0)} \;=\; \mathsf{Commit}(g(X;\mathbf{u}) + \rho\, g(X;\mathbf{u}')) = \langle \mathbf{s}(\mathbf{u}) + \rho\,\mathbf{s}(\mathbf{u}'),\, \mathbf{G}\rangle, \tag{36}$$

whose coefficient vector on the right anyone holding both challenge tuples can recompute. Proposition 7.5 shows that a false input survives the combination for at most one ρ. The step's

prover then opens the combined point at a fresh point, outside the circuit, and the circuit checks that opening succinctly as in item 1; the tuple the opening leaves behind is the new accumulator Acc_i (over a 2-cycle, its instance on that curve), exposed as a public input of the proof π_i the step produces. The paragraph "Closure: the prover-assisted step" below describes that opening.

Over a 2-cycle of curves (§7.4) the proofs alternate between the two curves, and the accumulator is a pair of instances, one per curve. The step that verifies a proof committed on one curve folds that proof's fresh tuple into the instance on the same curve, which was last updated two steps earlier and forwarded unchanged by the intervening step, and it forwards the other curve's instance in the same way, without point arithmetic. Figure 19 draws the chain. At its end a single multi-scalar multiplication of length d, run outside any circuit, checks the last accumulator and with it every opening in the history: for the scheme of Theorem 7.4 by Lemma 7.2.

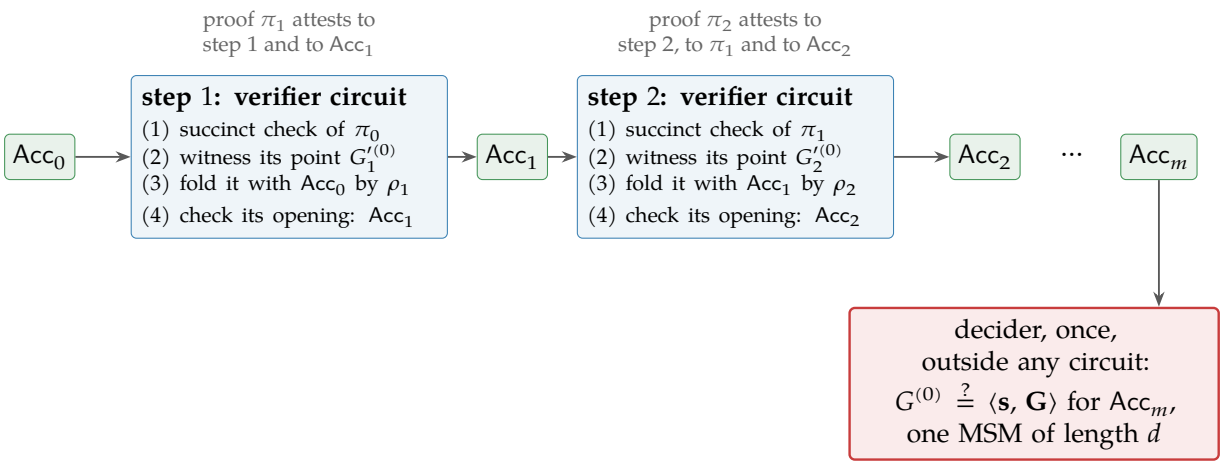


Figure 19: A chain of recursively composed proofs. Each step's verifier circuit (blue) runs the logarithmic part of the previous proof's opening check, witnesses the expensive point instead of computing it, combines the resulting tuple with the accumulator (green) it inherited by (36), and checks succinctly the opening of the combined point at a fresh point, whose tuple is the new accumulator, a public input of the step's own proof. Over a cycle of curves each Acc_i is a pair of instances, one per curve: each step folds into the instance on the curve of the proof it verifies and passes the other through unchanged. The decider (red) pays one length-d multi-scalar multiplication at the end of the chain, and no circuit ever contains one.

Why the fold is sound. The soundness of the fold (36) is one line of linear algebra in the group.

> **Proposition 7.5** (Soundness of the ρ-fold)**.** *Let $(G^{(0)}, \mathbf{u})$ and $(G'^{(0)}, \mathbf{u}')$ be two accumulator instances and ρ uniform in $\mathbb{F}$, drawn after both are fixed. If both instances are valid, the folded equation (36) holds for every ρ. If at least one is invalid, it holds for at most one value of ρ, hence with probability at most $1/|\mathbb{F}|$.*

Proof. Set $\Delta := G^{(0)} - \mathsf{Commit}(g(X;\mathbf{u}))$ and $\Delta' := G'^{(0)} - \mathsf{Commit}(g(X;\mathbf{u}'))$, the two errors. By linearity of the commitment (*Crypto Guide*, §"Pedersen vector commitments", the remark that linearity survives), $\mathsf{Commit}(g + \rho g') = \mathsf{Commit}(g) + [\rho]\mathsf{Commit}(g')$, so the left-hand side of (36) minus the right is $\Delta + [\rho]\,\Delta'$, and the equation holds exactly when

$$\Delta + [\rho]\,\Delta' = \mathcal{O}. \tag{37}$$

If both instances are valid, $\Delta = \Delta' = \mathcal{O}$ and the equation holds for all ρ. Otherwise two cases. If $\Delta' \neq \mathcal{O}$, then, $\mathbb{G}$ being cyclic of prime order, Δ' generates $\mathbb{G}$ and the map $\rho \mapsto [\rho]\Delta'$ is a bijection

$\mathbb{F} \to \mathbb{G}$ (*Math Guide*, §"The mixed inner product with group elements", the one-dimensional vector space $\mathbb{G}$ over $\mathbb{F}$), so exactly one ρ sends Δ' to $-\Delta$: the discrete logarithm of $-\Delta$ to the base Δ', which exists whether or not anyone can compute it. If $\Delta' = \mathcal{O}$, then $\Delta \neq \mathcal{O}$ and (37) holds for no ρ. A uniform ρ hits the at most one bad value with probability at most $1/|\mathbb{F}|$. □

Example 7.6 (The fold on the toy curve)**.** The fold is run on the 97-point curve and the four generators $G_0, \dots, G_3$ of §3.7, with $d = 4$ and the symmetric structured scalars of Theorem 3.3, on two challenge tuples, $(u_1, u_2) = (52, 44)$ with $\mathbf{s} = (80, 10, 68, 57)$ and $G^{(0)} = \langle \mathbf{s}, \mathbf{G} \rangle = (21, 10)$, and $(u'_1, u'_2) = (41, 39)$ with $\mathbf{s}' = (64, 11, 53, 47)$ and $G'^{(0)} = (23, 2)$, both valid by construction, for every $\rho \in \mathbb{F}_{97}$. With both instances valid the folded equation (36) holds for all 97 values of ρ. Spoiling an instance by adding a nonzero multiple of G_0 to its point makes its error Δ a nonzero point, and the surviving values of ρ are counted again. With the first instance false and the second valid, no ρ survives: this is the case $\Delta' = \mathcal{O} \neq \Delta$ of the proof. With the second false and the first valid, exactly one survives, $\rho = 0$, the value that discards the second instance altogether. With both false exactly one survives, $\rho = 50$, the discrete logarithm of $-\Delta$ to the base Δ' on this curve, where such logarithms are a short search. In every spoiled case the count is at most one of 97: the bound $1/|\mathbb{F}|$ of Proposition 7.5, attained.

Closure: the prover-assisted step. The folded claim (36) is not an accumulator instance. Its coefficient vector $\mathbf{s}(\mathbf{u}) + \rho\, \mathbf{s}(\mathbf{u}')$ is in general not the structured vector of any single challenge tuple, so carrying it means carrying both tuples, and after m folds all m of them: the decider would still pay one multiplication of length d, but the accumulator would grow with the chain, against the size bound that recursion requires (§7.2). The tuple type of Definition 7.3 is therefore *not* closed under verifier-side linear combination alone. It is closed under one more step, which the prover assists. Write $A := G^{(0)} + [\rho]\, G'^{(0)}$ for the combined point and $a(X) := g(X; \mathbf{u}) + \rho\, g(X; \mathbf{u}')$ for the combined polynomial, of degree below d. The combined claim says exactly that $A = \mathsf{Commit}(a)$. The step's prover opens A by the opening protocol of Construction 3.2 at a fresh point x', derived by the random oracle from A and both tuples, claiming the value $a(x')$, which the verifier computes for himself in $O(k)$ field operations from the two tuples, since $g(x'; \mathbf{u})$ and $g(x'; \mathbf{u}')$ are each a product of k factors. That opening does two things at once. First, it carries the combined claim up to a blinder.

> **Proposition 7.7** (Prover-assisted step, verifier-drawn challenges)**.** *Under the hypotheses of Theorem 3.5, let ρ, x' and the opening's challenges be drawn uniformly by the verifier, in this order, ρ after the two tuples are fixed. There is an extractor which, given rewindable access to a step prover accepted with non-negligible probability, runs in expected polynomial time and outputs either blinders δ, δ' with $G^{(0)} = \mathsf{Commit}(g(X; \mathbf{u});\, \delta)$ and $G'^{(0)} = \mathsf{Commit}(g(X; \mathbf{u}');\, \delta')$, or a nontrivial discrete-logarithm relation among $\mathbf{G}$, H, U, U'. Its knowledge error is $O(\log d/|\mathbb{F}|) + (d-1)/|\mathbb{F}| + 1/|\mathbb{F}|$.*

Proof. By Theorem 3.5 the opening yields a representation $A = \langle \mathbf{a}^*, \mathbf{G} \rangle + [r_A]H$, unique since a second one would be a discrete-logarithm relation among $\mathbf{G}$ and H, and certifies $a^*(x') = a(x')$ for the polynomial a^* with coefficient vector $\mathbf{a}^*$; two polynomials of degree below d that differ agree at a uniform x', drawn after A was fixed, with probability at most $(d-1)/|\mathbb{F}|$ (*Math Guide*, §"The Schwartz–Zippel lemma"), so except with that probability $\mathbf{a}^*$ is the coefficient vector of a and $A = \mathsf{Commit}(a;\, r_A)$. The blinder r_A is the prover's and need not be 0. Rewinding the choice of ρ and extracting again at a second value $\tilde{\rho} \neq \rho$ gives representations of A and of $\tilde{A} := G^{(0)} + [\tilde{\rho}]\, G'^{(0)}$, whose difference is $[\rho - \tilde{\rho}]\, G'^{(0)}$; solving yields $G'^{(0)} = \mathsf{Commit}(g(X; \mathbf{u}');\, \delta')$ and $G^{(0)} = \mathsf{Commit}(g(X; \mathbf{u});\, \delta)$ for blinders δ, δ' that the extractor computes. This is the argument of Proposition 7.5 applied to the $\mathbf{G}$-coordinates of the representations: a combination correct at two values of ρ has both errors zero. It is all the chain needs: a tuple whose point is $\mathsf{Commit}(g(X; \mathbf{u});\, \delta)$

still certifies the opening it came from, because, writing c for the final scalar that (15) calls a,

$$[c]\,\mathsf{Commit}(g;\,\delta) + [r^*]H = [c]\,\mathsf{Commit}(g) + [r^* + c\delta]H,$$

so that opening satisfies the honest check with the blinder $r^* + c\delta$ in place of r^*, and Theorem 3.5 applies to it; in (29) the offset joins the term $[f]H$ likewise. The chain thus carries the claim that each deferred point's $\mathbf{G}$-coordinates are $\mathbf{s}(\mathbf{u})$; the exact equality (35), blinder 0 included, is required only of the last accumulator, which the decider checks and an honest prover satisfies. An intermediate tuple may miss (35) by such a multiple $[\delta]H$ without any opening it serves being false. □

Second, the opening leaves behind a deferred claim of the atomic form: it ends, as every opening does, in a point $G''^{(0)}$ and challenges $\mathbf{u}''$ that the verifier circuit witnesses rather than computes. The accumulator after the step is again one tuple. This is the accumulation step of BCMS20 (Section 7.1), which likewise derives the combining challenge and the fresh point by the random oracle from the tuples and the combined point; there the accumulator is the opening itself, and its succinct part runs at the next step. With ρ, x' and the opening's challenges drawn by a verifier, the step's soundness is Proposition 7.7. With random-oracle challenges, the step's soundness is BCMS20's Theorem 7.1 for $\mathrm{PC}_{\mathrm{DL}}$ under its conjectured extractability (Theorem 7.4); for Construction 3.2 it is designed-but-unspecified.

What the accumulator carries. The deferred equation (35) is a claim of *commitment equality*, equivalently a claim about the value of one multi-scalar multiplication, for a coefficient vector $\mathbf{s}(\mathbf{u})$ that is public and structured: anyone holding the challenges can write it down. It is not itself an evaluation opening; no polynomial is being evaluated at any point when the decider runs, only a point recomputed and compared. The accumulation protocol reduces this claim, and carries it along the chain, through the prover-assisted opening just described, which turns each commitment equality into one evaluation claim at a fresh point and each opening back into one commitment equality. The Orchard verifier, once more, does none of this: it computes $G^{(0)}$ and compares.

7.4 A cycle of curves for efficient native recursion

The verifier circuit of §7.3 was "expressed over the scalar field of one Pasta curve and verifying a proof produced over the other". This subsection explains why the two curves are needed, what the arrangement buys, and where Orchard already uses half of it.

Two fields per curve, recalled. A curve E over a prime field has two fields attached to it (*Math Guide*, §"Base fields, scalar fields, and the Pasta cycle"): the *base field*, in which its points' coordinates lie and in which the group law's slopes, squarings and inversions are computed, and the *scalar field*, the integers modulo the prime order of its group, in which the scalars of $[k]P$ live. Write E_p for a curve over $\mathbb{F}_p$ and $q := \#E_p(\mathbb{F}_p)$ for its prime order, so that its base field is $\mathbb{F}_p$ and its scalar field $\mathbb{F}_q$. A PLONKish circuit of Definition 2.7 is defined over one prime field $\mathbb{F}$, its *native field*: its cells hold elements of $\mathbb{F}$ and its gates are polynomial identities over $\mathbb{F}$, so an addition or multiplication in $\mathbb{F}$ costs one cell or one gate. Arithmetic in any *other* prime field is *non-native*: an element of the other field must be carried as several $\mathbb{F}$-cells, each range-checked by the lookups of §2.7, and each of its operations becomes many gates with carries and reductions. Which field a circuit is native to is not a free choice. Its column polynomials are committed with the inner-product commitment of §3 on some curve, and a Pedersen commitment to a vector of $\mathbb{F}$-elements needs a group whose scalar field is $\mathbb{F}$; so *the native field of a circuit is the scalar field of the curve its proof is committed on*, exactly as the Action circuit, native to $\mathbb{F}_{p_{\mathsf{Pallas}}}$, is committed on Vesta, whose order is p_{Pallas} (§3.8).

Why one curve cannot host its own verifier. Take a proof committed on E_p. Its group elements, the column and quotient commitments, the cross terms (L_j, R_j) and the deferred point of its opening, are points of E_p with coordinates in $\mathbb{F}_p$. The in-circuit folding of §7.3 adds and scales such points, which is base-field arithmetic in $\mathbb{F}_p$, so a verifier circuit handles them natively only if its native field is $\mathbb{F}_p$. But a circuit committed on E_p is native to the *scalar* field $\mathbb{F}_q$: its challenges, its structured scalars and every field operation of the succinct part are $\mathbb{F}_q$-arithmetic. One curve would host both jobs only if $\mathbb{F}_p = \mathbb{F}_q$, that is, $\#E_p(\mathbb{F}_p) = p$. Curves with exactly p points exist; by Hasse's theorem (*Math Guide*, §"The group structure of $E(\mathbb{F}_p)$"), which writes $\#E_p(\mathbb{F}_p) = p+1-t$ with the trace t, they are the curves with $t = 1$, called *anomalous*. On an anomalous curve the discrete logarithm is computable in polynomial time, the attack of Smart, of Semaev and of Satoh and Araki, and the *Crypto Guide* lists "not anomalous" among the criteria a curve must meet to be used at all (§"Instantiation on elliptic curves; the Pasta curves", the anti-Smart criterion). So no *secure* curve has $\mathbb{F}_p = \mathbb{F}_q$, and no one curve can host both the coordinates of its proofs and its own verifier circuit natively.

The 2-cycle. The efficient resolution uses two curves that trade fields. A *2-cycle* of elliptic curves is a pair (E_p, E_q) with

$$\#E_p(\mathbb{F}_p) = q \qquad \text{and} \qquad \#E_q(\mathbb{F}_q) = p,$$

both primes: the scalar field of each curve is the base field of the other. Then:

1. a circuit committed on E_q has native field $\mathbb{F}_p$, the scalar field of E_q; and $\mathbb{F}_p$ is the coordinate field of E_p, so that circuit manipulates the points of a proof produced over E_p natively, and verifies such a proof at the cost §7.3 counted, the proof's scalars aside;
2. the proof that circuit produces is committed on E_q, and its points have coordinates in $\mathbb{F}_q$, the native field of a circuit committed on E_p; so a circuit over $\mathbb{F}_q$ handles its points natively in turn;
3. the proofs of a chain alternate between the two curves, each step handling its predecessor's points natively (Figure 20).

What the cycle does and does not buy. The cycle makes the *preceding* curve's coordinate arithmetic native; it does not identify the two scalar fields, which remain distinct primes. Challenges squeezed in the previous proof's field and the scalars derived from them are $\mathbb{F}_p$-elements that the $\mathbb{F}_q$-native circuit must handle: they need compatible encodings across the two fields, range checks where an element of the larger field is carried in the smaller, and, where a genuine $\mathbb{F}_p$ multiplication is unavoidable, non-native scalar arithmetic. Nor is the cycle a logical necessity. A single curve could verify its own proofs with non-native coordinate arithmetic throughout, at a cost of many gates per point operation multiplied by the $O(\log d)$ point operations of the succinct part; that is possible and substantially more expensive. The cycle is the efficient native design, not the only correct one.

The deployed instance. The Pasta cycle of the *Math Guide* (§"Base fields, scalar fields, and the Pasta cycle"; §"Pallas and Vesta assembled") is a 2-cycle: Pallas is defined over $\mathbb{F}_{p_{\mathsf{Pallas}}}$ with group order p_{Vesta}, and Vesta over $\mathbb{F}_{p_{\mathsf{Vesta}}}$ with group order p_{Pallas}, both curves $y^2 = x^3 + 5$ (protocol specification, § 5.4.9.6, "Pallas and Vesta"). The cycle follows from the specification's two primes and nothing else. With

$$p_{\mathsf{Pallas}} = \texttt{0x40000000 00000000 00000000 00000000 224698fc 094cf91b 992d30ed 00000001},$$
$$p_{\mathsf{Vesta}} = \texttt{0x40000000 00000000 00000000 00000000 224698fc 0994a8dd 8c46eb21 00000001},$$

the point $(-1, 2)$ lies on Pallas, $[p_{\mathsf{Vesta}}](-1, 2) = \mathcal{O}$ there, and exactly one multiple of p_{Vesta} lies in the Hasse interval $[\, p + 1 - 2\sqrt{p},\ p + 1 + 2\sqrt{p}\,]$ for $p = p_{\mathsf{Pallas}}$. That settles the order: the order

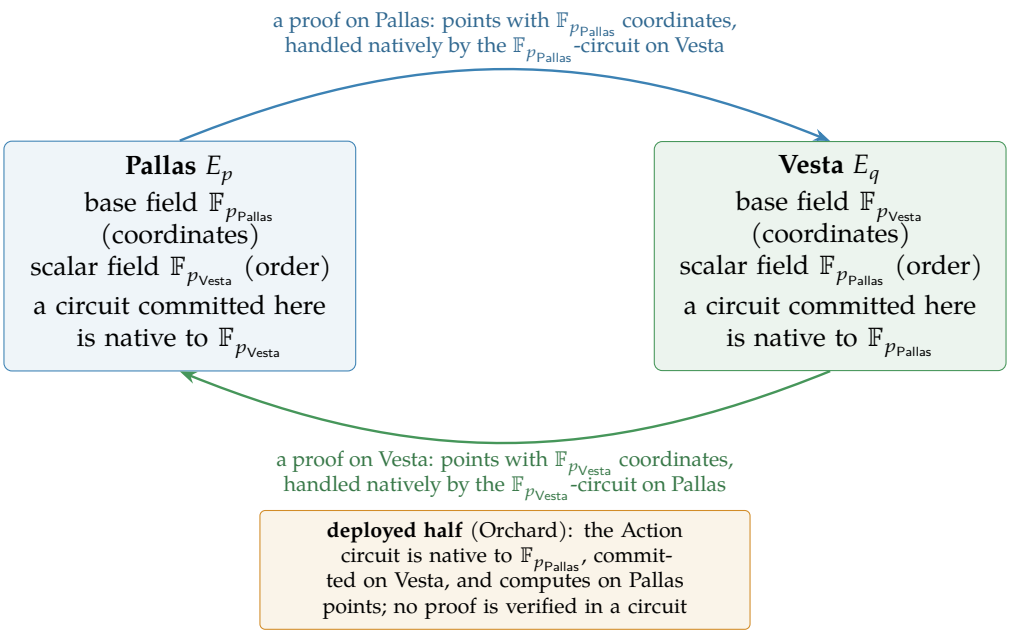


Figure 20: The Pasta 2-cycle. The base field of each curve (its coordinate field) is the scalar field of the other, so a circuit committed on Vesta (green) is native to $\mathbb{F}_{p_{\mathsf{Pallas}}}$ and handles Pallas points (blue) natively, and symmetrically. In a recursive chain the proofs alternate along the two arrows. Orchard (amber) uses one direction of the cycle only: Pallas arithmetic inside a Vesta-committed circuit, with no verifier circuit at all.

of the point $(-1, 2)$ divides the prime p_{Vesta} and is not 1, so it is p_{Vesta}, which therefore divides $\#\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$ by Lagrange's theorem (*Math Guide*, §"Cosets and Lagrange's theorem"); the group order lies in an interval of length $4\sqrt{p}$, far shorter than p_{Vesta}, which contains one multiple of p_{Vesta}; hence $\#\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}}) = p_{\mathsf{Vesta}}$. The symmetric check on Vesta shows that the Vesta group has order p_{Pallas}. The traces are

$$t_{\mathsf{Pallas}} = p_{\mathsf{Pallas}} + 1 - p_{\mathsf{Vesta}} = -86663725065984043395317759,$$
$$t_{\mathsf{Vesta}} = p_{\mathsf{Vesta}} + 1 - p_{\mathsf{Pallas}} = 86663725065984043395317761,$$

each far inside the Hasse bound and neither equal to 1: neither curve is anomalous, as the *Crypto Guide*'s criterion requires. The two traces sum to 2, as they must for any 2-cycle, since $(p+1-q)+(q+1-p) = 2$; the toy curve of §3.7, $y^2 = x^3+3$ over $\mathbb{F}_{79}$ with 97 points, has trace $79+1-97 = -17$, and the case excluded in that subsection, a curve over $\mathbb{F}_{97}$ with 97 points and so trace 1, is exactly the anomalous one.

The cycle in the accumulation chain. The Pasta cycle renders each step of §7.3 native. A proof committed on Vesta is produced by a circuit native to $\mathbb{F}_{p_{\mathsf{Pallas}}}$; that circuit verifies the previous proof of the chain, committed on Pallas, whose group elements, the column and quotient commitments, the pairs (L_j, R_j) and the point of the Pallas accumulator instance it forwards, have coordinates in exactly $\mathbb{F}_{p_{\mathsf{Pallas}}}$, so the point arithmetic of the succinct checks and of the fold (36) is native. The next proof, committed on Pallas by a circuit native to $\mathbb{F}_{p_{\mathsf{Vesta}}}$, verifies the Vesta proof by the symmetric argument. This cross-curve alternation avoids non-native emulation of the preceding curve's coordinate field; as "What the cycle does and does not buy" said, it does not eliminate the scalar-field conversions.

The half Orchard uses. This is the part of the cycle in consensus. The Action circuit is native to $\mathbb{F}_{p_{\mathsf{Pallas}}}$ and committed on Vesta (§3.8), and the statement it proves manipulates Pallas points: the

note commitment cm, a Pallas point whose x-coordinate cmx is a public input; the randomised verification key rk; and the net value commitment $\mathsf{cv}^{\mathsf{net}}$, the last two exposed by their coordinates in the instance column of §4.2. Their coordinates lie in $\mathbb{F}_{p_{\mathsf{Pallas}}}$, the circuit's native field, by the very coincidence the cycle is built on, so the circuit computes on them natively, with gates such as the incomplete-addition gate of §4.1.2. The protocol specification records the arrangement: Vesta for the proof system, Pallas for the application circuit, both curves designed to be efficiently implementable inside a circuit though only Pallas is so used (protocol specification, § 5.4.9.6, "Pallas and Vesta"); ZIP 224 says the same in the language of this subsection, that the proposal uses half of the cycle, Pallas being the curve embedded in Vesta's circuits, and that the full cycle is left for future use (ZIP 224, under "Curves"). One direction of Figure 20 is thus in consensus today; the return arrow, and everything in this section that depends on it, is not.

Part IV
Consensus Guide: Nakamoto consensus: the double-spend race, the arithmetic of confirmations, and the deployed most-work rule

Abstract

Assuming the *Math Guide*'s probability foundations and the hash and commitment foundations of the *Crypto Guide*, this volume of *The Zcash Arboretum* constructs Nakamoto consensus from first principles and quantifies the security it buys. It states the block tree and the most-work rule as the protocol specification defines them and as the deployed zebra implements them; builds, on the *Math Guide*'s foundations, the race-specific probability the analysis needs (memoryless clocks, the biased random walk, the negative binomial law); develops the exact first-seen-model double-spending analysis of Rosenfeld's *Analysis of hashrate-based double-spending* — the catch-up theorem, the confirmation formula, and the economics of the attack — with every derived number recomputed by script and the results instantiated for Zcash's proposed 25-second blocks; explains the draft NU7 transition to 25-second blocks and halving-preserving reissuance; and explains the protocol and implementation safety rails (difficulty adjustment, reorg windows, coinbase maturity, wallet confirmation policy) that frame the model's assumptions. Where the specification, the paper, and the deployed nodes disagree, the divergence is stated.

1 Scope, sources, and the two double-spends

The protocol layers above this one take a working ledger for granted. This volume documents the mechanism that selects that ledger: Nakamoto consensus — proof of work, the most-work rule, and the probabilistic finality they purchase. The treatment is quantitative throughout. Its spine is Meni Rosenfeld's *Analysis of hashrate-based double-spending* (11 December 2012; latest version 12 February 2014; arXiv:1402.2009), which gives an exact counterpart to the approximate analysis in Satoshi Nakamoto's whitepaper; we reprove its results in full, recompute its tables by script, and instantiate its conclusions for the proposed NU7 parameters.

Two distinct attacks are both called "double-spending", and this volume treats exactly one of them. Two spends of the same shielded note reveal the same nullifier, and the protocol specification's §"Nullifier Sets" forbids a repeated nullifier within a transaction or valid block chain; such a same-chain attempt is therefore invalid under the consensus rules. What consensus must prevent is the chain-level attack: paying a merchant on the chain everyone sees, then replacing that chain wholesale with another in which the payment never happened. No zero-knowledge proof (*Crypto Guide*, §"Interactive proofs, zero knowledge, and SNARKs") rules this out; under the model developed here, the proof-of-work race gives the attack its probabilistic bound.

The quantitative argument assumes the *Math Guide*'s probability section (§"Probability, asymptotics, and computation"), which constructs probability from its definition — finite spaces, conditioning, independence, random variables, expectation — and extends it to the three rungs used here: countable additivity, continuity along monotone events, and density-defined laws (§"Beyond finite spaces: countable additivity, limits, and densities"). The block and anchor vocabulary uses the *Crypto Guide*'s hash and commitment sections. On that base, Section 3 constructs the race-specific instruments no lower volume owns — memoryless clocks, random walks, the negative binomial law; the algebra used is elementary. Readers wanting only the results can read Sections 2, 6, and 8 and take the theorems on faith.

1.1 Sources and their standing

Rosenfeld's analysis supplies the probability and economic model, not consensus rules; its two printed tables are reproduced by exact arithmetic before instantiating the model here. The protocol specification and the relevant ZIPs define the protocol.

This volume targets the *draft NU7 rules* as of 23 September 2026: ZIP-259 selects ZIP-218, ZIP-237, and ZIP-2003. Activation heights remain unassigned, and some proposed limits and issuance rules are incomplete. These boundaries are stated where they affect a calculation; no unassigned constant is treated as a deployed rule.

Remark 1.1 (The paper's own condition). The arXiv v1 PDF of the paper is lightly broken: it contains exactly two citations, both rendered as "[?]" (in the abstract and in §1), and no References section at all. Both plainly intend the Nakamoto whitepaper, and we read them so. The paper numbers only two of its displayed equations — its equation (1) is our Theorem 5.2, its equation (2) our equation (3) — and sets its two model assumptions as a bare numbered list; the *Assumption* environments of Section 4 are ours.

2 The block tree and the most-work rule

Zcash transactions are grouped into blocks, and every block names its parent by including the parent header's hash (*Crypto Guide*, §"Hash functions and the random oracle model"); the genesis block alone has no parent. Blocks therefore form a tree rooted at genesis, and each root-to-leaf path — each *branch* — is one internally consistent version of history. Branches need not be consistent with one another: one branch can contain a transaction that conflicts with a transaction in a sibling branch, and that freedom is exactly the attack surface this volume quantifies.

2.1 Work, and the best chain

A single coherent history requires a rule every node can apply locally and use to converge once one valid branch is strictly preferred. The rule is not simply "longest": it weighs each block by the computational effort represented by its header target.

Definition 2.1 (Work of a block)**.** Let `ToTarget` be the conversion from the header's `nBits` field to its 256-bit target threshold (specification §7.7.4). The *work* of a block is

$$\mathsf{work} \;=\; \left\lfloor \frac{2^{256}}{\mathtt{ToTarget}(\mathtt{nBits}) + 1} \right\rfloor$$

(specification §7.7.5, "Definition of Work"). A lower target admits fewer valid headers, so a block meeting it certifies proportionally more expected hashing; work is that certificate made additive.

The deployed Zebra validator computes this quantity with a 256-bit in-word trick: `zebra` evaluates `(!expanded.0 / (expanded.0 + 1)) + 1`; the retired `zcashd` computed the identical expression on `arith_uint256`, with `~` spelling the complement. The form agrees with Definition 2.1 exactly: writing t for the target and $A = 2^{256}$, the complement is $\neg t = A - 1 - t$, and

$$\left\lfloor \frac{A - 1 - t}{t + 1} \right\rfloor + 1 = \left\lfloor \frac{A - (t + 1)}{t + 1} \right\rfloor + 1 = \left\lfloor \frac{A}{t + 1} \right\rfloor,$$

so the in-word computation equals the out-of-word quotient (checked by script over random targets).

Definition 2.2 (Best valid block chain)**.** Specification §3.3 ("The Block Chain"): a node "sums the work, as defined in §7.7.5, of all blocks in each valid block chain, and considers the valid block chain with greatest total work to be best. To break ties between leaf blocks, a node will prefer the block that it received first."

Rosenfeld's paper states the same rule for Bitcoin — "the branch representing the most proof of work", with first-seen tie-breaking — so the model uses the same selection rule. The shorthand "longest chain" is exact only while difficulty is constant, which is precisely the paper's Assumption 4.2 and not a property of the deployed chain (Section 8.2). Note also what the rule does *not* say: nothing marks a chain as permanently chosen. A node that learns of a heavier branch switches to it, however deep the fork point — subject only to the node-local rails of Section 8.4. Consensus is a standing competition, not an election with a winner.

Definition 2.3 (Confirmations)**.** A transaction has *n confirmations* if it is included in a block of the best chain and there are n blocks on the path from that block to the chain's leaf, inclusive. The block containing the transaction is its first confirmation.

Different nodes may briefly disagree on the best chain when two blocks of equal cumulative work arrive in different orders. Absent a concurrent extension of the other branch, a later block makes one branch strictly heavier, and nodes converge on it once that block propagates (Figure 1). Such ties are the benign case. The attack of Section 5 is the malicious one: a fork manufactured in private and released only when it wins.

(a) tied branches: nodes disagree (b) one more block makes the upper branch heavier

Figure 1: Branch selection. Arrows point from child to parent. In panel (a) two blocks extend the same parent and the branches carry equal work; each node keeps the one it saw first. In panel (b) a new block lands on the upper branch, which now has strictly more cumulative work; during propagation, every node that knows this state selects the upper branch, unless the lower branch has meanwhile been extended. At constant difficulty "more work" and "longer" coincide, which is how the paper, and this volume's figures, may draw block counts.

2.2 The attack

The double-spend attack, as the paper lays it out (§3):

1. broadcast a transaction paying the merchant;
2. secretly mine a branch from the last pre-transaction block, containing instead a conflicting transaction paying the attacker himself;
3. wait until the merchant's transaction has n confirmations and the merchant, satisfied, hands over the goods;
4. in the paper's first-seen tie model, keep extending the secret branch until it carries more work than the public one, then broadcast it. Every node that sees it switches to it, the payment to the merchant is no longer part of history, and the attacker has both the goods and the coins.

Nothing in step 4 violates any validity rule — the released branch is a perfectly well-formed chain, merely a different one. The only question is probabilistic: with what probability does the secret branch ever get ahead? Answering it exactly in that first-seen model is the business of Sections 4 and 5; the toolkit comes first.

3 A probability toolkit

The analysis ahead needs four tools: memoryless clocks (to model block discovery), a reduction from continuous time to a coin sequence, the negative binomial distribution (to count the attacker's progress), and one normalisation lemma about races. Their foundations are the *Math Guide*'s: probability spaces, conditioning, independence, random variables, and expectation are constructed there from the definition of probability onward (§"Probability, asymptotics, and computation"), and the extension beyond finite spaces — countable additivity, continuity along monotone events, density-defined laws, and infinite sequences of independent trials — is its §"Beyond finite spaces: countable additivity, limits, and densities". What no lower volume owns are the four instruments themselves; each is constructed here on that base, only as far as the critical path requires.

3.1 Memoryless clocks and the attribution sequence

Definition 3.1 (Exponential clock)**.** A random variable X is *exponential with rate* $\lambda > 0$ if $\Pr[X > t] = e^{-\lambda t}$ for all $t \geq 0$ — a law defined by the density $\lambda e^{-\lambda t}$ on $t \geq 0$ in the sense of the *Math Guide*'s density definition, with mean $1/\lambda$.

Lemma 3.2 (Memorylessness)**.** *An exponential X satisfies $\Pr[X > s + t \mid X > s] = \Pr[X > t]$ for all $s, t \geq 0$: having waited without success teaches nothing about the remaining wait.*

Proof. $\Pr[X > s + t \mid X > s] = e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t}$. □

Lemma 3.3 (Race of two clocks)**.** *Let X and Y be independent exponentials with rates λ and μ. Then*

$$\Pr[X < Y] \;=\; \frac{\lambda}{\lambda + \mu},$$

and the winning time $\min(X, Y)$ is exponential with rate $\lambda + \mu$.

Proof. Integrating over the time at which X fires,

$$\Pr[X < Y] = \int_0^\infty \lambda e^{-\lambda t} \Pr[Y > t]\, dt = \int_0^\infty \lambda e^{-(\lambda+\mu)t}\, dt = \frac{\lambda}{\lambda + \mu}.$$

For the minimum, $\Pr[\min(X, Y) > t] = \Pr[X > t]\Pr[Y > t] = e^{-(\lambda+\mu)t}$. Moreover, the joint density that X wins at time t factors as

$$\lambda e^{-(\lambda+\mu)t} = \frac{\lambda}{\lambda + \mu}\,(\lambda + \mu)e^{-(\lambda+\mu)t},$$

and similarly for Y. Thus the winner is independent of the winning time. □

Proposition 3.4 (The attribution sequence)**.** *Model the honest network and attacker as independent exponential clocks with rates p/T_0 and q/T_0, restarting when either fires (Section 4). Successive block attributions form an infinite sequence of independent, identically distributed Bernoulli trials in the* Math Guide*'s sense: attacker with probability q, honest network with probability p, independent of all earlier attributions and all elapsed times.*

Proof. By Lemma 3.3 the first block is the attacker's with probability $(q/T_0)/(p/T_0 + q/T_0) = q$. When a block is found, the finder's clock restarts by construction, and the loser's remaining wait is distributed as a fresh clock by Lemma 3.2; the state after each block is therefore probabilistically identical to the start, independent of the past. The density factorisation in Lemma 3.3 also makes each winner independent of that race's elapsed time. The claim follows by induction. □

Proposition 3.4 is the paper's bridge (its §3, with footnote 1 supplying the clock rates) from physical time to combinatorics: since the question "does the attacker ever get ahead?" concerns only the *order* of block discoveries, not their times, every result below is a statement about a p/q coin sequence, and the time constant T_0 disappears from the answers. Where the exponential model itself comes from — and how well Zcash's Equihash fits it — is a deployment question, taken up in Section 8.3.

3.2 The negative binomial law

Proposition 3.5 (Progress of the loser)**.** *In an attribution sequence with attacker probability $q = 1 - p$, let m be the number of attacker blocks found strictly before the honest network's n-th block. Then*

$$P(m) \;=\; \binom{m+n-1}{m} p^n q^m, \qquad m = 0, 1, 2, \ldots$$

— the negative binomial *distribution.*

Proof. The event fixes the first $m + n$ trials exactly this far: the $(m + n)$-th trial is honest (the n-th honest success), and among the first $m+n-1$ trials exactly m are the attacker's, in any of $\binom{m+n-1}{m}$ orders. Each such order has probability $p^{n-1}q^m \cdot p$. □

Lemma 3.6 (First-to-n race)**.** *In the same sequence, let H_n be the event that the honest network reaches n blocks before the attacker does, and A_n the reverse. Then H_n and A_n partition the sample space, and*

$$\Pr[H_n] = \sum_{m=0}^{n-1} \binom{m+n-1}{m} p^n q^m, \qquad \Pr[A_n] = \sum_{h=0}^{n-1} \binom{h+n-1}{h} q^n p^h,$$
$$\Pr[H_n] + \Pr[A_n] = 1.$$

Proof. Among the first $2n - 1$ trials one side already has n successes, so the race terminates; a tie is impossible because block counts advance one at a time. The event H_n says the n-th honest block arrives while the attacker holds $m \le n-1$; summing Proposition 3.5 over those m gives the first sum, and A_n is the same computation with the roles of p and q swapped. □

Remark 3.7 (From hash trials to exponential clocks). The exponential model is itself a limit, not an axiom. A miner's work is modelled as a stream of candidate evaluations, each an independent Bernoulli trial with a tiny success probability; the number of trials to the first success is geometric, and a geometric law with small success probability, viewed at the scale of its mean, converges to the exponential of Definition 3.1. The step that matters is *progress-freeness*: a candidate that

fails must carry no information usable by the next one. Whether Zcash's Equihash satisfies this is examined with the deployed parameters in Section 8.3.

4 Playing catch-up

The attacker of Section 2.2 finds himself, at the moment the merchant ships, some number of blocks behind the public chain, and wins if his branch ever overtakes it. This section computes the probability of that event as a function of the deficit; the next section supplies the distribution of the deficit itself.

Both computations run under the paper's two standing assumptions, stated here nearly verbatim (the environment form is ours, Remark 1.1):

Assumption 4.1 (Constant hashrates). The total hashrate of the honest network and the attacker is constant. Combined they have a hashrate of H, of which pH belongs to the honest network and qH to the attacker, where $0 < p, q < 1$ and $p + q = 1$.

Assumption 4.2 (Constant difficulty). The mining difficulty is constant, and such that with hashrate H the average time to find a block is T_0.

Under Assumption 4.2 every block carries equal work, so "most work" and "longest" coincide and the race can be scored in block counts. Section 8.2 explains how the deployed chain departs from these assumptions — difficulty adjusts after every block and hashrate drifts — and why the formulas should be read as an equal-difficulty baseline rather than an exact deployment model.

Definition 4.3 (The deficit walk)**.** Let z denote the honest chain's lead in blocks over the attacker's branch. By Proposition 3.4, each newly found block moves z up by 1 with probability p (honest block) or down by 1 with probability q (attacker block), independently of the past:

$$z_{i+1} = \begin{cases} z_i + 1 & \text{with probability } p, \\ z_i - 1 & \text{with probability } q. \end{cases}$$

The attack succeeds the moment z reaches -1: the secret branch is then strictly ahead, and releasing it rewrites history.

Theorem 4.4 (Catch-up probability; Rosenfeld §3)**.** *Let a_z be the probability that the walk of Definition 4.3, started at deficit z, ever reaches -1. Then*

$$a_z = \min(q/p,\, 1)^{\max(z+1,\,0)} = \begin{cases} 1 & \text{if } z < 0 \text{ or } q \geq p, \\ (q/p)^{z+1} & \text{if } z \geq 0 \text{ and } q < p. \end{cases} \tag{1}$$

Proof. For $z < 0$ the walk has already succeeded, so $a_z = 1$; assume $z \geq 0$. Fix an integer $N > z$ and absorb the walk at both -1 and N; let $h_N(z)$ be the probability of reaching -1 before N. Conditioning on the first step,

$$h_N(z) = p\, h_N(z+1) + q\, h_N(z-1) \qquad (0 \leq z \leq N-1), \qquad h_N(-1) = 1, \quad h_N(N) = 0,$$

a second-order linear recurrence. The trial solution $h_N(z) = t^z$ satisfies it exactly when t is a root of the *characteristic polynomial* $pt^2 - t + q = p(t-1)(t-q/p)$.

If $q \neq p$ the roots 1 and q/p are distinct, so $h_N(z) = A + B(q/p)^z$; the boundary conditions give

$$h_N(z) = \frac{(q/p)^z - (q/p)^N}{(q/p)^{-1} - (q/p)^N}.$$

If $q = p$ the root is double and $h_N(z) = A + Bz$, giving $h_N(z) = (N - z)/(N + 1)$.

Now let $N \to \infty$. The events $E_N = \{\text{reach } -1 \text{ before } N\}$ are nondecreasing in N, and their union is the event that the walk ever reaches -1: a walk that does so does it at a finite time, having visited finitely many states, so E_N holds for every N above the largest state visited. By continuity along monotone events (*Math Guide*, §"Beyond finite spaces: countable additivity, limits, and densities"), $a_z = \lim_{N\to\infty} h_N(z)$. For $q < p$ the term $(q/p)^N$ vanishes and the quotient tends to $(q/p)^{z+1}$; for $q = p$, $(N - z)/(N + 1) \to 1$; for $q > p$, dividing numerator and denominator by $(q/p)^N$ sends both to -1, so the quotient tends to 1. □

Corollary 4.5 (Half or more always catches up)**.** *When $q \geq p$, the attacker overtakes the public chain with probability 1 from any finite deficit. No confirmation count defends when the attacker controls at least half of the hashrate.*

Corollary 4.6 (Geometric decay in the deficit)**.** *When $q < p$, each additional block of deficit multiplies the catch-up probability by q/p. Concretely (script-computed): at $q = 10\%$, $a_0 = 1/9 \approx 0.1111$ and $a_5 \approx 1.88 \times 10^{-6}$; at $q = 30\%$, $a_0 = 3/7 \approx 0.4286$ and $a_5 \approx 6.20 \times 10^{-3}$.*

Corollary 4.7 (Blocks, not time)**.** *Equation (1) contains neither T_0 nor any other time scale: the security of a deficit is a function of block* counts *alone. Two chains with different block spacings but the same q offer identical security per confirmation — and therefore different security per minute (Section 6.1).*

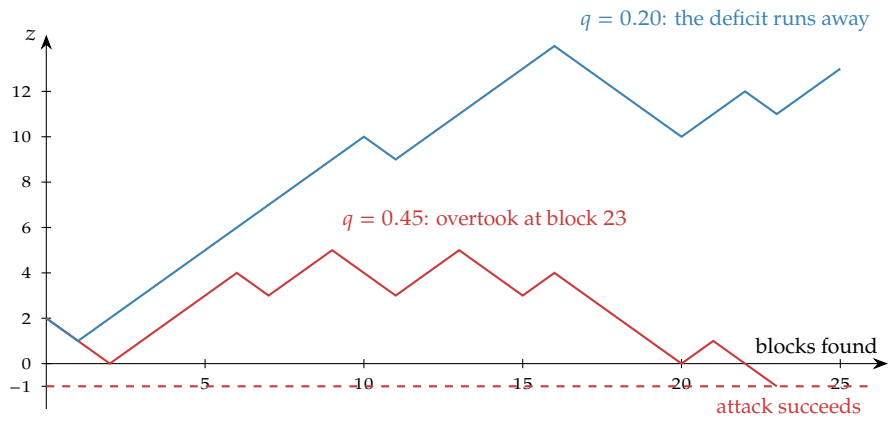


Figure 2: Two runs of the deficit walk from $z = 2$, simulated by script with fixed seeds. At $q = 0.45$ the drift is weak and this run reaches -1 after 23 blocks — the attack succeeds; at $q = 0.20$ the deficit grows roughly linearly, and by Theorem 4.4 the chance of ever returning from deficit 13 is $(1/4)^{14} \approx 3.7 \times 10^{-9}$.

5 Waiting for confirmations

The merchant's defence is patience: ship only after the paying transaction has n confirmations (Definition 2.3). This section computes exactly how much that patience buys in the paper's first-seen model. The paper's model of the attack timeline has one convention worth surfacing before the theorem.

Remark 5.1 (The pre-mined block). The paper assumes the attacker banks one block before the race is scored: "we assume one block was pre-mined by the attacker before commencing the attack" (§4). The attacker forks from the public leaf at the moment the merchant's transaction is broadcast, and a patient attacker times the broadcast to coincide with a private block he has just found but not announced. At the moment the honest chain reaches n confirmations, the attacker who has found m further blocks therefore holds $m+1$, and the deficit walk of Definition 4.3 starts at

$$z = n - m - 1.$$

The convention is a modelling choice, not a law: an attacker with no banked block starts at $z = n-m$, and every probability below shrinks accordingly. It is also, as Theorem 5.2 shows, precisely the choice that makes the final formula collapse into a symmetric race.

By Proposition 3.5, the number m of attacker blocks found while the honest network finds its n is negative binomial. Averaging the catch-up probability of Theorem 4.4 over m gives the central result for the specification and Rosenfeld first-seen tie rule. Section 8.1 explains the additional success opportunity supplied by Zebra's tie-break. The model is not an exact deployed rollback probability.

Theorem 5.2 (Double-spend success probability; Rosenfeld eq. (1))**.** *Let the merchant wait for $n \geq 1$ confirmations, under Assumptions 4.1 and 4.2 and the convention of Remark 5.1, with a private branch accepted only after it becomes strictly heavier. The probability that the attacker's branch eventually overtakes the public chain is $r(q,n) = 1$ if $q \geq p$, and otherwise*

$$r(q,n) = 1 - \sum_{m=0}^{n-1} \binom{m+n-1}{m} (p^n q^m - p^m q^n) = 2 \sum_{m=0}^{n-1} \binom{m+n-1}{m} p^m q^n. \tag{2}$$

Proof. Conditioning on m and applying Theorem 4.4 at $z = n - m - 1$,

$$r = \sum_{m=0}^{\infty} P(m)\, a_{n-m-1} = \underbrace{\sum_{m=0}^{n-1} P(m) \left(\frac{q}{p}\right)^{n-m}}_{\text{behind: must catch up}} + \underbrace{\sum_{m=n}^{\infty} P(m)}_{\text{already ahead}},$$

valid for $q < p$; when $q \geq p$ every $a_{n-m-1} = 1$ and $r = \sum_m P(m) = 1$ immediately. For the first sum, each term transforms exactly:

$$\binom{m+n-1}{m} p^n q^m \left(\frac{q}{p}\right)^{n-m} = \binom{m+n-1}{m} p^m q^n,$$

which by Lemma 3.6 is the probability that the attacker's n-th block arrives while the honest network holds m; summed over $m \leq n-1$, the first sum is $\Pr[A_n]$, the probability that the attacker wins the first-to-n race. The second sum is $\Pr[m \geq n] = 1 - \Pr[H_n] = \Pr[A_n]$ by the same lemma. Hence $r = 2\Pr[A_n]$, the right-hand form of (2); the middle form follows from $\Pr[A_n] = 1 - \Pr[H_n]$ applied to one of the two copies. The printed upper limit in the paper's equation (1) is n rather than $n-1$; the $m = n$ term is identically zero, so the forms agree. □

Remark 5.3 (The race doubling). The proof exposes a structure the paper's algebra passes through silently: under the one-pre-mined-block convention, the *catch-up half* — the honest chain reaches n first and the attacker later overtakes — has exactly the same probability as the *already-ahead half*, in which the attacker reaches n first. A first-to-n winner may have fallen behind and recovered along the way; neither half asserts otherwise. Thus the double-spend probability is twice the chance of winning a first-to-n race. The identity also absorbs the boundary case: at $q = p$ each race half is exactly $\frac{1}{2}$, giving $r = 1$ with no case split, continuously with the $q > p$ regime. The algebra here is the paper's; reading its two sums as the two halves of one race, and the observation that the pre-mine convention is what makes them equal, are this volume's presentational additions (verified by script over random (q, n); the two sums agree to floating-point precision).

Example 5.4 (A full race, exactly)**.** Take $q = 1/5$ (a twenty-percent attacker) and $n = 6$. The deficit starts at $z = 5 - m$, the catch-up factor is $(q/p)^{6-m} = 4^{-(6-m)}$, and every quantity below is an exact rational, printed to six decimals (script-computed):

m	$P(m)$	$a_{5-m} = 4^{-(6-m)}$	product
0	0.262144	0.000244	0.000064
1	0.314573	0.000977	0.000307
2	0.220201	0.003906	0.000860
3	0.117441	0.015625	0.001835
4	0.052848	0.062500	0.003303
5	0.021139	0.250000	0.005285
catch-up half (sum of products)			0.011654
already-ahead half, $\Pr[m \geq 6]$			0.011654
$r(0.2, 6)$			0.023308

The two halves agree to every printed digit — they are equal exactly, by Remark 5.3 — and the closed form (2) evaluates to the same 0.023308. A merchant facing a possible twenty-percent adversary who ships after six confirmations is assigned a 2.33% strict-overtake probability by this model.

Remark 5.5 (Satoshi's approximation). The whitepaper's §11 computes the same quantity with one approximation and two offsetting conventions: the attacker's progress is approximated as Poisson with mean $\lambda = nq/p$ (the law $\Pr[k] = e^{-\lambda}\lambda^k/k!$ on the nonnegative integers); no block is pre-mined, but success is scored at *breakeven* rather than strictly ahead, and those two conventions cancel exactly — the whitepaper's catch-up factor for m attacker blocks is $(q/p)^{n-m}$, term for term the factor of Remark 5.1. The Poisson approximation is the entire gap. In each of the following script-computed comparisons it pushes the estimate down: at $q = 10\%$, $n = 6$, the whitepaper's formula gives 0.0243% against the exact 0.0591% — an understatement by a factor of about 2.4; at $q = 30\%$, $n = 6$: 13.211% against 15.645%; at $q = 10\%$, $n = 10$: 0.0001% against 0.0008%. The direction is not uniform over all (q, n); the reason to prefer the negative binomial model is that it is exact under the stated assumptions, not that the approximation is always one-sided.

q	n=1	2	3	4	5	6	7	8	9	10
2%	4.000	0.237	0.016	0.0011	$8{\cdot}10^{-5}$			$< 10^{-5}$		
5%	10.000	1.450	0.232	0.039	0.0066	0.0012	$2.1{\cdot}10^{-4}$	$4{\cdot}10^{-5}$	$< 10^{-5}$	
10%	20.000	5.600	1.712	0.546	0.178	0.059	0.020	0.0067	0.0023	0.0008
20%	40.000	20.800	11.584	6.669	3.916	2.331	1.401	0.848	0.516	0.316
30%	60.000	43.200	32.616	25.207	19.762	15.645	12.475	10.003	8.055	6.511
40%	80.000	70.400	63.488	57.958	53.314	49.300	45.769	42.621	39.787	37.218
45%	90.000	85.050	81.375	78.342	75.716	73.375	71.252	69.299	67.487	65.793
48%	96.000	94.003	92.508	91.264	90.177	89.201	88.307	87.478	86.703	85.972
50%					100 for every n					

Table 1: The double-spend success probability $r(q, n)$, in percent, computed by script from Theorem 5.2. On the rows the paper's Table 1 also prints ($q \in \{2, 10, 20, 30, 40, 48, 50\}\%$), the script reproduces its values cell-for-cell; the 5% and 45% rows are this volume's additions.

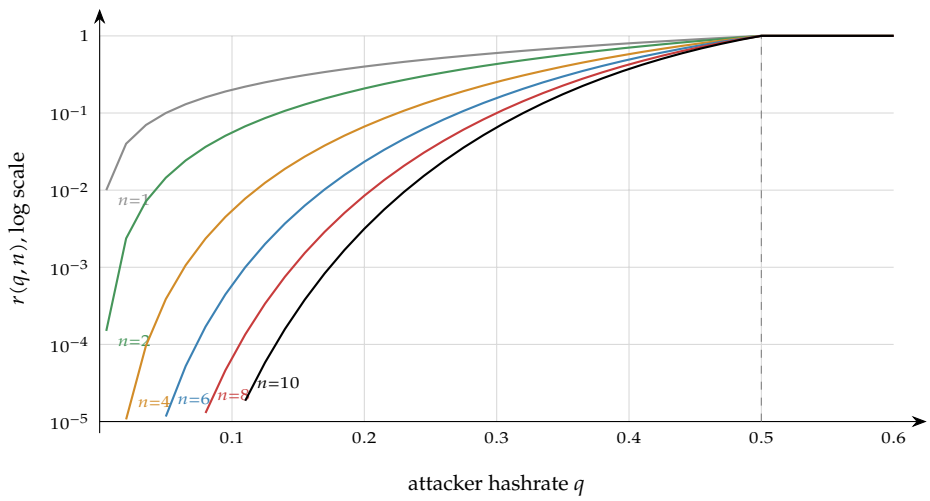


Figure 3: The success probability $r(q, n)$ against the attacker's hashrate share q, for $n \in \{1, 2, 4, 6, 8, 10\}$ confirmations, logarithmic vertical scale clipped at 10^{-5}; coordinates computed by script from Theorem 5.2. Every curve reaches 1 at $q = \frac{1}{2}$ (dashed) and stays there: beyond the majority line, confirmations are worthless. Below it, each added confirmation drops r by a roughly constant factor — the curves are near-equally spaced in log scale — and the factor improves as q falls.

6 What the numbers say

Theorem 5.2 compresses the security of Nakamoto consensus into one two-parameter function, and its qualitative lessons are all visible in Table 1 and Figure 3. This section states them precisely, then instantiates the one lesson that is specific to Zcash.

> **Proposition 6.1** (No finite count is final)**.** *For every $0 < q < p$ and every n, $r(q, n) \geq 2q^n > 0$.*

Proof. By Theorem 5.2, $r = 2\Pr[A_n]$, and one way for the attacker to win the first-to-n race is to find the first n blocks outright, an event of probability q^n. □

> **Proposition 6.2** (Divergence at the majority line)**.** *Fix a target $0 < \varepsilon < 1$. For $0 < q < \frac{1}{2}$, let $n_\varepsilon(q)$ be the least n with $r(q, n) < \varepsilon$. Then $n_\varepsilon(q) \to \infty$ as $q \uparrow \frac{1}{2}$.*

q	$r < 10\%$	$r < 1\%$	$r < 0.1\%$
2%	1	2	3
5%	2	3	4
8%	2	3	5
10%	2	4	6
15%	3	6	9
20%	4	8	13
25%	5	12	20
30%	9	20	32
35%	15	36	58
40%	34	82	133
45%	135	331	539

Table 2: The least number of confirmations pushing $r(q, n)$ below three targets, computed by script. The growth is logarithmic in the inverse target risk but explosive in q: a 10% attacker is answered by 6 confirmations, a 45% attacker by 539, and at 50% by no number at all.

Proof. First, $n_\varepsilon(q)$ exists. The event A_n that the attacker reaches n blocks first is the event that at least n of the first $2n-1$ block discoveries are the attacker's. Since $q < p$, each such length-$(2n-1)$ sequence has probability at most $q^n p^{n-1}$, and there are fewer than 2^{2n-1} sequences. Hence

$$r(q, n) = 2\Pr[A_n] \leq \frac{(4pq)^n}{p} \longrightarrow 0,$$

because $4pq < 1$.

For fixed n the function $q \mapsto r(q, n)$ is a polynomial on $[0, \frac{1}{2}]$, hence continuous, and $r(\frac{1}{2}, n) = 1$: at $q = p$ each half of the race of Remark 5.3 has probability exactly $\frac{1}{2}$ by Lemma 3.6 and symmetry. Given any N, for each $1 \leq n \leq N$ there is a $\delta_n > 0$ with $r(q, n) > \varepsilon$ whenever $q > \frac{1}{2} - \delta_n$. Taking $\delta = \min_{1 \leq n \leq N} \delta_n > 0$ excludes every $n \leq N$ throughout that neighbourhood, so $n_\varepsilon(q) > N$. Since N was arbitrary, the claimed divergence follows. □

Remark 6.3 (Nothing is special about six). The paper (§5): "There is nothing special about the default, often-cited figure of 6 confirmations. It was chosen based on the assumption that an attacker is unlikely to amass more than 10% of the hashrate, and that a negligible risk of less than 0.1% is acceptable. Both these figures are arbitrary" — six confirmations are "overkill for casual attackers, and at the same time powerless against more dedicated attackers". Table 2 is the honest replacement: pick the adversary you defend against and the risk you accept, and read off n; the pair (q, ε) is a policy, not a law of nature.

6.1 Zcash time: 25-second blocks

Corollary 4.7 says security is purchased in blocks. The wall-clock price is set by the target spacing: 25 seconds in ZIP-218. Since public discoveries arrive at rate p/T_0, waiting for n confirmations takes nT_0/p seconds in expectation in this model. Three confirmations cost a nominal 75 seconds and ten cost 250 seconds when all hashpower publishes on the public chain. With a 10% withholding attacker, the expected wait is instead divided by 0.9.

For example, a nominal thirty-minute wait buys 72 confirmations at 25-second spacing, compared with three at Bitcoin's 600-second spacing. The corresponding strict-overtake probabilities against a 10% attacker are approximately 9.35×10^{-34} and 1.712%, respectively; either count takes $30/0.9 \approx 33.3$ minutes in expectation under withholding. These are model calculations, not guarantees about either network.

The comparison holds q fixed. It contains no network and therefore does not price honest blocks racing one another during propagation. Quantifying that lost work requires a propagation

distribution and mining topology. Moreover, a policy waiting for a fixed amount of accumulated miner reward asks a different question: decreasing the reward per block in proportion to the target spacing requires proportionally more blocks to reach that amount. Ignoring transaction fees (value paid for transaction inclusion) and rounding, the shorter spacing then cancels out of the mean wait. The monetary and fixed-hashrate models are not interchangeable.

Remark 6.4 (Where the model bends). Three assumptions do not survive contact with the deployed chain; they determine how cautiously the calculations may be transferred to it. (i) *Difficulty is not constant*: Zcash retunes the target after every block (Section 8.2). The race is scored in work, not block counts, and q may be interpreted approximately as the attacker's share of work production while the branches' block weights remain close; once their difficulty schedules diverge, the equal-step random-walk formula is no longer exact. (ii) *Hashrate is not constant*: q in the theorems is whatever share the attacker sustains for the attack's duration; the paper's own caveat is that T_0 re-enters only if the attacker cannot sustain his hashrate long enough, which it judges unlikely for a serious attacker. (iii) *Propagation is not instantaneous*: honest self-orphaning wastes honest work and thereby raises the attacker's effective share; its size is not quantified here. A fourth gap is not the model's: the attacker of this volume follows the protocol's validity rules perfectly and attacks only the *choice* between valid histories. Attacks on the rules themselves are other volumes' subjects.

7 The economics of the attack

The probabilities of Section 5 answer the adversarial merchant, who assumes the customer is an attacker. The paper's §6 answers the realistic one: when is the attack worth mounting at all? Throughout this section $q < p$ — "[o]therwise, all bets are off" (§6) — and the model is deliberately stylised; its own caveats close the section.

The attacker targets k merchants simultaneously with payments of v each, all invalidated by the same secret branch; the goods are worth αv to him, $0 < \alpha \leq 1$; he gives up after mining o blocks in vain; and each of his blocks would earn the block income B if it ended up in the accepted chain. If the attack succeeds his blocks are all accepted (he keeps goods *and* coins *and* rewards); if it fails he has paid kv, keeps goods worth $k\alpha v$, and his o orphaned blocks forfeit oB. For an exact finite-stop profit calculation, r would have to be the success probability of that stopping policy. Following the paper, the tables below instead insert the unlimited-horizon $r(q, n)$ as a heuristic proxy. They are not an exact evaluation of a strategy that always stops after $o = 20$ private blocks.

Proposition 7.1 (Profitability; Rosenfeld §6). *Given success probability r and these stipulated payoffs, the attacker's expected profit over not attacking is*

$$k\alpha v - (1-r)(kv + oB) \;=\; kv(\alpha + r - 1) - (1-r)\,oB,$$

positive if and only if

$$v \;>\; \frac{(1-r)\,oB}{k(\alpha + r - 1)}$$

(when $\alpha + r > 1$; when $\alpha + r \leq 1$ the modeled attack is never strictly profitable). This modeled attack is not strictly profitable whenever

$$v \;\leq\; \frac{o(1-r)B}{k(\alpha + r - 1)} \;\overset{\alpha=1}{=}\; \frac{o}{k} \cdot \frac{B(1-r)}{r}, \tag{3}$$

which with the paper's choices $o = 20$, $k = 5$, $\alpha = 1$, $B = 25$ BTC is its equation (2), $v \leq 100(1/r - 1)$ BTC.

Proof. The gain $k\alpha v$ is certain (the goods are obtained either way); the loss $kv + oB$ is paid ex-

actly when the attack fails, with probability $1 - r$. Rearranging the positivity condition gives the threshold; substituting the paper's constants gives $20 \cdot 25 \cdot (1 - r)/(5r) = 100(1 - r)/r$. □

7.1 Reward-normalised exposure

The *block subsidy* is value issued by the protocol in a block, distinct from fees transferred by its transactions. Its *scheduled* component is base issuance determined by height, before any reserve payout. A *funding stream* is a required allocation of a share of that subsidy to a designated recipient. The *coinbase* is the block's first transaction, which distributes the available subsidy and fees according to the applicable rules. The *reserve* is the accounting balance of funds removed from circulation and awaiting reissuance.

The income B forfeited by an orphaned block is the amount its miner would have retained: miner subsidy plus retained transaction fees, not the subsidy allocated to other recipients. Under NU7 this depends on scheduled issuance, the prior reserve balance, and the applicable fee-removal rule (Section 8.8). A universal ZEC figure would therefore hide assumptions about both chain state and unresolved draft rules.

Instead, Table 3 measures transaction value in units of the chosen block income B. With the paper's $o = 20$, $k = 5$, and $\alpha = 1$, equation (3) becomes

$$\frac{v}{B} \leq 4\,\frac{1 - r}{r}.$$

Once a particular threat model supplies B, multiplying by it gives the corresponding monetary threshold. Treating B as constant over the attack remains a modelling assumption.

q	n=1	2	4	6	8	10
5%	36	271	10,327	344,743	10,931,506	336,737,801
10%	16	67	729	6,759	59,475	508,917
20%	6	15	55	167	467	1,262
30%	2	5	11	21	35	57
40%	1	1	2	4	5	6

Table 3: The largest whole-number multiple of block income B with non-positive heuristic expected profit: $\lfloor 4(1 - r)/r \rfloor$, computed by script. The table is dimensionless: it neither fixes a ZEC subsidy nor predicts future fee income. Values inherit every assumption of the model; the caveats of Remark 7.3 apply in full.

Within the model, a handful of confirmations makes the attack unprofitable against small attackers at any plausible payment size; against a 30% attacker, six confirmations make modeled expected profit non-positive only up to about 21 times the assumed block income B. The required count grows only logarithmically in the value at stake, by the geometric decay of Theorem 5.2. These are neither exact finite-stop profit calculations nor bounds on deployed rollback risk; the deployed tie-break, changing difficulty, and rollback rails require separate modelling.

Remark 7.2 (Majority is a different problem). At $q = p$, eventual strict overtake has probability 1, but the walk has zero drift; that fact alone does not give the attacker a lasting monopoly. When $q > p$, the economics change character entirely: the paper notes a strict majority attacker can double-spend at no cost beyond normal mining, reject every other miner's blocks to take the whole issuance, and exclude transactions at will, driving honest miners out and entrenching himself — so a majority attack "is best seen as an attempt to destroy" the currency, motivated from outside it (a short position, a competing system), not an attempt to profit inside it. Confirmation policy is not the defence at that point; the defence is the cost of assembling the majority, which lies outside this model.

Remark 7.3 (The model's own caveats). The paper flags each stylisation itself. The give-up point $o = 20$ "is a significant simplification" (an optimal attacker balances completion against compounding losses); the safe values "should be taken with a grain of salt, because of the many modeling assumptions"; and a model resting on mining *costs* rather than forfeited *rewards* would make the required confirmations linear, not logarithmic, in the transaction value — "very poor security, hence in a situation where this is relevant, we have already lost anyway" (§6). The bound's role is the shape it reveals — logarithmic patience buys exponential safety — not its third significant digit.

8 NU7: 25-second blocks and bounded shielded work

The ZIP-259 draft combines three changes: shorter blocks with limits on shielded work (ZIP-218), reissuance that preserves the halving schedule — periodic divisions of scheduled subsidy by two (ZIP-237) — and rejection of version 4 transactions (ZIP-2003). This section describes that proposal, not a deployed upgrade. Its activation height, denoted A, is unassigned on each network at the source boundary in Section 1.1.

NU7 introduces no new transaction format: versions 5 and 6 remain available. Their signatures nevertheless commit to a *consensus branch identifier*, a value identifying the network upgrade's rules. An unchanged encoding therefore does not make a pre-NU7 signature valid under NU7. Disabling version 4 also disables spending funds in Sprout, an older shielded pool, because the newer formats cannot represent its transfers; it does not transfer their value to a reserve or remove it from the supply.

8.1 Chain selection and the tie-break

The most-work rule remains Definition 2.2: compare the sum of the blocks' work, not their count. Changing target spacing changes neither the work definition nor this ordering.

Remark 8.1 (The implementation tie-break). The specification and Rosenfeld's model prefer the first-seen block at equal cumulative work. The examined Zebra implementation instead uses the larger tip hash. This distinction concerns local chain choice, not the validity of a block. The divergence matters to the attack model. At equal work, an attacker can compare its private tip hash with the public tip and reveal immediately when its hash wins the deployed deterministic tie-break; the strict-overtake boundary $z = -1$ used above does not count that success opportunity. Therefore $r(q, n)$ is exact for the stated strict-overtake model and is a lower bound if only its tie-break is replaced by Zebra's, keeping equal difficulty, unlimited rollback, and the other assumptions fixed. It is not thereby a lower bound on the full deployed process: changing difficulty and rollback limits also change the success event. Quantifying the extra probability, including any scope to choose among candidate private tips, requires a different stopping rule and is outside this volume.

8.2 Difficulty in motion

Assumption 4.2 freezes difficulty; Zcash retargets after every block. In ZIP-218, the NU7 target spacing is 25 seconds and the target-averaging window is 102 blocks. The timestamps determining the adjustment still come from the branch being evaluated, including on an attacker's private branch.

Write $N = 102$ for the target-averaging window and $T = N \cdot 25 = 2{,}550$ seconds for its nominal timespan.

The specification's §7.7.3 defines the rest of the calculation. For a block height h, let $m(h)$ be the median timestamp of the eleven preceding blocks: their sixth-smallest timestamp. Let $\bar{t}(h)$

be the mean target of the N blocks preceding h: the sum of their targets divided by N. Define the observed and damped timespans by

$$a(h) = m(h) - m(h - N),$$
$$d(h) = T + \mathrm{trunc}\left(\frac{a(h) - T}{4}\right),$$

where trunc discards the fractional part towards zero. The bounded timespan $c(h)$ is $d(h)$ clamped between $\lfloor 0.84T \rfloor$ and $\lfloor 1.32T \rfloor$: values below the lower endpoint are replaced by that endpoint, and similarly above the upper endpoint. The maximum target $\mathsf{PoWLimit}$ is $2^{243} - 1$ on Mainnet, the production network, and $2^{251} - 1$ on Testnet, the test network (§5.3). The target calculation is

$$t(h) = \min\left(\mathsf{PoWLimit},\ \left\lfloor \frac{\bar{t}(h)}{T} \right\rfloor c(h)\right),$$

followed by the compact encoding defined in the specification's §7.7.4. The order of the floor and multiplication matters. At NU7 the clamp endpoints are 2,142 and 3,366 seconds; the eleven-timestamp median, fourfold damping, and target cap are unchanged.

Activation does not reset the accumulated work or multiply the target by three in a single operation. The first NU7 blocks still look back over pre-NU7 history and adjust through the same bounded rule. Likewise, the timestamp rules do not shrink with the target spacing: a timestamp must exceed the median of the preceding eleven and be at most 90 minutes above it (§7.6). A further non-consensus receipt check rejects blocks more than two hours ahead of the validator's clock. Testnet's exceptional minimum-difficulty rule continues to use six target spacings: its delay threshold becomes $6 \cdot 25 = 150$ seconds. This exception does not apply to Mainnet.

Remark 8.2 (What per-block retargeting changes). The averaging window, damping, and clamp constrain the adjustment but do not prove that two forked branches retain close targets. Work per block moves inversely with the target; a branch's timestamps can change both its future block rate and the work carried by each block. Equal expected work production does not make the probability of an eventual overtake independent of these jump sizes. Consequently, the probability tables are an equal-work baseline, not an exact analysis of private branches whose difficulty schedules diverge.

8.3 Equihash and the memoryless model

Zcash's proof of work is Equihash with parameters $n = 200$, $k = 9$ (specification §7.7, §7.7.1; the algorithm is Biryukov and Khovratovich's, ePrint 2015/946), but the quantity the race counts is gated one step later: "[t]he difficulty filter is unchanged from Bitcoin, and is calculated using SHA-256d on the whole block header (including `solutionSize` and `solution`)", required to be at most `ToTarget(nBits)` (§7.7.2). Each candidate Equihash solution therefore faces a fresh SHA-256d lottery. Treating distinct headers as independent trials is the usual random-function model for SHA-256d, not a property asserted by the specification; under that model these are the Bernoulli trials of Remark 3.7.

The specification nowhere asserts progress-freeness — the argument is this volume's, not a citation. What Equihash changes is the granularity: candidates arrive in batches, one batch per memory-hard solver run. Progress within a run is discarded when a competing block arrives. Under independent identically distributed runs, the number of runs until an accepted header is geometric; replacing those discrete completion times by an exponential clock is a further approximation, coarse within one run and good only when a run is short relative to the target spacing: 25 seconds in the NU7 draft. Run duration is implementation- and hardware-dependent and is not quantified here. A proof-of-work whose solve time were comparable to the spacing would make block discovery visibly non-exponential and pull the attribution sequence away from Proposition 3.4.

8.4 Reorg rails, maturity, and the finality floor

Consensus does not impose a maximum reorganisation depth. ZIP-218 recommends a node-local 600-block rollback window: at the target spacing this is $600 \cdot 25 = 15{,}000$ seconds, or 250 minutes. Such a limit bounds the history that a node automatically reverses; it is not a proof that older blocks cannot be replaced by a higher-work valid chain. Following a branch outside that retained history requires different local state or operator intervention. Actual node implementations may choose a different window.

Remark 8.3 (What the rails buy). A finite rollback policy truncates some paths to eventual overtake, so the unlimited-rollback probability $r(q, n)$ is not the exact failure probability of a node enforcing that policy. A limit is also not global agreement: different nodes may stop on different sides of a long split. The specification's §3.3 adds a separate requirement for validators that risk Mainnet funds or display transaction information: their chain must include the activation block of the most recent settled network upgrade. That socially anchored checkpoint is distinct from probabilistic finality.

A *transparent output* records its value and destination in cleartext. The consensus maturity rule prevents spending such an output of a coinbase until the spending block is at least 100 heights later (§7.1.2). ZIP-218 keeps that count, a nominal $100 \cdot 25 = 2{,}500$ seconds. Coinbase outputs placed directly in a shielded pool do not have this transparent maturity restriction. A 600-block rollback window can therefore reverse an already mature coinbase: maturity permits spending, not finality.

8.5 Wallet confirmation policy

A wallet can distinguish *trusted* outputs that its own transaction handling can recreate after a rollback from an *untrusted* external payment that may require the sender to pay again. The examined wallet backend uses three confirmations for the former and ten for the latter, with zero-confirmation shielding of transparent funds allowed by its default policy. These choices are not consensus rules.

Table 1 quantifies their modelled trade-off: three confirmations assign a 10% attacker risk 1.712%, while ten assign the same attacker about 0.0008% and a 20% attacker about 0.316%. The count remains a (q, ε) policy, not a guarantee.

The *anchor* is the note-commitment-tree root against which a spend proves membership (*Crypto Guide*, §Privacy-preserving membership via zero-knowledge paths). ZIP-218 keeps the recommended anchor depth at three blocks. Its recommended transaction-expiry delta is 120 blocks, a nominal $120 \cdot 25 = 3{,}000$ seconds, without overriding an explicitly configured expiry delta. There is no instruction to divide every confirmation count by three.

8.6 A block budget for shielded work

A byte limit does not directly bound the computational work inside a block. The 2,000,000-byte limit must therefore be considered alongside the 25-second target spacing and the costs of shielded validation and wallet scanning. ZIP-218 adds limits on the shielded components that drive this work.

An Orchard *Action* pairs one spend slot with one output slot; dummy slots allow either side to carry no real transfer. Ironwood is the newer shielded pool using the same Action structure (ZIP-229). The older Sapling shielded pool counts spends and outputs separately; its cryptographic construction is not needed for this accounting.

Let O and I count Orchard and Ironwood Actions in a block, and let S count Sapling spends plus outputs. With the Ironwood coverage proposed in PR 1361, the complete shielded budget is

$$O \leq 330, \qquad I \leq 330, \qquad S \leq 300, \qquad O + I + S \leq 330.$$

The final inequality is shared across pools: their separate maxima cannot all be filled in one block. Sprout transfers cannot occur because NU7 rejects the version 4 format that represents them.

The Ironwood term remains under review rather than already specified by the merged ZIP-218 draft. For example, $O = 100$, $I = 200$, $S = 30$ exhausts the shared budget; increasing I to 201 violates it even though each per-pool limit still holds. At two Actions per transaction, a block devoted to one Orchard-protocol pool fits at most 165 such transactions under this budget, or a nominal $165/25 = 6.6$ transactions per second. This is a capacity calculation under a specified transaction shape, not a measured throughput guarantee. Transparent components remain subject to the byte limit rather than this shielded-action budget.

8.7 Scheduled issuance with shorter blocks

A *zatoshi* is the atomic currency unit: 1 ZEC equals 10^8 zatoshi. A *halving* divides the scheduled subsidy by two. Retaining its nominal wall-clock interval while tripling block frequency requires both three times as many blocks per interval and one third of the subsidy per block. ZIP-218 therefore changes the full interval from 1,680,000 to $L = 5{,}040{,}000$ blocks. These earlier intervals are needed to compute progress at activation, not to supply an alternative rule. The change preserves expected timing, not exact calendar dates, because actual block times remain random.

The progress already made towards the next halving must survive activation. Let H_B be the Blossom activation height: 653,600 on Mainnet and 584,000 on Testnet. Before Blossom, halvings were separated by 840,000 blocks; the initial slow-start schedule, which ramps up the subsidy during the first 20,000 blocks, contributes a shift of 10,000 blocks. These constants and the maximum subsidy $M = 1{,}250{,}000{,}000$ zatoshi are defined in the specification's §5.3 and §7.8. For a height $h \geq A$, the draft's halving count and scheduled subsidy s_h, measured in zatoshi, are

$$k(h) = \left\lfloor \frac{H_B - 10{,}000}{840{,}000} + \frac{A - H_B}{1{,}680{,}000} + \frac{h - A}{5{,}040{,}000} \right\rfloor, \tag{4}$$

$$s_h = \left\lfloor \frac{M}{2 \cdot 3 \cdot 2^{k(h)}} \right\rfloor. \tag{5}$$

The factors 2 and 3 account separately for Blossom's and NU7's spacing changes. If activation occurs during the second-halving era, $k(h) = 2$ gives $s_h = 52{,}083{,}333$ zatoshi, or 0.52083333 ZEC, until the next halving; $k(h) = 3$ gives 26,041,666 zatoshi.

The old Mainnet third-halving height $H_3 = 4{,}406{,}400$ is therefore not the NU7 third-halving height if $A < H_3$. Converting the remaining interval gives

$$H_3^{\mathrm{NU7}} = A + 3(H_3 - A) = 13{,}219{,}200 - 2A.$$

No numerical activation height is assigned here. Nor does retiming the halving function automatically retime a funding stream whose endpoint is specified as a literal block height. The draft does not provide such a change, so the funding-stream endpoints and the rescaled halving heights must be treated separately.

8.8 The NSM reserve and delayed reissuance

The *Network Sustainability Mechanism* (NSM) separates temporarily removed funds from the still-unissued scheduled subsidy. In the merged ZIP-237 draft, the *issued supply* V_h is the total value recorded in the chain's value pools after block h. Each pool is a separately tracked balance for one part of the protocol; tracking a shielded pool's total does not reveal the values of its individual notes. The *NSM reserve* R_h records value removed from circulation and not yet reissued. The reserve is public consensus state, not a spendable pool or an account with a spending key. In particular, it is not 21 million ZEC minus the issued supply: that larger difference also includes future scheduled issuance.

The initial reserve recovers the historical subsidy and fees that miners left unclaimed before NU6. Since ZIP-236, a coinbase must account for its full available value, so that historical shortfall is fixed. The draft defines the seed R_{A-1} as total scheduled issuance through the last pre-NU6 block minus issued supply there. The exact integer seeds for Mainnet and Testnet remain unspecified.

Let $D \geq A$ be the separately assigned height at which reissuance starts, and let d_h be the non-negative number of zatoshi removed from circulation in block h by whatever removal mechanisms have been deployed. With $L = 5{,}040{,}000$ blocks per halving interval, the draft selects

$$f = \frac{\lfloor 6{,}931{,}680{,}000/L \rfloor}{10^{10}} = \frac{1375}{10^{10}}.$$

Its additional subsidy u_h, total block subsidy b_h, and reserve update are

$$u_h = \begin{cases} 0, & h < D, \\ \lceil f R_{h-1} \rceil, & h \geq D, \end{cases} \tag{6}$$

$$b_h = s_h + u_h, \qquad R_h = R_{h-1} - u_h + d_h. \tag{7}$$

All amounts are integers in zatoshi. Reissuance uses the previous block's reserve, so a removal in the current block cannot finance its own additional subsidy. Before D, removals accumulate without payout. After D, a positive reserve pays at least one zatoshi: because R is an integer and $0 < f < 1$, one has $1 \leq \lceil fR \rceil \leq R$ for $R > 0$. The draft rejects a negative reserve; it does not silently clamp it to zero.

For a small exact example, take $R_{h-1} = 100{,}000{,}000$ zatoshi (1 ZEC) and $h \geq D$. The additional subsidy is $\lceil 13.75 \rceil = 14$ zatoshi. Without a new removal the reserve becomes 99,999,986; with a same-block removal of another 1 ZEC it becomes 199,999,986, but the current payout remains 14. This example does not use the unassigned historical seed.

Why this preserves supply and halvings. Transfers between ordinary value pools cancel in their total. A block increases that total by its subsidy and reduces it by its removals:

$$V_h - V_{h-1} = s_h + u_h - d_h, \qquad (V_h + R_h) - (V_{h-1} + R_{h-1}) = s_h.$$

The second identity follows by adding the reserve update; reissuance moves value between the two ledgers without creating a second copy. Scheduled issuance retains its halvings, with the reserve paying an additional amount on top. Funding streams, wherever active, take their prescribed fractions of the *total* subsidy b_h, not only s_h. The resulting subsidy depends on prior chain state as well as height.

To understand the chosen rate, temporarily ignore integer rounding and new removals. The recurrence becomes $R_{h+j} = (1-f)^j R_h$, so keeping about half the reserve after L blocks requires $f = 1 - 2^{-1/L} \approx (\ln 2)/L$. The draft implements an integer-scaled approximation with numerator 1375. Indeed, $(1 - 1375/10^{10})^{5{,}040{,}000} \approx 0.50007$. This is a half-life of the reserve without new contributions, not a new halving rule for the scheduled subsidy; small reserves depart from the smooth curve because every positive payout rounds up.

Unresolved deployment and fee details. ZIP-259 selects a reissuance date corresponding to February 2031, but the network-specific height D remains unassigned. The drafts do not yet specify a complete, consistent NU7 removal rule: ZIP-235 requires removal of at least 60% of fees through an explicit coinbase amount, while ZIP-259 describes a block-level contribution requiring no new transaction data. The reserve equations therefore leave d_h as a defined input. Neither a particular fee-removal mechanism nor an exact NU7 miner income is inferred from these incomplete rules.

9 Layer boundary

Every theorem above rests on the *Math Guide*'s probability section: the definition of probability, its extension beyond finite spaces, and the continuity theorem used in the catch-up proof. This volume supplies the best-chain model and the quantitative bounds in Tables 1, 2, and 3. Policies that decide which chain view to trust or when to treat a transaction as final lie above that boundary.

Part V
Ironwood Guide: The Zcash Orchard protocol and its Ironwood pool

Abstract

This volume of *The Zcash Arboretum* constructs the Orchard protocol for a shielded payment in its Ironwood pool: how the payment is constructed, delivered to its recipient and verified; why a valid payment cannot create value, consume a note twice or consume a note without the authority of its owner; and what its public data hide. Every public field of the payment is constructed, in dependency order, as a fixed function of private notes, keys and randomness; the relation that each Action of the payment proves is stated over those objects; and each security property is a theorem under named assumptions. The sealed Orchard pool enters only where a rule shared with the Ironwood pool, or a rule that distinguishes the two pools, is needed to state the Ironwood-pool payment correctly. The volume assumes the *Math Guide* and the *Crypto Guide* and cites the *Halo 2 Guide* for the proof system. It is non-normative: the Zcash Protocol Specification and the Zcash Improvement Proposals are authoritative.

1 Introduction

This section fixes the subject of the volume, its status and its vocabulary before any construction: the scope; in plain language, the structure of a shielded payment and the division of its verification; the Orchard protocol and its two value pools; and the requirements that a shielded payment must meet.

1.1 Scope and status

This volume of *The Zcash Arboretum* constructs the Orchard protocol for a shielded payment in the Ironwood pool, both terms being defined in §1.3: how the payment is constructed, delivered and verified; why a valid payment cannot create value, consume a note twice, or consume a note without the authority of its owner; and what it hides. The sealed Orchard pool enters only where a rule shared with the Ironwood pool, or a rule that distinguishes the two pools, is needed to state the Ironwood-pool payment correctly. The volume assumes the *Math Guide* and the *Crypto Guide*, and cites the *Halo 2 Guide* for the proof system. It is non-normative: the Zcash Protocol Specification, cited as the protocol specification with the title of a section, and the Zcash Improvement Proposals are authoritative.

Definition 1.1 (Block chain, transaction, consensus rules)**.** A *block chain* is a sequence of blocks that starts at the *genesis block*, the header of each block referring to the preceding block; the *height* of a block is its number of predecessors, so that the genesis block has height 0 (ZIP 200, Terminology; protocol specification, §"The Block Chain"). Each block contains an ordered sequence of one or more *transactions*, the units in which the chain records transfers of value (protocol specification, §"Transactions and Treestates"). The *consensus rules* are the validation rules that determine which block chains, and hence which blocks and transactions, are valid (ZIP 200, Terminology, "Consensus rule set").

Definition 1.2 (ZIP)**.** A *Zcash Improvement Proposal* (ZIP) is a design document that describes a feature of Zcash, or a change to it, with a concise technical specification and a rationale. Each ZIP carries a status. Two statuses are named in this volume: *Draft*, the status that every ZIP

receives on submission, and *Reserved*, the status of a ZIP number that the ZIP editors have reserved while no ZIP text has been published (ZIP 0, Abstract and "ZIP Status Field"). A deployed rule is cited to a section of the protocol specification by its title, or to a ZIP and its section.

Definition 1.3 (Network upgrade)**.** A *network upgrade* is an intentional change of the consensus rules that takes effect at a block height fixed for it, its *activation height*: blocks below that height are validated under the preceding rules, and blocks at or above it under the new rules. Network upgrades are named and activate in a fixed order (ZIP 200, Terminology and "Activation mechanism"). This volume names two: NU6.3, whose deployment ZIP is ZIP 258, and NU7, the upgrade that follows NU6.3, whose deployment ZIP is ZIP 259 (ZIP 258, Abstract; ZIP 259, Abstract and Rationale, "Rationale for no new transaction format").

The volume describes the protocol as specified for NU7, whose deployment ZIP has status Draft (ZIP 259, preamble). It names a network upgrade only where the name fixes which rule applies, gives no activation parameter, such as an activation height or a date, and describes no upgrade by what it adds or changes.

Definition 1.4 (Epistemic classes)**.** Where the status of a claim about the deployed protocol needs stating, the volume places it in one of exactly two classes. The claim is *specified* when normative text of the protocol specification or of a ZIP fixes it. It is *designed but unspecified* when the deployed construction is designed to satisfy it but neither the protocol specification, nor a ZIP, nor a published proof establishes it for the deployed instance. An example of each class is given where its object is introduced: knowledge soundness of the Action proof in §1.2, and the cross-address restriction in §1.3.

1.2 The structure of a shielded payment

This subsection names, in plain language, the objects of a shielded payment and the checks that verify it. Each term receives one defining clause here and its construction in the section named.

Definition 1.5 (Notes, note commitments, nullifiers)**.** A shielded payment transfers value held in *notes*. A note records, among other fields, an amount of value and the *address* of its recipient, a public value derived from the recipient's keys (§3.3); no note is published in the clear (§4.1). For each note that a payment creates, the chain records a commitment to the note in the sense of the *Crypto Guide*, §"Hiding and binding", its *note commitment*, in the extracted form constructed in §4.2. For each note that a payment consumes, the chain records a tag, the note's *nullifier*, which marks the note as consumed (§6.1).

Definition 1.6 (Value pool)**.** In this volume a *value pool* is a separately accounted collection of notes with its own record of note commitments, the *note commitment tree* (§5.1); its own record of nullifiers, the *nullifier set* (§6.2); and its own *chain value pool balance*, the total value that the chain accounts as held in the pool (§11.4).

Definition 1.7 (Action, dummy note)**.** A shielded payment is carried by *Actions*. An Action consumes one note of a pool and creates one note of the same pool. A side of an Action that

the payment does not use carries a *dummy note*, a note of value zero that pads the Action (§4.4; protocol specification, §"Action Transfers and their Descriptions" and §"Dummy Notes (Orchard)").

Definition 1.8 (Bundle, balancing value)**.** The Actions of one transaction in one pool, together with data common to them, form the transaction's *bundle* for that pool; a transaction carries at most one bundle per pool (§4.4; ZIP 229, Abstract and "Transaction Format"). The *balancing value* of a bundle is the net value, public, that the bundle moves out of its pool into the rest of the transaction; it is negative when value enters the pool (§8.2).

Definition 1.9 (Note ciphertext)**.** A created note reaches its recipient through the chain: the Action that creates it also carries a *note ciphertext*, an encryption of the note to the recipient's address, from which the recipient recovers the note (§10).

Remark 1.10 (Division of verification). A *verifier*, a party that checks transactions against the consensus rules, accepts a bundle, without learning its notes, on one proof per bundle, two kinds of signature and two checks against the state of the chain (protocol specification, §"Action Transfers and their Descriptions" and §"Action Descriptions").

(1) One zero-knowledge proof per bundle, the *Action proof* (*Crypto Guide*, §"SNARKs: succinct non-interactive arguments of knowledge"; *Halo 2 Guide*, §"The Halo 2 proof system"): an argument that the prover knows, for each Action, private data that together with the Action's public data satisfy one fixed relation, in the sense of the *Crypto Guide*, §"Languages, relations, and witnesses", namely the Action statement, defined in "The Action statement" (§9.2). The proof system has no notion of value or ownership; the relation supplies both.

(2) Per Action, a *spend-authorisation signature* (*Crypto Guide*, §"Syntax and security goal"), made with a signing key obtained from the *spend authorising key*, the signing secret associated with the consumed note's address (§3.1), by which the holder of that key authorises the consumption (§7).

(3) Per bundle, one *binding signature*, by which balance is checked: the total value of the notes that the bundle consumes, minus that of the notes it creates, equals its balancing value (§8.3).

(4) Two checks against the state of the chain. The proof refers to one earlier state of the pool's note commitment tree, through a digest of that state, the bundle's *anchor* (§5.2), and the verifier checks that the anchor is one that the chain recorded for an earlier block. The verifier also checks that every nullifier that the bundle publishes is new to the pool (§6.2).

Both kinds of signature sign one digest of the transaction, constructed in "Transaction digests and signatures" (§11.3). The two checks of (4) lie outside the proof because they concern the history of the whole chain, which a proof over the data of one Action cannot attest.

Knowledge soundness of the Action proof as deployed (*Crypto Guide*, §"SNARKs: succinct non-interactive arguments of knowledge"), the property under which an accepted proof establishes the knowledge claimed in (1), is designed but unspecified: the volume takes it as Assumption 9.11 (§9.3), not as a result.

1.3 The Orchard protocol and its two pools

Definition 1.11 (Orchard protocol)**.** The *Orchard protocol* is the cryptographic design shared by two value pools: its primitive instances, among them the Pallas and Vesta curves and the Sinsemilla hash (§2); its keys and addresses; its notes, note commitments and nullifiers; its note encryption; and the one relation that every Action proves, the Action statement, defined in "The Action statement" (§9.2), with the circuit designed to encode it (§9.3). ZIP 229 fixes this vocabulary (ZIP 229, Terminology), and the volume adopts it.

Definition 1.12 (Orchard pool, Ironwood pool)**.** The *Orchard pool* and the *Ironwood pool* are the two value pools of the Orchard protocol. Each has its own note commitment tree, and hence its own anchors, its own nullifier set and its own chain value pool balance (ZIP 229, Terminology, and Rationale, "Separate state").

The Ironwood pool admits net inflow, value moved into it by a negative balancing value (§11.4). The Orchard pool is *sealed*, in two respects (ZIP 258, "Consensus rules from NU6.3 activation"); each rule is stated where its object is constructed. First, no transaction increases the value held in the Orchard pool (§11.4). Second, every Orchard-pool Action creates its note at the address of the note it consumes (ZIP 229, Rationale, "`enableCrossAddress` polarity"), a rule made exact in "Bundle flags" (§9.1) and with the Action statement, defined in "The Action statement" (§9.2). ZIP 229, Abstract, states the effect of the second rule: outputs to the Orchard pool go to an address for which the creator of the transaction can authorise spends, which discourages transfers between users within the Orchard pool. Consensus does not prevent such transfers: any party able to authorise spends for an address can create an Orchard-pool note to that address, since the consumed side of the Action can be a dummy note at that address (ZIP 326, "Rationale for key-generation restrictions"; the witness of such an Action is exhibited in §9.2).

The restriction of an Action's created note to the address of its consumed note, in force for every Orchard-pool Action and for every Ironwood-pool Action whose bundle does not enable cross-address transfers (§9.1), is specified: the protocol specification, §"Action Descriptions", fixes the meaning of the flag that controls it, although ZIP 2006, which that section cites for it, has status Reserved (ZIP 2006, preamble).

Notes of the two pools have the same form and the same note commitment. They differ only in the note plaintext format, the encoding in which note encryption carries a note: an Ironwood-pool note uses the format whose first byte, the *lead byte* that names the format, is `0x03`, under which the note's *commitment trapdoor*, the randomness of its note commitment, is derived by hashing a random seed, which the plaintext carries, together with every field that the commitment binds (the format and the lead-byte rule in §4.3, the plaintext in §10.1; ZIP 258, "ZIP 2005 activation"). Actions of both pools prove the same relation, the Action statement, defined in "The Action statement" (§9.2), with one circuit and one *verifying key*, the public parameter against which a proof is checked (§9.3; protocol specification, §"Action Descriptions"; ZIP 258, "Consensus rules from NU6.3 activation"). The pools are distinguished by their note commitment trees, nullifier sets and chain value pool balances, and by the component of the transaction that carries their bundles (§11.2), not by separate circuits (ZIP 229, Abstract, and Rationale, "Reuse of the Orchard protocol with minimal changes").

Orchard-pool notes are spendable, in version-5 and version-6 transactions (§11.2), subject to the rules that seal the Orchard pool (ZIP 258, "Consensus rules from NU6.3 activation"; ZIP 259, Abstract).

The name "Orchard" alone qualifies the shared cryptography of the Orchard protocol; a statement that depends on the pool names the pool, the Orchard pool or the Ironwood pool.

1.4 Requirements on a shielded payment

Definition 1.13 (Transparent and shielded payments, observer)**.** A *transparent payment* is one whose consumed and created amounts and addresses are public. The goal of a shielded payment is to achieve the effects of a transparent payment, namely transfer of value, exclusion of *double-spending*, the consumption of one note twice, and conservation of value, while hiding the contents of its notes, and the links between the notes it consumes and the notes it creates, from an *observer*: any party that reads the chain but holds no key of the notes concerned. Requirements R1 to R5 below make this precise. Each of R1 to R4 has a *soundness part*, a property of every valid payment, and a *hiding part*, a bound on what the observer learns; R5 has a soundness part only.

Definition 1.14 (Requirements R1 to R5)**.**

R1. *Hidden representation of value.* Soundness: the chain's record of a created note binds that note, so that its owner can later prove possession of it and no party can present the record as the record of a different note. Hiding: the record reveals neither the recipient's address nor the value, and does not distinguish a real note from a dummy note.

R2. *Membership without identification.* Soundness: every consumed note of non-zero value was previously committed in the same pool, at the state of the pool's record of commitments to which the payment refers. Hiding: the payment does not identify which committed note it consumes. A note of value zero, such as a dummy note, transfers no value, and R2 does not require it to have been committed.

R3. *No second consumption.* Soundness: no note of a pool is consumed twice. Hiding: this holds without a public list of consumed notes; the data published when a note is consumed do not identify which committed note it is.

R4. *Conservation.* Soundness: in every valid bundle the total value of the consumed notes equals the total value of the created notes plus the balancing value, so that no payment creates value. Hiding: no note's value is disclosed; of the amounts, only the balancing value is public.

R5. *Spend authority*, a soundness part only. Only the holder of a note's spending authority, the spend authorising key associated with its address (§3.1), can consume the note; and no other party can alter what an authorised payment transfers, namely the notes it consumes and creates and the value it moves into or out of the pool, without invalidating it. Requirement R5 requires nothing of data that do not change what is transferred; which data the signatures fix is stated in "Transaction digests and signatures" (§11.3).

The requirement map, Table 5, given with the Action statement (§9.2), pairs the soundness and hiding parts of each requirement with the results that discharge them; "Security" (§12) states those results as theorems with their assumptions.

2 Notation and primitive instances

This section fixes the encodings and the primitive instances on which the later constructions rest, each with its Orchard parameters and domain separators. The underlying constructions belong to the *Math Guide* and the *Crypto Guide* and are cited, not re-derived; each named assumption is stated at the first construction whose security needs it.

2.1 Fields, groups, and encodings

The Pallas group $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$ is the group of points of the curve $y^2 = x^3 + 5$ over the prime field $\mathbb{F}_{p_{\mathsf{Pallas}}}$, with identity $\mathcal{O}$. It has prime order p_{Vesta} and cofactor 1, and $\mathbb{F}_{p_{\mathsf{Vesta}}}$ is both its scalar field and the base field of the Vesta curve (*Math Guide*, §"Base fields, scalar fields, and the Pasta cycle" and §"Pallas and Vesta assembled", Example "The Pasta curves"; *Crypto Guide*, §"Instantiation on elliptic curves; the Pasta curves", Definition "The Pasta curves"). The group, its order and the cycle with Vesta are cited, not re-derived; the moduli are those of the protocol specification, §"Pallas and Vesta". Table 1 maps this notation, once, to the names used by the specification and by the lower volumes. Those names appear in the table only; later text writes only the column headed "This volume". Maps defined on the Pallas group carry the specification's subscript $\mathbb{P}$, as in $\mathsf{Extract}_{\mathbb{P}}$.

Object	This volume	Specification	Lower volumes
Pallas base-field modulus	p_{Pallas}	$q_{\mathbb{P}}$	p
Pallas scalar-field modulus, Vesta base-field modulus, order of the Pallas group	p_{Vesta}	$r_{\mathbb{P}}$	q
Pallas base field and scalar field	$\mathbb{F}_{p_{\mathsf{Pallas}}}$, $\mathbb{F}_{p_{\mathsf{Vesta}}}$	$\mathbb{F}_{q_{\mathbb{P}}}$, $\mathbb{F}_{r_{\mathbb{P}}}$	$\mathbb{F}_p$, $\mathbb{F}_q$
Pallas group	$\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$	$\mathbb{P}$	$E_p(\mathbb{F}_p)$
k-bit little-endian encoding of an integer	LE_k	$\mathsf{I2LEBSP}_k$	—
Encoding of a point P	$P^\star$	$\mathsf{repr}_{\mathbb{P}}(P)$	—
Sinsemilla short commitment	$\mathsf{ShortCommit}$	$\mathsf{SinsemillaShortCommit}$	—
Spend validating key, as a point and as its x-coordinate	$\mathsf{ak}^{\mathbb{P}}$, ak	$\mathsf{ak}^{\mathbb{P}}$, ak	—

Table 1: Notation for the fields, the Pallas group and the encodings, with the names of the protocol specification and of the *Math* and *Crypto Guides*. The spend validating key is constructed in §3.1; the point $\mathsf{ak}^{\mathbb{P}}$ and the field element $\mathsf{ak} = \mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}})$ are kept apart throughout.

Scalar multiplication is written $[k]P$, for $k \in \mathbb{F}_{p_{\mathsf{Vesta}}}$ or an integer read modulo p_{Vesta}. The symbol $\|$ denotes concatenation of bit strings or of byte strings, and ε the empty string. Byte-string literals and tags are set in monospace, as in `z.cash:Orchard`, and single bytes as `0x`... The symbol $\perp$ is the failure value of a partial map; a map that may fail takes values in $X \cup \{\perp\}$ for its codomain X.

Both moduli have bit length 255, and $p_{\mathsf{Pallas}} < p_{\mathsf{Vesta}}$:

$$2^{254} < p_{\mathsf{Pallas}} < p_{\mathsf{Vesta}} < 2^{255}.$$

Two consequences are used later. First, the integer representative of an element of either field lies in $\{0, \dots, 2^{255} - 1\}$ and has a 255-bit encoding, which leaves the top bit of a 256-bit string free. Second, the integer representative of an element of $\mathbb{F}_{p_{\mathsf{Pallas}}}$ lies in $\{0, \dots, p_{\mathsf{Vesta}} - 1\}$; reading it as a Pallas scalar is therefore an injective, not a surjective, map $\mathbb{F}_{p_{\mathsf{Pallas}}} \to \mathbb{F}_{p_{\mathsf{Vesta}}}$. This map is used wherever a base-field element multiplies a point (§3.3, §6.1).

Definition 2.1 (Integer and byte encodings)**.** For $k \geq 0$ and an integer $m \in \{0, \dots, 2^k - 1\}$, the *k-bit little-endian encoding* $\mathrm{LE}_k(m) \in \{0,1\}^k$ is the bit sequence $b_0 b_1 \cdots b_{k-1}$ with $m = \sum_{i=0}^{k-1} b_i 2^i$. The map $\mathrm{LE}_k \colon \{0, \dots, 2^k - 1\} \to \{0,1\}^k$ is a bijection with inverse LE_k^{-1}. A field element is encoded through its integer representative in $\{0, \dots, p-1\}$, for p the modulus. In the conversion names below, I stands for integer, LE for little-endian, BS for bit string, and OS for octet string (a sequence of eight-bit bytes). The digit 2 is read "to", and P stands for primitive, a basic

conversion operation. The subscript k counts bits, not bytes. Three maps relate integers, bit strings and byte strings:

- the map $\mathsf{I2LEOSP}_k$ sends $m \in \{0, \dots, 2^k - 1\}$ to its $\lceil k/8 \rceil$-byte little-endian encoding;
- for k a multiple of 8, the map $\mathsf{LEOS2IP}_k$ sends a $k/8$-byte string $(s_0, \dots, s_{k/8-1})$ to the integer $\sum_i s_i 256^i \in \{0, \dots, 2^k - 1\}$;
- the map $\mathsf{LEBS2OSP}_k \colon \{0,1\}^k \to \{0, \dots, 255\}^{\lceil k/8 \rceil}$ pads its input on the right with zero bits to a multiple of 8 bits, packs each group of 8 bits into a byte, least significant bit first, and keeps the order of the groups.

Hence $\mathsf{LEBS2OSP}_k(\mathrm{LE}_k(m)) = \mathsf{I2LEOSP}_k(m)$. The byte form is needed wherever a bit string enters BLAKE2b or GroupHash (protocol specification, §"Integers, Bit Sequences, and Endianness").

The widths used later are LE_{10} (the Merkle layer tag, §5.1), LE_{64} (the note value, §4.2), LE_{88} (the diversifier index, §3.3), LE_{255} (field elements) and LE_{256} (the star encoding below; the key encoding of $\mathsf{r_{ivk}}$, §3.2).

Definition 2.2 (Coordinate extraction)**.** The map $\mathsf{Extract}_{\mathbb{P}} \colon \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}}) \to \mathbb{F}_{p_{\mathsf{Pallas}}}$ is

$$\mathsf{Extract}_{\mathbb{P}}(\mathcal{O}) := 0, \qquad \mathsf{Extract}_{\mathbb{P}}((x,y)) := x.$$

Its lifting $\mathsf{Extract}_{\mathbb{P}}^{\perp} \colon \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}}) \cup \{\perp\} \to \mathbb{F}_{p_{\mathsf{Pallas}}} \cup \{\perp\}$ maps $\perp$ to $\perp$ and agrees with $\mathsf{Extract}_{\mathbb{P}}$ on points (protocol specification, §"Coordinate Extractor for Pallas").

The map $\mathsf{Extract}_{\mathbb{P}}$ supplies a field element wherever a later construction needs one in place of a point: the field element ak (§3.1), the extracted note commitment cmx (§4.2), the node values of the note commitment tree (§5.1) and the nullifier nf (§6.1). It is two-to-one on non-identity points, by Lemma 2.4 and because $P \neq -P$ for $P \neq \mathcal{O}$ in a group of odd order; every binding argument over an extracted coordinate treats the opposite point explicitly.

Lemma 2.3 (Extract vanishes only at the identity)**.** *For $P \in \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$, $\mathsf{Extract}_{\mathbb{P}}(P) = 0$ if and only if $P = \mathcal{O}$.*

Proof. If $P = \mathcal{O}$, then $\mathsf{Extract}_{\mathbb{P}}(P) = 0$ by definition. A point $(0, y)$ would satisfy $y^2 = 5$ in $\mathbb{F}_{p_{\mathsf{Pallas}}}$, but 5 is a quadratic non-residue modulo p_{Pallas} (*Math Guide*, §"Quadratic residues and the Euler criterion", Example "5 is a non-square in the Pallas base field"). Hence no point other than $\mathcal{O}$ has extracted coordinate 0. □

Lemma 2.4 (Coordinate extraction identifies opposite points)**.** *For $P, P' \in \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$, $\mathsf{Extract}_{\mathbb{P}}(P) = \mathsf{Extract}_{\mathbb{P}}(P')$ if and only if $P' = P$ or $P' = -P$.*

Proof. Negation is $-\mathcal{O} = \mathcal{O}$ and $-(x,y) = (x,-y)$ (*Math Guide*, §"Explicit affine formulas", Definition "Negation"), so $\mathsf{Extract}_{\mathbb{P}}(-P) = \mathsf{Extract}_{\mathbb{P}}(P)$ for every P, which is the reverse implication. For the forward implication let $\mathsf{Extract}_{\mathbb{P}}(P) = \mathsf{Extract}_{\mathbb{P}}(P')$. If $P = \mathcal{O}$, then $\mathsf{Extract}_{\mathbb{P}}(P') = 0$, and Lemma 2.3 gives $P' = \mathcal{O} = -P$. If $P = (x,y)$, then $x \neq 0$ by Lemma 2.3, so $P' \neq \mathcal{O}$ and $P' = (x, y')$ with $y'^2 = x^3 + 5 = y^2$; hence $y' = \pm y$ and $P' = \pm P$. □

Definition 2.5 (Star encoding)**.** For $P \in \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$ the *star encoding* $P^\star \in \{0,1\}^{256}$ is

$$O^\star := \mathrm{LE}_{256}(0), \qquad (x,y)^\star := \mathrm{LE}_{256}(x + 2^{255}(y \bmod 2)),$$

with x and y read as integer representatives: the 255-bit x-coordinate fills the low bits and the parity of y the top bit. The partial inverse $\mathsf{abst}_{\mathbb{P}} \colon \{0,1\}^{256} \to \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}}) \cup \{\perp\}$ reads the first 255 bits of s as $\tilde{x} := \mathrm{LE}_{255}^{-1}(s_0 \cdots s_{254})$ and the last bit as the sign bit $\tilde{y} := s_{255}$, and returns

1. the failure value $\perp$ if $\tilde{x} \geq p_{\mathsf{Pallas}}$; otherwise let $x := \tilde{x} \in \mathbb{F}_{p_{\mathsf{Pallas}}}$;
2. the identity O if $x = 0$ and $\tilde{y} = 0$;
3. the failure value $\perp$ if $x^3 + 5$ is not a square in $\mathbb{F}_{p_{\mathsf{Pallas}}}$;
4. otherwise the point (x,y) with $y^2 = x^3 + 5$ and $y \bmod 2 = \tilde{y}$

(protocol specification, §"Pallas and Vesta").

The encoding is canonical: $\mathsf{abst}_{\mathbb{P}}(P^\star) = P$ for every point P, and $\mathsf{abst}_{\mathbb{P}}(s) \neq \perp$ implies $\mathsf{abst}_{\mathbb{P}}(s)^\star = s$. For the first claim, $O^\star$ decodes to O at step 2. A point (x,y) has $x \neq 0$ (Lemma 2.3), and $y \neq 0$ because a point with $y = 0$ has order 2, which a group of odd order does not contain (*Math Guide*, §"Torsion, the cofactor, and prime-order subgroups", Remark "No two-torsion on the Pasta curves"). The two square roots $\pm y$ are therefore non-zero and, p_{Pallas} being odd, of opposite parities, so step 4 returns (x,y). For the second claim, a string decoded at step 2 is $\mathrm{LE}_{256}(0) = O^\star$, and a string decoded at step 4 to (x,y) has low bits $\mathrm{LE}_{255}(x)$ and top bit $y \bmod 2$, so it equals $(x,y)^\star$. Consequently $P \mapsto P^\star$ is injective, each point has exactly one valid encoding, and every other string decodes to $\perp$, among them $\mathrm{LE}_{256}(2^{255})$ (whose x-part is 0 and sign bit 1) and every string whose x-part is at least p_{Pallas}. The star encoding is the form in which a point enters a hash or a commitment as a bit string.

2.2 GroupHash, domain separation, and nothing-up-my-sleeve generators

Construction 2.6 (GroupHash on Pallas)**.** For byte strings D (the domain separator) and M (the message), the point $\mathsf{GroupHash}(D,M) \in \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$ is computed as follows.

1. Let
$$\mathsf{DST} := D \,\|\, \texttt{-} \,\|\, \texttt{pallas} \,\|\, \texttt{_XMD:BLAKE2b_SSWU_RO_};$$
the map is defined when DST has at most 255 bytes, which holds for every domain separator of Table 2.
2. The map hash_to_field expands (M, DST) by expand_message_xmd over BLAKE2b-512 into two 64-byte strings. BLAKE2b runs with zero personalisation; the string DST, followed by a byte carrying its length, is part of the hashed input. Each string is read as a big-endian integer and reduced modulo p_{Pallas}, giving $u_0, u_1 \in \mathbb{F}_{p_{\mathsf{Pallas}}}$.
3. The simplified Shallue–van de Woestijne–Ulas map sends each u_i to a point Q_i of a curve 3-isogenous to Pallas.
4. The output is the image of $Q_0 + Q_1$ under the 3-isogeny onto Pallas.

No cofactor is cleared, since the Pallas group has prime order. The isogenous curve, the isogeny coefficients and the constant of the map are those of the protocol specification,

§"Group Hash into Pallas and Vesta", and are not reproduced. The map is deterministic, public and efficiently computable (*Crypto Guide*, §"Hashing to a field element", Construction "Expand-then-reduce"; §"Hashing to a curve point", Constructions "Simplified Shallue–van de Woestijne–Ulas map" and "The hash-to-curve pipeline", Remark "The isogeny detour forced by j-invariant 0").

Definition 2.7 (Domain separator and fixed generator)**.** A *domain separator* is a fixed ASCII byte string naming a use site; it is the first argument of GroupHash, so that distinct uses draw distinct generators. A *fixed generator* is the value of GroupHash on a public, fixed pair (D, M). It is nothing-up-my-sleeve: the public pair determines it, and no party chose it (*Crypto Guide*, §"Hashing to a curve point", Construction "Nothing-up-my-sleeve generators"). The strings are fixed by the protocol specification.

Assumption 2.8 (GroupHash as a random oracle). In security arguments GroupHash is modelled as a random oracle on pairs (D, M) with values in $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$ (*Crypto Guide*, §"The random oracle model", Definition "Random oracle", with the group in place of bit strings): distinct pairs receive independent uniform points. In particular the fixed generators of Table 2 are, in the model, independent uniform points. The assumption is a heuristic about the concrete map, supported by its design for indifferentiability from a random oracle (protocol specification, §"Group Hash into Pallas and Vesta", note); it is not a theorem. Every later binding, collision, uniqueness or unlinkability result that needs it names it.

Table 2 lists every GroupHash input of the Orchard protocol, with the object it defines and the section that uses it. The messages `G`, `K`, `v` and `r` are one-byte ASCII strings, and the Sinsemilla messages are the ASCII strings shown. The rows are those of the protocol specification, §"Spend Authorization Signature (Sapling and Orchard)", §"Computing ρ values and Nullifiers", §"DiversifyHash^{Sapling} and DiversifyHash^{Orchard} Hash Functions", §"Sinsemilla Hash Function", §"Sinsemilla commitments", §"MerkleCRH^{Orchard} Hash Function" and §"Homomorphic Pedersen commitments (Sapling and Orchard)".

Domain separator D	Message M	Object	Used in
`z.cash:Orchard`	`G`	$G^{\mathsf{SpendAuth}}$, spend-authorisation base	§3.1, §7.1
`z.cash:Orchard`	`K`	K^{Orchard}, nullifier base	§6.1
`z.cash:Orchard-gd`	$\mathsf{LEBS2OSP}_{88}(d)$	$\mathsf{g_d}$ for a diversifier d	§3.3
`z.cash:Orchard-gd`	ε	$\mathsf{g_d}$ if the row above gives $\mathcal{O}$	§3.3
`z.cash:SinsemillaQ`	`z.cash:Orchard-NoteCommit-M`	hash base of NoteCommit	§4.2
`z.cash:Orchard-NoteCommit-r`	ε	blinding base of NoteCommit	§4.2
`z.cash:SinsemillaQ`	`z.cash:Orchard-CommitIvk-M`	hash base of Commit^{ivk}	§2.4
`z.cash:Orchard-CommitIvk-r`	ε	blinding base of Commit^{ivk}	§2.4
`z.cash:SinsemillaQ`	`z.cash:Orchard-MerkleCRH`	hash base of MerkleCRH	§5.1
`z.cash:Orchard-cv`	`v`	V^{Orchard}, value base	§8.1
`z.cash:Orchard-cv`	`r`	R^{Orchard}, randomness base	§8.1
`z.cash:SinsemillaS`	$\mathsf{I2LEOSP}_{32}(j)$	table entry $S(j)$, $0 \le j < 2^{10}$	§2.4

Table 2: Every input of GroupHash in the Orchard protocol. The hash base of MerkleCRH is derived from its domain without the suffix `-M`, and MerkleCRH has no blinding base. The table entries $S(j)$ are shared by every Sinsemilla domain.

Remark 2.9 (Domain separation). The pairs of Table 2 are pairwise distinct, and the map from (D, M) to the input hashed by hash_to_field is injective, because DST is closed by a byte carrying its length. Under Assumption 2.8 the table's generators are therefore independent, and a

relation found in one context transfers to no other (*Crypto Guide*, §"Domain separation and personalisation", Proposition "Domain separation yields independent oracles"). The *Crypto Guide*'s inventory of Orchard tags and personalisations in the same section is cited, not repeated.

2.3 Pseudorandom functions and field reductions

Definition 2.10 (Pseudorandom function)**.** A keyed family $(F_k)_k$ of functions is a *pseudorandom function* (PRF) if no efficient adversary with adaptive oracle access distinguishes F_k, for a secret uniform key k, from a uniformly random function with the same domain and range, except with negligible advantage (*Crypto Guide*, §"Pseudorandom functions and permutations", Definition "Pseudorandom function").

Construction 2.11 (Expansion function)**.** For a key $k \in \{0,1\}^{256}$ and a byte string t,

$$\mathsf{PRFexpand}_k(t) := \text{BLAKE2b-512}(\texttt{Zcash_ExpandSeed}, \mathsf{LEBS2OSP}_{256}(k) \parallel t),$$

a 64-byte string, identified with $\{0,1\}^{512}$; here BLAKE2b-512(P, x) is unkeyed BLAKE2b with a 64-byte output, the 16-byte personalisation P and the input x. A key given as a 32-byte string enters as is. The key is thus carried in the message, not in BLAKE2b's native keyed mode (protocol specification, §"Pseudo Random Functions"), so the *Crypto Guide*'s PRF argument for native keying (§"PRFs from hash functions: length extension and keyed BLAKE2") does not cover this construction; Assumption 2.12 states the property directly. The key is written as a subscript; the leading byte of t names the derived quantity (protocol specification, §"Pseudo Random Functions").

Assumption 2.12 (Pseudorandomness of PRF expansion)*.* For k uniform on $\{0,1\}^{256}$, the function $t \mapsto \mathsf{PRFexpand}_k(t)$ from byte strings to $\{0,1\}^{512}$ is a PRF in the sense of the preceding definition. This is the security requirement that the protocol specification, §"Pseudo Random Functions", places on PRFexpand.

Definition 2.13 (Field reductions)**.** For $x \in \{0,1\}^{512}$, read as 64 bytes,

$$\begin{aligned} \mathrm{ToBase}(x) &:= \mathsf{LEOS2IP}_{512}(x) \bmod p_{\mathsf{Pallas}} \in \mathbb{F}_{p_{\mathsf{Pallas}}}, \\ \mathrm{ToScalar}(x) &:= \mathsf{LEOS2IP}_{512}(x) \bmod p_{\mathsf{Vesta}} \in \mathbb{F}_{p_{\mathsf{Vesta}}} \end{aligned}$$

(protocol specification, §"Orchard Key Components").

For x uniform on $\{0,1\}^{512}$, the statistical distance of $\mathrm{ToBase}(x)$ from the uniform distribution on $\mathbb{F}_{p_{\mathsf{Pallas}}}$ is at most $p_{\mathsf{Pallas}}/2^{512}$, and that of $\mathrm{ToScalar}(x)$ from the uniform distribution on $\mathbb{F}_{p_{\mathsf{Vesta}}}$ at most $p_{\mathsf{Vesta}}/2^{512}$ (*Math Guide*, §"Uniform sampling and the bias of modular reduction", Proposition "Bias of modular reduction", with $L = 512$). Both bounds are below 2^{-257}.

Construction 2.14 (Domain bytes of PRF expansion)**.** Every later use of PRFexpand derives one quantity as

$$y := f(\mathsf{PRFexpand}_\kappa(b \parallel t')),$$

where the lead byte b, the key κ, the remainder t' of the input and the reduction $f \in \{\mathrm{ToBase}, \mathrm{ToScalar}\}$, or no reduction, form one row of Table 3. Each row's objects are defined in the section it names (protocol specification, §"Orchard Key Components" for 0x06, 0x07, 0x08 and 0x82; §"Sending Notes (Orchard)" for 0x04, 0x09 and 0x0B).

b	Key κ	Remainder t'	f	Output	Section
0x06	sk	ε	ToScalar	ask	§3.1
0x07	sk	ε	ToBase	nk	§3.1
0x08	sk	ε	ToScalar	$\mathsf{r_{ivk}}$	§3.1
0x82	$\mathrm{LE}_{256}(\mathsf{r_{ivk}})$	$\mathsf{I2LEOSP}_{256}(\mathsf{ak}) \parallel \mathsf{I2LEOSP}_{256}(\mathsf{nk})$	none	$\mathsf{dk} \parallel \mathsf{ovk}$	§3.2
0x04	rseed	$\mathsf{I2LEOSP}_{256}(\rho)$	ToScalar	esk	§4.3
0x09	rseed	$\mathsf{I2LEOSP}_{256}(\rho)$	ToBase	ψ	§4.3
0x0B	rseed	$\mathsf{LEBS2OSP}_{256}(\mathsf{g_d}^{\star}) \parallel$ $\mathsf{LEBS2OSP}_{256}(\mathsf{pk_d}^{\star}) \parallel$ $\mathsf{I2LEOSP}_{64}(v) \parallel \mathsf{I2LEOSP}_{256}(\rho) \parallel$ $\mathsf{I2LEOSP}_{256}(\psi)$	ToScalar	rcm	§4.3

Table 3: Every lead byte of PRFexpand used in this volume. The row 0x82 applies no reduction: its 64-byte output splits into dk, the first 32 bytes, and ovk, the last 32 bytes.

Lemma 2.15 (Independence of domain-separated expansions)**.** *Let k be uniform on $\{0,1\}^{256}$ and used only as a* PRFexpand *key. Let an efficient algorithm choose, possibly adaptively, pairwise distinct inputs $t_1, \ldots, t_n$, in particular inputs with pairwise distinct leading bytes, and receive the answers. Under Assumption 2.12, the answers $(\mathsf{PRFexpand}_k(t_1), \ldots, \mathsf{PRFexpand}_k(t_n))$ are computationally indistinguishable from n independent uniform elements of $\{0,1\}^{512}$. Consequently, for reductions $f_i \in \{\mathrm{ToBase}, \mathrm{ToScalar}\}$, the answers $(f_i(\mathsf{PRFexpand}_k(t_i)))_i$ are computationally indistinguishable from independent uniform elements of the target fields, up to an additional statistical distance of at most the sum of the reduction distances. For a key within statistical distance δ of uniform, both conclusions hold with δ added.*

Proof. One hybrid replaces $\mathsf{PRFexpand}_k$ by a uniformly random function; the distinguishing advantage changes by at most the PRF advantage of the algorithm that runs the given one and forwards its queries to its own oracle (Assumption 2.12; adaptive queries are within the definition, *Crypto Guide*, §"Pseudorandom functions and permutations", Remark "Necessity of adaptivity"). A random function, sampled lazily, answers each new input with a fresh uniform value, so its answers on pairwise distinct inputs are independent and uniform, also when each input depends on earlier answers. For the reduced answers, replacing each f_i of a fresh uniform string by a fresh uniform field element changes the distribution of the whole interaction by at most the reduction distance of f_i; summing over i bounds the total by the triangle inequality, and the post-processing by the algorithm does not increase it (*Math Guide*, §"Statistical distance", Theorem "Properties of statistical distance"). A key within statistical distance δ of uniform changes the interaction, a function of the key and the algorithm's coins, by at most δ, by the same theorem. □

The hypothesis of Lemma 2.15 holds for the spending key sk, and for rseed towards an adversary that sees rseed only through PRFexpand outputs; the note plaintext, which contains rseed, is treated in §10.2. It does not hold for the row 0x82, whose key encodes the trapdoor $\mathsf{r_{ivk}}$, which is also used in $\mathsf{Commit}^{\mathsf{ivk}}$; that row is covered by Assumption 3.13 in §3.2. The lemma is used in §3.1, in Lemma "Rejection in key generation" (§3.2), in Propositions 3.18, 3.17, 4.8, 6.10 and 10.6, in Proposition "Confidentiality of the outgoing ciphertext" (§10.3), in Theorems 12.9 and 12.12, and in Remark "Hypothesis (H) for honest keys" (§11.3).

Remark 2.16 (Hash roles). Each hash is defined at its construction site. BLAKE2b-512 with personalisation `Zcash_ExpandSeed` serves PRFexpand (this subsection); BLAKE2b-512 without personalisation serves hash_to_field inside GroupHash (§2.2); personalised BLAKE2b-512 serves the RedPallas hash H (§7.1); BLAKE2b-256 serves the note-encryption key derivation and the outgoing cipher key (§10.2, §10.3); BLAKE2b serves the transaction and signature digests (§11.3); Sinsemilla serves MerkleCRH, NoteCommit and $\mathsf{Commit}^{\mathsf{ivk}}$ (§2.4); Poseidon serves the nullifier PRF, constructed with the nullifier (§6.1). The personalisations of PRFexpand, the note-encryption key derivation and the outgoing cipher key are those of the *Crypto Guide*'s inventory (§"PRFs from hash functions: length extension and keyed BLAKE2", Remark "Zcash's use of keyed and personalised BLAKE2"); every personalisation is stated at its construction.

2.4 Sinsemilla: hash, commitment, and short forms

Construction 2.17 (Sinsemilla hash on Pallas)**.** The *Crypto Guide*, §"Sinsemilla: an algebraic hash-based commitment", Construction "Sinsemilla hash", fixes the chunk width $k = 10$ (written w there, where k counts the pieces), the table and the domain base

$$S(j) := \mathsf{GroupHash}(\texttt{z.cash:SinsemillaS}, \mathsf{I2LEOSP}_{32}(j)) \qquad (0 \le j < 2^k),$$
$$Q(D) := \mathsf{GroupHash}(\texttt{z.cash:SinsemillaQ}, D)$$

for a byte string D, and the accumulator; it writes $\mathrm{LE}_{32}(j)$ for the four-byte encoding $\mathsf{I2LEOSP}_{32}(j)$. The Orchard instance adds the following (protocol specification, §"Sinsemilla Hash Function").

1. The *chunk bound* c is the largest integer with $2^c \le (p_{\mathsf{Vesta}} - 1)/2$; thus $c = 253$, and a message has at most $kc = 2530$ bits.

2. *Incomplete addition* $\dot{+}$ on $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}}) \cup \{\bot\}$ returns $\bot$ if either operand is $\bot$; for points it returns $\bot$ in an *exceptional case*, when an operand is $\mathcal{O}$ or the operands have equal x-coordinates (equal or opposite points), and their group sum otherwise (*Math Guide*, §"Explicit affine formulas", Remark "Complete versus incomplete addition").

3. For a byte string D and a bit string M of length at most kc, let $n := \lceil |M|/k \rceil \le c$, pad M with zero bits to nk bits, split the result into k-bit pieces, and let m_i be LE_k^{-1} of the i-th piece, $1 \le i \le n$. Then

$$\mathsf{Acc}_0 := Q(D), \qquad \mathsf{Acc}_i := (\mathsf{Acc}_{i-1} \dot{+} S(m_i)) \dot{+} \mathsf{Acc}_{i-1} \quad (1 \le i \le n),$$

   and $\mathsf{SinsemillaHashToPoint}(D, M) := \mathsf{Acc}_n$. The output is $\bot$ exactly when some incomplete addition meets an exceptional case.

4. The extracted hash is

$$\mathsf{SinsemillaHash}(D, M) := \mathsf{Extract}^{\bot}_{\mathbb{P}}(\mathsf{SinsemillaHashToPoint}(D, M)) \in \mathbb{F}_{p_{\mathsf{Pallas}}} \cup \{\bot\}.$$

Construction 2.18 (Sinsemilla commitment and short commitment)**.** For a byte string D, a bit string M of length at most kc and a trapdoor $r \in \mathbb{F}_{p_{\mathsf{Vesta}}}$, let

$$\hat{M} := \mathsf{SinsemillaHashToPoint}(D \,\|\, \texttt{-M}, M), \qquad H_D := \mathsf{GroupHash}(D \,\|\, \texttt{-r}, \varepsilon).$$

Then, with complete group addition,

$$\mathsf{SinsemillaCommit}_r(D, M) := \begin{cases} \hat{M} + [r]H_D & \text{if } \hat{M} \neq \bot, \\ \bot & \text{otherwise,} \end{cases}$$

$$\mathsf{ShortCommit}_r(D, M) := \mathsf{Extract}^{\bot}_{\mathbb{P}}(\mathsf{SinsemillaCommit}_r(D, M)) \in \mathbb{F}_{p_{\mathsf{Pallas}}} \cup \{\bot\}$$

(protocol specification, §"Sinsemilla commitments"). The domain D is routed to the hash as $D \,\|\, \texttt{-M}$ and to the blinding base as $D \,\|\, \texttt{-r}$ with the empty message. The *Crypto Guide* constructs the commitment with a generic independent blinding generator (§"Sinsemilla: an algebraic hash-based commitment", Construction "SinsemillaCommit"); the routing and the $\bot$ case are the Orchard additions.

The unblinded SinsemillaHash is deterministic and therefore not hiding; ShortCommit is its blinded counterpart. Proposition 2.24 and Lemma 2.26 state the security of both.

Construction 2.19 (Orchard instances)**.** The Orchard protocol uses three Sinsemilla instances, each with one fixed message length per domain, all within c chunks (Table 4): MerkleCRH, a SinsemillaHash instance whose layout is given in §5.1; NoteCommit, a SinsemillaCommit instance whose layout is given in Definition 4.4; and $\mathsf{Commit}^{\mathsf{ivk}}$, a ShortCommit instance defined next.

Instance	Form	Domain	Message bits	Chunks
MerkleCRH	SinsemillaHash	`z.cash:Orchard-MerkleCRH`	$10 + 2 \cdot 255 = 520$	52
NoteCommit	SinsemillaCommit	`z.cash:Orchard-NoteCommit`	$2 \cdot 256 + 64 + 2 \cdot 255 = 1086$	109
$\mathsf{Commit}^{\mathsf{ivk}}$	ShortCommit	`z.cash:Orchard-CommitIvk`	$2 \cdot 255 = 510$	51

Table 4: The Orchard Sinsemilla instances. Each domain hashes messages of the single length shown, of at most $c = 253$ chunks of $k = 10$ bits.

Definition 2.20 (The key commitment)**.** For $x, y \in \mathbb{F}_{p_{\mathsf{Pallas}}}$ and $\mathsf{r}_{\mathsf{ivk}} \in \mathbb{F}_{p_{\mathsf{Vesta}}}$,

$$\mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r}_{\mathsf{ivk}}}(x, y) := \mathsf{ShortCommit}_{\mathsf{r}_{\mathsf{ivk}}}(\texttt{z.cash:Orchard-CommitIvk},\ \mathrm{LE}_{255}(x) \,\|\, \mathrm{LE}_{255}(y)),$$

a 510-bit message (protocol specification, §"Sinsemilla commitments"). Its output is $\bot$ exactly when $\mathsf{SinsemillaHashToPoint}(\texttt{z.cash:Orchard-CommitIvk-M}, \cdot)$ returns $\bot$; it is 0 exactly when the commitment point is $\mathcal{O}$ (Lemma 2.3); otherwise it is the x-coordinate of a non-identity point, an element of $\{1, \ldots, p_{\mathsf{Pallas}} - 1\}$ and hence, since $p_{\mathsf{Pallas}} < p_{\mathsf{Vesta}}$, a non-zero Pallas scalar. The x-coordinates of non-identity points form a set of $(p_{\mathsf{Vesta}} - 1)/2$ elements, by Lemma 2.4 and because a group of odd prime order has no point with $y = 0$. The incoming viewing key of §3.2 is this instance applied to $(\mathsf{ak}, \mathsf{nk})$ with trapdoor $\mathsf{r}_{\mathsf{ivk}}$; the exclusion of the outputs 0 and $\bot$ by key generation is stated there.

Remark 2.21 (Cost inside a proof). Sinsemilla is used for its cost inside an arithmetic circuit with lookups, one table lookup and two incomplete additions over $\mathbb{F}_{p_{\mathsf{Pallas}}}$ per k-bit chunk (*Crypto Guide*, §"Sinsemilla: an algebraic hash-based commitment", Remark "Why Sinsemilla exists: cost inside a proof"; *Halo 2 Guide*, §"The lookup argument" and §"From statement to circuit: arithmetisation in practice"), and its collision resistance reduces to discrete logarithms on Pallas together with GroupHash modelled as a random oracle, not to discrete logarithms alone.

Assumption 2.22 (Discrete logarithms on Pallas). For every classical probabilistic polynomial-time algorithm, given a generator G of $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$ and $[s]G$ for s uniform in $\mathbb{F}_{p_{\mathsf{Vesta}}}$, the probability of outputting s is negligible (*Crypto Guide*, §"The discrete logarithm problem", Definition "Discrete logarithm problem, DLP", instantiated on Pallas; *Math Guide*, §"The elliptic-curve discrete logarithm problem"). Concretely, the best known classical attack costs about 2^{126} group operations (*Crypto Guide*, §"Instantiation on elliptic curves; the Pasta curves", Proposition "The Pasta curves against the criteria"); the level is cited, not recomputed.

Lemma 2.23 (Relations among fixed generators). *Under Assumptions 2.8 and 2.22, no efficient algorithm outputs, except with negligible probability, distinct inputs of* GroupHash *with values $P_1, \dots, P_t$ and coefficients $a_1, \dots, a_t \in \mathbb{F}_{p_{\mathsf{Vesta}}}$, not all zero, with $\sum_i [a_i]P_i = O$.*

Proof. In the model of Assumption 2.8 a reduction receiving a challenge $Y = [s]G$ answers each new GroupHash input with $[u]G + [w]Y$ for fresh uniform $u, w \in \mathbb{F}_{p_{\mathsf{Vesta}}}$, a uniform point and hence distributed as the oracle's answer. For every value of w exactly one u is consistent with the answer, so the w remain uniform and independent of the algorithm's view. A relation $\sum_i [a_i]P_i = O$ gives $c_0 + c_1 s = 0$ with $c_1 = \sum_i a_i w_i$, which is non-zero except with probability $1/p_{\mathsf{Vesta}}$, and then $s = -c_0 c_1^{-1}$ (*Crypto Guide*, §"Pedersen vector commitments", Theorem "Properties of the vector commitment", proof; §"Hashing to a curve point", Remark "Why NUMS generators are binding-safe"). □

Every later argument that ends in a non-trivial discrete-logarithm relation among values of GroupHash ends in this lemma.

Proposition 2.24 (Collision resistance of the Sinsemilla instances). *Consider an Orchard instance of Table 4 with message length l, hash domain D' (`z.cash:Orchard-MerkleCRH` for* MerkleCRH*; $D \parallel$ `-M` for a commitment instance with domain D) and, for a commitment instance, blinding base H_D. Under Assumptions 2.22 and 2.8, no efficient algorithm outputs, except with negligible probability:*

(*i*) *messages $M, M' \in \{0,1\}^l$ whose hash points $P :=$ SinsemillaHashToPoint(D', M) and $P' :=$ SinsemillaHashToPoint(D', M') are not $\perp$ and satisfy $P = P'$ with $M \neq M'$, or $P = -P'$; hence, by Lemma 2.4, no $M \neq M'$ with equal values of the extracted form* SinsemillaHash *other than $\perp$;*

(*ii*) *for a commitment instance, openings (M, r) and (M', r') in $\{0,1\}^l \times \mathbb{F}_{p_{\mathsf{Vesta}}}$ whose commitments $C :=$ SinsemillaCommit$_r(D, M)$ and $C' :=$ SinsemillaCommit$_{r'}(D, M')$ are not $\perp$ and satisfy $C = C'$ with $M \neq M'$, or $C = -C'$; hence no $M \neq M'$ with equal values of* ShortCommit *other than $\perp$;*

(*iii*) *a message in $\{0,1\}^l$ on which* SinsemillaHashToPoint$(D', \cdot)$ *returns $\perp$, equivalently on which the* SinsemillaHash *or commitment form returns $\perp$ for every trapdoor.*

Each such output yields efficiently a non-trivial discrete-logarithm relation among $Q(D')$, $S(0)$, ..., $S(2^k - 1)$ and, in (ii), H_D.

Proof. The proof is by citation, with the Orchard steps made in place. The cited arguments are, in the *Crypto Guide*, §"Sinsemilla: an algebraic hash-based commitment", Proposition "Collision resistance of Sinsemilla", with the paragraph after it on the extracted coordinate and the fixed length, and Construction "SinsemillaCommit"; and the Security argument of the protocol specification, §"Sinsemilla Hash Function": Lemma "An injectivity property for Sinsemilla", the theorem "Collision resistance of SinsemillaHash and SinsemillaHashToPoint" with its note extending

it to added terms with independent bases, which covers SinsemillaCommit and ShortCommit, and the theorem that a $\perp$ output of SinsemillaHashToPoint yields a non-trivial discrete-logarithm relation. These arguments write a non-$\perp$ output on the pieces $m = (m_1, \dots, m_n)$ as

$$\mathsf{SinsemillaHashToPoint}(D', M) = [2^n]Q(D') + \sum_{j=0}^{2^k-1} [\chi_j(m)]S(j), \tag{1}$$

where $\chi_j(m) := \sum_{i=1}^{n} 2^{n-i}\delta_{m_i,j}$ with δ the Kronecker delta, and with $[r]H_D$ added for a commitment. Equal outputs on distinct messages give

$$\textstyle\sum_j [\chi_j(m) - \chi_j(m')]S(j) + [r - r']H_D = \mathcal{O},$$

non-trivial because $m \mapsto \chi(m)$ is injective. Opposite outputs give

$$[2^{n+1}]Q(D') + \textstyle\sum_j [\chi_j(m) + \chi_j(m')]S(j) + [r + r']H_D = \mathcal{O},$$

non-trivial because its coefficient on $Q(D')$ is non-zero. An exceptional case at step i has the form $[\alpha]\mathsf{Acc}_{i-1} + S(m_i) = \mathcal{O}$ with $\alpha \in \{-1, 1, 2\}$; substituting (1) for Acc_{i-1} gives a relation with coefficient $\alpha\, 2^{i-1}$ on $Q(D')$. For the hash forms the terms in H_D are absent.

Three steps are made in place. (1) Each instance hashes one fixed length l per domain, and l is admissible: $n \le c$, with $n = 52$, 109 and 51 (Table 4). Admissibility is exactly the hypothesis under which no coefficient wraps modulo p_{Vesta}. Because $2^n \le 2^c \le (p_{\mathsf{Vesta}} - 1)/2$, each difference $\chi_j(m) - \chi_j(m')$ has absolute value below $2^n < p_{\mathsf{Vesta}}$ and vanishes modulo p_{Vesta} only if it vanishes as an integer, so uniqueness of binary expansions makes χ injective; and $0 < 2^{n+1} \le p_{\mathsf{Vesta}} - 1$ and $0 < |\alpha\, 2^{i-1}| \le 2^n$ keep the coefficient on $Q(D')$ non-zero. Padding to nk bits is injective on messages of the fixed length l, so $M \ne M'$ gives $m \ne m'$. (2) By the definition of incomplete addition, an instance returns $\perp$ exactly when some incomplete addition meets an exceptional case, the case the cited $\perp$ theorem treats. An operand $\mathcal{O}$ arises only if $Q(D')$ or some $S(j)$ is $\mathcal{O}$, itself a non-trivial relation, because an incomplete addition that meets no exceptional case never returns $\mathcal{O}$. The commitment forms return $\perp$ exactly when the point hash does, for every trapdoor, since the blinding addition is complete. (3) The generators $Q(D'), S(0), \dots, S(2^k - 1)$ and H_D are values of GroupHash on pairwise distinct inputs (Table 2), so the lemma on relations among fixed generators excludes each relation except with negligible probability. The consequences for the extracted forms follow from Lemma 2.4: equal extracted values other than $\perp$ come from equal or opposite points, which are the cases of (i) and (ii). □

Remark 2.25 (Fixed length). The fixed-length hypothesis cannot be dropped. The point hash SinsemillaHashToPoint zero-pads its input to a multiple of k bits, so for a message M whose length is not a multiple of k the messages M and $M \| 0$ collide with no discrete-logarithm relation. The specification requires collision resistance only between inputs of one fixed length for a given domain (protocol specification, §"Sinsemilla Hash Function"), and every Orchard domain hashes a single length.

Lemma 2.26 (Hiding and binding of SinsemillaCommit). *Let D be a byte string and $l \le kc$ a message length.*

(*i*) *Perfect hiding: if $H_D \ne \mathcal{O}$, then for every $M \in \{0,1\}^l$ with hash point $\hat{M} \ne \perp$ and r uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$ and independent of M, the commitment* $\mathsf{SinsemillaCommit}_r(D, M)$ *is uniform on $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$; hence the distribution of neither it nor* $\mathsf{ShortCommit}_r(D, M)$*, a fixed function of it, depends on M.*

(*ii*) *Computational binding: under Assumptions 2.22 and 2.8, no efficient algorithm outputs, except with negligible probability, openings (M, r) and (M', r') with $M \ne M'$ in $\{0,1\}^l$ whose*

> *commitments* $\mathsf{SinsemillaCommit}_r(D, M)$ *and* $\mathsf{SinsemillaCommit}_{r'}(D, M')$ *are equal and not* $\perp$.

Proof. The lemma is the *Crypto Guide*'s Construction "SinsemillaCommit" (§"Sinsemilla: an algebraic hash-based commitment") for the Orchard blinding base. (i) Since $H_D \neq \mathcal{O}$ and the group has prime order p_{Vesta}, the map $r \mapsto [r]H_D$ is a bijection from $\mathbb{F}_{p_{\mathsf{Vesta}}}$ onto the group, so $[r]H_D$ is uniform, and so is its translate by the fixed point $\hat{M}$ (*Crypto Guide*, §"The Pedersen commitment", Theorem "Perfect hiding"). (ii) Two such openings give the first relation in the proof of Proposition 2.24, whose coefficient vector on the $S(j)$ is non-zero for $n = \lceil l/k \rceil \leq c$ by step (1) of that proof. The only Orchard addition is the blinding base $H_D = \mathsf{GroupHash}(D \| \texttt{-r}, \varepsilon)$, whose input differs from those of $Q(D \| \texttt{-M})$ and of every $S(j)$, since its domain separator ends in `-r`; the lemma on relations among fixed generators excludes the relation. The specification states the same properties, with hiding conditional on no $\perp$ output (protocol specification, §"Sinsemilla commitments"). □

3 Keys and addresses

This section constructs the key components of the protocol specification, §"Orchard Key Components", from the instances of §2: the spending key and the spend-side secrets, the viewing keys, and the diversified addresses. It states each assumption that their security needs at its first use and proves, in Proposition 3.18, four one-way separations between key tiers.

3.1 The spending key and the spend-side secrets

Remark 3.1 (Scope of the key derivations). Every derivation of this section is that of a key generated with $\mathsf{use_qsk} = \mathsf{false}$, where $\mathsf{use_qsk}$ is the Boolean key-generation flag of ZIP 2005. ZIP 2005, §"Changes to the Protocol Specification", item §4.2.3, specifies, as a change to the specification's §"Orchard Key Components" and not in that section itself, an alternative source of $\mathsf{ak}^{\mathbb{P}}$ and $\mathsf{r}_{\mathsf{ivk}}$, with nk derived as here; it yields components $(\mathsf{ak}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$ of the same types, and $\mathsf{ak}^{\mathbb{P}}$ again of even y-coordinate. The volume does not construct the alternative and names none of its intermediate keys. Assumption 3.13, Propositions 3.18, 3.17, 6.10 and 10.6, Proposition "Confidentiality of the outgoing ciphertext" (§10.3), and the hypotheses on honest key generation of Theorems 12.9 and 12.12 are stated for $\mathsf{use_qsk} = \mathsf{false}$ only. For such keys every key component is a deterministic function of the spending key sk.

> **Definition 3.2** (Spending key and key generation)**.** The *spending key* sk is a uniformly random 256-bit string, identified with a 32-byte string (protocol specification, §"Constants"). Key generation draws sk, derives from it the components of this section, and discards sk and draws a new one when $\mathsf{ask} = 0$ (below), when $\mathsf{ivk} \in \{0, \perp\}$ (§3.2), or on the one further condition of the protocol specification, §"Orchard Key Components", which concerns the internal key of ZIP 32, a key this volume does not construct, and is stated in §3.2. A spending key that key generation does not discard is *accepted*; it determines every other component of a key generated with $\mathsf{use_qsk} = \mathsf{false}$.

> **Construction 3.3** (Spend-side secrets)**.** For a spending key sk, the *spend authorising key* ask, the

nullifier deriving key nk and the trapdoor $\mathsf{r_{ivk}}$ of $\mathsf{Commit}^{\mathsf{ivk}}$ are

$$\begin{aligned}\mathsf{ask} &:= \mathsf{ToScalar}(\mathsf{PRFexpand_{sk}}([\texttt{0x06}])) \in \mathbb{F}_{p_{\mathsf{Vesta}}},\\ \mathsf{nk} &:= \mathsf{ToBase}(\mathsf{PRFexpand_{sk}}([\texttt{0x07}])) \in \mathbb{F}_{p_{\mathsf{Pallas}}},\\ \mathsf{r_{ivk}} &:= \mathsf{ToScalar}(\mathsf{PRFexpand_{sk}}([\texttt{0x08}])) \in \mathbb{F}_{p_{\mathsf{Vesta}}}\end{aligned}$$

(protocol specification, §"Orchard Key Components"). The three lead bytes are rows of Table 3. Key generation discards sk if $\mathsf{ask} = 0$.

The scalars ask and $\mathsf{r_{ivk}}$ lie in $\mathbb{F}_{p_{\mathsf{Vesta}}}$, the scalar field of the Pallas group, because each multiplies a Pallas point: ask the generator $G^{\mathsf{SpendAuth}}$ below, $\mathsf{r_{ivk}}$ the blinding base of $\mathsf{Commit}^{\mathsf{ivk}}$ (§3.2). The key nk is an element of the base field $\mathbb{F}_{p_{\mathsf{Pallas}}}$.

The three inputs of $\mathsf{PRFexpand_{sk}}$ have distinct lead bytes. By Lemma 2.15, under Assumption 2.12, the triple $(\mathsf{ask}, \mathsf{nk}, \mathsf{r_{ivk}})$ computed from a uniform sk, before rejection and before the normalisation below, is therefore computationally indistinguishable from independent uniform elements of $\mathbb{F}_{p_{\mathsf{Vesta}}}$, $\mathbb{F}_{p_{\mathsf{Pallas}}}$ and $\mathbb{F}_{p_{\mathsf{Vesta}}}$, up to an additional statistical distance of three reduction distances, each below 2^{-257} (§2.3).

Construction 3.4 (Spend validating key and sign normalisation)**.** Let $G^{\mathsf{SpendAuth}} := \mathsf{GroupHash}(\texttt{z.cash:Orchard}, \texttt{G})$ (Table 2), the generator of the spend-authorisation signature scheme (protocol specification, §"Spend Authorization Signature (Sapling and Orchard)"). For $\mathsf{ask} \neq 0$:

1. compute the point $P := [\mathsf{ask}]\, G^{\mathsf{SpendAuth}}$;
2. if its y-coordinate is odd, that is, if bit 255 of the star encoding $P^\star$ is 1, replace ask by $-\mathsf{ask}$;
3. let $\mathsf{ak}^{\mathbb{P}} := [\mathsf{ask}]\, G^{\mathsf{SpendAuth}}$ for the updated ask, and $\mathsf{ak} := \mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}}) \in \mathbb{F}_{p_{\mathsf{Pallas}}}$.

The *spend validating key* is the point $\mathsf{ak}^{\mathbb{P}}$ together with its x-coordinate ak, a field element (protocol specification, §"Orchard Key Components").

After step 2 the point $\mathsf{ak}^{\mathbb{P}}$ has even y-coordinate. The point P is not $\mathcal{O}$, since $\mathsf{ask} \neq 0$ and $G^{\mathsf{SpendAuth}}$ generates a group of prime order, so $P = (x, y)$ with $y \neq 0$ (§2.1, on the star encoding). Replacing ask by $-\mathsf{ask}$ replaces P by $-P = (x, -y)$, and the representatives y and $p_{\mathsf{Pallas}} - y$ have opposite parities because p_{Pallas} is odd.

For an accepted key, $\mathsf{ak} \in \{1, \ldots, p_{\mathsf{Pallas}} - 1\}$, and ak determines $\mathsf{ak}^{\mathbb{P}}$. Indeed $\mathsf{ak}^{\mathbb{P}} \neq \mathcal{O}$, so $\mathsf{ak} \neq 0$ by Lemma 2.3. By Lemma 2.4 the points with x-coordinate ak are $\mathsf{ak}^{\mathbb{P}}$ and $-\mathsf{ak}^{\mathbb{P}}$, whose y-coordinates have opposite parities; the point $\mathsf{ak}^{\mathbb{P}}$ is the one with even y-coordinate, that is, $\mathsf{ak}^{\mathbb{P}} = \mathsf{abst}_{\mathbb{P}}(\mathrm{LE}_{256}(\mathsf{ak}))$, the *even-y lift* of ak. Theorem 12.9 cites this normalisation.

3.2 Viewing keys

Definition 3.5 (Full viewing key)**.** The *full viewing key* of an accepted key is the triple

$$\mathsf{fvk} := (\mathsf{ak}, \mathsf{nk}, \mathsf{r_{ivk}}) \in \{1, \ldots, p_{\mathsf{Pallas}} - 1\} \times \mathbb{F}_{p_{\mathsf{Pallas}}} \times \mathbb{F}_{p_{\mathsf{Vesta}}},$$

with ak the field element of §3.1; the point $\mathsf{ak}^{\mathbb{P}}$ is recomputed from ak by the even-y lift.

Construction 3.6 (Incoming viewing key scalar)**.** With the key commitment of §2.4 and $D :=$ `z.cash:Orchard-CommitIvk`,

$$\mathsf{ivk} := \mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r_{ivk}}}(\mathsf{ak}, \mathsf{nk}) = \mathsf{ShortCommit}_{\mathsf{r_{ivk}}}(D,\ \mathrm{LE}_{255}(\mathsf{ak}) \,\|\, \mathrm{LE}_{255}(\mathsf{nk})) \in \mathbb{F}_{p_{\mathsf{Pallas}}} \cup \{\perp\}.$$

Explicitly, let $\hat{M}$ be the Sinsemilla hash point of the 510-bit message $\mathrm{LE}_{255}(\mathsf{ak}) \,\|\, \mathrm{LE}_{255}(\mathsf{nk})$ under the domain `z.cash:Orchard-CommitIvk-M`, and let

$$H_D := \mathsf{GroupHash}(\texttt{z.cash:Orchard-CommitIvk-r}, \varepsilon)$$

be the blinding base; then $\mathsf{ivk} = \mathsf{Extract}_{\mathbb{P}}(\hat{M} + [\mathsf{r_{ivk}}]\, H_D)$ if $\hat{M} \neq \perp$, and $\mathsf{ivk} = \perp$ otherwise. Key generation discards sk when $\mathsf{ivk} \in \{0, \perp\}$, so an accepted key has $\mathsf{ivk} \in \{1, \dots, p_{\mathsf{Pallas}} - 1\}$. Because $p_{\mathsf{Pallas}} < p_{\mathsf{Vesta}}$, this integer is a non-zero element of $\mathbb{F}_{p_{\mathsf{Vesta}}}$, the scalar that multiplies the diversified base in §3.3.

Let $X_{\mathsf{Pallas}} := \{\mathsf{Extract}_{\mathbb{P}}(P) : P \in \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}}) \setminus \{O\}\} \subset \{1, \dots, p_{\mathsf{Pallas}} - 1\}$. The set X_{Pallas} has $(p_{\mathsf{Vesta}} - 1)/2$ elements (§2.4, Definition "The key commitment"), and an accepted ivk lies in it. The blinding base H_D is not O: the protocol specification, §"Sinsemilla commitments", states that the commitment is distributed as a uniform point for a uniform trapdoor, which holds only for a blinding base other than O.

Lemma 3.7 (Distribution of ivk)**.** *Let* $\mathsf{r_{ivk}}$ *be uniform on* $\mathbb{F}_{p_{\mathsf{Vesta}}}$ *and independent of* $(\mathsf{ak}, \mathsf{nk})$*, and let* $\hat{M} \neq \perp$*. Then* $\mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r_{ivk}}}(\mathsf{ak}, \mathsf{nk})$ *is* 0 *with probability* $1/p_{\mathsf{Vesta}}$ *and otherwise uniform on* X_{Pallas}*, independently of* $(\mathsf{ak}, \mathsf{nk})$*.*

Proof. By Lemma 2.26(i) the point $\hat{M} + [\mathsf{r_{ivk}}]\, H_D$ is uniform on $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$, whatever $\hat{M}$ is. It is O, of extracted coordinate 0, with probability $1/p_{\mathsf{Vesta}}$; each $x \in X_{\mathsf{Pallas}}$ is the extracted coordinate of exactly the two points $\pm P$ with $P \neq -P$ (Lemma 2.4), and so has probability $2/p_{\mathsf{Vesta}}$. □

In particular ivk is not uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$ (protocol specification, §"Orchard Key Components", note).

The further rejection condition of key generation concerns the internal key of ZIP 32, §"Orchard internal key derivation". That key has the same ak and nk and a trapdoor $t \in \mathbb{F}_{p_{\mathsf{Vesta}}}$ that is an efficiently computable function of $(\mathsf{ak}, \mathsf{nk}, \mathsf{r_{ivk}})$, and key generation discards sk when its incoming viewing key $\mathsf{Commit}^{\mathsf{ivk}}_{t}(\mathsf{ak}, \mathsf{nk})$ is 0 or $\perp$ (protocol specification, §"Orchard Key Components").

Lemma 3.8 (Rejection in key generation)**.** *Under Assumptions 2.22 and 2.8, a single draw of key generation from independent uniform* $(\mathsf{ask}, \mathsf{nk}, \mathsf{r_{ivk}})$ *is rejected with negligible probability. Under Assumptions 2.12, 2.22 and 2.8, the output of key generation is within negligible statistical distance of a single draw from a uniform* sk *without rejection.*

Proof. A single draw is rejected only if $\mathsf{ask} = 0$, of probability $1/p_{\mathsf{Vesta}}$; or $\hat{M} = \perp$; or $\mathsf{ivk} = 0$, of probability $1/p_{\mathsf{Vesta}}$ by Lemma "Distribution of ivk"; or $\mathsf{Commit}^{\mathsf{ivk}}_{t}(\mathsf{ak}, \mathsf{nk}) = 0$. The value $\mathsf{Commit}^{\mathsf{ivk}}_{t}(\mathsf{ak}, \mathsf{nk})$ commits to the message of ivk, so it is $\perp$ exactly when $\hat{M} = \perp$. The event $\hat{M} = \perp$ has negligible probability under Assumptions 2.22 and 2.8: an algorithm that samples $(\mathsf{ask}, \mathsf{nk})$ and outputs the message $\mathrm{LE}_{255}(\mathsf{ak}) \,\|\, \mathrm{LE}_{255}(\mathsf{nk})$ finds a $\perp$ input with that probability, which Proposition 2.24(iii) bounds. For $\hat{M} \neq \perp$, $\mathsf{Commit}^{\mathsf{ivk}}_{t}(\mathsf{ak}, \mathsf{nk}) = 0$ exactly when $\hat{M} + [t]\, H_D = O$ (Lemma 2.3). By (1), with $n = 51$ (Table 4), this is a relation among $Q(\texttt{z.cash:Orchard-CommitIvk-M})$, the $S(j)$ and H_D whose coefficient on the first is 2^{51}, non-zero modulo p_{Vesta}. An algorithm that samples $(\mathsf{ask}, \mathsf{nk}, \mathsf{r_{ivk}})$, computes t and outputs this

relation succeeds with the probability of the event, which the lemma on relations among fixed generators (§2.4) bounds under the same assumptions. The rejection event is efficiently decidable from the three outputs of $\mathsf{PRFexpand}_{\mathsf{sk}}$, so by Lemma 2.15 a first draw from a uniform sk is also rejected with negligible probability. Key generation outputs its first draw whenever that draw is accepted; its output is therefore within that negligible probability, in statistical distance, of a single draw without rejection. □

The proofs of this section analyse a single draw and add the statistical distance bounded by this lemma, under Assumptions 2.12, 2.22 and 2.8.

Construction 3.9 (Diversifier key and outgoing viewing key)**.** Let $K := \mathrm{LE}_{256}(\mathsf{r_{ivk}})$ and

$$R := \mathsf{PRFexpand}_K([\mathtt{0x82}] \,\|\, \mathsf{I2LEOSP}_{256}(\mathsf{ak}) \,\|\, \mathsf{I2LEOSP}_{256}(\mathsf{nk})),$$

a 64-byte string, where $\mathsf{I2LEOSP}_{256}$ encodes the field elements ak and nk as 32-byte little-endian integers. The *diversifier key* dk is the first 32 bytes of R and the *outgoing viewing key* ovk the last 32 bytes (protocol specification, §"Orchard Key Components"; row 0x82 of Table 3). One output of PRFexpand, keyed by the encoding of $\mathsf{r_{ivk}}$, yields both.

Definition 3.10 (Incoming and outgoing viewing keys)**.** The *incoming viewing key* is the pair (dk, ivk); the *outgoing viewing key* is ovk. The diversifier key dk derives diversifiers (§3.3). What the holders of ivk and of ovk can compute from note ciphertexts is stated in "Trial decryption and note acceptance" (§10.4) and "The outgoing ciphertext" (§10.3).

Proposition 3.11 (Binding of ivk to (ak, nk))**.** *Under Assumptions 2.22 and 2.8, no efficient algorithm outputs, except with negligible probability, triples* $(\mathsf{ak}, \mathsf{nk}, \mathsf{r_{ivk}}) \neq (\mathsf{ak}', \mathsf{nk}', \mathsf{r_{ivk}}')$ *with* $\mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r_{ivk}}}(\mathsf{ak}, \mathsf{nk}) = \mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r_{ivk}}'}(\mathsf{ak}', \mathsf{nk}') \neq \perp$.

Proof. Equal extracted values come from commitment points C, C' with $C = \pm C'$ (Lemma 2.4). If the messages differ, both cases are excluded by Proposition 2.24(ii), since LE_{255} is injective on integer representatives. If the messages agree, then $\mathsf{r_{ivk}} \neq \mathsf{r_{ivk}}'$, the case $C = C'$ gives $[\mathsf{r_{ivk}} - \mathsf{r_{ivk}}']\, H_D = \mathcal{O}$, impossible for $H_D \neq \mathcal{O}$ in a group of prime order, and the case $C = -C'$ is excluded by the same proposition. □

Hence ivk is bound to the pair (ak, nk), and so is every address of the key (§3.3): its transmission key $\mathsf{pk_d} = [\mathsf{ivk}]\,\mathsf{g_d}$ with $\mathsf{g_d} \neq \mathcal{O}$ determines ivk, an integer below p_{Vesta}. The binding is used in Lemma 12.5 ("Double-spend resistance", §12.2) and Theorem 12.9 ("Authorisation of spends", §12.3). Binding alone would follow from any collision-resistant hash of (ak, nk); the trapdoor $\mathsf{r_{ivk}}$ adds hiding.

Remark 3.12 (Hiding of ivk and the double use of $\mathsf{r_{ivk}}$). For $\mathsf{r_{ivk}}$ uniform and independent of every other quantity, ivk is independent of (ak, nk) by Lemma "Distribution of ivk". The derivation of (dk, ovk) is keyed by the same $\mathsf{r_{ivk}}$, so once dk or ovk is known that argument no longer applies, and modelling PRFexpand only as a PRF does not restore it: the key of row 0x82 is also the trapdoor of $\mathsf{Commit}^{\mathsf{ivk}}$, outside the hypothesis of Lemma 2.15 (§2.3; protocol specification, §"Orchard Key Components", note on the use of $\mathsf{r_{ivk}}$ as both trapdoor and key). The volume states the joint assumption on Pallas scalar multiplication and PRFexpand that the specification proposes as Assumption 3.13.

Assumption 3.13 (Joint hiding of the viewing keys). For keys with use_qsk = false: for every $(\mathsf{ak}, \mathsf{nk}) \in \{1, \dots, p_{\mathsf{Pallas}} - 1\} \times \mathbb{F}_{p_{\mathsf{Pallas}}}$, given to the distinguisher, and $\mathsf{r_{ivk}}$ uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$, the triple $(\mathsf{ivk}, \mathsf{dk}, \mathsf{ovk})$ computed from $(\mathsf{ak}, \mathsf{nk}, \mathsf{r_{ivk}})$ by the constructions of this subsection is computationally indistinguishable from

$$(\mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r_{ivk}}'}(\mathsf{ak}, \mathsf{nk}),\ u_1,\ u_2),$$

where $\mathsf{r_{ivk}}'$ is uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$ and u_1, u_2 are uniform 32-byte strings, all independent. It is a precise form of the joint assumption that the protocol specification proposes (§"Orchard Key Components", note); it is not perfect hiding. Results apply it after Lemma 2.15 (Assumption 2.12) has made $\mathsf{r_{ivk}}$ uniform and independent of $(\mathsf{ask}, \mathsf{nk})$, averaging over the key-generation distribution of $(\mathsf{ak}, \mathsf{nk})$. It is used by Propositions 3.18, 3.17, 6.10 and 10.6, by Proposition "Confidentiality of the outgoing ciphertext" (§10.3) and by Theorem 12.12.

3.3 Diversified addresses

Construction 3.14 (Diversified address)**.** For an index $j \in \{0, \dots, 2^{88} - 1\}$ chosen by the key holder:

1. the *diversifier* is $d := \mathsf{FF1\text{-}AES256}_{\mathsf{dk}}(\mathrm{LE}_{88}(j)) \in \{0,1\}^{88}$, the FF1 mode of NIST SP 800-38G over AES-256 with key dk, radix 2, length 88 and the empty tweak (*Crypto Guide*, §"Format-preserving encryption and FF1", Construction "The FF1 mode of NIST SP 800-38G"; protocol specification, §"Pseudo Random Permutations");

2. the *diversified base* is $\mathsf{g_d} := \mathsf{DiversifyHash}(d)$, where

$$T := \mathsf{GroupHash}(\texttt{z.cash:Orchard-gd},\ \mathsf{LEBS2OSP}_{88}(d))$$

   and

$$\mathsf{DiversifyHash}(d) := \begin{cases} T & \text{if } T \neq \mathcal{O}, \\ \mathsf{GroupHash}(\texttt{z.cash:Orchard-gd}, \varepsilon) & \text{otherwise} \end{cases}$$

   (protocol specification, §"$\mathsf{DiversifyHash}^{\mathsf{Sapling}}$ and $\mathsf{DiversifyHash}^{\mathsf{Orchard}}$ Hash Functions");

3. the *diversified transmission key* is $\mathsf{pk_d} := [\mathsf{ivk}]\,\mathsf{g_d}$, with ivk read as a scalar (§2.1).

The *address* is $(d, \mathsf{pk_d}) \in \{0,1\}^{88} \times (\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}}) \setminus \{\mathcal{O}\})$ (protocol specification, §"Orchard Key Components").

The map $\mathsf{LEBS2OSP}_{88}$ packs d into 11 bytes, eight bits at a time, least significant bit first (§2.1). Every index yields an address. The map GroupHash returns a point for every input, and DiversifyHash replaces the identity by the fixed fallback point, which is not the identity: the protocol specification types $\mathsf{DiversifyHash}^{\mathsf{Orchard}}$ into the non-identity points. The fallback case has negligible probability under Assumption 2.8. Hence $\mathsf{g_d} \neq \mathcal{O}$ for every d, and $\mathsf{pk_d} \neq \mathcal{O}$ because ivk is a non-zero scalar and the group has prime order. For each key dk the map $j \mapsto d$ is a permutation of the 88-bit strings (*Crypto Guide*, §"Format-preserving encryption and FF1", Proposition "FF1 is a keyed bijection on $\mathcal{X}$"), so distinct indices give distinct diversifiers: one incoming viewing key $(\mathsf{dk}, \mathsf{ivk})$ yields 2^{88} distinct addresses, all with transmission keys under the same ivk. No index is excluded.

Assumption 3.15 (FF1-AES256 as a pseudorandom permutation). For a uniform 256-bit key, FF1-AES256 with the empty tweak on $\{0,1\}^{88}$ is a pseudorandom permutation: no efficient ad-

versary with oracle access distinguishes it from a uniformly random permutation of $\{0,1\}^{88}$ (*Crypto Guide*, §"Pseudorandom functions and permutations", Definition "Pseudorandom permutation", and §"Format-preserving encryption and FF1", Definition "Format-preserving encryption"). It is the security requirement of the protocol specification, §"Pseudo Random Permutations", whose note records that the distinguishing attack of Dunkelman, Kumar, Lambooij and Sanadhya at this domain size and round count is no better than exhaustive search for the 256-bit key. It is used so that a diversifier reveals neither its index nor the key that produced it, in Proposition 3.17 ("Unlinkability of diversified addresses").

Assumption 3.16 (DDH on Pallas). Let P be a generator of $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$, let b and c be uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$ and a uniform on X_{Pallas} (§3.2), all independent. The triples $([a]P, [b]P, [ab]P)$ and $([a]P, [b]P, [c]P)$ are computationally indistinguishable. With a uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$ instead, this is the DDH assumption of the *Crypto Guide* (§"Diffie–Hellman: computational and decisional", Definition "Decisional Diffie–Hellman, DDH"). The volume assumes the form with a uniform on X_{Pallas} because ivk is so distributed, and claims no reduction between the two forms; the protocol specification expects the non-uniformity of ivk to cause no security issue (§"Orchard Key Components", note). It is used by Propositions 3.17 and 10.6 ("Confidentiality and key privacy of note encryption").

Proposition 3.17 (Unlinkability of diversified addresses)**.** *Let two keys be generated independently with* $\mathsf{use_qsk} = \mathsf{false}$. *An adversary holds* ak, nk *and* ovk *of both keys, and none of* $\mathsf{r_{ivk}}$, dk *and* ivk *of either. It chooses indices* j_0, j_1 *and* j' *with* $j' \notin \{j_0, j_1\}$, *and receives the address of key* 0 *at index* j_0, *the address of key* 1 *at index* j_1, *and the address of key* β *at index* j', *for a uniform hidden bit* β; *it outputs a bit* β'. *Conditioned on the event* E *that the three diversifiers are pairwise distinct, its advantage* $|\Pr[\beta' = 1 \mid \beta = 0] - \Pr[\beta' = 1 \mid \beta = 1]|$ *is negligible under Assumptions 2.12 (through Lemma 2.15), 3.13, 3.15, 2.8 and 3.16, together with Assumption 2.22 for Lemma "Rejection in key generation". This is the unlinkability requirement of the protocol specification, §"*$\mathsf{DiversifyHash}^{\mathsf{Sapling}}$ *and* $\mathsf{DiversifyHash}^{\mathsf{Orchard}}$ *Hash Functions", which stipulates distinct diversifiers.*

Proof. For each β the game is modified by a sequence of hops, each changing the adversary's output distribution by a negligible amount (*Crypto Guide*, §"The hybrid argument", Lemma "Hybrid lemma"). Each hop is justified by a reduction that runs the game; the event E is efficiently decidable from the diversifiers and has probability at least 1/2 in every game below, so conditioning on E at most doubles each cost.

1. Each key is replaced by a single draw without rejection (Lemma "Rejection in key generation", §3.2).

2. For each key in turn, $(\mathsf{ask}, \mathsf{nk}, \mathsf{r_{ivk}})$ is replaced by independent uniform elements of the three fields (Lemma 2.15); the reduction derives the remaining components and the addresses.

3. For each key in turn, $(\mathsf{ivk}, \mathsf{dk}, \mathsf{ovk})$ is replaced by $(\mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r_{ivk}}}(\mathsf{ak}, \mathsf{nk}), u_1, u_2)$ (Assumption 3.13); the reduction is given $(\mathsf{ak}, \mathsf{nk})$, generates the other key itself, and computes the game from the triple. Now dk and ovk are uniform and independent of all else, and, except on the negligible events $\mathsf{ivk} \in \{0, \bot\}$, the scalar ivk is uniform on X_{Pallas} and independent of $(\mathsf{ak}, \mathsf{nk})$ (Lemmas "Distribution of ivk" and "Rejection in key generation", §3.2).

4. For $b \in \{0,1\}$ in turn, $\mathsf{FF1\text{-}AES256}_{\mathsf{dk}_b}$ is replaced by a uniformly random permutation π_b of $\{0,1\}^{88}$ (Assumption 3.15); the reduction queries its oracle at the indices it needs. Then

$d' = \pi_\beta(\mathrm{LE}_{88}(j')) \neq d_\beta$ because $j' \neq j_\beta$, and on E the triple (d_0, d_1, d') is uniform on pairwise distinct triples, for either value of β. The event E fails only if $d_{1-\beta}$ equals d_β or d', of probability $2/2^{88}$.

5. On E the inputs $\mathsf{LEBS2OSP}_{88}(d)$ of GroupHash are distinct, since $\mathsf{LEBS2OSP}_{88}$ is injective, so under Assumption 2.8 the bases g_0, g_1, g' are independent uniform points, each O, which invokes the fallback, with probability $1/p_{\mathsf{Vesta}}$ only.

In the last game the adversary's view consists of ak, nk, ovk of both keys and (d_0, d_1, d'), independent of the rest and identically distributed for $\beta = 0, 1$, together with

$$(g_0, [\mathsf{ivk}_0]\, g_0), \qquad (g_1, [\mathsf{ivk}_1]\, g_1), \qquad (g', [\mathsf{ivk}_\beta]\, g'),$$

where ivk_0, ivk_1 are independent and uniform on X_{Pallas}. The first two pairs are public keys of the per-key-base ElGamal scheme with private keys ivk_0 and ivk_1. For $g_\beta \neq O$ the uniform point g' equals $[r]\, g_\beta$ for a uniform scalar r, so the third pair is $([r]\, g_\beta, [r]([\mathsf{ivk}_\beta]\, g_\beta))$, an encryption of O under key β (*Crypto Guide*, §"Key privacy", Construction "ElGamal encryption"). Distinguishing β is therefore the game of Definition "Key privacy under chosen-plaintext attack, IK-CPA" of the same subsection, for the message O. The proof of its Theorem "ElGamal is key-private under DDH" bounds the advantage by two DDH advantages. That proof plants the first exponent of the DDH triple as a private key; with private keys uniform on X_{Pallas}, the same proof, with the other key's private key drawn uniformly from X_{Pallas} (the x-coordinate of a uniform non-identity point), reduces to Assumption 3.16. Its reduction samples (d_0, d_1, d') itself and programs GroupHash at their three inputs with the planted bases before running the adversary. The *Crypto Guide*'s Remark "What key privacy buys upstairs" records this correspondence for diversified addresses. Summing the costs of the hops, the advantage is negligible. □

3.4 Key capabilities and scope

Figure 1 gives the derivation graph, each edge labelled by its operation, and each one-way edge also by the assumption that makes it one-way. Proposition 3.18 proves four separations: the full viewing key yields no ask, the incoming viewing key neither nk nor ask, the outgoing viewing key no ivk, and an address no ivk.

Proposition 3.18 (Capability separation). *Let a key be generated honestly with* use_qsk = false. *No efficient algorithm, given the public parameters and only the listed key material of the key, outputs the listed secret, except with negligible probability:*

(*a*) *given the full viewing key* $(\mathsf{ak}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$, *the scalar* ask*;*

(*b*) *given the incoming viewing key* $(\mathsf{dk}, \mathsf{ivk})$, *the key* nk *or the scalar* ask*;*

(*c*) *given* ovk, *the scalar* ivk*;*

(*d*) *given an address* $(d, \mathsf{pk}_\mathsf{d})$ *of the key, the scalar* ivk.

The hypotheses are: for (*a*)*, Assumptions 2.12* (*through Lemma 2.15*) *and 2.22; for* (*b*)*, those of* (*a*) *and, for* nk*, Assumption 3.13; for* (*c*)*, Assumptions 2.12 and 3.13; for* (*d*)*, Assumptions 2.12, 3.13, 2.8 and 2.22* (*"Discrete logarithms on Pallas"*)*. Every part also uses Lemma "Rejection in key generation"* (*§3.2*)*, under Assumptions 2.12, 2.22 and 2.8. These are the only separations claimed.*

Proof. By Lemma "Rejection in key generation" (§3.2), a single draw from a uniform sk replaces the key at a negligible cost. By Lemma 2.15, its triple $(\mathsf{ask}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$ is then replaced by independent uniform elements of $\mathbb{F}_{p_{\mathsf{Vesta}}}$, $\mathbb{F}_{p_{\mathsf{Pallas}}}$ and $\mathbb{F}_{p_{\mathsf{Vesta}}}$, at a further negligible cost: in each part the success of an algorithm is efficiently decidable from the three values, by deriving the remaining

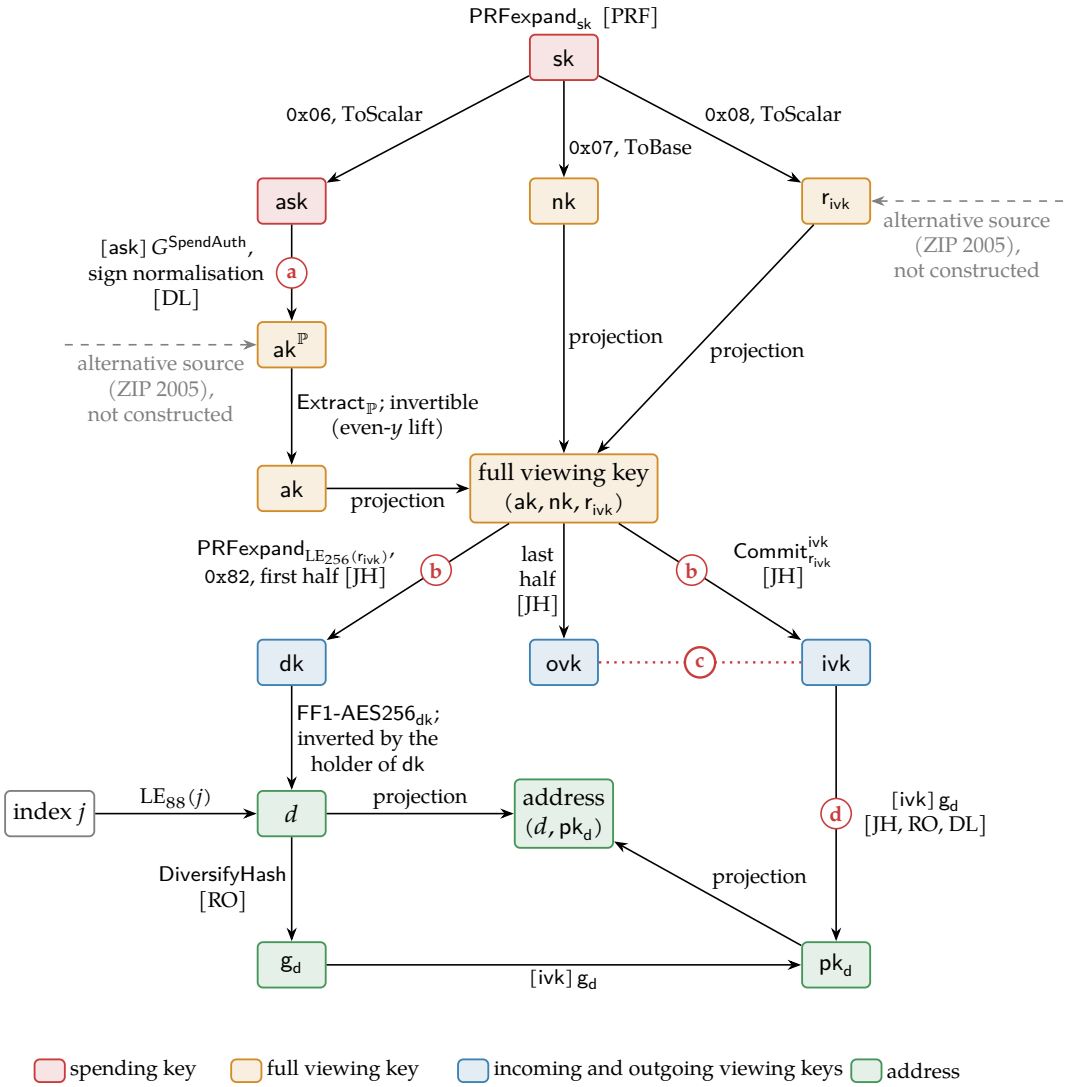


Figure 1: Derivation graph of a key generated with use_qsk = false, from the spending key sk to the address $(d, \mathsf{pk_d})$. Each edge carries its operation, and each one-way edge also carries, in brackets, the assumption under which it is one-way: PRF, Assumption 2.12; DL, Assumption 2.22; JH, Assumption 3.13; RO, Assumption 2.8. Each one-way edge is a PRF call, a scalar multiplication, the joint hiding assumption or GroupHash as a random oracle; bundling edges invert by projection, Extract$_{\mathbb{P}}$ by the even-y lift, and the FF1 edge is inverted by the holder of dk. The edges out of sk are one-way under Assumption 2.12, since an algorithm recovering sk from $(\mathsf{ask}, \mathsf{nk}, \mathsf{r_{ivk}})$ distinguishes PRFexpand$_{sk}$ from a random function. Colours give the lowest key tier that computes a node; the index j is uncoloured. The dashed edges into $\mathsf{ak}^{\mathbb{P}}$ and $\mathsf{r_{ivk}}$ mark the alternative source of ZIP 2005, not constructed. Markers (a) to (d) are the four separations that Proposition 3.18 proves; no other separation is claimed.

components, running the algorithm and comparing its output with the secret. The result is called the idealised key.

(a) Let an algorithm $\mathcal{A}$ output ask from $(\mathsf{ak}, \mathsf{nk}, \mathsf{r_{ivk}})$ with probability ϵ for the idealised key. An algorithm $\mathcal{B}$ receives $Y = [s]\, G^{\mathsf{SpendAuth}}$ for s uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$, draws nk and $\mathsf{r_{ivk}}$ uniformly, and runs $\mathcal{A}$ on $(\mathsf{Extract}_{\mathbb{P}}(Y), \mathsf{nk}, \mathsf{r_{ivk}})$. The input is distributed as for the idealised key, because the sign normalisation replaces ask by $\pm\mathsf{ask}$ and does not change ak (Lemma 2.4). A correct output a satisfies $[a]\, G^{\mathsf{SpendAuth}} = \mathsf{ak}^{\mathbb{P}}$, the even-$y$ lift of $\mathsf{Extract}_{\mathbb{P}}(Y)$, which is Y or $-Y$; $\mathcal{B}$ outputs whichever of a and $-a$ maps to Y. Hence $\mathcal{B}$ computes discrete logarithms to the generator $G^{\mathsf{SpendAuth}}$ with probability ϵ, and ϵ is negligible by Assumption 2.22.

(b) An algorithm that computes ask from $(\mathsf{dk}, \mathsf{ivk})$ computes it from $(\mathsf{ak}, \mathsf{nk}, \mathsf{r_{ivk}})$ by first deriving $(\mathsf{dk}, \mathsf{ivk})$, so (a) applies. Let $\mathcal{A}$ output nk from $(\mathsf{dk}, \mathsf{ivk})$ with probability ϵ. A distinguisher for Assumption 3.13, given $(\mathsf{ak}, \mathsf{nk})$ of the idealised key and a triple (x, y, z), runs $\mathcal{A}$ on (y, x) and outputs 1 exactly when $\mathcal{A}$ outputs nk. On the real triple it outputs 1 with probability ϵ. On the ideal triple $\mathcal{A}$ receives u_1 and $\mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r_{ivk}}'}(\mathsf{ak}, \mathsf{nk})$, which is independent of nk unless $\hat{M} = \bot$, a negligible event (Lemma "Rejection in key generation"); since nk is uniform on $\mathbb{F}_{p_{\mathsf{Pallas}}}$, $\mathcal{A}$ then succeeds with probability at most $1/p_{\mathsf{Pallas}}$ plus a negligible term. So ϵ exceeds the distinguisher's advantage by a negligible amount at most.

(c) Let $\mathcal{A}$ output ivk from ovk with probability ϵ. The distinguisher given $(\mathsf{ak}, \mathsf{nk})$ and (x, y, z) runs $\mathcal{A}$ on z and outputs 1 exactly when $\mathcal{A}$ outputs x. On the ideal triple, $z = u_2$ is independent of $x = \mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r_{ivk}}'}(\mathsf{ak}, \mathsf{nk})$, which, for $\hat{M} \neq \bot$, takes each value with probability at most $2/p_{\mathsf{Vesta}}$ by Lemma "Distribution of ivk". Hence ϵ is at most the advantage under Assumption 3.13 plus $2/p_{\mathsf{Vesta}}$ and a negligible term.

(d) Let $\mathcal{A}$ output ivk from the address at an index j, possibly of its choice, with probability ϵ. The distinguisher given $(\mathsf{ak}, \mathsf{nk})$ and (x, y, z) computes $d := \mathsf{FF1\text{-}AES256}_y(\mathrm{LE}_{88}(j))$, $\mathsf{g_d} := \mathsf{DiversifyHash}(d)$ and $\mathsf{pk_d} := [x]\, \mathsf{g_d}$, runs $\mathcal{A}$ on $(d, \mathsf{pk_d})$, and outputs 1 exactly when $\mathcal{A}$ outputs x. On the ideal triple, $\mathcal{A}$ succeeds with probability $\epsilon' \geq \epsilon - \delta$, for δ the advantage under Assumption 3.13; there, outside the negligible event $\hat{M} = \bot$, the value $\mathsf{ivk} := x$ is 0 with probability $1/p_{\mathsf{Vesta}}$ and otherwise uniform on X_{Pallas}, independently of $(\mathsf{ak}, \mathsf{nk})$ and of u_1, hence of d. An algorithm $\mathcal{B}$ receives a generator G and $Y = [s]\, G$ for s uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$. It draws t uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}} \setminus \{0\}$ and u_1 uniform, computes d from u_1 and j, programs $\mathsf{GroupHash}$ at $(\texttt{z.cash:Orchard-gd}, \mathsf{LEBS2OSP}_{88}(d))$ with the uniform non-identity point $[t]\, G$ (Assumption 2.8), answers other queries with fresh uniform points, and runs $\mathcal{A}$ on $(d, [t]\, Y) = (d, [s]([t]\, G))$. With probability $(p_{\mathsf{Vesta}} - 1)/(2p_{\mathsf{Vesta}})$ the exponent s lies in X_{Pallas}; it is then uniform on X_{Pallas}, and the view of $\mathcal{A}$ is that of the ideal triple conditioned on $\mathsf{ivk} \neq 0$, up to statistical distance $1/p_{\mathsf{Vesta}}$ for the programmed point. Algorithm $\mathcal{B}$ outputs the output of $\mathcal{A}$ and so computes s with probability at least

$$\frac{p_{\mathsf{Vesta}} - 1}{2p_{\mathsf{Vesta}}}\Big(\epsilon' - \frac{2}{p_{\mathsf{Vesta}}}\Big),$$

a loss factor of $2p_{\mathsf{Vesta}}/(p_{\mathsf{Vesta}} - 1)$, about 2, against Assumption 2.22. Hence ϵ', and with it ϵ, is negligible. □

Of the objects of this section only an address $(d, \mathsf{pk_d})$, with the base $\mathsf{g_d}$ computed from d, is public; sk, ask, $\mathsf{ak}^{\mathbb{P}}$, ak, nk, $\mathsf{r_{ivk}}$, the full viewing key, dk, ivk, ovk and the index j are secret key material. One incoming viewing key yields 2^{88} addresses (§3.3), unlinkable in the sense of Proposition 3.17 ("Unlinkability of diversified addresses").

What a holder of each key can compute from the objects of later sections is stated where those objects are constructed, each with a back-reference to Proposition 3.18: from nullifiers in "Nullifier uniqueness and unlinkability" (§6.4), and from note ciphertexts in "The outgoing ciphertext" (§10.3) and "Trial decryption and note acceptance" (§10.4). This section states no such power.

Keys and addresses are objects of the Orchard protocol, not of a pool: one key hierarchy and one address format serve both the Orchard pool and the Ironwood pool (ZIP 229, Abstract).

4 Notes and note commitments

This section defines the note, its commitment and its extracted commitment, and proves their hiding and binding; it constructs the derivation of all randomness of a created note from one seed under the lead byte `0x03`, and defines the Action and the bundle as units of a transaction.

4.1 Notes

Definition 4.1 (Unit of value)**.** The *zatoshi* is the smallest unit of value; one ZEC, the unit of the native currency of Zcash, is 10^8 zatoshi (protocol specification, §"Shielded Pools and Notes" and §"Mainnet and Testnet"). Every value in this volume, note values included, is an integer number of zatoshi.

Definition 4.2 (Note)**.** A *note* is a tuple $n = (\mathsf{addr}, v, \rho, \psi, \mathsf{rcm})$, where

- the address $\mathsf{addr} = (d, \mathsf{pk_d})$ of the recipient, of the type constructed in §3.3: a diversifier $d \in \{0,1\}^{88}$ and a transmission key $\mathsf{pk_d} \in \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}}) \setminus \{\mathcal{O}\}$;
- the value $v \in \{0, \dots, 2^{64} - 1\}$, in zatoshi;
- the elements $\rho \in \mathbb{F}_{p_{\mathsf{Pallas}}}$ and $\psi \in \mathbb{F}_{p_{\mathsf{Pallas}}}$;
- the commitment trapdoor $\mathsf{rcm} \in \mathbb{F}_{p_{\mathsf{Vesta}}}$

(protocol specification, §"Shielded Pools and Notes"; the lengths 88 and 64 are those of §"Constants"). The specification types rcm as an integer in $\{0, \dots, 2^{255} - 1\}$ (§"Commitment"). This volume types it in $\mathbb{F}_{p_{\mathsf{Vesta}}}$, because every derivation of rcm yields a reduced value and rcm acts only as the scalar of $[\mathsf{rcm}]$, which depends on rcm modulo p_{Vesta} alone; this typing makes the binding of Proposition 4.5 a statement about notes. For a created note, ψ and rcm are derived from its note seed, for an Ironwood-pool note in §4.3, and ρ is the nullifier of the note consumed in the same Action, a rule constructed in "Nullifier chaining" (§6.3).

A note is never published. A transaction that creates a note publishes its extracted note commitment cmx (Definition 4.4) and a ciphertext of its note plaintext for the recipient, constructed in "Encryption to the recipient" (§10.2) (protocol specification, §"Note Commitments" and §"Sending Notes (Orchard)"). Of the fields of a created note, only ρ appears on chain in the clear: it equals the nullifier that the same Action publishes for its consumed note ("Nullifier chaining", §6.3). This representation of value serves requirement R1 of §1.4; Propositions 4.5 and 4.8 state what the commitment hides.

Remark 4.3 (Roles of the fields)**.** The address and the value are the content of the note, its recipient and its amount; the trapdoor rcm randomises its commitment; the elements ρ and ψ, which also enter the commitment, are the note-specific inputs of its nullifier, constructed in "The nullifier" (§6.1).

4.2 Note commitments

Definition 4.4 (Note commitment)**.** For a note $n = ((d, \mathsf{pk_d}), v, \rho, \psi, \mathsf{rcm})$ let $\mathsf{g_d} := \mathsf{DiversifyHash}(d)$ (§3.3) and

$$M(n) := \mathsf{g_d}^{\star} \parallel \mathsf{pk_d}^{\star} \parallel \mathrm{LE}_{64}(v) \parallel \mathrm{LE}_{255}(\rho) \parallel \mathrm{LE}_{255}(\psi),$$

with ρ and ψ read as integers in $\{0, \dots, p_{\mathsf{Pallas}} - 1\}$; the string $M(n)$ has the fixed length 1086 of Table 4. With $D := \texttt{z.cash:Orchard-NoteCommit}$, the *note commitment* of n is

$$\mathsf{cm} := \mathsf{NoteCommit}_{\mathsf{rcm}}(M(n)) := \mathsf{SinsemillaCommit}_{\mathsf{rcm}}(D, M(n)) \in \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}}) \cup \{\perp\}.$$

By the routing of §2.4, with the hash point and the blinding base

$$\hat{M} := \mathsf{SinsemillaHashToPoint}(\texttt{z.cash:Orchard-NoteCommit-M}, M(n)),$$
$$H_D := \mathsf{GroupHash}(\texttt{z.cash:Orchard-NoteCommit-r}, \varepsilon),$$

the commitment is $\mathsf{cm} = \hat{M} + [\mathsf{rcm}]\, H_D$ if $\hat{M} \neq \perp$, and $\mathsf{cm} = \perp$ otherwise. The *extracted note commitment* is $\mathsf{cmx} := \mathsf{Extract}^{\perp}_{\mathbb{P}}(\mathsf{cm}) \in \mathbb{F}_{p_{\mathsf{Pallas}}} \cup \{\perp\}$ (§2.1). The specification's NoteCommit takes the five encoded fields as separate arguments, in the order of their encodings in $M(n)$ (protocol specification, §"Sinsemilla commitments", §"Sending Notes (Orchard)" and §"Coordinate Extractor for Pallas").

A sender never publishes $\perp$: it redraws the note seed (step (vi) of the derivation of §4.3). The extracted commitment cmx is the value an Action publishes for its created note, and the leaf of that note in the pool's note commitment tree of "The Merkle hash and the tree" (§5.1), whose nodes are elements of $\mathbb{F}_{p_{\mathsf{Pallas}}}$.

The blinding base H_D is not O: the protocol specification, §"Sinsemilla commitments", states that SinsemillaCommit is perfectly hiding for a uniform trapdoor, for every domain, which holds only for a blinding base other than O (as for $\mathsf{Commit}^{\mathsf{ivk}}$, §3.2).

Proposition 4.5 (Hiding and binding of note commitments)**.**

(*a*) *Perfect hiding. Let* rcm *be uniform on* $\mathbb{F}_{p_{\mathsf{Vesta}}}$. *For every note content* $(d, \mathsf{pk_d}, v, \rho, \psi)$ *whose message* $M(n)$ *has hash point* $\hat{M} \neq \perp$, *the commitment* cm *is uniform on* $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$. *Hence the distributions of* cm *and of* cmx *do not depend on* $(\mathsf{g_d}, \mathsf{pk_d}, v, \rho, \psi)$, *and* NoteCommit *is perfectly hiding on messages without a* $\perp$ *output, in the sense of the* Crypto Guide*'s Definition "Hiding game and hiding" (§"Hiding and binding").*

(*b*) *Binding of the extracted commitment. Under Assumptions 2.22 and 2.8, no efficient algorithm outputs, except with negligible probability, two notes* $n \neq n'$ *with* $\mathsf{cmx}(n) = \mathsf{cmx}(n') \neq \perp$. *In particular two distinct notes share no note commitment other than* $\perp$. *With Proposition 2.24(iii), which excludes messages with output* $\perp$, *part (b) gives computational binding in the sense of the* Crypto Guide*'s Definition "Binding game and binding" (same section), with the note content as the message and* rcm *as the randomness, stated for notes and for the extracted value.*

Proposition 4.8 states hiding for the trapdoor that a sender derives.

Proof. (a) This is Lemma 2.26(i) for the domain D and the message length 1086, whose hypothesis $H_D \neq O$ holds by the paragraph above; the value cmx is a fixed function of cm. Equal distributions for every pair of contents without a $\perp$ output are perfect hiding by the *Crypto Guide*'s Proposition "Distributional characterisation of hiding" (§"Hiding and binding").

(b) Step 1 reduces distinct notes to distinct openings. The map $(\mathsf{g_d}, \mathsf{pk_d}, v, \rho, \psi) \mapsto M(n)$ is injective: the string $M(n)$ concatenates fields of fixed lengths; the star encoding is injective (§2.1); the encoding LE_{64} is injective on $\{0, \dots, 2^{64} - 1\}$; and the encoding LE_{255} is injective on the representatives of $\mathbb{F}_{p_{\mathsf{Pallas}}}$, since $p_{\mathsf{Pallas}} < 2^{255}$ (§2.1). The map $n \mapsto (M(n), \mathsf{rcm})$ is therefore injective except on pairs of notes with $d \neq d'$ and $\mathsf{DiversifyHash}(d) = \mathsf{DiversifyHash}(d')$. Let T and T' be the values of GroupHash at $(\texttt{z.cash:Orchard-gd}, \mathsf{LEBS2OSP}_{88}(d))$ and at the corresponding input for d', distinct inputs because $\mathsf{LEBS2OSP}_{88}$ is injective. If $T, T' \neq O$, equal diversified bases give $T - T' = O$; otherwise $T = O$ or $T' = O$. Each is a non-trivial relation among values of GroupHash

on distinct inputs, which the lemma on relations among fixed generators (§2.4) excludes except with negligible probability.

Step 2 treats distinct openings. Let $\mathsf{cmx}(n) = \mathsf{cmx}(n') \neq \bot$. The lifting $\mathsf{Extract}^{\bot}_{\mathbb{P}}$ maps only $\bot$ to $\bot$, so cm and cm′ are points, and by Lemma 2.4 $\mathsf{cm}' = \mathsf{cm}$ or $\mathsf{cm}' = -\mathsf{cm}$. The pairs $(M(n), \mathsf{rcm})$ and $(M(n'), \mathsf{rcm}')$ are openings of the instance NoteCommit in $\{0,1\}^{1086} \times \mathbb{F}_{p_{\mathsf{Vesta}}}$. By Proposition 2.24(ii), outputs with $\mathsf{cm}' = -\mathsf{cm}$, or with $\mathsf{cm}' = \mathsf{cm}$ and $M(n) \neq M(n')$, have negligible probability (in the opposite case the relation in its proof has coefficient 2^{109+1} on the hash base, for a message of 109 chunks). In the remaining case $\mathsf{cm}' = \mathsf{cm}$ and $M(n) = M(n')$, and outside the event of Step 1 the openings differ, so $\mathsf{rcm} \neq \mathsf{rcm}'$ in $\mathbb{F}_{p_{\mathsf{Vesta}}}$; then $[\mathsf{rcm} - \mathsf{rcm}']\, H_D = \mathcal{O}$, impossible for $H_D \neq \mathcal{O}$ in a group of prime order p_{Vesta}. A note commitment other than $\bot$ shared by $n \neq n'$ gives equal extracted commitments other than $\bot$, which proves the particular case. □

4.3 The note seed: Ironwood derivations (lead byte 0x03)

Definition 4.6 (Note seed and lead byte)**.** A created note reaches its recipient as a *note plaintext*, the byte string constructed in "The note plaintext" (§10.1) that the sender encrypts. Its first byte, the *lead byte*, names the format of the plaintext and selects the derivation of the note's randomness. The sender draws one *note seed* rseed, a uniform 32-byte string, the key type of PRFexpand (§2.3), which the plaintext carries. The values ψ and rcm of the note and the ephemeral secret key $\mathsf{esk} \in \mathbb{F}_{p_{\mathsf{Vesta}}}$ of "Encryption to the recipient" (§10.2) are all derived from it (protocol specification, §"Note Plaintexts and Memo Fields" and §"Sending Notes (Orchard)").

Construction 4.7 (Derivations under lead byte 0x03)**.** The input is an address $(d, \mathsf{pk_d})$, a value $v \in \{0, \ldots, 2^{64} - 1\}$ and $\rho \in \mathbb{F}_{p_{\mathsf{Pallas}}}$. Let $\mathsf{g_d} := \mathsf{DiversifyHash}(d)$ and $\underline{\rho} := \mathsf{I2LEOSP}_{256}(\rho)$. For a point $P = (x, y)$ the 32-byte string $\mathsf{LEBS2OSP}_{256}(P^\star)$ equals $\mathsf{I2LEOSP}_{256}(x + 2^{255}(y \bmod 2))$ (§2.1). The sender

(i) checks that $\mathsf{pk_d} \neq \mathcal{O}$;

(ii) draws rseed uniformly from the 32-byte strings;

(iii) derives $\mathsf{esk} := \mathsf{ToScalar}(\mathsf{PRFexpand}_{\mathsf{rseed}}([\mathtt{0x04}] \,\|\, \underline{\rho})) \in \mathbb{F}_{p_{\mathsf{Vesta}}}$, and returns to (ii) if $\mathsf{esk} = 0$;

(iv) derives $\psi := \mathsf{ToBase}(\mathsf{PRFexpand}_{\mathsf{rseed}}([\mathtt{0x09}] \,\|\, \underline{\rho})) \in \mathbb{F}_{p_{\mathsf{Pallas}}}$;

(v) derives $\mathsf{rcm} := \mathsf{ToScalar}(\mathsf{PRFexpand}_{\mathsf{rseed}}(\mathsf{pre_rcm})) \in \mathbb{F}_{p_{\mathsf{Vesta}}}$, where

$$\begin{aligned}\mathsf{pre_rcm} := [\mathtt{0x0B}] \,\|\, \mathsf{LEBS2OSP}_{256}(\mathsf{g_d}^\star) \,\|\, \mathsf{LEBS2OSP}_{256}(\mathsf{pk_d}^\star) \\ \|\, \mathsf{I2LEOSP}_{64}(v) \,\|\, \underline{\rho} \,\|\, \mathsf{I2LEOSP}_{256}(\psi),\end{aligned}$$

a string of $1 + 32 + 32 + 8 + 32 + 32 = 137$ bytes;

(vi) computes cmx of the note $n := ((d, \mathsf{pk_d}), v, \rho, \psi, \mathsf{rcm})$ by Definition 4.4, and returns to (ii) if $\mathsf{cmx} = \bot$.

The output is the note n, its seed rseed and esk (protocol specification, §"Sending Notes (Orchard)"; ZIP 2005, "Changes to the Protocol Specification", §4.7.3).

The order is forced: the string pre_rcm contains ψ, so rcm is derived after ψ. The strings $\mathsf{g_d}^\star$ and $\mathsf{pk_d}^\star$ in pre_rcm are those of the message $M(n)$, packed into bytes, so rcm is a function of rseed

and of every committed field $(\mathsf{g_d}, \mathsf{pk_d}, v, \rho, \psi)$. The leading bytes `0x04`, `0x09` and `0x0B` are rows of Table 3.

An Ironwood-pool note has the type of Definition "Note" (§4.1) and is committed by the same NoteCommit (Definition 4.4). Every note created by an Ironwood-pool Action, a dummy created note included, has a plaintext with lead byte `0x03` and the derivations above; no note created by an Orchard-pool Action does. This is the only note-level distinction between the two pools (ZIP 229, "Transaction Format"; ZIP 258, "ZIP 2005 activation").

Proposition 4.8 (Hiding under the derived trapdoor)**.** *An adversary chooses two tuples* $(d^{(i)}, \mathsf{pk_d}^{(i)}, v^{(i)}, \rho^{(i)})$, $i \in \{0,1\}$, *of the input type of the construction above, with* $\mathsf{pk_d}^{(i)} \neq \mathcal{O}$. *The challenger draws a uniform bit b, runs steps (ii) to (vi) of the construction on tuple b, and returns* $(\mathsf{cm}_b, \mathsf{esk}_b)$, *the note commitment of the note it computes and its* esk*; the adversary receives no other function of* rseed *(the ciphertext that carries* rseed *is treated in Theorem 12.12). It outputs a bit* b'. *Let the number* ϵ_{PRF} *bound the advantage against Assumption 2.12 of each algorithm built from the adversary in the proof, each of which queries* $\mathsf{PRFexpand}_{\mathsf{rseed}}$ *on three inputs with pairwise distinct leading bytes; let* $\delta < 2^{-257}$ *bound the distance of* ToScalar *from uniform (§2.3); and let* $\epsilon_\perp$ *be the largest, over i, of the sum of two probabilities that one pass through steps (ii) to (vi) on tuple i returns to step (ii), one with* $\mathsf{PRFexpand}_{\mathsf{rseed}}$ *as specified and one with its three outputs replaced by independent uniform strings. Then*

$$|\Pr[b' = 1 \mid b = 0] - \Pr[b' = 1 \mid b = 1]| \leq 2\,(\epsilon_{\mathsf{PRF}} + \delta + \epsilon_\perp),$$

and ϵ_{PRF} *and* $\epsilon_\perp$ *are negligible under Assumptions 2.12, 2.22 and 2.8. The same bound holds with* cmx_b *in place of* cm_b.

Proof. Fix the adversary's tuples. One pass through steps (ii) to (vi) with a fresh rseed is a *trial*; its output is (cm, esk), and it is *accepted*, event A, when $\mathsf{esk} \neq 0$ and $\hat{M} \neq \perp$, since $\mathsf{cmx} = \perp$ exactly when $\hat{M} = \perp$. Trials are independent and identically distributed, and the challenger returns the output of the first accepted one; so its answer X_i on tuple i is distributed as the output Y_i of one trial conditioned on A. For a random variable Y and an event A with $\Pr[\neg A] = p < 1$, the statistical distance between Y and Y conditioned on A is at most p: for every set S,

$$\Pr[Y \in S] - \Pr[Y \in S \mid A] = \Pr[Y \in S,\, \neg A] - \Pr[Y \in S,\, A]\,\frac{p}{1-p},$$

and both terms lie in $[0, p]$.

Hybrid 1 replaces the outputs of $\mathsf{PRFexpand}_{\mathsf{rseed}}$ on $[\texttt{0x04}] \,\|\, \underline{\rho}$, $[\texttt{0x09}] \,\|\, \underline{\rho}$ and pre_rcm by independent uniform $u_1, u_2, u_3 \in \{0,1\}^{512}$, so that $\mathsf{esk} = \mathrm{ToScalar}(u_1)$, $\psi = \mathrm{ToBase}(u_2)$ and $\mathsf{rcm} = \mathrm{ToScalar}(u_3)$. The three inputs have distinct leading bytes, the input pre_rcm is chosen after the answer that determines ψ, and rseed is used only as a key of PRFexpand. By Lemma 2.15 (Assumption 2.12), the probability that any efficient test outputs 1 on the trial's output changes by at most ϵ_{PRF}. Hybrid 2 replaces rcm by a uniform element of $\mathbb{F}_{p_{\mathsf{Vesta}}}$, independent of all else, at statistical cost δ. The event A does not involve rcm, so its probability is the same in both hybrids.

In Hybrid 2 the values esk and ψ are independent, A is the intersection of an event on esk and an event on ψ, and rcm is uniform and independent of both. Conditioned on A, therefore, $\hat{M} \neq \perp$ and $\mathsf{cm} = \hat{M} + [\mathsf{rcm}]\,H_D$ is uniform on $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$ by Proposition 4.5(a), independently of esk, and esk is distributed as $\mathrm{ToScalar}(u_1)$ conditioned on being non-zero. This distribution W does not depend on i.

For each i, the adversary's output probability on X_i differs from that on W by at most the rejection probability of a trial as specified (X_i against Y_i), plus $\epsilon_{\mathsf{PRF}} + \delta$ (Y_i against its form in Hybrid 2), plus the rejection probability in Hybrid 2, which equals that with uniform strings (the unconditioned form in Hybrid 2 against W). The two rejection probabilities sum to at most $\epsilon_\perp$.

Adding the bounds for $i = 0$ and $i = 1$ gives the stated bound. For cmx_b, an adversary against cmx_b is one against cm_b that first applies the fixed map $\mathsf{Extract}^{\perp}_{\mathbb{P}}$.

Negligibility. With uniform strings, $\mathsf{esk} = 0$ has probability at most $1/p_{\mathsf{Vesta}} + \delta$. The event $\hat{M} = \perp$ has the probability with which an efficient algorithm finds a message of hash point $\perp$: the algorithm that runs the adversary's choice of tuples, draws u_2, and outputs $M(n)$ for tuple i with $\psi = \mathsf{ToBase}(u_2)$. That probability is negligible by Proposition 2.24(iii) under Assumptions 2.22 and 2.8. Rejection is efficiently decidable from the three outputs of $\mathsf{PRFexpand}_{\mathsf{rseed}}$, so the rejection probability as specified exceeds that with uniform strings by at most ϵ_{PRF}, negligible under Assumption 2.12. □

Remark 4.9 (Purpose of the hashed trapdoor). Because rcm is a hash of rseed and of every committed field, Ironwood-pool notes are recoverable in the sense of ZIP 2005. That ZIP proposes, with details subject to change, a Recovery Protocol for recovering Ironwood-pool funds if the Orchard protocol must be disabled because discrete logarithms become computable; a recovery protocol whose statement recomputes rcm by this derivation can bind a note even against an adversary able to compute discrete logarithms (ZIP 2005, Abstract; Rationale, "Proposed Recovery Protocol" and "Repairing note commitments"). Against such an adversary NoteCommit alone is not binding: the discrete-logarithm relations between its bases give a linear equation over $\mathbb{F}_{p_{\mathsf{Vesta}}}$ with several solutions, hence two distinct notes that open one commitment (ZIP 2005, Rationale, "Attacks against binding of note commitments"). What the deployed protocol checks of the derivation is stated with the Action statement, defined in "The Action statement" (§9.2).

The specification defines, for each pool, the set $\mathsf{allowedLeadBytes}^{pool}$ of lead bytes allowed in a note plaintext of that pool. For every transaction that contains Actions of the pool, the set is {0x03} for the Ironwood pool and {0x02} for the Orchard pool (protocol specification, §"Note Plaintexts and Memo Fields"; ZIP 2005, "Changes to the Protocol Specification", §3.2.1; ZIP 258, "ZIP 2005 activation"). The lead byte travels encrypted, and each decrypting party enforces the rule: decryption with an incoming viewing key, by the recipient, and decryption with an outgoing viewing key, by its holder, both return $\perp$ when the lead byte is not allowed (§"Decryption using an Incoming Viewing Key (Sapling and Orchard)" and §"Decryption using an Outgoing Viewing Key (Sapling and Orchard)"). For an Ironwood-pool output this is check (i) of the acceptance procedure of "Trial decryption and note acceptance" (§10.4). The one consensus check of the lead byte is stated in "Chain state and pool rules" (§11.4).

4.4 Actions, bundles, and dummy notes

Definition 4.10 (Action)**.** An *Action* of a pool consumes one note n^{old} and creates one note n^{new} of that pool; it is the specification's Action transfer, and every Action has both sides (protocol specification, §"Action Transfers and their Descriptions"). Its public data carry the extracted commitment cmx of the created note (Definition 4.4), the nullifier of the consumed note ("The nullifier", §6.1), a randomised validating key with a spend-authorisation signature ("Randomised validating keys", §7.2), a net value commitment ("Value commitments", §8.1), and the ciphertexts of "Encryption to the recipient" (§10.2). "Action descriptions and bundles" (§11.1) assembles them and redefines neither the Action nor the bundle. The relation that each Action must satisfy is the Action statement, defined in "The Action statement" (§9.2).

Definition 4.11 (Dummy notes)**.** Every Action has one consumed and one created note, so that its public fields are the same whether a side is real or not; a side that a payment does not use carries a *dummy note*, a note of value 0. A *dummy consumed note* is built as follows (protocol

specification, §"Dummy Notes (Orchard)"; ZIP 2005, "Changes to the Protocol Specification", §4.8.3):

(i) draw a fresh spending key as in §3.1, and derive its key components and an address $(d, \mathsf{pk_d})$ (§3.2, §3.3);

(ii) set $v := 0$, draw rseed uniformly, and set $\rho := \mathsf{Extract}_{\mathbb{P}}(P)$ for a uniform point P of $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$;

(iii) derive ψ and rcm under the lead byte chosen as for a created note of the pool (ZIP 2005, "Changes to the Protocol Specification", §4.8.3); in the Ironwood pool the lead byte is 0x03 and the derivations are steps (iv) and (v) of §4.3;

(iv) compute cm by Definition 4.4, and repeat from the draw of rseed with a fresh seed if $\mathsf{cm} = \perp$.

Its nullifier is computed as for any note ("The nullifier", §6.1). For step (iii) the two sources differ: the specification's section chooses the lead byte with the pool fixed to the Orchard pool, which gives 0x02 also in the Ironwood pool, whereas ZIP 2005, §4.8.3, makes the choice identical for dummy and non-dummy notes. The volume follows the ZIP. No condition of the Action statement checks the derivation of $\mathsf{rcm}^{\mathsf{old}}$ (Definition 9.2, Remark "What the statement does not check").

A *dummy created note* is a created note of value 0, built as any created note; in the Ironwood pool it therefore carries lead byte 0x03. Its address depends on the flag enableCrossAddress of the bundle ("Bundle flags", §9.1).

- When enableCrossAddress = 1, the dummy created note goes to a random address (protocol specification, §"Dummy Notes (Orchard)").
- When enableCrossAddress = 0, as in every Orchard-pool bundle, condition A10 of Definition 9.2 places the note that an Action creates at the expanded receiver, defined in "Bundle flags", of the note that the same Action consumes. A dummy created note then goes to the expanded receiver of that consumed note, real or dummy. In the Orchard pool, beside a real consumed note, the sender fills the component C_{enc} of the note ciphertext of the dummy created note ("Encryption to the recipient", §10.2) with random bytes, not with an encryption of its plaintext (ZIP 326, "Fabricated same-address outputs and randomized note ciphertexts"). A dummy consumed note beside a real created note lies at the expanded receiver of that created note: in place of step (i) it takes the created note's address and the key components of the key from which that address derives, and the Action's spend-authorisation signature requires that key's ask (Remark "Meaning of A10", §9.2). A fresh key as in step (i) remains admissible when both sides of the Action are dummies at its address.

That the public data do not reveal which sides of an Action are real is part of Theorem 12.12 (§12.4).

Definition 4.12 (Bundle)**.** The *bundle* of a pool in a transaction, which ZIP 229 calls the pool's component, is the sequence of that pool's Actions in the transaction together with the bundle-level data that later sections attach: one anchor ("Anchors", §5.2), one balancing value ("The balancing value", §8.2), one binding signature ("The binding signature", §8.3), the flags ("Bundle flags", §9.1) and one proof for all its Actions ("The Action circuit and the Halo 2 proof", §9.3). A transaction carries at most one bundle per pool (ZIP 229, "Transaction Format").

5 The note commitment tree

This section constructs the note commitment tree of a pool and its Merkle hash, defines the anchors, proves in Proposition 5.9 that a valid authentication path to an anchor exhibits a leaf of the anchored tree unless it yields a collision of the hash's Sinsemilla instance or an input on which that instance returns $\perp$, and proves in Lemma 5.15 that the authentication path of a leaf is a function of public data.

5.1 The Merkle hash and the tree

Remark 5.1 (Requirement R2). Requirement R2 of "Requirements on a shielded payment" (§1.4) asks the consumer of a note of non-zero value to show that the note was previously committed, without identifying it. In the Orchard protocol the claim to be shown is that the extracted commitment cmx of the note (Definition 4.4) is a leaf of the note commitment tree of the note's pool, and the leaf is not to be disclosed. Membership of a private leaf under a public root is provable in zero knowledge, with the root public and the leaf, its position and its authentication path private (*Crypto Guide*, §"Privacy-preserving membership via zero-knowledge paths", Theorem "Privacy-preserving membership"). We construct the tree, its roots, called anchors, and its authentication paths; the relation proved for each Action is the Action statement, defined in "The Action statement" (§9.2).

Definition 5.2 (Note commitment tree)**.** Let

$$\mathsf{MerkleDepth}^{\mathsf{Orchard}} := 32, \qquad \mathsf{Uncommitted}^{\mathsf{Orchard}} := 2 \in \mathbb{F}_{p_{\mathsf{Pallas}}}$$

(protocol specification, §"Constants"). Each pool of the Orchard protocol has its own *note commitment tree*, an append-only Merkle tree (*Crypto Guide*, §"Binary Merkle trees" and §"Append-only and incrementally updatable trees") of fixed depth $\mathsf{MerkleDepth}^{\mathsf{Orchard}} = 32$, hence with 2^{32} leaf positions. Its *layers* are numbered from layer 0, the root, to layer 32, the leaves; layer h has 2^h *nodes* $M_i^h \in \mathbb{F}_{p_{\mathsf{Pallas}}}$, $0 \le i < 2^h$.

1. *Appending.* Each Action of the pool in a transaction publishes the extracted commitment cmx of its created note (§4.4). When the transaction enters the chain, each such cmx is appended to the pool's tree: it occupies the leaf M_i^{32} for the least unused index i, and i is the *note position* pos of that commitment. Every pool's tree has no appended leaf before the first block of the chain.

2. *Leaves.* After n appends, the leaf M_i^{32} is the value appended at position i for $0 \le i < n$, and $M_i^{32} := \mathsf{Uncommitted}^{\mathsf{Orchard}} = 2$ for $n \le i < 2^{32}$.

3. *Internal nodes.* For $0 \le h < 32$ and $0 \le i < 2^h$,

$$M_i^h := \mathsf{MerkleCRH}(h, M_{2i}^{h+1}, M_{2i+1}^{h+1}),$$

   with $\mathsf{MerkleCRH}$ the Merkle hash constructed below; the *root* is M_0^0.

(Protocol specification, §"Note Commitment Trees", §"Merkle Path Validity", §"Note Commitments" and §"Transactions and Treestates".)

Construction 5.3 (Merkle hash). Let $D := \texttt{z.cash:Orchard-MerkleCRH}$. The Merkle hash

$$\mathsf{MerkleCRH}\colon \{0,\dots,31\} \times \mathbb{F}_{p_{\mathsf{Pallas}}} \times \mathbb{F}_{p_{\mathsf{Pallas}}} \to \mathbb{F}_{p_{\mathsf{Pallas}}}$$

is

$$\mathsf{MerkleCRH}(l, \mathit{left}, \mathit{right}) := \begin{cases} 0 & \text{if } \eta = \bot, \\ \eta & \text{otherwise,} \end{cases}$$

where

$$\eta := \mathsf{SinsemillaHash}(D,\ \mathrm{LE}_{10}(31-l) \,\|\, \mathrm{LE}_{255}(\mathit{left}) \,\|\, \mathrm{LE}_{255}(\mathit{right})) \in \mathbb{F}_{p_{\mathsf{Pallas}}} \cup \{\bot\},$$

with SinsemillaHash the extracted Sinsemilla hash of §2.4 and field elements encoded through their integer representatives. The protocol specification names the map $\mathsf{MerkleCRH}^{\mathsf{Orchard}}$ and the width 255 $\mathsf{MerkleHashLength}^{\mathsf{Orchard}}$ (§"$\mathsf{MerkleCRH}^{\mathsf{Orchard}}$ Hash Function" and §"Constants").

The prefix $\mathrm{LE}_{10}(31-l) = \mathrm{LE}_{10}(\mathsf{MerkleDepth}^{\mathsf{Orchard}} - 1 - l)$ is defined because $31 < 2^{10}$. It encodes 0 for the combination of two leaves into a node of layer $l = 31$, and 31 for the combination that yields the root, $l = 0$. Every message has the single length $10 + 255 + 255 = 520$ bits, 52 chunks of $k = 10$ bits, within the bound of $c = 253$ chunks of §2.4 (Table 4). The encoding LE_{255} is injective on $\mathbb{F}_{p_{\mathsf{Pallas}}}$ because $p_{\mathsf{Pallas}} < 2^{255}$ (§2.1), so for a fixed layer distinct argument pairs give distinct messages. The protocol specification requires that SinsemillaHash be collision resistant and that no input of 520 bits yielding $\bot$ can feasibly be found (§"$\mathsf{MerkleCRH}^{\mathsf{Orchard}}$ Hash Function", security requirements). For the domain D and 520-bit messages these are parts (i) and (iii) of Proposition 2.24, under Assumptions 2.22 and 2.8.

Remark 5.4 (Nodes as field elements). Because MerkleCRH ends in $\mathsf{Extract}_{\mathbb{P}}$, each layer's output has the type of its inputs, and one hash composes from the leaves to the root. Proofs of the Action statement, defined in "The Action statement" (§9.2), are Halo 2 proofs with the Vesta curve (protocol specification, §"Zero-Knowledge Proving System"), whose scalar field is $\mathbb{F}_{p_{\mathsf{Pallas}}}$ (*Math Guide*, §"Base fields, scalar fields, and the Pasta cycle"). A node that is one element of $\mathbb{F}_{p_{\mathsf{Pallas}}}$ is one value of that proof's arithmetic, whereas a point-valued node would carry two coordinates and a curve-equation condition at each of the 32 layers.

Remark 5.5 (Dropping the y-coordinate). A node serves only as a digest, never to recover a point, and $\mathsf{Extract}_{\mathbb{P}}(P) = \mathsf{Extract}_{\mathbb{P}}(P')$ only if $P' = P$ or $P' = -P$ (Lemma 2.4). A collision of $\mathsf{SinsemillaHash}(D, \cdot)$ on 520-bit messages is therefore either a collision of the underlying hash to a point or a pair of messages whose hash points are opposite. Proposition 2.24(i) excludes both, and part (iii) excludes a $\bot$ output, under Assumptions 2.22 and 2.8.

Remark 5.6 (Layer separation). The prefix $\mathrm{LE}_{10}(31-l)$ separates the inputs of different layers (protocol specification, §"$\mathsf{MerkleCRH}^{\mathsf{Orchard}}$ Hash Function", note). The 32 layer functions $\mathsf{MerkleCRH}(l, \cdot, \cdot)$ evaluate one instance $\mathsf{SinsemillaHash}(D, \cdot)$ on message sets with distinct prefixes. Every collision that the proof of Proposition 5.9 extracts therefore lies between two messages with the same prefix, and a node computed at one layer is accepted at another only through such a collision or a $\bot$ input (*Crypto Guide*, §"Layer and domain separation", Proposition "Separation forces within-domain collisions").

Lemma 5.7 (Uncommitted leaves). *No point P of the Pallas group satisfies* $\mathsf{Extract}_{\mathbb{P}}(P) = 2$. *Consequently* $\mathsf{Uncommitted}^{\mathsf{Orchard}} = 2$ *equals the extracted commitment* $\mathsf{cmx} = \mathsf{Extract}^{\bot}_{\mathbb{P}}(\mathsf{cm})$ *of no note, and a leaf position holding* 2 *holds the commitment of no note.*

Proof. By definition $\mathsf{Extract}_{\mathbb{P}}(\mathcal{O}) = 0 \neq 2$ in $\mathbb{F}_{p_{\mathsf{Pallas}}}$. A point $(x, y) \neq \mathcal{O}$ with $x = 2$ on the curve $y^2 = x^3 + 5$ over $\mathbb{F}_{p_{\mathsf{Pallas}}}$ would satisfy $y^2 = 2^3 + 5 = 13$. By Euler's criterion $13^{(p_{\mathsf{Pallas}}-1)/2} \equiv -1 \pmod{p_{\mathsf{Pallas}}}$ (*Math Guide*, §"Quadratic residues and the Euler criterion", Example "13 is a non-square in both Pasta fields"), so 13 is a non-square modulo p_{Pallas} and no such y exists. Only the base-field modulus p_{Pallas} enters. For the consequence, cmx is $\perp$ when $\mathsf{cm} = \perp$ and otherwise $\mathsf{Extract}_{\mathbb{P}}$ of a point, so it is not 2. The protocol specification states the consequence as the theorem "$\mathsf{Uncommitted}^{\mathsf{Orchard}}$ is not in the range of $\mathsf{NoteCommit}^{\mathsf{Orchard}}$" (§"Sinsemilla commitments"). □

The protocol specification, §"Note Commitment Trees", and ZIP 258, "Changes to the Protocol Specification", state the capacity rule: a block must not add Ironwood-pool note commitments that would make the Ironwood-pool note commitment tree exceed its capacity of $2^{\mathsf{MerkleDepth}^{\mathsf{Orchard}}} = 2^{32}$ leaves; the same rule binds the Orchard-pool tree. The Ironwood-pool tree is a tree of its own, built by the construction above with the same depth, the same value $\mathsf{Uncommitted}^{\mathsf{Orchard}}$ at unused positions and the same hash; only the values cmx published by Ironwood-pool Actions are appended to it.

5.2 Anchors

Definition 5.8 (Treestate and anchor)**.** A pool's *treestate* comprises its note commitment tree and its nullifier set, constructed in "Nullifier sets" (§6.2). Treestates are chained: the input treestate of the first block is empty; the input treestate of the first transaction of a block is the final treestate of the preceding block, and that of each later transaction the output treestate of the preceding transaction; the *final treestate* of a block is the output treestate of its last transaction (protocol specification, §"Transactions and Treestates"). The *anchor* of a block in a pool is the root M_0^0 of the pool's tree in the block's final treestate, an element of $\mathbb{F}_{p_{\mathsf{Pallas}}}$; every block has exactly one anchor per pool. A bundle carries one anchor field, $\mathsf{anchorIronwood}$ in the Ironwood pool and $\mathsf{anchorOrchard}$ in the Orchard pool, shared by all its Actions; an Action carries no anchor of its own ("Action descriptions and bundles", §11.1; protocol specification, §"Action Transfers and their Descriptions").

The protocol specification, §"Action Transfers and their Descriptions", and ZIP 258, "Changes to the Protocol Specification", state the anchor rule: whenever a transaction has Ironwood-pool Actions, its field $\mathsf{anchorIronwood}$ must refer to some earlier block's final Ironwood-pool treestate, that is, equal the Ironwood-pool anchor of a block that precedes the transaction's block in the chain. The field $\mathsf{anchorOrchard}$ is bound likewise to the Orchard pool. The rule sets no bound on how far back that block lies.

Proposition 5.9 (Membership soundness)**.** *Let rt be the anchor of a block in a pool, n the number of leaves appended to the pool's tree through that block, and $L_0, \dots, L_{2^{32}-1}$ the leaves of the tree that rt roots, so that $L_i = \mathsf{Uncommitted}^{\mathsf{Orchard}}$ for $i \geq n$. For $v, s_0, \dots, s_{31} \in \mathbb{F}_{p_{\mathsf{Pallas}}}$ and $\mathsf{pos} = \sum_{t=0}^{31} b_t 2^t$ with $b_t \in \{0, 1\}$, let*

$$a_0 := v, \qquad a_{t+1} := \begin{cases} \mathsf{MerkleCRH}(31-t,\, a_t,\, s_t) & \text{if } b_t = 0, \\ \mathsf{MerkleCRH}(31-t,\, s_t,\, a_t) & \text{if } b_t = 1 \end{cases} \qquad (0 \leq t < 32). \tag{2}$$

The triple $(v, \mathsf{pos}, (s_0, \dots, s_{31}))$ is a valid path to *rt if $a_{32} = rt$. This is the path verification of the* Crypto Guide *(§"Authentication paths and membership proofs", Definition "Path verification"), with level-t compression $\mathsf{MerkleCRH}(31-t, \cdot, \cdot)$ and the bits of pos as direction bits; "Authentication paths" (§5.3) constructs the path of an appended leaf.*

> (*i*) *From a valid path to rt with* $v \neq L_{\mathsf{pos}}$ *and the leaves* $L_0, \ldots, L_{n-1}$ *one computes efficiently either two distinct* 520*-bit messages with a common* 10*-bit prefix and equal values of* $\mathsf{SinsemillaHash}(D, \cdot)$*, for* $D =$ `z.cash:Orchard-MerkleCRH`*, or a* 520*-bit message on which* $\mathsf{SinsemillaHash}(D, \cdot)$ *returns* $\perp$*.*
>
> (*ii*) *Under Assumptions 2.22 and 2.8, an efficient adversary, even one that chooses the appended leaves, outputs a valid path to rt with* $v \neq L_{\mathsf{pos}}$ *with at most negligible probability.*
>
> (*iii*) *In particular, if* $v = \mathsf{Extract}_{\mathbb{P}}(P)$ *for a point P of the Pallas group, as the extracted commitment* $\mathsf{cmx} \neq \perp$ *of every note is, then, except with negligible probability,* $\mathsf{pos} < n$ *and v is the value appended at position* pos*, in the anchoring block or earlier.*

Proof. The argument is the *Crypto Guide*'s Theorem "Soundness of Merkle membership proofs" (§"Soundness: forging a path implies a collision"), with layer-dependent compressions.

(i) Let $j_t := \lfloor \mathsf{pos}/2^t \rfloor$, and write $j \oplus 1$ for the integer j with its lowest bit complemented. The genuine chain is $\bar{a}_t := M^{32-t}_{j_t}$ for $0 \leq t \leq 32$, with genuine siblings $\bar{s}_t := M^{32-t}_{j_t \oplus 1}$ for $0 \leq t < 32$. Since $j_t = 2j_{t+1} + b_t$, the node $\bar{a}_{t+1} = M^{31-t}_{j_{t+1}}$ is $\mathsf{MerkleCRH}(31 - t, \cdot, \cdot)$ of its two children, of which $\bar{a}_t$ is the left one if $b_t = 0$ and the right one if $b_t = 1$, the other being $\bar{s}_t$. The genuine chain therefore satisfies (2) with $\bar{s}_t$ in place of s_t, and $\bar{a}_0 = L_{\mathsf{pos}}$, $\bar{a}_{32} = M^0_0 = rt$ (*Crypto Guide*, §"Authentication paths and membership proofs", Proposition "Completeness", whose induction this is). The fold on the given path has $a_0 = v \neq \bar{a}_0$ and $a_{32} = rt = \bar{a}_{32}$. Let t^* be the least $t \geq 1$ with $a_t = \bar{a}_t$; it exists, and $a_{t^*-1} \neq \bar{a}_{t^*-1}$ by minimality. At level $t^* - 1$ both chains evaluate $\mathsf{MerkleCRH}(32 - t^*, \cdot, \cdot)$ with the running value on the side fixed by b_{t^*-1}: the first on a_{t^*-1} and s_{t^*-1}, the second on $\bar{a}_{t^*-1}$ and $\bar{s}_{t^*-1}$. The two 520-bit messages hashed there share the prefix $\mathrm{LE}_{10}(31 - (32 - t^*)) = \mathrm{LE}_{10}(t^* - 1)$, the layer tag of the *Crypto Guide*'s Proposition "Separation forces within-domain collisions" (§"Layer and domain separation"). They differ in the half that encodes the running value, LE_{255} being injective on $\mathbb{F}_{p_{\mathsf{Pallas}}}$, and their values under $\mathsf{MerkleCRH}$, $a_{t^*} = \bar{a}_{t^*}$, agree. If $\mathsf{SinsemillaHash}(D, \cdot)$ returns $\perp$ on either message, that message is the second output. Otherwise both values of $\mathsf{MerkleCRH}$ are values of $\mathsf{SinsemillaHash}(D, \cdot)$, and the pair is the first output. The genuine chain is computed from $L_0, \ldots, L_{n-1}$ by the recurrence of the tree. A node whose leaves all lie at positions at least n depends only on its layer, so the computation takes a number of evaluations of $\mathsf{MerkleCRH}$ linear in $n + 32$; the fold takes 32 more.

(ii) The extraction of (i) is deterministic and defined for every list of leaves, so it applies to leaves that an adversary chose. An adversary that outputs a valid path with $v \neq L_{\mathsf{pos}}$ with probability ϵ therefore yields an algorithm that outputs, with probability ϵ, a collision of $\mathsf{SinsemillaHash}(D, \cdot)$ on 520-bit messages or a 520-bit message with output $\perp$. Equal values of $\mathsf{SinsemillaHash}$ other than $\perp$ come from hash points that are equal on distinct messages or opposite (Lemma 2.4). Proposition 2.24, parts (i) and (iii), excludes both outputs except with negligible probability, under Assumptions 2.22 and 2.8; hence ϵ is negligible.

(iii) If $\mathsf{pos} \geq n$, then $L_{\mathsf{pos}} = 2$, and $v \neq 2$ by Lemma 5.7, so the path is one with $v \neq L_{\mathsf{pos}}$, which (ii) excludes except with negligible probability. Otherwise $\mathsf{pos} < n$, and outside the same event $v = L_{\mathsf{pos}}$, the value appended at position pos; the n leaves of the tree that rt roots are those appended through the anchoring block. □

Remark 5.10 (Old anchors). Appending leaves changes the final treestates of later blocks and of no earlier one. Once a block B is in the chain, its anchor therefore roots one fixed tree: a valid path to it stays valid after later appends, the anchor stays admissible under the anchor rule while B remains in the chain, and Proposition 5.9 applies to it at any later time. A valid path to the anchor of B for an extracted note commitment shows that the commitment was appended in B or earlier, which is what R2 requires. Excluding a second consumption of a note is not the task of the tree, which is append-only (protocol specification, §"Note Commitment Trees"), but that of the nullifier, constructed in "Nullifiers" (§6).

Definition 5.11 (Reorganisation)**.** A *reorganisation* is the replacement of the chain's most recent blocks by a competing branch.

With the height of a block as in §1.1, the *tip* is the last block of the chain. A reorganisation that removes the block whose final treestate a bundle's anchor refers to leaves the transaction in violation of the anchor rule on the new chain, unless a block of the new chain that precedes the transaction's block has the same anchor; to be included in the new chain, the transaction must then be rebuilt against an anchor of that chain. The block d blocks below a tip of height H, the block at height $H-d$, is removed only by a reorganisation that replaces at least $d+1$ blocks, since a reorganisation that replaces k blocks removes the blocks at heights $H-k+1$ to H (ZIP 315, "Rationale for anchor selection"; ZIP 218, "Anchor selection depth").

The *block target spacing*, the intended mean interval between consecutive blocks, is 25 seconds (ZIP 218, "Block target spacing").

Remark 5.12 (Recommended anchor depth). ZIP 315 recommends that a transaction built when the tip is at height H use as its anchor the final treestate of the block at height $H-3$, three blocks below the tip (ZIP 315, "Anchor selection"); ZIP 218 recommends the same depth of 3 blocks ("Anchor selection depth"). At the 25-second target spacing this depth is $3 \times 25 = 75$ seconds. The depth is a recommendation to the builder of a transaction, not a consensus rule: the anchor rule admits the final treestate of any earlier block.

Remark 5.13 (Anchor choice as public data). The anchor is published with the bundle. Call the set $\{0, \dots, n-1\}$ of positions of the leaves appended to the tree that an anchor roots the *candidate set* of the anchor; by Proposition 5.9(iii), a valid path to the anchor for an extracted note commitment has its position in this set, except with negligible probability. A later anchor roots a tree that contains every appended leaf of an earlier one, so its candidate set is at least as large. An anchor at a depth other than the one that other builders of transactions use distinguishes its builder. These two observations are not ZIP 315's stated rationale for a fixed depth, which is that too small a depth risks invalidation by a reorganisation, that too large a depth prevents recently received notes from being spent, and that a depth varying with the notes spent leaks information about them (ZIP 315, "Rationale for anchor selection").

5.3 Authentication paths

Construction 5.14 (Authentication path)**.** Let the pool's tree hold n leaves, namely the tree that the anchor rt of a block roots, for a block at or after the one that appended the note's commitment, so that the note position satisfies $\mathsf{pos} < n$. Write $\mathsf{pos} = \sum_{t=0}^{31} b_t 2^t$ with $b_t \in \{0,1\}$, and $j_t := \lfloor \mathsf{pos}/2^t \rfloor$. Levels $t = 0, \dots, 31$ count from the leaves: the *level-t route node* is $M_{j_t}^{32-t}$, a left child if $b_t = 0$ and a right child if $b_t = 1$. The *authentication path* of pos is

$$(s_0, \dots, s_{31}) \in \mathbb{F}_{p_{\mathsf{Pallas}}}^{32}, \qquad s_t := M_{j_t \oplus 1}^{32-t},$$

the sibling of the level-t route node in the tree of size n, with $j \oplus 1$ the integer j with its lowest bit complemented. Verification of $(v, \mathsf{pos}, (s_0, \dots, s_{31}))$ against rt is the fold (2): the level-t step evaluates $\mathsf{MerkleCRH}(31-t, \cdot, \cdot)$, whose message carries the prefix $\mathrm{LE}_{10}(t)$, with the running value on the left if $b_t = 0$ and on the right if $b_t = 1$; verification accepts if and only if $a_{32} = rt$, that is, if the triple is a valid path to rt in the sense of Proposition 5.9. In the protocol specification, §"Merkle Path Validity", the entry s_t is the node $M^h_{\mathsf{MerkleSibling}(h,\mathsf{pos})}$ with $h = 32 - t$.

The genuine path of an appended leaf is accepted: the fold on $(L_{\mathsf{pos}}, \mathsf{pos}, (s_0, \dots, s_{31}))$ reproduces the route nodes, $a_t = M_{j_t}^{32-t}$ for every t, and ends at $M_0^0 = rt$ (*Crypto Guide*, §"Authentication paths and membership proofs", Proposition "Completeness", whose induction applies

with the level-t compression $\mathsf{MerkleCRH}(31-t,\cdot,\cdot)$, as in the proof of Proposition 5.9). A note's commitment therefore has a valid path to the anchor of every block from the one that appended it onward.

Lemma 5.15 (Authentication paths from public data). *Let the tree hold n leaves, let* pos $< n$, *and let*

$$\mathsf{Empty}_0 := \mathsf{Uncommitted}^{\mathsf{Orchard}} = 2,$$
$$\mathsf{Empty}_{t+1} := \mathsf{MerkleCRH}(31-t,\ \mathsf{Empty}_t,\ \mathsf{Empty}_t) \qquad (0 \le t < 32).$$

For each level t the entry s_t of the authentication path of pos *is as follows.*

(*i*) *If $b_t = 1$, then s_t is the root of the complete subtree over the positions $[(j_t-1)\,2^t,\ j_t\,2^t)$, all below* pos*; it is a function of the leaves at those positions alone, and the same in every tree of size greater than* pos.

(*ii*) *If $b_t = 0$, then s_t is the root of the subtree over the positions $[(j_t+1)\,2^t,\ (j_t+2)\,2^t)$, a function of the leaves appended in that range, with* $\mathsf{Uncommitted}^{\mathsf{Orchard}}$ *at its unused positions; it equals* Empty_t *when $(j_t+1)\,2^t \ge n$.*

Hence the authentication path is a deterministic function of pos *and of the first n leaves, which are the values* cmx *published by the pool's Actions in the chain through the anchoring block. A holder of the note who has these leaves computes the path without communicating* pos*, and verification needs, besides the path, only the public anchor.*

Proof. By the recurrence of the tree, the node M_i^h is a function of the leaves at positions $[i\,2^{32-h},\ (i+1)\,2^{32-h})$, by induction on $32-h$. Apply this with $h = 32-t$ and $i = j_t \oplus 1$, which equals $j_t - 1$ when $b_t = 1$ and $j_t + 1$ when $b_t = 0$, since b_t is the parity of j_t. When $b_t = 1$ the range ends at $j_t\,2^t \le \mathsf{pos} < n$, so every position in it is appended, and its leaf is fixed once appended because the tree is append-only. When the range starts at or beyond n, every leaf in it is 2, and its root is Empty_t by induction on t: a level-$(t+1)$ node over an unused range is $\mathsf{MerkleCRH}(31-t,\cdot,\cdot)$ of two level-t nodes over unused ranges. Positions below n hold the appended values cmx, published in the chain. □

Figure 2 shows the construction and Lemma 5.15 on a tree of depth 3.

Remark 5.16 (Scope of hiding). The anchor is public. The position, the consumed note's cmx and its authentication path are not fields of the bundle, and the holder of the note computes the path locally (Lemma 5.15): obtaining the path of a particular position from another party would disclose pos to that party. The candidate set for the consumed note is the candidate set of the anchor (§5.2), the positions of the n appended leaves of the anchored tree. Leaves that an observer knows by other means to be commitments of dummy created notes (§4.4) or of notes already consumed, and other data of the transaction, reduce the effective set, so n alone does not measure it. What an Action reveals is stated in Theorem 12.12 ("Privacy", §12.4).

6 Nullifiers

This section constructs the nullifier of a note, the nullifier sets that consensus maintains per pool, and the rule that makes the nullifier of each consumed note the element ρ of the note created in the same Action. It proves two properties of the construction: uniqueness, from discrete logarithms on Pallas with GroupHash modelled as a random oracle; and, for keys generated with use_qsk = false, unlinkability, from these two assumptions, the pseudorandomness of Poseidon and of PRF expansion, and the joint hiding of the viewing keys. It also states what each key tier can compute from nullifiers.

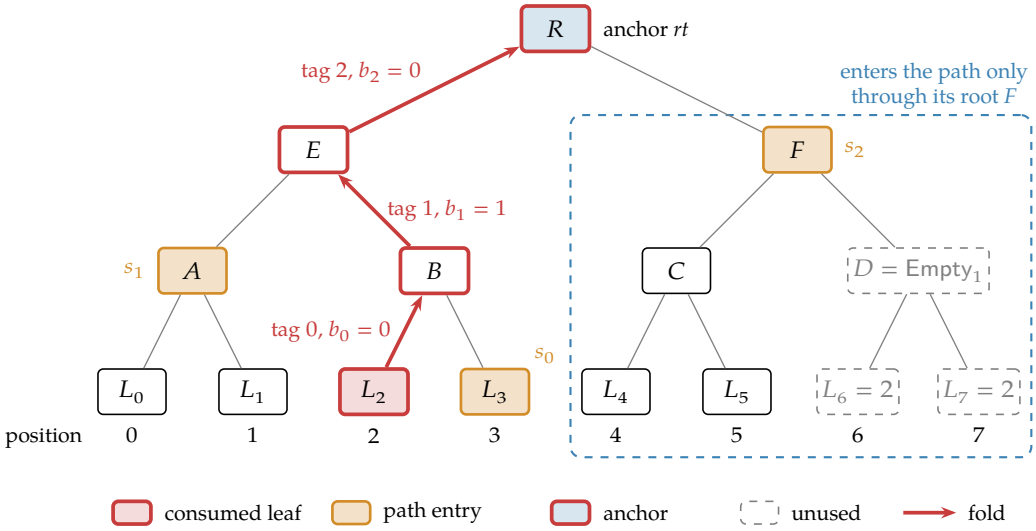


Figure 2: A depth-3 instance of the note commitment tree of §5.1; the protocol's tree has depth 32. Leaves L_0 to L_5 are appended ($n = 6$); positions 6 and 7 hold $\mathsf{Uncommitted}^{\mathsf{Orchard}} = 2$, so $D = \mathsf{Empty}_1$. The consumed note's leaf is L_2, at position 2 with bits $b_0 = 0$, $b_1 = 1$, $b_2 = 0$. Its authentication path is $(s_0, s_1, s_2) = (L_3, A, F)$, each entry the sibling of the route node at its level. The bold fold $L_2 \to B \to E \to R$ evaluates MerkleCRH at each level with the prefix $\mathrm{LE}_{10}(t)$, shown as tag t, and the running value on the side given by b_t; it ends at the root R, the anchor. By Lemma 5.15, the entry A, left of the route, is the root of a complete subtree and the same at every later anchor; the entry F, right of the route and the root over positions 4 to 7, is a function of the leaves appended at the anchor, with 2 at unused positions, and changes when L_6 or L_7 is appended; the entry L_3, right of the route but already appended, is fixed at every later anchor. The boxed subtree enters the path only through its root F. Every entry is computed from pos and the first n leaves.

6.1 The nullifier

Requirement R3 of "Requirements on a shielded payment" (§1.4) excludes a second consumption of a note without a public list of consumed notes. Each note therefore determines a *nullifier*, a tag published when the note is consumed, and a verifier (§1.2) rejects a nullifier already in the nullifier set of its pool (§6.2). The construction must make the nullifier

(a) *deterministic*: a function of the note and of the nullifier deriving key nk of the key that owns it, so that a second consumption publishes the same value; and

(b) *unlinkable*: to a party without nk, the published value reveals nothing about the note's commitment or address.

Construction 6.1 (Nullifier PRF)**.** Let $f\colon \mathbb{F}^3_{p_{\mathsf{Pallas}}} \to \mathbb{F}^3_{p_{\mathsf{Pallas}}}$ be the Poseidon permutation of width $t = 3$ over the Pallas base field, used as a sponge of rate 2 and capacity 1, with S-box $x \mapsto x^5$, 8 full and 56 partial rounds (*Crypto Guide*, §"Arithmetisation-friendly hashing: Poseidon", Constructions "The sponge construction" and "The Poseidon permutation", Example "The Orchard Poseidon instance"). Its round constants and MDS matrix are those of the protocol specification, §"PoseidonHash Function", and are not reproduced. In constant-input-length mode the hash $\mathsf{PoseidonHash}\colon \mathbb{F}_{p_{\mathsf{Pallas}}} \times \mathbb{F}_{p_{\mathsf{Pallas}}} \to \mathbb{F}_{p_{\mathsf{Pallas}}}$ is

$$\mathsf{PoseidonHash}(x, y) := \pi_1(f(x, y, 2^{65})),$$

where π_1 is the projection onto the first component; the capacity element 2^{65} encodes the input length 2, so no padding is needed (same citations). The *nullifier PRF* is

$$\mathsf{PRF}^{\mathsf{nfOrchard}}\colon \mathbb{F}_{p_{\mathsf{Pallas}}} \times \mathbb{F}_{p_{\mathsf{Pallas}}} \to \mathbb{F}_{p_{\mathsf{Pallas}}}, \qquad \mathsf{PRF}^{\mathsf{nfOrchard}}_{\mathsf{nk}}(\rho) := \mathsf{PoseidonHash}(\mathsf{nk}, \rho),$$

keyed by the nullifier deriving key $\mathsf{nk} \in \mathbb{F}_{p_{\mathsf{Pallas}}}$ of "The spending key and the spend-side secrets" (§3.1; protocol specification, §"Pseudo Random Functions").

Assumption 6.2 (Poseidon is a PRF). For nk uniform on $\mathbb{F}_{p_{\mathsf{Pallas}}}$, no efficient adversary with adaptive oracle access distinguishes $\rho \mapsto \mathsf{PoseidonHash}(\mathsf{nk}, \rho)$ from a uniformly random function $\mathbb{F}_{p_{\mathsf{Pallas}}} \to \mathbb{F}_{p_{\mathsf{Pallas}}}$, except with negligible advantage: this is the game of Definition "Pseudorandom function" (§2.3; *Crypto Guide*, §"Pseudorandom functions and permutations") with key, input and output sets $\mathbb{F}_{p_{\mathsf{Pallas}}}$. It is the security requirement that the protocol specification, §"Pseudo Random Functions", places on $\mathsf{PoseidonHash}$ keyed by its first argument. Its status is cryptanalytic: it is supported by the analysis of the Poseidon round function (protocol specification, §"PoseidonHash Function", notes) and has no reduction to a standard problem (*Crypto Guide*, Example "The Orchard Poseidon instance"). Of the results of this section only Proposition 6.10 uses it; Proposition 6.8 uses no property of $\mathsf{PoseidonHash}$, and the volume proves no unlinkability result for nullifiers without this assumption.

Definition 6.3 (Nullifier)**.** Let $K^{\mathsf{Orchard}} := \mathsf{GroupHash}(\texttt{z.cash:Orchard}, \texttt{K})$, the nullifier base, a fixed generator of Table 2. For $\mathsf{nk}, \rho, \psi \in \mathbb{F}_{p_{\mathsf{Pallas}}}$ and a point $\mathsf{cm} \in \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$,

$$\mathsf{DeriveNullifier}_{\mathsf{nk}}(\rho, \psi, \mathsf{cm}) := \mathsf{Extract}_{\mathbb{P}}([s]\, K^{\mathsf{Orchard}} + \mathsf{cm}) \in \mathbb{F}_{p_{\mathsf{Pallas}}},$$

where

$$s := (\mathsf{PRF}^{\mathsf{nfOrchard}}_{\mathsf{nk}}(\rho) + \psi) \bmod p_{\mathsf{Pallas}} \in \{0, \dots, p_{\mathsf{Pallas}} - 1\}:$$

the sum of the two field elements is taken as its representative in $\{0, \dots, p_{\mathsf{Pallas}} - 1\}$, and that integer multiplies K^{Orchard}. The *nullifier* of a note $n = (\mathsf{addr}, v, \rho, \psi, \mathsf{rcm})$ whose note commitment cm (Definition 4.4) is not $\perp$, under the nullifier deriving key nk of the key that owns addr, is

$$\mathsf{nf} := \mathsf{DeriveNullifier}_{\mathsf{nk}}(\rho, \psi, \mathsf{cm})$$

(protocol specification, §"Computing ρ values and Nullifiers").

Remark 6.4 (Reduction modulo p_{Pallas}). The protocol specification attaches two notes to the definition (§"Computing ρ values and Nullifiers"). First, the sum is reduced modulo p_{Pallas} intentionally, although the scalar field of the Pallas group is $\mathbb{F}_{p_{\mathsf{Vesta}}}$. Since $p_{\mathsf{Pallas}} < p_{\mathsf{Vesta}}$ (§2.1), the representative in $\{0, \dots, p_{\mathsf{Pallas}} - 1\}$ is a scalar without further reduction, by the injective reading $\mathbb{F}_{p_{\mathsf{Pallas}}} \to \mathbb{F}_{p_{\mathsf{Vesta}}}$ of §2.1, and the multiplier of K^{Orchard} ranges over p_{Pallas} of the p_{Vesta} scalars. Second, the inputs ρ and ψ must be those of the note committed to by cm.

Remark 6.5 (Determinism and masking). For a fixed nk the nullifier is a function of the fields ρ and ψ of the note and of its commitment cm, so a note has exactly one nullifier under the nk of its owner. That every accepted consumption of the note publishes this value is Lemma 12.5, proved in "Double-spend resistance" (§12.2) from the Action statement, defined in "The Action statement" (§9.2). The point $[s]\,K^{\mathsf{Orchard}}$ is added to cm before extraction. For notes of one key with pairwise distinct ρ, the key generated with $\mathsf{use_qsk} = \mathsf{false}$, the multipliers are jointly pseudorandom to an efficient party that holds the incoming viewing key and $\mathsf{ak}^{\mathbb{P}}$ but not nk, so the extracted values give it no usable information on the commitments or on the address; Proposition 6.10 states this exactly, under Assumption 6.2 and the further assumptions it names. The element ρ enters as the input of the PRF and ψ as an additive term of the multiplier. Both are fixed when the note is created: ρ by "Nullifier chaining" (§6.3), and ψ by the derivation of §4.3.

6.2 Nullifier sets

Each verifier maintains, for each shielded pool, a *nullifier set*, part of that pool's treestate ("Anchors", §5.2). Every Action publishes one nullifier, that of its consumed note, real or dummy (§4.4), and the nullifiers published by the Actions of valid transactions are inserted into the nullifier set of their pool (protocol specification, §"Nullifiers" and §"Nullifier Sets"). The consensus rule of §"Nullifier Sets" is that a nullifier must not repeat, either within a transaction or across transactions in a valid block chain; a nullifier is compared only with nullifiers of its own pool, so equal values in two pools do not repeat a nullifier (the specification treats the pools' nullifiers as disjoint).

Both pools use the one derivation of Definition 6.3, with the same PRF and the same nullifier base: the nullifier construction belongs to the Orchard protocol, not to a pool (ZIP 229, Terminology). The Orchard pool and the Ironwood pool keep separate nullifier sets, as they keep separate note commitment trees, anchors and value pool balances (ZIP 229, Rationale, "Separate state"; protocol specification, §"Nullifier Sets"). The nullifier of an Ironwood-pool Action is inserted into, and checked against, the nullifier set of the Ironwood pool only.

Remark 6.6 (Pool locality). Equal nullifier values in the two pools do not violate the consensus rule, so every uniqueness property derived from it holds within one pool. In particular the uniqueness of the elements ρ proved in "Double-spend resistance" (Corollary 12.7) is uniqueness within a pool, not across the two pools.

6.3 Nullifier chaining

Construction 6.7 (Chaining rule)**.** In every Action the sender sets the element ρ of the created

note to the nullifier that the same Action publishes for its consumed note,

$$\rho^{\mathsf{new}} := \mathsf{nf}^{\mathsf{old}} \in \mathbb{F}_{p_{\mathsf{Pallas}}},$$

and encodes it as the 32-byte string $\mathsf{I2LEOSP}_{256}(\rho^{\mathsf{new}})$. When the Action consumes no real note, its consumed side is a dummy note (Definition "Dummy notes", §4.4), and the nullifier of that dummy note is ρ^{new} (protocol specification, §"Sending Notes (Orchard)", §"Dummy Notes (Orchard)" and §"Faerie Gold attack and fix").

Every Action has a consumed side (Definition "Action", §4.4), so every note that an Action creates is constructed with ρ equal to the nullifier of that Action's consumed note. The leaves that enter a pool's note commitment tree are the extracted commitments of notes created by that pool's Actions ("The Merkle hash and the tree", §5.1), so the rule covers every note whose extracted commitment enters the tree. The consumed dummy note is not created by an Action: its ρ is $\mathsf{Extract}_{\mathbb{P}}$ of a uniformly random point of the Pallas group (§4.4).

The derivations of §4.3 take the chained ρ as input: with $\underline{\rho} := \mathsf{I2LEOSP}_{256}(\rho)$,

$$\mathsf{esk} = \mathsf{ToScalar}(\mathsf{PRFexpand}_{\mathsf{rseed}}([\mathtt{0x04}] \,\|\, \underline{\rho})), \qquad \psi = \mathsf{ToBase}(\mathsf{PRFexpand}_{\mathsf{rseed}}([\mathtt{0x09}] \,\|\, \underline{\rho})),$$

and rcm is derived from the string $\mathsf{pre_rcm}$, which contains $\underline{\rho}$ and the encoding of ψ. Hence esk and ψ are functions of rseed and of the nullifier that the note's own Action publishes, and rcm is a function of these and of the remaining note fields.

A *Faerie Gold attack* is one in which a sender creates several notes to one recipient that share a nullifier, so that only one of them can be spent (protocol specification, §"Faerie Gold attack and fix"). What a verifier can rely on, the uniqueness of the elements ρ within a pool and resistance to the Faerie Gold attack, is proved in "Double-spend resistance" (Corollary 12.7) from the nullifier-set rule and the Action statement, defined in "The Action statement" (§9.2). This subsection states the rule that an honest sender follows and draws no conclusion about accepted notes from it.

6.4 Nullifier uniqueness and unlinkability

Proposition 6.8 (Nullifier uniqueness)**.** *Under Assumptions 2.22 and 2.8, no efficient adversary outputs, except with negligible probability, two notes* $n = (\mathsf{addr}, v, \rho, \psi, \mathsf{rcm})$ *and* $n' = (\mathsf{addr}', v', \rho', \psi', \mathsf{rcm}')$ *and two nullifier deriving keys* $\mathsf{nk}, \mathsf{nk}' \in \mathbb{F}_{p_{\mathsf{Pallas}}}$*, all of its choice, whose note commitments* cm *and* cm' *are not* $\perp$ *and satisfy* $\mathsf{cm} \neq \mathsf{cm}'$ *and*

$$\mathsf{DeriveNullifier}_{\mathsf{nk}}(\rho, \psi, \mathsf{cm}) = \mathsf{DeriveNullifier}_{\mathsf{nk}'}(\rho', \psi', \mathsf{cm}').$$

The adversary may hold every nullifier deriving key, and $\mathsf{nk} = \mathsf{nk}'$ *and* $\rho = \rho'$ *are admitted. Since two distinct notes have distinct commitments except with negligible probability (Proposition 4.5(b)), the same holds for two distinct notes in place of two distinct commitments.*

Proof. The adversary supplies both openings, so no knowledge extraction is used. Let its output satisfy the conditions, and write

$$\begin{aligned} K &:= K^{\mathsf{Orchard}}, \\ Q &:= Q(\mathtt{z.cash:Orchard\text{-}NoteCommit\text{-}M}), \\ H_D &:= \mathsf{GroupHash}(\mathtt{z.cash:Orchard\text{-}NoteCommit\text{-}r}, \varepsilon) \end{aligned}$$

for the nullifier base and for the hash base and the blinding base of $\mathsf{NoteCommit}$ (Definition 4.4). A reduction that runs the adversary computes $s := (\mathsf{PRF}^{\mathsf{nfOrchard}}_{\mathsf{nk}}(\rho) + \psi) \bmod p_{\mathsf{Pallas}}$ and s' likewise; the pieces $m = (m_1, \ldots, m_{109})$ and m' of the messages $M(n)$ and $M(n')$, which have 109

chunks of $k = 10$ bits (Table 4); and rcm, rcm'. Every coefficient below is therefore known to it, the PRF outputs included, and no property of PoseidonHash is used.

Equal nullifiers are equal extracted coordinates, so by Lemma 2.4

$$[s]\,K + \mathsf{cm} = \theta([s']\,K + \mathsf{cm}') \qquad \text{for some } \theta \in \{1, -1\}.$$

Since $\mathsf{cm} \neq \bot$, its hash point is not $\bot$, and by the unrolled form (1) of the proof of Proposition 2.24,

$$\mathsf{cm} = [2^{109}]\,Q + \sum_{j=0}^{2^k-1} [\chi_j(m)]\,S(j) + [\mathsf{rcm}]\,H_D, \qquad \chi_j(m) = \sum_{i=1}^{109} 2^{109-i}\,\delta_{m_i,j},$$

and likewise for cm'. Substituting gives, with coefficients in $\mathbb{F}_{p_{\mathsf{Vesta}}}$ and s, s' read as scalars,

$$[s - \theta s']\,K + [(1-\theta)\,2^{109}]\,Q + \sum_j [\chi_j(m) - \theta\,\chi_j(m')]\,S(j) + [\mathsf{rcm} - \theta\,\mathsf{rcm}']\,H_D = \mathcal{O}. \tag{3}$$

The relation is non-trivial in each case.

- If $\theta = -1$, the coefficient of Q is 2^{110}, non-zero modulo p_{Vesta} because $0 < 2^{110} < 2^{253} \leq (p_{\mathsf{Vesta}} - 1)/2$.
- If $\theta = 1$ and $M(n) \neq M(n')$, then $m \neq m'$, and $\chi(m) \neq \chi(m')$ modulo p_{Vesta} by step (1) of the proof of Proposition 2.24 ($109 \leq c = 253$); some coefficient of an $S(j)$ is non-zero.
- If $\theta = 1$ and $M(n) = M(n')$, then $\mathsf{rcm} \neq \mathsf{rcm}'$, since equal messages and equal trapdoors give $\mathsf{cm} = \mathsf{cm}'$; the coefficient of H_D is non-zero.

The points $K, Q, S(0), \ldots, S(2^k - 1)$ and H_D are values of GroupHash on pairwise distinct inputs (Table 2). The reduction outputs these inputs with the coefficients of (3), a non-trivial relation with known coefficients. By the lemma on relations among fixed generators (§2.4), the reduction behind Proposition 2.24, this happens with negligible probability under Assumptions 2.22 and 2.8. The specification records that its Sinsemilla collision argument extends to additional terms with independent bases for this purpose (protocol specification, §"Sinsemilla Hash Function", Security argument, note; §"Faerie Gold attack and fix").

For distinct notes whose commitments are not $\bot$, equal commitments have negligible probability by Proposition 4.5(b), and distinct commitments are the case above. □

Remark 6.9 (Scope of Proposition 6.8). The proposition uses no pseudorandomness of PoseidonHash, and none could be used: the adversary holds the PRF keys, and the reduction knows every coefficient. It does not require distinct elements ρ, which the specification's security requirement assumes (§"Computing ρ values and Nullifiers"). Two notes with equal commitments are equal except with negligible probability (Proposition 4.5(b)) and so have equal nullifiers under one nk. Excluding equal commitments among the notes of a pool is the role of the chaining rule of §6.3, as Corollary 12.7 proves.

Proposition 6.10 (Nullifier unlinkability)**.** *Let a key be generated with* $\mathsf{use_qsk} = \mathsf{false}$ *from a uniform spending key, as in §3.1 and §3.2, with nullifier deriving key* nk*. An adversary receives the incoming viewing key* $(\mathsf{dk}, \mathsf{ivk})$ *and the point* $\mathsf{ak}^{\mathbb{P}}$ *of the key, but not* nk*. It then submits, adaptively, notes* $n_i = (\mathsf{addr}_i, v_i, \rho_i, \psi_i, \mathsf{rcm}_i)$*,* $1 \leq i \leq q$*, of its choice, with pairwise distinct* ρ_i *and note commitments* $\mathsf{cm}_i \neq \bot$*, and after each submission receives an answer:*

- *in the* real game, $\mathsf{nf}_i := \mathsf{DeriveNullifier}_{\mathsf{nk}}(\rho_i, \psi_i, \mathsf{cm}_i)$*;*
- *in the* ideal game, $\mathsf{Extract}_{\mathbb{P}}([u_i]\,K^{\mathsf{Orchard}} + \mathsf{cm}_i)$ *for* u_i *uniform on* $\mathbb{F}_{p_{\mathsf{Vesta}}}$*, independent of each other and of all else.*

> *It outputs a bit. Let* $\delta := (p_{\mathsf{Vesta}} - p_{\mathsf{Pallas}})/p_{\mathsf{Vesta}}$*, which is below* 2^{-167}*, the statistical distance between the uniform distribution on* $\{0, \dots, p_{\mathsf{Pallas}} - 1\}$*, read in* $\mathbb{F}_{p_{\mathsf{Vesta}}}$*, and the uniform distribution on* $\mathbb{F}_{p_{\mathsf{Vesta}}}$*. Then the difference of the probabilities that it outputs* 1 *in the two games is at most*
>
> $$2(\epsilon_{\mathsf{key}} + \epsilon_{\mathsf{expand}} + 3 \cdot 2^{-257} + \epsilon_{\mathsf{vk}}) + \epsilon_{\mathsf{Pos}} + q\,\delta,$$
>
> *where* $\epsilon_{\mathsf{expand}}$*,* ϵ_{vk} *and* ϵ_{Pos} *are the largest advantages against Assumptions 2.12 (through Lemma 2.15), 3.13 and 6.2 of the algorithms built from the adversary in the proof, and* ϵ_{key} *is the probability that a single draw of key generation from a uniform spending key is rejected plus the probability that* $\mathsf{ask} = 0$ *or* $\hat{M} = \perp$ *for independent uniform* $(\mathsf{ask}, \mathsf{nk})$*, with* $\hat{M}$ *the hash point of* $\mathsf{Commit}^{\mathsf{ivk}}$ *on* $(\mathsf{ak}, \mathsf{nk})$ *(§3.2). The term* ϵ_{key} *is negligible under Assumptions 2.12, 2.22 and 2.8 (Lemma "Rejection in key generation", §3.2); the bound is therefore negligible for every efficient adversary, whose number* q *of queries is polynomial, under these assumptions and Assumptions 3.13 and 6.2. When* $K^{\mathsf{Orchard}} \neq \mathcal{O}$*, the point* $[u_i]\,K^{\mathsf{Orchard}} + \mathsf{cm}_i$ *is uniform on* $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$ *whatever* cm_i *is, so the ideal answers are independent of the notes and of each other; in the model of Assumption 2.8,* $K^{\mathsf{Orchard}} = \mathcal{O}$ *has probability* $1/p_{\mathsf{Vesta}}$*.*

Proof. Write $K := K^{\mathsf{Orchard}}$. The proof passes from the real game H_0 through hybrid games $H_1, \dots, H_5$; each hop changes the probability of the output 1 by at most the amount stated (*Crypto Guide*, §"The hybrid argument", Lemma "Hybrid lemma"), and each reduction runs the adversary and the rest of the game.

Hybrid H_1*: idealised key.* The key is generated by a single draw from a uniform sk, without rejection, at the cost of the probability that the draw is rejected (Lemma "Rejection in key generation", §3.2). Then $(\mathsf{ask}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$ is replaced by independent uniform elements of $\mathbb{F}_{p_{\mathsf{Vesta}}}$, $\mathbb{F}_{p_{\mathsf{Pallas}}}$ and $\mathbb{F}_{p_{\mathsf{Vesta}}}$, from which the other components are derived as in §3.1 and §3.2. By Lemma 2.15, this costs at most $\epsilon_{\mathsf{expand}}$ plus the three reduction distances, each below 2^{-257} (§2.3): the three inputs of $\mathsf{PRFexpand}_{\mathsf{sk}}$ have distinct lead bytes, and the game is an efficient function of the triple, which derives $\mathsf{ak}^{\mathbb{P}}$, ivk and dk, runs the adversary and answers each query with the nullifier under nk.

Hybrid H_2*: joint hiding.* When $\mathsf{ask} \neq 0$, the pair $(\mathsf{ivk}, \mathsf{dk})$ is replaced by $(\mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r}_{\mathsf{ivk}}'}(\mathsf{ak}, \mathsf{nk}), u)$, for $\mathsf{r}_{\mathsf{ivk}}'$ uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$ and u a uniform 32-byte string, independent of all else. The cost is at most ϵ_{vk}. The reduction for Assumption 3.13 draws ask and nk itself; for $\mathsf{ask} \neq 0$ the pair $(\mathsf{ak}, \mathsf{nk})$ lies in the domain of the assumption (§3.1). It receives a triple for that pair, passes its first two components to the adversary as ivk and dk, together with $\mathsf{ak}^{\mathbb{P}}$, and answers every query with the nullifier under its own nk. Giving $\mathsf{ak}^{\mathbb{P}}$ costs nothing, because after H_1 the scalar ask is independent of $(\mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$ and the assumption holds for every fixed $(\mathsf{ak}, \mathsf{nk})$.

Hybrid H_3*: independent viewing keys.* The pair $(\mathsf{ivk}, \mathsf{dk})$ is replaced, whatever ask is, by $(\mathsf{Extract}_{\mathbb{P}}(W), u)$ for a uniform point W and a uniform 32-byte string u, independent of all else. When $\mathsf{ask} \neq 0$ and the hash point $\hat{M}$ of the message $\mathrm{LE}_{255}(\mathsf{ak}) \,\|\, \mathrm{LE}_{255}(\mathsf{nk})$ is not $\perp$, the commitment point $\hat{M} + [\mathsf{r}_{\mathsf{ivk}}']\,H_D$ is uniform on $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$ by Lemma 2.26(i), its blinding base H_D not being $\mathcal{O}$ (§3.2), so the two games are identically distributed conditioned on that event, which depends on $(\mathsf{ask}, \mathsf{nk})$ only. The cost is at most the probability that $\mathsf{ask} = 0$ or $\hat{M} = \perp$, the remaining part of ϵ_{key}. In H_3 the key nk enters the game only through the answers: dk, ivk and $\mathsf{ak}^{\mathbb{P}}$ are independent of it.

Hybrid H_4*: random function.* The function $\mathsf{PRF}^{\mathsf{nfOrchard}}_{\mathsf{nk}}$ is replaced by a uniformly random function $F\colon \mathbb{F}_{p_{\mathsf{Pallas}}} \to \mathbb{F}_{p_{\mathsf{Pallas}}}$. The cost is at most ϵ_{Pos}: the reduction for Assumption 6.2 runs H_3 with its oracle in place of $\mathsf{PRF}^{\mathsf{nfOrchard}}_{\mathsf{nk}}$, since nk, uniform on $\mathbb{F}_{p_{\mathsf{Pallas}}}$, is used nowhere else. In H_4 the i-th answer is $\mathsf{Extract}_{\mathbb{P}}([s_i]\,K + \mathsf{cm}_i)$ with $s_i := (F(\rho_i) + \psi_i) \bmod p_{\mathsf{Pallas}}$. The ρ_i are pairwise distinct, so, with F sampled lazily, $F(\rho_i)$ is uniform on $\mathbb{F}_{p_{\mathsf{Pallas}}}$ and independent of everything the adversary has received when it submits n_i. Translation by ψ_i is a bijection modulo p_{Pallas}, so s_i is uniform on $\{0, \dots, p_{\mathsf{Pallas}} - 1\}$ and independent of that view, whatever ψ_i is.

Hybrid H_5*: uniform scalars.* For $i = 1, \dots, q$ in turn, the scalar s_i, read in $\mathbb{F}_{p_{\mathsf{Vesta}}}$ (§2.1), is replaced

by u_i uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$. The two distributions differ by $1/p_{\mathsf{Pallas}} - 1/p_{\mathsf{Vesta}}$ on each of the p_{Pallas} residues and by $1/p_{\mathsf{Vesta}}$ on each of the other $p_{\mathsf{Vesta}} - p_{\mathsf{Pallas}}$ scalars, so their statistical distance is

$$\tfrac{1}{2}\Big(p_{\mathsf{Pallas}}\Big(\frac{1}{p_{\mathsf{Pallas}}} - \frac{1}{p_{\mathsf{Vesta}}}\Big) + \frac{p_{\mathsf{Vesta}} - p_{\mathsf{Pallas}}}{p_{\mathsf{Vesta}}}\Big) = \frac{p_{\mathsf{Vesta}} - p_{\mathsf{Pallas}}}{p_{\mathsf{Vesta}}} = \delta.$$

The rest of the game is a randomised function of the view before the i-th answer and of the replaced value, which is independent of that view; each step therefore costs at most δ, and the q steps at most $q\,\delta$ (*Math Guide*, §"Statistical distance", Theorem "Properties of statistical distance", parts (2) and (4)).

In H_5 the answers are those of the ideal game, and the key material is that of H_3. Neither in H_5 nor in the ideal game do the answers depend on the key, so the hops of H_3, H_2 and H_1, taken in reverse order with the answers of the ideal game, lead from H_5 to the ideal game at the same costs. Summing the costs of both passes gives the bound. The last statement of the proposition holds because, for $K \neq O$, the map $u \mapsto [u]\,K$ is a bijection from $\mathbb{F}_{p_{\mathsf{Vesta}}}$ onto the group of prime order p_{Vesta}, and translation by cm_i is a bijection of the group; under Assumption 2.8 the point K is uniform. □

Remark 6.11 (Design statement, not proved). The specification describes the nullifier as combining elliptic-curve cryptography with the Poseidon-based PRF so as to give privacy defence in depth against a weakness in either (protocol specification, §"Faerie Gold attack and fix"). It states that, under the weaker assumption that PoseidonHash preserves sufficient entropy from nk and ρ, the construction would remain unlinkable under a decisional Diffie–Hellman assumption on Pallas even if the Poseidon-based PRF were distinguishable from an ideal PRF (§"PoseidonHash Function", note; *Crypto Guide*, §"Diffie–Hellman: computational and decisional", Definition "Decisional Diffie–Hellman, DDH"; the form assumed in this volume is Assumption 3.16). The volume records this as the specification's design statement and derives no result from it; Proposition 6.10 relies on Assumption 6.2 instead.

For keys with use_qsk = false, the key tiers of Proposition 3.18 ("Capability separation") have the following powers over nullifiers. The full viewing key $(\mathsf{ak}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$ contains nk. Its holder computes $\mathsf{DeriveNullifier}_{\mathsf{nk}}(\rho, \psi, \mathsf{cm})$ for every note of the key whose fields it knows, and tests whether that value is in the nullifier set of the note's pool; that every accepted consumption of the note publishes this value is Lemma 12.5. It does so without spending authority, since the full viewing key yields no ask (Proposition 3.18(a)). A holder of only the incoming viewing key $(\mathsf{dk}, \mathsf{ivk})$ cannot link a nullifier to its note, for notes with pairwise distinct ρ, by Proposition 6.10.

Remark 6.12 (Limits). Proposition 6.10 gives nothing against a holder of nk, which includes every holder of the full viewing key: such a holder recomputes, and so links, the nullifier of every note of the key whose fields it knows. It does not concern links through other public data, such as timing or the structure of the transaction; the privacy of an Action, with its preconditions, is Theorem 12.12, in "Privacy" (§12.4).

7 Spend authorisation

This section instantiates RedDSA, the re-randomisable Schnorr signature scheme of the *Crypto Guide*, on the Pallas group; constructs the randomised validating key and the spend-authorisation signature of an Action; and proves two properties of them: the published key is, with the signature scheme's hash modelled as a random oracle, within negligible statistical distance of a key independent of the spend validating key; and a new valid signature under a randomised key yields the key's discrete logarithm. The signed message is a parameter throughout; "Transaction digests and signatures" (§11.3) fixes it.

7.1 The RedPallas signature scheme

Remark 7.1 (Spending authority and its signature). Requirement R5 of "Requirements on a shielded payment" (§1.4) admits only the holder of a note's spending authority as a party that consumes the note. For keys generated with $\mathsf{use_qsk} = \mathsf{false}$, the spending authority over the notes of every address derived from a spending key is knowledge of its spend authorising key ask of "The spending key and the spend-side secrets" (§3.1), the discrete logarithm of $\mathsf{ak}^{\mathbb{P}} = [\mathsf{ask}]\, G^{\mathsf{SpendAuth}}$ to the base $G^{\mathsf{SpendAuth}}$. No lower key tier yields ask, since every lower tier is computed from the full viewing key (Proposition 3.18(a)). Each Action therefore carries a signature that only a holder of ask can produce, under a validating key that does not link Actions consuming notes of the same key: a fresh re-randomisation of $\mathsf{ak}^{\mathbb{P}}$ (§7.2), never $\mathsf{ak}^{\mathbb{P}}$ itself. Theorem 12.9, in "Authorisation of spends" (§12.3), proves for keys with $\mathsf{use_qsk} = \mathsf{false}$ that no efficient party without ask produces an accepted Action that consumes a note of an honestly generated address and whose spend-authorisation signature is on a digest that the key holder did not sign under the Action's randomised key, except with negligible probability.

Definition 7.2 (RedPallas hash)**.** The *RedPallas hash* is the map H from byte strings to 64-byte strings

$$\mathsf{H}(z) := \text{BLAKE2b-512}(\texttt{Zcash_RedPallasH}, z),$$

with the notation of the expansion function (§2.3) and the 16-byte personalisation `Zcash_RedPallasH`. Its reduction to scalars is

$$\mathsf{H}^{\circledast}(z) := \mathsf{LEOS2IP}_{512}(\mathsf{H}(z)) \bmod p_{\mathsf{Vesta}} = \mathrm{ToScalar}(\mathsf{H}(z)) \in \mathbb{F}_{p_{\mathsf{Vesta}}}$$

(protocol specification, §"RedDSA, RedJubjub, and RedPallas" and §"BLAKE2 Hash Functions"). The symbol $\mathsf{H}^{\circledast}$ is that of the specification and of the *Crypto Guide*'s Remark "RedDSA as deployed" (§"RedDSA: re-randomisable Schnorr for Orchard"); it is distinct from the star encoding $P^{\star}$ of a point.

Assumption 7.3 (The RedPallas hash as a random oracle). In security arguments the RedPallas hash H is modelled as a random oracle with output length 512, that is, with range the 64-byte strings, queried by every party (*Crypto Guide*, §"The random oracle model", Definition "Random oracle"). Consequently $\mathsf{H}^{\circledast}$ is a random function into $\mathbb{F}_{p_{\mathsf{Vesta}}}$: its values at distinct inputs are independent, and each is distributed as $\mathrm{ToScalar}(U)$ for U uniform on the 64-byte strings. That distribution is at statistical distance at most

$$\delta_{512} := p_{\mathsf{Vesta}}/2^{512} < 2^{-257}$$

from the uniform distribution on $\mathbb{F}_{p_{\mathsf{Vesta}}}$ (*Math Guide*, §"Uniform sampling and the bias of modular reduction", Proposition "Bias of modular reduction", with $L = 512$), and it gives each scalar probability at most $1/p_{\mathsf{Vesta}} + 2^{-512}$, since each residue has at most $\lfloor 2^{512}/p_{\mathsf{Vesta}} \rfloor + 1$ preimages among the 2^{512} strings. The assumption is a heuristic about BLAKE2b, not a theorem. Every later result on RedPallas signatures or on the randomiser generator below names it.

Construction 7.4 (RedPallas)**.** RedPallas is the *Crypto Guide*'s Construction "RedDSA" (§"RedDSA: re-randomisable Schnorr for Orchard") on the Pallas group $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$, of prime order p_{Vesta}, with both hashes $\mathsf{H}^{\circledast}$ and with a base point $B \neq \mathcal{O}$ as parameter; the group

having prime order, B generates it. For a point P let $\underline{P} := \mathsf{LEBS2OSP}_{256}(P^\star)$, its 32-byte encoding; the map $P \mapsto \underline{P}$ is injective, as the star encoding is (§2.1) and $\mathsf{LEBS2OSP}_{256}$ is a bijection onto the 32-byte strings. Messages are byte strings.

1. *Keys.* A signing key is a scalar $x \in \mathbb{F}_{p_{\mathsf{Vesta}}}$; its validating key is $X := [x]\,B$.
2. *Signing* a message M under x: draw T uniformly from the 80-byte strings, where $80 = (512 + 128)/8$ is the specification's randomness length for a 512-bit hash; let

$$r := \mathsf{H}^{\circledast}(T \,\|\, \underline{X} \,\|\, M), \qquad R := [r]\,B,$$
$$c := \mathsf{H}^{\circledast}(\underline{R} \,\|\, \underline{X} \,\|\, M), \qquad S := r + c \cdot x \in \mathbb{F}_{p_{\mathsf{Vesta}}},$$

   and output the 64-byte signature $\sigma := \underline{R} \,\|\, \mathsf{I2LEOSP}_{256}(S)$.
3. *Validation* of a 64-byte string σ under a point X on M: let $\bar{R}$ be the first 32 bytes of σ and $\bar{S}$ the last 32; let $R := \mathsf{abst}_{\mathbb{P}}(\mathsf{LEBS2OSP}_{256}^{-1}(\bar{R}))$, $S := \mathsf{LEOS2IP}_{256}(\bar{S})$ and $c := \mathsf{H}^{\circledast}(\bar{R} \,\|\, \underline{X} \,\|\, M)$. Accept if and only if $R \neq \bot$, $S < p_{\mathsf{Vesta}}$ and

$$[S]\,B = R + [c]\,X. \tag{4}$$

(Protocol specification, §"RedDSA, RedJubjub, and RedPallas"; its validation equation carries the cofactor, which is 1 on Pallas.)

The partial inverse $\mathsf{abst}_{\mathbb{P}}$ returns $\bot$ on every 256-bit string that is not the star encoding of a point (§2.1); validation therefore rejects a non-canonical $\bar{R}$, and an accepted signature has $\bar{R} = \underline{R}$. An honest signature is accepted: its first half $\underline{R}$ decodes to R, its S is an integer representative below p_{Vesta}, the verifier recomputes the signer's c, and $[S]\,B = [r]\,B + [c \cdot x]\,B = R + [c]\,X$ (*Crypto Guide*, §"From identification to signature via the Fiat–Shamir transform", Proposition "Perfect correctness"). Both the nonce hash and the challenge hash take the encoded validating key $\underline{X}$: the scheme is *key-prefixed*. The message M is a parameter: the message of every RedPallas signature in this volume is the signature digest constructed in "Transaction digests and signatures" (§11.3), and no argument before that subsection depends on its value.

Definition 7.5 (Spend-authorisation and binding-signature instances)**.** The volume uses two instances of RedPallas, which share the signing and validation algorithms and differ in the base B and in the use of key re-randomisation (protocol specification, §"Spend Authorization Signature (Sapling and Orchard)" and §"Binding Signature (Sapling and Orchard)"):

1. the *spend-authorisation instance*, with key re-randomisation (§7.2) and base

$$B = G^{\mathsf{SpendAuth}} = \mathsf{GroupHash}(\texttt{z.cash:Orchard}, \texttt{G});$$

2. the *binding-signature instance*, without re-randomisation, used in "The binding signature" (§8.3), with base the value-commitment randomness base

$$B = R^{\mathsf{Orchard}} = \mathsf{GroupHash}(\texttt{z.cash:Orchard-cv}, \texttt{r}).$$

Both bases are rows of Table 2 (§2.2); the specification takes each as the generator parameter of RedDSA.

Definition 7.6 (Randomiser generator and key re-randomisation)**.** The *randomiser generator*

GenRandom draws T uniformly from the 80-byte strings and returns $\mathsf{H}^{\circledast}(T) \in \mathbb{F}_{p_{\mathsf{Vesta}}}$ (protocol specification, §"RedDSA, RedJubjub, and RedPallas"); its distance from uniform is bounded in §7.2. *Re-randomisation* by a randomiser $\alpha \in \mathbb{F}_{p_{\mathsf{Vesta}}}$ maps a key pair (x, X) to $(x + \alpha,\ X + [\alpha]\, B)$ (*Crypto Guide*, §"Key re-randomisation and unlinkability", Definition "Key re-randomisation").

Remark 7.7 (Key prefixing). Let σ be valid under X on M, with decoded components R and S and challenge $c = \mathsf{H}^{\circledast}(\underline{R} \,\|\, \underline{X} \,\|\, M)$, and let $X' := X + [\beta]\, B \neq \mathcal{O}$ for a scalar $\beta \neq 0$. The shifted string $\sigma' := \underline{R} \,\|\, \mathsf{I2LEOSP}_{256}(S + c\beta)$ satisfies $[S + c\beta]\, B = R + [c]\, X'$. Validation of σ' under X', however, recomputes the challenge as $c' := \mathsf{H}^{\circledast}(\underline{R} \,\|\, \underline{X'} \,\|\, M)$, the value of the oracle at an input other than that of c, since $\underline{X'} \neq \underline{X}$; and (4) holds for σ' under X' only if $[c - c']\, X' = \mathcal{O}$, that is, only if $c' = c$. Under Assumption 7.3, for each β the value c' is independent of c and equals it with probability at most $1/p_{\mathsf{Vesta}} + 2^{-512}$. A signature is therefore not transported from one key to a related key by shifting its response. With a challenge that omitted the key, $c' = c$ always, and the shift would turn every signature under X into one under X' (*Crypto Guide*, §"From identification to signature via the Fiat–Shamir transform", Remark "Two conventions"; §"Key re-randomisation and unlinkability", Remark "Unforgeability under re-randomised keys, and the key-prefixing subtlety"). The proof of Proposition 7.12, which admits randomisers chosen by the adversary, uses key prefixing in its steps (3) and (5).

7.2 Randomised validating keys

Construction 7.8 (Randomised validating key)**.** For each Action the spender draws a *spend-authorisation randomiser* $\alpha := \mathsf{GenRandom}()$, independently of every other draw, and sets

$$\mathsf{rsk} := \mathsf{ask} + \alpha \in \mathbb{F}_{p_{\mathsf{Vesta}}}, \qquad \mathsf{rk} := \mathsf{ak}^{\mathbb{P}} + [\alpha]\, G^{\mathsf{SpendAuth}} \in \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}}),$$

with ask the sign-normalised spend authorising key of the spending key of the consumed note and $\mathsf{ak}^{\mathbb{P}}$ its spend validating key (§3.1). Since $\mathsf{ak}^{\mathbb{P}} = [\mathsf{ask}]\, G^{\mathsf{SpendAuth}}$, it follows that $\mathsf{rk} = [\mathsf{rsk}]\, G^{\mathsf{SpendAuth}}$ (*Crypto Guide*, §"Key re-randomisation and unlinkability", Proposition "Re-randomisation consistency"), so $(\mathsf{rsk}, \mathsf{rk})$ is a key pair of the spend-authorisation instance. The Action publishes the *randomised validating key* rk and its *spend-authorisation signature*, the RedPallas signature under rsk on the signature digest constructed in "Transaction digests and signatures" (§11.3); it publishes none of ask, α, rsk, $\mathsf{ak}^{\mathbb{P}}$ and ak (protocol specification, §"Spend Authorization Signature (Sapling and Orchard)").

Lemma 7.9 (Distance of GenRandom from uniform)**.** *Under Assumption 7.3, let an algorithm, the observer, make at most q_h queries to* H *and, at one point, choose a byte string w and receive $\alpha := \mathsf{H}^{\circledast}(T \,\|\, w)$, for T uniform on the* 80*-byte strings and independent of all else. The pair formed by the observer's view and α is within statistical distance*

$$\delta_{512} + q_h \cdot 2^{-640} \tag{5}$$

of the pair obtained when α is replaced by a uniform element of $\mathbb{F}_{p_{\mathsf{Vesta}}}$ independent of all else. A randomiser from GenRandom *is the case $w = \varepsilon$; the nonce of a RedPallas signature under X on M is the case $w = \underline{X} \,\|\, M$. The first term of* (5) *is set by the* 512*-bit output of* H*, not by the* 640*-bit length of T; the length of T enters only the second term.*

Proof. Let E be the event that the observer queries H at $T \,\|\, w$. In a second experiment, $\mathsf{H}(T \,\|\, w)$ is replaced, in the computation of α only, by a uniform 64-byte string U independent of all else. In

the lazy realisation of the random oracle (*Crypto Guide*, §"The random oracle model", Definition "Random oracle") the answers at inputs other than $T \parallel w$ are independent of $\mathsf{H}(T \parallel w)$, so the two experiments are identical until E occurs; E has the same probability in both, and their outputs are within statistical distance $\Pr[E]$ (*Crypto Guide*, §"Security as a game", Remark "Game-hopping"). In the second experiment the observer's view is independent of T, so each query agrees with $T \parallel w$ in its first 80 bytes with probability at most 2^{-640}, and $\Pr[E] \leq q_h \cdot 2^{-640}$ by the union bound. There U is independent of H, of T and of the observer's coins, and the pair of view and α is a function of $\alpha = \mathsf{ToScalar}(U)$ and of them. Replacing α by a uniform scalar changes the pair by at most δ_{512} (Assumption 7.3; *Math Guide*, §"Statistical distance", Theorem "Properties of statistical distance", part (4)). The triangle inequality, part (2) of the same theorem, gives (5). □

Two consensus rules of the protocol specification, §"Action Descriptions", concern spend authorisation:

1. the randomised validating key rk of an Action must not be the identity $\mathcal{O}$;
2. the spend-authorisation signature of the Action must be valid under rk on the signature digest of "Transaction digests and signatures" (§11.3).

Validation rejects a non-canonical encoding of the signature's point component (§7.1). An honestly computed rk equals $\mathcal{O}$ only if $\alpha = -\mathsf{ask}$; for ask fixed before α is drawn, this event has probability at most $1/p_{\mathsf{Vesta}} + 2^{-512}$ (Assumption 7.3).

Proposition 7.10 (Unlinkability of randomised validating keys)**.** *Let $\mathsf{ak}_1^{\mathbb{P}}, \dots, \mathsf{ak}_n^{\mathbb{P}}$ be points of the Pallas group, equal or distinct, and let $\mathsf{rk}_i := \mathsf{ak}_i^{\mathbb{P}} + [\alpha_i]\, G^{\mathsf{SpendAuth}}$ for randomisers $\alpha_1, \dots, \alpha_n$ independent of each other and of the points.*

(*i*) *If each α_i is uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$, the tuple $(\mathsf{rk}_1, \dots, \mathsf{rk}_n)$ is uniform on $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})^n$, whatever the points $\mathsf{ak}_i^{\mathbb{P}}$, and hence independent of them. For each i and the scalar ask_i with $\mathsf{ak}_i^{\mathbb{P}} = [\mathsf{ask}_i]\, G^{\mathsf{SpendAuth}}$, the pair $(\mathsf{ask}_i + \alpha_i, \mathsf{rk}_i)$ is distributed exactly as a fresh key pair $(y, [y]\, G^{\mathsf{SpendAuth}})$ with y uniform, independently of ask_i. The distribution of the tuple is thus the same for every assignment of the points, and no observer, whatever its running time, distinguishes two assignments, for instance one in which two Actions share a key from one in which they do not, with non-zero advantage.*

(*ii*) *If each α_i is drawn by* $\mathsf{GenRandom}$*, then under Assumption 7.3, for an observer that receives the tuple and makes at most q_h queries to* H*, the pair of its view and the tuple is within statistical distance*

$$n(\delta_{512} + (q_h + n) \cdot 2^{-640})$$

of the pair obtained from a uniform tuple independent of H*; the pairs arising from two assignments of the points are therefore within statistical distance $2n(\delta_{512} + (q_h + n) \cdot 2^{-640})$.*

Proof. (i) For each i this is the *Crypto Guide*'s Theorem "Perfect unlinkability of re-randomised keys" (§"Key re-randomisation and unlinkability"). The base $G^{\mathsf{SpendAuth}}$ generates the Pallas group, of prime order p_{Vesta} (§7.1), so $\alpha \mapsto [\alpha]\, G^{\mathsf{SpendAuth}}$ is a bijection from $\mathbb{F}_{p_{\mathsf{Vesta}}}$ onto the group, and translation by $\mathsf{ak}_i^{\mathbb{P}}$ is a bijection of the group; hence rk_i is uniform whatever $\mathsf{ak}_i^{\mathbb{P}}$ is, and the independence of the α_i gives the product distribution. The scalar $\mathsf{ask}_i + \alpha_i$ is uniform for every fixed ask_i because α_i is, and $\mathsf{rk}_i = [\mathsf{ask}_i + \alpha_i]\, G^{\mathsf{SpendAuth}}$. Equal distributions give every distinguisher advantage zero (*Math Guide*, §"Statistical distance", Theorem "Properties of statistical distance", part (3)).

(ii) The randomisers are replaced by independent uniform scalars one at a time, for $i = 1, \dots, n$. At step i the lemma on the distance of $\mathsf{GenRandom}$ from uniform applies, with $w = \varepsilon$, to the

observer extended by the rest of the experiment: it holds the points and $\alpha_1, \dots, \alpha_{i-1}$, already uniform; it draws $T_{i+1}, \dots, T_n$ itself and queries H at them; it receives α_i, computes the tuple and runs the given observer. It makes at most $q_h + n - i$ queries, and the given observer's view and the tuple are a function of its view and α_i. By (5) and part (4) of the same theorem, step i changes the pair of view and tuple by at most $\delta_{512} + (q_h + n - i) \cdot 2^{-640}$. By the triangle inequality, the n steps together change it by at most

$$n\,\delta_{512} + \Big(nq_h + \frac{n(n-1)}{2}\Big)2^{-640} \le n(\delta_{512} + (q_h + n) \cdot 2^{-640}).$$

After the last step the α_i are uniform, independent and independent of H, so by (i) the tuple is uniform and independent of H, for every assignment of the points. The triangle inequality through that common distribution gives the factor 2 for two assignments. □

Remark 7.11 (Scope of Proposition 7.10). The proposition concerns rk alone, for randomisers drawn by the honest spender; a randomiser chosen by an adversary is the subject of Proposition 7.12 (§7.3). The spend-authorisation signature published with rk adds no information about $\mathsf{ak}^{\mathbb{P}}$: under Assumption 7.3 it is simulated from rk and the message M alone by programming H (*Crypto Guide*, §"Security in the random oracle model and the forking lemma", Lemma "Signature simulation"). The simulator draws u uniform on the 64-byte strings and S uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$, sets $c := \mathrm{ToScalar}(u)$ and $R := [S]\,G^{\mathsf{SpendAuth}} - [c]\,\mathsf{rk}$, programs H at $\underline{R} \,\|\, \underline{\mathsf{rk}} \,\|\, M$ to u, and outputs $\underline{R} \,\|\, \mathsf{I2LEOSP}_{256}(S)$. Beyond the abort probability of the lemma, a real signature differs from the simulated one only through its nonce: for a uniform nonce the two are identically distributed, since $S = r + c \cdot \mathsf{rsk}$ is then uniform given u and $R = [S]\,G^{\mathsf{SpendAuth}} - [c]\,\mathsf{rk}$. The difference is therefore at most the bound (5) per signature, with q_h there the number of all other queries to H in the experiment. The remaining public fields of an Action are treated in Theorem 12.12, in "Privacy" (§12.4).

7.3 RedPallas unforgeability

Proposition 7.12 (Unforgeability under re-randomisation). *Assume that discrete logarithms on Pallas are hard (Assumption 2.22) and that* H *is a random oracle (Assumption 7.3). Let x be uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$ and $\mathsf{ak}^{\mathbb{P}} := [x]\,G^{\mathsf{SpendAuth}}$. An adversary $\mathcal{A}$ receives $\mathsf{ak}^{\mathbb{P}}$, makes at most q_h queries to* H *and at most q_s queries to a signing oracle that answers a pair (α, M) of its choice, $\alpha \in \mathbb{F}_{p_{\mathsf{Vesta}}}$ and M a byte string, with the spend-authorisation signature under $x + \alpha$ on M, whose validating key is $\mathsf{ak}^{\mathbb{P}} + [\alpha]\,G^{\mathsf{SpendAuth}}$; it outputs a triple $(\mathsf{rk}^*, M^*, \sigma^*)$ with rk^* a point. It* forges *if σ^* is valid under rk^* on M^* and $(\mathsf{rk}^*, M^*, \sigma^*)$ is not the triple of validating key, message and signature of an oracle answer. Let ϵ be its forging probability, $Q := q_h + q_s + 1$ and*

$$\eta := Q\,\delta_{512} + q_s(q_h + 2q_s) \cdot 2^{-640}.$$

(*i*) Extraction. *There is an algorithm $\mathcal{D}$ that receives $\mathsf{ak}^{\mathbb{P}}$ but not x, runs $\mathcal{A}$ twice on shared coins, answers its signing queries without x, and outputs a scalar rsk^* with $\mathsf{rk}^* = [\mathsf{rsk}^*]\,G^{\mathsf{SpendAuth}}$, for the rk^* of the first run, with probability at least*

$$\frac{(\epsilon - \eta)^2}{Q} - \frac{Q}{p_{\mathsf{Vesta}}} - \frac{q_s(q_h + q_s)}{p_{\mathsf{Vesta}}} \qquad (\epsilon \ge \eta),$$

the bound of the Crypto Guide*'s Theorem "EUF-CMA security of Schnorr signatures in the ROM" (§"Security in the random oracle model and the forking lemma") with ϵ replaced by $\epsilon - \eta$ and the group order by p_{Vesta}.*

> (*ii*) Unforgeability under re-randomisation. *If $\mathcal{A}$ also outputs $\alpha^* \in \mathbb{F}_{p_{\mathsf{Vesta}}}$ with $\mathsf{rk}^* = \mathsf{ak}^{\mathbb{P}} + [\alpha^*]\, G^{\mathsf{SpendAuth}}$, of its own choice and possibly dependent on $\mathsf{ak}^{\mathbb{P}}$, then $\mathcal{D}$ outputs $x = \mathsf{rsk}^* - \alpha^*$, the discrete logarithm of $\mathsf{ak}^{\mathbb{P}}$, with the same probability. By Assumption 2.22, ϵ is then negligible for every efficient $\mathcal{A}$.*
>
> *With every randomiser zero, part (ii) is the existential unforgeability of RedPallas under chosen-message attack* (Crypto Guide, *§"Syntax and security goal", Definition "Existential unforgeability under chosen-message attack") in its strong form (Definition "Strong unforgeability" there), since freshness is required of the triple rather than of the message. Part (ii) implies the SURK-CMA requirement of the protocol specification, §"Signature with Re-Randomizable Keys", whose forgery is a message–signature pair not returned by the oracle, hence a triple not returned by it.*

Proof. The argument is route (ii) of the *Crypto Guide*'s Remark "Unforgeability under re-randomised keys, and the key-prefixing subtlety" (§"Key re-randomisation and unlinkability"), with the bound of its Theorem "EUF-CMA security of Schnorr signatures in the ROM" (§"Security in the random oracle model and the forking lemma"), which applies to RedPallas with the generator replaced by the base $G^{\mathsf{SpendAuth}}$ (Construction "RedDSA"). The Orchard steps are made in place; write $G := G^{\mathsf{SpendAuth}}$.

(1) *Idealisation.* First, the nonce r_i of each oracle answer is replaced by a uniform scalar. By the lemma on the distance of GenRandom from uniform, with $w = \underline{\mathsf{rk}_i} \,\|\, M_i$ and the rest of the experiment as observer, each of the at most q_s replacements costs at most $\delta_{512} + (q_h + 2q_s) \cdot 2^{-640}$: apart from this nonce query, the experiment queries H at most q_h times for $\mathcal{A}$, $q_s - 1$ times for other nonces, q_s times for challenges and once for validating the forgery. Second, the answers of H to the queries of $\mathcal{A}$ and to the validation query are replaced by uniform preimages, under ToScalar, of uniform scalars, so that their values under $\mathsf{H}^{\circledast}$ are uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$. Each of these at most $q_h + 1$ replacements costs at most δ_{512}, because in both distributions an answer, conditioned on its reduction, is uniform on the preimages of that reduction. The forging probability thus drops by at most η (*Math Guide*, §"Statistical distance", Theorem "Properties of statistical distance", part (3)).

(2) *Embedding.* The reduction receives a discrete-logarithm challenge $Y = [x]\, G$ with x uniform and gives $\mathcal{A}$ the key $\mathsf{ak}^{\mathbb{P}} := Y$.

(3) *Signing.* It answers the i-th query (α_i, M_i) without x, under $\mathsf{rk}_i := \mathsf{ak}^{\mathbb{P}} + [\alpha_i]\, G$, by the simulator of the *Crypto Guide*'s Lemma "Signature simulation" (§"Security in the random oracle model and the forking lemma"): it draws u_i uniform on the 64-byte strings and S_i uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$, sets $c_i := \mathrm{ToScalar}(u_i)$ and $R_i := [S_i]\, G - [c_i]\, \mathsf{rk}_i$, programs H at $\underline{R_i} \,\|\, \underline{\mathsf{rk}_i} \,\|\, M_i$ to u_i, aborting if that entry is already defined, and returns $\sigma_i := \underline{R_i} \,\|\, \mathsf{I2LEOSP}_{256}(S_i)$. After step (1) a real answer has a uniform nonce r_i and, at a fresh input, the challenge $c_i = \mathrm{ToScalar}(u_i)$ for a uniform u_i; then $S_i = r_i + c_i(x + \alpha_i)$ is uniform given u_i, and $R_i = [S_i]\, G - [c_i]\, \mathsf{rk}_i$, which is the simulated distribution (*Crypto Guide*, §"The Schnorr identification protocol", Theorem "Special honest-verifier zero knowledge"). The point R_i is uniform, so the reduction aborts with probability at most $q_s(q_h + q_s)/p_{\mathsf{Vesta}}$, as in the lemma. Key prefixing places $\underline{\mathsf{rk}_i}$ in every programmed input: entries programmed under distinct keys have distinct inputs, and the simulation needs no relation among the rk_i.

(4) *Critical query.* Let $(\mathsf{rk}^*, M^*, \sigma^*)$ be a forgery, with $\sigma^* = \bar{R}^* \,\|\, \bar{S}^*$ decoding to $R^* \neq \bot$ and $S^* < p_{\mathsf{Vesta}}$. If $\mathsf{rk}^* = \mathcal{O}$, the scalar $\mathsf{rsk}^* := 0$ satisfies (i), and in (ii) $x = -\alpha^*$; the reduction outputs these. Let $\mathsf{rk}^* \neq \mathcal{O}$. The validation input $\bar{R}^* \,\|\, \underline{\mathsf{rk}^*} \,\|\, M^*$ is not a programmed entry. Its first 32 bytes, its next 32 bytes and its remainder would otherwise give $\bar{R}^* = \underline{R_i}$, $\mathsf{rk}^* = \mathsf{rk}_i$ and $M^* = M_i$ for some i, the second by injectivity of $P \mapsto \underline{P}$, and the challenge would be c_i. Then $R^* = R_i$, and (4) gives $[S^*]\, G = R_i + [c_i]\, \mathsf{rk}_i = [S_i]\, G$. Since G generates the group, of prime order p_{Vesta}, and $S^*, S_i < p_{\mathsf{Vesta}}$, this forces $S^* = S_i$, so $\sigma^* = \sigma_i$ and the triple is an oracle answer. The input was queried by $\mathcal{A}$, except with probability at most $1/p_{\mathsf{Vesta}}$: otherwise its value under $\mathsf{H}^{\circledast}$ is a

uniform scalar drawn at validation, and, since $\mathsf{rk}^* \neq \mathcal{O}$ generates the group, at most one scalar satisfies (4). This guessing term is part of the term Q/p_{Vesta} of the cited theorem.

(5) *Fork.* The reduction wraps $\mathcal{A}$ as the algorithm of the *Crypto Guide*'s Theorem "General forking lemma" (§"Security in the random oracle model and the forking lemma"), with challenge set $\mathbb{F}_{p_{\mathsf{Vesta}}}$, Q prepared challenges $c_1, \dots, c_Q$, and coins that include those of $\mathcal{A}$ and of the simulator. The wrapper answers the j-th distinct unprogrammed query to H, of $\mathcal{A}$ or of validation, with a uniform preimage of c_j under ToScalar, and returns the index of the critical query with $\mathcal{A}$'s output. The two runs of the forking algorithm share the coins and the challenges before that index, so $\mathcal{A}$ makes the same critical query in both, with the same $\bar{R}^*$, the same rk^*, since key prefixing makes rk^* part of the critical input, and the same M^*. In (ii) they also share α^*, which rk^* determines because G generates the group. The runs receive distinct challenges $c \neq c'$ at the critical index and return responses S^* and S'^*.

(6) *Extraction.* The transcripts (R^*, c, S^*) and (R^*, c', S'^*) are accepting for rk^* with $c \neq c'$, so the *Crypto Guide*'s Theorem "2-special soundness" (§"The Schnorr identification protocol") gives

$$\mathsf{rsk}^* = (S^* - S'^*)(c - c')^{-1},$$

the discrete logarithm of rk^*, which is (i). For (ii) the reduction outputs $\mathsf{rsk}^* - \alpha^* = x$. The probability bound follows as in the cited theorem, from the forking bound

$$frk \geq acc\left(\frac{acc}{Q} - \frac{1}{p_{\mathsf{Vesta}}}\right),$$

with acc at least $\epsilon - \eta$ less the abort and guessing terms of steps (3) and (4). □

Remark 7.13 (Uses of Proposition 7.12). Proposition 7.12 holds for every message: it places no condition on M^* beyond the freshness of the triple. Theorem 12.9 applies part (i), and Proposition 11.10 applies part (ii) to a signature on a new digest.

8 Value commitments and the binding signature

This section constructs the net value commitment of an Action, defines the transparent transaction value pool, the balancing value of each pool and the fee, and constructs the binding signature. Proposition 8.10 keeps apart an algebraic identity, under which the binding validating key is a known multiple of the randomness base exactly when a pool balances modulo p_{Vesta}, and a knowledge statement, under which a valid binding signature forces balance modulo p_{Vesta}.

8.1 Value commitments

Remark 8.1 (Requirement R4). Requirement R4 of "Requirements on a shielded payment" (§1.4) is met through the additive homomorphism of the Pedersen commitment (*Crypto Guide*, §"The Pedersen commitment", Theorem "Additive homomorphism"). Each Action publishes a commitment to its net value; the group sum of a bundle's commitments commits to the sum of the net values under the sum of the trapdoors, and a verifier compares that sum with the bundle's public balancing value (§8.2) without learning any single value.

Definition 8.2 (Net value commitment)**.** Let an Action consume a note of value v^{old} and create a note of value v^{new}, both in $\{0, \dots, 2^{64}-1\}$, the value type of Definition "Note" (§4.1); a dummy side has value 0 (§4.4). The *net value* of the Action is the integer

$$v^{\mathsf{net}} := v^{\mathsf{old}} - v^{\mathsf{new}} \in \{-2^{64} + 1, \dots, 2^{64} - 1\}.$$

The *value base* and the *randomness base* are

$$V^{\mathsf{Orchard}} := \mathsf{GroupHash}(\texttt{z.cash:Orchard-cv}, \texttt{v}),$$
$$R^{\mathsf{Orchard}} := \mathsf{GroupHash}(\texttt{z.cash:Orchard-cv}, \texttt{r}),$$

the value-commitment rows of Table 2. For a trapdoor $\mathsf{rcv} \in \mathbb{F}_{p_{\mathsf{Vesta}}}$, the *net value commitment* of the Action is the point

$$\mathsf{cv}^{\mathsf{net}} := \mathsf{ValueCommit}^{\mathsf{Orchard}}_{\mathsf{rcv}}(v^{\mathsf{net}}) := [v^{\mathsf{net}} \bmod p_{\mathsf{Vesta}}]\, V^{\mathsf{Orchard}} + [\mathsf{rcv}]\, R^{\mathsf{Orchard}} \in \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$$

(protocol specification, §"Homomorphic Pedersen commitments (Sapling and Orchard)").

Both bases have order p_{Vesta}: the protocol specification types them as points of order p_{Vesta} (§"Balance and Binding Signature (Orchard)"), a set that excludes $\mathcal{O}$ (§"Pallas and Vesta"). The specification defines $\mathsf{ValueCommit}^{\mathsf{Orchard}}$ on the signed integers

$$\left\{-\frac{p_{\mathsf{Vesta}}-1}{2}, \dots, \frac{p_{\mathsf{Vesta}}-1}{2}\right\}$$

(§"Commitment"), and a signed integer enters the scalar multiplication as its residue modulo p_{Vesta} (§2.1). Reduction modulo p_{Vesta} is injective on this domain, since two of its elements differ by at most $p_{\mathsf{Vesta}} - 1$. The domain contains every net value, and every signed 64-bit integer, because $p_{\mathsf{Vesta}} > 2^{254}$ (§2.1) gives $(p_{\mathsf{Vesta}} - 1)/2 \geq 2^{253}$.

Construction 8.3 (Value-commitment trapdoor)**.** For each Action the author of the transaction draws rcv uniformly from $\mathbb{F}_{p_{\mathsf{Vesta}}}$, by $\mathsf{ValueCommit}^{\mathsf{Orchard}}.\mathsf{GenTrapdoor}$, independently of every other value (protocol specification, §"Sending Notes (Orchard)" and §"Homomorphic Pedersen commitments (Sapling and Orchard)"). Unlike rcm, ψ and esk, which are derived from the note seed rseed (§4.3), the trapdoor rcv is derived from no note, and no note field, note plaintext (§10.1) or transaction field (§11.2) contains it. The author retains every rcv of the bundle; together they determine the binding signing key (§8.3).

The specification types the trapdoor as an integer in $\{0, \dots, 2^{255} - 1\}$ (§"Commitment"). As for rcm (§4.1), this volume types it in $\mathbb{F}_{p_{\mathsf{Vesta}}}$, the set that GenTrapdoor samples, since rcv acts only as the scalar of [rcv].

Lemma 8.4 (Hiding and binding of net value commitments)**.** *With $G := V^{\mathsf{Orchard}}$ and $H := R^{\mathsf{Orchard}}$, the map $\mathsf{ValueCommit}^{\mathsf{Orchard}}$ is the Pedersen commitment of the* Crypto Guide *(§"The Pedersen commitment", Construction "Pedersen commitment"), with messages the residues modulo p_{Vesta}.*

(*i*) *Perfect hiding: for* rcv *uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$ and independent of v^{net}, the commitment $\mathsf{cv}^{\mathsf{net}}$ is uniform on $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$, whatever v^{net} is.*

(*ii*) *Computational binding: under Assumptions 2.22 and 2.8, no efficient algorithm outputs, except with negligible probability, pairs (v, rcv) and (v', rcv') of an integer and a trapdoor with $v \not\equiv v' \pmod{p_{\mathsf{Vesta}}}$ and $[v]\, V^{\mathsf{Orchard}} + [\mathsf{rcv}]\, R^{\mathsf{Orchard}} = [v']\, V^{\mathsf{Orchard}} + [\mathsf{rcv}']\, R^{\mathsf{Orchard}}$. On the domain of $\mathsf{ValueCommit}^{\mathsf{Orchard}}$ distinct integers are distinct residues, so the same bound excludes openings of one point to two distinct net values.*

Proof. (i) This is the *Crypto Guide*'s Theorem "Perfect hiding" (§"The Pedersen commitment"). Since $R^{\mathsf{Orchard}} \neq \mathcal{O}$ and the group has prime order, $\mathsf{rcv} \mapsto [\mathsf{rcv}]\, R^{\mathsf{Orchard}}$ is a bijection from $\mathbb{F}_{p_{\mathsf{Vesta}}}$

onto $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$; the uniform point $[\mathsf{rcv}]\,R^{\mathsf{Orchard}}$ stays uniform under translation by the fixed point $[v^{\mathsf{net}} \bmod p_{\mathsf{Vesta}}]\,V^{\mathsf{Orchard}}$. (ii) The double-opening algebra of the *Crypto Guide*'s Theorem "Computational binding under DLog" (same section) gives

$$[v - v']\,V^{\mathsf{Orchard}} - [\mathsf{rcv}' - \mathsf{rcv}]\,R^{\mathsf{Orchard}} = \mathcal{O},$$

a relation with non-zero coefficient $v - v'$ on V^{Orchard}. The two bases are values of GroupHash at distinct inputs; under Assumption 2.8 they replace the independent generators that the cited construction samples, and the lemma on relations among fixed generators (§2.4) excludes the relation except with negligible probability. The specification requires computational binding and states that the scheme is unconditionally hiding (protocol specification, §"Homomorphic Pedersen commitments (Sapling and Orchard)"). □

8.2 The balancing value

Definition 8.5 (Transparent transaction value pool)**.** *Transparent inputs* and *transparent outputs* are the cleartext value transfers of a transaction outside every shielded pool, with public amounts. Each transaction has a *transparent transaction value pool*: its transparent inputs add their values to it, and its transparent outputs remove theirs. The balancing value of each shielded pool, defined next for the two pools of the Orchard protocol, is also added to it (protocol specification, §"Transactions and Treestates").

Definition 8.6 (Balancing value)**.** For each pool of the Orchard protocol, the Orchard pool or the Ironwood pool, the *balancing value* $v^{\mathsf{balance}}_{pool}$ of a transaction is its declared net value, in zatoshi, of the notes of the pool that it consumes minus those that it creates: a signed integer in $\{-2^{63}, \ldots, 2^{63} - 1\}$, encoded in cleartext as the two's-complement 64-bit field `valueBalanceOrchard` or `valueBalanceIronwood`, of type `int64`. It is 0 when the transaction has no Action of the pool, and the field is then absent. A positive $v^{\mathsf{balance}}_{pool}$ takes net value out of the pool and adds it to the transparent transaction value pool; a negative one takes value from the transparent transaction value pool into the pool (protocol specification, §"Balance and Binding Signature (Orchard)"; ZIP 229, "Transaction Format").

In a correctly constructed transaction, $v^{\mathsf{balance}}_{pool}$ equals the sum of the net values v^{net} of the pool's Actions. A verifier cannot check this equation directly, since each v^{net} is hidden in its $\mathsf{cv}^{\mathsf{net}}$; the binding signature (§8.3) enforces it.

The protocol specification, §"Transactions and Treestates", states the consensus rule: the value remaining in the transparent transaction value pool must be non-negative. For a transaction other than a coinbase transaction (Definition "Coinbase transaction", §11.2) the remaining value is the *transaction fee* (ibid.); no transaction field encodes it (ZIP 229, "Transaction Format" and "Non-requirements"). The pool of a coinbase transaction also receives, as implicit inputs, the block subsidy and the fees of the block's other transactions, and its outputs must consume all of the value in it (protocol specification, §"Transactions and Treestates" and §"Transaction Consensus Rules"). ZIP 317, "Fee calculation", specifies the *conventional fee*, which wallets SHOULD pay and which is not a consensus requirement:

$$\begin{gathered} \mathit{marginal_fee} \cdot \max(\mathit{grace_actions}, \mathit{logical_actions}) \text{ zatoshi}, \\ \mathit{marginal_fee} = 5000, \qquad \mathit{grace_actions} = 2, \end{gathered}$$

where *logical_actions* is a sum of per-component contributions, in which each Ironwood-pool Action contributes one (the contribution *nActionsIronwood*). The fee of the worked payment is computed in Example 11.16 (§11.5).

Remark 8.7 (Disclosure by the balancing value). The balancing value is cleartext, so a negative one discloses the amount that enters the pool; the field names no receiving note or address.

8.3 The binding signature

Fix a pool of the Orchard protocol and a transaction whose bundle for that pool has $n \geq 1$ Actions, with net values v_i^{net}, trapdoors rcv_i and net value commitments $\mathsf{cv}_i^{\mathsf{net}}$ $(i = 1, \dots, n)$, and write v^{balance} for the pool's balancing value.

Definition 8.8 (Binding signing key). The *binding signing key* of the bundle is

$$\mathsf{bsk} := \mathsf{rcv}_1 + \cdots + \mathsf{rcv}_n \in \mathbb{F}_{p_{\mathsf{Vesta}}}$$

(protocol specification, §"Balance and Binding Signature (Orchard)"). By the additive homomorphism (*Crypto Guide*, §"The Pedersen commitment", Theorem "Additive homomorphism"),

$$\sum_{i=1}^{n} \mathsf{cv}_i^{\mathsf{net}} = \Big[\sum_{i=1}^{n} v_i^{\mathsf{net}}\Big] V^{\mathsf{Orchard}} + [\mathsf{bsk}]\, R^{\mathsf{Orchard}}, \tag{6}$$

the integer sum entering as its residue modulo p_{Vesta}.

Definition 8.9 (Binding validating key). The verifier computes, from public fields only, the *binding validating key*

$$\mathsf{bvk} := \Big(\sum_{i=1}^{n} \mathsf{cv}_i^{\mathsf{net}}\Big) - [v^{\mathsf{balance}}]\, V^{\mathsf{Orchard}},$$

where the subtracted point is $\mathsf{ValueCommit}_0^{\mathsf{Orchard}}(v^{\mathsf{balance}})$, the commitment with zero trapdoor, and v^{balance}, an element of the domain of $\mathsf{ValueCommit}^{\mathsf{Orchard}}$, enters as its residue modulo p_{Vesta}. No transaction field encodes bvk (protocol specification, §"Balance and Binding Signature (Orchard)").

An *opening* of a net value commitment $\mathsf{cv}_i^{\mathsf{net}}$ is a pair (v_i', r_i') of an integer v_i' and a scalar $r_i' \in \mathbb{F}_{p_{\mathsf{Vesta}}}$ with $\mathsf{cv}_i^{\mathsf{net}} = [v_i' \bmod p_{\mathsf{Vesta}}]\, V^{\mathsf{Orchard}} + [r_i']\, R^{\mathsf{Orchard}}$. In the next proposition a *binding signature* under a point X is a signature valid under X in the binding-signature instance of RedPallas, with base R^{Orchard} (§7.1); Construction "Binding signature" below fixes its use.

Proposition 8.10 (Balance and key knowledge). *For the bundle fixed above:*

(*i*) *Identity. If* $\mathsf{cv}_i^{\mathsf{net}} = \mathsf{ValueCommit}_{\mathsf{rcv}_i}^{\mathsf{Orchard}}(v_i^{\mathsf{net}})$ *for every* i *and* $\mathsf{bsk} := \sum_i \mathsf{rcv}_i$, *then*

$$\mathsf{bvk} - [\mathsf{bsk}]\, R^{\mathsf{Orchard}} = \Big[\sum_{i=1}^{n} v_i^{\mathsf{net}} - v^{\mathsf{balance}}\Big] V^{\mathsf{Orchard}};$$

hence $\mathsf{bvk} = [\mathsf{bsk}]\, R^{\mathsf{Orchard}}$ *if and only if* $\sum_i v_i^{\mathsf{net}} \equiv v^{\mathsf{balance}} \pmod{p_{\mathsf{Vesta}}}$.

(*ii*) *Knowledge. Let* (v_i', r_i') *be an opening of* $\mathsf{cv}_i^{\mathsf{net}}$ *for every* i, *and let* $b \in \mathbb{F}_{p_{\mathsf{Vesta}}}$ *satisfy* $\mathsf{bvk} =$

> $[b]\,R^{\mathsf{Orchard}}$. *If* $D := \sum_i v'_i - v^{\mathsf{balance}}$ *is non-zero modulo* p_{Vesta}, *then*
>
> $$\Big(b - \sum_{i=1}^{n} r'_i\Big) D^{-1} \bmod p_{\mathsf{Vesta}}$$
>
> *is the discrete logarithm of* V^{Orchard} *to the base* R^{Orchard}, *a relation between two values of* GroupHash.
>
> *Consequently, under Assumptions 2.22, 2.8 and 7.3, an efficient party that outputs the net value commitments and the balancing value of a bundle, openings* (v'_i, r'_i) *of every* $\mathsf{cv}_i^{\mathsf{net}}$ *and a binding signature under the resulting* bvk *on a message of its choice satisfies* $\sum_i v'_i \equiv v^{\mathsf{balance}} \pmod{p_{\mathsf{Vesta}}}$, *except with negligible probability. The proposition holds for every message.*

Proof. (i) Expanding each $\mathsf{cv}_i^{\mathsf{net}}$ by Definition 8.2 and applying the additive homomorphism gives (6); subtracting $[v^{\mathsf{balance}}]\,V^{\mathsf{Orchard}}$ gives the identity. Since V^{Orchard} has prime order p_{Vesta}, the point $[c]\,V^{\mathsf{Orchard}}$ is $\mathcal{O}$ if and only if $c \equiv 0 \pmod{p_{\mathsf{Vesta}}}$.

(ii) By the homomorphism, $\mathsf{bvk} = [D]\,V^{\mathsf{Orchard}} + [\sum_i r'_i]\,R^{\mathsf{Orchard}}$. Equating with $[b]\,R^{\mathsf{Orchard}}$ gives

$$[D]\,V^{\mathsf{Orchard}} = \Big[b - \sum_{i=1}^{n} r'_i\Big]\,R^{\mathsf{Orchard}},$$

and D is invertible modulo the prime p_{Vesta}. This is Part 2 of the proof of the *Crypto Guide*'s Theorem "The binding signature is a proof of knowledge of $\log_P B$" (§"Binding signatures as a proof of knowledge of a discrete logarithm"), with $P = R^{\mathsf{Orchard}}$ and the balancing value folded into the key as that section describes after the theorem.

Consequence. Let an efficient party $\mathcal{A}$ output, with probability ϵ, a bundle's net value commitments and balancing value, openings of every commitment, a message and a binding signature on it under the resulting bvk, with D non-zero modulo p_{Vesta}; this event is efficiently decidable from the output. If $\mathsf{bvk} = \mathcal{O}$, then $b := 0$ satisfies $\mathsf{bvk} = [b]\,R^{\mathsf{Orchard}}$. Otherwise Part 1 of the proof of the same theorem, special soundness with forking in the random-oracle model for H (Assumption 7.3), rewinds $\mathcal{A}$ to two accepting transcripts with a common first component and distinct challenges and extracts b with $\mathsf{bvk} = [b]\,R^{\mathsf{Orchard}}$. In this volume the extraction is steps (1) and (4) to (6) of the proof of Proposition 7.12, with no signing queries ($q_s = 0$), the base R^{Orchard} in place of $G^{\mathsf{SpendAuth}}$ and bvk in place of rk^*. Key prefixing places the encoding of bvk in the critical query, so both runs sign under the same bvk, and the openings and the balancing value of the first run satisfy the hypothesis of part (ii) for it. If ϵ is non-negligible, the extraction succeeds with non-negligible probability, and part (ii) then outputs the discrete logarithm of V^{Orchard} to the base R^{Orchard}: a non-trivial relation between values of GroupHash at distinct inputs, which the lemma on relations among fixed generators (§2.4) excludes except with negligible probability under Assumptions 2.8 and 2.22. No argument uses the value of the message, and no concrete bound is claimed. The security argument of the protocol specification, §"Balance and Binding Signature (Orchard)", is the deployed counterpart. □

The openings are obtained, and the congruence is promoted to an equality of integers, in Lemma 12.3 and Theorem 12.4 (§12.1).

Construction 8.11 (Binding signature)**.** The binding signature scheme $\mathsf{BindingSig}^{\mathsf{Orchard}}$ is the binding-signature instance of RedPallas (§7.1): RedPallas on the base R^{Orchard}, without key re-randomisation; its base differs from the spend-authorisation base $G^{\mathsf{SpendAuth}}$. The author of the transaction computes bsk from the trapdoors it retains and signs with it; the verifier recomputes bvk from the public fields and validates the signature under it (protocol specification, §"Binding Signature (Sapling and Orchard)" and §"Balance and Binding Signature

(Orchard)"). The signed message is the signature digest SigHash constructed in "Transaction digests and signatures" (§11.3); which transaction data the signatures fix is stated there, in Proposition 11.10.

For an honestly constructed bundle with $\sum_i v_i^{\mathsf{net}} = v^{\mathsf{balance}}$, Proposition 8.10(i) gives $\mathsf{bvk} = [\mathsf{bsk}]\, R^{\mathsf{Orchard}}$, so $(\mathsf{bsk}, \mathsf{bvk})$ is a key pair of the instance and the honest signature is accepted (§7.1). Under Assumption 7.3 a valid signature is a proof of knowledge of the discrete logarithm of bvk to the base R^{Orchard} (*Crypto Guide*, §"Binding signatures as a proof of knowledge of a discrete logarithm", Theorem "The binding signature is a proof of knowledge of $\log_P B$"), so, for a party that also outputs openings of every net value commitment, the consequence stated in Proposition 8.10, under Assumptions 2.22, 2.8 and 7.3, makes it certify balance modulo p_{Vesta} without taking any net value as input: it is computed from bsk, the message and fresh randomness. What the public data of a bundle hide is stated in Theorem 12.12 (§12.4).

The construction runs once for each pool of the Orchard protocol. A transaction carries a balancing value and a binding signature, `bindingSigOrchard` or `bindingSigIronwood`, for each pool whose bundle has at least one Action. Each pool's bvk sums only that pool's net value commitments and subtracts only that pool's balancing value; both pools use $\mathsf{ValueCommit}^{\mathsf{Orchard}}$ with the bases V^{Orchard} and R^{Orchard}, and $\mathsf{BindingSig}^{\mathsf{Orchard}}$. The consensus rule of the protocol specification, §"Transaction Consensus Rules" (ZIP 229, "Consensus Rules"), is: if a pool's bundle has Actions, its binding signature must be a valid signature of SigHash under that pool's bvk. Validation rejects a non-canonical encoding of the signature's point component (§7.1; protocol specification, §"RedDSA, RedJubjub, and RedPallas").

9 The Action statement and its proof

This section states the relation that is proved for every Action. It fixes the bundle flags that enter the relation as public inputs, states the Action statement over the objects constructed in §2 to §8, pairs each condition with the requirement it serves and with the check outside the proof that completes it, and states the proof system together with the two assumptions on the Action proof that "Security" (§12) uses.

9.1 Bundle flags

Each bundle carries one *flags byte*, present exactly when the bundle has at least one Action: flagsOrchard for the Orchard-pool bundle of a version 5 or version 6 transaction, and flagsIronwood for the Ironwood-pool bundle of a version 6 transaction (ZIP 229, "Transaction Format"). Bits are numbered from the least significant, bit 0. In both bytes bit 0 is enableSpends and bit 1 is enableOutputs. The consensus rules of ZIP 229, "Consensus Rules", fix the remaining bits:

1. in flagsIronwood, bit 2 is enableCrossAddress, and bits 3 to 7 are reserved and must be 0;
2. in flagsOrchard, bits 2 to 7 are reserved and must be 0, in version 5 and version 6 transactions alike.

The format table of ZIP 229, "Transaction Format", lists enableCrossAddress at bit 2 of flagsOrchard; this contradicts the consensus rules of the same ZIP, which reserve that bit, and the volume follows the consensus rules. The two texts differ in the role of the bit, not in its value in a valid transaction: ZIP 258, "Consensus rules from NU6.3 activation", requires every Orchard-pool Action to be created with enableCrossAddress = 0.

Definition 9.1 (Flag values and expanded receiver)**.** Let a bundle have flags byte $f = \sum_{i=0}^{7} f_i\, 2^i$

with $f_i \in \{0,1\}$. Its *flag values*, elements of $\{0,1\}$, are

$$\mathsf{enableSpends} := f_0, \qquad \mathsf{enableOutputs} := f_1,$$
$$\mathsf{enableCrossAddress} := \begin{cases} f_2 & \text{for an Ironwood-pool bundle,} \\ 0 & \text{for an Orchard-pool bundle,} \end{cases}$$

and $\mathsf{disableCrossAddress} := 1 - \mathsf{enableCrossAddress}$. For an Orchard-pool bundle bit 2 is reserved and 0, so every Orchard-pool Action has $\mathsf{enableCrossAddress} = 0$ by encoding. The *expanded receiver* of a payment address $(d, \mathsf{pk_d})$ is the pair $(\mathsf{g_d}, \mathsf{pk_d})$ with $\mathsf{g_d} = \mathsf{DiversifyHash}(d)$, the diversified base of "Diversified addresses" (§3.3); the expanded receiver of a note is that of its address. The flags have the following meaning (protocol specification, §"Action Descriptions"):

- the value $\mathsf{enableSpends} = 1$ enables non-zero-valued spends in the bundle's Actions;
- the value $\mathsf{enableOutputs} = 1$ enables non-zero-valued outputs in the bundle's Actions;
- the value $\mathsf{enableCrossAddress} = 1$ enables the expanded receiver of an Action's created note to differ from that of the note the same Action consumes.

The Action statement takes disableCrossAddress as its eighth primary input (same section). No normative text fixes the position of this input in the encoded instance, and none is asserted here.

ZIP 229 calls $\mathsf{enableCrossAddress} = 1$ the normal case for Ironwood-pool Actions (Rationale, "`enableCrossAddress` polarity"); the value is not required. No consensus rule requires bit 2 of flagsIronwood to be 1. A bundle in which that bit is 0 is admitted, and each of its Actions is then restricted, by condition A10 of Definition 9.2, to create its note at the expanded receiver of the note it consumes.

The flags are properties of the bundle. Encoded once in its flags byte, they are the same for every Action of the bundle (protocol specification, §"Action Descriptions", note on the encoding of these components once per pool). The verifier reads them from the bundle and supplies enableSpends, enableOutputs and disableCrossAddress as public inputs of each Action; they are not computed inside the proof. They enter the relation only through conditions A8, A9 and A10 of Definition 9.2.

9.2 The Action statement

The Action statement is a relation in the sense of the *Crypto Guide*'s Definition "Language, relation, witness" (§"Languages, relations, and witnesses"), with the primary input as instance and the auxiliary input as witness, each a tuple of the typed objects below, encoded as bit strings by the encodings of §2.1. Every condition is an equation over objects constructed in §2 to §8.

Definition 9.2 (Orchard Action statement)**.** In this definition $\mathcal{E}$ abbreviates the Pallas group $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$, with identity $\mathcal{O}$, and $\mathcal{E}^* := \mathcal{E} \setminus \{\mathcal{O}\}$. For $P = (x, y) \in \mathcal{E}^*$ let $x(P) := x$ and $y(P) := y$, and let $x(\mathcal{O}) := y(\mathcal{O}) := 0$, so that $\mathsf{Extract}_{\mathbb{P}}(P) = x(P)$ for every $P \in \mathcal{E}$ (protocol specification, §"Coordinate Extractor for Pallas"). An integer that multiplies a point acts as its residue modulo p_{Vesta} (§2.1).

The relation $\mathcal{R}_{\mathsf{Action}}$ is the set of pairs $(\mathsf{pub}, \mathsf{aux})$ of a *primary input*

$$\mathsf{pub} = (rt,\ \mathsf{cv^{net}},\ \mathsf{nf^{old}},\ \mathsf{rk},\ \mathsf{cmx},\ \mathsf{enableSpends},\ \mathsf{enableOutputs},\ \mathsf{disableCrossAddress})$$

and an *auxiliary input*

$$\mathsf{aux} = (\mathsf{path}, \mathsf{pos}, \mathsf{g_d}^{\mathsf{old}}, \mathsf{pk_d}^{\mathsf{old}}, v^{\mathsf{old}}, \rho^{\mathsf{old}}, \psi^{\mathsf{old}}, \mathsf{rcm}^{\mathsf{old}}, \mathsf{cm}^{\mathsf{old}}, \alpha, \mathsf{ak}^{\mathbb{P}}, \mathsf{nk}, \mathsf{r_{ivk}}, \mathsf{g_d}^{\mathsf{new}}, \mathsf{pk_d}^{\mathsf{new}}, v^{\mathsf{new}}, \psi^{\mathsf{new}}, \mathsf{rcm}^{\mathsf{new}}, \mathsf{rcv}, \mathsf{c})$$

that have the following types and satisfy conditions A1 to A10 below.

- Primary input: $rt, \mathsf{nf}^{\mathsf{old}}, \mathsf{cmx} \in \{0, \dots, p_{\mathsf{Pallas}} - 1\}$, read as elements of $\mathbb{F}_{p_{\mathsf{Pallas}}}$; $\mathsf{cv}^{\mathsf{net}}, \mathsf{rk} \in \mathcal{E}$; and $\mathsf{enableSpends}, \mathsf{enableOutputs}, \mathsf{disableCrossAddress} \in \{0, 1\}$.
- Auxiliary input: $\mathsf{path} = (s_0, \dots, s_{31})$ with every $s_t \in \{0, \dots, p_{\mathsf{Pallas}} - 1\}$; $\mathsf{pos} \in \{0, \dots, 2^{32} - 1\}$; $\mathsf{g_d}^{\mathsf{old}}, \mathsf{pk_d}^{\mathsf{old}}, \mathsf{g_d}^{\mathsf{new}}, \mathsf{pk_d}^{\mathsf{new}}, \mathsf{ak}^{\mathbb{P}} \in \mathcal{E}^*$; $v^{\mathsf{old}}, v^{\mathsf{new}} \in \{0, \dots, 2^{64} - 1\}$; $\rho^{\mathsf{old}}, \psi^{\mathsf{old}}, \psi^{\mathsf{new}}, \mathsf{nk} \in \mathbb{F}_{p_{\mathsf{Pallas}}}$; $\mathsf{cm}^{\mathsf{old}} \in \mathcal{E}$; $\mathsf{rcm}^{\mathsf{old}}, \mathsf{rcm}^{\mathsf{new}}, \mathsf{r_{ivk}}, \mathsf{rcv}, \alpha \in \{0, \dots, 2^{255} - 1\}$; and $\mathsf{c} = (c_0, \dots, c_{63}) \in \{0, 1\}^{64}$.

Let $\rho^{\mathsf{new}} := \mathsf{nf}^{\mathsf{old}} \in \mathbb{F}_{p_{\mathsf{Pallas}}}$, let $\delta := \mathsf{disableCrossAddress}$, and let

$$M^{\mathsf{old}} := (\mathsf{g_d}^{\mathsf{old}})^\star \parallel (\mathsf{pk_d}^{\mathsf{old}})^\star \parallel \mathrm{LE}_{64}(v^{\mathsf{old}}) \parallel \mathrm{LE}_{255}(\rho^{\mathsf{old}}) \parallel \mathrm{LE}_{255}(\psi^{\mathsf{old}}),$$
$$M^{\mathsf{new}} := (\mathsf{g_d}^{\mathsf{new}})^\star \parallel (\mathsf{pk_d}^{\mathsf{new}})^\star \parallel \mathrm{LE}_{64}(v^{\mathsf{new}}) \parallel \mathrm{LE}_{255}(\rho^{\mathsf{new}}) \parallel \mathrm{LE}_{255}(\psi^{\mathsf{new}}),$$

the 1086-bit messages of Definition 4.4, with the diversified bases witnessed. The conditions, all equations in $\mathbb{F}_{p_{\mathsf{Pallas}}}$ or in $\mathcal{E}$, are:

A1. Old note commitment integrity: $\mathsf{NoteCommit}_{\mathsf{rcm}^{\mathsf{old}}}(M^{\mathsf{old}}) \in \{\mathsf{cm}^{\mathsf{old}}, \perp\}$.

A2. New note commitment integrity: $\mathsf{Extract}^{\perp}_{\mathbb{P}}(\mathsf{NoteCommit}_{\mathsf{rcm}^{\mathsf{new}}}(M^{\mathsf{new}})) \in \{\mathsf{cmx}, \perp\}$.

A3. Merkle path validity: $v^{\mathsf{old}} \cdot (root - rt) = 0$ or some evaluation of SinsemillaHash in the fold, the root included, returns $\perp$. Here $root := a_{32}$ is the result of the fold (2) of Proposition 5.9 with $a_0 := \mathsf{Extract}_{\mathbb{P}}(\mathsf{cm}^{\mathsf{old}})$, siblings $s_0, \dots, s_{31}$ and the bits of pos as direction bits, in which the evaluation of MerkleCRH at level t encodes its left argument x as $\mathrm{LE}_{255}(\tilde{x})$, where $\tilde{x} := x + c_{2t}\, p_{\mathsf{Pallas}}$ if this integer is below 2^{255} and $\tilde{x} := x$ otherwise, and its right argument likewise with c_{2t+1}.

A4. Value commitment integrity:

$$\mathsf{cv}^{\mathsf{net}} = \mathsf{ValueCommit}^{\mathsf{Orchard}}_{\mathsf{rcv}}(v^{\mathsf{old}} - v^{\mathsf{new}}) = [v^{\mathsf{old}} - v^{\mathsf{new}}]\, V^{\mathsf{Orchard}} + [\mathsf{rcv}]\, R^{\mathsf{Orchard}}.$$

A5. Nullifier integrity: $\mathsf{nf}^{\mathsf{old}} = \mathsf{DeriveNullifier}_{\mathsf{nk}}(\rho^{\mathsf{old}}, \psi^{\mathsf{old}}, \mathsf{cm}^{\mathsf{old}})$, that is,

$$\mathsf{nf}^{\mathsf{old}} = \mathsf{Extract}_{\mathbb{P}}([(\mathsf{PRF}^{\mathsf{nfOrchard}}_{\mathsf{nk}}(\rho^{\mathsf{old}}) + \psi^{\mathsf{old}}) \bmod p_{\mathsf{Pallas}}]\, K^{\mathsf{Orchard}} + \mathsf{cm}^{\mathsf{old}}).$$

A6. Spend authority: $\mathsf{rk} = \mathsf{ak}^{\mathbb{P}} + [\alpha]\, G^{\mathsf{SpendAuth}}$.

A7. Diversified address integrity: with $\mathsf{ivk} := \mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r_{ivk}}}(\mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}}), \mathsf{nk})$, either $\mathsf{ivk} = \perp$ or $\mathsf{pk_d}^{\mathsf{old}} = [\mathsf{ivk}]\, \mathsf{g_d}^{\mathsf{old}}$.

A8. Enable spend flag: $v^{\mathsf{old}} \cdot (1 - \mathsf{enableSpends}) = 0$.

A9. Enable output flag: $v^{\mathsf{new}} \cdot (1 - \mathsf{enableOutputs}) = 0$.

A10. Cross-address restriction: the four equations

$$\delta \cdot (x(\mathsf{g_d}^{\mathsf{old}}) - x(\mathsf{g_d}^{\mathsf{new}})) = 0, \qquad \delta \cdot (y(\mathsf{g_d}^{\mathsf{old}}) - y(\mathsf{g_d}^{\mathsf{new}})) = 0,$$
$$\delta \cdot (x(\mathsf{pk_d}^{\mathsf{old}}) - x(\mathsf{pk_d}^{\mathsf{new}})) = 0, \qquad \delta \cdot (y(\mathsf{pk_d}^{\mathsf{old}}) - y(\mathsf{pk_d}^{\mathsf{new}})) = 0.$$

(Protocol specification, §"Action Statement (Orchard)", for the typed inputs and conditions A1 to A9; §"Action Descriptions" for the eighth primary input and the meaning of enableCrossAddress that A10 formalises.)

The anchor rt is the bundle's anchor ("Anchors", §5.2). The rule $\mathsf{rk} \neq \mathcal{O}$ of "Randomised validating keys" (§7.2) is checked outside the statement. The typing is part of the relation: the spend validating key $\mathsf{ak}^{\mathbb{P}}$, both diversified bases and both transmission keys are non-identity points (protocol specification, §"Action Statement (Orchard)", note on types). The trapdoors $\mathsf{rcm}^{\mathsf{old}}$, $\mathsf{rcm}^{\mathsf{new}}$, $\mathsf{r_{ivk}}$ and rcv and the randomiser α act only as scalars on points of order p_{Vesta}, and the statement does not require them to be below p_{Vesta} (same note). The created note's element ρ is not an auxiliary input; condition A2 fixes it. The relation is an NP-relation in the sense of the cited definition: its instances and witnesses have fixed length, and membership is decided by evaluating the typing and the ten conditions.

Points of the primary input enter the proof by their coordinates. The specification's encoding note fixes the first seven primary inputs as the nine elements

$$rt,\ x(\mathsf{cv}^{\mathsf{net}}),\ y(\mathsf{cv}^{\mathsf{net}}),\ \mathsf{nf}^{\mathsf{old}},\ x(\mathsf{rk}),\ y(\mathsf{rk}),\ \mathsf{cmx},\ \mathsf{enableSpends},\ \mathsf{enableOutputs}$$

of $\mathbb{F}_{p_{\mathsf{Pallas}}}$, in this order (protocol specification, §"Action Statement (Orchard)", note on the encoding of primary inputs); the convention $x(\mathcal{O}) = y(\mathcal{O}) = 0$ encodes $\mathsf{cv}^{\mathsf{net}} = \mathcal{O}$. The eighth input, disableCrossAddress, is one further element of $\mathbb{F}_{p_{\mathsf{Pallas}}}$ (§"Action Descriptions"); no normative text fixes its position, and the volume asserts none.

Conditions and the checks that complete them. Each condition is paired below with the requirement of "Requirements on a shielded payment" (§1.4) that it serves and, where one exists, with the check outside the proof that completes it; Table 5 collects the pairing and Figure 3 draws it.

Conditions A1 and A2 bind the consumed and the created note to the commitments $\mathsf{cm}^{\mathsf{old}}$ and cmx under NoteCommit, the Sinsemilla commitment with domain `z.cash:Orchard-NoteCommit` (Definition 4.4), whose binding is Proposition 4.5; they serve R1. Condition A2 fixes the created note's ρ to the nullifier that the same Action publishes for the note it consumes, the chaining rule of "Nullifier chaining" (§6.3). Its consequences for the uniqueness of ρ and for the Faerie Gold attack are proved in "Double-spend resistance" (Corollary 12.7).

Condition A3 serves R2. Since $0 \leq v^{\mathsf{old}} < 2^{64} < p_{\mathsf{Pallas}}$ (§2.1), the value v^{old} is zero in $\mathbb{F}_{p_{\mathsf{Pallas}}}$ only if it is the integer 0, and a field has no zero divisors. Hence, outside its $\perp$ case and with canonical encodings ($\mathsf{c} = 0$), A3 holds if and only if $v^{\mathsf{old}} = 0$ or $root = rt$, the specification's form: either $v^{\mathsf{old}} = 0$, or $(\mathsf{Extract}_{\mathbb{P}}(\mathsf{cm}^{\mathsf{old}}), \mathsf{pos}, \mathsf{path})$ is a valid path to rt in the sense of Proposition 5.9, with the layer-tagged hash of "The Merkle hash and the tree" (§5.1) and the paths of depth 32 of "Authentication paths" (§5.3). A consumed note of value zero is therefore exempt from membership. The exemption lets an Action whose consumed side is a dummy note ("Actions, bundles, and dummy notes", §4.4) create a real note without consuming a committed one. For $v^{\mathsf{old}} \neq 0$ the condition is membership of $\mathsf{Extract}_{\mathbb{P}}(\mathsf{cm}^{\mathsf{old}})$ under rt, up to the two relaxations below; Proposition 5.9 gives its soundness together with the anchor rule of "Anchors", the check outside the proof that completes A3.

The two relaxations are the circuit's: it does not check that the encoding of each layer's input is canonical, and its path check may pass when a hash on the path, the root included, is 0, which occurs only when SinsemillaHash returns $\perp$ (protocol specification, §"Merkle Path Validity", notes on Orchard; §"Action Statement (Orchard)", notes on the Merkle path validity check). Remark "$\perp$-weakened conditions" below treats the $\perp$ case. The bits c, absent from the specification's auxiliary input, record the encodings that the circuit witnesses. Since p_{Pallas} is odd and

has bit length 255 (§2.1), $2p_{\mathsf{Pallas}} > 2^{255}$, so the integers below 2^{255} congruent to an argument $x \in \{0, \dots, p_{\mathsf{Pallas}} - 1\}$ are x and, when it is below 2^{255}, $x + p_{\mathsf{Pallas}}$: the bits range over every encoding the circuit admits, and $\mathsf{c} = 0$ gives the canonical ones. Non-canonical encodings leave the extraction in the proof of Proposition 5.9(i) valid. At the least $t^* \geq 1$ at which the fold agrees with the genuine chain of that proof, the running values at level $t^* - 1$ differ as elements of $\mathbb{F}_{p_{\mathsf{Pallas}}}$, so no representative of the one equals a representative of the other, and the two 520-bit messages hashed there remain distinct with a common prefix. Parts (ii) and (iii) of the proposition therefore hold for the fold of A3.

Condition A4 serves R4. It is Definition 8.2: the integer $v^{\mathsf{old}} - v^{\mathsf{new}} \in \{-2^{64} + 1, \dots, 2^{64} - 1\}$, written as *sign* · *magnitude* with *magnitude* $\in \{0, \dots, 2^{64} - 1\}$ and *sign* $\in \{-1, +1\}$, acts as a scalar modulo p_{Vesta}. The specification requires the scalar multiplication to be correct on this whole signed range, which differs from the range of the balancing value (protocol specification, §"Action Statement (Orchard)", note on the signed range). The binding signature of "The binding signature" (§8.3) is the check outside the proof that completes value conservation.

Condition A5 serves R3. It is Definition 6.3; the multiplier of K^{Orchard} is the integer in $\{0, \dots, p_{\mathsf{Pallas}} - 1\}$ obtained by reducing the sum modulo p_{Pallas}. The nullifier-set rule of "Nullifier sets" (§6.2) is the check outside the proof that completes it.

Conditions A6 and A7 serve R5. Condition A6 is the re-randomisation of "Randomised validating keys" (§7.2), with $G^{\mathsf{SpendAuth}} = \mathsf{GroupHash}(\texttt{z.cash:Orchard}, \mathsf{G})$. Condition A7 recomputes the incoming viewing key as the $\mathsf{Commit}^{\mathsf{ivk}}$ instance of "Viewing keys" (§3.2), under the domain `z.cash:Orchard-CommitIvk`, from the field element $\mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}})$, not from the point; it ties the witnessed key components $\mathsf{ak}^{\mathbb{P}}$, nk and $\mathsf{r_{ivk}}$ to the expanded receiver $(\mathsf{g_d}^{\mathsf{old}}, \mathsf{pk_d}^{\mathsf{old}})$ of the consumed note. Outside its $\perp$ case it forces $\mathsf{ivk} \neq 0$, since $\mathsf{pk_d}^{\mathsf{old}} \neq \mathcal{O}$, and the receiver determines ivk, an integer below $p_{\mathsf{Pallas}} < p_{\mathsf{Vesta}}$, since $\mathsf{g_d}^{\mathsf{old}}$ is a non-identity point of a group of prime order p_{Vesta}. The tie is the binding of ivk to $(\mathsf{ak}, \mathsf{nk})$ (Proposition "Binding of ivk to (ak, nk)", §3.2): under Assumptions 2.22 and 2.8, an efficient algorithm outputs two witnesses that satisfy A7 outside its $\perp$ case for one receiver, with distinct triples $(\mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}}), \mathsf{nk}, \mathsf{r_{ivk}} \bmod p_{\mathsf{Vesta}})$, only with negligible probability. The tie fixes the point $\mathsf{ak}^{\mathbb{P}}$ only up to sign (Lemma 2.4); Lemma 12.5 and Theorem 12.9 use it. The spend-authorisation signature under rk, verified outside the proof (§7.2), completes both conditions.

Conditions A8 and A9 have the product form of A3. The values are below $2^{64} < p_{\mathsf{Pallas}}$ and the flags are bits, so A8 holds if and only if $v^{\mathsf{old}} = 0$ or $\mathsf{enableSpends} = 1$, and A9 if and only if $v^{\mathsf{new}} = 0$ or $\mathsf{enableOutputs} = 1$, the specification's forms. They tie the hidden values to the public flags of "Bundle flags" (§9.1).

Condition A10 also has the product form of A3: each of its equations is a public field element times a difference of witnessed values. The four points lie in $\mathcal{E}^*$, where the affine coordinates determine the point. For $\delta = 1$, A10 therefore holds if and only if $(\mathsf{g_d}^{\mathsf{new}}, \mathsf{pk_d}^{\mathsf{new}}) = (\mathsf{g_d}^{\mathsf{old}}, \mathsf{pk_d}^{\mathsf{old}})$, equality of the expanded receivers as points and not only of their x-coordinates; for $\delta = 0$ it is vacuous. Condition A10 formalises the meaning of $\mathsf{enableCrossAddress}$ stated in the protocol specification, §"Action Descriptions". Every Orchard-pool Action has $\delta = 1$; an Ironwood-pool Action has $\delta = 1 - \mathsf{enableCrossAddress}$, normally 0 (ZIP 229, Rationale, "`enableCrossAddress` polarity"). Conditions A8 to A10 serve no requirement of their own.

Remark 9.3 (Meaning of A10). Condition A10 constrains each Action separately: when $\mathsf{disableCrossAddress} = 1$, the note an Action creates is at the expanded receiver of the note that same Action consumes. The consumed note may be a dummy of value 0, which A3 exempts from membership. Hence a bundle with $\mathsf{disableCrossAddress} = 1$ can still create a note at any expanded receiver $(\mathsf{g_d}, \mathsf{pk_d})$ for which the transaction creator produces a dummy-spend Action in the same bundle. Such an Action has a witness: let its consumed note be a dummy of value 0 at $(\mathsf{g_d}, \mathsf{pk_d})$ and its created note, of any value v, be at the same receiver, and let $(\mathsf{ak}^{\mathbb{P}}, \mathsf{nk}, \mathsf{r_{ivk}})$ be the key components of that receiver. Condition A3 holds because $v^{\mathsf{old}} = 0$, A8 for either value of $\mathsf{enableSpends}$, A10 with $\delta = 1$ because the two receivers agree, and A9 with $\mathsf{enableOutputs} = 1$

when $v \neq 0$. Conditions A1, A2, A4 and A5 hold when $\mathsf{cm}^{\mathsf{old}}$, cmx, $\mathsf{cv}^{\mathsf{net}}$ and $\mathsf{nf}^{\mathsf{old}}$ are computed from the witness by their definitions, and A6 and A7 hold for the key components of the receiver, whose address satisfies $\mathsf{pk_d} = [\mathsf{ivk}]\,\mathsf{g_d}$ (§3.3). The Action also carries a valid spend-authorisation signature under its rk. By Theorem 12.9, in "Authorisation of spends", for an honestly generated address with $\mathsf{use_qsk} = \mathsf{false}$ this requires that address's spend authorising key ask, except with negligible probability. Consensus enforces no more than this per-Action restriction (ZIP 326, "Rationale for key-generation restrictions").

Remark 9.4 (Effect on the Orchard pool). For the Orchard pool, whose flag encoding forces $\mathsf{disableCrossAddress} = 1$, the pool-level effect is the one that ZIP 229, Abstract, states: outputs to the Orchard pool go to an address for which the transaction creator can authorise spends, which discourages economic activity between users within that pool. By the dummy-spend construction of the preceding remark, consensus does not prevent transfers between users within the Orchard pool (ZIP 326, "Rationale for key-generation restrictions"; ZIP 258, "Consensus rules from NU6.3 activation", for the rule on Orchard-pool Actions).

Remark 9.5 (Status). The expanded receiver $(\mathsf{g_d}, \mathsf{pk_d})$, the meaning of $\mathsf{enableCrossAddress}$ that A10 formalises, and the eight-input primary input with $\mathsf{disableCrossAddress} = 1 - \mathsf{enableCrossAddress}$ are specified in the protocol specification, §"Action Descriptions"; the flag encoding in ZIP 229; the Orchard-pool rule and the selection of the verifying key by upgrade in ZIP 258. The specification's §"Action Statement (Orchard)" and the proof-validity rule of §"Action Descriptions" list the first seven primary inputs and conditions A1 to A9 only; Definition 9.2 follows §"Action Descriptions" for the rest. The specification names the conditions; the numbering A1 to A10 is this volume's. ZIP 2006, to which the specification refers for the restriction, is a reserved number without content and is not cited for content.

Remark 9.6 (Sign of $\mathsf{ak}^{\mathbb{P}}$). The statement does not constrain the sign of the witnessed point $\mathsf{ak}^{\mathbb{P}}$. Condition A7 reads only $\mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}})$, which is the same for $\mathsf{ak}^{\mathbb{P}}$ and $-\mathsf{ak}^{\mathbb{P}}$ (Lemma 2.4), so the even-y normalisation of key generation ("The spending key and the spend-side secrets", §3.1) is not enforced by the proof (protocol specification, §"Action Statement (Orchard)", note on the sign of the spend validating key). Theorem 12.9, in "Authorisation of spends", treats both signs.

Remark 9.7 ($\perp$-weakened conditions). Conditions A1, A2, A3 and A7 are the specification's $\perp$-weakened forms: each holds when the Sinsemilla instance it evaluates, $\mathsf{NoteCommit}$, the hash of $\mathsf{MerkleCRH}$ or $\mathsf{Commit}^{\mathsf{ivk}}$, returns $\perp$ on the witnessed input, for A3 on some input of the fold, because the circuit evaluates Sinsemilla with incomplete addition and cannot exclude its exceptional cases (protocol specification, §"Action Statement (Orchard)", note on $\perp$ outputs and note on the Merkle path validity check; §"Merkle Path Validity", notes on Orchard). The relation is therefore the disjunction, not the plain conjunction. The weakening costs a negligible term. A witness that satisfies A1 or A2 only through $\perp$ contains a 1086-bit message, M^{old} or M^{new}, on which $\mathsf{SinsemillaHashToPoint}(\texttt{z.cash:Orchard-NoteCommit-M}, \cdot)$ returns $\perp$; one that satisfies A3 only through $\perp$ contains a 520-bit message on which $\mathsf{SinsemillaHash}(\texttt{z.cash:Orchard-MerkleCRH}, \cdot)$ returns $\perp$; one that satisfies A7 only through $\perp$ contains the 510-bit message $\mathrm{LE}_{255}(\mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}})) \,\|\, \mathrm{LE}_{255}(\mathsf{nk})$ with the same property for the domain $\texttt{z.cash:Orchard-CommitIvk-M}$ (§2.4). An efficient algorithm that runs an adversary together with the extractor of Assumption 9.11 and outputs such a message succeeds with negligible probability by Proposition 2.24(iii), which turns the message into a non-trivial discrete-logarithm relation among the $\mathsf{GroupHash}$ generators, under Assumptions 2.22 and 2.8. For the witnesses extracted from accepted proofs, A1, A2, A3 and A7 therefore hold in their unweakened forms, without the $\perp$ case, except with negligible probability.

Remark 9.8 (What the statement does not check).

(i) No condition involves ask. The point $\mathsf{ak}^{\mathbb{P}} = [\mathsf{ask}]\,G^{\mathsf{SpendAuth}}$ is computed at key generation, outside the statement, and ask is not a witness. Conditions A6 and A7 relate rk and

the address to the witnessed $\mathsf{ak}^{\mathbb{P}}$, and the spend-authorisation signature under rk supplies knowledge of the signing key; Theorem 12.9 combines the two.

(ii) The trapdoors $\mathsf{rcm}^{\mathsf{old}}$ and $\mathsf{rcm}^{\mathsf{new}}$ and the elements ψ^{old} and ψ^{new} are free witnesses: no condition checks that rcm and ψ were derived from rseed as in "The note seed" (§4.3). The sender computes that derivation; the recipient checks it, in "Trial decryption and note acceptance" (§10.4), and consensus checks it only for coinbase outputs ("Chain state and pool rules", §11.4). Hence, in the deployed protocol, the hashed rcm adds no binding: against an adversary able to compute discrete logarithms among the bases of $\mathsf{NoteCommit}$, $\mathsf{NoteCommit}$ is not binding, and such an adversary can open one commitment to two distinct notes (ZIP 2005, Rationale, "Attacks against binding of note commitments"). The volume claims nothing further in that setting.

(iii) The statement reads none of the Action's note-encryption fields, which "Note encryption" (§10) constructs; the specification records that the absence of a check on the ephemeral key is intentional (§"Action Statement (Orchard)", note).

Remark 9.9 (Key source). The statement takes the key components $(\mathsf{ak}^{\mathbb{P}}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$ as witnesses and checks them only through A5, A6 and A7. It therefore applies equally to key components whose $\mathsf{ak}^{\mathbb{P}}$ and $\mathsf{r}_{\mathsf{ivk}}$ come from the alternative source of ZIP 2005 ("Changes to the Protocol Specification", §4.2.3 "Orchard Key Components") cited in "The spending key and the spend-side secrets" (§3.1). The key-derived results of the volume, namely Proposition 3.18, Assumption 3.13, Propositions 3.17, 6.10 and 10.6, and the hypotheses on honest key generation of Theorems 12.9 and 12.12, remain restricted to keys with $\mathsf{use_qsk} = \mathsf{false}$.

Table 5 is the requirement map of the volume. For each requirement of "Requirements on a shielded payment" (§1.4) it lists the soundness content, with the conditions, the checks outside the proof and the results that discharge it, and the hiding content, with the results that discharge it.

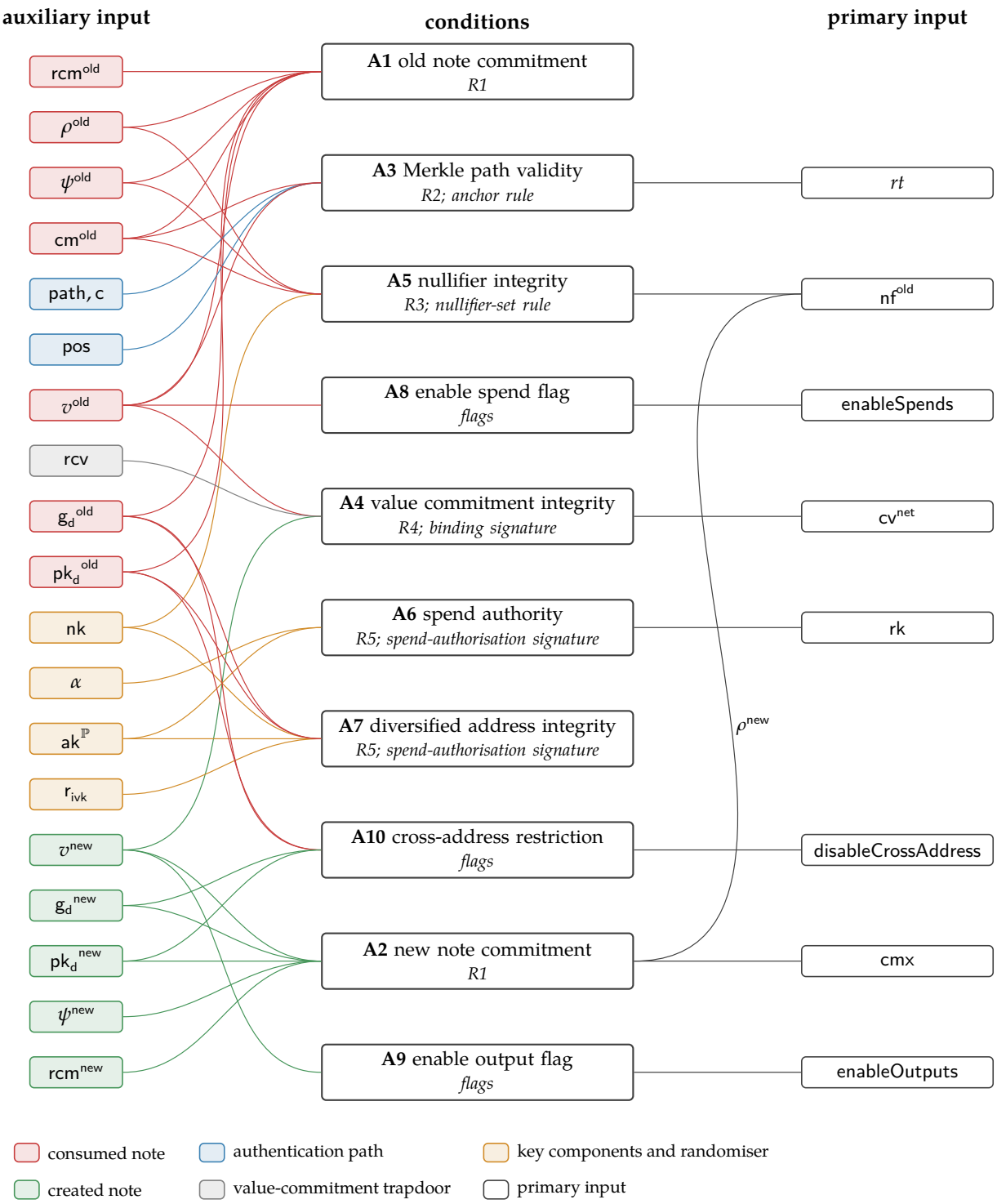


Figure 3: The Action statement of Definition 9.2. The auxiliary input is on the left, coloured by group as in the legend; the primary input is on the right; conditions A1 to A10 are in the centre, ordered by the side of the Action they constrain rather than by number. Each condition has an edge to every variable it reads; the edge from $\mathsf{nf}^{\mathsf{old}}$ to A2 carries $\rho^{\mathsf{new}} := \mathsf{nf}^{\mathsf{old}}$. Each condition is tagged with the requirement of §1.4 that it serves, or with "flags" for A8 to A10, and, where one exists, with the check outside the proof that completes it: the anchor rule for A3, the nullifier-set rule for A5, the binding signature for A4, and the spend-authorisation signature for A6 and A7.

Requirement	Soundness	Hiding
R1, hidden representation of value	A1 and A2 bind the consumed and the created note to their commitments (Proposition 4.5)	commitment hiding (Propositions 4.5 and 4.8); value-commitment hiding (§8.1); indistinguishability of real and dummy Actions (Theorem 12.12)
R2, membership without identification	A3 with the anchor rule of "Anchors" (Proposition 5.9)	zero knowledge of the proof (Assumption 9.14) with a locally computed authentication path (Lemma 5.15), composed in Theorem 12.12
R3, no second consumption	A5 with the nullifier-set rule (Theorem 12.6)	nullifier unlinkability (Proposition 6.10)
R4, conservation	A4 with the binding signature (Theorem 12.4)	value-commitment hiding (Theorem 12.12)
R5, spend authority	A6 and A7 with the spend-authorisation signature (Theorem 12.9) and bundle binding (Proposition 11.10)	none: R5 has a soundness part only
Flags	A8 to A10 serve no requirement of their own: they tie the hidden values and receivers to the public flags of §9.1	—

Table 5: The requirement map: for each requirement of §1.4, the conditions of Definition 9.2, the checks outside the proof and the results that discharge its soundness part, and the results that discharge its hiding part. It is the only requirement map of the volume.

9.3 The Action circuit and the Halo 2 proof

The Action statement of Definition 9.2 is a relation and prescribes no computation. The *post-NU6.3 Action circuit* is one fixed PLONKish circuit (*Halo 2 Guide*, §"PLONKish arithmetisation") whose constraints are designed to encode the typing and conditions A1 to A10; that every satisfying assignment yields a witness of $\mathcal{R}_{\mathsf{Action}}$ is part of Assumption 9.11 below. The same circuit serves every Action. Its instance is the encoding of the eight primary inputs of Definition 9.2, bound to the proof as in the *Halo 2 Guide*, §"Instance columns and binding the public statement"; this instance is the only Orchard-specific part of the proof system constructed in this volume. Proofs are Halo 2 proofs (*Halo 2 Guide*, §"The Halo 2 proof system" and §"The complete Halo 2 protocol"; protocol specification, §"Zero-Knowledge Proving System"; ZIP 229, Abstract): a PLONKish polynomial interactive oracle proof compiled with the inner-product polynomial commitment and made non-interactive by the Fiat–Shamir transform, with no trusted setup.

The circuit's native field is $\mathbb{F}_{p_{\mathsf{Pallas}}}$, the base field of the Pallas group, so the Pallas point operations of A1 to A10 are arithmetic in the native field. The proof's polynomial commitments lie in the Vesta group $\mathbb{G}_{\mathsf{Vesta}}$, the group of points of the curve $y^2 = x^3 + 5$ over $\mathbb{F}_{p_{\mathsf{Vesta}}}$, which has prime order p_{Pallas} (§2.1; *Math Guide*, §"Base fields, scalar fields, and the Pasta cycle"); the committed polynomials therefore have coefficients in $\mathbb{F}_{p_{\mathsf{Pallas}}}$. The Pallas group carries the protocol's keys, commitments, signatures and note encryption; the Vesta group carries only the proof's commitments.

The circuit has 2^{11} rows and a constraint-degree bound of 9 across its gates and its permutation and lookup identities (*Halo 2 Guide*, §"From statement to circuit: arithmetisation in practice"). Both are cited constants; in this volume they enter only the Schwartz–Zippel term of the first supporting analysis below.

Every Action of either pool, whatever it consumes or creates, dummy sides included, is an instance of the same relation and circuit. The proofs of a bundle's Actions are aggregated into one

proof, encoded once per bundle, which the verifier checks against the primary inputs of all the bundle's Actions and the verifying key; each Action carries its own spend-authorisation signature (protocol specification, §"Action Transfers and their Descriptions" and §"Action Descriptions", note on proof aggregation; ZIP 229, "Transaction Format"). The verification procedure is stated in "Verification of a transaction" (§11.6).

Both pools use the post-NU6.3 circuit and its proving and verifying keys; the pools are distinguished by their note commitment trees, nullifier sets, value pool balances and component position in the transaction, not by circuits (ZIP 229, Abstract, and Rationale, "Reuse of the Orchard protocol with minimal changes"). Consensus selects the Action verifying key by network upgrade (protocol specification, §"Action Descriptions", consensus rules). The key in force for every Action, in either pool and in version 5 and version 6 transactions alike, is the post-NU6.3 verifying key, named by upgrade and not by pool (ZIP 258, "Consensus rules from NU6.3 activation").

Assumption 9.10 (Discrete logarithms on Vesta). For every classical probabilistic polynomial-time algorithm, given a generator G of $\mathbb{G}_{\mathsf{Vesta}}$ and $[s]G$ for s uniform in $\mathbb{F}_{p_{\mathsf{Pallas}}}$, the probability of outputting s is negligible (*Crypto Guide*, §"Standing assumptions", Remark "Standing assumptions for the sequel", item 1, for the Vesta curve). Concretely, the best known classical attack costs about 2^{126} group operations (*Crypto Guide*, §"Instantiation on elliptic curves; the Pasta curves", Proposition "The Pasta curves against the criteria"); the level is cited, not recomputed. The volume uses this assumption only in the extraction analysis of the inner-product commitment that supports Assumption 9.11 (the second supporting analysis below).

Assumption 9.11 (Knowledge soundness of the Action proof). Let the hash of the Fiat–Shamir transform of the proof system be modelled as a random oracle. For every efficient adversary $\mathcal{A}$ that outputs primary inputs $\mathsf{pub}_1, \dots, \mathsf{pub}_n$ and an aggregate proof π, there is an efficient extractor $\mathcal{E}$, with access to $\mathcal{A}$ and to its random-oracle queries, that outputs auxiliary inputs $\mathsf{aux}_1, \dots, \mathsf{aux}_n$ such that the probability of the event

$$\pi \text{ is accepted for } (\mathsf{pub}_1, \dots, \mathsf{pub}_n) \text{ under the post-NU6.3 verifying key}$$
$$\text{and } (\mathsf{pub}_i, \mathsf{aux}_i) \notin \mathcal{R}_{\mathsf{Action}} \text{ for some } i$$

is negligible. This is the knowledge-soundness clause of the *Crypto Guide*'s Definition "SNARK" (§"SNARKs: succinct non-interactive arguments of knowledge") for the relation $\mathcal{R}_{\mathsf{Action}}$ of Definition 9.2. It includes that every satisfying assignment of the post-NU6.3 circuit yields a witness of $\mathcal{R}_{\mathsf{Action}}$, for which the specification requires canonical decompositions of the scalar of A5 and of the scalar ivk in $[\mathsf{ivk}]\,\mathsf{g_d}^{\mathsf{old}}$ (protocol specification, §"Action Statement (Orchard)", note on canonical scalar decompositions).

Assumption 9.11 is an assumption, not a proved result. Its status is designed but unspecified: bounds exist for the analysed protocol, and no theorem carries them to the deployed Fiat–Shamir-compiled protocol (*Halo 2 Guide*, §"The algebraic group model"). The volume assumes the property, proves no bound and asserts no loss factor. The two supporting analyses that follow are cited, not re-proved, and not combined into a bound.

Remark 9.12 (Supporting analysis (i): a Schwartz–Zippel term). A non-zero gap polynomial D over $\mathbb{F}_{p_{\mathsf{Pallas}}}$, fixed by the prover's commitments before the challenge is drawn, vanishes at a uniform challenge with probability at most $\deg D / p_{\mathsf{Pallas}}$ (*Halo 2 Guide*, §"Why checking one random point is a proof"). At the Orchard parameters, 2^{11} rows and constraint-degree bound 9, the constraint expression has degree below $9 \cdot 2^{11}$, and so does the committed quotient, of 8 chunks of degree below 2^{11}, times the vanishing polynomial of degree 2^{11} (*Halo 2 Guide*, §"The complete Halo 2 protocol", Phase 4). Hence $\deg D < 9 \cdot 2^{11} = 18432$, and the term is below

$18432/p_{\mathsf{Pallas}} < 2^{-239}$. The term bounds one check at one challenge and is not a security level of the proof: the complete protocol has further randomised checks, and under the Fiat–Shamir transform an adversary may retry challenges, one hash query per retry (*Halo 2 Guide*, §"Removing the verifier: the Fiat–Shamir transform").

Remark 9.13 (Supporting analysis (ii): extraction for the inner-product commitment). The inner-product polynomial commitment, whose commitments lie in $\mathbb{G}_{\mathsf{Vesta}}$, has an interactive opening protocol that is knowledge-sound under the discrete-logarithm assumption in that group, Assumption 9.10, with an expected-polynomial-time tree extractor (*Halo 2 Guide*, §"The inner-product polynomial commitment", Theorem "Knowledge soundness of the opening protocol"). The Fiat–Shamir-compiled argument inherits the interactive extraction bound only up to a loss in the adversary's random-oracle query count (§"From the interactive extractor to the non-interactive argument"), and the tighter analyses in the algebraic group model are proved for the analysed protocol, not for the deployed transcript (§"The algebraic group model"). No loss factor is asserted.

Assumption 9.14 (Zero knowledge of the Action proof). Let the hash of the Fiat–Shamir transform be modelled as a programmable random oracle. There is an efficient simulator that, given only the primary inputs $\mathsf{pub}_1, \ldots, \mathsf{pub}_n$ of a bundle and programming the random oracle, outputs an aggregate proof whose distribution is statistically indistinguishable from that of an honestly generated proof for any $\mathsf{aux}_1, \ldots, \mathsf{aux}_n$ with $(\mathsf{pub}_i, \mathsf{aux}_i) \in \mathcal{R}_{\mathsf{Action}}$ for every i: every distinguisher that makes polynomially many random-oracle queries, whatever its running time, has negligible advantage. This is zero knowledge of statistical grade in the non-interactive, random-oracle form of the *Crypto Guide*'s Definition "Zero knowledge" (§"The simulation paradigm and zero knowledge").

Assumption 9.14 is an assumption, not a proved result. The *Halo 2 Guide*, §"Zero knowledge: hiding the witness in Halo 2", records honest-verifier zero knowledge of the interactive protocol and zero knowledge of its Fiat–Shamir compilation in the programmable random-oracle model, subject to its composition and Fiat–Shamir hypotheses, without reproving it. Assumptions 9.11 and 9.14 are the two assumptions on the Action proof that "Security" (§12) composes with the propositions of the preceding sections.

10 Note encryption

This section constructs the in-band delivery of a created note: its plaintext, its encryption to the recipient by one-shot Diffie–Hellman on the diversified base, the outgoing ciphertext keyed from ovk and the Action's own public fields, and the trial decryption and acceptance checks by which a recipient obtains a note consistent with the published cmx. It states the two assumptions that note encryption adds, proves decryption correctness, confidentiality and key privacy of the encryption to the recipient, confidentiality of the outgoing ciphertext, and the consistency of accepted notes, and states what each key tier can decrypt.

10.1 The note plaintext

Every Action carries, beside the extracted commitment cmx of its created note, a *note ciphertext* $(\mathsf{epk}^\star, C_{\mathsf{enc}}, C_{\mathsf{out}})$. The component C_{enc} encrypts the plaintext of the created note so that the holder of the recipient's incoming viewing key ivk recovers the note and its memo from the chain alone (protocol specification, §"In-band secret distribution (Sapling and Orchard)" and §"Action Descriptions"). The component C_{out} lets the sender, or any party that holds the outgoing viewing key ovk under which it was formed, recover the same plaintext (§10.3). The fields $\mathsf{epk}^\star$, C_{enc}

and C_{out} are not inputs of the Action statement (Definition 9.2), which therefore does not constrain them; the checks that make a decrypted note consistent with cmx are made by the recipient (§10.4).

Definition 10.1 (Note plaintext)**.** Let a created note have recipient address $(d, \mathsf{pk_d})$ and value $v \in \{0, \dots, 2^{64} - 1\}$. Its *note plaintext* is the tuple $\mathsf{np} := (\mathsf{leadByte}, d, v, \mathsf{rseed}, \mathsf{memo})$, encoded as the byte string

$$\mathsf{leadByte} \parallel \mathsf{LEBS2OSP}_{88}(d) \parallel \mathsf{I2LEOSP}_{64}(v) \parallel \mathsf{rseed} \parallel \mathsf{memo},$$

where $\mathsf{leadByte}$ is one byte, the format version of "The note seed" (§4.3), equal to $\mathtt{0x03}$ for an Ironwood-pool output; d is the 88-bit diversifier of the address (11 bytes); v occupies 8 bytes; rseed is the 32-byte note seed; and memo is a 512-byte string chosen by the sender, whose use is by agreement between sender and recipient (protocol specification, §"Note Plaintexts and Memo Fields" and §"Encodings of Note Plaintexts and Memo Fields"). The encoding has the fixed length $1 + 11 + 8 + 32 + 512 = 564$ bytes, so the length of a ciphertext carries no information about the plaintext.

The plaintext omits $\mathsf{pk_d}$, ρ, ψ and rcm. The recipient computes $\mathsf{pk_d} = [\mathsf{ivk}]\,\mathsf{g_d}$, takes $\rho = \mathsf{nf}^{\mathsf{old}}$ from the same Action ("Nullifier chaining", §6.3), and derives ψ and rcm from rseed (§4.3).

For each output the sender draws rseed uniformly from the 32-byte strings, independently of every other output. The ephemeral secret key of this section is

$$\mathsf{esk} = \mathsf{ToScalar}(\mathsf{PRFexpand}_{\mathsf{rseed}}([\mathtt{0x04}] \parallel \mathsf{I2LEOSP}_{256}(\mathsf{nf}^{\mathsf{old}}))),$$

derived in "The note seed" (§4.3) together with ψ and rcm. The sender redraws rseed when $\mathsf{esk} = 0$ or $\mathsf{cmx} = \perp$, so $\mathsf{esk} \in \mathbb{F}^*_{p_{\mathsf{Vesta}}}$ (protocol specification, §"Sending Notes (Orchard)"; ZIP 2005, "Changes to the Protocol Specification", §4.7.3).

10.2 Encryption to the recipient

Construction 10.2 (Encryption to the recipient)**.** Let $(d, \mathsf{pk_d})$ be the recipient's address, $\mathsf{g_d} = \mathsf{DiversifyHash}(d)$ its diversified base ("Diversified addresses", §3.3), and esk as in §10.1. A star encoding $P^\star$ enters a byte string as the 32 bytes $\mathsf{LEBS2OSP}_{256}(P^\star)$ (§2.1), with which it is identified below. The sender computes the *ephemeral public key*, the *shared secret* and the *encryption key*

$$\mathsf{epk} := [\mathsf{esk}]\,\mathsf{g_d}, \qquad s := [\mathsf{esk}]\,\mathsf{pk_d},$$
$$K_{\mathsf{enc}} := \mathsf{KDF}(s, \mathsf{epk}) := \text{BLAKE2b-256}(\mathtt{Zcash_OrchardKDF},\ s^\star \parallel \mathsf{epk}^\star),$$

with BLAKE2b-256(P, x) unkeyed BLAKE2b with a 32-byte output and the 16-byte personalisation P, and the ciphertext $C_{\mathsf{enc}} := \mathsf{Enc}_{K_{\mathsf{enc}}}(\mathsf{np})$. The pair Enc_K, Dec_K denotes ChaCha20-Poly1305 encryption and decryption under the 256-bit key K with the all-zero 96-bit nonce and empty associated data; decryption returns $\perp$ when the tag does not verify (*Crypto Guide*, §"The ChaCha20-Poly1305 AEAD", Construction "ChaCha20-Poly1305"). The Action publishes the 32-byte field $\mathsf{epk}^\star$ and C_{enc}, of $564 + 16 = 580$ bytes, the plaintext and the 16-byte tag (protocol specification, §"Encryption (Sapling and Orchard)", §"Orchard Key Agreement", §"Orchard Key Derivation" and §"Symmetric Encryption").

The key agreement is Diffie–Hellman on Pallas: public keys are non-identity points, private keys lie in $\mathbb{F}^*_{p_{\mathsf{Vesta}}}$, and both the derivation of a public key from a base and the agreement are scalar multiplication (protocol specification, §"Orchard Key Agreement"). A consensus rule requires

$\mathsf{epk}^\star$ to be the canonical encoding of a non-identity Pallas point (protocol specification, §"Action Descriptions"). The fixed nonce is safe because every key encrypts once: K_{enc} depends on the fresh esk of its output, and the key of C_{out} is derived separately (§10.3). The scheme is the *Crypto Guide*'s Construction "DH-based hybrid public-key encryption" (§"Hybrid public-key encryption") over the per-address base $\mathsf{g_d}$ in place of a fixed generator, as the same volume records for deployed note delivery (§"Shielded-note delivery as deployed hybrid encryption").

Proposition 10.3 (Decryption correctness)**.** *Let* $(d, \mathsf{pk_d})$ *be an address of the incoming viewing key* ivk*, so* $\mathsf{pk_d} = [\mathsf{ivk}]\,\mathsf{g_d}$ *(§3.3), and let* $\mathsf{epk} = [\mathsf{esk}]\,\mathsf{g_d}$ *with* $\mathsf{esk} \in \mathbb{F}^*_{p_{\mathsf{Vesta}}}$*. Then*

$$[\mathsf{ivk}]\,\mathsf{epk} = [\mathsf{esk}]\,\mathsf{pk_d} = s.$$

Hence the holder of ivk *computes* $K_{\mathsf{enc}} = \mathsf{KDF}([\mathsf{ivk}]\,\mathsf{epk}, \mathsf{epk})$ *from the published* $\mathsf{epk}^\star$ *and obtains* $\mathsf{Dec}_{K_{\mathsf{enc}}}(C_{\mathsf{enc}}) = \mathsf{np}$*, and a party that knows* $(\mathsf{pk_d}, \mathsf{esk})$ *obtains the same shared secret as* $[\mathsf{esk}]\,\mathsf{pk_d}$*.*

Proof. In the cyclic Pallas group $[a]([b]\,P) = [ab]\,P = [b]([a]\,P)$, so $[\mathsf{ivk}]([\mathsf{esk}]\,\mathsf{g_d}) = [\mathsf{esk}]([\mathsf{ivk}]\,\mathsf{g_d}) = [\mathsf{esk}]\,\mathsf{pk_d}$. The recipient decodes $\mathsf{epk} = \mathsf{abst}_{\mathbb{P}}(\mathsf{epk}^\star)$, and because the star encoding is canonical (§2.1) it re-encodes to the published bytes. The function KDF is deterministic and both parties apply it to $(s^\star, \mathsf{epk}^\star)$, so they derive the same K_{enc}; correctness of ChaCha20-Poly1305 gives $\mathsf{Dec}_{K_{\mathsf{enc}}}(\mathsf{Enc}_{K_{\mathsf{enc}}}(\mathsf{np})) = \mathsf{np}$. This is the *Crypto Guide*'s Theorem "Correctness of hybrid encryption" (§"Hybrid public-key encryption") over the base $\mathsf{g_d}$. □

Assumption 10.4 (The note-encryption hashes as random oracles). In security arguments the key-derivation function KDF (BLAKE2b-256 personalised `Zcash_OrchardKDF`), the function $\mathsf{PRF}^{\mathsf{ock}}$ (BLAKE2b-256 personalised `Zcash_Orchardock`, constructed in §10.3), and PRFexpand on inputs with leading byte `0x04`, the derivation of esk from rseed, are modelled as independent random oracles (*Crypto Guide*, §"The random oracle model", Definition "Random oracle"). The PRF assumption (Assumption 2.12) does not suffice for the last: it requires a key unknown to the adversary, whereas the plaintext that K_{enc} encrypts contains rseed, the key from which esk, and hence K_{enc}, is derived. On its other inputs PRFexpand is used only under Assumption 2.12. The assumption is a heuristic about BLAKE2b, not a theorem; every result that uses KDF, $\mathsf{PRF}^{\mathsf{ock}}$ or the derivation of esk names it.

Assumption 10.5 (Confidentiality and integrity of ChaCha20-Poly1305). For a uniform 256-bit key that encrypts a single plaintext, ChaCha20-Poly1305 with the all-zero nonce and empty associated data is

(a) *confidential*: the encryptions of any two plaintexts of equal length are computationally indistinguishable; and

(b) *integral*: no efficient adversary without the key outputs a ciphertext, other than one it was given, that decrypts to a value other than $\perp$.

The *Crypto Guide*'s Theorem "Security of ChaCha20-Poly1305" (§"The ChaCha20-Poly1305 AEAD") derives both, with explicit bounds, from the pseudorandomness of ChaCha20. The protocol specification requires one-time INT-CTXT and IND-CPA security of the scheme, properties (b) and (a) above (*Crypto Guide*, Definitions "Ciphertext integrity, INT-CTXT" and "CPA security", §"Authenticated encryption with associated data" and §"Chosen-plaintext security"; protocol specification, §"Symmetric Encryption"). The scheme is not key-committing: a party that chooses two keys can construct one ciphertext that decrypts

validly under both (*Crypto Guide*, §"The ChaCha20-Poly1305 AEAD", Remark "Wrong-key rejection and key commitment"). No result of this volume relies on the tag against a sender who chooses the key.

Proposition 10.6 (Confidentiality and key privacy of note encryption)**.** *Let a key be generated honestly with* use_qsk = false*. An efficient adversary receives the address* $(d, \mathsf{pk_d})$ *of the key at an index of its choice and data* z *that an efficient algorithm computes independently of the key and of* rseed*; it chooses a value* v*, a memo and* $\rho \in \mathbb{F}_{p_{\mathsf{Pallas}}}$*. An output to* $(d, \mathsf{pk_d})$ *with these values and* $\rho = \mathsf{nf}^{\mathsf{old}}$ *is formed as in §10.1 and the construction above, with* rseed *uniform, and* cmx *is the extracted commitment of its note. Then the triple* $(\mathsf{epk}^\star, C_{\mathsf{enc}}, \mathsf{cmx})$ *is computationally indistinguishable from* $(E^\star, C', \mathsf{cmx})$*, where* E *is a uniform non-identity Pallas point and* C' *is the encryption of a fixed* 564*-byte string under an independent uniform key. The hypotheses are Assumptions 3.16, 10.4, 10.5, 2.8 (for* $\mathsf{g_d}$*), 3.13 (for the joint distribution of* ivk *and* d*) and 2.12 (for the dependence of* cmx *on* rseed*), together with Assumption 2.22 for Lemma "Rejection in key generation" (§3.2).*

Since E and C' are computed from neither the plaintext nor the address, the pair $(\mathsf{epk}^\star, C_{\mathsf{enc}})$ reveals neither the plaintext, rseed included, nor the recipient's address: given two addresses of independent keys, the data z may contain the second, and the ciphertext is indistinguishable from one independent of both. The value cmx is included so that the result applies within an Action. Of the hypotheses only Assumption 3.16, used through the computational Diffie–Hellman problem that it implies, concerns the one-shot key agreement between esk and $\mathsf{pk_d}$; Assumptions 10.4 and 10.5 concern the hash functions and the cipher. The adversary is passive: the volume does not claim indistinguishability of encryptions against an adversary that may also obtain decryptions of ciphertexts of its choice other than the challenge, before and after receiving it (IND-CCA2), which the protocol specification, §"Key Derivation", requires of the scheme.

Proof. The proof is a sequence of games (*Crypto Guide*, §"Security as a game", Remark "Game-hopping"); each reduction below runs the adversary and the rest of the game. Let Q be the event that the adversary queries the oracle of Assumption 10.4 for PRFexpand at an input keyed by rseed.

Preliminary step. As in the proof of Proposition 3.18, the key is replaced by a single draw without rejection and its triple $(\mathsf{ask}, \mathsf{nk}, \mathsf{r_{ivk}})$ by independent uniform elements (Lemma 2.15, Assumption 2.12; Lemma "Rejection in key generation", §3.2, Assumptions 2.22 and 2.8). Then $(\mathsf{ivk}, \mathsf{dk})$ is replaced by $(\mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r_{ivk}}'}(\mathsf{ak}, \mathsf{nk}), u_1)$ (Assumption 3.13; the reduction draws ask and nk itself). Now $d = \mathsf{FF1\text{-}AES256}_{u_1}(\mathrm{LE}_{88}(j))$, and ivk is, independently of d, 0 with probability $1/p_{\mathsf{Vesta}}$ and otherwise uniform on the set X_{Pallas} of x-coordinates of non-identity points, of $(p_{\mathsf{Vesta}} - 1)/2$ elements, except on the negligible event $\hat{M} = \perp$ (Lemma "Distribution of ivk", §3.2). The redraw of rseed is replaced by a single trial, at the cost of the redraw probability, negligible as in the proof of Proposition 4.8.

(1) By Assumption 10.4, esk is ToScalar of a random-oracle output at an input keyed by rseed. It is replaced by an independent uniform element of $\mathbb{F}^*_{p_{\mathsf{Vesta}}}$. The oracle's value at that input is used nowhere else, so the games are identical until Q occurs; they differ by at most $\Pr[Q]$ in the new game plus the reduction distance of ToScalar, below 2^{-257}, and $1/p_{\mathsf{Vesta}}$ for the value 0 (*Crypto Guide*, §"Security as a game", Remark "Game-hopping"; §2.3).

(2) The key K_{enc} is replaced by an independent uniform key. The games differ only on the event C that the adversary queries the KDF oracle at $s^\star \parallel \mathsf{epk}^\star$, and $s = [\mathsf{esk}]\,\mathsf{pk_d}$ solves the computational Diffie–Hellman instance $(\mathsf{g_d}, \mathsf{pk_d}, \mathsf{epk})$. This is the game hop of the *Crypto Guide*'s Theorem "Passive security of hybrid encryption" (§"Hybrid public-key encryption"), with loss factor the number q_K of KDF queries, run over $\mathsf{g_d}$. A distinguisher $\mathcal{D}$ for Assumption 3.16 receives $([a]\,P, [b]\,P, [c]\,P)$ with a uniform on X_{Pallas} and b, c uniform. It answers each new query

of GroupHash under the domain `z.cash:Orchard-gd` with $[t]P$ for a fresh uniform t, recorded, a uniform point as in the model of Assumption 2.8, so that $\mathsf{g_d} = [t]P$ with t known. It sets $\mathsf{pk_d} := [t]([a]P) = [a]\mathsf{g_d}$, which embeds a as ivk, and $\mathsf{epk} := [t]([b]P) = [b]\mathsf{g_d}$, which embeds b as esk; it draws u_1 and rseed, computes d, cmx and C_{enc} under a uniform key itself, and answers the other oracles lazily. It then decodes the first 32 bytes of a uniformly chosen KDF query to a point S and outputs 1 exactly when $S = [t]([c]P)$. The simulation differs from the game only when $\mathsf{ivk} = 0$ in the game, or $b = 0$ or $t = 0$ in the simulation, of total probability at most $3/p_{\mathsf{Vesta}}$. On a Diffie–Hellman triple $\mathcal{D}$ therefore outputs 1 with probability at least $\Pr[C]/q_K - 3/p_{\mathsf{Vesta}}$; on a random triple the point $[t]([c]P)$ is, for $t \neq 0$, uniform and independent of the view, and $\mathcal{D}$ outputs 1 with probability at most $4/p_{\mathsf{Vesta}}$. An algorithm that computes Diffie–Hellman secrets on Pallas thus decides DDH, and $\Pr[C] \leq q_K\,(\epsilon_{\mathcal{D}} + 7/p_{\mathsf{Vesta}})$ for the advantage $\epsilon_{\mathcal{D}}$ of $\mathcal{D}$. The exponent embedded as ivk is uniform on X_{Pallas} in Assumption 3.16. Against the computational Diffie–Hellman problem with a uniform exponent on $\mathbb{F}_{p_{\mathsf{Vesta}}}$, which lies in X_{Pallas} with probability $(p_{\mathsf{Vesta}} - 1)/(2p_{\mathsf{Vesta}})$ and on those instances gives an exact simulation, the same reduction loses a factor $2p_{\mathsf{Vesta}}/(p_{\mathsf{Vesta}} - 1)$, about 2 (proof of Proposition 3.18, part (d)).

(3) Now $\mathsf{epk} = [\mathsf{esk}]\,\mathsf{g_d}$ with esk uniform on $\mathbb{F}^*_{p_{\mathsf{Vesta}}}$ is a uniform non-identity point for every generator $\mathsf{g_d}$, and the KDF context contains no key of the recipient (*Crypto Guide*, §"Key privacy", Remark "What key privacy buys upstairs"). The ciphertext C_{enc} is an encryption under a uniform key used once, and Assumption 10.5(a) replaces np by the fixed string; the reduction obtains C_{enc} from its challenger and computes the rest. The last game outputs $(E^\star, C', \mathsf{cmx})$.

(4) In the last game rseed enters the view only through cmx, that is through $\mathsf{PRFexpand}_{\mathsf{rseed}}$ under the leading bytes `0x09` and `0x0B`. A distinguisher with oracle access to $\mathsf{PRFexpand}_{\mathsf{rseed}}$ or to a random function computes cmx from its oracle, runs the adversary, and outputs 1 when some key k of a query at a leading byte `0x04` satisfies $\mathsf{PRFexpand}_k([\texttt{0x09}] \,\|\, \mathsf{I2LEOSP}_{256}(\rho))$ equal to its oracle's answer at that input. It outputs 1 with probability at least $\Pr[Q]$ in the first case. In the second the view depends on that answer only through cmx, and the trapdoor rcm, ToScalar of the independent answer at the `0x0B` input, is within 2^{-257} of uniform; by Proposition 4.5(a), cmx is therefore independent of that answer up to $2^{-257} + \epsilon_\perp$, for $\epsilon_\perp$ the probability that the hash point of $M(n)$ is $\perp$, negligible as in the proof of Proposition 4.8. The distinguisher thus outputs 1 with probability at most $q \cdot 2^{-512} + 2^{-257} + \epsilon_\perp$, for q such queries, and $\Pr[Q]$ in the last game is negligible under Assumption 2.12 (with Lemma 2.15). Every reduction of hops (2) and (3) draws rseed itself and detects Q, so each hop changes the probability of Q by at most its own cost. Hence $\Pr[Q]$ is negligible in the game of hop (1) as well, and the sum of the costs of all hops is negligible.

The DDH hybrid of the *Crypto Guide*'s Theorem "ElGamal is key-private under DDH" (§"Key privacy") is not used: as stated it embeds a uniform exponent of $\mathbb{F}_{p_{\mathsf{Vesta}}}$ as the recipient's key, whereas ivk is not uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$; and the factor-2 argument above transfers a search problem such as computational Diffie–Hellman, not a decision problem, since on instances outside X_{Pallas} a distinguisher's output is unconstrained. □

10.3 The outgoing ciphertext

Construction 10.7 (Outgoing ciphertext)**.** The sender chooses an outgoing viewing key ovk of its own, such as that of the address whose note the Action consumes ("Viewing keys", §3.2), or $\mathsf{ovk} = \perp$. If $\mathsf{ovk} \neq \perp$, the *outgoing cipher key* is

$$\begin{aligned}\mathsf{ock} &:= \mathsf{PRF}^{\mathsf{ock}}_{\mathsf{ovk}}(\mathsf{cv^{net}}, \mathsf{cmx}, \mathsf{epk}) \\ &:= \text{BLAKE2b-256}(\texttt{Zcash_Orchardock},\ \mathsf{ovk} \,\|\, (\mathsf{cv^{net}})^\star \,\|\, \mathsf{I2LEOSP}_{256}(\mathsf{cmx}) \,\|\, \mathsf{epk}^\star),\end{aligned}$$

where $\mathsf{cv^{net}}$ is the Action's net value commitment (Definition 8.2) and cmx its extracted note

commitment (Definition 4.4), and the *outgoing plaintext* is the $32 + 32 = 64$-byte string $\mathsf{op} := \mathsf{pk_d}^\star \parallel \mathsf{I2LEOSP}_{256}(\mathsf{esk})$. If $\mathsf{ovk} = \perp$, the sender draws ock uniformly from the 256-bit strings and op uniformly from the 64-byte strings. In both cases $C_{\mathsf{out}} := \mathsf{Enc}_{\mathsf{ock}}(\mathsf{op})$, of $64 + 16 = 80$ bytes, and the Action publishes it; with $\mathsf{ovk} = \perp$ no outgoing viewing key recovers it (protocol specification, §"Encryption (Sapling and Orchard)", §"Sending Notes (Orchard)" and §"Pseudo Random Functions").

Construction 10.8 (Recovery with ovk)**.** For an Ironwood-pool output, the holder of ovk

(i) computes ock from ovk and the Action's public $\mathsf{cv^{net}}$, cmx and $\mathsf{epk}^\star$, and decrypts C_{out} under ock to the outgoing plaintext op, rejecting $\perp$;

(ii) parses op as a 32-byte string b_1 followed by a 32-byte string b_2, and reads esk as the integer $\mathsf{LEOS2IP}_{256}(b_2)$, rejecting $\mathsf{esk} \geq p_{\mathsf{Vesta}}$, and $\mathsf{pk_d}$ as the point $\mathsf{abst}_{\mathbb{P}}(b_1)$, rejecting $\perp$ and $\mathcal{O}$;

(iii) computes $s := [\mathsf{esk}]\,\mathsf{pk_d}$, $K_{\mathsf{enc}} := \mathsf{KDF}(s, \mathsf{epk})$ and $\mathsf{np} := \mathsf{Dec}_{K_{\mathsf{enc}}}(C_{\mathsf{enc}})$, rejecting $\perp$;

(iv) rejects unless $\mathsf{leadByte} = \mathtt{0x03}$;

(v) with $\rho := \mathsf{nf^{old}}$ of the same Action, rejects unless

$$\mathsf{esk} = \mathsf{ToScalar}(\mathsf{PRFexpand}_{\mathsf{rseed}}([\mathtt{0x04}] \parallel \mathsf{I2LEOSP}_{256}(\rho)));$$

(vi) computes $\mathsf{g_d} := \mathsf{DiversifyHash}(d)$, then ψ, then rcm under $\mathtt{0x03}$ from $\mathsf{g_d}$, $\mathsf{pk_d}$, v, ρ and ψ, by steps (iv) and (v) of the derivation of §4.3;

(vii) rejects if the note commitment of $((d, \mathsf{pk_d}), v, \rho, \psi, \mathsf{rcm})$ (Definition 4.4) is $\perp$ or its x-coordinate differs from cmx, or if $([\mathsf{esk}]\,\mathsf{g_d})^\star \neq \mathsf{epk}^\star$;

(viii) otherwise returns the note $((d, \mathsf{pk_d}), v, \rho, \psi, \mathsf{rcm})$ and memo.

The order $\mathsf{g_d}$ and $\mathsf{pk_d}$, then ψ, then rcm is forced because rcm depends on all three. The specification's decryption sections still use the legacy derivation of rcm; ZIP 2005 specifies the reordered procedure (protocol specification, §"Decryption using an Outgoing Viewing Key (Sapling and Orchard)"; ZIP 2005, "Changes to the Protocol Specification", §4.20.2 and §4.20.3). The specification's further check that b_1 re-encodes $\mathsf{pk_d}$ holds for every string that $\mathsf{abst}_{\mathbb{P}}$ decodes, the star encoding being canonical (§2.1). For $\mathsf{pk_d}$ the specification's procedure rejects only $\perp$; the rejection of $\mathcal{O}$ in step (ii) makes the returned $\mathsf{pk_d}$ a transmission key of Definition "Note" (§4.1).

Remark 10.9 (Transplant resistance). The key ock is a function of the Action's own $\mathsf{cv^{net}}$, cmx and epk. A ciphertext C_{out} that a party without ovk copies into an Action whose $(\mathsf{cv^{net}}, \mathsf{cmx}, \mathsf{epk})$ differ is decrypted by the holder of ovk under a key that, by Assumption 10.4, is uniform and independent of the copied ciphertext; by Assumption 10.5(b) that decryption returns $\perp$ except with negligible probability. A copied C_{out} is therefore rejected, not attributed to the other Action.

Proposition 10.10 (Confidentiality of the outgoing ciphertext)**.** *Let the sending key be generated honestly with* $\mathsf{use_qsk} = \mathsf{false}$. *To an efficient party that holds neither its* ovk *nor its* $\mathsf{r_{ivk}}$, *and may hold its* ak, nk, ivk *and* dk, *each ciphertext* C_{out} *that the holder of the key forms is computationally indistinguishable, jointly with every other public field of its Action, from the encryption of a fixed* 64-*byte string under an independent uniform key. The hypotheses are Assumptions 2.12, 3.13, 10.4, 10.5,*

> *9.14, 2.22 and 2.8.*

Proof. As in the preliminary step of the proof of Proposition 10.6, the key is replaced by a single draw without rejection and its triple $(\mathsf{ask}, \mathsf{nk}, \mathsf{r_{ivk}})$ by independent uniform elements (Lemma 2.15, Assumption 2.12; Lemma "Rejection in key generation", §3.2, Assumptions 2.22 and 2.8). After this step, Assumption 3.13 makes ovk indistinguishable from a uniform 32-byte string independent of ivk and dk; the reduction draws ask and nk itself, obtains $(\mathsf{ivk}, \mathsf{dk}, \mathsf{ovk})$ from its challenger, computes every other field, and simulates the Action proof, whose witness contains $\mathsf{r_{ivk}}$, under Assumption 9.14. Then ock is a random-oracle output (Assumption 10.4) at an input that the party queries with negligible probability, hence uniform, and distinct Actions give distinct inputs, their epk differing except with negligible probability, so each ock encrypts once; Assumption 10.5(a) then applies. With $\mathsf{ovk} = \perp$, ock is uniform by construction. The ciphertext C_{out} thus reveals nothing about $(\mathsf{pk_d}, \mathsf{esk})$. □

A holder of ovk recovers, for every output sent under that ovk, the pair $(\mathsf{pk_d}, \mathsf{esk})$, hence the shared secret $[\mathsf{esk}]\,\mathsf{pk_d}$ (Proposition 10.3), K_{enc}, the note plaintext and the recipient's address $(d, \mathsf{pk_d})$; in this way the sender, or a party to whom the sender discloses ovk, reads outgoing notes (protocol specification, §"Decryption using an Outgoing Viewing Key (Sapling and Orchard)"). By this route it reads only outputs sent under that ovk. Detecting the notes sent to the key's own addresses by trial decryption requires ivk (§10.4), and ovk yields no ivk (Proposition 3.18, "Capability separation", part (c), for keys generated with $\mathsf{use_qsk} = \mathsf{false}$).

10.4 Trial decryption and note acceptance

Construction 10.11 (Trial decryption)**.** No public field of an Action names its recipient, so the holder of an incoming viewing key attempts every output: it computes $\mathsf{epk} := \mathsf{abst}_{\mathbb{P}}(\mathsf{epk}^{\star})$, rejecting $\perp$, which consensus excludes, then $s := [\mathsf{ivk}]\,\mathsf{epk}$, $K_{\mathsf{enc}} := \mathsf{KDF}(s, \mathsf{epk})$ and $\mathsf{np} := \mathsf{Dec}_{K_{\mathsf{enc}}}(C_{\mathsf{enc}})$ (protocol specification, §"Decryption using an Incoming Viewing Key (Sapling and Orchard)"). A result other than $\perp$ is a *candidate*, not an accepted note.

The same ivk serves the outputs of both pools, since keys and addresses are defined for the Orchard protocol, not for a pool (§3.4), and an address does not name a pool (ZIP 326, "Scanning for Ironwood-pool and Orchard-pool notes"); the pool of an output is that of the bundle that carries it. For an honestly produced ciphertext under a key independent of the trial key, decryption returns $\perp$ except with negligible probability (Assumption 10.5(b); *Crypto Guide*, §"The ChaCha20-Poly1305 AEAD", Remark "Wrong-key rejection and key commitment"); against a sender who chooses the keys the tag gives no such guarantee.

Construction 10.12 (Acceptance of an Ironwood-pool output)**.** Given a candidate $\mathsf{np} = (\mathsf{leadByte}, d, v, \mathsf{rseed}, \mathsf{memo})$ of an Ironwood-pool output, the recipient accepts only if every check passes:

(i) $\mathsf{leadByte} = \mathtt{0x03}$, the recipient's enforcement of the lead-byte rule of "The note seed" (§4.3);

(ii) with $\rho := \mathsf{nf}^{\mathsf{old}}$ of the same Action and $\underline{\rho} := \mathsf{I2LEOSP}_{256}(\rho)$, the note is re-derived in the order

$$\begin{aligned} \mathsf{g_d} &:= \mathsf{DiversifyHash}(d), \qquad \mathsf{pk_d} := [\mathsf{ivk}]\,\mathsf{g_d}, \\ \psi &:= \mathsf{ToBase}(\mathsf{PRFexpand}_{\mathsf{rseed}}([\mathtt{0x09}] \,\|\, \underline{\rho})), \\ \mathsf{rcm} &:= \mathsf{ToScalar}(\mathsf{PRFexpand}_{\mathsf{rseed}}(\mathsf{pre_rcm})), \end{aligned}$$

with the 137-byte string pre_rcm of §4.3 formed from $\mathsf{g_d}$, $\mathsf{pk_d}$, v, ρ and ψ;

(iii) $([\mathsf{esk}]\,\mathsf{g_d})^\star = \mathsf{epk}^\star$ for

$$\mathsf{esk} := \mathrm{ToScalar}(\mathsf{PRFexpand}_{\mathsf{rseed}}([\mathtt{0x04}] \,\|\, \underline{\rho})),$$

a check that ZIP 2005 places directly after the computation of $\mathsf{g_d}$;

(iv) the note commitment cm' of the re-derived note (Definition 4.4) is not $\perp$ and $\mathsf{Extract}_{\mathbb{P}}(\mathsf{cm}') = \mathsf{cmx}$.

The accepted note is $n = ((d, \mathsf{pk_d}), v, \rho, \psi, \mathsf{rcm})$, with memo (ZIP 2005, "Changes to the Protocol Specification", §4.20.2 and §4.20.3; protocol specification, §"Decryption using an Incoming Viewing Key (Sapling and Orchard)" and §"Note Plaintexts and Memo Fields").

The order in (ii) is forced because rcm depends on $\mathsf{g_d}$, $\mathsf{pk_d}$ and ψ; the recovery with ovk of §10.3 follows the same order. The specification's decryption sections still use the legacy derivation of rcm; ZIP 2005 specifies both reordered procedures. The specification's check $\mathsf{rcm} < p_{\mathsf{Vesta}}$ holds for every output of ToScalar.

Remark 10.13 (Role of check (iii)). The Action statement (Definition 9.2) does not constrain epk. Check (iii) makes the recipient accept only an epk formed on the base $\mathsf{g_d}$ of the diversifier in the plaintext. Without it a sender could agree the key with one address of a recipient while naming the diversifier of another, and link the two addresses by observing whether the recipient accepts (ZIP 212, "Motivation").

Remark 10.14 (Where the Ironwood derivation is checked). The Action statement takes ψ and rcm as witnesses and does not check their derivation from rseed (Definition 9.2). An accepted Ironwood-pool note has, by checks (i), (ii) and (iv), ψ and rcm derived under 0x03 and the commitment cmx. For a coinbase output, consensus runs the recovery with ovk of §10.3 under the all-zero ovk ("Chain state and pool rules", §11.4), which checks the lead byte and the whole 0x03 derivation. For every other output, these recipient checks are the only check of the 0x03 note format.

Orchard-pool outputs carry lead byte 0x02 and use the legacy trapdoor derivation of the protocol specification, §"Sending Notes (Orchard)"; the recipient decrypts them with the same ivk under that byte and derivation (§"Note Plaintexts and Memo Fields"), on which this volume does not rely.

Proposition 10.15 (Consistency of accepted notes)**.** *Let the acceptance procedure accept an Ironwood-pool output with note $n = ((d, \mathsf{pk_d}), v, \rho, \psi, \mathsf{rcm})$.*

(*a*) *The note n is addressed to $(d, [\mathsf{ivk}]\,\mathsf{g_d})$, an address of the recipient's key; it has $\rho = \mathsf{nf}^{\mathsf{old}}$ of the same Action; and its ψ and* rcm *are derived from its* rseed *under* 0x03.

(*b*) *The extracted commitment of n satisfies* $\mathsf{Extract}_{\mathbb{P}}(\mathsf{NoteCommit}_{\mathsf{rcm}}(M(n))) = \mathsf{cmx}$.

(*c*) *For every efficient sender, which may choose $\mathsf{epk}^\star$, C_{enc}, C_{out} and the keys under which they are formed, the probability that the recipient accepts n while the sender outputs a note $n' \neq n$ with $\mathsf{cmx}(n') = \mathsf{cmx}$ is negligible, under the binding of note commitments (Proposition 4.5), hence under Assumptions 2.22 and 2.8. In particular, under Assumption 9.11, the committed values and the trapdoor of the created note that an extractor recovers from the Action proof, which satisfy condition A2 of the Action statement (Definition 9.2), are $(\mathsf{g_d}, \mathsf{pk_d}, v, \rho, \psi)$ and* rcm *modulo p_{Vesta}, those of n, except with negligible probability.*

Proof. Parts (a) and (b) hold by construction: acceptance requires checks (i), (ii) and (iv), and check (iv) recomputes the commitment of n deterministically and compares its x-coordinate with

cmx, whatever the ciphertext and whichever key decrypted it. The address $(d, [\mathsf{ivk}]\,\mathsf{g_d})$ is an address of the key because $\mathsf{FF1\text{-}AES256_{dk}}$ is a permutation of the 88-bit strings, so every d is the diversifier of one index (§3.3).

(c) Consider the algorithm that generates the recipient's key, runs the sender, runs the acceptance procedure on its output, and outputs n and the sender's n'. When the recipient accepts, n opens $\mathsf{cmx} \neq \perp$ by (b), so an $n' \neq n$ with $\mathsf{cmx}(n') = \mathsf{cmx}$ gives two distinct notes with equal extracted commitments other than $\perp$, which Proposition 4.5(b) excludes except with negligible probability. For the particular case, the extractor of Assumption 9.11 run on the sender is such a sender-side algorithm, with an opening of the committed values and a trapdoor in place of a note. Outside the negligible $\perp$-weakened case of condition A2, which yields an input on which NoteCommit returns $\perp$ (Proposition 2.24(iii)), its opening (M', rcm') satisfies $\mathsf{Extract}_{\mathbb{P}}(\mathsf{NoteCommit}_{\mathsf{rcm}'}(M')) = \mathsf{cmx}$. Step 2 of the proof of Proposition 4.5(b), which uses Proposition 2.24(ii) and the fact that $H_D \neq O$, then gives $M' = M(n)$ and $\mathsf{rcm}' \equiv \mathsf{rcm} \pmod{p_{\mathsf{Vesta}}}$ except with negligible probability, and $M(n)$ determines $(\mathsf{g_d}, \mathsf{pk_d}, v, \rho, \psi)$ because the encoding is injective. The proof uses neither the integrity of ChaCha20-Poly1305 nor key commitment, which the scheme lacks (Assumption 10.5): a ciphertext valid under two keys yields under each a candidate that must still pass check (iv). □

On acceptance the recipient records n with its note position, the index of cmx among the leaves of the note commitment tree of the output's pool, fixed by the order in which the chain appends extracted commitments ("The Merkle hash and the tree", §5.1; protocol specification, §"Note Commitment Trees"). The position determines the authentication path, which the recipient computes from public data (Lemma 5.15) and which, with the opening n of the leaf cmx (Proposition 10.15), a later Action consuming n requires (conditions A1 and A3 of Definition 9.2).

The key tiers of Proposition 3.18 ("Capability separation") have the following powers over note ciphertexts, for keys generated with use_qsk = false.

Incoming tier. A holder of the incoming viewing key (dk, ivk), of which trial decryption and acceptance use only ivk, detects and reads every note sent to any address of the key, value and memo included. It cannot compute the nullifiers of these notes, which require nk, and (dk, ivk) yields neither nk nor ask (Proposition 3.18(b)). It cannot link a published nullifier to one of them, for notes with pairwise distinct ρ (Proposition 6.10), so the nullifier sets alone do not show it which are consumed; and it cannot authorise a spend.

Full-viewing tier. A holder of the full viewing key $(\mathsf{ak}, \mathsf{nk}, \mathsf{r_{ivk}})$ derives (dk, ivk) and ovk ("Viewing keys", §3.2). It detects and reads incoming notes as above, and reads the outputs sent under its ovk (§10.3), not those sent under $\mathsf{ovk} = \perp$ or another outgoing viewing key. With nk it computes the nullifier $\mathsf{nf} = \mathsf{DeriveNullifier_{nk}}(\rho, \psi, \mathsf{cm})$ of each accepted note (Definition 6.3) and detects its consumption by the membership of nf in the nullifier set of the note's pool ("Nullifier sets", §6.2; protocol specification, §"Decryption using an Incoming Viewing Key (Sapling and Orchard)", notes). It cannot spend, since it yields no ask (Proposition 3.18(a)). For each unspent note the owner thus holds the note, its position, its authentication path and its nullifier, the consumed-note inputs of an Action (Definition 9.2); spending authority is ask.

11 Transactions and consensus rules

This section assembles the public data of an Action and of a bundle, states the transaction format that carries them, constructs the digest that both kinds of signature sign and proves in Proposition 11.10 which data the signatures fix, states the chain state and the rules of each pool, and gives the procedures of the sender and of the verifier, with a worked two-Action payment.

Field	Type enforced on parsing	Encoding	Bytes	Constructed in
$\mathsf{cv}^{\mathsf{net}}$	$\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$, the identity admitted	$(\mathsf{cv}^{\mathsf{net}})^\star$	32	Definition 8.2
nf	$\{0, \dots, p_{\mathsf{Pallas}} - 1\}$	$\mathsf{I2LEOSP}_{256}(\mathsf{nf})$	32	Definition 6.3
rk	$\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$; the identity excluded by a separate rule	$\mathsf{rk}^\star$	32	§7.2
cmx	$\{0, \dots, p_{\mathsf{Pallas}} - 1\}$	$\mathsf{I2LEOSP}_{256}(\mathsf{cmx})$	32	Definition 4.4
epk	$\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}}) \setminus \{O\}$	$\mathsf{epk}^\star$	32	§10.2
C_{enc}	580-byte string	itself	580	§10.2
C_{out}	80-byte string	itself	80	§10.3
Action description			820	

Table 6: The fields of an Action description, in the order of their encoding, with the type that parsing enforces and the byte length of the encoding.

11.1 Action descriptions and bundles

The objects of this subsection are constructed in §4 to §10. The subsection fixes their encodings and their grouping; it redefines neither the Action nor the bundle (Definitions "Action" and "Bundle", §4.4). As in §10.2, a star encoding $P^\star$ enters a byte string as the 32 bytes $\mathsf{LEBS2OSP}_{256}(P^\star)$ and is identified with them.

Definition 11.1 (Action description)**.** The *Action description* of an Action is the byte string

$$(\mathsf{cv}^{\mathsf{net}})^\star \parallel \mathsf{I2LEOSP}_{256}(\mathsf{nf}) \parallel \mathsf{rk}^\star \parallel \mathsf{I2LEOSP}_{256}(\mathsf{cmx}) \parallel \mathsf{epk}^\star \parallel C_{\mathsf{enc}} \parallel C_{\mathsf{out}}$$

of $5 \cdot 32 + 580 + 80 = 820$ bytes, whose fields, in this order, are those of Table 6: the net value commitment of Definition 8.2, the nullifier of the consumed note (Definition 6.3), the randomised validating key of "Randomised validating keys" (§7.2), the extracted commitment of the created note (Definition 4.4), and the ephemeral public key and the two ciphertexts of "Encryption to the recipient" (§10.2) and "The outgoing ciphertext" (§10.3) (protocol specification, §"Action Descriptions" and §"Action Description Encoding and Consensus").

Every field of an Action description must be the canonical encoding of its type (protocol specification, §"Action Descriptions"; for cmx also §"Action Description Encoding and Consensus"). Hence nf and cmx are integers below p_{Pallas}, and neither is checked to be the x-coordinate of a Pallas point; epk is a non-identity point by its type; rk is a point, and the rule of "Randomised validating keys" (§7.2) excludes the identity; and $\mathsf{cv}^{\mathsf{net}}$ may be the identity. The Action's spend-authorisation signature, of 64 bytes ("The RedPallas signature scheme", §7.1), and its proof are not part of the Action description. The bundle carries the signatures, one per Action, and one aggregate proof, whose per-Action instances are in the order of the Action descriptions (ZIP 229, "Consensus Rules").

A bundle of one pool with $n \geq 1$ Actions publishes:

(i) its n Action descriptions;

(ii) one anchor rt ("Anchors", §5.2), shared by all n Actions: no Action description carries an anchor, and the transaction carries one anchor field for each pool with Actions, anchorOrchard and anchorIronwood;

(iii) one flags byte ("Bundle flags", §9.1), whose bit 0 is enableSpends and bit 1 enableOutputs, and whose bit 2 is enableCrossAddress in flagsIronwood and reserved and 0 in flagsOrchard;

(iv) the balancing value v^{balance} of "The balancing value" (§8.2);

(v) one aggregate proof over its n Actions ("The Action circuit and the Halo 2 proof", §9.3);

(vi) n spend-authorisation signatures, one per Action;

(vii) one binding signature ("The binding signature", §8.3).

The anchor and the flags are common to all Actions of the bundle and are encoded once (protocol specification, §"Action Transfers and their Descriptions" and §"Action Descriptions", note). A transaction carries at most one bundle per pool, hence one binding signature for each pool with Actions (ZIP 229, "Transaction Format" and "Consensus Rules"). Figure 4 shows each public field with the checks that consume it.

A bundle that consumes s real notes and creates o real notes has $n \geq \max(s, o)$ Actions, since every Action consumes exactly one note and creates exactly one; its $n - s$ consumed sides and $n - o$ created sides without a real note carry dummy notes of value 0 (Definition "Dummy notes", §4.4; protocol specification, §"Dummy Notes (Orchard)"). When the consumed value exceeds the payment and the fee, the difference is *change*: an ordinary created note to an address of the sender, constructed as every other created note.

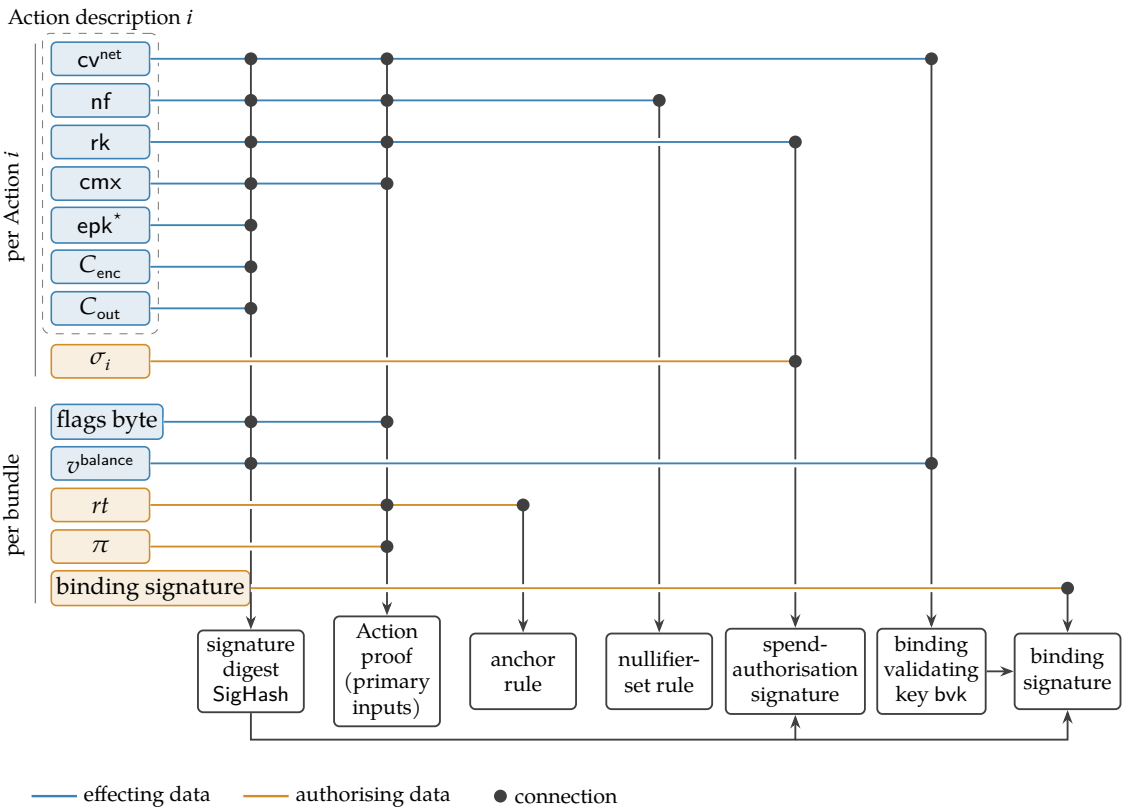


Figure 4: The public data of one bundle and the checks that consume them. Rows are the fields: per Action i, the net value commitment $\mathsf{cv}^{\mathsf{net}}$, the nullifier nf, the randomised validating key rk, the extracted commitment cmx, the ephemeral key $\mathsf{epk}^{\star}$ and the ciphertexts C_{enc} and C_{out} of the Action description, and the spend-authorisation signature σ_i; per bundle, the flags byte, the balancing value v^{balance}, the anchor rt, the aggregate proof π and the binding signature. Each column ends in the check or derived key that consumes it. A dot joins a field to a column; a row that crosses a column without a dot is not joined to it. Blue rows are effecting data and amber rows authorising data ("Transaction digests and signatures", §11.3). Every effecting field, the ciphertexts included, joins the signature digest SigHash, which feeds every spend-authorisation signature and the binding signature; the anchor and the proof do not join it (Proposition 11.10). The fields nf, cmx, $\mathsf{cv}^{\mathsf{net}}$, rk, the flags and rt are primary inputs of the Action proof; rt also feeds the anchor rule, nf the nullifier-set rule, rk the spend-authorisation signature, and $\mathsf{cv}^{\mathsf{net}}$ and v^{balance} the binding validating key bvk.

11.2 The transaction format

The transaction formats in force are version 5, specified in ZIP 225, and version 6, specified in ZIP 229; a transaction of any other version, version 4 included, is invalid (protocol specification, §"Transaction Consensus Rules", as amended by ZIP 2003, "Specification"; ZIP 259, Abstract). Of the two Orchard-protocol bundles a version-5 transaction can carry only the Orchard-pool bundle; a transaction with Ironwood-pool Actions is of version 6 (protocol specification, §"Action Descriptions"). The Orchard-pool rules of "Chain state and pool rules" (§11.4) apply to Orchard-pool Actions in both versions (ZIP 229, "Consensus Rules"). *Sprout*, Zcash's first shielded protocol, spends its notes only through *JoinSplit descriptions*, its transfer descriptions, which only transaction versions 2 to 4 carry (protocol specification, §"Transaction Encoding and Consensus"); Sprout funds are therefore unspendable, and their value remains counted as issued, in the sense of the total issued supply defined in "Chain state and pool rules" (§11.4) (ZIP 2003, Abstract and "Interaction with the proposed Network Sustainability Mechanism"; ZIP 259, Abstract). This subsection describes version 6.

Construction 11.2 (Version-6 transaction)**.** A version-6 transaction consists of the fields of the version-5 format up to and including the Orchard component (ZIP 225), followed by the Ironwood component. Its first five fields form the header:

(i) header, whose bit 31, the fOverwintered flag, is set and whose bits 0 to 30 hold the version, 6;

(ii) nVersionGroupId, which must be 0xD884B698;

(iii) nConsensusBranchId, which must equal the consensus branch identifier used by the signature digest of "Transaction digests and signatures" (§11.3), the identifier of the consensus branch under which the transaction is validated, where a *consensus branch* is a sequence of consecutive blocks under one consensus rule set whose first block is the genesis block or follows a block validated under older rules, and each network upgrade fixes for the branch that it begins a globally unique non-zero 32-bit *consensus branch identifier* (ZIP 200, Terminology and "Specification");

(iv) lock_time, the transparent lock time, which no rule of this volume uses;

(v) nExpiryHeight, the *expiry height*.

In version 6, bits 0 and 1 of flagsOrchard are named enableSpends and enableOutputs, as in flagsIronwood, and bits 2 to 7 of flagsOrchard are reserved and must be 0, in version 5 and version 6 alike ("Bundle flags", §9.1). The Ironwood component has the structure of the Orchard component, with its own flags, balancing value, anchor, proof and signatures; both encode each Action as the 820-byte Action description of §11.1 (ZIP 229, "Transaction Format" and "Consensus Rules"; protocol specification, §"Transaction Consensus Rules").

Definition 11.3 (Coinbase transaction)**.** A *coinbase transaction* is the first transaction of a block; it collects the block subsidy, the value newly issued in that block, and the fees of the block's other transactions. Every block has exactly one (protocol specification, §"Coinbase Transactions").

The field nExpiryHeight is 0, which disables expiry, or a block height. The *expiry rule* is: for a transaction other than a coinbase transaction, nExpiryHeight is at most 499 999 999, and such a transaction with nExpiryHeight $\neq 0$ is invalid in every block of height greater than nExpiryHeight; the nExpiryHeight of a coinbase transaction equals the height of its block (protocol specification,

Field	Bytes	Content
nActionsIronwood	compactSize	the count n
vActionsIronwood	$820\,n$	the n Action descriptions of §11.1
flagsIronwood	1	bit 0 enableSpends, bit 1 enableOutputs, bit 2 enableCrossAddress, bits 3 to 7 reserved and 0
valueBalanceIronwood	8	the balancing value, signed 64-bit little-endian
anchorIronwood	32	the anchor rt
sizeProofsIronwood	compactSize	the proof length, $2720 + 2272\,n$
proofsIronwood	$2720 + 2272\,n$	the aggregate proof
vSpendAuthSigsIronwood	$64\,n$	one spend-authorisation signature per Action, in Action order
bindingSigIronwood	64	the binding signature

Table 7: The Ironwood component of a version-6 transaction with n Ironwood-pool Actions (ZIP 229, "Transaction Format"). The fields after the count are present if and only if $n > 0$.

§"Transaction Consensus Rules"; ZIP 203, "Specification"; ZIP 229, "Transaction Format"). The sender's choice of the expiry height is stated in "Construction of a transaction" (§11.5).

Definition 11.4 (Ironwood component)**.** Let n be the number of Ironwood-pool Actions of a version-6 transaction. The *Ironwood component* is the sequence of the fields of Table 7, in that order. The fields after nActionsIronwood are present if and only if $n > 0$, and an absent valueBalanceIronwood is taken to be 0. A count and a proof length are encoded as *compactSize*, the variable-length integer encoding of Bitcoin (ZIP 229, "Transaction Format").

The consensus rules on the component are: $n < 2^{16}$; if $n > 0$, at least one of enableSpends and enableOutputs of flagsIronwood is 1; bits 3 to 7 of flagsIronwood are 0; valueBalanceIronwood lies in $\{-\mathsf{MAX_MONEY}, \ldots, \mathsf{MAX_MONEY}\}$, where $\mathsf{MAX_MONEY} = 2.1 \cdot 10^{15}$ zatoshi is a protocol constant (protocol specification, §"Constants"); and proofsIronwood has exactly $2720 + 2272\,n$ bytes (ZIP 229, "Consensus Rules"; protocol specification, §"Transaction Consensus Rules" and §"Action Description Encoding and Consensus"). The flags byte differs from flagsOrchard only in bit 2, which is enableCrossAddress in flagsIronwood and reserved and 0 in flagsOrchard ("Bundle flags", §9.1). The proof length is the one that ZIP 229 fixes; the *Halo 2 Guide*, §"Halo 2's specifics: parameters, generators from a hash, the deferred multi-scalar multiplication, and the proof bytes", paragraph "The bytes", describes what its constant and its per-Action term cover. For one Action the proof has $2720 + 2272 = 4992$ bytes.

11.3 Transaction digests and signatures

Write $\mathsf{h}(P, x) :=$ BLAKE2b-256(P, x), unkeyed BLAKE2b with a 32-byte output, the 16-byte personalisation P and the input x (§10.2); the personalisation is mixed into the initialisation vector (*Crypto Guide*, §"Domain separation and personalisation"). For a byte string C and $0 \le a \le b \le |C|$, the string $C[a{:}b]$ consists of the bytes a to $b - 1$ of C, counted from 0. Every field enters a digest in its encoding of Table 6 or Table 7, and β denotes the 4-byte little-endian encoding of the consensus branch identifier.

Definition 11.5 (Transaction identifier)**.** The *transaction identifier* of a version-6 transaction is

$$\mathsf{txid} := \mathsf{h}(\texttt{ZcashTxHash_} \,\|\, \beta,\ d_{\mathsf{hdr}} \,\|\, d_{\mathsf{tr}} \,\|\, d_{\mathsf{Sap}} \,\|\, d_{\mathsf{Orc}} \,\|\, d_{\mathsf{Irw}}),$$

the Ironwood digest last, where:

(i) the *header digest* d_{hdr} is h under `ZTxIdHeadersHash` of the 4-byte little-endian encodings of header, nVersionGroupId, nConsensusBranchId, lock_time and nExpiryHeight;

(ii) the *transparent digest* d_{tr} and the *Sapling digest* d_{Sap} are those of ZIP 244, "T.2: transparent_digest" and "T.3: sapling_digest", the latter with the version-6 node of ZIP 229; they cover the transparent inputs and outputs and the Sapling component, the component of Zcash's second shielded protocol;

(iii) for $n \geq 1$ Ironwood-pool Actions, with the fields of Action i indexed by i, the *Ironwood digest* is

$$d_{\mathsf{Irw}} := \mathsf{h}(\texttt{ZTxIdIronwd_H_v6},\ d_{\mathsf{C}} \,\|\, d_{\mathsf{M}} \,\|\, d_{\mathsf{N}} \,\|\, \mathsf{flagsIronwood} \,\|\, \mathsf{valueBalanceIronwood}),$$
$$d_{\mathsf{X}} := \mathsf{h}(P_{\mathsf{X}},\ r_1^{\mathsf{X}} \,\|\, \cdots \,\|\, r_n^{\mathsf{X}}) \qquad (\mathsf{X} \in \{\mathsf{C}, \mathsf{M}, \mathsf{N}\}),$$

with the *compact*, *memo* and *non-compact* records and personalisations

$$r_i^{\mathsf{C}} := \mathsf{nf}_i \,\|\, \mathsf{cmx}_i \,\|\, \mathsf{epk}_i^{\star} \,\|\, C_{\mathsf{enc},i}[0{:}52], \qquad P_{\mathsf{C}} := \texttt{ZTxIdIrnActCH_v6},$$
$$r_i^{\mathsf{M}} := C_{\mathsf{enc},i}[52{:}564], \qquad P_{\mathsf{M}} := \texttt{ZTxIdIrnActMH_v6},$$
$$r_i^{\mathsf{N}} := (\mathsf{cv}_i^{\mathsf{net}})^{\star} \,\|\, \mathsf{rk}_i^{\star} \,\|\, C_{\mathsf{enc},i}[564{:}580] \,\|\, C_{\mathsf{out},i}, \qquad P_{\mathsf{N}} := \texttt{ZTxIdIrnActNH_v6};$$

for $n = 0$ it is $d_{\mathsf{Irw}} := \mathsf{h}(\texttt{ZTxIdIronwd_H_v6}, \varepsilon)$;

(iv) the *Orchard digest* d_{Orc} has the same structure over the Orchard-pool Actions, with flagsOrchard and valueBalanceOrchard, under the personalisation `ZTxIdOrchardH_v6`, with the Action digests personalised

$$P_{\mathsf{C}} = \texttt{ZTxIdOrcActCHash}, \quad P_{\mathsf{M}} = \texttt{ZTxIdOrcActMHash}, \quad P_{\mathsf{N}} = \texttt{ZTxIdOrcActNHash};$$

without Orchard-pool Actions it is $\mathsf{h}(\texttt{ZTxIdOrchardH_v6}, \varepsilon)$.

(ZIP 244, "Digests", "txid_digest", "T.1: header_digest" and "T.4: orchard_digest"; ZIP 229, "Ironwood component digest", "Anchor commitment (version 6)" and "Summary of the resulting digest structure".)

The ciphertext C_{enc} is the ChaCha20 encryption of the 564-byte note plaintext, byte for byte, followed by the 16-byte tag (*Crypto Guide*, §"The ChaCha20-Poly1305 AEAD", Construction "ChaCha20-Poly1305"; "The note plaintext", §10.1). The offsets therefore split it into the encryption of the 52-byte prefix of the plaintext, the lead byte, d, v and rseed ($1 + 11 + 8 + 32 = 52$), the encryption of the 512-byte memo, ending at $52 + 512 = 564$, and the tag. The empty Ironwood digest differs from the empty Orchard digest, since the two personalisations differ. Neither pool digest contains the anchor.

Definition 11.6 (Authorising-data digest; effecting and authorising data)**.** The *authorising-data digest* of a version-6 transaction is

$$\mathsf{auth} := \mathsf{h}(\texttt{ZTxAuthHash_} \,\|\, \beta,\ a_{\mathsf{tr}} \,\|\, a_{\mathsf{Sap}} \,\|\, a_{\mathsf{Orc}} \,\|\, a_{\mathsf{Irw}}),$$

where a_{tr} is the transparent scripts digest and a_{Sap} the Sapling authorising-data digest (ZIP 244, "Authorizing Data Commitment"; ZIP 229), and, for $n \geq 1$ Ironwood-pool Actions,

$$a_{\mathsf{Irw}} := \mathsf{h}(\texttt{ZTxAuthIrnwdH_v6},\ \mathsf{proofsIronwood} \,\|\, \mathsf{vSpendAuthSigsIronwood} \,\|\, \mathsf{bindingSigIronwood} \,\|\, \mathsf{anchorIronwood}),$$

and $a_{\mathsf{Irw}} := \mathtt{h}(\mathtt{ZTxAuthIrnwdH_v6}, \varepsilon)$ for $n = 0$; the Orchard digest a_{Orc} has the same structure under `ZTxAuthOrchaH_v6`. Each Orchard-protocol anchor is committed if and only if its component has Actions. The *effecting data* of a transaction are the data under the transaction identifier digest; its *authorising data* are the proofs, the signatures, the anchors and the transparent scripts, the data under the authorising-data digest (ZIP 229, "Ironwood component digest" and "Anchor commitment (version 6)"; ZIP 244, "Authorizing Data Commitment").

ZIP 244, "Authorizing Data Commitment", describes the pair of the transaction identifier and the authorising-data digest as a commitment to all data of the transaction. In version 6 the anchors are authorising data so that a transaction can be signed before its anchor is chosen and its proofs are computed; replacing a bundle's anchor and proof together does not change the transaction identifier (ZIP 229, Motivation and "Anchor commitment (version 6)"). Below the two roots, a node whose directly hashed input differs between versions 5 and 6 has a version-6 personalisation of its own; the roots keep theirs and hash one more child in version 6. The nodes that omit or include an anchor in version 6 are personalised `ZTxIdSSpendNH_v6`, `ZTxAuthSapliH_v6`, `ZTxIdOrchardH_v6` and `ZTxAuthOrchaH_v6` (ZIP 229, "Ironwood component digest" and "Anchor commitment (version 6)").

Definition 11.7 (Signature digest)**.** The *signature digest* of a transaction is its digest not associated with a transparent input, under the hash type SIGHASH_ALL, which covers all transparent inputs and outputs:

$$\mathsf{SigHash} := \mathtt{h}(\mathtt{ZcashTxHash_} \parallel \beta,\ d_{\mathsf{hdr}} \parallel s_{\mathsf{tr}} \parallel d_{\mathsf{Sap}} \parallel d_{\mathsf{Orc}} \parallel d_{\mathsf{Irw}}),$$

whose header and shielded children are those of txid and whose child s_{tr} is the transparent signature digest of ZIP 244, "S.2: transparent_sig_digest". For a coinbase transaction and for a transaction without transparent inputs, $s_{\mathsf{tr}} = d_{\mathsf{tr}}$, so $\mathsf{SigHash} = \mathsf{txid}$; otherwise s_{tr} commits in addition to the values and scripts of the coins that the transparent inputs spend. The digest SigHash is the message of every spend-authorisation signature and of each pool's binding signature, the message left as a parameter in "The RedPallas signature scheme" (§7.1) (ZIP 244, "Signature Digest"; ZIP 229, "Transaction Identifiers, Auth Digests, and Signature Digests"; protocol specification, §"Action Descriptions" and §"Balance and Binding Signature (Orchard)").

Proposition 8.10, which holds for every message, applies to the binding signature on SigHash.

Remark 11.8 (Version 5). In version 5 the root of the transaction identifier has four children, the header, transparent, Sapling and Orchard digests, under the same root personalisation. The Orchard digest, under `ZTxIdOrchardHash`, hashes

$$d_{\mathsf{C}} \parallel d_{\mathsf{M}} \parallel d_{\mathsf{N}} \parallel \mathsf{flagsOrchard} \parallel \mathsf{valueBalanceOrchard} \parallel \mathsf{anchorOrchard},$$

with the Action digests of version 6, and the Orchard authorising-data digest, under `ZTxAuthOrchaHash`, covers the proofs and the signatures only. In version 5 the Orchard anchor is therefore effecting data and fixed by the signatures; the proofs are authorising data in both versions. A version-5 transaction carries Orchard-pool Actions only (ZIP 244, "txid_digest", "T.4: orchard_digest" and "A.3: orchard_auth_digest").

Figure 5 draws both trees. The binding of the signatures needs one further assumption, stated at its first use.

Assumption 11.9 (Collision resistance of BLAKE2b). No efficient algorithm outputs, except with negligible probability, two distinct pairs (P, x) and (P', x'), each of a 16-byte personalisation and an input string, with $\mathtt{h}(P, x) = \mathtt{h}(P', x')$. The notion is the *Crypto Guide*'s Defi-

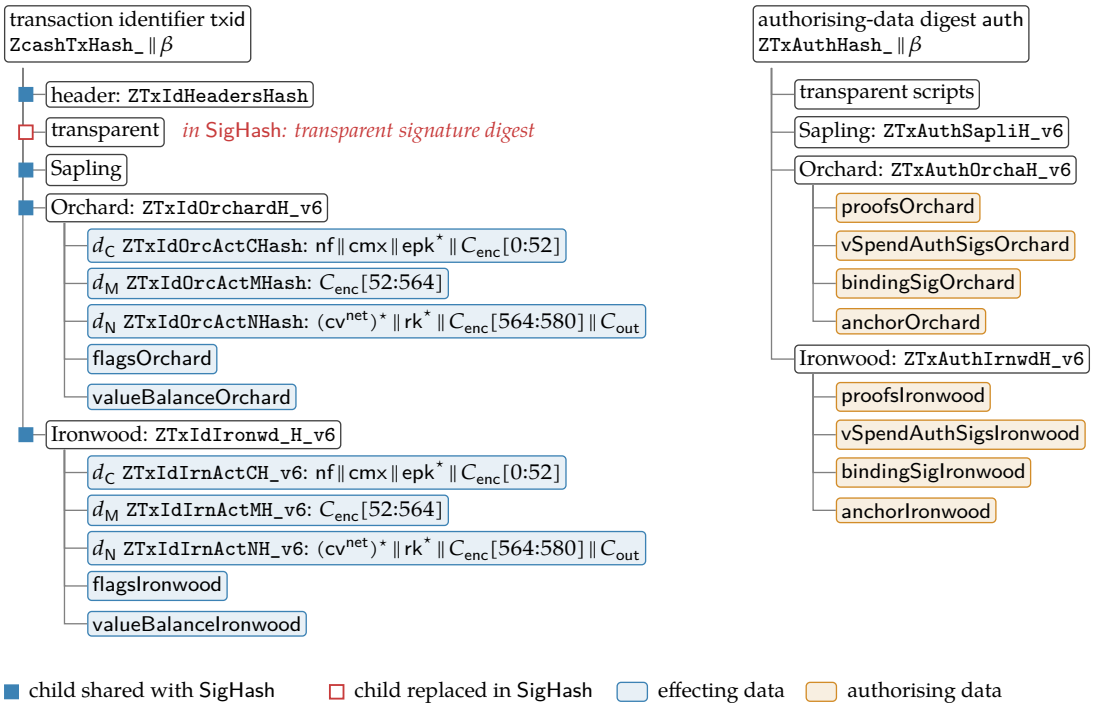


Figure 5: The version-6 transaction identifier (left) and authorising-data digest (right) as trees of personalised BLAKE2b-256 nodes, with the Orchard and Ironwood components expanded (ZIP 244; ZIP 229). Each node hashes the concatenation of its children, in order, under the personalisation shown; the string β encodes the consensus branch identifier. The Action digests d_{C}, d_{M} and d_{N} hash the listed fields of every Action of their component, Action after Action; the byte flagsIronwood is the Ironwood component's own flags byte. The header, transparent and Sapling digests are not expanded. The signature digest SigHash has the root personalisation and the children of txid (filled squares), except the transparent child (open square), which it replaces by the transparent signature digest. The anchors and the proofs lie under the authorising-data digest only. By Proposition 11.10, the signatures fix every field under txid and no field under the authorising-data digest.

nition "Collision resistance" (§"Security notions: preimage, second-preimage, and collision resistance"), read for the unkeyed deployment as that volume's Remark "Convention: keyed definitions, unkeyed deployments" (§"Hash functions and the random oracle model", §"Syntax") prescribes; the personalisation is mixed into the initialisation vector (§"Domain separation and personalisation"), so the pair (P, x) is the hashed input. It is used by Proposition 11.10 and by the consensus-branch binding below.

Proposition 11.10 (Bundle binding)**.** *Let T be a version-6 transaction, and for a bundle of T, of either pool, let $\mathsf{rk}_1, \ldots, \mathsf{rk}_n$ be the randomised validating keys of its Actions, $\mathsf{rk}_i = [\mathsf{rsk}_i]\, G^{\mathsf{SpendAuth}}$ with $\mathsf{rsk}_i = \mathsf{ask}_i + \alpha_i$ as in "Randomised validating keys" (§7.2), where ask_i is the spend authorising key of the spending key that authorises Action i. Assume:*

- (H) *each ask_i is uniform on the non-zero scalars a for which $[a]\, G^{\mathsf{SpendAuth}}$ has even y-coordinate (§3.1), and every value of the experiment other than a signature by the holder of ask_i is computed from $\mathsf{ak}_i^{\mathbb{P}} = [\mathsf{ask}_i]\, G^{\mathsf{SpendAuth}}$ and from values drawn independently of ask_i;*
- (S) *the holder of ask_i signs under rsk_i only $\mathsf{SigHash}(T)$, and signs any other message only under $\mathsf{ask}_i + \alpha$ for a fresh randomiser α drawn by $\mathsf{GenRandom}$.*

An adversary receives T and all other public data, and obtains on request, for messages of its choice, signatures of the key holders under fresh randomisers as in (S); it receives no ask_i and no rsk_i. Under Assumptions 11.9, 2.22 and 7.3, the probability that it outputs an accepted transaction T' that contains an Action description with validating key rk_i for some i and whose effecting data differ from those of T is negligible.

The signatures thus fix the effecting data: every Action's nf, cmx, $\mathsf{cv}^{\mathsf{net}}$, rk, epk, C_{enc} and C_{out}, the number and order of the Actions, each bundle's flags and balancing value, the header, with the consensus branch identifier and the expiry height, and the transparent data. They fix neither the anchor nor the proof of a bundle: for a pair (rt', π') such that π' verifies on the primary inputs formed with rt', replacing (rt, π) by (rt', π') changes neither the transaction identifier nor any signed message, so every signature remains valid, and π' binds rt' as a primary input (Definition 9.2).

Proof. Let E be the event that the adversary outputs such a T'. On E, let i be the least index for which T' contains an Action description with key rk_i, and let σ' be the spend-authorisation signature that T' carries for it. Since T' is accepted, σ' is valid under rk_i on $m' := \mathsf{SigHash}(T')$ ("Randomised validating keys", §7.2). Let $m := \mathsf{SigHash}(T)$, and split E into $E_=$, where $m' = m$, and $E_{\neq}$, where $m' \neq m$.

Case $E_=$: a collision. The digest $\mathsf{SigHash}$ is computed by the tree of its definition from the effecting data and, when the transaction has transparent inputs, from the values and scripts of the coins that they spend. Each node hashes a string from which its children and its fields are recovered uniquely: its input consists of fixed-length fields and digests, of concatenations of fixed-length per-Action records whose number the length determines, or, for transparent outputs, of records whose variable-length script carries its length; the two nodes that share a personalisation, the transparent digest and the transparent signature digest, hash strings of different lengths (ZIP 244, "TxId Digest" and "Signature Digest"). Compare the trees of T and T' from the root, whose outputs are equal. At a node with equal outputs, either the two (personalisation, input) pairs differ, a collision of h, or they agree, and then the node's children have equal outputs and its fields are equal. The effecting data of T and T' differ, so some field, or some number of records, differs, and descending along the nodes that cover it ends in a collision. An algorithm that runs the experiment, which it can since it samples every key, and performs this descent outputs a collision with probability $\Pr[E_=]$, which is negligible by Assumption 11.9.

Case $E_{\neq}$: a forgery. An algorithm $\mathcal{B}$ plays the adversary of Proposition 7.12. It receives $Y = [x]\, G^{\mathsf{SpendAuth}}$ for x uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$, with access to H and to the signing oracle, and draws i^*

uniformly from $\{1, \ldots, n\}$. If $Y = \mathcal{O}$ or Y has odd y-coordinate, it stops; otherwise, an event of probability $(p_{\mathsf{Vesta}} - 1)/(2p_{\mathsf{Vesta}})$, the point Y has the distribution that (H) gives to $\mathsf{ak}^{\mathbb{P}}_{i^*}$, and $\mathcal{B}$ sets $\mathsf{ak}^{\mathbb{P}}_{i^*} := Y$. It computes every other value of the experiment as (H) permits, in particular the proof of T, whose witness (Definition 9.2) contains $\mathsf{ak}^{\mathbb{P}}$ and α but not ask; it draws every randomiser by GenRandom itself; it obtains each signature by the holder of ask_{i^*}, on m and on the adversary's requests, as the oracle's answer on the pair (α, M); and it produces the other signatures with keys it holds. The adversary's view is then that of the experiment. On $E_{\neq}$ with $i = i^*$, $\mathcal{B}$ outputs $(\mathsf{rk}_{i^*}, m', \sigma')$ with the randomiser α_{i^*}, of its own choice, as part (ii) of Proposition 7.12 admits. The oracle answered under the key rk_{i^*} only on m, except when a randomiser drawn for a request equals α_{i^*}, which has negligible probability by the lemma on the distance of GenRandom from uniform (§7.2); outside that event the output is a forgery. The index i^* is independent of the adversary's view, so $\mathcal{B}$ forges with probability at least

$$\frac{p_{\mathsf{Vesta}} - 1}{2p_{\mathsf{Vesta}}} \cdot \frac{\Pr[E_{\neq}]}{n}$$

less a negligible term, where $n < 2^{16}$. By Proposition 7.12(ii), under Assumptions 2.22 and 7.3, $\Pr[E_{\neq}]$ is negligible.

Anchor and proof. By the definitions above, the anchor and the proof of each bundle are inputs of the authorising-data digest and of no node of txid or SigHash. Replacing them changes neither, and every signature, whose message is SigHash, remains valid. The verifier checks the bundle's proof against the primary inputs formed with the bundle's anchor field (protocol specification, §"Action Descriptions"), now rt', so the modified bundle is accepted only if π' verifies for rt', and the anchor rule then applies to rt'. □

Remark 11.11 (Hypothesis (H) for honest keys). For keys generated with use_qsk = false from uniform spending keys, hypothesis (H) holds up to a negligible change in the probability of every efficiently decidable event, in particular of the event of Proposition 11.10, which an algorithm holding the keys decides by verifying T' and comparing its effecting data with those of T. By Lemma 2.15 (Assumption 2.12) the triple $(\mathsf{ask}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$ is indistinguishable from independent uniform elements; the sign normalisation of "The spending key and the spend-side secrets" (§3.1) maps a uniform non-zero ask to the uniform distribution on the set of (H); and rejection, by Lemma "Rejection in key generation" (§3.2), costs a negligible term under Assumptions 2.12, 2.22 and 2.8. A dummy spend uses such a key (Definition "Dummy notes", §4.4).

Remark 11.12 (Consensus-branch binding). The consensus branch identifier enters SigHash twice, in the root personalisation and in the header digest, and a consensus rule requires nConsensusBranchId to equal the identifier of the consensus branch under which the transaction is validated (protocol specification, §"Transaction Consensus Rules"; ZIP 244, "txid_digest" and "T.1: header_digest"). Under any other branch a signed transaction T either fails that rule or carries another nConsensusBranchId, hence other effecting data, and Proposition 11.10 excludes valid signatures on such a transaction that keeps an Action of T, except with negligible probability. A signed transaction is therefore valid under one consensus branch only and cannot be carried across the activation of a network upgrade (ZIP 259, "Backward compatibility"). That the anchors and the proofs are authorising data does not affect this binding, since the header is effecting data.

11.4 Chain state and pool rules

The Orchard pool and the Ironwood pool each have their own chain state: a note commitment tree ("The Merkle hash and the tree", §5.1) with its anchors ("Anchors", §5.2), a nullifier set ("Nullifier sets", §6.2) and a chain value pool balance, defined below. An Action is checked only against the state of its own pool (ZIP 229, Terminology and Rationale, "Separate state"; protocol specification, §"Nullifier Sets" and §"Note Commitment Trees").

Definition 11.13 (Chain value pool balances)**.** The *Ironwood chain value pool balance* of a block chain is the negation of the sum of the valueBalanceIronwood fields of all its transactions, and the *Orchard chain value pool balance* the negation of the sum of their valueBalanceOrchard fields, an absent field counting as 0. The *total issued supply* is the sum of the balances of all chain value pools of the protocol specification, §"Chain Value Pool Balances", the transparent pool and the other shielded pools included, computed without overflow (ZIP 209, Terminology and "Specification").

The consensus rules on the balances are: a block that would make the chain value pool balance of any pool negative is invalid, and a block that would make the total issued supply exceed MAX_MONEY (§11.2) is invalid (ZIP 209, "Specification"; protocol specification, §"Chain Value Pool Balances").

Remark 11.14 (Outflow bound). Fix a pool P and write each of its balancing values as $v = v^+ - v^-$ with $v^+ := \max(v, 0)$ and $v^- := \max(-v, 0)$. The chain value pool balance of P is $\sum v^- - \sum v^+$, summed over the transactions of the chain, so the rule that it is non-negative at every block states that at every block the net value that has left P through positive balancing values, $\sum v^+$, is at most the net value that entered it through negative ones, $\sum v^-$. Value created inside P by a failure of soundness can therefore leave P only up to the value recorded as having entered it (ZIP 209, "Specification").

The following Orchard-pool rules apply to every transaction, whatever its version (ZIP 258, "Consensus rules from NU6.3 activation"; ZIP 229, "Consensus Rules").

(i) Cross-address transfers are disabled for every Orchard-pool Action: bits 2 to 7 of flagsOrchard are 0, so every Orchard-pool Action has enableCrossAddress = 0 and primary input disableCrossAddress = 1 ("Bundle flags", §9.1), and the verifying key that consensus selects for every Action enforces condition A10 of Definition 9.2. The meaning of A10 and its pool-level effect are those stated at A10 in "The Action statement" (§9.2).

(ii) For every transaction, valueBalanceOrchard ≥ 0: net value may leave the Orchard pool and never enter it, so the Orchard chain value pool balance never increases.

Of the two Orchard-protocol pools, therefore, only the Ironwood pool accepts net inflow: a *shielding transaction*, one whose transparent inputs fund a negative balancing value, can place that negative balancing value, among the Orchard-protocol pools, only in its Ironwood component (ZIP 258, same section; protocol specification, §"Shielded Pools and Notes").

The coinbase transaction of "The transaction format" (§11.2) is subject to the following rules on the shielded path (protocol specification, §"Transaction Consensus Rules"; ZIP 229, "Consensus Rules"; ZIP 258, "Consensus rules from NU6.3 activation").

(iii) A coinbase transaction has no Orchard-pool Actions.

(iv) The bit enableSpends is 0 in flagsOrchard of a coinbase transaction of version 5 or 6 and in flagsIronwood of a coinbase transaction of version 6. By condition A8 of Definition 9.2, under Assumption 9.11, every Action of a coinbase transaction has, except with negligible probability, an extracted witness with $v^{\mathsf{old}} = 0$: a coinbase transaction consumes no note of non-zero value. The rule concerns coinbase transactions only; every other transaction may spend Ironwood-pool notes.

(v) Every Ironwood-pool output of a coinbase transaction must decrypt to a note plaintext by the recovery with ovk of "The outgoing ciphertext" (§10.3) under the all-zero outgoing viewing key, the string of 32 zero bytes, so its plaintext is public and its lead byte is 0x03, the only lead byte that recovery accepts (ZIP 213, "Specification"; ZIP 258, "Changes to the Protocol Specification").

Rule (v) is the only check by consensus of the lead byte of an Orchard-protocol output. Every Ironwood-pool output uses the note format of lead byte 0x03 ("The note seed", §4.3), and the Action statement does not check the derivation of rcm (Definition 9.2, Remark "What the statement does not check"); for every other output the lead byte is encrypted and the recipient enforces it, decryption returning $\perp$ when the byte is not allowed for the pool ("Trial decryption and note acceptance", §10.4; protocol specification, §"Decryption using an Incoming Viewing Key (Sapling and Orchard)").

The rules of ZIP 218, "Shielded pool action limits", bound per block the number of Orchard-pool Actions and a shielded budget that counts them together with the descriptions of the other shielded protocols; they do not mention Ironwood-pool Actions, which a version-6 transaction counts in the separate field nActionsIronwood (ZIP 229, "Transaction Format"). These are block-level rules, outside the Action statement.

11.5 Construction of a transaction

The following procedure extends the single spend of the *Crypto Guide*'s Example "An Orchard shielded spend" (§"A shielded spend, end to end") to a bundle with signatures and balance. Its steps follow the order of the preceding sections.

Construction 11.15 (Ironwood-pool bundle)**.** The input is the sender's keys; the Ironwood-pool notes to consume, with their leaf positions; each recipient's address, which its holder derives from an incoming viewing key ("Diversified addresses", §3.3) and hands to the sender; the amounts; and the fee. The sender performs the following steps.

(1) *Pairing.* For s consumed and o created notes it forms $n \geq \max(s, o)$ Actions (§11.1). A dummy consumed note comes from a fresh uniformly random spending key, with value 0, a uniform rseed, $\rho = \mathsf{Extract}_{\mathbb{P}}$ of a uniform Pallas point and an arbitrary authentication path; a dummy created note has value 0 and a random address (Definition "Dummy notes", §4.4).

(2) *Anchor.* It takes as rt the root of the Ironwood tree's final treestate at a recent block, at the depth that "Anchors" (§5.2) recommends, and computes the authentication path of each real consumed note from public data (Lemma 5.15).

(3) *Created notes.* For each Action it computes $\mathsf{nf}^{\mathsf{old}}$ of the consumed note by Definition 6.3, sets $\rho^{\mathsf{new}} := \mathsf{nf}^{\mathsf{old}}$ ("Nullifier chaining", §6.3), draws a fresh uniform 32-byte rseed, derives esk, ψ^{new} and $\mathsf{rcm}^{\mathsf{new}}$ under lead byte 0x03 as in "The note seed" (§4.3), redrawing rseed when $\mathsf{esk} = 0$ or $\mathsf{cmx} = \perp$, and computes $\mathsf{cm}^{\mathsf{new}}$ by Definition 4.4 and $\mathsf{cmx} := \mathsf{Extract}_{\mathbb{P}}(\mathsf{cm}^{\mathsf{new}})$.

(4) *Value commitments.* For each Action it draws rcv uniformly from $\mathbb{F}_{p_{\mathsf{Vesta}}}$ and sets $\mathsf{cv}^{\mathsf{net}} := [v^{\mathsf{old}} - v^{\mathsf{new}}]\, V^{\mathsf{Orchard}} + [\mathsf{rcv}]\, R^{\mathsf{Orchard}}$ (Definition 8.2).

(5) *Randomised keys.* For each Action it draws α by GenRandom and sets $\mathsf{rsk} := \mathsf{ask} + \alpha$ and $\mathsf{rk} := \mathsf{ak}^{\mathbb{P}} + [\alpha]\, G^{\mathsf{SpendAuth}}$ ("Randomised validating keys", §7.2), with ask and $\mathsf{ak}^{\mathbb{P}}$ of the consumed note's spending key, the dummy key for a dummy spend.

(6) *Flags and balance.* It sets $\mathsf{enableSpends} = \mathsf{enableOutputs} = 1$, and $\mathsf{enableCrossAddress} = 1$ whenever some Action's created note pays an expanded receiver other than that of its consumed note (condition A10 of Definition 9.2); it sets $v^{\mathsf{balance}} := \sum (v^{\mathsf{old}} - v^{\mathsf{new}})$ over the Actions, and chooses the created values so that the fee is the value left in the transparent transaction value pool ("The balancing value", §8.2).

(7) *Proof.* It computes one Halo 2 proof over the n instances

$$(rt, \mathsf{cv}^{\mathsf{net}}, \mathsf{nf}^{\mathsf{old}}, \mathsf{rk}, \mathsf{cmx}, \mathsf{enableSpends}, \mathsf{enableOutputs}, 1 - \mathsf{enableCrossAddress})$$

with the witnesses of Definition 9.2 ("The Action circuit and the Halo 2 proof", §9.3). By Assumption 9.11 a prover of an accepted proof knows, for each Action, a witness satisfying A1 to A10; under Assumption 9.14 the proof reveals nothing beyond the instances.

(8) *Encryption.* For each Action it computes $\mathsf{epk} := [\mathsf{esk}]\,\mathsf{g_d}$, C_{enc} and C_{out} as in "Encryption to the recipient" (§10.2) and "The outgoing ciphertext" (§10.3).

(9) *Header.* It sets the version to 6, nConsensusBranchId to the identifier of the consensus branch in force, and nExpiryHeight (§11.2).

(10) *Digest.* It computes SigHash ("Transaction digests and signatures", §11.3).

(11) *Spend authorisation.* For each Action it computes the spend-authorisation signature under rsk on SigHash.

(12) *Binding signature.* It sets $\mathsf{bsk} := \sum \mathsf{rcv} \bmod p_{\mathsf{Vesta}}$ and $\mathsf{bvk} := \sum \mathsf{cv}^{\mathsf{net}} - [v^{\mathsf{balance}}]\,V^{\mathsf{Orchard}}$, checks $\mathsf{bvk} = [\mathsf{bsk}]\,R^{\mathsf{Orchard}}$, which the protocol specification recommends to the signer, and computes the binding signature under bsk on SigHash ("The binding signature", §8.3).

(Protocol specification, §"Sending Notes (Orchard)", §"Dummy Notes (Orchard)", §"Spend Authorization Signature (Sapling and Orchard)" and §"Balance and Binding Signature (Orchard)".)

Steps (10) to (12) depend on neither rt nor the proof, which are authorising data, so signing can precede proving (ZIP 229, Motivation).

The sender sets nExpiryHeight. ZIP 218, "Default expiry delta", recommends to node implementations that create transactions a default expiry height 120 blocks above the current height (ZIP 203, "Specification"), which at the 25-second target spacing stated in "Anchors" (§5.2) is $120 \times 25 = 3000$ seconds, 50 minutes. The consensus rule is the expiry rule of "The transaction format" (§11.2).

Example 11.16 (A two-Action payment)**.** The example is numeric only in amounts, fees, value differences, the balancing value, sizes and counts, each the output of a script; every curve point, field element, hash, ciphertext and signature is named and its derivation shown, not evaluated. The subscripts S, R and D mark the sender's key, the recipient's key and the dummy spending key.

Amounts. The sender holds one Ironwood-pool note n_1^{old} of 7 ZEC, 700 000 000 zatoshi, and pays 5 ZEC, 500 000 000 zatoshi, to a recipient's address $\mathsf{addr_R}$. One real consumed note and two created notes need $\max(1, 2) = 2$ Actions, hence two logical actions, and the conventional fee of "The balancing value" (§8.2) is $5000 \cdot \max(2, 2) = 10\,000$ zatoshi. The change is $700\,000\,000 - 500\,000\,000 - 10\,000 = 199\,990\,000$ zatoshi. The following table pairs the notes into Actions.

	Action 1	Action 2
consumed note	n_1^{old}, the sender's	n_2^{old}, a dummy
spending key	the sender's: $\mathsf{ask_S}$, $\mathsf{ak_S^{\mathbb{P}}}$, $\mathsf{nk_S}$	the dummy key: $\mathsf{ask_D}$, $\mathsf{ak_D^{\mathbb{P}}}$, $\mathsf{nk_D}$
v_i^{old}	700 000 000	0
created note	n_1^{new} to $\mathsf{addr_R}$	n_2^{new}, the change, to $\mathsf{addr_S}$
v_i^{new}	500 000 000	199 990 000
$v_i^{\mathsf{old}} - v_i^{\mathsf{new}}$	200 000 000	−199 990 000

The net values sum to $\mathsf{valueBalanceIronwood} = 200\,000\,000 + (-199\,990\,000) = 10\,000$ zatoshi, the fee. The transaction has no transparent inputs or outputs, so the fee is the whole value remaining in its transparent transaction value pool.

Public fields. The recipient derives $\mathsf{addr_R} = (d_\mathsf{R}, \mathsf{pk_{d,R}})$ from its incoming viewing key ("Diversified addresses", §3.3) and gives it to the sender; $\mathsf{addr_S}$ is an address of the sender's own key. Write $(\mathsf{g_{d,1}}, \mathsf{pk_{d,1}})$ and $(\mathsf{g_{d,2}}, \mathsf{pk_{d,2}})$ for the expanded receivers of $\mathsf{addr_R}$ and $\mathsf{addr_S}$, and $\mathsf{nk}_1 := \mathsf{nk_S}$, $\mathsf{nk}_2 := \mathsf{nk_D}$. For $i = 1, 2$ the sender computes, from the consumed note's fields $(\rho_i^{\mathsf{old}}, \psi_i^{\mathsf{old}}, \mathsf{cm}_i^{\mathsf{old}})$,

$$\begin{aligned} \mathsf{nf}_i^{\mathsf{old}} &:= \mathsf{DeriveNullifier}_{\mathsf{nk}_i}(\rho_i^{\mathsf{old}}, \psi_i^{\mathsf{old}}, \mathsf{cm}_i^{\mathsf{old}}) && \text{(Definition 6.3)}, \\ \rho_i^{\mathsf{new}} &:= \mathsf{nf}_i^{\mathsf{old}} && \text{(§6.3)}, \end{aligned}$$

draws a uniform rseed_i, derives esk_i, ψ_i^{new} and $\mathsf{rcm}_i^{\mathsf{new}}$ from rseed_i under lead byte $\mathtt{0x03}$ ("The note seed", §4.3), and sets

$$\begin{aligned} \mathsf{cm}_i^{\mathsf{new}} &:= \mathsf{NoteCommit}_{\mathsf{rcm}_i^{\mathsf{new}}}(\mathsf{g}_{\mathsf{d},i}^{\star} \,\|\, \mathsf{pk}_{\mathsf{d},i}^{\star} \,\|\, \mathrm{LE}_{64}(v_i^{\mathsf{new}}) \,\|\, \mathrm{LE}_{255}(\rho_i^{\mathsf{new}}) \,\|\, \mathrm{LE}_{255}(\psi_i^{\mathsf{new}})), \\ \mathsf{cmx}_i &:= \mathsf{Extract}_{\mathbb{P}}(\mathsf{cm}_i^{\mathsf{new}}) \end{aligned}$$

(Definition 4.4). With rcv_i uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$ and α_i drawn by $\mathsf{GenRandom}$,

$$\begin{aligned} \mathsf{cv}_1^{\mathsf{net}} &:= [200\,000\,000]\, V^{\mathsf{Orchard}} + [\mathsf{rcv}_1]\, R^{\mathsf{Orchard}}, & \mathsf{rk}_1 &:= \mathsf{ak}_\mathsf{S}^{\mathbb{P}} + [\alpha_1]\, G^{\mathsf{SpendAuth}}, \\ \mathsf{cv}_2^{\mathsf{net}} &:= [-199\,990\,000]\, V^{\mathsf{Orchard}} + [\mathsf{rcv}_2]\, R^{\mathsf{Orchard}}, & \mathsf{rk}_2 &:= \mathsf{ak}_\mathsf{D}^{\mathbb{P}} + [\alpha_2]\, G^{\mathsf{SpendAuth}}, \end{aligned}$$

the negative difference read in $\mathbb{F}_{p_{\mathsf{Vesta}}}$ (Definition 8.2; "Randomised validating keys", §7.2). The anchor rt is the anchor of a recent Ironwood-pool block, with the authentication path of $\mathsf{Extract}_{\mathbb{P}}(\mathsf{cm}_1^{\mathsf{old}})$; the path of the dummy is arbitrary, since A3 exempts $v_2^{\mathsf{old}} = 0$. The flags are $\mathsf{enableSpends} = \mathsf{enableOutputs} = \mathsf{enableCrossAddress} = 1$, because both created notes pay expanded receivers other than those of their Actions' consumed notes (A10). One proof π covers the instances $(rt, \mathsf{cv}_i^{\mathsf{net}}, \mathsf{nf}_i^{\mathsf{old}}, \mathsf{rk}_i, \mathsf{cmx}_i, 1, 1, 0)$ for $i = 1, 2$. Finally $\mathsf{epk}_i := [\mathsf{esk}_i]\, \mathsf{g_{d,i}}$, $C_{\mathsf{enc},i}$ encrypts the plaintext of n_i^{new} to its receiver, and $C_{\mathsf{out},i}$ is formed under the sender's ovk ("Encryption to the recipient", §10.2; "The outgoing ciphertext", §10.3).

Signatures, balance and size. The transaction has no transparent inputs, so $\mathsf{SigHash}$ equals its transaction identifier ("Transaction digests and signatures", §11.3). The spend-authorisation signatures are σ_1 under $\mathsf{rsk}_1 = \mathsf{ask_S} + \alpha_1$ and σ_2 under $\mathsf{rsk}_2 = \mathsf{ask_D} + \alpha_2$, both on $\mathsf{SigHash}$. The binding signing key is $\mathsf{bsk} := \mathsf{rcv}_1 + \mathsf{rcv}_2 \bmod p_{\mathsf{Vesta}}$, and

$$\begin{aligned} \mathsf{bvk} &= \mathsf{cv}_1^{\mathsf{net}} + \mathsf{cv}_2^{\mathsf{net}} - [10\,000]\, V^{\mathsf{Orchard}} \\ &= [200\,000\,000 - 199\,990\,000 - 10\,000]\, V^{\mathsf{Orchard}} + [\mathsf{bsk}]\, R^{\mathsf{Orchard}} = [\mathsf{bsk}]\, R^{\mathsf{Orchard}}; \end{aligned}$$

$\mathsf{bindingSigIronwood}$ is the binding signature under bsk on $\mathsf{SigHash}$. By Table 7 the Ironwood component has

$$1 + 2 \cdot 820 + 1 + 8 + 32 + 3 + (2720 + 2272 \cdot 2) + 2 \cdot 64 + 64 = 9141$$

bytes: the count, the Action descriptions, the flags, the balancing value, the anchor, the proof length, the proof of 7264 bytes, the spend-authorisation signatures and the binding signature.

The two views. The verifier receives only the public data: for each i the Action description $(\mathsf{cv}_i^{\mathsf{net}}, \mathsf{nf}_i^{\mathsf{old}}, \mathsf{rk}_i, \mathsf{cmx}_i, \mathsf{epk}_i, C_{\mathsf{enc},i}, C_{\mathsf{out},i})$ and σ_i, and rt, the flags byte with bits 0, 1 and 2 set, $\mathsf{valueBalanceIronwood} = 10\,000$, π and $\mathsf{bindingSigIronwood}$. It runs the procedure of "Verification of a transaction" (§11.6). What these data reveal is the subject of Theorem 12.12 in "Privacy" (§12.4). The recipient trial-decrypts every output with its incoming viewing key. On output 1 it recovers the plaintext $(\mathtt{0x03}, d_\mathsf{R}, 500\,000\,000, \mathsf{rseed}_1, \mathsf{memo})$, re-derives the note with $\rho = \mathsf{nf}_1^{\mathsf{old}}$ of the same Action, checks epk_1 and cmx_1, and records the leaf position of cmx_1 ("Trial decryption and note acceptance", §10.4); the sender recovers the change note of output 2 in the same way. Either note is later consumed by the procedure of this subsection.

11.6 Verification of a transaction

Each step of the following procedure is tagged with the requirements whose soundness part it serves and with the conditions of Definition 9.2 that it completes, in the pairing of the requirement map, Table 5.

Construction 11.17 (Verification)**.** Given the chain state of each pool, a transaction T in a block is accepted, as far as its Orchard-protocol components are concerned, if and only if the following checks pass.

(1) *Parsing.* The version of T is 5 or 6, with the version group identifier of its version, and $\mathsf{nConsensusBranchId}$ equals the identifier of the consensus branch in force. Every field of every Action description is the canonical encoding of its type (Table 6): $\mathsf{cv}^{\mathsf{net}}$ and rk are points, epk is a non-identity point, and nf and cmx are integers below p_{Pallas}; $\mathsf{rk} \neq \mathcal{O}$; every signature encoding is canonical as RedPallas validation requires (§7.1); and each component satisfies the rules of "The transaction format" (§11.2) on its count, its reserved flag bits, its flags, the range of its balancing value and the length of its proof. The transaction T has a transparent input, a Spend description in its Sapling component or an Orchard-protocol bundle with $\mathsf{enableSpends} = 1$, and it has a transparent output, an Output description in its Sapling component or an Orchard-protocol bundle with $\mathsf{enableOutputs} = 1$, where Spend and Output descriptions are the Sapling component's descriptions of a consumed and of a created note.

(2) *Proof.* For each pool with Actions, the aggregate proof verifies under the post-NU6.3 verifying key of "The Action circuit and the Halo 2 proof" (§9.3) on the per-Action primary inputs

$$(rt, \mathsf{cv}^{\mathsf{net}}, \mathsf{nf}, \mathsf{rk}, \mathsf{cmx}, \mathsf{enableSpends}, \mathsf{enableOutputs}, 1 - \mathsf{enableCrossAddress}),$$

with rt and the flags read from the bundle. [R1 to R5, through A1 to A10.]

(3) *Anchor rule.* Each bundle's rt is the root of its pool's final treestate at an earlier block of the chain ("Anchors", §5.2). [R2, with A3.]

(4) *Nullifier rule.* No nullifier repeats within T or across the transactions of the chain; the nullifiers of each pool are checked against that pool's nullifier set ("Nullifier sets", §6.2). [R3, with A5.]

(5) *Spend authorisation.* Each Action's spend-authorisation signature is valid under its rk on $\mathsf{SigHash}$ ("Randomised validating keys", §7.2). [R5, with A6 and A7.]

(6) *Balance.* For each pool with Actions, the verifier computes $\mathsf{bvk} := \sum \mathsf{cv}^{\mathsf{net}} - [v^{\mathsf{balance}}]\, V^{\mathsf{Orchard}}$ over the pool's Actions and checks the binding signature under bvk on $\mathsf{SigHash}$ ("The binding signature", §8.3; *Crypto Guide*, §"Binding signatures as a proof of knowledge of a discrete logarithm"); and the fee, the value remaining in the transparent transaction value pool, is non-negative ("The balancing value", §8.2), a coinbase transaction satisfying instead the coinbase rule of the protocol specification, §"Transaction Consensus Rules", which balances its outputs against the block subsidy and the fees of the block's other transactions. [R4, with A4.]

(7) *Pool rules.* The rules of "Chain state and pool rules" (§11.4), the expiry rule of "The transaction format" (§11.2) and the capacity rule of each pool's note commitment tree ("The Merkle hash and the tree", §5.1) hold.

(Protocol specification, §"Transaction Consensus Rules", §"Action Descriptions", §"Action Description Encoding and Consensus", §"RedDSA, RedJubjub, and RedPallas", §"Action Transfers and their Descriptions", §"Nullifier Sets", §"Note Commitment Trees", §"Balance and Binding Signature (Orchard)" and §"Transactions and Treestates"; ZIP 229, "Consensus Rules"; ZIP 2003, "Specification"; ZIP 259, Abstract.) These are the consensus checks on the Orchard-protocol components of T; the remaining rules of §"Transaction Consensus Rules" concern its other components.

Remark 11.18 (Division of the checks). Under Assumption 9.11 the proof yields, for each Action, a witness that satisfies Definition 9.2 for its primary input. Whether rt is a root that the chain recorded and whether nf is new are facts about the chain's history, which no proof over one Action's data attests; the verifier checks them against its state, in steps (3) and (4). Beyond the parsing of step (1), the fee rule of step (6) and the rules of step (7), validity is thus the conjunction of one proof per bundle, two kinds of signature and these two checks against chain state; apart from the coinbase rule (v) of §11.4, no step decrypts or inspects a note.

Construction 11.19 (State update)**.** On accepting a block, for each pool: every nullifier of the pool's Actions enters the pool's nullifier set, so that a later Action that publishes the same nullifier fails step (4); every cmx is appended to the pool's note commitment tree, in the order of the transactions and, within a transaction, of the Actions, so that the note's holder can later prove membership against an anchor whose tree contains it; and the chain value pool balance of the pool decreases by each transaction's balancing value (protocol specification, §"Nullifier Sets", §"Transactions and Treestates" and §"Chain Value Pool Balances").

The verifier holds no viewing key and decrypts no Orchard-protocol ciphertext, except for the decryption of Ironwood-pool coinbase outputs under the all-zero outgoing viewing key that "Chain state and pool rules" (§11.4) requires, whose plaintexts are public by that rule (protocol specification, §"Transaction Consensus Rules"; ZIP 213, "Specification"). Its input is the transaction's public data and the chain state; what those data reveal about the private inputs is the subject of Theorem 12.12 in "Privacy" (§12.4).

12 Security

This section composes the propositions of the preceding sections with the two assumptions on the Action proof, Assumptions 9.11 and 9.14. Theorem 12.4 states conservation per pool; its proof uses condition A4, the binding signature and Assumption 9.11. Lemma 12.5 and Theorem 12.6 exclude a second consumption of a note, Corollary 12.7 the Faerie Gold attack and Theorem 12.9 a consumption without the spend authorising key; Theorem 12.12 states what the public data of a bundle hide. Theorem 12.14 collects, under one list of assumptions, membership soundness, Theorems 12.6, 12.4 and 12.9 with Proposition 11.10, and Theorem 12.12.

Efficient adversaries and negligible functions are those of the *Crypto Guide*, §"Adversaries and the security parameter", and computational indistinguishability is that of its Definition "Perfect, statistical, and computational indistinguishability" (§"Distribution ensembles and indistinguishability"). An adversary interacts with the honest parties, namely key holders, senders and signers, and with the random oracles of Assumptions 2.8, 7.3, 9.11, 9.14 and 10.4, which are independent of one another; every reduction simulates the honest parties and the random oracles. Every result of the section holds except with negligible probability, and every result that uses a property of key generation is claimed for keys generated with $\mathsf{use_qsk} = \mathsf{false}$ ("The spending key and the spend-side secrets", §3.1).

Definition 12.1 (Pools, accepted Actions and witnesses).

(a) A *pool* is the Orchard pool or the Ironwood pool. A statement made for each pool uses that pool's note commitment tree, anchors, nullifier set and bundle, its balancing value, valueBalanceOrchard or valueBalanceIronwood, and its binding signature, bindingSigOrchard or bindingSigIronwood.

(b) An adversary outputs a *block chain*: a sequence of blocks each of whose transactions passes the verification of "Verification of a transaction" (§11.6) against the chain state that the state update of that subsection produces from the preceding transactions. An Action, or a bundle, of a transaction of that chain is *accepted*.

(c) The *witness* of an accepted Action is the auxiliary input that the extractor of Assumption 9.11 outputs for it, applied to the Action's bundle as stated below. The Action *consumes* the note $(\mathsf{g_d}^{\mathsf{old}}, \mathsf{pk_d}^{\mathsf{old}}, v^{\mathsf{old}}, \rho^{\mathsf{old}}, \psi^{\mathsf{old}}, \mathsf{rcm}^{\mathsf{old}})$ of its witness and *creates* the note $(\mathsf{g_d}^{\mathsf{new}}, \mathsf{pk_d}^{\mathsf{new}}, v^{\mathsf{new}}, \rho^{\mathsf{new}}, \psi^{\mathsf{new}}, \mathsf{rcm}^{\mathsf{new}})$, with $\rho^{\mathsf{new}} = \mathsf{nf}^{\mathsf{old}}$ (Definition 9.2); the values v^{old} and v^{new} are read from the witness.

(d) In this section a note is given by its expanded receiver $(\mathsf{g_d}, \mathsf{pk_d})$ (Definition "Flag values and expanded receiver", §9.1) in place of its address, and its trapdoor rcm acts through its residue modulo p_{Vesta}; two notes are equal when these six components are. Its note commitment is $\mathsf{NoteCommit}_{\mathsf{rcm}}(M)$ and its extracted commitment is $\mathsf{Extract}^{\perp}_{\mathbb{P}}(\mathsf{NoteCommit}_{\mathsf{rcm}}(M))$, with M the message of Definition 4.4 formed from $(\mathsf{g_d}, \mathsf{pk_d}, v, \rho, \psi)$.

(e) A *committed note* of a pool is a note whose extracted commitment is a leaf of that pool's note commitment tree.

Assumption 9.11 is stated for one aggregate proof. For a block chain it is applied to each bundle, to the algorithm that runs the adversary, with every honest party and random oracle of the experiment, and outputs that bundle's primary inputs and proof; the extractors of distinct bundles are run on the same execution. An oracle that a reduction does not simulate itself, such as the random oracle H of Assumption 7.3 or a signing oracle, receives each new query of a rewound run, and a query repeated on the same prefix of a run is answered as before. By the union bound over the polynomially many bundles, every witness satisfies Definition 9.2, except with negligible probability. By Remark "⊥-weakened conditions" (§9.2), conditions A1, A2, A3 and A7 then hold in their unweakened forms, without the ⊥ case, except with negligible probability, under Assumptions 2.22 and 2.8. The results below include these events in their negligible terms.

Remark 12.2 (Binding for notes given by expanded receivers). Step 1 of the proof of Proposition 4.5(b) shows that the map $(\mathsf{g_d}, \mathsf{pk_d}, v, \rho, \psi) \mapsto M$ is injective, and Step 2 reads a note only through M and its trapdoor. For notes in the sense of (d) the two steps therefore show, under Assumptions 2.22 and 2.8: no efficient algorithm outputs, except with negligible probability, two distinct notes with equal extracted commitments other than ⊥. In particular a note that opens an extracted commitment is the only opening that an efficient algorithm finds. The proof of Proposition 6.8 likewise reads a note only through M, rcm, ρ and ψ, and applies to such notes.

12.1 Value conservation

Lemma 12.3 (Modular balance is integer balance). *Let $n < 2^{16}$; for $1 \le i \le n$ let v_i^{old} and v_i^{new} be integers in $\{0, \dots, 2^{64} - 1\}$; and let v^{balance} be an integer in $\{-\mathsf{MAX_MONEY}, \dots, \mathsf{MAX_MONEY}\}$.*

If

$$\sum_{i=1}^{n}(v_i^{\mathsf{old}} - v_i^{\mathsf{new}}) \equiv v^{\mathsf{balance}} \pmod{p_{\mathsf{Vesta}}},$$

then the two sides are equal as integers. The hypotheses hold for every accepted bundle of either pool: the bound on n and the range of the balancing value are consensus rules ("The transaction format", §11.2; protocol specification, §"Transaction Consensus Rules", for both pools), and v_i^{old} and v_i^{new} are 64-bit values of the auxiliary input of Definition 9.2.

Proof. Let $v^{\mathsf{sum}} := \sum_i (v_i^{\mathsf{old}} - v_i^{\mathsf{new}}) - v^{\mathsf{balance}}$. Each difference has absolute value at most $2^{64} - 1$, and $\mathsf{MAX_MONEY} = 2.1 \cdot 10^{15}$ (protocol specification, §"Constants"), so

$$|v^{\mathsf{sum}}| \leq (2^{16} - 1)(2^{64} - 1) + \mathsf{MAX_MONEY} = 1\,208\,907\,374\,970\,555\,465\,089\,025 < 2^{80}.$$

The hypothesis makes v^{sum} a multiple of p_{Vesta}, and $p_{\mathsf{Vesta}} > 2^{254}$ (§2.1); the only such multiple of absolute value below 2^{80} is 0. □

The security argument of the protocol specification, §"Balance and Binding Signature (Orchard)", bounds the balancing value by its signed 64-bit encoding instead, which places v^{sum} in

$$\{-1\,208\,916\,596\,242\,592\,319\,864\,832, \ldots, 1\,208\,916\,596\,242\,592\,319\,864\,833\};$$

that bound is also below 2^{80}, so the conclusion holds without the range rule.

Theorem 12.4 (Balance)**.** *Under Assumptions 9.11, 2.22, 2.8 and 7.3, for each pool, the Ironwood pool included, every accepted bundle of that pool output by an efficient adversary, with n Actions and balancing value v^{balance}, satisfies*

$$\sum_{i=1}^{n}(v_i^{\mathsf{old}} - v_i^{\mathsf{new}}) = v^{\mathsf{balance}}$$

as integers, the values read from the Actions' witnesses, except with negligible probability. The binding signature checked is that of the same pool, under the binding validating key bvk *recomputed from that bundle's net value commitments and balancing value; both pools use the bases V^{Orchard} and R^{Orchard}.*

Proof. Fix a pool and an efficient adversary $\mathcal{A}$. The adversary outputs polynomially many bundles, and the union bound reduces the claim to the j-th bundle of the pool, for each j. Let $\mathcal{B}_j$ be the algorithm that runs $\mathcal{A}$ with the extractors of Assumption 9.11 and outputs that bundle's net value commitments $\mathsf{cv}_1^{\mathsf{net}}, \ldots, \mathsf{cv}_n^{\mathsf{net}}$, its balancing value, the pairs

$$(v_i^{\mathsf{old}} - v_i^{\mathsf{new}},\ \mathsf{rcv}_i \bmod p_{\mathsf{Vesta}}) \qquad (1 \leq i \leq n)$$

read from the witnesses, the digest SigHash of its transaction and the pool's binding signature. Algorithm $\mathcal{B}_j$ is efficient and forwards the queries of $\mathcal{A}$ to H.

(1) *Openings.* By condition A4 of Definition 9.2, $\mathsf{cv}_i^{\mathsf{net}} = [v_i^{\mathsf{old}} - v_i^{\mathsf{new}}]\, V^{\mathsf{Orchard}} + [\mathsf{rcv}_i]\, R^{\mathsf{Orchard}}$, so each output pair is an opening of $\mathsf{cv}_i^{\mathsf{net}}$ in the sense of "The binding signature" (§8.3), except with the negligible probability that a witness fails the definition.

(2) *Signature.* The bundle is accepted, so its binding signature is valid on SigHash under

$$\mathsf{bvk} = \sum_{i=1}^{n} \mathsf{cv}_i^{\mathsf{net}} - [v^{\mathsf{balance}}]\, V^{\mathsf{Orchard}}$$

in the binding-signature instance of RedPallas, with base R^{Orchard} (step (6) of the verification of §11.6). This key belongs to the pool's own bundle: each pool's bvk sums only that pool's net value commitments and subtracts only its balancing value (§8.3).

(3) *Congruence.* Algorithm $\mathcal{B}_j$ is therefore an efficient party of the consequence of Proposition 8.10: it outputs the net value commitments and the balancing value of a bundle, openings of every commitment, and a binding signature under the resulting bvk on a message of its choice. The consequence is proved in two steps. The extraction of Proposition 7.12, in the random-oracle model for H (Assumption 7.3), yields b with $\mathsf{bvk} = [b]\, R^{\mathsf{Orchard}}$; if the openings did not balance modulo p_{Vesta}, part (ii) of Proposition 8.10 would turn the residual V^{Orchard} component into the discrete logarithm of V^{Orchard} to the base R^{Orchard}, which Assumptions 2.22 and 2.8 exclude. Hence $\sum_i (v_i^{\mathsf{old}} - v_i^{\mathsf{new}}) \equiv v^{\mathsf{balance}} \pmod{p_{\mathsf{Vesta}}}$, except with negligible probability.

(4) *Integers.* The bundle is accepted, so it has $n < 2^{16}$ Actions, its balancing value lies in the range $\{-\mathsf{MAX_MONEY}, \ldots, \mathsf{MAX_MONEY}\}$, and the witnessed values lie in $\{0, \ldots, 2^{64} - 1\}$. Lemma 12.3 turns the congruence into an equality of integers.

The argument uses no property of the pool beyond its own bundle and binding signature, and applies to the Ironwood pool as to the Orchard pool. □

12.2 Double-spend resistance

Lemma 12.5 (One nullifier per note)**.** *Under Assumptions 9.11, 2.22 and 2.8, except with negligible probability: whenever two accepted Actions, of either pool, have witnesses whose consumed notes have the same extracted commitment* $\mathsf{Extract}_{\mathbb{P}}(\mathsf{cm}^{\mathsf{old}})$*, the two witnesses consume the same note with the same nullifier deriving key* nk*, and the two Actions publish the same nullifier* $\mathsf{nf}^{\mathsf{old}}$*. The lemma uses neither* v^{old} *nor condition A3.*

Proof. Let w and w' be the two witnesses, primed components belonging to w'. Outside the negligible events of the paragraph after Definition "Pools, accepted Actions and witnesses", both satisfy Definition 9.2 with A1 and A7 unweakened. Every step below is carried out by the efficient algorithm that runs the adversary with the extractors and searches the accepted Actions for such a pair.

(1) *The note.* By A1, $\mathsf{cm}^{\mathsf{old}} = \mathsf{NoteCommit}_{\mathsf{rcm}^{\mathsf{old}}}(M^{\mathsf{old}}) \neq \perp$, and likewise for w'. The two consumed notes thus have equal extracted commitments other than $\perp$, and by Remark "Binding for notes given by expanded receivers" they are equal, except with negligible probability. Equal notes have equal messages and trapdoors modulo p_{Vesta}, hence $\mathsf{cm}^{\mathsf{old}} = \mathsf{cm}'^{\mathsf{old}}$.

(2) *The key.* By A7, ${\mathsf{pk_d}}^{\mathsf{old}} = [\mathsf{ivk}]\, {\mathsf{g_d}}^{\mathsf{old}}$ with $\mathsf{ivk} := \mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r_{ivk}}}(\mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}}), \mathsf{nk}) \neq \perp$, and, the notes being equal, ${\mathsf{pk_d}}^{\mathsf{old}} = [\mathsf{ivk}']\, {\mathsf{g_d}}^{\mathsf{old}}$ likewise. Both values of $\mathsf{Commit}^{\mathsf{ivk}}$ are integers below $p_{\mathsf{Pallas}} < p_{\mathsf{Vesta}}$, read as scalars (§2.1), and ${\mathsf{g_d}}^{\mathsf{old}}$ is a non-identity point of a group of prime order p_{Vesta}; hence $\mathsf{ivk} = \mathsf{ivk}'$. By the binding of ivk to $(\mathsf{ak}, \mathsf{nk})$ (Proposition "Binding of ivk to $(\mathsf{ak}, \mathsf{nk})$", §3.2), under Assumptions 2.22 and 2.8,

$$(\mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}}),\ \mathsf{nk},\ \mathsf{r_{ivk}} \bmod p_{\mathsf{Vesta}}) = (\mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}'^{\mathbb{P}}),\ \mathsf{nk}',\ \mathsf{r_{ivk}}' \bmod p_{\mathsf{Vesta}}),$$

except with negligible probability; in particular $\mathsf{nk} = \mathsf{nk}'$.

(3) *The nullifier.* By A5, $\mathsf{nf}^{\mathsf{old}} = \mathsf{DeriveNullifier}_{\mathsf{nk}}(\rho^{\mathsf{old}}, \psi^{\mathsf{old}}, \mathsf{cm}^{\mathsf{old}})$ (Definition 6.3), a function of its four arguments, which are equal for w and w' by (1) and (2). The two Actions therefore publish the same nullifier. No step uses v^{old} or A3. □

Theorem 12.6 (No double-spending)**.** *Under Assumptions 9.11, 2.22 and 2.8, for each pool, except with negligible probability, no committed note of non-zero value is consumed by two accepted Actions of that pool. The guarantee is per pool: the two pools' nullifiers are checked against separate sets.*

Proof. Let two distinct accepted Actions of one pool consume the same note. Their consumed notes have the same extracted commitment, so by Lemma 12.5, except with negligible probability,

the two Actions publish the same nullifier. Both belong to transactions of the adversary's block chain, and the nullifier rule of "Nullifier sets" (§6.2), step (4) of the verification of §11.6, rejects a nullifier of the pool that repeats within a transaction or across the transactions of the chain. The event therefore lies in the negligible event of the lemma. The argument uses neither the value nor the membership of the note and holds for every note; the case of a committed note of non-zero value is the one that moves value. The nullifier sets of the two pools are separate (§6.2), so the argument concerns two Actions of one pool only. □

Corollary 12.7 (Faerie Gold resistance)**.** *Under Assumptions 9.11, 2.22 and 2.8, except with negligible probability:*

(*a*) *the notes created by distinct accepted Actions of one pool have pairwise distinct elements ρ, and are therefore pairwise distinct;*

(*b*) *two distinct notes have distinct nullifiers, whatever their nullifier deriving keys: no efficient algorithm outputs two distinct notes, with note commitments other than $\perp$, and two keys $\mathsf{nk}, \mathsf{nk}'$, equal or not, under which the two notes have equal nullifiers.*

Hence the Faerie Gold attack, as defined in "Nullifier chaining" (§6.3), fails within a pool: (a) excludes two Actions creating the same note, which would carry one nullifier, and (b) excludes distinct notes sharing a nullifier, so that consuming one note accepted by a recipient never blocks another. The uniqueness of ρ is pool-local: the pools' nullifier sets are separate, so one value of ρ may occur once in each pool.

Proof. (a) The note that the witness of an accepted Action creates has $\rho^{\mathsf{new}} = \mathsf{nf}^{\mathsf{old}}$ of that Action: the Action statement fixes $\rho^{\mathsf{new}} := \mathsf{nf}^{\mathsf{old}}$ and binds the created note to the published cmx by A2 (Definition 9.2), so the chaining rule of §6.3 is enforced through Assumption 9.11. Distinct accepted Actions of one pool publish distinct nullifiers, by the nullifier rule (§6.2). Their created notes therefore have distinct ρ, and notes with a distinct component are distinct. The created note is the only opening of cmx that an efficient algorithm finds (Remark "Binding for notes given by expanded receivers"); a recipient that accepts a note from the Action obtains that note (Proposition 10.15(c)) and checks $\rho = \mathsf{nf}^{\mathsf{old}}$ itself (Proposition 10.15(a)).

(b) Two distinct notes have distinct note commitments, except with negligible probability, by the same remark; Proposition 6.8, which admits every pair of nullifier deriving keys, then excludes equal nullifiers.

Faerie Gold. Let a recipient accept a note n' from an accepted Action of a pool, and let nf' be its nullifier under the recipient's nk. By (a) no other accepted Action of the pool creates n', so the notes that the recipient accepts from distinct Actions are distinct. An accepted Action of the pool whose witness consumes a note $n \neq n'$ publishes, by A5, the nullifier of n under the witnessed key, which differs from nf' by (b). The value nf' therefore enters the pool's nullifier set only through an Action that consumes n' itself, and the consumption of another note does not make n' unspendable.

Pool locality. The argument of (a) uses the nullifier set of one pool. An Orchard-pool Action and an Ironwood-pool Action may publish the same nullifier and create notes with the same ρ; a consumption in one pool inserts its nullifier into that pool's set only, and so blocks no note of the other pool (Remark "Pool locality", §6.2). □

12.3 Authorisation of spends

Definition 12.8 (Spend-authority experiment)**.** A key holder generates a spending key sk with

use_qsk = false and derives $(\mathsf{ask}, \mathsf{nk}, \mathsf{r_{ivk}})$, $\mathsf{ak}^{\mathbb{P}}$ and $\mathsf{ak} = \mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}})$ as in "The spending key and the spend-side secrets" (§3.1), and ivk as in "Viewing keys" (§3.2). The adversary receives the full viewing key $(\mathsf{ak}, \mathsf{nk}, \mathsf{r_{ivk}})$ but not ask, sees every public field, may create notes to any address of the key, and may obtain spend-authorisation signatures: on a query (α, m), with $\alpha \in \mathbb{F}_{p_{\mathsf{Vesta}}}$ and m a byte string, it receives a RedPallas signature on m under $\mathsf{rsk} = \mathsf{ask} + \alpha$ in the spend-authorisation instance, the signing model of Proposition 7.12. A *note of the key* is a note whose expanded receiver satisfies $\mathsf{pk_d} = [\mathsf{ivk}]\,\mathsf{g_d}$ for the key's ivk. The adversary *wins* if it outputs a block chain with an accepted Action whose witness consumes a note of the key, of any value, a dummy of value zero included, and whose spend-authorisation signature is valid under its rk on a signature digest m, where no query (α, m) had $\mathsf{rk} = \mathsf{ak}^{\mathbb{P}} + [\alpha]\,G^{\mathsf{SpendAuth}}$.

Theorem 12.9 (Spend authority)**.** *For keys generated with* use_qsk = false*, and under Assumptions 9.11, 2.22, 2.8 (through the binding of* $\mathsf{Commit}^{\mathsf{ivk}}$*), 7.3 and 2.12, no efficient adversary wins the spend-authority experiment, except with negligible probability.*

Proof. Let an efficient adversary $\mathcal{A}$ win with probability ϵ, and write $G := G^{\mathsf{SpendAuth}}$.

(1) *Idealised key.* By Lemma "Rejection in key generation" (§3.2), the key is replaced by a single draw from a uniform sk without rejection, at a cost negligible under Assumptions 2.12, 2.22 and 2.8. By Lemma 2.15 (Assumption 2.12), its triple $(\mathsf{ask}, \mathsf{nk}, \mathsf{r_{ivk}})$, before the sign normalisation, is then replaced by independent uniform elements of $\mathbb{F}_{p_{\mathsf{Vesta}}}$, $\mathbb{F}_{p_{\mathsf{Pallas}}}$ and $\mathbb{F}_{p_{\mathsf{Vesta}}}$, at a further negligible cost. Both hops apply because the event that $\mathcal{A}$ wins is efficiently decidable from the triple: an algorithm derives the key's components, runs $\mathcal{A}$, answers its signing queries with ask, runs the extractors on its output and checks the winning condition. After this step nk and $\mathsf{r_{ivk}}$ are independent of ask, and, when $\mathsf{ask} \neq 0$, the sign normalisation makes $\mathsf{ak}^{\mathbb{P}}$ a uniform non-identity point of even y-coordinate (Remark "Hypothesis (H) for honest keys", §11.3). This independence lets the reduction below simulate the full viewing key from $\mathsf{ak}^{\mathbb{P}}$ alone.

(2) *Reduction.* An algorithm $\mathcal{B}$ plays the adversary of Proposition 7.12. It receives $Y = [x]\,G$ for x uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$, with access to H and to the signing oracle. If $Y = \mathcal{O}$ or Y has odd y-coordinate, it stops. Otherwise, an event of probability $(p_{\mathsf{Vesta}} - 1)/(2p_{\mathsf{Vesta}})$, since exactly half of the $p_{\mathsf{Vesta}} - 1$ non-identity points have even y-coordinate (§2.1, on the star encoding), the point Y is distributed as $\mathsf{ak}^{\mathbb{P}}$ in step (1) conditioned on $\mathsf{ask} \neq 0$. Algorithm $\mathcal{B}$ sets $\mathsf{ak}^{\mathbb{P}} := Y$ and $\mathsf{ak} := \mathsf{Extract}_{\mathbb{P}}(Y)$, draws nk and $\mathsf{r_{ivk}}$ uniformly, computes ivk, and gives $\mathcal{A}$ the full viewing key $(\mathsf{ak}, \mathsf{nk}, \mathsf{r_{ivk}})$. It answers each query (α, m) of $\mathcal{A}$ with the oracle's signature under $x + \alpha$, forwards the queries of $\mathcal{A}$ to H, and simulates every other honest party and random oracle. When $\mathcal{A}$ outputs a block chain, $\mathcal{B}$ runs the extractors of Assumption 9.11, as stated after Definition "Pools, accepted Actions and witnesses", and searches for an Action that meets the winning condition, all of whose parts it can check; failing one, it stops. For the Action found, with witness w, it determines $\theta \in \{1, -1\}$ with $\mathsf{ak}^{\mathbb{P}}_w = [\theta]\,Y$, stopping if there is none, and outputs the Action's triple (rk, m, σ) together with the witnessed randomiser α_w and θ.

(3) *The witnessed key.* Except with negligible probability, $\mathsf{ak}^{\mathbb{P}}_w \in \{Y, -Y\}$. By A7, in its unweakened form, $\mathsf{pk_d}^{\mathsf{old}} = [\mathsf{ivk}_w]\,\mathsf{g_d}^{\mathsf{old}}$ with $\mathsf{ivk}_w := \mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r_{ivk}}w}(\mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}}_w), \mathsf{nk}_w) \neq \bot$, and the consumed note is a note of the key, so $\mathsf{pk_d}^{\mathsf{old}} = [\mathsf{ivk}]\,\mathsf{g_d}^{\mathsf{old}}$. As in step (2) of the proof of Lemma 12.5, $\mathsf{ivk}_w = \mathsf{ivk}$, and the binding of ivk to $(\mathsf{ak}, \mathsf{nk})$ gives $\mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}}_w) = \mathsf{ak} = \mathsf{Extract}_{\mathbb{P}}(Y)$, under Assumptions 2.22 and 2.8. By Lemma 2.4, $\mathsf{ak}^{\mathbb{P}}_w \in \{Y, -Y\}$. The statement does not constrain the sign of the witnessed point (Remark "Sign of $\mathsf{ak}^{\mathbb{P}}$", §9.2), and the proof treats both signs. By A6,

$$\mathsf{rk} = \mathsf{ak}^{\mathbb{P}}_w + [\alpha_w]\,G = [\theta]\,Y + [\alpha_w]\,G.$$

(4) *Forgery.* An answer of the signing oracle to a query (α, m') has validating key $Y + [\alpha]\,G$ and message m'. The winning condition excludes a query (α, m) with $\mathsf{rk} = Y + [\alpha]\,G$, so the

output triple is not the triple of an oracle answer: it is a forgery in the sense of Proposition 7.12. Algorithm $\mathcal{B}$ is efficient, and forges with probability at least

$$\frac{p_{\mathsf{Vesta}} - 1}{2p_{\mathsf{Vesta}}}(\epsilon - \nu)$$

for a negligible ν, the costs of step (1) and of the events excluded in step (3).

(5) *Discrete logarithm.* By Proposition 7.12(i), an algorithm $\mathcal{D}$ runs $\mathcal{B}$ twice on shared coins and outputs, with at least the probability that part states for the forging probability of $\mathcal{B}$, a scalar rsk^* with $\mathsf{rk} = [\mathsf{rsk}^*]\,G$ for the rk of the first run. It outputs $x' := \theta\,(\mathsf{rsk}^* - \alpha_w)$, with θ and α_w of the first run. By step (3), $[\theta]\,Y = [\mathsf{rsk}^* - \alpha_w]\,G$, so $x' = x$. For $\theta = 1$ this is part (ii) of the proposition with $\alpha^* = \alpha_w$; for $\theta = -1$ the same relation gives $-x = \mathsf{rsk}^* - \alpha_w$, so a witnessed point $-\mathsf{ak}^{\mathbb{P}}$ is no easier than $\mathsf{ak}^{\mathbb{P}}$: either sign yields ask, the discrete logarithm of the honest $\mathsf{ak}^{\mathbb{P}}$. By Assumption 2.22 the probability that $\mathcal{D}$ outputs x is negligible, hence so is the forging probability of $\mathcal{B}$, and with it ϵ; the loss factor $2p_{\mathsf{Vesta}}/(p_{\mathsf{Vesta}} - 1)$ is about 2. □

Remark 12.10 (Authority without linkability). An Action publishes rk and a signature under $\mathsf{rsk} = \mathsf{ask} + \alpha$, never $\mathsf{ak}^{\mathbb{P}}$ or ak. For α drawn by $\mathsf{GenRandom}$ afresh per Action, Proposition 7.10 (Assumption 7.3) makes the keys rk of n Actions independent of their spend validating keys up to statistical distance $n\,(\delta_{512} + (q_h + n) \cdot 2^{-640})$, with $\delta_{512} < 2^{-257}$, and makes $(\mathsf{rsk}, \mathsf{rk})$ distributed as a fresh key pair; the signature is a function of rsk, the digest and fresh randomness. The keys rk of distinct spends of one key are therefore unlinkable, while Theorem 12.9 shows that an accepted Action with a valid signature under its rk still demonstrates knowledge of ask: from an adversary that produces one for an honestly generated key without its ask, the reduction of its proof computes ask.

12.4 Privacy

Definition 12.11 (Public leakage and privacy preconditions)**.** The *public leakage* of a bundle is its pool, its number of Actions, its anchor, its flags and its balancing value; the rest of the transaction, namely its transparent components, the other pool's bundle and its header with the expiry height, is public as well. The *private inputs* of a bundle are, per Action, its witness (Definition 9.2), the plaintext of its created note ("The note plaintext", §10.1), the outgoing viewing key under which its C_{out} is formed, or $\perp$, and whether each side is real or a dummy. The *privacy preconditions* are:

(P1) every key involved, namely the key of each consumed note, of each recipient, and the key whose ovk the sender uses, is generated honestly with $\mathsf{use_qsk} = \mathsf{false}$;

(P2) the bundle is an Ironwood-pool bundle generated by the procedure of "Construction of a transaction" (§11.5): the seed rseed of each created note fresh and uniform, the trapdoor rcv uniform, the randomiser α drawn by $\mathsf{GenRandom}$, each real consumed note a note of the pool that the procedure of "Trial decryption and note acceptance" (§10.4) accepted from an output of an accepted Action, and dummies as in the protocol specification, §"Dummy Notes (Orchard)": a dummy consumed note has a random spending key, value 0, ρ the x-coordinate of a uniform point, a uniform rseed and an unchecked path, and a dummy created note is an ordinary note of value 0 to an address of a freshly generated key;

(P3) the authentication path is computed locally from public data (Lemma 5.15);

(P4) the adversary holds no viewing key of a key involved (no nk of a spender, no ivk of a recipient, no ovk of the sender), and no output is a coinbase output.

> Within these preconditions the adversary may choose which notes are consumed and the recipients' addresses, and may know the consumed notes' contents.

Under (P2) the real consumed notes of one key have pairwise distinct elements ρ. An accepted note has $\rho = \mathsf{nf}^{\mathsf{old}}$ of the Action whose output carries it (Proposition 10.15(a)); one output yields at most one note under one ivk; and distinct Actions of the pool publish distinct nullifiers (§6.2). The experiment of the next theorem imposes this property directly.

> **Theorem 12.12** (Privacy of an Action). *Assume preconditions (P1) to (P4), and let two sequences of private inputs of one Ironwood-pool bundle have equal public leakage, both satisfy Definition 9.2 with the common anchor and flags, and each Action consume either a leaf of the tree at that anchor not consumed before or a dummy, and create either a note to an honestly generated address or a dummy. Under Assumptions 9.14, 2.12, 3.13, 6.2, 3.16, 2.8, 10.4, 10.5 and 7.3, with Assumption 2.22 for Lemma "Rejection in key generation" (§3.2) and the redraw of the note seed (§4.3), the public data of the two bundles (per Action* $\mathsf{cv}^{\mathsf{net}}$, nf, rk, cmx, $\mathsf{epk}^{\star}$, C_{enc}, C_{out} *and the spend-authorisation signature; the proof and the binding signature) are computationally indistinguishable. In particular, beyond its public leakage an Action reveals nothing about its private inputs, and an Action whose spend or output side is a dummy is indistinguishable from one whose sides are real.*
>
> *Precisely, in the following experiment every efficient adversary has negligible advantage. Keys are generated as in (P1), and the adversary receives addresses of them at indices of its choice. It outputs the leaves of the Ironwood-pool tree through an anchoring block, the rest of a transaction, flags, and two specifications of n Actions with equal balancing values. A specification gives, per Action, a consumed side, either "dummy" or a note at one of these addresses whose extracted commitment is a leaf; a created side, either "dummy" or one of these addresses with a value and a memo; and the sender's* ovk*, that of one of the keys or* $\perp$*. The consumed leaves of a specification are pairwise distinct, the consumed notes of one key have pairwise distinct* ρ*, and the construction of (P2) yields witnesses of Definition 9.2 with the anchor and flags. A challenger builds the bundle of specification* β*, for a uniform bit* β*, by that construction and gives the adversary its public data; the adversary outputs a bit* β'*, and its advantage is* $|\Pr[\beta' = 1 \mid \beta = 0] - \Pr[\beta' = 1 \mid \beta = 1]|$.

Proof. For $\beta \in \{0, 1\}$ let G_0^β be the experiment with specification β. Each of the games G_1^β to G_7^β below changes the probability that the adversary outputs 1 by a negligible amount (*Crypto Guide*, §"The hybrid argument", Lemma "Hybrid lemma"; each game is reached in polynomially many steps, one key or one Action at a time), and G_7^0 and G_7^1 are identically distributed. Every reduction runs the experiment with the adversary and samples every value that its hop does not concern. The keys involved are those of (P1), the spending keys of the dummy consumed notes and the keys of the dummy created notes.

Game G_1 (*proof*). The aggregate proof is replaced by the output of the simulator of Assumption 9.14 on the primary inputs, which are public; the witnesses satisfy Definition 9.2 by hypothesis. The cost is the statistical distance of that assumption. From G_1 on, a witness enters the game only through the public fields computed from its components.

Game G_2 (*keys*). Each key involved is replaced in turn, as in the preliminary step of the proof of Proposition 10.6 and hybrids H_1 to H_3 of the proof of Proposition 6.10: by a single draw without rejection (Lemma "Rejection in key generation", §3.2; Assumptions 2.12, 2.22 and 2.8); its triple $(\mathsf{ask}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$ by independent uniform elements (Lemma 2.15); and its triple $(\mathsf{ivk}, \mathsf{dk}, \mathsf{ovk})$ by $(\mathsf{Extract}_{\mathbb{P}}(W), u_1, u_2)$, for a uniform point W and uniform 32-byte strings u_1 and u_2 (Assumption 3.13, whose reduction draws ask and nk itself, and the negligible event that $\mathsf{ask} = 0$ or that the hash point of $\mathsf{Commit}^{\mathsf{ivk}}$ is $\perp$). The hops apply because after G_1 the trapdoor $\mathsf{r}_{\mathsf{ivk}}$ enters the game only through $(\mathsf{ivk}, \mathsf{dk}, \mathsf{ovk})$. From G_2 on, the components ask, nk, ivk, dk and ovk of each key are independent, and dk and ovk are uniform.

Game G_3 (*outgoing ciphertexts*). Each C_{out} is replaced by the encryption of a fixed 64-byte

string under an independent uniform key, by Proposition "Confidentiality of the outgoing ciphertext" (§10.3). For $\mathsf{ovk} \neq \perp$, the key ock is a value of the random oracle $\mathsf{PRF}^{\mathsf{ock}}$ (Assumption 10.4) at an input that begins with the uniform secret ovk, which q queries of the adversary hit with probability at most $q \cdot 2^{-256}$. Distinct Actions give distinct inputs, their epk differing except with negligible probability, so each ock is uniform and encrypts once, and Assumption 10.5(a) replaces the plaintext. For $\mathsf{ovk} = \perp$, the key is uniform by construction and Assumption 10.5(a) applies directly.

Game G_4 (*randomised keys*). Each randomiser α_i is replaced by a uniform scalar, at a cost of at most $\delta_{512} + q_h \cdot 2^{-640}$ each, $\delta_{512} < 2^{-257}$, for q_h queries to H (Lemma "Distance of GenRandom from uniform", §7.2; Assumption 7.3). Then $y_i := \mathsf{rsk}_i = \mathsf{ask} + \alpha_i$ is uniform and independent of ask and of every other value, and $\mathsf{rk}_i = [y_i]\, G^{\mathsf{SpendAuth}}$ (Proposition 7.10(i)); the signature σ_i is computed under y_i. From G_4 on, no ask enters the game.

Game G_5 (*nullifiers*). For each key that consumes a note in the bundle, real or dummy, the key nk enters G_4 only through the nullifiers of those notes: G_1 removed it from the proof and G_2 from the viewing keys. As in hybrids H_4 and H_5 of the proof of Proposition 6.10, $\mathsf{PRF}^{\mathsf{nfOrchard}}_{\mathsf{nk}}$ is replaced by a uniformly random function (Assumption 6.2), and each multiplier of K^{Orchard} by a uniform scalar $u_i \in \mathbb{F}_{p_{\mathsf{Vesta}}}$, at a statistical cost below 2^{-167} each; the elements ρ of the consumed notes of one key are pairwise distinct by the definition of the experiment, and a dummy consumed note has a key of its own. Each nullifier is then $\mathsf{Extract}_{\mathbb{P}}([u_i]\, K^{\mathsf{Orchard}} + \mathsf{cm}_i)$, the x-coordinate of a uniform point whatever cm_i is, since $K^{\mathsf{Orchard}} \neq \mathcal{O}$ except with probability $1/p_{\mathsf{Vesta}}$ (Assumption 2.8).

Game G_6 (*ephemeral keys and note ciphertexts*). For each created note in turn, real or dummy, epk is replaced by a uniform non-identity point and C_{enc} by the encryption of a fixed 564-byte string under an independent uniform key, by steps (1) to (4) of the proof of Proposition 10.6 (Assumptions 10.4, 3.16, 10.5, 2.8 and 2.12), which keep cmx. The other public data, the data z of that proposition, depend after G_2 to G_5 on the recipient's key only through the addresses of that key given to the adversary and the outputs to them. The reduction of step (2) of that proof embeds the exponent a as the ivk of the recipient's key in every such address, computing its transmission key as $[t]([a]\, P)$ for the diversified base $[t]\, P$ that it programs, and computes every other output to that key from its own esk; the proof is otherwise as written.

Game G_7 (*note commitments*). For each created note in turn, cmx is replaced by the x-coordinate of a uniform point. After G_6 the seed rseed of the note enters the game only through ψ and rcm, that is through cm: step (1) of G_6 replaced esk, and the ciphertexts no longer depend on the plaintext. The proof of Proposition 4.8 then places cm within $\epsilon_{\mathsf{PRF}} + \delta + \epsilon_{\perp}$ of a uniform point, with $\delta < 2^{-257}$ and ϵ_{PRF}, $\epsilon_{\perp}$ negligible under Assumptions 2.12, 2.22 and 2.8.

Value commitments and the binding signature. In G_7^{β} every field other than the net value commitments and the binding signature is distributed independently of β: the nullifiers and the extracted commitments are x-coordinates of uniform points, each epk is a uniform non-identity point, the ciphertexts encrypt fixed strings under fresh keys, each rk_i and σ_i come from a fresh key y_i, the proof is simulated from the primary inputs, and the rest of the transaction is common. The net values v_i^{β} of specification β enter only

$$\mathsf{cv}_i^{\mathsf{net}} = [v_i^{\beta}]\, V^{\mathsf{Orchard}} + [\mathsf{rcv}_i]\, R^{\mathsf{Orchard}}, \qquad \mathsf{bsk} = \sum_{i=1}^{n} \mathsf{rcv}_i,$$

and, through these, the digest, the primary inputs of the simulator and bvk. Let λ be the discrete logarithm of V^{Orchard} to the base R^{Orchard}, which exists because R^{Orchard} generates the group (§8.1). The map

$$(\mathsf{rcv}_i)_i \mapsto (\mathsf{rcv}_i + (v_i^0 - v_i^1)\, \lambda)_i$$

is a bijection of $\mathbb{F}^n_{p_{\mathsf{Vesta}}}$. It carries the net value commitments of specification 0 to those of specification 1, and it preserves $\sum_i \mathsf{rcv}_i$, because $\sum_i v_i^0 = \sum_i v_i^1$, the common balancing value (step (6) of

the construction of §11.5). The rcv_i being uniform and independent, the tuple $(\mathsf{cv}_1^{\mathsf{net}}, \ldots, \mathsf{cv}_n^{\mathsf{net}}, \mathsf{bsk})$ has the same distribution for both β, and G_7^0 and G_7^1 are identically distributed. This step is perfect and needs no reduction; the game is not required to compute λ.

The advantage of the adversary is therefore at most the sum of the costs of the hops for $\beta = 0$ and $\beta = 1$, which is negligible. In G_7 a dummy side differs from a real side only through the value in $\mathsf{cv}^{\mathsf{net}}$, so two sequences that differ in whether a side is a dummy, with equal public leakage, are covered by the same argument. □

Remark 12.13 (Non-claims). Theorem 12.12 hides nothing of the public leakage: each bundle's balancing value, so the net amount that each transaction moves across a pool's boundary is public; each pool's chain value pool balance, the negation of the sum of its balancing values ("Chain state and pool rules", §11.4), so a pool hides which notes are held by whom, never the aggregate value it holds; the number of Actions and the presence of each component; the flags; the transparent components; the fee; the anchor, which confines the consumed note to the leaves of the anchored tree ("Authentication paths", §5.3); and the transaction's timing and expiry height. Per Action the chain gains one nullifier and one extracted note commitment, which by the theorem reveal neither sender, recipient nor amount. The theorem claims nothing for coinbase outputs, whose outgoing ciphertexts decrypt under the all-zero ovk (§11.4); nothing against a holder of a viewing key of a key involved, whose powers are stated in "Nullifier uniqueness and unlinkability" (§6.4), "The outgoing ciphertext" (§10.3) and "Trial decryption and note acceptance" (§10.4); nothing for an Orchard-pool bundle, whose created notes use lead byte `0x02` and a derivation on which the volume does not rely (§10.4); and nothing about linkage through information outside the transaction, such as an address given to several parties, voluntary disclosures, or wallet and network metadata.

12.5 The end-to-end theorem

Theorem 12.14 (End-to-end security). *Assume, listed once: Assumption 2.22; Assumptions 9.11 and 9.14, with Assumption 9.10 in the supporting analysis of Assumption 9.11; Assumption 3.16; Assumptions 2.12 and 6.2; Assumption 3.13; Assumptions 10.4 and 10.5; Assumption 11.9; and Assumptions 2.8 and 7.3, for* GroupHash *and* H *as random oracles. Clauses* (*iv*) *and* (*v*) *are key-derived and are claimed for keys generated with* use_qsk = false*; clauses* (*i*) *to* (*iii*) *use no property of key generation. For each pool, no efficient adversary achieves any of the following, except with negligible probability:*

(*i*) *an accepted Action for which the extractor of Assumption 9.11 outputs no witness satisfying Definition 9.2; or a second opening, other than its witnessed created note, of the* cmx *of an accepted Action; or an accepted Action whose witnessed* v^{old} *is non-zero and whose consumed note is not a committed note of the pool, a leaf of the tree at the Action's anchor* (*membership soundness*) [*R1 and R2 soundness*]*;*

(*ii*) *two accepted Actions of the pool consuming one committed note of non-zero value* [*R3 soundness*]*;*

(*iii*) *an accepted bundle whose balancing value differs, as an integer, from the sum of its Actions'* $v^{\mathsf{old}} - v^{\mathsf{new}}$ [*R4 soundness*]*;*

(*iv*) *winning the spend-authority experiment of "Authorisation of spends", that is, an accepted Action consuming a note of an honestly generated key without a signature of the key holder, the adversary holding the key's full viewing key but not its* ask*; or an accepted transaction that changes the effecting data of a transaction T while retaining one of its Actions, where the Actions of T are*

> *authorised by holders of honestly generated keys who sign as hypothesis (S) of Proposition 11.10 states [R5];*
>
> (v) *for Ironwood-pool bundles, distinguishing the public data of two bundles with equal public leakage under preconditions (P1) to (P4) of "Privacy", in particular a real Action from a dummy one [R1 hiding; R2 hiding, without identifying the note; R3 hiding, without a public list of consumed notes; R4 hiding, without disclosed amounts].*
>
> *The bracketed tags refer to the requirement map of "The Action statement" (§9.2, Table 5), which is not repeated.*

Proof. Each clause is a result of this section or of an earlier one.

(i) The first part is Assumption 9.11, applied to each bundle as stated after Definition "Pools, accepted Actions and witnesses", with Remark "⊥-weakened conditions" (§9.2) for the unweakened forms of A1, A2, A3 and A7. By A2 the created note opens cmx, and by Remark "Binding for notes given by expanded receivers" it is the only opening that an efficient algorithm finds. For $v^{\mathsf{old}} \neq 0$, condition A3, outside its ⊥ case, makes the fold of A3 on $(\mathsf{Extract}_{\mathbb{P}}(\mathsf{cm}^{\mathsf{old}}), \mathsf{pos}, \mathsf{path})$, with the witnessed encodings c, reach the bundle's anchor rt, and the anchor rule, step (3) of the verification of §11.6, makes rt the anchor of an earlier block in the pool. By A1, $\mathsf{cm}^{\mathsf{old}}$ is a point, so Proposition 5.9(iii), which applies to that fold (§9.2), gives, under Assumptions 2.22 and 2.8, that $\mathsf{Extract}_{\mathbb{P}}(\mathsf{cm}^{\mathsf{old}})$ is the leaf appended at position pos, in the anchoring block or earlier: the consumed note is a committed note of the pool.

(ii) Theorem 12.6.

(iii) Theorem 12.4, which adds Assumption 7.3.

(iv) The first part is Theorem 12.9. The second is Proposition 11.10, under Assumptions 11.9, 2.22 and 7.3; its hypothesis (H) holds for honestly generated keys up to a negligible change in the probability of every efficiently decidable event, the event of that proposition included, under Assumptions 2.12, 2.22 and 2.8 (Remark "Hypothesis (H) for honest keys", §11.3).

(v) Theorem 12.12.

Clauses (i) to (iii) use Assumptions 9.11, 2.22, 2.8 and 7.3 only, and no property of key generation; clauses (iv) and (v) use honest key generation with $\mathsf{use_qsk} = \mathsf{false}$. Each assumption of the list is used by some clause, and Assumption 9.10 enters only through the supporting analysis of Assumption 9.11 ("The Action circuit and the Halo 2 proof", §9.3). □

Remark 12.15 (Nature of the guarantees). Beyond Assumption 9.11, every soundness reduction ends in a discrete-logarithm problem on Pallas: either a non-trivial relation among values of $\mathsf{GroupHash}$, modelled as a random oracle (Assumption 2.8), namely Sinsemilla collisions and ⊥ outputs (Proposition 2.24), with membership soundness (Proposition 5.9), the binding of $\mathsf{NoteCommit}$ and of $\mathsf{Commit}^{\mathsf{ivk}}$, nullifier uniqueness (Proposition 6.8) and the independence of V^{Orchard} and R^{Orchard} (Proposition 8.10); or the discrete logarithm of a given point, in the spend-authority reduction through the RedPallas extraction of Proposition 7.12. Every extraction from a RedPallas signature, the binding signature included, holds in the random-oracle model for H (Assumption 7.3). The digest step of bundle binding ends instead in a BLAKE2b collision (Assumption 11.9). Soundness is therefore computational and conditional on Assumption 9.11, which "The Action circuit and the Halo 2 proof" (§9.3) classifies as designed but unspecified, with discrete logarithms on Vesta entering only its supporting analysis; no concrete bound is asserted, since that assumption carries no loss factor. In clause (v), the contribution of the proof is statistical (Assumption 9.14; *Halo 2 Guide*, §"Zero knowledge: hiding the witness in Halo 2"); that of the value commitments with the binding signature is perfect (the final step of the proof of Theorem 12.12); that of rk and the spend-authorisation signatures is statistical in the random-oracle model for H (game G_4 there); and the unlinkability of every other published field is computational. Assumption 6.2 is cryptanalytic, with no reduction to a standard problem.

Part VI
Wallet Guide: The Zcash wallet layer: keys, transactions, and light-client synchronisation

Abstract

This volume of *The Zcash Arboretum* states the protocols that a Zcash wallet runs: the derivation of its keys and addresses from one seed; the discovery of its notes from the chain through an untrusted light-client server, and what that server learns; and the construction, authorisation and submission of its payments. Each protocol is stated as a definition, construction or rule, and its correctness, security or privacy guarantee as a lemma, proposition or theorem under named assumptions. The subject is the wallet of a user of the Orchard protocol who holds funds in the Ironwood pool, the sealed Orchard pool, the Sapling pool and the transparent pool; the light-client service is an abstract query and response interface, and no implementation is described. The volume assumes the *Math Guide*, the *Crypto Guide*, the *Halo 2 Guide* and the *Consensus Guide*, and cites the *Ironwood Guide* for every construction of the shielded protocol. It is non-normative: the Zcash Protocol Specification and the Zcash Improvement Proposals are authoritative.

1 Introduction

This section fixes the subject of the volume, its status and the vocabulary that every later section uses.

1.1 Scope and status

This volume states the protocols by which a wallet, in the sense of "Parties, pools, and classifications" (§1.2), performs three tasks. It derives and holds keys and addresses: hierarchical key derivation (ZIP 32), and unified addresses and unified viewing keys (ZIP 316). It discovers and tracks its notes: the compact-block format (ZIP 307) as an abstract message format, trial decryption, nullifier tracking, the retention of the note commitment tree information that its witnesses need, and what an untrusted light-client server learns. It pays: anchors and confirmation depths (ZIP 315), expiry (ZIP 203), the conventional fee (ZIP 317), transaction construction (note selection, change, dummy Actions), partially created transactions (ZIP 374), the transaction lifecycle from submission to a terminal state, and payment requests (ZIP 321).

The subject is the wallet of a user of the Orchard protocol who holds funds in the Ironwood pool, the sealed Orchard pool, the Sapling pool and the transparent pool. Sapling constructions and transparent scripts are cited to the protocol specification and to the BIPs where a wallet protocol needs them, and are never constructed; constructions of the shielded protocol are cited to the *Ironwood Guide* by section title, and are never re-derived. The light-client service is an abstract query and response interface, and tree storage is the information that a wallet retains; no implementation is described. The volume is non-normative: the protocol specification and the ZIPs are authoritative.

The volume describes the protocols as specified for NU7, whose deployment ZIP, ZIP 259, has status Draft.

ZIP 307 specifies the compact format for the Sapling pool only: a compact block of height, block hash and transactions, and a compact transaction of index, transaction identifier, and compact Sapling spends and outputs. The parent hash, the tree sizes, the compact Actions of both Orchard-protocol pools, the transparent part and pool selection are designed but unspecified in the sense of the *Ironwood Guide*, Definition "Epistemic classes" ("Scope and status"), and the volume states them abstractly in "Compact blocks (ZIP 307)" (§4).

Rules of the Ironwood pool and of NU6.3 are cited to the NU6.3 edition of the protocol specification, labelled "NU6.3 proposal", whose section numbers equal those of the default edition;

the text on use_qsk for protocol specification, §"Orchard Key Components", is in no edition and is cited to ZIP 2005, "Changes to the Protocol Specification", directly.

Table 1 gives the status, and only the status, of each ZIP that the volume cites. Every later citation of these ZIPs relies on the table for status. ZIP 258 activates ZIP 2005 with NU6.3 (ZIP 258, "ZIP 2005 activation").

ZIP	Title	Status
32	Shielded Hierarchical Deterministic Wallets	Final
48	Transparent Multisig Wallets	Draft
173	Bech32 Format	Final
203	Transaction Expiry	Final
211	Disabling Addition of New Value to the Sprout Chain Value Pool	Final
212	Allow Recipient to Derive Ephemeral Secret from Note Plaintext	Final
218	25-second Block Target Spacing	Draft
225	Version 5 Transaction Format	Final
227	Issuance of Zcash Shielded Assets	Draft
229	Version 6 Transaction Format	Draft
231	Memo Bundles	Draft
244	Transaction Identifier Non-Malleability	Final
248	Extensible Transaction Format	Draft
256	Deployment of Consensus Bug Fixes Between NU6.1 and NU6.2	Final
258	Deployment of the NU6.3 Network Upgrade	Draft
259	Deployment of the NU7 Network Upgrade	Draft
302	Standardized Memo Field Format	Draft
307	Light Client Protocol for Payment Detection	Draft
312	FROST for Spend Authorization Multisignatures	Draft
314	Privacy upgrades to the Zcash light client protocol	Reserved
315	Best Practices for Wallet Implementations	Draft
316	Unified Addresses and Unified Viewing Keys	Revision 0 Active; Revision 1 Withdrawn; Revision 2 Draft
317	Proportional Transfer Fee Mechanism	Revision 0 Active; Revisions 1 and 2 Draft
318	Orchard to Ironwood Migration	Draft
320	Defining an Address Type to which funds can only be sent from Transparent Addresses	Active
321	Payment Request URIs	Active
326	NU6.3 Consequences for Wallets	Draft
374	Partially Created Zcash Transaction Format	Revision 0 Draft
401	Addressing Mempool Denial-of-Service	Active
2003	Disallow version 4 transactions	Draft
2005	Ironwood Quantum Recoverability	Proposed

Table 1: Status of the ZIPs cited, as given in the preamble of each ZIP.

1.2 Parties, pools, and classifications

A *wallet* is a party that holds a seed and the keys derived from it, and no copy of the chain. A *light-client server* is a party that answers the queries of a wallet from its own view of the best chain, the best valid block chain of the *Consensus Guide*, "Work, and the best chain" (Definition "Best valid block chain"). The *network* is the set of the nodes that relay transactions and blocks and of the miners that produce blocks. The trust placed in the server is not fixed here; it is an assumption of "Architecture and authority" (§5.1).

A wallet handles funds in four value pools. The first is the *transparent pool*. The second is the *Sapling pool* (protocol specification, §"Chain Value Pool Balances"), whose constructions are

cited, never constructed. The third and fourth are the *Ironwood pool* and the sealed *Orchard pool*, the two value pools of the Orchard protocol (*Ironwood Guide*, "The Orchard protocol and its two pools", Definitions "Orchard protocol" and "Orchard pool, Ironwood pool"; ZIP 229, Terminology); each has its own note commitment tree, its own nullifier set and its own chain value pool balance. The rules that seal the Orchard pool are cited there (ZIP 258, "Consensus rules from NU6.3 activation"), and are stated for wallets in "Payments to pools" (§11.2).

Heights are counted as in the *Ironwood Guide*, "Scope and status" (Definition "Block chain, transaction, consensus rules"), the genesis block having height 0. The *tip* T is the height of the last block of the wallet's view of the best chain. Every other height (the target height, the anchor height, the expiry height) is defined where it is first used.

Definition 1.1 (Classes of claim)**.** The classes *specified* and *designed but unspecified* are those of the *Ironwood Guide*, Definition "Epistemic classes" ("Scope and status"), applied here to wallet rules as well as to claims about the deployed protocol. A wallet rule is *specified* when normative text of a ZIP or of the protocol specification fixes it, the status of the ZIP being that of Table 1. A wallet rule is *designed but unspecified* when wallets are designed to follow it and no normative text fixes it. This volume adds a third class: a question is an *open problem* when neither a ZIP nor a stated design settles it.

Every design-stage statement of the volume carries one class of Definition 1.1, and every deployed rule is cited at its use to the section of the protocol specification or of the ZIP that fixes it. A specified rule keeps the normative level of its source (MUST, SHOULD, MAY).

2 Hierarchical key derivation (ZIP 32)

This section derives every key of a wallet from one seed. It fixes the seed and its fingerprint, the hardened-only derivation of Orchard spending keys and the standard account path, and the passage from an account spending key to the key components that the *Ironwood Guide* constructs from a spending key, including the alternative branch of ZIP 2005. It then constructs the internal keys of an account and proves the separations between the two scopes, bounds the probability that an account key is invalid, and closes with the domains of key derivation.

2.1 Wallet seeds

Definition 2.1 (Wallet seed)**.** A *wallet seed* is a byte string S with $32 \leq \mathrm{length}(S) \leq 252$, where $\mathrm{length}(S)$ is its length in bytes (MUST). The method that generates S MUST give it at least 256 bits of entropy; the requirement extends to any mnemonic or input randomness from which S is obtained (ZIP 32, "Specification: Wallet seeds"). Every key of every account of the wallet is derived from S: the account spending key in §2.3, and what recovery needs beyond S in §2.4.

Every key of every account is a deterministic function of S, so a seed space of lower entropy is searched once for all accounts, and across many wallets as a multi-target search that matches the first derived addresses or viewing keys against a set of observed ones (ZIP 32, "Rationale for the 256-bit entropy requirement").

Construction 2.2 (Seed fingerprint)**.** For a wallet seed S, the *seed fingerprint* is

$$\mathtt{SeedFP}(S) := \text{BLAKE2b-256}(\mathtt{Zcash_HD_Seed_FP},\ [\mathrm{length}(S)] \parallel S),$$

unkeyed BLAKE2b with a 32-byte output and the 16-byte personalisation shown, over the

single length byte of S followed by S; the length fits one byte because $\text{length}(S) \leq 252$. It MAY be used to identify the seed of a hierarchical deterministic wallet. Its string form is Bech32m (BIP 350) with human-readable part `zip32seedfp` and the 32 fingerprint bytes as data; no short tag is defined (ZIP 32, "Seed Fingerprints"). A partially created transaction carries the fingerprint together with a derivation path, so that a signing party locates its key (§12.3).

A partially created transaction relies on collision resistance of BLAKE2b-256 under this personalisation (*Crypto Guide*, §"Security notions: preimage, second-preimage, and collision resistance", Definition "Collision resistance"); Lemma 2.5 additionally models it as a random oracle, so that the fingerprint reveals nothing of the keys. The fingerprint of a full viewing key is constructed with the key components, in §2.4.

2.2 Hardened-only derivation

ZIP 32 instantiates its hardened-only derivation for Orchard with the master-key domain `ZcashIP32Orchard`, a 16-byte BLAKE2b personalisation that SHOULD be disjoint from every other BLAKE2b domain separator of Zcash, and the child-key domain `0x81`, a lead byte of $\mathsf{PRFexpand}$ (ZIP 32, "Specification: Hardened-only key derivation", "Instantiation", and "Specification: Orchard key derivation"). An *Orchard extended spending key* is a pair $(\mathsf{sk}, \mathsf{c})$ of 32-byte strings, where sk is a spending key in the sense of the *Ironwood Guide*, §"The spending key and the spend-side secrets", and c is the *chain code* (ZIP 32, "Orchard extended keys"). Only hardened derivation is defined for Orchard: child derivation fails at every index $i < 2^{31}$. ZIP 32 defines no extended full viewing key for Orchard; a full viewing key has no child-derivation capability for which a chain code would serve. For $0 \leq i < 2^{31}$ write $i' := i + 2^{31}$.

Construction 2.3 (Orchard master key generation)**.** For a wallet seed S (Definition 2.1), let

$$I := \text{BLAKE2b-512}(\texttt{ZcashIP32Orchard}, S),$$

unkeyed BLAKE2b with a 64-byte output, and let $\mathsf{sk}_m := I[0..32)$ and $\mathsf{c}_m := I[32..64)$, the first and the last 32 bytes. The *master extended spending key* is $m_{\mathsf{Orchard}} := (\mathsf{sk}_m, \mathsf{c}_m) =: \mathsf{MKG}(S)$ (ZIP 32, "Hardened-only master key generation" and "Orchard master key generation").

Construction 2.4 (Orchard child key derivation)**.** For a parent extended spending key $(\mathsf{sk}_{\mathsf{par}}, \mathsf{c}_{\mathsf{par}})$ and an index $i \in \{0, \ldots, 2^{32} - 1\}$: if $i < 2^{31}$, derivation fails; otherwise

$$I := \mathsf{PRFexpand}_{\mathsf{c}_{\mathsf{par}}}([\texttt{0x81}] \,\|\, \mathsf{sk}_{\mathsf{par}} \,\|\, \mathsf{I2LEOSP}_{32}(i)),$$

a 64-byte string, and $\mathsf{CKD}((\mathsf{sk}_{\mathsf{par}}, \mathsf{c}_{\mathsf{par}}), i) := (I[0..32), I[32..64))$. Here $\mathsf{I2LEOSP}_{32}(i)$ is the 4-byte little-endian encoding of i, and $\mathsf{PRFexpand}_k(t) = \text{BLAKE2b-512}(\texttt{Zcash_ExpandSeed}, k \,\|\, t)$ is the *Ironwood Guide*, §"Pseudorandom functions and field reductions", Construction "Expansion function" (ZIP 32, "Hardened-only child key derivation" and "Orchard child key derivation"). The parent chain code is the key of the expansion; the parent spending key is message data.

Lemma 2.5 (Pseudorandomness of derived spending keys)**.** *Assume the following.*

(*i*) *BLAKE2b-512 under the personalisation* `ZcashIP32Orchard` *and BLAKE2b-256 under the personalisation* `Zcash_HD_Seed_FP` *are modelled as random oracles* (Crypto Guide, §*"The*

random oracle model", Definition "Random oracle"), independent of each other and of every other personalised use of BLAKE2b (Crypto Guide, *§"Domain separation and personalisation", Proposition "Domain separation yields independent oracles"), and of every other derivation from S that is modelled as a random oracle.*

(*ii*) *The seed S has min-entropy at least* 256 *bits, the reading of the entropy requirement of Definition 2.1 that the lemma uses.*

(*iii*) *The* Ironwood Guide*'s Assumption "Pseudorandomness of PRF expansion"* (*§"Pseudorandom functions and field reductions"*) *holds.*

Let N be a fixed finite set of distinct nodes of the Orchard derivation tree rooted at m_{Orchard}, each reached by Construction 2.4 along hardened indices, and let T be the subtree spanned by the paths from the root to the nodes of N. Then the spending keys of the nodes of N, the chain codes of those nodes of N that are not a proper ancestor of another node of N, and $\mathsf{SeedFP}(S)$ *are jointly computationally indistinguishable from independent uniform* 32*-byte strings. For an adversary making q random-oracle queries, the distinguishing advantage is at most $q \cdot 2^{-256}$ plus one PRF advantage for each node of T whose chain code keys a derivation in T; there are at most as many such nodes as T has edges.*

Proof. A hybrid argument (*Crypto Guide*, §"The hybrid argument", Lemma "Hybrid lemma"). Game 0 is the real experiment. Game 1 answers every oracle query whose input is determined by S (the master query of Construction 2.3, the input of SeedFP and the inputs of the other modelled derivations) by fresh uniform strings, independent of S. The two games differ only if the adversary itself queries such an input. In Game 1 its view is independent of S until that happens, so each of its q queries does so with probability at most 2^{-256} by hypothesis (ii), and the games differ by at most $q \cdot 2^{-256}$. In Game 1 the pair $(\mathsf{sk}_m, \mathsf{c}_m)$ and $\mathsf{SeedFP}(S)$ are independent uniform strings.

Then visit the nodes of T that have a child in T, each parent before its children. At such a node v the chain code c_v is, by the earlier hops, a uniform string that is used only as a $\mathsf{PRFexpand}$ key and is not released, because v is a proper ancestor of a node of N. The hop replaces $\mathsf{PRFexpand}_{\mathsf{c}_v}$ by a uniformly random function. The children of v in T have distinct indices, hence distinct inputs $[\mathtt{0x81}] \| \mathsf{sk}_v \| \mathsf{I2LEOSP}_{32}(i)$, so their 64-byte outputs become fresh uniform strings (*Ironwood Guide*, §"Pseudorandom functions and field reductions", Lemma "Independence of domain-separated expansions", first conclusion, applied to a uniform key; its statistical-distance clause is not used). The hop costs the PRF advantage of a reduction that samples every other node itself and forwards the queries of v to its oracle. In the last game every released value is a fresh uniform string; the chain codes of proper ancestors, which are not fresh relative to their children, are exactly those the lemma does not release. □

Remark 2.6 (The spending key as a leaf capability). In Construction 2.4 the parent chain code keys the expansion and the parent spending key is message data, so deriving children requires the pair $(\mathsf{sk}, \mathsf{c})$. A 32-byte spending key alone derives no child: by Lemma 2.5, without c the children are pseudorandom (ZIP 32, "Orchard child key derivation"). A wallet that keeps $\mathsf{sk}_{\mathsf{acct}}$ and not $\mathsf{c}_{\mathsf{acct}}$ keeps no power to derive keys below the account.

2.3 The standard account path

Definition 2.7 (Account spending key)**.** For a wallet seed S, a coin type t and an account index $a \in \{0, \dots, 2^{31} - 1\}$,

$$(\mathsf{sk}_{\mathsf{acct}}, \mathsf{c}_{\mathsf{acct}}) := \mathsf{CKD}(\mathsf{CKD}(\mathsf{CKD}(\mathsf{MKG}(S), 32'), t'), a'),$$

the path $m_{\mathsf{Orchard}}/\mathit{purpose}'/\mathit{coin_type}'/\mathit{account}'$. The purpose index is $32' = \texttt{0x80000020}$, following BIP 43; the coin type is the SLIP 44 value, 133 on Mainnet and 1 on Testnet, every test network sharing the value 1; accounts are numbered sequentially from 0, as in BIP 44. All three levels are hardened. The *account spending key* is $\mathsf{sk}_{\mathsf{acct}}$; no construction of this volume uses $\mathsf{c}_{\mathsf{acct}}$ (Remark 2.6). A wallet implementing Orchard ZIP 32 derivation MUST support this path for every account in $\{0, \ldots, 2^{31} - 1\}$, MUST support generating the default payment address, at diversifier index 0, of every account it supports, and MAY generate further diversified addresses of an account (ZIP 32, "Key path levels" and "Orchard key path").

The path ends at the account. Addresses are not path children: an account has 2^{88} diversified addresses per scope, the images of the diversifier indices (*Ironwood Guide*, §"Diversified addresses", Construction "Diversified address"). There is no change level. A shielded output does not show its address on chain, so returning change to the originating address is indistinguishable from using a change address; the wallet separates change from received funds by the internal scope below the account (§2.5; ZIP 32, "Key path levels"). Figure 1 shows the derivation from the seed to the addresses of both scopes.

Example 2.8 (The account path of a fixed seed)**.** Let S be the 32 bytes $\texttt{0x00}, \texttt{0x01}, \ldots, \texttt{0x1f}$, on Mainnet (coin type 133), account 0. The pairs $(\mathsf{sk}, \mathsf{c})$ along the path, in hexadecimal, are:

Node		Value
m	sk	`7eee3c1017870990a3dd6891b82f80be8976c1e7dc20d60817a5e88e8b2cd4b8`
	c	`ab8b7a00509ef20e469b5292b61d474b7cffcb1657924cda720250ae40526677`
$m/32'$	sk	`93af0a87c2e4704d8825e145557c6a5d4dcbc9016170e276a2080a2dc9eb0f87`
	c	`6e1e808d410d6d1f84e62e67c8b60e80e5ee4de5021d575c0433a625d609c45c`
$m/32'/133'$	sk	`149fcee594a2a03de277ab919f22611d0ef6aa8e8e9a0bef32f01a0bcb0e8c4c`
	c	`07d13fdeccd21da16b467f8e785edc32f74ceeb541bede0394647f10fdb7114f`
$m/32'/133'/0'$	sk	`b67d8d87cab9189500afb45dbca9f92c924c1de9d9ee1451be4a78313cb223b4`
	c	`7ff0a6392e144d4c8f45bca21fb7827b508446ac069890aefbebbe388b7a84c0`

The last spending key is $\mathsf{sk}_{\mathsf{acct}}$, and it is valid in the sense of §2.6; the last chain code is $\mathsf{c}_{\mathsf{acct}}$, which no construction uses. The master spending key agrees with the first Orchard test vector of ZIP 32, "Test Vectors".

2.4 Orchard key components of an account

The *Ironwood Guide* constructs from a spending key sk the components ask, nk and $\mathsf{r}_{\mathsf{ivk}}$ (§"The spending key and the spend-side secrets"), the spend validating key $\mathsf{ak}^{\mathbb{P}}$ and its coordinate ak (same subsection), the full viewing key $\mathsf{fvk} = (\mathsf{ak}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$, the incoming viewing key scalar $\mathsf{ivk} = \mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r}_{\mathsf{ivk}}}(\mathsf{ak}, \mathsf{nk})$, and $(\mathsf{dk}, \mathsf{ovk})$ as the two halves of

$$\mathsf{PRFexpand}_{\mathrm{LE}_{256}(\mathsf{r}_{\mathsf{ivk}})}([\texttt{0x82}] \,\|\, \mathsf{I2LEOSP}_{256}(\mathsf{ak}) \,\|\, \mathsf{I2LEOSP}_{256}(\mathsf{nk}))$$

(§"Viewing keys"), and the diversified addresses (§"Diversified addresses"). These constructions are the $\mathsf{use_qsk} = \mathsf{false}$ branch (*Ironwood Guide*, §"The spending key and the spend-side secrets", Remark "Scope of the key derivations"; protocol specification, §"Orchard Key Components"). They are cited, not restated; this subsection supplies $\mathsf{sk} := \mathsf{sk}_{\mathsf{acct}}$ and the $\mathsf{use_qsk} = \mathsf{true}$ branch.

Lemma 2.9 (Account keys as uniform spending keys)**.** *Let R be a result of the* Ironwood Guide *about a key generated with* $\mathsf{use_qsk} = \mathsf{false}$*, stated as a bound on the probability of a win event W in a game G whose challenger receives the output* sk *of key generation* (Ironwood Guide, *§"The spending key and the spend-side secrets", Definition "Spending key and key generation"*)*, runs in time polynomial given* sk*, and decides W. Let G_{acct} be G with* $\mathsf{sk} := \mathsf{sk}_{\mathsf{acct}}$ *at a fixed account path, the adversary $\mathcal{A}$ also receiving* $\mathsf{SeedFP}(S)$ *and the spending keys at a fixed finite set of other account paths, and let*

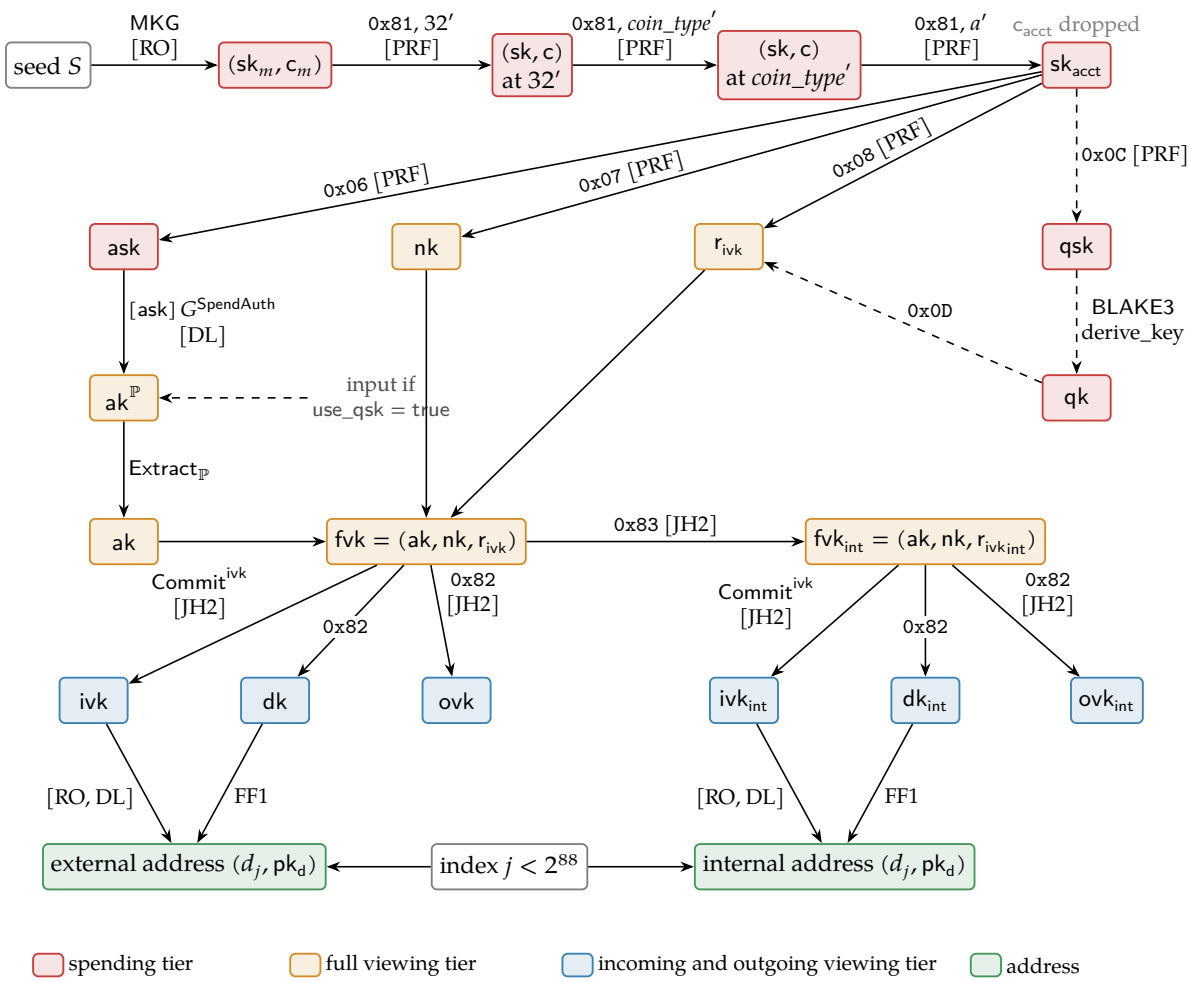


Figure 1: Derivation of the keys of one account, from the seed S through the master key and the hardened levels 32′, *coin_type*′ and a' to the account spending key $\mathsf{sk}_{\mathsf{acct}}$, whose chain code is dropped; then the key components, the full viewing key fvk, the internal full viewing key $\mathsf{fvk}_{\mathsf{int}}$, the incoming and outgoing viewing keys of each scope, and the addresses at each index j. Solid edges are those of a key with use_qsk = false; dashed edges are the use_qsk = true sources of $\mathsf{ak}^{\mathbb{P}}$, an input, and of $\mathsf{r}_{\mathsf{ivk}}$, through qsk and qk. Each derivation edge carries its expansion byte or operation; unlabelled edges are projections into a tuple or inputs. Each one-way edge also carries the assumption under which it is one-way: PRF, the *Ironwood Guide*'s Assumption "Pseudorandomness of PRF expansion" (§"Pseudorandom functions and field reductions"); RO, a random oracle (BLAKE2b under `ZcashIP32Orchard` for MKG, GroupHash for the diversified base); DL, its Assumption "Discrete logarithms on Pallas"; JH2, the joint hiding assumption of both scopes stated in §2.5. Colours give the lowest key tier that computes a node. The separations the figure depicts are proved in §2.5.

V be the event that $\mathsf{sk}_{\mathsf{acct}}$ passes the discard conditions of that Definition (made exact in §2.6). Let $\mathcal{B}$ be the adversary of G that samples those extra inputs as independent uniform 32-byte strings, answers the random-oracle queries of $\mathcal{A}$ by lazy sampling, and runs $\mathcal{A}$. Then

$$|\Pr[W \text{ in } G_{\mathsf{acct}} \mid V] - \Pr[\mathcal{B} \text{ wins } G]| \leq \frac{2\delta}{1-\rho-\delta},$$

where δ is the advantage bound of Lemma 2.5 for the distinguisher of the proof and ρ is the probability that a single draw from a uniform sk is discarded, negligible by the Ironwood Guide*'s Lemma "Rejection in key generation" (§"Viewing keys"); the bound assumes $\rho + \delta < 1$. Every key-side result of the* Ironwood Guide *therefore holds for $\mathsf{sk}_{\mathsf{acct}}$ up to this loss.*

Proof. The key $\mathsf{sk}_{\mathsf{acct}}$ is only computationally close to uniform, so the statistical-distance clause of the *Ironwood Guide*'s Lemma "Independence of domain-separated expansions" (§"Pseudorandom functions and field reductions") does not apply; the proof is a reduction. Both V and W are decided efficiently by the challenger. A distinguisher for Lemma 2.5 receives either the real values ($\mathsf{sk}_{\mathsf{acct}}$, the other spending keys and $\mathsf{SeedFP}(S)$) or independent uniform strings, runs the challenger and $\mathcal{A}$ on them, and outputs 1 exactly on $W \wedge V$; a second distinguisher outputs 1 exactly on V. Write $a := \Pr[W \wedge V]$ and $v := \Pr[V]$ for the real values, and a_u, v_u for the uniform ones. Then $|a - a_u| \leq \delta$ and $|v - v_u| \leq \delta$, with $v_u = 1 - \rho$. With uniform inputs, the extra inputs are independent of the key and distributed as $\mathcal{B}$ samples them. Key generation outputs a single uniform draw conditioned on acceptance, so $\Pr[\mathcal{B} \text{ wins } G] = a_u/v_u$. Hence

$$\left|\frac{a}{v} - \frac{a_u}{v_u}\right| \leq \frac{|a - a_u|}{v} + \frac{a_u}{v_u} \cdot \frac{|v_u - v|}{v} \leq \frac{2\delta}{v} \leq \frac{2\delta}{1-\rho-\delta},$$

using $a_u \leq v_u$ and $v \geq v_u - \delta$. □

Construction 2.10 (Key components under use_qsk)**.** The flag $\mathsf{use_qsk} \in \{\mathsf{false}, \mathsf{true}\}$ is chosen at key generation. For a spending key sk, in both branches

$$\mathsf{nk} := \mathsf{ToBase}(\mathsf{PRFexpand}_{\mathsf{sk}}(\mathtt{[0x07]})).$$

1. If $\mathsf{use_qsk} = \mathsf{false}$: ask, $\mathsf{ak}^{\mathbb{P}}$, ak and $\mathsf{r}_{\mathsf{ivk}}$ are those of the *Ironwood Guide*, and $\mathsf{qsk} = \mathsf{qk} = \bot$.
2. If $\mathsf{use_qsk} = \mathsf{true}$: a spend validating key $\mathsf{ak}^{\mathbb{P}} \in \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}}) \setminus \{\mathcal{O}\}$ with even y-coordinate, that is, with bit 255 of its star encoding equal to 0, is an input, obtained by a method outside this construction, for example a directly generated ask with $\mathsf{ak}^{\mathbb{P}} = [\mathsf{ask}]\, G^{\mathsf{SpendAuth}}$, or FROST distributed key generation. Then

$$\begin{aligned}
\mathsf{ak} &:= \mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}}),\\
\mathsf{qsk} &:= \mathsf{truncate}_{32}(\mathsf{PRFexpand}_{\mathsf{sk}}(\mathtt{[0x0C]})),\\
\mathsf{qk} &:= \mathsf{BLAKE3.derive_key}(\mathtt{Zcash\ ZIP\ 2005\ qk\text{-}derivation\ v1}, \mathsf{qsk}, 32),\\
\mathsf{r}_{\mathsf{ivk}} &:= \mathsf{ToScalar}(\mathsf{PRFexpand}_{\mathsf{qk}}(\mathtt{[0x0D]} \,\|\, \mathsf{I2LEOSP}_{256}(\mathsf{ak}) \,\|\, \mathsf{I2LEOSP}_{256}(\mathsf{nk}))),
\end{aligned}$$

   where $\mathsf{truncate}_{32}$ keeps the first 32 bytes and $\mathsf{BLAKE3.derive_key}(\textit{context}, \textit{material}, \textit{length})$ is the function that ZIP 2005 specifies, cited to it and not constructed; no property of it is assumed.

The components ivk, dk, ovk and every address are computed from $(\mathsf{ak}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$ by the *Ironwood Guide*'s constructions in both branches, so they depend on $\mathsf{use_qsk}$ (ZIP 2005, "Changes to the

Protocol Specification", §4.2.3 "Orchard Key Components", and "Usage with FROST"). No edition of the protocol specification carries this branch.

Wallet rules. A wallet MUST generate every key of one ZIP 32 account with the same value of use_qsk (ZIP 326, "Wallet key-generation restrictions"). The rule of the same section on Orchard-pool funds and use_qsk = true keys is stated once, with routing (§11.2). The value of use_qsk is not part of the recovery information, so a wallet restoring from a seed scans with both values (ZIP 326, "Scanning when restoring from seed"; §6.2).

Class (Definition 1.1): how an account whose keys are to be recoverable from the seed with use_qsk = true derives $\mathsf{ak}^{\mathbb{P}}$ from $\mathsf{sk}_{\mathsf{acct}}$ is fixed by no ZIP; it is designed but unspecified.

Remark 2.11 (Scope of the key results). Lemma 2.9 and every later key-side result of this volume (separations, hiding, validity bounds) is claimed for use_qsk = false only. For use_qsk = true the trapdoor $\mathsf{r}_{\mathsf{ivk}}$ is an expansion keyed by qk, and a result would need BLAKE3 in its key-derivation mode modelled as a PRF, which is not assumed, and a hypothesis on the source of $\mathsf{ak}^{\mathbb{P}}$. Constructive statements, which state what is computed from what, hold in both branches.

Proposition 2.12 (Key recoverability)**.** *For* use_qsk = false*, every spending, full viewing, incoming viewing and outgoing viewing key and every address of an account is a deterministic function of the seed, the coin type, the account index, the scope and the diversifier index; the seed with the account index recovers all of them. For* use_qsk = true *the same holds of* nk, qsk *and* qk*, and, given* ak*, of* $\mathsf{r}_{\mathsf{ivk}}$, ivk, dk, ovk *and every address; recovery needs* ak *in addition, and, to spend, the key material that signs under* ak*, held collectively by the signers under FROST. In both cases the wallet must know the account's value of* use_qsk *or try both. The function is partial: it is undefined at an account index whose key is invalid* (*§2.6*)*.*

Proof. Composition of Constructions 2.3 and 2.4 and Definition 2.7, which map the seed to $\mathsf{sk}_{\mathsf{acct}}$; of Construction 2.10, which maps $\mathsf{sk}_{\mathsf{acct}}$ and use_qsk, together with $\mathsf{ak}^{\mathbb{P}}$ when use_qsk = true, to $(\mathsf{ak}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$; and of the *Ironwood Guide*'s constructions from $(\mathsf{ak}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$ (§"Viewing keys" and §"Diversified addresses"). The keys of the internal scope are a function of fvk (§2.5), and the composition only adds that function. ZIP 2005, "Changes to the Protocol Specification", §4.2.3 "Orchard Key Components", notes, and "Usage with FROST" state the same requirements for recovery. □

Construction 2.13 (Full viewing key fingerprint and tag)**.** For a full viewing key $(\mathsf{ak}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$ of either branch,

$$\mathsf{FVFP}(\mathsf{ak}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}}) := \text{BLAKE2b-256}(\texttt{ZcashOrchardFVFP}, \mathsf{I2LEOSP}_{256}(\mathsf{ak}) \parallel \mathsf{I2LEOSP}_{256}(\mathsf{nk}) \parallel \mathsf{I2LEOSP}_{256}(\mathsf{r}_{\mathsf{ivk}})),$$

over the 96-byte string shown, which is the raw encoding of an Orchard full viewing key (protocol specification, §"Orchard Raw Full Viewing Keys"). It MAY be used to identify the key uniquely; its uses rely on collision resistance, cited as for Construction 2.2; Lemma 2.24 models it as a random oracle. The *tag* is $\mathsf{FVFP}[0..4)$, its first 4 bytes; it serves key lookup and MUST NOT be assumed unique, and nothing is claimed of it (ZIP 32, "Orchard Full Viewing Key Fingerprints and Tags"). In ZIP 32's encoding of an extended spending key, a key names its parent by this tag, the master key's parent tag being four zero bytes (ZIP 32, "Orchard extended spending keys").

2.5 Internal keys: change and auto-shielding

With no change level, the separation between externally visible receipt and wallet-internal operations runs below the account. The internal operations are change and *auto-shielding*, the automatic transfer of received transparent funds to the most recent shielded pool (ZIP 315, "Terminology"; its rules in §11.3). The derivation applies to keys at the account level (ZIP 32, "Orchard internal key derivation").

Construction 2.14 (Internal full viewing key)**.** For a full viewing key $\mathsf{fvk} = (\mathsf{ak}, \mathsf{nk}, \mathsf{r_{ivk}})$ let $K := \mathrm{LE}_{256}(\mathsf{r_{ivk}})$, written $\mathsf{I2LEBSP}_{256}(\mathsf{r_{ivk}})$ in ZIP 32, and

$$\mathsf{r_{ivk_{int}}} := \mathsf{ToScalar}(\mathsf{PRFexpand}_K([\mathtt{0x83}] \parallel \mathsf{I2LEOSP}_{256}(\mathsf{ak}) \parallel \mathsf{I2LEOSP}_{256}(\mathsf{nk}))) \in \mathbb{F}_{p_{\mathsf{Vesta}}}.$$

The *internal full viewing key* is $\mathsf{fvk_{int}} := (\mathsf{ak}, \mathsf{nk}, \mathsf{r_{ivk_{int}}})$. The internal keys are $\mathsf{ivk_{int}} := \mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r_{ivk_{int}}}}(\mathsf{ak}, \mathsf{nk})$ and $(\mathsf{dk_{int}}, \mathsf{ovk_{int}})$, the two halves of the $\mathtt{0x82}$ expansion keyed by $\mathrm{LE}_{256}(\mathsf{r_{ivk_{int}}})$ over the same message. The input of the $\mathtt{0x83}$ expansion has the key and message body of the $\mathtt{0x82}$ expansion and another lead byte. The internal expanded spending key is $(\mathsf{ask}, \mathsf{nk}, \mathsf{r_{ivk_{int}}})$; no 32-byte internal spending key exists (ZIP 32, "Orchard internal key derivation"; protocol specification, §"Orchard Key Components").

Hence $\mathsf{ak_{int}} = \mathsf{ak}$ and $\mathsf{nk_{int}} = \mathsf{nk}$, as the protocol specification, §"Orchard Key Components", notes: spend authorisation and nullifier derivation serve both scopes, and ivk, dk and ovk are derived again from $\mathsf{r_{ivk_{int}}}$. The construction does not force the keys of the two scopes to differ; their equality is an event of negligible probability (Proposition 2.17), not an excluded case. For Sapling, ZIP 32 keeps the spend authorising key and derives again the key that enters the proof and the nullifier, the outgoing viewing key and the diversifier key, so the two scopes share spend authority and differ in the key that enters the proof and the nullifier (ZIP 32, "Sapling internal key derivation"); this is cited, not constructed. Payments are sent from an account and a scope, not from an address: no outgoing viewing key per diversified address exists (ZIP 32, "Orchard diversifier and OVK derivation"; ZIP 316, "Usage of Outgoing Viewing Keys").

Proposition 2.15 (Capability of a full viewing key)**.** *From* fvk *alone one computes, for both scopes,* ivk, dk, ovk *and the address at every diversifier index* $j \in \{0, \ldots, 2^{88} - 1\}$. *For a key with* $\mathsf{use_qsk} = \mathsf{false}$, *no efficient algorithm given* fvk *outputs* ask, *and no efficient algorithm given the keys of one account outputs the spending key of another account, except with negligible probability. Recovering the index and scope of a given address is stated with the index (§3.1).*

Proof. Computation: $\mathsf{fvk_{int}}$ is a function of fvk (Construction 2.14), and each address is a function of $(\mathsf{dk}, \mathsf{ivk})$ and j (*Ironwood Guide*, §"Diversified addresses", Construction "Diversified address"). No ask: the *Ironwood Guide*, §"Key capabilities and scope", Proposition "Capability separation", part (a), under its Assumptions "Pseudorandomness of PRF expansion" and "Discrete logarithms on Pallas", transferred to $\mathsf{sk_{acct}}$ by Lemma 2.9. No other account: the spending key of another account lies at a distinct path, and only hardened derivation exists; by Lemma 2.5 the two keys are jointly indistinguishable from independent uniform strings, so an algorithm given every key of this account outputs the other key with probability at most 2^{-256} plus the advantage bound of that lemma. □

Assumption 2.16 (Joint hiding of the viewing keys of both scopes). For keys with $\mathsf{use_qsk} = \mathsf{false}$: for every $(\mathsf{ak}, \mathsf{nk}) \in \{1, \ldots, p_{\mathsf{Pallas}} - 1\} \times \mathbb{F}_{p_{\mathsf{Pallas}}}$, given to the distinguisher, and $\mathsf{r_{ivk}}$ uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$, the six keys $(\mathsf{ivk}, \mathsf{dk}, \mathsf{ovk}, \mathsf{ivk_{int}}, \mathsf{dk_{int}}, \mathsf{ovk_{int}})$, computed by the *Ironwood Guide*'s

constructions and Construction 2.14, are computationally indistinguishable from

$$(\mathsf{Commit}_r^{\mathsf{ivk}}(\mathsf{ak}, \mathsf{nk}),\ u_1,\ u_2,\ \mathsf{Commit}_{r'}^{\mathsf{ivk}}(\mathsf{ak}, \mathsf{nk}),\ u_3,\ u_4),$$

where r and r' are uniform on $\mathbb{F}_{p_{\mathsf{Vesta}}}$ and $u_1, \ldots, u_4$ are uniform 32-byte strings, all independent. It extends the *Ironwood Guide*'s Assumption "Joint hiding of the viewing keys" (§"Viewing keys"), which covers the external triple only, to the 0x83 row, whose key $\mathrm{LE}_{256}(\mathsf{r}_{\mathsf{ivk}})$ is also the $\mathsf{Commit}^{\mathsf{ivk}}$ trapdoor of the external ivk and so lies outside the hypothesis of the *Ironwood Guide*'s Lemma "Independence of domain-separated expansions" (§"Pseudorandom functions and field reductions") (Remark "Hiding of ivk and the double use of $\mathsf{r}_{\mathsf{ivk}}$" (§"Viewing keys")); the PRF assumption does not give it (protocol specification, §"Orchard Key Components", note on $\mathsf{r}_{\mathsf{ivk}}$ as trapdoor and key). It is applied after Lemma 2.9, averaging over the key-generation distribution of $(\mathsf{ak}, \mathsf{nk})$.

Proposition 2.17 (Separation of scopes under trial decryption)**.** *Let an account key have* use_qsk = false*, with external and internal incoming viewing keys outside* $\{0, \perp\}$*. An output honestly addressed to an internal-scope address of the account is accepted under the account's external* ivk *only with negligible probability, under the* Ironwood Guide*'s Assumptions "Discrete logarithms on Pallas" (§"Sinsemilla: hash, commitment, and short forms") and "GroupHash as a random oracle" (§"GroupHash, domain separation, and nothing-up-my-sleeve generators"), Assumption 2.16 and the hypotheses of Lemma 2.9. Consequently an external incoming viewing key reveals no change or auto-shielding output. Since* $\mathsf{fvk}_{\mathsf{int}}$ *is computable from* fvk*, a full viewing key reveals both scopes, and the boundary between the scopes lies at the incoming viewing tier.*

Proof. Acceptance under a key ivk derives the transmission key again as $[\mathsf{ivk}]\,\mathsf{g_d}$ from the decrypted diversifier and checks the note commitment of the derived note against cmx (*Ironwood Guide*, §"Trial decryption and note acceptance", Construction "Acceptance of an Ironwood-pool output", check (iv); for an Orchard-pool output the check of the protocol specification, §"Decryption using an Incoming Viewing Key (Sapling and Orchard)"). The honest output commits to a note n with transmission key $[\mathsf{ivk}_{\mathsf{int}}]\,\mathsf{g_d}$. If the note n' derived under $\mathsf{ivk}_{\mathsf{ext}}$ differs from n, acceptance exhibits two distinct notes with one extracted commitment, which happens with negligible probability (*Ironwood Guide*, §"Note commitments", Proposition "Hiding and binding of note commitments", part (b)). If $n' = n$, then $[\mathsf{ivk}_{\mathsf{ext}}]\,\mathsf{g_d} = [\mathsf{ivk}_{\mathsf{int}}]\,\mathsf{g_d}$; since $\mathsf{g_d} \neq \mathcal{O}$ in a group of prime order and both scalars lie in $\{1, \ldots, p_{\mathsf{Pallas}} - 1\}$ with $p_{\mathsf{Pallas}} < p_{\mathsf{Vesta}}$, this forces $\mathsf{ivk}_{\mathsf{ext}} = \mathsf{ivk}_{\mathsf{int}}$ (*Ironwood Guide*, §"Viewing keys"). By Lemma 2.9 and Assumption 2.16 the pair is replaced by $(\mathsf{Commit}_r^{\mathsf{ivk}}(\mathsf{ak}, \mathsf{nk}), \mathsf{Commit}_{r'}^{\mathsf{ivk}}(\mathsf{ak}, \mathsf{nk}))$ with independent uniform r and r', at the cost of the distinguishing advantage, since equality is efficiently decidable. By the *Ironwood Guide*'s Lemma "Distribution of ivk" (§"Viewing keys"), the second value equals any fixed value with probability at most $2/p_{\mathsf{Vesta}}$, independently of the first, so the two are equal with probability at most $2/p_{\mathsf{Vesta}}$. □

Construction 2.18 (Transparent outgoing viewing keys)**.** A P2PKH (pay-to-public-key-hash) output pays to the hash of one secp256k1 public key (*Crypto Guide*, §"The transparent layer's schemes: ECDSA and BIP-340 over secp256k1"). P2PKH keys have no outgoing viewing key, yet auto-shielding outputs need one. For the BIP 32 account-level extended public key $(\mathsf{c}, \mathsf{pk})$ of a transparent account, at path $m/44'/\mathit{coin_type}'/\mathit{account}'$ (BIP 44), let

$$I_{\mathsf{ovk}} := \mathsf{PRFexpand}_{\mathsf{c}}([\mathtt{0xD0}] \,\|\, \mathsf{ser}_P(\mathsf{pk})),$$

with the chain code c used as a 32-byte key and $\mathsf{ser}_P(\mathsf{pk})$ the 33-byte compressed encoding of BIP 32; then $\mathsf{ovk}_{\mathsf{external}} := I_{\mathsf{ovk}}[0..32)$ and $\mathsf{ovk}_{\mathsf{internal}} := I_{\mathsf{ovk}}[32..64)$ (ZIP 316, "Deriving Internal Keys"). The external transparent incoming viewing key is the extended public key of the non-hardened child 0 of $(\mathsf{c}, \mathsf{pk})$ and the internal one that of the non-hardened child 1 (ZIP 316, "Deriving a UIVK from a UFVK"; BIP 44), each obtained by BIP 32 public-to-public derivation, which is HMAC-SHA512 keyed by c (BIP 32, "Public parent key → public child key"; *Crypto Guide*, §"PRFs from hash functions: length extension and keyed BLAKE2").

Assumption 2.19 (Hiding of the transparent account key). For an honestly generated account-level extended public key $(\mathsf{c}, \mathsf{pk})$, the extended public keys of its non-hardened children 0 and 1 and the two halves of I_{ovk} are jointly computationally indistinguishable from two independently generated extended public keys, each a uniform 32-byte chain code with the public key of a uniform secret key, and two independent uniform 32-byte strings. The chain code c keys both HMAC-SHA512 in BIP 32 non-hardened derivation and $\mathsf{PRFexpand}$ in the $\mathtt{0xD0}$ row, so this is not an instance of the *Ironwood Guide*'s Lemma "Independence of domain-separated expansions" (§"Pseudorandom functions and field reductions"); it is assumed.

Proposition 2.20 (Capability separation of internal keys)**.** *The following hold, where "derives" means outputs the key with non-negligible probability.*

(*i*) *A holder of a full viewing key derives the external and internal incoming viewing keys and the external and internal outgoing viewing keys.*

(*ii*) *A holder of the external incoming viewing key derives neither the internal incoming viewing key nor any outgoing viewing key.*

(*iii*) *A holder of the external outgoing viewing key derives neither the internal outgoing viewing key nor any incoming viewing key.*

For Orchard the statements hold for keys with $\mathsf{use_qsk} = \mathsf{false}$ *under Lemma 2.9 and Assumption 2.16; for Sapling they are the properties that ZIP 316, "Deriving Internal Keys", states for the construction of ZIP 32, "Sapling internal key derivation", and are cited, not proved; for transparent P2PKH keys under Assumption 2.19 (ZIP 316, "Deriving Internal Keys").*

Proof. (i) Constructive: Construction 2.14 and the *Ironwood Guide*'s constructions; for transparent keys, Construction 2.18 from $(\mathsf{c}, \mathsf{pk})$.

(ii) and (iii), Orchard: by Lemma 2.9 and Assumption 2.16 the six keys are replaced by the ideal distribution, at the cost of the distinguishing advantage, since success is decided by comparing the output with the key. In the ideal distribution every key not computed from the held one is independent of it: either a uniform 32-byte string, output with probability 2^{-256}, or a $\mathsf{Commit}^{\mathsf{ivk}}$ value under an independent trapdoor, output with probability at most $2/p_{\mathsf{Vesta}}$ (*Ironwood Guide*, §"Viewing keys", Lemma "Distribution of ivk"). Transparent: the same argument in the ideal distribution of Assumption 2.19, where an independently generated extended public key has a uniform chain code, output with probability 2^{-256}. Sapling: cited to ZIP 316, "Deriving Internal Keys", which asserts them from the one-wayness of $\mathsf{PRFexpand}$ and $\mathsf{CRH}^{\mathsf{ivk}}$; not proved here. □

2.6 Valid account keys

Definition 2.21 (Valid account key)**.** For use_qsk = false, an account spending key $\mathsf{sk}_{\mathsf{acct}}$ is *valid* if and only if $\mathsf{ask} \neq 0$, $\mathsf{ivk} \notin \{0, \bot\}$ and $\mathsf{ivk}_{\mathsf{int}} \notin \{0, \bot\}$. The first two conditions are discard conditions of the *Ironwood Guide*, §"The spending key and the spend-side secrets", Definition "Spending key and key generation"; the third is the specification's condition on the internal key of Construction 2.14, which that Definition names without constructing (protocol specification, §"Orchard Key Components"; ZIP 32, "Orchard internal key derivation"). ZIP 2005 amends the derivation for use_qsk = true ("Changes to the Protocol Specification", §4.2.3), for which this volume states no validity condition. Only keys a wallet uses need be valid; intermediate path nodes are never expanded into components.

Remark 2.22 (Derivation failure)*.* ZIP 32 notes that a child spending key yields an invalid external or internal full viewing key "with small probability" and prescribes no wallet behaviour ("Orchard child key derivation"). The specification's instruction to discard the key and repeat with a new sk applies to a freshly random key, not to a fixed path, where nothing is drawn again: an account index whose key is invalid has no key. Class (Definition 1.1): the wallet's handling of such an index is an open problem; Proposition 2.23 bounds its probability.

Proposition 2.23 (Probability of an invalid account key)**.** *Model the Sinsemilla generators, the initial point $Q := Q(\texttt{z.cash:Orchard-CommitIvk-M})$ and the points $S(j)$ for $0 \leq j < 2^{10}$, as independent uniform points of the Pallas group, independent of every other quantity* (Ironwood Guide, *§"GroupHash, domain separation, and nothing-up-my-sleeve generators", Assumption "GroupHash as a random oracle"*)*. Let*

$$\varepsilon := \frac{3\lceil 2^{512}/p_{\mathsf{Vesta}}\rceil}{2^{512}} + \frac{408}{p_{\mathsf{Vesta}}} < \frac{412}{p_{\mathsf{Vesta}}} = 2^{-245.31}.$$

For use_qsk = false *and a fixed account path,* $\Pr[\mathsf{sk}_{\mathsf{acct}}$ *is invalid*$] \leq \varepsilon + \delta$*, where δ is the sum of the advantages of efficient reductions, each deciding validity, against Lemma 2.5, the* Ironwood Guide*'s Assumption "Pseudorandomness of PRF expansion"* (*§"Pseudorandom functions and field reductions"*) (*rows* 0x06, 0x07 *and* 0x08) *and Assumption 2.16* (*row* 0x83)*, plus the statistical distance of* ToScalar *from uniform, below* 2^{-257} (Ironwood Guide, *§"Pseudorandom functions and field reductions"*)*. For the n accounts a wallet uses,* $\Pr[$*some key is invalid*$] \leq n\varepsilon + \delta_n$*, where δ_n is the corresponding sum for the n paths jointly; the union runs over the keys used, not over path nodes. The bound averages over generators modelled as random and is not a frequency for the fixed deployed generators.*

Proof. Replace $\mathsf{sk}_{\mathsf{acct}}$ by a uniform string (Lemma 2.5); the 512-bit outputs of rows 0x06, 0x07 and 0x08 by independent uniform strings (the *Ironwood Guide*'s Lemma "Independence of domain-separated expansions" (§"Pseudorandom functions and field reductions"), first conclusion); $\mathsf{r}_{\mathsf{ivk}}$ by a uniform element of $\mathbb{F}_{p_{\mathsf{Vesta}}}$ (the reduction distance); and, on the event $\mathsf{ask} \neq 0$, the pair $(\mathsf{ivk}, \mathsf{ivk}_{\mathsf{int}})$ by $(\mathsf{Commit}^{\mathsf{ivk}}_{r}(\mathsf{ak}, \mathsf{nk}), \mathsf{Commit}^{\mathsf{ivk}}_{r'}(\mathsf{ak}, \mathsf{nk}))$ with independent uniform r and r' (Assumption 2.16). Each replacement changes the probability of invalidity by at most one term of δ, since validity is efficiently decidable. Then, by the union bound (*Math Guide*, §"The union bound and a birthday calculation", Theorem "Union bound / Boole's inequality"), the key is invalid only on one of the following events.

(a) $\mathsf{ask} = 0$. The sign normalisation preserves 0, so this happens exactly when a uniform 512-bit integer is divisible by p_{Vesta}, with probability $\lceil 2^{512}/p_{\mathsf{Vesta}}\rceil/2^{512}$ (*Math Guide*, §"Uniform sampling and the bias of modular reduction").

(b) The Sinsemilla hash point $\hat{M}$ of the message $\mathrm{LE}_{255}(\mathsf{ak}) \,\|\, \mathrm{LE}_{255}(\mathsf{nk})$ is $\bot$. The message has 510 bits, 51 chunks of 10 bits, so the hash performs 102 incomplete additions, two per round,

$\mathsf{Acc}_i := (\mathsf{Acc}_{i-1} \dot{+} S(m_i)) \dot{+} \mathsf{Acc}_{i-1}$ (protocol specification, §"Sinsemilla Hash Function"; *Ironwood Guide*, §"Sinsemilla: hash, commitment, and short forms", Construction "Sinsemilla hash on Pallas"). An incomplete addition $A \dot{+} B$ of points is exceptional exactly when $A = \mathcal{O}$, $B = \mathcal{O}$, $A = B$ or $A = -B$. At the first exceptional addition every accumulator so far equals its formal value, an integer combination of Q and the $S(j)$, so one of four linear relations among the generators holds. Each relation is nontrivial: it either states that the single generator $S(m_i)$ is $\mathcal{O}$, or gives Q a coefficient 2^t or 2^{t+1}, non-zero modulo the odd prime p_{Vesta}. The message depends only on $(\mathsf{ask}, \mathsf{nk})$ and $G^{\mathsf{SpendAuth}}$, which are independent of Q and the $S(j)$, so each relation holds with probability $1/p_{\mathsf{Vesta}}$, and the 102 additions cost at most $408/p_{\mathsf{Vesta}}$. Both scopes commit to the same message, so this event is counted once.

(c) $\mathsf{ivk} = 0$ with $\hat{M} \neq \perp$. The addition of $[r]H_D$ is complete, and $\mathsf{ivk} = 0$ exactly when $\hat{M} + [r]H_D = \mathcal{O}$ (*Ironwood Guide*, §"Fields, groups, and encodings", Lemma "Extract vanishes only at the identity"), which, since $H_D \neq \mathcal{O}$, holds for a single value of r: probability $1/p_{\mathsf{Vesta}} \leq \lceil 2^{512}/p_{\mathsf{Vesta}} \rceil / 2^{512}$.

(d) $\mathsf{ivk}_{\mathsf{int}} = 0$ with $\hat{M} \neq \perp$: likewise, under the independent r', with the same bound.

The sum is ε. Negligibility alone is the *Ironwood Guide*'s Lemma "Rejection in key generation" (§"Viewing keys"). For n accounts, apply the replacements to the n paths jointly and the union bound over the n keys; path nodes that are not account keys are never expanded into components and do not enter the union. □

2.7 The PRFexpand lead-byte registry

Domain separation of PRFexpand is per key: under one key, distinct lead bytes give jointly pseudorandom outputs (*Ironwood Guide*, §"Pseudorandom functions and field reductions", Lemma "Independence of domain-separated expansions"), while expansions under different keys may reuse a byte. Byte conventions: the key that ZIP 32 writes $\mathsf{I2LEBSP}_{256}(\mathsf{r}_{\mathsf{ivk}})$ is $\mathrm{LE}_{256}(\mathsf{r}_{\mathsf{ivk}})$; the bytes of ak are $\mathsf{I2LEOSP}_{256}(\mathsf{ak})$ of the field element $\mathsf{ak} = \mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}})$, which for the sign-normalised $\mathsf{ak}^{\mathbb{P}}$ equals the byte form $\mathsf{LEBS2OSP}_{256}$ of its star encoding, bit 255 being 0 (*Ironwood Guide*, §"The spending key and the spend-side secrets"); the key nk enters as $\mathsf{I2LEOSP}_{256}(\mathsf{nk})$ (protocol specification, §"Orchard Key Components").

Table 2 lists the expansions and personalisations that this volume constructs. The expansions of the key components and of notes are the rows of the *Ironwood Guide*'s Construction "Domain bytes of PRF expansion" (§"Pseudorandom functions and field reductions") and are not restated. The list of expansion bytes that the protocol specification maintains, as amended by ZIP 2005, is §"Pseudo Random Functions" (ZIP 2005, "Changes to the Protocol Specification", §4.1.2); bytes this volume does not use are not listed.

Lemma 2.24 (Independence of the key-derivation domains)**.** *Under the* Ironwood Guide*'s Assumption "Pseudorandomness of PRF expansion" (§"Pseudorandom functions and field reductions"), with BLAKE2b under distinct personalisations modelled as independent random oracles, the expansions of rows* 0x81 *and* 0x0C*, whose keys (a chain code of the Orchard tree, and* $\mathsf{sk}_{\mathsf{acct}}$*) are pseudorandom by Lemma 2.5 and enter no function other than* PRFexpand*, and the three personalised BLAKE2b functions of Table 2 are jointly indistinguishable from independent random functions. The lemma excludes the other rows, each outside the hypothesis of the* Ironwood Guide*'s Lemma "Independence of domain-separated expansions" (§"Pseudorandom functions and field reductions"): the row* 0x83*, whose key is also the* $\mathsf{Commit}^{\mathsf{ivk}}$ *trapdoor, is covered by Assumption 2.16; the row* 0xD0*, whose key also keys HMAC-SHA512, by Assumption 2.19; the row* 0xDD*, keyed by the BLAKE3 output* qk *of which nothing is assumed (Remark 2.11), carries no claim.*

Byte	Key	Message	Output	Constructed in
0x81	$\mathsf{c_{par}}$	$\mathsf{sk_{par}} \parallel \mathsf{I2LEOSP}_{32}(i)$	$(\mathsf{sk}_i, \mathsf{c}_i)$	Constr. 2.4
0x83	$\mathrm{LE}_{256}(\mathsf{r_{ivk}})$	$\mathsf{I2LEOSP}_{256}(\mathsf{ak}) \parallel \mathsf{I2LEOSP}_{256}(\mathsf{nk})$	$\mathsf{r_{ivk_{int}}}$ (ToScalar)	Constr. 2.14
0x0C	sk	ε (empty)	qsk ($\mathsf{truncate}_{32}$)	Constr. 2.10
0x0D	qk	$\mathsf{I2LEOSP}_{256}(\mathsf{ak}) \parallel \mathsf{I2LEOSP}_{256}(\mathsf{nk})$	$\mathsf{r_{ivk}}$ if use_qsk = true (ToScalar)	Constr. 2.10
0xD0	transparent chain code c	$\mathsf{ser}_P(\mathsf{pk})$	$\mathsf{ovk_{external}} \parallel \mathsf{ovk_{internal}}$	Constr. 2.18
Personalisation		**Function**	**Output**	**Constructed in**
ZcashIP32Orchard		BLAKE2b-512 of S	$(\mathsf{sk}_m, \mathsf{c}_m)$	Constr. 2.3
Zcash_HD_Seed_FP		BLAKE2b-256 of $[\mathrm{length}(S)] \parallel S$	$\mathsf{SeedFP}(S)$	Constr. 2.2
ZcashOrchardFVFP		BLAKE2b-256 of the 96-byte encoding of fvk	FVFP	Constr. 2.13

Table 2: Domains of key derivation. Upper part: each expansion $\mathsf{PRFexpand}_{\mathrm{key}}(b \parallel \mathrm{message})$ that this volume constructs, with its lead byte b, key, message and output; the rows 0x0C and 0x0D are those of ZIP 2005. Lower part: the 16-byte BLAKE2b personalisations of key derivation, with the function, output and constructing environment of each.

Proof. For each key, the *Ironwood Guide*'s Lemma "Independence of domain-separated expansions" (§"Pseudorandom functions and field reductions"), applied after the hybrid of Lemma 2.5 has made the keys uniform; this is a reduction, as in Lemma 2.9, not the statistical-distance clause of that lemma. For the three personalisations, the *Crypto Guide*'s Proposition "Domain separation yields independent oracles" (§"Domain separation and personalisation"): they are pairwise distinct 16-byte strings, and the two BLAKE2b-256 uses and the BLAKE2b-512 use differ also in the output length, a parameter of BLAKE2b. The proof of Lemma 2.5 uses the same two citations directly. □

3 Addresses and viewing keys (ZIP 316)

This section passes from the keys of an account to the addresses and viewing keys that a wallet hands out. It defines the diversifier index and the recovery of an index from an address, the raw encodings of receivers and keys, the unified container with its jumbled string encoding and its viewing-key form, the unified address at an index of a viewing key, the extensions of Revision 2 of ZIP 316, and the recipient addresses a sender handles.

3.1 Diversifier indices

The address layer meets two requirements.

(R1) Each payee receives an address unlinkable to every other address of the same holder, while all of them are spendable by one key and recoverable from one seed (protocol specification, §"$\mathsf{DiversifyHash}^{\mathsf{Sapling}}$ and $\mathsf{DiversifyHash}^{\mathsf{Orchard}}$ Hash Functions", security requirements; ZIP 32; ZIP 316, "Rationale for address derivation"). Each shielded protocol meets (R1) with diversified addresses.

(R2) Transparent, Sapling and Orchard receivers coexist, each with its own format. A recipient publishes one string, from which the sender selects the most preferred receiver type it supports, and whose characters a sender can check (ZIP 316, "Motivation" and "Require-

ments"). ZIP 316 meets (R2) above the protocols, with unified containers (§3.3) and their jumbled string encoding (§3.4).

The *Ironwood Guide* constructs the Orchard key tiers, the diversifier map and the diversified base (§"Viewing keys" and §"Diversified addresses"). This subsection adds the diversifier index, shared by Sapling and Orchard and used by transparent derivation.

The Orchard address at index j of an incoming viewing key $(\mathsf{dk}, \mathsf{ivk})$ is that of the *Ironwood Guide*, §"Diversified addresses", Construction "Diversified address", cited and not re-derived: $d := \mathsf{FF1\text{-}AES256}_{\mathsf{dk}}(\mathrm{LE}_{88}(j))$, $\mathsf{g_d} := \mathsf{DiversifyHash}^{\mathsf{Orchard}}(d)$, which is $\mathsf{GroupHash}(\mathtt{z.cash:Orchard\text{-}gd}, \mathsf{LEBS2OSP}_{88}(d))$ unless that is the identity and the fallback point $\mathsf{GroupHash}(\mathtt{z.cash:Orchard\text{-}gd}, \varepsilon)$ otherwise, and $\mathsf{pk_d} := [\mathsf{ivk}]\,\mathsf{g_d}$; the address is $(d, \mathsf{pk_d})$. For each scope $s \in \{\mathsf{ext}, \mathsf{int}\}$ of an account (Construction 2.14) the pair $(\mathsf{dk}_s, \mathsf{ovk}_s)$ is the first and the second 32 bytes of $\mathsf{PRFexpand}_{\mathrm{LE}_{256}(\mathsf{r_{ivk}}_s)}([\mathtt{0x82}] \,\|\, \mathsf{I2LEOSP}_{256}(\mathsf{ak}) \,\|\, \mathsf{I2LEOSP}_{256}(\mathsf{nk}))$ (*Ironwood Guide*, §"Viewing keys"), and ivk_s is the incoming viewing key of the scope. The address at index 0 is the *default address* of the scope (protocol specification, §"Orchard Key Components"; ZIP 32, "Orchard diversifier and OVK derivation").

Definition 3.1 (Diversifier index)**.** For a 32-byte diversifier key dk and an index $j \in \{0, \ldots, 2^{88} - 1\}$, the *diversifier at index j* is

$$d_j := \mathsf{FF1\text{-}AES256.Encrypt}(\mathsf{dk}, \mathtt{""}, \mathrm{LE}_{88}(j)) \in \{0,1\}^{88},$$

the FF1 mode of NIST SP 800-38G over AES-256 with key dk, radix 2, length 88 and the empty tweak (*Crypto Guide*, §"Format-preserving encryption and FF1", Construction "The FF1 mode of NIST SP 800-38G"; protocol specification, §"Pseudo Random Permutations"). The map is the same in both shielded protocols; only the key differs.

1. For Orchard, $\mathsf{dk} = \mathsf{dk}_s$ of the scope s, as above.
2. For Sapling, dk is the diversifier-key component of the account's Sapling extended spending key: $\mathsf{dk}_m = \mathsf{truncate}_{32}(\mathsf{PRFexpand}_{\mathsf{sk}_m}([\mathtt{0x10}]))$ at the master key and $\mathsf{dk}_i = \mathsf{truncate}_{32}(\mathsf{PRFexpand}_{I_L}([\mathtt{0x16}] \,\|\, \mathsf{dk_{par}}))$ for a hardened child, with I_L the left half of the child's derivation output (ZIP 32, "Sapling master key generation" and "Sapling child key derivation"); Sapling key derivation is cited and not constructed.

The diversifier d_j, and the index j, are *valid* for a protocol if and only if that protocol's $\mathsf{DiversifyHash}$ does not return $\perp$ at d_j.

3. Orchard: $\mathsf{DiversifyHash}^{\mathsf{Orchard}}$ is total, so every index is valid, a scope has exactly 2^{88} addresses, and the default diversifier is d_0 (ZIP 32, "Orchard diversifier and OVK derivation").
4. Sapling: $\mathsf{DiversifyHash}^{\mathsf{Sapling}}(d) := \mathsf{GroupHash}^{\mathbb{J}(r)*}_{\mathsf{URS}}(\mathtt{Zcash_gd}, \mathsf{LEBS2OSP}_{88}(d))$, a hash into the points of order $r_{\mathbb{J}}$ of the Jubjub curve that returns $\perp$ on some inputs (protocol specification, §"$\mathsf{DiversifyHash}^{\mathsf{Sapling}}$ and $\mathsf{DiversifyHash}^{\mathsf{Orchard}}$ Hash Functions", §"Jubjub" and §"Group Hash into Jubjub", cited and not constructed). Approximately half of all indices are valid (ZIP 32, "Sapling diversifier derivation"); the probability under the specification's random-oracle model is derived in §3.6. The default Sapling diversifier is d_j for the least valid j (same section).

Totality of $\mathsf{DiversifyHash}^{\mathsf{Orchard}}$ removes the search for a valid index. The Sapling failure rate matters when one index must serve several protocols at once (§3.6).

Index-choice rules, specified (protocol specification, §"Orchard Key Components" and

§"Sapling Key Components", notes). Orchard diversifier indices SHOULD NOT be chosen at random; ZIP 32 specifies their use in hierarchical deterministic wallets. Address generators MAY encode information in the diversifier index, recoverable by a holder of dk (Lemma 3.3). For Sapling, generators MAY encode information in the diversifier itself, and it is RECOMMENDED to encode it in the ZIP 32 diversifier index instead, so that only a holder of dk reads it.

Proposition 3.2 (Distinct diversifiers). (*a*) *For every diversifier key* dk *the map* $j \mapsto d_j$ *is a bijection from* $\{0, \dots, 2^{88} - 1\}$ *onto* $\{0,1\}^{88}$. *In particular distinct indices give distinct diversifiers, and a wallet issues the indices* $0, 1, 2, \dots$ *over the whole index space without a repeated diversifier.*

(*b*) *If instead* k *diversifiers are drawn independently and uniformly from* $\{0,1\}^{88}$, *the probability* P *that two coincide satisfies*

$$1 - e^{-k(k-1)/2^{89}} \le P \le \frac{k(k-1)}{2^{89}},$$

so a collision becomes likely once k *is of order* $2^{44} = \sqrt{2^{88}}$; *at* $k = 2^{44}$ *the bounds are* 0.393 *and* 0.500.

Proof. (a) The map LE_{88} is a bijection from $\{0, \dots, 2^{88} - 1\}$ onto $\{0,1\}^{88}$, and $\mathsf{FF1\text{-}AES256}$ under a fixed key and tweak is a bijection of $\{0,1\}^{88}$ (*Crypto Guide*, §"Format-preserving encryption and FF1", Proposition "FF1 is a keyed bijection on $\mathcal{X}$"); the composition is a bijection. (b) The *Math Guide*, §"The union bound and a birthday calculation", Proposition "Birthday bound", with $N = 2^{88}$. □

The specification notes that an 88-bit diversifier is not unguessable at the 128-bit security level and that randomly chosen diversifiers collide near 2^{44} choices; the keyed permutation removes the collisions, provided indices do not repeat (protocol specification, §"$\mathsf{DiversifyHash}^{\mathsf{Sapling}}$ and $\mathsf{DiversifyHash}^{\mathsf{Orchard}}$ Hash Functions", notes). One further reason for a keyed permutation is ZIP 32's: it makes the diversifier sequence of each account pseudorandom and uncorrelated with that of any other account, so issued diversifiers do not reveal how many addresses have been generated (ZIP 32, "Sapling diversifier derivation" and "Orchard diversifier and OVK derivation"). The *Ironwood Guide* supplies Assumption "FF1-AES256 as a pseudorandom permutation" (§"Diversified addresses") and the unlinkability proposition only.

In Orchard, dk_s is derived from the full viewing key of scope s, so a holder of the account's Orchard full viewing key computes dk_s for both scopes, hence the index of any diversifier of the account (Lemma 3.3; ZIP 32, "Orchard diversifier and OVK derivation"). In Sapling the same capability belongs to the extended full viewing key, of which dk is a component. A holder of the incoming viewing key $(\mathsf{dk}, \mathsf{ivk})$ of one scope has this capability for that scope only.

Lemma 3.3 (Index recovery and ownership). *Let* $(d, \mathsf{pk_d})$ *be an Orchard address and* $(\mathsf{dk}, \mathsf{ivk})$ *the incoming viewing key of some scope.*

(*i*) *The value* $j := \mathrm{LE}_{88}^{-1}(\mathsf{FF1\text{-}AES256.Decrypt}(\mathsf{dk}, \texttt{""}, d))$ *is defined for every* d *and every* dk, *and* $d_j = d$. *Decryption alone therefore proves nothing about ownership.*

(*ii*) *The address* $(d, \mathsf{pk_d})$ *is the address of* $(\mathsf{dk}, \mathsf{ivk})$ *at some index if and only if* $\mathsf{pk_d} = [\mathsf{ivk}]\,\mathsf{DiversifyHash}^{\mathsf{Orchard}}(d)$; *the index is then unique and equals* j.

(*iii*) *A holder of an account's full viewing key decides whether* $(d, \mathsf{pk_d})$ *belongs to the account, and at which scope and index, by applying* (*ii*) *with* $(\mathsf{dk_{ext}}, \mathsf{ivk_{ext}})$ *and then with* $(\mathsf{dk_{int}}, \mathsf{ivk_{int}})$. *Both*

> *tests succeed only if* $\mathsf{ivk_{ext}} = \mathsf{ivk_{int}}$*, which has negligible probability for keys with* $\mathsf{use_qsk} = \mathsf{false}$ *under Assumption 2.16 and the hypotheses of Lemma 2.9.*
>
> *The Sapling case of (i) and (ii) is the same with* $\mathsf{DiversifyHash}^{\mathsf{Sapling}}$*, the test of (ii) failing when it returns* $\perp$*. For (iii), both tests succeed only if the two Sapling incoming viewing keys are equal; this volume does not bound that probability for Sapling.*

Proof. (i) FF1 decryption is the two-sided inverse of encryption under every key and tweak (*Crypto Guide*, §"Format-preserving encryption and FF1", Proposition "FF1 is a keyed bijection on $\mathcal{X}$"), and LE_{88} is a bijection. (ii) If $(d, \mathsf{pk_d})$ is the address at index j', then $d = d_{j'}$, so $j' = j$ by (i) and Proposition 3.2 (a), and $\mathsf{pk_d} = [\mathsf{ivk}]\,\mathsf{DiversifyHash}^{\mathsf{Orchard}}(d)$ by the *Ironwood Guide*'s Construction "Diversified address" (§"Diversified addresses"). Conversely, if $\mathsf{pk_d} = [\mathsf{ivk}]\,\mathsf{DiversifyHash}^{\mathsf{Orchard}}(d)$, the address at index j is $(d_j, [\mathsf{ivk}]\,\mathsf{DiversifyHash}^{\mathsf{Orchard}}(d_j)) = (d, \mathsf{pk_d})$. (iii) If both tests succeed, then $[\mathsf{ivk_{ext}}]\,\mathsf{g_d} = [\mathsf{ivk_{int}}]\,\mathsf{g_d}$ with $\mathsf{g_d} \neq \mathcal{O}$ in a group of prime order p_{Vesta}, and both scalars lie in $\{1, \ldots, p_{\mathsf{Pallas}} - 1\}$ with $p_{\mathsf{Pallas}} < p_{\mathsf{Vesta}}$, so $\mathsf{ivk_{ext}} = \mathsf{ivk_{int}}$. Its probability is bounded as in the proof of Proposition 2.17: after the replacement of Assumption 2.16, at most $2/p_{\mathsf{Vesta}}$ by the *Ironwood Guide*'s Lemma "Distribution of ivk" (§"Viewing keys"), plus the distinguishing advantage, equality being efficiently decidable. □

Unlinkability of the addresses of one key is cited with its assumptions. For Orchard it is the *Ironwood Guide*, §"Diversified addresses", Proposition "Unlinkability of diversified addresses": for two keys generated independently with $\mathsf{use_qsk} = \mathsf{false}$, an adversary that holds ak, nk and ovk of both keys and chooses indices j_0, j_1 and $j' \notin \{j_0, j_1\}$ distinguishes which key produced the third address with negligible advantage, conditioned on the three diversifiers being pairwise distinct. Its assumptions are the *Ironwood Guide*'s "Pseudorandomness of PRF expansion", "Joint hiding of the viewing keys", "FF1-AES256 as a pseudorandom permutation", "GroupHash as a random oracle", "DDH on Pallas" and "Discrete logarithms on Pallas". Account keys are admitted as such keys by Lemma 2.9, the win event being decidable by the challenger; the addresses of one key at distinct indices have distinct diversifiers by Proposition 3.2 (a). For Sapling the protocol specification argues the same property under DDH on the prime-order subgroup of Jubjub, with $\mathsf{GroupHash}^{\mathbb{J}^{(r)*}}$ modelled as a random oracle, through IK-CPA key privacy of ElGamal (§"$\mathsf{DiversifyHash}^{\mathsf{Sapling}}$ and $\mathsf{DiversifyHash}^{\mathsf{Orchard}}$ Hash Functions", notes; *Crypto Guide*, §"Key privacy"). This volume does not prove the Sapling case.

Remark 3.4 (Chosen-diversifier oracle). Suppose that the holder of an incoming viewing key gives an adversary the address of that key at any diversifier the adversary chooses. The adversary then breaks unlinkability outright: given an address $(d', \mathsf{pk_d}')$, it requests the address at d' and compares the returned transmission key with $\mathsf{pk_d}'$. The two are equal if and only if the key is the same, the base $g_{d'}$ being a point other than the identity in a group of prime order. No other privacy property is affected. Implementations SHOULD avoid providing such a chosen-diversifier oracle (protocol specification, §"$\mathsf{DiversifyHash}^{\mathsf{Sapling}}$ and $\mathsf{DiversifyHash}^{\mathsf{Orchard}}$ Hash Functions", notes). An interface returning the address at a requested *index* gives no chosen diversifier (Lemma 3.5), but it returns the issued addresses themselves: it links every address whose index the requester queries, and under sequential issuance (Proposition 3.2) small indices reach every issued address.

> **Lemma 3.5** (Index queries give no chosen diversifier)**.** *Let* dk *be either diversifier key of an Orchard key with* $\mathsf{use_qsk} = \mathsf{false}$*. Under Assumption 2.16, the hypotheses of Lemma 2.9, and the* Ironwood Guide*'s Assumption "FF1-AES256 as a pseudorandom permutation" (§"Diversified addresses"), an adversary without* dk *that obtains the diversifiers at q indices of its choice obtains a given d, fixed independently of* dk*, with probability at most $q/2^{88}$ plus its distinguishing advantages against the two*

> *assumptions and, for an account key, the loss of Lemma 2.9. This volume does not prove the Sapling case.*

Proof. Replace dk by a uniform 32-byte string independent of the other keys; by Assumption 2.16, applied after Lemma 2.9, the probability changes by at most that distinguishing advantage and the loss of that Lemma, the win event being decidable. Then replace FF1-AES256 keyed by dk with a uniform permutation of $\{0,1\}^{88}$; the probability changes by at most the distinguishing advantage. Repeated indices add nothing, so let the adversary query $k \le q$ distinct indices. Each image is uniform on the values not yet returned, so the i-th image is the first to equal d with probability

$$\prod_{j=1}^{i-1} \frac{2^{88}-j}{2^{88}-j+1} \cdot \frac{1}{2^{88}-i+1} = 2^{-88};$$

summing over $i \le k$, d is returned with probability $k \cdot 2^{-88} \le q \cdot 2^{-88}$. □

3.2 Raw encodings

The items of unified containers carry the byte encodings below. Each is stated with the validity condition that a decoder applies.

Definition 3.6 (Raw receiver and key encodings). (a) An *Orchard raw payment address* has 43 bytes, $\mathsf{LEBS2OSP}_{88}(d) \parallel \mathsf{pk_d}^{\star}$, where $\mathsf{pk_d}^{\star} = \mathsf{LEBS2OSP}_{256}(\mathsf{repr}_{\mathbb{P}}(\mathsf{pk_d}))$ is the 32-byte compressed encoding of the Pallas point $\mathsf{pk_d}$ (*Ironwood Guide*, §"Fields, groups, and encodings"). A decoder MUST treat the address as invalid if $\mathsf{abst}_{\mathbb{P}}$ returns $\perp$ or the identity (protocol specification, §"Orchard Raw Payment Addresses").

(b) A *Sapling raw payment address* has 43 bytes, $\mathsf{LEBS2OSP}_{88}(d) \parallel \mathsf{LEBS2OSP}_{256}(\mathsf{repr}_{\mathbb{J}}(\mathsf{pk_d}))$, with the compressed Jubjub encoding. It MUST be considered invalid if $\mathsf{abst}_{\mathbb{J}}$ returns $\perp$, if the encoding of $\mathsf{pk_d}$ is non-canonical, or if $\mathsf{pk_d}$ is not a point of order $r_{\mathbb{J}}$, the identity included (protocol specification, §"Sapling Payment Addresses").

(c) An *Orchard raw full viewing key* has 96 bytes,

$$\mathsf{I2LEOSP}_{256}(\mathsf{ak}) \parallel \mathsf{I2LEOSP}_{256}(\mathsf{nk}) \parallel \mathsf{I2LEOSP}_{256}(\mathsf{r_{ivk}}).$$

It MUST be considered invalid if ak, nk or $\mathsf{r_{ivk}}$ is not the canonical encoding of an element of its field ($\mathbb{F}_{p_{\mathsf{Pallas}}}, \mathbb{F}_{p_{\mathsf{Pallas}}}, \mathbb{F}_{p_{\mathsf{Vesta}}}$), if ak is not the x-coordinate of a point of $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$ other than the identity, or if the external or the internal incoming viewing key derived from it is 0 or $\perp$ (protocol specification, §"Orchard Raw Full Viewing Keys").

(d) An *Orchard raw incoming viewing key* has 64 bytes, $\mathsf{dk} \parallel \mathsf{I2LEOSP}_{256}(\mathsf{ivk})$. It MUST be considered invalid unless $\mathsf{ivk} \in \{1, \ldots, p_{\mathsf{Pallas}} - 1\}$ (protocol specification, §"Orchard Raw Incoming Viewing Keys").

(e) String forms. No Bech32 or Bech32m encoding exists for an individual Orchard payment address, incoming viewing key or full viewing key; the unified encodings of §3.4 are their only strings. A Sapling payment address keeps a standalone Bech32 string, with human-readable part `zs` on Mainnet and `ztestsapling` on Testnet (same sections).

(f) Transparent receivers. A *P2PKH* (pay-to-public-key-hash) address pays to the 20-byte $\mathsf{hash160}$ of a 33-byte compressed ECDSA validating key; a *P2SH* (pay-to-script-hash) address pays to the 20-byte $\mathsf{hash160}$ of a redeem script that the spender reveals and satisfies (*Crypto Guide*, §"The transparent layer's hashes"). Their raw encoding prefixes two

version bytes to the hash and is written in Base58Check (protocol specification, §"Transparent Addresses"). Inside a unified container a transparent receiver is the bare 20-byte hash, without the two version bytes (ZIP 316, "Requirements for both Unified Addresses and Unified Viewing Keys").

One Orchard raw address serves both pools of the Orchard protocol (§3.3).

3.3 Unified addresses

ZIP 316 defines Revision 0, the format of this subsection and of §3.4, and Revision 2, which adds metadata items, splits the address prefix into a shielded-only and a transparent-enabled form, and renames the viewing-key prefixes (§3.7); Revision 1 is withdrawn (ZIP 316, "Revisions"). The revision of a container is read from its human-readable part. The status of ZIP 316 is that of Table 1; everything Revision 2 adds is specified in the sense of Definition 1.1.

Definition 3.7 (Unified container)**.** For $n \in \{0, \dots, 2^{64} - 1\}$, compactSize$(n)$ is the variable-length integer of Bitcoin that ZIP 316 cites:

$$\text{compactSize}(n) := \begin{cases} [n] & \text{if } n < 253, \\ [\texttt{0xFD}] \,\|\, \mathsf{I2LEOSP}_{16}(n) & \text{if } 253 \le n < 2^{16}, \\ [\texttt{0xFE}] \,\|\, \mathsf{I2LEOSP}_{32}(n) & \text{if } 2^{16} \le n < 2^{32}, \\ [\texttt{0xFF}] \,\|\, \mathsf{I2LEOSP}_{64}(n) & \text{otherwise,} \end{cases}$$

the range of n fixing the form. An *item* is a pair (τ, v) of a *typecode* $\tau \in \mathbb{N}$ and a byte string v. The *raw encoding* of a container with items $(\tau_1, v_1), \dots, (\tau_k, v_k)$, where $\tau_1 < \cdots < \tau_k$, is

$$\mathsf{raw} := \bigparallel_{i=1}^{k} \text{compactSize}(\tau_i) \,\|\, \text{compactSize}(\text{length}(v_i)) \,\|\, v_i.$$

Both τ_i and length(v_i) MUST be at most `0x2000000`; since `0x2000000` $\in [2^{16}, 2^{32})$, each takes at most 5 bytes and an item header at most 10.

Typecodes `0x00` and `0x01` are *transparent*, `0x02` and `0x03` *shielded*, and `0xC0` to `0xFC` designate *metadata items*, items that are neither receivers nor viewing keys (§3.7). A *unified address* (UA) is a container whose non-metadata items are receivers, in the raw encodings of Definition 3.6 under the typecodes of Table 3. A *unified full viewing key* (UFVK) and a *unified incoming viewing key* (UIVK) are containers whose non-metadata items are full, respectively incoming, viewing keys (§3.5); UVK denotes either. No typecode exists for Sprout.

Validity. Every container holds at least one non-metadata item. A Revision 0 UA, and a Revision 0 UVK, holds at least one shielded item. A Revision 2 shielded-only UA holds no transparent item; a Revision 2 transparent-enabled UA may consist of a single P2PKH receiver, and a Revision 2 UVK need hold no shielded item. A Consumer MUST reject a container

(i) that violates the containment rule of its revision;

(ii) in which a typecode repeats, or which holds both `0x00` and `0x01`;

(iii) which holds only metadata items;

(iv) whose items are not in ascending typecode order;

(v) which has bytes after the last item;

(vi) any of whose items fails the validation of its encoding (Definition 3.6; §3.5).

Typecode	Item	Receiver	UFVK item	UIVK item
0x00	transparent P2PKH	20	65	65
0x01	transparent P2SH	20	—	—
0x02	Sapling	43	128	64
0x03	Orchard	43	96	64

Table 3: Typecodes and item lengths of Revision 0, in bytes. In Revision 0 Producers MUST NOT include typecode 0x01 in a UFVK or UIVK, and Consumers MUST treat it as unrecognised there. The Priority List of preference is 0x03, then 0x02, then 0x01 or 0x00 (at most one of the two is present), the reverse of the encoding order. Receiver lengths are those of Definition 3.6; the viewing-key lengths are those of §3.5: $65 = 32 + 33$ (chain code and compressed key), 128 for $(\mathsf{ak}, \mathsf{nk}, \mathsf{ovk}, \mathsf{dk})$, 64 for $(\mathsf{dk}, \mathsf{ivk})$, and 96 for $(\mathsf{ak}, \mathsf{nk}, \mathsf{r_{ivk}})$ (ZIP 316, "Encoding of Unified Addresses" and "Encoding of Unified Full/Incoming Viewing Keys").

> A Consumer MUST ignore items with typecodes it does not recognise, with the exception stated in §3.7; a Consumer therefore parses a container that holds unrecognised items.
>
> Same-account rule. Within one UA or UVK, all receivers, full viewing keys and incoming viewing keys derived hierarchically SHOULD belong to the same account at the ZIP 32 or BIP 44 account level, or, for ZIP 48 and FROST UVKs, to one set of signers able to generate independent addresses for that spending authority. FROST key material MUST be derived according to ZIP 312 and ZIP 2005, in particular ZIP 2005, "Usage with FROST".
>
> Sources: ZIP 316, "Encoding of Unified Addresses", "Requirements for both Unified Addresses and Unified Viewing Keys" and "Receivers".

The ascending order makes the raw encoding a function of the set of items, so a representation that discards the order reproduces the same encoding, provided it retains unrecognised items (ZIP 316, "Rationale for Item ordering"). This order is not the priority order below.

Receiver selection. The Sender of a payment to a UA MUST use the receiver of the most preferred type that it supports, by the Priority List Orchard (0x03), Sapling (0x02), then P2SH (0x01) or P2PKH (0x00); an unrecognised item is treated as absent, with the exception of §3.7. A wallet MAY let its users configure which receiver types it can send to, and MUST NOT let them change the order of the Priority List except by opting into an experiment, whose specification MAY add to the list and SHOULD keep the preference for more recent shielded protocols (ZIP 316, "Encoding of Unified Addresses", "Transfers" and "Experimental Usage"). The Orchard receiver belongs to the Orchard protocol, not to a pool: one ivk trial-decrypts both Orchard-pool and Ironwood-pool note ciphertexts (ZIP 326, "Scanning for Ironwood-pool and Orchard-pool notes"), and the Ironwood pool changes no address structure or encoding (ZIP 229, "Abstract"). The pool in which a payment to an Orchard receiver is made is fixed by the routing rule of §11.2, which carries the prohibition of ZIP 326, "Wallet key-generation restrictions", on paying external receivers in the Orchard pool; the Priority List is unchanged.

3.4 String encoding: F4Jumble and Bech32m

The Bech32m checksum does not cascade: a user who checks only the beginning and the end of a long string can miss an adversarial substitution of an inner receiver. ZIP 316 therefore passes the whole payload through F4Jumble, an unkeyed four-round Feistel construction that approximates a random permutation of the entire byte string, so that every output character depends on every input byte. It states two goals (ZIP 316, "Jumbling").

(i) *Near second-preimage resistance.* Given q honestly generated UAs or UVKs, an adversary does not produce a valid one that it controls, knowing at least one constituent private key, whose string matches one of the targets on a chosen set of characters, for concreteness the first n and the last m.

(ii) *Nonmalleability.* Without any private key, an adversary does not produce a valid UA or UVK usefully different from a target, for example with its shielded receiver removed, that matches the target on the checked characters.

Generic attacks bound what is achievable: about $w\,c^{n+m}/q$ for (i) and w/p for (ii), where w is the work to produce and verify one UA or UVK, c the size of the character set and p the probability that a random decoding is valid and useful (ZIP 316, "Discussion"). Jumbling is not authentication. F4Jumble is unkeyed, so anyone can jumble and checksum a wholly substituted address, and an address is still obtained or verified through a trusted channel; jumbling raises only the cost of partial-match and malleation attacks against a genuine string.

Construction 3.8 (F4Jumble)**.** Let $\ell_H := 64$ and $\ell_{\mathsf{MAX}} := (2^{16}+1)\,\ell_H = 4194368$. For $i \in \{0,1\}$ and byte strings u, with the first argument of BLAKE2b the personalisation,

$$H_i(u) := \text{BLAKE2b-}(8\ell_L)(\texttt{UA_F4Jumble_H} \,\|\, [i,0,0],\ u),$$

$$G_i(u) := \text{the first } \ell_R \text{ bytes of } \bigparallel_{j=0}^{\lceil \ell_R/\ell_H \rceil - 1} \text{BLAKE2b-512}(\texttt{UA_F4Jumble_G} \,\|\, [i] \,\|\, \mathsf{I2LEOSP}_{16}(j),\ u),$$

an ℓ_L-byte hash and an extendable-output function in the sense of the *Crypto Guide*, §"Hash functions and the random oracle model", Definition "Extendable-output function", for the lengths ℓ_L and ℓ_R below. For a byte string M of length ℓ_M with $38 \le \ell_M \le \ell_{\mathsf{MAX}}$, let

$$\ell_L := \min(\ell_H, \lfloor \ell_M/2 \rfloor), \qquad \ell_R := \ell_M - \ell_L,$$

split $M = a \,\|\, b$ with $\text{length}(a) = \ell_L$, and let

$$x := b \oplus G_0(a), \qquad y := a \oplus H_0(x),$$
$$d := x \oplus G_1(y), \qquad c := y \oplus H_1(d),$$

and $\mathsf{F4Jumble}(M) := c \,\|\, d$ (ZIP 316, "Solution"). Figure 2 shows the four rounds.

Each personalisation has 16 bytes: 13+3 for H_i and 13+1+2 for G_i. They are pairwise distinct, differing at byte 12 between the H and the G family and at bytes 13 to 15 within a family, and distinct from the three personalisations of key derivation in Table 2 and from that of PRFexpand, `Zcash_ExpandSeed`. Being of one length and distinct, they are prefix-free tags, so BLAKE2b under these personalisations is modelled as independent random oracles (*Crypto Guide*, §"Domain separation and personalisation", Proposition "Domain separation yields independent oracles").

Proposition 3.9 (F4Jumble is a permutation)**.** *For every ℓ_M with $38 \le \ell_M \le \ell_{\mathsf{MAX}}$,* F4Jumble *is well defined and is a bijection of the ℓ_M-byte strings. Its inverse maps $c \,\|\, d$, with* $\text{length}(c) = \ell_L$*, to $a \,\|\, b$ computed by*

$$y := c \oplus H_1(d), \quad x := d \oplus G_1(y), \quad a := y \oplus H_0(x), \quad b := x \oplus G_0(a).$$

Proof. Well defined. From $\ell_M \ge 38$, $\ell_L \ge 19$; from $\ell_L \le 64$, the output length $8\ell_L$ of H_i is admissible for BLAKE2b. If $\ell_M \le 128$ then $\ell_R \le 64$ and one block of G_i suffices. Otherwise $\ell_L = 64$ and $\ell_R = \ell_M - 64 \le 2^{16} \cdot 64$, so $\lceil \ell_R/\ell_H \rceil \le 2^{16}$ and every counter j fits $\mathsf{I2LEOSP}_{16}$; this is the reason for $\ell_{\mathsf{MAX}} = (2^{16}+1)\,\ell_H$.

Bijection. Each of the four steps replaces one half by its XOR with a function of the other half, which it leaves unchanged, and $r = p \oplus q$ implies $p = r \oplus q$. Each step is therefore inverted by recomputing the same function of the unchanged half, and the composition of the four inverses in reverse order is the stated map, a two-sided inverse. This is the argument of the *Crypto Guide*,

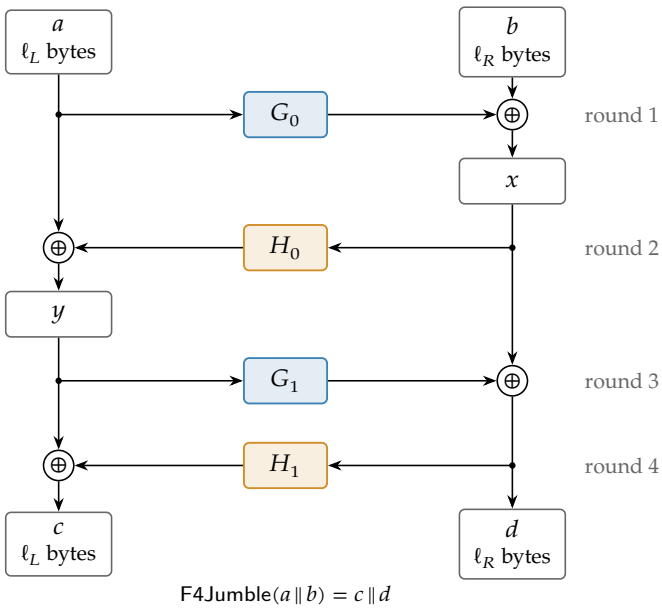


Figure 2: The four rounds of F4Jumble (Construction 3.8) on the unequal halves a (ℓ_L bytes) and b (ℓ_R bytes). Each round replaces one half by its XOR with a function of the other half, which it leaves unchanged: G_0 of a into b gives x, H_0 of x into a gives y, G_1 of y into x gives d, and H_1 of d into y gives c. Blue boxes are the extendable-output functions G_i, with ℓ_R output bytes; amber boxes are the hashes H_i, with ℓ_L output bytes. The names are those of the cells of Example 3.11. The structure is the content of Proposition 3.9.

§"Format-preserving encryption and FF1", Proposition "Feistel is a bijection regardless of the round functions". Unequal halves change nothing, since G_i maps into ℓ_R bytes and H_i into ℓ_L bytes, the lengths of the halves they are XORed into. □

ZIP 316, "Heuristic analysis", motivates the number of rounds.

Construction 3.10 (Unified string encoding)**.** Human-readable parts of Revision 0: `u` for a UA, `uview` for a UFVK and `uivk` for a UIVK on Mainnet; on Testnet `test` is appended (`utest`, `uviewtest`, `uivktest`). Let raw be the raw encoding of Definition 3.7, *hrp* the human-readable part and padding the human-readable part in US-ASCII right-padded with zero bytes to 16 bytes.

1. Producer. It MUST ensure length(raw) $\leq \ell_{\mathsf{MAX}} - 16 = 4194352$, and outputs Bech32m(*hrp*, F4Jumble(raw ‖ padding)), the Bech32m encoding of BIP 350 with its length restriction ignored.

2. Consumer. It MUST decode as follows, rejecting the string at the first failure:

   (1) decode with Bech32m, rejecting an incorrect checksum;

   (2) reject a decoded byte string shorter than 38 or longer than ℓ_{MAX} bytes;

   (3) apply $\mathsf{F4Jumble}^{-1}$ (Proposition 3.9);

   (4) reject a result that does not end in the 16-byte padding of the string's human-readable part, and remove that padding;

   (5) parse the rest as a raw encoding, rejecting the whole string if any rule of Definition 3.7 fails.

> UFVKs and UIVKs use the same procedure under their own human-readable parts (ZIP 316, "Encoding of Unified Addresses", "Encoding of Unified Full/Incoming Viewing Keys" and "Usage for Unified Addresses, UFVKs, and UIVKs"; BIP 350).

The padding supplies the redundancy that defeats the generic nonmalleability attack, since a random decoding must end in the exact 16 bytes, and gives the human-readable part the same protection against malleation as the payload, which may help against cross-network substitution and against confusion of an address with a viewing key (ZIP 316, "Usage for Unified Addresses, UFVKs, and UIVKs" and "Heuristic analysis").

Length rules (ZIP 316, "Rationale for length restrictions" and "Efficiency"). The 38-byte minimum is Revision 2's: one P2PKH item of 22 bytes and the padding. Revision 0 specified 48 bytes. No set of items valid in Revision 0 has an encoding of 22 to 31 bytes (the smallest Revision 0 input is $45 + 16 = 61$ bytes), so the difference is unobservable except in the error reported, and a Consumer supporting Revision 2 MAY apply either minimum to Revision 0 strings. A Consumer that cannot support strings up to ℓ_{MAX} MUST support those with ℓ_M of at least 256 bytes; a full implementation supports ℓ_{MAX}.

Display rule, specified (ZIP 316, "Requirements for both Unified Addresses and Unified Viewing Keys"). If the string of a UA or UVK is shown in an abridged form, at least its first 20 characters MUST be shown. The rationale is the ZIP's ("Rationale for showing at least the first 20 characters"). After the longest Mainnet human-readable part of Revision 0, `uview`, and the separator `1`, 14 characters remain, 70 bits of jumbled data at 5 bits per character, below the target of 2^{125} against classical attacks, and multi-target attacks are possible; 20 is an absolute minimum and a wallet may show more. Only a prefix counts, because displays that show different substrings let a user compare only the characters common to all of them, and because checksum characters complicate the analysis of partial matching.

Item sizes. A Sapling or Orchard receiver item has $1 + 1 + 43 = 45$ bytes (typecode, length, 11-byte diversifier and 32-byte transmission key), a P2PKH receiver item $1 + 1 + 20 = 22$ bytes, and the padding 16 bytes. Hence a UA with Sapling and Orchard receivers has $\ell_M = 106$, and one with P2PKH, Sapling and Orchard receivers $\ell_M = 128$ (ZIP 316, "Efficiency").

Example 3.11 (A unified address, end to end)**.** Let the seed be the 32 bytes `0x00`, `0x01`, ..., `0x1f`, on Mainnet, with the Orchard account 9 at $m_{\mathsf{Orchard}}/32'/133'/9'$ (Definition 2.7), the external scope and the index $j = 1$. Every value below is computed by script from the seed.

Receiver. The diversifier key dk is the first half of the `0x82` expansion of the account's full viewing key, and FF1 encryption of $\mathrm{LE}_{88}(1)$ gives d_1 (Definition 3.1):

dk	`82cc9d79742fe5ae9a142b9336a98677b154fe20401eb18998dbed915b0453ce`
$\mathsf{LEBS2OSP}_{88}(\mathrm{LE}_{88}(1))$	`01000000000000000000`
d_1	`3fadf8edb20a3301e8260a`
$\mathsf{pk_d}^{\star}$	`a311f4cbd54d7d6a76baac88c244b0b121c6dc22a8bcce15898e267829fc1e01`

Decryption $\mathsf{FF1\text{-}AES256.Decrypt}(\mathsf{dk}, \texttt{""}, d_1)$ returns $\mathrm{LE}_{88}(1)$, so $j = 1$, and the ownership test $\mathsf{pk_d} = [\mathsf{ivk}]\,\mathsf{DiversifyHash}^{\mathsf{Orchard}}(d_1)$ of Lemma 3.3 holds for the external ivk. It fails for the internal $\mathsf{ivk_{int}}$, although $\mathsf{dk_{int}}$ also decrypts d_1, to the index 184814530803446824004762235: recovery alone proves nothing, and the test on $\mathsf{pk_d}$ decides. The neighbouring diversifiers are $d_0 =$ `e340636542ece1c81285ed` and $d_2 =$ `987fd74a2256c596a66f83`; that consecutive indices give unrelated strings illustrates the *Ironwood Guide*'s Assumption "FF1-AES256 as a pseudorandom permutation" (§"Diversified addresses") and is not evidence for it.

Container. The UA has one item, so its raw encoding (Definition 3.7) is the header $\mathsf{compactSize}(\texttt{0x03}) \parallel \mathsf{compactSize}(43) = \texttt{032b}$ followed by the 43-byte receiver $d_1 \parallel \mathsf{pk_d}^{\star}$, 45 bytes in all. The padding is `75` (`u`) followed by fifteen zero bytes, so M has $\ell_M = 61$, $\ell_L = 30$ and $\ell_R = 31$. The cells of Construction 3.8:

$a = M[0..30)$	`032b3fadf8edb20a3301e8260aa311f4cbd54d7d6a76baac88c244b0b121`
$b = M[30..61)$	`c6dc22a8bcce15898e267829fc1e017500000000000000000000000000000000`
$x = b \oplus G_0(a)$	`e6f1529b03a0ad0a4209da8e4365ca14a5988bf1d03084a1c326f929c574ca`
$y = a \oplus H_0(x)$	`37b358eb784bb2b315fd35c595628cfa9aa8045ef271728dd0cbf84dc81a`
$d = x \oplus G_1(y)$	`ad26d7fb043e09cbf18e8f4d80d2b423ecb01719170463d644be4c76daab6b`
$c = y \oplus H_1(d)$	`98979f9a7d2dfce7b1616a8d2b9975f87878171b6fcf33b303f770b4aa64`

The header, the diversifier and the padding are legible in a and b, and in neither c nor d.

String. The 61 bytes of $c \| d$ regroup into 98 five-bit symbols, the last padded with two zero bits. With the human-readable part `u`, the separator `1` and the 6-symbol checksum, the string has $1 + 1 + 98 + 6 = 106$ characters:

```
u1nztelxna9h7w0vtpd2xjhxt4lpu8s9cmdl8n8vcr7actf2ny45nd07cy8cyuhuvw3axcp545y0ktq9cezuzx84jyhex8dk4tdvwhu4dl
```

Decoding reverses each stage of Construction 3.10, and $\mathsf{F4Jumble}^{-1}(c \| d) = M$. The string equals the unified-address test vector for account 9 and index 1 of the zcash-test-vectors suite, which ZIP 32, "Test Vectors", names.

3.5 Unified viewing keys

Definition 3.12 (Unified full and incoming viewing keys)**.** A UFVK and a UIVK are containers of Definition 3.7 under the string encoding of Construction 3.10 with their own human-readable parts; a Consumer MUST use the decoding procedure of Construction 3.10. Their items, at the receiver typecodes, are the following.

1. Orchard, `0x03`: the 96-byte raw full viewing key, respectively the 64-byte raw incoming viewing key, of Definition 3.6.

2. Sapling, `0x02`: in a UFVK the 128 bytes

$$\begin{aligned}&\mathsf{EncodeExtFVKParts}(\mathsf{ak}, \mathsf{nk}, \mathsf{ovk}, \mathsf{dk})\\ &\quad := \mathsf{LEBS2OSP}_{256}(\mathsf{repr}_{\mathbb{J}}(\mathsf{ak})) \| \mathsf{LEBS2OSP}_{256}(\mathsf{repr}_{\mathbb{J}}(\mathsf{nk})) \| \mathsf{ovk} \| \mathsf{dk}\end{aligned}$$

   of ZIP 32, "Sapling helper functions", which SHOULD be derived from the account-level extended full viewing key; in a UIVK the 64 bytes $\mathsf{dk} \| \mathsf{I2LEOSP}_{256}(\mathsf{ivk})$, valid only if $\mathsf{ivk} \in \{1, \ldots, 2^{251} - 1\}$ (protocol specification, §"Sapling Incoming Viewing Keys"; ZIP 316 notes the exclusion of 0). The Sapling UIVK item carries dk beyond the specification's Sapling incoming viewing key, which is ivk alone (protocol specification, §"Sapling Incoming Viewing Keys"); dk maps an index to a diversifier and back, so a UIVK yields addresses at indices (Construction 3.16) and recovers the index of a diversifier (Lemma 3.3).

3. Transparent P2PKH, `0x00`: the 65 bytes $\mathsf{c} \| \mathsf{ser}_P(\mathsf{pk})$ of a 32-byte BIP 32 chain code and a 33-byte compressed public key, the last 65 bytes of the BIP 32 extended public key serialisation (BIP 32, "Serialization format"). The UFVK item is the key at $m/44'/coin_type'/a'$, the account level of BIP 44, and the UIVK item its external (non-change) child at $m/44'/coin_type'/a'/0$.

The current viewing-key types reuse the receiver typecodes; future types MAY use others. The Priority List does not govern viewing keys: a Consumer normally takes the union of the information of all items (ZIP 316, "Encoding of Unified Full/Incoming Viewing Keys").

The transparent item omits the BIP 32 depth, parent fingerprint and child number; the format is ZIP 316's. That the child number of the account-level key would reveal the account index is this volume's inference, not a statement of the ZIP. The Sapling UFVK item carries no chain code, so a UFVK supports no non-hardened Sapling child derivation below the account.

Construction 3.13 (Attenuation of a UFVK to a UIVK)**.** Item by item, with the typecode preserved:

1. a Sapling full viewing key to its incoming viewing key (protocol specification, §"Sapling Key Components"), with dk carried over;
2. an Orchard full viewing key $(\mathsf{ak}, \mathsf{nk}, \mathsf{r_{ivk}})$ to the external-scope incoming viewing key $(\mathsf{dk}, \mathsf{ivk})$ (*Ironwood Guide*, §"Viewing keys");
3. a transparent P2PKH account key to its child at non-hardened index 0 by BIP 32 public-to-public derivation (BIP 32, "Public parent key → public child key").

Items with unrecognised typecodes, and items of experiments the user has not opted into, MUST be dropped (ZIP 316, "Deriving a UIVK from a UFVK"). The treatment of metadata items is stated with them, in §3.7.

Assumption 3.14 (Independence of the per-protocol key trees). For a wallet seed S (Definition 2.1) and an account index a, the Orchard account key of Definition 2.7, the Sapling extended spending key at $m_{\mathsf{Sapling}}/32'/coin_type'/a'$ of ZIP 32 and the BIP 32 extended private key at $m/44'/coin_type'/a'$ are jointly computationally indistinguishable from three independently generated keys of the respective protocols: a uniform Orchard spending key with a uniform chain code, a uniform Sapling master-derived key and a uniform BIP 32 key.

The assumption has a heuristic basis and no proof. It follows if BLAKE2b-512 under `ZcashIP32Sapling` (the Sapling master key) and HMAC-SHA512 under the key `Bitcoin seed` (the BIP 32 master key) are modelled as random oracles independent of BLAKE2b-512 under `ZcashIP32Orchard`, and the Sapling and BIP 32 hardened child derivations as PRFs keyed by the parent chain code (HMAC as a PRF: *Crypto Guide*, §"PRFs from hash functions: length extension and keyed BLAKE2"; ZIP 32, "Sapling master key generation" and "Sapling child key derivation"; BIP 32, "Master key generation" and "Private parent key → private child key"). Lemma 2.5 proves the Orchard component and nothing more.

Proposition 3.15 (Capabilities of a unified viewing key)**.** *Let the keys of one account be generated from a seed, with* $\mathsf{use_qsk} = \mathsf{false}$ *for the Orchard component, and let Assumption 3.14 hold.*

(*a*) *From a UFVK the following are computed: for Orchard, the incoming and outgoing viewing keys of both scopes and* nk*, hence receipt, spend and outgoing detection (Proposition 2.15); for Sapling, the same through ZIP 32's internal key derivation; for transparent keys, every external and internal address and the transparent outgoing viewing keys (Construction 2.18). No secret that is infeasible to compute from its own component becomes feasible from the whole UFVK: in particular the Orchard* ask (Ironwood Guide, *§"Key capabilities and scope", Proposition "Capability separation", part (a)), the Sapling spend authorising key and a transparent private key.*

(*b*) *A UIVK gives receipt detection for the external scope in each shielded protocol it carries, and no spend detection for its shielded items:* nk *is infeasible to compute from an Orchard* $(\mathsf{dk}, \mathsf{ivk})$ (Ironwood Guide, *§"Key capabilities and scope", Proposition "Capability separation", part (b)), and internal-scope receipt is neither detected (Proposition 2.17) nor derivable (Proposition 2.20). A transparent UIVK item yields the external addresses $m/44'/coin_type'/a'/0/j$ and not the internal ones. Spends of transparent funds are public on the transparent chain, so their visibility is a property of the chain, not of the key.*

Metadata items confer no viewing authority (§3.7), so the statements hold for containers of either revision.

Proof. Componentwise. The computations of (a) and (b) are those of the cited constructions applied to each item. For the separations: by Assumption 3.14 the components are indistinguishable from independently generated keys, so a reduction given one component generates the others itself. An algorithm that outputs a secret of one protocol from the whole container thus yields an algorithm that outputs it from that protocol's component alone, with the advantage of Assumption 3.14 added; the success of either is efficiently decidable, by recomputing the public key or viewing key from the output. The per-component statements are Propositions 2.15, 2.17 and 2.20 with the *Ironwood Guide*, §"Key capabilities and scope", Proposition "Capability separation", under the assumptions named there, for Orchard; ZIP 32 and the protocol specification, §"Sapling Key Components", cited, for Sapling; and Assumption 2.19 with BIP 32, "Security", for the transparent keys. The transparent spend clause is a property of transparent inputs, which name the outpoints they spend in the clear. □

3.6 Deriving unified addresses

Construction 3.16 (Unified address at an index)**.** Let U be a UIVK. Choose one index $j \in \{0, \dots, 2^{88} - 1\}$, which MUST be valid for every item of U:

1. for an Orchard item, always;
2. for a Sapling item, if and only if $\mathsf{DiversifyHash}^{\mathsf{Sapling}}(d_j) \neq \perp$, with d_j the diversifier of the item's dk at j (Definition 3.1);
3. for a transparent P2PKH item, if and only if $0 \leq j \leq 2^{31} - 1$ and BIP 32 derivation of the child at index j succeeds; it fails with probability below 2^{-127} (BIP 32, "Public parent key → public child key").

Each receiver is derived from its item at the same j: a Sapling or Orchard receiver as the address at index j of $(\mathsf{dk}, \mathsf{ivk})$ (protocol specification, §"Sapling Key Components" and §"Orchard Key Components"; *Ironwood Guide*, §"Diversified addresses", Construction "Diversified address"), and a transparent P2PKH receiver as the $\mathsf{hash160}$ of the non-hardened BIP 32 public child j of the external key, so that its path is $m/44'/coin_type'/a'/0/j$ (BIP 44). Items with unrecognised typecodes, and items of experiments the user has not opted into, MUST be dropped; metadata items are treated in §3.7 (ZIP 316, "Deriving a Unified Address from a UIVK").

Proposition 3.17 (Indices valid for a unified key)**.** *Let $V(U)$ be the set of indices valid for every item of a UIVK U.*

(a) *If U has only an Orchard item, $V(U) = \{0, \dots, 2^{88} - 1\}$.*

(b) *If U has a transparent item, $V(U) \subseteq \{0, \dots, 2^{31} - 1\}$, a fraction $2^{31}/2^{88} = 2^{-57}$ of the index space, so U has at most 2^{31} unified addresses.*

(c) *Let U have a Sapling item, and let* BLAKE2s-256 *in* $\mathsf{GroupHash}^{\mathbb{J}(r)*}$ *be modelled as a random oracle, the model of the protocol specification, §"Group Hash into Jubjub", notes. The events that*

the indices are valid for the Sapling item are independent over distinct indices, each of probability

$$\pi := \frac{h_{\mathbb{J}}\,(r_{\mathbb{J}} - 1)}{2^{256}} = \frac{8\,(r_{\mathbb{J}} - 1)}{2^{256}} \approx 0.4528,$$

with $h_{\mathbb{J}} = 8$ the cofactor and $r_{\mathbb{J}}$ the prime subgroup order of Jubjub (protocol specification, §"Jubjub"); an index is invalid with probability $1 - \pi \approx 0.5472$. ZIP 32's "approximately half" and ZIP 316's probability $1/2$ are this value rounded.

(d) *With a Sapling and a transparent item, $V(U)$ is the set of indices below 2^{31} that are Sapling-valid and at which BIP 32 derivation succeeds; its expected size lies within $2^{31} \cdot 2^{-127} = 2^{-96}$ of $\pi \cdot 2^{31}$.*

For the Sapling key of account 9 of the seed of Example 3.11, 4527 of the indices $0, \ldots, 9999$ are valid, a fraction 0.4527, and index 0 is valid.

Proof. The set $V(U)$ is the intersection of the per-item valid sets of Construction 3.16, which gives (a) and (b), distinct indices giving distinct receivers.

(c) Distinct indices give distinct diversifiers (Proposition 3.2 (a)), hence distinct inputs to BLAKE2s-256, whose outputs h are then independent and uniform on $\{0,1\}^{256}$. The group hash returns a point if and only if $P := \mathsf{abst}_{\mathbb{J}}(h) \neq \perp$ and $[h_{\mathbb{J}}]\,P \neq \mathcal{O}$ (protocol specification, §"Group Hash into Jubjub"). The map $\mathsf{abst}_{\mathbb{J}}$ reads h as a 255-bit integer $v < q_{\mathbb{J}}$ and a sign bit $\tilde{u}$, rejecting $v \geq q_{\mathbb{J}}$ and v with no curve point (protocol specification, §"Jubjub"). Each point with $u \neq 0$ is the image of exactly one h, since u and $q_{\mathbb{J}} - u$ have different parities, and each of the two points $(0, \pm 1)$ is the image of two. The curve has $h_{\mathbb{J}} r_{\mathbb{J}}$ points and its $h_{\mathbb{J}}$-torsion subgroup, the kernel of $[h_{\mathbb{J}}]$, has 8 points: $(0,1)$, $(0,-1)$ and six with $u \neq 0$. The strings h for which the hash succeeds are thus in bijection with the points with $u \neq 0$ outside that subgroup, of which there are $h_{\mathbb{J}} r_{\mathbb{J}} - 2 - 6 = 8\,(r_{\mathbb{J}} - 1)$, so $\pi = 8\,(r_{\mathbb{J}} - 1)/2^{256}$.

(d) Parts (b) and (c) and Construction 3.16 (3) combined by linearity of expectation over the 2^{31} indices below 2^{31}, each of which fails BIP 32 derivation with probability below 2^{-127}. □

Remark 3.18 (Default addresses). ZIP 316 fixes neither an order in which a wallet searches for a valid index nor a default unified address. The only normative defaults are ZIP 32's per-protocol default diversifiers: for Sapling d_j with j the least valid index, for Orchard d_0 (ZIP 32, "Sapling diversifier derivation" and "Orchard diversifier and OVK derivation"). Issuing the unified address at the least index valid for every item is designed but unspecified (Definition 1.1).

Index 0 and recovery across wallets, specified (ZIP 316, "Deriving a Unified Address from a UIVK"). Index 0 is invalid for a Sapling item with probability $1 - \pi$ (Proposition 3.17), so a wallet that requires a Sapling receiver often issues its first unified address at a later index, while some wallets, earlier than ZIP 316 or without a Sapling component in their unified addresses, always generate a transparent P2PKH address at index 0. Therefore every Zcash wallet, whether or not it supports unified addresses, MUST assume that transparent funds may exist at diversifier index 0 of each ZIP 32 account, even where it would never generate a unified address at that index; reliable recovery of funds when key material moves between wallets depends on this. Its use in restoration is stated in §6.2.

3.7 Revision 2 containers

Definition 3.19 (Metadata items)**.** The following is Revision 2 of ZIP 316, specified (Definition 1.1).

1. Human-readable parts. A UA is either *shielded-only*, with human-readable part `zu`, which MUST NOT hold a transparent item (typecode `0x00` or `0x01`), a Consumer

rejecting any that does, or *transparent-enabled*, with `tu`. A UFVK uses `uvf` and a UIVK `uvi`; on Testnet `test` is appended. The human-readable part distinguishes a Revision 0 container (`u`, `uview`, `uivk`) from a Revision 2 one.

2. Metadata items, typecodes `0xC0` to `0xFC` (Definition 3.7), MAY affect the interpretation of the container, MUST NOT be selected as receivers and MUST NOT confer viewing authority.

3. MUST-understand items. Typecodes `0xE0` to `0xFC` are MUST-understand: a Consumer that does not recognise such an item MUST regard the whole container as unsupported and not process it further. This is the exception to the ignore rule of Definition 3.7.

4. Registry. Typecodes `0x04` to `0xBF` are unassigned; `0xC0` to `0xDF` are metadata that need not be understood, none assigned; `0xE0` is the Address Expiry Height and `0xE1` the Address Expiry Time, permitted in `zu` and `tu` addresses and not in Revision 0; `0xE2` to `0xFC` are unassigned MUST-understand metadata; `0xFD` to `0xFFF9` are unassigned; and `0xFFFA` to `0xFFFF` are experimental. Producers MUST NOT include items whose typecodes are unassigned or not permitted for the type of the container.

5. Address expiry items. An item of typecode `0xE0` holds the *Address Expiry Height*, an unsigned 32-bit little-endian integer; a UA carrying it MUST be considered expired when the height of the chain exceeds that value. An item of typecode `0xE1` holds the *Address Expiry Time*, an unsigned 64-bit little-endian Unix time; a UA carrying it MUST be considered expired when the current time is after that value. A Producer MAY include either, both or neither.

6. Retroactive rule. A Revision 0 container MUST NOT hold a MUST-understand metadata item, and a Consumer MUST reject one that does.

7. Derivation. To derive a UIVK from a UFVK, or a UA from a UFVK or UIVK, a Consumer MUST understand every MUST-understand item of the source. The derived container MUST be Revision 2 if the source holds any metadata item, and SHOULD be Revision 2 if the source is. A UIVK derived from a UFVK MUST retain its `0xE0` and `0xE1` items unmodified. A UA derived from a UFVK or UIVK that carries `0xE0` (respectively `0xE1`) MUST carry `0xE0` (respectively `0xE1`) with a value at most that of the viewing key. Metadata items that need not be understood and are unrecognised are dropped under Constructions 3.13 and 3.16.

8. Containment. Revision 2 drops the rule that a UVK holds a shielded item, and admits a `tu` UA that consists of one P2PKH receiver (Definition 3.7). Revision 2 also admits a P2SH viewing-key item, which lies outside this volume; Table 3 is that of Revision 0.

Sources: ZIP 316, "Revisions", "Metadata Items", "Address Expiration Metadata", "Typecode Registry", "Requirements for both Unified Addresses and Unified Viewing Keys", "Deriving a UIVK from a UFVK" and "Deriving a Unified Address from a UIVK".

A Consumer that predates Revision 2 does not recognise the Revision 2 human-readable parts. Requiring MUST-understand items to appear only under those parts therefore enforces the MUST-understand rule even against such a Consumer (ZIP 316, "Rationale for making Revision 0 UA/UVKs with MUST-understand Typecodes invalid").

Sender rules for expiring addresses, specified (ZIP 316, "Address Expiration Metadata" and its "Rationale"). The expiry height of a transaction is its field nExpiryHeight (*Ironwood Guide*, §"The transaction format"). A Sender supporting Revision 2 MUST set a nonzero nExpiryHeight in a transaction to a UA with an Address Expiry Height, and MUST NOT send the transaction if the nExpiryHeight it would otherwise construct exceeds that height; lowering nExpiryHeight to the

address's value instead would reveal the destination's expiry on chain. With only an Address Expiry Time, it SHOULD choose nExpiryHeight so that the transaction expires no more than 24 hours after the current time. With both items both restrictions apply, and with several recipient UAs in one transaction all their constraints apply. A wallet MUST NOT send to an address it knows to have expired, and its error SHOULD suggest obtaining a current address. The choice of nExpiryHeight is stated in §9.3.

Linkability through metadata (ZIP 316, "Address Expiration Metadata"). Diversified UAs derived from one UIVK that hold only Sapling or Orchard receivers and no metadata are unlinkable (§3.1); with metadata items present this no longer holds, and it is RECOMMENDED that the expiry height and time of distinct derived addresses vary. Re-issuing the same receivers with different expiry metadata yields a visually distinct address that is directly linkable to the original, since the receivers are shared.

3.8 Recipient addresses

Definition 3.20 (Recipient address)**.** A *recipient address* is one of the following:

1. a transparent address, P2PKH or P2SH, in its Base58Check encoding (Definition 3.6 (f));
2. a Sapling payment address in its Bech32 encoding (Definition 3.6 (e));
3. a unified address (Definition 3.7, Construction 3.10 and Definition 3.19);
4. a *TEX address* (transparent-source-only address): the Bech32m encoding, with human-readable part `tex` on Mainnet and `textest` on Testnet, of the 20-byte validating-key hash of a P2PKH address. It is obtained from a Base58Check P2PKH address whose decoding has 22 bytes beginning with the lead bytes `[0x1C, 0xB8]` (Testnet `[0x1D, 0x25]`) by removing them, and it is parsed by requiring the human-readable part and exactly 20 decoded bytes (ZIP 320, "Specification").

Sprout addresses are not recipient addresses: no transaction adds value to the Sprout pool, and no typecode exists for them (§14.2; ZIP 316, "Requirements for both Unified Addresses and Unified Viewing Keys"). A Sender paying a TEX address, as any output of a transaction, MUST spend only transparent (P2SH or P2PKH) outputs in that transaction, and SHOULD send only to transparent outputs (ZIP 320, "Specification"; §11.3).

Two predicates on recipient addresses, defined here, are used by payment construction and by ZIP 321 (ZIP 321, "memo"; §14.2).

(i) An address is *memo-capable* if and only if it is a Sapling payment address or a unified address that contains a Sapling or Orchard receiver, since a memo travels only in a note plaintext (*Ironwood Guide*, §"The note plaintext"). A memo reaches the recipient of a unified address only when the Sender selects a shielded receiver.

(ii) An address is *transparent-only* if and only if it is a transparent or TEX address, or a unified address whose non-metadata items are all transparent receivers, which is possible only as a Revision 2 `tu` address.

4 Compact blocks (ZIP 307)

This section states the compact block as an abstract message format: the records of a block and of a transaction, the compact outputs, spends and Actions and the fields they omit, the compact variant of trial decryption with the soundness of accepting a plaintext that no tag authenticates, and the placement of notes in the note commitment trees from per-block tree sizes. The service

that delivers compact blocks is the subject of "The light-client service" (§5).

4.1 The compact block and the compact transaction

A *compact block* carries only the data of a block that a wallet needs to detect payments to its addresses, to detect spends of its notes and to maintain the witnesses of its notes (ZIP 307, "Compact Stream Format"); for the transparent pool it carries, on request, the data needed to detect spends of the wallet's transparent coins.

Remark 4.1 (Normative status of the compact format). ZIP 307, "Compact Stream Format", "Output Compression" and "Spend Compression", specifies the compact format for the Sapling pool only: a compact block of height, block hash and transactions, and a compact transaction of index, transaction identifier, spends and outputs. It names neither the Orchard pool nor the Ironwood pool, nor per-block tree sizes, nor transparent data. ZIP 318 and ZIP 326 address light wallets and specify no compact format. The remainder of the format below (the parent hash of a block, the tree sizes, the compact Actions of both Orchard-protocol pools, the transparent part and pool selection) is designed but unspecified (Definition 1.1). The volume states the format as an abstract message format, not as a serialisation, and states no field number, field width or encoded size. The status of each ZIP cited is that of Table 1.

Definition 4.2 (Compact block). A *compact block* of height h is a tuple

$$CB = (h, \mathsf{hash}, \mathsf{prevHash}, \mathsf{hdr}, \mathsf{vtx}, S)$$

where

1. $h \in \mathbb{N}$ is the block height;
2. hash and prevHash are the 32-byte hashes of the block and of its parent block (protocol specification, §"Block Header Encoding and Consensus");
3. hdr is either the encoded block header or empty; the abstract format does not guarantee that it is present;
4. $\mathsf{vtx} = (ctx_0, \ldots, ctx_{m-1})$ is the sequence of compact transactions (Definition 4.3) of the block, in block order;
5. $S = (S^{\mathsf{Sapling}}, S^{\mathsf{Orchard}}, S^{\mathsf{Ironwood}})$ gives, for each shielded pool pool, the number S^{pool} of leaves of that pool's note commitment tree after every transaction of the block has been appended, written $S^{\mathsf{pool}}_{\mathsf{end}}(h)$.

Each S^{pool} is an integer in $[0, 2^{32}]$, where 2^{32} is the leaf capacity of a tree of depth $\mathsf{MerkleDepth}^{\mathsf{Sapling}} = \mathsf{MerkleDepth}^{\mathsf{Orchard}} = 32$ (*Ironwood Guide*, §"The Merkle hash and the tree", Definition "Note commitment tree"; protocol specification, §"Constants"). The width of any serialisation of S^{pool} lies outside the abstract format.

Definition 4.3 (Compact transaction). A *compact transaction* is a tuple

$$ctx = (i, \mathsf{txid}, SSp, SOut, A^{\mathsf{Orchard}}, A^{\mathsf{Ironwood}}, T)$$

where

1. $i \in \mathbb{N}$ is the index of the transaction in its block;

2. txid is the 32-byte transaction identifier in protocol byte order, the form in which it is computed, not the byte-reversed form used for display (*Ironwood Guide*, §"Transaction digests and signatures", Definition "Transaction identifier"; protocol specification, §"Transaction Identifiers");

3. *SSp* and *SOut* are the sequences of compact Sapling spends and compact Sapling outputs, and A^{Orchard} and A^{Ironwood} the sequences of compact Actions of the Orchard pool and of the Ironwood pool (Definition 4.5), each in the order of the transaction's descriptions, on which note positions depend (§4.4);

4. T is an optional *transparent part* $(O_{\mathsf{in}}, O_{\mathsf{out}})$: O_{in} is the sequence of outpoints (txid', n) spent by the transaction's transparent inputs, in input order, and O_{out} the sequence of pairs $(\mathsf{value}, \mathsf{script})$ of its transparent outputs, in output order, with value in zatoshi.

The coinbase transaction of a block is its first transaction and the one whose single transparent input has a null previous outpoint (protocol specification, §"Coinbase Transactions"); it spends no outpoint, its O_{in} is empty, and it is recognised by $i = 0$. No fee field is part of the abstract format.

Definition 4.4 (Pool selection)**.** Let $\mathcal{P} \subseteq \{\mathsf{transparent}, \mathsf{Sapling}, \mathsf{Orchard}, \mathsf{Ironwood}\}$. The compact block *restricted to* $\mathcal{P}$ is obtained from a compact block by keeping, in each compact transaction, only the components of the pools in $\mathcal{P}$ (*SSp* and *SOut* for Sapling, the sequence A^{pool} for an Orchard-protocol pool, and T only when $\mathsf{transparent} \in \mathcal{P}$), and by omitting every compact transaction left with no component. The tuple S is kept whole, and a compact block is returned for every block of a requested range, possibly with empty vtx. The default is $\mathcal{P} = \{\mathsf{Sapling}, \mathsf{Orchard}, \mathsf{Ironwood}\}$. The range query of "The query interface" (§5.2) takes $\mathcal{P}$ as an argument. Pool selection is designed but unspecified (Definition 1.1); ZIP 307 includes by default only the transactions that have Sapling spends or outputs (ZIP 307, "Block header validation").

For every shielded pool in $\mathcal{P}$ the restriction keeps every commitment of that pool in its order, since an omitted compact transaction has no component of the pool. The per-pool counts and offsets of Definition 4.11 are therefore the same on a restricted block as on the unrestricted one.

4.2 Compact outputs, spends, and Actions

Definition 4.5 (Compact spend, output, and Action)**.** Let C_{enc} be the note ciphertext of a Sapling output or of an Action: 580 bytes, the encryption of the 564-byte note plaintext followed by the 16-byte tag. Its split at the offsets 52 and 564 is that of the *Ironwood Guide*, §"Transaction digests and signatures", Definition "Transaction identifier".

(a) A *compact Sapling spend* is the 32-byte nullifier nf of a Spend description.

(b) A *compact Sapling output* is the triple $(\mathsf{cmu}, \mathsf{ephemeralKey}, C_{\mathsf{enc}}[0{:}52])$, a payload of $32 + 32 + 52 = 116$ bytes.

(c) A *compact Action*, of the Orchard pool and of the Ironwood pool alike, is the tuple $(\mathsf{nf}, \mathsf{cmx}, \mathsf{ephemeralKey}, C_{\mathsf{enc}}[0{:}52])$, a payload of $32 + 32 + 32 + 52 = 148$ bytes.

Parts (a) and (b) are specified (ZIP 307, "Output Compression" and "Spend Compression"); part (c) is designed but unspecified (Definition 1.1). The fields are those of the protocol specification, §"Spend Description Encoding and Consensus", §"Output Description Encoding

and Consensus" and §"Action Description Encoding and Consensus". Only payload sizes are stated; the size of any serialisation lies outside the abstract format.

The nullifier nf of a compact Action has two roles. It detects a spend of the note that the Action consumes, and it is the element ρ of the note that the Action creates (*Ironwood Guide*, §"Nullifier chaining", Construction "Chaining rule"), an input of esk, ψ and rcm and hence of the commitment that acceptance recomputes (Definition 4.9). A compact Action without nf therefore cannot be accepted.

A compact block omits the memo $C_{\mathsf{enc}}[52{:}564]$, the tag $C_{\mathsf{enc}}[564{:}580]$, the outgoing ciphertext C_{out}, value commitments, randomised validating keys, anchors, proofs and signatures (ZIP 307, "Output Compression", "Spend Compression" and "Transaction privacy"). Recovery of a memo or of an outgoing plaintext requires the full transaction, fetched by its identifier ("Leakage of transaction queries", §8.4).

Proposition 4.6 (Sufficiency of the compact fields)**.** *Let CB be a compact block of height h, and let a wallet hold incoming viewing keys and the nullifiers of its notes.*

(*i*) *Receipt detection, that is compact decryption and acceptance of an output (§4.3), is a function of the key, of h, and of the record* $(\mathsf{cmu}, \mathsf{ephemeralKey}, C_{\mathsf{enc}}[0{:}52])$ *of a Sapling output or* $(\mathsf{nf}, \mathsf{cmx}, \mathsf{ephemeralKey}, C_{\mathsf{enc}}[0{:}52])$ *of an Action.*

(*ii*) *Spend detection is a function of the fields* nf *of SSp,* A^{Orchard} *and* A^{Ironwood}.

(*iii*) *Note placement is a function of the order of the commitments in* vtx *and of S (§4.4).*

No field omitted by the compact block enters (i) to (iii).

Proof. (i) Trial decryption and acceptance read the ephemeral key, the ciphertext prefix that encrypts the lead byte, diversifier, value and note seed, the commitment, the height through the admissible lead bytes, and, for an Action, nf as ρ (*Ironwood Guide*, §"Trial decryption and note acceptance", Constructions "Trial decryption" and "Acceptance of an Ironwood-pool output"; ZIP 307, "Scanning for relevant transactions" for Sapling); the memo and the tag enter only the full decryption. (ii) A spend of a note is detected by membership of its nullifier among the published ones (ZIP 307, "Detecting spends"). (iii) The chain appends each pool's commitments in block, transaction and description order to the least unused leaf (*Ironwood Guide*, §"The Merkle hash and the tree", Definition "Note commitment tree"; protocol specification, §"Note Commitment Trees"), so positions follow from that order and a tree size. □

The omitted fields serve the verification of the chain, which the client delegates to the server; that delegation and its assumption are stated in "Architecture and authority" (§5.1).

4.3 Compact trial decryption

Trial decryption and acceptance of full outputs are those of the *Ironwood Guide*, §"Trial decryption and note acceptance" (Constructions "Trial decryption" and "Acceptance of an Ironwood-pool output"), cited and not re-derived. This subsection states the compact variant.

Definition 4.7 (Compact note plaintext)**.** The *compact note plaintext* is the tuple $\mathsf{np_c} = (\mathsf{leadByte}, d, v, \mathsf{rseed})$, encoded as

$$\mathsf{leadByte} \parallel \mathsf{LEBS2OSP}_{88}(d) \parallel \mathsf{I2LEOSP}_{64}(v) \parallel \mathsf{rseed},$$

of $1 + 11 + 8 + 32 = 52$ bytes: the note plaintext of the *Ironwood Guide*, §"The note plaintext",

Definition "Note plaintext", without its memo (protocol specification, §"Note Plaintexts and Memo Fields" and §"Encodings of Note Plaintexts and Memo Fields"). The AEAD encryption is a stream-cipher encryption followed by a tag, so $C_{\mathsf{enc}}[0{:}52]$ is the encryption of exactly these 52 bytes, which are the contents and the opening of the note commitment (ZIP 307, "Output Compression").

Construction 4.8 (Compact decryption of an output)**.** Input: an incoming viewing key ivk of the pool's protocol and a compact output or compact Action with fields $\mathsf{ephemeralKey}$ and $c = C_{\mathsf{enc}}[0{:}52]$.

1. Compute $\mathsf{epk} := \mathsf{abst}(\mathsf{ephemeralKey})$ on the pool's curve, $\mathsf{abst}_{\mathbb{J}}$ on Jubjub for Sapling and $\mathsf{abst}_{\mathbb{P}}$ on Pallas for either Orchard-protocol pool; return $\perp$ if $\mathsf{epk} = \perp$.
2. Compute $K := \mathsf{KDF}(\mathsf{KA.Agree}(\mathsf{ivk}, \mathsf{epk}), \mathsf{ephemeralKey})$ with the key agreement and key derivation of the pool's protocol.
3. Compute $\mathsf{np_c} := c \oplus \mathrm{KS}_1(K, 0^{96}, 52)$, where $\mathrm{KS}_1(K, 0^{96}, 52)$ is the first 52 bytes of the ChaCha20 keystream under key K and the all-zero nonce 0^{96} of the protocol specification, §"Symmetric Encryption", starting at block counter 1, the counter from which ChaCha20-Poly1305 encrypts (block 0 yields the Poly1305 key; *Crypto Guide*, §"The ChaCha20-Poly1305 AEAD", Construction "ChaCha20-Poly1305"; RFC 8439, §2.8); equivalently, bytes 64 to 115 of the keystream from counter 0.
4. Return $\mathsf{np_c}$ parsed as a compact note plaintext (Definition 4.7).

The tag is never checked. The result is a *candidate*, not an accepted note (ZIP 307, "Scanning for relevant transactions").

Definition 4.9 (Acceptance of a compact candidate)**.** Let $\mathsf{np_c} = (\mathsf{leadByte}, d, v, \mathsf{rseed})$ be the candidate returned by Construction 4.8 under ivk for an output of pool pool in a compact block of height h. Let H_{C} be the Canopy activation height and $\mathsf{ZIP212GracePeriod} = 32{,}256$ blocks (protocol specification, §"Constants"). The set of admissible lead bytes is

$$\mathsf{allowedLeadBytes}^{\mathsf{pool}}(h) := \begin{cases} \{\mathtt{0x01}\} & \text{if } h < H_{\mathsf{C}}, \\ \{\mathtt{0x01}, \mathtt{0x02}\} & \text{if } H_{\mathsf{C}} \le h < H_{\mathsf{C}} + \mathsf{ZIP212GracePeriod}, \\ \{\mathtt{0x02}\} & \text{otherwise, if } \mathsf{pool} \neq \mathsf{Ironwood}, \\ \{\mathtt{0x03}\} & \text{otherwise} \end{cases}$$

(protocol specification, §"Note Plaintexts and Memo Fields"), so that an Orchard-pool note has lead byte $\mathtt{0x02}$ and an Ironwood-pool note $\mathtt{0x03}$. The wallet accepts $\mathsf{np_c}$ if and only if every check passes:

(i) $\mathsf{leadByte} \in \mathsf{allowedLeadBytes}^{\mathsf{pool}}(h)$; for Sapling, moreover, $\mathsf{rcm} < r_{\mathbb{J}}$ and $\mathsf{g_d} := \mathsf{DiversifyHash}(d) \neq \perp$, with $\mathsf{rcm} := \mathsf{rseed}$ read as an integer for lead byte $\mathtt{0x01}$;

(ii) with $\mathsf{pk_d} := [\mathsf{ivk}]\,\mathsf{g_d}$, and with $\rho := \mathsf{nf}$ of the same Action for either Orchard-protocol pool, the note commitment recomputed from $(\mathsf{g_d}, \mathsf{pk_d}, v, \rho, \psi, \mathsf{rcm})$, where ψ and rcm are derived from rseed as the pool and the lead byte prescribe (ρ and ψ absent for Sapling), is not $\perp$ and extracts to the record's cmu (Sapling) or cmx (either Orchard-protocol pool);

(iii) for every lead byte other than 0x01, the field ephemeralKey equals the encoding of $[\mathsf{esk}]\,\mathsf{g_d}$, with esk re-derived from rseed, and from ρ for either Orchard-protocol pool.

The accepted note is $n = ((d, \mathsf{pk_d}), v, \rho, \psi, \mathsf{rcm})$ without memo, with ρ and ψ absent for a Sapling note. The derivations of ψ, rcm and esk are cited and not restated: for Sapling and the Orchard pool to the protocol specification, §"Decryption using an Incoming Viewing Key (Sapling and Orchard)" (ZIP 212, "Changes to the process of sending Sapling or Orchard notes"); for the Ironwood pool to the *Ironwood Guide*, §"Trial decryption and note acceptance", Construction "Acceptance of an Ironwood-pool output", whose checks (i), (ii) with (iv), and (iii) are checks (i), (ii) and (iii) here (ZIP 2005, "Changes to the Protocol Specification"). The method of comparison is not part of the rule.

Check (iii) is a step of the specification's decryption procedure and of ZIP 307, not an addition (ZIP 212, "Changes to the process of receiving Sapling or Orchard notes"). For an Ironwood-pool note it is made directly after the computation of $\mathsf{g_d}$, in the order of ZIP 2005. Without it a sender could agree the key with one address of a recipient while naming the diversifier of another, and link the two addresses by observing acceptance (*Ironwood Guide*, §"Trial decryption and note acceptance", Remark "Role of check (iii)"; ZIP 212, "Motivation"). A Sapling note with lead byte 0x01, created under the rules before ZIP 212, skips check (iii), since its esk is not derivable from rseed.

For a note n the *commitment input* is the pair of committed values and trapdoor from which check (ii) computes its commitment: $((\mathsf{g_d}, \mathsf{pk_d}, v), \mathsf{rcm})$ for Sapling and $((\mathsf{g_d}, \mathsf{pk_d}, v, \rho, \psi), \mathsf{rcm})$ for either Orchard-protocol pool. An *address of* ivk is a pair $(d, [\mathsf{ivk}]\,\mathsf{DiversifyHash}(d))$ with $\mathsf{DiversifyHash}(d) \neq \bot$.

Proposition 4.10 (Soundness of compact acceptance)**.** *Let cm* be the commitment,* cmu *or* cmx*, that a compact output or compact Action carries.*

(*a*) *Every note n accepted by Definition 4.9 is an opening of cm*. For every efficient sender, which chooses* ephemeralKey*,* C_{enc} *and the keys under which they are formed, the probability that the wallet accepts n while the sender outputs a note n′ whose commitment input differs from that of n and whose commitment extracts to cm* is negligible, under the binding of the pool's note commitment scheme. The tag of the AEAD gives no integrity against the sender, who holds K, so its omission loses none.*

(*b*) *For an output honestly formed for a note n′ whose address is not an address of* ivk*, the probability that* ivk *accepts a candidate is negligible under the same assumption.*

Whether cm is the commitment of an output of the best chain is not a property of decryption; it is settled under the assumption of "Architecture and authority" (§5.1).*

Proof. (a) Check (ii) recomputes the commitment of n deterministically from its commitment input and compares it with cm^*, whatever the ciphertext and whichever key decrypted it, so n opens cm^*. Consider the algorithm that generates the wallet's key, runs the sender and the compact decryption and acceptance, and outputs the commitment inputs of n and n'. When the event of (a) occurs it outputs two distinct commitment inputs whose commitments extract to the same $cm^* \neq \bot$. For either Orchard-protocol pool this is excluded, except with negligible probability, by the argument of the *Ironwood Guide*, §"Trial decryption and note acceptance", Proposition "Consistency of accepted notes", part (c), under Proposition "Hiding and binding of note commitments" (§"Note commitments"). That proof uses neither the integrity of ChaCha20-Poly1305 nor key commitment, so it applies unchanged to a candidate decrypted from the 52-byte prefix; and binding holds for every rcm, hence also for the trapdoor derivation of the Orchard pool. For Sapling the named assumption is the computational binding of $\mathsf{NoteCommit}^{\mathsf{Sapling}}$, the security

requirement of the protocol specification, §"Windowed Pedersen commitments", together with the injectivity of $\mathsf{Extract}_{\mathbb{J}^{(r)}}$ on $\mathbb{J}^{(r)}$ (protocol specification, §"Coordinate Extractor for Jubjub").

(b) Let n be accepted. By check (ii) its address is $(d, [\mathsf{ivk}]\,\mathsf{g_d})$ with $\mathsf{g_d} = \mathsf{DiversifyHash}(d) \neq \perp$, an address of ivk. Suppose the commitment inputs of n and n' were equal. Then the diversified base $\mathsf{DiversifyHash}(d')$ of the address $(d', \mathsf{pk_d}')$ of n' equals $\mathsf{g_d}$, and $\mathsf{pk_d}' = [\mathsf{ivk}]\,\mathsf{g_d} = [\mathsf{ivk}]\,\mathsf{DiversifyHash}(d')$, so $(d', \mathsf{pk_d}')$ is an address of ivk, contrary to the hypothesis. The commitment inputs therefore differ, and the honest sender outputs n', whose commitment extracts to cm^*; the honest sender is an efficient sender of (a), which bounds the probability. □

4.4 Note positions and tree sizes

For each shielded pool, every note commitment of a block, not only those of the wallet's notes, is appended to the wallet's copy of that pool's note commitment tree in block order, transaction order and description order: cmx of each Action of the pool and cmu of each Sapling output (ZIP 307, "Creating and updating note witnesses"; *Ironwood Guide*, §"The Merkle hash and the tree", Definition "Note commitment tree"; protocol specification, §"Note Commitment Trees"). The leaf of each accepted note is marked for witness retention. The part of the tree a wallet must retain is fixed in "Commitment-tree synchronisation" (§7).

Definition 4.11 (Note position)**.** Let pool be a shielded pool, b a block height with compact block CB, and $n_{\mathsf{pool}}(b)$ the number of commitments of pool in CB. The *start size* of b is

$$S^{\mathsf{pool}}_{\mathsf{start}}(b) := \begin{cases} S^{\mathsf{pool}}_{\mathsf{end}}(b-1) & \text{if the wallet knows it,} \\ 0 & \text{otherwise, if } b \text{ lies below the activation of pool,} \\ S^{\mathsf{pool}}_{\mathsf{end}}(b) - n_{\mathsf{pool}}(b) & \text{otherwise,} \end{cases}$$

the wallet knowing $S^{\mathsf{pool}}_{\mathsf{end}}(b-1)$ when it holds the compact block of height $b-1$ or a tree state of pool at the end of $b-1$; in the third case the value is taken from CB alone, and the block is rejected if it is negative. For the j-th output o of pool, counted from $j = 0$, of a compact transaction ctx of b,

$$\mathsf{position}(o) := S^{\mathsf{pool}}_{\mathsf{start}}(b) + \mathsf{off}(ctx) + j,$$

where $\mathsf{off}(ctx)$ is the number of commitments of pool in the compact transactions of b before ctx. Positions are per pool. A block scanned out of order thus places its notes from its own compact block. Note positions are designed but unspecified (Definition 1.1).

For a pool outside the pool set $\mathcal{P}$ of the query that returned b, the count $n_{\mathsf{pool}}(b)$ is not served, and neither position nor check is defined.

The position of a note is the leaf index of its commitment (*Ironwood Guide*, §"The Merkle hash and the tree", Definition "Note commitment tree", item 1). Every spend needs it for the authentication path. A Sapling nullifier depends on it, through $\rho = \mathsf{MixingPedersenHash}(\mathsf{cm}, \mathsf{pos})$ (protocol specification, §"Computing ρ values and Nullifiers"), while the nullifier of an Orchard-protocol note does not.

Lemma 4.12 (Consistency of tree sizes)**.** *Let the wallet reject a block b unless, for each shielded pool of the pool set $\mathcal{P}$ of the query that returned b,*

$$S^{\mathsf{pool}}_{\mathsf{end}}(b) \geq n_{\mathsf{pool}}(b) \quad \textit{and} \quad S^{\mathsf{pool}}_{\mathsf{start}}(b) + n_{\mathsf{pool}}(b) = S^{\mathsf{pool}}_{\mathsf{end}}(b).$$

(*a*) *If every reported size is the true size, no block is rejected, and every position of Definition 4.11 is the leaf index of the commitment in the pool's tree.*

(b) *A reported* $S^{\mathsf{pool}}_{\mathsf{end}}(b)$ *that is inconsistent with the block's own commitment count, or with the known end size of* $b-1$*, is rejected.*

(c) *A block with* $n_{\mathsf{pool}}(b) > 0$ *and reported size* 0 *is rejected. A size read as zero therefore yields a position only where* 0 *is consistent with the block, and the check needs no means of telling an absent size from zero.*

(d) *A size that is false but consistent with both conditions is not detected; its exclusion is clause* (a) *of the server assumption of "Architecture and authority"* (§5.1).

Proof. (a) The chain appends the commitments of a pool in block, transaction and description order, each to the least unused index (*Ironwood Guide*, §"The Merkle hash and the tree", Definition "Note commitment tree"). The true start size of b is thus the true end size of $b-1$, which equals $S^{\mathsf{pool}}_{\mathsf{end}}(b) - n_{\mathsf{pool}}(b) \geq 0$, so both conditions hold in every case of the start size; and the j-th commitment of ctx lands at index $S^{\mathsf{pool}}_{\mathsf{start}}(b) + \mathsf{off}(ctx) + j$. (b) A size $S^{\mathsf{pool}}_{\mathsf{end}}(b) < n_{\mathsf{pool}}(b)$ fails the first condition. With a known start size, a size that differs from $S^{\mathsf{pool}}_{\mathsf{start}}(b) + n_{\mathsf{pool}}(b)$ fails the second. (c) A size 0 with $n_{\mathsf{pool}}(b) > 0$ fails the first condition. (d) Let the wallet not know the end size of $b-1$, and let b lie at or above the activation of pool, so that its start size is that of the third case of Definition 4.11. Adding one integer $\delta \neq 0$ to $S^{\mathsf{pool}}_{\mathsf{end}}(b)$ and to the end sizes of all later scanned blocks, such that every shifted size stays in $[0, 2^{32}]$ and at least the count of its block, preserves both conditions for b, whose start size is computed from its own end size, and for every later block, whose start and end sizes shift by δ together. □

Comparison of the running position with $S^{\mathsf{pool}}_{\mathsf{end}}(b)$ identifies the last commitment of pool in b, at position $S^{\mathsf{pool}}_{\mathsf{end}}(b) - 1$. The tree after that commitment, or for $n_{\mathsf{pool}}(b) = 0$ the tree at the end of $b-1$, is the tree state at the end of b. Its root is the anchor that a later spend against height b uses (*Ironwood Guide*, §"Anchors"; protocol specification, §"Transactions and Treestates"), and the wallet records it as a candidate anchor.

5 The light-client service

This section states the server from which a light wallet obtains chain data. It fixes the architecture and the one assumption on the server that the volume uses, states which checks a light client can make on the server's responses and what the server can fabricate or omit undetected, and defines the service as an abstract interface of typed queries on the chain, the note commitment trees, transactions, submission and transparent addresses.

5.1 Architecture and authority

Definition 5.1 (Light-client architecture)**.** A *light-client system* has three components:

1. a *full node*, which validates the chain and is the root of trust for chain state;
2. a *proxy*, which extracts from the node's view of the best chain the data of compact blocks (Definition 4.2) and answers queries;
3. a *light client*, which holds keys and no copy of the chain, queries the proxy, and maintains its own view of chain state and of its notes.

This volume calls the node and the proxy together the *server*. The wallet of "Parties, pools, and classifications" (§1.2) is the light client. The best chain is that of the *Consensus Guide*, §"Work, and the best chain" (ZIP 307, "High-Level Design").

Assumption 5.2 (Honest-but-curious server). The server satisfies two clauses.

(a) *Integrity.* It answers every query from a correct view of the current best chain, each response being the value that the interface of "The query interface" (§5.2) assigns to the query on that chain, and it transmits queries and responses faithfully.

(b) *Curiosity.* It records every query with its arguments, order and timing, every response, and the transport metadata of every connection, and computes arbitrarily on that record.

Every correctness result of this volume that relies on the server cites clause (a); the privacy results of §8 take clause (b) as the power of the adversary. No other server model is used. ZIP 307, "Security Model", states the same model: the node and proxy together are honest but curious, and are trusted to provide a correct view of the current best chain and to transmit queries and responses faithfully. Its goal is payment-detection privacy: the proxy does not learn which transactions are addressed to a given wallet and, given network privacy provided separately, does not link different connections or queries to one wallet; the goal is defined in "Payment-detection privacy" (§8.1). The ZIP does not address private spending.

Proposition 5.3 (Light-client checks)**.** *Let a wallet receive responses of the service (§5.2), with compact blocks that carry no header, which the abstract format does not guarantee (Definition 4.2). Then every check the wallet can make compares two responses, or a response with a value the wallet computes from responses. The relations among response fields that the format fixes are the following:*

(*i*) *height continuity and* prevHash *linkage of consecutive compact blocks, and of the first block of a range with the block the wallet has stored, the checks of ZIP 307, "Client-server interaction", phases B and D, and "Block header validation";*

(*ii*) *consistency of the per-block tree sizes with the commitments served (Lemma 4.12);*

(*iii*) *recomputation of the note commitment of each accepted candidate (Definition 4.9, Proposition 4.10);*

(*iv*) *agreement of a served tree node (a frontier, a subtree root, a root) with its recomputation from commitments the wallet has scanned;*

(*v*) *agreement of a fetched transaction with its queried identifier and with the compact records of that identifier (Remark "Checks of a fetched transaction");*

(*vi*) *equality of a height, block hash or transaction identifier that several responses report.*

No check relates a response to a block header or to proof of work: the wallet verifies no tree root, nullifier, transaction identifier or block hash against a header.

Proof. The fields of a compact block and of its elements (Definitions 4.2, 4.3 and 4.5) and the responses of the service are values of the chain or tree data derived from it; without hdr no response carries a header, so no field, not even hash or prevHash, which the header determines, can be checked against one. A check is a predicate on the wallet's inputs, which are these responses and its keys, so it compares responses with one another or with values computed from them. Items (i) to (vi) relate pairs of fields that the format makes functions of one another: the heights and hashes of adjacent blocks; a block's commitment count and its sizes; a record's commitment and its plaintext; a tree node and the leaves below it; a fetched transaction and its identifier and compact records; and a height, hash or identifier repeated across responses. Proof of work is a relation between a header and its solution and target (*Consensus Guide*, §"Equihash and the memoryless model"), which no response carries. □

ZIP 307, "Block header validation", proposes checks against a header when a compact block carries one; the abstract format does not guarantee a header, and no result of this volume uses them.

Corollary 5.4 (Fabrication of compact outputs).

(*a*) *Under clause* (*a*) *of Assumption 5.2, every compact candidate accepted by Definition 4.9 is a note of the best chain: its commitment is a leaf of its pool's note commitment tree at the position that Definition 4.11 assigns. The fields a compact record omits* (*proofs, signatures, value commitments, anchors, full ciphertexts*) *are checked by consensus for every transaction of the best chain, so delegating their verification to the server loses no check a light client could make.*

(*b*) *Without clause* (*a*)*, no check authenticates the compact fields against the chain. A server that knows one of the wallet's addresses can construct a note to it, compute its commitment and note ciphertext, choose the remaining compact fields* (nf *of an Action, which is* ρ *of the note*)*, and serve the result as a compact output or Action, with tree sizes consistent with it; the result passes Definition 4.9. Unless a transaction of the chain appends the same commitment, no authentication path from it to an anchor of the chain can be found except with negligible probability, so the wallet displays a note whose every spend is rejected.*

Proof. (a) By Proposition 4.10(a) an accepted candidate opens the commitment of its compact record. By clause (a) that record is the corresponding field of a transaction of the best chain, whose commitments are appended to the pool's tree in the order that fixes the position (*Ironwood Guide*, §"The Merkle hash and the tree", Definition "Note commitment tree"; Lemma 4.12(a)). The omitted fields are validated by the consensus rules for every transaction of the chain (*Ironwood Guide*, §"Transaction digests and signatures" and §"Anchors").

(b) Acceptance checks only the lead byte, the field ranges, the commitment recomputed from the plaintext and $\mathsf{epk} = [\mathsf{esk}]\,\mathsf{g_d}$ (Definition 4.9), and an honestly constructed note to an address of the wallet satisfies all of them; the tree-size checks of Lemma 4.12 pass on sizes the server chooses consistently, by part (d) of that lemma. A spend proves a valid path from the commitment to an anchor of the chain (*Ironwood Guide*, §"Authentication paths"), and a valid path from a value that is not the leaf at its position yields a collision of the Merkle hash, found with at most negligible probability (*Ironwood Guide*, §"Anchors", Proposition "Membership soundness"; protocol specification, §"Merkle Path Validity", for Sapling). ZIP 307, "Output Compression", grounds the commitment check on the same trust, in the entity that assembles transactions. □

Remark 5.5 (Omission is undetected). Completeness of the served data is trusted, not verified. The size check of Lemma 4.12 detects the omission of a commitment only while the served sizes are correct. A server that omits compact outputs or Actions and serves tree sizes, frontiers, subtree roots and roots consistent with the omission, or omits or alters revealed nullifiers, of which a compact block carries no count, passes every check of Proposition 5.3. Omitted outputs hide receipts. Omitted nullifiers hide spends: the wallet shows spent notes as spendable, and a respend of such a note is rejected for a repeated nullifier (*Ironwood Guide*, §"Nullifier sets") while it discloses to every party that receives it that it spends what an earlier transaction spent, the leak of repeated spends of ZIP 315, "Obtaining user consent for information leakage". Only clause (a) of Assumption 5.2 excludes omission (ZIP 307, "Security Model").

Remark 5.6 (Malformed responses). A response with a field outside its type (a nullifier not of 32 bytes, a non-canonical encoding of a field element, a malformed compact output or Action, a block outside the requested range) is an error of the response: the results of the query are not applied, and the query may be repeated. It is not a continuity error and triggers no rollback ("Reorganisation detection and recovery", §7.6). This rule is designed but unspecified (Definition 1.1).

Remark 5.7 (Checks of a fetched transaction). A full transaction fetched by its identifier SHOULD be checked against the corresponding compact outputs, in addition to verification of its signatures (ZIP 307, "Block header validation"). The checks are: the identifier computed from the fetched transaction equals the queried one (*Ironwood Guide*, §"Transaction digests and signatures", Definition "Transaction identifier"; protocol specification, §"Transaction Identifiers"); its outputs and Actions agree with the compact records; and its signatures verify over its signature digest (*Ironwood Guide*, §"Transaction digests and signatures", Definition "Signature digest"). They bind the full transaction to the compact data and its authorisation to its effects. They bind neither to the chain: compact blocks carry too little to check an identifier against the block's transaction tree, and that binding rests on clause (a) of Assumption 5.2.

5.2 The query interface

A wallet holds keys and no chain state (Definition 5.1), so the interface delivers what the protocols below consume: every compact output and Action of every shielded pool, for trial decryption (*Ironwood Guide*, §"Trial decryption and note acceptance"); every revealed nullifier, for spend detection (*Ironwood Guide*, §"Nullifier sets"); for each pool, tree sizes, frontiers and the roots of complete subtrees, for positions and authentication paths (*Ironwood Guide*, §"Authentication paths"); the full transactions that concern the wallet; submission of its transactions and their status; and, for transparent receivers, the transparent outputs paid to its addresses.

Definition 5.8 (Subtree root and completing height). Let $\mathsf{pool} \in \{\mathsf{Sapling}, \mathsf{Orchard}, \mathsf{Ironwood}\}$, whose note commitment tree has depth 32 and layers numbered from layer 0, the root, to layer 32, the leaves (*Ironwood Guide*, §"The Merkle hash and the tree", Definition "Note commitment tree"; protocol specification, §"Note Commitment Trees", for Sapling). For $0 \leq i < 2^{16}$, *subtree* i of the tree is the subtree whose leaves are the positions $[\, i \cdot 2^{16},\ (i+1) \cdot 2^{16})$; its *root* is the node M_i^{16}, sixteen layers above the leaves. When the tree holds n leaves, subtree i is *complete* if $n \geq (i+1) \cdot 2^{16}$, after which its root no longer changes; the complete subtrees are those with $i < \lfloor n/2^{16} \rfloor$. The *completing height* of a complete subtree i is the height of the block whose transactions append position $(i+1) \cdot 2^{16} - 1$.

Definition 5.9 (Transaction status). The *status* of a transaction identifier txid on the server's best chain, with tip height T, is one of:

1. $\mathsf{unknown}$: the server knows no transaction with identifier txid;
2. $\mathsf{unmined}$: the server knows the transaction, in its mempool or in a block off the best chain, and no block of the best chain contains it;
3. $\mathsf{mined}(h)$: the block of the best chain at height $h \leq T$ contains it.

The encoding of the status lies outside the abstract interface. The status is designed but unspecified (Definition 1.1).

Definition 5.10 (Light-client query interface). The *light-client service* is the set of typed queries of Table 4, each answered from the server's best chain under clause (a) of Assumption 5.2. A *block identifier* is a height or a 32-byte block hash. A *pool set* is a set $\mathcal{P}$ of the pool selection of "The compact block and the compact transaction" (§4.1): shielded pools, with transparent data selectable alongside. The service serves every pool set and transparent data. A query

Query	Arguments	Response
Chain		
Tip()	none	(h, hash) of the tip of the best chain; the tip input of every confirmation count (§9)
Block(b)	block identifier b	the compact block of b, with all shielded pools and transparent data
Blocks($s, e, \mathcal{P}$)	heights $s \leq e$; pool set $\mathcal{P}$	the compact blocks at heights $s, s+1, \ldots, e$ in ascending order, restricted to $\mathcal{P}$; every height of $[s, e]$ present, a block left with no transaction carrying an empty list
Trees		
TreeState(b)	block identifier b	height and hash of b, and for each shielded pool the size n and the frontier of its tree in the final treestate of b; an empty frontier denotes the empty tree
SubtreeRoots(pool, i)	shielded pool pool; start index i	in index order, the pair (root, completing height) of every complete subtree of the tree of pool with index at least i
Transactions		
Tx(txid)	32-byte identifier	the full serialised transaction
Submit(*bytes*)	serialised transaction	accepted(txid) if the server's node admits it to its mempool, else rejected
Status(txid)	32-byte identifier	the status of Definition 5.9
Transparent		
Utxos(A)	set A of transparent addresses	the unspent outputs of the best chain paying to A, each with outpoint, value, script and height

Table 4: Queries, arguments, and responses of the light-client service (Definition 5.10). Each response is the value on the server's best chain; $\perp$ answers an argument that names nothing on it.

> whose argument names no block of the best chain, or a transaction that the server does not know, is answered with $\perp$. Queries of the mempool and of server metadata are not part of the interface.

The frontier in a response of TreeState is that of the *Crypto Guide*, §"Append-only and incrementally updatable trees", Definition "Frontier": for a tree of n leaves, the left-sibling nodes along the path of the next leaf, one for each set bit of n, from which the root and every later append are computed. The empty frontier is the state before the pool's first commitment. The final treestate of a block is that of the protocol specification, §"Transactions and Treestates" (*Ironwood Guide*, §"Anchors", Definition "Treestate and anchor"). The serialisation of the frontier is fixed by no ZIP and is not restated. ZIP 307, "Proxy operation", also identifies a transaction by its block and its index in the block; the interface uses the identifier alone.

Remark 5.11 (Classification of the queries). ZIP 307, "Proxy operation", specifies the queries Tip, Block, Blocks and Tx. No ZIP specifies TreeState, SubtreeRoots, Submit, Status or the transparent query Utxos, and ZIP 307, "Client-server interaction", leaves the acquisition of a starting tree state open: "Or, it has to set X to the block height at which Sapling activated, so as to be sent the entire commitment tree. [TODO: Decide which to specify.]" These queries, and pool selection in Blocks, are designed but unspecified (Definition 1.1).

Remark 5.12 (Minimal query set). A shielded synchronisation uses Tip, SubtreeRoots for each shielded pool, Blocks and TreeState; the full transactions that concern the wallet add Tx. A wallet that watches transparent receivers adds Utxos, whose argument names its addresses in the clear ("Key independence of queries", §8.2). This set is the requirement of the synchronisation protocols of "Note discovery" (§6) and "Commitment-tree synchronisation" (§7), not a deployed fact.

6 Note discovery

This section states the protocol by which a wallet reconstructs its notes from the chain. It fixes the chain state and the account birthday that bound the scan, the keys an account is scanned with under the rules of ZIP 326, the note state of a wallet and the scan of one compact block that updates it, the continuity predicate and the conditions under which a scan in any order is correct, the theorems of sound and complete discovery and of recovery from the seed, and the work that scanning costs. The tree information a wallet retains for its witnesses is the subject of "Commitment-tree synchronisation" (§7).

Throughout, P ranges over the shielded pools Sapling, Orchard and Ironwood; H_{S} is the Sapling activation height, and $H_{\mathsf{NU6.3}}$ the NU6.3 activation height of ZIP 258. For a compact block b, $n_P(b)$ and $S^P_{\mathsf{end}}(b)$ are the commitment count and the end size of Definition 4.11.

6.1 The account birthday

Definition 6.1 (Chain state)**.** For a height h, the *chain state* at h is

$$C(h) := (h,\ \mathsf{hash}_h,\ (m_P(h), F_P(h))_{P\in\{\mathsf{Sapling},\mathsf{Orchard},\mathsf{Ironwood}\}}),$$

where hash_h is the 32-byte hash of block h; $m_P(h) \in [0, 2^{32}]$ is the number of leaves of the note commitment tree of pool P in the final treestate of block h; and $F_P(h)$ is the frontier of that tree (*Crypto Guide*, §"Append-only and incrementally updatable trees", Definition "Frontier"): for each set bit t of $m_P(h)$, the root of the completed left subtree of height t on the rightmost filled path. The empty tree has $m_P(h) = 0$ and the empty frontier, and the full tree, $m_P(h) = 2^{32}$, has its root as frontier. The three trees are distinct trees of depth 32 ($\mathsf{MerkleDepth}^{\mathsf{Sapling}} = \mathsf{MerkleDepth}^{\mathsf{Orchard}} = 32$; protocol specification, §"Constants" and §"Note Commitment Trees"): the Sapling tree of the specification, and the Orchard-pool and Ironwood-pool trees, each built with the Orchard Merkle hash (*Ironwood Guide*, §"The Merkle hash and the tree", Definition "Note commitment tree"). The chain state $C(h)$ is the response of $\mathsf{TreeState}(h)$ (Table 4).

By the *Crypto Guide*'s Definition "Frontier" (§"Append-only and incrementally updatable trees"), the pair $(m_P(h), F_P(h))$ determines the root of the tree of P after block h and suffices to append every later leaf without any leaf appended before block $h+1$. That it also yields every later authentication path is shown in "Frontiers as scan seeds" (§7.3).

Definition 6.2 (Account birthday)**.** The *birthday* of an account is a height B, known to the wallet, that is a lower bound on the height of the first block in which the account could have received funds (ZIP 326, "Terminology"). With the birthday the wallet stores the chain state $C(B-1)$, which summarises every note commitment of every pool before B. The scan of the account starts at block B, and no block below B is scanned for it. The birthday of a wallet is the minimum of the birthdays of its accounts.

(i) *Added accounts.* Let an account with birthday B' be added while the greatest height of a block the wallet has scanned exceeds B'. Every block of $[B', \text{that height}]$ is then scanned again with the scanning keys of the new account (§6.2), since earlier scans did not try them; the results of the other accounts are unchanged.

(ii) *No recorded birthday.* A wallet that restores a seed without a recorded birthday takes $B := H_{\mathsf{S}}$: no shielded address of a light client can have received a relevant transaction before Sapling activation (ZIP 307, "Client-server interaction", "Importing a pre-existing

seed"). The bound concerns shielded receipts only; the transparent history of the account's addresses may lie below it.

Remark 6.3 (Choosing a birthday). Let the tip be at height T when an account is created, and let d be at least the wallet's assumed maximum reorganisation depth. The birthday of the new account is taken as $B \leq T - d + 1$, with stored chain state $C(B-1)$ at height $T - d$ or below. A reorganisation that replaces at most d blocks re-mines no transaction below height $T-d+1$; with B closer to the tip, such a reorganisation could re-mine a transaction paying the account at a height below B, which the wallet never scans, and the payment would be missed. Consensus imposes no maximum reorganisation depth (*Consensus Guide*, §"Reorg rails, maturity, and the finality floor"), so d is a wallet parameter, not a guarantee. The rollback window of 600 blocks that ZIP 218, "Block-count-based constants", recommends concerns nodes and bounds no wallet's choice of d. An earlier birthday is always safe for discovery and costs scanning work (§6.5).

6.2 Scanning keys

Definition 6.4 (Scanning keys of an account)**.** Let an account a have Sapling keys and Orchard-protocol keys derived as in "Hierarchical key derivation (ZIP 32)" (§2), the latter with the flag $\mathsf{use_qsk}$ of Construction 2.10. Scopes are $s \in \{\mathsf{ext}, \mathsf{int}\}$.

(i) *Sapling.* For each scope s, the pair $(\mathsf{ivk}^{\mathsf{S}}_s, \mathsf{nk}^{\mathsf{S}}_s)$ of incoming viewing key and nullifier deriving key (ZIP 32, "Sapling internal key derivation"). Sapling outputs are scanned from B.

(ii) *Orchard protocol.* For each scope s and each $u \in \{\mathsf{false}, \mathsf{true}\}$, the incoming viewing key $\mathsf{ivk}_{s,u}$ of scope s computed with $\mathsf{use_qsk} = u$, and the nullifier deriving key nk, the same in both branches of Construction 2.10 and in both scopes (Construction 2.14).

A receiver and its ivk are scoped to the Orchard protocol, not to a pool: the same ivk trial-decrypts Orchard-pool and Ironwood-pool note ciphertexts (ZIP 326, "Scanning for Ironwood-pool and Orchard-pool notes"); the pool of an Action is that of the bundle that carries it, and Definition 4.9 admits lead byte `0x02` in the Orchard pool and `0x03` in the Ironwood pool. The *use_qsk set* U_a is $\{u\}$ when the wallet knows the account's value u and $\{\mathsf{false}, \mathsf{true}\}$ when it does not.

The following rules of ZIP 326, with their normative levels, fix which ivk is tried on which output.

(R1) *Restoration.* A wallet restoring from a seed MUST take $U_a = \{\mathsf{false}, \mathsf{true}\}$ and scan with both values until rule (R2) narrows it, because $\mathsf{use_qsk}$ is not part of the recovery information ("Scanning when restoring from seed").

(R2) *Narrowing.* Once the wallet sees funds received in either Orchard-protocol pool under a value u, it SHOULD stop scanning with the other value, $U_a := \{u\}$. The evidence stands if the transaction that provided it is later reorganised away, since it was valid on some chain ("Scan ranges").

(R3) *Ironwood pool.* The Ironwood pool is scanned from $\max(B, H_{\mathsf{NU6.3}})$ to the tip with $\mathsf{ivk}_{s,u}$ for $u \in U_a$, hence with up to four keys ($\{\mathsf{ext}, \mathsf{int}\} \times \{\mathsf{false}, \mathsf{true}\}$) while $\mathsf{use_qsk}$ is unknown, subject to rule (R4). The value $u = \mathsf{true}$ is kept for an account whose birthday precedes NU6.3 activation, since a recorded birthday may be conservatively early ("Scan ranges").

(R4) *First funds.* ZIP 326 describes this rule as an optimisation and does not require it ("First-funds scanning in the Ironwood pool"). While the wallet knows that the account has received no Ironwood-pool funds in any block below the one scanned, only the external keys $\mathsf{ivk}_{\mathsf{ext},u}$ are needed for an output, except in a transaction whose valueBalanceIronwood is negative, where both scopes are used. The knowledge is kept only for the contiguous prefix of heights from $\max(B, H_{\mathsf{NU6.3}})$ scanned without a receipt, and a reorganisation whose last unchanged block is at height h_f cuts the prefix back to $\max(h_f, H_{\mathsf{NU6.3}})$. A compact transaction carries no value balance (Definition 4.3), so a scan of compact blocks applies the exception only where it obtains valueBalanceIronwood otherwise; without it, both scopes are used.

(R5) *Orchard pool* ("Scan ranges"). Orchard-pool note ciphertexts SHOULD NOT be scanned with use_qsk = true keys. For an account with birthday after NU6.3 activation they SHOULD NOT be scanned at all. For an account with birthday before NU6.3 activation they SHOULD NOT be scanned with the external use_qsk = false key after NU6.3 activation. If the account's Orchard-pool balance is known to be zero at a block at or after NU6.3 activation, they SHOULD NOT be scanned in any block that descends from it.

(R6) *Order.* A wallet MAY scan blocks in any order ("First-funds scanning in the Ironwood pool").

(R7) *Outgoing notes.* Outgoing notes are detected with the ovk of the same use_qsk value, which adds no key dimension ("Outgoing-note detection").

Under (R5) the Orchard pool is scanned with use_qsk = false only: with $\mathsf{ivk}_{\mathsf{int,false}}$ from B until the Orchard-pool balance is known to be zero after NU6.3 activation, and with $\mathsf{ivk}_{\mathsf{ext,false}}$ from B to $H_{\mathsf{NU6.3}} - 1$. The *scanning keys* $K_a(P, b, t)$ of account a for pool P, block b and transaction t are the triples $(\mathsf{ivk}, \mathsf{nk}, s)$ that items (i) and (ii) and rules (R1) to (R5) select. Every ivk is labelled (a, s), which attributes each accepted note to its account and scope.

Transparent receipts are detected at the transparent addresses the wallet watches, which MUST include index 0 of each account ("Deriving unified addresses", §3.6; ZIP 316, "Deriving a Unified Address from a UIVK"). The Ironwood pool is scanned as a pool of its own: its outputs are accepted under its lead byte, its revealed nullifiers are matched only against nullifiers of Ironwood-pool notes, and its tree is its own (*Ironwood Guide*, §"Nullifier sets", Remark "Pool locality").

Remark 6.5 (Omitted keys and conformant senders). Rules (R2), (R4) and (R5) omit a key only where the restrictions of ZIP 326, "Wallet key-generation restrictions", exclude an output of the account under it. Every key of an account has one use_qsk value (MUST). No wallet sends Orchard-pool funds to use_qsk = true keys, or to any external receiver after NU6.3 activation (MUST NOT). Internal receivers are exposed to no party outside the wallet (ZIP 326, "Scanning for Ironwood-pool and Orchard-pool notes"), so before the account's first Ironwood-pool funds a note at its internal Ironwood receiver arises only from the account's own transaction, funded by a negative valueBalanceIronwood ("First-funds scanning in the Ironwood pool"). No value enters the Orchard pool after NU6.3 activation, so a zero Orchard-pool balance stays zero ("Scan ranges"). An output created in violation of these restrictions by a non-conformant sender may be missed; this is the cost of the SHOULD NOT rules, and it is part of the statement of Theorem 6.14.

Proposition 6.6 (Spend detection requires the nullifier key)**.** *The nullifier of an Orchard-protocol note is* $\mathsf{DeriveNullifier}_{\mathsf{nk}}(\rho, \psi, \mathsf{cm})$ (Ironwood Guide, *§"The nullifier", Definition "Nullifier"), and that of a Sapling note is* $\mathsf{PRF}^{\mathsf{nfSapling}}_{\mathsf{nk}^\star}(\rho^\star)$ *with* $\rho^\star = \mathsf{repr}_{\mathbb{J}}(\mathsf{MixingPedersenHash}(\mathsf{cm}, \mathsf{pos}))$*, a function of the nullifier deriving key and of the note's position* (*protocol specification, §"Computing* ρ *values and*

Nullifiers").

(*a*) *Let an Orchard-protocol account key have* use_qsk = false. *No efficient algorithm that receives the incoming viewing key* (dk, ivk) *of one scope and* $\mathsf{ak}^{\mathbb{P}}$, *but not* nk, *and is then given notes of its choice with pairwise distinct* ρ *outputs the nullifier of one of them, except with negligible probability.*

(*b*) *Assume that* $\mathsf{PRF}^{\mathsf{nfSapling}}$, *keyed by* $\mathsf{nk}^{\star}$ *of a Sapling key generated as the specification prescribes, is pseudorandom to an efficient algorithm that holds the incoming viewing key and the spend validating key of that key: the specification's requirement that* $\mathsf{PRF}^{\mathsf{nfSapling}}$ *be a pseudorandom function* (§*"Pseudo Random Functions"*), *applied to this key distribution. Then the statement of* (*a*) *holds for the Sapling key.*

A holder of (dk, ivk) *therefore detects receipts, since trial decryption needs only* ivk (*Construction 4.8*), *and cannot recognise a revealed nullifier as that of its note. A wallet that holds only incoming viewing keys has an empty tracked-nullifier set, and the set of notes it considers unspent is a superset of the true set. Links through other public data are outside the statement* (Ironwood Guide, §*"Nullifier uniqueness and unlinkability", Remark "Limits"*). *For* use_qsk = true *no claim is made* (*Remark 2.11*).

Proof. (a) Let $\mathcal{C}$ be an algorithm that outputs the nullifier of a submitted note with probability ε. It yields a distinguisher for the *Ironwood Guide*'s Proposition "Nullifier unlinkability" (§"Nullifier uniqueness and unlinkability"): the distinguisher runs $\mathcal{C}$ on its inputs (dk, ivk) and $\mathsf{ak}^{\mathbb{P}}$, submits the notes $\mathcal{C}$ is given, and outputs 1 exactly when an answer equals the value $\mathcal{C}$ outputs for that note. In the real game it outputs 1 with probability at least ε. In the ideal game each answer is $\mathsf{Extract}_{\mathbb{P}}$ of a point uniform on $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$, independent of the notes, except with probability $1/p_{\mathsf{Vesta}}$; it equals a given value with probability at most $2/\#\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$, since $\mathsf{Extract}_{\mathbb{P}}$ is at most two-to-one. Hence ε is bounded by the advantage bound of that proposition plus a negligible term, under the assumptions it names. For the internal scope the proof of that proposition applies with its hybrid H_2 changed: the pair $(\mathsf{ivk}_{\mathsf{int}}, \mathsf{dk}_{\mathsf{int}})$ is replaced by $(\mathsf{Commit}^{\mathsf{ivk}}_{r'}(\mathsf{ak}, \mathsf{nk}), u_3)$ by Assumption 2.16, whose reduction, like that of H_2, draws ak and nk itself and passes on the fourth and fifth components of its tuple. The cost is the advantage against that assumption in place of the term for the *Ironwood Guide*'s Assumption "Joint hiding of the viewing keys" (§"Viewing keys"), and the later hybrids are unchanged, since the game after the hop is that of H_2. That proposition concerns a key generated from a uniform spending key; for an account key the bound holds up to the loss of Lemma 2.9. (b) An algorithm that outputs $\mathsf{PRF}^{\mathsf{nfSapling}}_{\mathsf{nk}^{\star}}(\rho^{\star})$ for a submitted $\rho^{\star}$ not queried before distinguishes the function from a random function, which outputs that value with probability 2^{-256}, contradicting the assumption. □

6.3 Receipt and spend detection

Definition 6.7 (Note state of a wallet)**.** The *note state* of a wallet consists of:

(i) per account a and pool P, the set of *detected notes*, each with its value, scope, position pos (Definition 4.11), creating transaction, its nullifier when derivable, its memo once fetched, and a *spent mark*: either unspent or the triple (height, index in block, txid) of the transaction that revealed its nullifier;

(ii) per pool P, the *tracked set* W_P of the nullifiers of detected notes of P not marked spent, and the set R_P of *retained unlinked nullifiers*, each with its revealing transaction;

(iii) for each wallet-relevant transaction, its mined height, or $\perp$ if it is unmined;

(iv) the chain state (Definition 6.1) of the last block of each maximal interval of scanned heights; for a scan in increasing height, that of the last scanned block.

A transaction is *wallet-relevant* if and only if it creates a detected note or reveals the nullifier of one. A light wallet stores no copy of the chain; its note state is reconstructed from compact blocks (Definition 4.2) and fetched full transactions alone, supplied by a server that the wallet does not trust for privacy (Assumption 5.2, clause (b)). The information retained for witnesses is adjoined in "The information a wallet retains" (§7.4).

Construction 6.8 (Scanning a block)**.** Let b be a compact block that continues the wallet's chain, in the sense fixed in "Scan order" (§6.4), scanned for a set A of accounts. For each pool P, in block, transaction and description order, the start size of P in b (Lemma 4.12) and Definition 4.11 give each compact output or Action its position.

1. *Receipts.* For each compact Sapling output (for $P = \mathsf{Sapling}$) and each compact Action of P (for an Orchard-protocol pool), in transaction t, and each account $a \in A$, run Construction 4.8 and Definition 4.9 under every $(\mathsf{ivk}, \mathsf{nk}, s) \in K_a(P, b, t)$ (Definition 6.4). Record an accepted note with its value, scope s, position and transaction. When nk is held, compute its nullifier nf (for Sapling from its position), and: if $\mathsf{nf} \in R_P$, mark the note spent at the retained transaction and remove nf from R_P; otherwise add nf to W_P.

2. *Spends.* For each nullifier revealed in P by b (each compact Sapling spend, and the nf of each compact Action of P): if it lies in W_P, mark the matching note spent at $(\mathrm{height}(b), \mathrm{index}, \mathsf{txid})$ and remove the nullifier from W_P; otherwise retain it in R_P with its revealing transaction.

3. *Chain state.* Compute $C(\mathrm{height}(b))$ from the stored $C(\mathrm{height}(b) - 1)$ and b: the hash of b, and for each pool the size $S^P_{\mathsf{end}}(b)$ and the frontier obtained by appending the commitments of P in b, in order, to $F_P(\mathrm{height}(b) - 1)$. It replaces $C(\mathrm{height}(b) - 1)$ in the note state.

Step 1 precedes step 2, so a receipt and its spend in one block, or in one range scanned in increasing height, are both found in one pass. A nullifier is compared only with nullifiers of its own pool (*Ironwood Guide*, §"Nullifier sets"). The retention period of R_P is fixed by Lemma 6.12 (§6.4).

Enhancement. For each wallet-relevant transaction the wallet may fetch the full transaction by $\mathsf{Tx}(\mathsf{txid})$ (phase C of ZIP 307, "Client-server interaction"), which yields the memo, by decryption of the full C_{enc}, and the outgoing plaintexts recoverable under the account's ovk, at the privacy cost stated in "Leakage of transaction queries" (§8.4).

Class (Definition 1.1). Trial decryption, nullifier matching in increasing height and enhancement are specified (ZIP 307, "Local processing", "Detecting spends" and "Client-server interaction"); the scanning keys are those of ZIP 326. The retained set R_P and the chain state of step 3 are designed but unspecified.

6.4 Scan order

Remark 6.9 (The interaction of ZIP 307). ZIP 307, "Client-server interaction", divides the interaction between client and server into four phases.

(A) *First start.* The client obtains the tip height X and the commitment tree state at X, or sets $X := H_{\mathsf{S}}$; the ZIP leaves the choice open. For an imported seed $X := H_{\mathsf{S}}$ ("Importing a pre-existing seed").

(B) *First update.* The client obtains the tip Y and the compact blocks $X, \ldots, Y$. It checks that the hash of block X equals its stored copy, a mismatch meaning that X was orphaned by a reorganisation, and discards block X without further processing. For each later block it validates the header if one is present, trial-decrypts for its notes, creates and updates witnesses of detected notes, and scans for their nullifiers.

(C) *Transactions.* At user direction, the client fetches full transactions.

(D) *Later updates.* The client obtains the tip Z and the blocks $Y, \ldots, Z$, rechecks block Y as in (B), updates witnesses, marks the notes whose nullifiers appear as spent at that height, and scans for new notes as in (B).

In the queries of Table 4 the phases use Tip, TreeState or Block, Blocks and Tx.

Scanning a range $[h, h']$ requires the chain state $C(h-1)$: its hash for the continuity check of block h, and each pool's size and frontier for positions and authentication paths. The wallet holds it after scanning block $h-1$, or obtains it by $\mathsf{TreeState}(h-1)$; a server-supplied frontier is accepted under clause (a) of Assumption 5.2 and is checked against no header (Proposition 5.3).

Definition 6.10 (Continuity predicate)**.** Let the wallet hold the chain state $C(h) = (h, \mathsf{hash}_h, (m_P(h), F_P(h))_P)$. A compact block b *continues the wallet's chain at* h if and only if

(i) $\mathsf{height}(b) = h + 1$;

(ii) $\mathsf{prevHash}(b) = \mathsf{hash}_h$; and

(iii) for every pool P, the start size of P in b, $S^P_{\mathsf{end}}(b) - n_P(b)$ (Lemma 4.12), equals $m_P(h)$.

Within a range each block is checked against the chain state computed from its predecessor, and the first block against the stored chain state. When a range is scanned directly below an already scanned block $h' + 1$, that block is checked again against the chain state $C(h')$ that the new range computes. This extends ZIP 307's single hash comparison at the start of each phase to every block and to both sides of every range boundary.

Rule (*verify first*). The scan results of a block are applied to the note state only after the block is checked to continue the wallet's chain, and the results of a range only after each of its blocks is. Failure of the predicate is a *continuity error*, handled by rollback ("Reorganisation detection and recovery", §7.6). A malformed or insufficient field is an error of the response, not a continuity error ("Architecture and authority", §5.1). The predicate beyond ZIP 307's comparison is designed but unspecified (Definition 1.1).

Definition 6.11 (Scanned heights)**.** For a wallet with birthday B, the *maximum scanned height* is the greatest height of a scanned block, and the *fully scanned height* f is the greatest $h \geq B - 1$ such that every block of $[B, h]$ has been scanned. Under out-of-order scanning f may lie below the maximum scanned height.

Lemma 6.12 (Out-of-order spend detection)**.** *Fix a chain. Let a note of an account in pool P be created at height h_n and its nullifier revealed at height $h_s \geq h_n$, and let the wallet hold the note's nullifier deriving key. If each nullifier retained in R_P (Construction 6.8) that was revealed at height h is kept until the fully scanned height is at least h, the note is marked spent at the revealing transaction whichever of the blocks h_n and h_s is scanned first. The retention bound is the least that guarantees this:*

if a nullifier revealed at h is discarded while $f < h$, some block below h is unscanned, and a note of the account created there and spent at h is never marked spent.

Proof. If block h_n is scanned no later than block h_s, the note's nullifier is in W_P when block h_s is scanned (steps 1 and 2 of Construction 6.8, in that order when $h_n = h_s$), and it is matched on arrival. Otherwise block h_s is scanned first; the nullifier matches no tracked note and is retained in R_P. Since $B \leq h_n \leq h_s$ (Definition 6.2), f reaches h_s only after block h_n is scanned, so the nullifier is still in R_P when step 1 finds the note, and the note is marked spent. Minimality: block h has been scanned, since it revealed the nullifier, so $f < h$ means that some $h' \in [B, h-1]$ is unscanned; a note created at h' and spent at h has then lost the only record of its spend. □

Lemma 6.13 (Order independence of scanning)**.** *Fix a chain and a range $[B, T]$. Let a schedule scan every block of $[B, T]$ once, in any order and in ranges of any lengths, each block b with key sets $K_a(P, b, t)$, and let*

- (a) *the results of each range be applied only after continuity is verified (Definition 6.10), and*
- (b) *unlinked nullifiers be retained as Lemma 6.12 requires.*

Then the resulting notes, positions, nullifiers and spent marks equal those of a scan of $[B, T]$ in increasing height that applies to each block the same key sets, and the chain state held for height T is $C(T)$. The key sets of Definition 6.4 depend on the order only through rules (R2), (R4) and the last clause of (R5), whose omitted keys accept no output of the account under the restrictions of ZIP 326; under those restrictions every order of Definition 6.4 yields the note state of the scan in increasing height. The witness half of order independence is stated in "Witness readiness" (§7.5).

Proof. Receipt detection of an output is local to its block: it depends only on the compact output and the key set (Construction 4.8, Definition 4.9). Positions are local: by Lemma 4.12 the start size of each pool in b follows from the compact block b alone, and (a) makes it agree with the end size of the predecessor. Nullifiers are functions of the note, the key and, for Sapling, the position. Spent marks follow from Lemma 6.12 in either order. The chain state at T is computed by step 3 of Construction 6.8 from the chain state below the last range that ends at T, which (a) makes the true one, by induction over the ranges. For the last clause, a key omitted by (R2), (R4) or the last clause of (R5) in one order and tried in another accepts no output of the account (the Remark on omitted keys above; for (R5), no value enters the Orchard pool after NU6.3 activation, so a zero Orchard-pool balance stays zero), so both orders accept the same outputs. □

Theorem 6.14 (Soundness and completeness of note discovery)**.** *Assume clause (a) of Assumption 5.2, and let the server's best chain be unchanged while an account a with birthday B is scanned by Construction 6.8 with the scanning keys of Definition 6.4, under conditions (a) and (b) of Lemma 6.13. Let f be the fully scanned height. For every $h \leq f$, restricted to notes created at heights at most h and to spent marks set by nullifiers revealed at heights at most h:*

- (i) *the note state holds a note of a in pool P at position p if and only if the best chain has, at or below h, an output of P at position p that Definition 4.9 accepts under a key of $K_a(P, b, t)$ for its block b and transaction t;*
- (ii) *when the wallet holds the note's nullifier deriving key, the note is marked spent if and only if its nullifier is revealed in pool P of the best chain at or below h, and, except with negligible probability, a revealed nullifier equal to it marks a spend of that note and of no other.*

Completeness is over accepted outputs under the keys that Definition 6.4 applies, not over all outputs addressed to the account: an output that its sender encrypted malformed, one that acceptance rejects, and one that a non-conformant sender created under a key omitted by the rules of ZIP 326 are excluded. Ranges scanned out of order above f add notes and spent marks above h, which the statement does not cover. A holder of incoming viewing keys alone obtains (i) only (Proposition 6.6). The assumptions of (ii) are those named in the results cited in the proof.

Proof. By clause (a) the scanned compact blocks are those of the best chain, and every output of each is trial-decrypted under the applicable keys (Construction 6.8). Soundness of acceptance is Proposition 4.10, and membership of the accepted note in the best chain is Corollary 5.4(a). Positions are exact by Lemma 4.12(a) and Definition 6.10. The result does not depend on the order, by Lemma 6.13; this gives (i).

For (ii), spent status is nullifier matching within P, with spends found in either order by Lemma 6.12. For Orchard-protocol notes, one nullifier per note and distinct nullifiers for distinct notes are the *Ironwood Guide*'s Lemma "One nullifier per note" (§"Double-spend resistance") and Proposition "Nullifier uniqueness" (§"Nullifier uniqueness and unlinkability"), under the assumptions they name. For Sapling notes they are the Spend statement, which fixes the nullifier of the consumed note (protocol specification, §"Spend Statement (Sapling)"), and the requirement that $\mathsf{PRF}^{\mathsf{nfSapling}}$ be collision-resistant across all keys (§"Pseudo Random Functions"), with $\rho^\star$ distinct for distinct positions (§"Computing ρ values and Nullifiers"). Without clause (a), Remark 5.5 defeats completeness and Corollary 5.4(b) defeats soundness. □

Theorem 6.15 (Recovery from the seed)**.** *Let an account have* $\mathsf{use_qsk} = \mathsf{false}$ *and a valid account key ("Valid account keys", §2.6), so that its keys are determined by the seed S and the account index (Proposition 2.12). Under clause (a) of Assumption 5.2, scanning the best chain from a lower bound on the account's birthday, or from H_S when none is recorded (Definition 6.2), with the scanning keys of Definition 6.4, in particular with $U_a = \{\mathsf{false}, \mathsf{true}\}$ by (R1) until (R2) narrows it, reconstructs the note state of the account up to the fully scanned height, as characterised by Theorem 6.14. Memos, and the outgoing plaintexts of outputs that the account encrypted under its* ovk*, are recovered once the wallet-relevant transactions are fetched. Transparent funds are recovered at the addresses the wallet watches: index 0 of each account by obligation (§3.6); how far beyond index 0 a wallet looks is fixed by no ZIP and is designed but unspecified (Definition 1.1). Data that never reached the chain, such as the memos of transactions never mined, is not recoverable. An account with* $\mathsf{use_qsk} = \mathsf{true}$*, whose keys S does not determine, is outside the theorem, although Definition 6.4 scans with both values when restoring.*

Proof. Proposition 2.12 gives the keys, hence the scanning keys. Definition 6.2 makes the scan start at or below the first receipt. Theorem 6.14 gives the note state. Memos and outgoing plaintexts are functions of the fetched full transactions and the keys, and under clause (a) Tx returns the transactions of the best chain. □

Remark 6.16 (A reference interaction). The following interaction of a wallet that synchronises by ranges out of order is designed but unspecified (Definition 1.1). Per synchronisation the wallet runs:

1. $\mathsf{SubtreeRoots}(P, 0)$ for each pool, storing the roots of the complete subtrees;
2. $\mathsf{Tip}()$;
3. selection of ranges: those near the tip, and those completing the subtrees of found notes (§7.5), before historic ranges;

4. for each range $[s, e]$: $\mathsf{TreeState}(s-1)$ unless $C(s-1)$ is held, then $\mathsf{Blocks}(s, e, \mathcal{P})$, the continuity check and the scan;
5. a return to step 2 when a scan changes the selection: a newly found note, or a rollback after a continuity error.

The order is a wallet policy fixed by no ZIP, and it is observable by the server ("Leakage of range queries", §8.3).

6.5 The cost of scanning

Trial decryption costs one key agreement per compact output or Action per candidate ivk, for every block from the birthday to the tip, independently of the wallet's own activity. Compaction (Definition 4.5) reduces the bandwidth, not the number of trials. Restoring an account repeats this work from its birthday. Retrieval of full transactions and the maintenance of authentication paths scale with the wallet's own notes, the latter linearly in its unspent notes.

> **Proposition 6.17** (Scanning work per target day). *At NU7's target spacing of* 25 *seconds* (*ZIP 218, "Block target spacing"*; Consensus Guide, *§"Zcash time: 25-second blocks"*), *a target day of* 86,400 *seconds holds* 3456 *blocks. A block holds at most* 330 *Orchard-pool Actions* (*ZIP 218, "Shielded pool action limits"*), *so a target day holds at most* 1,140,480 *compact Orchard-pool Actions, with at most* 168,791,040 *bytes of their compact payload at* 148 *bytes each* (*Definition 4.5*), *and requires at most* 1,140,480 *trial decryptions of Orchard-pool Actions per candidate* ivk; *the number of candidate keys of an account is fixed by Definition 6.4. The same bounds hold for Orchard-pool Actions, Sapling spends and Sapling outputs together, since the shared budget of the same section limits their sum to* 330 *per block, and a compact Sapling output* (116 *bytes*) *or spend* (32 *bytes*) *is smaller than a compact Action. The figures count target blocks: the number of blocks mined in a day varies, and the target spacing is not a rate limit.*
>
> *The bound covers no Ironwood-pool Action: ZIP 218 counts none against any limit, so the specified text bounds neither their number nor the trial decryptions they cost. A proposed amendment would charge them to the shared budget of* 330 *units, under which the same figures would bound the Orchard-protocol total* (Consensus Guide, *§"A block budget for shielded work"*). *The Ironwood term is an open problem* (*Definition 1.1*).

Proof. The counts are $86{,}400/25 = 3456$, $330 \cdot 3456 = 1{,}140{,}480$ and $148 \cdot 1{,}140{,}480 = 168{,}791{,}040$. The per-block limits are $\mathsf{OrchardBlockActionLimit} = 330$ and $\mathsf{GlobalShieldedBudget} = 330$ of ZIP 218, "Shielded pool action limits". Each compact Action, and each compact Sapling output, costs one trial decryption per candidate key (Construction 4.8); a compact Sapling spend costs none. □

7 Commitment-tree synchronisation

This section states the tree information that a wallet retains in order to spend its notes. It shows that the witness maintenance of ZIP 307 forces a scan from each note to its anchor; decomposes each note commitment tree at level 16 into subtrees and a cap built from downloaded subtree roots; shows that a frontier seeds every later witness; defines the information a wallet retains and proves it sufficient; states exactly when a witness is computable; and states the detection of reorganisations and the recovery from them, with the bound on the rollback height, the safety theorem and the correctness theorem of synchronisation.

Throughout, P ranges over the shielded pools Sapling, Orchard and Ironwood, with the chain state $C(h)$ of Definition 6.1, the birthday B of Definition 6.2, and the commitment count $n_P(b)$ and end size $S^P_{\mathsf{end}}(b)$ of a compact block b (Definition 4.11). Assumption 5.2 is the server assumption; only its clause (a) is used here.

7.1 Witness maintenance by linear scanning

Remark 7.1 (What a spend needs). A detected note is not yet spendable. A spend needs, besides the note, its position p and the authentication path of p to the anchor of a block at or after the note's block (*Ironwood Guide*, §"Authentication paths", Construction "Authentication path"; anchors as in §"Anchors", Definition "Treestate and anchor"). By the *Ironwood Guide*'s Lemma "Authentication paths from public data" (§"Authentication paths"), that path is a function of p and of the first n leaves of the pool's tree, n being the tree size at the anchor. Path construction and anchor selection are cited, not re-derived; the anchor height is fixed in "Anchors, confirmations, and expiry" (§9). The lemma is stated for the trees of the Orchard protocol; its proof uses only the recurrence of the tree and applies unchanged to the Sapling tree, which has the same depth 32 and the same layer convention (protocol specification, §"Note Commitment Trees" and §"Merkle Path Validity"). Everything below applies to each pool P separately.

An *incremental witness* of a note at position p, after $m > p$ leaves have been appended to the pool's tree, is the information from which the authentication path of p in the tree of size m is computed: the position p, the siblings of the path whose ranges lie below m, and the frontier of the leaves appended after p (ZIP 307, "Local processing", "Creating and updating note witnesses"; *Crypto Guide*, §"Append-only and incrementally updatable trees", Definition "Frontier").

Construction 7.2 (Linear witness maintenance). ZIP 307, "Local processing", "Creating and updating note witnesses", states the protocol for the Sapling pool and its commitments cmu; it applies to each pool with that pool's commitments, cmx for Actions. Compact blocks are processed in increasing height, with the order of transactions and outputs preserved. The wallet holds its copy of the pool's tree as a frontier and one incremental witness per tracked note. For each note commitment c of the block, in order:

(i) append c to the tree;

(ii) append c to every held witness;

(iii) if the output of c is accepted by Definition 4.9, create a witness of it from the current tree.

A witness updated through block X yields the authentication path of its note to the anchor of block X. No ZIP specifies any other tree-maintenance protocol.

Lemma 7.3 (Linear maintenance consumes every commitment). *Under linear witness maintenance, let a note be created in block h_n and let $a \geq h_n$. The wallet holds the authentication path of the note to the anchor of block a only after it has processed, in order, every note commitment of the pool in every block from h_n through a. It creates the note's witness only after it holds the tree state at the end of block $h_n - 1$, obtained either by processing every earlier block or from a chain state (Definition 6.1). In particular, a wallet restored from a seed spends no note created at h_n before it has scanned forward from h_n to the anchor.*

Proof. By induction on the processed commitments. Steps (i) and (ii) are the only operations on the tree and the witnesses, and each consumes the next commitment in chain order; step (iii) creates a witness from the current tree, which after the commitment of the note is the tree state at the end of block $h_n - 1$ followed by the commitments of h_n up to that of the note. A witness therefore reflects block X only after every commitment through X has passed through step (ii), and it yields no path to the anchor of a before it reflects block a. □

Remark 7.4 (The retained-information method). The method of "Subtree roots and the cap" (§7.2) to "Witness readiness" (§7.5) removes the constraint of Lemma 7.3. It replaces per-note witnesses

by subtree roots, obtained through the query SubtreeRoots (Definition 5.10), and a frontier, so that the path of a note needs the commitments of its own subtree only. The method is designed but unspecified (Definition 1.1): no ZIP specifies it or the subtree-root query.

7.2 Subtree roots and the cap

Definition 7.5 (Subtrees and the cap)**.** Fix a pool P and its note commitment tree of depth 32, with nodes M_i^h in layer h counted from the root (*Ironwood Guide*, §"The Merkle hash and the tree", Definition "Note commitment tree"; protocol specification, §"Merkle Path Validity"). Let $\mathsf{MerkleCRH}_P$ be its Merkle hash, $\mathsf{MerkleCRH}^{\mathsf{Sapling}}$ for Sapling and $\mathsf{MerkleCRH}^{\mathsf{Orchard}}$ for either Orchard-protocol pool, and $\mathsf{Uncommitted}_P$ its unused-leaf value, $\mathsf{Uncommitted}^{\mathsf{Sapling}}$ or $\mathsf{Uncommitted}^{\mathsf{Orchard}} = 2$ (protocol specification, §"Constants"). For $0 \le l \le 32$ and $0 \le i < 2^{32-l}$, the node at *level* l and index i is

$$\mathsf{node}(l, i) := M_i^{32-l},$$

the root over the positions $[\, i\,2^l,\ (i+1)\,2^l)$; its *range* is that interval. The nodes satisfy the first line below, and the second defines the values $\mathsf{Empty}_{P,l}$:

$$\mathsf{node}(l+1, i) = \mathsf{MerkleCRH}_P(31 - l,\ \mathsf{node}(l, 2i),\ \mathsf{node}(l, 2i+1)),$$
$$\mathsf{Empty}_{P,0} := \mathsf{Uncommitted}_P, \qquad \mathsf{Empty}_{P,l+1} := \mathsf{MerkleCRH}_P(31 - l,\ \mathsf{Empty}_{P,l},\ \mathsf{Empty}_{P,l})$$

for $0 \le l < 32$. Each $\mathsf{Empty}_{P,l}$ is the root over a range of unused leaves (*Ironwood Guide*, §"Authentication paths", Lemma "Authentication paths from public data", for Orchard-protocol trees; the same recurrence for Sapling). *Subtree* i is the node $\mathsf{node}(16, i)$ with its range, and its completeness, root and completing height are those of Definition 5.8. There are $2^{32}/2^{16} = 2^{16} = 65536$ subtrees, with indices 0 to $2^{16} - 1$, each over 65536 positions. The subtree of a position p is $\sigma(p) := \lfloor p/2^{16} \rfloor$. The *cap* is the tree formed by the nodes $\mathsf{node}(l, i)$ with $16 \le l \le 32$; its leaves are the 2^{16} level-16 nodes, and its root is the root $\mathsf{node}(32, 0)$ of the tree. The same geometry holds for the Sapling, Orchard-pool and Ironwood-pool trees.

Construction 7.6 (The cap over level-16 values)**.** For a pool P and a list $r_0, \ldots, r_{2^{16}-1}$ of level-16 values, the *cap over* (r_i) is the append-only Merkle tree of depth 16 (*Crypto Guide*, §"Append-only and incrementally updatable trees", Definition "Append-only Merkle tree") whose cap-level-0 labels are the r_i, whose empty-leaf value is $\bot_0 := \mathsf{Empty}_{P,16}$, and whose combination of two cap-level-u labels into a cap-level-$(u+1)$ label is

$$\mathsf{NodeHash}_{u+1} := \mathsf{MerkleCRH}_P(15 - u,\ \cdot,\ \cdot) \qquad (0 \le u < 16),$$

the tree's Merkle hash at level $u + 16$, that is, in layer $31 - (u + 16)$. Complete subtree roots obtained from $\mathsf{SubtreeRoots}(P, i)$ enter at cap positions $i, i+1, \ldots$ in index order.

Lemma 7.7 (Cap root equals the tree root)**.** *Let the tree of P hold n leaves, and for each i let $r_i := \mathsf{node}(16, i)$ of that tree: the complete root when $(i+1)\,2^{16} \le n$; the root over the appended leaves of subtree i padded with $\mathsf{Uncommitted}_P$ when $i\,2^{16} < n < (i+1)\,2^{16}$; and $\mathsf{Empty}_{P,16}$ when $i\,2^{16} \ge n$. Then the cap-level-u node of index i of the cap over (r_i) equals $\mathsf{node}(16 + u, i)$ for $0 \le u \le 16$. In*

> *particular the root of the cap equals* $\mathsf{node}(32, 0)$*, the anchor of the pool for that tree* (Ironwood Guide, *§"Anchors", Definition "Treestate and anchor"*).

Proof. Induction on u. For $u = 0$ the claim is the definition of r_i. For the step, the cap combines the cap-level-u nodes of indices $2i$ and $2i + 1$ with $\mathsf{MerkleCRH}_P(15 - u, \cdot, \cdot)$; by the induction hypothesis they are $\mathsf{node}(16 + u, 2i)$ and $\mathsf{node}(16 + u, 2i + 1)$, and the recurrence of the tree gives $\mathsf{node}(17 + u, i) = \mathsf{MerkleCRH}_P(31 - (16 + u), \dots)$, in the same layer $15 - u$. The value $r_i = \mathsf{Empty}_{P,16}$ for an unused subtree follows from the leaf rule, by which unused positions hold $\mathsf{Uncommitted}_P$ (*Ironwood Guide*, §"The Merkle hash and the tree", Definition "Note commitment tree", item "Leaves"), and from the induction in the proof of the *Ironwood Guide*'s Lemma "Authentication paths from public data" (§"Authentication paths"). The empty labels of the cap, $\bot_{u+1} := \mathsf{NodeHash}_{u+1}(\bot_u, \bot_u)$, are then $\mathsf{Empty}_{P,16+u}$ by the same induction, so the cap's labels of empty ranges agree with the tree's. □

> **Lemma 7.8** (Path decomposition at level 16)**.** *Let the tree hold n leaves, let $p < n$ and $\sigma := \sigma(p)$, and let $(s_0, \dots, s_{31})$ be the authentication path of p* (Ironwood Guide, *§"Authentication paths", Construction "Authentication path"*)*, $s_t = \mathsf{node}(t, j_t \oplus 1)$ with $j_t := \lfloor p/2^t \rfloor$.*
>
> (*i*) *For $t < 16$, the range of s_t lies inside subtree σ, and s_t is a function of the leaves of subtree σ below n; it is $\mathsf{Empty}_{P,t}$ if its range starts at or beyond n.*
>
> (*ii*) *For $t \geq 16$, s_t is the cap node of cap level $t - 16$ and index $j_t \oplus 1$, a function of the level-16 nodes r_i with i in its range and $i \neq \sigma$. These are complete subtree roots; the root over the incomplete subtree $\lfloor (n-1)/2^{16} \rfloor$ when n is not a multiple of 2^{16}, a function of the frontier of the tree after its n-th leaf; and $\mathsf{Empty}_{P,16}$.*
>
> *Hence the leaves of subtree σ, the roots of the complete subtrees other than σ and the frontier of the tree of size n suffice for the path, and the wallet needs the commitments of no subtree other than σ.*

Proof. The range of s_t is $[\,(j_t \oplus 1)\,2^t,\ ((j_t \oplus 1) + 1)\,2^t)$, and s_t is a function of the leaves in its range (*Ironwood Guide*, §"Authentication paths", Lemma "Authentication paths from public data", whose proof shows this for every node). For $t < 16$ the range lies inside $[\,\sigma\,2^{16},\ (\sigma+1)\,2^{16})$, since j_t and $j_t \oplus 1$ have the same quotient by 2^{16-t}, which is σ. For $t \geq 16$ the range is a union of whole subtree ranges, and it excludes σ because $j_t \oplus 1 \neq j_t$ and $\lfloor \sigma/2^{t-16} \rfloor = j_t$; Lemma 7.7 computes s_t from the level-16 nodes of its range. The root of the incomplete subtree is the fold, with the $\mathsf{Empty}_{P,\cdot}$ values as right siblings, of the carries below level 16 of the frontier of the tree of size n, one for each set bit of $n \bmod 2^{16}$ (*Crypto Guide*, §"Append-only and incrementally updatable trees", Proposition "Incremental append in logarithmic time", proof). □

> **Construction 7.9** (Acquisition of subtree roots)**.** At each synchronisation the wallet queries $\mathsf{SubtreeRoots}(P, 0)$ for each pool $P \in \{\mathsf{Sapling}, \mathsf{Orchard}, \mathsf{Ironwood}\}$ (Table 4) and records, for each complete subtree i, the pair (root, completing height) of Definition 5.8; the roots enter the cap at index i (Construction 7.6). The roots are trusted under clause (a) of Assumption 5.2: no root is checked against a header, and a root is verified only if the wallet also computes it from scanned commitments (Proposition 5.3, item (iv)). The acquisition is designed but unspecified (Definition 1.1).

7.3 Frontiers as scan seeds

Lemma 7.10 (A frontier suffices to continue the tree)**.** *Let F be the frontier of the tree of pool P after block $h-1$* (Crypto Guide, *§"Append-only and incrementally updatable trees", Definition "Frontier"*)*, holding m leaves, with carries f_t for the set bits t of m, and let $c_m, c_{m+1}, \ldots$ be the note commitments of P in blocks $h, h+1, \ldots$ in chain order. From F and these commitments alone the wallet computes*

(*i*) *the root of the tree after every block $b \geq h-1$, and*

(*ii*) *for every position $p \geq m$ appended by a block at most b, the authentication path of p to that root.*

The commitments at positions below m are not needed. Consequently the chain state $C(B-1)$ stored with the birthday B of an account (*Definitions 6.1 and 6.2*) *seeds every later witness of the account.*

Proof. (i) Repeated application of the *Crypto Guide*'s Proposition "Incremental append in logarithmic time" (§"Append-only and incrementally updatable trees"), each append producing the next frontier and root.

(ii) By the *Ironwood Guide*'s Lemma "Authentication paths from public data" (§"Authentication paths"), a right sibling of p ($b_t = 0$) has a range above $p \geq m$, hence over later leaves or unused positions, and is computed from the c_q with $\mathsf{Empty}_{P,\cdot}$ values. A left sibling ($b_t = 1$) has the range $[\,(j_t - 1)\,2^t,\ j_t\,2^t)$. If the range starts at or above m, it is computed from later leaves. If it lies below m, then $j_t\,2^t \leq m$ and $j_t = \lfloor p/2^t \rfloor \geq \lfloor m/2^t \rfloor$ give $j_t = \lfloor m/2^t \rfloor$, which is odd since $b_t = 1$; so bit t of m is set and the range is the block of the frontier at level t, whose root is the carry f_t. If the range contains m, its part below m is the union of the frontier's blocks at the set bits of m below t, since the range is aligned at level t; its root is the fold of those carries with the later leaves of the range. □

Definition 7.11 (Seeded batch)**.** A batch of consecutive blocks $b_0, \ldots, b_k$, of heights increasing by one, is *seeded* with the chain state $C(h_s)$ of the block preceding b_0 (Definition 6.1), with frontiers $F_P(h_s)$ of sizes $m_P(h_s)$. Before any retained information is modified, the wallet accepts the batch only if b_0 continues the seed in the sense of Definition 6.10: $\mathsf{height}(b_0) = h_s + 1$, $\mathsf{prevHash}(b_0) = \mathsf{hash}_{h_s}$, and $m_P(h_s) + n_P(b_0) = S^P_{\mathsf{end}}(b_0)$ for each pool P (Definition 4.2, Lemma 4.12); each later block is checked against its predecessor in the same way. A violation is a continuity error (Definition 6.10), handled in "Reorganisation detection and recovery" (§7.6). Under clause (a) of Assumption 5.2, the root of each seed frontier $F_P(h_s)$ equals the anchor of block h_s in P; nothing else checks it, since the abstract format does not guarantee a header (Proposition 5.3). The per-block sizes are designed but unspecified (Definition 4.2, Remark 4.1).

7.4 The information a wallet retains

Definition 7.12 (Retained tree information)**.** The *retained tree information* of a wallet is adjoined to its note state (Definition 6.7). It comprises a finite set $\mathcal{H}$ of heights, the *retained block boundaries*, chosen by the wallet and the same for every pool, which contains $B-1$, and for each pool P:

(a) for each unspent note of the wallet in P, its position p and its leaf, and each sibling s_t with $t < 16$ of the path of p (Lemma 7.8(i)) from the moment its range lies wholly below the tree size; such a node is a function of appended leaves and does not change thereafter (*Ironwood Guide*, §"Authentication paths", Lemma "Authentication paths from public data", items (i) and (ii) and proof);

(b) the complete subtree roots with their completing heights (Definition 5.8), and the frontier of the tree after the last scanned block;

(c) for each $r \in \mathcal{H}$, the chain state $C(r)$, whose frontiers are the tree states after block r: the anchors of the blocks in $\mathcal{H}$ are the wallet's candidate anchors, and the heights in $\mathcal{H}$ are its rollback points.

The *rollback window* is the span of $\mathcal{H} \setminus \{B-1\}$. Because $\mathcal{H}$ is common to the pools, a rollback restores all trees to one block. Any other node may be discarded once no retained path needs it. An anchor older than the window is retained only by explicit choice: ZIP 318, "Anchor-height bucketing and cohorts", requires the anchor of each migration transaction to be the tree state at a shared network-wide boundary height, so a wallet that follows it retains those boundaries; any other retention of old anchors is wallet policy. The definition fixes information, not a representation, and is designed but unspecified (Definition 1.1).

Proposition 7.13 (Sufficiency of the retained information)**.** *Fix a pool P.*

(*i*) *Let p be the position of a retained note, $\sigma := \sigma(p)$, and $A \in \mathcal{H}$ a retained block boundary with tree size $n > p$. If the blocks that hold the commitments of subtree σ below n and at or above the birthday seed are scanned, the authentication path of p to the anchor of A is computable from the retained information.*

(*ii*) *For every height r at which the wallet holds a chain state $C(r)$ for all pools, with $n_r(P) := m_P(r)$ the size of the tree of P after r, discard every retained node over a range that meets $[n_r(P), 2^{32})$, every note position at or above $n_r(P)$, every subtree root with completing height above r and every $C(r')$ with $r' > r$, and take the frontier of $C(r)$ as the frontier after the last scanned block. The result is a function of the blocks at heights at most r, and it is information that a wallet which scanned only those blocks retains.*

Proof. (i) By Lemma 7.8, each sibling s_t is of one of the following sorts. A sibling whose range lies below n is the same node at every later tree state: for $t < 16$ it is in item (a), or, where its range lies below the birthday seed, it is a carry of the seed frontier (Lemma 7.10); for $t \geq 16$ it is a cap node over complete subtree roots of item (b) (Lemma 7.7). The sibling ranges are pairwise disjoint, so at most one contains $n-1$ without lying below n. That sibling is the root over its range, whose part below n is the union of the blocks of the frontier of $C(A)$ at the set bits of n below its level; it is the fold of those carries with $\mathsf{Empty}_{P,\cdot}$ values, and the frontier of $C(A)$ is in item (c). A sibling whose range starts at or beyond n is $\mathsf{Empty}_{P,t}$.

(ii) After the discards, every retained node lies over a range below $n_r(P)$ and is a function of the leaves appended through block r (*Ironwood Guide*, §"Authentication paths", Lemma "Authentication paths from public data"); every retained subtree root has completing height at most r, and is likewise such a function; the retained chain states are those at heights at most r; and the frontier after r is that of $C(r)$, which the wallet holds for every pool at the same r. A wallet that scanned only the blocks at most r can hold each of these items, and item (a) is satisfied, since a sibling whose range is not below $n_r(P)$ is not yet due. □

7.5 Witness readiness

Theorem 7.14 (Witness from retained information)**.** *Let A be a retained block boundary with tree size n in pool P, let p be a position, $\sigma := \sigma(p)$, and let $(s_0, \ldots, s_{31})$ be the authentication path of p to the anchor of A. If $p \geq n$, the* Ironwood Guide*'s Construction "Authentication path" (§"Authentication paths") defines no path, and by its Proposition "Membership soundness" (§"Anchors"), item (iii),*

no valid path to that anchor with position p exists, except with negligible probability, for the Orchard-protocol pools; for Sapling, by the same argument under the collision resistance of PedersenHash *that the protocol specification requires of* $\mathsf{MerkleCRH}^{\mathsf{Sapling}}$ *(§"*$\mathsf{MerkleCRH}^{\mathsf{Sapling}}$ *Hash Function"), since* $\mathsf{Uncommitted}^{\mathsf{Sapling}}$ *is not in the range of* $\mathsf{NoteCommit}^{\mathsf{Sapling}}$ *(§"Windowed Pedersen commitments"). Otherwise the path is computable from the wallet's information if every s_t is of one of the following kinds:*

(a) *$t < 16$, and s_t is computable from the commitments of subtree σ below n that the wallet has scanned or holds summarised in a carry, whose range lies below n, of a seed frontier (Lemma 7.10) or in the frontier of $C(A)$;*

(b) *$t = 16$, and s_t is a complete subtree root that the wallet holds;*

(c) *$t \geq 16$, and s_t is a cap node computed from held complete roots, carries at levels at least 16 of the frontier of $C(A)$ or, with range below n, of a seed frontier, $\mathsf{Empty}_{P,16}$ and, where its range meets the incomplete subtree, the root over that subtree's leaves below n, computed from the frontier of $C(A)$;*

(d) *the range of s_t starts at or beyond n, and $s_t = \mathsf{Empty}_{P,t}$;*

(e) *$t \geq 16$, and s_t is a carry of the frontier of $C(A)$ or a carry, whose range lies below n, of a seed frontier.*

Proof. Kinds (a) to (d) are the sorts of sibling in the proof of Proposition 7.13(i), extended in kind (c) by frontier carries, and a sibling of kind (e) is a retained node. □

Example 7.15 (Kinds of the siblings of one path)**.** Let $p = 65537$, so $\sigma = 1$, and let the anchor's tree hold $n = 471098 = 7 \cdot 65536 + 12346$ leaves, so that subtrees 0 to 6 are complete and subtree 7 is incomplete. The siblings $s_0, \ldots, s_{15}$ are of kind (a); s_{16}, the root of subtree 0, is of kind (b); s_{17}, over subtrees 2 and 3, and s_{18}, over subtrees 4 to 7, which meets the incomplete subtree, are of kind (c); and $s_{19}, \ldots, s_{31}$, whose ranges start at $8 \cdot 65536$ or beyond, are of kind (d).

Definition 7.16 (Block extent of a subtree)**.** For pool P and subtree s, let $c_P(s)$ be the completing height of subtree s (Definition 5.8), and let $c_P(-1)$ be the activation height of P. The *block extent* of subtree s is the interval of heights $[\max(c_P(s-1), B),\ c_P(s)]$, both ends included, the part below B being summarised by the chain state $C(B-1)$ (Lemma 7.10); for the incomplete last subtree the interval ends at the anchor height. After detecting a note at position p, the wallet scans the extent of subtree $\sigma(p)$ beyond the ranges it has already scanned. The incomplete subtree of every pool is covered by the range from $\max(B, h_c)$ to the tip, where h_c is the least, over the pools, of the completing height of the pool's last complete subtree, or of the pool's activation height if it has none.

The start of an extent is inclusive because the completing block of subtree $s-1$ may also hold the first commitments of subtree s; the positions within that block are fixed by Lemma 4.12.

Proposition 7.17 (Readiness of a witness)**.** *Let a note of pool P lie at position p, $\sigma := \sigma(p)$, and let the anchor block A, of height a, have tree size $n > p$ and $\kappa := \lfloor n/2^{16} \rfloor$ complete subtrees. The witness of p at A is computable if*

(1) *the blocks of subtree σ are scanned from the start of its extent (Definition 7.16) up to a, or through its completing height if $\sigma < \kappa$;*

(2) *the wallet holds the root of every complete subtree $i < \kappa$ with $i \neq \sigma$, including those after σ; and*

(3) *the frontier of $C(A)$ is retained, which scanning the extent of subtree κ through a yields when n is not a multiple of 2^{16}.*

Condition (2) can be weakened through frontier carries: every complete subtree $i \neq \sigma$ lies under a sibling of the path of p, and the root of subtree i is not needed when its range lies inside a carry, at a level at least 16 and at most that of the sibling, of the frontier of $C(A)$ or, provided the carry's range lies below n, of a seed frontier. Note eligibility, anchor choice and confirmation depth are separate conditions ("Spendable notes", §9.5).

Proof. Each condition supplies one kind of Theorem 7.14: condition (1) kind (a); condition (2) kind (b) and the complete part of kind (c); condition (3) the incomplete part of kind (c). Kind (d) needs nothing. For condition (3), the carries of the frontier at levels $t \geq 16$ are cap nodes over complete roots, and those below level 16 are functions of the leaves of subtree κ below n, appended in its extent. For the last claim, the ranges of $s_{16}, \ldots, s_{31}$ partition the subtree indices other than σ; if the range of subtree i lies inside such a carry, which is the same node at A, its range lying below n, the sibling over it is that carry, of kind (e), or a cap node computed from it, of kind (c), without the root of subtree i. □

Remark 7.18 (Scheduling). Scanning the tip range and the extents of the subtrees of found notes before historic ranges is a priority rule of wallet policy, designed but unspecified (Definition 1.1). The order of range requests is observable by the server ("Leakage of range queries", §8.3).

Corollary 7.19 (Witnesses under any scan order)**.** *Under the hypotheses of Lemma 6.13, a schedule of ranges that covers $[B, T]$ in any order yields, for every note and every retained anchor, the witness of a scan in increasing height from B, provided a note enters a witness only once the extent of its subtree (Definition 7.16) is scanned up to the anchor and the conditions of Proposition 7.17 hold.*

Proof. By the *Ironwood Guide*'s Lemma "Authentication paths from public data" (§"Authentication paths"), the path is a function of the position and of the leaves below n, which do not depend on the scan order. Positions are local to their blocks by Lemma 4.12, and the detection of the note is order-independent by Lemma 6.13. Once the stated extents are scanned, Theorem 7.14 makes the path computable. □

Remark 7.20 (Cost of a restored wallet). A restored wallet seeds each tree from the chain state $C(B-1)$ (Lemma 7.10), holds one root per complete subtree (Construction 7.6), and scans for witnesses only the extents of the subtrees of its notes and the tip range, instead of replaying the tree from the pool's first block. By Proposition 7.17, the witness of a found note is ready once those ranges are scanned. The trial decryption of every block from B onward (Proposition 6.17) remains, and so does the scanning of the extents.

7.6 Reorganisation detection and recovery

Remark 7.21 (What a reorganisation invalidates). A reorganisation (*Ironwood Guide*, §"Anchors", Definition "Reorganisation") makes three kinds of wallet information wrong: the transactions of the wallet mined in replaced blocks become unmined; the anchors of replaced blocks vanish; and tree information over positions appended in replaced blocks no longer describes the chain.

Construction 7.22 (Detection of a reorganisation)**.** Each synchronisation after the first requests blocks from its previously synchronised height X inclusive and checks that the hash of the received block X equals the stored one; a mismatch implies that block X was orphaned by a reorganisation (ZIP 307, "Client-server interaction", phases B and D). This volume extends

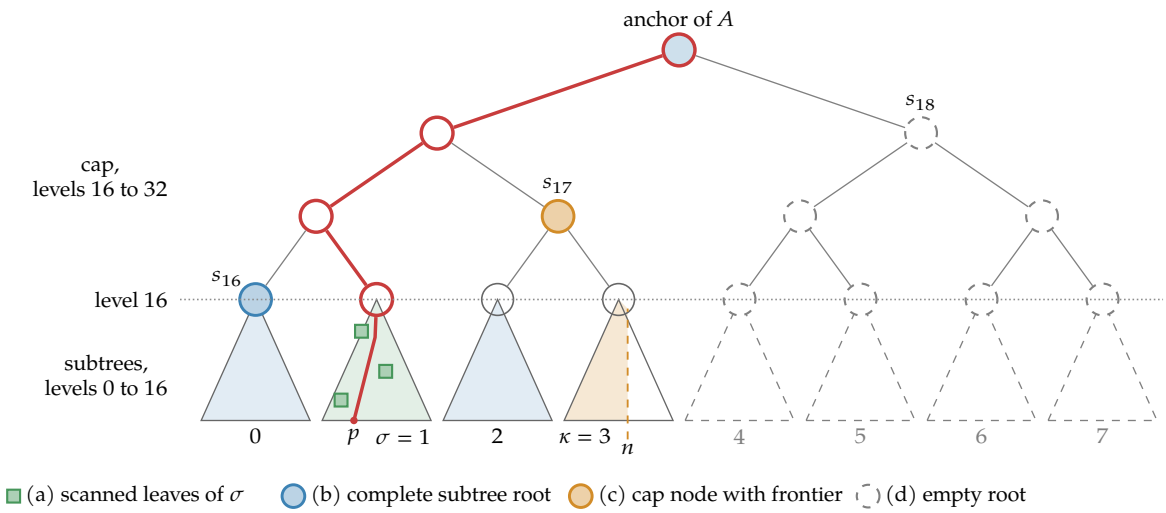


Figure 3: A note commitment tree cut at level 16, drawn with eight subtrees in place of 2^{16} and the cap above them compressed to three levels. Subtrees 0 and 2 are complete, subtree $\sigma = 1$ holds the note at position p, subtree $\kappa = 3$ is incomplete at the anchor's tree size n, and subtrees 4 to 7 are unused. The route of p to the anchor is drawn in red. Its siblings are marked by the kind of Theorem 7.14 and the source that supplies them: the siblings $s_0, \ldots, s_{15}$ inside subtree σ, from its scanned leaves (a); s_{16}, a downloaded complete subtree root (b); s_{17}, a cap node over a complete root and the root of subtree κ computed from the frontier after A (c); and s_{18}, the empty root beyond n (d). The split of the path at level 16 is Lemma 7.8.

the check to every block and to both sides of every range boundary (Definition 6.10). Failure of the continuity predicate is a *continuity error* and triggers recovery (Construction 7.24). A malformed field, or a tree size that is absent, outside $[0, 2^{32}]$ or below the block's own commitment count (Lemma 4.12(b)), is an error of the response and triggers none. The extension beyond ZIP 307's comparison is designed but unspecified (Definition 1.1).

Definition 7.23 (Rollback to a height)**.** *Rollback* to a height r at which the wallet holds a chain state $C(r)$ consists of the following changes.

(i) The chain state of the wallet becomes $C(r)$, and each pool's retained tree information becomes that of Proposition 7.13(ii) for r.

(ii) Every transaction mined above r is marked unmined. The spend marks that such transactions set, on shielded notes and on transparent outputs, are cleared, and the nullifier of each note so unmarked returns to the tracked set W_P of its pool.

(iii) Scan results above r are discarded: the unlinked nullifiers of R_P revealed above r, and the scanned ranges above r, so that the scanned heights of Definition 6.11 are at most r. A note whose creating transaction is now unmined loses its position and, for Sapling, its nullifier, which depends on the position; both are derived again if a later scan detects the note.

(iv) The knowledge of rule (R4) of Definition 6.4 is cut back as that rule states; the evidence of rule (R2) stands.

(v) Local-only data is kept: sent and received notes stay recorded, since the memos and transactions that the wallet created may be unrecoverable from the chain, while a note

whose transaction is unmined is excluded from balances and from spendability ("Spendable notes", §9.5).

Construction 7.24 (Recovery from a continuity error)**.** Let $k \geq 2$ be an offset, a wallet parameter. On a continuity error at height e, that is, when a block of height e fails Definition 6.10 against the wallet's chain state $C(e-1)$, the wallet runs the following steps.

(1) If $e - k < B - 1$, it obtains $C'(B-1) := \mathsf{TreeState}(B-1)$, replaces its stored chain state at $B-1$ by $C'(B-1)$, rolls back to $B-1$ (Definition 7.23), and goes to step (4).

(2) Otherwise let r be the greatest height with $B - 1 \leq r \leq e - k$ that is a retained block boundary ($r \in \mathcal{H}$) or at which the wallet has recorded the hash of its scanned block. The height $B-1$ qualifies, since $B - 1 \in \mathcal{H}$.

(3) If $r \in \mathcal{H}$, the wallet rolls back to r. Otherwise it obtains $C'(r) := \mathsf{TreeState}(r)$ (Definition 6.1), trusted under clause (a) of Assumption 5.2. If the hash of $C'(r)$ equals the recorded hash, the wallet takes the frontiers of $C'(r)$ as its tree states after r, discards every retained node over a position at or above their sizes, and rolls back to r. If the hashes differ, its block r is off the server's chain, and it repeats step (1) with $e := r + 1$.

(4) It restarts synchronisation from the query $\mathsf{Tip}()$.

A fetched chain state extends the depth to which the wallet recovers beyond its rollback window, but only at a height whose hash the wallet has recorded: without that comparison, a fetched chain state cannot show that the wallet's blocks at most r lie on the server's chain, and results from replaced blocks below r would survive. The construction is designed but unspecified (Definition 1.1).

Proposition 7.25 (Recovery reaches the fork)**.** *Assume clause (a) of Assumption 5.2, let the server's best chain be unchanged during recovery, and exclude collisions of the block hash. Let the wallet's scanned heights below e form the interval $[B, e-1]$, as in a scan in increasing height. The wallet's block at a height $h \in [B-1, e-1]$ is the block it scanned at h, or for $h = B-1$ the block of its stored $C(B-1)$. The* fork point *h_f is the greatest $h \in [B-1, e-1]$ at which the wallet's block is the server's block at h, and $h_f < B-1$ if there is none. Agreement at a height implies agreement at every lower height of $[B-1, e-1]$: the wallet's blocks are linked by the checks of Definition 6.10, and each header contains the hash of its parent (protocol specification, §"Block Header Encoding and Consensus").*

(i) A continuity error at e implies $h_f \leq e - 2$.

(ii) Repeated recovery by Construction 7.24 strictly decreases the rollback height r until $r \leq h_f$, or until step (1) replaces the stored chain state at $B-1$; after either, no continuity error recurs.

(iii) No offset below 2 guarantees progress: a rollback to $e-1$ keeps the orphaned block, and the next scan fails again at e.

Proof. (i) Under clause (a) the server's block e has the requested height, so the failing clause of Definition 6.10 is (ii) or (iii). The server's block e has as parent its chain's block $e-1$, and its sizes are functions of that chain's prefix (Lemma 4.12(a)). If the wallet's block $e-1$ were the server's block $e-1$, its hash and its sizes, which the wallet computed from the same prefix, would satisfy both clauses. So the chains disagree at $e-1$, and $h_f \leq e-2$.

(ii) Consider one recovery, with rollback height $r \leq e - k \leq e - 2$. If r is a retained boundary and $r > h_f$, the wallet's block r is off the server's chain, so the server's block $r+1$, whose prevHash

is the hash of the server's block r, fails continuity at $e' = r + 1$, and the next rollback height is at most $e' - 2 < r$. If r is a recorded height and the hashes differ, step (3) repeats with $e := r + 1$ and the next height is at most $r + 1 - k < r$. If the hashes agree, $r \leq h_{\mathsf{f}}$ by the definition of the fork point. Heights are bounded below by $B - 1$, and a rollback height that would lie below it is replaced by step (1). Once $r \leq h_{\mathsf{f}}$, the wallet's block r is on the server's chain, and every later block of that chain continues its predecessor. After step (1) the wallet's chain state at $B - 1$ is the server's and every result above it is discarded, so the same holds.

(iii) After a rollback to $e-1$ the wallet's block $e-1$ is unchanged, and by (i) it is off the server's chain; the server's block e fails against it again. □

Theorem 7.26 (Reorganisation safety)**.** *Under the hypotheses of Proposition 7.25, recovery by Construction 7.24 ends at a height r such that every block of the wallet at most r is a block of the server's chain: either a retained boundary or a recorded height $r \leq h_{\mathsf{f}}$, the latter with the chain state* $\mathsf{TreeState}(r)$*, or $r = B - 1$ with the chain state* $\mathsf{TreeState}(B - 1)$*. Scanning the server's chain from $r + 1$ then yields the note state and the retained tree information of a scan of that chain from the birthday, with local-only data kept.*

Proof. Proposition 7.25 gives the end height r and its property. After the rollback, the retained information is a function of the blocks at most r (Proposition 7.13(ii), Definition 7.23), and those blocks are shared by both chains; where a fetched chain state is used, its frontiers are those of the server's chain under clause (a). Theorem 6.14 and Lemma 7.10 then apply to the server's chain from $r + 1$, seeded with $C(r)$. □

Theorem 7.27 (Correctness of synchronisation)**.** *Assume clause (a) of Assumption 5.2 and exclude collisions of the block hash. Let the server's best chain be unchanged from some time on, the* final chain*, and let T be the height of a tip of the final chain that the wallet obtains after that time. Let the wallet*

(1) *check continuity before applying the results of a range (Definition 6.10), on both sides of every range boundary;*

(2) *retain unlinked nullifiers as Lemma 6.12 requires;*

(3) *take positions from consistent sizes (Lemma 4.12);*

(4) *recover from every continuity error by Construction 7.24, scanning a height again only after a rollback below it;*

(5) *enter a note into a witness only once the extent of its subtree is scanned up to the anchor (Definition 7.16);*

(6) *scan the range that contains T after obtaining T, with continuity checked at its lower boundary (Definition 6.10), and discard scan results above T; and*

(7) *hold, as its complete subtree roots, those returned by a* $\mathsf{SubtreeRoots}$ *query made after obtaining T, restricted to completing height at most T.*

If, after its last rollback, its scanned ranges cover $[B, T]$, in any order, the wallet holds the note state and the retained tree information of a scan of the final chain in increasing height from B.

Proof. The theorem composes earlier results and adds no mechanism. By (6) the range that contains T was requested after the tip query that returned T, so under clause (a) the wallet's block

T is that of the final chain. By (1), each block of $[B, T]$ was checked against the wallet's chain state of its predecessor, and by (4) no height below it was scanned again afterwards without a rollback that also discarded the block; so the wallet's blocks of $[B-1, T]$ are linked by prevHash, and by the hash chain of the headers each of them is the final chain's block at its height. By (4) a height is scanned again only after a rollback below it, which discards the earlier result at that height (Definition 7.23 (iii)), so every kept result of $[B, T]$ is that of the wallet's current block; by (6) no result above T is kept. The scan of $[B, T]$ is therefore one of the final chain without a continuity error. By (7) and clause (a), the retained subtree roots are those of the final chain with completing height at most T. Lemma 6.13 gives the note state of the scan in increasing height, Corollary 7.19 the witnesses, and Theorem 6.14 characterises the notes found. □

Remark 7.28 (The rollback window). Consensus bounds no reorganisation depth, and node rollback windows are local policy (*Consensus Guide*, §"Reorg rails, maturity, and the finality floor"). ZIP 307, "Local processing", advises caching tree and witness state for 100 recent blocks, on the ground that no full node rolls back the chain by more than 100 blocks. That ground no longer holds: ZIP 218, "Block-count-based constants", recommends a node rollback limit of 600 blocks at its 25-second target spacing and keeps coinbase maturity at 100 blocks, and neither figure bounds the window of a wallet. The depth that a wallet repairs from its retained information is the span of its retained block boundaries, a wallet parameter; a chain state fetched by TreeState at a height whose hash the wallet has recorded extends it under clause (a) of Assumption 5.2 (Construction 7.24).

8 Leakage to the light-client server

This section states what the server of "The light-client service" (§5) learns about a light wallet. It defines payment-detection privacy and unlinkability as indistinguishability games against the server of Assumption 5.2 and the view on which the server decides; proves that the view is a deterministic function of the chain, a tuple of schedule parameters and transport metadata, never of the wallet's keys; derives the leakage of range queries, transaction queries, transparent-address queries and submission as instances of that theorem; and closes with a table of the server's observables and the class of each mitigation.

8.1 Payment-detection privacy

The server is the node and the proxy together (Definition 5.1), under Assumption 5.2: clause (a) makes the response to each query a function of the server state and the query, and clause (b) is the power of the adversary. No second adversary is defined. The *server state* Σ_t at time t is the server's best chain at t together with the state of its node's transaction pool; the latter enters only the responses of Submit and Status (Definition 5.9). An *evolution* is a family $\Sigma = (\Sigma_t)_{t \le t^\star}$ up to a horizon $t^\star$. A *wallet configuration* is a pair (K, u): K is the wallet's key material, the spending, full viewing, incoming viewing, outgoing viewing and nullifier deriving keys of every account; u is every other input of the wallet, namely the birthdays, the user's actions and their times, the scheduling policy's configuration, any state retained from earlier sessions, and the law from which the network endpoint of each connection is drawn.

Definition 8.1 (Server view). The *view* V of a wallet is the record of clause (b) of Assumption 5.2: the transcript

$$((c_j, q_j, a_j, t_j, r_j))_{j=1,\ldots,J}$$

of the wallet's queries in the order issued, each with its connection c_j, its query name q_j of Definition 5.10, its argument tuple a_j, its time t_j and its response r_j; and, for each connection

c, the *transport metadata* $M_c = (E_c, t_c^{\mathsf{open}}, t_c^{\mathsf{close}})$, the network endpoint that the server sees and the open and close times. Times are recorded at a finite resolution, and a run up to a horizon issues finitely many queries, so every view ranges over a countable set.

Definition 8.2 (Payment-detection privacy game)**.** Let π be a scheduling policy, a public randomised rule that emits the wallet's queries, made precise in "Key independence of queries" (§8.2). The *payment-detection privacy game* PD_π is the following indistinguishability game, with the advantage of a distinguishing game of the *Crypto Guide*, §"Security as a game", Definition "Security game and advantage"; the adversary is not restricted to polynomial time.

1. The adversary fixes an evolution Σ, whose chain may contain outputs to either key set, and two configurations (K_0, u) and (K_1, u) with the same u.
2. The challenger draws a uniform bit $b \in \{0, 1\}$, runs the wallet (K_b, u) under π against a server with evolution Σ up to $t^\star$, and hands the adversary the view V of that run (Definition 8.1).
3. The adversary outputs $b' \in \{0, 1\}$ and wins if and only if $b' = b$; its advantage is $\mathrm{Adv} = |\Pr[b' = b] - 1/2|$.

Network privacy holds when the endpoint E_c of each connection is drawn afresh, independently of everything else, from a law that does not depend on which wallet opens the connection. The *unlinkability game* UL_π, under network privacy, differs in two points: the adversary fixes Σ and two configurations (K_0, u_0) and (K_1, u_1); and the challenger runs two consecutive synchronisation sessions, both of wallet 0 when $b = 0$, the second resuming from the state in which the first ends, and a session of wallet 0 followed by a session of wallet 1 when $b = 1$. The adversary receives the views of both sessions. Policy π provides *payment-detection privacy* (respectively *unlinkability*) if every adversary has advantage zero in PD_π (respectively UL_π), and provides it *within* ε if every advantage is at most ε.

Remark 8.3 (The goal of ZIP 307). ZIP 307, "Security Model", states the goal informally and with a lowercase "should": the proxy should not learn which transactions received from the chain are addressed to a given light wallet, and, assuming network privacy ("via Tor or similar"), should not be able to link different connections or queries as deriving from the same wallet. It takes the node and proxy combination to be honest but curious, trusted to provide a correct view of the current best chain and to transmit queries and responses faithfully. Game PD_π formalises the first sentence and game UL_π the second; the trust clause is clause (a) of Assumption 5.2, an assumption distinct from the goal. The chain is common to both worlds of each game. What the chain itself reveals to a party without keys is the subject of the *Ironwood Guide*, §"Privacy", Theorem "Privacy of an Action", whose Remark "Non-claims" excludes wallet and network metadata, the subject of this section.

Remark 8.4 (Scope and status). ZIP 307, "Security Model", does not address how to spend notes privately. Submission through the same server is nevertheless part of the view of Definition 8.1 and is analysed in "The submission channel" (§8.5). The status of ZIP 307 and of ZIP 314 is that of Table 1. ZIP 314, titled "Privacy upgrades to the Zcash light client protocol", carries no specification text, so a specified successor to the privacy treatment of ZIP 307 is an open problem (Definition 1.1).

ZIP 307 makes two statements about its interaction of phases A to D ("Scan order", §6.4). First, since synchronisation proceeds over a broadcast medium, it leaks no information about which transactions interest the client ("Transaction privacy"). Second, the start of each of phases B and D reveals the height to which the client last synchronised, an information leak under the security model, assuming network privacy ("Block privacy via bucketing"). The two statements hold

together: the first concerns the content of the responses, the second the bounds of the queries, two components of the view that Theorem 8.8 separates. The second is a leak in UL_π, since the resume height links a session to the session that reached it; both are made precise in "Key independence of queries" (§8.2) and "Leakage of range queries" (§8.3).

8.2 Key independence of queries

A light wallet delegates the storage and filtering of the chain to the server and keeps its keys; its note state (Definition 6.7) is rebuilt on the device. What reaches the server is its query schedule, recorded as in Definition 8.1.

Lemma 8.5 (Query arguments carry no key material)**.** *Every argument of every query of Definition 5.10 is one of the following: a block identifier, a height or a block hash* (Block, TreeState)*; a pair of heights $s \leq e$ with a pool set $\mathcal{P}$* (Blocks)*; a shielded pool with a subtree index* (SubtreeRoots)*; a transaction identifier* (Tx, Status)*; a set A of transparent addresses* (Utxos)*; the bytes of a transaction* (Submit)*. Query* Tip *takes none. No argument is a spending, full viewing, incoming viewing, outgoing viewing or nullifier deriving key, and no argument other than that of* Submit *is a nullifier of a note that the wallet holds unspent. The argument of* Submit *carries the nullifiers of the notes it spends; it is a component of L ("transactions submitted") whose law depends on K, so submission is not covered by PD_π ("The submission channel", §8.5). Against adversaries restricted to polynomial time, what the public data of its Ironwood-pool bundle reveal of its keys is bounded by the* Ironwood Guide*, §"Privacy", Theorem "Privacy of an Action", under the preconditions and assumptions named there; that theorem claims nothing for an Orchard-pool bundle (its Remark "Non-claims") and does not treat Sapling components. A transparent address is a receiver, disclosed by declaration (the Remark "Transparent boundary" below). Which arguments the wallet chooses may depend on its keys; that dependence is carried by the schedule parameters of Theorem 8.8.*

Proof. By inspection of the argument types of Definition 5.10 (Table 4). □

Lemma 8.6 (Local detection and witness assembly)**.** *Trial decryption and acceptance (Construction 4.8, Definition 4.9), nullifier matching and the search of retained unlinked nullifiers (Construction 6.8), and the assembly of authentication paths from retained information (Definition 7.12, Proposition 7.13) take as inputs the wallet's keys, responses already received and retained data, and issue no query.*

Proof. By inspection of the cited definitions. The underlying procedures are the *Ironwood Guide*'s, §"Trial decryption and note acceptance" (Construction "Trial decryption") and §"Authentication paths" (Lemma "Authentication paths from public data"), both computed from data already held. □

Definition 8.7 (Schedule parameters)**.** A *scheduling policy* π is a public randomised rule that, from the responses received so far and a tuple L, emits the next query with its argument, its connection and its time. The *schedule parameters* L collect every other input on which the emitted schedule depends: the wallet birthday (Definition 6.2) and the resume heights; the bounds, the pool sets and the order of range queries; the subtree extents requested (Definition 7.16); the transaction identifiers fetched or queried; the transactions submitted; the sets of transparent addresses queried; and the policy's random coins, a finite bit string that fixes times, orders and connection choices. Every component takes values in a countable set. Keys enter the schedule only through L (Lemmas 8.5 and 8.6). Given Σ, the law of L is induced

by the configuration (K, u). The endpoints $E = (E_c)_c$ are drawn from the law that u fixes, independently of L; without network privacy that law may be a single point.

For random variables X and Y with values in a countable set, $\Delta(X, Y) := \frac{1}{2}\sum_s |\Pr[X = s] - \Pr[Y = s]|$ is their statistical distance.

Theorem 8.8 (The server's view of a light wallet)**.** *Assume Assumption 5.2, and let π be a scheduling policy. There is a deterministic function F_π with*

$$V = F_\pi(\Sigma, L, E).$$

Hence:

(*i*) *two wallets with equal L and equal endpoints produce the same view, whatever their keys;*

(*ii*) *in PD_π, with L_b the schedule parameters of wallet b and $V_b := F_\pi(\Sigma, L_b, E)$, every adversary that fixes Σ has*

$$\mathrm{Adv} \le \tfrac{1}{2}\,\Delta(V_0, V_1) \le \tfrac{1}{2}\,\Delta(L_0, L_1),$$

and the first bound is attained by an adversary that is not restricted in time; in particular π provides payment-detection privacy if and only if, for every Σ, K_0, K_1 and u, the view has the same law for K_0 and K_1, which holds whenever the law of L given Σ is the same for both;

(*iii*) *the bounds of (ii) hold for UL_π with the pair of the two sessions' parameter tuples in place of L, the endpoints being independent of b under network privacy.*

Privacy holds exactly when the view has the same law for both key sets, and in particular whenever the law of L given Σ does not depend on the wallet's receipts, its detected notes and spends, or the transparent addresses it watches; each leak of "Leakage of range queries" (§8.3) to "The submission channel" (§8.5) is an instance.

Proof. Induction on the transcript. Query j, with its name, argument, connection and time, is emitted by π from L and responses $r_1, \dots, r_{j-1}$. Response r_j is, by clause (a) of Assumption 5.2, a function of Σ_{t_j} and query j. The metadata M_c is E_c with open and close times that the schedule fixes. So V is a function F_π of the three inputs, which gives (i).

For (ii), fix Σ. An adversary is, after its choice, a randomised decision rule; write $D(v) \in [0, 1]$ for the probability that it outputs 1 on view v. By the *Crypto Guide*'s Definition "Security game and advantage" (§"Security as a game"), $\mathrm{Adv} = \frac{1}{2}|\Pr[b' = 1 \mid b = 1] - \Pr[b' = 1 \mid b = 0]|$, and

$$\Big|\sum_v (\Pr[V_1 = v] - \Pr[V_0 = v])D(v)\Big| \le \sum_{v \in S} (\Pr[V_1 = v] - \Pr[V_0 = v]) = \Delta(V_0, V_1),$$

where S is the set of v with $\Pr[V_1 = v] > \Pr[V_0 = v]$ (or its complement, for the opposite sign); equality holds for D the indicator of S. For the second bound, E is independent of L_b with the same law in both worlds, so $\Delta((L_0, E), (L_1, E)) = \Delta(L_0, L_1)$; and for a deterministic f, $\Delta(f(X), f(Y)) \le \Delta(X, Y)$, since $\Pr[f(X) = s]$ is the sum of $\Pr[X = x]$ over the preimage of s and the triangle inequality applies term by term. An adversary that chooses Σ at random has an advantage at most the maximum over its choices. The characterisation follows, since the advantage of the optimal adversary is $\frac{1}{2}\Delta(V_0, V_1)$. For (iii), apply the same argument to the pair of sessions; under network privacy E is independent of b and of the parameter tuples. □

Corollary 8.9 (Compact responses reveal no receipt)**.** *The response to* $\mathsf{Blocks}(s, e, \mathcal{D})$, $\mathsf{Block}(b)$, $\mathsf{TreeState}(b)$ *and* $\mathsf{SubtreeRoots}(\mathsf{pool}, i)$ *is the same for every wallet that issues the same query at*

the same time. Let a policy's shielded synchronisation consist of Tip, SubtreeRoots(pool, 0) *for each shielded pool,* TreeState($B-1$), Blocks *queries that cover* $[B, T]$ *in increasing height with bounds fixed by* B*, the resume heights, the returned tips and the policy's coins and one pool set* $\mathcal{P}$ *in every* Blocks *query, fixed independently of receipts, and the* TreeState *queries of Construction 7.24, with the retained boundaries* $\mathcal{H}$ *and the heights at which block hashes are recorded fixed by* B*, the resume heights, the responses and the policy's coins; and let its query times be fixed by its coins and the responses, not by the duration of local computation. Its* L *is then a function of* B*, the resume heights, the pool set and the coins, none a function of receipts, so its compact-block phase leaks no receipt. The commitment recomputation that replaces the discarded AEAD tag as the integrity check of compact decryption (Definition 4.9, Proposition 4.10) is local by Lemma 8.6. Requesting every complete subtree root from index* 0 *is independent of the positions of the wallet's notes; requesting a particular extent is not (Proposition 8.15). A policy that drops the Orchard pool from* $\mathcal{P}$ *once rule (R5) of Definition 6.4 stops its scanning reveals the height at which the Orchard-pool balance became known to be zero; keeping the pool in* $\mathcal{P}$ *and omitting only the trial decryption avoids this.*

Proof. The first sentence is clause (a) of Assumption 5.2. The rollback heights of Construction 7.24 are functions of $\mathcal{H}$, the stored block hashes and the responses, hence, by the hypothesis on $\mathcal{H}$ and on the recorded heights, of Σ, B, the resume heights and the coins. The remainder is Theorem 8.8(i) and (ii) with the listed L, whose law given Σ does not depend on K. □

The hypothesis on times is necessary: a policy that issues its next query when the processing of a range ends, when that processing takes longer for a range with detected notes, puts a function of receipts into the times of L.

Remark 8.10 (Transparent boundary). Query Utxos(A) puts the addresses of A into L and into the view in the clear, by declaration, not by inference, and a query with $|A| \geq 2$ presents the members of A as one requester's. The addresses are derived from K, so in PD_π a policy that queries its watched transparent addresses has advantage $\frac{1}{2}$ against the adversary that reads A. The guarantee of Corollary 8.9 is a property of the shielded layer only.

Proposition 8.11 (Transparent-address queries)**.** *Let a wallet watch* n *transparent addresses and query them one address per query, in rounds that visit every address once in a uniformly random order, at times whose successive gaps are independent draws of law* $\mathrm{Exp}(1/T)$ *with* $T = 86400/n$ *seconds, of mean* T*, one expected query per address per day* (Math Guide, *§"Beyond finite spaces: countable additivity, limits, and densities", Example "A jittered schedule"*)*. Then no query argument contains two addresses, and the server cannot link two addresses by co-occurrence in one query. It can still link them through the connection and its endpoint when their queries share either, and through the times in* L*: per-address randomised timing does not unlink addresses queried over one connection. The schedule removes one linking channel, not linkability. The schedule is designed but unspecified (Definition 1.1); ZIP 318, "Transfer scheduling", specifies an analogous schedule, a uniformly random shuffle and independent exponential delays, for its migration transactions.*

Proof. The arguments are singletons by construction. The remaining channels are the connection, endpoint and time components of the view, which Theorem 8.8 retains unchanged. The law of the gaps is cited, not derived. □

8.3 Leakage of range queries

Proposition 8.12 (Birthday bound from range queries)**.** *Let* m *be the least height of any block that the wallet requests, where* Blocks($s, e, \mathcal{P}$) *requests the heights of* $[s, e]$, Block(h) *the height* h*, and* TreeState(h) *the height* $h+1$ *whose chain state below it is fetched. A policy that requests no block*

below the wallet birthday B (Definition 6.2, the minimum of its accounts' birthdays) reveals $B \leq m$, and $B = m$ when it requests a range that starts at B, which need not be its first range under a prioritised order. Since B lower-bounds the first block in which any account could have received funds (ZIP 326, "Terminology"), the server learns the upper bound m on the height below which the wallet expects no receipt; once the wallet requests a range that starts at B, so that $m = B$, it learns that bound exactly. Adding an account with an earlier birthday lowers m later. A policy that starts at Sapling activation, as ZIP 307 prescribes for a seed that the client did not generate ("Importing a pre-existing seed"), and as this volume does for a seed restored without a recorded birthday (Definition 6.2), reveals only that constant. The birthday is a component of u, common to both worlds of PD_π; its disclosure is a disclosure of u and, where birthdays differ, a linking component in UL_π.

Proof. The birthday B is a component of L and m a function of the view; the statement is an instance of Theorem 8.8. □

Proposition 8.13 (Range bounds and scan state)**.** *In the interaction of ZIP 307, "Client-server interaction", the range of phase B starts at the height X that the previous synchronisation reached, and that of phase D at the height Y that phase B reached. Each start therefore reveals the previous synchronisation height, the tip at the time of the previous session, and so links consecutive sessions: in UL_π an adversary that chooses u_1 with a resume height other than the tip returned to the first session wins with probability 1. Under a prioritised policy the bounds are those of the policy's ranges (verification ranges near the stored tip, extents of subtrees, historic ranges); they reveal the scan state, the maximum and fully scanned heights of Definition 6.11, to the extent that the ranges are functions of it, and need not equal the previous synchronisation height.*

Proof. The bounds are components of L, and the statement is an instance of Theorem 8.8(ii) and (iii). For the adversary: when $b = 0$ the second session starts at the tip returned to the first, when $b = 1$ at the resume height of u_1, and the two differ with probability 1. □

Remark 8.14 (Order of range queries). The order of range queries is a component of L. A policy that orders its ranges by a wallet-dependent priority ("Scan order", §6.4, the reference interaction; "Witness readiness", §7.5, scheduling) reveals that order, by Theorem 8.8. Every such priority is designed but unspecified (Definition 1.1).

Proposition 8.15 (Out-of-order subtree completion)**.** *Let a policy, on detecting a note at position p of pool P, always request the block extent of subtree $s = \lfloor p/2^{16} \rfloor$ (Definition 7.16) as a query that the in-order schedule would not issue at that time. The extent starts at $\max(c_P(s-1), B)$ and ends at the completing height $c_P(s)$, or at the anchor height for the incomplete last subtree; the server knows every completing height from the responses to $\mathsf{SubtreeRoots}(P, 0)$, so it learns s and that the wallet holds a detected note among the at most 2^{16} leaves of subtree s. Against such a policy, an adversary in PD_π that places an output to K_0 alone in a subtree outside both wallets' in-order schedule at that time, and chooses K_1 with no receipt in Σ, wins with probability 1. The statement is an inference over the abstract policy of "Witness readiness" (§7.5), not a measurement of any deployment; no mitigation is specified, and one is an open problem (Definition 1.1).*

Proof. The requested extent is a function of p, a receipt, and a component of L; the map from extent bounds to s is public. Wallet 0 issues the extent query and wallet 1 does not, so the adversary that outputs $b' = 0$ exactly when the view contains that query wins with probability 1, an instance of Theorem 8.8(ii). □

Construction 8.16 (Bucketing)**.** Fix a bucket size $N \geq 1$ and let $\lfloor X \rfloor_N := X - (X \bmod N)$. Phase B requests $\mathsf{Blocks}(\lfloor X \rfloor_N, Y, \mathcal{P})$ in place of $\mathsf{Blocks}(X, Y, \mathcal{P})$, and phase D requests $\mathsf{Blocks}(\lfloor Y \rfloor_N, Z, \mathcal{P})$ in place of $\mathsf{Blocks}(Y, Z, \mathcal{P})$. The blocks from $\lfloor X \rfloor_N$ to X are checked, height by height, to carry the block hash of the wallet's chain view, and are then discarded without further processing. On a mismatch at height $Q > \lfloor X \rfloor_N$ the wallet rolls back (Definition 7.23) to $Q - 1$ if it holds $C(Q-1)$. Otherwise it obtains $C'(Q-1) := \mathsf{TreeState}(Q-1)$, compares hashes as in step (3) of Construction 7.24, takes its frontiers and rolls back to $Q - 1$. Both chains then agree at $Q - 1$, and it sets $X := Q - 1$. A mismatch at $Q = \lfloor X \rfloor_N$ goes to Construction 7.24 with $e := Q$. In phase D, Y replaces X throughout this check. Most reorganisations are thereby handled within the same range. The construction is specified, as the modification of phases B and D in ZIP 307, "Block privacy via bucketing", which the ZIP describes as a proposed enhancement.

Proposition 8.17 (Leakage under bucketing)**.** *Under Construction 8.16 the start of a range of phase B or D reveals $\lfloor X \rfloor_N$ and no more of the resume height X: two wallets whose resume heights share a bucket and whose parameter tuples agree elsewhere issue ranges with equal starts. The ends of the ranges, their order and their timing are unchanged, and so is their leakage. The gain is in the set of resume heights consistent with a start, from one height to one bucket of N heights; it is not an advantage bound of UL_π, in which an adversary that chooses u_1 with a resume height in another bucket still wins with probability* 1.

Proof. The start is $\lfloor X \rfloor_N$, a function of X that is constant on buckets; the other components of L are untouched (Theorem 8.8). The last clause is the argument of the proof of Proposition 8.13 with starts in distinct buckets. □

8.4 Leakage of transaction queries

Proposition 8.18 (Uncovered transaction queries)**.** *A query* $\mathsf{Tx}(\mathsf{txid})$ *or* $\mathsf{Status}(\mathsf{txid})$ *issued without cover puts that transaction into L and reveals it as one that the wallet is fetching or tracking: a received transaction whose memo or outgoing plaintext the wallet recovers (the enhancement of Construction 6.8), a sent one, or a transparent dependency. This is the set that payment-detection privacy protects. Against a policy that fetches exactly its wallet-relevant transactions, an adversary in PD_π that places an output to K_0 alone, at a height that the common schedule scans before the horizon, and chooses K_1 with no receipt in Σ, wins with probability* 1. *A transaction identifier queried in two sessions links them in UL_π.*

Proof. The fetched set is a function of receipts and a component of L. Wallet 0 fetches the transaction of the output and wallet 1 does not; the statement is an instance of Theorem 8.8(ii) and (iii). □

Remark 8.19 (Rules of ZIP 307 for full transactions). ZIP 307, "Transaction privacy", states the following interoperability rules. A light client needs full transactions for the memo fields of received notes and for outgoing ciphertexts, to recover spend information when a seed is imported. It SHOULD obscure the exact transactions of interest by also downloading numerous uninteresting transactions; it SHOULD download all transactions of any block from which it fetches a single full transaction; and it MUST convey to the user that fetching full transactions reduces privacy. The MUST is unconditional, not a fallback for omitting the SHOULDs. Under the second SHOULD the fetched set reveals the blocks that hold a transaction of interest, not the transactions. The rules are specified (Definition 1.1).

No query of Definition 5.10 returns the full transactions of a block, since Block returns the compact block; the second SHOULD therefore costs one Tx query per transaction of the block.

Remark 8.20 (Timing of transaction queries). A Tx query issued at a time fixed by the completion of the range in which its transaction was detected is correlated in time with that range; under network privacy with separate connections, the times in L still link the two (Theorem 8.8(iii)). A delay drawn independently of that completion time would remove the correlation. No ZIP specifies one for transaction queries, and no stated design fixes its law: an open problem (Definition 1.1).

8.5 The submission channel

Proposition 8.21 (Submission through the server). *Let the wallet submit a transaction by* Submit *on a connection c to its synchronisation server. The server then holds the complete transaction before it is broadcast, at a time in L, on a connection with endpoint E_c. Without separation at the network layer it links the transaction to that endpoint and to the other queries of the same connection or endpoint, among them the birthday bound, the range bounds and the transaction queries of the session. The disclosure lies outside the goal of ZIP 307 and inside the view of Theorem 8.8.*

Proof. The transaction, its time and its connection are components of the view; the statement is an instance of Theorem 8.8(i). □

The server's operator terminates the transport, so channel encryption such as TLS protects queries from third parties and not from the server. Every connection exposes to the server its endpoint E_c and its open and close times, the endpoint being the wallet's network address or that of an exit of an anonymity network or of a proxy. Without network privacy E_c is a component common to all sessions of one wallet, which is why the unlinkability goal of ZIP 307, "Security Model", assumes network privacy.

Proposition 8.22 (Separate circuits). *Let submission and synchronisation use connections whose endpoints are drawn independently, as separate anonymity-network circuits give; or let submission go to a server that does not share its view with the synchronisation server, in which case the* Submit *query is absent from that server's view and the separation rests on that non-collusion assumption. Then the endpoint no longer links the* Submit *query to the synchronisation session. Correlation remains through the times in L; through content, the anchor and expiry heights of the transaction against the range bounds of the session; and through application-level identifiers, such as a later* Status *query for the submitted transaction identifier on the synchronisation connection. For payments the separation is designed but unspecified (Definition 1.1). For migration transactions ZIP 318 specifies the network-privacy step ("Network-layer privacy": the wallet MUST offer Tor and, where available, Nym, and MAY broadcast through a server other than the synchronisation server) and the separation in time of synchronisation from broadcast ("Decoupling synchronization from broadcast").*

Proof. The endpoint components of the two connections are independent, so their joint law is the same whether or not the connections belong to one wallet; every other component of L is unchanged, and the remaining channels are those components (Theorem 8.8(iii)). In the second case the synchronisation server's view contains no Submit query, and under non-collusion the other server's view is not joined to it. □

8.6 Summary of server observables

Table 5 lists the observables of the server, what each reveals, its mitigation and the class of that mitigation (Definition 1.1). The interface of Definition 5.10 omits mempool queries, so no row concerns the mempool.

Observable	Reveals	Mitigation	Class
Network endpoint and connection times	network location and activity windows	network anonymity (Proposition 8.22)	designed but unspecified; specified for migration broadcasts (ZIP 318)
Least requested height	an upper bound on the birthday (Proposition 8.12)	a start at Sapling activation, at the cost of Proposition 6.17	specified for an imported seed (ZIP 307)
Range bounds and order	resume height and scan state (Proposition 8.13)	bucketing of the start (Construction 8.16)	specified (ZIP 307)
Out-of-order completion ranges	the subtree of a detected note (Proposition 8.15)	none	open problem; the leak is inferred
Transaction queries	the transactions of interest (Proposition 8.18)	decoys and whole-block fetches (SHOULD)	specified (ZIP 307)
Submission	the transaction before broadcast, linked to its session (Proposition 8.21)	separate circuits	designed but unspecified
Transparent-address queries	the addresses, by declaration (§8.2)	one address per query at exponential times in random order (Proposition 8.11); linkage by connection and timing remains	designed but unspecified
Pool sets of range queries	if narrowed by wallet state, the height at which the Orchard-pool balance became known to be zero (Corollary 8.9)	one pool set fixed independently of receipts; only trial decryption omitted	designed but unspecified

Table 5: Observables of the light-client server, what each reveals about the wallet, the mitigation of each, and the class of that mitigation (Definition 1.1). Every row is a component of the schedule parameters or of the transport metadata of Theorem 8.8.

Every row of Table 5 is a component of L or of the transport metadata (Theorem 8.8); none is a function of the content of compact blocks (Corollary 8.9). The size of each leak depends on policies that the protocol leaves open: beyond the rules of ZIP 307 and the migration schedule of ZIP 318, no ZIP fixes the scheduling policy of a wallet.

9 Anchors, confirmations, and expiry

This section fixes the heights and the set of notes from which a wallet builds a transaction. It states the confirmation policy of ZIP 315; the target height and the anchor height, with the lemma that ties the anchor depth to the trusted confirmation depth; the expiry rule of ZIP 203 and the expiry height a wallet sets; the record a wallet keeps of the transactions it creates and the notes those transactions lock; and the spendability categories of ZIP 315, with the spendable set defined from the wallet's state.

9.1 The confirmation policy

An output mined in a block of the best chain leaves the chain when a reorganisation replaces that block (*Ironwood Guide*, §"Anchors", Definition "Reorganisation"), so a wallet spends an output only once it has a number of confirmations fixed by a policy. The rules of this subsection are those of ZIP 315 and are specified (Definition 1.1); the status of ZIP 315 is that of Table 1.

Definition 9.1 (Confirmation policy)**.** A *confirmation policy* is a triple $(c_{\mathrm{tr}}, c_{\mathrm{untr}}, z)$ of integers $1 \leq c_{\mathrm{tr}} \leq c_{\mathrm{untr}}$, the *trusted* and the *untrusted confirmation depth*, and a flag $z \in \{\mathsf{true}, \mathsf{false}\}$ that permits zero-confirmation shielding. Confirmations are those of the *Consensus Guide*, §"Work, and the best chain", Definition "Confirmations": at a tip of height T, an output mined in the block of height $m \leq T$ of the best chain has $T - m + 1$ confirmations, its own block being the first; an output of a transaction that no block of the best chain contains has 0. ZIP 315 RECOMMENDS the depths $c_{\mathrm{tr}} = 3$ and $c_{\mathrm{untr}} = 10$ ("Trusted and untrusted TXOs") and permits, without recommending, $z = \mathsf{true}$ ("Transparent UTXOs", MAY); a wallet SHOULD have a policy that it communicates to the user in a clear form ("Trusted and untrusted TXOs").

Definition 9.2 (Trusted and untrusted outputs)**.** A *TXO* is an output, transparent coin or shielded note, of a transaction of the best chain or of a mempool visible to the wallet (ZIP 315, "Terminology"). A TXO received by the wallet is *trusted* if it is an output of a transaction created by the wallet itself, which a wallet SHOULD treat as trusted, or of an external transaction that the user has marked trusted, a marking that a wallet MAY offer; every other received TXO is *untrusted* (ZIP 315, "Trusted and untrusted TXOs"; ZIP 315, "Prompt accessibility of funds"). A trusted TXO requires c_{tr} confirmations and an untrusted TXO c_{untr}. The output of a shielding transaction is not trusted by virtue of its internal receiver (ZIP 315, "Transparent UTXOs"); its depth rule is stated below.

The rationale of the default depths is ZIP 315's and is cited, not derived (ZIP 315, "Rationale for the given numbers of confirmations"): the trusted depth is 3 because reorganisations are anecdotally shorter than three blocks, and the two depths differ because the value of a trusted TXO is recoverable after a rollback, the wallet re-creating its transaction, whereas an untrusted TXO may need the sender to pay again. The modelled double-spend risk of a given count is that of the *Consensus Guide*, §"Waiting for confirmations" and §"Wallet confirmation policy": three confirmations give an attacker with 10% of the hash rate a success probability of 1.712%, and ten about 0.0008%. A confirmation count bounds the modelled double-spend probability only for a

stated attacker share q of the hash rate and a tolerated probability ε (*Consensus Guide*, §"Wallet confirmation policy"); it is not a guarantee.

Confirmation counts are in blocks, not in time. At the 25-second target spacing (ZIP 218, "Block target spacing"; *Consensus Guide*, §"Zcash time: 25-second blocks"), 10 blocks are $10 \times 25 = 250$ s and 3 blocks are $3 \times 25 = 75$ s. ZIP 218 does not argue for reducing confirmation counts (ZIP 218, "Motivation") and keeps the anchor depth at 3 blocks (ZIP 218, "Anchor selection depth"). Neither an equal count nor an equal elapsed time establishes an equal risk, whose model is that of the *Consensus Guide*.

Two kinds of transaction enter the rules below; both are constructed in "TEX transfers and shielding transactions" (§11.3). A *shielding transaction* spends transparent UTXOs of the wallet and pays their value, less the fee, to an internal shielded receiver of the spending account, with no other recipient. An *ephemeral UTXO* is the transparent output that the first transaction of a transfer to a TEX address creates and its second transaction spends (ZIP 320, "Design considerations for Senders"; ZIP 315, "Trusted and untrusted TXOs").

The following rules are specified. *Floor*: wallets SHOULD NOT permit the spending of TXOs with fewer than 3 confirmations, except (i) transparent UTXOs spent in shielding transactions and (ii) ephemeral UTXOs of a ZIP 320 transfer, whose second transaction spends an output that may be unmined (ZIP 315, "Trusted and untrusted TXOs"). *Zero-confirmation shielding*: wallets MAY spend received transparent UTXOs in wallet-internal shielding transactions with zero confirmations (ZIP 315, "Transparent UTXOs"); the flag z of Definition 9.1 records whether the wallet does. *Shielding outputs*: a shielded output of a wallet-internal shielding transaction meets its depth requirement if and only if every transparent UTXO that the transaction spent has at least $c_{\text{untr}} - c_{\text{tr}}$ confirmations and the shielding transaction has at least c_{tr} (ZIP 315, "Shielded TXOs"). With h_s the greatest mining height of the spent UTXOs and T the tip, the first condition is $T - h_s + 1 \geq c_{\text{untr}} - c_{\text{tr}}$.

Remark 9.3 (Two readings of the shielding rule). ZIP 315 states the rule for shielding outputs twice, and the two statements differ. The paragraph "Shielded TXOs" requires $c_{\text{untr}} - c_{\text{tr}}$ confirmations of the source UTXOs; the paragraph "Transparent UTXOs" requires c_{untr} confirmations of the transaction that produced them. Both require c_{tr} confirmations of the shielding transaction. Let the sources be mined at greatest height h_s and the shielding transaction at $h_s + d$ with $d \geq 0$. At the policy $(3, 10, \mathsf{true})$, the first reading is met at every tip

$$T \geq h_s + \max(6,\ 2 + d),$$

and the second at every tip $T \geq h_s + \max(9,\ 2 + d)$; the two agree for $d \geq 7$. Indeed, a block at height m has $T - m + 1$ confirmations at tip T (Definition 9.1), and

$$\begin{aligned} T - h_s + 1 \geq 7 &\iff T \geq h_s + 6, \\ T - h_s + 1 \geq 10 &\iff T \geq h_s + 9, \\ T - (h_s + d) + 1 \geq 3 &\iff T \geq h_s + 2 + d. \end{aligned}$$

The rule stated above is the first reading, the one inside ZIP 315's definition of a confirmed-spendable TXO. Which reading the ZIP intends is an open problem (Definition 1.1).

9.2 Target and anchor heights

Definition 9.4 (Target height)**.** Let T be the height of the tip of the wallet's view of the chain, the height returned by $\mathsf{Tip}()$ (Definition 5.10). The *target height* of a transaction built at that view is $H := T + 1$, the height of the first block that could include it. Every height-dependent rule of construction is fixed by H: the consensus branch, the fee rule and the expiry height are

those of a block of height H, and confirmation counts are taken at the tip $T = H - 1$.

Definition 9.5 (Anchor height)**.** Let $(c_{\mathrm{tr}}, c_{\mathrm{untr}}, z)$ be a confirmation policy (Definition 9.1) and $H \geq c_{\mathrm{tr}}$ a target height. The *anchor height* of a transaction with target height H is

$$A := H - c_{\mathrm{tr}},$$

a function of H and the policy alone. One anchor height serves every anchor that the transaction carries, including that of an Orchard-protocol bundle whose Actions spend no note. For a pool $P \in \{\mathrm{Sapling}, \mathrm{Orchard}, \mathrm{Ironwood}\}$, the anchor of the transaction's bundle of P is the root of P's note commitment tree in the final treestate of the block at height A (*Ironwood Guide*, §"Anchors", Definition "Treestate and anchor"), and the authentication path of every note that the transaction spends from P is computed to that root (*Ironwood Guide*, §"Authentication paths"). Since $c_{\mathrm{tr}} \geq 1$, the block at height A precedes the block at height H, as the consensus anchor rules require (*Ironwood Guide*, §"Anchors"; protocol specification, §"Action Transfers and their Descriptions" and §"Spend Transfers, Output Transfers, and their Descriptions"). A transaction therefore carries, for each pool it touches, its own anchor and authentication paths, the Orchard pool and the Ironwood pool of the Orchard protocol separately.

Lemma 9.6 (Anchor height and confirmations)**.** *Let T be the tip, $H = T + 1$ and $A = H - c_{\mathrm{tr}}$. For a note mined at height $m \leq T$ of the best chain, the following are equivalent:*

(*i*) *the note's commitment is a leaf of the tree rooted by the anchor of its pool at A;*

(*ii*) *$m \leq A$;*

(*iii*) *the note has at least c_{tr} confirmations at T.*

The anchor depth thus equals the trusted confirmation depth: a note with exactly c_{tr} confirmations lies in the block at height A. At $c_{\mathrm{tr}} = 3$ the anchor height is $A = T - 2$, and a note with at least $c_{\mathrm{untr}} = 10$ confirmations is mined at a height at most $T - 9 < A$.

Proof. The final treestate of the block at height A holds exactly the commitments appended in the blocks at heights $0, \ldots, A$ (*Ironwood Guide*, §"Anchors", Definition "Treestate and anchor" and Remark "Old anchors"), so (i) and (ii) are equivalent. By Definition 9.1 the note has $T - m + 1$ confirmations, and $T - m + 1 \geq c_{\mathrm{tr}}$ if and only if $m \leq T + 1 - c_{\mathrm{tr}} = A$, so (ii) and (iii) are equivalent. The instances follow with $c_{\mathrm{tr}} = 3$, $A = T + 1 - 3$, and $T - m + 1 \geq 10$ if and only if $m \leq T - 9$. □

The rule for the anchor is specified: a wallet SHOULD choose an anchor a number of blocks back from the head of the chain equal to the trusted confirmation depth, the ZIP's instance being the final treestate of the block at height $H_{315} - 3$ when the current block is at height H_{315} (ZIP 315, "Anchor selection"). ZIP 218 recommends that the depth stay 3 blocks at the 25-second spacing, $3 \times 25 = 75$ s (ZIP 218, "Anchor selection depth"). The rule is a recommendation to the builder, not a consensus rule: the anchor rules admit the final treestate of any earlier block.

Remark 9.7 (The anchor convention). ZIP 315 counts the anchor depth back from the head of the chain, and ZIP 218, "Anchor selection depth", reads it as the tip, so that a 4-block reorganisation is needed to remove the anchor; the *Ironwood Guide*, §"Anchors", Remark "Recommended anchor depth", follows that reading, which puts the anchor at $T - 3$ with $c_{\mathrm{tr}} + 1$ confirmations. This volume's convention counts from the block being built, $H = T + 1$, one block shallower, which gives Lemma 9.6: the anchor block has exactly c_{tr} confirmations. Both satisfy the consensus anchor rules.

The anchor depth is c_{tr} whatever the trust of the notes spent. The eligibility of each note is decided by its own depth (Definition 9.2), so an untrusted note spent at depth c_{untr} is proved against the anchor at depth c_{tr}, whose tree contains it by Lemma 9.6. ZIP 315 gives three reasons for the rule, specified and cited (ZIP 315, "Rationale for anchor selection"): a rollback past the anchor block invalidates the transaction, whose revealed nullifiers then link it to any later transaction spending the same notes; too deep an anchor keeps recently received notes from being spent; and a fixed depth, rather than one depending on whether the notes spent are trusted, does not reveal their trust. The anchor is published data (*Ironwood Guide*, §"Anchors", Remark "Anchor choice as public data").

Every Orchard-protocol bundle with at least one Action carries an anchor, which must refer to the final treestate of an earlier block in its pool, whether or not any Action spends a note of nonzero value (protocol specification, §"Action Transfers and their Descriptions"; consensus rule). Giving such a bundle the anchor at A makes it indistinguishable by its anchor from a bundle that spends (*Ironwood Guide*, §"Anchors", Remark "Anchor choice as public data").

9.3 Expiry (ZIP 203)

Definition 9.8 (Transaction expiry)**.** Every transaction of version at least 3 carries a field $\mathsf{nExpiryHeight} \in \{0, \ldots, 2^{32} - 1\}$, a block height and never a timestamp; for a non-coinbase transaction, $\mathsf{nExpiryHeight} \leq 499{,}999{,}999$ (consensus rule). A non-coinbase transaction with $N := \mathsf{nExpiryHeight} \neq 0$ is *expired* at height h if and only if $h > N$, and it MUST NOT be mined at a height greater than N (consensus rule), so the block at height N is the last that may include it; the value $N = 0$ disables expiry (protocol specification, §"Transaction Encoding and Consensus"; ZIP 203, "Specification"). Coinbase transactions, which a wallet does not create, are outside the definition.

Lemma 9.9 (Includability at the next block)**.** *Let t be a non-coinbase transaction with expiry height N, and T the tip. The expiry rule permits t in the block at the target height $H = T + 1$ if and only if $N = 0$ or $T < N$; when $N \neq 0$, it permits t in no block of height greater than N. Inclusion requires the other consensus rules as well; the lemma concerns expiry only.*

Proof. By Definition 9.8, the rule permits t at height H if and only if $N = 0$ or $H \leq N$, and $H = T + 1 \leq N$ if and only if $T < N$. The second claim is the rule itself. □

Construction 9.10 (Expiry height of a wallet transaction)**.** A wallet sets $\mathsf{nExpiryHeight} := H + \delta$, where H is the target height (Definition 9.4) and $\delta \geq 1$ is an *expiry delta* chosen by the wallet. Reading ZIP 203's "current height" as the target height is designed but unspecified (Definition 1.1). Every later rule that depends on the delta, the locking of notes (§9.4) and the polling and re-sending of transactions (§13), uses the δ actually chosen.

ZIP 315 specifies (SHOULD) that a wallet create transactions using the default expiration height of ZIP 203, which it gives as 40 blocks from the current height (ZIP 315, "Expiration height"). ZIP 218 changes the default of node implementations that create transactions to $\mathsf{NU7PoWTargetSpacingRatio} \times 40 = 120$ blocks (SHOULD), to keep an expiry of about 50 minutes, and keeps an explicitly configured delta (ZIP 218, "Default expiry delta"); ZIP 315 is not updated for the 25-second spacing. At that spacing, ZIP 315's SHOULD therefore has two readings: the literal 40 blocks, about $40 \times 25 = 1000$ s, and the ZIP 203 default as ZIP 218 changes it, 120 blocks, about $120 \times 25 = 3000$ s. Which one a wallet uses is an open problem

(Definition 1.1). Either time is an estimate, since block intervals vary, and not a guaranteed period of validity.

The choice of delta has a privacy consequence, specified. ZIP 315 lists the expiry height, with the anchor position, among the data that reveal to which height the creator was synchronised and some of its policy (ZIP 315, "Kinds of information leakage"), and it specifies a common default (ZIP 315, "Expiration height"). An expiry delta other than the common one therefore distinguishes its creator, as a fee other than the conventional fee does (§10).

The expiry height is fixed by the transaction identifier: nExpiryHeight is effecting data, an input of the header digest (ZIP 244, "T.1: header_digest"), so by Lemma 13.1(ii) a party matches a transaction, and rejects it once expired, by its identifier alone.

When a transaction expires, nodes remove it from their mempools together with every transaction that depends on it (ZIP 203, "Specification"). This is node policy layered on the consensus rule, which constrains only the contents of blocks.

9.4 The transaction record and locked notes

Definition 9.11 (Transaction record)**.** The *transaction record* of a wallet holds, for each transaction t that the wallet creates:

(i) the identifier $\mathsf{txid}(t)$ and the encoding of t, once the wallet holds it;

(ii) its fee, its target height H_t and its expiry height N_t;

(iii) the set of its inputs: the notes it spends in each shielded pool and the transparent outpoints it spends;

(iv) its mined height, or $\perp$ while it is unmined;

(v) the earliest height h_t^{obs} at which it was observed: initially H_t for a transaction the wallet builds, and the height of the block in which it was first seen mined for a record entered from compact blocks;

(vi) the latest tip at which the server reported it unmined.

A record entered without the encoding of the transaction has N_t unknown; such is the record of a transaction learnt from compact blocks, whose compact transaction carries no expiry height (Definition 4.3), and later unmined by a rollback (Definition 7.23). The transitions of the record are those of "Status tracking and terminal states" (§13.2). The record is information that the wallet retains, not a storage layout.

Definition 9.12 (Locked note)**.** Fix an expiry delta $\delta \geq 1$, a parameter of the definition, whose default at the 25-second spacing is the open problem of §9.3. At target height H, a note is *locked* if and only if some recorded transaction t that spends it satisfies one of:

(a) t is mined, necessarily at a height below H;

(b) $N_t = 0$;

(c) $N_t \geq H$;

(d) N_t is unknown and $h_t^{\mathrm{obs}} + \delta \geq H$.

A locked note is excluded from selection.

Proposition 9.13 (Locks follow includability)**.** *Let t be a recorded transaction with known expiry height N_t that spends a note. At target height $H = T + 1$, the conditions of Definition 9.12 hold for t if and only if t is mined or the expiry rule permits t at H (Lemma 9.9). Hence, for a note that no mined recorded transaction spends and whose recorded spending transactions all have known expiry heights, being unlocked is the clause of ZIP 315 that the TXO is not committed to be spent in another unexpired transaction created by the wallet (ZIP 315, "Categories of TXOs according to spendability"); a mined spend, which condition (a) locks, makes the note spent, and ZIP 315's clause that the TXO is unspent excludes it; and an unmined t with $N_t > 0$ releases its inputs, as far as t is concerned, at every target height $H > N_t$.*

Proof. For known N_t, condition (d) of Definition 9.12 is void. By Lemma 9.9, the expiry rule permits t at H if and only if $N_t = 0$ or $H \leq N_t$, which are conditions (b) and (c); with condition (a) this gives the first claim. By Definition 9.8, t is unexpired at H if and only if $N_t = 0$ or $H \leq N_t$, so for an unmined t the conditions hold exactly when t is an unexpired transaction of the wallet that spends the note, and a note is unlocked exactly when no such transaction exists, which is ZIP 315's clause (ZIP 203, "Specification"). For $H > N_t > 0$ and t unmined, each of (a) to (d) fails. □

Remark 9.14 (Unknown expiry). Condition (d) of Definition 9.12 assumes that t was created with the delta δ at a target height at most h_t^{obs}, so that $N_t \leq h_t^{\text{obs}} + \delta$. If $N_t = 0$, or its creator used a larger delta with $N_t > h_t^{\text{obs}} + \delta$, the note is released from $H = h_t^{\text{obs}} + \delta + 1$ while t can still be mined (for $H \leq N_t$, or at every H if $N_t = 0$), and a re-spend conflicts with the late mining of t. If its creator used a smaller delta, the note stays locked longer than necessary. Either misjudgement lasts until the wallet sees t mined, or obtains its encoding and hence N_t by the query Tx (Definition 5.10).

Definition 9.15 (Watched nullifiers)**.** The *watched nullifiers* are the set $W := \bigcup_P W_P$ of the tracked sets of Definition 6.7: the nullifiers of the wallet's detected notes not marked spent. A lock (Definition 9.12) removes a note from selection and never removes its nullifier from W. A nullifier leaves W only when Construction 6.8 marks its note spent, that is, when a transaction revealing it is mined, whatever that transaction's expiry height, $N = 0$ included; or when a rollback unmines the creating transaction of a Sapling note, whose nullifier is then discarded until a later scan derives it again (Definition 7.23(iii)). A rollback that clears the spent mark returns the nullifier to W.

The mining of any transaction that spends a note of the wallet is therefore detected by scanning through W, independently of status queries (*Ironwood Guide*, §"Nullifier sets"). A rule that removed from W the nullifiers spent by transactions with $N = 0$, combined with status polling that ends at the first report of an unmined transaction, would leave such a transaction, when it has no output to the wallet, mined and never observed as mined: no block would reveal a watched nullifier, and no note of the wallet would be created by it.

9.5 Spendable notes

Definition 9.16 (Spendability categories)**.** Relative to a chain and a wallet state (ZIP 315, "Categories of TXOs according to spendability"):

(i) a TXO is *known-spendable* if and only if (1) it is unspent at the wallet's view of the tip and that view is reasonably up to date, (2) it is not committed to be spent in another unexpired transaction created by the wallet (Proposition 9.13), and (3) the wallet can effect the spend, by holding its spending key or by mediating interaction with a hardware

device or another offline signing protocol;

(ii) a shielded TXO is *confirmed-spendable* if and only if it is known-spendable, the wallet can construct its witness (Theorem 7.14), and either it is not an output of a wallet-internal shielding transaction and has c_{tr} confirmations if trusted, c_{untr} if untrusted (Definition 9.2), or it is such an output and meets the rule for shielding outputs of §9.1 (ZIP 315, "Shielded TXOs");

(iii) a known-spendable TXO that is not confirmed-spendable is *unconfirmed-spendable*;

(iv) a TXO is *watch-only* if and only if the wallet holds its full viewing key, or for a transparent TXO its address, but not its spending key;

(v) a transparent UTXO is *confirmed-shieldable* if and only if it is known-spendable and the wallet can construct a correct redeem script for it (ZIP 315, "Transparent UTXOs").

The following rule is specified: a wallet MUST NOT attempt to spend, in a user-initiated transaction, a TXO that is not confirmed-spendable. ZIP 315 notes that the confirmed-spendable balance is 0 until the wallet has synchronised at least the nullifier set to the tip (ZIP 315, "Reporting of balances").

Definition 9.17 (Spendable note at a target height)**.** Let T be the tip, $H = T + 1$, $A = H - c_{\text{tr}}$ (Definition 9.5), and f the fully scanned height (Definition 6.11). A shielded note n of pool P, mined at height m, is *spendable at* H if and only if:

(a) the note n is in the note state (Definition 6.7), it is not marked spent, and $f \geq T$;

(b) it is not locked at H (Definition 9.12);

(c) the wallet holds the spending key of the account of n, or mediates a signer for it;

(d) it meets its depth: $T - m + 1 \geq c_{\text{tr}}$ or $T - m + 1 \geq c_{\text{untr}}$ according to its trust (Definition 9.2), or, if n is an output of a wallet-internal shielding transaction, the rule for shielding outputs of §9.1;

(e) the authentication path of the position of n to the anchor of P at A is computable from retained information (Theorem 7.14, Proposition 7.17).

The *spendable set* at H is the set of notes spendable at H, each with its value, pool, position and authentication path at A.

Condition (d) implies $m \leq A$ in each case, since $c_{\text{untr}} \geq c_{\text{tr}}$ and the shielding transaction has at least c_{tr} confirmations, so n lies in the tree of the anchor by Lemma 9.6. The spendable set at H is the set of confirmed-spendable shielded notes of Definition 9.16, with ZIP 315's "reasonably up-to-date" instantiated as $f \geq T$; the instantiation is designed but unspecified (Definition 1.1). Clause (a) is clause (1) of known-spendability: with $f \geq T$, every block up to the tip has been scanned, and a note is marked spent exactly when its nullifier is revealed on the best chain at or below T (Theorem 6.14, item (ii)). Clause (b) is clause (2) by Proposition 9.13, with condition (d) of Definition 9.12 in place of the expiry height where that is unknown; clause (c) is clause (3); clauses (e) and (d) are the witness and depth clauses of confirmed-spendability. The spendable set is the input state of construction ("Transaction construction", §11): it is reconstructed by synchronisation (§6, Theorem 6.14) and taken as given by fee computation and note selection.

Remark 9.18 (Spending before the nullifier set is complete). A policy that admits a note once the block extent of its own subtree is scanned up to A (Definition 7.16), before every range up to

T is scanned, departs from clause (a) of Definition 9.17 and from ZIP 315, whose clause on an up-to-date view carries an open TODO in the ZIP. The departure fails as follows. A note whose spend was mined in a range not yet scanned passes the policy; a transaction that spends it again is rejected by the consensus rule that a nullifier must not repeat (*Ironwood Guide*, §"Nullifier sets"), and its repeated nullifier links it to the earlier spend for every party that sees it. The departure is designed but unspecified (Definition 1.1).

10 Fees (ZIP 317)

This section states the fee a wallet pays. It fixes the fee as a consensus quantity, defines the conventional fee of ZIP 317 over logical actions and bounds it before signing; it states the padded Action counts on which the fee is computed; and it states the relay, mempool and block-template rules that apply to a transaction paying less.

10.1 The conventional fee

For a transaction other than a coinbase transaction, the *transaction fee* is the value remaining in its transparent transaction value pool: the sum of its transparent input values and of the balancing values of its shielded pools, minus the sum of its transparent output values (protocol specification, §"Transactions and Treestates"; *Ironwood Guide*, Definition "Transparent transaction value pool" and Definition "Balancing value", in "The balancing value"). Consensus requires it to be non-negative. No transaction field encodes it, and every term of the sum is public, so the fee is public.

A fee rule for a shielded protocol meets the requirements of ZIP 317, "Requirements": the fee is computable from the public data of the transaction alone, since a fee that depended on private construction data, such as which notes were spent or what change was returned, would disclose that data; it scales linearly with the numbers of inputs and outputs, which makes transaction-rate denial of service costly (ZIP 317, "Motivation" and "Denial of Service"); it neither penalises padding for privacy nor favours any pool; and it lets a small number of inputs of value below the marginal fee be spent. A fee other than the one every wallet pays distinguishes the wallet that chose it: non-standard fees may reveal specific users, wallets or wallet versions (ZIP 317, "Security and Privacy considerations").

The conventional fee (Definition 10.2) is not a consensus rule. ZIP 317, "Fee calculation", states that it is not a consensus requirement that fees follow the formula, and that wallets SHOULD create transactions that pay it, to reduce information leakage, unless overridden by the user. It binds through relay and block-template policy only (§10.3). ZIP 259, "NU7 consensus changes", changes neither the formula nor its parameters.

In what follows, $\lceil x \rceil$ is the least integer $\geq x$ and $\lfloor x \rfloor$ the greatest integer $\leq x$, for real x.

Definition 10.1 (Logical actions)**.** Let a transaction of version 5 or version 6 have the following public fields of its final encoding (protocol specification, §"Transaction Encoding and Consensus"; ZIP 229, "Transaction Format"), each a non-negative integer:

- *tx_in_total_size* and *tx_out_total_size*, the byte lengths of the encoded fields `tx_in` and `tx_out`, the concatenations of the encoded transparent inputs and outputs;
- *nJoinSplit*, the number of Sprout JoinSplit descriptions;
- *nSpendsSapling* and *nOutputsSapling*, the numbers of Sapling spend and output descriptions;

- *nActionsOrchard*, the number of Orchard-pool Actions;
- *nActionsIronwood*, the number of Ironwood-pool Actions, 0 in a version 5 transaction.

With the parameters *p2pkh_standard_input_size* := 150 and *p2pkh_standard_output_size* := 34, in bytes, the number of *logical actions* of the transaction is

$$\begin{aligned} \mathit{logical_actions} := {} & \max(\lceil \mathit{tx_in_total_size}/150 \rceil, \lceil \mathit{tx_out_total_size}/34 \rceil) \\ & + 2\,\mathit{nJoinSplit} + \max(\mathit{nSpendsSapling}, \mathit{nOutputsSapling}) \\ & + \mathit{nActionsOrchard} + \mathit{nActionsIronwood}, \end{aligned}$$

the sum of the transparent, Sprout, Sapling, Orchard and Ironwood contributions, in that order (ZIP 317, "Fee calculation", the contributions of Revision 0 and Revision 1).

The Sprout contribution $2\,\mathit{nJoinSplit}$ is 0 in every valid transaction: transactions of version 5 and version 6 have no JoinSplit field, and transactions of version 4 are invalid under NU7 (ZIP 2003; ZIP 259, "NU7 consensus changes"). The term is kept so that the definition is that of ZIP 317. The two divisors are the maximal encoded sizes of a P2PKH input and output (ZIP 317, "Rationale for the chosen parameters"):

$$\begin{aligned} 150 &= 36 \text{ (outpoint: 32-byte txid, 4-byte index)} + 110 \text{ (script)} + 4 \text{ (sequence)}, \\ 110 &= 1 \text{ (length)} + 1 \text{ (signature length)} + 72 \text{ (signature)} + 1 \text{ (sighash type)} \\ &\quad + 1 \text{ (key length)} + 33 \text{ (compressed key)} + 1 \text{ (margin)}, \\ 34 &= 8 \text{ (value)} + 1 \text{ (script length)} + 25 \text{ (P2PKH script)}. \end{aligned}$$

Charging transparent parts by size rather than by count charges a large script in proportion to its size; a 2-of-3 P2SH multisig input, about 70% larger than a P2PKH input, is counted so (same section).

Definition 10.2 (Conventional fee)**.** Let *marginal_fee* := 5000 zatoshi per logical action and *grace_actions* := 2. The *conventional fee* of a transaction with *logical_actions* as in Definition 10.1 is

$$\mathit{conventional_fee} := \mathit{marginal_fee} \cdot \max(\mathit{grace_actions}, \mathit{logical_actions}) \text{ zatoshi}$$

(ZIP 317, "Fee calculation").

The parameter values and their rationale are ZIP 317's ("Rationale for the chosen parameters").

Remark 10.3 (Revisions of the fee rule). ZIP 317, "Fee calculation", defines *logical_actions* as the sum of the contributions of the applicable revision, and a contribution defined by a revision is included in every later one (ZIP 317, "Revisions"). Revision 0 defines the transparent, Sprout, Sapling and Orchard contributions. Revision 1 adds the Ironwood contribution *nActionsIronwood*: Ironwood-pool Actions have the Action structure of Orchard-pool Actions and are counted identically. Revision 2 adds a contribution for memo bundles (ZIP 231), defined on the transaction format of ZIP 248, which NU7 does not introduce (ZIP 259, "Rationale for no new transaction format"). Definition 10.1 is therefore Revision 0 with the Revision 1 term. The statuses of these ZIPs are those of Table 1.

Corollary 10.4 (The minimum fee)**.** *Every transaction with logical_actions* ≤ 2 *has conventional fee marginal_fee*$\cdot$*grace_actions* $= 5000 \cdot 2 = 10{,}000$ *zatoshi, and no transaction has a smaller conventional fee.*

Proof. By Definition 10.2, since $\max(2, \mathit{logical_actions}) = 2$ when $\mathit{logical_actions} \leq 2$, and $\max(2, \mathit{logical_actions}) \geq 2$ always. □

A wallet computes the fee before it signs, and the encoded size of a transparent input depends on its signature. The following proposition bounds the fee from the unsigned transaction.

Proposition 10.5 (Fee estimate before signing)**.** *Let a wallet fix, before signing, every field of a transaction except the scripts of its transparent inputs $1, \ldots, k$, and assign to each input i a bound $b_i \in \mathbb{N}$ on its encoded size after signing, as follows:*

(*i*) $b_i = 150$ *for a P2PKH input spending a public key with a compressed 33-byte encoding, such as each of the wallet's own transparent keys;*

(*ii*) *otherwise any b_i known to bound the encoded size of the signed input.*

Let $\mathit{fee}_{\mathsf{est}}$ be the conventional fee computed with tx_in_total_size replaced by $\sum_{i=1}^{k} b_i$. Then $\mathit{fee}_{\mathsf{est}}$ is at least the conventional fee of the signed transaction.

Proof. Signatures enter only the scripts of the transparent inputs, so the field `tx_out` and every shielded count of Definition 10.1 are the same in the unsigned and the signed transaction. A DER-encoded signature has at most 72 bytes, so the script of a P2PKH input with a compressed key has at most $1 + 1 + 72 + 1 + 1 + 33 = 109$ bytes and the input at most $36 + 109 + 4 = 149 \leq 150$ bytes; under clause (ii) the bound holds by hypothesis. The signed *tx_in_total_size* is therefore at most $\sum_i b_i$. The maps $x \mapsto \lceil x/150 \rceil$, $(a, b) \mapsto \max(a, b)$, the addition of the unchanged remaining contributions, and $n \mapsto 5000 \cdot \max(2, n)$ are each non-decreasing in every argument, so their composition is, and $\mathit{fee}_{\mathsf{est}}$ is at least the conventional fee of the signed transaction. □

Hypothesis (i) does not extend to a P2PKH input with an uncompressed 65-byte key encoding, which has up to 181 bytes and needs $b_i \geq 181$ under (ii). For an input with no known bound on its signed size, the fee is unknown before signing.

10.2 The fee on padded counts

The fields of Definition 10.1 are those of the final encoded transaction: every bundle padded and every change output included (ZIP 317, "Fee calculation"). A wallet therefore prices a transaction by predicting its padded shape (Definition 10.6) and its change before it selects inputs; a fee computed on unpadded counts underpays whenever padding adds a logical action. This rule is ZIP 317's and is not a consensus rule.

Definition 10.6 (Padded Action count)**.** Let a bundle of the Orchard protocol, in the Orchard pool or the Ironwood pool, have $s \in \mathbb{N}$ real spends and $o \in \mathbb{N}$ real outputs, and let $\mathsf{enableCrossAddress} \in \{0, 1\}$ be its flag value (*Ironwood Guide*, §"Bundle flags", Definition "Flag values and expanded receiver"). Its *requested Action count* is

$$r(s, o) := \begin{cases} \max(s, o) & \text{if } \mathsf{enableCrossAddress} = 1, \\ s + o & \text{if } \mathsf{enableCrossAddress} = 0, \end{cases}$$

and its *padded Action count* is $n(s, o) := 0$ if $s = o = 0$ and $n(s, o) := \max(r(s, o), 2)$ otherwise. The counts *nActionsOrchard* and *nActionsIronwood* of Definition 10.1 are the padded Action counts of the two bundles. A non-empty Sapling bundle with s real spends and o real outputs has $\mathit{nSpendsSapling} = s$ and $\mathit{nOutputsSapling} = \max(o, 2)$.

The count $s + o$ is that of the pairing a wallet performs when cross-address transfers are disabled: the note an Action creates has the expanded receiver of the note it consumes, so each real spend occupies an Action with a fabricated zero-valued output at its own receiver, and each real output an Action with a fabricated zero-valued spend at its receiver (ZIP 326, "Fabricated same-address outputs and randomized note ciphertexts"; ZIP 374, "Fabricated same-address outputs (NU6.3)"), respectively. Every Orchard-pool bundle has $\mathsf{enableCrossAddress} = 0$ (*Ironwood Guide*, "Chain state and pool rules", rule (i); ZIP 258, "Consensus rules from NU6.3 activation"); an Ironwood-pool bundle may set it to 1.

The floor 2 and the Sapling padding are a padding policy, designed but unspecified in the sense of Definition 1.1. Their only general ZIP anchor is non-normative: ZIP 317, "Requirements", notes that standard builders pad to two inputs and two outputs for each shielded pool, and "Rationale for the chosen parameters" calls padding to two actions the standard approach; ZIP 318 fixes the padding of its migration and note-preparation transactions only (below). The Sapling rule departs from ZIP 317's remark by adding no spend beyond those needed. Dummy Sapling spends, which this policy does not use, are constructed as in protocol specification, §"Dummy Notes (Sapling)"; the dummy Actions of the Orchard protocol are those of §11.7. Logical actions count an Action once: a charge per input plus per output would penalise the padding that the Action structure forces, since each Action consumes one note and creates one, and the maximum in each contribution favours no pool (ZIP 317, "Rationale for logical actions").

ZIP 318 specifies two exceptions. Under "Canonical migration transaction structure", a canonical migration transaction has one Orchard-pool spend in an Orchard-pool bundle of exactly two Actions, one spend Action and one output Action that carries either change or a zero-valued dummy output (ZIP 318, "Rationale for the canonical transaction shape"), and one Ironwood-pool output in an unpadded Ironwood-pool bundle of one Action. Its *logical_actions* is 3 and its conventional fee 15,000 zatoshi. The "canonical fee (provisionally the ZIP 317 minimum fee)" of ZIP 318 is read here as this conventional fee: read as 10,000 zatoshi, it leaves one unpaid action (Definition 10.8). A note-preparation transaction SHOULD have an Orchard-pool bundle of exactly 16 Actions, padded as needed (ZIP 318, "Note preparation transactions"), so its count is 16, not $n(s, o)$.

Proposition 10.7 (Fee of a single-pool payment)**.** *Consider a transaction with no transparent inputs or outputs, one input note and one payment output, with or without one change output, whose bundles are padded as in Definition 10.6.*

(*a*) *If the input note and every output lie in the Ironwood pool, in a bundle with* $\mathsf{enableCrossAddress} = 1$*, or all lie in the Sapling pool, the conventional fee is* 10,000 *zatoshi.*

(*b*) *If the input is a single Orchard-pool note, the payment output lies in the Ironwood pool, and the change output, if any, lies in the Orchard pool or the Ironwood pool, then the conventional fee is* 20,000 *zatoshi.*

Proof. (a) In the Ironwood pool $s = 1$ and $o \in \{1, 2\}$, so $r = \max(1, o) \leq 2$ and $n = 2$. In the Sapling pool the contribution is $\max(1, \max(o, 2)) = 2$. In either case *logical_actions* $= 2$, and Corollary 10.4 gives 10,000 zatoshi.

(b) No value enters the Orchard pool, since $\mathsf{valueBalanceOrchard} \geq 0$ for every transaction (*Ironwood Guide*, "Chain state and pool rules", rule (ii); ZIP 258, "Consensus rules from NU6.3 activation"), and an Orchard-pool output lies at the receiver of the note consumed in its Action; so a payment to another party cannot lie in the Orchard pool, and the full routing rule is that of §11.2. The payment lies in the Ironwood pool by hypothesis. The Orchard-pool bundle has $s = 1$ and $o \in \{0, 1\}$, so $r = s + o \leq 2$ and $n = 2$. The Ironwood-pool bundle has $s = 0$ and $o \in \{1, 2\}$, so $n = 2$. Hence *logical_actions* $= 4$ and the conventional fee is $5000 \cdot 4 = 20{,}000$ zatoshi. □

10.3 Relay, mempool, and block template

The conventional fee binds through three mechanisms, none of them a consensus rule. Nodes that normally relay transactions are expected to relay those paying at least the conventional fee, unless other reasons of robustness or denial-of-service mitigation apply (ZIP 317, "Transaction relaying"). Mempool size limiting adds a penalty *low_fee_penalty* to the eviction weight of a transaction paying less than the conventional fee (ZIP 401; ZIP 317, "Mempool size limiting"). Block-template construction limits the unpaid actions of a block (Definition 10.8).

Definition 10.8 (Unpaid actions)**.** For a transaction tx with fee $tx.fee$ as in §10.1 and $tx.logical_actions$ as in Definition 10.1, the number of *unpaid actions* is $unpaid_actions(tx) := 0$ if tx is a coinbase transaction, and otherwise

$$unpaid_actions(tx) := \max(0, \max(grace_actions, tx.logical_actions) - \lfloor tx.fee/marginal_fee \rfloor).$$

The unpaid actions of a block are the sum over its transactions. The recommended block-template algorithm adds a transaction that pays less than the conventional fee only while that sum stays at most $block_unpaid_action_limit = 50$ (ZIP 317, "Recommended algorithm for block template construction").

Lemma 10.9 (Zero unpaid actions)**.** *A transaction other than a coinbase transaction has* $unpaid_actions = 0$ *if and only if its fee is at least its conventional fee.*

Proof. Let $m := \max(grace_actions, logical_actions)$, an integer. By Definition 10.8, $unpaid_actions = 0$ if and only if $\lfloor fee/marginal_fee \rfloor \geq m$. For an integer m and real x, $\lfloor x \rfloor \geq m$ if and only if $x \geq m$; and since $marginal_fee > 0$, $fee/marginal_fee \geq m$ if and only if $fee \geq marginal_fee \cdot m$, which is the conventional fee of Definition 10.2. □

A transaction with more than $block_unpaid_action_limit = 50$ unpaid actions is never included by the recommended algorithm, and nodes MAY drop it, or any transaction with more unpaid actions than a configurable limit (ZIP 317, "Transaction relaying"). A relay policy may set that limit to 0; by Lemma 10.9 it then drops exactly the transactions that pay less than the conventional fee. The conventional fee is therefore the least fee that passes every unpaid-action limit, whatever its value. For example, the migration transaction of §10.2 with $logical_actions = 3$ has one unpaid action at a fee of 10,000 zatoshi.

A logical action is not the unit of the block limits of ZIP 218, "Shielded pool action limits": these count the sum $nSpendsSapling + nOutputsSapling$ of a block's Sapling descriptions, not the maximum that Definition 10.1 charges, so paying the conventional fee does not ensure that a transaction fits a given block (*Consensus Guide*, "A block budget for shielded work"). The limits add no transaction field. ZIP 259, "Rationale for no new transaction format", refers to a block-level fee contribution that needs no transaction field; no published ZIP specifies that calculation, so it is designed but unspecified (Definition 1.1), and the conventional fee does not depend on it.

11 Transaction construction

This section states how a wallet turns payments into transactions. It defines the transaction plan and its balance equation; the pool in which each payment and each change output lies; transfers to TEX addresses and shielding transactions; input selection as a fixed point of the fee; change; the outgoing viewing key of every output; the assembly of Orchard-protocol bundles with dummy

Actions in uniform order; and the authorisation of a built transaction, with the theorem that the result pays what the plan states. The input state is the spendable set of Definition 9.17; the fee is the conventional fee of §10.

11.1 The transaction plan

Definition 11.1 (Payment and payment request)**.** A *payment* is a transfer of funds implemented by a single output of a transaction: a triple (a, v, m) of a recipient address a (Definition 3.20), a value $v \in \mathbb{N}$ in zatoshi and an optional memo m, a memo being admitted only when a is memo-capable. A *payment request* is a finite map R from indices to payments; it asks for one transaction that pays every payment in its range (ZIP 321, "Terminology"). ZIP 321 encodes payment requests as URIs (§14.1); the label and message parameters of that encoding are display data and enter no transaction.

Construction has two phases. *Planning* maps a payment request and the wallet state, its spendable notes (Definition 9.17) and its tip, to a plan exact to the zatoshi, and uses no spending key. *Execution* builds, proves and authorises the transactions of the plan and records them (§13); submission is a separate act. Every choice of inputs, fee and change is fixed in the plan before a spending key is used. No ZIP defines plans: the two-phase structure is designed but unspecified (Definition 1.1).

Definition 11.2 (Transaction plan)**.** A *transaction plan* is a non-empty sequence $P = (S_1, \dots, S_\ell)$ of *steps*. Each step describes one transaction by:

(i) its target height H (Definition 9.4), anchor height A (Definition 9.5) and expiry height N (Definition 9.8);

(ii) its *spending account*, the account of the wallet to which every input of the step belongs;

(iii) its inputs, partitioned by pool (transparent, Sapling, Orchard, Ironwood), each either a note of the spending account spendable at H (Definition 9.17), a transparent UTXO of that account that meets its depth (§9.1), or a reference (j, i) to output i of an earlier step S_j;

(iv) its payment outputs, each a payment of the request together with the pool assigned to it (§11.2);

(v) its change outputs, each a value and a pool;

(vi) its fee $f \in \mathbb{N}$, computed under the conventional fee rule (Definition 10.2).

A transparent output of a step may be marked *ephemeral*: it pays a fresh transparent address of the wallet and exists only to be spent by a later step (§11.3). Each step satisfies the *balance equation*

$$\sum_{\text{inputs}} v = \sum_{\text{payments}} v + \sum_{\text{change}} v + f,$$

the sums taken over all pools, ephemeral outputs counted among the payments of their step. In the built transaction f is the value remaining in the transparent transaction value pool: the sum of the transparent inputs, minus the transparent outputs, plus every shielded balancing value (*Ironwood Guide*, §"The balancing value", Definitions "Transparent transaction value pool" and "Balancing value").

Clause (ii) restricts this section to steps funded by one account; ZIP 316 also admits transfers whose sending account is undetermined, a case that enters only the choice of outgoing viewing

key (§11.6).

Definition 11.3 (Well-formed plan)**.** A plan $(S_1, \dots, S_\ell)$ is *well formed* if and only if

(i) every reference (j, i) among the inputs of step S_k has $j < k$;

(ii) every output of an earlier step that a step spends is ephemeral; no step spends a change output of an earlier step;

(iii) no output, of the wallet or of a step, is spent twice within the plan;

(iv) every ephemeral output of a step is spent by exactly one later step.

Proposition 11.4 (Chaining through transparent outputs)**.** *Let step S_k of a plan spend output i of an earlier step S_j, and let t_j be the transaction built for S_j.*

(*a*) *If the output is transparent, every field of the transaction of S_k is computable once t_j is built, before t_j is mined.*

(*b*) *If the output is a shielded note of non-zero value, the proof that covers its spend (its Sapling spend proof, or the proof of its Orchard-protocol bundle) is not computable before t_j is mined.*

(*c*) *If the output is an Orchard-protocol note and the transaction of S_k has version 6, its signature digest, and hence every spend-authorisation and binding signature of it, is computable once t_j is built; only proving waits for the mining of t_j.*

(*d*) *If the output is a Sapling note, the signature digest of the transaction of S_k is not computable before t_j is mined.*

A plan whose steps are all completed before any of them is submitted therefore passes value between steps only through transparent outputs, which Definition 11.3 (ii) and the ephemeral outputs of Definition 11.2 realise.

Proof. (a) A transparent input names the outpoint $(\mathsf{txid}(t_j), i)$ and is signed over a digest that commits to the value and script of the spent coin (ZIP 244, "S.2: transparent_sig_digest"). The identifier is a function of the effecting data of t_j (*Ironwood Guide*, §"Transaction digests and signatures", Definition "Transaction identifier"), and value and script are fixed by S_j; none depends on the mining of t_j.

(b) A proof for a spend of a note of non-zero value needs the note's authentication path to the anchor (*Ironwood Guide*, §"Authentication paths"; condition A3 of the Action statement, which exempts only a consumed note of value 0). The path is a function of the note's position in its tree, which is fixed only when t_j is mined (Definition 4.11).

(c) The version-6 signature digest commits to no anchor and to no proof (ZIP 229, "Anchor commitment (version 6)"; *Ironwood Guide*, §"Transaction digests and signatures", Definition "Authorising-data digest; effecting and authorising data" and Definition "Signature digest"). The effecting data of an Orchard-protocol spend are computable before mining: its nullifier is $\mathsf{DeriveNullifier}_{\mathsf{nk}}(\rho, \psi, \mathsf{cm})$, a function of note data and not of the position (*Ironwood Guide*, §"The nullifier"), and ρ of the note is fixed by the Action of t_j that creates it; the key rk and the commitment $\mathsf{cv}^{\mathsf{net}}$ depend on keys, values and fresh randomness. ZIP 374, "Anchors and pre-authorization", states this workflow, and ZIP 318, "Note preparation transactions", uses it.

(d) A Sapling nullifier is computed from $\rho = \mathsf{MixingPedersenHash}(\mathsf{cm}, \mathsf{pos})$, where pos is the note's position (protocol specification, §"Computing ρ values and Nullifiers"), and it is effecting data, committed by the signature digest (ZIP 244, "Signature Digest").

For the final claim: by (b) and (d), a step that spends a shielded note of non-zero value of an earlier step cannot be completed before that step's transaction is mined, so a plan completed as a unit passes value only through transparent outputs, which (a) admits; a zero-valued Orchard-protocol note carries no value. □

The partially created form of (c) is that of §12.5. Fee and change are fixed jointly: the change outputs enter the padded counts of Definition 10.6 and hence the conventional fee, and the fee determines the change value; §11.5 resolves the dependence.

Execution is not deterministic. The note seeds rseed, hence esk, ψ and rcm, the value-commitment trapdoors rcv, the randomisers α, the proofs and the permutations of §11.7 are drawn fresh (*Ironwood Guide*, §"Construction of a transaction"), so one plan executed twice yields transactions with different effecting data and different transaction identifiers. A plan fixes values and shape, not the transaction identifier.

11.2 Payments to pools

Construction 11.5 (Output pool of a payment)**.** The output of a payment (a, v, m) is determined by a:

(i) a transparent address yields a transparent output to it;

(ii) a TEX address yields a transparent P2PKH output of the second transaction of a ZIP 320 transfer (§11.3);

(iii) a Sapling address yields a Sapling output;

(iv) a unified address yields an output to its receiver of the most preferred type in the Priority List of §3.3 that the sender supports, which ZIP 316 makes a MUST (ZIP 316, "Encoding of Unified Addresses"); an Orchard receiver is paid by an Ironwood-pool output;

(v) an address with no receiver that the sender supports is refused, and so is a payment with a memo whose output is transparent.

Case (iv) is sound because both pools of the Orchard protocol share keys, addresses and receivers (*Ironwood Guide*, §"The Orchard protocol and its two pools"; ZIP 326, "Scanning for Ironwood-pool and Orchard-pool notes" and "Receiving funds into the Ironwood pool").

Orchard-pool rules. Consensus admits no value into the Orchard pool: for every transaction, valueBalanceOrchard ≥ 0; and every Orchard-pool Action has cross-address transfers disabled, so the note it creates lies at the expanded receiver of the note it consumes (ZIP 258, "Consensus rules from NU6.3 activation"; *Ironwood Guide*, §"Chain state and pool rules", rules (i) and (ii), and §"Bundle flags"). A wallet MUST NOT send funds to any external receiver, its own included, in the Orchard pool (ZIP 326, "Wallet key-generation restrictions"), and MUST NOT use its knowledge of the internal addresses of all its accounts to send funds from one account to an internal address of another (ZIP 326, "Scanning for Ironwood-pool and Orchard-pool notes"). A wallet that generates use_qsk = true keys MUST NOT send Orchard-pool funds to them at any point (ZIP 326, "Wallet key-generation restrictions"). This volume reads "external receiver" as every receiver that is not an internal-scope receiver (Construction 2.14) of an account of the wallet.

Change. A change output lies at an internal-scope address of the spending account (§11.5) in the Ironwood pool, in the Sapling pool, or, only as Proposition 11.6 allows, in the Orchard pool. The choice among these is wallet policy, designed but unspecified (Definition 1.1); a change output in a pool other than that of the inputs moves value between pools, and the moved amount is published in the balancing values.

Proposition 11.6 (Orchard-pool change)**.** *In a transaction built by a wallet that follows Construction 11.5, every Orchard-pool output of non-zero value lies at the expanded receiver of the note consumed by its own Action, a real note or a fabricated zero-valued note, and that receiver is an internal-scope receiver of the spending account. Hence no payment output lies in the Orchard pool; non-zero change remains in the Orchard pool only at internal-scope receivers of the spending account, with total at most the value of the Orchard-pool notes spent; any other change lies in the Ironwood or the Sapling pool.*

Proof. Condition A10 of the Action statement with enableCrossAddress = 0 (*Ironwood Guide*, §"Bundle flags" and §"Chain state and pool rules", rule (i)) places each Orchard-pool output at the expanded receiver of the note consumed in its Action. By the first rule of ZIP 326 under the reading of Construction 11.5, an output of non-zero value lies at an internal-scope receiver of an account of the wallet, and by the second that account is the spending account. A payment output is routed by cases (i) to (iv) of Construction 11.5, none of which yields an Orchard-pool output. Let V_{in} and V_{out} be the totals of the consumed and created Orchard-pool notes; the balancing value of the Orchard pool is $V_{\text{in}} - V_{\text{out}} \geq 0$ (rule (ii)), so the Orchard-pool change, a part of V_{out}, is at most V_{in}. Every other change output lies, by Construction 11.5, in the Ironwood or the Sapling pool. □

The following consequences are cited later rather than restated. Orchard-pool bundles are priced by Definition 10.6 with the requested count $s + o$; Orchard-pool fabricated outputs and spends are assembled in §11.7; and the canonical migration and note-preparation transactions of ZIP 318 are the exceptions to the padding floor stated in §10.2.

One Orchard-protocol spending key authorises spends in both Orchard-protocol pools. Ironwood-pool and Orchard-pool spends use the same full viewing key and the same spend authorising key ask, each pool's bundle with its own note commitment tree, anchor and nullifier set (*Ironwood Guide*, §"Key capabilities and scope", §"The Orchard protocol and its two pools" and §"Chain state and pool rules"); the two bundles of one transaction are built, proved and signed side by side.

Migration is specified. ZIP 2005 states that wallets SHOULD move all funds they control, transparent, Sprout and Sapling funds included, into recoverable Ironwood-pool notes as soon as practically possible, and that other wallets are REQUIRED to be able to receive them (ZIP 2005, "Proactive movement of funds to recoverable notes"). ZIP 318 specifies a scheduled migration from the Orchard pool to the Ironwood pool; its scheduling rules and privacy analysis are cited, not restated (ZIP 318, "Specification" and "Privacy Implications"). Its block-count constants are stated for a 75-second block spacing; their treatment under the 25-second spacing of ZIP 218, which NU7 deploys (ZIP 259, "NU7 consensus changes"), is an open problem (Definition 1.1). The privacy of the choice of source pool is stated in §11.4.

11.3 TEX transfers and shielding transactions

Construction 11.7 (Transfer to a TEX address)**.** A sender paying one or more TEX addresses from shielded funds MUST create two transactions (ZIP 320, "Design considerations for Senders"), realised as two steps of one plan.

(1) The first transaction tr_0 spends shielded notes only and pays, in place of the n TEX payments of values $v_1, \ldots, v_n$, one ephemeral transparent output of value

$$v_{\mathsf{eph}} := \sum_{i=1}^{n} v_i + fee(tr_1)$$

to a fresh ephemeral transparent address of the wallet.

(2) The second transaction tr_1 spends exactly that output and pays the n TEX payments as P2PKH outputs (*Crypto Guide*, §"The transparent layer's hashes"), with no memo and no change, so that its input equals its outputs plus its fee.

Here $fee(tr_1)$ is the conventional fee of one P2PKH input priced at 150 bytes (Proposition 10.5 (i), the ephemeral key being a compressed key of the wallet) and n P2PKH outputs of 34 bytes each:

$$fee(tr_1) = 5000 \cdot \max(2, \max(\lceil 150/150 \rceil, \lceil 34n/34 \rceil)) = 5000 \cdot \max(2, n) \text{ zatoshi},$$

which is 10,000 zatoshi for $n = 1$ and $n = 2$ and 15,000 zatoshi for $n = 3$. A single transaction that pays a TEX address and spends shielded inputs is refused: a sender to a TEX address MUST spend only transparent (P2SH or P2PKH) UTXOs in that transaction, and SHOULD pay only transparent outputs in it (ZIP 320, "Specification"). ZIP 320 does not require that those UTXOs come from a single transparent address.

The plan of Construction 11.7 is well formed (Definition 11.3): its one reference points from tr_1 to the ephemeral output of tr_0, which is spent exactly once. By Proposition 11.4 (a) both transactions are completed before either is mined, and ZIP 315 exempts the ephemeral UTXO from the confirmation floor (§9.1).

Ephemeral addresses are specified. A wallet based on ZIP 32 SHOULD choose ephemeral addresses so that their private keys are recoverable from the seed, SHOULD NOT choose them so that they can be linked between transactions without the seed or the relevant transparent viewing keys, and so SHOULD avoid collisions with the addresses of earlier outputs, change outputs included; a wallet MUST recognise, and be able to spend, funds that a recipient returns to an ephemeral address (ZIP 320, "Design considerations for Senders"). An ephemeral address reserved for a construction that fails is not reused, since reuse would link the two transactions; this rule is designed but unspecified (Definition 1.1).

Construction 11.8 (Shielding transaction)**.** A *shielding transaction* is the transaction of a one-step plan with an empty payment request. It spends transparent UTXOs of the spending account and pays their total, less the conventional fee, to change outputs at an internal-scope shielded receiver of that account (Construction 2.14), with no other recipient and no transparent output. Its outputs lie in the Ironwood pool, since no value enters the Orchard pool (Construction 11.5) and ZIP 2005 asks for recoverable Ironwood-pool notes (ZIP 2005, "Proactive movement of funds to recoverable notes"); its balancing value in that pool is negative. Its UTXOs may be spent with zero confirmations (§9.1). The value threshold at which a wallet shields is wallet policy, designed but unspecified (Definition 1.1).

Shielding rules are specified. A wallet that receives transparent funds SHOULD auto-shield them by default; it SHOULD NOT auto-shield from several transparent addresses in one transaction, and SHOULD NOT use *opportunistic shielding*, the shielding of received transparent funds within a user-initiated transaction, which would show its recipients the link between the transparent addresses and the payment (ZIP 315, "Long-term storage of funds"). Users MUST be able to disable auto-shielding (ZIP 315, "Auto-shielding"). Spent transparent UTXOs SHOULD be sent only to an internal shielded receiver of the wallet, except the ephemeral outputs of a ZIP 320 transfer (ZIP 315, "Linkability of transactions or addresses").

Transparent inputs are public, so a transaction that spends UTXOs of different transparent receivers links those receivers. ZIP 315 states that UTXOs received on different transparent receivers SHOULD NOT be shielded in one transaction, and that shielding several UTXOs of one address together leaks nothing further (ZIP 315, "Linkability of transactions or addresses"). The same holds for ephemeral addresses, so a transaction spends the ephemeral output of at most one ZIP 320 transfer.

11.4 Input selection

Definition 11.9 (Covering selection)**.** Fix a step with payments of total *out*, target height H, anchor height A and a number k of change outputs (§11.5 fixes k). Let U be the finite set of inputs of the spending account that Definition 11.2 (iii) admits at (H, A), each $u \in U$ with value $v(u)$, and let $\mathcal{S} \subseteq 2^U$ be the family of *admissible* selections, those that meet the pool rules stated at the end of this subsection. For $I \in \mathcal{S}$ let $shape(I)$ be the public counts of Definition 10.1 of the step funded by I, with its payments and k change outputs, after padding (Definition 10.6), and let $fee(I) := conventional_fee(shape(I))$ (Definition 10.2). A selection $I \in \mathcal{S}$ is *covering* if and only if

$$\sum_{u \in I} v(u) \geq out + fee(I).$$

Adding an input never lowers a count of Definition 10.1, so *fee* is non-decreasing under inclusion, and a covering selection is a fixed point of the fee: its total must cover a fee that depends on it. A plan requires a covering selection for each step; any procedure that returns one conforms. The following construction is one such procedure.

Construction 11.10 (Fee escalation)**.** Set $I_0 := \emptyset$ and $t_0 := 0$. At round $k \geq 0$:

(1) compute the balance of the step on I_k (§11.5 gives its exact form);

(2) if I_k is covering, return I_k;

(3) otherwise let $r_k := out + fee(I_k) - t_k > 0$ be the shortfall, and choose $I_{k+1} \in \mathcal{S}$ with total $t_{k+1} := \sum_{u \in I_{k+1}} v(u) \geq t_k + r_k$; if no such I_{k+1} exists, fail with insufficient funds.

The choice of I_{k+1} among the admissible sets that step (3) allows is the wallet's.

Proposition 11.11 (Termination of fee escalation)**.** *Fee escalation terminates after finitely many rounds. On success the returned selection is covering, and the fee of the step is the conventional fee of its final padded shape. On failure at round k, no admissible selection has total at least* $out + fee(I_k)$*. No bound on the number of rounds is claimed beyond this finiteness.*

Proof. A round that does not stop has $t_{k+1} \geq t_k + r_k > t_k$, so the totals increase strictly. Every t_k lies in the finite set $\{\sum_{u \in I} v(u) : I \in \mathcal{S}\} \cup \{0\}$, which admits no infinite strictly increasing sequence. On success the returned I_k is covering by step (2), and its fee is $fee(I_k)$, the conventional fee of $shape(I_k)$ by Definition 11.9. On failure, step (3) found no admissible set of total at least $t_k + r_k = out + fee(I_k)$. □

A failure does not show that no covering selection exists: an admissible set J with $fee(J) < fee(I_k)$ and total in $[out + fee(J), out + fee(I_k))$ covers. Fee escalation is therefore sound, returning only covering selections, and complete only for choice rules that exclude this case.

Definition 11.12 (Uneconomic input)**.** An input u is *uneconomic* if and only if $v(u) < marginal_fee$, where $marginal_fee = 5000$ zatoshi (Definition 10.2). The strict inequality is ZIP 317's wording, "below the marginal fee" (ZIP 317, "Requirements" and "Rationale for the chosen parameters").

Lemma 11.13 (Cost of an additional input)**.** *Let L be the number of logical actions of a step (Definition 10.1) and L' that after adding one input.*

(*i*) *A P2PKH input priced at* 150 *bytes gives $L' - L \in \{0, 1\}$.*

(*ii*) *A shielded spend added to a non-empty bundle with s real spends and o real outputs, $s + o \geq 1$, gives $L' - L \in \{0, 1\}$, and leaves the bundle's padded count unchanged if and only if $s + 1 \leq \max(o, 2)$ in an Orchard-protocol bundle with cross-address transfers enabled, $s + o = 1$ in an Orchard-pool bundle, and $s + 1 \leq \max(o, 2)$ in a Sapling bundle.*

(*iii*) *A shielded spend that opens an empty bundle gives $L' - L = 2$, the padded count going from* 0 *to* 2 *for every kind of bundle.*

(*iv*) *The fee rises by $\mathit{marginal_fee} \cdot (\max(2, L') - \max(2, L)) \leq \mathit{marginal_fee} \cdot (L' - L)$, and not at all while $L' \leq \mathit{grace_actions} = 2$.*

Hence an uneconomic input of positive value adds positive net value to the step if and only if $\max(2, L') = \max(2, L)$: when padding absorbs its spend, or while the step stays within the grace window.

Proof. (i) For every $x \in \mathbb{N}$, $\lceil (x + 150)/150 \rceil = \lceil x/150 \rceil + 1$; the transparent contribution is the maximum of this term and the unchanged output term, so it rises by 0 or 1.

(ii) and (iii). By Definition 10.6 the contribution of a non-empty bundle is $\max(\max(s, o), 2)$ with cross-address transfers enabled, $\max(s + o, 2)$ in an Orchard-pool bundle, and $\max(s, \max(o, 2))$ for Sapling, whose outputs are padded to $\max(o, 2)$; an empty bundle contributes 0. Each non-empty form is $\max(x, c)$ with $c \geq 2$ and a count x that rises by exactly one, so it rises by 0 or 1, and by 0 if and only if $x + 1 \leq c$: that is $s + 1 \leq \max(o, 2)$ in the first and third forms, and $s + o + 1 \leq 2$, which with $s + o \geq 1$ is $s + o = 1$, in the second. A single spend in an empty bundle gives $\max(1, 2) = 2$ with cross-address transfers enabled, $\max(1 + 0, 2) = 2$ in the Orchard pool and $\max(1, \max(0, 2)) = 2$ for Sapling. No other contribution changes.

(iv) From Definition 10.2, since $n \mapsto \max(2, n)$ is non-decreasing and rises by at most the rise of n. For the final claim: if $\max(2, L') = \max(2, L)$ the fee is unchanged and the input adds $v(u) > 0$; otherwise the fee rises by at least $\mathit{marginal_fee} > v(u)$.

The instances $(s, o,$ rise of the padded count$)$ are, with cross-address transfers enabled and for Sapling, $(0, 1, 0)$, $(1, 1, 0)$, $(1, 2, 0)$ and $(2, 2, 1)$; in an Orchard-pool bundle $(0, 1, 0)$, $(1, 1, 1)$, $(1, 2, 1)$ and $(2, 2, 1)$; and $(0, 0, 2)$ for the empty bundle of every kind. □

ZIP 315 states that automatic shielding and automatic or opportunistic migration SHOULD NOT be applied to inputs whose cost of shielding or migrating exceeds their economic value (ZIP 315, "Auto-shielding"); under the conventional fee, Lemma 11.13 decides which inputs these are.

The choice of source pool is constrained by privacy. Value that crosses pools is published in the balancing values (*Ironwood Guide*, §"The balancing value", Remark "Disclosure by the balancing value"), so selecting inputs in the pool of the payment outputs keeps the amount private, and ZIP 315 states the aim of minimising pool crossing (ZIP 315, "Linkability of transactions or addresses"). Wallets MUST NOT automatically combine funds across pools to satisfy a transfer, since that could reveal the total the user holds in a pool (ZIP 315, "Information leakage for transfers between pools"). One policy serving these rules is designed but unspecified (Definition 1.1): order the shielded pools with the pool of the outputs first, then the other Orchard-protocol pool, then Sapling, or, for Sapling outputs, Sapling, then Ironwood, then Orchard; remove the pools that the user's policy forbids; and spend from the first pool whose notes alone cover the payment.

Transparent sources and recipients are specified. A wallet MUST NOT send funds to a transparent address unless all of the source funds come from shielded pools, and this SHOULD be a single shielded pool (ZIP 315, "Linkability of transactions or addresses"); the ZIP 320 transfer

of Construction 11.7 meets this rule, its first transaction spending only shielded notes. A wallet SHOULD NOT spend funds from a transparent address in a transaction with an external recipient unless the user gives explicit consent for that transaction, and MAY restrict transfers further (ZIP 315, "Allowed transfers"). The admissible family $\mathcal{S}$ of Definition 11.9 consists of the selections that meet the rules of this and the preceding paragraph, those of §11.3, and the wallet's policy.

11.5 Change

Construction 11.14 (Balance and change)**.** *Change* is the excess of a step's inputs over its payments and fee, returned to the wallet in $k \geq 0$ outputs whose pool follows Construction 11.5. Let *in* be the total of the inputs, over all pools and ephemeral inputs included, *out* the total of the payment outputs, and f_k the conventional fee of the padded shape with k change outputs, which is non-decreasing in k.

(1) If $in < out + f_0$, fail with shortfall $out + f_0 - in$, the requirement fed back to selection (Construction 11.10).

(2) If $in = out + f_0$, the step has no shielded input or output and no change memo is requested, set $k := 0$ and fee f_0.

(3) Otherwise let k be the least number of change outputs, 1, or a larger number chosen by the wallet to split change across notes, a choice designed but unspecified (Definition 1.1), subject to $in \geq out + f_k$. The change is $c := in - out - f_k \geq 0$, divided among the k outputs, each carrying the change memo if one is requested. If $out + f_0 \leq in < out + f_1$, no $k \geq 1$ qualifies: the step fails with requirement $out + f_1 - in$, or, if $in > out + f_0$, may instead set $k := 0$ and fold the positive remainder $in - out - f_0$ into the fee; the plan records which.

Balance invariant. Before building, the transparent, Sapling, Orchard and Ironwood balances of the step, each the value of the step's inputs in that pool minus the value of its outputs in that pool, sum to the planned fee. A sum above the fee means that change was omitted; a sum below it, insufficient funds.

Change goes to an internal-scope address of the spending account (ZIP 32, "Orchard internal key derivation"; Construction 2.14; ZIP 316, "Deriving Internal Keys"), its diversifier index a wallet policy; non-zero change remains in the Orchard pool only at an internal-scope receiver of the spending account, the receiver of the note consumed in its Action (Proposition 11.6). A zero-valued fabricated output at the receiver of a spent note is not change.

The balance invariant is the balance equation of Definition 11.2 with every term moved to one side, summed pool by pool.

Proposition 11.15 (Zero-valued change)**.** *Let a step have a shielded input or output, and let Construction 11.14 complete it with $k \geq 1$ change outputs, of total 0 when $in = out + f_k$.*

(*i*) *The public counts of the step are the same function of its inputs, payments and k whether the change is 0 or positive.*

(*ii*) *If change were instead omitted at exact balance, the counts at exact balance would differ from those with change exactly when one more output of the change pool raises that pool's padded count or opens a bundle (Definition 10.6). An observer who knows the payment outputs, such as a party that receives every payment, and sees the counts would then infer that the inputs sum exactly to the payments plus the fee.*

(*iii*) *In the Ironwood pool, bundles with equal public leakage that differ only in whether the change is* 0 *or positive are computationally indistinguishable.*

The zero-valued output also carries a requested change memo. Where padding already absorbs the change output, as in the two-Action Orchard-pool bundle of ZIP 318, the omission of (*ii*) *reveals nothing* (*ZIP 318, "Canonical migration transaction structure"*).

Proof. (i) and (ii) follow from Construction 11.14 and Definition 10.6: the counts depend on k, which is at least 1 whatever the value of c, and omission replaces k by 0 at exact balance only. (iii) is *Ironwood Guide*, §"Privacy", Theorem "Privacy of an Action", under its preconditions (P1) to (P4) and its assumptions: Actions whose private inputs have equal public leakage are indistinguishable, and a created note's value is a private input, not public leakage. No claim is made for Sapling outputs beyond (i). □

A zero-valued note needs no witness. Condition A3 of the Action statement, $v^{\mathsf{old}} \cdot (root - rt) = 0$, checks the authentication path only of a consumed note of non-zero value (*Ironwood Guide*, §"The Action statement"), so a zero-valued change note can later be spent without tracking its path.

Only a step with no shielded input or output and no change memo omits change at exact balance. The second transaction tr_1 of Construction 11.7 is the instance, and a change memo is not carried there. A change note of value $v < marginal_fee$ is an uneconomic input when it is later spent (Definition 11.12), and adds positive net value only where Lemma 11.13 gives no rise of the fee. The wallet's policy for such change is designed but unspecified (Definition 1.1), constrained by the following proposition.

Proposition 11.16 (Fee uniformity)**.** *The fee of a transaction is public, the value remaining in its transparent transaction value pool, and its conventional fee is a function of public fields* (*Definitions 10.1 and 10.2*)*, so every observer computes both. A transaction whose fee differs from the conventional fee of its shape is distinguished from every transaction of that shape that pays the conventional fee. Folding a remainder into the fee* (*Construction 11.14, case* (*3*)) *produces exactly such a fee whenever the remainder is positive. The folding threshold is the wallet's policy, designed but unspecified* (*Definition 1.1*)*.*

Proof. The fee is the sum of the transparent input values and the balancing values, minus the transparent output values, each a public field (*Ironwood Guide*, §"The balancing value"); the transparent sizes and the shielded counts of Definition 10.1 are public fields. The predicate "the fee differs from the conventional fee of the shape" is therefore a function of the public data; it holds for a transaction with a folded remainder $in - out - f_0 > 0$, whose fee exceeds f_0, and fails for every transaction of the same shape that pays the conventional fee. ZIP 317 states that non-standard fees may reveal specific users or wallets (ZIP 317, "Security and Privacy considerations"). □

11.6 Outgoing viewing keys

Definition 11.17 (Outgoing viewing key of an output)**.** For a transfer, the sender determines:

(a) the *sending account*: the account that the transfer names, if it names one; otherwise, if every fund used comes from addresses of one account, that account, a rule that ZIP 316 makes a SHOULD; otherwise the sending account is undetermined;

(b) the *preferred sending protocol*, the most preferred receiver type (the Priority List of §3.3) among the funds used: Orchard if any Orchard-protocol note, of either pool, is spent, else Sapling, else transparent.

The *outgoing viewing key of an output* is, with the sending account determined, that account's external or internal ovk of the preferred protocol at the account level, external for a payment and internal for a wallet-internal output such as change or a shielding output (SHOULD); with the sending account undetermined, the external or internal ovk of the full viewing key of one address of the preferred protocol from which funds are sent, for example the first (SHOULD) (ZIP 316, "Usage of Outgoing Viewing Keys"). The Orchard external ovk is that of the *Ironwood Guide*, Construction "Diversifier key and outgoing viewing key" in §"Viewing keys", and the internal one that of Construction 2.14; the transparent pair is that of Construction 2.18, the two halves of $I_{\mathsf{ovk}} = \mathsf{PRFexpand_c}(\mathtt{[0xD0]} \| \mathsf{ser}_P(\mathsf{pk}))$; the Sapling keys are those of ZIP 32, "Sapling internal key derivation". The choice governs the outgoing ciphertext C_{out} of every payment and change output. Its construction, which also admits $\mathsf{ovk} = \perp$ with a uniformly random ock, is cited (*Ironwood Guide*, §"The outgoing ciphertext", Construction "Outgoing ciphertext"; protocol specification, §"Sending Notes (Sapling)" and §"Sending Notes (Orchard)").

For change the rule gives the internal ovk. The specification also permits no ovk: with $\mathsf{ovk} = \perp$ the sender draws ock and the outgoing plaintext uniformly, no viewing key recovers the outgoing plaintext of the change, and the wallet finds its change by trial decryption under the internal ivk (*Ironwood Guide*, §"Trial decryption and note acceptance"; protocol specification, §"Sending Notes (Orchard)"). Encrypting change with no ovk departs from the SHOULD of ZIP 316.

The holder of an account's external ovk therefore recovers the recipient, value and memo of the account's Ironwood-pool payment outputs (*Ironwood Guide*, §"The outgoing ciphertext", Construction "Recovery with ovk"), and those of its change only with the internal ovk.

11.7 Padding, dummies, and shuffling

Construction 11.18 (Assembly of an Orchard-protocol bundle)**.** Fix an Orchard-protocol bundle with s real spends and o real outputs, and let n be its padded Action count (Definition 10.6), or $n = 1$ for the Ironwood-pool bundle of a canonical migration transaction, or $n = 16$ for the Orchard-pool bundle of a note-preparation transaction (§10.2). The wallet sets $\mathsf{enableSpends} = \mathsf{enableOutputs} = 1$, the flags that ZIP 315 notes depend under normal circumstances only on whether a transaction is a coinbase transaction (ZIP 315, "Kinds of information leakage").

(A) *Cross-address transfers enabled*, an Ironwood-pool bundle with $\mathsf{enableCrossAddress} = 1$, the normal case of ZIP 229 (ZIP 229, "`enableCrossAddress` polarity"), which the wallet uses for every Ironwood-pool bundle. Pad the list of real spends with $n - s$ dummy consumed notes and the list of real outputs with $n - o$ dummy created notes (*Ironwood Guide*, §"Actions, bundles, and dummy notes", Definition "Dummy notes"), each at an independent fresh random address; draw independent uniform permutations σ and τ of $\{1, \ldots, n\}$; place spend j in Action $\sigma(j)$ and output j in Action $\tau(j)$.

(B) *Orchard pool*, cross-address transfers disabled (Construction 11.5). Form

  (i) for each real spend, one Action whose output is a fabricated zero-valued note at the spent note's address, with its C_{enc} filled with random bytes (MUST), since a real encryption would let a holder of that receiver's ivk link the spend's nullifier to the address (ZIP 326, "Fabricated same-address outputs and randomized note ciphertexts");

  (ii) for each real output, one Action whose spend is a fabricated zero-valued note at that output's address, a real spend signed with the account's ask for that receiver, carrying no dummy spending key (ZIP 374, "Fabricated same-address outputs (NU6.3)");

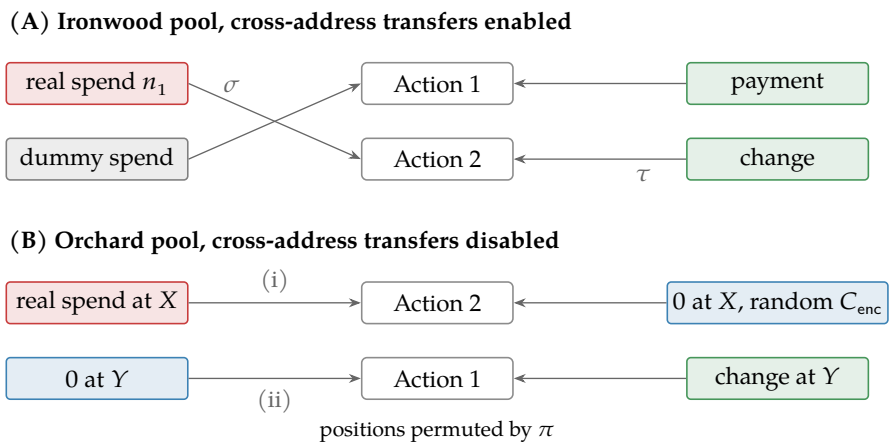


Figure 4: Assembly of Orchard-protocol bundles with $n = 2$ Actions (Construction 11.18). (A) An Ironwood-pool bundle with one real spend and two real outputs: the padded spend list and the output list are permuted independently by σ and τ, so any spend may pair with any output. (B) An Orchard-pool bundle with one real spend at receiver X and one change output at an internal receiver Y of the spending account: each real side is paired with a fabricated zero-valued side at its own receiver, the one beside the real spend with a random C_{enc}, and only the order of the Actions is permuted, by π. Red marks real spends, green real outputs, blue fabricated sides and grey dummies.

(iii) $n-s-o$ fully dummy Actions, whose dummy spend and dummy output share one common fresh random address, the only Actions that carry a dummy spending key;

then draw a uniform permutation π of $\{1, \dots, n\}$ and place the j-th Action so formed at position $\pi(j)$.

The second Action of the canonical Orchard-pool bundle of ZIP 318 is of kind (B)(iii) when that bundle carries no change, and of kind (B)(ii) when it does (ZIP 318, "Canonical migration transaction structure"). A dummy output at an address independent of its Action's consumed note exists only in bundles with cross-address transfers enabled; in kind (B)(iii) the dummy output shares the fresh random address of its dummy spend. A dummy spend coexists with any anchor, since condition A3 of the Action statement checks no path for a zero-valued consumed note. Per Action, rcv, α and the created note's rseed are drawn fresh, and ρ of the created note is the Action's $\mathsf{nf}^{\mathsf{old}}$ (*Ironwood Guide*, §"Construction of a transaction"). A Sapling bundle samples rcv and α per spend and rcv per output, pads its outputs as in Definition 10.6, and its binding signing key is the sum of the spends' rcv minus the sum of the outputs' (protocol specification, §"Balance and Binding Signature (Sapling)"). The builder keeps the permutations, which are not part of the transaction, so that later roles (§12) attach per-output data to the right Action.

Proposition 11.19 (Uniform order and pairing of Actions)**.** *In a bundle with cross-address transfers enabled built by Construction 11.18, the position of each spend and the pairing of spends with outputs are uniform and independent of the order of the plan: the map from spend index to Action position is a uniform permutation, and the spend-to-output pairing is a uniform permutation independent of it. In an Orchard-pool bundle the order of the Actions is a uniform permutation independent of the order of the plan. Positions therefore carry no information about the plan beyond the Action count; for*

> *an Ironwood-pool bundle, the public data of the Actions reveal no more than their public leakage.*

Proof. With σ and τ independent uniform permutations of $\{1, \dots, n\}$ applied to the padded spend and output lists, Action i pairs spend $\sigma^{-1}(i)$ with output $\tau^{-1}(i)$, so spend j is paired with output $\pi(j)$ for $\pi := \tau^{-1} \circ \sigma$. The map $(\sigma, \tau) \mapsto (\sigma, \pi)$ is a bijection of pairs of permutations, since $\tau = \sigma \circ \pi^{-1}$; it maps the uniform distribution to the uniform distribution, so π is uniform and independent of σ. The plan's order enters only as the indexing of the lists, which both permutations randomise. The Orchard-pool case is the single uniform permutation π. The last clause is *Ironwood Guide*, §"Privacy", Theorem "Privacy of an Action", under its preconditions (P1) to (P4) and its assumptions, applied to the bundle as a whole. No indistinguishability is claimed between the kinds (B)(i), (B)(ii) and (B)(iii) of Orchard-pool Actions, whose bundles that theorem does not cover. □

11.8 Authorisation of a transaction

The transaction carries the consensus branch identifier in force at its target height H (*Ironwood Guide*, §"The transaction format"). Transaction versions 5 and 6 are valid and version 4 is not (ZIP 2003, "Specification"; ZIP 259, "Abstract"); a transaction with an Ironwood-pool component is of version 6. The lock time is specified: the field nLockTime is public and may distinguish a transaction if used, and SHOULD be zero (ZIP 315, "Kinds of information leakage"); the wallet sets it to 0, and ZIP 318 lists lock_time = 0 among the SHOULDs of its canonical migration transaction structure (ZIP 318, "Canonical migration transaction structure"). The reason is the one proved for the fee in Proposition 11.16: a public field set otherwise than by every other wallet distinguishes the transaction. The expiry height is that of §9.3.

Construction 11.20 (Authorising a built transaction)**.** Let a transaction be built from one step by Construction 11.18, every field of its effecting data fixed.

(1) Compute its signature digest SigHash over the effecting data (ZIP 244, "Signature Digest", as amended by ZIP 229, "Anchor commitment (version 6)", which omits the anchors from the version-6 digest; *Ironwood Guide*, §"Transaction digests and signatures"). Every spend-authorisation signature and every binding signature signs SigHash.

(2) For each Orchard-protocol Action of either pool, with the Action's randomiser α, set $\mathsf{rsk} := \mathsf{ask} + \alpha$ and $\mathsf{spendAuthSig} := \mathsf{Sign}_{\mathsf{rsk}}(\mathsf{SigHash})$ (*Ironwood Guide*, §"Randomised validating keys"). Every Action carries one: a fully dummy Action's spend is signed with its own random key, and a fabricated zero-valued spend beside a real Orchard-pool output with the account's ask, not with a dummy key. Each Sapling spend likewise signs SigHash under its randomised key (protocol specification, §"Spend Authorization Signature (Sapling and Orchard)").

(3) For each bundle, set $\mathsf{bsk} := \sum \mathsf{rcv} \bmod p_{\mathsf{Vesta}}$ over its Actions, check it against bvk, and sign SigHash under bsk (*Ironwood Guide*, §"The binding signature"); the Sapling bsk is that of Construction 11.18.

(4) Sign each transparent input under the key of its address, by the scheme of the *Crypto Guide*, §"The transparent layer's schemes: ECDSA and BIP-340 over secp256k1", over the signature digest of that input (ZIP 244, "S.2: transparent_sig_digest").

Keys. One Orchard-protocol ask serves both scopes and both pools (Construction 2.14; ZIP 32, "Orchard internal key derivation"). One Sapling ask serves both scopes, the internal scope differing in nsk, which enters the proof and the nullifier, not the signature (ZIP 32, "Deriving

a Sapling internal spending key"), so a Sapling spend is added under the scope of its note. Each transparent input needs the key of its own address.

Lemma 11.21 (Scope of the signature digest)**.** *The signature digests of version-5 and version-6 transactions commit to the effecting data and, for transparent inputs, to the values and scripts of the coins spent (ZIP 244, "S.2: transparent_sig_digest"), excluding proofs and signatures, and the version-6 digest also excludes the anchors. Hence steps (2) to (4) of Construction 11.20 need no proof; proofs and signatures can be computed in either order; in version 6, signing can precede the choice of anchor; and the transaction identifier is unchanged by proofs, by signatures and, in version 6, by anchors.*

Proof. By the *Ironwood Guide*, §"Transaction digests and signatures", Definition "Signature digest", SigHash has the header and shielded children of txid and the transparent signature digest as its children, all computed over effecting data and, for the transparent signature digest, the values and scripts of the coins spent; proofs, signatures and transparent scripts are authorising data, under the authorising-data digest (Definition "Authorising-data digest; effecting and authorising data"). ZIP 229, "Anchor commitment (version 6)", moves the anchors to the authorising data in version 6; in version 5 they are effecting data (ZIP 244, "Signature Digest"). A step of Construction 11.20 computes a signature over SigHash from keys and randomisers, and the binding key from the rcv values, none of which is a proof, so the order of proving and signing is free. The transaction identifier is a function of the effecting data alone (Definition "Transaction identifier"). The multi-party form of the lemma is that of §12.9. □

Theorem 11.22 (Payment integrity)**.** *Let a well-formed plan (Definition 11.3) be executed by Construction 11.18 and Construction 11.20, with Orchard-protocol keys generated with* use_qsk = false. *In each resulting transaction:*

(*i*) *each payment is paid by exactly one output, in the pool that Construction 11.5 assigns, to the routed receiver, of the payment's value and with its memo, none for a transparent output;*

(*ii*) *every other output is a change output of Construction 11.14, an ephemeral output of the step, or a zero-valued dummy or fabricated output;*

(*iii*) *the fee equals the planned fee: the conventional fee of the final shape, or, where Construction 11.14 folds a remainder into it, that fee plus the remainder.*

Moreover, for a version-6 transaction whose signing keys meet hypotheses (H) and (S) of Ironwood Guide, *§"Transaction digests and signatures", Proposition "Bundle binding", and under the assumptions of that proposition and of its Theorem "Spend authority" (§"Authorisation of spends"), no other party produces, except with negligible probability, an accepted transaction that carries any of the transaction's Orchard-protocol randomised validating keys* rk *with different effecting data. The theorem claims content, not inclusion: inclusion further requires that no spent nullifier be revealed first, that the anchor block remain on the best chain and that inclusion occur by the expiry height, none of which the wallet controls (§13).*

Proof. (i) and (ii). By Definition 11.2 each payment is one output of its step, placed by Construction 11.5, and the only other outputs of a step are its change and ephemeral outputs. Construction 11.18 adds only notes of value 0: dummy created notes, fabricated outputs and the outputs of fully dummy Actions. The padding therefore creates no output of non-zero value, and each real output keeps the receiver, value and memo that the plan gives it.

(iii) In the built transaction the fee is the value remaining in the transparent transaction value pool, the transparent inputs minus the transparent outputs plus the balancing value of each shielded pool, and each balancing value is the value consumed minus the value created in its pool. Since dummy and fabricated notes have value 0, the remaining value is the total of the step's inputs minus the total of its payment, ephemeral and change outputs, which is the planned fee by the balance equation of Definition 11.2 and the balance invariant of Construction 11.14; the planned fee is f_k, or f_0 plus the folded remainder.

Binding. Every signature of the transaction signs SigHash (Construction 11.20), which covers the effecting data (Lemma 11.21). By *Ironwood Guide*, §"Transaction digests and signatures", Proposition "Bundle binding", an accepted transaction that carries some rk_i of the transaction carries the same effecting data, except with negligible probability; Theorem "Spend authority" excludes a valid signature under a key of the wallet on a digest that the wallet did not sign. For transparent and Sapling signatures the commitment to the effecting data is that of ZIP 244; no lower volume proves the unforgeability of those schemes, and none is claimed here. □

The wallet records each of its shielded outputs from the data it built the output with: the address, the value, rseed and the memo, and, for an Orchard-protocol output, ρ, the nullifier of the spend of its Action. For such an output,

$$\mathsf{esk} = \mathrm{ToScalar}(\mathsf{PRFexpand}_{\mathsf{rseed}}([\mathtt{0x04}] \parallel \mathsf{I2LEOSP}_{256}(\rho))),$$

and the note plaintext follow without any viewing key (*Ironwood Guide*, §"Encryption to the recipient", and §"The outgoing ciphertext", Construction "Recovery with ovk"). A fabricated zero-valued Orchard-pool output at a spent note's receiver carries a random C_{enc}, which trial decryption under any ivk rejects except with negligible probability (*Ironwood Guide*, §"Encryption to the recipient", Assumption "Confidentiality and integrity of ChaCha20-Poly1305"; ZIP 326, "Fabricated same-address outputs and randomized note ciphertexts"). Non-zero Orchard-pool change lies only at internal-scope receivers (Proposition 11.6), so scanning finds it under the internal ivk (Definition 6.4), and no external ivk is needed for it (ZIP 326, "Wallet key-generation restrictions").

12 Partially created transactions (PCZT)

This section states the protocol by which the construction, proving and authorisation of one transaction are divided among several parties (ZIP 374). It defines the partially created transaction, its versions and the kinds of its fields; the roles that act on it and the modification state they maintain; its fields, its canonical empty bundles and its encoding; the version-2 rules for deferred anchors and the Ironwood bundle; construction and IO finalisation; signing, with the theorem that a Signer's signatures authorise exactly what it checked; proving; combining and redaction, with the order independence of proving and signing and the data a PCZT discloses; and spend finalisation and extraction.

12.1 The PCZT and its requirements

Definition 12.1 (Partially created transaction)**.** A *partially created transaction* (PCZT) of one transaction is a value of the typed record of §12.3 for one PCZT version, exchanged among the parties to the creation of that transaction. It carries the data each role needs for its step (adding inputs and outputs, proving, signing, extracting) and accumulates proofs and signatures until the transaction is complete; since it carries all data a signer needs, a signer can be offline. A PCZT consists of a *Global* record and one *bundle* per protocol: transparent, Sapling, Orchard and, from version 2, Ironwood. Every bundle is present in every PCZT, empty bun-

dles included, because metadata of a protocol may have to be held before it is known whether that protocol has inputs or outputs (ZIP 374, "Abstract", "Specification" and "Structure").

The PCZT protocol is specified (Definition 1.1) in ZIP 374, Revision 0, whose status is that of Table 1; no consensus rule depends on it. It covers transactions of versions 5 and 6 only: it supports neither Sprout nor version 4 or earlier, since version 4 predates the ZIP 244 digest and several of the role separations of this section, the independence of proving from signing in particular, fail for it (ZIP 374, "Specification"). The protocol serves signing devices that hold spending keys but can neither compute proofs nor hold a whole transaction; proof delegation, the prover needing note data and the full viewing key but no spending key; transactions with several signers (threshold signing, the transparent multisignature wallets of ZIP 48) or several contributors of inputs, whose parallel contributions are merged; and, for version 6, pre-authorisation, in which the signatures precede the choice of anchors, the witnesses and the proofs (ZIP 374, "Motivation").

ZIP 374, "Requirements", states six requirements, recorded here as the ZIP's and not proved as such:

(R1) every intermediate state of the creation of a version-5 or version-6 transaction is representable, and adding inputs and outputs, finalising the input and output set, attaching lookup data, proving, signing, merging parallel contributions and extracting are each performable by distinct entities, in an order constrained only by data dependencies;

(R2) no spending key material is required, the keys of dummy spends excepted, which authorise nothing of value and are removable before any Signer sees the PCZT;

(R3) a Signer determines from the PCZT alone what it authorises (amounts, recipients, and the correspondence between the fields it signs and the values shown to its user), and data a Signer does not need is removable;

(R4) merging two PCZTs of the same transaction is deterministic and fails on any conflict;

(R5) for version 6, signatures can be produced while anchors, witnesses and proofs are absent, and supplying them later leaves the signatures valid;

(R6) the encoding is compact and strictly versioned, so that no data is read under another version's rules.

Requirement (R3) is made a theorem in §12.7, (R5) a proposition in §12.5 and (R4) a construction in §12.9.

Definition 12.2 (PCZT versions)**.** Version 1 represents transactions of version 5 (ZIP 225); version 2 represents transactions of versions 5 and 6 (ZIP 229). The PCZT version does not fix the transaction version: a version-2 PCZT carries a transaction of version 5 or 6, subject to the per-field rules, and a version-1 PCZT only one of version 5. Relative to version 1, version 2 adds the Ironwood bundle; a note version in $\{V2, V3\}$ per Orchard-protocol bundle; anchors that may be absent; and an encoding that writes a bundle equal to its slot's canonical empty bundle as absent (ZIP 374, "Versioning"). The fields are made exact in §12.3 and the encoding in §12.4.

No field can be added within a version: the encoding is positional, and a parser cannot skip a field it does not know. A PCZT has no non-proprietary key–value maps, so every field of a version is known in advance and a Signer's view of a PCZT is complete; the proprietary maps are the only extension point within a version, and MUST be used for internal processes only. The format evolves by new versions, which the four-byte version of the header distinguishes. Implementations SHOULD parse every specified version and SHOULD emit the oldest version able to

represent a PCZT's state when interoperating with entities that may lack the newest; conversion between versions is lossless exactly when the target version represents the state (ZIP 374, "Extensibility" and "Proprietary Use fields"). The states that version 1 cannot represent are listed in §12.4.

Definition 12.3 (Field kinds of a PCZT)**.** Every field of a PCZT is of one of three kinds (ZIP 374, "Structure"):

(i) *effecting data*, required fields that are part of the final transaction and are always present, committed to by the signature digest (*Ironwood Guide*, §"Transaction digests and signatures");

(ii) *authorising data*, initially absent and filled as roles act: proofs, signatures, binding signing keys and, in a transaction of version 6, the anchors;

(iii) *context data*, which enables roles to act and never enters the final transaction: note openings, keys, witnesses, derivation records and address strings.

In a transaction of version 5 the anchors are effecting data (ZIP 374, "Anchors and pre-authorization").

The signature digest commits to every non-transparent effecting field as soon as any input is signed, so no shielded analogue of the transparent hash types that leave inputs or outputs open exists. The transparent and the shielded modifiable flags of §12.2 are therefore kept apart, so that a purely transparent PCZT behaves as a partially signed Bitcoin transaction of BIP 174 and BIP 370 does (ZIP 374, "Separate Shielded Modifiable Flag from Transparent Modifiable Flags").

12.2 Roles and the modification state

Definition 12.4 (Roles)**.** A *role* is a partial map from PCZTs to PCZTs, undefined where the role fails; a role that draws randomness is a family of such maps indexed by its random coins, and the Combiner maps a finite set of PCZTs to one. The roles, with the parties that may execute them (ZIP 374, "Specification" and "Roles"), are:

(1) *Creator* (a single entity): chooses the transaction version and consensus branch identifier, and sets the Global fields and the empty bundles;

(2) *Constructor*: adds transparent inputs and outputs, Sapling spends and outputs, and Orchard-protocol Actions;

(3) *IO Finalizer* (anyone): declares the input and output set final, computes the binding signing keys and signs dummy spends;

(4) *Updater* (anyone): adds data later roles need, such as derivation records, full viewing keys, witnesses and, for version 6, anchors;

(5) *Redactor* (anyone): removes data later roles do not need;

(6) *Prover* (a holder of the note data): adds proofs;

(7) *Signer* (a holder of a spending key): adds signatures;

(8) *Combiner* (anyone): merges PCZTs;

(9) *Spend Finalizer* (anyone): assembles transparent unlocking scripts;

(10) *Transaction Extractor* (anyone): checks completeness and produces the transaction.

The lifecycle runs the Creator, the Constructors, the IO Finalizer and the Updaters, then the Provers and the Signers, the Combiners and the Redactors, and last the Spend Finalizer and the Transaction Extractor. A *Verifier* is not a role: it is an inspection of a PCZT's internal consistency, for example the relations of §12.7, performed on behalf of an entity, such as a hardware signer, that trusts it and cannot perform the checks itself (ZIP 374, "Verifier").

Definition 12.5 (Modification state)**.** The Global field $\mathsf{tx_modifiable} \in \{0,1\}^8$, bits numbered from the least significant, has the following bits (ZIP 374, "Global" and "Combiner"):

(i) bit 0, transparent inputs modifiable, and bit 1, transparent outputs modifiable: each set by the Creator, checked, and possibly cleared, by a Constructor before it adds an input or output of its kind, cleared by the IO Finalizer when shielded spends or outputs exist, and merged by AND;

(ii) bit 2, a signature with hash type SIGHASH_SINGLE is present: cleared by the Creator, set by a Signer that adds such a signature, read by a Constructor, which then preserves the pairing of each such input with its output, and merged by OR;

(iii) bits 3 to 6: zero, which ZIP 374 requires in version 1 and the Combiner requires of every PCZT it merges;

(iv) bit 7, shielded modifiable: set by the Creator, checked, and possibly cleared, by a Constructor before it adds a shielded spend or output, cleared by the IO Finalizer when shielded spends or outputs exist and by every Signer, and merged by AND.

The Creator's value is therefore 1000 0011 in binary.

Signing freezes the effects. After adding a shielded signature a Signer clears bits 0, 1 and 7: every shielded signature, spend-authorisation and binding alike, is on the one signature digest, which commits to all effecting data (*Ironwood Guide*, §"Transaction digests and signatures", Proposition "Bundle binding"), and no shielded hash type leaves an input or output open. After adding a transparent signature it clears bit 0 unless the hash type has ANYONECANPAY, clears bit 1 unless its base type is NONE, sets bit 2 if its base type is SINGLE, with or without ANYONECANPAY, and clears bit 7 (ZIP 374, "Global" and "Signer").

A wallet executing a plan (Definition 11.2) acts as Creator, Constructor, IO Finalizer and Updater. It builds the bundles by Construction 11.18 and then, as Updater, attaches to each non-dummy Orchard-pool, Ironwood-pool and Sapling spend the derivation record of §12.3, with the seed fingerprint of Construction 2.2 and the path $(32', \mathit{coin_type}', \mathit{account}')$ of Definition 2.7; to each transparent input the BIP 44 path $(44', \mathit{coin_type}', \mathit{account}', \mathit{scope}, \mathit{index})$, the last two components non-hardened; and to each output that pays a payment of the plan (Definition 11.1) that payment's address string as user_address, which a Signer checks (§12.7) (ZIP 374, "Updater" and "Zip32Derivation"; ZIP 32; BIP 44).

12.3 Fields

Definition 12.6 (PCZT fields)**.** *Types.* The types u8, u32 and u64 are the unsigned integers of that width, i128 the signed integers of 128 bits and $\mathsf{bool} = \{\mathsf{false}, \mathsf{true}\}$; $\mathbb{B}^n$ is the set of byte strings of length n, T^n the sequences of n elements of T, and $T_1 \times T_2$ the pairs; $\mathsf{Opt}(T)$ is T together with an absent value $\perp$; $\mathsf{List}(T)$ is the finite sequences over T, $\mathsf{Map}(K, V)$ the finite

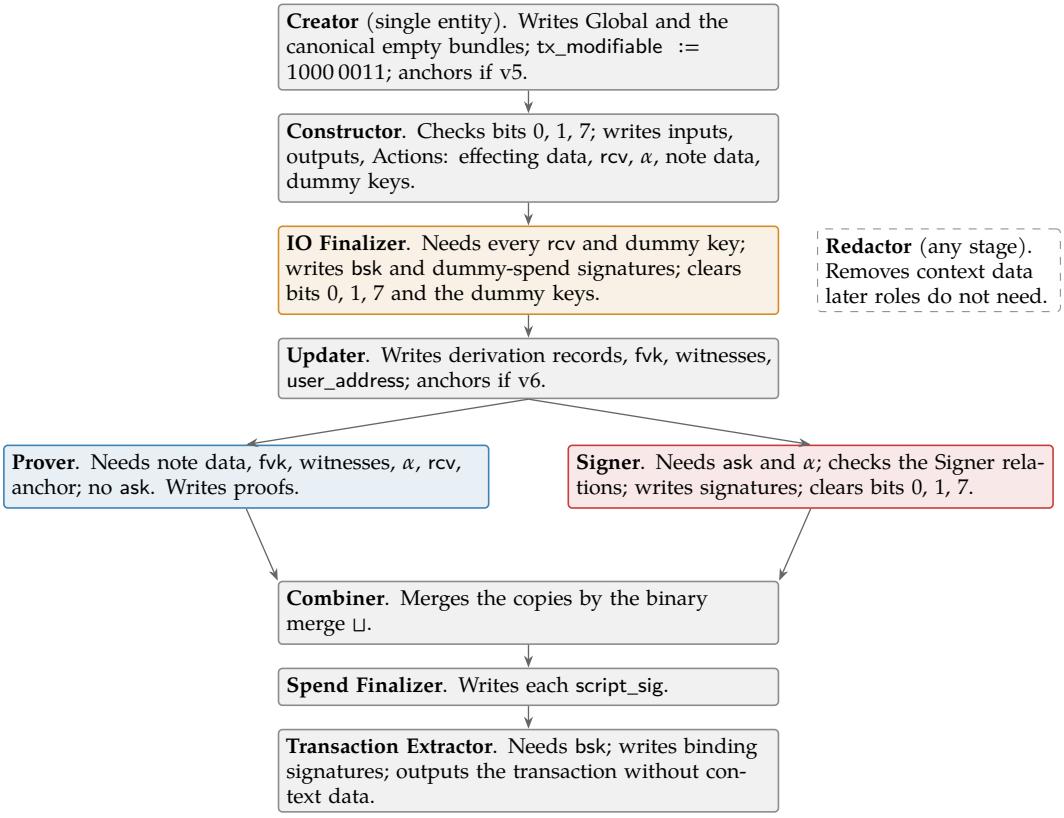


Figure 5: The roles of a PCZT (Definition 12.4), with the fields each writes, the secrets it needs and the bits of tx_modifiable it clears (Definition 12.5). Red marks the one role that needs the spend authorising key ask, amber the role that needs the value-commitment trapdoors and the dummy keys, blue the role that needs the note data and fvk but no ask, and grey the roles that need no secret. The Prover and the Signer run in either order or in parallel on copies joined by the Combiner; what each recipient learns is stated in §12.9.

maps from K to V, and Str the UTF-8 strings; $\mathsf{Enum}\{A_0, A_1, \dots\}$ is a set of tags, A_i with index i. A *derivation record* is a value of $D := \mathbb{B}^{32} \times \mathsf{List}(\mathsf{u32})$, a seed fingerprint (Construction 2.2) and a derivation path, an index with bit 31 set being hardened. Let $P := \mathsf{Map}(\mathsf{Str}, \mathsf{List}(\mathsf{u8}))$, the type of a proprietary map. Each record below lists its fields in order, the order of the encoding (§12.4); a field marked (v1) or (v2) is present only in that version (ZIP 374, "Structure" and the sections of each record).

Global. tx_version $\in$ u32, which is 5 in version 1 and 5 or 6 in version 2; version_group_id $\in$ u32, consistent with it (ZIP 225, ZIP 229); consensus_branch_id $\in$ u32, which for tx_version = 6 names an upgrade at which version 6 is supported, NU6.3 or later, that of NU7 being `0x77190AD9` (ZIP 259, "NU7 network constants"); fallback_lock_time $\in \mathsf{Opt}(\mathsf{u32})$, absence meaning 0; expiry_height $\in$ u32 (ZIP 203); coin_type $\in$ u32, the SLIP 44 coin type, which is not part of the transaction and against which roles check network-specific data; tx_modifiable $\in$ u8 (Definition 12.5); proprietary $\in P$.

Transparent bundle. inputs $\in \mathsf{List}(\mathrm{TI})$ and outputs $\in \mathsf{List}(\mathrm{TO})$. An input TI: prevout_txid $\in \mathbb{B}^{32}$; prevout_index $\in$ u32; sequence $\in \mathsf{Opt}(\mathsf{u32})$, absence meaning $2^{32} - 1$; required_time_lock_time $\in \mathsf{Opt}(\mathsf{u32})$, at least 500,000,000; required_height_lock_time $\in \mathsf{Opt}(\mathsf{u32})$, in $\{1, \dots, 499{,}999{,}999\}$; script_sig $\in \mathsf{Opt}(\mathsf{List}(\mathsf{u8}))$; value $\in$ u64; script_pubkey $\in \mathsf{List}(\mathsf{u8})$; redeem_script $\in \mathsf{Opt}(\mathsf{List}(\mathsf{u8}))$; partial_signatures $\in \mathsf{Map}(\mathbb{B}^{33}, \mathsf{List}(\mathsf{u8}))$; sighash_type $\in$ u8; bip32_derivation $\in \mathsf{Map}(\mathbb{B}^{33}, D)$; four preimage maps, for RIPEMD160, SHA256, HASH160 and HASH256, of types $\mathsf{Map}(\mathbb{B}^{20}, \mathsf{List}(\mathsf{u8}))$, $\mathsf{Map}(\mathbb{B}^{32}, \mathsf{List}(\mathsf{u8}))$, $\mathsf{Map}(\mathbb{B}^{20}, \mathsf{List}(\mathsf{u8}))$ and $\mathsf{Map}(\mathbb{B}^{32}, \mathsf{List}(\mathsf{u8}))$ in that order; proprietary $\in P$. An output TO: value $\in$ u64; script_pubkey $\in \mathsf{List}(\mathsf{u8})$; redeem_script $\in \mathsf{Opt}(\mathsf{List}(\mathsf{u8}))$; bip32_derivation $\in \mathsf{Map}(\mathbb{B}^{33}, D)$; user_address $\in \mathsf{Opt}(\mathsf{Str})$; proprietary $\in P$.

Sapling bundle. spends $\in \mathsf{List}(\mathrm{SS})$; outputs $\in \mathsf{List}(\mathrm{SO})$; value_sum $\in$ i128, the net value of the spends minus the outputs; anchor $\in \mathbb{B}^{32}$ (v1) or $\mathsf{Opt}(\mathbb{B}^{32})$ (v2); bsk $\in \mathsf{Opt}(\mathbb{B}^{32})$. A spend SS: effecting cv, nf, rk $\in \mathbb{B}^{32}$; zkproof $\in \mathsf{Opt}(\mathbb{B}^{192})$; spend_auth_sig $\in \mathsf{Opt}(\mathbb{B}^{64})$; recipient $\in \mathsf{Opt}(\mathbb{B}^{43})$; value $\in \mathsf{Opt}(\mathsf{u64})$; rcm $\in \mathsf{Opt}(\mathbb{B}^{32})$ and rseed $\in \mathsf{Opt}(\mathbb{B}^{32})$, at most one set, by whether the note predates ZIP 212; rcv $\in \mathsf{Opt}(\mathbb{B}^{32})$; the proof generation key $(\mathsf{ak}, \mathsf{nsk}) \in \mathsf{Opt}(\mathbb{B}^{32} \times \mathbb{B}^{32})$; witness $\in \mathsf{Opt}(\mathsf{u32} \times (\mathbb{B}^{32})^{32})$; $\alpha \in \mathsf{Opt}(\mathbb{B}^{32})$; a derivation record in $\mathsf{Opt}(D)$; dummy_ask $\in \mathsf{Opt}(\mathbb{B}^{32})$; proprietary $\in P$. An output SO: effecting cv, cmu, epk $\in \mathbb{B}^{32}$ and the ciphertexts $C_{\mathsf{enc}}, C_{\mathsf{out}} \in \mathsf{List}(\mathsf{u8})$, whose lengths depend on the transaction version; zkproof $\in \mathsf{Opt}(\mathbb{B}^{192})$; recipient $\in \mathsf{Opt}(\mathbb{B}^{43})$; value $\in \mathsf{Opt}(\mathsf{u64})$; rseed, rcv, ock $\in \mathsf{Opt}(\mathbb{B}^{32})$; a derivation record in $\mathsf{Opt}(D)$; user_address $\in \mathsf{Opt}(\mathsf{Str})$; proprietary $\in P$. The Sapling constructions are those of the protocol specification and are cited, not restated.

Orchard bundle. actions $\in \mathsf{List}(\mathrm{OA})$; flags $\in$ u8, with bit 0 enableSpends, bit 1 enableOutputs, bit 2 enableCrossAddress (v2; zero in version 1 and in the Orchard bundle of every transaction mined under NU6.3 or later, the rule of Construction 11.5), and bits 3 to 7 zero; value_sum $\in \mathsf{u64} \times \mathsf{bool}$; anchor $\in \mathbb{B}^{32}$ (v1) or $\mathsf{Opt}(\mathbb{B}^{32})$ (v2); note_version $\in \mathsf{Enum}\{\mathsf{V2}, \mathsf{V3}\}$ (v2), where V2 is the note plaintext of ZIP 212, lead byte `0x02`, and V3 that of ZIP 2005, lead byte `0x03`, which MUST be V2 in the Orchard bundle and is implicitly V2 in version 1; zkproof $\in \mathsf{Opt}(\mathsf{List}(\mathsf{u8}))$, one proof for the bundle; bsk $\in \mathsf{Opt}(\mathbb{B}^{32})$. An Action OA: $\mathsf{cv}^{\mathsf{net}} \in \mathbb{B}^{32}$; a spend OS; an output OO; rcv $\in \mathsf{Opt}(\mathbb{B}^{32})$, one per Action, since an Action has one net value commitment (*Ironwood Guide*, §"Value commitments"). A spend OS: effecting nf, rk $\in \mathbb{B}^{32}$; spend_auth_sig $\in \mathsf{Opt}(\mathbb{B}^{64})$; recipient $\in \mathsf{Opt}(\mathbb{B}^{43})$, a raw Orchard address; value $\in \mathsf{Opt}(\mathsf{u64})$; ρ, rseed $\in \mathsf{Opt}(\mathbb{B}^{32})$; fvk $\in \mathsf{Opt}(\mathbb{B}^{96})$, the encoding of $(\mathsf{ak}, \mathsf{nk}, \mathsf{r}_{\mathsf{ivk}})$; witness $\in \mathsf{Opt}(\mathsf{u32} \times (\mathbb{B}^{32})^{32})$, a position and a 32-level authentication path; $\alpha \in \mathsf{Opt}(\mathbb{B}^{32})$; a derivation record in $\mathsf{Opt}(D)$; dummy_sk $\in \mathsf{Opt}(\mathbb{B}^{32})$; proprietary $\in P$. In a transaction of

version 6 the witness may be absent until proving (§12.5). An output OO: effecting $\mathsf{cmx}, \mathsf{epk} \in \mathbb{B}^{32}$ and $C_{\mathsf{enc}}, C_{\mathsf{out}} \in \mathsf{List}(\mathsf{u8})$; $\mathsf{recipient} \in \mathsf{Opt}(\mathbb{B}^{43})$; $\mathsf{value} \in \mathsf{Opt}(\mathsf{u64})$; $\mathsf{rseed}, \mathsf{ock} \in \mathsf{Opt}(\mathbb{B}^{32})$; a derivation record in $\mathsf{Opt}(D)$; $\mathsf{user_address} \in \mathsf{Opt}(\mathsf{Str})$; $\mathsf{proprietary} \in P$.

Ironwood bundle (v2). The fields of the Orchard bundle, with $\mathsf{note_version} = \mathsf{V3}$ and its anchor a root of the Ironwood note commitment tree (§12.5).

Which role sets, requires or clears each field is as ZIP 374 states per field; this section restates it where it defines a role.

Every transparent input's value and script_pubkey are required, not optional, because the signature digest commits to the value and script of every coin a transaction spends (ZIP 244, as amended by ZIP 229; *Ironwood Guide*, §"Transaction digests and signatures"), and the binding signatures are on that digest (ZIP 374, "TransparentInput").

Remark 12.7 (The sign of an Orchard-protocol value sum). The pair $\mathsf{value_sum} = (m, b)$ denotes the net value of the bundle's spends minus its outputs, but ZIP 374 does not state the meaning of b: a gap in the specification. The reference implementation of ZIP 374 reads $b = \mathsf{true}$ as the net value $-m$ and $b = \mathsf{false}$ as $+m$, and the canonical empty bundle has $(0, \mathsf{false})$. This volume adopts that reading, and writes v^{balance} for the integer that a bundle's value_sum denotes; for Sapling it is the i128 value itself.

A Signer MUST parse the user_address of each output that carries one and confirm that it contains the output's recipient, directly or as a receiver of a unified address (ZIP 374, "TransparentOutput", "SaplingOutput" and "OrchardOutput"; Definition 3.7 and Construction 3.10). The anchors, witnesses and commitments that these fields carry are those of the *Ironwood Guide*, §"The note commitment tree", with its subsections "Anchors" and "Authentication paths"; they are cited, not rebuilt.

Definition 12.8 (Canonical empty bundles). The *canonical empty bundle* of each slot is (ZIP 374, "v2 Encoding"):

(i) transparent: empty input and output lists;

(ii) Sapling: empty spend and output lists, $\mathsf{value_sum} = 0$, $\mathsf{anchor} = \bot$, $\mathsf{bsk} = \bot$;

(iii) Orchard: no Actions, $\mathsf{flags} = 0000\,0011$ in binary (spends and outputs enabled), $\mathsf{value_sum} = (0, \mathsf{false})$, $\mathsf{anchor} = \bot$, $\mathsf{note_version} = \mathsf{V2}$, $\mathsf{zkproof} = \mathsf{bsk} = \bot$;

(iv) Ironwood: no Actions, $\mathsf{flags} = 0000\,0111$ in binary (spends, outputs and cross-address transfers enabled), $\mathsf{value_sum} = (0, \mathsf{false})$, $\mathsf{anchor} = \bot$, $\mathsf{note_version} = \mathsf{V3}$, $\mathsf{zkproof} = \mathsf{bsk} = \bot$.

A bundle that differs from it in any field, its flags, note version or anchor included, is not canonical empty.

12.4 Encoding

Construction 12.9 (PCZT encoding). The encoding of a PCZT of version n is

$$\texttt{50 43 5a 54} \,\|\, \mathsf{I2LEOSP}_{32}(n) \,\|\, \mathit{body}_n,$$

the first four bytes being the ASCII string `PCZT`. A string that does not begin with this 8-byte header is not a PCZT encoding, and a parser that meets an unknown version MUST NOT

parse the remainder (ZIP 374, "Encoding"). The body is built from the following encodings of the types of Definition 12.6 (ZIP 374, "Data type encodings"):

(i) u8: one raw byte;

(ii) u32 and u64: $\mathsf{varint}(x)$, the digits of x in base 128, least significant first, one per byte, with bit 7 set on every byte but the last. Encoders MUST emit the shortest form. A k-bit integer takes at most $\lceil k/7 \rceil$ bytes, 5 for u32 and 10 for u64, and in an encoding of that length the last byte contributes only the high k mod 7 bits, so the fifth byte of a u32 is at most `0x0f`; decoders MUST reject encodings that exceed either bound and MAY accept non-minimal encodings within them;

(iii) i128: $\mathsf{varint}(\mathsf{zigzag}(x))$, with $\mathsf{zigzag}(x) = 2x$ for $x \geq 0$ and $2|x| - 1$ for $x < 0$;

(iv) bool: the byte `0x00` for false or `0x01` for true, other bytes rejected;

(v) $\mathbb{B}^n$, T^n and pairs: the concatenation of the elements, with no length prefix, so that a witness path is 1024 raw bytes;

(vi) $\mathsf{Opt}(T)$: the byte `0x00` for $\perp$, and `0x01` followed by the value otherwise, other tag bytes rejected;

(vii) $\mathsf{List}(T)$ and Str: the varint of the number of elements, of bytes for a string, followed by the elements; invalid UTF-8 is rejected;

(viii) $\mathsf{Map}(K, V)$: the varint of the number of entries, followed by each key and its value. Encoders MUST emit the entries in strictly ascending lexicographic byte order of the unencoded keys, without duplicates; decoders MAY accept other orders and duplicates;

(ix) Enum: the varint of the tag's index;

(x) a record: the concatenation of its fields in the order of Definition 12.6, without tags or padding, a field absent from version n being omitted entirely.

Encoders MUST NOT emit data after the last field; decoders MAY ignore or reject it. *Layout* (ZIP 374, "v1 Encoding" and "v2 Encoding"): $body_1$ is the Global record followed by the transparent, Sapling and Orchard bundles, each Action encoded as $\mathsf{cv}^{\mathsf{net}}$, spend, output, rcv. The body $body_2$ is the Global record followed by the four bundles, transparent, Sapling, Orchard and Ironwood, each as a value of Opt; a bundle equal to its slot's canonical empty bundle (Definition 12.8) MUST be encoded as absent and any other as present, and a parser MUST restore the canonical empty bundle from absence. Encoding in version 1 MUST fail when $\mathsf{tx_version} \neq 5$, when the Ironwood bundle is not canonical empty, when the Orchard note version is not V2, or when an anchor required before signing a transaction of version 5 is absent: these are the states version 1 cannot represent.

Lemma 12.10 (Canonicity of the encoding)**.** *For every version n and every PCZT P representable in version n, exactly one byte string is an output for P permitted to an encoder that obeys the MUST rules of Construction 12.9, and distinct PCZTs have distinct encodings. Canonicity is a property of encoder output: since decoders may accept non-minimal varints, unordered or repeated map entries and trailing data, several strings may decode to one PCZT.*

Proof. By induction on the type structure. *Uniqueness.* The permitted encoding of each primitive is a function of its value: the shortest base-128 representation is unique, its last byte non-zero unless the value is 0; the map zigzag is a bijection from the integers onto the naturals; the tags of options, booleans and enumerations are fixed; the counts of lists and strings are determined

Offset	Bytes	Field
0	50 43 5a 54	header, PCZT
4	02 00 00 00	version 2
8	06	tx_version = 6
9	98 ed 92 c4 0d	version_group_id
14	d9 95 e4 b8 07	consensus_branch_id
19	00	fallback_lock_time = ⊥
20	80 ad e2 04	expiry_height
24	85 01	coin_type = 133
26	83	tx_modifiable, bits 0, 1, 7
27	00	empty proprietary map
28 to 31	00 00 00 00	four absent bundles

Table 6: The minimal version-2 PCZT of Example 12.11: offset, bytes and field, in the encoding of Construction 12.9.

by the value. A finite map admits exactly one strictly ascending order of its entries without duplicates, since lexicographic order on byte strings is total. A record concatenates uniquely encoded fields in a fixed order. In version 2 the presence of each bundle is determined by equality with Definition 12.8, and no trailing data is permitted. *Injectivity.* Each encoding is self-delimiting given its type, being of fixed length or a count followed by self-delimiting elements, so a decoder that reads the fields in order recovers every value; an absent version-2 bundle is restored as the canonical empty bundle, the only bundle encoded as absent. The header fixes n, so encodings of distinct versions differ as well. □

The canonical-empty rule is exact equality so that two copies of one PCZT encode identically whatever software produced them, and re-encoding cannot move a bundle between its present and absent forms (ZIP 374, "Omitting empty bundles in PCZT v2").

Example 12.11 (A minimal version-2 PCZT)**.** The Creator's PCZT (§12.6) for a transaction of version 6, with version group identifier `0xD884B698` (ZIP 229), consensus branch identifier `0x77190AD9` (NU7, ZIP 259), expiry height 10,000,000, coin type 133 (Mainnet), no fallback lock time, no proprietary entries and every bundle canonical empty, encodes in version 2 as the 32 bytes of Table 6. With coin type 1 the coin type takes the one byte `01`, the total is 31 bytes, and every later field moves one byte earlier: the position of a field depends on the values before it, which is why no field can be added within a version.

12.5 PCZT v2: deferred anchors and the Ironwood bundle

In version 2 the Ironwood bundle carries the Ironwood-pool component of a transaction of version 6, with the structure, field meaning and role interactions of the Orchard bundle, the Ironwood pool being a second value pool of the Orchard protocol (ZIP 229; *Ironwood Guide*, §"The Orchard protocol and its two pools"). It differs in three respects: its enableCrossAddress bit is meaningful and normally 1; its anchor is a root of the Ironwood note commitment tree, distinct from the Orchard tree; and its note version MUST be V3. Every role MUST reject a PCZT with tx_version = 5 whose Ironwood bundle is not canonical empty (Definition 12.8) (ZIP 374, "IronwoodBundle (PCZT v2)").

Definition 12.12 (Anchors and witnesses in version 2)**.** The following rules hold (ZIP 374, "Anchors and pre-authorization").

(i) For tx_version = 5, every shielded bundle's anchor is set when the bundle's first spend

is added and never changes, since the version-5 digest commits to it.

(ii) For tx_version = 6, which requires version 2, an anchor MAY be absent until proving; an Updater MAY set an absent anchor, or replace a set one, at any time before the Prover runs for that bundle; setting or replacing an anchor MUST clear every proof already in that bundle, the Sapling spends' proofs or the Orchard-protocol bundle's proof, since a proof takes the anchor as a primary input; and the witness of a Sapling or Orchard-protocol spend MAY be absent until proving, and is set by an Updater once known.

(iii) Whenever a bundle has both an anchor and witnesses, the Updater that sets either MUST check that every witness of a spend of non-zero value roots to the anchor, and the Prover MUST check the same before proving.

Deferral is governed by the transaction version, not by the PCZT version: a transaction of version 5 in a version-2 PCZT fixes its anchors as in version 1. Absence is an absent option, not an all-zero string, because zero is an element of $\mathbb{F}_{p_{\mathsf{Pallas}}}$ and so a syntactically valid anchor (ZIP 374, "Optional anchors in PCZT v2").

Proposition 12.13 (Deferred anchors)**.** *Let P be a PCZT of a transaction of version 6.*

(*i*) *If every spend of P is signed, replacing the anchor rt of a bundle and its proof π by any anchor rt' and a proof π' valid for it leaves every signature valid and the transaction identifier unchanged: re-anchoring needs only re-proving.*

(*ii*) *A spend of an Orchard-protocol note whose creating transaction is not yet mined can be fully signed before that note is in the tree, and its signatures remain valid once an anchor, a witness and a proof are supplied: every field that the signature digest commits to is computable from the note data, and the anchor is not among them.*

Neither holds for a transaction of version 5, and (ii) fails for a Sapling spend in version 6.

Proof. (i) The version-6 digest and transaction identifier omit the anchors, which are authorising data (ZIP 229, "Anchor commitment (version 6)"), and by *Ironwood Guide*, §"Transaction digests and signatures", Proposition "Bundle binding", replacing (rt, π) by (rt', π') changes neither the transaction identifier nor any signed message, so every signature remains valid.

(ii) The effecting fields of an Orchard-protocol spend are nf, rk and its Action's $\mathsf{cv}^{\mathsf{net}}$. The nullifier is $\mathsf{DeriveNullifier}_{\mathsf{nk}}(\rho, \psi, \mathsf{cm})$ (*Ironwood Guide*, §"The nullifier"), a function of the key and of note data, with ρ fixed by the creating Action and ψ and cm derived from rseed, ρ, the address and the value; it does not depend on the note's position. The key rk and the commitment $\mathsf{cv}^{\mathsf{net}}$ depend on keys, values and fresh randomness only.

For version 5 the digest commits to the anchors (ZIP 244, "Signature Digest"), so replacing an anchor invalidates every signature, which refutes (i); and an anchor fixed before the note is in the tree is the root of a tree state that does not contain the note, for which no authentication path of the note exists except with negligible probability (*Ironwood Guide*, §"Anchors", Proposition "Membership soundness"), so a spend of non-zero value signed then admits no valid proof, so no signature made before the note is in the tree survives completion, which refutes (ii). A Sapling nullifier depends on the note's position in the tree (protocol specification, §"Computing ρ values and Nullifiers"), which is unknown until the note is mined, and is effecting data; for Sapling only (i) holds. □

ZIP 374 calls the two workflows re-anchoring and the spending of unmined notes; Proposition 11.4 (c) is the latter for the steps of a plan.

12.6 Construction and IO finalisation

The transaction of a PCZT is of version 5, with version group identifier `0x26A7270A` (protocol specification, §"Transaction Consensus Rules"), or of version 6, with `0xD884B698` (ZIP 229), whose consensus branch identifier names NU6.3 or a later upgrade; version 4 is not representable, and is invalid from NU7 on (ZIP 2003, deployed by ZIP 259). An Ironwood component requires version 6, the mixed case being rejected by the rule of §12.5. The consensus branch identifier fixes the rules in force, and a role that does not recognise it cannot check the PCZT against them (ZIP 374, "Global").

Creator. The Creator chooses $\mathsf{tx_version}$, $\mathsf{version_group_id}$ and $\mathsf{consensus_branch_id}$, sets the other Global fields with $\mathsf{tx_modifiable} = 1000\,0011$ in binary, and sets every bundle to its canonical empty bundle (Definition 12.8); for $\mathsf{tx_version} = 5$ it sets the Sapling and Orchard anchors, and for $\mathsf{tx_version} = 6$ it MAY leave them to an Updater (Definition 12.12) (ZIP 374, "Creator").

Constructor. Before adding anything the Constructor MUST check the bit of $\mathsf{tx_modifiable}$ for the kind added (Definition 12.5), and it updates the bundle's $\mathsf{value_sum}$. Per Orchard-protocol Action, in the notation of the *Ironwood Guide*, it sets $\mathsf{cv}^{\mathsf{net}} := \mathsf{ValueCommit}^{\mathsf{Orchard}}_{\mathsf{rcv}}(v^{\mathsf{old}} - v^{\mathsf{new}})$ for a fresh rcv (§"Value commitments"); the nullifier $\mathsf{nf}^{\mathsf{old}}$ of the consumed note (§"The nullifier"); a uniform α and $\mathsf{rk} := \mathsf{ak}^{\mathbb{P}} + [\alpha]\, G^{\mathsf{SpendAuth}}$ (§"Randomised validating keys"); and the created note with $\rho^{\mathsf{new}} := \mathsf{nf}^{\mathsf{old}}$, its cmx (§"Note commitments"), epk and C_{enc} (§"Encryption to the recipient") and C_{out} (§"The outgoing ciphertext"); together with the context fields of the spend and the output. Dummy and fabricated Actions and their order are those of Construction 11.18, only fully dummy Actions carrying $\mathsf{dummy_sk}$. Spends and outputs MUST agree with the bundle flags. When the bundle's anchor is set, each spend added carries a witness rooting to it, or, for a note of value zero, whose path the Action statement does not check, any placeholder; for $\mathsf{tx_version} = 6$ a spend MAY be added without a witness, even of a note not yet in the tree; for $\mathsf{tx_version} = 5$ every non-dummy spend is of a note in the tree, with a witness to the anchor. A transparent input with a lock-time requirement is added only if its lock type is compatible with those of the other inputs and, when inputs are already signed, it leaves the lock time unchanged (§12.10) (ZIP 374, "Constructor" and "OrchardBundle").

Construction 12.14 (IO finalisation)**.** Once the input and output set is final, the IO Finalizer performs the following steps, failing where stated (ZIP 374, "IO Finalizer").

(1) If shielded spends or outputs exist, it clears bits 0, 1 and 7 of $\mathsf{tx_modifiable}$.

(2) For each Orchard-protocol bundle with Actions $1, \ldots, n$, it requires every rcv_i, sets $\mathsf{bsk} := \sum_i \mathsf{rcv}_i \in \mathbb{F}_{p_{\mathsf{Vesta}}}$, computes

$$\mathsf{bvk} := \textstyle\sum_i \mathsf{cv}^{\mathsf{net}}_i - [v^{\mathsf{balance}}]\, V^{\mathsf{Orchard}},$$

with v^{balance} the integer that the bundle's $\mathsf{value_sum}$ denotes (§12.3), and fails unless $[\mathsf{bsk}]\, R^{\mathsf{Orchard}} = \mathsf{bvk}$ (*Ironwood Guide*, §"The binding signature"). The Orchard and the Ironwood bundles are finalised separately, each by this computation on its own Actions.

(3) For the Sapling bundle, it sets bsk to the sum of the spends' rcv minus the sum of the outputs' rcv, checked in the same way against the Sapling binding validating key (protocol specification, §"Balance and Binding Signature (Sapling)").

(4) It computes the signature digest over the effecting data (*Ironwood Guide*, §"Transaction digests and signatures"), signs each dummy spend under the spend authorising key derived from its $\mathsf{dummy_sk}$ or given by its $\mathsf{dummy_ask}$, randomised by the spend's α, and clears that key.

(5) For each bundle with enableCrossAddress = 0, it MUST verify that each Action's output recipient equals its spend recipient, and fail otherwise.

The condition of step (5) is condition A10 of the Action statement (*Ironwood Guide*, §"The Action statement"), so no valid proof exists for an Action that violates it; the step reports the violation at the role responsible for it.

Lemma 12.15 (Value-sum consistency)**.** *Let an Orchard-protocol bundle have Actions with* $\mathsf{cv}_i^{\mathsf{net}} = \mathsf{ValueCommit}^{\mathsf{Orchard}}_{\mathsf{rcv}_i}(v_i^{\mathsf{net}})$*, and let* v^{balance} *be the integer that its* value_sum *denotes. Step* (2) *of Construction 12.14 passes if and only if* $\sum_i v_i^{\mathsf{net}} \equiv v^{\mathsf{balance}} \pmod{p_{\mathsf{Vesta}}}$*, and* bsk *is then a binding signing key whose validating key is* bvk*. The check reads no value* v_i^{net}*, which may therefore have been redacted.*

Proof. With $\mathsf{bsk} = \sum_i \mathsf{rcv}_i$,

$$\mathsf{bvk} - [\mathsf{bsk}]\, R^{\mathsf{Orchard}} = \left[\sum_i v_i^{\mathsf{net}} - v^{\mathsf{balance}}\right] V^{\mathsf{Orchard}},$$

which is $\mathcal{O}$ if and only if the coefficient vanishes modulo p_{Vesta}, since V^{Orchard} is a non-identity point of the group of prime order p_{Vesta}. This is *Ironwood Guide*, §"The binding signature", Proposition "Balance and key knowledge", part (i). No independence of V^{Orchard} and R^{Orchard} is used. The check computes bvk from the commitments and value_sum alone. □

The binding signing key cannot wait for the Transaction Extractor, since every rcv would then travel to it and open every value commitment to every intermediate party; nor can Constructors maintain it incrementally, since serial Constructors would have to update it and parallel ones, whose results a Combiner merges, must not. One role run once the set is final removes both problems, and the same role signs dummy spends so that no Signer parses spending-key material (ZIP 374, "Adding an IO Finalizer role").

12.7 Signing

Digests. The Signer derives the effecting data from the PCZT and computes the shielded signature digest SigHash itself (ZIP 244 for version 5, as amended by ZIP 229 for version 6; *Ironwood Guide*, §"Transaction digests and signatures"), neither proofs nor signatures entering it; for each transparent input it signs, it computes that input's digest under the input's sighash_type (ZIP 244, "S.2: transparent_sig_digest") and signs it by the scheme of the *Crypto Guide*, §"The transparent layer's schemes: ECDSA and BIP-340 over secp256k1". A Signer that cannot sign under an input's hash type MUST NOT sign that input (ZIP 374, "TransparentInput").

Orchard-protocol spends. For an Action it signs, with α present, the Signer sets $\mathsf{rsk} := \mathsf{ask}+\alpha$, fails unless $\mathsf{rk} = [\mathsf{rsk}]\, G^{\mathsf{SpendAuth}}$, and sets spend_auth_sig to the RedPallas signature on SigHash under rsk (*Ironwood Guide*, §"Randomised validating keys"); Ironwood-pool Actions are signed in the same way in their own bundle. A signature produced elsewhere, for example by a hardware device, is attached if and only if it is valid under the Action's rk on SigHash. The Signer then updates tx_modifiable (Definition 12.5) (ZIP 374, "Signer" and "OrchardSpend").

Definition 12.16 (Signer relations)**.** For an Orchard-protocol Action whose explicit data are present, with v^{old} and v^{new} the values of its spend and its output, the *Signer relations* are, in the notation of the *Ironwood Guide*:

(a) $\mathsf{cv}^{\mathsf{net}} = \mathsf{ValueCommit}^{\mathsf{Orchard}}_{\mathsf{rcv}}(v^{\mathsf{old}} - v^{\mathsf{new}})$ (§"Value commitments");

(b) the consumed note rebuilt from (recipient, v^{old}, ρ, rseed) has nullifier nf under the nk

of fvk (§"The nullifier"), and fvk owns the recipient: $\mathsf{pk_d} = [\mathsf{ivk}]\,\mathsf{g_d}$ for the ivk that fvk determines (§"Diversified addresses");

(c) $\mathsf{rk} = \mathsf{ak}^{\mathbb{P}} + [\alpha]\,G^{\mathsf{SpendAuth}}$ for the ak of fvk (§"Randomised validating keys");

(d) cmx is the commitment of the created note rebuilt from (recipient, v^{new}, rseed) with $\rho^{\mathsf{new}} = \mathsf{nf}$ (§"Note commitments");

(e) for an output that is not fabricated, esk derived from rseed and ρ^{new} gives epk and a C_{enc} that decrypts to a plaintext with the explicit recipient, value and rseed (§"Encryption to the recipient"), and, when ock is present, C_{out} decrypts under it to $(\mathsf{pk_d}, \mathsf{esk})$ (§"The outgoing ciphertext").

A fabricated zero-valued output at a spent note's receiver carries a random C_{enc} (Construction 11.18) and is exempt from (e): a Signer MUST NOT reject it on that ground alone, and SHOULD classify it as a tolerable dummy, checking (d) if it wishes (ZIP 374, "Signer", "Verifier" and "Fabricated same-address outputs (NU6.3)").

Obligations before signing (ZIP 374, "Signer"). A Signer MUST reject a PCZT that carries any dummy_sk or dummy_ask: the IO Finalizer signs dummy spends and clears these keys, so their presence asks the Signer to parse spending-key material. It MUST, for each output with a user_address, parse it and confirm that it contains the output's recipient (§12.3). It SHOULD verify that the explicit values and note data it shows its user are consistent with the effecting data it signs, through the relations of Definition 12.16; a field absent because it was redacted is not an inconsistency, and only a present field that contradicts the effecting data is. The relations need the value and note commitments, BLAKE2b and the functions built on it, nullifier derivation, key agreement, the key derivation and the AEAD of note encryption, the signature digest and RedPallas signing (RedJubjub for Sapling), and no proof system.

Theorem 12.17 (Signer safety). *Let a Signer hold the* ask *of an Orchard-protocol key generated with* use_qsk = false, *and suppose that for every PCZT P it signs it computes $m_P := \mathsf{SigHash}(T_P)$ from the effecting data T_P of P, performs the obligations above, verifies relations (a) to (e) of Definition 12.16, with all their fields present, for every Action whose spend it signs, displays the values, recipients and address strings so checked, and signs under keys* $\mathsf{ask} + \alpha$ *no message other than such digests m_P. Under the assumptions of* Ironwood Guide, *§"Authorisation of spends", Theorem "Spend authority", and Assumption "Collision resistance of BLAKE2b" (§"Transaction digests and signatures"), except with negligible probability:*

(*i*) *every accepted Action that consumes a note of the key carries a signature on some m_P;*

(*ii*) *every accepted transaction that carries an Action with a randomised validating key* rk *under which the Signer signed has the effecting data T_Q of some PCZT Q that the Signer signed with an Action of key* rk*; in it the Actions whose spends the Signer signed consume and create the notes, with the values and recipients, that the Signer displayed for Q, and every other effecting field, balancing values and transparent data included, equals that of Q.*

The theorem concerns content, not inclusion.

Proof. (i) The Signer is the signing oracle of the *Ironwood Guide*, §"Authorisation of spends", Definition "Spend-authority experiment", queried with (α, m_P) for α taken from the PCZT and so possibly chosen by an adversary, which that experiment admits; the full viewing key that the PCZT discloses is given to the adversary there. Theorem "Spend authority" bounds by a negligible function the probability of an accepted Action that consumes a note of the key without a queried digest.

(ii) Let T' be accepted and carry an Action with key rk and a signature σ' valid under rk on $\mathsf{SigHash}(T')$. The reduction of the proof of Theorem "Spend authority", which simulates the key holder from $\mathsf{ak}^{\mathbb{P}}$ and answers queries with adversarial randomisers, turns a triple $(\mathsf{rk}, \mathsf{SigHash}(T'), \sigma')$ that is not a triple of an answer of the Signer into a forgery in *Ironwood Guide*, §"RedPallas unforgeability", Proposition "Unforgeability under re-randomisation"; hence, except with negligible probability, $\mathsf{SigHash}(T') = m_Q$ for a PCZT Q that the Signer signed under rk. The collision case of the proof of Proposition "Bundle binding" uses only that each node of the digest tree hashes a string from which its children and fields are recovered uniquely, which holds equally for the version-5 tree (ZIP 244, "TxId Digest" and "Signature Digest"), where the anchors are among the hashed fields and are effecting data (Definition 12.3); it therefore applies to both transaction versions and gives $T' = T_Q$ as effecting data, except with negligible probability under Assumption "Collision resistance of BLAKE2b". The digest covers the value and script of each transparent input (§12.3), so a false value in Q yields a digest that validation does not compute and a signature that is not accepted. For each Action whose spend the Signer signed, relations (a) and (d) make the displayed data an opening of $\mathsf{cv}^{\mathsf{net}}$ and of cmx, and by Lemma "Hiding and binding of net value commitments" and Proposition "Hiding and binding of note commitments" of the *Ironwood Guide* no other opening is feasible; relation (b) identifies the consumed note, whose commitment fixes its value; relation (e) makes the created note decryptable by its recipient (Proposition "Decryption correctness"); and the check of $\mathsf{user_address}$ ties each recipient to the displayed string. □

Pre-authorisation (ZIP 374, "Signer" and "Anchors and pre-authorization"). For a transaction of version 6 a Signer MAY sign while the shielded anchors, witnesses or proofs are absent, since signatures commit to none of them (Proposition 12.13). A Signer asked to do so SHOULD be aware, and where applicable make its user aware, that a spend without a witness may be of a note that never exists on chain: the signature authorises the spend whether or not the note is created. For a transaction of version 5 the anchors are in the digest and MUST be present before signing.

> **Lemma 12.18** (Binding key after IO finalisation)**.** *If IO finalisation (Construction 12.14) succeeded for an Orchard-protocol bundle, and its commitments* $\mathsf{cv}_i^{\mathsf{net}}$ *and its* $\mathsf{value_sum}$ *are unchanged afterwards, then a binding signature under* bsk *on the signature digest is valid under the* bvk *that validation derives from the transaction's* $\mathsf{cv}_i^{\mathsf{net}}$ *and balancing value* v^{balance}*.*

Proof. Validation computes bvk from the same commitments and balancing value (*Ironwood Guide*, §"The binding signature", Definition "Binding validating key"), and IO finalisation checked that it equals $[\mathsf{bsk}]\,R^{\mathsf{Orchard}}$ (Lemma 12.15). Correctness of RedPallas in the binding-signature instance (*Ironwood Guide*, §"The binding signature", Construction "Binding signature") gives validity. □

The hypothesis of Lemma 12.18 holds for every PCZT reached by the roles after finalisation: $\mathsf{cv}^{\mathsf{net}}$ is required effecting data, which no later role changes and the merge of §12.9 keeps equal, and a finalised bundle merges only with a bundle of equal Action count and $\mathsf{value_sum}$.

12.8 Proving

Per Orchard-protocol Action the Prover requires the spend's recipient, value, ρ, rseed, fvk, witness and α, the output's recipient, value and rseed, and the Action's rcv; the bundle's anchor MUST be set, and before proving the Prover MUST verify that the path of the witness of every spend of non-zero value roots to it, and fail otherwise. For Sapling it requires per spend the recipient, the value, rcm or rseed, rcv, the proof generation key and the witness, and per output the recipient,

the value, rseed and rcv. The Prover needs no spending key; a delegated Prover learns the note data of the bundle but gains no spending authority (§12.9) (ZIP 374, "Prover").

Construction 12.19 (Proof creation from a PCZT)**.** For an Orchard-protocol bundle with no Actions the Prover does nothing. Otherwise, for each Action i, it rebuilds the consumed note from its explicit data and the created note from its explicit data with $\rho_i^{\mathsf{new}} = \mathsf{nf}_i$; forms the primary input

$$(rt,\ \mathsf{cv}_i^{\mathsf{net}},\ \mathsf{nf}_i,\ \mathsf{rk}_i,\ \mathsf{cmx}_i,\ \mathsf{enableSpends},\ \mathsf{enableOutputs},\ \mathsf{disableCrossAddress})$$

with rt the bundle's anchor, and the auxiliary input from the witness, the two notes, α_i, the components of fvk_i and rcv_i (*Ironwood Guide*, §"The Action statement"); and fails if the pair does not satisfy the Action statement, in particular if fvk_i does not own the consumed note or cmx_i differs from the commitment of the rebuilt created note. It then creates one Halo 2 proof for all Actions of the bundle (*Ironwood Guide*, §"The Action circuit and the Halo 2 proof"; *Halo 2 Guide*, §"The Halo 2 proof system") and sets zkproof. The Ironwood bundle is proved in the same way with its own anchor. The primary input consists of effecting data and the anchor only.

An Action whose rk is the identity point $\mathcal{O}$ is invalid (protocol specification, §"Action Descriptions"; ZIP 256, "Orchard identity rk"), and the Transaction Extractor MUST refuse it (ZIP 374, "Transaction Extractor"). A Prover that refuses it as well spares a proof of a transaction that cannot be valid; ZIP 374 does not state that check for the Prover, and it is designed but unspecified (Definition 1.1).

12.9 Combining, redaction, and privacy

Construction 12.20 (Binary merge)**.** The *merge* $P \sqcup Q$ of two PCZTs is defined, or fails, as follows (ZIP 374, "Combiner").

(1) *Global.* The fields tx_version, version_group_id, consensus_branch_id, fallback_lock_time, expiry_height and coin_type MUST be equal. The field tx_modifiable merges bitwise, bits 0, 1 and 7 by AND and bit 2 by OR, bits 3 to 6 being required zero on both sides.

(2) *Lists* of inputs, outputs, spends and Actions. If either side's bundle is IO-finalised, its bsk being present, both sides have equally many entries and equal value_sum. Otherwise the entries at the positions common to both lists merge pairwise, and the extra entries of the longer list are appended provided the shorter side's tx_modifiable permits additions of that kind, the merge failing otherwise. The merged shielded bundle's value_sum is that of the side each of whose lists of the bundle is at least as long as the other side's, the merge failing if neither side is; with lists of equal lengths the two MUST be equal. ZIP 374 leaves this unstated, so it is designed but unspecified (Definition 1.1).

(3) *Required fields* of merged entries, and the bundle flags, note_version and, in version 1, anchor, MUST be equal.

(4) *Optional fields* merge as $x \sqcup \bot = \bot \sqcup x = x$, and $x \sqcup y = x$ if $x = y$, failing otherwise; in version 2 this rule covers the anchors.

(5) *Maps* merge by union of their keys, the values of a common key being required equal.

> The Combiner of k PCZTs merges them into one that contains every field of each, or fails. It MUST NOT combine PCZTs that do not represent the same transaction, that is, PCZTs not derived by roles from a common ancestor, which rules (1) to (3) check structurally. Once the input and output set is final, the effecting data determine the transaction identifier, which then identifies the PCZT; for version 6, re-anchored copies keep that identifier.

BIP 174 lets a combiner keep one of two conflicting values and discard the other. The PCZT merge requires equality instead, which is always possible because every field is typed and known in advance, so every conflict fails at the Combiner and none diverges later (ZIP 374, "Combiner rejecting non-identical duplicates").

> **Lemma 12.21** (Commutativity of the merge)**.** *For PCZTs P and Q, the merge $P \sqcup Q$ is defined if and only if $Q \sqcup P$ is, and then the two are equal.*

Proof. By cases on the rules of Construction 12.20. The equality tests of rules (1) and (3) are symmetric; AND and OR are commutative; the rule for optional fields and the union of maps with equal values on common keys are symmetric; the condition on IO-finalised bundles is symmetric; in rule (2) the longer list and the side whose permission is consulted are determined by the lengths, not by the order of the arguments, and so is the merged value_sum, which is required equal at equal lengths; and the pairwise merges of common entries commute by induction on the structure of the entries. □

ZIP 374 requires further that combining SHOULD be functionally order-independent, $\mathrm{Combine}(f_A(p), f_B(p)) = f_A(f_B(p)) = f_B(f_A(p))$ for participants that apply f_A and f_B to p. That is a requirement on roles, not a property proved here for all of them; Proposition 12.22 proves it for the Prover and the Signer.

> **Proposition 12.22** (Order independence of proving and signing)**.** *Let p be an IO-finalised PCZT on which the Prover (Construction 12.19) and a Signer (§12.7) are both defined, the anchors being set for proving and, for a transaction of version 5, fixed before signing (Definition 12.12). Fix the random coins of both roles. Then* $\mathrm{Prove}(\mathrm{Sign}(p))$ *and* $\mathrm{Sign}(\mathrm{Prove}(p))$ *are defined and both equal* $\mathrm{Prove}(p) \sqcup \mathrm{Sign}(p)$*: the two roles run in either order, or in parallel on copies whose outputs are merged by Construction 12.20. For a transaction of version 6 signing may moreover precede the setting of the anchors (Proposition 12.13). If a copy is redacted before either role, and retains the fields that role reads, the equality holds up to the redacted fields, which the merge restores from the other side.*

Proof. Signing. The signed digest is computed from effecting data only, excluding proofs and signatures (Lemma 11.21), and the relations of Definition 12.16 read effecting and context data; Prove writes neither, so Sign reads the same values on p and on $\mathrm{Prove}(p)$ and, with its coins fixed, writes the same signatures and bits. *Proving.* The primary input of the Action statement consists of effecting data and the anchor, and the auxiliary input of context data (Construction 12.19; *Ironwood Guide*, §"The Action statement"); Sign writes none of them, and the Prover reads no spending key, so Prove writes the same proofs on p and on $\mathrm{Sign}(p)$. *Merge.* The role Prove sets only proof fields, absent in p, and Sign only signatures, absent in p, and bits of tx_modifiable, which it only clears, or sets in the case of bit 2; the bundle of p is finalised on both sides with equal counts and value sums. Neither role writes a required field, so rule (3) holds, and a transparent signature is a new key of partial_signatures, merged by rule (5). Rules (1) to (5) of Construction 12.20 therefore give $\mathrm{Prove}(p) \sqcup \mathrm{Sign}(p) = \mathrm{Prove}(\mathrm{Sign}(p)) = \mathrm{Sign}(\mathrm{Prove}(p))$. ZIP 374, "Prover", states the same independence. □

Redactor (ZIP 374, "Redactor" and the field descriptions). Anyone may clear fields that later roles do not need: per Orchard-protocol Action the spend's spend_auth_sig, recipient, value, ρ,

rseed, fvk, witness, α, derivation record, dummy key and proprietary entries, the output's recipient, value, rseed, ock, derivation record, user_address and proprietary entries, and rcv; the corresponding Sapling fields; and the transparent and Global lookup data. A PCZT with inputs of several independent Signers is sent to each with only what that Signer needs, for example without the other Signers' values of α.

Proposition 12.23 (Disclosure by a PCZT)**.** *A PCZT carries strictly more than its transaction: for each shielded spend or output it may carry the counterparty address, the value, the note randomness,* rcv, *which opens* $\mathsf{cv}^{\mathsf{net}}$, *the full viewing key of the spending account, derivation records,* ock, *which opens* C_{out}, *and address strings. Every recipient of a PCZT at a stage learns all data present at that stage; a delegated Prover learns the note data of the spends and outputs but gains no spending authority; the extracted transaction contains none of these data. Participants SHOULD send a PCZT only to entities needed for the remaining roles and SHOULD redact what those do not need, and wallets SHOULD protect stored PCZTs as they protect their note records* (*ZIP 374, "Privacy Implications"*).

Proof. The listed fields are context data (Definition 12.3), which extraction omits (§12.10); a recipient reads every field present. The Prover's data (Construction 12.19) contain fvk but no ask, and an accepted spend of a note of the key without a signature under $\mathsf{ask} + \alpha$ on a digest that the key holder signed is excluded, except with negligible probability, by *Ironwood Guide*, §"Authorisation of spends", Theorem "Spend authority", under its assumptions, whose adversary holds the full viewing key. □

12.10 Spend finalisation and extraction

Spend Finalizer (ZIP 374, "Spend Finalizer" and "TransparentInput"; BIP 11; BIP 16). For each transparent input with enough data, the Spend Finalizer builds script_sig: for P2PKH, the push of the one signature, with its hash-type byte, under the key that hashes to the address, followed by the push of that key; for P2SH multisignature, the opcode OP_0, the pushes of m signatures in the key order of the redeem script and the push of the redeem script, failing with fewer than m signatures. It MUST fail on a signature whose hash type is not the input's sighash_type. It SHOULD then clear the input's redeem script, partial signatures, derivation records and preimages. It keeps value and script_pubkey, required fields that the digest needs, and sequence and the lock-time requirements, which the digest and Definition 12.24 read at extraction; ZIP 374's "all other optional data" would include these last fields, a gap in the specification.

Definition 12.24 (Lock-time determination)**.** If no transparent input has a required time or height lock, the lock time is fallback_lock_time, or 0 if it is absent. Otherwise the lock type is one supported by every input that states a requirement, an input that states both types supporting both; an input that states neither supports both. If both types remain possible, the height type is chosen; if none remains, the requirements are unsatisfiable. The lock time is the maximum required value of the chosen type (ZIP 374, "Determining Lock Time").

Construction 12.25 (Transaction extraction)**.** The Transaction Extractor checks, failing on any violation (ZIP 374, "Transaction Extractor"), that

(i) tx_version, version_group_id and consensus_branch_id name a supported format, and the lock-time requirements are satisfiable (Definition 12.24);

(ii) for tx_version = 5 the Ironwood bundle is canonical empty;

(iii) every script_sig is set;

(iv) every Sapling spend has its proof and signature, and every Sapling output its proof;

(v) every Orchard-protocol bundle with $n \geq 1$ Actions has a zkproof of the canonical length of $2720 + 2272\,n$ bytes (*Ironwood Guide*, §"The transaction format"), and every Action a spend_auth_sig;

(vi) every non-empty shielded bundle has its anchor and bsk;

(vii) no rk is $\mathcal{O}$.

It then sets each bundle's balancing value to v^{balance}, failing if it lies outside the signed 64-bit range of the transaction field; sets the lock time by Definition 12.24; computes SigHash; for each bundle derives bvk from its value commitments and balancing value, fails unless it matches bsk, and signs SigHash under bsk, by RedPallas for the Orchard and Ironwood bundles and by RedJubjub for Sapling (Lemma 12.18); and outputs the transaction, without any context data (ZIP 229). It SHOULD then verify every proof and signature of that transaction, each spend_auth_sig under its rk on SigHash included, so that an invalid transaction is found before submission.

A party that holds an Orchard-protocol output's recipient, value and rseed, with ρ^{new} the nf of its Action's spend, recovers the full note plaintext, memo included, without ock or any viewing key, by the derivation of esk stated with Theorem 11.22 (§11.8; *Ironwood Guide*, §"Encryption to the recipient", Proposition "Decryption correctness"; ZIP 212). The random C_{enc} of a fabricated output decrypts to nothing, as the construction intends.

13 The transaction lifecycle

This section states what a wallet does with a transaction between its creation and a terminal state: the encoding it submits and the identifier by which it tracks the transaction; the rule that a transaction is recorded before it is released, with the completeness of the record that follows; submission and rebroadcast; the status observations that update the record; and the terminal states, re-sending and the consensus branch. A transaction is *created* when it is built, proved and authorised from one step of a plan (Definition 11.2; Constructions 11.18 and 11.20). The record is that of Definition 9.11.

13.1 Submission and the transaction identifier

The wallet serialises a created transaction in the encoding of its version: version 5 (ZIP 225, "Transaction Format") or version 6 (ZIP 229, "Transaction Format"), the only versions in force; a transaction with Ironwood-pool Actions has version 6 (*Ironwood Guide*, §"The transaction format", Construction "Version-6 transaction"; protocol specification, §"Transaction Encoding and Consensus"). The argument of Submit (Definition 5.10) is this byte string. The encoding is cited, not restated.

The wallet tracks a transaction t by its *transaction identifier* $\mathsf{txid}(t)$: the root of the BLAKE2b-256 digest tree of ZIP 244 over the effecting data of t,

$$\mathsf{txid}(t) = \mathsf{h}(\texttt{ZcashTxHash_} \,\|\, \beta,\ d_{\mathsf{hdr}} \,\|\, d_{\mathsf{tr}} \,\|\, d_{\mathsf{Sap}} \,\|\, d_{\mathsf{Orc}} \,\|\, d_{\mathsf{Irw}}),$$

where $\mathsf{h}(P, x)$ is BLAKE2b-256 of x under the 16-byte personalisation P, β is the 4-byte little-endian consensus branch identifier, and the child d_{Irw} is absent in version 5 (*Ironwood Guide*, §"Transaction digests and signatures", Definition "Transaction identifier", for version 6, and Remark "Version 5"; ZIP 244, "txid_digest"; ZIP 229, "Transaction Identifiers, Auth Digests, and

Signature Digests"; protocol specification, §"Transaction Identifiers"). The header digest d_{hdr} hashes, under `ZTxIdHeadersHash`, the 4-byte little-endian encodings of header, nVersionGroupId, nConsensusBranchId, lock_time and nExpiryHeight (ZIP 244, "T.1: header_digest"), so the expiry height N of Definition 9.8 is effecting data. The wallet computes $\mathsf{txid}(t)$ from t and needs no response of the server to know it.

Lemma 13.1 (Non-malleability of the identifier)**.** *Let t be a transaction of version* 5 *or* 6*.*

(*i*) *Replacing the authorising data of t, namely its proofs, signatures and transparent scripts and, in version* 6*, its anchors* (anchorSapling, anchorOrchard *and* anchorIronwood*, ZIP 229, "Anchor commitment (version 6)"*; Ironwood Guide*, §"Transaction digests and signatures", Definition "Authorising-data digest; effecting and authorising data"), leaves* $\mathsf{txid}(t)$ *unchanged.*

(*ii*) *Under the* Ironwood Guide*'s Assumption "Collision resistance of BLAKE2b"* (*§"Transaction digests and signatures"), no efficient algorithm outputs, except with negligible probability, two transactions with different effecting data and equal identifiers.*

(*iii*) *Under the same assumption, a transaction built for one consensus branch and the same payment built for another have different identifiers, except with negligible probability.*

Proof. (i) Authorising data lie under the authorising-data digest and under no node of the tree whose root is txid (*Ironwood Guide*, §"Transaction digests and signatures", Definition "Authorising-data digest; effecting and authorising data"). (ii) Each node of the tree hashes an injective encoding (fixed-size fields and length-prefixed scripts, concatenated in order) of effecting data or of the digests of its children (ZIP 244, "txid_digest"). If the effecting data differ, the inputs of some node differ. A node whose inputs differ either has equal digests, which is a collision, or passes different digests to its parent, whose inputs then differ; at the root, equal identifiers from different inputs are a collision. (iii) The branch identifier enters both the root personalisation `ZcashTxHash_` $\|\, \beta$ and the header digest through nConsensusBranchId, so the two root pairs differ, and equal identifiers are a collision (*Ironwood Guide*, §"Transaction digests and signatures", Remark "Consensus-branch binding"). □

By (i) and (ii), the identifier tracks a pending transaction from its creation on, before and after any re-signing or re-proving. The identifier of versions up to 4, the double SHA-256 of the encoding (protocol specification, §"Transaction Identifiers"), is malleable; it belongs to no version in force and is not used.

Construction 13.2 (Recording and submission)**.** The following rules are designed but unspecified (Definition 1.1).

(L1) *Recording before release.* The wallet enters each transaction t it creates into its transaction record (Definition 9.11), which locks the notes t spends (Definition 9.12), in storage that survives restarts and rollbacks (Definition 7.23, item (v)), before it submits t or releases its bytes to any other party. It never removes an entry of the record.

(L2) *Multi-step plans.* For a plan of $\ell > 1$ steps (Definitions 11.2 and 11.3), the wallet creates every step before it records any, records all ℓ transactions in one update of the record, and submits them after that update, in step order. A construction that fails at some step leaves no recorded, hence by (L1) no transmittable, prefix. Every step can be created before any is mined, since a well-formed plan chains steps only through transparent outputs (Proposition 11.4); the transfer to a TEX address of Construction 11.7 is the instance (ZIP 320, "Design considerations for Senders").

(L3) *Submission.* The wallet submits t by Submit(*bytes*), whose response is accepted(txid) or rejected (Definition 5.10). Neither response changes the record. It submits at a tip T that leaves a margin of blocks, a wallet parameter, before the last includable height N (Definition 9.8), rather than at $T = N - 1$.

(L4) *Rebroadcast.* While t is pending and unmined (Definition 13.8), the wallet resubmits it at intervals whenever, at its tip T, $N = 0$ or $T < N$ (Lemma 9.9) and no transaction included in its view of the chain spends an input of t. For $N = 0$ this continues until t is mined or an input of t is spent by a mined transaction. The rule is motivated by ZIP 203, "Specification", under which nodes drop a transaction from their mempools at its expiry.

The semantics of the two responses of Submit are those of the server's node and are fixed by no ZIP: accepted means that the node admitted t to its mempool, rejected that it refused it. Under clause (a) of Assumption 5.2, acceptance implies neither propagation to other nodes nor inclusion in a block, since relay policy is local to each node and may refuse a transaction that consensus still admits in the next block (Lemma 9.9); and rejection does not imply that t is invalid. Rule (L3) keeps a margin for this reason. Each submission is an observation of the server (Proposition 8.21).

Proposition 13.3 (Completeness of the transaction record)**.** *Let the wallet follow rule (L1), and let no party other than the wallet hold the bytes of a created transaction t before the wallet submits them. Then, if a block of any chain includes a transaction with the effecting data of t, the record contains t, and the included transaction has identifier* txid(t)*, except with negligible probability. The converse fails: a recorded transaction that the network never received is stale, not lost, and, for $N \neq 0$, reaches the terminal state expired unmined of Definition 13.8; for $N = 0$ it stays pending by item (c) until the end of its consensus branch is known, and then reaches expired unmined with the bound N^* ("Consensus-branch bound" below).*

Proof. A block includes a transaction only if the party that assembled the block held its bytes. The first party other than the wallet to hold bytes of t received them by the wallet's submission, by hypothesis; by (L1) the wallet recorded t before that event, and the entry survives restarts and rollbacks and is never removed. A valid transaction with the effecting data of t carries, for each spend of t other than a dummy, a signature under a key of the wallet on the signature digest those data fix; except with negligible probability no party obtains such a signature except from bytes the wallet released (*Ironwood Guide*, §"RedPallas unforgeability", Proposition "Unforgeability under re-randomisation", for Orchard-protocol spends; for transparent and Sapling spends the unforgeability of those schemes is assumed, as in §11.8, where no lower volume proves it). Bytes derived from those of t by another party with the effecting data of t differ from t at most in authorising data, so the included transaction has identifier txid(t) by part (i) of the lemma above. The converse fails because recording does not imply delivery. □

Where a partially created transaction is extracted by another party ("Spend finalisation and extraction", §12.10), that party holds the bytes of t before any submission, and the hypothesis is arranged differently: the wallet records txid(t), N and the inputs of t, locking them, before it releases the partially created transaction to any party that may extract it, and at a point where the effecting data of t are final. After IO finalisation (Construction 12.14) of a partially created transaction with shielded spends or outputs, or, for one that is transparent-only, once signatures without SIGHASH_ANYONECANPAY and without SIGHASH_NONE have cleared bits 0 and 1 of tx_modifiable (ZIP 374, "Global"), the inputs and outputs are final, the lock time is determined by Definition 12.24, and the proofs and signatures still to be added are authorising data; by part (i) of the lemma the extracted transaction then has the recorded identifier, and the proof above applies

with release in place of submission. The entry acquires the encoding of t when the wallet obtains it.

Remark 13.4 (Retained data). From creation until it is mined, t exists only in the wallet's record, in remote mempools and, for an extracted partially created transaction, at the extracting party. Once the wallet holds the encoding of t (Definition 9.11, item (i)) the record keeps it, and rule (L4) needs it; before that the wallet does not rebroadcast t. It may hold data that the chain never will: a transaction that ends expired and unmined lies on no chain that agrees with the server's up to N (Lemma 13.7), and its entry may be the only trace of it.

13.2 Status tracking and terminal states

Remark 13.5 (Scope of scanning). Scanning detects a transaction only through a nullifier of the watched set W (Definition 9.15) or an output accepted under the wallet's scanning keys (Construction 6.8). A recorded transaction that spends no note of W and has no such output, every fully transparent transaction among them, is invisible to scanning. Its fate is learnt only by queries $\mathsf{Status}(\mathsf{txid})$ (Definition 5.9), each of which puts its identifier into the view of the server (Proposition 8.18).

For a recorded transaction t write N for its expiry height, h_t for its mined height ($\perp$ while unmined), h_t^{obs} for its earliest observation height, and u_t for the latest tip at which the server reported it unmined ($\perp$ initially): items (ii), (iv), (v) and (vi) of Definition 9.11. A *status observation* (s, T) is a response s of $\mathsf{Status}(\mathsf{txid}(t))$ received while the wallet's view of the chain has tip T.

Construction 13.6 (Status transitions). The record entry of t changes only as follows.

(i) On an observation (s, T) with $s \in \{\mathsf{unknown}, \mathsf{unmined}\}$: $u_t := T$.

(ii) On an observation $(\mathsf{mined}(h'), T)$: $h_t := h'$, $h_t^{\mathrm{obs}} := \min(h_t^{\mathrm{obs}}, h')$ and $u_t := \perp$.

(iii) Scanning a block of height h' in which t is wallet-relevant (Construction 6.8) has the effect of (ii).

(iv) A rollback to r (Definition 7.23) sets $h_t := \perp$ if $h_t > r$, and $u_t := \min(u_t, r)$ if $u_t \neq \perp$, since a report at a tip above r concerned blocks that the rollback discards.

Query termination. The wallet issues $\mathsf{Status}(\mathsf{txid}(t))$ for a pending transaction t with $h_t = \perp$: if $N \neq 0$, until $u_t \geq N$, that is, until t is reported unmined at a tip of at least N; if $N = 0$, for as long as t is pending and unmined. No rule stops at a first report of $\mathsf{unmined}$ or at a fixed depth after first observation. A rollback that lowers u_t below N, or sets $h_t := \perp$, resumes the queries. The transitions and the termination rule are designed but unspecified (Definition 1.1); the termination rule rests on the consensus rule of Definition 9.8 (ZIP 203, "Specification"). For an entry with N_t unknown (Definition 9.11), the wallet obtains the encoding by $\mathsf{Tx}(\mathsf{txid})$ (Definition 5.10); until then it queries as for $N = 0$, does not rebroadcast, and its inputs are locked by condition (d) of Definition 9.12.

Lemma 13.7 (Expired unmined is final on its chain). *Assume clause (a) of Assumption 5.2. Let t be a transaction created by the wallet, with expiry height $N \neq 0$, and let (s, T) be an observation with $s \in \{\mathsf{unknown}, \mathsf{unmined}\}$ and $T \geq N$, answered from a best chain C of the server* (Consensus Guide, *§"Work, and the best chain") whose blocks at heights at most N are those of the wallet's view. Then no chain whose blocks at heights at most N are those of C includes t.*

Proof. The wallet's view has tip $T \geq N$, so C has blocks at every height up to N. By clause (a), s is the status of $\mathsf{txid}(t)$ on C (Definition 5.9): no block of C contains t, so no block of height at most N of a chain that agrees with C up to N does. The transaction t is not a coinbase transaction, and by Lemma 9.9 the expiry rule permits it in no block of height greater than N (protocol specification, §"Transaction Consensus Rules"; ZIP 203, "Specification"). □

A reorganisation with fork point below N reopens the question, since the new chain need not agree with C up to N. The wallet meets it as a continuity error (Definition 6.10), and the rollback of Construction 7.24 to some $r < N$ sets $u_t \leq r < N$ by transition (iv), which resumes the queries.

Let B be the birthday of the wallet (Definition 6.2) and $\mathcal{H}$ its retained block boundaries (Definition 7.12). Write $r_{\min}$ for the least element of $\mathcal{H} \setminus \{B-1\}$, the lowest rollback point of the rollback window, and $r_{\min} := B - 1$ when that set is empty.

Definition 13.8 (Terminal states)**.** A recorded transaction t is *pending* from its recording until it reaches one of the following states.

(a) *Mined*: $h_t \neq \bot$ and $h_t \leq r_{\min}$, so that no rollback to a point of the window unmines t. The state is terminal relative to the rollback window only, since consensus bounds no reorganisation depth (Remark 7.28); a deeper rollback, to $B-1$ or to a fetched chain state in steps (1) and (3) of Construction 7.24, returns t to pending by transition (iv).

(b) *Expired unmined*: $N \neq 0$, $h_t = \bot$ and $u_t \geq N$, which gives the conclusion of Lemma 13.7 when the server's chain agrees with the wallet's view up to N; a disagreement surfaces as a continuity error (Definition 6.10), whose rollback to $r < N$ returns t to pending by transition (iv). Its inputs are released (Proposition 9.13, every later target height exceeding N), its entry is kept, and the user is notified.

(c) For $N = 0$, a value chosen explicitly: t remains pending until it is mined, rebroadcast by rule (L4), or, once the end of its consensus branch is known, until it reaches state (b) with the bound N^* of that branch ("Consensus-branch bound" below). Its mining is detected by scanning through W, from which a spend, $N = 0$ included, removes a nullifier only once mined (Definition 9.15), or by continued status queries.

The states and their transitions are designed but unspecified (Definition 1.1). The notification is specified: the wallet should notify the user of expired transactions that must be re-sent, and user interfaces and services must never rely on zero-confirmation transactions (ZIP 203, "Wallet behavior and UI").

Remark 13.9 (Re-sending). Re-sending an expired payment usually spends the same TXOs, and ZIP 315 lists which spends of given TXOs are repeated across transactions among the kinds of information leakage, the case arising when a previous transaction expired (ZIP 315, "Kinds of information leakage"): the repeated spends link the expired transaction to its replacement. Selecting other notes avoids that link, but the nullifiers that the expired transaction revealed still link it to any later transaction that spends those notes (ZIP 315, "Rationale for anchor selection"). A wallet that re-sends accepts one of the two links. ZIP 315, "Note selection", records as a TODO, not as a rule, that such revealed nullifiers be spent in a note-management transaction; how to avoid both links is an open problem (Definition 1.1).

Consensus-branch bound. The consensus branch bounds a pending transaction as well as its expiry height does. An authorisation binds the consensus branch identifier through β and $\mathsf{nConsensusBranchId}$, which must equal the identifier of the branch under which t is validated, so a transaction authorised for one branch is invalid under another whatever its expiry height (*Ironwood Guide*, §"Transaction digests and signatures", Remark "Consensus-branch binding", under its Assumption "Collision resistance of BLAKE2b"; ZIP 244, "txid_digest" and "T.1:

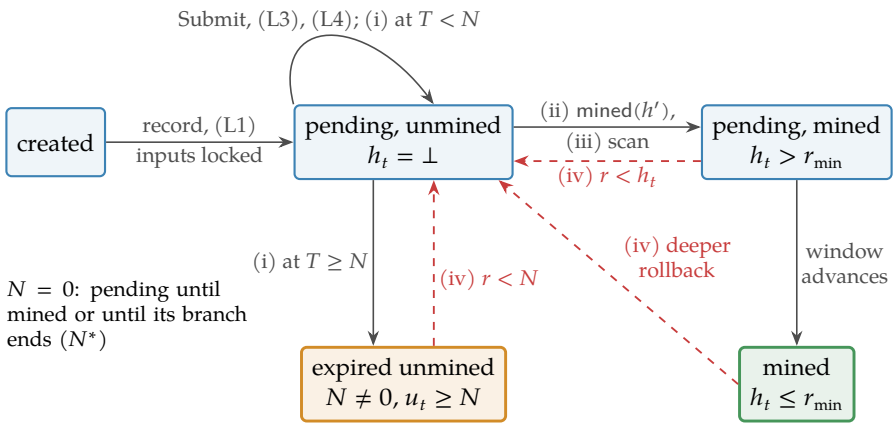


Figure 6: The states of a recorded transaction t (Definition 13.8). Solid edges are transitions of the record: recording by rule (L1), which locks the inputs of t; submission, rebroadcast and reports unmined or unknown at a tip below N, which leave t pending; a report mined(h') or the scan of a block containing t, transitions (ii) and (iii); burial below the lowest rollback point $r_{\min}$ of the window; and, for $N \neq 0$, a report of an unmined t at a tip $T \geq N$, after which Lemma 13.7 excludes t from every chain that agrees with the server's up to N. Dashed red edges are rollbacks, transition (iv): below h_t from either mined state, from the terminal one only by a rollback deeper than the window, and below N from expired unmined. The green and amber states are terminal; a transaction with $N = 0$ stays pending until mined, unless its consensus branch ends.

header_digest"; ZIP 259, "Backward compatibility"; protocol specification, §"Transaction Consensus Rules"). A pending transaction built for a branch that ends at a height X, with $N = 0$ or $X \leq N$, can be mined only below X. Once X is known, write $N^* := X - 1$ if $N = 0$ or $N \geq X$, and $N^* := N$ otherwise. Rule (L4), the query termination of Construction "Status transitions", state (b) of Definition 13.8 and conditions (b) and (c) of Definition 9.12 apply with N^* in place of N, and Lemma 13.7 holds with N^*, by the branch binding in place of the expiry rule. Thus t becomes expired unmined from a report at a tip of at least N^*, and its inputs are released for the rebuild at every target height $H \geq X$. This rule is designed but unspecified (Definition 1.1). To pay across the activation of the next branch, the wallet rebuilds and re-authorises the payment for the branch in force at its target height (Definition 9.4): a new transaction, with a new identifier by part (iii) of the lemma of §13.1 and a new entry of the record.

14 Payment requests (ZIP 321)

This section states the encoding by which a payee states payments to a payer. It gives the URI grammar of ZIP 321, the grouping of parameters into payments and the conditions under which a parser accepts a URI; then the meaning of the address, amount, memo, label and message parameters, the capacity of one request, and the passage from an accepted request to a transaction plan.

14.1 The URI grammar

A payment request precedes planning (§11.1): the payee states to the payer, in a form that a wallet reads mechanically, the recipient addresses, amounts and memos of the payments it asks for, and ZIP 321 fixes this encoding as a URI scheme. Its terms *payment* and *payment request* (ZIP 321, "Terminology") are those of Definition 11.1. Everything that follows once the recipients are fixed is constructed in the earlier sections of this volume and in the *Ironwood Guide*, §"Note encryption", §"Trial decryption and note acceptance", §"Authentication paths", §"Anchors" and

§"Transaction digests and signatures"; this section re-derives none of it. The status of ZIP 321 is that of Table 1.

Definition 14.1 (Payment request URI)**.** Let L be the set of strings derived from `zcashurn` in the following grammar in ABNF (RFC 5234), reproduced from ZIP 321, "URI Syntax", with its longest rule continued on an indented line:

```
zcashurn       = "zcash:" ( zcashaddress [ "?" zcashparams ]
                 / "?" zcashparams )
zcashaddress   = 1*( ALPHA / DIGIT )
zcashparams    = zcashparam [ "&" zcashparams ]
zcashparam     = [ addrparam / amountparam / assetparam / memoparam
                 / messageparam / labelparam / otherreqparam / otherparam ]
NONZERO        = %x31-39
DIGIT          = %x30-39
paramindex     = "." NONZERO 0*3DIGIT
addrparam      = "address"   [ paramindex ] "=" zcashaddress
amountparam    = "amount"    [ paramindex ] "=" 1*DIGIT [ "." 1*8DIGIT ]
assetparam     = "req-asset" [ paramindex ] "=" *base64url
labelparam     = "label"     [ paramindex ] "=" *qchar
memoparam      = "memo"      [ paramindex ] "=" *base64url
messageparam   = "message"   [ paramindex ] "=" *qchar
paramname      = ALPHA *( ALPHA / DIGIT / "+" / "-" )
otherreqparam  = "req-" paramname [ paramindex ] [ "=" *qchar ]
otherparam     = paramname [ paramindex ] [ "=" *qchar ]
qchar          = unreserved / pct-encoded / allowed-delims / ":" / "@"
allowed-delims = "!" / "$" / "'" / "(" / ")" / "*" / "+" / "," / ";"
```

Here `ALPHA`, `unreserved` and `pct-encoded` are those of RFC 3986, Appendix A, and `base64url` is the alphabet of RFC 4648, Section 5, in which the values 62 and 63 are written `-` and `_`, with padding omitted. The grammar is read as follows: a parameter whose name is `address`, `amount`, `label`, `memo`, `message` or `req-asset` is derived only by its own production, and `otherreqparam` and `otherparam` derive only parameters with other names. A *payment request URI* is a string of L that meets the acceptance conditions (a) to (f) stated below in this subsection.

Without the reading, `otherparam` would derive `amount=1%30`, with name `amount`, no index and a `qchar` value; ZIP 321, "Invalid Examples", rejects a URI with this parameter.

Surface forms. The grammar derives no `//` after the scheme, so a payment request URI has no authority component and is not hierarchical in the sense of RFC 3986, Section 1.2.3; ZIP 321, "Invalid Examples", rejects `zcash://` followed by an address. A URI `zcash:`A`?`P, with A a `zcashaddress`, MUST be treated as `zcash:?address=`A`&`P (ZIP 321, "URI Semantics"). A request for a single payment SHOULD place its address in the hier-part; several payments are written with `addrparam` at distinct indices (same section). By this equivalence and condition (b) below, a URI `zcash:`A`?address=`B`&` ... is not accepted: after the rewriting it carries `address` twice at the empty index.

Indices and grouping. Identify the empty index with 0 and let $J := \{0, 1, \ldots, 9999\}$. The *rendering* ι maps 0 to the empty string and $i \geq 1$ to `.` $\|$ $\mathrm{dec}(i)$, where $\mathrm{dec}(i)$ is the decimal representation of i without leading zeros. The strings that match `[ paramindex ]` are the empty string and the strings `.` $d_1 \cdots d_k$ with $1 \leq k \leq 4$, $d_1 \in \{1, \ldots, 9\}$ and $d_2, \ldots, d_k$ decimal digits; these are exactly the decimal representations without leading zeros of the integers 1 to 9999. Hence ι is a bijection from J onto the index suffixes, and its inverse is parsing. The strings `.0`, `.01` and `.10000` match no production, and ZIP 321, "Invalid Examples", rejects `address.0=` and `amount.0=`. Parameters whose suffixes denote the same index $i \in J$, the empty index included, belong to one payment P_i; the order of the parameters is insignificant, and indices need not be consecutive (ZIP 321, "URI

Semantics"). A URI therefore decodes to a finite map $i \mapsto P_i$ on a subset of J, and carries at most $|J| = 10{,}000$ payments: 9999 at indices 1 to 9999 and one at the empty index.

Lexical rules. Percent encoding (RFC 3986, Section 2.1) occurs only inside `qchar`, that is, only in the values of `label`, `message`, `otherreqparam` and `otherparam`. Addresses, amounts, indices and parameter names are literal, so `amount=1%30`, `%61mount=1` and an address beginning with `%74` are not in L (ZIP 321, "URI Syntax" and "Invalid Examples"). The values of `memo` and `req-asset` are `base64url` without padding, and implementations MUST NOT accept the characters `+`, `/` and `=`, which occur only in the standard base64 alphabet (ZIP 321, "URI Syntax"; RFC 4648, Section 5).

Acceptance. A parser accepts a string u as a payment request URI only if all of the following hold (ZIP 321, "URI Syntax", "URI Semantics" and "Forward compatibility"):

(a) $u \in L$ under the reading of Definition 14.1; a purported ZIP 321 URI that cannot be so parsed MUST NOT be accepted;

(b) after the hier-part address is rewritten as `address` at the empty index, no pair of a parameter name and an index occurs twice (MUST NOT);

(c) every index that carries a parameter other than `address` carries an `address` (MUST);

(d) u names no parameter with prefix `req-` that the parser does not recognise, since such a parameter makes the entire URI invalid (MUST); other unrecognised parameters are ignored (SHOULD);

(e) the address, amount and memo rules of §14.2 hold;

(f) u decodes to at least one payment, a payment request containing one or more payments (ZIP 321, "Terminology").

The prefix `req-` is part of the name of a parameter defined by a later revision of ZIP 321, not a modifier of an arbitrary parameter; the parameters `address`, `amount`, `label`, `memo` and `message` never carry it, since every conformant parser is required to understand them (ZIP 321, "Forward compatibility"). An accepted URI asks for one transaction: implementations SHOULD construct a single transaction that pays every address of the request (ZIP 321, "URI Semantics").

Custom assets. The parameter `req-asset` is specified in ZIP 321, "Custom Assets", with an encoding defined by ZIP 227, which the protocol described in this volume does not include. A parser of this protocol does not recognise `req-asset`, so by condition (d) a URI that carries it is invalid to that parser, and the further rule that `amount` and `req-asset` exclude each other at one index (ZIP 321, "URI Semantics") never applies.

Remark 14.2 (Case of literals). Quoted strings in ABNF are case-insensitive (RFC 5234, Section 2.3), so L contains strings beginning `ZCASH:` and strings with the parameters `AMOUNT=1` and `REQ-ASSET=`; the value ranges `NONZERO` and `DIGIT`, given as `%x` ranges, and the `base64url` alphabet are case-sensitive. ZIP 321 does not address case: it states neither whether such URIs are to be accepted nor whether `amount` and `AMOUNT` at one index are the same parameter for condition (b). The volume states the grammar as written and resolves neither question; no parser's behaviour is normative. Class: open problem (Definition 1.1).

14.2 Parameter semantics

Addresses. Implementations SHOULD check that each `zcashaddress` is a valid encoding of an address other than a Sprout address under protocol specification, §"Encodings of Addresses and Keys": a transparent address (§"Transparent Addresses", Base58Check), a Sapling payment address (§"Sapling Payment Addresses", Bech32, ZIP 173) or a unified address (§"Unified Payment Addresses and Viewing Keys" and ZIP 316, Bech32m, BIP 350). Address formats later added to that section SHOULD be supported; where the network of the request is known, each address

SHOULD be checked to belong to it; and every requirement of ZIP 316 applies to a unified address (ZIP 321, "URI Semantics"). Sprout addresses MUST NOT be supported (same section). ZIP 321 gives as its reason the restriction of ZIP 211 on adding value to the Sprout pool; in the protocol described, version-4 transactions are invalid, and Sprout transfers exist only in transactions of versions 2 to 4, so Sprout funds are unspendable (ZIP 2003, "Abstract"; ZIP 259, "Abstract"; *Ironwood Guide*, §"The transaction format"). A TEX address (ZIP 320) is in no section of the specification, and ZIP 321 names none; its reading as the recipient address of item 4 of Definition 3.20 is designed but unspecified (Definition 1.1). The addresses of an accepted request are thus recipient addresses in the sense of Definition 3.20.

Construction 14.3 (Amount codec)**.** Let $\mathsf{COIN} = 10^8$ zatoshi per ZEC (*Ironwood Guide*, Definition "Unit of value" in §"Notes") and $\mathsf{MAX_MONEY} = 2.1 \cdot 10^{15}$ zatoshi (*Ironwood Guide*, §"The transaction format"), so that 21,000,000 ZEC is MAX_MONEY zatoshi. For a string x of decimal digits, let $\mathrm{int}(x)$ be the natural number it denotes, leading zeros ignored, and let 0^k be the string of k characters `0`.

(1) *Parse.* Let s match `1*DIGIT [ "." 1*8DIGIT ]`, and write $s = c$ or $s = c \,\|\, . \,\|\, f$ with c its whole part and f its fraction. Set $z := \mathrm{int}(f \,\|\, 0^{8-|f|})$, or $z := 0$ without a fraction, and

$$v := \mathrm{int}(c) \cdot \mathsf{COIN} + z.$$

Accept s if and only if $v \leq \mathsf{MAX_MONEY}$; an accepted s denotes v zatoshi, and $\mathrm{parse}(s) := v$.

(2) *Render.* For $v \in \{0, \ldots, \mathsf{MAX_MONEY}\}$ write $v = c \cdot \mathsf{COIN} + z$ with $0 \leq z < \mathsf{COIN}$. Let $\mathrm{render}(v)$ be $\mathrm{dec}(c)$ if $z = 0$, and otherwise $\mathrm{dec}(c) \,\|\, . \,\|\, f$, where f is the 8-digit zero-padded decimal representation of z with its trailing characters `0` removed.

Leading zeros of the whole part and trailing zeros of the fraction are ignored, the bound MAX_MONEY is ZIP 321's "MUST NOT be greater than 21000000 ZEC", and a string with a comma or other grouping separator, with an empty whole part (`.5`), an empty fraction (`50.`) or more than 8 fraction digits matches no production (ZIP 321, "ZEC Transfer Amount"). ZIP 321 fixes only the value that an amount denotes; the rendering is canonical, not required.

Lemma 14.4 (Round trip of amounts)**.** *For every* $v \in \{0, \ldots, \mathsf{MAX_MONEY}\}$*, the string* $\mathrm{render}(v)$ *matches* `1*DIGIT [ "." 1*8DIGIT ]` *and* $\mathrm{parse}(\mathrm{render}(v)) = v$*. The map* parse *is not injective. For every accepted* s*,* $\mathrm{render}(\mathrm{parse}(s))$ *is the unique string of value* $\mathrm{parse}(s)$ *whose whole part has no leading zero unless it is the single character* `0`*, whose fraction has no trailing zero, and which has no fraction when* $\mathrm{parse}(s)$ *is a multiple of* COIN.

Proof. Write $v = c \cdot \mathsf{COIN} + z$ with $0 \leq z < \mathsf{COIN}$. The string $\mathrm{dec}(c)$ is non-empty. If $z = 0$, then $\mathrm{render}(v) = \mathrm{dec}(c)$ matches the production, and parse returns $c \cdot \mathsf{COIN} = v$. If $z \neq 0$, the 8-digit form of z has a non-zero digit, so removing its trailing zeros leaves a fraction f with $1 \leq |f| \leq 8$, and the production matches. Appending $8 - |f|$ zeros to f restores the 8-digit form of z, so parse returns $c \cdot \mathsf{COIN} + z = v$, and accepts since $v \leq \mathsf{MAX_MONEY}$. The strings `50`, `050` and `50.00` all denote 5,000,000,000 zatoshi, so parse is not injective. Finally, an accepted s determines $(\mathrm{int}(c), z)$, which are the quotient and remainder of $\mathrm{parse}(s)$ by COIN since $z < \mathsf{COIN}$. A string of the stated form is fixed by these two numbers: its whole part is $\mathrm{dec}(\mathrm{int}(c))$, and its fraction is absent if $z = 0$ and otherwise the 8-digit form of z without trailing zeros. That string is $\mathrm{render}(\mathrm{parse}(s))$. □

Examples. On the examples of ZIP 321, "ZEC Transfer Amount" and "Examples", Construction 14.3 gives the following. The parameter `amount=1` denotes 100,000,000 zatoshi,

`amount=123.456` denotes 12,345,600,000 zatoshi, and `amount.1=0.789` denotes 78,900,000 zatoshi for the payment at index 1. The values `50`, `050` and `50.00` denote 5,000,000,000 zatoshi, and `0.5` and `00.500` denote 50,000,000 zatoshi. The value `21000000` denotes MAX_MONEY = 2,100,000,000,000,000 zatoshi and is accepted, while `21000000.00000001` exceeds MAX_MONEY and is rejected. The values `50,000.00`, `50,00`, `50.`, `.5` and `0.123456789` are invalid. The parameters `address.0=` and `amount.0=` are invalid, their index having a leading zero, and so is any parameter with the five-digit suffix `.10000`.

Absent amount. ZIP 321 gives a payment a value only "if an `amount` parameter is provided" (ZIP 321, "ZEC Transfer Amount"). A payment without one has no value under ZIP 321, while Definition 11.1 requires a value, so the payer fixes the value before a plan is formed. Class: designed but unspecified (Definition 1.1).

Construction 14.5 (Memo codec)**.** A *memo* is a 512-byte string M, the memo field of a note plaintext (*Ironwood Guide*, §"The note plaintext"; protocol specification, §"Encodings of Note Plaintexts and Memo Fields"). Let b64u be the encoding of RFC 4648, Section 5, with padding omitted, and let $\mathrm{strip}(M)$ be M with its trailing `0x00` bytes removed.

(1) *Encode.* $\mathrm{Enc}(M) := \mathrm{b64u}(\mathrm{strip}(M))$, a string of at most $\lceil 512 \cdot 8/6 \rceil = 683$ characters.

(2) *Decode.* Reject a value that contains a character outside the `base64url` alphabet, in particular `+`, `/` or `=`, or whose length is congruent to 1 modulo 4, a length to which no octet string encodes. Otherwise let b be its decoding under RFC 4648, Section 5, and require $|b| \leq 512$ (MUST); then $\mathrm{Dec}(\mathit{value}) := b \,\|\, \texttt{0x00}^{512-|b|}$ (ZIP 321, "Query Keys").

A payment without a `memo` parameter has no memo (Definition 11.1); its output, if shielded, carries the memo field that ZIP 302 designates as supplying no memo, $\texttt{0xF6} \,\|\, \texttt{0x00}^{511}$, whose encoding is `9g` (ZIP 302, "Specification"). A parameter `memo=` with the empty value differs from an absent parameter: it decodes to $\texttt{0x00}^{512}$, which a reader under ZIP 302 reads as the empty text. If the address at the index of a `memo` parameter does not permit memos, that is, is not memo-capable in the sense of predicate (i) of Definition 3.20, as for a transparent or TEX address, parsers MUST consider the entire URI invalid (ZIP 321, "Query Keys"), since a memo travels only in a note plaintext.

Lemma 14.6 (Round trip of memos)**.** *For every memo M,* $\mathrm{Dec}(\mathrm{Enc}(M)) = M$.

Proof. The string $\mathrm{Enc}(M)$ uses only the `base64url` alphabet, and an unpadded encoding of n octets has length $\lceil 8n/6 \rceil$, never congruent to 1 modulo 4, so decoding does not reject it. The encoding b64u is injective with inverse the decoding of RFC 4648, so $b = \mathrm{strip}(M)$, and $|b| \leq 512$. The bytes that strip removes are zeros, so appending $512 - |b|$ zeros to b returns M. □

ZIP 321 does not say whether a value whose final character carries non-zero unused bits (RFC 4648, Section 3.5) is accepted; the encoding Enc produces none. Stripping trailing zeros gives the shortest encoding and is not required: $\mathrm{b64u}(M)$ decodes to the same memo. As an instance, ZIP 321's example value `VGhpcyBpcyBhIHNpbXBsZSBtZW1vLg`, of 30 characters, decodes to the 22 bytes of the text "This is a simple memo.", padded with zeros to 512 bytes.

Labels and messages. The `label` at an index names its address, for example by the name of the recipient; a client that renders the payment for inspection SHOULD display the label together with the address, and a displayed label MUST be identifiable as distinct from the address. The `message` at an index is descriptive text that a client may display for that payment (ZIP 321, "Query Keys"). Both values are `qchar` strings, percent-decoded to octet strings; an octet outside `qchar` reaches them only by percent encoding. Neither enters the transaction (Definition 11.1).

Capacity. A request carries at most 10,000 payments (§14.1), and the one transaction it asks for is bounded further by consensus; the bound that binds is the least of the following.

(i) *Size.* A block has at most 2,000,000 bytes (protocol specification, §"Block Header Encoding and Consensus"), and a Sapling output contributes at least 948 bytes to a transaction, so a transaction has at most $\lfloor 2{,}000{,}000/948 \rfloor = 2109$ Sapling outputs (protocol specification, §"Balance and Binding Signature (Sapling)"). This is the bound that ZIP 321, "URI Semantics", states.

(ii) *Shielded budget.* ZIP 218, part of NU7, requires of every block, with S the number of Sapling spends plus outputs, A the number of Orchard-pool Actions and n_{JS} the number of JoinSplit descriptions, each summed over the block's transactions,

$$S \leq 300, \qquad A \leq 330, \qquad A + S + 2\,n_{\mathsf{JS}} \leq 330$$

(ZIP 218, "Shielded pool action limits"). Versions 5 and 6 carry no JoinSplit description (*Ironwood Guide*, §"The transaction format"), so $n_{\mathsf{JS}} = 0$. Each of these sums over a block is at least the term of any one of its transactions. A transaction that pays n payments by Sapling outputs has $o \geq n$ Sapling outputs; with s Sapling spends and a Orchard-pool Actions it is mined only if $s + o \leq 300$ and $a + s + o \leq 330$, hence only if

$$n \leq o \leq \min(300,\, 330 - a) - s.$$

The all-Sapling bound is therefore $\min(2109, 300) = 300$ outputs, fewer with spends or Orchard-pool Actions, and the bound of (i) never binds; one transaction carries at most 330 units of the shared budget.

(iii) *Orchard pool.* No payment output lies in the Orchard pool (Construction 11.5, Proposition 11.6), so the Orchard-pool Actions of the transaction only consume the shared budget of (ii).

(iv) *Ironwood pool and transparent outputs.* ZIP 218 as written counts neither Ironwood-pool Actions nor transparent components, so payments by Ironwood-pool or transparent outputs are bounded by the size limit alone. Whether Ironwood-pool Actions enter the shared budget is an open proposal (*Consensus Guide*, §"A block budget for shielded work"). Class of (iv): open problem (Definition 1.1).

From request to plan. Each payment $P_i = (a, v, m)$ of an accepted request, its value fixed by the payer if absent, is a payment in the sense of Definition 11.1 and becomes one output of the plan: its pool is that of Construction 11.5, a TEX address is paid by Construction 11.7, and its memo, if any, is the memo field of that output's note plaintext (Construction 14.5). A TEX payment funded from shielded notes needs the two transactions of Construction 11.7, so the MUST of ZIP 320, "Design considerations for Senders", and not the SHOULD of ZIP 321 that asks for a single transaction, fixes the shape of such a plan. From there construction is that of §11.1 and Constructions 11.18 and 11.20: an Orchard-protocol output is sealed as in the *Ironwood Guide*, §"Encryption to the recipient", a Sapling output as in the protocol specification, §"Sending Notes (Sapling)", and the *Ironwood Guide*, §"Transaction digests and signatures", gives the digest that every signature signs.

Part VII
FlyClient Guide: Merkle mountain ranges, the chain-history commitment, and the FlyClient sampling protocol

Abstract

This volume of *The Zcash Arboretum* states the protocol by which a light client decides which of several claimed chains carries the most work while receiving a number of block headers logarithmic in the chain length. It constructs Merkle mountain ranges under the two bagging orders in use, with inclusion paths, prefix roots, position binding and aggregate binding; fixes the security model of the FlyClient paper of Bünz, Kiffer, Luu and Zamani; constructs the FlyClient protocol from the header chain commitment and the per-sample check through the optimal sampling distribution to the non-interactive protocol and its security theorem, with every hypothesis named; and states the chain-history commitment of ZIPs 221, 244 and 258, to which every Zcash block header after the Heartwood activation block commits. Each component of a Zcash FlyClient is classified as specified, designed but unspecified, or an open problem. The volume describes the protocol as specified for NU7, whose deployment ZIP, ZIP 259, has status Draft. It assumes the *Math Guide*, the *Crypto Guide* and the *Consensus Guide*, and cites the *Ironwood Guide* for network upgrades, ZIP statuses and transaction-level objects. It is non-normative: the Zcash Protocol Specification and the Zcash Improvement Proposals are authoritative.

1 Introduction

This section fixes the subject of the volume, its status, its sources and its notation before any construction: the scope; the named network upgrades and ZIP statuses on which later rules depend; the three classes in which the deployment boundary places each component of a Zcash FlyClient; the primary paper and the policy for its printed defects; the normative sources; and one table of conventions closing with a symbol index.

1.1 Scope and status

This volume of *The Zcash Arboretum* states three objects and the relation between them. The first is the FlyClient protocol of Bünz, Kiffer, Luu and Zamani (ePrint 2019/226). The second is the chain-history commitment of ZIP 221 (Final), to which every block header after the Heartwood activation block (Definition 1.1) commits: directly, as the header field hashLightClientRoot, in blocks validated under the Heartwood or Canopy rules, and, from NU5, as the first input of the hash hashBlockCommitments of ZIP 244 (Final). The header fields are named in "The header consensus rules" (§6.7), and their values are defined in "The epoch tree and its root commitment" (§6.4) and "The block commitment hash" (§6.6). The third is the boundary between the protocol and the commitment: each component of a Zcash FlyClient is classified as specified, designed but unspecified, or an open problem (Definition 1.3). The volume is non-normative: the Zcash Protocol Specification, cited as the protocol specification with the title of a section, and the Zcash Improvement Proposals are authoritative.

The volume assumes the following, each cited by italic volume name and section title. From the *Math Guide*: probability, densities and asymptotics (§"Probability, asymptotics, and computation", with §"Beyond finite spaces: countable additivity, limits, and densities" and §"Asymptotic notation"). From the *Crypto Guide*: hash functions and their security notions (§"Hash functions and the random oracle model", with §"Security notions: preimage, second-preimage, and collision resistance" and §"The random oracle model"); the provable-security

model (§"The provable-security model", with §"Adversaries and the security parameter", §"Security as a game" and §"Reductions"); interactive arguments (§"Interactive protocols, proofs, and arguments"); the Fiat–Shamir transform (§"The Fiat–Shamir transform: from interactive to non-interactive"); and the binding and inclusion soundness of Merkle trees (§"Merkle trees and commitments to sets", with its subsections §"Binary Merkle trees", §"Authentication paths and membership proofs", §"Soundness: forging a path implies a collision", §"Layer and domain separation", §"Append-only and incrementally updatable trees" and §"Vector commitments"). From the *Consensus Guide*: work and the most-work rule (§"The block tree and the most-work rule" and §"Work, and the best chain"), memoryless clocks (§"Memoryless clocks and the attribution sequence"), the retarget rule under NU7 (§"Difficulty in motion") and Equihash (§"Equihash and the memoryless model"). From the *Ironwood Guide*, only for the objects it constructs: block chain, height, network upgrades, ZIPs and the status Draft (§"Scope and status"); the Orchard pool and the Ironwood pool (§"The Orchard protocol and its two pools"); the encoding LEBS2OSP (§"Fields, groups, and encodings"); pool roots, final treestates and anchors (§"The note commitment tree" and §"Anchors"); branch IDs and the valid transaction formats, version 5 and version 6 under NU7 (§"The transaction format"; ZIP 2003 (Draft); ZIP 259 (Draft)); and transaction identifiers, the personalised hash h and the branch-ID encoding β (§"Transaction digests and signatures"). The light-client model is cited directly to ZIP 307 (Draft), "Security Model" and "Block header validation".

Definition 1.1 (Named network upgrades)**.** A *network upgrade*, its *activation height* and the fixed order in which network upgrades activate are those of the *Ironwood Guide*, §"Scope and status", Definition "Network upgrade" (ZIP 200 (Final), "Terminology" and "Activation mechanism"). This volume names the following network upgrades, in activation order, each with its deployment ZIP and the status of that ZIP:

Upgrade	ZIP	Status	Upgrade	ZIP	Status
Sapling	205	Final	NU6	253	Final
Blossom	206	Final	NU6.1	255	Final
Heartwood	250	Final	NU6.2	257	Final
Canopy	251	Final	NU6.3	258	Draft
NU5	252	Final	NU7	259	Draft

An upgrade is named only where its name fixes which header rule or tree-node rule applies. The *Heartwood activation block* is the block whose height is the activation height of Heartwood. No activation height, date or branch ID value is given here; the branch IDs of the upgrades from Heartwood on, whose upgrade epochs bear an epoch tree ("Upgrade epochs and epoch trees", §6.2), are tabulated in Table 3.

Definition 1.2 (ZIP statuses)**.** A *ZIP* and its status *Draft*, the status of every ZIP on submission, are those of the *Ironwood Guide*, §"Scope and status", Definition "ZIP". Two further statuses occur in this volume (ZIP 0 (Active), "ZIP Status Field"):

(i) *Final*, the status of a Consensus or Standards ZIP that is both implemented and activated on the Zcash network;

(ii) *Active*, the status, typically of a Process or Informational ZIP, reached once rough consensus on it has been reached; ZIP 0 itself has this status.

The status is a property of the document and does not decide whether its rules are in force. The

rules of ZIP 258 (Draft) and ZIP 229 (Draft) apply from NU6.3 activation: ZIP 258, "NU6.3 deployment", lists ZIP 229 among the primary sources of the NU6.3 consensus changes, and ZIP 258, "Changes to ZIP 221", amends the history-tree node of ZIP 221. Every ZIP is cited with its status at its first citation in the volume.

The volume describes the protocol as specified for NU7, whose deployment ZIP, ZIP 259, has status Draft (ZIP 259, preamble). NU6.3 is active.

Definition 1.3 (Epistemic classes)**.** A *component* of a Zcash FlyClient is a data structure, rule, check or proof format of the protocol, or a hypothesis of its security theorem. A component is

(i) *specified* when normative text of the protocol specification or of a ZIP fixes it, whatever the status of that ZIP (Definition 1.2);

(ii) *designed but unspecified* when a published design with an analysis exists but no ZIP or section of the protocol specification fixes it for Zcash;

(iii) an *open problem* when no design exists.

A property proved in this volume of a specified object is classified by its object: the object is specified, and the property is proved here.

The class *specified* is that of the *Ironwood Guide*, §"Scope and status", Definition "Epistemic classes"; that volume applies *designed but unspecified* to a property that a deployed construction is designed to satisfy but that no specification, ZIP or proof establishes, whereas this volume applies it to a component that exists only as a published design; the two uses share the condition that no normative text fixes the object, and the sense of the *Ironwood Guide* is not used here.

1.2 Sources and conventions

The primary source is B. Bünz, L. Kiffer, L. Luu and M. Zamani, "FlyClient: Super-Light Clients for Cryptocurrencies", IACR ePrint 2019/226, in its sixth revision, the last revision on the ePrint page and the revision of record; the paper also appeared at the IEEE Symposium on Security and Privacy. Every statement taken from the paper is cited by the section, definition, theorem, lemma, algorithm or printed page of that revision. A section of the paper is written "Section 5.3", without the section sign, which this volume reserves for sections of the protocol specification and of the *Arboretum*.

Where a statement of this volume rests on a printed statement of the paper that is defective, the volume states the corrected form at that point and names the departure. Only defects that touch a stated result are mentioned, and there is no separate register of them. Claims of the paper that this volume does not prove are stated as the paper's, with their gaps: for example Assumption 5.11 ("Transition rewriting") and Remark 4.11 ("Lemma 1 of the paper").

The normative sources are the following. The protocol specification, in the sections §"The Block Chain", §"Integers, Bit Sequences, and Endianness", §"Constants", §"BLAKE2 Hash Functions", §"Transaction Encoding and Consensus", §"Block Header Encoding and Consensus", §"Difficulty filter", §"Difficulty adjustment", §"nBits conversion" and §"Definition of Work". ZIP 0 (Active), which defines the ZIP process and its statuses; ZIP 200 (Final); ZIP 204 (Draft); ZIPs 205, 206, 250, 251, 252, 253, 255 and 257 (Final), the deployment ZIPs of the upgrades named in Definition 1.1 before NU6.3; ZIP 218 (Draft); ZIP 221 (Final); ZIP 229 (Draft); ZIP 244 (Final); ZIP 258 (Draft); ZIP 259 (Draft); ZIP 307 (Draft); and ZIP 2003 (Draft). A ZIP is cited by its number and the title of a section, the title following the number in the same sentence; the protocol specification is cited as protocol specification, §"Title".

The conventions of the volume and the symbols of later sections are fixed in Table 1. The paper overloads several symbols: n denotes both the honest participation and the chain length;

k the persistence depth, the samples per interval and $\log_c \delta$; λ both the security parameter and a statistical exponent; δ both the weight fraction checked in full and the backbone parameter for which $\mu = 1 - \delta$ bounds the adversary's mining power; L both the fork threshold and the number of blocks of the checked suffix; and c both the weight bound and a ratio of mining power. The table assigns one meaning to each symbol, and the index that closes it records where each is defined.

2 The verification problem

This section fixes the object a light client verifies, the Zcash block header, and the baseline procedure that verifies a chain of headers in full. It states the trust model of the light-client protocol specified for Zcash and the problem the volume solves, with each term of its solution defined in one clause and constructed in a later section.

2.1 Header-only verification

Definition 2.1 (Light client)**.** A *light client* is a client that is not a full participant in the network of Zcash peers. It can send and receive payments, but does not store or validate a copy of the block chain (ZIP 221, "Terminology").

The verifier of the FlyClient paper is a light client given the additional inputs fixed in Definition 4.16 ("Security and succinctness of chain proofs", §4.4).

Byte strings and integers are related by the maps of the *Ironwood Guide*, §"Fields, groups, and encodings", Definition "Integer and byte encodings": the map $\mathsf{I2LEOSP}_k$ sends an integer in $\{0, \dots, 2^k - 1\}$ to its $\lceil k/8 \rceil$-byte little-endian encoding, and the map $\mathsf{LEOS2IP}_k$ sends a $k/8$-byte string to the integer it encodes in little-endian order.

Definition 2.2 (CompactSize)**.** The *CompactSize encoding* of an integer $v \in \{0, \dots, 2^{64} - 1\}$ is the byte string

$$\mathsf{CompactSize}(v) := \begin{cases} \mathsf{I2LEOSP}_8(v) & \text{if } 0 \le v \le 252, \\ \texttt{0xFD} \,\|\, \mathsf{I2LEOSP}_{16}(v) & \text{if } 253 \le v \le \texttt{0xFFFF}, \\ \texttt{0xFE} \,\|\, \mathsf{I2LEOSP}_{32}(v) & \text{if } \texttt{0x10000} \le v \le \texttt{0xFFFFFFFF}, \\ \texttt{0xFF} \,\|\, \mathsf{I2LEOSP}_{64}(v) & \text{if } \texttt{0x100000000} \le v \le \texttt{0xFFFFFFFFFFFFFFFF}, \end{cases}$$

of length 1, 3, 5 or 9 bytes respectively. Only this encoding, the shortest, is valid for v: a byte string that decodes to v under one of the other three prefixes MUST be rejected (ZIP 204, "CompactSize").

ZIP 204 has status Draft; the minimal-encoding rule binds block headers independently, through the protocol specification, §"Block Header Encoding and Consensus", which requires the header field solutionSize to be encoded with the minimum number of bytes. That rule prevents a miner from testing several encodings of one Equihash solution against the difficulty filter.

The upgrade epochs delimited by the network upgrades of Definition 1.1 are fixed in "Upgrade epochs and epoch trees" (§6.2).

Definition 2.3 (Block header)**.** A Zcash *block* consists of a block header and a sequence of transactions. A *block header* hdr is the concatenation, in the following order, of the fields

Conventions	
Logarithms	The logarithm $\log x$ is natural and $\log_2 x$ is to base 2 (*Math Guide*, §"Symbols and how to say them"). For real $b > 0$, $b \neq 1$, $\log_b x := \log x / \log b$, as in $\log_{1/c}$: a real-base logarithm, distinct from the discrete logarithm $\log_g h$, whose base is a group element.
Floor, ceiling	The floor $\lfloor x \rfloor$ and ceiling $\lceil x \rceil$ of a real x (*Math Guide*, §"Symbols and how to say them"; §"The division algorithm").
Asymptotics	The symbols O and Θ and negligible functions are as in the *Math Guide*, §"Asymptotic notation" and §"Polynomial, exponential, and negligible functions"; and $\mathrm{polylog}(N) := (\log N)^{O(1)}$. With several parameters, n or N grows while λ, σ, c and L are fixed; the budget R is either fixed or, as in the paper, $R = c\,n$, and each bound states which.
Three parameters	λ the security parameter (*Math Guide*, §"The security parameter"); σ the statistical exponent of a failure target $2^{-\sigma}$; κ the output length in bits of the paper's hash H. The paper writes λ for all three.
Node fields	Fields of a node v of the difficulty Merkle mountain range are sans-serif and accessed with a dot, as $v.\mathsf{w}$, $v.\mathsf{t}$.
Personalised hash	The hash h and the encoding β are those of the *Ironwood Guide*, §"Transaction digests and signatures"; the hash is always written with its personalisation and input in full.
Two epochs	The *retarget epoch* ("Retargeting and the difficulty-raising attack") and the *upgrade epoch* ("Upgrade epochs and epoch trees") are distinct objects; "epoch" is not used alone where it is ambiguous.
Unused symbols	The *Crypto Guide*'s c (completeness error) and L (language) do not occur.

Symbol index (pointers to the defining subsection; nothing is defined here)	
Header-only verification	N the length of a claimed chain; η a block height in the sense of the *Ironwood Guide*, §"Scope and status"; $\omega(C)$ the total work of a chain C; hdr a block header; every chain-length quantity before "Merkle mountain ranges" is written N or "the chain length", never n.
Merkle mountain ranges (section preamble)	the hash $H\colon \{0,1\}^* \to \{0,1\}^\kappa$ and its output length κ.
The recursive definition	n the leaf count of a Merkle mountain range, hence of the range a head commits to, never a block height; j a leaf index and x_j the value of leaf j; M_n the range over n values and rt_n its root; $\mathrm{alt}(v)$ the altitude of a node.
Peaks and bagging	$P_1, \dots, P_d$ the peaks, $d = \mathrm{popcount}(n)$; Bag_R and Bag_L the two bagging orders.
Aggregating ranges	ser the serialisation of a node tuple; W the cumulative weight before a leaf.
The variable-difficulty backbone model	D a difficulty level (larger is harder); T a target (smaller is harder); ω, restated, never written D; q the queries per player per round; r a round; ν_r and t_r the honest and adversarial players of round r; (γ, s) the participation parameters; μ the adversary's power bound; ℓ the persistence depth.
Retargeting and the difficulty-raising attack	E the retarget-epoch length; Δ the rounds a retarget epoch takes; τ the dampening factor; f the expected blocks per round.
The (c, L)-adversary	c and L the parameters of the adversary; ρ the power ratio of the paper's evaluation.
The difficulty Merkle mountain range	y and z the level exponent and the retarget-epoch count of the feasibility bounds; the dotted node fields.
The sampling distribution	δ the weight fraction checked in full; k the catch parameter, $\delta = c^k$, its only meaning; a a fork point; ξ a point of $[0,1]$; φ a density on $[0,1]$; K the constant of the indifference density; g the FlyClient density and G its distribution function; p the single-draw catch probability; Q the sample count; u the samples per dyadic interval; U a uniform variable on $[0,1]$.
The non-interactive protocol	R the head-recomputation budget.
Header chain commitments and per-sample checks	B_η the block at height η; heights are written η, never i.
Upgrade epochs and epoch trees	A an activation height; A_H that of Heartwood, distinct from the hash H; $\alpha(\eta)$ the greatest activation height below η; $\mathcal{T}_\eta$ the epoch tree committed by B_η, distinct from a target T, with $n = \eta - \alpha(\eta)$ leaves.
The epoch tree and its root commitment	$b(\mathcal{T})$ the branch ID of an epoch tree $\mathcal{T}$.
The header consensus rules	A_{NU5} the activation height of NU5.
The committed range	A_{NU7} the activation height of NU7.

Bytes	Field	Content
4	nVersion	signed 32-bit integer; the consensus rules require nVersion ≥ 4
32	hashPrevBlock	the block hash of the previous block
32	hashMerkleRoot	the root of the transaction Merkle tree over the identifiers of the block's transactions, stated in "The authorising-data root" (§6.5)
32	commitment field	named as below
4	nTime	unsigned 32-bit integer, the block timestamp
4	nBits	unsigned 32-bit integer, an encoding of the target threshold
32	nNonce	32 arbitrary bytes
3	solutionSize	CompactSize(1344)
1344	solution	an Equihash solution

with integers encoded little-endian. The commitment field is named hashReserved before Sapling, hashFinalSaplingRoot in Sapling and Blossom, hashLightClientRoot in Heartwood and Canopy, and hashBlockCommitments from NU5; its value in each upgrade epoch is fixed in "The header consensus rules" (§6.7). The *block hash* of hdr is the 32-byte string

$$\mathsf{BlockHash}(\mathrm{hdr}) := \mathrm{SHA256d}(\mathrm{hdr}), \qquad \mathrm{SHA256d}(x) := \mathrm{SHA256}(\mathrm{SHA256}(x)),$$

taken over the whole header serialisation, solutionSize and solution included, in internal byte order, the order in which the hash function outputs its bytes. Let $\mathsf{ToTarget}\colon \{0, \dots, 2^{32}-1\} \to \mathbb{N}$ be the conversion of protocol specification, §"nBits conversion". The header hdr *satisfies proof of work* when solution is a valid Equihash solution for hdr and

$$\mathsf{LEOS2IP}_{256}(\mathsf{BlockHash}(\mathrm{hdr})) \leq \mathsf{ToTarget}(\mathsf{nBits})$$

(protocol specification, §"Difficulty filter"). The consensus rules further require, for a block at height η, that $\mathsf{nBits} = \mathsf{ThresholdBits}(\eta)$, the value the difficulty-adjustment rule of protocol specification, §"Difficulty adjustment" assigns to that height.

The definition follows protocol specification, §"Block Header Encoding and Consensus" for the field table and its consensus rules, and ZIP 221, "Block header semantics and consensus rules" and ZIP 244 for the renaming of the commitment field. The hash SHA256d is registered in the *Crypto Guide*, §"The transparent layer's hashes". The Equihash condition is cited, not reconstructed: *Consensus Guide*, §"Equihash and the memoryless model". The *work* of a block whose header carries nBits is

$$\mathsf{work}(\mathrm{hdr}) := \left\lfloor \frac{2^{256}}{\mathsf{ToTarget}(\mathsf{nBits}) + 1} \right\rfloor,$$

and the *total work* $\omega(C)$ of a chain C is the sum of the work of its blocks; a node considers the valid chain of greatest total work to be best, breaking ties between leaf blocks in favour of the block received first (*Consensus Guide*, §"The block tree and the most-work rule", and its subsection §"Work, and the best chain"; protocol specification, §"The Block Chain" and §"Definition of Work").

The fixed-width fields of Definition 2.3 occupy $4+32+32+32+4+4+32 = 140$ bytes. Since $253 \leq 1344 \leq$ 0xFFFF, the field solutionSize occupies 3 bytes, and with the 1,344-byte solution a Zcash block header is 1,487 bytes, of which the solution is 1344/1487, about 90.4 per cent.

Definition 2.4 (Simplified payment verification)**.** A *claimed chain* of length $N \geq 1$ is a sequence of block headers $C = (\mathrm{hdr}_0, \dots, \mathrm{hdr}_{N-1})$ in which hdr_0 is the header of the genesis block;

header hdr_η is the header at height η. Given one or more claimed chains, a *simplified-payment-verification* (SPV) client downloads every header of every claimed chain and checks, for each chain C:

(S1) *linkage*: for $1 \le \eta \le N-1$, the field hashPrevBlock of hdr_η equals $\mathsf{BlockHash}(\mathrm{hdr}_{\eta-1})$;

(S2) *proof of work*: for $0 \le \eta \le N-1$, header hdr_η satisfies proof of work in the sense of Definition 2.3, and its field nBits equals $\mathsf{ThresholdBits}(\eta)$, computed from $\mathrm{hdr}_0, \ldots, \mathrm{hdr}_{\eta-1}$.

It rejects every chain that fails a check and accepts, among the rest, the chain C of greatest total work $\omega(C)$, under the best-chain rule of the *Consensus Guide*, §"Work, and the best chain".

The definition is that of ePrint 2019/226, Section 1, p. 2, instantiated with the header checks of protocol specification, §"Block Header Encoding and Consensus". For a claimed chain of N blocks the client receives N headers, 1,487 N bytes, so its communication is $\Theta(N)$. A block is at most 2,000,000 bytes (same section), so the saving over downloading full blocks is at most a constant factor, not an asymptotic one.

An SPV client checks neither the validity of transactions nor any consensus rule beyond those of the header, and therefore relies on the following assumption.

Assumption 2.5 (SPV assumption). "The chain with the most PoW solutions follows the rules of the network and will eventually be accepted by the majority of miners." (ePrint 2019/226, Section 1, p. 2, Assumption 1.)

The paper formalises the property an SPV client checks as Definition 4.14, and strengthens Assumption 2.5 quantitatively as Assumption 4.8 ("The (c, L)-adversary", §4.3).

2.2 The light-client model of ZIP 307

ZIP 307 (Draft) specifies a light-client protocol in which a light client obtains, from a Zcash node and a proxy, *compact blocks*: per-block messages that may carry the block header. The trust model is stated in ZIP 307, "Security Model": the node and proxy combination is assumed to be "honest but curious" and "is trusted to provide a correct view of the current best chain state and to faithfully transmit queries and responses".

ZIP 307, "Block header validation" describes, as a proposed enhancement that the ZIP records as only partially implemented (only check (Z2) is made), checks of the SPV kind on each compact block that carries a header. With the fields named as in Definition 2.3, the checks are:

(Z1) the field nVersion is at least the minimum block version;

(Z2) the field hashPrevBlock equals the block hash of the previous compact block;

(Z3) the Equihash solution is valid;

(Z4) the target $\mathsf{ToTarget}(\mathsf{nBits})$ is non-zero and at most the proof-of-work limit PoWLimit (protocol specification, §"Constants");

(Z5) when the last 27 compact blocks all carry headers, the field nBits is correct under the difficulty-adjustment rule;

(Z6) the block hash, read as a little-endian integer, is at most $\mathsf{ToTarget}(\mathsf{nBits})$.

In that proposal, a compact block that fails a check MUST be discarded. The same section of ZIP 307 further describes a check that the commitment field equals the root of the Sapling note commitment tree recomputed from the compact block; from Heartwood that check no longer applies, since the field changes meaning (Definition 2.3; ZIP 221, "Block header semantics and

consensus rules"). Checks (Z1) to (Z6), where made on every header, cost communication linear in the chain length N, as in Definition 2.4.

Of the trust that ZIP 307 places in the node and proxy, a FlyClient verifier would replace only the trust in which header chain carries the most work, under Assumption 4.8 ("The (c, L)-adversary"). It would replace the linear header checks by the logarithmic sampling of "The FlyClient protocol" (§5); it does not replace faithful transmission of queries and of compact-block contents. Such a verifier is designed but unspecified in the sense of Definition 1.3; its status is fixed in "A Zcash verifier" (§7.4).

2.3 Statement of the problem

The problem is the following. Under the SPV assumption (Assumption 2.5), strengthened quantitatively as Assumption 4.8 in "The (c, L)-adversary" (§4.3), a light client is to accept the head of the chain of greatest total work after communication sublinear in the chain length N. The inputs and the chain to be selected are those of Definition 2.4; the assumption is strengthened to Assumption 4.8, the selection holds except with a failure probability fixed in "The FlyClient protocol" (§5), and the communication falls from $\Theta(N)$ to sublinear.

A client that does not check every proof of work "can be tricked into accepting an invalid chain by a malicious prover who can precompute a sufficiently-long chain using its limited computational power" (ePrint 2019/226, Section 1, p. 3); the adversary is fixed in "The FlyClient security model" (§4).

The solution uses four objects. A *Merkle mountain range* is a binary hash tree over a sequence that admits appending a leaf without changing its peaks or any node beneath them ("Merkle mountain ranges", §3). A *header chain commitment* is a root, carried in every header after the genesis header, of such a range over all blocks strictly before that header ("Header chain commitments and per-sample checks", §5.1). A *work-weighted sample* is a block drawn at a random point of cumulative work and checked against the commitment of the head ("The sampling distribution", §5.4). A *suffix* of the chain is downloaded and checked in full, its extent fixed in the same subsection. *FlyClient* is the protocol that verifies a proof-of-work chain by drawing logarithmically many such samples and checking each against the commitment of the head (ePrint 2019/226, Section 1). The commitment of the paper covers the chain from the genesis block, whereas the commitment of ZIP 221 covers one upgrade epoch: the blocks from the activation height of the preceding network upgrade up to the committing block, exclusive (ZIP 221, "Specification"; "Upgrade epochs and epoch trees", §6.2).

3 Merkle mountain ranges

This section constructs the commitment on which the FlyClient protocol and the chain-history commitment of ZIP 221 rest: a binary hash tree over a sequence of values whose shape is fixed by the length of the sequence, extended by one value at logarithmic cost, and opened at one position j by a path from which the root over the prefix of length $j-1$ is recomputed, and, under the right fold, also the root over the prefix of length j. The construction is parametrised by the order in which its perfect subtrees are joined: the right fold of ePrint 2019/226 and the left fold of ZIP 221, "Background". The fixed-depth trees of the *Crypto Guide*, §"Binary Merkle trees" and §"Append-only and incrementally updatable trees", are the special case of a sequence whose length is a power of two.

Throughout, $H\colon \{0,1\}^* \to \{0,1\}^\kappa$ is a hash function drawn from a family that is collision resistant in the sense of the *Crypto Guide*, Definition "Collision resistance" (§"Security notions: preimage, second-preimage, and collision resistance"); ePrint 2019/226, Definition 8, states the same assumption. A statement holds *barring a collision* when a deterministic procedure, given any input on which the statement fails, outputs two distinct strings with equal images under H, using the number of evaluations of H stated with it. Such a procedure is a tight reduction in

the sense of the *Crypto Guide*, Definition "Reduction loss and tightness" (§"Reductions"), so a probabilistic polynomial-time adversary makes the statement fail with negligible probability.

3.1 The recursive definition

Definition 3.1 (Merkle mountain range)**.** Let $n \geq 1$ and $x_1, \dots, x_n \in \{0,1\}^\kappa$. The *Merkle mountain range* M_n over $(x_1, \dots, x_n)$ is the rooted ordered binary tree, every internal node of which has exactly two children, a left and a right one, with a value in $\{0,1\}^\kappa$ at every node, defined by recursion on n:

1. if $n = 1$, M_1 is the single leaf with value x_1;
2. if $n > 1$, let $i = \lfloor \log_2(n-1) \rfloor$, so that $2^i < n \leq 2^{i+1}$ and 2^i is the largest power of two strictly below n; the root has as left child the range M_{2^i} over $(x_1, \dots, x_{2^i})$ and as right child the range M_{n-2^i} over $(x_{2^i+1}, \dots, x_n)$, where $1 \leq n - 2^i \leq 2^i$.

Every internal node with children of values v_ℓ and v_r has value $H(v_\ell \| v_r)$, the image of the 2κ-bit concatenation of its children's values. The leaves are ordered from left to right and leaf j holds x_j. The *root* rt_n is the value of the root node; *range* abbreviates Merkle mountain range.

The *depth* of a node is the number of edges from the root to it, and the depth of a tree the maximal depth of a leaf. A *perfect subtree of altitude a* is a subtree with 2^a leaves, all at depth a below its root; its root node v has *altitude* $\mathrm{alt}(v) = a$. The *right spine* is the sequence of nodes that starts at the root and continues from each internal node to its right child, ending at a leaf.

ZIP 221, "Terminology", calls a Merkle mountain range a binary hash tree that admits appends of new leaves without changing the value of existing nodes; Theorem 3.11 and Proposition 3.12 below establish it for Definition 3.1 for the peaks and every node beneath them. An append rehashes only the right-spine nodes above the first perfect one, the counterparts of the bagging nodes of ZIP 221, "Background". Definition 3.1 is ePrint 2019/226, Definition 11, with the base case $n = 1$, implicit there, made explicit. Every node of M_n is the root of a range over a run of consecutive values, by the recursion. Its depth is fixed in Lemma 3.2.

Lemma 3.2 (Left children are perfect)**.** *Let $n \geq 1$.*

1. *A range over 2^a values is a perfect subtree of altitude a.*
2. *Every left child in M_n is perfect: the left child of a node over $m \geq 2$ leaves is the perfect range over $2^{\lfloor \log_2(m-1) \rfloor}$ leaves.*
3. *Descend the right spine of M_n from the root, stopping at the first node whose leaf count is a power of two. The left children of the nodes passed, followed by the node where the descent stops, are perfect subtrees of pairwise distinct power-of-two sizes, decreasing from the root downwards, covering consecutive runs of leaves from left to right; their sizes are the terms of the binary expansion of n.*
4. *The range M_n has depth exactly $\lceil \log_2 n \rceil$, which is 0 for $n = 1$.*

Proof. (1) By induction on a. For $a = 0$ the range is a leaf. For $a \geq 1$, $\lfloor \log_2(2^a - 1) \rfloor = a - 1$, so the root has two children over 2^{a-1} values each, perfect of altitude $a - 1$ by hypothesis; all leaves lie at depth a.

(2) Every node over $m \geq 2$ leaves is a range M_m, whose left child is a range over $2^{\lfloor \log_2(m-1) \rfloor}$ values, perfect by (1).

(3) Let a right-spine node have m leaves. If m is a power of two, the node is perfect by (1) and the descent stops. Otherwise $2^{\lfloor \log_2 m \rfloor} < m$, hence $m - 1 \geq 2^{\lfloor \log_2 m \rfloor}$ and $\lfloor \log_2(m-1) \rfloor = \lfloor \log_2 m \rfloor$:

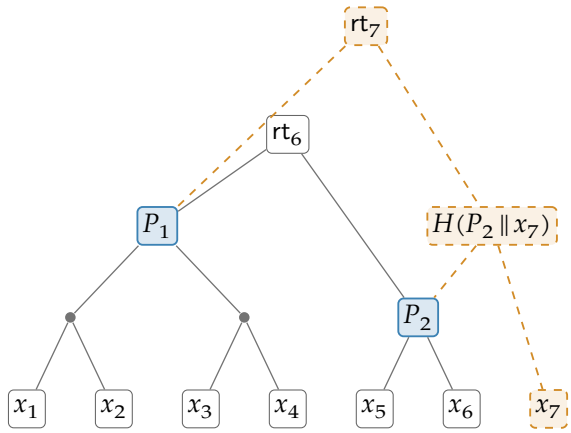


Figure 1: The range M_6 over $x_1, \dots, x_6$ in single-rooted form (solid): the root rt_6 has left child P_1, the perfect range over $x_1, \dots, x_4$, and right child $P_2 = H(x_5 \| x_6)$, the perfect range over x_5, x_6. The nodes P_1 and P_2 are the peaks of Definition 3.4; P_2 is the first perfect node on the right spine of M_6, whose last node is the leaf x_6. Dashed: appending x_7 creates only the nodes $H(P_2 \| x_7)$ and $\mathsf{rt}_7 = H(P_1 \| H(P_2 \| x_7))$; the nodes P_1, P_2 and every node beneath them are shared by M_6 and M_7.

the left child takes the highest term $2^{\lfloor \log_2 m \rfloor}$ of the binary expansion of m and the right child the remaining terms. By induction along the spine, the left children passed take the terms of the expansion of n from the highest downwards, and the descent stops when one term is left. The runs are consecutive because each left child precedes its sibling's leaves.

(4) By induction on n. For $n = 1$ the depth is 0. For $n > 1$ let $i = \lfloor \log_2(n-1) \rfloor$. The left child is perfect over 2^i leaves, of depth i; the right child has $n - 2^i \le 2^i$ leaves and depth $\lceil \log_2(n - 2^i) \rceil \le i$ by hypothesis. The depth of M_n is therefore $i + 1$, which equals $\lceil \log_2 n \rceil$ since $2^i < n \le 2^{i+1}$. □

Example 3.3 (Six and seven leaves)**.** For $n = 6$, $i = \lfloor \log_2 5 \rfloor = 2$ and $6 = 4 + 2$. The left child is the perfect range P_1 over $x_1, \dots, x_4$; the right child has two leaves, is perfect, and has value $P_2 = H(x_5 \| x_6)$; hence $\mathsf{rt}_6 = H(P_1 \| P_2)$. For $n = 7$, $i = \lfloor \log_2 6 \rfloor = 2$ and $7 = 4 + 3$; the three-leaf right child splits as $3 = 2 + 1$, so $7 = 4 + (2 + 1)$ and

$$\mathsf{rt}_7 = H(P_1 \| H(H(x_5 \| x_6) \| x_7)).$$

3.2 Peaks and bagging

For $n \ge 1$, $\mathrm{popcount}(n)$ is the number of ones in the binary expansion of n.

Definition 3.4 (Peaks)**.** Let $n \ge 1$ and write $n = 2^{a_1} + \cdots + 2^{a_d}$ with $a_1 > \cdots > a_d \ge 0$, so that $d = \mathrm{popcount}(n)$, and let $s_i = \sum_{l<i} 2^{a_l}$ for $1 \le i \le d$. The *peaks* of the sequence $(x_1, \dots, x_n)$ are $P_1, \dots, P_d$, where P_i is the perfect range over $(x_{s_i+1}, \dots, x_{s_i+2^{a_i}})$, of altitude $\mathrm{alt}(P_i) = a_i$. There is one peak per set bit of n, of altitude that bit's index; the altitudes strictly decrease from the left, and the peaks cover consecutive runs of leaves. The symbol P_i denotes both the subtree and its root value.

The peaks are the mountains of ZIP 221, "Background", whose leftmost peak is the highest. The same section numbers nodes in the order of their creation; that numbering enters no committed value and no check, and the root is a function of the leaf sequence and the bagging order of Definition 3.5 alone. No node numbering is used in this volume.

Definition 3.5 (Bagging orders)**.** Let $P_1, \dots, P_d$ be the peaks of $(x_1, \dots, x_n)$. The *right fold* and the *left fold* of the peaks are

$$\begin{aligned}\mathrm{Bag}_R(P_1, \dots, P_d) &= H(P_1 \,\|\, H(P_2 \,\|\, \cdots H(P_{d-1} \,\|\, P_d) \cdots)),\\ \mathrm{Bag}_L(P_1, \dots, P_d) &= H(\cdots H(H(P_1 \,\|\, P_2) \,\|\, P_3) \cdots \|\, P_d);\end{aligned}$$

a single peak is its own bag under either fold. The left fold is the order of ZIP 221, "Background", which repeatedly joins the two leftmost peaks. For $b \in \{R, L\}$ the tree M_n^b consists of the d perfect peak subtrees joined by the $d-1$ internal *bagging nodes* of Bag_b, and rt_n^b is its root value. Both are determined by n, b and the leaf sequence.

Proposition 3.6 (Peaks of the recursive form)**.** *For every $n \geq 1$ the peaks of $(x_1, \dots, x_n)$ are exactly the left children of the right-spine nodes of M_n, from the root down to the first perfect right-spine node, followed by that node. Hence $M_n = M_n^R$ and $\mathrm{rt}_n = \mathrm{Bag}_R(P_1, \dots, P_d)$.*

Proof. By Lemma 3.2(3) the nodes named are perfect subtrees whose sizes are the terms of the binary expansion of n, largest first, covering consecutive runs of leaves from the left: they are $P_1, \dots, P_d$ in order. Along the spine, the t-th node passed has value $H(P_t \,\|\, v_{t+1})$ with v_{t+1} the value of the next spine node, and the last spine node considered is P_d; unwinding gives the right fold. □

Lemma 3.7 (Path lengths under the two baggings)**.** *Let $n \geq 1$ have peaks $P_1, \dots, P_d$ of altitudes $a_1 > \cdots > a_d$, let $b \in \{R, L\}$, and let leaf j lie in P_i. The depth $\mathrm{depth}_b(n, j)$ of leaf j in M_n^b is*

$$\mathrm{depth}_R(n, j) = \begin{cases} a_i + i & i < d, \\ a_d + d - 1 & i = d, \end{cases} \qquad \mathrm{depth}_L(n, j) = \begin{cases} a_1 + d - 1 & i = 1, \\ a_i + d - i + 1 & i \geq 2. \end{cases}$$

Consequently the depth of M_n^R is $\lceil \log_2 n \rceil$; the depth of M_n^L is $\lfloor \log_2 n \rfloor + \mathrm{popcount}(n) - 1 \leq 2\lfloor \log_2 n \rfloor$, attained by the leaves of P_1; and the two depths differ exactly when $d \geq 3$. For $n = 7$ they are 3 and 4; for $n = 127$ they are 7 and 12. Leaf depths are functions of (b, n, j); for $d \geq 2$ they are not all equal.

Proof. The leaf lies at depth a_i below the root of P_i; it remains to find the depth of that root in the bag. Under Bag_R the t-th bagging node lies at depth $t-1$ and has left child P_t for $t \leq d-1$, and the last bagging node has right child P_d; so P_i lies at depth i for $i < d$ and P_d at depth $d-1$. Under Bag_L the root has right child P_d and left child $\mathrm{Bag}_L(P_1, \dots, P_{d-1})$, and inductively P_i lies at depth $d - i + 1$ for $i \geq 2$ while P_1 shares depth $d-1$ with P_2; for $d = 1$ the single peak lies at depth 0.

Maxima. Under R, $a_i \leq a_1 - (i-1)$ gives $a_i + i \leq a_1 + 1$ and $a_d + d - 1 \leq a_1$; the maximum is $a_1 + 1$ for $d \geq 2$ and a_1 for $d = 1$, which is $\lceil \log_2 n \rceil$ in both cases, as n is a power of two exactly when $d = 1$. Under L, for $i \geq 2$, $a_i + d - i + 1 \leq a_1 + d - 2i + 2 \leq a_1 + d - 2$, so the maximum is $a_1 + d - 1$, attained by P_1; with $a_1 = \lfloor \log_2 n \rfloor$ and $d = \mathrm{popcount}(n) \leq \lfloor \log_2 n \rfloor + 1$ this is the stated value and bound. The maxima $a_1 + 1$ and $a_1 + d - 1$ coincide for $d = 2$, both equal a_1 for $d = 1$, and differ for $d \geq 3$. For $d \geq 2$ a leaf of P_1 and a leaf of P_d differ in depth under R ($a_1 + 1$ against at most a_1), and a leaf of P_1 and a leaf of P_2 under L ($a_1 \neq a_2$). □

A *cycle* of H is a value v and a nonempty sequence of evaluations of H that, starting from inputs that include v, recomputes v.

Theorem 3.8 (Fold order)**.** *Let $n \geq 1$ have d peaks. The shapes M_n^R and M_n^L coincide if and only if $d \leq 2$. For $d \geq 3$:*

1. *the roots rt_n^R and rt_n^L are distinct hash expressions over the same peaks, and from values of the peaks for which $\mathsf{rt}_n^R = \mathsf{rt}_n^L$ a collision or a cycle of H is computed;*
2. *for every leaf j, the sequence of sides of the siblings on leaf j's path above its peak differs between M_n^R and M_n^L;*
3. *the peaks and the perfect subtrees beneath them coincide, and so do the asymptotic costs of append, of storage of the peaks and of a path, each logarithmic in n; the leaf depths differ as in Lemma 3.7, so the bound $\lceil \log_2 n \rceil$ on the path length of ePrint 2019/226 does not hold for M_n^L.*

Proof. Definition 3.1 splits $n = 2^{\lfloor \log_2(n-1) \rfloor} + (\text{rest})$ with the first peak on the left and recurses on the rest, which is the right fold (Proposition 3.6). For $d = 1$ both folds are P_1 and for $d = 2$ both are $H(P_1 \| P_2)$. For $d \geq 3$ the root's left child is P_1 under R and a bagging node under L, so the shapes differ.

(1) The root inputs are $P_1 \| \mathrm{Bag}_R(P_2, \dots, P_d)$ and $\mathrm{Bag}_L(P_1, \dots, P_{d-1}) \| P_d$. If the root values agree and these 2κ-bit strings differ, they are a collision. Otherwise $P_1 = \mathrm{Bag}_L(P_1, \dots, P_{d-1})$ and $P_d = \mathrm{Bag}_R(P_2, \dots, P_d)$, each a value recomputed from itself by $d - 2 \geq 1$ evaluations of H: a cycle.

(2) Above P_1 the path has under R the single sibling $\mathrm{Bag}_R(P_2, \dots, P_d)$ and under L the $d-1 \geq 2$ siblings $P_2, \dots, P_d$. Above P_d it has under R the $d - 1 \geq 2$ siblings $P_{d-1}, \dots, P_1$ and under L the single sibling $\mathrm{Bag}_L(P_1, \dots, P_{d-1})$. Above P_i with $2 \leq i < d$ the first sibling is on the right under R, namely $\mathrm{Bag}_R(P_{i+1}, \dots, P_d)$, and on the left under L, namely $\mathrm{Bag}_L(P_1, \dots, P_{i-1})$.

(3) Both trees are built on the peaks of Definition 3.4, which depend on the leaf sequence alone; there are $d \leq \lfloor \log_2 n \rfloor + 1$ of them. Appending x_{n+1} joins the last peaks of equal altitude with x_{n+1}, at most $\lfloor \log_2 n \rfloor + 1$ evaluations of H, and re-bagging the at most $\lfloor \log_2(n+1) \rfloor + 1$ new peaks costs one evaluation fewer than their number, under either fold. A path has at most $2\lfloor \log_2 n \rfloor$ siblings under either fold by Lemma 3.7, whose depths are the stated ones. □

Collision resistance alone does not exclude a cycle, so part (1) does not assert that the two roots differ numerically. No later statement of this volume uses such an inequality; the later sections use the shapes and path lengths of part (2) and Lemma 3.7.

Example 3.9 (Eleven leaves)**.** The eleven-leaf tree of ZIP 221, "Background", has $11 = 8 + 2 + 1$ and peaks P_1, P_2, P_3 of altitudes 3, 1 and 0, joined by two bagging nodes. The left fold that ZIP 221 commits and the right fold are

$$\mathsf{rt}_{11}^L = H(H(P_1 \| P_2) \| P_3), \qquad \mathsf{rt}_{11}^R = H(P_1 \| H(P_2 \| P_3)).$$

A leaf of P_1, P_2, P_3 has path length 4, 3, 2 under Bag_R and 5, 3, 1 under Bag_L. Figure 2 shows both trees.

3.3 Appending a leaf

Construction 3.10 (Append)**.** Every node v of a range carries its leaf count m. Procedure $\mathrm{Append}(v, x)$ takes the root v of a range over m leaves and a value $x \in \{0,1\}^\kappa$ and returns the root of a range over $m + 1$ leaves:

1. if m is a power of two (v is perfect; $m = 1$ included), return a new node with left child v, right child the leaf x, value $H(v \| x)$ and leaf count $m + 1$;
2. otherwise, with children v_ℓ and v_r, replace v_r by $\mathrm{Append}(v_r, x)$, set the value of v to

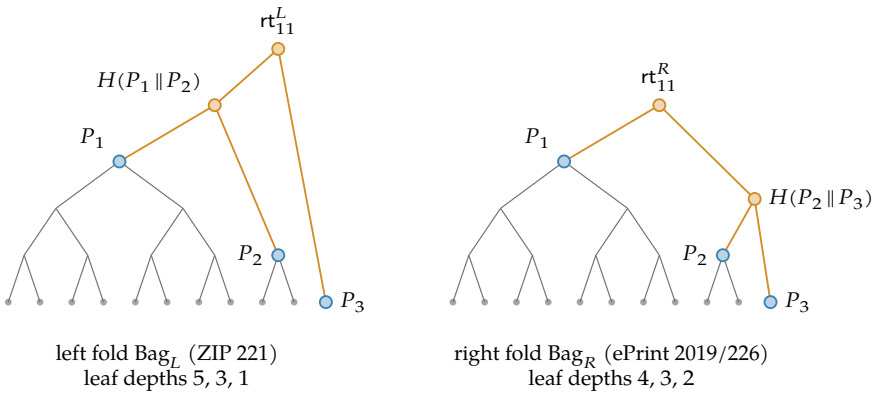


Figure 2: The eleven-leaf tree of Example 3.9, without node numbering. Both panels carry the same peaks P_1, P_2, P_3 (blue) of altitudes 3, 1, 0 over the same leaves. Left: the left fold $\mathsf{rt}^L_{11} = H(H(P_1 \| P_2) \| P_3)$ that ZIP 221 commits. Right: the right fold $\mathsf{rt}^R_{11} = H(P_1 \| H(P_2 \| P_3))$ of the recursion of Definition 3.1. Bagging nodes are amber. The leaves of P_1 lie one level deeper under the left fold; the depths below each panel are those of a leaf of P_1, P_2, P_3.

$H(v_\ell \| v_r)$ with the new v_r, set its leaf count to $m + 1$, and return v.

The construction is ePrint 2019/226, Algorithm 5. The order $H(v \| x)$ in step 1 is that of the algorithm and of Definition 3.1; the base case of the paper's proof of its Theorem 6 writes the reverse order.

Theorem 3.11 (Correctness of append)**.** *For $n \geq 1$, $\mathrm{Append}(M_n, x_{n+1})$ is M_{n+1} over $(x_1, \dots, x_{n+1})$: the leaves of M_n followed by x_{n+1} as the rightmost leaf.*

Proof. By induction on n, following ePrint 2019/226, Theorem 6. If $n = 2^i$, Definition 3.1 for $n + 1$ has $\lfloor \log_2 n \rfloor = i$ and right part 1: left child M_n, right child the leaf x_{n+1}, which is step 1. Otherwise $n = 2^i + m$ with $1 \leq m < 2^i$, and $n + 1 = 2^i + (m + 1)$ with the same i, since $m + 1 \leq 2^i$. The left child M_{2^i} is unchanged, and the right child is Append applied to the range M_m over $(x_{2^i+1}, \dots, x_n)$, which is M_{m+1} over $(x_{2^i+1}, \dots, x_{n+1})$ by hypothesis; step 2 rehashes the root. □

Proposition 3.12 (Cost of append)**.** *Procedure $\mathrm{Append}(M_n, x_{n+1})$ creates or rehashes only right-spine nodes of M_n and one new node, with at most $\lceil \log_2 n \rceil + 1$ evaluations of H. Every left child of M_n, and the perfect right-spine node at which the recursion stops, is a subtree of M_{n+1} with unchanged values.*

Proof. The recursion descends the right spine until the first node whose leaf count is a power of two, at depth at most $\lceil \log_2 n \rceil$ (Lemma 3.2(4)), evaluates H once to create the new node, and once more at each spine node above it. Left children and the node where the descent stops are never modified. □

Figure 1 shows the case $n = 6$: two evaluations of H create $H(P_2 \| x_7)$ and rt_7.

3.4 Inclusion paths and prefix roots

Definition 3.13 (Inclusion path)**.** Let $b \in \{R, L\}$, $n \geq 1$, $1 \leq j \leq n$, and $m = \mathrm{depth}_b(n, j)$ (Lemma 3.7). Let $u_0, u_1, \dots, u_m$ be the nodes from leaf j to the root of M^b_n. The *inclusion path*

of leaf j is $\pi = (\pi_1, \dots, \pi_m)$, where π_t is the value of the sibling of u_{t-1}, listed bottom-up. The *side* $\sigma_t \in \{\text{left}, \text{right}\}$ records whether that sibling is the left or the right child of u_t. The sides and m are functions of (b, n, j) through the shape of M_n^b.

The verifier $\text{Verify}_b(\mathsf{rt}, n, j, x, \pi)$, on $\mathsf{rt}, x \in \{0,1\}^\kappa$ and a sequence π of κ-bit values:

1. rejects unless $|\pi| = \text{depth}_b(n, j)$;
2. sets $y_0 = x$ and, for $t = 1, \dots, m$, $y_t = H(\pi_t \| y_{t-1})$ if $\sigma_t = \text{left}$ and $y_t = H(y_{t-1} \| \pi_t)$ otherwise;
3. accepts if and only if $y_m = \mathsf{rt}$.

By construction Verify_b accepts (x_j, π) with π the inclusion path of leaf j against rt_n^b. Under Bag_R every path has at most $\lceil \log_2 n \rceil$ siblings (Lemma 3.7); ePrint 2019/226, Definition 10, counts at most $\log_2 n + 1$ hashes, the leaf pair included. For $d \geq 2$, path lengths differ between leaves (Lemma 3.7). The paper's Algorithm 4 requires $|\pi| = \lceil \log_2(n-1) \rceil$ and so rejects valid paths: 1,225 of the 8,384 leaf paths of the ranges M_n, $2 \leq n \leq 129$, among them both leaves of M_2 and leaf 5 of M_6. The verifier above fixes the length from (b, n, j) instead. For n a power of two, M_n^b is a perfect tree, and the inclusion path and its verifier coincide with the *Crypto Guide*'s Definition "Authentication path" and Definition "Path verification" (§"Authentication paths and membership proofs") at index $j - 1$, with the bits b_t fixed by j instead of carried in the path.

Theorem 3.14 (Prefix roots from one path)**.** *For $1 \leq j \leq n$, from x_j and the inclusion path of leaf j in M_n a verifier computes rt_j and, for $j \geq 2$, rt_{j-1}, the roots of the ranges over $(x_1, \dots, x_j)$ and $(x_1, \dots, x_{j-1})$.*

Proof. For rt_j, by induction on n, following ePrint 2019/226, Theorem 7, with siblings bottom-up. If $j = n$, $\mathsf{rt}_j = \mathsf{rt}_n$ is the recomputed root y_m. Otherwise $n > 1$; let 2^i be the largest power of two strictly below n, so that the root's children are M_{2^i} and a range over $n - 2^i$ values.

1. If $j = 2^i$, rt_j is the value of the root's left child, which is y_{m-1}.
2. If $j < 2^i$, the path without its top sibling is the inclusion path of leaf j in the left child M_{2^i}, and the hypothesis applies.
3. If $2^i < j < n$, then $j - 1 \geq 2^i$ and $j < 2^{i+1}$, so 2^i is also the largest power of two strictly below j. The top sibling is P, the value of M_{2^i}, and the path without it is the inclusion path of leaf $j - 2^i$ in the right child; by hypothesis it yields the root r' over $(x_{2^i+1}, \dots, x_j)$, and $\mathsf{rt}_j = H(P \| r')$ by Definition 3.1.

For rt_{j-1}, the bottom-up fold of Lemma 3.15, whose proof does not use this theorem. The paper's proof lists the path top-down with the path nodes included, while its Algorithm 3 outputs the siblings bottom-up; the bottom-up format of Definition 3.13 is the one fixed here. □

Lemma 3.15 (Left siblings cover the prefix)**.** *Let $b \in \{R, L\}$, $1 \leq j \leq n$, and consider the inclusion path of leaf j in M_n^b. Its left siblings are subtrees whose leaf runs partition $\{1, \dots, j-1\}$, higher siblings covering earlier leaves; its right siblings cover $\{j+1, \dots, n\}$ and enter none of the folds below. Let $j \geq 2$ and let leaf j lie in peak P_i.*

1. *Under Bag_R the left siblings, from the top down, are the peaks of $(x_1, \dots, x_{j-1})$. The* bottom-up fold *(set r to the lowest left sibling, then $r \leftarrow H(\pi \| r)$ for each further left sibling π upwards) yields rt_{j-1}; started instead from $r = x_j$ and applied to every left sibling, it yields rt_j.*

2. *Under* Bag_L*, for* $i \geq 2$ *the topmost left sibling is* $\mathsf{Bag}_L(P_1, \dots, P_{i-1})$ *and the others lie in* P_i*, with sizes the terms of the binary expansion of* $j - 1 - s_i$*; for* $i = 1$ *all left siblings lie in* P_1*. The* top-down fold (*set* r *to the topmost left sibling, then* $r \leftarrow H(r \parallel \pi)$ *for each further left sibling* π *downwards*) *yields* rt^L_{j-1}.

Proof. A leaf $l < j$ leaves the path of leaf j at their lowest common ancestor, in whose left child l lies; that left child is a left sibling on the path. Distinct ancestors give disjoint runs, higher ancestors earlier runs. The same argument for $l > j$ gives the right siblings. Inside a perfect subtree the left siblings of the leaf at offset o from its first leaf are perfect subtrees whose sizes are the terms of the binary expansion of o, largest highest.

Leaf j lies at offset $o = j - 1 - s_i$ in P_i, and $j - 1 = s_i + o$, so the peaks of $(x_1, \dots, x_{j-1})$ are $P_1, \dots, P_{i-1}$ followed by the left siblings inside P_i, top-down. Under R the siblings $P_{i-1}, \dots, P_1$ lie above P_i as left children of bagging nodes (proof of Lemma 3.7); the bottom-up fold is then the right fold of the peaks of $(x_1, \dots, x_{j-1})$, which is rt_{j-1} by Proposition 3.6. Started from x_j, it is the right fold of those peaks followed by x_j. The largest power of two strictly below j is the highest term of $j - 1$, the size of the first of these peaks, and the recursion of Definition 3.1 on j splits off the peaks of $j-1$ in turn and ends at the single leaf x_j; this fold is therefore rt_j. Under L with $i \geq 2$ the root of P_i is the right child of the bagging node $H(\mathsf{Bag}_L(P_1, \dots, P_{i-1}) \parallel P_i)$, and every higher node on the path is a left child; the top-down fold is the left fold of the peaks of $(x_1, \dots, x_{j-1})$. For $i = 1$ the root of P_1 is a left child and every higher node on the path too. □

Construction 3.16 (Prefix root)**.** Procedure $\mathsf{PrefixRoot}_b(n, j, \pi)$ takes $b \in \{R, L\}$, $2 \leq j \leq n$ and a sequence π of κ-bit values with $|\pi| = \mathsf{depth}_b(n, j)$, with sides σ_t as in Definition 3.13. It selects the π_t with $\sigma_t =$ left and outputs their bottom-up fold for $b = R$ and their top-down fold for $b = L$ (Lemma 3.15).

Correctness of the construction is Lemma 3.15. The construction corrects the printed pseudocode of ePrint 2019/226, Algorithm 6, which uses an undefined index, ends in a vestigial boolean comparison, and has a call signature that does not match its call in Algorithm 2, and it fixes the siblings bottom-up.

Theorem 3.17 (Position binding)**.** *Let* $b \in \{R, L\}$*,* $n \geq 1$ *and* $1 \leq j \leq n$*. From any* $\mathsf{rt} \in \{0,1\}^\kappa$*, chosen by the adversary, and two pairs* $(x, \pi) \neq (x', \pi')$ *both accepted by* $\mathsf{Verify}_b(\mathsf{rt}, n, j, \cdot, \cdot)$*, a collision of* H *is computed with at most* $2\,\mathsf{depth}_b(n, j)$ *evaluations of* H*. In particular* $x \neq x'$ *yields a collision, and the siblings of an accepted path and every value it recomputes are functions of* (rt, b, n, j)*, barring a collision.*

Proof. Both chains $y_0, \dots, y_m$ and $y'_0, \dots, y'_m$ of Definition 3.13 have the length $m = \mathsf{depth}_b(n, j)$ and the sides fixed by (b, n, j), and $y_m = y'_m = \mathsf{rt}$. Since $(x, \pi) \neq (x', \pi')$, some step has differing inputs; let t be the highest step with $(y_{t-1}, \pi_t) \neq (y'_{t-1}, \pi'_t)$. Then $y_t = y'_t$: for $t = m$ both equal rt, and for $t < m$ the inputs of step $t + 1$ agree. The two 2κ-bit inputs of step t place their operands on the same sides and differ, and H maps both to y_t: a collision. For $n = 1$ the path is empty and Verify_b accepts only $x = \mathsf{rt}$, so no two distinct pairs are accepted. The extractor is deterministic and the reduction tight in the sense of the *Crypto Guide*, §"Reductions"; it is the first-level-of-agreement argument of the *Crypto Guide*, Proposition "Binding of the root" (§"Binary Merkle trees") and §"Soundness: forging a path implies a collision". □

The theorem extends the *Crypto Guide*'s Theorem "Merkle trees are succinct position-binding vector commitments" (§"Vector commitments") from perfect trees to the shapes fixed by n and the bagging order. It is stated for a root chosen by the adversary, not for a fixed honest root as in

the *Crypto Guide*'s Remark "Scope: a fixed honest root": in FlyClient the root of the head comes from the prover.

Corollary 3.18 (Prefix binding and root sensitivity)**.** *Let $b \in \{R, L\}$ and $1 \le j \le n$.*

1. *If $\mathsf{rt} = \mathsf{rt}^b_n$ over $(x_1, \dots, x_n)$ and Verify_b accepts (x, π) at position j, then $x = x_j$ and, for $j \ge 2$, $\mathrm{PrefixRoot}_b(n, j, \pi) = \mathsf{rt}^b_{j-1}$ over $(x_1, \dots, x_{j-1})$, barring a collision: one accepted path binds $x_1, \dots, x_j$ as a prefix of the committed sequence.*

2. *If $(x'_1, \dots, x'_m)$ differs from $(x_1, \dots, x_m)$ at some $i \le m$, then $\mathsf{rt}^b_m(x') \ne \mathsf{rt}^b_m(x)$, barring a collision: changing x_i changes every root rt^b_m with $m \ge i$.*

When x_i is the hash of block i, binding the blocks themselves requires in addition collision resistance of that hash.

Proof. (1) The honest pair (x_j, π^*) is accepted. By Theorem 3.17 the pairs (x, π) and (x_j, π^*) are equal barring a collision, so $x = x_j$ and the fold of Construction 3.16 on π is the fold on π^*, which is rt^b_{j-1} by Lemma 3.15. (2) The two trees have the same shape M^b_m. The honest paths of leaf i in both are accepted against the common root, with $x_i \ne x'_i$; Theorem 3.17 yields a collision. □

Part (1) is ePrint 2019/226, Corollary 3, and part (2) its Corollary 4, stated for both bagging orders and with the collision named.

3.5 Aggregating ranges

A verifier that samples leaves by cumulative weight needs each path to bind the sampled leaf's weight interval, and a verifier that constrains unsampled leaves needs aggregates over their runs; both follow from metadata propagated up the tree (ePrint 2019/226, Section 2 and Section 6).

Definition 3.19 (Aggregating range)**.** An *aggregating range* extends Definition 3.1, after ZIP 221, "Specification", which propagates metadata "by either summing the metadata of both children, or inheriting the metadata of a specific child". It is given by:

1. a set of *node tuples* $v = (v.\mathsf{h}, v.f_1, \dots, v.f_r)$ with $v.\mathsf{h} \in \{0,1\}^\kappa$;

2. a *serialisation* ser from tuples to bit strings that is injective and prefix-free, no value of ser being a proper prefix of another, so that $\mathrm{ser}(\ell) \,\|\, \mathrm{ser}(r)$ determines the pair (ℓ, r);

3. a *leaf map* from a leaf's data to its tuple;

4. for each field f_i one rule: inherit from the left child, inherit from the right child, or sum, in $\mathbb{Z}$ or in $\mathbb{Z}/2^{256}\mathbb{Z}$;

5. the *parent map* $\mathrm{par}(\ell, r)$, whose hash field is $H(\mathrm{ser}(\ell) \,\|\, \mathrm{ser}(r))$, over the children's full serialisations, and whose other fields follow the rules.

A labelling of the shape M^b_n with leaf tuples at the leaves is *consistent* when every internal node is par of its children; its *root* is the root's tuple. Inclusion paths and Verify_b are those of Definition 3.13 with tuples in place of values and par in place of $H(\cdot \,\|\, \cdot)$; Verify_b accepts if and only if the recomputed root tuple equals the given one.

A serialisation of fixed width is prefix-free. Injectivity alone does not suffice: with $\mathrm{ser}(x) = 0$, $\mathrm{ser}(y) = 01$ and $\mathrm{ser}(z) = 10$, the pairs (x, z) and (y, x) have the same concatenation 010.

Lemma 3.20 (Aggregate binding). *Let* ser *be injective and prefix-free, $b \in \{R, L\}$ and $n \geq 1$.*

1. *From two consistent labellings of M_n^b with equal root tuples that differ at some node, a collision of H is computed; barring a collision, equal roots imply agreement on every field of every node.*
2. *From any root tuple, chosen by the adversary, and two distinct accepted paths (leaf tuple and sibling tuples) at one position j, a collision of H is computed: an accepted path binds the full tuple of every node it recomputes and of every sibling it carries.*
3. *For $2 \leq j \leq n$ and the inclusion path of leaf j in a consistent labelling, the folds of Lemma 3.15 run with* par *in place of $H(\cdot \| \cdot)$ yield the root tuple of the consistent labelling of M_{j-1}^b over the first $j-1$ leaf tuples.*

Proof. Unique decoding: if $\mathrm{ser}(\ell) \| \mathrm{ser}(r) = \mathrm{ser}(\ell') \| \mathrm{ser}(r')$, one of $\mathrm{ser}(\ell)$, $\mathrm{ser}(\ell')$ is a prefix of the other, hence they are equal, and $\ell = \ell'$ by injectivity; then $\mathrm{ser}(r) = \mathrm{ser}(r')$ and $r = r'$.

(1) Take a node at which the labellings differ and whose parent they agree on; one exists because they agree at the root. The parent tuples are equal, so their hash fields are, while the child pairs differ; by unique decoding the two hash inputs differ: a collision.

(2) As in the proof of Theorem 3.17, with tuples: at the highest step whose inputs differ, the outputs are equal tuples, hence have equal hash fields, and the inputs are distinct pairs, hence by unique decoding distinct strings: a collision.

(3) In a consistent labelling the tuple of every subtree, a bagging node included, is determined by its leaf tuples through par. By Lemma 3.15 the left siblings are, under R, the peak tuples of the consistent labelling over the first $j-1$ leaves, and so are they under L with $i \leq 2$. Under L with $i \geq 3$, the topmost is the tuple of the bagging node $\mathrm{Bag}_L(P_1, \ldots, P_{i-1})$ of that labelling, and the others are its peak tuples inside P_i. The folds of that lemma, run with par, therefore yield the root tuple of M_{j-1}^b. The engine of (1) and (2) is the *Crypto Guide*, Proposition "Binding of the root" (§"Binary Merkle trees"). □

Lemma 3.21 (Cumulative-weight interval of a leaf). *Let* ser *be injective and prefix-free, let w be a summed field, w_j the w-field of the tuple of leaf j, and W the sum of the w-fields of the left siblings on an accepted path of leaf j.*

1. *Binding, for any root tuple chosen by the adversary: barring a collision, W and w_j are functions of the root, b, n and j. One accepted path therefore binds the interval $[W, W + w_j)$ of leaf j.*
2. *Meaning, for the path of a consistent labelling: $W = \sum_{i<j} w_i$, under either bagging order. For a sum taken modulo 2^{256}, this holds as an identity of integers when the sum of w over all n leaves is below 2^{256}.*

Part (2) is not claimed for a root chosen by the adversary, whose siblings off the path are never recomputed from their children.

Proof. (1) Lemma 3.20(2). (2) In a consistent labelling the summed field of a node is the sum over its leaves, by induction on height; with the total below 2^{256} no reduction occurs. The left siblings partition $\{1, \ldots, j-1\}$ by Lemma 3.15, a property of any binary tree. □

Two instances follow. The difficulty Merkle mountain range of ePrint 2019/226, whose nodes aggregate difficulty and time, is Definition 5.6 in §5.3. The node vector of ZIP 221, whose nodes aggregate work, heights and pool data, is Definition 6.3 in §6.3.

4 The FlyClient security model

This section fixes the model in which the FlyClient theorem is stated: the variable-difficulty backbone with its unit of work, the adversary and persistence; the retarget rule, the attack it excludes and the limit of that exclusion for a sampling verifier; the (c, L) assumption; and the definitions of security and succinctness for chain proofs.

4.1 The variable-difficulty backbone model

The model of this section is the one ePrint 2019/226 adopts in Section 3.1 and Section 3.2, after Garay, Kiayias and Leonardos (the variable-difficulty backbone, CRYPTO 2017), and it is stated as the paper adopts it. Whether a Zcash chain satisfies its hypotheses is classified in §7, not here.

Definition 4.1 (Variable-difficulty backbone model)**.** Fix the security parameter λ (*Math Guide*, §"The security parameter") and a hash length κ. The execution proceeds in rounds $r = 1, 2, \ldots$. The proof-of-work function is a random oracle $H\colon \{0,1\}^* \to \{0,1\}^\kappa$ (*Crypto Guide*, §"The random oracle model"), its outputs read as integers in $[0, 2^\kappa)$. In round r there are ν_r honest and t_r adversarial *active players*, and each active player makes q sequential queries to H per round. A block B declares a *target* $T \in [1, 2^\kappa]$ in its header, and B carries a *valid proof of work* at T if and only if $H(B) < T$. A message broadcast in round r is delivered to every player in round $r + 1$, in an order the adversary chooses. The honest participation sequence $(\nu_r)_r$ is (γ, s)*-respecting*: for every set S of at most s consecutive rounds,

$$\max_{r \in S} \nu_r \le \gamma \min_{r \in S} \nu_r.$$

Each block also records the round in which it was mined, which Definition 4.5 reads.

The paper's paraphrase in Section 3.1 omits "consecutive"; read literally, for $s \ge 3$ it bounds participation over the whole execution.

The paper writes λ for the oracle output length and n_r for the honest player count; here the length is κ, λ being the security parameter, and the count is ν_r, n being a leaf count (§1.2).

Definition 4.2 (Difficulty level and total work)**.** A single query to H meets target T with probability $T/2^\kappa$. The *difficulty level* of T is

$$D(T) = 2^\kappa / T,$$

the expected number of queries per block valid at T. A harder target is a smaller T and a larger $D(T)$: the level is a work-like quantity inversely proportional to the target and is never the target itself. The *total work* of a finite sequence of blocks $B_1, \ldots, B_m$ with declared targets $T_1, \ldots, T_m$ is

$$\omega = \sum_{i=1}^{m} D(T_i),$$

the paper's *cumulative difficulty*. A chain is *heavier* than another when its total work is larger.

The paper also calls a level a "target" in its Definition 7; in this volume D denotes only a level and T only a target. The Zcash analogue, the work of a block computed from its compact target, is defined in the *Consensus Guide*, §"Work, and the best chain", from protocol specification, §"Definition of Work". It is a different normalisation of the same ordering and is not used by the model.

Definition 4.3 (Adversary)**.** Fix μ with $0 < \mu < 1$. The *adversary* is a PPT algorithm (*Math Guide*, §"Algorithms, running time, and PPT"; *Crypto Guide*, §"Adversaries and the security parameter") that controls the t_r adversarial players of each round r, with $t_r < \mu\nu_r$, so that it makes at most $t_r q$ queries to H in round r. It is *rushing*: in each round it receives every honest message of the round before choosing its own, and it orders the delivery of all messages. Its goal is to make a light client accept a chain whose total work is at least that of the honest chain.

The prior knowledge of the verifier is fixed once, in Definition 4.16.

Assumption 4.4 (Persistence and liveness). The backbone parameters $(\mu, \gamma, s, \tau, E, f)$, with τ, E and f the constants of Definition 4.5, are such that there is a *persistence depth* ℓ with, except with probability negligible in λ:

1. (persistence) for all honest players P_1, P_2 and rounds $r_1 \leq r_2$, the chain of P_1 at round r_1 with its last ℓ blocks removed is a prefix of the chain of P_2 at round r_2;
2. (liveness) a transaction given to every honest player for a number of consecutive rounds fixed by the parameters lies in that stable prefix of every honest chain.

ePrint 2019/226, Section 3.1 (p. 7), adopts this as a parameter choice, citing Theorems 26 and 27 of its reference [22], Garay, Kiayias and Leonardos, "The Bitcoin backbone protocol: analysis and applications" (EUROCRYPT 2015), a fixed-difficulty analysis; it names no theorem of its reference [23] (CRYPTO 2017), the source of the variable-difficulty model. No cited theorem establishes the assumption in the model of Definition 4.1.

The tips of honest chains need not coincide: persistence gives agreement only on all but the last ℓ blocks. The paper's sentence that persistence and liveness give "a single chain adopted by all honest full nodes" overstates this; the assumption states the corrected form. The persistence depth ℓ is distinct from the catch parameter k of §5.

4.2 Retargeting and the difficulty-raising attack

Definition 4.5 (Target recalculation with dampening)**.** Fix constants $E \geq 1$ (the *retarget-epoch length*, in blocks), $\tau > 1$ (the *dampening factor*), an initial participation estimate $\nu_0 > 0$ and an initial target T_0. Targets are constant within each *retarget epoch* of E consecutive blocks. At the end of an epoch whose E blocks were mined at target T in Δ rounds, read from the rounds the blocks record, the *participation estimate* is

$$\hat{\nu}(T, \Delta) = \frac{2^\kappa E}{qT\Delta},$$

and the target of the next epoch is

$$\mathrm{Retarget}(T, \Delta) = \begin{cases} T/\tau & \text{if } (\nu_0/\hat{\nu}(T,\Delta))\, T_0 < T/\tau, \\ \tau T & \text{if } (\nu_0/\hat{\nu}(T,\Delta))\, T_0 > \tau T, \\ (\nu_0/\hat{\nu}(T,\Delta))\, T_0 & \text{otherwise,} \end{cases}$$

that is, the value $(\nu_0/\hat{\nu})\, T_0$ clamped to $[T/\tau, \tau T]$. Let $f = q\nu_0 T_0/2^\kappa$, the expected number of blocks per round of ν_0 players at T_0. A chain *respects* the rule when every epoch's target is the image under Retarget of the previous epoch's target and round span.

The unclamped value is the *raw retarget*

$$\frac{\nu_0}{\hat{\nu}(T,\Delta)}\,T_0 = \frac{\nu_0 T_0\, qT\Delta}{2^\kappa E} = \frac{T\Delta f}{E},$$

which restores f blocks per round at the estimated participation. The current target T is an input of the rule; the paper's list of constants omits it. The definition is ePrint 2019/226, Definition 1 (Section 3.1, p. 7), after Garay, Kiayias and Leonardos, Definition 2. The function is written Retarget rather than the paper's $D(n_0, T_0)$, because D is a difficulty level, and ν_0 and T_0 are the initial constants of the cited definition, not the previous-epoch quantities of the gloss in the paper's Definition 1. The retarget epoch is distinct from the upgrade epoch of §6.

Under variable difficulty the total work of a chain, not its length, decides between chains. An adversary may therefore mine fewer blocks at higher levels and use the larger variance of a sum of few heavy terms to exceed a more powerful honest chain: the *difficulty-raising attack* of Bahack (arXiv:1312.7013), recalled in ePrint 2019/226, Section 3.1.

Proposition 4.6 (Success of the difficulty-raising attack)**.** *Model block discovery by exponential clocks* (Consensus Guide, *§"Memoryless clocks and the attribution sequence"*)*: honest queries arrive at rate Λ per unit time and adversarial queries at rate $\Lambda/3$, one third of the honest power; each query succeeds independently with probability $1/D$ at level D, so that the successes of a party querying at rate Λ' at level D form an exponential clock of rate Λ'/D. Fix a time $t_0 > 0$ and let $W = \Lambda t_0$.*

(*i*) *The expected total work of the blocks the honest players find in $[0, t_0]$ is W, whatever their targets, so the expected total work of the honest chain at t_0 is at most W.*

(*ii*) *Suppose the adversary may mine one block at an arbitrary level, in violation of Definition 4.5, and sets its level to $D = W$. Then it finds that block, whose work alone equals W, before t_0 with probability*

$$1 - e^{-1/3} \approx 0.2835,$$

about 28.3 *per cent, a constant independent of λ.*

Proof. (i) An honest query at target T yields a block of work $D(T)$ with probability $T/2^\kappa = 1/D(T)$, hence has expected work 1 whatever the target. By linearity of expectation the expected work found in $[0, t_0]$ equals the expected number of honest queries in $[0, t_0]$, which is $\Lambda t_0 = W$ (*Math Guide*, §"The clock toolkit: exponential clocks, Poisson counts, and the Chernoff method", Theorem "Poisson arrival counts"). The honest chain carries at most the work found.

(ii) At level $D = W$ the adversary's successes form an exponential clock of rate $(\Lambda/3)/W = 1/(3t_0)$. Its first arrival exceeds t_0 with probability $e^{-t_0/(3t_0)} = e^{-1/3}$ (*Math Guide*, same section, Lemma "Tail, mean, and memorylessness"), so it occurs before t_0 with probability $1 - e^{-1/3}$. □

In the discrete query model the same statement holds as an inequality: if $W/3$ is an integer, the adversary's $W/3$ queries at level W all fail with probability $(1-1/W)^{W/3} \le e^{-1/3}$, since $1-x \le e^{-x}$, so it succeeds with probability at least $1 - e^{-1/3}$. Since $\log(1-x) \ge -x - x^2$ for $0 \le x \le 1/2$, $(1-1/W)^{W/3} \ge e^{-1/3-1/(3W)} \ge e^{-1/3}(1-1/(3W))$, so for every such $W \ge 3$ the two probabilities differ by at most $e^{-1/3}/(3W) < 1/W$. The paper states "roughly 28%" without derivation (ePrint 2019/226, Section 3.1, p. 7); the comparison, as in the paper, is with the expected honest work.

Remark 4.7 (The clamp as a defence). Definition 4.5 excludes the attack of Proposition 4.6: a target changes only at epoch ends and by at most the factor τ, so no block may declare an arbitrary level. The paper, citing Garay, Kiayias and Leonardos, states that this rule suffices against difficulty-raising attacks for full nodes, which see every transition and reject a chain that violates it (ePrint 2019/226, Section 3.1, p. 7); that sufficiency is the cited authors' result and is not proved here. A verifier that samples few blocks sees few transitions. The clamp constrains the transitions it does not see only through data that the chain commits to, which §5.3 constructs; no conclusion

about that construction is drawn here. The Zcash retarget rule is a different function, a per-block rule over an averaging window, defined in protocol specification, §"Difficulty adjustment", and described in the *Consensus Guide*, §"Difficulty in motion"; it is not an instance of Definition 4.5.

4.3 The (c, L)-adversary

Fix an honest player and let C be its chain at round r, the *honest chain*. A *fork* of a chain C' against C has *fork point* a, the last block common to C and C'; its blocks are those of C' after a. A block of the fork is *validly mined* if it carries a valid proof of work at its declared target (Definition 4.1), and *invalid* otherwise; its *validly mined work* is the total work of its validly mined blocks.

Assumption 4.8 ((c, L)-adversary). The printed form (ePrint 2019/226, Assumption 2, Section 3.2, p. 8) reads: "There exists no adversary in the variable-difficulty model that respects the target recalculation function from Definition 1 and can, with non-negligible probability, produce a fork that contains more than L blocks such that a c fraction of the difficulty weight in these blocks is honest."

The operative form, for $0 < c < 1$ and $L \geq 1$: for every PPT adversary respecting Definition 4.5 in the model of Definition 4.1, except with probability negligible in λ, every fork it produces that has more than L blocks after its fork point a, and total work after a at least the total work of C after a at the same round, has validly mined work less than c times its total work after a. Such a fork is therefore at least a $(1 - c)$ fraction invalid by weight.

The operative form departs from the printed one in three stated respects. The word "honest" in the printed form means validly mined. The length condition is "more than L blocks", the printed assumption's own wording, where the paper's preceding sentence (Section 3.2, p. 8) says "of length L or longer"; the choice is consistent with checking the last L blocks in full (Remark 4.10). The operative form quantifies only over forks whose work after the fork point reaches that of the honest chain, following the same sentence ("a c fraction of the honest forks weight"): the unrestricted reading fails for every $\mu > 0$, since the adversary's own fully valid private fork of more than L blocks has valid fraction 1, and only forks that reach the honest weight can win under Definition 4.16.

Remark 4.9 (Two readings of c). In Assumption 4.8 the parameter c bounds the validly mined fraction of a competitive long fork's weight. The paper's evaluation (ePrint 2019/226, Section 7.1, p. 20, and Section 7.2) reads it instead as a power ratio ρ, adversarial to honest mining power, so that the adversary's share of total power is $\rho/(1 + \rho)$: one third at $\rho = 1/2$, and $9/19 \approx 0.4737$, about 47.4 per cent, at $\rho = 0.9$, where the paper prints 47.3 per cent by truncation. The identification of c with ρ is not part of the assumption; it needs a bridge from mining power to fork weight, which Remark 4.13 classifies.

Remark 4.10 (Short forks and the checked suffix). Assumption 4.8 constrains only forks of more than L blocks, by design: over a short window the variance of mining lets even an adversary with small power produce a heavier short fork with non-negligible probability (ePrint 2019/226, Section 3.2, p. 8). A verifier relying on the assumption must therefore check in full a suffix of the claimed chain containing at least its last L blocks. At constant difficulty the paper takes $L = \delta n$; under variable difficulty it states the checked suffix as a weight fraction δ (Section 6, p. 17), leaving its relation to the block count L implicit. The size of the suffix under variable difficulty is fixed in §5.4 (Remark 5.24); no conclusion about it is drawn here.

Remark 4.11 (Lemma 1 of the paper). The paper's only derivation of Assumption 4.8 from mining power is its Lemma 1 with Corollary 1 (ePrint 2019/226, pp. 8–9), stated here to register their defect. Lemma 1 as printed: in the constant-difficulty backbone, if X is the number of blocks any adversary mines while the honest chain adopts L blocks, and the adversary's block rate is at most

μ times the honest adoption rate, then for $c > \mu$,

$$\Pr[X \geq cL] \leq e^{L(c-\mu)}(c/\mu)^{-cL}.$$

Corollary 1: for $L = \Theta(\lambda)$ and every μ there is $c < 1$ for which the assumption holds at constant difficulty.

The printed bound is the Chernoff bound of a Poisson variable of mean μL (*Math Guide*, §"The clock toolkit: exponential clocks, Poisson counts, and the Chernoff method", Corollary "Chernoff bound for a Poisson variable"). The proof asserts without proof that such a variable models X, and it switches to the undefined symbols $1 - \delta$ and n, a footnote setting $\mu = 1 - \delta$. Under independent exponential block clocks the count X is negative binomial (same section, Theorem "Negative-binomial arrival count"), and the printed expression is not an upper bound for it in general: the *Math Guide*'s Example "The Poisson expression is not a bound for the arrival count", in the same section, exhibits $\mu = 1/2$, $c = 3/4$, $L = 200$ and $L = 400$ with the exact tail above the printed expression. Lemma 1 and Corollary 1 are therefore unproved as printed; Proposition 4.12 replaces them in the clock model.

The clock model identifies, at constant difficulty, honest adoption and adversarial block discovery after a fork point a with independent exponential clocks started at a, of rates 1 and μ, the rates normalised to the honest adoption rate (*Consensus Guide*, §"Memoryless clocks and the attribution sequence"). For $M \geq 1$ let $X_{a,M}$ be the number of adversarial arrivals strictly before the M-th honest arrival after a. The count $X_{a,M}$ is nondecreasing in M.

Proposition 4.12 (Constant-difficulty bound)**.** *In the clock model, count as validly mined in a fork from a only blocks the adversary finds after a. Let* $0 < \mu < c < 1$ *and*

$$\beta = \Big(\frac{1+c}{1+\mu}\Big)^{1+c}\Big(\frac{\mu}{c}\Big)^{c}, \qquad \theta = \Big(\frac{c(1+\mu)}{\mu(1+c)}\Big)^{c}.$$

Then:

(*i*) *for every fork point a and* $M \geq 1$, $\Pr[X_{a,M} \geq cM] \leq \beta^M$, *and* $\beta < 1$;

(*ii*) *for any set A of candidate fork points, the probability that for some* $a \in A$ *a fork from a exists at some time with more than L blocks, total work after a at least that of the honest chain, and validly mined fraction at least c, is at most*

$$|A|\,\frac{(1+\theta)\,\beta^{L+1}}{1-\beta}.$$

For $|A|$ *polynomial in* λ *and* $L = \Theta(\lambda)$ *this is negligible in* λ, *so Assumption 4.8 holds in the clock model, which recovers Corollary 1 there. The proposition says nothing about the variable-difficulty backbone.*

Proof. (i) By memorylessness the clocks restart at a, and the bound is the *Math Guide*'s Theorem "Chernoff bound for the arrival count", with the law of Theorem "Negative-binomial arrival count" (§"The clock toolkit: exponential clocks, Poisson counts, and the Chernoff method"), which also shows $\beta < 1$. That proof attains β at $e^{t^*} = c(1+\mu)/(\mu(1+c)) > 1$ as $e^{-t^*c}\Phi_1(t^*) = \beta$, where $\Phi_M(t) = (1+\mu-\mu e^t)^{-M}$ is the moment generating function of $X_{a,M}$; note that $\theta = e^{t^*c}$.

(ii) Fix $a \in A$. At constant difficulty the work of a block is constant, so work is block count. Consider a time at which the honest chain has m blocks after a, and a fork from a at that time with L'' blocks, total work at least that of the honest chain and validly mined fraction at least c. Then $L'' \geq \max(m, L+1)$. The validly mined blocks were found by the adversary after a and before the $(m+1)$-th honest arrival, so there are at most $X_{a,m+1}$ of them, and the fraction condition gives

$$X_{a,m+1} \geq cL'' \geq c\max(m, L+1).$$

If $m \le L$, then $X_{a,L+1} \ge X_{a,m+1} \ge c(L+1)$, an event of probability at most β^{L+1} by (i). If $m = M-1$ with $M \ge L+2$, the event is $X_{a,M} \ge cM - c$; the Chernoff method (*Math Guide*, same section, Theorem "The Chernoff method") at t^* bounds its probability by

$$e^{-t^*(cM-c)}\Phi_M(t^*) = e^{t^*c}\left(e^{-t^*c}\Phi_1(t^*)\right)^M = \theta\beta^M.$$

The union bound (*Math Guide*, §"The union bound and a birthday calculation") over these events gives, for the fork point a,

$$\beta^{L+1} + \theta \sum_{M \ge L+2} \beta^M \le \beta^{L+1} + \frac{\theta\beta^{L+2}}{1-\beta} \le \frac{(1+\theta)\,\beta^{L+1}}{1-\beta},$$

and the union over A contributes the factor $|A|$. For negligibility, β and θ are constants with $\beta < 1$, so for $L = \Theta(\lambda)$ the bound is a polynomial in λ times a function exponentially small in λ, which is negligible (*Math Guide*, §"Polynomial, exponential, and negligible functions"). □

The source omits step (ii). The identification of the fork experiment with the clock model is the paper's (proof of Corollary 1) and is not derived from Definition 4.1; the counting hypothesis on validly mined blocks is returned to in §5.2.

Remark 4.13 (Status of the (c, L) assumption). The paper conjectures that (c, L) can be derived from the backbone parameters $(\mu, \gamma, s, \tau, E, f)$ and leaves the derivation out of scope ("not trivial and out of the scope of this paper", ePrint 2019/226, Section 3.2, p. 8). No such derivation exists: in the classes of Definition 1.3 it is an open problem. Proposition 4.12 covers only the constant-difficulty clock model and does not connect the variable-difficulty backbone to Assumption 4.8. Every result of this volume that uses the assumption is conditional on it.

4.4 Security and succinctness of chain proofs

Definition 4.14 (SPV predicate)**.** Let $C = (B_0, \dots, B_\eta)$ be a chain with genesis block B_0 and head B_η. The *SPV predicate* of C is 1 if and only if every block $B_{\eta'}$ carries a valid proof of work at the target that Definition 4.5 assigns to its position, and for $1 \le \eta' \le \eta$ the hash of $B_{\eta'-1}$ is contained in the header of $B_{\eta'}$. The statement a prover proves at round r is that it knows a chain ending in B_η, with total work ω (Definition 4.2), whose SPV predicate is 1.

The definition is ePrint 2019/226, Definition 2 (Section 3.3, p. 9). The paper writes B_n and D for the head and the cumulative difficulty; here η is a height and ω the total work, n and D being reserved.

Definition 4.15 (Equivalence up to the last blocks)**.** Two chains are *equivalent up to the last ℓ blocks*, with ℓ the persistence depth of Assumption 4.4, if removing the last ℓ blocks of each leaves one a prefix of the other. The verifier treats valid proofs of equivalent chains as proofs of one chain and takes the one of greatest total work as its representative; no predicate is evaluated on the last ℓ blocks.

The paper writes k for this depth (Section 3.3, p. 9); here ℓ, k being the catch parameter.

Definition 4.16 (Verifier setting)**.** The verifier is a light client (Definition 2.1) whose only prior knowledge of the chain is the genesis block B_0, and which has oracle access to H to check proofs of work. It receives a set $\mathcal{P}$ of proofs from several provers, at least one of which is honest, and accepts the SPV predicate of the proof whose head carries the greatest total

work, up to Definition 4.15.

The honest-prover hypothesis places eclipse attacks, in which every prover is adversarial, outside the model (ePrint 2019/226, Section 3.3, p. 9).

Definition 4.17 (Security of a chain proof protocol)**.** A *chain proof protocol* is a pair (P, V) of algorithms with oracle access to H: the prover P maps a chain to a proof, and the verifier V maps a set of proofs to an output (B, ω), a head and a total work, or to rejection. The *security game*, for an environment, a PPT adversary $\mathcal{A}$ as in Definition 4.3 and a round r, is as follows.

1. The backbone execution of Definition 4.1 runs to round r, with $\mathcal{A}$ controlling the adversarial players.
2. At the start of round r the verifier receives a set $\mathcal{P}$ containing at least one proof that P computes on the chain of an honest full node; $\mathcal{A}$ chooses the other proofs.
3. The adversary wins if either of the following clauses fails.
   (a) At the end of round r the verifier outputs the SPV predicate of greatest total work for some head B: a pair (B, ω) such that the SPV predicate holds for a chain ending in B of total work ω, and no proof in $\mathcal{P}$ of a chain whose SPV predicate holds carries more total work, up to Definition 4.15.
   (b) Every honest full node at round r holds a chain of which the chain committed by B, with its last ℓ blocks removed, is a prefix.

The protocol is *secure* if for every environment, every PPT $\mathcal{A}$ and every round r the probability that $\mathcal{A}$ wins is negligible in λ.

The definition states both clauses of ePrint 2019/226, Definition 3 (Section 3.3, p. 9). The game idiom is that of the *Crypto Guide*, §"Security as a game"; the multi-prover game is defined here.

Definition 4.18 (Succinctness)**.** A chain proof protocol (P, V) is *succinct* if for every PPT prover and every round r, every proof produced at round r has size $O(\mathrm{polylog}(N))$ bits, where N is the number of blocks of the honest chain, $\mathrm{polylog}(N) = (\log N)^{O(1)}$ as fixed in §1.2, and the asymptotics are in N with λ, c and L fixed.

The definition is ePrint 2019/226, Definition 4 (p. 9), after Kiayias, Miller and Zindros, "Non-interactive proofs of proof-of-work", Definition 4.

Definition 4.19 (Non-interactive proof of proof of work)**.** A *non-interactive proof of proof of work* is a chain proof protocol for the SPV predicate in which each prover sends a single message and no message passes from verifier to prover, and which is secure (Definition 4.17) and succinct (Definition 4.18).

The term names this class of protocols. Kiayias, Miller and Zindros use the same name for their superblock construction, a different member of the class.

5 The FlyClient protocol

This section constructs the protocol of ePrint 2019/226 in the model of §4, each object before its first use: the header chain commitment and the check applied to one sampled block, two lemmas on forks, the difficulty Merkle mountain range with its node checks, the sampling distribution,

the interactive and the non-interactive protocol, the security theorem with its hypotheses and status, and verification after acceptance. Throughout, $H\colon \{0,1\}^* \to \{0,1\}^\kappa$ is the hash function of §3, which is also the proof-of-work oracle of Definition 4.1; log is the natural logarithm and $\mathrm{Adv}_H^{\mathrm{col}}(\mathcal{B})$ the collision advantage of an algorithm $\mathcal{B}$ (*Crypto Guide*, Definition "Collision resistance", §"Security notions: preimage, second-preimage, and collision resistance"), the security parameter left implicit.

5.1 Header chain commitments and per-sample checks

Blocks are written B_0 (the genesis block), B_1, B_2, ..., with B_η the block at height η and $\mathrm{hdr}(B)$ the header of B.

Construction 5.1 (Header chain commitment)**.** Fix an aggregating range (Definition 3.19) under the bagging Bag_R (Definition 3.5) whose leaf map sends a block B to a node tuple $\mathrm{leaf}(B)$ with hash field $\mathrm{leaf}(B).\mathsf{h} = H(\mathrm{hdr}(B))$. Leaf j of every range is $\mathrm{leaf}(B_{j-1})$, so leaf j is the block at height $j-1$. For every height $\eta \geq 1$ let M_η be the range over the η leaves $\mathrm{leaf}(B_0), \dots, \mathrm{leaf}(B_{\eta-1})$ and rt_η its root tuple. The producer of B_η obtains M_η from $M_{\eta-1}$ by Construction 3.10 applied to $\mathrm{leaf}(B_{\eta-1})$, with the parent map par in place of $H(\cdot \| \cdot)$ (M_1 is the single leaf $\mathrm{leaf}(B_0)$), and records the hash field $\mathsf{rt}_\eta.\mathsf{h}$ in the header of B_η. The genesis header records no root. A full node accepts B_η only if its header records $\mathsf{rt}_\eta.\mathsf{h}$.

The construction has the following invariant: the header of B_η commits, through one root, to exactly the blocks strictly before it, and a head B_η commits to $n = \eta$ leaves, n denoting this leaf count below. The root's fields other than its hash field are not recorded; a verifier recomputes the root tuple from its two children, which the hash field binds (Lemma 3.20). The range of ePrint 2019/226, Section 2, is the instance whose tuples carry the hash field alone; the paper's difficulty Merkle mountain range is the instance of Definition 5.6. The one further header field is the paper's only consensus change (Section 4.2). The paper indexes the committing block and its range inconsistently; the invariant fixes one indexing, and the paper's is not carried. In the classes of Definition 1.3, this commitment from the genesis block is the paper's design and no Zcash rule specifies it; the specified commitment is the epoch-scoped instance of "The chain-history commitment" (§6), from which nothing is concluded here.

One sample at index $j \in \{1, \dots, n\}$ against a head B_η must establish three facts: that the sampled block is leaf j under the head's root, that the leaf's tuple is the one its header determines, and that the root in the sampled header commits to exactly the first $j-1$ of the same leaves.

Construction 5.2 (Per-sample check)**.** The inputs are a root tuple rt with leaf count n and an index $1 \leq j \leq n$. The prover sends the header hdr_j of the block at height $j-1$, the tuple v_j of leaf j, and its inclusion path π in M_n (Definition 3.13) with every sibling as a full tuple. The verifier performs the following checks.

(a) *Leaf*: $v_j.\mathsf{h} = H(\mathrm{hdr}_j)$, and every other field of v_j that the leaf map determines from the header alone equals its value computed from hdr_j.

(b) *Inclusion*: $\mathrm{Verify}_R(\mathsf{rt}, n, j, v_j, \pi)$ of Definition 3.13, in the tuple form of Definition 3.19, accepts; the side of each sibling and the path length $\mathrm{depth}_R(n, j)$ are fixed by (j, n).

(c) *Prefix*: for $j \geq 2$, the hash field of $\mathrm{PrefixRoot}_R(n, j, \pi)$ (Construction 3.16), run with par in place of $H(\cdot \| \cdot)$, equals the root recorded in hdr_j; for $j = 1$, hdr_1 equals the genesis header, which the verifier knows.

(d) *Proof of work*: hdr_j carries a valid proof of work at the target T it declares, $H(\mathrm{hdr}_j) < T$ (Definition 4.1).

Any failure rejects.

ePrint 2019/226, Algorithm 2, lists checks (b) and (c); checks (a) and (d) come from its Definition 5 and Section 6.1. Check (b) uses the length rule of Definition 3.13, not that of the paper's Algorithm 4.

Proposition 5.3 (Soundness of the per-sample check)**.** *Let* rt *be any root tuple, possibly chosen by the adversary, with leaf count n, and let $1 \le j \le n$. There is an explicit reduction $\mathcal{B}$ such that, for every adversary $\mathcal{A}$ that outputs samples at position j against (rt, n), the probability that every sample passes Construction 5.2 while one of the following fails is at most $\mathrm{Adv}_H^{\mathrm{col}}(\mathcal{B})$:*

(i) any two passing samples at position j carry the same leaf tuple and the same header, so the header is bound to position j of the sequence that rt *commits;*

(ii) every header-determined field of that leaf tuple is its value computed from that header;

(iii) for $j \ge 2$, the root recorded in that header is the hash field of a prefix-root tuple that is the same for every passing sample at j, hence a function of (rt, n, j); if rt *is the root of a consistent labelling over $v_1, \dots, v_n$, it is the root of the consistent labelling over $v_1, \dots, v_{j-1}$.*

Proof. The reduction $\mathcal{B}$ runs $\mathcal{A}$ and inspects two passing samples (hdr, v, π) and $(\mathrm{hdr}', v', \pi')$ at position j. (i) If $(v, \pi) \neq (v', \pi')$, Lemma 3.20(2), the tuple form of Theorem 3.17 under the prefix-free serialisation of Definition 3.19, yields a collision at the highest step whose inputs differ. If $v = v'$ but $\mathrm{hdr} \neq \mathrm{hdr}'$, check (a) gives $H(\mathrm{hdr}) = v.\mathsf{h} = H(\mathrm{hdr}')$, a collision. (ii) is check (a). (iii) Barring the collisions of (i), $\pi = \pi'$, so the fold of check (c) over the left siblings of π is the same tuple for every passing sample, and check (c) equates its hash field with the recorded root. For a consistent labelling, Lemma 3.15 with Lemma 3.20(3) identifies the fold with the root over $v_1, \dots, v_{j-1}$, as Theorem 3.14 does for plain values. □

5.2 Forks against a committed chain

A claimed chain C *forks* from the honest chain at its fork point, the last common block (§4.3). Its head records the root of a range over the leaf tuples $v'_1, \dots, v'_n$ of its blocks, and C *places* a block B at leaf position j when $v'_j.\mathsf{h} = H(\mathrm{hdr}(B))$.

Lemma 5.4 (Honest blocks after the fork point)**.** *Let the head of C record the hash field of the root of a consistent labelling over $v'_1, \dots, v'_n$. Let B be a block of the honest chain at height η above the fork point, whose header records the hash field of the honest root over the honest leaf tuples $v_1, \dots, v_\eta$, and let C place B at leaf position $j \ge 2$. Suppose that*

1. *the sequences $(v'_1.\mathsf{h}, \dots, v'_{j-1}.\mathsf{h})$ and $(v_1.\mathsf{h}, \dots, v_\eta.\mathsf{h})$ differ, possibly in length; and*
2. *no block header equals $\mathrm{ser}(\ell) \,\|\, \mathrm{ser}(r)$ for node tuples ℓ, r (leaf–internal separation).*

Then from a sample of B that passes Construction 5.2 against the head of C an explicit reduction computes a collision of H, or finds a leaf position $i < j$ of C whose tuple v'_i has hash field $H(\mathrm{ser}(\ell) \,\|\, \mathrm{ser}(r))$ for node tuples ℓ, r. A sample at i passes check (a) only on a header hashing to that value, which by hypothesis 2 is a collision. Consequently, barring a collision, either a sample on B rejects or leaf i does, and that block or leaf belongs, in the analysis of §5.4, to the invalid set of the fork.

Proof. By Proposition 5.3(iii) a passing sample makes the root recorded in $\mathrm{hdr}(B)$ equal to the hash field of the root of the consistent labelling of M_{j-1} over $v'_1, \ldots, v'_{j-1}$. That value is also the hash field of the honest root of M_η. Descend both labellings from their roots, keeping a pair of nodes with equal hash fields whose leaf subsequences differ in their hash fields. At a pair of internal nodes, equal hash fields $H(\mathrm{ser}(\ell) \,\|\, \mathrm{ser}(r)) = H(\mathrm{ser}(\ell') \,\|\, \mathrm{ser}(r'))$ with $(\ell, r) \neq (\ell', r')$ are a collision by unique decoding (proof of Lemma 3.20); otherwise the children agree as tuples, the leaf subsequences of the left children or of the right children differ, and the descent continues there. Two leaves with equal hash fields do not have different hash-field sequences, so the descent ends at a leaf opposite an internal node. If the leaf is honest, its hash field is $H(\mathrm{hdr})$ for the header hdr of an honest block, and $H(\mathrm{hdr}) = H(\mathrm{ser}(\ell) \,\|\, \mathrm{ser}(r))$ with distinct inputs by hypothesis 2 is a collision. If the leaf is a leaf v'_i of C, $i \leq j-1$, opposite an honest internal node with children ℓ, r, then $v'_i.\mathrm{h} = H(\mathrm{ser}(\ell) \,\|\, \mathrm{ser}(r))$, which is the second outcome. The descent takes at most $\max(\lceil \log_2(j-1) \rceil, \lceil \log_2 \eta \rceil)$ steps. □

The lemma is ePrint 2019/226, Section 5.1 with Corollary 4 (here Corollary 3.18), extended to a block placed at a height other than its own, where the two roots have different leaf counts and hypothesis 2 is needed. It concerns honest blocks placed after the fork point and states nothing about blocks the adversary mines; those are the subject of the next lemma.

Lemma 5.5 (No reuse of work across the fork point)**.** *Model H as a random oracle with κ-bit outputs* (Crypto Guide, *§"The random oracle model"*)*, and let at most q_H queries to H be made in the experiment, by $\mathcal{A}$, by the verification of its output, and by the computation of the prefix root at every position of the output fork after its fork point. Call a query $H(x)$* reused *if x is the header of a block B of the output fork after its fork point, x records the hash field of the prefix root of the fork at B's position, and the query precedes the first query whose answer is that recorded value. Then*

$$\Pr[\mathcal{A} \text{ outputs a fork with a block mined by a reused query}] \leq q_H^2\, 2^{-\kappa} + \mathrm{Adv}_H^{\mathrm{col}}(\mathcal{B})$$

for an explicit reduction $\mathcal{B}$. Consequently, except with that probability, only adversarial work done after the fork's prefix is fixed counts against the honest chain.

Proof. By Construction 5.1 the root recorded in a header after the fork point is the hash field of a tuple over the fork's prefix; by Proposition 5.3(iii) that tuple is a function of the fork's root, position and leaf count, barring a collision, which accounts for the term $\mathrm{Adv}_H^{\mathrm{col}}(\mathcal{B})$. Its hash field is the answer of a query, made at the latest during that computation. A reused query fixes a recorded value r before any query has answer r. The first query with answer r has an input not queried before, since a repeated input returns an earlier answer; its answer is uniform on $\{0,1\}^\kappa$ and independent of all earlier queries, so it equals one of the at most q_H values recorded in earlier query inputs with probability at most $q_H\, 2^{-\kappa}$. The union bound over the at most q_H queries (*Math Guide*, §"The union bound and a birthday calculation") gives $q_H^2\, 2^{-\kappa}$. □

The paper asserts the statement in one sentence (ePrint 2019/226, Section 3.2); the proof above is this volume's. The bound is in the number of oracle queries, not in the mining rate.

5.3 The difficulty Merkle mountain range

A sampling verifier sees few difficulty transitions, so the transitions it does not see must be constrained by committed aggregates.

Definition 5.6 (Difficulty Merkle mountain range)**.** The *difficulty Merkle mountain range* (*diffi-*

culty MMR) is the instance of Definition 3.19 whose node v is the tuple

$$v = (v.\mathsf{h},\ v.\mathsf{w},\ v.\mathsf{t},\ v.\mathsf{D}_{\mathsf{start}},\ v.\mathsf{D}_{\mathsf{next}},\ v.\mathsf{n}),$$

with a serialisation that concatenates injective fixed-width encodings of the six fields, hence prefix-free. For a block B whose header declares target T, the leaf map sets

- $v.\mathsf{h} = H(\mathrm{hdr}(B))$;
- $v.\mathsf{w} = v.\mathsf{D}_{\mathsf{start}} = D(T)$, the difficulty level of T (Definition 4.2);
- $v.\mathsf{t}$ the number of rounds between the round recorded by the predecessor of B and the round recorded by B (0 for the genesis block);
- $v.\mathsf{D}_{\mathsf{next}}$ the level of the target that Definition 4.5 assigns to the block after B;
- $v.\mathsf{n} = 1$.

The parent of ℓ and r has hash field $H(\mathrm{ser}(\ell) \parallel \mathrm{ser}(r))$, over both children's complete tuples; its fields w, t and n are the sums of the children's; it inherits $\mathsf{D}_{\mathsf{start}}$ from ℓ and $\mathsf{D}_{\mathsf{next}}$ from r.

The fields h, w, $\mathsf{D}_{\mathsf{start}}$ and n of a leaf are determined by its header alone and are checked by item (a) of Construction 5.2; the fields t and $\mathsf{D}_{\mathsf{next}}$ depend on neighbouring blocks and are constrained only by Construction 5.8. The definition makes three precisions against ePrint 2019/226, Definition 7. The paper writes $H(lc, rc)$ without fixing its input; hashing the children's hash fields alone would leave their other fields unbound. The paper's "timestamp" t is an elapsed time, and its "targets" w, $\mathsf{D}_{\mathsf{start}}$ and $\mathsf{D}_{\mathsf{next}}$ are difficulty levels. The paper's checks name fields t_{start}, t_{end} and D_{end} that its Definition 7 never declares; the declared aggregate t suffices for every timing check below, so the field set is the declared one. The paper's data field is omitted, since no check uses it. No encoding of the fields is fixed by the paper.

Lemma 5.7 (Weight binding for the difficulty MMR)**.** *Let a sample at leaf j pass Construction 5.2 against a root rt with leaf count n, the shape being fixed by n under Bag_R. Except with probability $\mathrm{Adv}_H^{\mathrm{col}}(\mathcal{B})$ for an explicit reduction $\mathcal{B}$:*

1. *the leaf weight $v_j.\mathsf{w}$ is the difficulty level $D(T)$ of the target T at which the header's proof of work is valid;*
2. *the interval $[W, W + v_j.\mathsf{w})$, with W the sum of the fields w of the left siblings on the path, is a function of (rt, n, j).*

Proof. (1) Check (a) sets $v_j.\mathsf{w} = D(T)$ for the declared target T, check (d) validates the proof of work at that T, and Proposition 5.3(i) and (ii) bind the header and its fields to position j. (2) Lemma 3.21(1). Whether W equals the true prefix weight is part (2) of that lemma, which holds for a consistent labelling only and is not claimed here. □

Construction 5.8 (Node checks)**.** Assume, with ePrint 2019/226, Section 6.1, that the retarget-epoch length E of Definition 4.5 and the leaf count n are powers of two, so that the retarget epochs are the perfect subtrees of altitude $\log_2 E$. For every parent p on a sample's path, with children ℓ and r (one of them a carried sibling), the verifier checks:

1. $p = \mathrm{par}(\ell, r)$ under Definition 5.6;

2. $\ell.\mathsf{D}_{\mathsf{next}} = r.\mathsf{D}_{\mathsf{start}}$;
3. for each of ℓ and r, internal consistency:
   (a) a node v of altitude below $\log_2 E$ lies inside one retarget epoch and has $v.\mathsf{w} = v.\mathsf{n} \cdot v.\mathsf{D}_{\mathsf{start}}$, and $v.\mathsf{D}_{\mathsf{next}} = v.\mathsf{D}_{\mathsf{start}}$ unless v contains the last block of its epoch;
   (b) a node v of altitude $\log_2 E$ is one retarget epoch, has $v.\mathsf{w} = E \cdot v.\mathsf{D}_{\mathsf{start}}$, and $v.\mathsf{D}_{\mathsf{next}} = D(\mathrm{Retarget}(2^\kappa / v.\mathsf{D}_{\mathsf{start}}, v.\mathsf{t}))$, the level that Definition 4.5 assigns after an epoch of $v.\mathsf{t}$ rounds at level $v.\mathsf{D}_{\mathsf{start}}$;
   (c) a node v above altitude $\log_2 E$ is *feasible*: its $(v.\mathsf{t}, v.\mathsf{w}, v.\mathsf{D}_{\mathsf{start}}, v.\mathsf{D}_{\mathsf{next}}, v.\mathsf{n})$ is attained by some sequence of clamped retargets (Propositions 5.9 and 5.10).

At the sampled leaf the checks (a) and (d) of Construction 5.2 apply; their effect on w and $\mathsf{D}_{\mathsf{start}}$ is Lemma 5.7. *Weight-position check*: when a leaf is requested at a point x of cumulative weight, $0 \le x < \mathsf{rt.w}$, the returned leaf j satisfies $W \le x < W + v_j.\mathsf{w}$, with W the sum of the fields w of the left siblings on its path; this interval is bound to (rt, n, j) by Lemma 3.21(1). Any failure rejects.

The construction is ePrint 2019/226, Section 6.1 ("Difficulty MMR Verification"); the paper states the weight-position check only informally (Section 2).

Proposition 5.9 (Feasibility envelope). *Let consecutive retarget epochs $0, \dots, z-1$ have levels $D_0 = \mathsf{D}_{\mathsf{start}}, D_1, \dots, D_{z-1}$, and let $D_z = \mathsf{D}_{\mathsf{next}} = \tau^y \mathsf{D}_{\mathsf{start}}$, where every retarget obeys the clamp of Definition 4.5, so that $D_{e+1}/D_e \in [1/\tau, \tau]$. Then:*

1. *$|y| \le z$ and, for every e,*
$$\mathsf{D}_{\mathsf{start}} \max(\tau^{-e}, \tau^{y-z+e}) \le D_e \le \mathsf{D}_{\mathsf{start}} \min(\tau^e, \tau^{y+z-e});$$
2. *for integer y with $z-y$ even both envelopes are attained by clamped sequences, so the total weight $E \sum_{e<z} D_e$ is maximal when the difficulty is raised by τ per epoch and then lowered by τ per epoch to $\mathsf{D}_{\mathsf{next}}$, and minimal when it is lowered first and then raised;*
3. *an epoch at target T that lowers the difficulty maximally (new target τT) lasts $\Delta \ge \tau E/f$ rounds, and one that raises it maximally (new target T/τ) lasts $\Delta \le E/(f\tau)$ rounds.*

Proof. (1) Since $|\log_\tau D_{e+1} - \log_\tau D_e| \le 1$, induction on e from epoch 0 gives $|\log_\tau(D_e/\mathsf{D}_{\mathsf{start}})| \le e$, and from epoch z gives $|\log_\tau(D_e/\mathsf{D}_{\mathsf{start}}) - y| \le z - e$; at $e = 0$ the second bound is $|y| \le z$. (2) For $z - y$ even the pointwise upper envelope has exponent steps $+1$ up to $e = (y+z)/2$ and -1 after, and the lower envelope steps -1 up to $e = (z-y)/2$ and $+1$ after; both are clamped, and the weight is increasing in each D_e. (3) The raw retarget is $T\Delta f/E$ (§4.2); the clamp is reached from above when $T\Delta f/E \ge \tau T$ and from below when $T\Delta f/E \le T/\tau$. A script checks the sums against brute force over exponent sequences, and the two thresholds. □

Proposition 5.10 (Feasibility bounds). *In the setting of Proposition 5.9, let y be an integer with $y \ge 0$ and $z - y$ even. If a node spanning the z epochs is attained by a clamped sequence, then*

1. *$y \le z$ (Proposition 5.9(1));*
2. $\mathsf{w} \le E\, \mathsf{D}_{\mathsf{start}} \dfrac{(\tau+1)\,\tau^{(y+z)/2} - \tau^{1+y} - 1}{\tau - 1}$;

3. $\mathsf{w} \geq E\,\mathsf{D}_{\mathsf{start}}\,\dfrac{\tau^y + \tau - (\tau+1)\,\tau^{-(z-y)/2}}{\tau - 1}$;
4. *if* $y = z$, *then* $\mathsf{t} \leq zE/(f\tau)$.

Proof. Bounds (2) and (3) are the geometric sums of the envelopes of Proposition 5.9(2); at $y = z$ both reduce to $(\tau^z - 1)/(\tau - 1)$. For (4), $y = z$ forces a maximal raise in each of the z epochs, each of at most $E/(f\tau)$ rounds by Proposition 5.9(3). A script confirms that the closed forms equal the brute-force extremes in all 123 cases with $\tau \in \{2, 4, 3/2\}$, $1 \leq z \leq 11$, $0 \leq y \leq z$ and $z - y$ even; at $\tau = 4$, $y = 0$, $z = 2$ they give 5 and 5/4 per unit of E. □

The paper states these checks with k and n in place of y and z, which here are the catch parameter and the leaf count (ePrint 2019/226, Section 6.1, pp. 18–19). Three departures are made. The printed lower bound has exponent $(y+z)/2$ in place of $-(z-y)/2$ and is then negative, -5 against the corrected 5/4 at $\tau = 4$, $y = 0$, $z = 2$. The printed bounds count one level per epoch, whereas the leaf rule $\mathsf{w} = \mathsf{D}_{\mathsf{start}}$ sums per block, so both scale by E. The printed upper summation overcounts by τ^z at $y = z$, where its closed form is right. The paper further requires that t cover the epochs in which the difficulty is lowered, without a formula, and calls the cases of non-integer and negative y "similar" without stating them. The conditions above are necessary, not sufficient: the general feasibility predicate that item 3(c) of Construction 5.8 requires is not given by the paper.

Assumption 5.11 (Transition rewriting). In the model of Definition 4.1, for every adversary $\mathcal{A}$ that with non-negligible probability p outputs a chain C accepted by the node checks on every sample, some of whose difficulty transitions violate Definition 4.5, there is an adversary $\mathcal{A}'$ that makes the same number of oracle queries, respects the retarget rule, and with probability at least $p - \mathsf{negl}(\kappa)$ outputs a chain C' whose validly mined blocks occupy the same positions as those of C.

The assumption is the statement of ePrint 2019/226, Lemma 4, taken as a hypothesis because its proof is a sketch. It handles subtrees whose aggregates are inconsistent with their children, which Lemma 3.21(2) does not cover. The paper's proof strategy is as follows. Let x' be the highest node on a claimed path at which the node checks fail. Every path through x', or carrying it as a sibling, rejects, so barring a collision every leaf of the subtree M' under the parent of x' is unsampleable without rejection. The adversary $\mathcal{A}'$ replaces the transitions inside M' by clamped ones consistent with the aggregates $(\mathsf{w}, \mathsf{t}, \mathsf{D}_{\mathsf{start}}, \mathsf{D}_{\mathsf{next}}, \mathsf{n})$ of the root of M', which is possible because that node passes the feasibility check and the leaves of M', being invalid, need no proof of work; this is repeated until no invalid transition remains. The paper writes "consistency with x_0" where the parent of x' is meant.

Remark 5.12 (Gaps in the rewriting argument). (1) The feasibility step presumes a complete feasibility predicate, which the paper gives only in the simplified setting of Proposition 5.10. (2) Rewriting the leaves of M' changes every later root and hence every later header, so $\mathcal{A}'$ must fix the rewrite before it mines the later valid blocks, and must mine them anew; the paper asserts without a simulation argument that this succeeds with probability at least $p - \mathsf{negl}(\kappa)$. (3) "The same valid blocks" holds for positions, not for header contents. Assumption 5.11 is therefore the paper's claim, not a result of this volume.

5.4 The sampling distribution

Uniform sampling fails on forks near the tip: almost every uniform sample lands before a late fork point, where the two chains agree. If the fork point were known, each sample after it would be invalid with probability at least $1 - c$ under Assumption 4.8.

Construction 5.13 (Dyadic-suffix sampling)**.** At constant difficulty, over a claimed chain of n blocks and with u samples per interval, the verifier draws, for every integer i with $0 \le i < \log_2 n$, u samples uniformly and independently from the last $n/2^i$ blocks, and checks every one of the last $\min(L, u)$ blocks.

The samples do not depend on the prover's answers, so the protocol is one-shot and public-coin (*Crypto Guide*, §"Interactive protocols, proofs, and arguments"). The construction is ePrint 2019/226, Section 5.3, with u in place of the paper's k.

Lemma 5.14 (Dyadic miss probability)**.** *Under Assumption 4.8, at constant difficulty, for every fork point a and every fork from a of more than L blocks whose total work after a is at least that of the honest chain, the probability that the $u \log_2 n$ samples of Construction 5.13 include no invalid block of the fork is at most $((1+c)/2)^u$, except with probability negligible in λ.*

Proof. Let the fork point be a and choose i with $(2^i - 1)n/2^i \le a < (2^{i+1} - 1)n/2^{i+1}$. The interval of the last $n' = n/2^i$ blocks then contains a in its first half, so the fork length $l = n - a$ satisfies $l > n'/2$, and by Assumption 4.8, at constant difficulty, the interval holds at least $(1-c)l > (1-c)n'/2$ invalid blocks. Each of its u independent uniform samples misses them with probability at most $1 - (1-c)/2 = (1+c)/2$, and the samples of the other intervals are discarded. A script checks the bound for $c \in \{0.25, 0.5, 0.9\}$, $u \in \{1, 5, 20\}$ and intervals of 8, 64 and 1,024 blocks. □

The lemma is ePrint 2019/226, Lemma 2. A fork of at most $\min(L, u)$ blocks lies in the checked suffix. When $u < L$, a fork of l blocks with $u < l \le L$ is covered by neither clause, since Assumption 4.8 does not constrain it; the construction is sound for $u \ge L$, or with the last L blocks checked in full.

Definition 5.15 (Continuum model and catch probability)**.** The claimed chain is the interval $[0, 1]$ in relative cumulative work: a block whose cumulative-work interval is $[W, W + w)$ within total committed work ω occupies $[W/\omega, (W + w)/\omega)$; genesis is at 0 and the head at 1. A *fork point* is a point $a \in [0, 1)$. By Assumption 4.8 in its operative form, the adversary designates an *invalid set* $S \subseteq [a, 1]$ of Lebesgue measure at least $(1-c)(1-a)$; every such S is *admissible*. A *sampling density* is a probability density φ on $[0, 1]$ (*Math Guide*, §"Beyond finite spaces: countable additivity, limits, and densities"). When a suffix $[1-\delta, 1]$ is checked in full, a set S that meets it in positive measure is caught with probability 1, and any other S is caught by one draw with probability $\int_S \varphi$. The *single-draw catch probability* at a is the infimum $p_\varphi(a)$ of these probabilities over admissible S. For Q independent draws (*Math Guide*, §"Conditional probability and independence") against a set S fixed before the draws, all miss with probability $(1 - \int_S \varphi)^Q \le (1 - p_\varphi(a))^Q$. The *design problem* is to choose φ to maximise $\min_a p_\varphi(a)$.

Lemma 5.16 (Best response)**.** *Let φ be non-decreasing on an interval $[a, b]$ with $b \le 1$, and let $0 \le m \le b - a$. Among measurable $S \subseteq [a, b]$ of measure m, the integral $\int_S \varphi$ is minimal for $S = [a, a + m]$. Hence against a density non-decreasing on $[a, 1-\delta]$, with $[1-\delta, 1]$ checked in full, an adversary forking at $a \le 1 - \delta/c$ places its valid weight last, and its invalid set is $[a, 1 - (1-a)c]$.*

Proof. With $I = [a, a + m]$, $\int_S \varphi - \int_I \varphi = \int_{S \setminus I} \varphi - \int_{I \setminus S} \varphi$; the two sets have equal measure, $S \setminus I$ lies right of $a + m$ and $I \setminus S$ left of it, and φ is non-decreasing, so the difference is non-negative. A set meeting the suffix in positive measure is caught with certainty, so the adversary's best set lies in $[a, 1-\delta]$ and has the least admissible measure $(1-c)(1-a)$; it ends at $a + (1-c)(1-a) = 1 - (1-a)c \le 1 - \delta$. □

Construction 5.17 (Indifference density)**.** Equal catch probability at every fork point, that is $\Phi(1-c(1-a))-\Phi(a)$ constant in a for an antiderivative Φ of φ, gives on differentiation

$$\varphi(a) = c\,\varphi(1-c(1-a)).$$

The function $\varphi(\xi) = K/(1-\xi)$ solves it. With $K = (1-c)/c$ every invalid segment $[a, 1-c(1-a)]$ has mass $(c-1)\log(c)/c$, which is $\log 2 \approx 0.693147$ at $c = 1/2$. The equation fixes φ only up to a factor periodic in $\log(1-\xi)$ with period $\log(1/c)$. The function is not integrable on $[0,1]$, its mass on $[0, 1-\epsilon]$ being $K\log(1/\epsilon)$, so it is not a density there.

The segment mass is $K(\log(1-a) - \log(c(1-a))) = -K\log c$; a script checks it for $c \in \{0.3, 0.5, 0.9\}$ and $a \in \{0, 0.2, 0.7\}$, and the divergence. The construction is ePrint 2019/226, Section 5.4, p. 16.

Construction 5.18 (FlyClient sampling density)**.** Fix $0 < c < 1$ and $\delta \in (0, c]$; the suffix $[1-\delta, 1]$ is checked in full. The *FlyClient sampling density* is

$$g(\xi) = \begin{cases} \dfrac{1}{(\xi-1)\log\delta} & 0 \le \xi \le 1-\delta, \\ 0 & \text{otherwise,} \end{cases} \qquad G(\xi) = \frac{\log(1-\xi)}{\log\delta} \quad (0 \le \xi \le 1-\delta).$$

Both factors of the denominator are negative, so $g > 0$ on its support, and g is increasing there; G is its distribution function, with $G(1-\delta) = 1$. For $a \le (c-\delta)/c$ the invalid segment $[a, 1-(1-a)c]$ lies in $[0, 1-\delta]$ and has catch probability

$$G(1-(1-a)c) - G(a) = \frac{\log c}{\log\delta},$$

which is $1/k$ at $\delta = c^k$. For $a > (c-\delta)/c$ every admissible invalid set meets the checked suffix in positive measure, and the fork is caught with probability 1.

Normalisation, G and the catch probability are direct integrations. For the last clause, $[a, 1-\delta]$ has measure $1-\delta-a < (1-c)(1-a)$ exactly when $a > 1-\delta/c = (c-\delta)/c$. A script checks normalisation, G and the catch probability $1/k$ for $c \in \{0.25, 0.3, 0.5, 0.75, 0.9\}$ and $2 \le k \le 9$. The density is ePrint 2019/226, Section 5.4, p. 16; Figure 3 shows it.

Definition 5.19 (Sampling map)**.** Let the head's root tuple rt carry committed work rt.w, with leaf count n. The *sampling map* sends a uniform $U \in [0,1]$ to the point $\xi = 1-\delta^U$, the inverse of G at U, so that ξ has density g; it sends ξ to the point $x = \xi \cdot \mathrm{rt.w}$ of cumulative work, and x to the leaf j whose interval $[W, W + v_j.\mathrm{w})$ contains x. That interval is bound to (rt, n, j) by Lemma 3.21(1) and is checked by the weight-position check of Construction 5.8. The *checked suffix* is the set of blocks whose interval meets $[(1-\delta)\,\mathrm{rt.w}, \mathrm{rt.w})$, together with the last L blocks and the head.

Assumption 5.20 (Discretisation). For every chain C output by an adversary respecting Definition 4.5, and for the fork point a of C against the honest chain, if C has more than L blocks after a and total work after a at least that of the honest chain, then one draw of Definition 5.19 returns an invalid block, or a block on which the per-sample check fails, with probability at least $p_g(a)$ of Definition 5.15 minus a function negligible in κ; the Q draws are independent.

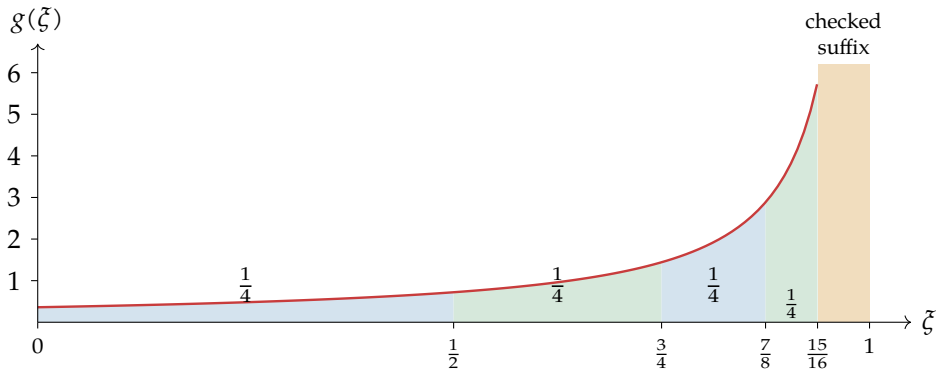


Figure 3: The FlyClient sampling density g of Construction 5.18 for $c = 1/2$ and $k = 4$, so $\delta = 1/16$ and $g(\xi) = 1/((1-\xi)\log 16)$ on $[0, 15/16]$. The amber strip is the suffix $[15/16, 1]$, checked in full, on which g vanishes. The four shaded segments $[1-c^i, 1-c^{i+1}]$, $i = 0, \dots, 3$, are the invalid sets of an adversary forking at $1-c^i$ with its valid weight last; each has mass $1/4$ under g, and together they tile the support, which is the proof of Theorem 5.21(1).

The paper asserts that the continuum analysis "still produces a good distribution for the discrete case" (ePrint 2019/226, Section 5.4) and gives no proof; the hypothesis is named here.

Theorem 5.21 (Optimal sampling distribution)**.** *Let $0 < c < 1$, let $k \geq 1$ be real and $\delta = c^k$. Suppose that the suffix $[1-\delta, 1]$ is checked in full and that the adversary's valid weight after any fork point a is at most a c fraction. Then:*

1. *if k is an integer, for every sampling density φ, $\min_a p_\varphi(a) \leq 1/k$;*
2. *the density g of Construction 5.18 attains $\min_a p_g(a) = 1/k$, against every placement of the invalid weight.*

Proof. (1) Tiling. At the fork points $a_i = 1 - c^i$, $i = 0, \dots, k-1$, the placement with valid weight last is available to the adversary against every φ, and leaves the invalid segment $[1-c^i, 1-c^{i+1}]$, of measure $c^i(1-c) = (1-c)(1-a_i)$, inside $[0, 1-\delta]$. These k segments tile $[0, 1-c^k] = [0, 1-\delta]$, so $1 \geq \sum_i \int_{[1-c^i, 1-c^{i+1}]} \varphi \geq k \min_i p_\varphi(a_i)$. No monotonicity of φ is used. (2) The density g is increasing on $[0, 1-\delta]$, so for $a \leq (c-\delta)/c$ Lemma 5.16 makes $[a, 1-(1-a)c]$ the adversary's best set, whose catch probability is $1/k$ by Construction 5.18; later fork points meet the suffix and are caught with probability 1. A script checks the k tiling masses. □

The theorem is ePrint 2019/226, Theorem 2, pp. 16–17. The paper prints $c \in (0, 1]$, where $\log \delta = 0$ at $c = 1$ and g is undefined; the domain here is $0 < c < 1$. The paper's proof sums over $k+1$ segments, $i = 0, \dots, k$, the last of which lies in the checked suffix; the range is $i = 0, \dots, k-1$. The paper's Lemma 3, a monotonicity lemma, is not needed.

Proposition 5.22 (Sample count)**.** *Let $k > 1$. Then Q independent draws from g all miss an optimally placed invalid set with probability at most $(1-1/k)^Q$, and*

$$(1-1/k)^Q \leq 2^{-\sigma} \iff Q \geq \frac{\sigma}{\log_2(1/(1-1/k))}.$$

At constant difficulty with the last $L = c^k n$ blocks checked, $k = \log_{1/c}(n/L)$ and the least such Q is

> $\Theta(\sigma k)$, *which is* $O(\sigma \log_{1/c} n)$; *for constant* L *it satisfies*
>
> $$\lim_{n\to\infty} \frac{Q}{\sigma \log_c(1/2)\ \log n} = 1.$$
>
> *Here* σ *is the statistical exponent, not the security parameter.*

Proof. The miss bound is Definition 5.15 with Theorem 5.21(2); the equivalence follows on taking logarithms. Since $\log_2(1/(1-1/k)) = 1/(k\log 2) + O(1/k^2)$, the least Q satisfies $Q \sim \sigma k \log 2 = \sigma \log_c(1/2)\log(n/L)$, and $\log(n/L)/\log n \to 1$ for constant L. A script cross-checks the least Q against exact rational arithmetic for $2 \le k \le 9$ at $\sigma = 50$. □

The proposition is ePrint 2019/226, Section 5.4 ("Optimizing the Proof Size"), p. 17.

Remark 5.23 (Domain of the catch parameter). At $k = 1$, $\delta = c$: one draw suffices in the continuum model, and $\log_2(1/(1-1/k))$ is undefined, so Proposition 5.22 is stated for $k > 1$. For $k < 1$, $\delta > c$ and the checked suffix catches every adversary; the paper assumes $k \ge 1$ without loss of generality. For non-integer $k > 1$, Construction 5.18 still gives catch probability $\log c/\log\delta = 1/k$ and hence the sample bound. The tiling of part (1) of Theorem 5.21 needs integral k.

Remark 5.24 (Length of the checked suffix). Under variable difficulty the suffix checked in full is the weight fraction δ, and a sound instantiation makes it contain at least the last L blocks, as Definition 5.19 does (Remark 4.10). The length L is bounded below by the L of the (c, L) assumption in force and trades against Q: a larger L means a larger δ, a smaller k and fewer samples. The asymptotic size bound holds for constant L; within it, L may be chosen numerically. Which (c, L) holds for Zcash is classified in "The deployment boundary" (§7).

Example 5.25 (Sample counts). At $c = 1/2$ and $\sigma = 50$, with $k = \log_2(n/L)$, the least Q with $(1-1/k)^Q \le 2^{-50}$ is as follows.

n	L	k	bound	Q
2^{20}	100	13.36	445.3	446
2^{22}	100	15.36	514.7	515
2^{22}	1,000	12.03	399.5	400
2^{24}	100	17.36	584.0	585
2^{26}	100	19.36	653.4	654

Every k here is non-integer, so Remark 5.23 applies: the counts are sufficient, and their optimality is not claimed.

5.5 The interactive protocol

> **Construction 5.26** (Interactive FlyClient). The parties are a verifier $\mathcal{V}$ in the setting of Definition 4.16, which knows the genesis block B_0, and provers $\mathcal{P}_1, \dots, \mathcal{P}_m$, at least one of them honest. The parameters are c, L, a real $k > 1$, $\delta = c^k$ and Q. The ranges are difficulty MMRs (Definition 5.6) committed by Construction 5.1.
>
> 1. Each prover sends its head B_η, whose header records $\mathsf{rt}_n.\mathsf{h}$ over $B_0, \dots, B_{\eta-1}$, together with the two children of the root tuple (for $n = 1$, the single leaf tuple). The verifier recomputes the root tuple rt with par, rejects unless rt.h equals the recorded value, and reads $n = \mathsf{rt.n}$ and the claimed total work $\omega = \mathsf{rt.w} + D(T_{\text{head}})$, where T_{head} is the target the head declares.
>
> 2. If all heads are identical, $\mathcal{V}$ accepts that head.
>
> 3. Otherwise $\mathcal{V}$ orders the claims by claimed total work, heaviest first, and for each prover

draws Q independent uniforms, maps each to a point of cumulative work by Definition 5.19, and requests the leaf at each point.

4. For each requested point the prover returns the header, the leaf tuple and its inclusion path with full node tuples. The sample must pass Construction 5.2, whose checks (a) and (d) require a proof of work valid at a target that the difficulty-MMR fields declare, and Construction 5.8 on every node of its path, including the weight-position check that the drawn point lies in the returned leaf's interval.

5. The prover sends the checked suffix of Definition 5.19, and $\mathcal{V}$ checks it in full: every block passes the per-sample and node checks, and consecutive blocks, the head included, satisfy the linkage and target conditions of Definition 4.14, the target of each block being the one that the field $\mathsf{D}_{\mathsf{next}}$ of its predecessor's leaf declares.

6. Any failure rejects that prover. Among the surviving claims $\mathcal{V}$ accepts the heaviest, outputting its head and claimed total work; claims equivalent under Definition 4.15 count as one claim.

The construction is ePrint 2019/226, Algorithms 1 and 2 with Definition 5 and Sections 5.3, 5.4 and 6.1. Step 5 belongs to the protocol by Sections 5.3 and 5.4 of the paper, although its Algorithm 1 omits it. For claims of different weight the paper delegates the comparison to the generic verifier of Kiayias, Miller and Zindros ("Non-interactive proofs of proof-of-work"), which checks the longest claim first; under variable difficulty the order is by claimed total work, as in step 3.

5.6 The non-interactive protocol

Construction 5.27 (Non-interactive FlyClient)**.** Construction 5.26 is public-coin and one-round: the only message of $\mathcal{V}$ is its sampling coins, which do not depend on the provers' answers. The non-interactive protocol applies the Fiat–Shamir transform (*Crypto Guide*, §"The Fiat–Shamir transform: from interactive to non-interactive"), the challenge hashing the instance. Let $\mathcal{O}$ be a random oracle (*Crypto Guide*, §"The random oracle model"); the uniforms are $U_i = \mathcal{O}(H(\mathrm{hdr}(B_\eta)), i)$, $i = 1, \dots, Q$, read as points of $[0, 1)$, seeded by the proof-of-work output of the head. Through rt_n the head commits to the whole claimed chain.

1. *Prover.* It computes $U_1, \dots, U_Q$ from its head and the sampled leaves by Definition 5.19, and sends one message: the head with the root's children, for each i the sampled header, leaf tuple and inclusion path with full node tuples, and the checked suffix.

2. *Verifier.* For each proof received it recomputes $U_1, \dots, U_Q$ from the head and runs steps 1 and 4 to 6 of Construction 5.26 on the points they determine.

The paper suggests a standard hash (ePrint 2019/226, Section 6.2) and fixes no domain separation, no map from hash outputs to points and no proof encoding. In the classes of Definition 1.3 the construction is designed but unspecified for Zcash.

Assumption 5.28 (Head-recomputation budget). During one execution the adversary produces at most R heads that carry a valid proof of work.

Assumption 4.8 bounds the validly mined weight of a fork, not the number of head recomputations, and does not imply Assumption 5.28. The paper's budget $R = cn$ ("our security assumption gives a concrete bound on the number of PoW puzzles the adversary can solve, which is cn", ePrint 2019/226, Section 6.2) is asserted, not derived.

Proposition 5.29 (Grinding bound)**.** *Assume Assumptions 5.28, 5.20, 5.11 and 4.8, let $\mathcal{O}$ be a random oracle, let H be the random oracle of Definition 4.1, with κ-bit outputs, and let $k > 1$. Let at most $q_{\mathcal{O}}$ queries to $\mathcal{O}$ and q_H queries to H be made. The probability that an adversary makes Construction 5.27 accept a claimed chain whose fork is not caught is at most*

$$R\,(1 - 1/k)^Q + q_{\mathcal{O}}\, q_H\, 2^{-\kappa} + \mathrm{Adv}_H^{\mathrm{col}}(\mathcal{B}) + \mathsf{negl}(\kappa)$$

for an explicit reduction $\mathcal{B}$. Hence

$$Q > \frac{\sigma + \log_2 R}{\log_2(1/(1 - 1/k))}$$

suffices for failure probability below $2^{-\sigma}$ plus negligible terms. This Q is $\Theta((\sigma + \log R)\,k)$ and, at constant difficulty with constant L, $\Theta((\sigma + \log R)\log_{1/c} n)$; it exceeds the interactive count of Proposition 5.22 by the $\log_2 R$ bits of grinding.

Proof. Each head with a valid proof of work fixes, through rt_n, the claimed chain and its invalid set before its challenge $\mathcal{O}(H(\mathrm{hdr}(B_\eta)), \cdot)$ is known, barring a collision of H (Proposition 5.3). The challenge of a header cannot be computed before $H(\mathrm{hdr})$ is queried, and that same query decides validity. Since $\mathcal{O}$ is independent of H, the challenges of the at most R valid heads are uniform and independent of which heads are valid, except when $\mathcal{O}$ was queried at a value before it arose as an H-output, which has probability at most $q_{\mathcal{O}}\, q_H\, 2^{-\kappa}$. So by Assumption 5.20 and Theorem 5.21 the Q samples of each valid head all miss with probability at most $(1 - 1/k)^Q$ plus a negligible term. A head without a valid proof of work fails step 5 of Construction 5.26. By Assumption 5.28 at most R heads qualify, and the union bound (*Math Guide*, §"The union bound and a birthday calculation") gives the bound. The threshold follows on taking logarithms, and its order as in the proof of Proposition 5.22. □

The bound is proved here because the *Crypto Guide* proves Fiat–Shamir security only for Sigma protocols (§"The Fiat–Shamir transform: from interactive to non-interactive"); the argument above uses only that the challenge is a random-oracle image of the proof-of-work output of a header committing to the whole instance. Seeded by the header itself, the challenge could be evaluated before the proof of work is attempted, and an adversary could mine only heads whose samples miss.

Example 5.30 (Grinding cost)**.** At $c = 1/2$, $\sigma = 50$, $L = 100$ and $n = 2^{22}$, $k = \log_2(n/L) \approx 15.36$. Proposition 5.22 gives $Q \geq 514.7$, so 515 samples. With the paper's $R = cn = 2^{21}$ the budget adds 21 bits, and Proposition 5.29 gives $Q > 730.8$, so 731 samples. At the other (n, L) of Example 5.25, namely $(2^{20}, 100)$, $(2^{22}, 1{,}000)$, $(2^{24}, 100)$ and $(2^{26}, 100)$, the non-interactive counts are 615, 568, 853 and 981.

Proposition 5.31 (Transferability)**.** *A proof of Construction 5.27 is publicly verifiable: its verification uses only the proof, the genesis block and the public oracles, so any party can relay it unchanged, and it convinces every verifier that the original convinces. For a given head the sampled points are deterministic; under a canonical proof encoding there is therefore one proof per head, which one party can compute for every verifier.*

Proof. The verifier's decision is a deterministic function of the proof and public data, and the points are a function of the head. □

The paper's statement that "there only exists a single valid non-interactive proof" (ePrint 2019/226, Section 8.1) holds only under a canonical encoding, which no ZIP specifies. ZIP 221,

"Security and Privacy Considerations" records that, because FlyClient proofs are non-interactive and publicly verifiable, "they could be shared among many light clients after the initial server interaction".

5.7 The security theorem

Theorem 5.32 (FlyClient security and succinctness)**.** *Assume:*

1. $0 < c < 1$, *a real* $k > 1$ *and* $\delta = c^k$;
2. *the variable-difficulty backbone model* (*Definition 4.1*), *in which every adversary is a* (c, L)-*adversary* (*Assumption 4.8*);
3. H *collision resistant and, as in Definition 4.1, a random oracle, and the sampling oracle* $\mathcal{O}$ *a random oracle;*
4. *Assumptions 5.28, 5.20, 5.11 and 4.4;*
5. *a verifier that knows only the genesis block, and at least one honest prover* (*Definition 4.16*);
6. *no block header equals* $\mathrm{ser}(\ell) \parallel \mathrm{ser}(r)$ *for node tuples* ℓ, r (*leaf–internal separation, hypothesis 2 of Lemma 5.4*).

Then an adversary wins the game of Definition 4.17 against Construction 5.27 with probability at most $R\,(1 - 1/k)^Q$ *plus negligible terms, which is at most* $2^{-\sigma}$ *plus negligible terms when* Q *satisfies Proposition 5.29. Proofs have*

$$O(L_\delta + (\sigma + \log R)\, k \log n)$$

headers and hashes, where L_δ *is the number of blocks of the checked suffix. The construction is succinct* (*Definition 4.18*) *whenever* L_δ, k *and* $\log R$ *are* $O(\mathrm{polylog}(n))$, *in particular at constant difficulty with constant* L *and* $\delta = L/n$, *where* $L_\delta = L$ *and* $k = \log_{1/c}(n/L)$; *there, with the paper's* $R = cn$, *the size is* $O(L + (\sigma + \log n) \log_{1/c}(n) \log n)$.

Proof structure. The proof composes the obligations O1 to O6 of Remark 5.33, following the five steps of ePrint 2019/226, Section 6.3. (0) The honest prover's proof passes every check: its labelling is consistent, its transitions are clamped, and Lemma 3.21(2) makes each weight interval its true prefix weight. (1) Assumption 5.11 replaces an adversary with invalid transitions by one that respects the retarget rule, to which Assumption 4.8 applies (O3). (2) O1 pins every sampled answer to one committed chain; O2 makes every block after the fork point adversarial, or places a leaf of the fork in its invalid set, and makes its validly mined weight post-fork; O3 binds each sampled leaf's weight interval to the root, while the agreement of that interval with the chain's true prefix weight rests on the node checks and Assumption 5.11. (3) O4: a fork heavier than the honest chain and longer than L blocks is caught by one draw with probability at least $1/k$, and a fork within the checked suffix with certainty. (4) O5: the Fiat–Shamir step with the grinding bound. (5) O6: the size count. Agreement of honest full nodes on the prefix up to the last ℓ blocks, the second clause of Definition 4.17, is Assumption 4.4. □

The theorem is ePrint 2019/226, Theorem 1 (p. 10, restated p. 20); ZIP 221, "Security and Privacy Considerations" restates its honest-connection and genesis hypotheses. The paper states the size as $O(L + \lambda \log_{1/c}(n) \log n)$. That equals the constant-difficulty bound above with $R = cn$ only when $\log n = O(\lambda)$, for instance for n polynomial in λ; under the convention that n grows with σ fixed, the theorem states the $\log R$ form and no stronger bound.

Remark 5.33 (Proof obligations). Each obligation is listed with the results that discharge it.

O1 Binding: Theorem 3.17, Corollary 3.18, Lemma 3.20 and Proposition 5.3.

O2 Fork structure: Lemmas 5.4 and 5.5.

O3 Weight binding and transitions: Lemma 5.7, Construction 5.8 and, for subtrees whose aggregates are inconsistent, Assumption 5.11.

O4 Catch probability and sample count: Theorem 5.21, Definition 5.19, Assumption 5.20 and Proposition 5.22.

O5 Grinding: Proposition 5.29, which fixes the non-interactive Q.

O6 Size: Corollary 5.34.

Corollary 5.34 (Proof size). *For the interactive protocol with Q from Proposition 5.22, constant L and constant difficulty, a proof has size $O(\sigma \log n \log_{1/c} n + L)$: $Q = \Theta(\sigma \log_{1/c} n)$ samples, each a header and a path of at most $\lceil \log_2 n \rceil$ nodes under Bag_R (Lemma 3.7), and the L suffix blocks. For the non-interactive protocol, with Q from Proposition 5.29, σ is replaced by $\sigma + \log_2 R$.*

Proof. Multiply Q by the cost of one sample. The suffix blocks form a contiguous run of leaves ending at leaf n, so the union of their paths has $O(L + \log n)$ nodes and is sent once. □

The corollary is ePrint 2019/226, Corollary 2 (Section 5.4), which states Θ; the matching lower bound is not proved here.

Remark 5.35 (Status of the security theorem). Theorem 5.32 is the paper's conditional claim, not an independently completed proof. Deriving the (c, L) assumption from the backbone parameters is open (Remark 4.13); the paper's Lemma 1 is invalid as printed (Remark 4.11); its Lemma 4 is a sketch with recorded gaps and enters as Assumption 5.11; the step from the continuum to blocks is Assumption 5.20; and the budget $R = cn$ is asserted. Proved in this volume are binding (O1), fork structure (O2) under leaf–internal separation, the binding of sampled weight intervals to the root, optimality, the sample count and the grinding bound.

5.8 Verification after acceptance

Proposition 5.36 (Inclusion after acceptance). *Let a head B_η with root tuple rt_n be accepted. Membership of the block at height $j - 1$ in the accepted chain is established by one inclusion path of leaf j against rt_n, and membership of a transaction in that block by one further Merkle path against the transaction root its header carries; no intermediate header is needed. Both are binding barring a collision.*

Proof. For the block, Proposition 5.3(i); for the transaction path, *Crypto Guide*, §"Soundness: forging a path implies a collision". □

The proposition is ePrint 2019/226, Sections 1 and 4.2.

Proposition 5.37 (Verification against a fresh head). *A client may run the protocol against the freshest heads and store one recent block B^* of its previously accepted chain lying more than 2ℓ blocks below that chain's head. Let the height of B^* be less than that of the newly accepted head minus ℓ. Under the hypotheses of Theorem 5.32, except with its soundness error, the accepted head's committed range contains B^* at its height, which the client checks by one inclusion path (Proposition 5.36); a head is rejected if a sample exposes an invalid block.*

Proof. By Theorem 5.32 both accepted chains are equivalent up to their last ℓ blocks to chains of honest full nodes. A head B_η commits to heights 0 to $\eta - 1$, so clause (b) of Definition 4.17 at the

earlier round places B^* in every honest chain of that round with its last ℓ blocks removed, and Assumption 4.4 places it in every later honest chain. The height of B^* is below that of the new head minus ℓ, so B^* lies in the part of the new committed chain that clause (b) makes a prefix of an honest chain. □

The paper states without qualification that the range of a fresh head "cannot contain invalid blocks and it must contain all stable blocks" (ePrint 2019/226, Section 4.2, "Unstable Blocks"); that sentence is not carried.

> **Proposition 5.38** (Subchain proofs, the paper's claim). *Assume the hypotheses of Theorem 5.32. Suppose a verifier accepted a valid proof for a chain with head B_η when the honest chain had that head, and later, when the honest chain has head $B_{\eta'}$, $\eta' > \eta$, receives a subproof for the blocks from B_η to $B_{\eta'}$, sampled as an instance of the protocol with B_η in place of genesis, together with one inclusion path showing that B_η lies in the range committed by $B_{\eta'}$. Then, except with the soundness error of Theorem 5.32 for the instance from B_η, it accepts no chain that a verifier given a full proof for height η' would reject. The instance from B_η, which samples only leaves after B_η of a range committed from B_0, is the paper's design (ePrint 2019/226, Section 8.2) and is not constructed here.*

Proof idea. This is ePrint 2019/226, Theorem 3 (Section 8.2), with the paper's proof idea. A fork after B_η faces a fresh instance of the protocol with B_η as genesis. A fork before B_η fails the inclusion path, since the verifier's B_η is not in the adversary's range, barring a collision (Proposition 5.36). □

The proposition concerns one commitment range from the genesis block.

6 The chain-history commitment

This section states the commitment that Zcash block headers carry, each rule cited to the ZIP or the protocol specification section that fixes it, and each object before its first use: the upgrade epochs over which it ranges, the node vector of its Merkle mountain range, the hashing of the tree and its root commitment, the authorising-data root and the block commitment hash of ZIP 244 that wraps the root from NU5, the header consensus rules, and the range of blocks to which each header commits.

6.1 The deployed chain commitment

ZIP 221 (Final), "FlyClient - Consensus-Layer Changes", changes the semantics of the block header and the consensus rules from Heartwood. The 32-byte header field that held the root of the Sapling note commitment tree instead commits to the root of a Merkle mountain range (Definition 3.1; ZIP 221, "Terminology": a binary hash tree that admits appends of new leaves "without changing the value of existing nodes") whose nodes carry aggregated metadata. Its leaves are the blocks from the last activation height below the committing block up to the parent of that block: the *epoch tree* of Definition 6.2. ZIP 221 thus deploys the invariant of Construction 5.1 restricted to one *upgrade epoch*, the heights from one activation height up to the next (Definition 6.1), and not from the genesis block. The tree and its node fields 1 to 14 are ZIP 221's; fields 15 to 17 are ZIP 258's (Draft, in force from NU6.3 by Definition 1.2); and from NU5 the root is committed as the first input of the *block commitment hash*, a personalised hash of the root together with the block's authorising-data root (ZIP 244; "The block commitment hash", §6.6). Seven node fields serve reasoning about difficulty without matching the node of Definition 5.6 (ePrint 2019/226, Definition 7) field for field (Remark 6.4). Each term of this paragraph carries here only its defining clause; its definition is in the subsection named.

The purpose of the commitment, after ZIP 221, "Motivation", is to let a light client verify, with a number of headers logarithmic rather than linear in the chain length and correctly with high probability, the validity of a chain received from a full node, the inclusion of a block in it, and metadata of any block or range of blocks; the commitment also admits subtree proofs, which check only blocks later than the most recently verified one. Transaction inclusion within a block is left to ZIP 307 (Draft). No claim of this paragraph is proved in this section; the protocol that would realise it is that of "The FlyClient protocol" (§5), and its relation to the deployed commitment is classified in "The deployment boundary" (§7).

6.2 Upgrade epochs and epoch trees

Definition 6.1 (Upgrade epoch)**.** Let $0 = A_0 < A_1 < A_2 < \cdots$ be the distinct activation heights of the network upgrades of a chain (ZIP 200, "Activation mechanism"), height 0 counted as an activation height (ZIP 221, "Specification"). The named upgrades and their order are those of Definition 1.1; the list of activation heights includes those of upgrades that definition does not name. The *upgrade epoch* beginning at A_i is the interval of heights $[A_i, A_{i+1})$, unbounded for the last activation height. For $i \geq 1$ its blocks are validated under one consensus rule set and one *consensus branch ID*, the 32-bit identifier of the last upgrade that activates at A_i (ZIP 200, "Specification"; *Ironwood Guide*, §"The transaction format"). The upgrade epoch is distinct, by name and by object, from the retarget epoch of Definition 4.5; the word "epoch" is not used alone. The Heartwood activation height is written A_H, and B_{A_H} is the Heartwood activation block.

Definition 6.2 (Epoch tree)**.** For a block height $\eta > 0$, the *preceding activation height* $\alpha(\eta)$ is the largest activation height $A_i < \eta$, the "preceding network upgrade height" of ZIP 221, "Specification"; $\alpha(0)$ is undefined. For $\eta > A_H$ the *epoch tree* $\mathcal{T}_\eta$ is the Merkle mountain range M_n^L under the left fold Bag_L (Definition 3.5), bagging nodes included as internal nodes, with the

$$n = \eta - \alpha(\eta) \geq 1$$

leaves $B_{\alpha(\eta)}, \ldots, B_{\eta-1}$ in height order: leaf j is $B_{\alpha(\eta)+j-1}$. Since $\alpha(\eta)$ is the last activation height below η, all its leaves lie in the one upgrade epoch beginning at $\alpha(\eta)$. Its node contents are fixed in "The node vector" (§6.3) and its hashing in "The epoch tree and its root commitment" (§6.4). No epoch tree is defined for $\eta \leq A_H$.

Which header commits to $\mathcal{T}_\eta$ is not part of Definition 6.2: it is Definition 6.22 in "The header consensus rules" (§6.7), and its consequences for the committed ranges are stated in "The committed range" (§6.8).

6.3 The node vector

Definition 6.3 (Node vector)**.** The epoch tree $\mathcal{T}_\eta$ is an instance of Definition 3.19 under Bag_L. Every node v of it, leaf, internal node, bagging node or root, carries one *node vector* of fields, serialised canonically by the map ser of Definition 6.5; that serialisation is used whenever a parent commitment is formed. The fields are numbered as in ZIP 221, "Tree Node specification". For a block $B_{\eta'}$ with header hdr', the *leaf map* sets both members of each earliest/latest pair from $B_{\eta'}$ itself:

(1) field $\mathsf{hashSubtreeCommitment}$ is $\mathsf{BlockHash}(\mathrm{hdr}')$ (Definition 2.3), in internal byte order

and unpersonalised;

(2), (3) fields nEarliestTimestamp and nLatestTimestamp are the field nTime of hdr';

(4), (5) fields nEarliestTargetBits and nLatestTargetBits are the field nBits of hdr' (the semantics of nTime and nBits: *Consensus Guide*, §"Difficulty in motion");

(6), (7) fields hashEarliestSaplingRoot and hashLatestSaplingRoot are $\mathsf{LEBS2OSP}_{256}(\mathsf{rt})$ for rt the root of the Sapling note commitment tree of the final Sapling treestate of $B_{\eta'}$ (protocol specification, §"Block Header Encoding and Consensus"), an opaque 32-byte value in this volume;

(8) field nSubTreeTotalWork is the work $\lfloor 2^{256}/(\mathsf{ToTarget}(\mathsf{nBits})+1)\rfloor$ of $B_{\eta'}$ (*Consensus Guide*, §"Work, and the best chain", Definition "Work of a block");

(9), (10) fields nEarliestHeight and nLatestHeight are the height η';

(11) field nSaplingTxCount is the number of transactions of $B_{\eta'}$ in which vSpendsSapling or vOutputsSapling is non-empty;

(12), (13) fields hashEarliestOrchardRoot and hashLatestOrchardRoot are the Orchard-pool anchor of $B_{\eta'}$, the root of the Orchard pool's note commitment tree in the final treestate of $B_{\eta'}$ (*Ironwood Guide*, §"Anchors", Definition "Treestate and anchor");

(14) field nOrchardTxCount is the number of transactions of $B_{\eta'}$ in which vActionsOrchard is non-empty (*Ironwood Guide*, §"The transaction format");

(15)–(17) fields hashEarliestIronwoodRoot, hashLatestIronwoodRoot and nIronwoodTxCount are the Ironwood-pool anchor of $B_{\eta'}$ and the number of its transactions in which vActionsIronwood is non-empty, computed as fields 12 to 14.

The *parent rule*: an internal node with left child ℓ and right child r takes each earliest field (2, 4, 6, 9, 12, 15) from ℓ and each latest field (3, 5, 7, 10, 13, 16) from r; it sums the additive fields, field 8 modulo 2^{256} and the per-pool counts 11, 14 and 17 in $\mathbb{Z}$; and it sets field 1 by Construction 6.10, the personalised hash of $\mathsf{ser}(\ell) \parallel \mathsf{ser}(r)$. Fields 12 to 14 are ZIP 221's own, added at NU5 (ZIP 252). Fields 15 to 17 are ZIP 258's, "Changes to ZIP 221" (Draft), whose rules apply from NU6.3 activation (Definition 1.2).

For field 8, ZIP 221, "Tree Node specification", states that computations modulo 2^{256} are fine because cumulative chain work is assumed to be at most 2^{256}, an aggregate effort of 2^{256} or more being infeasible; under that assumption, with the total below 2^{256}, the field of a node is the integer sum of the work of its leaves (Lemma 3.21(2)). Table 2 summarises the definition.

Remark 6.4 (The FlyClient fields). ZIP 221, "FlyClient Requirements and Recommendations", designates seven fields, 2 to 5 and 8 to 10, which "enable FlyClient in the variable-difficulty model" by letting a light client reason about the application of the difficulty adjustment over a range of blocks. They serve the purpose of Definition 5.6 without matching it field for field. The leaf count of a node, the field n of the difficulty MMR, is nLatestHeight − nEarliestHeight + 1 (Lemma 6.6). The field w of the difficulty MMR sums difficulty levels (Definition 4.2), whereas nSubTreeTotalWork sums work, a different normalisation. The field nEarliestTargetBits is a compact target, ordered inversely to $\mathsf{D}_{\mathsf{start}}$. No field carries the elapsed time t; the timestamps of the first and last block stand in its place. No field carries $\mathsf{D}_{\mathsf{next}}$: the next target of a Zcash block is a function of an averaging window of preceding blocks (*Consensus Guide*, §"Difficulty in motion"), which the aggregates of a node do not determine.

Fields 6, 7 and 11 to 14 are ZIP 221's "Non-FlyClient Commitments", chosen "primarily to improve light client security guarantees", with no counterpart in ePrint 2019/226. Field 1 "forms

	Field	Type	Leaf value for $B_{\eta'}$	Internal node	ZIP
1	hashSubtreeCommitment	char[32]	BlockHash(hdr$'$)	hash	221
2	nEarliestTimestamp	uint32	nTime	inherit left	221
3	nLatestTimestamp	uint32	nTime	inherit right	221
4	nEarliestTargetBits	uint32	nBits	inherit left	221
5	nLatestTargetBits	uint32	nBits	inherit right	221
6	hashEarliestSaplingRoot	char[32]	final Sapling root	inherit left	221
7	hashLatestSaplingRoot	char[32]	final Sapling root	inherit right	221
8	nSubTreeTotalWork	uint256	work of the block	sum mod 2^{256}	221
9	nEarliestHeight	CompactSize	height η'	inherit left	221
10	nLatestHeight	CompactSize	height η'	inherit right	221
11	nSaplingTxCount	CompactSize	transactions with Sapling spends or outputs	sum	221
12	hashEarliestOrchardRoot	char[32]	Orchard-pool anchor	inherit left	221
13	hashLatestOrchardRoot	char[32]	Orchard-pool anchor	inherit right	221
14	nOrchardTxCount	CompactSize	transactions with Orchard-pool Actions	sum	221
15	hashEarliestIronwoodRoot	char[32]	Ironwood-pool anchor	inherit left	258
16	hashLatestIronwoodRoot	char[32]	Ironwood-pool anchor	inherit right	258
17	nIronwoodTxCount	CompactSize	transactions with Ironwood-pool Actions	sum	258

Table 2: The fields of the node vector of Definition 6.3: serialised type, value at the leaf for the block $B_{\eta'}$ with header hdr$'$, rule at an internal node ("hash" is Construction 6.10), and source: ZIP 221, "Tree Node specification", for fields 1 to 14, and ZIP 258, "Changes to ZIP 221", for fields 15 to 17.

the structure of the commitment tree" (ZIP 221, "Tree nodes"), and fields 15 to 17 are the ZIP 258 counterparts for the Ironwood pool. The earliest roots let a client check the treestate transitions over a range of blocks without recomputing the roots from genesis. ZIP 221, "Non-FlyClient Commitments", states that hashLatestSaplingRoot serves the purpose of the former header field hashFinalSaplingRoot, one block delayed (Proposition 6.23). The per-pool counts follow the rule of the same section that a new shielded pool adds a count of its transactions, applied at NU5 (fields 12 to 14) and at NU6.3 (fields 15 to 17, ZIP 258).

Definition 6.5 (Field vector by upgrade epoch)**.** The fields are encoded as follows: a char[32] field as its 32 bytes; a uint32 or uint256 field little-endian, as $\mathsf{I2LEOSP}_{32}$ or $\mathsf{I2LEOSP}_{256}$ (protocol specification, §"Integers, Bit Sequences, and Endianness"); a height or count as its CompactSize encoding in minimal form (Definition 2.2); and each pool-root field as $\mathsf{LEBS2OSP}_{256}$ of the root, as the specification encodes the Orchard anchor (protocol specification, §"Transaction Encoding and Consensus"). For a node v, $\mathrm{ser}(v)$ is the concatenation of the encodings of its fields in field order. The *field vector* of an upgrade epoch is:

1. fields 1 to 11 in the Heartwood and Canopy upgrade epochs;
2. fields 1 to 14 in the NU5, NU6, NU6.1 and NU6.2 upgrade epochs;
3. fields 1 to 17 in the NU6.3 and NU7 upgrade epochs.

The vectors are named by their fields. ZIP 221, "Tree Node specification", omits fields 12 to 14 "before NU5 activation", and ZIP 258, "Changes to ZIP 221", omits fields 15 to 17 "before NU6.3 activation"; neither says whether the height of a leaf's block or that of the committing block governs, and the two differ only for the tree committed by an activation block. *Adopted reading*, resolving that ambiguity: every node of $\mathcal{T}_\eta$ uses the field vector of the upgrade epoch beginning at $\alpha(\eta)$, the upgrade epoch of its leaves.

The reading is adopted for the following reason. Under the other reading the tree $\mathcal{T}_A$ committed by B_A, at an activation height A that adds fields, would serialise every node of the previous upgrade epoch with the added fields; every internal node would change value, and $\mathcal{T}_A$ would not be an append of $\mathcal{T}_{A-1}$, contrary to the property by which ZIP 221, "Terminology", defines a Merkle mountain range. The ZIP's pseudocode, which takes a leaf's Orchard fields from the leaf's own block and serialises them only when present, is consistent with the reading. Two consequences rest on it: $\mathcal{T}_A$ uses the previous upgrade epoch's vector, and $\mathcal{T}_{A+1}$ is the first tree with the new one; under the other reading $\mathcal{T}_A$ would carry the new vector. For NU7, no ZIP among the sources of NU7 consensus changes listed in ZIP 259 (Draft), "NU7 consensus changes", amends ZIP 221 or ZIP 244, so the nodes of the trees of the NU7 upgrade epoch carry fields 1 to 17 under either reading.

The fixed-width fields occupy 144, 208 and 272 bytes in the three vectors, and the three, four and five CompactSize fields 1 to 9 bytes each, so a serialised node has 147 to 171, 212 to 244 and 277 to 317 bytes, the ranges that ZIP 221 and ZIP 258 state. A toy chain computed by script illustrates the reading, with an activation block that commits a tree of the previous vector and the next block the first to commit a tree of the new one; it establishes nothing.

Lemma 6.6 (Leaf count and shape). *Every node v of $\mathcal{T}_\eta$, bagging nodes included, spans a run of consecutive leaves $B_{\eta_1}, \dots, B_{\eta_2}$ with v.nEarliestHeight $= \eta_1$ and v.nLatestHeight $= \eta_2$, so its leaf count is* nLatestHeight − nEarliestHeight + 1. *The node v roots a perfect subtree of $\mathcal{T}_\eta$ if and only if that count is a power of two. In particular a leaf is the only kind of node with count* 1.

Proof. By induction on the parent rule of Definition 6.3. A leaf has both heights equal to its block's. An internal node inherits nEarliestHeight from its left child and nLatestHeight from its right child, and in a range under Bag_L the run of the left child immediately precedes that of the right child, the heights of the leaves of $\mathcal{T}_\eta$ being consecutive by Definition 6.2. A perfect subtree of altitude a spans 2^a leaves (Definition 3.1), and every node inside a peak roots a perfect subtree. A bagging node of the left fold spans the peaks $P_1, \dots, P_i$ for some $i \geq 2$ (Definition 3.5), whose sizes are distinct powers of two, set bits of n (Definition 3.4); a sum of two or more distinct powers of two has at least two set bits and is not a power of two. An internal node spans at least two leaves. A script checks the statement for $1 \leq n \leq 1{,}024$. □

Lemma 6.7 (Leaves and header linkage). *For $\alpha(\eta) \leq \eta' \leq \eta - 1$, field* 1 *of the leaf of $\mathcal{T}_\eta$ for $B_{\eta'}$ equals the field* hashPrevBlock *of the header of $B_{\eta'+1}$, which for $\eta' = \eta - 1$ is the committing block B_η. Consequently a received header is matched to its leaf by computing its block hash and comparing it with field* 1*, and the leaves add no hashing beyond the block hashes.*

Proof. By Definition 6.3, field 1 of the leaf is BlockHash of the header of $B_{\eta'}$ in internal byte order; by Definition 2.3, the field hashPrevBlock of $B_{\eta'+1}$ is that same hash. ZIP 221, "Tree Node specification", states the identity. □

Remark 6.8 (Non-monotone timestamps). Consensus bounds the field nTime of a block below, strictly above the median of the timestamps of the preceding eleven blocks, and above, by that median plus 90 minutes (protocol specification, §"Block Header Encoding and Consensus"; *Consensus Guide*, §"Difficulty in motion"). It orders two timestamps only at a distance: the median-time-past never decreases and rises by at least one second every eleven blocks, so a block at least $11 \cdot 5{,}400 = 59{,}400$ heights after another has the strictly later timestamp. By the same count, nLatestTimestamp $\geq$ nEarliestTimestamp holds for every node with nLatestHeight − nEarliestHeight $\geq 11 \cdot 5{,}399 = 59{,}389$, and for a leaf, whose two timestamps are equal; for smaller spans this argument gives no order. ZIP 221, "Tree Node specification", says

that the reverse order "may occur within subtrees smaller than PoWMedianBlockSpan blocks"; this is an example, not a bound. A block whose timestamp lies up to 90 minutes ahead of the median exceeds the timestamps of the later blocks whose timestamps track real time, until real time passes it, and 90 minutes spans more than eleven target spacings (protocol specification, §"Constants"; from NU7, ZIP 218).

6.4 The epoch tree and its root commitment

The tree is defined from the prose of ZIP 221, "Background", "Specification", "Tree Node specification" and "Block header semantics and consensus rules", with ZIP 258, "Changes to ZIP 221". That prose is ambiguous at two points, both at an activation block: the field vector, settled by the adopted reading of Definition 6.5, and the branch ID of the personalisation, settled by Definition 6.9. The pseudocode of ZIP 221 is cited only as corroboration of the two readings; it carries editorial defects, among them in the serialisation of the Orchard fields, none of which bears on either reading.

Definition 6.9 (Branch ID of a tree)**.** ZIP 221, "Tree Node specification", states that the personalisation encodes "the consensus branch ID for the epoch of the block containing the commitment", and continues: "Which is to say, each node in the tree commits to the consensus branch that produced it." The two sentences agree at every $\eta > A_H$ that is not an activation height, since B_η and the leaves of $\mathcal{T}_\eta$ then lie in one upgrade epoch. At an activation height $A > A_H$ the first names the upgrade epoch beginning at A and the second the upgrade epoch of the leaves $B_{\alpha(A)}, \dots, B_{A-1}$. At A_H no epoch tree is defined (Definition 6.2). *Adopted reading*, resolving an ambiguity of specified text: the branch ID $b(\mathcal{T})$ of an epoch tree $\mathcal{T}$ is the consensus branch ID of the upgrade epoch whose blocks are the leaves of $\mathcal{T}$, and every internal node and the root commitment of $\mathcal{T}$ are hashed under it.

The reading is adopted for three reasons. (1) "Which is to say" introduces the second sentence as a restatement of the first, and it is the sentence that assigns a branch ID to each node. (2) Under the first reading the tree committed by B_A would not be an append of the tree committed by B_{A-1}, every internal node changing value, contrary to the property by which ZIP 221, "Terminology", defines a Merkle mountain range (the reason for which Definition 6.5 adopts the leaves' vector); and the whole previous upgrade epoch would be rehashed in one block, the cost that ZIP 221, "Header Format Change", calls impractical and avoids by a special case at the Heartwood activation block and at no later one. (3) The pseudocode of ZIP 221 hashes each parent under the branch ID of its left child and the root commitment under that of the root node, each leaf carrying the branch ID of its own block. Every later statement about the tree committed by an activation block rests on this reading and cites it; under the first reading each holds with the branch ID of the new upgrade epoch in place of $b(\mathcal{T}_A)$.

Construction 6.10 (Hashing of the epoch tree)**.** Let $\mathsf{h}(P, x)$ be BLAKE2b-256 with the 16-byte personalisation P on input x (*Ironwood Guide*, §"Transaction digests and signatures"; *Crypto Guide*, §"Domain separation and personalisation"), and let β be the 4-byte little-endian encoding of $b(\mathcal{T}_\eta)$. The personalisation is the 12 ASCII bytes `ZcashHistory` followed by β, 16 bytes in all. The tree $\mathcal{T}_\eta$ is the range M_n^L whose nodes are node vectors (Definition 6.3) under the field vector of Definition 6.5. Its bagging nodes, which join the two leftmost remaining peaks until one remains,

$$(\cdots((P_1 \parallel P_2) \parallel P_3) \cdots) \parallel P_d,$$

are ordinary internal nodes, and a single peak is the root node. For an internal node with

Upgrade epoch	Branch ID	Fixed by	Bytes 13 to 16 of the personalisation	Fields
Heartwood	`0xF5B9230B`	ZIP 250	`0B 23 B9 F5`	1 to 11
Canopy	`0xE9FF75A6`	ZIP 251	`A6 75 FF E9`	1 to 11
NU5	`0xC2D6D0B4`	ZIP 252	`B4 D0 D6 C2`	1 to 14
NU6	`0xC8E71055`	ZIP 253	`55 10 E7 C8`	1 to 14
NU6.1	`0x4DEC4DF0`	ZIP 255	`F0 4D EC 4D`	1 to 14
NU6.2	`0x5437F330`	ZIP 257	`30 F3 37 54`	1 to 14
NU6.3	`0x37A5165B`	ZIP 258 (Draft)	`5B 16 A5 37`	1 to 17
NU7	`0x77190AD9`	ZIP 259 (Draft)	`D9 0A 19 77`	1 to 17

Table 3: The branch IDs entering the personalisation $\texttt{ZcashHistory} \| \beta$ of Construction 6.10: one row per upgrade epoch that bears an epoch tree, each tree hashed under its own personalisation, with the ZIP that fixes the branch ID (ZIP 258, "NU6.3 deployment"; ZIP 259, "NU7 network constants"), the little-endian bytes β and the field vector of Definition 6.5.

children ℓ and r, field 1 is

$$\mathsf{h}(\texttt{ZcashHistory} \| \beta,\ \mathrm{ser}(\ell) \| \mathrm{ser}(r)),$$

over the children's full serialisations; its other fields follow the parent rule of Definition 6.3. The branch ID is not a node field: it enters only through the personalisation, and whoever recomputes a node hash supplies $b(\mathcal{T}_\eta)$ by Definition 6.9, the upgrade epoch being fixed by the node's heights. For the Heartwood branch ID `0xF5B9230B` the last four bytes of the personalisation are `0B 23 B9 F5`.

Definition 6.11 (Chain-history root)**.** For $\eta > A_H$ the *chain-history root* of $\mathcal{T}_\eta$ is

$$\mathsf{hashChainHistoryRoot}(\mathcal{T}_\eta) := \mathsf{h}(\texttt{ZcashHistory} \| \beta,\ \mathrm{ser}(\text{root node of } \mathcal{T}_\eta)),$$

with β encoding $b(\mathcal{T}_\eta)$ (ZIP 221, "Tree Node specification"). It is not field 1 of the root node: the committed value covers the root's aggregates, namely the total work modulo 2^{256}, the first and last timestamps, target bits and heights, the boundary pool roots and the per-pool counts. When $\alpha(\eta) = \eta - 1$ the tree has one leaf, which is its root node, and the value is defined there too. ZIP 258, "Changes to ZIP 221", leaves the definition unchanged apart from the added fields.

Definition 6.12 (History input of a block)**.** For $\eta \geq A_H$ the *history input* of B_η is

$$\mathsf{hashLightClientRoot}(\eta) := \begin{cases} \text{the string of 32 zero bytes} & \text{if } \eta = A_H, \\ \mathsf{hashChainHistoryRoot}(\mathcal{T}_\eta) & \text{if } \eta > A_H. \end{cases}$$

The value at A_H is the sentinel that ZIP 221, "Block header semantics and consensus rules", requires in the Heartwood activation block and states "MUST NOT be interpreted as a root hash". The history input is a total function of the height from A_H on, and it is the value that ZIP 244, "Block Header Changes", names "hashLightClientRoot (as described in ZIP 221)".

Proposition 6.13 (Binding of the chain-history root)**.** *Let $\mathcal{T}$ and $\mathcal{T}'$ be epoch trees hashed under one branch ID b. If* $\mathsf{hashChainHistoryRoot}(\mathcal{T}) = \mathsf{hashChainHistoryRoot}(\mathcal{T}')$*, then either $\mathcal{T}$ and $\mathcal{T}'$ agree on every field of every node, or a collision of* $\mathsf{h}(\texttt{ZcashHistory} \| \beta, \cdot)$ *is computed from them efficiently. If in addition SHA-256d is collision resistant, the leaves' block hashes bind the headers of the leaf blocks.*

Proof. The proposition is the ZIP 221 instance of Lemma 3.20(1), whose hypotheses are checked as follows. *Serialisation.* The branch ID b fixes one upgrade epoch and hence one field vector for both trees (Table 3; Definitions 6.9 and 6.5). Its field order is fixed, and each field is of fixed width or a minimal CompactSize, whose first byte fixes its length (Definition 2.2). A serialised node therefore parses field by field: ser is injective, and no serialised node is a proper prefix of another, since the two parse to the same fields up to the end of the shorter and the number of fields is fixed; minimality makes ser a function of the node. *Root and shape.* If the root commitments are equal, then the two root serialisations are equal, or they are distinct inputs with equal images, a collision. Equal root nodes have equal nEarliestHeight and nLatestHeight, hence equal leaf count n by Lemma 6.6, hence the same shape M_n^L. The shape fixes which positions are leaves, so no leaf of one tree is compared with an internal node of the other, although the tree carries no layer tags (ZIP 221, "Tree nodes", identifies leaves by their block range of 1). *Extraction.* Lemma 3.20(1) then yields a collision at the first level of disagreement. *Headers.* Field 1 of each leaf is the SHA-256d block hash of its header (Definition 2.3; *Crypto Guide*, §"The transparent layer's hashes"); equal leaves with distinct headers are a collision of SHA-256d. □

Proposition 6.14 (Binding of range transaction counts)**.** *Let a node v of $\mathcal{T}_\eta$ be authenticated by an inclusion path (Definition 3.13, in the tuple form of Definition 3.19, siblings as full node vectors) accepted against the chain-history root of the epoch tree of a chain the client trusts. Barring a collision as in Proposition 6.13, v.*nSaplingTxCount*, and where present v.*nOrchardTxCount *and v.*nIronwoodTxCount*, equal the numbers of transactions in the blocks of the height range of v in which* vSpendsSapling *or* vOutputsSapling, vActionsOrchard *and* vActionsIronwood*, respectively, are non-empty. Hence a client that obtains the transactions of those blocks, each authenticated against its block's transaction root, detects the withholding of any such transaction by a count mismatch, and a range whose count is zero contains none and may be skipped; the count does not identify the missing transaction.*

Proof. By Lemma 3.20(2), an accepted path binds the full vector of every node it recomputes and of every sibling it carries, so v is the node of the trusted tree at its position, barring a collision; the serialisation is prefix-free by the proof of Proposition 6.13. In that tree the count fields are consistent, and by the parent rule of Definition 6.3 and Lemma 6.6 the value of each count field of v is its sum over the leaves of v's height range. The reading as detection of withholding is ZIP 221, "Non-FlyClient Commitments". □

Proposition 6.15 (Cross-branch separation)**.** *For branch IDs $b \neq b'$ with encodings $\beta \neq \beta'$ the personalisations $\texttt{ZcashHistory} \| \beta$ and $\texttt{ZcashHistory} \| \beta'$ differ. Model $\mathsf{h}(P, \cdot)$, for each personalisation P, as an independent random oracle with 256-bit output. An adversary making q oracle queries, the queries of the verification of its output included, outputs x, x' with*

$$\mathsf{h}(\texttt{ZcashHistory} \| \beta, x) = \mathsf{h}(\texttt{ZcashHistory} \| \beta', x')$$

with probability at most $q^2/2^{256}$. Hence no internal-node hash or root commitment computed under b equals one computed under b', except with that probability. Leaves are unpersonalised block hashes,

identical in every branch, and are not separated.

Proof. The modelling hypothesis is that of the *Crypto Guide*, §"Domain separation and personalisation", Proposition "Domain separation yields independent oracles": distinct tags give independent random oracles. Both values must be answers to queries, one to each oracle; each of the at most q^2 pairs of such queries agrees with probability 2^{-256}, and the union bound (*Math Guide*, §"The union bound and a birthday calculation") gives the bound. ZIP 221, "Tree nodes", states the design intent: "valid nodes from one consensus branch cannot be used to make false statements about parallel consensus branches". □

6.5 The authorising-data root

The header field hashMerkleRoot (Definition 2.3) is the root of Bitcoin's SHA-256d transaction Merkle tree over the transaction identifiers of the block (protocol specification, §"Block Header Encoding and Consensus"). The identifier of a version-5 or version-6 transaction excludes its authorising data (ZIP 244; *Ironwood Guide*, §"Transaction digests and signatures", Definition "Authorising-data digest; effecting and authorising data"), so for such transactions hashMerkleRoot does not commit to the authorising data; the identifier of a version-4 transaction covers its whole encoding. The version-4 format is valid until NU7, from which ZIP 2003 (Draft), deployed by ZIP 259 (Draft), makes it invalid.

Construction 6.16 (Authorising-data root)**.** Let a block have the transactions $t_1, \dots, t_m$, where $m \geq 1$ because of the coinbase transaction. Leaf i is the *authorising-data commitment* of t_i: for version 5, the digest auth_digest of ZIP 244, "Authorizing Data Commitment"; for version 6, the digest of the *Ironwood Guide*'s Definition "Authorising-data digest; effecting and authorising data" (ZIP 229 (Draft), "Block Commitments": hashAuthDataRoot takes the version-6 digest "with no further structural change"); and for a transaction of a version before 5, the 32 bytes `0xFF`, a placeholder used only in this tree and occurring only in blocks before NU7. The leaves are padded with leaves of 32 zero bytes to $2^{\lceil \log_2 m \rceil}$, and hashAuthDataRoot is the root of the binary Merkle tree of height $\lceil \log_2 m \rceil$ over them, each non-leaf node being $\mathtt{h}(\mathtt{ZcashAuthDatHash}, \mathit{left} \| \mathit{right})$, under a 16-byte personalisation without branch ID. The insertion order MUST equal that of the transaction identifiers in the tree under hashMerkleRoot, so that one path identifies the same transaction in both trees (ZIP 244, "Block Header Changes").

Proposition 6.17 (Joint binding of the two transaction roots)**.** *Assume, with ZIP 244, "Authorizing Data Commitment", that the pair of a transaction's identifier and its authorising-data commitment is binding: distinct transactions with equal pairs form a collision of that pair map. Let two lists of m transactions have equal* hashMerkleRoot *and equal* hashAuthDataRoot*. Then the lists are equal, or a collision of SHA-256d, of* $\mathtt{h}(\mathtt{ZcashAuthDatHash}, \cdot)$ *or of the pair map is computed from them. The binding is claimed for lists of the given length only: the identifier tree is Bitcoin's transaction Merkle tree and inherits its duplicate-last-node caveat, and the header commits to no transaction count.*

Proof. For fixed m, Bitcoin's duplication rule and the zero padding of Construction 6.16 are fixed maps of the list, so the path argument of the *Crypto Guide*, §"Binary Merkle trees", Proposition "Binding of the root", applies to each tree: the two lists of identifiers agree, else a collision of SHA-256d is extracted, and the two lists of authorising-data commitments agree, else a collision of $\mathtt{h}(\mathtt{ZcashAuthDatHash}, \cdot)$. The common insertion order aligns the two leaf sequences, so each position carries equal pairs, and distinct transactions there form a collision of the pair map. ZIP 244, "Authorizing Data Commitment", states that the pair "constitutes a commitment to all the

data of a serialized transaction"; for a transaction of a version before 5 the identifier covers all data and the second member is the constant placeholder. The caveat is the *Crypto Guide*, §"Layer and domain separation", Remark "Unbalanced trees, padding, and a deployed failure". □

6.6 The block commitment hash

In this subsection 0^{32} denotes the string of 32 zero bytes.

Definition 6.18 (Block commitment hash)**.** For a block B_η at a height in or after the NU5 upgrade epoch, the *block commitment hash* is

$$\mathsf{hashBlockCommitments}(B_\eta) := \mathsf{h}(\texttt{ZcashBlockCommit},\ \mathsf{hashLightClientRoot}(\eta) \,\|\, \mathsf{hashAuthDataRoot}(B_\eta) \,\|\, 0^{32}),$$

with $\mathsf{hashLightClientRoot}(\eta)$ the history input of Definition 6.12, $\mathsf{hashAuthDataRoot}(B_\eta)$ the root of Construction 6.16 for B_η, and 0^{32} the *terminator* (ZIP 244, "Block Header Changes"). The input has $32 + 32 + 32 = 96$ bytes, and the 16-byte personalisation carries no branch ID. The history input is a function of the blocks before B_η; the authorising-data root is a function of B_η itself.

Proposition 6.19 (Reopening the block commitment)**.** *From NU5 the chain-history root appears in no header. Let c be the value* $\mathsf{hashBlockCommitments}(B_\eta)$ *of a block in or after the NU5 upgrade epoch. A verifier given a claimed history input r, or a claimed root node vector from which it computes $r = \mathsf{h}(\texttt{ZcashHistory} \,\|\, \beta, \mathrm{ser}(root))$ with β encoding the branch ID of Definition 6.9, and a 32-byte value a, accepts r if and only if*

$$c = \mathsf{h}(\texttt{ZcashBlockCommit},\ r \,\|\, a \,\|\, 0^{32});$$

only the 32-byte value of $\mathsf{hashAuthDataRoot}$ *is needed, not its tree. Two distinct accepted pairs $(r, a) \neq (r', a')$ exhibit a collision of $\mathsf{h}(\texttt{ZcashBlockCommit}, \cdot)$. Hence an accepted r is the block's history input except with that collision probability, and an inclusion path in the epoch tree is checked against a header through it.*

Proof. The pair $(\mathsf{hashLightClientRoot}(\eta), \mathsf{hashAuthDataRoot}(B_\eta))$ is accepted by Definition 6.18. The input $r \,\|\, a \,\|\, 0^{32}$ has fixed length, so distinct pairs give distinct inputs, and two accepted pairs with equal image c are a collision. □

Remark 6.20 (Extension of the block commitment hash). ZIP 244, "Block Header Changes", treats the value as "a linked list of hashes terminated by arbitrary data": a later upgrade may replace the terminator by the hash of a new structure that ends in its own terminator. The same section states two interoperability rules: fully validating nodes MUST always use the entire structure defined by the latest activated protocol version they support, and a third-party server that supports a later protocol version SHOULD provide a value that can be used in place of the all-zero terminator, so that a light client can validate the parts it needs. No upgrade through NU7 replaces the terminator: ZIP 258, "Changes to ZIP 221", leaves $\mathsf{hashBlockCommitments}$ "unaffected" beyond the added node fields; ZIP 229, "Block Commitments", changes no structure; and no source of NU7 consensus changes listed in ZIP 259, "NU7 consensus changes", amends ZIP 244.

6.7 The header consensus rules

Definition 6.21 (The commitment field by upgrade epoch)**.** The 32-byte header field that follows hashMerkleRoot (Definition 2.3) is named hashReserved before Sapling, hashFinalSaplingRoot in the Sapling and Blossom upgrade epochs, hashLightClientRoot in the Heartwood and Canopy upgrade epochs, and hashBlockCommitments from NU5 (protocol specification, §"Block Header Encoding and Consensus"; ZIP 221, "Block header semantics and consensus rules"; ZIP 244, "Block Header Changes"). The header byte format and version are unchanged by each renaming.

Definition 6.22 (Header commitment rule)**.** For a block B_η the commitment field satisfies the following rules.

(i) For $\eta < A_H$ it follows the earlier rules of the protocol specification, §"Block Header Encoding and Consensus", not restated here: in the Sapling and Blossom upgrade epochs it is $\mathsf{LEBS2OSP}_{256}$ of the root of the Sapling note commitment tree of the block's final Sapling treestate.

(ii) For $A_H \leq \eta$ in the Heartwood or Canopy upgrade epoch it MUST equal the history input $\mathsf{hashLightClientRoot}(\eta)$ of Definition 6.12, which supplies the sentinel at A_H.

(iii) For η in or after the NU5 upgrade epoch, the NU5 activation block included and without sentinel, it MUST equal $\mathsf{hashBlockCommitments}(B_\eta)$ of Definition 6.18.

(ZIP 221, "Block header semantics and consensus rules"; ZIP 244, "Block Header Changes"; protocol specification, §"Block Header Encoding and Consensus".) The value is undefined for the genesis block, which no case covers.

In the NU5 activation block $B_{A_{\mathrm{NU5}}}$, A_{NU5} the NU5 activation height, the history input is $\mathsf{hashChainHistoryRoot}(\mathcal{T}_{A_{\mathrm{NU5}}})$, the root of the completed tree of the Canopy upgrade epoch. It is hashed under the Canopy branch ID `0xE9FF75A6` (ZIP 251) by the adopted reading of Definition 6.9, the first sentence of ZIP 221 giving the NU5 branch ID instead, and it is built from nodes of fields 1 to 11 by the adopted reading of Definition 6.5. By Definition 6.22 every valid block after the Heartwood activation block commits to the root of its epoch tree: directly in the Heartwood and Canopy upgrade epochs, and from NU5 as the first input of the block commitment hash.

6.8 The committed range

Proposition 6.23 (One-block delay)**.** *For $\eta > A_H$ the tree to which B_η commits (Definition 6.22) has the leaves $B_{\alpha(\eta)}, \ldots, B_{\eta-1}$. No header commits to a tree that contains its own block, and $B_{\eta-1}$ first enters a committed tree at B_η. Consequently the final Sapling root of a block at a height of at least A_H is committed one block later, as the field* hashLatestSaplingRoot *of the root node committed by the next block. ZIP 221, "Non-FlyClient Commitments", accepts the delay because "using the most recent treestate as a transaction anchor is already unsafe in reorgs".*

Proof. By Definition 6.2, η lies outside $[\alpha(\eta), \eta - 1]$. Since $\eta - 1 \geq A_H$ and $\alpha(\eta)$ is the largest activation height below η, $\alpha(\eta) \leq \eta - 1$, so $B_{\eta-1}$ is the last leaf of $\mathcal{T}_\eta$; it is no leaf of $\mathcal{T}_{\eta'}$ for $\eta' \leq \eta - 1$, whose leaves lie below η'. Then Definition 6.22. The latest-root field of the root node is that of its last leaf, by the parent rule of Definition 6.3. □

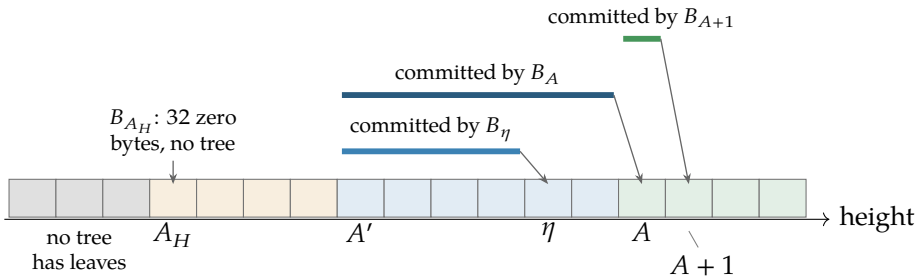


Figure 4: Committed ranges on a height axis, one cell per block, with the Heartwood activation height A_H and two later activation heights $A' < A$, consecutive, and $A' < \eta < A$. Grey: heights below A_H, a leaf of no epoch tree (Corollary 6.25). The Heartwood activation block carries the sentinel of Definition 6.12 and commits no tree. Each bar is the leaf range of the tree to which the header at the arrow's head commits: B_η commits $B_{A'}, \dots, B_{\eta-1}$; the activation block B_A commits the whole upgrade epoch $B_{A'}, \dots, B_{A-1}$ (Theorem 6.24(1)); and B_{A+1} commits the single leaf B_A (Theorem 6.24(2)). No bar reaches below A_H.

Theorem 6.24 (Reset at each activation height)**.** *Let $A > A_H$ be an activation height and $A' = \alpha(A)$ the preceding one. Then:*

1. *B_A commits to the complete tree $\mathcal{T}_A$ over $B_{A'}, \dots, B_{A-1}$, the whole upgrade epoch beginning at A', hashed under that upgrade epoch's branch ID $b(\mathcal{T}_A)$ by the adopted reading of Definition 6.9 and built with that upgrade epoch's field vector by the adopted reading of Definition 6.5; the commitment field equals* hashChainHistoryRoot$(\mathcal{T}_A)$ *before NU5, and* hashBlockCommitments(B_A) *with it as first input from NU5;*
2. *B_{A+1} commits to the single-leaf tree over B_A;*
3. *for $A' < \eta < A$, the leaves of $\mathcal{T}_\eta$ are a proper prefix of those of $\mathcal{T}_A$, so every ended upgrade epoch from Heartwood on is committed completely by exactly one header, that of the next activation block;*
4. *the reset occurs at every upgrade, including those that leave the field vector unchanged (Canopy, NU6, NU6.1, NU6.2 and NU7).*

Proof. By the definition of α, $\alpha(A) = A'$ and $\alpha(A+1) = A$. For $A' < \eta < A$, $\alpha(\eta) = A'$, so $\mathcal{T}_\eta$ has the leaves $B_{A'}, \dots, B_{\eta-1}$, a proper prefix of those of $\mathcal{T}_A$; only $\mathcal{T}_A$ has the full leaf set, and for $\eta > A$ no tree has the leaf $B_{A'}$, since $\alpha(\eta) \geq A$. The commitment fields follow from Definition 6.22. Part (4) holds because α depends on the activation heights alone. A script illustrates (1) to (3) on a toy chain, a one-block upgrade epoch included; it establishes nothing. □

Figure 4 shows the ranges of the theorem.

Corollary 6.25 (No commitment below the Heartwood activation height)**.** *The Heartwood activation block B_{A_H} commits to no tree, its history input being the sentinel of Definition 6.12. For $\eta > A_H$, $\alpha(\eta) \geq A_H$, so no committed tree has a leaf below A_H. Below A_H the commitment field holds a Sapling root or is reserved (Definition 6.21).*

Proof. The height A_H is an activation height below every $\eta > A_H$, and $\alpha(\eta)$ is the largest such; then Definition 6.22, item (ii). □

Corollary 6.26 (The NU7 upgrade epoch). *Let A_{NU7} be the NU7 activation height. The block $B_{A_{\mathrm{NU7}}}$ commits, as the first input of its block commitment hash, to the completed tree of the NU6.3 upgrade epoch, hashed under the NU6.3 branch ID* `0x37A5165B` *by the adopted reading of Definition 6.9. The block $B_{A_{\mathrm{NU7}}+1}$ commits to a new tree over $B_{A_{\mathrm{NU7}}}$, hashed under the NU7 branch ID* `0x77190AD9` *(ZIP 259, Draft) with the field vector of Definition 6.5 for the NU7 upgrade epoch.*

Proof. The instance $A = A_{\mathrm{NU7}}$ of Theorem 6.24, NU6.3 being the upgrade that precedes NU7 (Definition 1.1), with the constants of ZIP 259, "NU7 network constants", and ZIP 258, "NU6.3 deployment". □

7 The deployment boundary

This section classifies every component of a Zcash FlyClient in the classes of Definition 1.3. It opens with one table that places each hypothesis of Theorem 5.32 and each obligation of Remark 5.33 in its class, with the cited fact that places it there. The subsections that follow elaborate the rows: the specified commitment and the absence of a specified consumer, the differences between the paper's commitment and the deployed one, the conditions that the ZIPs impose on any sample check of a Zcash verifier, and the open problems.

7.1 Classification of the security hypotheses

Remark 7.1 (Classification of the hypotheses). The classes of Definition 1.3 apply to components, not to properties. A property proved in this volume of a specified object is recorded as "the object is specified, and the property is proved here", never as specified. Table 4 classifies every hypothesis of Theorem 5.32 and every obligation O1 to O6 of Remark 5.33, together with two components of the deployed commitment, the chain commitment (row (1)) and its composition across upgrade epochs (row (4)); O1 and O2 share row (3). Each row gives its class and the cited fact that places it there or, for an assumption on the client's environment, the source that records it. Hypothesis 1 of the theorem fixes the parameters c, k and δ, whose values an instantiation chooses; rows (7) and (9) classify what those values rest on. The rows read as follows.

(1) *Chain commitment*: specified. It is the epoch-scoped, metadata-bearing, branch-separated epoch tree $\mathcal{T}_\eta$ (Definition 6.2, Construction 6.10), to whose chain-history root every header after the Heartwood activation block commits under Definition 6.22, enforced by consensus (ZIPs 221 and 244; ZIP 258, Draft, whose rules are in force from NU6.3 by Definition 1.2).

(2) *Collision resistance of the hash functions*: the functions are specified, namely personalised BLAKE2b-256 h for tree nodes, the chain-history root, the authorising-data root and the block commitment hash (ZIPs 221 and 244; protocol specification, §"BLAKE2 Hash Functions"), and SHA-256d for block hashes, hence for leaves (protocol specification, §"Block Header Encoding and Consensus"). Their collision resistance is a standard assumption (*Crypto Guide*, §"Security notions: preimage, second-preimage, and collision resistance").

(3) *O1, binding within one upgrade epoch*: the commitment is specified (Definition 6.22), and the property is proved here of the specified object (Proposition 6.13); no normative text fixes the property. *O2, fork structure for ZIP 221 trees*: open, since it is conditional on a sufficient sample check for those trees, which no specification or result supplies (Proposition 7.4); Lemmas 5.4 and 5.5 are stated for the paper's construction only. Hypothesis 6 of Theorem 5.32, leaf–internal separation, enters only through Lemma 5.4 and shares the class of O2: it is stated for the paper's construction, and for ZIP 221 trees no specification or result supplies its counterpart.

(4) *Composition across upgrade epochs*: open. Each upgrade epoch has its own tree (Theorem 6.24), and ZIP 221 states no rule that chains them.

(5) *The verifier's anchor*, the genesis block in the paper: open (Corollary 6.25).

(6) *O3, weight binding*: the node vector is specified (Definition 6.3), and the binding of the field-8 interval of a leaf to the root and its position within one upgrade epoch is proved here of the specified object (Lemma 3.21(1) with Proposition 6.13). The link from a sampled header to its leaf's weight is conditional on the same sufficient sample check, and so open. Sampling by the chain's total work is open, field 8 of a root summing one upgrade epoch (Definition 6.3; Proposition 7.8).

(7) *Assumption 4.8, the transition checks of O3* (*Construction 5.8*) *and Assumption 5.11 under the Zcash retarget rule*, in both of its regimes: open (protocol specification, §"Difficulty adjustment"; ZIP 218, Draft).

(8) *Assumption 5.20*: open for Zcash; no analysis relates the work of individual blocks to the invalid segments of a Zcash fork.

(9) *O4, catch probability and sample count*: designed but unspecified, and proved in the paper's model under its hypotheses (Theorem 5.21; Proposition 5.22).

(10) *The random-oracle hypothesis on the sampling hash*: open. No ZIP or specification section fixes a sampling hash for Zcash, so the hypothesis has no object; the model is that of the *Crypto Guide*, §"The random oracle model".

(11) *O5, grinding*: the transform of Construction 5.27 is designed but unspecified. Its Zcash transcript (header bytes, hash, domain separation, and map from output to weight points across upgrade epochs) and a derivation of the budget R of Assumption 5.28 are open.

(12) *O6, size*: the bound of Corollary 5.34 is designed but unspecified; a Zcash proof format is open.

(13) *At least one honest prover*: an assumption on the client's connectivity, not a component, so no class applies. ZIP 221, "Security and Privacy Considerations", summarises the analysis of the paper, not the setting of ZIP 221 itself, as follows: "The security analysis assumes adversary mining power is bounded by a known fraction of combined mining power of honest nodes, and cannot drop or tamper with messages between client and full nodes. It also assumes the client is connected to at least one full node and knows the genesis block." The bound on mining power is row (7); the absence of dropping or tampering and the connection to at least one full node are this row; knowledge of the genesis block is row (5).

Open problems (§7.5) derives rows (4), (5), (7), (11) and (12), and the total work of row (6), from cited rules; O2 of row (3), with the header-to-weight link of row (6), and rows (8) and (10) are open for the reasons stated above.

The verifier as a whole is designed but unspecified. The protocol of "The FlyClient protocol" (§5) exists with the paper's theorem in the paper's model; no ZIP specifies a Zcash instantiation; and the paper's model does not cover Zcash's retarget rule. The paper proves security only for its model of retargeting once per retarget epoch (Definition 4.5), places rules that retarget from a moving average outside it, and credits them with heuristic guarantees only (ePrint 2019/226, Section 3.1, pages 7–8); it names Ethereum there and does not name Zcash. ZIP 221, "Security and Privacy Considerations", names Zcash among the chains "with rapidly adjusting difficulty" that the paper leaves open.

	Hypothesis or obligation	Class	Placed by
(1)	chain commitment $\mathcal{T}_\eta$	specified	Definition 6.22; ZIPs 221, 244, 258
(2)	collision resistance of h and SHA-256d	functions specified; resistance assumed	ZIPs 221, 244; *Crypto Guide*
(3)	O1: binding within one upgrade epoch	commitment specified; property proved here	Proposition 6.13
	O2: fork structure for ZIP 221 trees; hypothesis 6 (leaf–internal separation)	open problem	Proposition 7.4; sufficiency not claimed
(4)	composition across upgrade epochs	open problem	Theorem 6.24; ZIP 221 silent
(5)	anchor of the verifier	open problem	Corollary 6.25
(6)	O3: weight binding	node vector specified; interval binding proved here; header-to-weight link and total work open	Lemma 3.21(1); Proposition 7.8
(7)	(c, L), transition checks, transition rewriting	open problem	retarget rule of the specification; ZIP 218
(8)	discretisation	open problem	no analysis for Zcash
(9)	O4: catch probability, sample count	designed but unspecified	Theorem 5.21; Proposition 5.22
(10)	random oracle for sampling	open problem	no sampling hash specified
(11)	O5: grinding	transform designed but unspecified; transcript and R open	Construction 5.27; Assumption 5.28
(12)	O6: size	designed but unspecified; format open	Corollary 5.34
(13)	at least one honest prover	environment; no class	ZIP 221, "Security and Privacy Considerations"

Table 4: The components of a Zcash FlyClient, covering every hypothesis of Theorem 5.32 and every obligation of Remark 5.33, with its class under Definition 1.3 and the result or normative text that places it there. Rows are numbered as in Remark 7.1, which states each in full.

7.2 The specified commitment and its consumers

Row (1) of Remark 7.1 is the following. ZIP 221 (Final) specifies the tree, with FlyClient as its stated motivation (ZIP 221, "Motivation"). ZIP 221, "FlyClient Requirements and Recommendations", designates seven fields for reasoning about the difficulty adjustment over a range of blocks: nEarliestTimestamp, nLatestTimestamp, nEarliestTargetBits, nLatestTargetBits, nEarliestHeight, nLatestHeight and nSubTreeTotalWork (Remark 6.4). The pool roots and transaction counts are its "Non-FlyClient Commitments". No ZIP or specification section specifies a consumer of the tree, a proof object or a verifier. The specified class depends on no implementation: Definition 1.2 places the fields of ZIP 258 (Draft) in force from NU6.3.

Remark 7.2 (Recomputation against proof verification). The consensus rule of Definition 6.22 is an equality between the commitment field of a header and a value that the validator recomputes from the chain it holds: $\mathsf{hashChainHistoryRoot}(\widehat{\mathcal{T}}_\eta)$ in the Heartwood and Canopy upgrade epochs, and the block commitment hash from NU5 (protocol specification, §"Block Header Encoding and Consensus"; ZIP 221, "Block header semantics and consensus rules"; ZIP 244, "Block Header Changes"). No ZIP or specification section defines an inclusion proof for the tree, a proof object or a verification algorithm. The operation that the paper's verifier performs, Construction 5.2, is therefore unspecified for Zcash.

No specified light-client protocol consumes the commitment. ZIP 307 (Draft) does not reference ZIP 221, while ZIP 221, "Motivation", defers the verification of transaction inclusion to ZIP 307: it "follows the usual reference protocol".

7.3 The paper's commitment and the deployed commitment

Remark 7.3 (Deltas from the paper's commitment). Against Construction 5.1 and Definition 5.6, the deployed commitment differs in five respects.

(1) *Scope.* It is epoch-scoped, not rooted at the genesis block (Theorem 6.24). The one-block delay of Proposition 6.23 is shared with the paper, whose header also commits to exactly the blocks strictly before it.

(2) *Metadata.* The seven FlyClient fields serve the purpose of Definition 5.6 (ePrint 2019/226, Definition 7) without matching it field for field (Remark 6.4); the pool roots and counts have no counterpart.

(3) *Hashing.* Every parent is hashed by the personalised, branch-separated BLAKE2b-256 $\mathtt{h}(\mathtt{ZcashHistory} \| \beta,\ \mathrm{ser}(\ell) \| \mathrm{ser}(r))$ over full serialisations, the root commitment hashChainHistoryRoot covering the root's aggregates, and every leaf carries the SHA-256d block hash; the paper hashes with one generic collision-resistant H (Propositions 6.13 and 6.15).

(4) *Placement.* The root enters through the repurposed commitment field (Definition 6.21), and from NU5 only as the first input of the block commitment hash (Proposition 6.19), whereas the paper adds a header field that can replace the previous block hash (ePrint 2019/226, Section 4.2, page 12).

(5) *Bagging.* The deployed tree bags its peaks by the left fold Bag_L, the paper by the right fold Bag_R (ePrint 2019/226, Definition 11). Path tops, prefix folds and the maximal path length therefore differ whenever the leaf count has three or more set bits (Theorem 3.8; Lemma 3.7).

ZIP 221, "Tree Node specification", states: "Other than the metadata commitments, the MMR tree's construction is standard." Items (1), (3), (4) and (5) record the respects in which the deployed construction also differs from the paper's.

7.4 A Zcash verifier

The verifier is designed but unspecified (Remark 7.1). The conditions below are obligations that ZIPs 221 and 244 impose on any sample check for ZIP 221 trees; they are not a construction of this volume, and no Zcash verifier is specified here. A *sample check for ZIP 221 trees* is a deterministic algorithm that takes the header of a head, an index j and a prover's message, and accepts or rejects; its purpose is to establish, for the head's epoch tree, the conclusions of Proposition 5.3 and Lemma 5.7. A condition is *necessary* when some message that meets every other condition and violates it also violates such a conclusion, so that a check that does not reject that message fails the conclusion.

Proposition 7.4 (Necessary conditions of a Zcash sample check)**.** *Let the head B_η, $\eta > A_H$, commit to $\mathcal{T}_\eta$ with $n = \eta - \alpha(\eta)$ leaves, let β encode $b(\mathcal{T}_\eta)$, and let a sample be leaf j of $\mathcal{T}_\eta$, $1 \le j \le n$, the block at height $\alpha(\eta) + j - 1$, with received header hdr_j. Each of the following conditions, the height clause of (ii) and the parsing clause of (iv) excepted, is necessary for a sample check for ZIP 221 trees. Sufficiency is not claimed. The height clause of (ii) is required for the leaf's stated heights to match its position (Lemma 6.6); parsing under the field vector is the hypothesis under which Proposition 6.13 and Lemma 3.20(2) apply; the necessity of neither is claimed.*

(i) *Field* 1 *of the leaf equals* $\mathsf{BlockHash}(\mathrm{hdr}_j)$*, the SHA-256d hash of the full received header, Equihash solution included (Lemma 6.7).*

(ii) *Fields* 2 *and* 3 *of the leaf equal the field* nTime *of* hdr_j*, fields* 4 *and* 5 *its field* nBits*; field* 8 *equals the work* $\lfloor 2^{256}/(\mathsf{ToTarget}(\mathsf{nBits}) + 1) \rfloor$ *of that* nBits*; and fields* 9 *and* 10 *equal* $\alpha(\eta) + j - 1$*, the height that the position fixes, the header carrying no height (Definition 2.3).*

(iii) *The header* hdr_j *satisfies proof of work in the sense of Definition 2.3: a valid Equihash solution, and the difficulty filter at the target* $\mathsf{ToTarget}(\mathsf{nBits})$ *it declares.*

(iv) *The sampled leaf of (i) and (ii) and the path siblings are received as full node serialisations, parsed under the field vector of Definition 6.5 for the upgrade epoch beginning at $\alpha(\eta)$. Starting from that leaf, every parent is recomputed: field* 1 *as* $\mathsf{h}(\texttt{ZcashHistory} \| \beta,\ \mathrm{ser}(\ell) \| \mathrm{ser}(r))$ *(Construction 6.10), the other fields by the parent rule of Definition 6.3. The recomputed root node yields* $\mathsf{hashChainHistoryRoot}(\mathcal{T}_\eta)$*, which is compared with the head's commitment field under Definition 6.22: directly in the Heartwood and Canopy upgrade epochs, and from NU5 through Proposition 6.19 with the head's* hashAuthDataRoot.

(v) *The sides and the length of the path are those that j and n fix under* Bag_L*, the length being* $\mathrm{depth}_L(n, j)$ *(Lemma 3.7).*

(vi) *For $j \ge 2$: the root node ρ output by* $\mathrm{PrefixRoot}_L(n, j, \pi)$ *(Construction 3.16), run on the path π with the parent rule in place of $H(\cdot \| \cdot)$, is the claimed root node of $\mathcal{T}_{\alpha(\eta)+j-1}$, and the value*

$$r = \mathsf{h}(\texttt{ZcashHistory} \| \beta,\ \mathrm{ser}(\rho))$$

is accepted against the commitment field of hdr_j *under Definition 6.22: r equals that field in the Heartwood and Canopy upgrade epochs, and from NU5 r is accepted by Proposition 6.19 with the* hashAuthDataRoot *of the sampled block.*

Proof. For each condition, a message that meets the other five and violates it also violates the conclusion named.

(i) An honest header with valid proof of work, paired with a leaf whose field 1 is the hash of a different, unmined block of the claimed tree. The proof of work checked is not that of the committed block, contrary to Proposition 5.3(i) and Lemma 5.7(1).

(ii) An honest header paired with a leaf of inflated field 8, in a tree that the adversary builds over such leaves under a head that it mines, so that every path verifies. The header-determined fields are not those of the header, contrary to Proposition 5.3(ii), and the interval that Lemma 3.21(1) binds belongs to a forged weight.

(iii) A tree built over fabricated headers without valid solutions. Every other condition is met by construction, and unmined headers pass, contrary to Lemma 5.7(1).

(iv) If a node is taken in any other form, the fields that do not enter the recomputation are unbound. With the siblings supplied as field-1 digests and the path checked on field 1 alone, the siblings' field 8, which fixes the interval of Lemma 3.21, and the leaf's pool fields (6, 7 and, where the vector carries them, 11 to 17), which its header does not determine, are chosen freely; two passing samples at j then carry distinct leaf tuples, contrary to Proposition 5.3(i). If the recomputation starts from a node other than the leaf matched in (i) and (ii), the matched header is bound to nothing. Proposition 6.13 binds full serialisations only.

(v) The single-block tuple of a leaf placed at a position where the shape M_n^L has an internal node: a tree with fewer than n leaves then passes, and a tuple committed at another depth is accepted as leaf j. Theorem 3.17, which assumes the shape fixed by n and the bagging order, does not apply.

(vi) Honest blocks, mined on the honest chain after the fork point, appended as leaves of the fork's tree. They pass (i) to (v), while their own commitments bind the honest prefix, not the fork's first $j-1$ leaves, contrary to Proposition 5.3(iii). This is the case that Lemma 5.4 excludes by the prefix check.

No condition is shown sufficient. □

The conditions are the ZIP 221 counterparts of the checks of Construction 5.2: (i) and (ii) of check (a), (iii) of check (d), (iv) and (v) of check (b), and (vi) of check (c). Three further statements bound their reach. For $j = 1$ the sample is the block $B_{\alpha(\eta)}$. For $\alpha(\eta) > A_H$ it is an activation block, which commits to the completed tree of the previous upgrade epoch (Theorem 6.24); for $\alpha(\eta) = A_H$ it carries the sentinel of Definition 6.12 (Corollary 6.25). In both cases its prefix check belongs to epoch composition, which is open. The consensus condition $\mathsf{nBits} = \mathsf{ThresholdBits}(\alpha(\eta) + j - 1)$ of Definition 2.3 depends on preceding headers and is not decided by one sample; it is the object of the transition checks of row (7), which are open. And condition (iv) makes Proposition 6.13 apply, but the fork structure of row (3) additionally needs a sufficient check, which is open.

Example 7.5 (Per-sample size)**.** The size is evaluated at the scale of one upgrade epoch, since n is the leaf count of one epoch tree and never the length of the chain. The message of Proposition 7.4 for one sample consists of the 1,487-byte header (§2.1), the sampled leaf's own serialisation of at most 317 bytes (Definition 6.5), at most $\lfloor \log_2 n \rfloor + \mathrm{popcount}(n) - 1$ siblings of at most 317 bytes each (Lemma 3.7), and the 32 bytes of the sampled block's $\mathsf{hashAuthDataRoot}$. For $n = 2^{20}$ a path has at most 20 siblings, and a sample at most

$$1{,}487 + 317 + 20 \cdot 317 + 32 = 8{,}176 \text{ bytes},$$

of which the header is 18.2 per cent. For $n = 2^{20} - 1$, with 20 peaks, a path has at most 38 siblings, and a sample at most

$$1{,}487 + 317 + 38 \cdot 317 + 32 = 13{,}882 \text{ bytes},$$

of which the header is 10.7 per cent. The leaf's fields fixed by its header and its position, 1 to 5 and 8 to 10, could be omitted by a proof format, which is open. The figures are computed by script.

7.5 Open problems

The following open items, each a row of Remark 7.1 except the serving interface, are derived below from cited rules: epoch composition, the anchor below the Heartwood activation height, total work, the proof format, the serving interface, the Fiat–Shamir transcript, and the security analysis under Zcash's retarget rule with its NU7 regime. Composition with compact-block scanning follows. Availability of prover data is not a protocol question.

Remark 7.6 (Epoch composition). Epoch trees are hashed under distinct branch IDs, and each complete root is committed only by the next activation header (Theorem 6.24). No specified rule chains them, and ZIP 221, "Specification", is silent. Composition is needed for prefix proofs and for difficulty checks: the averaging window and the median-time span of the first blocks of an upgrade epoch reach into the previous upgrade epoch (protocol specification, §"Difficulty adjustment"), so a transition check at an upgrade boundary needs two epoch trees. Proposition 5.38 needs an inclusion path of the accepted head against the later head's root, and so applies within one epoch tree. Across epoch trees it does not apply: no tree of a later upgrade epoch contains a leaf of an earlier one (Definition 6.2), and only the activation header commits to the completed earlier tree. No design exists: the item is open.

Remark 7.7 (Anchor). The anchor below the Heartwood activation height is open, by Corollary 6.25.

Proposition 7.8 (No committed node aggregates total work). *Let $\eta > A_H$.*

(*a*) *If SHA-256d is collision resistant, the header of B_η commits through* hashPrevBlock *to the headers of $B_0, \dots, B_{\eta-1}$, hence to their fields* nBits *and to the total work ω of that chain (§2.1; the model's counterpart is Definition 4.2).*

(*b*) *For every node v of $\mathcal{T}_\eta$, field 8 is the sum modulo 2^{256} of the work of the blocks at heights v.*nEarliestHeight *to v.*nLatestHeight*, all in $[\alpha(\eta), \eta - 1]$; field 8 of no node of $\mathcal{T}_\eta$ depends on $B_0, \dots, B_{\alpha(\eta)-1}$.*

(*c*) *Hence no inclusion path against the root to which B_η commits opens ω. Sampling by the chain's cumulative work (Definition 5.19) and the acceptance rule need field 8 of the root of every completed epoch tree $\mathcal{T}_A$, $A_H < A \le \alpha(\eta)$ an activation height, each committed only by B_A (Theorem 6.24), together with the work of the blocks below A_H, which no tree commits (Corollary 6.25).*

No ZIP addresses the item; it is open.

Proof. (a) The block hash covers the whole header, nBits and hashPrevBlock included (Definition 2.3); induction down the chain, a divergence yielding a collision of SHA-256d. (b) Field 8 of a leaf is its block's work and that of an internal node the sum of its children's (Definition 6.3); the leaves of $\mathcal{T}_\eta$ are $B_{\alpha(\eta)}, \dots, B_{\eta-1}$ (Definition 6.2); the height range of v is that of Lemma 6.6. (c) A path opens only fields of nodes of $\mathcal{T}_\eta$, and by (b) the only work-valued field, field 8, sums work over leaves of $\mathcal{T}_\eta$; the pool-root fields depend on earlier blocks, but through treestates, from which work is not computed. The work of each block is at least 1, since ToTarget(nBits) $+1 \le 2^{256}$, so ω exceeds the work of the leaves of $\mathcal{T}_\eta$ whenever $\alpha(\eta) > 0$. The remaining work is committed only as stated, by Theorem 6.24 and Corollary 6.25. ZIP 221, "Tree Node specification", assumes that cumulative chain work stays below 2^{256}, which makes each modular sum exact (Lemma 3.21(2)). □

Part (b) does not contradict the statement of (a) that every header commits to the total work: that commitment is through the linkage of every header, which a sampling verifier does not receive.

Remark 7.9 (Proof format). The proof format is open; no ZIP or specification section defines one. From NU5 any format carries, for every header whose commitment it opens, the 32-byte hashAuthDataRoot needed to reopen the block commitment hash (Proposition 6.19).

Remark 7.10 (Serving interface). The serving interface is open: no ZIP or specification section defines a message for requesting or serving tree nodes, inclusion paths or assembled proofs. Availability of prover data is not a protocol question. Every node, and hence every inclusion path, of $\mathcal{T}_\eta$ is a function of the chain up to its last leaf $B_{\eta-1}$, not of the blocks of the upgrade epoch alone, since the pool-root fields (6, 7, 12, 13, 15 and 16) are roots of final treestates, which depend on every note commitment since the activation of the pool (Definition 6.3; Construction 6.10; ZIP 221, "Tree Node specification"; ZIP 258, "Changes to ZIP 221"). Any party that holds that chain computes them; what is unspecified is the message.

Remark 7.11 (Transcript). The transcript is open. Construction 5.27 derives the sampling randomness from a hash of the head (ePrint 2019/226, Section 6.2). For Zcash no ZIP or specification section fixes the header bytes, the hash, the domain separation or the map from output to weight points across upgrade epochs, so Proposition 5.29 has no Zcash instance.

Remark 7.12 (Security analysis). The security analysis is open. Zcash retargets after every block from an averaging window with median-time spans, damping and asymmetric clamps, which is not Definition 4.5. The blocks of the upgrade epochs from Heartwood through NU6.3 are governed by a 17-block window at a 75-second target spacing (protocol specification, §"Difficulty adjustment"); NU7 as specified uses 102 blocks at 25 seconds (ZIP 218, Draft, "Averaging window"; *Consensus Guide*, §"Difficulty in motion"). Neither regime is covered by Theorem 5.32, and the paper contains no derivation of (c, L) from either (Remark 4.13). The published analysis of a Zcash instantiation (Perazzo and Capecchi, arXiv 2604.26736, Section IV-A, Definition IV.1) assumes its adversarial work budget rather than deriving it from the retarget rule, so it does not close the gap; it is recorded here because the open classification must account for it. ZIP 221, "Security and Privacy Considerations", records that the paper leaves such chains with only heuristic guarantees, that "The use of these fields has not been analysed in the academic security literature", and that "A more in-depth security analysis of FlyClient should be performed before designing a FlyClient-based light client ecosystem for Zcash". No ZIP addresses the gap.

Remark 7.13 (NU7). ZIP 218 (Draft) changes the target spacing from 75 to 25 seconds, the averaging window from 17 to 102 blocks and the expected stale rate (ZIP 218, "Block target spacing", "Averaging window", "Stale block rate"). These bear on the (c, L) that an instantiation could assume, not on Theorem 5.21, which is stated in c and δ. Transition checks must use the height-dependent spacing and window, span two regimes across one more upgrade boundary (Corollary 6.26), and meet first adjustments that ZIP 218, "Effect on difficulty adjustment", expects to be limited by PoWMaxAdjustDown (*Consensus Guide*, §"NU7: 25-second blocks and bounded shielded work").

Remark 7.14 (Compact-block scanning). Composition with compact-block scanning is open. A FlyClient replaces trust in the header chain, not in the channel through which a wallet detects its transactions (ZIP 307, Draft).

Part VIII
ZSA Guide: Zcash Shielded Assets: issuance, transfer, and the counterfeiting boundary

Abstract

Assuming the Orchard protocol of the *Ironwood Guide* and transaction construction from the *Wallet Guide*, this volume of *The Zcash Arboretum* documents Zcash Shielded Assets: per-asset value bases and asset identity, transparent issuance and finalization (ZIP-227), shielded transfer and burn (ZIP-226), and the split-note construction that guards the counterfeiting boundary. The upstream ZIPs are in flux — their transaction carrier was withdrawn and their fee terms removed upstream — so every claim is classified as specified, designed-but-unspecified, or open. Where ZIP text, fork text, and the implementation disagree, all three readings are stated and the divergence is flagged.

1 Scope, status, and sources

Zcash Shielded Assets (ZSA) is the proposed extension of Orchard that carries user-issued assets in the shielded pool. Two consensus ZIPs define it: ZIP 226, Transfer and Burn, and ZIP 227, Issuance, both drafts. The combined protocol, OrchardZSA, enables issuance (ZIP 227) and transfer and burn (ZIP 226) of Custom Assets on the Zcash chain; ZIP 227 must only be implemented in conjunction with ZIP 226, and issuance is deliberately transparent — issuance outputs are unencrypted — so that every asset's circulating supply can be tracked on chain (`zip-0226.rst` lines 42–43; `zip-0227.rst` lines 47, 54).

Terminology is ZIP 227's: an *Asset* is a type of note transferable on the Zcash blockchain; ZEC is the default — and currently the only defined — Asset on mainnet, TAZ on testnet; a *Custom Asset* is any Asset other than ZEC and TAZ (`zip-0227.rst` lines 35–39).

The design reduces to one alteration of Orchard's value algebra. The fixed value base V^{Orchard} of the homomorphic Pedersen value commitment (*Ironwood Guide*, §"Value commitments") is replaced, per Asset, by an *Asset Base* witnessed inside the circuit; the same Asset Base must appear in the old and the new note commitments and as the base of the Action's value commitment. Under the discrete-log binding assumption for independently derived bases, the binding signature then enforces balance separately per Asset Identifier without revealing which Assets — or how many distinct ones — a transaction transfers (`zip-0226.rst` lines 59–63). The transparent protocol is unchanged: unshielding is impossible for Custom Assets, burn is the only supply-reduction mechanism and sends to no address, and the Native Asset cannot be burnt (`zip-0226.rst` lines 206–230).

1.1 Scope and deferrals

This volume develops only the delta over Orchard: the per-asset note type and note commitment; the Asset Base and its derivation from an Asset Identifier; the per-asset value commitment, the burn mechanism, and the burn-aware balance equation; split notes, the mechanism that replaces dummy spends for Custom Assets; the changes to the Action circuit; the issuance layer of ZIP 227 — issuance keys, issue notes, reference notes, and the node-side issuance state — and the note-plaintext extension.

Everything the delta rests on is constructed in earlier volumes and is cited, never re-derived: notes and note commitments (*Ironwood Guide*, §"Notes and note commitments"); value commitments, the bases V^{Orchard} and R^{Orchard}, and the binding signature (*Ironwood Guide*, §"Value commitments and the binding signature"); the Action statement and dummy spends (*Ironwood Guide*, §"The Action statement", §"Actions, bundles, and dummy notes"); nullifiers (*Ironwood Guide*,

§"Nullifiers"); Sinsemilla (*Ironwood Guide*, §"Sinsemilla: hash, commitment, and short forms"); and, at the wallet layer, transaction construction, PCZT, and fees (*Wallet Guide*, §"Transaction construction", §"Partially created transactions (PCZT)", §"Fees (ZIP-317)").

Deployment status is stated once, in §1.2. Subsequent sections construct their objects without requalifying each of them with it.

1.2 Status

ZSAs are not part of NU7. Its deployment draft, ZIP 259, introduces no transaction format and retains versions 5 and 6 without a ZSA carrier. ZIPs 226 and 227 remain draft proposals for a future upgrade; their mechanisms are not rules an NU7 validator enforces.

The essential distinction is between a cryptographic construction and its *carrier*: the transaction fields that encode it, together with the digests signed by its participants. ZIP 226 assumes the recoverability changes of ZIP 2005, but its carrier references the withdrawn ZIPs 230 and 246. The deployed version 6 format is ZIP 229, which does not carry ZSAs. The extensible version 7 draft, ZIP 248, lists potential ZSA bundles without specifying their encodings or digests. The experimental format below is therefore not a standardised ZSA transaction format.

Two other assignments remain incomplete: ZIP 226 leaves the note-plaintext lead byte unassigned, and the conventional fee rule in ZIP 317 defines no ZSA issuance or asset-creation terms. The fork-specific choices below are identified where used; neither is silently promoted to consensus.

1.3 Sources

ZIPs 226 and 227 define the proposed constructions. The experimental implementation is not identical to those texts; Table 1 fixes which versions supply the fork-specific formulas below. Current upgrade scope follows ZIP 259, not the experimental code's feature names.

Artifact	Pin / date	Role
`zcash/zips` (upstream)	`69610984`, 2026-07-09	primary ZIP ground truth
`QED-it/orchard`, branch `zsa1`	`cf801a5d`, 2026-06-29	crate `orchard` 0.14.0
`QED-it/librustzcash`, branch `zsa1`	`3f9b53fc`, 2026-04-17	v6 carrier, fee rule
`QED-it/zips`, branch `zsa1`	`fd71419a`, 2026-02-11	fork baseline for diffing

Table 1: Primary ZSA sources, pinned.

Upstream versus fork. Upstream ZIP 226 diverged from the fork text at `fd71419a` in three substantive ways: it added the ZIP 2005 dependency paragraph; it changed the split-note ψ^{nf} from "sampled uniformly at random on $\mathbb{F}_{p_{\mathsf{Pallas}}}$" to a $\mathsf{PRFexpand}$ derivation whose domain-separator byte `0x0A` is reserved by ZIP 2005; and it replaced the lead byte `0x03` with the `{{ZSALEADBYTE}}` placeholder. ZIP 227 upstream differs from the fork only in punctuation and one reference title.

Code versus ZIP. Where the pinned code disagrees with the pinned ZIP text, this volume presents the ZIP formula and flags the divergence inline. Two content divergences head the list. First, the split-note nullifier randomizer: ZIP 226 derives ψ^{nf} with domain byte `0x0A`, while `orchard-zsa @ cf801a5d` computes it with $\mathsf{PRFexpand}$ domain `0x09` over a random per-note seed (`src/circuit.rs` lines 316–319) — the older fork-era derivation. Second, the note-plaintext lead byte: a placeholder upstream, but `0x03` in code (`src/primitives/zcash_note_encryption_domain.rs` lines 46–49) — the very value ZIP 2005 now specifies differently. A third divergence is one of status rather than content: the digest personalisations in `orchard-zsa` match the withdrawn ZIP 246 exactly, and `librustzcash-zsa`

implements `TxVersion::V6` with the withdrawn ZIP 230 wire format — the entire carrier layer the code implements is specified only by withdrawn ZIPs (§1.2). Further code-versus-text divergences confirmed at the same pins — the omitted explicit-fee digest field, the missing memo branch and its non-spec substitute digest, the empty-action issuance rule, and the valueBurn bound wording — are inventoried in §6.6, §3.7, and §4.4.

The two library pins are separate snapshots, not a single deployed stack. Checks of the implementation below establish library behaviour, not node-side enforcement of ZIP 227's issuance state.

1.4 Method: the three bins for ZSA

This volume classifies every design-stage claim as *specified*, *designed but unspecified*, or an *open problem*, and keeps the bin visible in the prose. Bin (i) holds the constructions and consensus rules that the live ZIP texts state directly: the per-asset note and commitment, Asset Base derivation, value commitments and balance equation, the split-note circuit delta, issuance keys, issue and reference notes, the issuance state machine, and burn. Bin (ii) holds both the gate-level realization of that circuit delta and the carrier layer: the v6 wire format, digest tree and sighash, note-plaintext lead byte, and ZSA fee terms. Those details are fixed only by implementation, an externally hosted unversioned circuit description, withdrawn ZIPs, or the QED-it fork texts (§1.2). Bin (iii) holds assignment to a future network upgrade and activation, the eventual lead-byte assignment, and the carrier format's future home. Formulas below identify whether their source is ZIP text, pinned code, or an explicitly labelled derivation in this volume.

2 Asset identity

An OrchardZSA note denominates value in one of many assets sharing a single shielded pool, so the protocol must attach to every note an asset identity satisfying two demands. First, uniqueness: "The Asset identification should be unique (among all shielded pools) and different issuer public keys should not be able to generate the same Asset Identifier" (`zip-0227.rst`:76). Second, usability inside the circuit: the identity must ultimately be a Pallas point, because it serves as the asset-specific base of the value commitment (§2.4). ZIP 227 meets both demands with one derivation chain,

$$\texttt{asset_desc} \longrightarrow \mathsf{assetDescHash};$$
$$(\mathsf{issuer}, \mathsf{assetDescHash}) \longrightarrow \mathsf{AssetId} \longrightarrow \mathsf{AssetDigest} \longrightarrow \mathsf{AssetBase},$$

which the ZIP itself presents as a relation diagram (`zip-0227.rst`:210–224): the issuer-chosen description string is hashed, the hash paired with the issuer key into an identifier, the identifier hashed into a digest, and the digest mapped onto the Pallas curve. Every Asset issued under ZIP 227 is scoped to the issuer that issued it. Its Asset Identifier is intended to be globally unique and is "used across all Zcash protocols that support ZSAs" (`zip-0227.rst`:227–240); for two descriptions under one issuer, uniqueness is computational and rests on BLAKE2b-256 collision resistance.

This section constructs the chain end to end: the issuance key pair at its root (§2.1); the hash steps from description to identifier (§2.2); the group hash from identifier to base point (§2.3); the native asset's special position outside the chain (§2.4); and the parts of the chain consensus sees on the wire versus recomputes (§2.5). Deployment status for the whole protocol is fixed once, in §1.2, and not restated per object.

2.1 The issuance key pair

The OrchardZSA protocol adds exactly two keys to the key material of Orchard: the *issuance authorizing key* isk, which authorizes the issuance of Asset Identifiers by a given issuer and is

used only by that issuer, and the *issuance validating key* ik, which validates issuance transactions (`zip-0227.rst`:84–91). Neither key touches the spend path. The ZIP's rationale: the issuance key structure is independent of the original key tree but derived analogously via ZIP 32, which keeps issuance details and Asset Identifiers consistent across multiple shielded pools, and separates issuance authority from spend authority — allowing a potential transfer of issuance authority without compromising spend authority (`zip-0227.rst`:524).

Definition 2.1 (Issuance authorization signature scheme)**.** The scheme IssueAuthSig is instantiated as a BIP-340 Schnorr signature over the secp256k1 curve, with constants set per the secp256k1 standard parameters as described in BIP 340. Its associated types are: messages are arbitrary byte sequences; signatures are 64-byte sequences or $\perp$; public keys are 32-byte sequences or $\perp$; private keys are 32-byte sequences (`zip-0227.rst`:122–135). Signing fixes the BIP-340 auxiliary data to $a = [\texttt{0x00}]^{32}$: the signature is $\sigma := \mathsf{Sign}(\mathsf{isk}, M)$ with auxiliary data a, or $\perp$ if Sign fails; its scheme-level encoding is $\texttt{issueAuthSig} = [\texttt{0x00}] \parallel \sigma$ (`zip-0227.rst`:190–207).

In the implementation, type `ZSASchnorr` realizes the scheme with `ALGORITHM_BYTE = 0x00`, secret keys as `secp256k1::SecretKey`, public keys as `XOnlyPublicKey`, and signing via `sign_schnorr_with_aux_rand` with the all-zero 32-byte auxiliary array.

Construction 2.2 (Issuance key derivation)**.** The issuance authorizing key is generated by ZIP 32's hardened-only key derivation — the maps MKGh and CKDh of ZIP 32, "Specification: Hardened-only key derivation", whose Orchard instance is the *Wallet Guide*'s §"Hardened-only derivation" — instantiated in a dedicated Issuance context (`zip-0227.rst`:139–166). Let S be a seed of at least 32 and at most 252 bytes with at least 256 bits of entropy, as ZIP 32's "Specification: Wallet seeds" requires.

(1) *Context.* The issuance context sets Issuance.MKGDomain = `ZcashSA_Issue_V1` and Issuance.CKDDomain = `0x81` (`zip-0227.rst`:148–149).

(2) *Master key.* Set $m_{\mathsf{Issuance}} := \mathsf{MKGh}^{\mathsf{Issuance}}(S)$.

(3) *Child derivation.* Set $\mathsf{CKDsk}((\mathsf{sk}_{\mathsf{par}}, c_{\mathsf{par}}), i) := \mathsf{CKDh}^{\mathsf{Issuance}}((\mathsf{sk}_{\mathsf{par}}, c_{\mathsf{par}}), i)$. The two-argument call fixes the general form's *lead* = 0 and *tag* = [ ], exactly the case whose suffix the ZIP-32 encoding omits (`zip-0032.rst`:449–462).

(4) *Path.* Derive along m_{Issuance} / purpose′ / coin_type′ / account′, with purpose fixed to 227 (`0xe3`; hardened `0x800000e3` per BIP 43), coin_type as in the ZIP 32 key path levels, and account fixed to index 0 (`zip-0227.rst`:160–164).

(5) *Key extraction.* From the generated pair (sk, c), set $\mathsf{isk} := \mathsf{sk}$ (`zip-0227.rst`:164–166).

The implementation realizes the context as a struct `Issuance` implementing the `zip32` crate's hardened-only `Context` trait, with `MKG_DOMAIN` set to `ZcashSA_Issue_V1` and with `CKD_DOMAIN` reusing the Orchard byte, constant `PrfExpand::ORCHARD_ZIP32_CHILD`; the seed-level constructor `from_zip32_seed` derives along $[227', \text{coin_type}', 0']$ and rejects any non-zero account with error `NonZeroAccount`. The `zip32` crate 0.2.0 implements MKGh and CKDh exactly, and its `derive_child(index)` calls `ckdh_internal(index, 0, [])` — *lead* = 0, *tag* = [ ], omitted from the PRF message.

Remark 2.3 (Domain separation between the Orchard and Issuance trees). The byte Issuance.CKDDomain = `0x81` equals Orchard.CKDDomain (`zip-0032.rst`:476), and the implementation reuses `PrfExpand::ORCHARD_ZIP32_CHILD` (`zcash_spec src/prf_expand.rs`:119 @ `d5e84264`).

Separation between the two ZIP-32 trees therefore rests entirely on the differing MKGDomain: distinct master secret keys and chain codes feed every subsequent PRFexpand call. Neither ZIP states whether the byte reuse is deliberate; ZIP 32 asks only that CKDDomain values be tracked as part of the global set of domains defined for PRFexpand. The derivation is *specified*; we flag the shared byte because the separation argument is easy to misattribute to it.

Definition 2.4 (Issuance validating key and issuer identifier)**.** The issuance validating key is $\mathsf{ik} := \mathsf{PubKey}(\mathsf{isk})$, where PubKey is the BIP-340 algorithm and the output is in big-endian order as defined in BIP 340; the derivation returns $\perp$ if PubKey fails, which happens with very low probability, in which case the keys are discarded and derivation repeated with a different isk (`zip-0227.rst`:168–183). Its encoding prepends a signature-scheme byte, which MUST be `0x00`, indicating BIP 340:

$$\mathsf{ik_encoding} = [\texttt{0x00}] \,\|\, \mathsf{ik} \qquad (33 \text{ bytes}),$$

and the issuer identifier is defined by $\mathsf{issuer} = \mathsf{ik_encoding}$ (`zip-0227.rst`:103–110, 178–179). The encoding currently appears on chain only in the `issuer` field of an issuance bundle (`zip-0227.rst`:178–180).

Remark 2.5 (Permanence of the issuer binding). ZIP 227 notes that "the equality of `issuer` and `ik_encoding` might not hold in future when key rotation is specified for issuance key pairs" (`zip-0227.rst`:103–110). No rotation mechanism exists today: an asset's identity is permanently bound to the one ik under which it was issued, and the ZIP's remedy for key compromise is to finalize the asset and reissue under a new Asset Identifier: "If an issuer identifier is compromised, the finalize boolean for all the Assets issued with that key should be set to 1; then the issuer should change to a new issuance authorizing key (hence a new issuer identifier), and issue new Assets, each with a new AssetId" (`zip-0227.rst`:640–645, Security and Privacy Considerations, "Issuance Key Compromise"). This is a *stated limitation* of the specified design; the rotation mechanism itself is an open design area.

2.2 From description to identifier

An issuer names each of its assets with an *asset description* `asset_desc`: a non-empty byte sequence, which SHOULD be a well-formed UTF-8 code unit sequence according to Unicode 15.0.0 or later (`zip-0227.rst`:237–241). Within the scope of an issuer, Asset Identifier uniqueness is obtained by way of the asset description (`zip-0227.rst`:237–240). The current ZIP imposes no maximum length — its rationale notes that a description "can be longer than 512 bytes without incurring chain costs", the 512-byte cap being an earlier design (`zip-0227.rst`:528–539). The description is also global: ZIP 226 states that "Every Asset is defined by an Asset description, `asset_desc`, which is a global byte string (scoped across all future versions of Zcash)", so that in a future upgrade the same string can carry over to a different shielded pool, with nodes transforming the identifier chain between pools without balance violations (`zip-0226.rst`:93–103).

Definition 2.6 (Asset Identifier)**.** The description is hashed,

$$\mathsf{assetDescHash} := \text{BLAKE2b-256}(\texttt{ZSA-AssetDescCRH}, \texttt{asset_desc})$$

(`zip-0227.rst`:243–247), and the Asset Identifier is the pair

$$\mathsf{AssetId} := (\mathsf{issuer}, \mathsf{assetDescHash})$$

(`zip-0227.rst`:249–253), with the canonical encoding

$$\mathsf{EncodeAssetId}(\mathsf{AssetId}) := [\texttt{0x00}] \parallel \mathsf{issuer} \parallel \mathsf{assetDescHash} \qquad (66 = 1 + 33 + 32 \text{ bytes}),$$

where the initial `0x00` is a version byte enabling future issuance protocols (`zip-0227.rst`:255–261). The version byte is not to be confused with the byte that follows it — the first byte of issuer, which indicates the signature scheme (Definition 2.4); every current encoding happens to begin with two `0x00` bytes of distinct meaning.

The pair structure carries the uniqueness argument. Because the issuer identifier is a component of AssetId, two distinct issuer-key encodings cannot produce the same pair; in the ZIP's words, the Custom Asset "is described via a combination of the issuer identifier and an asset description string, to preclude the possibility of two different issuers creating colliding Custom Assets" (`zip-0227.rst`:525). Within one issuer, distinct descriptions separate assets via assetDescHash only subject to BLAKE2b-256 collision resistance; for every new Asset there MUST be a new and unique Asset Identifier (`zip-0226.rst`:93–96).

Remark 2.7 (Why a hash, not the string). ZIP 227 replaces the direct description string of an earlier design with a collision-resistant hash for five stated reasons (`zip-0227.rst`:528–559): a fixed 32 bytes per Issue Action costs less bandwidth than up to 512 bytes; descriptions can exceed 512 bytes without chain cost; carrying the byte string on chain would not make the user-visible description consensus-visible anyway (the string could itself be a hash of an off-chain description); absent key rotation, marking an issuer as trusted is insufficient — each Asset Identifier needs independent out-of-band verification that conveys the description; and an on-chain description is an "attractive nuisance" — hashing forces wallets to obtain the description from a trusted party, a registry, or direct user approval. Consistent with the last point, wallets MUST NOT display just the `asset_desc` string as the name of the Asset; the ZIP suggests a petname system mapping names to Asset Identifiers via client configuration, or the Asset Digest as a compact identifying byte sequence, e.g. in QR codes (`zip-0227.rst`:263–266).

Implementation. The identifier is the enum variant `AssetId::V0`, holding the validating key and the 32-byte description hash; `encode_asset_id` emits the version byte `0x00`, the 33-byte key encoding, and the hash — 66 bytes, matching EncodeAssetId. The function `compute_asset_desc_hash` enforces non-emptiness through its argument type `NonEmpty<u8>` and *panics* if the description is not well-formed UTF-8 (`orchard-zsa src/issuance.rs`:135–154 @ `cf801a5d`). The specification says SHOULD; the implementation enforces a hard MUST at its API boundary. Since only the 32-byte hash is consensus-visible, neither variant is consensus-enforceable: we classify the UTF-8 constraint as an issuer-side convention — *specified* as a SHOULD, with the stricter behaviour an implementation choice.

2.3 From identifier to base point

Two further steps land the identifier on the Pallas curve. The full chain, collected:

$$\begin{aligned}
\mathsf{assetDescHash} &:= \text{BLAKE2b-256}(\texttt{ZSA-AssetDescCRH}, \texttt{asset_desc}),\\
\mathsf{AssetId} &:= (\mathsf{issuer}, \mathsf{assetDescHash}),\\
\mathsf{AssetDigest}_{\mathsf{AssetId}} &:= \text{BLAKE2b-512}(\texttt{ZSA-Asset-Digest}, \mathsf{EncodeAssetId}(\mathsf{AssetId})),\\
\mathsf{AssetBase}_{\mathsf{AssetId}} &:= \mathsf{ZSAValueBase}(\mathsf{AssetDigest}_{\mathsf{AssetId}}),
\end{aligned}$$

where the digest is defined at `zip-0227.rst`:268–278, the base at `zip-0227.rst`:280–290, and the OrchardZSA instantiation of the final step is

$$\mathsf{ZSAValueBase}(\mathsf{AssetDigest}) := \mathsf{GroupHash}^{\mathsf{Pallas}}(\texttt{z.cash} : \texttt{OrchardZSA}, \mathsf{AssetDigest})$$

(`zip-0227.rst`:292–301). The subscript AssetId may be omitted when clear from context (`zip-0227.rst`:224). The group hash is the protocol specification's $\mathsf{GroupHash}^{\mathbb{P}}$ (§5.4.9.8, "Group Hash into Pallas and Vesta"), whose construction — `expand_message_xmd` over BLAKE2b, two field elements through the simplified SWU map, summed and carried through the isogeny — the *Ironwood Guide* develops in "GroupHash, domain separation, and nothing-up-my-sleeve generators"; we reuse it as a deterministic, public, nothing-up-my-sleeve map into the Pallas group and do not re-derive it here.

ZIP 226 types the result as a *non-trivial* point: AssetBase is "the unique element of the Pallas group that identifies each Asset in the Orchard protocol ... a valid group element that is not the identity and is not $\perp$" (`zip-0226.rst`:111–117), a non-identity point of $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})$ in our notation. Here uniqueness can only mean that the derivation is deterministic for a given identifier, not that the map is injective: a 512-bit digest is mapped into a roughly 255-bit group, so collisions exist in principle. Asset separation at the base and global-state levels is computational and requires that collisions in the digest/group-hash chain be infeasible to find. Its byte form is

$$\texttt{asset_base} := \mathrm{LEBS2OSP}_{\ell_{\mathbb{P}}}(\mathrm{repr}_{\mathbb{P}}(\mathsf{AssetBase})),$$

the 32-byte representation stored in note plaintexts and burn fields (`zip-0226.rst`:114; `zip-0230.rst`:540–545). ZIP 226 notes the encoding assumes canonicity, "true for the Pallas group, but may not hold for future shielded protocols" (`zip-0226.rst`:111–117). Table 2 collects the sizes along the chain.

Object	Bytes	Form	Source
ik	32	x-only BIP-340 key, big-endian	`zip-0227.rst`:130–133
ik_encoding = issuer	33	[0x00] ‖ ik	`zip-0227.rst`:178–179
assetDescHash	32	BLAKE2b-256 output	`zip-0227.rst`:243–247
EncodeAssetId(AssetId)	66	[0x00] ‖ issuer ‖ assetDescHash	`zip-0227.rst`:255–259
AssetDigest	64	BLAKE2b-512 output	`zip-0227.rst`:268–278
`asset_base`	32	$\mathrm{LEBS2OSP}_{\ell_{\mathbb{P}}}(\mathrm{repr}_{\mathbb{P}}(\cdot))$	`zip-0230.rst`:540–545

Table 2: Byte sizes along the asset-identity derivation chain.

Why the base must be a genuine group-hash output. The chain's final step is not a formality. In a transfer, the output note of a split Action cannot be allowed to carry an arbitrary Pallas point as its base: "We must enforce it to be an actual output of a GroupHash computation ... Without this enforcement the prover could input a multiple (or linear combination) of an existing Asset Base, and thereby attack the network by overflowing the ZEC value balance and hence counterfeiting ZEC funds" (`zip-0226.rst`:299–307). The circuit therefore ensures the value base points of all output notes are actual GroupHash outputs "as they originate in the Issuance protocol which is publicly verified", and ZIP 226 adds: "We prevent a potential malleability attack on the Asset Identifier by ensuring the output notes receive an Asset Base that exists on the global state" (`zip-0226.rst`:104–106, 299–307). The enforcement mechanism belongs to the transfer sections of this volume (§5).

Remark 2.8 (The identity-point corner case). The protocol specification's $\mathsf{GroupHash}^{G}(D, M)$ has the full group as codomain, so its output can in principle be the zero point (`protocol.tex` §5.4.9.8; the *Ironwood Guide*, §"GroupHash, domain separation, and nothing-up-my-sleeve generators", constructs the hash-to-curve pipeline). ZIP 227's ZSAValueBase never states what happens in that event; the non-identity requirement comes from ZIP 226's typing of the note field (`zip-0226.rst`:111–117) and from the implementation, which panics inside `AssetBase::custom` ("this will happen with negligible probability"), rejects the identity in `AssetBase::from_bytes`, and carries an `Error::AssetBaseCannotBeIdentityPoint`

check in `IssueAction::verify` that the panic renders unreachable (`orchard-zsa src/note/asset_base.rs`:98–138, `src/issuance.rs`:216–244 @ `cf801a5d`). Whether the intended consensus behaviour is "reject the transaction" or "the issuer must pick a different description" is unspecified in the ZIP text. The event has negligible probability; the specification gap is nevertheless real, and we classify the corner case as *open*.

2.4 The native asset

ZEC is the default — and currently the only defined — Asset for mainnet, TAZ for testnet; a *Custom Asset* is any Asset other than ZEC and TAZ (`zip-0227.rst`:35–39). The native asset does not pass through the derivation chain at all. Its base is fixed by definition: "the Asset Base of the ZEC Asset will be kept as the original value base point, V^{Orchard}" (`zip-0226.rst`:98); for ZEC,

$$\mathsf{AssetBase}_{\mathsf{AssetId}} := V^{\mathsf{Orchard}},$$

"so that the value commitment for ZEC notes is computed identically to the Orchard protocol deployed in NU5. As such $\mathsf{ValueCommit}^{\mathsf{Orchard}}_{\mathsf{rcv}}(v)$ as defined in ZIP 224 is used as $\mathsf{ValueCommit}^{\mathsf{OrchardZSA}}_{\mathsf{rcv}}(V^{\mathsf{Orchard}}, v)$" (`zip-0226.rst`:182–194). The point $V^{\mathsf{Orchard}} := \mathsf{GroupHash}^{\mathsf{Pallas}}$(`z.cash : Orchard-cv, "v"`) is already constructed, together with its companion R^{Orchard}, in the *Ironwood Guide*, "Value commitments", Definition "Net value commitment"; we cite it and do not re-derive it.

Two consequences follow. First, indistinguishability: ZIP 230 states that "An Orchard note is essentially equivalent to an OrchardZSA note with $\mathsf{AssetBase} = V^{\mathsf{Orchard}}$; in particular, they have the same note commitment ... and need not be distinguished in note plaintexts" (`zip-0230.rst`:531–538), and ZIP 226's privacy discussion states that OrchardZSA note commitments for the native asset are indistinguishable from those for non-native Assets (`zip-0226.rst`:83–84). Second, exclusion from burning: the first consensus rule for `assetBurn` requires $\mathsf{AssetBase} \neq V^{\mathsf{Orchard}}$ for every $(\mathsf{AssetBase}, v) \in$ `assetBurn` (`zip-0226.rst`:220–223) — the native asset cannot be burnt.

Implementation. The native base is `AssetBase::zatoshi()`, computed as the Pallas hash-to-curve over personalization `"z.cash:Orchard-cv"` of the message `"v"` — exactly V^{Orchard}; `AssetBase::is_zatoshi()` is a constant-time equality with that point; and `ValueCommitment::derive` computes $[v]\,V + [\mathsf{rcv}]\,R$ with $V =$ `asset.cv_base()` and R the fixed base R^{Orchard}, so a ZEC note's value commitment is bit-identical to NU5 Orchard's.

2.5 Identity on the wire

Of the whole chain, consensus sees exactly two links on the issuance side: the issuer and the description hash. The v6 transaction's issuance bundle carries `issuerLength` (compactSize), `issuer` (the issuer identifier per ZIP 227), `nIssueActions` (compactSize), `vIssueActions`, and `issueAuthSig` — the signature bytes preceded by a `0x00` byte indicating BIP 340. The first four fields are always present; `issueAuthSig` is present iff `nIssueActions` > 0, and if `nIssueActions` $= 0$ then `issuerLength` MUST be 0 (`zip-0230.rst`:166–179, 215–217, 430–443). Each Issue Action carries `assetDescHash` as `byte[32]`, `nNotes` (compactSize, possibly zero; for an existing Asset, this permits finalization without further issuance), `vNotes` (115 bytes per note description: 43-byte recipient, 8-byte value, 32-byte ρ, 32-byte rseed), and `flagsIssuance` with finalize in the least significant bit; the burn encoding carries `assetBase` as `byte[32]` (`zip-0230.rst`:367–383, 446–494). On the transfer side, the v6 Orchard note plaintext inserts the asset between rseed and the memo key (the 32-byte key that replaces the in-band 512-byte memo under ZIP-231 and from which the recipient derives the key decrypting its memo in the

transaction-level memo bundle; §4.7),

$$[\texttt{leadByte}] \parallel \mathrm{LEBS2OSP}_{88}(d) \parallel \mathrm{I2LEOSP}_{64}(v) \parallel \mathsf{rseed} \parallel \texttt{asset_base} \parallel \mathsf{K}^{\mathsf{memo}}$$

(zip-0230.rst:523–553).

The consensus rules bind identity to validation: the ik_encoding used to validate issueAuthSig MUST be taken from the issuer field and MUST start with byte 0x00, and for each Issue Action, the Issue Note's AssetBase is derived from the assetDescHash field of the action and the issuer field of the bundle (zip-0227.rst:483–491). The implementation makes the recomputation structural. Function read_bundle decodes the issuer via IssueValidatingKey::decode, checking the algorithm byte; function read_action reads the 32-byte hash and recomputes the base as AssetBase::custom of AssetId::new_v0(ik, asset_desc_hash). The Asset Base is never carried on the wire for issuance, always re-derived. Redundantly with construction, IssueAction::verify checks that every note's base equals the one derived from the validating key and the description hash, failing with error IssueBundleIkMismatchAssetBase otherwise. The Asset Identifier is therefore not serialized as a standalone field: its two components are consensus-visible, and the Asset Base is a recomputable function of them, anchored to the issuer key that signs the bundle.

Classification. The entire derivation chain of this section is *specified*: every formula above appears in the ZIP texts at rigor and is realized by the pinned implementation forks at the pinned commits (orchard-zsa @ cf801a5d, librustzcash-zsa @ 3f9b53fc). Both ZIPs are Status Draft, Category Consensus, owned by Pablo Kogan, Vivek Arte, Daira-Emma Hopwood, and Jack Grigg, created 2022-05-01 and discussed in zcash/zips issue #618 and PR #680 (zip-0226.rst:1–18, zip-0227.rst:1–18); deployment status is stated in the scope section. Three reference gaps qualify the bin. First, ZIP 227's Test Vectors and Reference Implementation sections read "LINK TBD" and its Deployment section "TBD" (zip-0227.rst:661–675); the de-facto assertion-checked vectors live in the fork: file test_vectors/asset_base.rs, generated from the zcash-hackworks test-vectors script orchard_zsa_asset_base.py, is replayed by a test that recomputes the full chain from (ik_encoding, asset_desc) to asset_base (src/note/asset_base.rs:236–261 @ cf801a5d), with companion vectors for the signature scheme and the ZIP-32 derivation in issuance_auth_sig.rs and zip32.rs. The vectors' fixed [u8; 512] description arrays are a relic of the dropped 512-byte cap and imply no bound. Second, the corner cases flagged above remain: the identity-point behaviour is *open* (Remark 2.8), the UTF-8 constraint is a SHOULD with a stricter implementation (§2.2), and key rotation is an open design area (Remark 2.5). Third, upstream zcash/zips @ 69610984 is the text of record; the QEDit zips fork (zips-zsa @ fd71419a) is essentially identical for the asset-identity material, and the differences — upstream's later additions of a ZIP-2005 dependency, a nullifier ψ^{nf} derivation, and a lead-byte rule — do not touch this chain.

3 Issuance and finalization (ZIP-227)

The transfer mechanism below moves Custom Assets between shielded notes; this section brings them into existence. ZIP-227 specifies issuance on the Zcash chain: a per-transaction *issuance bundle* in which a publicly named issuer creates new notes, a per-Asset *finalize* switch that ends issuance forever, and a node-maintained global supply map.

Issuance is deliberately *not* shielded. The ZIP's Motivation states: "This ZIP only enables *transparent* issuance. As a first step, transparency will allow for proper testing of the applications that will be most used in the Zcash ecosystem, and will enable the supply of Assets to be tracked" (ZIP-227, *Motivation*). The issuer, Asset, issued amounts, finalization status, raw recipient addresses, ρ values, and random seeds are serialized in the clear. For an ordinary recipient, the address does not reveal the nullifier key, so a later spend is designed to be unlinkable to the

public issue-note description. The mandatory all-zero-key reference note is an explicit exception (§3.3).

Encoding boundary. The issuance mechanism follows ZIP 227. Its encodings and digests below follow the experimental ZIP 230/246 carrier, not ZIP 229's deployed v6 format (§1.2). Fork-only fee rules and differences between the ZIP and the pinned libraries are marked where they affect a construction.

3.1 Issuance keys and the issuer identifier

Issuance authority is a new capability, deliberately disjoint from spend authority. The rationale in the ZIP: the issuance key tree is independent of the spending key tree but derived analogously (via ZIP 32), so an organization can delegate issuance without exposing funds, and a future scheme can replace the issuance leaf with a multisignature without touching spending keys (ZIP-227, *Rationale*, and *Issuance Key Derivation*).

The key material itself is constructed once, in §2.1: the BIP-340 signature scheme (Definition 2.1), the hardened-only ZIP-32 derivation of isk in the $\mathsf{Issuance}$ context (Construction 2.2), and the validating key with the issuer identifier, $\mathsf{ik} := \mathit{PubKey}(\mathsf{isk})$ and $\mathsf{issuer} = \mathsf{ik_encoding} = \mathtt{[0x00]} \| \mathsf{ik}$ (Definition 2.4). We restate only the parameters the issuance layer consumes: seed S of 32 to 252 bytes, $\mathsf{Issuance.MKGDomain} = \mathtt{ZcashSA_Issue_V1}$, $\mathsf{Issuance.CKDDomain} = \mathtt{0x81}$, hardened path $m_{\mathsf{Issuance}} / 227' / \mathit{coin_type}' / 0'$, and $\mathsf{isk} := \mathsf{sk}$ at the leaf (ZIP-227, *Issuance Key Derivation*). No rotation exists today: the on-chain issuer name is the validating key itself, and the ZIP notes the equality might not hold in future if key rotation is ever specified (Remark 2.5; §3.8).

3.2 Asset identity: description, digest, and Asset Base

Every Custom Asset is named by its issuer plus an issuer-chosen description, and consumed by the protocol as a Pallas point. The full derivation chain — $\mathsf{assetDescHash}$, $\mathsf{AssetId}$, $\mathsf{EncodeAssetId}$ with its version byte, $\mathsf{AssetDigest}$, and $\mathsf{AssetBase} = \mathsf{ZSAValueBase}(\mathsf{AssetDigest})$ — is constructed once, in §2.2 and §2.3 (Definition 2.6); the hash-not-string rationale is Remark 2.7, and the UTF-8 SHOULD-versus-panic delta between ZIP and implementation is recorded in §2.2. The native Asset bypasses the chain entirely: its base is the Orchard value-commitment base V^{Orchard} (§2.4). ZIP 226 excludes the Pallas identity from the Asset-Base type; `AssetBase::custom` panics after hashing to it, while `AssetBase::from_bytes` rejects it on parse (Remark 2.8). What the issuance layer adds to the chain is consensus-side recomputation: validators re-derive every Issue Note's $\mathsf{AssetBase}$ from the bundle's `issuer` and the action's `assetDescHash`, never reading it off the wire (§2.5; consensus rules, §3.7).

3.3 Issue notes and the derived ρ

Definition 3.1 (Issue note)**.** An *Issue Note* is a tuple $(\mathsf{d}, \mathsf{pk_d}, \mathsf{v}, \mathsf{AssetBase}, \rho, \mathsf{rseed})$,

$$\mathsf{Note}^{\mathsf{Issue}} := \{0,1\}^{\ell_{\mathsf{d}}} \times \mathsf{KA}^{\mathsf{Orchard}}.\mathsf{Public} \times \{0\,..\,2^{\ell_{\mathsf{value}}} - 1\} \times \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})^* \times \mathbb{F}_{p_{\mathsf{Pallas}}} \times \{0,1\}^{8\cdot 32},$$

a diversifier and diversified transmission key as in Orchard, a value v with $\ell_{\mathsf{value}} = 64$, a non-identity Asset Base, the nullifier-input field element ρ, and a 32-byte rseed that MUST be sampled uniformly at random by the issuer (ZIP-227, *Issue Note*; the ZIP writes the fourth and fifth components $\mathbb{P}^*$ and $\mathbb{F}_{q_{\mathbb{P}}}$, our $\mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})^*$ and $\mathbb{F}_{p_{\mathsf{Pallas}}}$).

Issue-note commitments are not serialized: the wire format carries the note fields themselves in the clear (§3.4), and each validator computes the commitment — by the OrchardZSA $\mathsf{NoteCommit}$ of §4 — and appends it to the Orchard note commitment tree. For an ordinary issue

note, the recipient address does not reveal the nullifier key; under the nullifier PRF assumptions, a later nullifier is computationally unlinkable to that public description. This privacy claim does not apply to the reference note, whose all-zero spending key is public (ZIP-227, *Issue Note*, *Reference Notes*, and *Issuance Protocol*). Issuance is transparent at creation; later transfers of ordinary issue notes are shielded.

Computation of ρ. An Orchard note takes $\rho := \mathsf{nf}_{\mathrm{old}}$ from the Action that creates it (the *Ironwood Guide*, §"Nullifier chaining"); an issue note has no such Action, so ZIP-227 derives ρ from the transaction's OrchardZSA transfer bundle instead:

$$\mathsf{DeriveIssuedRho}(\mathsf{nf}, i_{\mathsf{A}}, i_{\mathsf{N}}) := \mathrm{ToBase}(\mathsf{PRFexpand}_{\mathsf{nf}}(\mathtt{0x84} \,\|\, \mathrm{I2LEOSP}_{32}(i_{\mathsf{A}}) \,\|\, \mathrm{I2LEOSP}_{32}(i_{\mathsf{N}}))),$$

$$\rho := \mathsf{DeriveIssuedRho}(\mathsf{nf}_{0,0},\ \mathsf{index}_{\mathrm{Action}},\ \mathsf{index}_{\mathrm{Note}}),$$

with the PRF keyed by nf as 32 little-endian bytes ($\mathrm{I2LEOSP}_{256}$), where $\mathsf{nf}_{0,0}$ is the nullifier of the input note in the *first* Action of the *first* Action Group of the OrchardZSA bundle (the v6 format partitions a bundle's Actions into one or more Action Groups, each carrying its own anchor and proof; §6.2), and the two indices are the zero-based positions of the Issue Action within the issuance bundle and of the note within its action (ZIP-227, *Computation of ρ*). The domain-separator byte `0x84` extends the PRFexpand lead-byte registry (the *Wallet Guide*, §"The PRFexpand lead-byte registry"); the patched `zcash_spec` carries it as `ORCHARD_DERIVED_ISSUE_RHO` (QED-it `zcash_spec`, `src/prf_expand.rs` line 107 @ `d5e8426`). The implementation computes the identical expression — PRFexpand keyed by the 32-byte little-endian nullifier over `0x84` followed by two 4-byte little-endian indices, reduced by ToBase. This derivation is why an issuance bundle cannot travel alone: requiring at least one Action Group both supplies $\mathsf{nf}_{0,0}$ and blocks replay — a transaction containing only an issuance bundle would have a SIGHASH computed solely from that bundle, so a duplicated bundle would share the SIGHASH (ZIP-227, *Rationale*).

In the builder, notes are created with ρ unset, via `new_issue_note`; once the transfer bundle's first nullifier is known, `update_rho_for_issuance_note` sets ρ and resamples rseed until both the derived ephemeral key is valid and the note commitment is not $\perp$[1] (`orchard-zsa`, `src/note.rs` lines 457–489 @ `cf801a5d`).

Reference notes. The first time an Asset is issued, the issuer MUST place a *reference note* as the first note of that Issue Action: a note of value $\mathsf{v} = 0$ whose recipient $(\mathsf{d}, \mathsf{pk}_{\mathsf{d}})$ is the default diversified payment address (diversifier index 0) derived from the *all-zero* Orchard spending key (ZIP-227, *Reference Notes*). The implementation hard-codes the resulting 43-byte raw address as `RAW_REFERENCE_RECIPIENT`, byte-identical to the array in the ZIP, with `ReferenceKeys::sk() = SpendingKey::from_bytes([0; 32])`; the predicate `is_reference_note` requires zero value and exactly this recipient, `get_reference_note` accepts only the *first* note of an action, and the builder inserts the reference note itself when told the issuance is a first issuance — adding a recipient for an Asset that already has notes fails with `CannotBeFirstIssuance`. The reference note is retained in global state as $\mathsf{note}_{\mathsf{ref}}$ (§3.7).

The ZIP mandates reference notes and stores them in consensus state, but states no rationale for their existence. The evident function — that every issued Asset Base is guaranteed at least one note in the commitment tree, available as a ZIP-226 split-input source for a first spend — is our reading, stated nowhere in the ZIP text or its rationale sections; we classify the mechanism itself as *specified* and this purpose as *designed but unspecified*.

[1] Strictly, these negligible-probability rejection steps condition rseed, a technical deviation from the ZIP's "sampled uniformly at random" (ZIP-227, *Issue Note*).

3.4 The issuance bundle and its encoding

An *Issuance Bundle* carries the issuer identifier, the list of Issue Actions, and one authorizing signature: issuer, which lets validators verify that each AssetId is properly associated with this issuer; vIssueActions, an array of Issue Actions; and issueAuthSig, the encoding of a signature over the transaction SIGHASH by the issuance authorizing key isk (ZIP-227, *Issuance Bundle*). An *Issuance Action* groups the notes issued for one Asset: the 32-byte assetDescHash, the note list, and a flags byte carrying finalize (ZIP-227, *Issuance Action*). Table 3 gives the v6 serialization; the encoding rigor caveat of the section introduction applies.

Field	Type	Notes
`issuerLength`	`compactSize`	MUST be 0 if `nIssueActions` = 0
`issuer`	`byte[issuerLength]`	= ik_encoding, 33 bytes when present
`nIssueActions`	`compactSize`	
`vIssueActions`	`IssueAction[nIssueActions]`	
`issueAuthSig`	`IssueAuthSignature`	present iff `nIssueActions` > 0
Per `IssueAction`:		
`assetDescHash`	`byte[32]`	
`nNotes`	`compactSize`	may be 0 (existing-Asset finalization)
`vNotes`	`IssueNoteDescription[nNotes]`	115 bytes each
`flagsIssuance`	`byte`	finalize at LSB; other bits MUST be 0
Per `IssueNoteDescription` (43 + 8 + 32 + 32 = 115 bytes):		
`recipient`	`byte[43]`	$\mathrm{LEBS2OSP}_{88}(\mathsf{d}) \parallel \mathsf{pk_d}^{\star}$
`value`	`uint64`	little-endian
`rho`	`byte[32]`	as for Orchard notes
`rseed`	`byte[32]`	as for Orchard notes

Table 3: ZSA issuance fields of the v6 transaction format (fork ZIP-230, *Transaction Format*, *Issuance Action Description*, and *Issue Note Description* @ zips-zsa `fd71419`; upstream text identical apart from the Withdrawn status).

Three encoding facts deserve emphasis. First, the `recipient` field is exactly an Orchard raw payment address (the *Wallet Guide*, §"Raw encodings"), and together with the bundle's issuer and the action's assetDescHash, the four fields of an `IssueNoteDescription` determine the Issue Note completely (ZIP-230, *Issue Note Description*; ZIP-227, *Issue Note*) — the note commitment is deliberately absent, since validators recompute it (§3.3). Second, the encoding permits `nNotes` = 0; for an existing Asset this lets the issuer finalize without issuing more (ZIP-230, *Issuance Action Description*; the new-Asset corner case is recorded in §3.7). Third, the flags byte is strict: `IssuanceFlags::from_byte` returns `None` whenever any bit other than the LSB is set, and deserialization fails on such bytes, enforcing ZIP-230's "remaining bits are set to 0" at parse time.

The `librustzcash-zsa` serializer pins down the absent-bundle corner cases: `read_bundle` reads `issuer` as a length-prefixed vector and returns `None` only for an empty issuer with zero actions, erroring on an empty issuer with non-empty actions and on a non-empty issuer with zero actions; `write_bundle` encodes an absent bundle as two zero `compactSizes`. The authorization serializes as a length-prefixed `sighashInfo` followed by the length-prefixed signature encoding, and the parser accepts exactly `[0x00]` as `sighashInfo` (`transaction/components/issuance.rs` lines 20–191 and `sighash_versioning.rs` @ `3f9b53fc`; the meaning of `[0x00]` is §3.5).

In the `orchard-zsa` object model, the type `IssueBundle<T: IssueAuth>` holds the validating key `ik` (an `IssueValidatingKey<ZSASchnorr>`), a non-empty vector of actions, and an authorization typestate `T`; an `IssueAction` holds the 32-byte `asset_desc_hash`, its `Vec<Note>`, and its `IssuanceFlags`.

3.5 Authorization: BIP-340 over the transaction sighash

The issuance authorization signature scheme IssueAuthSig is instantiated as a BIP-340 Schnorr signature over secp256k1 with the standard parameters, not RedPallas; ZIP-227 gives no rationale for that curve choice. Batch verification MAY be used, and precomputation MAY be used if and only if it produces equivalent results (ZIP-227, *Issuance Authorization Signature Scheme*). The associated types (same section):

$$\mathsf{Message} = \mathbb{B}^{\mathbb{Y}^{[\mathbb{N}]}}, \quad \mathsf{Signature} = \mathbb{B}^{\mathbb{Y}^{[64]}} \cup \{\bot\}, \quad \mathsf{Public} = \mathbb{B}^{\mathbb{Y}^{[32]}} \cup \{\bot\}, \quad \mathsf{Private} = \mathbb{B}^{\mathbb{Y}^{[32]}},$$

that is, arbitrary byte-sequence messages, 64-byte signatures, 32-byte x-only public keys, and 32-byte secret keys, with $\bot$ as the failure value. Signing fixes the BIP-340 auxiliary data to the all-zero block:

$$a := [\texttt{0x00}]^{32}, \qquad \sigma := \mathit{Sign}(\mathsf{isk}, M) \text{ with auxiliary data } a, \qquad \texttt{issueAuthSig} := [\texttt{0x00}] \,\|\, \sigma,$$

returning $\bot$ on failure; validation returns 0 if $\sigma = \bot$, else 1 exactly when the BIP-340 $\mathit{Verify}(\mathsf{ik}, M, \sigma)$ succeeds[2] (ZIP-227, *Issuance Authorization Signing and Validation*). With the scheme byte, the two wire encodings are 33 bytes (ik_encoding) and 65 bytes (`issueAuthSig`).

What is signed. The message is the transaction SigHash as defined by protocol specification §4.10 under the ZIP-226 modifications, with sighash type SIGHASH_ALL, and the consensus rule requires IssueAuthSig.Validate(ik, SigHash, `issueAuthSig`) = 1 (ZIP-227, *Specification: Consensus Rule Changes*). Under ZIP-246 the signature digest includes branch S.5, the issuance_digest, identical to the txid branch — so the issuance signature commits to *all* effecting data of the transaction, including the issuance bundle's own effecting data, though never to any signature (ZIP-246, *Signature Digest*). ZIP-246 also introduces sighash algorithm versioning: each bundle carries sighashInfo = [sighashVersion] ‖ associatedData, versions are numbered from 0 per transaction version, version 0 is by convention the "commit to all effecting data" algorithm, and for v6 only version 0 is defined, with empty associated data for the Transparent, Sapling, Orchard, and Issuance bundles — hence the constant `[0x00]` the parser demands (ZIP-246, *Sighash versioning*).

Implementation. The `ZSASchnorr` scheme wraps the `secp256k1` crate: signing builds a keypair from isk and calls `sign_schnorr_with_aux_rand(msg, keypair, &[0u8; 32])` — the ZIP's $a = [\texttt{0x00}]^{32}$ — ik is the x-only public key, and `encode()` of key and signature each prepend `ALGORITHM_BYTE = 0x00`, with `decode()` rejecting any other lead byte. Bundle construction is a typestate chain — `AwaitingNullifier` → (`update_rho`) → `AwaitingSighash` → (`prepare(sighash)`) → `Prepared` → (`sign(isk)`) → `Signed` — so a bundle cannot be signed before its notes' ρ values and the sighash exist. On the transaction side, `v6_signature_hash` includes the txid issuance_digest, and the builder signs the issue bundle with `prepare(*shielded_sig_commitment)` where that commitment is `signature_hash(tx, SignableInput::Shielded, txid_parts)` — the same SIGHASH_ALL shielded digest the Sapling and Orchard signatures sign. The builder also realizes the $\mathsf{nf}_{0,0}$ rule constructively: `first_nullifier` returns the first Action's nullifier only for an OrchardZSA bundle, and building with an issuance builder but no such bundle fails with `BundleTypeNotSatisfiable`.

A divergence to flag. ZIP-246's txid and signature digests include a sixth branch, T.6/S.6 memo_digest (ZIP-246, *TxId Digest* and *Signature Digest*), but `librustzcash-zsa`'s `to_hash` hashes only the header, transparent, Sapling, Orchard, and issuance branches — the fork does not implement ZIP-231 memo bundles (`transaction/txid.rs` lines 415–460 @ `3f9b53fc`). The

[2] The ZIP's Validate definition writes $\mathit{Verify}(\mathsf{key}, M, \sigma)$ for the parameter it names ik — an editorial slip we flag when citing (ZIP-227, *Issuance Authorization Signing and Validation*).

digest the fork code signs is therefore not byte-identical to the digest ZIP-246 specifies; whichever is corrected, the issuance signature's coverage claim above should be read against the fork's five-branch digest today.

3.6 Transaction digests

The v6 digest tree extends the ZIP-244 structure the *Ironwood Guide* presents in §"Transaction digests and signatures": the txid digest is BLAKE2b-256 with personalization `ZcashTxHash_` ‖ CONSENSUS_BRANCH_ID over six branches T.1–T.6 (header, transparent, Sapling, Orchard, issuance, memo), and the signature digest has the same six-branch structure with S.5 identical to T.5; a signature digest is produced for each transparent input, Sapling input, OrchardZSA Action, and additionally for each Issuance Action (ZIP-246, *TxId Digest* and *Signature Digest*). The issuance branches are three nested hashes and one authorizing hash, each over the wire encodings of Table 3 (ZIP-246, *txid_digest* T.5 and *A.4: issuance_auth_digest*):[3]

$$\begin{aligned}
\mathsf{issuance_digest} &= \text{BLAKE2b-256}(\texttt{ZTxIdSAIssueHash}, \\
&\qquad \texttt{issuerLength} \,\|\, \texttt{issuer} \,\|\, \mathsf{issue_actions_digest}), \\
\mathsf{issue_actions_digest} &= \text{BLAKE2b-256}(\texttt{ZTxIdIssuActHash}, \\
&\qquad \|_{\text{actions}}\ \texttt{assetDescHash} \,\|\, \mathsf{issue_notes_digest} \,\|\, \texttt{flagsIssuance}), \\
\mathsf{issue_notes_digest} &= \text{BLAKE2b-256}(\texttt{ZTxIdIAcNoteHash}, \\
&\qquad \|_{\text{notes}}\ \texttt{recipient} \,\|\, \texttt{value} \,\|\, \texttt{rho} \,\|\, \texttt{rseed}), \\
\mathsf{issuance_auth_digest} &= \text{BLAKE2b-256}(\texttt{ZTxAuthZSAOrHash}, \texttt{issueAuthSig} \text{ field encoding}),
\end{aligned}$$

where the A.4 preimage is the ZIP-230 `IssueAuthSignature` composite

$$\texttt{sizeSighashInfo} \,\|\, \texttt{sighashInfo} \,\|\, \texttt{sizeSignature} \,\|\, \texttt{signature}$$

(ZIP-230, *Issuance Authorization Signature*), and a transaction without issuance components hashes the empty string under the same personalizations.

The code computes exactly these issuance-specific hashes. On the txid side, `hash_issue_bundle_txid_data` hashes

$$\text{compactSize}(|\mathsf{ik_encoding}|) \,\|\, \mathsf{ik_encoding} \,\|\, \mathsf{issue_actions_digest}$$

under `ZTxIdSAIssueHash`; the actions digest hashes, per action, the 32-byte hash, the notes digest, and the flags byte; the notes digest hashes, per note, the 43-byte recipient, the 8-byte little-endian value, and the two 32-byte fields. On the authorizing side, `hash_issue_bundle_auth_data` hashes

$$\text{compactSize}(|\mathsf{sighashInfo}|) \,\|\, \mathsf{sighashInfo} \,\|\, \text{compactSize}(65) \,\|\, [\texttt{0x00}] \,\|\, \sigma,$$

asserting the 65-byte length and the `0x00` lead. The empty-bundle variants hash the empty string, and all four personalizations match ZIP-246; both digests live in `src/bundle/commitments/issuance.rs` lines 11–86 (`orchard-zsa` @ `cf801a5d`). For the wiring into transaction identifiers, `TxIdDigester::digest_issue` maps the bundle to its `commitment()`; `to_hash` appends the digest only when the version `has_orchard_zsa()`; and `BlockTxCommitmentDigester` takes the authorizing commitment (`librustzcash-zsa`, `transaction/txid.rs` lines 381–383, 445–452, 595–598 @ `3f9b53fc`).

Deterministic regression digests are pinned in the tests — for a seeded two-asset bundle,

[3] Two editorial defects in ZIP-246 to note when citing: the T.5 child list labels `issuerLength/issuer/issue_actions_digest` as T.5a/T.5b/T.5c, while the following section headings call `issue_actions_digest` "T.5a" and `issue_notes_digest` "T.5a.i"; and the A.4 empty case reads "In the case that the transaction has no Orchard Actions" where "no Issuance Actions" is evidently intended.

issuance_digest = `ee70e3b61674fd0428ac0020cc4fc5819386e39c4eb3c63357c84c998195bcdb`,
issuance_auth_digest = `6df77af7b5323d99376336b770e4c5b06ffc195de81ac7692d1b08b6eb19534d`

— and cross-implementation test vectors ship in `src/test_vectors/` (`asset_base.rs`, `issuance_auth_sig.rs`), exercised by the asset-base and authorization tests (`orchard-zsa`, `src/bundle/commitments/issuance.rs` lines 156–190 and `src/test_vectors/` @ `cf801a5d`). The upstream ZIP-227 Test Vectors section, by contrast, still reads LINK TBD.

3.7 Global issuance state and the consensus rules

Supply integrity is the one property issuance cannot delegate to cryptography: no transaction field simultaneously witnesses everything issued and everything burnt for an Asset, so — along the lines of the ZIP-209 pool turnstile — every node maintains a per-Asset circulation record (ZIP-227, *Rationale for Global Issuance State*).

Definition 3.2 (Global issuance state)**.** The map

$$\begin{aligned} \mathsf{issued_assets} : \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}})^* &\to \{0\,..\,\mathsf{MAX_ISSUE}\} \times \{0,1\} \times \mathsf{Note}^{\mathsf{Issue}}, \\ \mathsf{AssetBase} &\mapsto (\mathsf{balance}, \mathsf{final}, \mathsf{note}_{\mathsf{ref}}), \end{aligned}$$

records, for every issued Asset, the amount in circulation (issued minus burnt), the finalization status — which can never change from 1 to 0 — and the Asset's reference note. A missing entry ($\mathsf{issued_assets}(\mathsf{AssetBase}) = \bot$) is read as $\mathsf{balance} = 0$, $\mathsf{final} = 0$, $\mathsf{note}_{\mathsf{ref}} = \bot$; the constant $\mathsf{MAX_ISSUE} = 2^{64} - 1$ caps the recorded circulating balance of any Custom Asset at any one state (ZIP-227, *Global Issuance State*).

The map threads through the chain deterministically: it MUST be updated by a node during the processing of any transaction that contains burn information or an issuance bundle; the input state for the OrchardZSA activation block is the empty map; the input state of a block's first transaction is the final state of the preceding block; each subsequent transaction's input state is the preceding transaction's output state; and the block's final state is its last transaction's output state (ZIP-227, *Management of the Global Issuance State*).

Per-transaction rules. For a v6 transaction (ZIP-227, *Specification: Consensus Rule Changes*):

(T1) `nActionGroupsOrchard` MUST be 0 or 1, and `nAGExpiryHeight` MUST be 0; the v6 parser of `librustzcash-zsa` enforces both at deserialization, in `transaction/components/orchard.rs` lines 107–132 @ `3f9b53fc`;

(T2) the output state $\mathsf{issued_assets}_{\mathsf{OUT}}$ MUST be initialized to the input state $\mathsf{issued_assets}_{\mathsf{IN}}$;

(T3) the assetBurn set MUST satisfy the ZIP-226 rules (§4: tuples $(\mathsf{AssetBase}, \mathsf{v})$ with the Native Asset excluded, $0 < \mathsf{v} \leq \mathsf{MAX_BURN_VALUE} = 2^{63} - 1$, and no duplicate AssetBase; ZIP-226, burn rules); and

(T4) for every $(\mathsf{AssetBase}, \mathsf{v}) \in \mathsf{assetBurn}$, the node MUST check $\mathsf{balance} \geq \mathsf{v}$ for the entry of AssetBase in $\mathsf{issued_assets}_{\mathsf{OUT}}$ and subtract v from that balance, *prior to* processing the issuance bundle.

Issuance-bundle rules. For a transaction carrying an issuance bundle (ZIP-227, *Specification: Consensus Rule Changes*):

(I1) the transaction MUST also contain at least one OrchardZSA Action Group (the ρ/replay rationale of §3.3);

(I2) the encoding ik_encoding MUST be taken from the `issuer` field and MUST start with the byte `0x00` (BIP-340);

(I3) the signature MUST satisfy IssueAuthSig.Validate(ik, SigHash, `issueAuthSig`) = 1;

(I4) for each Issue Action: every `IssueNoteDescription` MUST be a valid ZIP-230 encoding; the AssetBase is derived from assetDescHash and issuer (§3.2); and the Asset's final bit in $\mathsf{issued_assets}_{\mathsf{OUT}}$ MUST NOT be 1;

(I5) if $\mathsf{note}_{\mathsf{ref}} = \perp$, the first Issue Note MUST be a reference note (§3.3), and the node MUST set $\mathsf{note}_{\mathsf{ref}}$ to it;

(I6) for every Issue Note note_j: its ρ MUST equal the value computed per §3.3; the supply update

$$\mathsf{issued_assets}_{\mathsf{OUT}}.\mathsf{balance} + \mathsf{v}_j \leq \mathsf{MAX_ISSUE}, \qquad \mathsf{issued_assets}_{\mathsf{OUT}}.\mathsf{balance} \mathrel{+}= \mathsf{v}_j$$

MUST hold and be applied; and the node MUST compute the note commitment per ZIP-226;

(I7) if finalize = 1 in `flagsIssuance`, the node MUST set final = 1.

The commitments enter the tree in a fixed order: for each Action Group of the OrchardZSA bundle, the commitment of every new note per Action; then, for each Issue Action of the issuance bundle, the commitment of every Issue Note (ZIP-227, *Addition to the Note Commitment Tree*).

The rationale ties the pieces together: balances of issued Assets must never go negative, MAX_ISSUE is a practical limit that also lets an issuer create an Asset's complete supply in a single transaction, and the same map records finalization so that further issuance of a finalized Asset is rejected (ZIP-227, *Rationale for Global Issuance State*). Beyond consensus, the ZIP RECOMMENDS that full validators keep a mutable `issuanceSupplyInfoMap` from AssetId to (`totalSupply`, `finalize`) for wallets and applications tracking total supply (ZIP-227, *Implementing Zcash Nodes*).

The library realization. Function `verify_issue_bundle` takes the bundle, the sighash, a caller-supplied view of global state (`get_global_records`), and the first nullifier. It rejects any sighash kind other than `AllEffecting`, verifies the BIP-340 signature over the 32-byte sighash with the bundle's ik, folds the actions over a `BTreeMap<AssetBase, AssetRecord>`, and returns the updated records — `amount`, `is_finalized`, `reference_note` — for the caller to persist. The fold realizes the per-action and per-note rules as typed errors:

- `IncorrectRhoDerivation` — a note's ρ differs from the recomputed derivation of §3.3;
- `MissingReferenceNoteOnFirstIssuance` — no record exists for the Asset, and the action's first note is not a reference note;
- `IssueActionPreviouslyFinalizedAssetBase` — reissuance against a finalized record;
- `IssueBundleIkMismatchAssetBase` and `AssetBaseCannotBeIdentityPoint` — a note's Asset Base differs from the one `IssueAction::verify` derives from (ik, assetDescHash), or the derived point is the Pallas identity. The latter variant is intended but unreachable at this pin because `AssetBase::custom` panics first (Remark 2.8);
- `ValueOverflow` — the checked `u64` value sum overflows, the code's realization of $\mathsf{MAX_ISSUE} = 2^{64} - 1$.

A signature-skipping variant, `check_issue_bundle_without_sighash`, exists for verifying historical blocks from a trusted checkpoint (`src/issuance.rs` lines 647–705 @ `cf801a5d`).

Three boundary observations, each load-bearing for anyone re-deriving consensus from these libraries:

- *Empty-action divergence.* The implementation rejects an action with no notes unless it is finalized (`IssueActionWithoutNoteNotFinalized`, `src/issuance.rs` lines 216–219), but ZIP-227's consensus-rule list contains no such MUST, and ZIP-230 merely notes that `nNotes` may be zero "to only finalize". For an already-existing Asset, the written consensus rules admit an empty, non-finalizing action that the reference implementation rejects; for a new Asset, the ZIP's instruction to take the first note as the reference note is undefined when the list is empty.
- *First-issuance detection.* The ZIP keys the reference-note requirement on $\mathsf{note}_{\mathsf{ref}} = \perp$; the code keys it on the complete absence of an `AssetRecord` (`get_global_records` returning `None`). The two are equivalent under the invariant that $\mathsf{note}_{\mathsf{ref}}$ is set exactly on first issuance — rule (I5) — and no rule ever clears it, so an entry exists if and only if its $\mathsf{note}_{\mathsf{ref}}$ is set.
- *Node-side enforcement locus.* Neither `orchard-zsa` nor `librustzcash-zsa` contains the issued_assets chain-state maintenance or the consensus caller of `verify_issue_bundle`: the libraries return updated records for a node to persist, and the QEDit node integration is not among our local ground-truth repositories. The global-state rules are therefore *specified and library-supported*, with their node enforcement unverified locally.

3.8 Finalization

The finalize boolean signals that no further issuance of the specific Custom Asset is possible: every issuance action carries it in `flagsIssuance` (a requirement of the ZIP's Requirements section), setting it sets the Asset's final bit — rule (I7) — and transactions attempting to issue further amounts of a previously finalized Asset are rejected — rule (I4) (ZIP-227, *Requirements* and *Issuance Action*). The bit is monotone: consensus state never changes final from 1 to 0 (Definition 3.2).

Finalization is the ZIP's whole lifecycle policy, and its Concrete Applications read as patterns over one bit (ZIP-227, *Concrete Applications*):

- *One-time issuance*: set finalize = 1 on the first action; the entire supply exists from one transaction — MAX_ISSUE is sized to permit this (§3.7).
- *Stopping supply*: issue with finalize = 1 in a final issuance transaction, or in one containing a single note of value 0 (the wire format admits note-less finalizing actions as well, Table 3; the library insists on the finalizing variant, §3.7).
- *NFTs*: issue multiple actions in one transaction, each with value = 1 at the Asset's fundamental unit and finalize = 1 — each action defines a distinct finalized Asset Identifier with supply one, subject to the collision-resistance assumptions of §2.2.

Finalization is also the entirety of the compromise story. No key rotation exists for issuance keys; on compromise of isk, the ZIP's guidance is to set finalize = 1 for all Assets issued with that key, switch to a new isk — hence a new issuer identifier, since issuer = ik_encoding — and issue new Assets with new AssetIds (ZIP-227, *Issuance Key Compromise*). The old Assets keep circulating and burning; they simply never grow again.

3.9 Fees

The fee treatment of issuance is the sharpest fork/upstream split in the ZSA stack, and we bin it explicitly. The QEDit fork of ZIP-317 adds two contributions to the logical-action count of the conventional fee (the *Wallet Guide*, §"Fees (ZIP-317)", for the surrounding machinery: *marginal_fee* = 5000 zatoshis, *grace_actions* = 2):

$$\begin{aligned} \mathit{contribution}_{\mathsf{ZSAIssuance}} &= \mathit{nZSAIssueNotes}, \\ \mathit{contribution}_{\mathsf{ZSACreation}} &= \mathit{creation_cost} \cdot \mathit{nAssetCreations}, \end{aligned}$$

where *nZSAIssueNotes* counts `IssueNote` outputs, *nAssetCreations* counts Custom Assets newly added to the global issuance state, and *creation_cost* = 100 logical actions (fork ZIP-317, *Fee calculation* @ zips-zsa `fd71419`). The stated rationale: every new Asset adds a row that validators track in perpetuity (§3.7), and the two-orders-of-magnitude creation cost disincentivizes junk assets.

ZIP 317 defines neither contribution. These formulas are *fork-specified*: they describe the experimental ZSA fee policy, not the conventional fee rule for deployed transactions.

4 Transfer and burn (ZIP-226)

ZIP-226 (*Transfer and Burn of Zcash Shielded Assets*; Status: Draft; owners Pablo Kogan and Vivek Arte of QEDit, Daira-Emma Hopwood, and Jack Grigg) defines, jointly with the issuance protocol of ZIP-227 (§3), the OrchardZSA transfer and burn protocol. We develop it as four deltas against the Orchard baseline, each citing the construction it modifies in the *Ironwood Guide* rather than re-deriving it: the note gains one field, the *Asset Base* (§4.1); the note commitment becomes a case split that keeps ZEC commitments identical to NU5 (§4.2); the value commitment trades its fixed value base for the per-asset base, which turns the binding signature into a computational per-asset balance check and gives burn its consensus meaning (§4.3, §4.4); and dummy spends give way to *Split Inputs* for Custom Assets (§5). We then state the extended Action statement (§4.6) and the carrier format with its digests and fees (§4.7).

ZIP 226 assumes ZIP 2005's recoverability changes, whereas the experimental fork does not implement them. Where the difference affects a construction — the split-note ψ^{nf} (§5) and the lead byte (§4.7) — we flag it in place.

4.1 One new field: the OrchardZSA note

Definition 4.1 (OrchardZSA note)**.** An OrchardZSA note extends the Orchard note — the tuple $(\mathsf{addr}, v, \rho, \psi, \mathsf{rcm})$ of the *Ironwood Guide*, "Notes" — by exactly one field:

$$\mathbf{n} := (\mathsf{addr},\ v,\ \mathsf{AssetBase},\ \rho,\ \psi,\ \mathsf{rcm}),$$

where $\mathsf{AssetBase} \in \mathcal{E}^* := \mathcal{E}(\mathbb{F}_{p_{\mathsf{Pallas}}}) \setminus \{O\}$ is "a valid group element that is not the identity and is not bottom" (ZIP-226; the ZIP writes $\mathbb{P}^*$ for $\mathcal{E}^*$). All other components are as in protocol specification §3.2. Formally,

$$\begin{aligned}\mathsf{Note}^{\mathsf{OrchardZSA}} := {} & \{0,1\}^{88} \times \mathsf{KA}^{\mathsf{Orchard}}.\mathsf{Public} \times \{0,\dots,2^{64}-1\} \times \mathcal{E}^* \\ & \times \mathbb{F}_{p_{\mathsf{Pallas}}} \times \mathbb{F}_{p_{\mathsf{Pallas}}} \times \mathsf{NoteCommit}^{\mathsf{Orchard}}.\mathsf{Trapdoor},\end{aligned}$$

where the ZIP's $\mathbb{F}_{q_P}$ is this series' $\mathbb{F}_{p_{\mathsf{Pallas}}}$ and $\ell_{\mathsf{value}} = 64$ (ZIP-226).

The byte encoding of the new field is the star-encoding of the series' conventions: ZIP-226 sets $\mathsf{asset_base} := \mathsf{LEBS2OSP}_{\ell_P}(\mathsf{repr}_{\mathbb{P}}(\mathsf{AssetBase}))$, which is $\mathsf{AssetBase}^\star \in \{0,1\}^{256}$ in 32 bytes.

In a transfer, an Asset Base is not an arbitrary point chosen by the transactor. Every Custom Asset's base descends from the ZIP-227 issuance chain — constructed in §3, reproduced here because the soundness argument of §5 depends on its shape:

$$\begin{aligned}\mathsf{assetDescHash} &:= \mathsf{BLAKE2b\text{-}256}(\texttt{"ZSA-AssetDescCRH"},\ \mathsf{asset_desc}),\\ \mathsf{AssetId} &:= (\mathsf{issuer},\ \mathsf{assetDescHash}),\\ \mathsf{EncodeAssetId}(\mathsf{AssetId}) &:= \texttt{0x00} \,\|\, \mathsf{issuer} \,\|\, \mathsf{assetDescHash},\\ \mathsf{AssetDigest} &:= \mathsf{BLAKE2b\text{-}512}(\texttt{"ZSA-Asset-Digest"},\ \mathsf{EncodeAssetId}(\mathsf{AssetId})),\\ \mathsf{AssetBase} &:= \mathsf{GroupHash}^{\mathsf{Pallas}}(\texttt{z.cash}:\texttt{OrchardZSA},\ \mathsf{AssetDigest})\end{aligned}$$

(ZIP-227, which names the last map ZSAValueBase). For the native Asset (ZEC/TAZ) the Asset Base is instead *defined* to be

$$V^{\mathsf{Orchard}} := \mathsf{GroupHash}^{\mathsf{Pallas}}(\texttt{z.cash} : \texttt{Orchard-cv}, \texttt{"v"}),$$

the fixed value-commitment base of the *Ironwood Guide*, "Value commitments" (ZIP-226). This single identification is what keeps every ZEC computation below identical to NU5: ZIP-226 uses $\mathsf{ValueCommit}^{\mathsf{Orchard}}_{\mathsf{rcv}}(v)$ as $\mathsf{ValueCommit}^{\mathsf{OrchardZSA}}_{\mathsf{rcv}}(V^{\mathsf{Orchard}}, v)$. The implementation realizes it as `AssetBase::zatoshi() = hash_to_curve("z.cash:Orchard-cv")(b"v")`.

4.2 The note commitment: a case split

Definition 4.2 (OrchardZSA note commitment)**.** For an OrchardZSA note,

$$\begin{aligned}&\mathsf{NoteCommit}^{\mathsf{OrchardZSA}}_{\mathsf{rcm}}(\mathsf{g_d}^\star, \mathsf{pk_d}^\star, v, \rho, \psi, \mathsf{AssetBase})\\ &\quad := \begin{cases}\mathsf{NoteCommit}^{\mathsf{Orchard}}_{\mathsf{rcm}}(\mathsf{g_d}^\star, \mathsf{pk_d}^\star, v, \rho, \psi) & \text{if } \mathsf{AssetBase} = V^{\mathsf{Orchard}},\\ \mathsf{cm_{ZSA}} & \text{otherwise,}\end{cases}\end{aligned}$$

where

$$\begin{aligned}\mathsf{cm_{ZSA}} := {}&\mathsf{Sinsemilla}_{\texttt{z.cash:ZSA-NoteCommit-M}}(\mathsf{g_d}^\star \parallel \mathsf{pk_d}^\star \parallel \mathrm{LE}_{64}(v) \parallel \mathrm{LE}_{255}(\rho) \parallel \mathrm{LE}_{255}(\psi) \parallel \mathsf{AssetBase}^\star)\\ &\dotplus\ [\mathsf{rcm}]\,\mathsf{GroupHash}^{\mathsf{Pallas}}(\texttt{z.cash} : \texttt{Orchard-NoteCommit-r}, \texttt{""})\end{aligned}$$

(ZIP-226; the ZIP writes I2LEBSP for the series' LE_k). The signature is

$$\begin{aligned}\mathsf{NoteCommit}^{\mathsf{OrchardZSA}} : {}&\mathsf{NoteCommit}^{\mathsf{Orchard}}.\mathsf{Trapdoor} \times \{0,1\}^{256} \times \{0,1\}^{256} \times \{0,\ldots,2^{64}-1\}\\ &\times \mathbb{F}_{p_{\mathsf{Pallas}}} \times \mathbb{F}_{p_{\mathsf{Pallas}}} \times \mathcal{E}^* \ \rightarrow\ \mathsf{NoteCommit}^{\mathsf{Orchard}}.\mathsf{Output},\end{aligned}$$

with trapdoor and output types those of $\mathsf{NoteCommit}^{\mathsf{Orchard}}$, unchanged (ZIP-226).

The two branches carry the design's two promises. In the native branch the Asset Base is not hashed at all: a ZEC note commitment under OrchardZSA *is* the NU5 commitment, bit for bit. In the ZSA branch the message appends $\mathsf{AssetBase}^\star$ (256 bits) after ψ, against the baseline layout of the *Ironwood Guide*, "Note commitments":

$$\mathsf{g_d}^\star \parallel \mathsf{pk_d}^\star \parallel \mathrm{LE}_{64}(v) \parallel \mathrm{LE}_{255}(\rho) \parallel \mathrm{LE}_{255}(\psi) \ \longmapsto\ \cdots \parallel \mathrm{LE}_{255}(\psi) \parallel \mathsf{AssetBase}^\star,$$

under the fresh hash domain `z.cash:ZSA-NoteCommit-M` but the *same* randomness base $\mathsf{GroupHash}^{\mathsf{Pallas}}(\texttt{z.cash} : \texttt{Orchard-NoteCommit-r}, \texttt{""})$. ZIP-226's rationale says that "the nested structure of the Sinsemilla-based commitment allows us to add the Asset Base as a final recursive step", and calls the output "still indistinguishable from the original Orchard ZEC note commitments". More precisely, the common output type lets ZSA commitments inhabit the same note commitment tree, while their computational indistinguishability rests on the hiding and hash assumptions of the randomized Sinsemilla commitment (ZIP-226). One caveat bounds the claim: Orchard (v5) note commitments and OrchardZSA (v6) note commitments are contextually distinguishable when observed in transactions, because the two protocols use different transaction versions (ZIP-226).

Remark 4.3 (Implementation). Function `NoteCommitment::derive` computes *both* commitments. For ZEC it calls `CommitDomain::new` on the prefix `"z.cash:Orchard-NoteCommit"`; for ZSA it calls `new_with_separate_domains` on the M-prefix `"z.cash:ZSA-NoteCommit"` and the shared r-prefix `"z.cash:Orchard-NoteCommit"`. It then selects between them in constant time on `asset.is_zatoshi()`. The sinsemilla crate appends the `-M` and `-r` suffixes to these prefixes.

The nullifier of an ordinary (non-split) OrchardZSA note is generated exactly as in Orchard (protocol specification §4.16; *Ironwood Guide*, "The nullifier"); only Split Inputs use the modified, randomized formula of §5 (ZIP-226).

4.3 Per-asset value: the variable-base value commitment

Definition 4.4 (OrchardZSA net value commitment)**.** Each Action commits to its net value change in its Asset:

$$\mathsf{cv}^{\mathsf{net}} := \mathsf{ValueCommit}^{\mathsf{OrchardZSA}}_{\mathsf{rcv}}(\mathsf{AssetBase}, v^{\mathsf{net}}) = [\, v^{\mathsf{net}} \,]\, \mathsf{AssetBase} + [\mathsf{rcv}]\, R^{\mathsf{Orchard}},$$
$$v^{\mathsf{net}} := v_{\mathrm{old}} - v_{\mathrm{new}}, \qquad R^{\mathsf{Orchard}} := \mathsf{GroupHash}^{\mathsf{Pallas}}(\texttt{z.cash} : \texttt{Orchard-cv}, \texttt{"r"}),$$

where AssetBase is the Action's Asset Base and the randomness base R^{Orchard} is unchanged (ZIP-226).

Exactly one thing changes against the baseline $\mathsf{cv}^{\mathsf{net}} = [v_{\mathrm{old}} - v_{\mathrm{new}}]\, V^{\mathsf{Orchard}} + [\mathsf{rcv}]\, R^{\mathsf{Orchard}}$ of the *Ironwood Guide*, "Value commitments": the fixed-base multiplication by V^{Orchard} becomes a variable-base multiplication by the Action's AssetBase, while the randomness term keeps its fixed base (ZIP-226). For ZEC, $\mathsf{AssetBase} = V^{\mathsf{Orchard}}$ and the commitment reduces to the NU5 one (§4.1).

The prover still signs the SIGHASH with $\mathsf{bsk} = \sum_{i=1}^{n} \mathsf{rcv}_i$ over the n Actions of a transfer (ZIP-226); the verifier's key acquires burn terms — one zero-trapdoor commitment subtracted per assetBurn entry, each a bare multiple of its base — displayed in this volume as (2) (§5.1). The verifier checks one Pallas-group equation, not independent algebraic coordinates for the assets. Since Pallas is cyclic, every pair of non-identity bases has some discrete-log relation; the separation is computational. After the randomness terms cancel, any nonzero per-asset imbalance that still verifies yields a useful relation among the independently group-hashed Asset Bases, V^{Orchard}, and R^{Orchard}. Under the discrete-log and hash-to-curve assumptions, finding such a relation is infeasible. Subject to those assumptions, the binding signature therefore enforces that each Custom Asset's net Action value equals its assetBurn entry (or 0 if absent), while the ZEC component equals $v^{\mathsf{Orchard}}_{\mathsf{balance}}$. This is the computational content of ZIP-226's "per Asset Base" claim; its wording that Pedersen commitments add homomorphically only for the same base is not a literal group-theoretic statement. The binding-signature mechanics — bsk/bvk cancellation and RedPallas on the base R^{Orchard} — are those of the *Ironwood Guide*, "The binding signature", and we do not restate them.

Remark 4.5 (Implementation: derive_bvk)*.* Function `derive_bvk_raw` computes the ZIP equation verbatim: $\sum \mathsf{cv}^{\mathsf{net}}$ minus the zero-trapdoor commitment of `value_balance` at `AssetBase::zatoshi()` minus, over the burn set, the zero-trapdoor commitment of each `value` at its `asset`. Function `ValueCommitment::derive` computes $V \cdot \mathsf{value} + R \cdot \mathsf{rcv}$ with V the point `asset.cv_base()` and R the point `hash_to_curve("z.cash:Orchard-cv")` applied to `b"r"`, the value signed per the sign of the value sum.

4.4 Burn

Burning is the value-balance mechanism read as an exit door. The mechanism "does NOT send Assets to a non-spendable address, it simply reduces the total number of units of a given Custom Asset in circulation"; it is enforced at consensus level as an extension of the value-balance mechanism, and it "makes it globally provable that a given amount of a Custom Asset has been destroyed". The Native Asset (ZEC/TAZ) cannot be burnt this way (ZIP-226).

ZIP-226 attaches three consensus rules to the burn set, whose cardinality it denotes L:

1. for every $(\mathsf{AssetBase}, v) \in \mathsf{assetBurn}$: $\mathsf{AssetBase} \neq V^{\mathsf{Orchard}}$;

2. $v > 0$ and $v \leq \mathsf{MAX_BURN_VALUE} := 2^{63} - 1$;[4]

3. every AssetBase has at most one entry in assetBurn.

Burn data lives *per Action Group*: the ActionGroupDescription carries nAssetBurn (compactSize) and vAssetBurn, an array of 40-byte entries — asset_base (byte[32]) followed by valueBurn (uint64, little-endian) — so that parties assembling separate Action Groups can each burn their own Assets (withdrawn ZIP-230; §4.7).

The transaction-level field valueBalanceOrchard (int64) remains the net *ZEC-only* flow, "the net value of Orchard spends minus outputs" (withdrawn ZIP-230), and the transparent protocol is unchanged, so a Custom Asset cannot be unshielded by transfer: "All Custom Assets are contained within the shielded pool" (ZIP-226). Burning is the only supply-reduction path. What a burn reveals on the wire is exactly its assetBurn entry: the 32-byte Asset Base and amount — what ZIP-226 calls "the amount and identifier of the Asset being burnt". In a transfer, "the overall balance is checked without revealing which (or how many distinct) Assets are being transferred"; under the note-encryption and commitment-hiding assumptions, the transferred amounts and Asset Bases remain private (ZIP-226).

A burn also updates ZIP-227's global issuance state: issued_assets(AssetBase).balance $\in \{0, \ldots, \mathsf{MAX_ISSUE}\}$ is "computed as the amount of the Asset that has been issued less the amount of the Asset that has been burnt", and the map MUST be updated during processing of any transaction containing burn information (ZIP-227). Supply sufficiency — that a burn not exceed the circulating supply — is therefore a state-level rule of the issuance protocol, not one of ZIP-226's three per-transaction rules above; we construct the state machine in §3. The implementation enforces both layers side by side: `validate_bundle_burn` returns the per-transaction errors `ZatoshiAsset`, `ZeroAmount`, `InvalidAmount`, and `DuplicateAsset`, and the state-level errors `AssetNotFoundInState` and `InsufficientSupply`.

4.5 Split Inputs

Padding is where the asset dimension bites. An Orchard transfer with more outputs than inputs pads with *dummy* spends: zero-value notes whose Merkle membership check is gated off by v_{old} (constraint A3 of the *Ironwood Guide*, "The Action statement"). For Custom Assets that padding is unsound, because the circuit forces the output note's Asset Base to equal the input note's (§4.6) — and a dummy input's Asset Base is chosen freely by the prover. ZIP-226 states the attack that must be excluded: without enforcing that every Custom Asset input exists in the note commitment tree, "the prover could input a multiple (or linear combination) of an existing Asset Base, and thereby attack the network by overflowing the ZEC value balance and hence counterfeiting ZEC funds". The fix: dummy notes remain in use for ZEC notes only; Custom Assets always prove Merkle membership, padding instead with Split Inputs, which force the output's Asset Base to equal a genuinely issued, GroupHash-derived base (ZIP-226).

The construction is developed once, in §5: the *Split Input* — a copy of a previously issued, tree-resident note of the target Asset, spent with split_flag = 1 and its value zeroed in the value-commitment computation only (Definition 5.1); its randomized nullifier with the offset point L^{Orchard}, (3); the three-way divergence over the derivation of ψ^{nf} (§5.6); and the builder mechanics of padding per asset, dummy notes for ZEC and split spends for Custom Assets (§5.8). The three wallet strategies ZIP-226 offers for sourcing the copied note are enumerated in §5.2 (`zip-0226.rst`:287–291). Burn entry points reject duplicate Assets and amounts that do not fit in 63 bits (`add_burn`).

[4] ZIP-230's AssetBurn table instead says valueBurn is "checked by consensus to be non-zero, and less than MAX_BURN_VALUE". The implementation rejects only amounts strictly greater than MAX_BURN_VALUE, i.e. permits equality, agreeing with ZIP-226 (orchard-zsa, `src/bundle/burn_validation.rs`, which defines $\mathsf{MAX_BURN_VALUE} = (1 \ll 63) - 1$). A specification-text defect, not a consensus divergence.

4.6 The Action statement, extended

A bin statement first. ZIP-226's "Circuit Statement" section specifies the constraint delta in prose and formulas, while deferring "all modifications in the Circuit" to the externally hosted, unversioned document *OrchardZSA circuit*. That document describes the public and private inputs, intermediate values, and constraints, but the ZIP does not embed or pin a complete statement at the rigor of the *Ironwood Guide*'s Definition "Orchard Action statement". The pinned implementation is therefore the reproducible source for the exact gate realization, and every gate-level claim below cites it (orchard-zsa, `src/circuit/circuit_zsa.rs`). Bin: *externally described and implemented, but not fixed by a versioned upstream specification*.

ZIP-226 states the deltas against A1–A9 as follows:

(a) the Asset Base is witnessed once as an auxiliary input, and both note commitment integrity constraints (A1, A2) switch to $\mathsf{NoteCommit}^{\mathsf{OrchardZSA}}$ with that same witness — one variable simultaneously enforces input/output Asset equality and commitment correctness;

(b) an auxiliary boolean is_native_asset is constrained to equal 1 exactly when $\mathsf{AssetBase} = V^{\mathsf{Orchard}}$;

(c) $\mathsf{enableZSA} = 0$ forces $\mathsf{is_native_asset} = 1$;

(d) the value commitment's fixed-base multiplication (A4) becomes a variable-base multiplication by the witnessed Asset Base;

(e) a boolean split_flag replaces the spent value inside the value commitment: the committed net value becomes $v' - v_{\text{new}}$, with $v' = 0$ if $\mathsf{split_flag} = 1$ and $v' = v_{\text{old}}$ otherwise; $\mathsf{split_flag} = 1$ forces $\mathsf{is_native_asset} = 0$;

(f) the Merkle path constraint (A3) becomes $(v_{\text{old}} = 0 \wedge \mathsf{is_native_asset} = 1) \vee \text{ValidMerklePath}$ — the dummy skip survives only for the native Asset;

(g) nullifier integrity is randomized for split notes (§5).

The circuit realizes these deltas in a single gate, `q_orchard`, whose twelve named constraints are stated once, in §5.5, in the code's names (the code names the ZIP's is_native_asset as is_zatoshi_asset); we do not restate them here.

Two gadgets carry the arithmetic weight. The in-circuit value commitment computes $[\text{magnitude}]\,\mathsf{asset_base}$ by variable-base long scalar multiplication on the witnessed non-identity point, applies mul_sign for the sign, and adds the fixed-base $[\mathsf{rcv}]\,R^{\mathsf{Orchard}}$ (`src/circuit/value_commit_orchard.rs`). The in-circuit note commitment shares one Sinsemilla evaluation between the two branches of Definition 4.2: it muxes the initial point Q between Q_{ZEC} (domain `z.cash:Orchard-NoteCommit-M`) and Q_{ZSA} (domain `z.cash:ZSA-NoteCommit-M`; the hard-coded point is test-asserted in `src/constants/sinsemilla.rs`) on is_zatoshi_asset, hashes the common message prefix once, hashes the two suffixes (ZEC: the zero-padded piece h_2; ZSA: the Asset Base bits), muxes the resulting hash point, and adds the shared blinding term $[\mathsf{rcm}]\,R$ — the shared randomness base of Definition 4.2 is what makes this sharing possible (`src/circuit/note_commit.rs`). The Asset Base message pieces: $h_{\text{ZSA}} = (\text{bits 249–253 of } \psi) \,\|\, (\text{bit 254 of } \psi) \,\|\, (\text{bits 0–3 of } x(\mathsf{asset}))$; $i = \text{bits 4–253 of } x(\mathsf{asset})$; $j = (\text{bit 254 of } x(\mathsf{asset})) \,\|\, (\tilde{y}(\mathsf{asset})) \,\|\, 8 \text{ zero bits}$ (ibid.).

The public instance grows from 9 to 10 field elements:

$$(\mathsf{anchor},\ \mathsf{cv}^{\mathsf{net}}.x,\ \mathsf{cv}^{\mathsf{net}}.y,\ \mathsf{nf}_{\text{old}},\ \mathsf{rk}.x,\ \mathsf{rk}.y,\ \mathsf{cmx},\ \mathsf{enable_spends},\ \mathsf{enable_outputs},\ \mathsf{enable_zsa}),$$

with the new enable_zsa last. In the vanilla flavor enable_zsa is always false; the instance-construction API pads the missing tenth element with zero, preserving the vanilla flavor's

public-input semantics. The crate carries two circuit flavors, `OrchardVanilla` and `OrchardZSA` (orchard-zsa, `src/circuit.rs`, `src/flavor.rs`). At the bundle level the enableZSAs flag is bit 2 of flagsOrchard (LSB order, after enableSpendsOrchard and enableOutputsOrchard; the implementation constant is `FLAG_ZSA_ENABLED = 0b0000_0100`), and coinbase transactions MUST set enableSpendsOrchard and enableZSAs to 0 (withdrawn ZIP-230).

4.7 The carrier: transaction format, digests, and fees

The following encodings belong to the experimental ZIP 230/246 carrier, not ZIP 229's deployed v6 format (§1.2). The transfer and burn constructions are draft-specified; their transaction encoding remains *designed but unspecified* for deployment.

Note plaintext. The OrchardZSA note plaintext adds asset_base (32 bytes) after rseed. Per withdrawn ZIP-230 the v6 Orchard note plaintext is

$$\mathsf{np} := [\mathsf{leadByte}] \,\|\, \mathrm{LEBS2OSP}_{88}(d) \,\|\, \mathrm{I2LEOSP}_{64}(v) \,\|\, \mathsf{rseed} \,\|\, \mathsf{asset_base} \,\|\, K^{\mathsf{memo}},$$

with a 32-byte memo key K^{memo} replacing the 512-byte memo field, giving encCiphertext of 132 bytes and an OrchardZSAAction of 372 bytes (cv 32 + nullifier 32 + rk 32 + cmx 32 + ephemeralKey 32 + encCiphertext 132 + outCiphertext 80). ZIP-230 notes that an Orchard note "is essentially equivalent to an OrchardZSA note with AssetBase $= V^{\mathsf{Orchard}}$" and that the two need not be distinguished in plaintexts.

Remark 4.6 (The lead byte is unassigned upstream). Upstream ZIP-226 requires leadByte to be the unresolved placeholder `{{ZSALEADBYTE}}` when the protocol's note-creation sections (§4.7.3 "Sending Notes", §4.8.3 "Dummy Notes") are invoked in the computation of ρ and ψ for OrchardZSA notes — a sentence with no counterpart in the fork's ZIP-226 and no implementation counterpart. Upstream ZIP-230 replaced its original `0x03` with the placeholder `{{LEADBYTE}}` because ZIP-2005 separately claims lead byte `0x03`. The QEDit fork ZIPs still say `0x03`, and the implementation hard-codes NOTE_VERSION_BYTE_V3 = `0x03` with MEMO_SIZE = 512 and no K^{memo} — the pre-memo-bundle layout, giving COMPACT_NOTE_SIZE_ZSA = 52 + 32 = 84 and a 612-byte encCiphertext, diverging from ZIP-230's 132 bytes (orchard-zsa, `src/primitives/zcash_note_encryption_domain.rs`, `src/primitives/orchard_primitives.rs`). No lead byte is currently reserved for ZSA notes upstream. Bin: *open*.

Digests. Withdrawn ZIP-246 extends the ZIP-244 digest tree — the baseline is the *Ironwood Guide*, "Transaction digests and signatures" — with a per-Action-Group orchard_burn_digest (node T.4a.vi) under orchard_action_groups_digest (personalization `ZTxIdOrcActGHash`), beside the compact and non-compact action digests, flagsOrchard, anchorOrchard, and nAGExpiryHeight. The compact digest covers nullifier, cmx, ephemeralKey, and encCiphertext (personalization `ZTxId6OActC_Hash`); the non-compact digest covers cv, rk, and outCiphertext (`ZTxId6OActN_Hash`). The burn digest is BLAKE2b-256 with personalization `ZTxIdOrcBurnHash` over, per assetBurn tuple,

$$\mathsf{assetBase} \,\|\, \mathsf{valueBurn} \quad \text{(64-bit unsigned little-endian)},$$

the empty burn set hashing to BLAKE2b-256(`ZTxIdOrcBurnHash`, `[]`) (ZIP-246). One disambiguation: the header-digest field T.1g burn_amount is ZIP-233's network-sustainability burn, unrelated to asset burn (ZIP-246, T.1; ZIP-230, Transaction Format).

Parsing and bundle types. Function `read_v6_bundle` (gated by `zcash_unstable = "nu7"`) reads nActionGroups, which must be 0 or exactly 1 ("A V6 transaction must contain at most one action group"), then the actions, flags, anchor, nAGExpiryHeight (must be 0), the burn vector

(32-byte Asset Base plus note value), the proof, per-action versioned signatures, valueBalance, and a versioned binding signature; the write path is symmetric via `write_burn` (librustzcash-zsa, `zcash_primitives/src/transaction/components/orchard.rs`). The bundle type carries the burn set as the field `burn: Vec<(AssetBase, NoteValue)>` beside `value_balance`, and `binding_validating_key()` returns `derive_bvk(actions, value_balance, burn)`; the builder collects burns in a `BTreeMap<AssetBase, NoteValue>`, canonically ordered and duplicate-free by construction (orchard-zsa, `src/bundle.rs`, `src/builder.rs`). One open seam: ZIP-230 permits nActionGroupsOrchard > 1 with per-group burn fields (the multi-party rationale of §4.4), the implementation rejects more than one group, and ZIP-226's bvk equation is written over a single flat action/burn set; how multi-group aggregation would interact with the binding signature is unexercised. Bin: *open*.

Tree ordering. Per Action Group, every new OrchardZSA note commitment is appended to the note commitment tree per Action; afterwards, Issue Note commitments are appended per Issue Action (ZIP-227, "Addition to the Note Commitment Tree"; §3).

Fees. The QEDit fork of ZIP-317 adds contribution_ZSAIssuance and contribution_ZSACreation terms — issue notes and asset creations respectively — to the conventional-fee formula; it adds no burn term, so a burn contributes to fees only through the Actions that fund it. Upstream ZIP-317 contains no ZSA changes. The conventional-fee baseline is the *Wallet Guide*, "Fees (ZIP-317)".

5 Split notes and the counterfeiting boundary

An Orchard bundle pairs inputs and outputs into Actions by padding the shorter side with *dummy notes*: an Action with no real input spends a zero-valued note under a freshly sampled key, and the membership constraint waves it through because the check is gated by the spent value — constraint A3 of the Action statement, $v_{\text{old}} \cdot (\mathsf{root} - \mathsf{anchor}) = 0$, which exempts a consumed note of value zero from membership so that an Action with a dummy consumed note can create a real note (the *Ironwood Guide*, "The Action statement"). The construction is protocol §4.8.3, implemented in `orchard-zsa`, `SpendInfo::dummy`: a random spending key, a zero-valued note, and a random Merkle path that is never checked. The exemption is sound there because every Orchard value commitment rides the *fixed* base V^{Orchard} (*Ironwood Guide*, "Value commitments"): a dummy spend can move no value, whatever path it fails to prove. ZIP-226 replaces the fixed base with a witnessed one, and the same exemption becomes a mint. This section develops the attack that forces the change, the *split note* that ZIP-226 substitutes for the dummy note on custom assets, the ZIP-specified circuit delta, and the implemented gate that draws the counterfeiting boundary.

Ground truth for this section: ZIP-226 and ZIP-227 at `zcash/zips` commit `69610984` (2026-07-09); the pinned implementation, crate `orchard` of the QED-it fork (`orchard-zsa`, branch `zsa1`) at commit `cf801a5d` (2026-06-29); the QED-it `zips` fork at `fd71419a` where it diverges from upstream. Split-note logic lives entirely in the `orchard` crate; the `librustzcash` fork consumes its bundle API and contains no split-note code of its own (verified by search at `3f9b53fc`).

5.1 The counterfeit construction the boundary excludes

ZIP-226 generalises the net value commitment from the fixed base V^{Orchard} to a per-asset base (ZIP-226, "Value Commitment"):

$$\begin{aligned} \mathsf{cv}^{\mathsf{net}} &:= \mathsf{ValueCommit}^{\mathsf{OrchardZSA}}_{\mathsf{rcv}}(\mathsf{AssetBase}_{\mathsf{AssetId}}, v^{\mathsf{net}}_{\mathsf{AssetId}}) \\ &= [\, v^{\mathsf{net}}_{\mathsf{AssetId}} \,]\, \mathsf{AssetBase}_{\mathsf{AssetId}} + [\mathsf{rcv}]\, R^{\mathsf{Orchard}}, \end{aligned} \tag{1}$$

with $v^{\mathsf{net}} := v_{\mathrm{old}} - v_{\mathrm{new}}$ and R^{Orchard} the randomness base of the *Ironwood Guide*'s net value commitment ("Value commitments"), unchanged. The native asset keeps the old base: "the Asset Base of the ZEC Asset will be kept as the original value base point $\mathcal{V}^{\mathsf{Orchard}}$" (ZIP-226, "Asset Identifiers"; in code `AssetBase::zatoshi()` is $\mathsf{GroupHash}^{\mathsf{Pallas}}$(`z.cash:Orchard-cv, "v"`)). Verification subtracts the declared native balance and the declared burns (ZIP-226, "Value Balance Verification"; implemented in `derive_bvk_raw`):

$$\begin{aligned}\mathsf{bvk} = \Big(\sum_{i=1}^{n} \mathsf{cv}_i^{\mathsf{net}}\Big) &- \mathsf{ValueCommit}_0^{\mathsf{OrchardZSA}}(V^{\mathsf{Orchard}}, v^{\mathsf{balanceOrchard}}) \\ &- \sum_{(\mathsf{AssetBase},\, v) \in \mathsf{assetBurn}} \mathsf{ValueCommit}_0^{\mathsf{OrchardZSA}}(\mathsf{AssetBase}, v),\end{aligned} \tag{2}$$

and the binding signature under bvk proves knowledge of $\mathsf{bsk} = \sum_i \mathsf{rcv}_i$ exactly as in Orchard.

Suppose now that the dummy exemption survived unchanged: a padding Action proves no membership, and — since the base of (1) is a private witness — may witness *any* point A as its asset base. Choose $A := [-1]\, V^{\mathsf{Orchard}}$, witness $v_{\mathrm{old}} = 0$ and $v_{\mathrm{new}} = v$:

$$\mathsf{cv}_1^{\mathsf{net}} = [\,0 - v\,]\, A + [\mathsf{rcv}_1]\, R^{\mathsf{Orchard}} = [\,v\,]\, V^{\mathsf{Orchard}} + [\mathsf{rcv}_1]\, R^{\mathsf{Orchard}}.$$

Pair it with a second Action that outputs a genuine v-valued ZEC note,

$$\mathsf{cv}_2^{\mathsf{net}} = [\,0 - v\,]\, V^{\mathsf{Orchard}} + [\mathsf{rcv}_2]\, R^{\mathsf{Orchard}},$$

and declare $v^{\mathsf{balanceOrchard}} = 0$ with no burns. The V^{Orchard}-components cancel in (2),

$$\mathsf{bvk} = \mathsf{cv}_1^{\mathsf{net}} + \mathsf{cv}_2^{\mathsf{net}} = [\mathsf{rcv}_1 + \mathsf{rcv}_2]\, R^{\mathsf{Orchard}},$$

the attacker knows $\mathsf{bsk} = \mathsf{rcv}_1 + \mathsf{rcv}_2$, the binding signature verifies — and a v-valued ZEC note now exists that no one paid for. Suitably chosen known relations among accepted bases can be exploited analogously. (*Classification:* the construction is assembled here from the specified formulas (1) and (2); ZIP-226 states the attack only in words.) The ZIP's own statement, verbatim: the output note of split Actions "cannot contain just any Asset Base. We must enforce it to be an actual output of a GroupHash computation... Without this enforcement the prover could input a multiple (or linear combination) of an existing Asset Base, and thereby attack the network by overflowing the ZEC value balance and hence counterfeiting ZEC funds" (ZIP-226, "Split Notes", rationale).

The boundary to draw is therefore precise: every asset base that enters a value commitment must either be the fixed native base or descend from an accepted GroupHash output. Security then assumes that no useful discrete-log relation among that base, V^{Orchard}, and R^{Orchard} can be found. The one place an unproven base could otherwise enter is an Action with no real input.

5.2 The split input

Definition 5.1 (Split Input, Split Action)**.** ZIP-226 defines a *Split Input* as "an Action input used to ensure that the output note of that Action is of a validly issued AssetBase when there is no corresponding real input note, in situations where the number of outputs are larger than the number of inputs", and a *Split Action* as an Action containing a Split Input (ZIP-226, "Terminology").

Mechanically, a Split Input is a *copy of a previously issued note* — one whose commitment is already in the note commitment tree — with exactly two changes: a boolean witness split_flag is set to 1, and the note's value is replaced by 0 *in the value-commitment computation only* (ZIP-226, "Split Notes"). The copied note's commitment $\mathsf{cm}_{\mathrm{old}}$ still opens with the true value: in

the circuit the value is zeroed solely in the $\mathsf{cv}^{\mathsf{net}}$ equation, while the old-note commitment check consumes v_{old} unchanged (`orchard-zsa`, `src/circuit/circuit_zsa.rs`: the gate zeroes v_{old} via the $(1-\mathsf{split_flag})$ factor in the value equation, lines 127–131, whereas the NoteCommit call at lines 637–655 takes v_{old} as witnessed). The Action therefore neither moves value from the copied note nor publishes that note's ordinary nullifier; instead it publishes the randomized split nullifier of §5.6. The copy drags the original note's asset base into the proof, where an equality constraint passes it to the output note.

The split Action participates in the same computational per-asset binding argument as §4.3: a successful imbalance would require a useful discrete-log relation among the independently derived bases. For a correct transaction the custom-asset commitments satisfy

$$\sum_{j \in S_{\mathsf{CA}}} \mathsf{cv}_j^{\mathsf{net}} - \sum_{(\mathsf{AssetBase},\, v) \in \mathsf{assetBurn}} [\, v \,]\, \mathsf{AssetBase} = \sum_{j \in S_{\mathsf{CA}}} [\mathsf{rcv}_j]\, R^{\mathsf{Orchard}},$$

so the split Action's $\mathsf{cv}^{\mathsf{net}} = [\, 0 - v_{\text{new}} \,]\, \mathsf{AssetBase} + [\mathsf{rcv}]\, R^{\mathsf{Orchard}}$ cancels against the real spending Actions of the same asset. The builder implements exactly this: `ActionInfo::value_sum` uses a spent value of 0 when split_flag is set, and `derive_bvk_raw` computes (2) over the resulting commitments.

For the note to copy, ZIP-226 offers the wallet three options: (1) another note of the same Asset Base being spent in the same transaction; (2) a different unspent note of that Asset Base, which is *not* thereby spent; (3) an already spent note of that Asset Base — the zeroed value prevents double spending (ZIP-226, "Split Notes"). The ZIP adds that "we do not care about whether the previously issued note copied to create a Split Input is owned by the sender, or whether it was nullified before"; §5.7 examines how far that sentence actually holds against the constraint system.

5.3 Membership as well-formedness: the issuance anchor

The fix ZIP-226 specifies, verbatim: "for Custom Assets we enforce that every input note to an OrchardZSA Action must be proven to exist in the set of note commitments in the note commitment tree. We then enforce this real note to be unspendable in the sense that its value will be zeroed in split Actions and the nullifier will be randomized... the proof itself ensures that the output note is of the same Asset Base as the input note. In the circuit, the split note functionality will be activated by a boolean private input (aka the split_flag boolean)" (ZIP-226, "Split Notes", rationale).

Why does tree membership prove that an asset base is a GroupHash output? Because for custom assets the tree is fed exclusively from two sources that both enforce it. Transfer outputs inherit their base from a proven input by the in-circuit equality constraint (§5.4). Issuance outputs are checked publicly: a consensus rule obliges validators to recompute each Issue Note's AssetBase from the issuance bundle's issuer key and the IssueAction's assetDescHash (ZIP-227, "Specification: Consensus Rule Changes"), along the derivation chain constructed in §2.3 (ZIP-227, "Asset Bases"), and then appends the issue-note commitments to the *same* Orchard note commitment tree — transfer-Action commitments first, then issuance notes (ZIP-227, "Addition to the Note Commitment Tree"). Hence, under note-commitment binding, a custom-asset note commitment resident in the tree commits, by induction along the two rules, to a genuine GroupHash output. ZIP-226 states the conclusion: "This ensures that the value base points of all output notes of a transfer are actual outputs of a GroupHash, as they originate in the Issuance protocol which is publicly verified" ("Split Notes", rationale); and, on the identifier side, "We prevent a potential malleability attack on the Asset Identifier by ensuring the output notes receive an Asset Base that exists on the global state" ("Rationale for Asset Identifiers").

One clarification the ZIP's sentence invites: "every input note to an OrchardZSA Action" is scoped to *custom-asset* inputs. Native-asset (ZEC) Actions in a v6 bundle retain the dummy-note exemption unchanged — "The ZEC-based Actions will still include dummy input notes, whereas

the OrchardZSA Actions will include split input notes and will not include dummy input notes" (ZIP-226, "Backward Compatibility"; the "Split Notes" rationale states the same). The constraint-level form of this scoping is the disjunction in §5.4.

5.4 The specified circuit delta

We state ZIP-226's circuit changes ("Circuit Statement") as a delta against the Action statement A1–A9 of the *Ironwood Guide* ("The Action statement"). Two new boolean witnesses appear: split_flag, and is_native_asset, constrained to equal 1 exactly when the witnessed base is the native one, $\mathsf{AssetBase} = V^{\mathsf{Orchard}}$. One public input appears: the bundle-level enableZSA flag (a bit of the v6 flagsOrchard field), with the constraint $\mathsf{enableZSA} = 0 \implies \mathsf{is_native_asset} = 1$ — switching the flag off collapses the circuit to native-asset behaviour and, transitively via the split constraint below, disables split notes (ZIP-226, "Overview" and "Circuit Statement", The enableZSA Flag; in code the flag is the tenth public input, taken from `Flags::zsa_enabled()`, `orchard-zsa`, `src/circuit.rs`, lines 476–494, 560).

- **A1, A2 (note commitment integrity).** Both commitment checks replace $\mathsf{NoteCommit}^{\mathsf{Orchard}}$ with $\mathsf{NoteCommit}^{\mathsf{OrchardZSA}}$, which takes the witnessed AssetBase as an additional final argument; the *same* once-witnessed base feeds the old-note check, the new-note check, and the value commitment (ZIP-226, "Circuit Statement", Asset Base Equality). For the native asset $\mathsf{NoteCommit}^{\mathsf{OrchardZSA}}$ coincides with $\mathsf{NoteCommit}^{\mathsf{Orchard}}$; otherwise it is the Sinsemilla commitment over the extended message ending in the asset-base encoding, in the hash domain `"z.cash:ZSA-NoteCommit-M"` with the blinding base shared with Orchard (`"z.cash:Orchard-NoteCommit-r"`) (ZIP-226, "Note Structure and Commitment"). The full $\mathsf{NoteCommit}^{\mathsf{OrchardZSA}}$ construction is Definition 4.2; its in-circuit mux is described in §4.6.

- **A3 (membership).** The value gate becomes a disjunction (ZIP-226, "Circuit Statement", Asset Identifier Consistency for Split Actions — Merkle Path Validity):

$$(v_{\text{old}} = 0 \ \wedge\ \mathsf{is_native_asset} = 1) \ \vee\ (\text{Valid Merkle Path}),$$

  so the dummy skip survives *only* for the native asset. Every custom-asset input — split or not, zero-valued or not — must prove membership.

- **A4 (value commitment).** Two changes. First, "the fixed-base multiplication constraint between the value and the value base point of the value commitment, cv, is replaced with a variable-base multiplication between the two" (ZIP-226, "Circuit Statement", Value Commitment Correctness), the base being the witnessed AssetBase. Second, the input value is replaced by a generic v' (ZIP-226, "Circuit Statement", Asset Identifier Consistency for Split Actions):

$$\mathsf{cv}^{\mathsf{net}} = \mathsf{ValueCommit}^{\mathsf{OrchardZSA}}_{\mathsf{rcv}}(\mathsf{AssetBase}, v' - v_{\text{new}}), \qquad v' = \begin{cases} 0 & \text{if } \mathsf{split_flag} = 1, \\ v_{\text{old}} & \text{otherwise,} \end{cases}$$

  together with $\mathsf{split_flag} = 1 \implies \mathsf{is_native_asset} = 0$: split notes exist only for custom assets, and a zero-valued native dummy cannot masquerade as one.

- **A5 (nullifier).** The derivation changes for split notes "to prevent the identification of notes" (ZIP-226, "Circuit Statement", Asset Identifier Consistency for Split Actions); §5.6 develops it.

- **A6, A7 (spend authority, address integrity).** Untouched by the ZIP's change list — and, decisively for §5.7, they remain *unconditional*: no split_flag factor relaxes them.

5.5 The implemented gate

A naming note, once: the ZIP's is_native_asset is `is_zatoshi_asset` throughout the implementation, assigned 1 exactly when the witnessed asset equals `AssetBase::zatoshi()` (`orchard-zsa`, `src/circuit/gadget.rs`, `circuit_zsa.rs`). We keep the ZIP name in prose and the code name inside code-derived formulas.

The implemented circuit concentrates the delta in a single gate, `q_orchard`, which carries twelve named constraints where the vanilla circuit's corresponding gate has four (`src/circuit/circuit_zsa.rs`, lines 68–185, versus `src/circuit/circuit_vanilla.rs`, lines 59–100). In the code's names:

$$\mathsf{bool_check}(\mathsf{split_flag}), \qquad \mathsf{bool_check}(\mathsf{is_zatoshi_asset}),$$
$$v_{\text{old}} \cdot (1 - \mathsf{split_flag}) - v_{\text{new}} = \text{magnitude} \cdot \text{sign},$$
$$(v_{\text{old}} + 1 - \mathsf{is_zatoshi_asset}) \cdot (\mathsf{root} - \mathsf{anchor}) = 0,$$
$$v_{\text{old}} = 0 \ \vee\ \mathsf{enable_spends} = 1, \qquad v_{\text{new}} = 0 \ \vee\ \mathsf{enable_outputs} = 1,$$
$$\mathsf{is_zatoshi_asset} \cdot (\mathsf{asset}_x - \mathsf{zatoshi}_x) = 0, \qquad \mathsf{is_zatoshi_asset} \cdot (\mathsf{asset}_y - \mathsf{zatoshi}_y) = 0,$$
$$(1 - \mathsf{is_zatoshi_asset}) \cdot (\Delta_x \cdot \Delta_x^{\mathsf{inv}} - 1) \cdot (\Delta_y \cdot \Delta_y^{\mathsf{inv}} - 1) = 0,$$
$$(1 - \mathsf{split_flag}) \cdot (\psi_{\text{old}} - \psi^{\mathsf{nf}}) = 0, \qquad \mathsf{split_flag} \cdot \mathsf{is_zatoshi_asset} = 0,$$
$$(1 - \mathsf{enable_zsa}) \cdot (1 - \mathsf{is_zatoshi_asset}) = 0,$$

where Δ_x, Δ_y abbreviate the code's `diff_asset_x`, `diff_asset_y` — the witnessed asset's coordinates minus the native base's — and ψ^{nf} is the code's `psi_nf` (§5.6). Four points merit unpacking.

The folded value equation. The single constraint $v_{\text{old}} \cdot (1 - \mathsf{split_flag}) - v_{\text{new}} = \text{magnitude} \cdot \text{sign}$ folds the ZIP's case-split v' into one expression; the witness generator supplies the matching net value, $-v_{\text{new}}$ for split spends and $v_{\text{old}} - v_{\text{new}}$ otherwise (`circuit_zsa.rs`, lines 462–501).

The Merkle factor. The disjunction of A3 is encoded as the single product $(v_{\text{old}} + 1 - \mathsf{is_zatoshi_asset}) \cdot (\mathsf{root} - \mathsf{anchor}) = 0$. The code comment argues the encoding safe from wrap-around: is_zatoshi_asset is boolean-checked and v_{old} is range-checked to 64 bits in the note-commitment evaluation, so over $\mathbb{F}_{p_{\mathsf{Pallas}}}$ the first factor vanishes exactly when $v_{\text{old}} = 0$ and $\mathsf{is_zatoshi_asset} = 1$ (`circuit_zsa.rs`, lines 132–139). The consequence bears repeating: for every custom-asset input the factor is nonzero, so $\mathsf{root} = \mathsf{anchor}$ is mandatory — a strictly stronger membership requirement than vanilla Orchard's v_{old}-gate, applying even to zero-valued non-split custom-asset inputs.

Proving inequality. The assignment $\mathsf{is_zatoshi_asset} = 0$ must itself be sound: the circuit must verify that the witnessed base *differs* from V^{Orchard}. The prover witnesses field inverses $\Delta_x^{\mathsf{inv}}, \Delta_y^{\mathsf{inv}}$ (assigned 0 when the corresponding difference vanishes), and the triple product above forces at least one difference to be exhibited invertible (`circuit_zsa.rs`, lines 160–170 for the gate, 792–841 for the assignment).

The variable-base value commitment. With the base witnessed, the vanilla circuit's fixed-base *short* multiplication $[\,v^{\mathsf{net}}\,]\,V^{\mathsf{Orchard}}$ — whose 64-bit range check is implicit in the short-multiplication decomposition — is no longer available. The ZSA gadget makes the range check explicit: the magnitude is decomposed against the Sinsemilla lookup table as six 10-bit windows plus one 4-bit short check (64 bits), then promoted to a full-width scalar (`ScalarVar::from_base`) and fed to the variable-base long multiplication — the code notes it is "optimized for ~255-bit scalar" and leaves a TODO to specialise it for 64 bits — after which the sign is applied and the blind $[\mathsf{rcv}]\,R^{\mathsf{Orchard}}$ added (`src/circuit/value_commit_orchard.rs`,

lines 38–127). Two supporting details: the ZSA circuit uses the extended lookup configuration `PallasLookupRangeCheck4_5BConfig` (an additional lookup-table column for 4- and 5-bit short checks), and it refuses to synthesise under `CircuitVersion::InsecureUnanchoredBase`, the pre-NU6.2 `halo2_gadgets` variable-base scalar-multiplication version whose base point was not anchored — directly load-bearing here, because the entire counterfeiting boundary rests on the multiplication being by the *witnessed* AssetBase and nothing else (`circuit_zsa.rs`, lines 22, 51, 190–197, 334–337; `circuit.rs`, lines 151–185).

The flag interactions are tested asymmetrically: the three legal combinations of

$$(\mathsf{is_zatoshi_asset}, \mathsf{split_flag})$$

— $(1,0)$, $(0,1)$, and $(0,0)$ — are proven passing, while the $(1,1)$ combination is excluded at witness-generation time by the test helper's assertion ("We cannot create a split note with a zatoshi asset"), so the constraint $\mathsf{split_flag} \cdot \mathsf{is_zatoshi_asset} = 0$ is not itself exercised by a failing-proof test (`circuit_zsa.rs`, lines 1174–1312). Documented cost: the ZSA proof measures 5120 bytes for one Action and 7392 for two, against the vanilla 4992 and 7264, at the shared circuit size $K = 11$; the enableZSA bit is the tenth public input, at index 9 (the proof-size tests of `circuit_zsa.rs` and `circuit_vanilla.rs`; `circuit.rs`, lines 68–80, 536–563).

5.6 The split nullifier

A split input copies a note that may sit unspent in someone's wallet; its Action must nevertheless publish *some* nullifier. Publishing the note's true nullifier would be harmful in both directions: it would mark an unspent note spent (bricking it), and it would collide with the honest nullifier if the same note were spent for real in the same transaction. ZIP-226 therefore randomises the split note's nullifier ("Split Notes"):

$$\mathsf{nf}_{\mathrm{old}} = \mathsf{Extract}(\,[\,F_{\mathsf{nk}}(\rho_{\mathrm{old}}) + \psi^{\mathsf{nf}}\,]\,G^{\mathsf{nf}} \;+\; \mathsf{cm}_{\mathrm{old}} \;+\; L^{\mathsf{Orchard}}\,), \tag{3}$$

where F_{nk} and G^{nf} abbreviate the nullifier PRF $\mathsf{PRF}^{\mathsf{nfOrchard}}_{\mathsf{nk}}$ and the nullifier base K^{Orchard} of condition A5 of the *Ironwood Guide*'s Definition "Orchard Action statement" (ZIP-226 writes $\mathcal{K}^{\mathsf{Orchard}}$ for G^{nf} and computes the bracketed sum in the Pallas base field). Two deltas against A5: the note's ψ_{old} is replaced by a fresh ψ^{nf}, and a fixed offset point

$$L^{\mathsf{Orchard}} := \mathsf{GroupHash}^{\mathsf{Pallas}}(\texttt{z.cash} : \texttt{Orchard}, \texttt{"L"})$$

is added (ZIP-226, "Split Notes"; the constant is pinned and test-asserted against the GroupHash evaluation in `orchard-zsa`, `src/constants/nullifier_l.rs`). The gate constraint $(1 - \mathsf{split_flag}) \cdot (\psi_{\mathrm{old}} - \psi^{\mathsf{nf}}) = 0$ of §5.5 makes the substitution split-only: non-split OrchardZSA nullifiers coincide with the Orchard derivation, as the ZIP states ("Note Structure and Commitment": the non-split nullifier "is generated in the same manner as in the Orchard protocol").

In circuit, the nullifier gadget computes the standard point $\mathsf{cm}_{\mathrm{old}} + [\,F_{\mathsf{nk}}(\rho_{\mathrm{old}}) + \psi^{\mathsf{nf}}\,]\,G^{\mathsf{nf}}$ from the witnessed ψ^{nf}, forms the offset variant by adding the constant L^{Orchard}, and muxes the two on split_flag before extraction (`src/circuit/derive_nullifier.rs`, lines 54–111). Out of circuit, `Nullifier::derive` mirrors the computation and selects the offset variant in constant time exactly when the note carries a `rseed_split_note` (`src/note/nullifier.rs`, lines 73–90; `src/note.rs`, lines 415–426).

Deriving ψ^{nf}: a three-way divergence. The value of ψ^{nf} is always constrained through the A5 equation to the published nullifier. Only the equality $\psi^{\mathsf{nf}} = \psi_{\mathrm{old}}$ is gated off in the split case; no constraint derives ψ^{nf} from a seed. How a wallet samples it for a split note is therefore wallet-level, not consensus-level. The three extant texts disagree; per the volume's policy (§1.3), the

upstream ZIP formula comes first. Upstream ZIP-226 — written against the ZIP-2005 baseline, "expected to deploy before any ZSA activation" — specifies

$$\psi^{\mathsf{nf}} := \mathrm{ToBase}(\mathsf{PRFexpand}_{\mathsf{rseed_nf}}(\texttt{0x0A} \,\|\, \rho_{\mathrm{old}}))$$

with rseed_nf uniform on 32 bytes and the domain byte `0x0A` "reserved by ZIP 2005 for this use" (`zip-0226.rst`:297 @ `69610984`). The QED-it `zips` fork specifies neither derivation: there ψ^{nf} "is sampled uniformly at random" on the Pallas base field (fork `zip-0226.rst`:293 @ `fd71419a`). The implementation derives

$$\psi^{\mathsf{nf}} := \mathrm{ToBase}(\mathsf{PRFexpand}_{\mathsf{rseed_split_note}}(\texttt{0x09} \,\|\, \rho_{\mathrm{old}})),$$

the ordinary ψ domain byte keyed by a fresh random 32-byte `rseed_split_note` (`orchard-zsa`, `src/circuit.rs`:316–319, `src/note.rs`:114–116 and 417–425; domain byte `PSI = 0x09` per the QED-it `zcash_spec` fork, `src/prf_expand.rs`:89 @ `d5e84264`). Direct field sampling is uniform; the two PRF-and-reduction constructions are intended to be pseudorandom and have at most the negligible modular-reduction bias of ToBase. The implementation matches neither ZIP text exactly. (*Classification:* each derivation is specified in its own text; the divergence is consensus-invisible.)

Why the offset L^{Orchard}? The randomised ψ^{nf} is intended to make the split nullifier unlinkable to the note's true nullifier. ZIP-226 states no rationale for additionally adding L^{Orchard}. A plausible purpose is robustness against a prover who *chooses* $\psi^{\mathsf{nf}} = \psi_{\mathrm{old}}$ — the circuit does not forbid it in the split case — and would thereby publish the note's true nullifier from a zero-valued split Action, bricking the original note or colliding with an honest spend of it elsewhere in the transaction. The constant offset makes the pre-extraction group point different even under that choice. It is not an absolute noncollision proof for the extracted nullifier, since elliptic-curve points with opposite signs have the same extracted x-coordinate. (*Classification:* designed-but-unspecified; the preceding sentence is our inference, stated in neither ZIP nor code.)

5.7 Ownership, in fact and in prose

Recall the ZIP's claim that "we do not care about whether the previously issued note copied to create a Split Input is owned by the sender" (§5.2). Both the proof and the Action signature narrow that claim. Spend-authority consistency (A6) and address integrity (A7) are unchanged and unconditional in the ZSA circuit: the proof establishes $\mathsf{rk} = [\alpha]\, G^{\mathsf{SpendAuth}} + \mathsf{ak}$ and $\mathsf{ivk} = \mathsf{Commit}^{\mathsf{ivk}}_{\mathsf{r_{ivk}}}(\mathsf{Extract}(\mathsf{ak}), \mathsf{nk})$ with $\mathsf{pk_{d_{old}}} = [\mathsf{ivk}]\, \mathsf{g_{d_{old}}}$ — for split inputs too (`orchard-zsa`, `src/circuit/circuit_zsa.rs`, lines 554–623; the old-note commitment equality at lines 625–659). Because the copied note's $\mathsf{cm_{old}}$ must open to the actual tree leaf (Sinsemilla binding plus the mandatory Merkle check of §5.5), the witnessed $\mathsf{pk_{d_{old}}}$ is the original recipient's, and A7 then forces the prover to know $(\mathsf{ak}, \mathsf{nk}, \mathsf{r_{ivk}})$ consistent with it — the full-viewing-key components or an equivalent witness. More decisively, every Action must carry a RedPallas spend authorization signature under rk. A split spend has `dummy_sk = None`; preparation auto-signs only true dummy spends, and `apply_signatures` finishes a split Action only after receiving a `SpendAuthorizingKey` whose validating key matches ak, otherwise returning `MissingSignatures` (`src/builder.rs`, lines 308–315 and 1347–1432, `SpendInfo::create_split_spend`, `Bundle::prepare`, `Bundle::apply_signatures`, and `Bundle::sign`). Thus constructing a valid transaction requires the copied note's spend-authorizing key, absent a signature forgery or discrete-log break. The ZIP's sentence holds for *spentness* — a nullified note can still be copied because its value is zeroed — but not for ownership. The builder consequently copies the first real spend of that Asset in the same bundle, option (1) of §5.2 (`src/builder.rs`, `pad_spend`). (*Classification:* the circuit and signature behavior are implemented; the ZIP rationale overstates the available choices.)

One genuine exception exists, and neither ZIP states it. ZIP-227 requires the first note of any asset's first issuance to be a *reference note*: value 0, recipient the default diversified address of the all-zero Orchard spending key — a key anyone can derive (ZIP-227, "Reference Notes"; `orchard-zsa`, `src/constants/reference_keys.rs` pins the all-zero spending key and the 43-byte raw address). Because that spending key is public, anyone can satisfy A6/A7 and produce the Action signature for the reference note, which makes it a universally available split input for its asset — the one case in which "ownership does not matter" holds literally. (*Classification:* inference; the connection between reference notes and split inputs is stated in neither ZIP nor the implementation.)

5.8 Builder mechanics: padding, per asset

The two padding regimes coexist per ZIP-226's rule quoted in §5.3: dummy inputs for ZEC Actions, split inputs for custom-asset Actions. The builder partitions spends and outputs by asset, pads each side of each partition to equal length, shuffles within each asset, then shuffles the assembled Actions across assets (`orchard-zsa`, `src/builder.rs`, lines 1081–1164). The padding spend is asset-dependent (`pad_spend`, lines 928–944): for the native asset, `SpendInfo::dummy` exactly as in Orchard; for a custom asset, `create_split_spend` on the first real spend of that asset in the bundle. If the bundle spends none, `pad_spend` produces `BuildError::NoSplitNoteAvailable`; the current per-asset iterator unwraps that result and panics with the error rather than returning it from `bundle_for_version`. Unlike a dummy, a split input cannot be conjured from nothing. Output-side padding uses dummy outputs of the padding asset in both regimes.

The copy itself is minimal. Function `Note::create_split_note` duplicates the note and sets exactly one new field, $\mathsf{rseed_split_note}$ — a fresh random 32-byte seed, drawn by `RandomSeed::random` and resampled until the derived esk is valid — asserting the asset is not the native one; `SpendInfo::create_split_spend` keeps the original note's full viewing key and Merkle path, sets no dummy signing key, and raises $\mathsf{split_flag}$ (`src/note.rs`, lines 436–442, 95–103; `src/builder.rs`, lines 302–318). Recipient, value, asset, ρ, rseed — hence $\mathsf{cm}_{\mathrm{old}}$ — are identical to the tree-resident original; the only witness-level differences downstream are $\mathsf{split_flag} = 1$ and the ψ^{nf} drawn from the new seed. Witness assembly routes exactly that: `Witnesses::from_action_context_unchecked` takes ψ^{nf} from $\mathsf{rseed_split_note}$ when present and from rseed otherwise — so $\psi^{\mathsf{nf}} = \psi_{\mathrm{old}}$ for every non-split spend — and packs $(\psi^{\mathsf{nf}}, \mathsf{asset}, \mathsf{split_flag})$ into the ZSA witness extension that the vanilla flavour leaves unknown (`src/circuit.rs`, lines 199–244, 300–345).

The counterfeiting boundary, restated once in full: a custom-asset output note's base equals its Action's input-note base (the A1/A2 equality on the shared witness); that input note is a tree leaf (the mandatory Merkle factor); under note-commitment binding, every custom-asset tree leaf descends from a publicly recomputed $\mathsf{GroupHash}$ output (the issuance anchor of §5.3); and the copy that makes padding possible moves no value (the zeroed v'). Those links enforce the counterfeiting boundary of §5.1. Independently, the randomized nullifier (3) avoids publishing the original note's ordinary nullifier except with negligible collision probability under honest randomisation; it is not an absolute noncollision guarantee (§5.6).

6 Transaction integration and outlook

The preceding sections construct the OrchardZSA objects — asset bases (§2), notes and Actions (§4, §5), issuance (§3), and burn (§4.4). This section places them in a transaction: the v6 wire format (ZIP-230), the digest tree that yields transaction identifiers, sighashes, and the authorizing-data commitment (ZIP-246), the transaction-level consensus rules of ZIP-226/227, the fee treatment (ZIP-317), the state of the QED-it implementation stack at the pins in Table 1, and the obli-

gations the format imposes on wallets. The format is experimental; §1.2 states its deployment boundary.

6.1 Experimental and deployed carriers

The carrier below is the experimental OrchardZSA format of ZIP 230, with digests from ZIP 246. Both ZIPs are withdrawn. Its version number 6 must not be confused with ZIP 229's deployed version 6 format, which carries Ironwood and no ZSA bundles. The two byte formats are incompatible despite the shared number.

We use the experimental format to explain how the note, issuance, burn, and signature constructions compose into one transaction. Its fields and digests are *designed but unspecified* for deployment; the remaining specification obligations are in §6.8.

6.2 The experimental wire format (ZIP-230)

Common and transparent fields. The common header carries `header` (`fOverwintered` and the version), `nVersionGroupId`, `nConsensusBranchId`, `lock_time`, `nExpiryHeight`, an explicit 8-byte `fee` (`uint64`, per ZIP-2002), and an 8-byte `zip233Amount` (`uint64`, per ZIP-233) (ZIP-230, Transaction Format). Making the fee an explicit field replaces its derivation as the residual difference between transaction input and output value and lets clients read it without resolving prior transparent outputs (ZIP-2002). Separately, signatures become versioned below. The transparent fields add `vSighashInfo`: one `TransparentSighashInfo` (a `compactSize` length followed by `sighashInfo` bytes) per transparent input (ZIP-230, Transaction Format).

Orchard fields: Action Groups. The Orchard component becomes a vector of *Action Groups*: `nActionGroupsOrchard`, `vActionGroupsOrchard`, then a transaction-level `valueBalanceOrchard` (`int64`) and `bindingSigOrchard`. The balance and binding signature are present iff `nActionGroupsOrchard` > 0; an absent `valueBalanceOrchard` means $v^{\mathsf{Orchard}}_{\mathsf{balance}} = 0$ (ZIP-230, Transaction Format). Each `ActionGroupDescription` contains, in order (ZIP-230, OrchardZSA Action Group Description):

- `nActionsOrchard` (`compactSize`, MUST be > 0) and `vActionsOrchard`, at 372 bytes per `OrchardZSAAction`;
- `flagsOrchard`, LSB-first: `enableSpendsOrchard`, `enableOutputsOrchard`, `enableZSAs`, remaining bits zero;
- `anchorOrchard` (32 bytes) — each group carries its own anchor;
- `nAGExpiryHeight` (`uint32`);
- `nAssetBurn` and `vAssetBurn`, at 40 bytes per `AssetBurn` entry;
- `sizeProofsOrchard` and `proofsOrchard`, one aggregated proof for the group;
- `vSpendAuthSigsOrchard`, one `OrchardSignature` per Action.

Field `nAGExpiryHeight` exists for forward compatibility with ZSA atomic swaps. ZIP-228 is now a Draft that uses this field to expire swap orders, but it still depends on the withdrawn ZIP-230/246 carrier; for the core OrchardZSA proposal the field is zero by consensus (ZIP-230, Rationale for `nAGExpiryHeight`; §6.4). This restriction concerns `nAGExpiryHeight`, not the transaction-level expiry field `nExpiryHeight`. Burn fields sit inside each Action Group rather than at transaction level so that, in a future multi-party Action Group world, each party can burn assets within its own group (ZIP-230, Rationale for including Burn fields inside OrchardZSA Action Groups).

Action and burn encodings. An `OrchardZSAAction` occupies 372 bytes: `cv` (32), `nullifier` (32), `rk` (32), `cmx` (32), `ephemeralKey` (32), `encCiphertext` (132), and `outCiphertext` (80) (ZIP-230, OrchardZSA Action Group Description). The 132-byte ciphertext is the memo-bundle-era layout: the note plaintext gains the 32-byte asset base and replaces the in-band 512-byte memo with a 32-byte memo key K^{memo} (ZIP-231), the memo chunks themselves moving to a transaction-level memo bundle (fields `nMemoChunks`, `pruned`, `vMemoChunks`, present iff `fAllPruned` = 0) whose pruned entries are replaced by their digests (ZIP-230, Transaction Format; §6.3). The v6 note plaintext is

$$\texttt{leadByte} \parallel \mathrm{LEBS2OSP}_{88}(d) \parallel \mathrm{I2LEOSP}_{64}(v) \parallel \mathsf{rseed} \parallel \mathsf{asset_base} \parallel K^{\mathsf{memo}},$$

with $\mathsf{asset_base} = \mathrm{LEBS2OSP}_{\ell_P}(\mathrm{repr}_P(\mathsf{AssetBase}))$, and the lead byte mandatory for all v6 note plaintexts (ZIP-230, note plaintext encoding). Upstream, the lead byte is the unassigned placeholder `{{LEADBYTE}}`: the original assignment `0x03` was taken by ZIP-2005, and no replacement is assigned (§6.6). For the same K^{memo} reason the Sapling `OutputDescriptionV6` shrinks `encCiphertext` to 100 bytes (ZIP-230, Sapling Output Description). An `AssetBurn` entry is 40 bytes: `AssetBase` (32, the encoding of $\mathsf{AssetBase}^{\mathsf{Orchard}}$) and `valueBurn` (`uint64`). ZIP-230 says the amount is non-zero and strictly less than MAX_BURN_VALUE, but ZIP-226's consensus rule and the pinned code permit equality (§4.4).

Issuance bundle. The issuance component carries five fields: `issuerLength` (`compactSize`), `issuer`, `nIssueActions`, `vIssueActions`, and `issueAuthSig`. The signature is present iff `nIssueActions` > 0; the first four fields are always present, and if `nIssueActions` = 0 then `issuerLength` MUST be 0 — an empty issuance bundle is exactly two zero `compactSize` values (ZIP-230, Transaction Format). An `IssueAction` carries `assetDescHash` (32 bytes), `nNotes` and `vNotes` at 115 bytes per `IssueNoteDescription`, and `flagsIssuance` with `finalize` in the LSB; `nNotes` may be zero, so a finalize-only action for an existing Asset is expressible; the new-Asset corner case remains undefined in ZIP 227 (§3.7). An `IssueNoteDescription` is `recipient` (43 bytes, the raw Orchard payment-address encoding), `value` (`uint64`), `rho` (32), `rseed` (32). The `IssueAuthSignature` is `sizeSighashInfo` and `sighashInfo`, then `sizeSignature` and a signature "preceded by a `0x00` byte indicating a BIP 340 signature" (ZIP-230, issuance field encodings) — the same `0x00` scheme byte that prefixes the issuer key encoding (ZIP-227, Issuance Authorization Signature Scheme).

Versioned signatures. The `SaplingSignature` and `OrchardSignature` types prepend a `compactSize`-prefixed `sighashInfo` to the 64-byte RedJubjub or RedPallas signature (ZIP-230, signature encodings). ZIP-246 defines the versioning:

$$\mathsf{sighashInfo} := [\ \mathsf{sighashVersion}\ ] \parallel \mathsf{associatedData},$$

where version 0 is the "commit to all effecting data" algorithm; for v6, only version 0 is defined, with empty `associatedData`, for all four of Transparent, Sapling, Orchard, and Issuance (ZIP-246, Sighash versioning). The machinery buys the ability to introduce — and, critically, to retire — sighash algorithms per transaction version without a new format.

Coinbase. For coinbase transactions the `enableSpendsOrchard` and `enableZSAs` bits MUST be zero (ZIP-230, consensus rules).

6.3 Digests: txid, sighash, and authorizing data (ZIP-246)

ZIP-246 extends the ZIP-244 digest tree — constructed for vanilla Orchard in the *Ironwood Guide*, §"Transaction digests and signatures" — from four top-level branches to six:

$$\mathsf{txid} := \text{BLAKE2b-256}(\texttt{ZcashTxHash_} \parallel \mathsf{branchid},\ d_{\mathrm{hdr}} \parallel d_{\mathrm{transp}} \parallel d_{\mathrm{Sap}} \parallel d_{\mathrm{Orch}} \parallel d_{\mathrm{issue}} \parallel d_{\mathrm{memo}}),$$

the nodes T.1–T.6 of the ZIP, each itself a personalized BLAKE2b-256 (ZIP-246, TxId Digest). New relative to ZIP-244 are the fee and burn fields in the header digest, the Sapling output digests, the entire Orchard Action-Group subtree, the issuance subtree, and the memo subtree. The header digest T.1 is

$$d_{\text{hdr}} := \text{BLAKE2b-256}(\texttt{ZTxIdHeadersHash},\ \mathsf{version} \,\|\, \mathsf{versionGroupId} \,\|\, \mathsf{branchid} \,\|\, \mathsf{lockTime} \,\|\, \mathsf{expiryHeight} \,\|\, \text{LE}_{64}(\mathsf{fee}) \,\|\, \text{LE}_{64}(\mathsf{burnAmount}))$$

(ZIP-246, T.1). The Orchard branch T.4 hashes the concatenated per-group digests and the balance, $d_{\text{Orch}} = \text{BLAKE2b-256}(\texttt{ZTxIdOrchardHash}, d_{\text{AGs}} \,\|\, \text{LE}_{64}(v^{\text{Orchard}}_{\text{balance}}))$, with a defined empty case; each group digest (T.4a) covers a compact digest (nf, cmx, `ephemeralKey`, `encCiphertext`), a non-compact digest (cv, rk, `outCiphertext`), `flagsOrchard`, `anchorOrchard`, `nAGExpiryHeight`, and a burn digest over the (AssetBase, valueBurn) tuples (ZIP-246, T.4a). The issuance branch T.5 hashes `issuerLength` ‖ `issuer` and a per-action digest (per action: `assetDescHash`, a per-note digest over (`recipient`, `value`, ρ, rseed), `flagsIssuance`), each level with a defined empty case (ZIP-246, T.5). The memo branch T.6 hashes a nonce and the concatenation of per-chunk digests; the v6 pruned memo bundle stores exactly these `memo_chunk_digest` / `memo_digest` values in place of pruned chunks, so pruning preserves the txid (ZIP-246, T.6; ZIP-230, Transaction Format). Table 4 collects the personalization strings; the implementation defines the same constants (§6.6).

Node	Personalization	Content
T.1	`ZTxIdHeadersHash`	header, incl. fee and burn amount
T.3b.i	`ZTxId6SOutC_Hash`	Sapling v6 outputs, compact
T.3b.ii	`ZTxId6SOutN_Hash`	Sapling v6 outputs, non-compact
T.4	`ZTxIdOrchardHash`	Action-Group digests, balance
T.4a	`ZTxIdOrcActGHash`	one Action Group
T.4a.i	`ZTxId6OActC_Hash`	nf, cmx, `ephemeralKey`, `encCiphertext`
T.4a.ii	`ZTxId6OActN_Hash`	cv, rk, `outCiphertext`
T.4a.vi	`ZTxIdOrcBurnHash`	burn tuples
T.5	`ZTxIdSAIssueHash`	issuer, issue-actions digest
—	`ZTxIdIssuActHash`	per action: desc. hash, notes, flags
—	`ZTxIdIAcNoteHash`	per note: recipient, value, ρ, rseed
T.6	`ZTxIdMemo___Hash`	nonce, chunks digest
—	`ZTxIdMemoCksHash`, `ZTxIdMemoCk_Hash`	memo chunk digests
A.1	`ZTxAuthTransHash`	scripts, now incl. per-input `sighashInfo`
A.3	`ZTxAuthOrchaHash`	group auth digests, `bindingSigOrchard`
A.3a	`ZTxAuthOrcAGHash`	`proofsOrchard`, spend-auth digest
A.3a.ii	`ZTxAuthOrSASHash`	`spendAuthSig` encodings
A.4	`ZTxAuthZSAOrHash`	`issueAuthSig` encoding

Table 4: ZIP-246 digest personalizations (ZIP-246, TxId Digest and Authorizing Data Commitment). Unnumbered rows are children whose labels the ZIP text assigns inconsistently; see the closing remark of this subsection.

Signature digest. The sighash reuses the tree with the transparent branch replaced by an input-specific variant:

$$\mathsf{sighash} := \text{BLAKE2b-256}(\texttt{ZcashTxHash_} \,\|\, \mathsf{branchid},\ d_{\text{hdr}} \,\|\, d^{S}_{\text{transp}} \,\|\, d_{\text{Sap}} \,\|\, d_{\text{Orch}} \,\|\, d_{\text{issue}} \,\|\, d_{\text{memo}}),$$

produced per transparent input, per Sapling input, per OrchardZSA Action, and — new in v6 — per Issue Action; the Sapling, Orchard, issuance, and memo branches are identical to their txid-side counterparts, so the issuance bundle and the memo bundle enter

every sighash (ZIP-246, Signature Digest). On this digest ZIP-227 hangs its authorization rule: the SigHash is the §4.10 SIGHASH transaction hash with the ZIP-226 modifications, computed with `SIGHASH_ALL`, and the issuance authorization signature MUST satisfy $\mathsf{IssueAuthSig.Validate}(\mathsf{ik}, \mathsf{SigHash}, \texttt{issueAuthSig}) = 1$, where ik is decoded from the `issuer` field and its encoding is required to begin with `0x00`, the BIP-340 scheme byte (ZIP-227, Specification: Consensus Rule Changes).

Authorizing-data commitment. The commitment to authorizing data follows the same pattern: A.1 covers transparent scripts together with the new per-input `TransparentSighashInfo`; A.2 keeps the ZIP-244 structure with field encodings altered by `sighashInfo`; A.3 commits to the per-group proofs and spend-authorization signatures and to `bindingSigOrchard`; and a new A.4 commits to the `issueAuthSig` field encoding, with a defined empty case (ZIP-246, Authorizing Data Commitment).

Remark 6.1 (The reference text is itself unfinished). ZIP-246 is internally incomplete even as withdrawn text: the authorizing-data section carries "TODO: Update the Authorizing Data Commitment below for the `sighashInfo` changes to the tx format," Rationale and Reference implementation are TBD, and the subtree labels are inconsistent — `issue_actions_digest` is listed as T.5c but headed T.5a with children T.5a.i, and the burn-digest children are labelled T.4b.i/T.4b.ii under node T.4a.vi (`zip-0246.rst`). Anyone respecifying the format inherits these defects as open editorial work; the implementation, not the ZIP, is currently the more precise artifact.

6.4 Transaction-level consensus rules

One Action Group. Although the format is a vector of Action Groups, ZIP-227 collapses it for OrchardZSA: per transaction, `nActionGroupsOrchard` MUST be 0 or 1, and `nAGExpiryHeight` MUST be 0 (ZIP-227, Specification: Consensus Rule Changes). The vector form is used by draft ZIP-228's proposed ZSA swaps, which would relax the one-group rule for swap bundles; it is not active in the core ZIP-227 rules.

Issuance requires an Action Group. A transaction with an issuance bundle MUST also contain at least one OrchardZSA Action Group (ZIP-227, Specification: Consensus Rule Changes). The rule is load-bearing twice over. First, issue-note ρ derivation consumes the first Action's nullifier: each issued note's ρ is computed by the function ZIP-227 names $\mathsf{DeriveIssuedRho}$ — constructed in §3.3 — keyed by $\mathsf{nf}_{0,0}$, the nullifier of the input note of the first Action in the first Action Group, over the Issue Action and note indices (ZIP-227, Computation of ρ). The indices domain-separate notes within the transaction, while the nullifier-keyed PRF makes collisions across transactions negligible under its pseudorandomness assumption; it does not give literal global uniqueness. Second, without a spend in the transaction the issuance-only SIGHASH would be replayable; the mandatory Action Group ties the issuance signature to a nullifier that consensus consumes (ZIP-227, Rationale).

Global issuance state. Nodes maintain a map issued_assets from asset bases to triples $(\mathsf{balance}, \mathsf{final}, \mathsf{note_ref})$, chained empty map → per-block → per-transaction. Burns decrement balance, requiring $\mathsf{balance} \geq v$, *before* the transaction's issuance bundle is processed; issuance requires $\mathsf{final} \neq 1$, increments balance subject to the overflow bound $\mathsf{MAX_ISSUE} = 2^{64} - 1$, and sets final when `finalize` $= 1$. The first issuance of an asset MUST include a *reference note*: value 0, paid to the default address of the all-zero Orchard spending key (ZIP-227, Global Issuance State and Reference Notes). The 43-byte raw encoding of that address is fixed in the ZIP and reproduced as a constant in the implementation (§6.6).

Burn rules and the extended balance equation. For every $(\mathsf{AssetBase}, v)$ in `assetBurn`: $\mathsf{AssetBase} \neq V^{\mathsf{Orchard}}$ (ZEC cannot be burned this way), $0 < v \leq \mathsf{MAX_BURN_VALUE}$, and no two entries share an AssetBase; the bound $\mathsf{MAX_BURN_VALUE} := 2^{63} - 1$ is chosen for compatibility with signed 64-bit value-balance fields (ZIP-226, Burn Mechanism). Burns enter value balance through the binding key. Where vanilla Orchard computes bvk by subtracting the committed declared balance from the sum of net value commitments (the *Ironwood Guide*, §"The binding signature"), ZIP-226 subtracts one zero-randomness commitment per burn entry as well:

$$\begin{aligned}\mathsf{bvk} := \Big(\sum_{i=1}^{n} \mathsf{cv}^{\mathsf{net},i}\Big) &- \mathsf{ValueCommit}_0^{\mathsf{OrchardZSA}}(V^{\mathsf{Orchard}}, v_{\mathsf{balance}}^{\mathsf{Orchard}}) \\ &- \sum_{(\mathsf{AssetBase},\, v) \,\in\, \mathtt{assetBurn}} \mathsf{ValueCommit}_0^{\mathsf{OrchardZSA}}(\mathsf{AssetBase}, v),\end{aligned}$$

and the prover signs the SIGHASH with $\mathsf{bsk} = \sum_{i=1}^{n} \mathsf{rcv}_i$ (ZIP-226, Burn Mechanism). Under the unknown-discrete-log assumptions made explicit in §4.3, the signature computationally enforces that every non-ZEC asset nets to exactly its declared burns.

Commitment-tree order. Note commitments enter the global tree in a fixed order: first, per Action Group, the commitments of the new OrchardZSA notes of each Action; then, per Issue Action, the issue-note commitments (ZIP-227, Specification: Consensus Rule Changes). Wallets and nodes must agree on this order to agree on note positions.

6.5 The experimental fee policy

The fee treatment chains through three documents: ZIP-226 (Transaction Fees) defers to ZIP-317 "as described in ZIP 227"; ZIP-227 (Changes to ZIP 317) states that the conventional fee is altered for the presence of issuance actions and the creation of new Custom Assets, pointing at ZIP-317's Fee calculation section. Fees remain payable in ZEC only (ZIP-227, rationale). All variants share the conventional-fee form documented in the *Wallet Guide*, §"Fees (ZIP-317)":

$$\mathsf{conventionalFee} := \mathsf{marginalFee} \cdot \max(\mathsf{graceActions},\ \mathsf{logicalActions}),$$

with $\mathsf{marginalFee} = 5000$ zatoshis and $\mathsf{graceActions} = 2$ (ZIP-317, Fee calculation).

ZIP 317 contains no ZSA fee contributions. The experimental fork adds a charge for creating an asset, to discourage permanent state growth from unused assets: the parameter $\mathsf{creationCost} = 100$ logical actions, and two contributions summed into logicalActions alongside the base terms of the deployed rule:

$$\begin{aligned}\mathsf{contribution}_{\mathsf{ZSAIssuance}} &:= \mathsf{nZSAIssueNotes}, \\ \mathsf{contribution}_{\mathsf{ZSACreation}} &:= \mathsf{creationCost} \cdot \mathsf{nAssetCreations},\end{aligned}$$

where nZSAIssueNotes counts issue-note outputs and nAssetCreations counts Custom Assets newly added to the global issuance state (fork ZIP-317, Fee calculation). The rationale is a one-time charge for state growth: each new asset adds an issued_assets row tracked in perpetuity, while subsequent issuance, finalization, and burn only mutate the row. The implementation uses the same contributions. As the *Wallet Guide* states for the upstream rule (§"The conventional fee"), paying the conventional fee is not a consensus requirement in any ZIP-317 variant. Classification: the ZSA fee rule is *fork-specified*; whether it returns upstream as a ZIP-317 revision, as Ironwood's did, is undecided.

6.6 Implementation status at the pinned commits

Serialization. The experimental reader parses the header, one `vSighashInfo` per transparent input (required to equal the version-0 encoding), the Sapling v6 bundle, the Orchard v6 bundle, and the issuance bundle (`zcash_primitives`, `transaction/mod.rs`). The Orchard reader rejects more than one Action Group ("A V6 transaction must contain at most one action group"), requires at least one Action, and rejects a nonzero `nAGExpiryHeight`; the writer emits exactly one group with `nAGExpiryHeight` = 0 (`transaction/components/orchard.rs`) — the ZIP-227 consensus rules hard-coded at the serialization layer. Issuance-bundle serialization matches ZIP-230, including the two-zero-`compactSize` empty bundle and rejection of the inconsistent issuer/action combinations. The sighash implementation extends the v5 hash with the issuance digest for ZSA-capable versions, with sighash-info constants `ORCHARD_SIGHASH_INFO_V0` = `ISSUE_SIGHASH_INFO_V0` = `[0x00]` (`sighash_v6.rs`, `sighash_versioning.rs`). The orchard fork defines the ZIP-246 personalizations of Table 4, asserting the `0x00` BIP-340 byte when hashing the 65-byte issuance signature encoding.

Verification code. Function `derive_bvk_raw` implements the extended balance equation of §6.4 verbatim; `validate_bundle_burn` enforces the ZIP-226 burn rules — non-ZEC asset, non-zero amount, amount $\leq 2^{63} - 1$, no duplicates — plus presence in and sufficient supply under the global issuance state; and `verify_issue_bundle` checks the BIP-340 signature over the 32-byte sighash, the per-note ρ against DeriveIssuedRho, non-finalization, u64-checked supply addition, and the reference note on first issuance, whose 43-byte `RAW_REFERENCE_RECIPIENT` constant equals the array in ZIP-227. The `0x84` domain separator for issued-note ρ lives in the QED-it `zcash_spec` fork (rev `d5e8426`). Proof sizes record the circuit cost: the ZSA aggregated proof has base size 2848 bytes and 2272 bytes per Action, against a vanilla base of 2720 (`primitives/orchard_primitives_zsa.rs`, `orchard_primitives_vanilla.rs`).

Divergences from the withdrawn ZIP text. Four are material. The first three block byte-compatibility with ZIP-230/246 as written; the fourth changes private witness construction but is not enforced by the circuit.

1. *No explicit fee field.* The v6 reader serializes no fee, and `zip233Amount` only behind the `zip-233` cargo feature; the header txid digest correspondingly omits ZIP-246's T.1f fee field, with the comment "TODO: Factor this out ... when implementing ZIP 246 in full" (`transaction/mod.rs`, `txid.rs`).
2. *No memo bundles.* The forks implement the pre-ZIP-231 layout: OrchardZSA notes carry a full 512-byte memo, giving a 612-byte `encCiphertext` with an 84-byte compact prefix (52 vanilla bytes plus the 32-byte asset base), and Sapling v6 outputs are read in the v5 580-byte format (`primitives/zcash_note_encryption_domain.rs`, `transaction/components/sapling.rs`). No memo fields and no T.6 are implemented; instead the action-group txid digest inserts a non-spec memos digest (personalization `ZTxIdOrcActMHash`) between the compact and non-compact digests and hashes only the 84-byte compact slice into the compact digest, with TODOs "Remove once new Memo Bundles are implemented (ZIP-231)" (`primitives/orchard_primitives_zsa.rs`).
3. *Lead byte.* The implementation hard-codes `NOTE_VERSION_BYTE_V3` = `0x03`; upstream ZIP-230 withdrew that claim after ZIP-2005 took `0x03`, leaving the ZSA lead byte an unassigned placeholder. The fork's choice predates upstream ZIP-2005 assigning `0x03` to Ironwood notes and directing future proposals to pick a new value; the collision is fork-only context, with no format ambiguity on Zcash mainnet. Reassignment is a hard prerequisite for any deployment: the lead byte selects the note-plaintext parser.

4. *Split-note* ψ^{nf}. Upstream ZIP-226 now derives the split-note nullifier randomizer as $\psi^{\mathsf{nf}} = \mathsf{ToBase}(\mathsf{PRFexpand}_{\mathsf{rseed}_{\mathsf{nf}}}(\mathtt{0x0A} \| \rho^{\mathsf{old}}))$ with $\mathsf{rseed}_{\mathsf{nf}}$ sampled uniformly at random, the first-byte domain `0x0A` "reserved by ZIP 2005 for this use" (ZIP-226, split-note nullifier); the fork at `cf801a5d` realizes the older fork text (uniformly random ψ^{nf}) via a separate `rseed_split_note` under the ordinary ψ domain `0x09` (`note.rs`, `circuit.rs`).

6.7 Wallet integration

The withdrawn ZIP-230 and ZIP-226 design imposes two obligations. First, all wallets SHOULD switch to sending only its v6 transactions once permitted: a v5 transaction reveals that all its Orchard notes are ZEC, and the design's v6 OrchardZSA output format incorporates the quantum-recoverability changes (ZIP-230, Implications for Wallets; ZIP-226, Backwards Compatibility with ZEC Notes). This is a claim about the withdrawn carrier, not the different v6 format now assigned to ZIP-229. Second, wallets MUST support parsing and trial-decrypting v6 outputs — the new lead byte and the memo-bundle changes — by activation, or rescan once support lands: funds sent to an un-upgraded wallet are temporarily inaccessible, because the new lead byte selects different rcm/ψ derivations and a v5-decryption wallet rejects the note (ZIP-230, Implications for Wallets). Payment requests have a specified extension path: ZIP-321's `req-asset` parameter carries an Asset Identifier and amount (the *Wallet Guide*, §"The URI grammar").

Multi-party and hardware signing is the open end. No PCZT structure exists for OrchardZSA bundles or issuance bundles: the librustzcash fork's transaction extractor panics with "PCZT support for ZSA is not implemented" when handed an OrchardZSA bundle, the orchard fork's PCZT module is written against the vanilla flavor only, and the released PCZT v2 targets ZIP-229's Ironwood v6, not ZSA (the *Wallet Guide*, §"Partially created transactions (PCZT)" and §"PCZT v2: deferred anchors and the Ironwood bundle"). Classification: *open* — hardware-wallet and multi-party signing for ZSAs is unspecified in any draft.

6.8 Outlook

Between the pinned implementation and any deployment stand three cross-cutting respecification problems.

1. *A carrier.* Transaction version 6 now belongs to NU6.3's Ironwood format (ZIP-229), whose non-requirements exclude the ZSA machinery (§6.1). The designated successor for what ZIP-230/246 deferred — an extensible transaction format — is ZIP-248, now a merged Draft for V7. Its registry lists OrchardZSA with a provisional variant and ZSA issuance without an allocated bundle type, under "Potential Future Bundle Types"; it does not specify either bundle's encoding or digest. The ZSA-specific transaction format therefore remains incomplete: before a future ZSA deployment, the byte format of §6.2 must be mapped or respecified for V7, and §6.3 re-anchored to the resulting digest rules.

2. *A base protocol.* ZIP 226 assumes ZIP 2005's recoverability changes, but the experimental implementation does not. No ZIP specifies whether OrchardZSA targets the restricted Orchard pool or is rebased onto Ironwood. Classification: *open*.

3. *Reconvergence of spec and code.* The divergence list of §6.6 — fee field, memo bundles, lead byte, ψ^{nf} — plus the orphaned fee treatment of §6.5 must land on one side or the other: either the respecified ZIPs adopt what the code does, or the code moves. Each divergence is identified, but resolving it still requires protocol choices and implementation work.

The summary judgment this volume can defend: a ZSA transaction layer is designed and implemented in the pinned forks, but the deployable protocol is *currently unspecified* in the sense this series requires — there is no complete merged carrier specification under which a mainnet

node could implement it. The byte-level facts of this section are fixed by withdrawn ZIP text and pinned fork code, and they will remain the diff base for any respecification; the remaining open items are enumerated above and in the scope section.

Part IX
FROST Guide: Threshold RedPallas: FROST and its Zcash integration surfaces

Abstract

Assuming Schnorr signatures from the *Crypto Guide*, RedPallas from the *Ironwood Guide*, PCZT from the *Wallet Guide*, and the finalizer model from the *Crosslink Guide*, this volume of *The Zcash Arboretum* documents threshold signing for Zcash. It constructs FROST as specified in RFC 9591, then its re-randomised form for Zcash spend authorisation and recoverability custody. The re-randomised Zcash stack is outside the completed cryptographic-library audits identified here; each design-stage claim is classified by the rigour of its source.

1 Scope, sources, and deployment surfaces

This volume documents threshold custody of Zcash spend authority: how t of n parties jointly produce, without reconstructing the signing key during signing, a signature that consensus accepts as an ordinary RedPallas spend-authorisation signature. Three layers carry the construction. The base scheme is FROST, the two-round threshold Schnorr protocol of Komlo and Goldberg, specified in RFC 9591 (§2). The Zcash layer is Re-Randomized FROST (Gouvêa–Komlo, IACR ePrint 2024/436), which shifts every share by a common randomizer so that the aggregate signature verifies under the per-Action randomised key; ZIP 312 specifies it for Zcash, and the `frost-rerandomized` and `reddsa` crates ship it (§3). The deployment picture — the shipped stack, the proposed uses, and the gaps — closes the volume (§4).

Three custody surfaces anchor the treatment, one per rigour bin (§1.3): spend-authorisation custody for Orchard, where the construction is specified and shipped; custody of the ZIP-2005 quantum spending key `qsk`, where a Proposed consensus ZIP constrains a FROST ceremony whose recovery protocol it defers; and Crosslink finalizer keys, for which no threshold design was identified in the cited Crosslink corpus. The surfaces and their bins are fixed in §1.4.

1.1 Scope

The object under study is a t-of-n signing protocol whose output has the same format and verification rule as single-party spend authorisation. ZIP 312 states the requirements directly: the signatures must verify as Sapling or Orchard spend-authorisation signatures, and should meet the protocol specification's "Signature with Re-Randomizable Keys" criteria. Plain FROST fails the first clause — it signs for a fixed group key, and consensus verifies only per-Action randomised keys — and closing that gap is the volume's central construction (§3).

The volume develops RedPallas only, per its title. ZIP 312 itself defines two ciphersuites — FROST(Jubjub, BLAKE2b-512) for Sapling RedJubjub and FROST(Pallas, BLAKE2b-512) for Orchard RedPallas — and we cite the Jubjub suite where the ZIP fixes both at once, but develop only the Pallas suite (§3.5).

ZIP 312's stated non-requirements bound the scope from below: it does not remove the Coordinator role; it defines no key-generation procedure (out of scope there, exactly as in RFC 9591); and it provides no network privacy. Key generation nevertheless appears twice in this volume, both times forced by sources: once as published with the base scheme (§2.2), and once as the unspecified-but-shipped DKG surface of the `reddsa` crate, binned in §1.3.

One altitude note governs every deployment claim. Availability is library-level, not consensus-level: no consensus rule names FROST, and none needed to change. The verifier accepts the aggregate because it is a valid RedPallas signature; as ZIP 312's rationale puts it, the method by which the commitment R was generated is irrelevant to the verifier.

1.2 The source ladder

Table 1 pins the primary sources used for the full review. Repository heads were rechecked on 2026-09-05: the FROST-facing protocol text is unchanged at `zcash/zips e753a6a3`, and the algorithms and Zcash ciphersuite behaviour described below remain unchanged at `ZcashFoundation/frost 0966bd1` and `ZcashFoundation/reddsa 63c6ac0`. Local copies of the papers and RFC are archived with the volume's drafts.

Artefact	Pin / date	Evidence status
RFC 9591 (IRTF CFRG)	June 2024	Informational; CFRG consensus
ePrint 2024/436 (Gouvêa–Komlo)	recv. 2024-03-13	preprint; no venue
ZIP 312	`6961098410`, 2026-07-09	Draft (Wallet category)
ZIP 2005	`6961098410`, 2026-07-09	Proposed (Consensus category)
`ZcashFoundation/frost`	`2016e44ba4`, 2026-04-23	shipped code; workspace 3.0.0
`ZcashFoundation/reddsa`	`3792daa95e`, 2026-05-22	shipped code; crate 0.5.2
`ZcashFoundation/frost-tools`	`06c0dbdb`, 2026-08-12	demos; not for production
ePrint 2020/852; 2022/833	SAC 2020; CRYPTO 2022	peer-reviewed antecedents

Table 1: Primary FROST sources used for the full review. Current heads rechecked 2026-09-05.

The RFC. RFC 9591, "The Flexible Round-Optimized Schnorr Threshold (FROST) Protocol for Two-Round Schnorr Signatures" (D. Connolly, Zcash Foundation; C. Komlo, University of Waterloo and Zcash Foundation; I. Goldberg, University of Waterloo; C. A. Wood, Cloudflare; IRTF CFRG stream, Category Informational, June 2024), records CFRG consensus on the base two-round protocol and leaves key generation out of scope. It explicitly says that it is not an Internet Standard and that IRTF results might not be suitable for deployment. Two of its four authors are Zcash Foundation, so the RFC and Zcash implementation share provenance. Its deltas against the SAC 2020 paper are catalogued in §2.8.

The paper. "Re-Randomized FROST", by Gouvêa and Komlo (both Zcash Foundation), is IACR ePrint 2024/436 (`https://eprint.iacr.org/2024/436`): received 2024-03-13, approved 2024-03-15, publication info "Preprint", no peer-reviewed venue as of 2026-09-05. We read it in full and work from it throughout §3. Its headline Theorem 1 claims unforgeability in the random-oracle model under AOMDL. That game restricts oracle queries to declared algebraic combinations, unlike the OMDL game used for plain FROST in BCKMTZ. The printed reduction has material gaps documented in §3.3; this volume therefore does not treat that claim as an established security theorem.

The ZIP. ZIP 312, "FROST for Spend Authorization Multisignatures" (owners Conrado Gouvêa, Chelsea Komlo, and Deirdre Connolly; Status Draft, Category Wallet; Discussions-To zcash/zips#382, pull request zcash/zips#662), is the Zcash specification of the construction.[5] It is visibly unfinished — its Created field still reads `2022-08-dd`, day unfilled — so we cite it as a moving object, by repository commit (`zcash/zips @ 6961098410`, 2026-07-09) and Draft status, never as a stable specification. Its threat model trusts the Coordinator and every key-share holder with transaction privacy and unlinkability, but its security requirement says that a rogue Coordinator must not be able to create a signed transaction without the approval of `MIN_PARTICIPANTS` participants. What it specifies — the unlinkability requirement, the randomizer flow, and the two ciphersuites — is developed in §3.4 and §3.5.

[5] The FROST ZIP is ZIP 312 alone. ZIP 313, its numeric neighbor, is "Reduce Conventional Transaction Fee to 1000 zatoshis" — Status Obsolete, obsoleted by ZIP 317 — and has nothing to do with FROST.

The crates. Repository `ZcashFoundation/frost` (@ `2016e44ba4`, 2026-04-23) is a Rust workspace at version 3.0.0 (`rust-version` 1.81): crate `frost-core` carries the ciphersuite-generic protocol; five ciphersuite crates instantiate the RFC (ed25519, ed448, P-256, ristretto255, and secp256k1), while the separate `frost-secp256k1-tr` crate implements a Taproot-oriented suite; `frost-rerandomized` carries the re-randomised layer, and `gencode` generates the per-suite code. The repository also vendors an audit report dated 2021-03-23 (`zcash-frost-audit-report-20210323.pdf`) and the source of the ZF FROST book (published at `frost.zfnd.org`), whose "FROST with Zcash" chapters (zcash-devtool demo, Ywallet demo, FROST server) and serialisation-format page — the page ZIP 312 cites for `encode_signing_package` — feed §4. Repository `ZcashFoundation/reddsa` (@ `3792daa95e`, 2026-05-22; crate version 0.5.2) holds the Zcash ciphersuites: its `frost` feature enables `frost-rerandomized` 3.0.0, and `src/frost/redpallas.rs` defines FROST(Pallas, BLAKE2b-512) as struct `PallasBlake2b512`, ZIP 312's reference implementation surface (§3.6). Crate `reddsa` also ships machinery ZIP 312 does not specify — DKG hash personalisations and even-Y normalisation hooks — binned in §1.3. Its README marks the `frost` feature experimental while ZIP 312 remains Draft and explicitly excludes that feature from the crate's SemVer and MSRV guarantees.

The antecedent papers. FROST itself is Komlo–Goldberg, IACR ePrint 2020/852 (SAC 2020); the security analysis the re-randomised extraction argument extends is Bellare–Crites–Komlo–Maller–Tessaro–Zhu, "Better than Advertised Security for Non-Interactive Threshold Signatures" (CRYPTO 2022; ePrint 2022/833). They are the paper's references [10] and [1] respectively; §2 works from both.

1.3 The three-bin rule

The series rule is binary for implementation behaviour and ternary for design. A claim about implementation behaviour cites the implementation — crate and file — and the specification section it realises; a claim about a design-stage protocol is classified *specified*, *designed but unspecified*, or an *open problem*, and the bin stays visible in the prose. This volume sits astride the boundary: one surface is shipped code, one is a consensus proposal, one is an acknowledged gap. Applied once, here, for the whole volume:

1. Re-Randomized FROST for spend-authorisation custody is *specified*: RFC 9591 published June 2024, ZIP 312 in Draft, ePrint 2024/436, and shipped code — `frost-rerandomized` 3.0.0 and `reddsa` 0.5.2. The latter's FROST feature is explicitly experimental. Availability is library-level, not consensus-level (§1.1).

2. Custody of the ZIP-2005 quantum spending key is specified at the ZIP level (Status Proposed, Category Consensus), but the Recovery Protocol it exists to serve is "to be introduced, potentially, in a future upgrade" — *designed but unspecified*.

3. Crosslink finalizer threshold custody is an *open problem*: the *Crosslink Guide*'s own classification, which we adopt.

One surface refuses the clean split, and we bin it here once. No ZIP specifies FROST key generation: ZIP 312 declares it out of scope (as does RFC 9591), and ZIP 271's May-2025 note confirms the gap — ZIP 312's missing key-generation text is its stated reason not to couple disbursement of the *lockbox* (the in-protocol reserve that has accrued a portion of each block subsidy since NU6, per ZIP 2001) to that specification. Yet crate `reddsa` ships DKG support: hash personalisations `FROST_RedPallasD` (HDKG) and `FROST_RedPallasI` (HID), beyond ZIP 312's H1–H5 and HR, plus even-Y normalisation hooks (`post_generate/post_dkg`, applying `into_even_y` to secret shares and the public key package) so that the group verifying key meets the encoding constraint of Orchard's spend validating key. Deployed code with no specification above it: we

classify the DKG surface *designed but unspecified* and cite the crate at commit `3792daa95e` wherever we rely on it.

1.4 The three deployment surfaces

Spend-authorisation custody (specified). The product ZIP 312 promises is signatures "indistinguishable from RedPallas Orchard Spend Authorization Signatures". The single-party object being imitated is settled in the *Ironwood Guide*: §"Randomised validating keys" constructs $\mathsf{rsk} := \mathsf{ask} + \alpha$ and $\mathsf{rk} := \mathsf{ak} + [\alpha]\, G^{\mathsf{SpendAuth}} = [\mathsf{rsk}]\, G^{\mathsf{SpendAuth}}$, with GenRandom hashing 80 uniform bytes through $H^{\circledast}$, and §"RedPallas unforgeability" proves EUF-CMA from discrete log in the random-oracle model. Per the layering rule we cite both and re-prove neither; this volume's work begins where the single signer ends, at splitting ask into shares that a signing coalition need not reassemble (§3). The wallet-side integration path is likewise on the record: the *Wallet Guide*, §"Partially created transactions (PCZT)", records ZIP 374's motivation (iii), threshold signing (for example FROST, RFC 9591) — the multiparty transaction format FROST signing rides in §4.3.

Custody of the quantum spending key (designed but unspecified). ZIP 2005, "Ironwood Quantum Recoverability" (owners Daira-Emma Hopwood and Jack Grigg; Status Proposed, Category Consensus; created 2025-03-31), writes FROST into its requirements twice: the design should be fully compatible with FROST threshold multisignatures (citing ZIP 312), and should lose no pre-quantum security when FROST and/or hardware wallets are used. Its "Usage with FROST" section then constrains the ceremony: the point ak is derived jointly, via the FROST DKG, from shares of ask; the participants must privately agree on a value sk and use it with $\mathsf{use_qsk} = \mathsf{true}$ to derive nk, qsk, qk, and $\mathsf{rivk_ext} = \mathsf{H}^{\mathsf{rivk_ext}}_{\mathsf{qk}}(\mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}), \mathsf{nk})$; all participants must obtain the same extracted key $\mathsf{Extract}_{\mathbb{P}}(\mathsf{ak})$, nk, and rivk_ext. Subject to the specified derivations, the ZIP says that this also implies equal qsk and qk under collision resistance of H^{qk} and $\mathsf{H}^{\mathsf{rivk_ext}}$. Each participant must hold qsk at least as securely as ask. The outlined Recovery Protocol would require a RedDSA signature under ak — preserving t-of-n for as long as discrete log on Pallas stands — together with a proof $\mathsf{SoK}^{\mathsf{qsk}}$ of knowledge of a qsk with $\mathsf{H}^{\mathsf{qk}}(\mathsf{qsk}) = \mathsf{qk}$. The asymmetry is the point to retain: the key qsk is held by *every* participant, so a quantum adversary with qsk and the ability to solve discrete logarithms on Pallas no longer faces a threshold; the ZIP accordingly recommends moving funds to a fully post-quantum threshold protocol once one is available. The deferred Recovery Protocol is what parks this surface in the middle bin; the full treatment is §4.7.

Crosslink finalizer keys (open problem). The *Crosslink Guide*, §"Finalizer keys", records the prototype state: one ed25519 key per finalizer (`ed25519-zebra`), a 76-byte vote format signed by appending a 64-byte ed25519 signature over those 76 bytes, and — by negative search over the four Crosslink repositories (tfl-book, crosslink-protocol-guide, crosslink-spec, zebra-crosslink), dated 2026-07-15 there — no FROST or threshold-signing design for finalizer keys sourced anywhere in the corpus. That guide's classification bins threshold custody, key rotation, revocation, and initial-roster formation as open problems, and observes that FROST's presence in the PCZT flow makes the absence "a visible gap rather than an exotic ask". We adopt the classification and return to the surface in §4.8.

Adjacent surfaces, cited but not developed. Three further ZIPs touch FROST and mark how wide the surface is growing. ZIP 271 ("Dev Fund Extension and One-Time Disbursement", Proposed) anticipates spending lockbox one-time-disbursement chunks "to a FROST address", and its May-2025 note that ZIP 312 does not specify key generation is its stated reason not to couple the two specifications. ZIP 326 ("NU6.3 Consequences for Wallets", Draft, created 2026-06-30)

cites ZIP 2005's "Usage with FROST" section. ZIP 374 ("Partially Created Zcash Transaction Format", Draft, created 2024-12-09) lists FROST per RFC 9591 under its multiparty-transactions motivation. Each appears in this volume exactly where it bears (§4); none is developed as a surface of its own.

2 FROST

The base object of this volume is the threshold Schnorr scheme FROST (*Flexible Round-Optimized Schnorr Threshold* signatures) of Komlo and Goldberg, published at SAC 2020 and maintained as IACR ePrint 2020/852; we work from the final revision of 22 December 2020. The scheme lets any t of n parties, each holding a Shamir share of a signing key s, produce a signature (R, z) that verifies under the ordinary single-party Schnorr equation for the group public key $Y = g^s$. The signature carries no threshold marker, and a verifier applies the same equation as it would to a single-signer signature. That compatibility is what lets the scheme use Zcash's existing RedDSA validators: the consensus rules do not observe the threshold ceremony. The deployment surfaces this volume targets are spend-authorisation custody over RedPallas (the *Ironwood Guide*, §"The RedPallas signature scheme"), custody of the ZIP-2005 quantum spending key (whose "Usage with FROST" section constrains the key generation), and Crosslink finalizer keys (the *Crosslink Guide*, §"Finalizer keys").

Throughout, $\mathbb{G}$ is a group of prime order q with generator g, and H, H_1, H_2 are hash functions with outputs in $\mathbb{Z}_q^*$, modelled as random oracles. The paper fixes the single-party scheme it must match — the Schnorr signature scheme of the *Crypto Guide*, §"From identification to signature via the Fiat–Shamir transform" — restated here in the paper's multiplicative notation: a Schnorr signature over m under secret key $s \in \mathbb{Z}_q$ and public key $Y = g^s \in \mathbb{G}$ is produced by sampling a nonce $k \xleftarrow{\$} \mathbb{Z}_q$, setting $R = g^k$, $c = H(R, Y, m)$, $z = k + s \cdot c$, and outputting $\sigma = (R, z)$; verification recomputes c and accepts iff $R = g^z \cdot Y^{-c}$. FROST's whole design problem is to compute the pair (R, z) jointly, without reconstructing the nonce or signing key during signing and without opening the concurrency attacks described in §2.4.

2.1 Shamir sharing and share conversion

Two secret-sharing mechanisms cooperate in FROST, and keeping them distinct is the key to reading the signing equation.

Shamir sharing (the long-lived key). A dealer — or, in §2.2, every participant acting as a dealer simultaneously — selects $t-1$ random coefficients $a_1, \dots, a_{t-1}$ and forms the polynomial of degree at most $t-1$

$$f(x) = s + \sum_{i=1}^{t-1} a_i x^i, \qquad f(0) = s,$$

giving participant P_i the share $(i, f(i))$, with distinct nonzero identifiers in $\mathbb{Z}_q$ and $1 \le t \le n < q$. Coefficients are sampled independently and uniformly in $\mathbb{Z}_q$. Any t shares determine f by Lagrange interpolation and hence $s = f(0)$; fewer than t reveal nothing. For a signing set $S = \{p_1, \dots, p_\alpha\}$ of α participant identifiers ($t \le \alpha \le n$), the interpolation weight of participant i at zero is the Lagrange coefficient

$$\lambda_i = \prod_{j=1, j \ne i}^{\alpha} \frac{p_j}{p_j - p_i},$$

computable from the identifiers alone — the coalition can be chosen *after* shares are dealt, which is what makes FROST's "any t of n, decided at signing time" flexibility possible.

Additive sharing and conversion (the per-signature nonce). A sum $s = \sum_i s_i$ with each summand chosen locally is an additive sharing: it needs no dealer and no interaction. The Cramer–Damgård–Ishai share conversion observation, used by FROST in its simplest case, is that additive shares convert locally to Shamir form: if $s = \sum_{i \in S} s_i$, then the values s_i/λ_i are Shamir shares of the same s (evaluations of a degree-$(\alpha - 1)$ polynomial interpolating them). FROST uses exactly this to make the group nonce k: each signer contributes an additively-shared summand non-interactively, and the λ_i weights in the signing equation perform the conversion so that the response aggregates by plain summation. No DKG runs at signing time.

2.2 Key generation: Pedersen's DKG with a knowledge proof

FROST generates its long-lived shares with a variant of Pedersen's DKG — n parallel executions of Feldman's verifiable secret sharing, one with each participant as dealer — hardened with a proof of knowledge of each dealer's free coefficient. The paper's Figure 1, reproduced:

Round 1.

1. Every participant P_i samples t random values $(a_{i0}, \dots, a_{i(t-1)}) \xleftarrow{\$} \mathbb{Z}_q$ and defines the polynomial $f_i(x) = \sum_{j=0}^{t-1} a_{ij} x^j$ of degree at most $t - 1$.

2. Every P_i computes a proof of knowledge of a_{i0}: with $k \xleftarrow{\$} \mathbb{Z}_q$, $R_i = g^k$, $c_i = H(i, \Phi, g^{a_{i0}}, R_i)$, $\mu_i = k + a_{i0} \cdot c_i$, set $\sigma_i = (R_i, \mu_i)$, where Φ is a context string preventing replay across ceremonies.

3. Every P_i computes the public commitment $\vec{C}_i = \langle \phi_{i0}, \dots, \phi_{i(t-1)} \rangle$ with $\phi_{ij} = g^{a_{ij}}$.

4. Every P_i broadcasts $\vec{C}_i, \sigma_i$.

5. On receiving $\vec{C}_\ell, \sigma_\ell$, participant P_i verifies $R_\ell \stackrel{?}{=} g^{\mu_\ell} \cdot \phi_{\ell 0}^{-c_\ell}$ with $c_\ell = H(\ell, \Phi, \phi_{\ell 0}, R_\ell)$, *aborting on failure*; on success all σ_ℓ are deleted.

Round 2.

1. Each P_i securely sends each other participant P_ℓ the share $(\ell, f_i(\ell))$, then deletes f_i and every share except its own $(i, f_i(i))$.

2. Each P_i verifies each received share against the dealer's commitment:

$$g^{f_\ell(i)} \stackrel{?}{=} \prod_{k=0}^{t-1} \phi_{\ell k}^{i^k \bmod q},$$

   aborting if the check fails.

3. Each P_i computes its long-lived signing share $s_i = \sum_{\ell=1}^{n} f_\ell(i)$, stores it securely, and deletes the $f_\ell(i)$.

4. Each P_i computes its public verification share $Y_i = g^{s_i}$ and the group public key $Y = \prod_{j=1}^{n} \phi_{j0}$.

Three design decisions deserve comment. First, the *proof of knowledge of a_{i0}* is the delta over plain Pedersen DKG: it addresses rogue-key attacks in the $t \geq n/2$ regime, where a participant could otherwise choose its commitment as a function of others' to bias the aggregate key (the fix was suggested to the authors by Shlomovits and Turkel; the paper's changelog dates it to the 2020-07-08 revision). Second, the *complaint handling* is abort-only: where Gennaro et al. add

a complaint/disqualification phase to make the DKG robust, FROST aborts on any failed share check and expects the cause to be investigated out of band; the paper notes that implementations wanting robustness can substitute the Gennaro et al. construction. Third, a *consistent view* of the broadcast commitments $\vec{C}_i$ across participants is assumed, not provided — the paper defers to whatever broadcast mechanism the deployment supplies. Each participant ends holding (i, s_i), with $Y_i = g^{s_i}$ public for share verification during signing and Y the single key the outside world sees.

2.3 Preprocessing and the two-round signing protocol

Signing splits into a message-independent *preprocess* stage and a single online round (equivalently, run as one two-round protocol with the commitment exchange inlined). A semi-trusted *signature aggregator* SA coordinates. The paper trusts it for publication and correct misbehaviour reporting, not for unforgeability, and claims in its security model that a malicious SA can deny service or falsely accuse but cannot learn the private key or cause an unapproved message to be signed.

Preprocess(π) (paper Figure 2). Each participant P_i generates π single-use nonce pairs and publishes their commitments: for $1 \le j \le \pi$,

$$(d_{ij}, e_{ij}) \xleftarrow{\$} \mathbb{Z}_q^* \times \mathbb{Z}_q^*, \qquad (D_{ij}, E_{ij}) = (g^{d_{ij}}, g^{e_{ij}}),$$

appending (D_{ij}, E_{ij}) to a list L_i published to a predetermined location, and storing the secret halves for later.

Sign(m) (paper Figure 3). The aggregator SA selects a signing set S of α participants ($t \le \alpha \le n$), fetches each member's next unused commitment pair, and forms the ordered list $B = \langle (i, D_i, E_i) \rangle_{i \in S}$, sending (m, B) to every $P_i, i \in S$. Each participant then:

1. validates m (a signer signs only messages it is willing to sign) and checks $D_\ell, E_\ell \in \mathbb{G}^*$ for every commitment in B, aborting on failure;
2. computes the *binding values*
$$\rho_\ell = H_1(\ell, m, B), \quad \ell \in S,$$
the group commitment and challenge
$$R = \prod_{\ell \in S} D_\ell \cdot (E_\ell)^{\rho_\ell}, \qquad c = H_2(R, Y, m);$$
3. computes its response, converting its Shamir share to the additive aggregation domain via λ_i (over the set S):
$$z_i = d_i + (e_i \cdot \rho_i) + \lambda_i \cdot s_i \cdot c;$$
4. securely deletes $((d_i, D_i), (e_i, E_i))$, then returns z_i to SA.

Finally SA recomputes ρ_i, $R_i = D_i \cdot (E_i)^{\rho_i}$, $R = \prod_{i \in S} R_i$ and c, and verifies every share against the public data:

$$g^{z_i} \stackrel{?}{=} R_i \cdot Y_i^{c \cdot \lambda_i},$$

identifying and reporting the misbehaving participant and aborting if any check fails; otherwise it publishes $\sigma = (R, z)$ with $z = \sum_{i \in S} z_i$.

Correctness. The proof is two interpolations (paper §6.1). The key polynomial $F_1(x) = \sum_{j=1}^{n} f_j(x)$ from key generation has $s_i = F_1(i)$ and $s = F_1(0)$. The nonce values $(i, (d_i + e_i \cdot \rho_i)/\lambda_i)$ define a degree-$(\alpha - 1)$ polynomial $F_2(x)$ with $F_2(0) = \sum_{i \in S}(d_i + e_i \cdot \rho_i) = k$. Setting $F_3(x) = F_2(x) + c \cdot F_1(x)$, each response is $z_i = \lambda_i F_3(i)$, so $z = \sum_{i \in S} z_i$ interpolates $F_3(0) = k + c \cdot s$; with $R = g^k$ and $c = H_2(R, Y, m)$, the pair (R, z) is a correct Schnorr signature on m.

2.4 Why the binding factor exists: Drijvers and ROS

The one non-obvious ingredient is $\rho_i = H_1(i, m, B)$. The binding factor is FROST's answer to concurrency attacks that affected or constrained the earlier two-round schemes discussed by the paper.

The Drijvers attack. Against two-round Schnorr multisignatures in which the adversary may open T parallel sessions, Drijvers et al. showed how to forge by finding a hash output that is a sum of others: $c^* = H(R^*, Y, m^*)$ with $c^* = \sum_{j=1}^{T} H(R_j, Y, m_j)$, using Wagner's k-tree algorithm for the generalised birthday problem in time $O(\kappa \cdot b \cdot 2^{b/(1+\lg \kappa)})$ for $\kappa = T + 1$ and b the bitlength of the group order. The attack transfers from the n-of-n multisignature setting to a t-of-n threshold scheme with an adversary controlling up to $t - 1$ signers, and it is why the Gennaro et al. scheme must cap its signing parallelism. Benhamouda et al.'s subsequent polynomial-time ROS solver (for $\ell > \lg q$ parallel sessions) turned the same structural weakness from subexponential to practical against schemes including two-round MuSig and GJKR-DKG-based threshold Schnorr; their paper concludes that "the current version of" FROST is *not* affected.

How binding addresses it. Because each signer derives $\rho_i = H_1(i, m, B)$ before responding, its share z_i is bound to this message, this signing set, and this exact commitment list. Except with random-oracle collisions, changing those inputs alters some ρ_ℓ, hence R and c, so directly transplanting a response fails verification. The paper's own generalisation (its Equation 1, with linear combinations γ_j) states the residual attack surface precisely: the forger must find

$$H_2(R^*, Y, m^*) = \sum_j \gamma_j \cdot H_2(R_j, Y, m_j),$$

where $R^* = g^{r^*} \cdot \prod_j \widehat{R}_j^{\gamma_j}$ is itself a function of every input on the right-hand side. Wagner's algorithm needs the two sides of the collision to vary independently; here every candidate combination on the right moves the left-hand target, degrading the generalised birthday collision into multiple preimage attacks. The paper's footnote 7 concedes this step is heuristic: sufficiency of the condition is immediate, necessity is not proved.

2.5 Nonce hygiene

The preprocessed pairs (d_{ij}, e_{ij}) are one-time keys in the strictest sense. The paper's stipulations:

- each pair is used at most once, and the signer *securely deletes* it in the same signing step that returns z_i (Figure 3, step 6);
- an accidentally reused (d_{ij}, e_{ij}) can expose the long-term share s_i;
- a SA that maliciously replays a commitment pair cannot elicit another response from an honest participant; the replay causes an abort and can therefore deny service;
- the commitment store (if a server) is trusted only for availability and freshness; possession of the public (D_{ij}, E_{ij}) lists grants no signing power.

See §2.8 for RFC 9591's sharpening of nonce generation (hedged sampling, and an explicit warning that deterministic nonces enable complete key recovery in the multi-party setting).

2.6 The security theorem as stated

The paper gives an EUF-CMA reduction, in the random-oracle model, for an intermediate scheme under the *discrete logarithm problem* in $\mathbb{G}$ — deliberately not to one-more discrete log (the adversary must return the discrete logarithms of $\ell+1$ challenges after at most ℓ queries to a discrete-log oracle), so that the negative result of Drijvers et al., which rules out security reductions from that assumption for a family of two-round Schnorr multisignatures, does not apply. The proof is for an intermediate scheme, *FROST-Interactive*, in which the binding value is produced by a one-time verifiable random function $\rho_i = a_{ij} + b_{ij} \cdot H_\rho(m, B)$ with committed keys.

> **Theorem 6.1** (ePrint 2020/852). *If the discrete logarithm problem in $\mathbb{G}$ is (τ', ϵ')-hard, then the FROST-Interactive signature scheme over $\mathbb{G}$ with n signing participants, a threshold of t, and a preprocess batch size of π is $(\tau, n_h, n_p, n_s, \epsilon)$-secure whenever*
>
> $$\epsilon' \leq \frac{\epsilon^2}{2n_h + (\pi+1)n_p + 1} \quad \text{and}$$
>
> $$\tau' = 4\tau + (30\pi n_p + (4t-2)n_s + (n+t-1)t + 6) \cdot t_{\text{exp}} + O(\pi n_p + n_s + n_h + 1),$$
>
> *such that t_{exp} is the time of an exponentiation in $\mathbb{G}$, assuming the number of participants compromised by the adversary is less than the threshold t.*

Here n_h, n_p, n_s bound the forger's random-oracle, preprocess, and signing queries. The architecture is Bellare–Neven generalised forking: the DL challenge ω is embedded as the honest participant's free-coefficient commitment ϕ_{t0} in a simulated KeyGen, the simulator is run twice on the same tape with oracle answers diverging from index J, and two forgeries yield $\text{dlog}(Y) = (z' - z)/(h'_J - h_J)$. The VRF is what lets the simulator win the programming race: it can compute R before the forger does and program $H_2(R, Y, m)$ with success probability 1.

The step from FROST-Interactive to plain FROST (replace the VRF by $\rho_i = H_1(i, m, B)$) is argued *heuristically* in the paper's §6.2, not proven — the Equation-1 gap of §2.4. This gap is what the later literature closed, differently and under a different assumption (§2.7).

2.7 The corrected and extended analyses

Three later results matter for how this volume cites FROST's security.

Crites–Komlo–Maller (ePrint 2021/1375). This work, "How to Prove Schnorr Assuming Schnorr: Security of Multi- and Threshold Signatures" (IACR ePrint 2021/1375, received 2021-10-12, last revised 2022-08-03), introduced the optimisation now called *FROST2*: a single binding factor $d = h_1(vk, lr)$ shared by all signers, replacing the original per-signer $d_i = h_1(vk, lr, i)$ (that original is retroactively *FROST1*) and reducing the signers' scalar multiplications from linear in the number of signers to constant.

Bellare–Crites–Komlo–Maller–Tessaro–Zhu (CRYPTO 2022). The paper "Better than Advertised Security for Non-interactive Threshold Signatures" — its FROST/BLS part maintained as Bellare–Tessaro–Zhu, ePrint 2022/833, June 2022 — gave the analysis now cited by RFC 9591. The authors define a hierarchy TS-UF-0 $\leq \cdots \leq$ TS-UF-4 (and strong variants TS-SUF-2/3/4) grading what counts as a trivial forgery. Their results, proven in the ROM under the *one-more discrete logarithm* (OMDL) assumption, without the AGM, and in the trusted-dealer setting (the DKG is explicitly not modelled):

- FROST1 is TS-SUF-3-secure, and is *not* TS-UF-4-secure;
- FROST2 is TS-SUF-2-secure, and is *not* TS-UF-3-secure — the shared binding factor loses the guarantee that the signer set that committed is the signer set that signed;

- concretely (their Theorem 5.1, verbatim bound), for a TS-SUF-2 adversary A making at most q_s preprocessing and q_h random-oracle queries there is an OMDL adversary B making at most $2q_s + ns$ challenge queries (ns their number of parties) with

$$\mathrm{Adv}^{\text{ts-suf-2}}_{\mathrm{FROST2}[\mathbb{G}]}(A) \le \sqrt{q \cdot (\mathrm{Adv}^{\mathrm{omdl}}_{\mathbb{G}}(B) + 3q^2/p)}, \qquad q = q_s + q_h + 1,$$

  and an analogous Theorem 5.4 for FROST1 at TS-SUF-3. In these source bounds, p denotes the group order and q the query count, replacing the earlier section's use of q for the group order.

The BCKMTZ analysis for the RFC's FROST1 signing layer is therefore based on OMDL in the ROM, static corruptions below t, and trusted-dealer keys. It does not establish composition with the DKG shipped by the implementation, and its basis differs from the original paper's DL reduction for the interactive variant plus a heuristic bridge.

Gouvêa–Komlo (ePrint 2024/436). The preprint "Re-Randomized FROST" (received 2024-03-13) is the design ZIP 312 builds on: FROST unchanged, plus algorithms GenRand ($\alpha \xleftarrow{\$} \mathbb{Z}_p$), RandShare ($\bar{x}_i \leftarrow x_i + \hat{\alpha}$ with $\hat{\alpha} = H_r(\alpha, \mathrm{aux})$, aux an optional external-protocol transcript contribution), RandPKShare ($\overline{PK}_i = PK_i \cdot g^{\hat{\alpha}}$) and RandPK ($\overline{PK} = PK \cdot g^{\hat{\alpha}}$). Correctness rests on the $\sum_{i \in S} \lambda_i = 1$ identity, giving $z = r + c(x + \hat{\alpha})$. Its Theorem 1 *claims* unforgeability in the ROM under *algebraic* OMDL (AOMDL). Unlike BCKMTZ's OMDL game, AOMDL restricts discrete-log-oracle queries to declared algebraic combinations, so these are not the "same security assumptions" claimed in the preprint's abstract. The stated bound (its Equation 3) is

$$\mathrm{Adv}^{\mathrm{sec}}_{TS,A} \le \sqrt{q \cdot \mathrm{Adv}^{\mathrm{aomdl}}_{D}(\lambda) + 2q^2/p - \mathrm{negl}(\lambda)}\,, \qquad q = q_h + q_s,$$

via the local forking lemma. The intended extraction uses the public randomizer to strip $c^* H_r(\alpha^*, \mathrm{aux}^*)$ from z^*, but the printed proof has material defects recorded in §3.3; the claimed bound is not independently established here. The coordinator chooses the randomizer and is trusted for privacy and unlinkability only, not for unforgeability. The full treatment of this paper belongs to the re-randomisation chapter (§3).

2.8 RFC 9591 against the paper: what the RFC changed

RFC 9591 (IRTF CFRG, Informational, June 2024; Connolly, Komlo, Goldberg, Wood) is the specification implemented by the `frost-core` 3.0.0 workspace this volume cites. Reconciling it against ePrint 2020/852:

- *Two rounds only.* The RFC specifies the two-round protocol; the paper's single-round-with-preprocessing variant (batch parameter π, commitment server) is declared out of scope.
- *No DKG.* The paper's Figure-1 KeyGen is dropped; key generation is out of scope, with *trusted-dealer* generation given in Appendix C only. The BCKMTZ analysis the RFC cites matches this framing (it, too, models a trusted dealer).
- *Binding factor inputs widened.* The paper binds $\rho_i = H_1(i, m, B)$. The RFC computes, per participant,

$$\texttt{binding_factor}_i = H_1(\texttt{PK_enc} \,\|\, H_4(\texttt{msg}) \,\|\, H_5(\texttt{encoded commitment list}) \,\|\, \texttt{enc}(i)),$$

  adding the *group public key* to the hash and pre-hashing the message and commitment list. The per-participant structure means RFC 9591 specifies FROST1: the RFC's §7.2 rules the single-binding-factor (FROST2) optimisation NOT RECOMMENDED, citing the loss of the started-set-equals-signed-set guarantee (the TS-SUF-3 vs TS-SUF-2 separation of §2.7).

- *Hedged nonces.* The paper samples (d, e) uniformly; the RFC hedges: each nonce is $H_3(r \| \texttt{enc}(sk_i))$ with r a fresh 32-byte random string, so a weak RNG is backstopped by the key share. It warns that fully deterministic derivation enables complete key recovery in the multi-party setting.
- *Five hash roles.* H_1 (binding factor), H_2 (challenge), H_3 (nonce), H_4 (message pre-hash), and H_5 (commitment-list pre-hash) are domain-separated with a per-ciphersuite `contextString`, except that FROST(Ed25519, SHA-512) deliberately leaves H_2 unprefixed for RFC 8032 compatibility. The challenge input order `group_comm_enc` ‖ `PK_enc` ‖ `msg` equals the paper's $H_2(R, Y, m)$.
- *Roles renamed and channels stated.* The signature aggregator becomes the *Coordinator*; the channel is assumed authenticated. Reliable delivery is also assumed for protocol completion; confidentiality is not required. The Coordinator SHOULD verify the aggregate signature and MAY verify individual shares — the paper's per-share check $g^{z_i} = R_i \cdot Y_i^{c\lambda_i}$ survives as verify_signature_share — with identifiable abort, as in the paper, and no robustness.
- *Ciphersuites.* The RFC ships FROST(Ed25519, SHA-512), FROST(ristretto255, SHA-512) (RECOMMENDED), FROST(Ed448, SHAKE256), FROST(P-256, SHA-256), and FROST(secp256k1, SHA-256); no Zcash suite appears in the RFC itself. The two Zcash suites, FROST(Jubjub, BLAKE2b-512) and FROST(Pallas, BLAKE2b-512), are instead defined by ZIP 312, whose H_2 personalisation `Zcash_RedPallasH` makes the FROST challenge equal RedDSA's $\mathsf{H}^{\circledR}$ — the precise sense in which the RFC protocol plugs into the spend-authorisation verifier of the *Ironwood Guide*, §"The RedPallas signature scheme".
- *Security claim.* The RFC claims EUF-CMA per BCKMTZ, citing both the original paper and BCKMTZ — i.e., the specification's claim rests on the OMDL/ROM analysis, not the paper's DL-based theorem for the interactive variant.

Classification: FROST as in RFC 9591 and its Zcash ciphersuites as in ZIP 312 are *specified*; ZIP 312 itself remains Draft status (zips commit 6961098410, 2026-07-09), with its key-generation procedure explicitly out of scope and PedPoP DKG for Zcash deployment *designed-but-unspecified* (implemented in `frost-core`'s `dkg` module but specified in no ZIP or RFC).

3 Re-Randomized FROST for RedPallas

The FROST of §2, specified in RFC 9591 (§2.8), constructs its aggregate relative to a *fixed* group public key PK. Zcash instead supplies a freshly randomised validating key for every Sapling Spend and Orchard Action, and consensus checks the spend-authorisation signature against that per-Action key — a key to which, as §3.1 makes precise, the unchanged FROST shares do not interpolate. This chapter presents the design that closes the gap: the Re-Randomized FROST of Gouvêa and Komlo (IACR ePrint 2024/436), its Zcash specification in ZIP 312, the ciphersuite FROST(Pallas, BLAKE2b-512), the published implementation in the `frost-rerandomized` and `reddsa` crates, and the custody surfaces the construction serves.

A rigour note before the material. RFC 9591 is an Informational RFC recording CFRG consensus, not an Internet Standard. The Gouvêa–Komlo paper is a preprint — received 2024-03-13, no peer-reviewed venue — and ZIP 312 carries Draft status (zips commit 6961098410, 2026-07-09). Following the series rule, every claim beyond the RFC is binned as *specified*, *designed but unspecified*, or an *open problem*; the closing classification is in §3.7.

3.1 The mismatch: consensus verifies only randomised keys

The protocol specification (§4.15, spend authorization signatures) prescribes, for every Sapling Spend and every Orchard Action, the following signer-side flow. The signer chooses a fresh

spend-auth randomizer α and:

1. samples $\alpha \leftarrow \mathsf{SpendAuthSig.GenRandom}()$;
2. derives $\mathsf{rsk} = \mathsf{SpendAuthSig.RandomizePrivate}(\alpha, \mathsf{ask})$;
3. derives $\mathsf{rk} = \mathsf{SpendAuthSig.DerivePublic}(\mathsf{rsk})$;
4. proves the Spend or Action statement with α in the auxiliary (secret) input and rk in the primary input;
5. signs: $\mathsf{spendAuthSig} = \mathsf{SpendAuthSig.Sign}_{\mathsf{rsk}}(\mathsf{SigHash})$.

Consensus therefore verifies the signature under $\mathsf{rk} = \mathsf{ak} + [\alpha]\, G^{\mathsf{SpendAuth}}$. This differs from ak except in the possible identity-randomizer case $\alpha = 0$.

The randomisation algorithms are RedDSA's, from the protocol specification (§5.4.7):

$$\mathsf{RedDSA.RandomizePrivate}(\alpha, \mathsf{sk}) := \mathsf{sk} + \alpha \pmod{r_{\mathbb{G}}},$$
$$\mathsf{RedDSA.RandomizePublic}(\alpha, \mathsf{vk}) := \mathsf{vk} + [\alpha]\, G,$$

with identity randomizer 0; RedDSA.GenRandom chooses T uniformly among byte sequences of length $(\ell_H + 128)/8$ — 80 bytes at $\ell_H = 512$ — and returns $H^{\circledast}(T)$, where $H^{\circledast}(B) := \mathrm{LEOS2IP}_{\ell_H}(H(B)) \bmod r_{\mathbb{G}}$. Signing draws its nonce as $r = H^{\circledast}(T \,\|\, \mathsf{vk}^{\star} \,\|\, M)$ for fresh uniform T, sets $R = [r]\, G$ and $S = (r + H^{\circledast}(R^{\star} \,\|\, \mathsf{vk}^{\star} \,\|\, M) \cdot \mathsf{sk}) \bmod r_{\mathbb{G}}$, and outputs $\sigma = R^{\star} \,\|\, S^{\star}$. Validation recomputes $c = H^{\circledast}(R^{\star} \,\|\, \mathsf{vk}^{\star} \,\|\, M)$ and accepts iff $R \neq \bot$, $S < r_{\mathbb{G}}$, $R^{\star}$ is canonical (post-ZIP-216), and $[h_{\mathbb{G}}](-[S]\, G + R + [c]\, \mathsf{vk}) = 0_{\mathbb{G}}$. RedPallas specialises the template with $\mathbb{G}$ the Pallas group, $\ell_H = 512$, and $H(x) = \text{BLAKE2b-512}(\texttt{Zcash_RedPallasH}, x)$; $\mathsf{SpendAuthSig}^{\mathsf{Orchard}}$ uses the generator $G^{\mathsf{SpendAuth}} = \mathsf{GroupHash}^{\mathsf{Pallas}}(\texttt{z.cash}:\texttt{Orchard}, \texttt{"G"})$ and "is instantiated as RedPallas with key re-randomization". The security notion the protocol requires of the scheme is SURK-CMA (Strong Unforgeability with Re-randomized Keys), adapted from Fleischhacker et al. (PKC 2016, §3), whose oracle answers queries of the form (m, α) — message and adversary-chosen randomizer together.

The construction and the cited single-party EUF-CMA result are settled in the *Ironwood Guide*: §"Randomised validating keys" builds $\mathsf{rsk} := \mathsf{ask} + \alpha$ and $\mathsf{rk} := \mathsf{ak} + [\alpha]\, G^{\mathsf{SpendAuth}}$, and §"RedPallas unforgeability" proves EUF-CMA under DL in the ROM *including* adversary-chosen α, by key-prefixed forking with $\mathsf{ask} = \mathsf{rsk}^{*} - \alpha^{*}$. Per the layering rule we cite those results and do not re-prove them. Because its Proposition "Unforgeability under re-randomisation" requires only a fresh key–message–signature triple, it also implies the protocol specification's strong SURK-CMA notion, for a single signer.

The threshold problem is now visible. A FROST sharing of ask gives shares sk_i with $\sum_{i \in S} \lambda_i \mathsf{sk}_i = \mathsf{ask}$ for every signing set S; an honest plain-FROST execution constructs a signature under ak. For a nonzero Action randomizer, however, the verifier's key is rk. The noninteractive construction below shifts every participant by the same effective randomizer. A separately generated Shamir sharing of a randomizer could also shift the group secret, but would be a different, interactive construction.

3.2 Re-Randomized FROST: the construction

The design is due to Gouvêa and Komlo (both Zcash Foundation), "Re-Randomized FROST", IACR ePrint 2024/436, received 2024-03-13; we work from the full text and page renders of its construction and equations. The transcription below normalises the source-level slips identified after the construction. As in §2 we quote the paper in its own multiplicative notation (g^x for $[x]\, G$, $PK \cdot g^{\hat{\alpha}}$ for $PK + [\hat{\alpha}]\, G$); the additive RedPallas form returns with ZIP 312 in §3.4.

Definition 3.1 (Perfectly rerandomizable threshold scheme; ePrint 2024/436, Definition 5)**.** A threshold signature scheme

$$TS = (\mathsf{Setup}, \mathsf{KeyGen}, (\mathsf{Sign}, \mathsf{Sign}'), \mathsf{Combine}, \mathsf{Verify})$$

is *perfectly re-randomizable* if there exist additional PPT algorithms GenRand, RandShare, RandPKShare, and RandPK, and a randomness distribution X, such that the Rand^* algorithms take a randomizer $\alpha \in X$ and an auxiliary string $\mathsf{aux} \in \{0,1\}^*$.

Remark 3.2 (What "perfectly" specifies). The printed Definition 5 requires only the existence, inputs, and outputs of four algorithms. It states neither a correctness relation nor equality of any distributions. The adjective "perfectly" therefore adds no formal property in that definition. Functional correctness of the concrete scheme comes from Equation (1); no distributional guarantee follows from Definition 5 alone.

The paper's Remark 1 explains aux: it models an optional contribution from an external protocol transcript, so that a new session changes the re-randomised key even under a repeated α.

Construction 3.3 (Re-Randomized FROST; ePrint 2024/436, Figure 5)**.** The additions to plain FROST — the paper's grey-boxed lines — are exactly four algorithms:

$$\begin{aligned}
\mathsf{GenRand}() &: \alpha \xleftarrow{\$} \mathbb{Z}_p; \\
\mathsf{RandShare}(x_i, \alpha, \mathsf{aux}) &: \hat{\alpha} \leftarrow H_r(\alpha, \mathsf{aux}); \quad \bar{x}_i \leftarrow x_i + \hat{\alpha}; \\
\mathsf{RandPKShare}(PK_i, \alpha, \mathsf{aux}) &: \hat{\alpha} \leftarrow H_r(\alpha, \mathsf{aux}); \quad \overline{PK}_i \leftarrow PK_i \cdot g^{\hat{\alpha}}; \\
\mathsf{RandPK}(PK, \alpha, \mathsf{aux}) &: \hat{\alpha} \leftarrow H_r(\alpha, \mathsf{aux}); \quad \overline{PK} \leftarrow PK \cdot g^{\hat{\alpha}}.
\end{aligned}$$

The FROST core runs unchanged on the shifted inputs. Algorithm Sign() samples $(r_k, s_k) \xleftarrow{\$} \mathbb{Z}_p^2$ and commits $(R_k, S_k) = (g^{r_k}, g^{s_k})$. Algorithm Sign$'$, given the randomised share $\bar{x}_k$ and randomised key $\overline{PK}$ in place of x_k and PK, derives each binding factor a_i by H_{non} from the participant identifier, the message, $\overline{PK}$, and the full commitment list $\{(i, R_i, S_i)\}_{i \in S}$, then computes

$$R \leftarrow \prod_{i \in S} R_i \cdot S_i^{a_i}, \qquad c \leftarrow H_{\text{sig}}(\overline{PK}, R, m), \qquad z_k \leftarrow r_k + s_k a_k + c \lambda_k \bar{x}_k.$$

Algorithm Combine outputs $z \leftarrow \sum_{i \in S} z_i$, $\sigma \leftarrow (R, z)$; $\mathsf{Verify}(\overline{PK}, m, \sigma)$ recomputes the challenge and accepts iff $R \cdot \overline{PK}^c = g^z$.[a]

[a] The paper's own argument order for H_{sig} is inconsistent: Sign$'$ and Combine write $H_{\text{sig}}(PK, R, m)$, while Verify and the reduction's table write $H_{\text{sig}}(PK, m, R)$ — and both differ from the order RFC 9591 (§4.6) and RedDSA fix, $c = H_2(R^\star \| PK^\star \| m)$, which ZIP 312 and the released `reddsa` crate use. We read the paper's orderings as a presentational slip with no normative weight; the implemented order is RedDSA's. The paper also twice refers to "Figure 4" where Figure 5 is meant.

The displayed construction also normalises several evident errors in Figure 5. Its Setup lists only H_{non} and H_{sig} even though every randomisation algorithm calls H_r. In Sign$'$, its formal argument names the local randomised share $\bar{x}_i$ although the algorithm and its body use signer index k; we consistently write $\bar{x}_k$. Its loop also prints k as the first input to H_{non} for every a_i; the iterated index must be i, as above, for the commitment $R = \prod_i R_i S_i^{a_i}$ and response term $s_k a_k$ to use the same per-participant binding factors. The paper's preliminary Shamir definition has two further typographical slips: its polynomial omits the factor Z from the $a_1 Z$ term and its prose says that t coefficients are sampled while listing $a_1, \ldots, a_{t-1}$. Its subsequent evaluation formula is the standard degree-$(t-1)$ sharing used here.

The paper is explicit that this is the whole design: "We do not make any changes to the plain FROST protocol." Its signing equations are unchanged; the added algorithms supply the shifted share and public-key inputs.

Correctness is one identity. The paper's Equation (2):

$$z = \sum_{i\in S} r_i + \sum_{i\in S} s_i a_i + c \sum_{i\in S} \lambda_i \bar{x}_i = \sum_{i\in S} r_i + \sum_{i\in S} s_i a_i + c\Big(\sum_{i\in S} \lambda_i x_i + \hat{\alpha} \sum_{i\in S} \lambda_i\Big) = r + c\,(x + \hat{\alpha}) = r + c\,\hat{x}, \quad (1)$$

"because x is Shamir secret shared" and "$\sum_{i\in S} \lambda_i = 1$ since it is the interpolation of the constant polynomial $f(X) = 1$". The output is a regular re-randomised signature for $\overline{PK} = PK \cdot g^{\hat{\alpha}}$ with $\hat{\alpha} = H_r(\alpha, \mathrm{aux})$.

The identity deserves emphasis, because it is the entire reason the construction is this simple. Every participant adds the *same* $\hat{\alpha}$ to its share — no dealing, no per-participant shares of the randomizer — and the Lagrange weights, summing to one over any interpolating set, carry the common shift through interpolation so that the group secret moves by $\hat{\alpha}$ rather than by $|S|\cdot\hat{\alpha}$. Had the weights summed to anything else, or varied with S in a way the signers could not predict, the shifted shares would interpolate to a key nobody computed.

3.3 Unforgeability: TRUF, AOMDL, and Theorem 1

The game. The paper's notion (Definition 6, game TRUF in Figure 4) allows adversarial randomizers and static corruption of fewer than t parties. Its signing oracle records only the queried messages Q, and a forgery wins only when $m^* \notin Q$. It is therefore an *existential* notion: a new signature for a previously signed message is not a winning forgery. The protocol specification instead requires SURK-CMA, whose oracle records message–signature pairs and accepts any fresh pair, including a different signature on a previously queried message. The paper therefore uses a weaker freshness condition, and Theorem 1 does not, as printed, establish the protocol specification's strong notion. The game pseudocode also assigns $\mathsf{RandShare}(x_k, \alpha, \mathrm{aux})$ to $\bar{x}_i$ but later passes $\bar{x}_k$, and its call to Sign' omits the message argument that the construction requires. Those are interface errors in the printed game; the discussion here uses the evident k index and includes m.

There is a second mismatch in the unrandomised branch. The winning predicate also accepts a forgery under PK, but every signing-oracle answer first applies $H_r(\alpha, \mathrm{aux})$; the adversary has no ordinary signing oracle under PK. Contrary to the paper's prose, setting its outer randomizer α to zero does not produce the identity shift. The construction's randomizer domain is $\mathbb{Z}_p$ and $H_r(0, \mathrm{aux})$ need not be zero. Thus the separate verification branch does not embed the ordinary EUF-CMA game either.

The assumption. Intended AOMDL is the *algebraic one-more discrete logarithm* game: the adversary receives $\ell + 1$ challenges, may query a discrete-log oracle only on declared linear combinations of environment-generated elements, and should make at most ℓ queries. The algebraic restriction is absent from §2.7's OMDL. Since it reduces the adversary's allowed oracle queries, AOMDL hardness is formally a weaker hardness assumption than OMDL hardness, obtained at the cost of an algebraic-adversary restriction; it is neither literally the same assumption nor a stronger hardness assumption. Figure 1 is not a well-defined rendition of that game: its oracle accepts $\{\beta_i\}_{i=1}^{\ell}$ but its return expression uses β_0 as well, and it never checks that the queried element X equals the declared combination $g^{\alpha} \prod_i X_i^{\beta_i}$. After the natural repair of supplying coefficients for every challenge, a second defect remains: the game initialises its query counter $q \leftarrow 0$ but never increments it. Its final test $q < \ell + 1$ then imposes no query limit, so an adversary can request every challenge logarithm. We interpret AOMDL by its intended one-more definition below, not by that malformed pseudocode.

The theorem claim. The preprint states the following (Theorem 1 with Equation (3); the radical was checked against the rendered page):

> **Theorem 1** (ePrint 2024/436). *Rerandomized-FROST is unforgeable in the random oracle model, assuming the AOMDL assumption holds in* $\mathbb{G}$, concretely
>
> $$\mathrm{Adv}^{\mathrm{sec}}_{TS,\mathcal{A}} \le \sqrt{q \cdot \mathrm{Adv}^{\mathrm{aomdl}}_{D}(\lambda) + 2q^2/p - \mathrm{negl}(\lambda)}\,, \qquad q = q_h + q_s, \tag{2}$$
>
> with q_h random-oracle and q_s signing queries.

Remark 3.4 (Placement of the negligible term). The printed bound places $-\mathrm{negl}(\lambda)$ *inside* the square root; we transcribe it as printed. The customary form of such forking-lemma bounds — including the BCKMTZ bound quoted in §2.7 — carries additive slack outside the radical, and a negative term inside can make the radicand negative. Neither the theorem statement nor its proof explains how that expression is to be interpreted in that case.

Intended proof shape, and why it is not verified. The reduction $\mathcal{B}$ receives $2q_s + 1$ AOMDL challenges, sets $PK = X_0$, and simulates verification shares in the exponent. Even the displayed public-share simulation is not well formed: its product shadows the output index i and leaves both j and k unbound. If C is the set of corrupted identifiers and $L_j(i)$ is the Lagrange basis coefficient for node $j \in \{0\} \cup C$ evaluated at the honest identifier i, the required exponent relation is

$$PK_i = PK^{L_0(i)} \prod_{j \in C} (g^{x_j})^{L_j(i)}.$$

The printed expression does not establish that simulated public shares interpolate to PK.

Round-one answers consume challenge pairs. In round two, the response equation

$$z_k = r_k + s_k a_k + c\lambda_k \bar{x}_k$$

requires the reduction to query the logarithm of

$$Z_k = R_k S_k^{a_k} \overline{PK}_k^{c\lambda_k}. \tag{3}$$

The preprint instead prints $R_i S_i^{a_k} \overline{PK}_k^{c}$: it mixes i and k and omits the Lagrange exponent λ_k. Its claimed "perfect" signing simulation therefore does not follow from the displayed query.

The random-oracle simulation also is not a consistent lazy sampler as printed. Its H_r table stores only the queried pair (α, aux), not the sampled output, so a repeat query has no associated value to return. Its H_{non} query signature omits m although its lookup and table include m; when it samples a_j for each other participant, it stores a_k rather than a_j; and its new-query branch returns the sampled H_{sig} value c rather than the requested binding value a_k. The proof later invokes an unrandomised case $\alpha = \perp$, although neither the construction's domain $\mathbb{Z}_p$ nor the TRUF oracle defines that sentinel.

The forking step is underspecified too. It says to reprogram $H_{\mathrm{sig}}(PK', R^*, m^*)$ without defining PK', then asserts a second forgery with the same R^* but potentially different (α, aux). It does not show that the reprogrammed query is the challenge query for both accepted forgeries when their re-randomised public keys differ. Nor does the proof derive the concrete probability bound in Equation (2); it concludes only with a qualitative "non-negligible probability" invocation of the local forking lemma.

The first fork-extraction identity is algebraically sound if both forgeries share the same commitment R, verify under re-randomised keys, and have $c_0 \neq c_1$. Writing $h_b = H_r(\alpha_b, \mathrm{aux}_b)$, it removes the public shifts and obtains

$$x = \frac{(z_0^* - c_0 h_0) - (z_1^* - c_1 h_1)}{c_0 - c_1}.$$

The subsequent nonce extraction is not sound as printed. If two response shares have the form

$$u_b = r_i + s_i a_b + c_b \lambda_i (x_i + h_b),$$

then for $a_0 \neq a_1$ the corresponding identity is

$$s_i = \frac{u_0 - u_1 - c_0 \lambda_i x_i - c_0 \lambda_i h_0 + c_1 \lambda_i x_i + c_1 \lambda_i h_1}{a_0 - a_1}.$$

The printed numerator gives the two c_1 terms the wrong signs and uses the outer randomizers in place of their H_r images. More seriously, the proof claims that when $c_0 = c_1$, $a_0 = a_1$, and $h_0 \neq h_1$, the reduction can still recover both (r_i, s_i). After subtracting the known randomizer difference, however, the transcripts give only one linear equation in those two nonce scalars. They are underdetermined.

Together with Figure 1's missing query-counter increment, these are material gaps, not merely notation. A repair may be possible, but this volume does not supply a complete replacement reduction. It therefore records Theorem 1 as a preprint claim and does not use Equation (2) as a verified assurance statement.

Scope even under a repaired proof. The claimed result is only existential unforgeability in the paper's weaker game; no cited theorem establishes threshold SURK-CMA. The preprint contains no formal unlinkability definition or theorem. Corruptions are static, and the threshold-scheme game uses centralised key generation; the paper says it "can [be] adapted" to a DKG, but gives no composition proof. Finally, the paper's §5.1 sketches how a DKG-style contribution round could remove the trusted-for-privacy coordinator (all parties contribute to the randomizer, none learns it) at the cost of extra communication rounds; ZIP 312 does not adopt this. The cost of re-randomisation itself is one fixed-base scalar multiplication per participant and per coordinator; the paper's Table 1 benchmarks (Pallas suite, Ryzen 9 5900X) put the signing overhead at 83% with 2 signers, 45% at 7, and 8% at 67.

3.4 ZIP 312: the specification

ZIP 312, "FROST for Spend Authorization Multisignatures" (Status Draft, Category Wallet, created 2022-08; owners Gouvêa, Komlo, Connolly; zcash/zips PR #662), specifies Re-Randomized FROST as a delta against RFC 9591. Its target property beyond unforgeability is *unlinkability*, which the ZIP defines as statistical independence of signatures from the signers' long-term keys: for uniform randomizers and absent metadata leakage, it is "impossible to determine whether two transactions were generated by the same party".

The signing-flow changes to RFC 9591.

- *Randomizer generation.* A new helper `randomizer_generate(msg, commitment_list)`: draw `rng_randomizer` ← `random_bytes(Ns)`, form

  ```
  randomizer_input = rng_randomizer ‖
                     encode_signing_package(commitment_list, msg),
  ```

  and return H_R(`randomizer_input`). Implementations MAY use another encoding "as long as all values (the message, and the identifier, binding nonce and hiding nonce for each participant) are unambiguously encoded".

- *Round one* is unchanged.

- *Round two.* The Coordinator generates the randomizer and sends it to each signer "over a confidential and authenticated channel" — plain FROST requires reliability and authentication but not confidentiality — together with the message and the commitment list. Each participant then runs `sign` with sk_i + randomizer in place of its share and PK + [randomizer] G in place of the group public key.
- *Aggregation* shifts the group key the same way: PK += [randomizer] G.
- *Share verification* shifts both keys: PK_i += [randomizer] G and PK += [randomizer] G in `verify_signature_share`.

The ZIP's "randomizer" is the paper's *effective* randomizer $\hat{\alpha}$: per the paper's §7 implementation notes, GenRand and H_r are folded into the single hash call above, and the value distributed to signers is the already-hashed scalar.

The algebra, in RedDSA notation. The ZIP's Rationale restates the correctness identity of (1) additively, and it is worth displaying because it is the entire consensus-facing content of the design. A valid RedDSA response must satisfy $z = r + c \cdot \mathsf{sk}$. Each FROST share contributes

$$z_i = (\texttt{hiding_nonce}_i + \texttt{binding_nonce}_i \cdot \rho_i) + \lambda_i \cdot c \cdot \mathsf{sk}_i,$$

and under re-randomisation the key terms become $c \cdot (\mathsf{sk} + \text{randomizer})$ on the single-signer side and, summed over the coalition,

$$\sum_{i \in S} \lambda_i \, c \, (\mathsf{sk}_i + \text{randomizer}) = c\Big(\sum_{i \in S} \lambda_i \mathsf{sk}_i + \text{randomizer} \cdot \sum_{i \in S} \lambda_i\Big) = c \, (\mathsf{sk} + \text{randomizer}),$$

since $\sum_i \lambda_i \mathsf{sk}_i = \mathsf{sk}$ by Shamir and $\sum_i \lambda_i = 1$ (the ZIP proves the latter in its `sum-lambda-proof` footnote, citing math.SE 1325342). Every participant adds the same randomizer to its share; the weights-sum-to-one identity is exactly why the group secret shifts by α and not by $|S| \cdot \alpha$.

Who learns the randomizer. The threat model is the ZIP's sharpest departure from RFC 9591. The Coordinator generates the randomizer and is "trusted with the privacy of the transaction (which includes the unlinkability property)" but *not* with unforgeability: "a rogue Coordinator ... should not be able to create signed transactions without the approval of `MIN_PARTICIPANTS` participants". All key-share holders receive the randomizer and are likewise trusted with privacy only. The role assignment is natural: the Coordinator fits the transaction builder, who already creates the zero-knowledge proof — and the proof needs α as auxiliary input (step 4 of the flow in §3.1). The confidentiality requirement is not optional hygiene: anyone learning $\hat{\alpha}$ can unrandomise the on-chain key, $\mathsf{ak} = \mathsf{rk} - [\hat{\alpha}]\, G$, so the paper advises encrypting the randomizer "so that eavesdroppers can't break unlinkability". Non-requirements are stated with equal care: the coordinator-less variant of RFC 9591 §7.5 is unsupported;[6] share holders can always link the signing operation to the eventual on-chain transaction; key generation is out of scope; network privacy is out of scope.

The message is a hash. The signed message is the SigHash, which by itself does not expose enough transaction semantics for a signer to know what is being authorised. Signers MUST therefore check the SigHash against transaction data sent over the same channel, or recompute it themselves; the mechanism is out of the ZIP's scope. The same authorisation principle applies to any external signer: a signer who cannot see what it signs authorises whatever the builder chose.

[6] ZIP 312's reference for the variant mislabels it as RFC 9591 "Section 7.3: Removing the Coordinator Role"; the section is 7.5 (§7.3 is "Nonce Reuse Attacks"), and the reference's URL anchor targets the correct section.

Unlinkability, and where it is proven. Neither the paper nor ZIP 312 supplies a formal threshold-unlinkability proof: the paper attempts only an existential-unforgeability proof (§3.3), and the ZIP argues unlinkability in its Rationale by matching distributions — "`randomizer_generate` generates `randomizer` uniformly at random as required by `RedDSA.GenRandom`", and the signing flow is compatible with RedDSA.RandomizePrivate, RedDSA.RandomizePublic, RedDSA.Sign, and RedDSA.Validate — while its Requirements defer the formal notion to the protocol specification's SURK-CMA definition. More precisely, reducing a 512-bit random-oracle output modulo the group order is statistically close to uniform, not exactly uniform; the same qualification applies to RedDSA.GenRandom. The distribution-matching argument does connect to a proved statement: the *Ironwood Guide*'s Proposition "Unlinkability of randomised validating keys" shows honest uniform re-randomisations are distributed as fresh Schnorr keys. What no source supplies is a threshold-specific unlinkability theorem. Nor does the preprint's weaker existential game establish the threshold SURK-CMA requirement. Bin: the mechanism is *specified* (ZIP 312, Draft); its unlinkability guarantee rests on a distribution argument, and its requested strong-unforgeability guarantee has no matching threshold theorem among the cited sources.

3.5 The ciphersuite FROST(Pallas, BLAKE2b-512)

ZIP 312 defines the ciphersuite in RFC 9591's template. The group is Pallas with base point $G^{\mathsf{SpendAuth}}$ as in protocol §5.4.7.1, of order p_{Vesta}; `RandomScalar` is uniform in $[0, \texttt{Order} - 1]$; `SerializeElement` is the point compression $\mathrm{repr}_{\mathsf{Pallas}}$ ($P^\star$ in series notation) and `DeserializeElement` is $\mathrm{abst}_{\mathsf{Pallas}}$, erroring on $\perp$ *and on the identity element*; `SerializeScalar` is little-endian over 32 bytes and `DeserializeScalar` rejects values $\geq$ the order. The hash is BLAKE2b-512 with 16-byte personalisation strings and $N_h = 64$; scalar-valued hashes interpret the 64 output bytes as a little-endian integer reduced mod the order. Table 2 lists the oracles.

Oracle	Role	Personalisation
H_1	binding factor	`FROST_RedPallasR`
H_2	challenge	`Zcash_RedPallasH`
H_3	nonce	`FROST_RedPallasN`
H_4	message pre-hash	`FROST_RedPallasM`
H_5	commitment-list pre-hash	`FROST_RedPallasC`
H_R	randomizer	`FROST_RedPallasA`

Table 2: FROST(Pallas, BLAKE2b-512) hash personalisations (ZIP 312). The challenge hash reuses RedPallas's own personalisation; the FROST additions use the `FROST_` prefix.

The single load-bearing row is H_2: ZIP 312 states it is "equivalent to $H^{\circledR}(m)$" of RedPallas, so the FROST challenge *is* the on-chain challenge, and signature verification for the suite is RedPallas validation itself. The ZIP's compatibility rationale completes the argument that a FROST aggregate is a plain RedPallas signature: the (R, S) output split matches RedDSA.Validate's parsing; FROST and Zcash use the same challenge input order (R, public key, message); and the fact that r and R are FROST-generated rather than produced by RedDSA.Sign's nonce derivation is irrelevant to a verifier, which only parses R and checks the validation equation. A parallel suite FROST(Jubjub, BLAKE2b-512) — personalisations `FROST_RedJubjub{R,N,M,C,A}` and `Zcash_RedJubjubH` — serves Sapling spend authorisation. ZIP 312 defines that suite separately; this volume develops only the Pallas suite.

3.6 Published library implementation: `frost-rerandomized` and `reddsa`

Claims in this subsection cite ZcashFoundation/frost at commit `2016e44` (2026-04-23, workspace version 3.0.0) and ZcashFoundation/reddsa at commit `3792daa` (2026-05-22, crate version

0.5.2), rechecked against current heads on 2026-09-05. The pins are published releases — tag `frost-core/v3.0.0` (crates.io 2026-04-23) and `reddsa` 0.5.2 — so main-at-commit and the released crates coincide. ZIP 312 names `reddsa` as the reference implementation of both ciphersuites. A current-head check on 2026-09-05 found frost `0966bd1` and reddsa `63c6ac0`; their post-release changes update the Rust edition, minimum compiler, dependencies, benchmarks, and CI, not the re-randomized protocol derived below.

The generic layer. Crate `frost-rerandomized` implements Construction 3.3 over any `frost-core` ciphersuite. Trait impl `Randomize<KeyPackage>` computes the randomised signing share as `signing_share` + `randomizer` and the randomised verifying share by adding the `randomizer_element`; struct `RandomizedParams` carries the `randomizer` ("also called alpha" in its documentation), `randomizer_element` = `[randomizer]` G, and

$$\texttt{randomized_verifying_key} = PK + \texttt{randomizer_element}.$$

The current flow differs instructively from a literal reading of the ZIP: the Coordinator sends a randomizer *seed*, not the randomizer, and each participant recomputes the randomizer from that seed and the full round-one commitment map in its signing package. The derivation, its divergence from the draft text — the message is no longer a hash input — and the reconciliation in flight in zips PR #895 are treated once, in §4.2.

The RedPallas suite. Module `src/frost/redpallas.rs` in `reddsa` implements the `frost-core` `Ciphersuite` trait for `PallasBlake2b512` (contextual ID `"FROST(Pallas, BLAKE2b-512)"`). Its `generator()` is `orchard::SpendAuth::basepoint()`, whose constant bytes are reproducible as `pallas::Point::hash_to_curve("z.cash:Orchard")(b"G")` — the protocol's $G^{\mathsf{SpendAuth}}$. Oracles $H_1/H_3/H_4/H_5$ carry the `FROST_RedPallas{R,N,M,C}` personalisations of Table 2; H_2 goes through the crate's `HStar` default, `Zcash_RedPallasH`; the `RandomizedCiphersuite::hash_randomizer` impl uses `FROST_RedPallasA` (the ZIP's H_R); scalar reduction is the 64-byte-wide `from_bytes_wide`, whose Pallas implementation delegates to pasta's `from_uniform_bytes`. Beyond ZIP 312, the crate defines two further personalisations — `HDKG` = `FROST_RedPallasD` and `HID` = `FROST_RedPallasI`, the DKG and identifier hashes — for key generation, which the ZIP does not specify at all.

Even-Y normalisation. One deployment detail appears in neither the paper nor the ZIP 312 body: the `reddsa` hooks `post_generate/post_dkg` normalise fresh key material to the even-Y representative the protocol specification requires of ak; the full mechanism is §4.1. The *randomised* key rk need not satisfy the even-Y condition — only ak does.

The RedPallas re-randomised API. Module `src/frost/redpallas/rerandomized.rs` re-exports the re-randomised flow for the suite — `sign_with_randomizer_seed()`, `aggregate()`, and `aggregate_custom()` — together with the type aliases `Randomizer` and `RandomizedParams` over `PallasBlake2b512`. The module README's worked example runs the full trusted-dealer plus re-randomised signing flow and ends by verifying the aggregate under the `randomized_verifying_key()` of its `RandomizedParams` — the executable form of this chapter's claim that a FROST aggregate is a plain RedPallas signature under rk.

3.7 Custody surfaces and classification

(a) Spend-authorisation custody. The primary surface: a t-of-n sharing of ask signing Orchard Actions (and, via the Jubjub twin suite, Sapling Spends). FROST replaces step 5 of the signer flow in §3.1; steps 1–4 move to the Coordinator-as-builder, who draws the randomizer, proves the Action statement with α as auxiliary input, and publishes rk — the circuit and consensus

rules are untouched, per the *Ironwood Guide*, §"Randomised validating keys". Bin: *specified* (ZIP 312, Draft), with key generation explicitly out of the ZIP's scope; DKG-based key generation for Zcash remains *designed but unspecified* (implemented in `frost-core`, specified nowhere), as §2.8 already concluded for the base protocol. Related anticipations elsewhere in the ZIP corpus: ZIP 316 requires FROST unified viewing key material to follow ZIPs 312 and 2005, and ZIP 271 anticipates lockbox disbursement to "a FROST address", noting (as of May 2025) that ZIP 312 does not specify key generation.

(b) ZIP 2005 quantum spending key custody. ZIP 2005's "Usage with FROST" section constrains threshold deployments of its quantum-recovery hierarchy: the key ak is derived jointly, by FROST DKG, from participants' shares of ask, and the participants privately agree on a value sk from which nk, qsk, qk, and $\mathsf{rivk}_{\mathsf{ext}}$ derive. The full ceremony, its derivations, and its asymmetric threat model — every participant holds qsk, so the threshold property does not survive an adversary that can also solve Pallas discrete logarithms — are developed once, in §4.7. Bin: the FROST-facing requirements are *specified* (ZIP 2005's requirements list calls for full compatibility with ZIP 312 FROST, hardware wallets, and their combination); the combined DKG-plus-shared-sk key generation is *designed but unspecified* — no source specifies the protocol by which participants "privately agree" on sk.

(c) Crosslink finalizer keys. The *Crosslink Guide*, §"Finalizer keys", bins threshold custody of finalizer keys as an *open problem*, and we adopt that bin. One precision for this volume: the applicable protocol there is *plain* FROST — the finalizer design has no key re-randomisation, so the natural suite is RFC 9591's FROST(Ed25519, SHA-512), not either ZIP 312 suite. The prototype state and this observation are developed once, in §4.8.

(d) PCZT signing. The multiparty transaction flow that would host the Coordinator role is the PCZT of the *Wallet Guide*, §"Partially created transactions (PCZT)": ZIP 374 (Partially Created Zcash Transaction Format, Draft, owner Jack Grigg) names threshold signing ("for example FROST", per RFC 9591) in its Motivation among the multiparty use cases its Signer role must accommodate.

Classification. In the series' three bins: base FROST and its five ciphersuites are *specified* in RFC 9591, an Informational RFC recording CFRG consensus (§2.8). Re-Randomized FROST, its threat model, and both Zcash ciphersuites are *specified* by ZIP 312 at Draft status. The preprint's claimed existential-unforgeability proof has the gaps in §3.3, no cited theorem establishes threshold SURK-CMA, and unlinkability rests on a distribution argument rather than a theorem. Key generation — FROST DKG for Zcash, and ZIP 2005's shared-sk agreement — is *designed but unspecified*; threshold custody of Crosslink finalizer keys is an *open problem*. Implementation-behaviour claims (seed-based randomizer derivation, even-Y normalisation, the extra DKG personalisations) are implementation facts of the pinned commits, cited as such.

4 Deployment and open questions

The preceding sections dealt in designs and proofs; this one deals in what ships. We pin the exact code paths available to a prospective Zcash threshold-custody deployment, reconcile them against the draft ZIP text where the two have diverged, then walk the three custody surfaces this volume targets — spend authorisation through PCZT, the ZIP-2005 quantum spending key, and Crosslink finalizer keys — and close by sorting every artefact into the series' three bins (*specified*, *designed-but-unspecified*, *open problem*), each with the concrete trigger on which its classification must be revisited.

4.1 The shipped library stack

Two Zcash Foundation repositories carry the implementation. The `frost` workspace (commit `2016e44ba4`, 2026-04-23, tagged `frost-core/v3.0.0`; workspace version 3.0.0, MSRV 1.81) provides the generic `frost-core`, the five RFC 9591 ciphersuite crates `frost-ristretto255`, `-ed25519`, `-ed448`, `-p256`, and `-secp256k1`, the separate `frost-secp256k1-tr` Taproot suite, and the generic re-randomisation layer `frost-rerandomized`. Beyond RFC 9591's two-round protocol the workspace implements trusted-dealer key generation (the RFC's appendix), a variant of the original paper's Pedersen-with-knowledge-proof DKG that omits its context string (ePrint 2020/852, §2.2 — not in the RFC), Repairable-Threshold-Scheme share recovery (ePrint 2017/1155), share refresh, and Re-Randomized FROST (ePrint 2024/436, §3). The README declares the crates "considered stable and feature complete", with SemVer guarantees.

The `frost-rerandomized` crate is `no_std`; its `Randomize` implementations add the randomizer group element to every verifying share of a `PublicKeyPackage` and the randomizer scalar to the signing share of a `KeyPackage`, whose `randomize` documentation warns: "You MUST NOT reuse the randomized key package for more than one signing." Its README notes that "the main ciphersuite crates do not re-expose the rerandomization functions... The exceptions are the Zcash ciphersuites in `reddsa` which do expose the randomized functionality" — the Zcash ciphersuite modules are the shipped top-level suites that re-export those generic functions.

The `reddsa` crate (commit `3792daa95e`, 2026-05-22, version 0.5.2) supplies those ciphersuites: its feature `frost = ["frost-rerandomized", "alloc"]` pulls `frost-rerandomized` 3.0.0, and its modules `reddsa::frost::redpallas` and `reddsa::frost::redjubjub` implement the two ZIP-312 ciphersuites (`PallasBlake2b512`, `JubjubBlake2b512`), each with a `keys` module (trusted-dealer, DKG, repairable) and a `rerandomized` submodule re-exporting `sign_with_randomizer_seed`, `aggregate`, `aggregate_custom`, `Randomizer`, and `RandomizedParams`. Function `hash_randomizer` instantiates ZIP 312's H_R as $H^{\circledast}$ (the code's `HStar`) under the personalisation `FROST_RedPallasA` (respectively `FROST_RedJubjubA`), matching §3.5. The crate's README calls this feature experimental because ZIP 312 is still Draft and states that neither its SemVer nor its MSRV guarantee applies to the feature.

The RedPallas module additionally enforces the even-y requirement that ZIP 2005's amended protocol-spec §4.2.3 places on `ak` (§4.7): trait `keys::EvenY` negates the group verifying key, every verifying share, the signing shares, and the VSS commitment coefficients whenever the serialised key has $\tilde{y} = 1$ (the code's evenness test is `serialized[31] & 0x80 == 0`), and the `post_generate` and `post_dkg` ciphersuite hooks call `into_even_y` on fresh key material, so trusted-dealer and DKG outputs alike land on the even-y representative. Negation commutes with Shamir sharing because it is linear. The RedJubjub module carries no such machinery; the constraint is Pallas/Orchard-specific.

4.2 The randomizer flow: draft ZIP against shipped crate

ZIP 312 ("FROST for Spend Authorization Multisignatures"; owners Gouvea, Komlo, Connolly; Status Draft; created 2022-08) specifies, at zips commit `6961098410`, the randomizer generation

$$\texttt{signing_package_enc} = \texttt{encode_signing_package(commitment_list,msg)},$$
$$\texttt{randomizer} = H_R(\texttt{random_bytes}(N_s) \,\|\, \texttt{signing_package_enc}),$$

with H_R for FROST(Pallas, BLAKE2b-512) being BLAKE2b-512 under the personalisation `FROST_RedPallasA`, output reduced to a Pallas scalar (an element of $\mathbb{F}_{p_{\mathsf{Vesta}}}$); H_2 is `Zcash_RedPallasH`, i.e. exactly RedPallas's $H^{\circledast}$ (§3.5). For Jubjub the personalisations are `FROST_RedJubjub{R,N,M,C,A}` and `Zcash_RedJubjubH`, with base point G^{Sapling}. Round 2 then proceeds as plain FROST with three substitutions,

$$sk_i \leftarrow sk_i + \texttt{randomizer}, \qquad PK_i \leftarrow PK_i + \texttt{ScalarBaseMult(randomizer)},$$
$$group_public_key \leftarrow group_public_key + \texttt{ScalarBaseMult(randomizer)},$$

the second applying in share verification. The Coordinator sends the randomizer over a "confidential and authenticated channel" — adding confidentiality to plain FROST's reliable, authenticated-channel assumption, because the randomizer links the ceremony to the on-chain rk.

The released crate differs from this draft text. Crate `frost-rerandomized` 3.0.0 deprecates `Randomizer::new(rng, signing_package)` — the earlier, message-inclusive API. Even that API is not byte-for-byte the draft flow: it samples and serialises a field scalar before hashing, whereas the ZIP starts with N_s unconstrained random bytes. The recommended API is instead `Randomizer::new_from_commitments`: the Coordinator draws an N_s-byte `randomizer_seed`, sends the *seed*, and each participant regenerates

```
randomizer = hash_randomizer(seed‖encode_group_commitments(commitments))
```

locally via `regenerate_from_seed_and_commitments`. The message is no longer an input. The crate documentation says that hashing the complete commitment map hedges the Coordinator's RNG because the map includes signer commitments. This is its implementation rationale, not a property proved by the cited RFC, ZIP, or preprint; in particular, the Coordinator receives the commitments before generating the seed and remains trusted for unlinkability. In the paper's terms (ePrint 2024/436, Definition 5 and Remark 1), the commitment encoding plays the role of the optional contributory string aux, which models "the transcript of some external protocol"; the paper's own §7 implementation notes describe the older design, with the message still included, in the same terms.

The gap between draft and crate is being closed in zips PR #895 ("[ZIP 312] Specify key generation, change randomizer handling"; opened August 2024, 21 commits, still open and mergeable as of 2026-09-05, with its latest review comments dated 2026-08-28). It adds a key-generation specification, revises the randomizer handling (the randomizer "couldn't possibly be generated using the message as an input"), integrates the ZIP-2005 qsk material, renames the scheme consistently to "Re-Randomized FROST", and adds Daira-Emma Hopwood as a ZIP owner. Its commit history also retires the draft's serialisation reference — the ZF FROST book's moving-target serialisation-format page — in favour of RFC 9591's serialisation function. Until it merges, this volume presents the crate flow — message omission included — as the observed fact of the pinned commit and the ZIP text as a divergent Draft specification. No conformance conclusion follows while that text remains in flight.

The trust geometry is worth stating exactly. The paper's §5.1 holds that the central randomizer-choosing party is "trusted with preserving privacy of the scheme" but "not trusted with security", and that removing it would require a DKG-style contributory sub-protocol at extra rounds and complexity. ZIP 312's threat model matches: the Coordinator is trusted with transaction privacy and unlinkability, while the requirements say it must not be able to forge without `MIN_PARTICIPANTS` approvals; every key-share holder is trusted with transaction privacy. Non-requirements are equally explicit: ZIP 312 does not remove the Coordinator role, does not (at this commit) provide key generation, and places network privacy out of scope.

4.3 PCZT as the integration surface

ZIP 374's Motivation names "multiparty transactions — threshold signing (for example FROST [RFC 9591])" among the problems the PCZT format solves; the *Wallet Guide*, §"Partially created transactions (PCZT)", documents the format and its role graph. The insertion point for FROST is the Signer role (*Wallet Guide*, §"Signing"): to sign an Orchard spend the action's re-randomisation scalar α must be present, the local-key path computes $\mathsf{rsk} := \mathsf{ask} + \alpha$ — the re-randomisation of the *Ironwood Guide*, §"Randomised validating keys" — and checks $\mathsf{VerificationKey}(\mathsf{rsk}) = \mathsf{rk}$, and the *external* path, `apply_orchard_signature`, accepts a signature if and only if it verifies under the action's rk over the shielded sighash (`InvalidExternalSignature` otherwise). A FROST-aggregated RedPallas signature is exactly such an external signature;

for it to verify, the effective FROST randomizer must equal the PCZT action's α, so that the re-randomised group key RandPK produces equals that action's rk.

PCZT v2 has since shipped in pczt 0.8.0. Current pczt 0.9.3 carries separate Orchard and Ironwood bundles and tags an externally produced `SpendAuthSignature` with its value pool and action index; `apply_orchard_spend_auth_signature` dispatches to the corresponding bundle and verifies under that action's rk. The invariant in the preceding paragraph — one shared shielded sighash and the action's own α — is unchanged for v6 Ironwood spends (librustzcash `91f448b5`, `pczt/src/roles/signer/mod.rs`).

The end-to-end flow has been demonstrated (ZF FROST book, `zcash/devtool-demo.md`):

1. `zcash-devtool pczt create` produces the PCZT;
2. the `zcash-sign` tool (from ZcashFoundation/frost-zcash-demo, now `frost-tools`) prints a SIGHASH and a `Randomizer` from the transaction plan;
3. `frost-client` runs the FROST protocol over `frostd` with that SIGHASH as the message and that randomizer;
4. `zcash-sign sign` writes the aggregated signature into `frost_pczt.signed`;
5. `zcash-devtool pczt combine` merges the signed and proven PCZTs;
6. `zcash-devtool pczt send` broadcasts.

The current `frost-tools` head (`06c0dbdb`, 2026-08-12) does not exercise the version-3 seed flow described in §4.2. Its lockfile selects `frost-core` 2.2.0, `frost-rerandomized` 2.1.0, and an older `reddsa` 0.5.1 revision. For PCZT, the Coordinator accepts each Action's already-generated scalar randomizer and sends that value to participants through the legacy raw-randomizer API. This is a dated implementation boundary: the demo validates the PCZT external-signature path, but it is not evidence for integration of the pinned version-3 library stack or its seed flow.

That distinction creates an ordering requirement for any version-3 integration. The recommended seed API derives the effective randomizer from round-one commitments, so it cannot in general be made equal to an α already fixed inside a PCZT Action. The transaction builder must instead use the derived randomizer as that Action's α before constructing its proof and rk, or use a separately justified compatibility path. Neither the legacy demo nor a specification cited here demonstrates that reordered end-to-end flow.

Here `zcash-devtool` builds on `zcash_keys` with the features `"orchard"`, `"unstable"`, and `"unstable-frost"`. The `unstable-frost` feature (= `["orchard"]`) exposes the constructor `UnifiedFullViewingKey::from_orchard_fvk` — a UFVK from an Orchard FVK alone, since a FROST wallet has no unified spending key — and the `zcash_keys` changelog (0.3.0, 2024-08-19) frames the flag as existing "to temporarily expose API features that are needed specifically when creating FROST threshold signatures", to be removed "once key derivation for FROST has been fully specified and implemented". Repository `librustzcash` itself, that is, classifies FROST key derivation as not fully specified.

One check remains outside every specification: ZIP 312 requires that signers "MUST check that the given SIGHASH matches the data sent from the Coordinator, or compute the SIGHASH themselves", and explicitly leaves the mechanism out of scope. In the demonstrated flow the SIGHASH travels from `zcash-sign` to the participants through `frost-client`; by what mechanism a share-holding participant gains independent confidence in it is unspecified.

4.4 Wallet-integration constraints

The ZF FROST book's technical-details chapter records the constraints a FROST-capable Zcash wallet inherits. The documented Zcash integration cannot authorise transparent inputs, which require ECDSA; the book accordingly suggests supporting Orchard only ("Orchard is simpler to

handle"). That is historical integration advice rather than a current-pool limit: the page predates the Ironwood pool. With a DKG-generated ak, the remaining Orchard key components derive from ak plus either an sk used only for nk and rivk, or independently generated nk and rivk. Under this key-generation flow, a seed phrase alone does not restore the wallet: recovery also needs the key share, all verifying shares, the participant identifiers, and the FVK components. The book stores these in participant configuration files; the library also provides the Repairable Threshold Scheme (§4.1).

The Foundation's status report ("The State of FROST for Zcash", 2025-05-20) named wallet integration the missing piece ("The missing piece is for wallets to integrate FROST"): at publication time it reported no mainstream wallet integration, while "technical-inclined users can use FROST with Zcash today using our `frost-client` command line tool". The same post lists the "lack of a standardized key generation specification for FROST" as the interoperability limiter, flags metadata protection (e.g. Tor) as needing work, and offers ZF as a candidate operator of a production `frostd`.

4.5 Transport: frostd and frost-client

The reference transport is `frostd`, a JSON-over-HTTPS session server: clients authenticate with key pairs, the Coordinator creates sessions naming the participants' public keys, and all FROST messages are end-to-end encrypted. Thus the server need not be trusted with message contents or signing authority, although it remains an availability dependency. The CLI `frost-client` drives it; both live in ZcashFoundation/frost-tools (renamed from `frost-zcash-demo`). Least Authority, as Zcash Ecosystem Security Lead, audited `frost-client` and `frostd` (announcement 2025-05-01) with no high-severity findings, all findings addressed on `main`. No ZIP specifies this transport; ZIP 312 places network privacy out of scope (§4.2). The tools repository labels the command-line programs as demos and says they should not be used in production. The `frost-client` README also warns that it stores secrets unencrypted in its configuration file and recommends protecting that file or building a separate tool from the client as a base.

4.6 Audit coverage, precisely

Two cryptographic-library audits are relevant, and neither covers the code path Zcash custody would run. NCC Group audited the v0.6.0 release (commit `5fa17ed`) of `frost-core`, `frost-ed25519`, `frost-ed448`, `frost-p256`, `frost-secp256k1`, and `frost-ristretto255` — including trusted-dealer and DKG key generation and FROST signing — with all findings addressed and reviewed by NCC (public report 2023-10-23). The repository README is explicit: "This does not include `frost-secp256k1-tr` and rerandomized FROST." Earlier, the 2021 "Audit of FROST implementation" (JP Aumasson and Adrien Hamelink, Taurus; Omer Shlomovits, ZenGo; dated 2021-03-23) covered the pre-rewrite implementation in the RedJubjub repository's `src/frost.rs` (404 source lines by the report's count, at commit `2ebc08f`, 2021-02-25) — two-round FROST with a trusted dealer only, RedJubjub only — concluding "We did not find any major security issue." The report also records that the implementation omitted the paper's secure deletion of nonces and commitments after signing.

The composition, therefore: `frost-rerandomized` plus the `reddsa` RedPallas/RedJubjub ciphersuites — the exact stack of §4.1 — is not covered by any completed third-party audit identified in the repository and audit announcements rechecked 2026-09-05. Of those audits, only the 2021 review touched a Zcash FROST ciphersuite, and it predates the `frost-core` rewrite. The separate Least Authority audit covers the transport tools, not this cryptographic composition.

4.7 Custody of the ZIP-2005 quantum spending key

ZIP 2005 ("Ironwood Quantum Recoverability"; owners Daira-Emma Hopwood and Jack Grigg; Status Proposed, Category Consensus; activation at the NU6.3 activation height) is the second

custody surface. Its "Usage with FROST" section derives ak jointly, from the participants' shares of ask, via FROST DKG. It then says that participants "MUST privately agree on a value sk" and use it with use_qsk = true to derive the remaining material. The protocol must give every participant the same extracted validating-key value ak, nk, and rivk_ext; the specified derivations then imply equal qsk and qk under the ZIP's stated collision-resistance assumptions. In this series, $\mathsf{ak} = [\mathsf{ask}]G^{\mathsf{SpendAuth}}$ denotes the Pallas point. ZIP 2005 writes $\mathsf{ak}^{\mathbb{P}}$ for that point and $\mathsf{ak} := \mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}})$ for its extracted field element. With that point/field-element distinction explicit, its derivations are

$$\begin{aligned}
\mathsf{H}^{\mathsf{qsk}}(\mathsf{sk}) &= \mathrm{truncate}_{32}(\mathsf{PRFexpand}_{\mathsf{sk}}(\texttt{[0x0C]})),\\
\mathsf{H}^{\mathsf{qk}}(\mathsf{qsk}) &= \mathrm{BLAKE3.derive_key}(\texttt{"Zcash ZIP 2005 qk-derivation v1"}, \mathsf{qsk}, 32),\\
\mathsf{H}^{\mathsf{rivk_ext}}_{\mathsf{qk}}(\mathsf{ak}, \mathsf{nk}) &= \mathrm{ToScalar}^{\mathsf{Orchard}}(\mathsf{PRFexpand}_{\mathsf{qk}}(\texttt{[0x0D]} \,\|\, \mathrm{I2LEOSP}_{256}(\mathsf{ak}) \,\|\, \mathrm{I2LEOSP}_{256}(\mathsf{nk}))).
\end{aligned}$$

The displayed key derivations use the PRFexpand domain-separator bytes 0x0C and 0x0D; ZIP 2005 also reserves 0x0A and 0x0B for note-related uses. In the use_qsk = true branch of the amended protocol-spec §4.2.3, the wallet obtains the point $\mathsf{ak}^{\mathbb{P}}$ "by any suitably secure method that ensures the last bit of $\mathrm{repr}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}})$ is 0" — the requirement the reddsa EvenY machinery of §4.1 enforces — and sets the ZIP's field element $\mathsf{ak} = \mathsf{Extract}_{\mathbb{P}}(\mathsf{ak}^{\mathbb{P}})$, $\mathsf{qsk} = \mathsf{H}^{\mathsf{qsk}}(\mathsf{sk})$, $\mathsf{qk} = \mathsf{H}^{\mathsf{qk}}(\mathsf{qsk})$, and $\mathsf{rivk} = \mathsf{H}^{\mathsf{rivk_ext}}_{\mathsf{qk}}(\mathsf{ak}, \mathsf{nk})$. The ZIP's list of admissible generation methods for $\mathsf{ak}^{\mathbb{P}}$ explicitly includes "FROST distributed key generation [zip-0312-key-generation] in which the corresponding spend-authorization material is *t*-of-*n* shared among the participants".

The threat model shifts accordingly. The host wallet in a multi-signer FROST setup is the ZIP-312 Coordinator; the ZIP gives it nk, the extracted validating-key value ak, and qk. With FROST and hardware wallets, spend authorisation breaks only by (1) possession of qsk *plus* finding discrete logarithms on Pallas — and "the adversary is an authorized FROST signer — they hold a copy of qsk by construction" is the ZIP's variant 1.a — or (2) breaking RedDSA, FROST, the DKG, or the proving system. Each participant MUST treat qsk as a secret held at least as securely and reliably as ask: it is not needed until the Recovery Protocol, but losing it can lose funds once the current Orchard protocol is disabled. A quantum adversary holding qsk — which every participant holds — and able to solve discrete logarithms on Pallas may steal funds, so the ZIP RECOMMENDS that once a fully post-quantum threshold multisignature protocol exists, all FROST-held funds migrate to its pool. FROST's key advantage — signing with shares without reconstructing ask during signing — would be retained in the outlined Recovery Protocol, which would still require a RedDSA signature.

One tension is unresolved. FROST DKG exists precisely to avoid a commonly-known secret, yet the ceremony above requires the participants to "privately agree on a value sk" from which nk and qsk derive — and neither ZIP 2005 nor the crates give a protocol for that agreement. ZIP 2005's Deployment section points at PR #895 ("There is no significant existing deployment of FROST, so we can write this into ZIP 312 from the start"), so the resolution is expected there: as of 2026-09-05 the PR's commit history drafts a contributory sk-generation section (2026-03-05), still unmerged, so the shared-sk agreement stays *designed but unspecified* with #895's merge as its trigger.

4.8 Crosslink finalizer keys

The third surface is prospective only. The *Crosslink Guide*, §"Finalizer keys", records that the prototype uses single ed25519 keys per finalizer (ed25519-zebra), with 76-byte votes signed by appending a 64-byte ed25519 signature over those 76 bytes, and that no FROST or threshold-signing design for finalizer keys is sourced anywhere in the Crosslink corpus (negative search over tfl-book, crosslink-protocol-guide, crosslink-spec, and zebra-crosslink, dated 2026-07-15 in that guide); key rotation, revocation, threshold custody, and initial-roster formation are classified open problems in every design source.

We record one observation of our own, clearly marked as such: finalizer votes are plain ed25519, with no key re-randomisation, so the fitting ciphersuite would be RFC 9591's FROST(Ed25519, SHA-512) via the `frost-ed25519` crate, whose v0.6.0 predecessor was covered by the NCC audit, with no re-randomisation layer — finalizer keys are deliberately long-lived roster identities, not unlinkable one-time keys, so the entire apparatus of §3 is unnecessary there. No published design says this; it is this volume's inference from the vote format and the audit table.

4.9 Classification and revisit triggers

Per the series' three-way scheme:

- *Specified.* FROST as in RFC 9591 (§2.8); Re-Randomized FROST for spend-authorisation custody, per ZIP 312 at Draft status — PR #895's in-flight randomizer and key-generation changes are its named revisit trigger and do not move the bin (§4.2); the `frost-core`/`frost-rerandomized`/`reddsa` API surface at the pinned versions (§4.1); the PCZT Signer external-signature path (§4.3).
- *Designed-but-unspecified.* FROST key generation for Zcash (out of scope at the pinned ZIP-312 commit; the crates implement the PedPoP DKG of ePrint 2020/852, but no ZIP specifies it); the ZIP-2005 use_qsk FROST key-generation constraints (ZIP Proposed, referencing the unmerged #895; the sk-agreement step has no protocol, §4.7); the transport (`frostd` exists and is audited, but no ZIP covers it, §4.5).
- *Open problem.* Crosslink finalizer threshold custody (§4.8); post-quantum threshold signing for the qsk-era migration (§4.7); hardware-wallet FROST share custody — ZIP 2005's threat model assumes hardware-wallet FROST signers, but none was identified in the device integrations checked for this audit.

Two former revisit triggers have fired without changing these bins. NU6.3 is active, but `zcash_keys` still gates `from_orchard_fvk` behind `unstable-frost` pending a specified FROST derivation. PCZT v2 is released, and the current signer preserves the α and external-signature surface for Ironwood as described in §4.3. The remaining classifications expire on the following identifiable events, in rough order of expected impact:

- *PR #895 merges.* The randomizer flow, the key-generation specification, the sk-agreement protocol, and the ZIP-312/ZIP-2005 integration all re-derive from the merged text; the serialisation reference it pins replaces the ZF-FROST-book moving target (§4.2).
- *An audit of the re-randomised stack is announced.* The "no completed third-party audit" claim of §4.6 must be rechecked immediately before any printing in any case.
- *A finalizer key-management design is published.* The Crosslink bin moves, and the `frost-ed25519` observation is either confirmed or discarded (§4.8).
- *A post-quantum threshold multisignature protocol exists.* ZIP 2005's RECOMMENDED migration of FROST-held funds becomes actionable (§4.7).